

Econometric Modelling with Time Series

Specification, Estimation and Testing

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Preface

This book provides a general framework for specifying, estimating and testing time series econometric models. Special emphasis is given to estimation by maximum likelihood, but other methods are also discussed including quasi-maximum likelihood estimation, generalized method of moments, nonparametrics and estimation by simulation. An important advantage of adopting the principle of maximum likelihood as the unifying framework for the book is that many of the estimators and test statistics proposed in econometrics can be derived within a likelihood framework, thereby providing a coherent vehicle for understanding their properties and interrelationships.

In contrast to many existing econometric textbooks, which deal mainly with the theoretical properties of estimators and test statistics through a theorem-proof presentation, this book is very concerned with implementation issues in order to provide a fast-track between the theory and applied work. Consequently many of the econometric methods discussed in the book are illustrated by means of a suite of programs written in GAUSS and MATLAB®.¹ The computer code emphasizes the computational side of econometrics and follows the notation in the book as closely as possible, thereby reinforcing the principles presented in the text. More generally, the computer code also helps to bridge the gap between theory and practice by enabling the reproduction of both theoretical and empirical results published in recent journal articles. The reader, as a result, may build on the code and tailor it to more involved applications.

Organization of the Book

Part ONE of the book is an exposition of the basic maximum likelihood framework. To implement this approach, three conditions are required: the probability distribution of the stochastic process must be known and specified correctly, the parametric specifications of the moments of the distribution must be known and specified correctly, and the likelihood must be tractable. The properties of maximum likelihood estimators are presented and three fundamental testing procedures – namely, the Likelihood Ratio test, the Wald test and the Lagrange Multiplier test – are discussed in detail. There is also a comprehensive treatment of iterative algorithms to compute maximum likelihood estimators when no analytical expressions are available.

Part TWO is the usual regression framework taught in standard econometric courses but presented within the maximum likelihood framework.

¹ GAUSS is a registered trademark of Aptech Systems, Inc. <http://www.aptech.com/> and MATLAB® is a registered trademark of The MathWorks, Inc. <http://www.mathworks.com/>.

Both nonlinear regression models and non-spherical models exhibiting either autocorrelation or heteroskedasticity, or both, are presented. A further advantage of the maximum likelihood strategy is that it provides a mechanism for deriving new estimators and new test statistics, which are designed specifically for non-standard problems.

Part THREE provides a coherent treatment of a number of alternative estimation procedures which are applicable when the conditions to implement maximum likelihood estimation are not satisfied. For the case where the probability distribution is incorrectly specified, quasi-maximum likelihood is appropriate. If the joint probability distribution of the data is treated as unknown, then a generalized method of moments estimator is adopted. This estimator has the advantage of circumventing the need to specify the distribution and hence avoids any potential misspecification from an incorrect choice of the distribution. An even less restrictive approach is not to specify either the distribution or the parametric form of the moments of the distribution and use nonparametric procedures to model either the distribution of variables or the relationships between variables. Simulation estimation methods are used for models where the likelihood is intractable arising, for example, from the presence of latent variables. Indirect inference, efficient methods of moments and simulated methods of moments are presented and compared.

Part FOUR examines stationary time series models with a special emphasis on using maximum likelihood methods to estimate and test these models. Both single equation models, including the autoregressive moving average class of models, and multiple equation models, including vector autoregressions and structural vector autoregressions, are dealt with in detail. Also discussed are linear factor models where the factors are treated as latent. The presence of the latent factor means that the full likelihood is generally not tractable. However, if the models are specified in terms of the normal distribution with moments based on linear parametric representations, a Kalman filter is used to rewrite the likelihood in terms of the observable variables thereby making estimation and testing by maximum likelihood feasible.

Part FIVE focusses on nonstationary time series models and in particular tests for unit roots and cointegration. Some important asymptotic results for nonstationary time series are presented followed by a comprehensive discussion of testing for unit roots. Cointegration is tackled from the perspective that the well-known Johansen estimator may be usefully interpreted as a maximum likelihood estimator based on the assumption of a normal distribution applied to a system of equations that is subject to a set of

cross-equation restrictions arising from the assumption of common long-run relationships. Further, the trace and maximum eigenvalue tests of cointegration are shown to be likelihood ratio tests.

Part SIX is concerned with nonlinear time series models. Models that are nonlinear in mean include the threshold class of model, bilinear models and also artificial neural network modelling, which, contrary to many existing treatments, is again addressed from the econometric perspective of estimation and testing based on maximum likelihood methods. Nonlinearities in variance are dealt with in terms of the GARCH class of models. The final chapter focusses on models that deal with discrete or truncated time series data.

Even in a project of this size and scope, sacrifices have had to be made to keep the length of the book manageable. Accordingly, there are a number of important topics that have had to be omitted.

- (i) Although Bayesian methods are increasingly being used in many areas of statistics and econometrics, no material on Bayesian econometrics is included. This is an important field in its own right and the interested reader is referred to recent books by Koop (2003), Geweke (2005), Koop, Poirier and Tobias (2007) and Greenberg (2008), *inter alia*. Where appropriate, references to Bayesian methods are provided in the body of the text.
- (ii) With great reluctance a chapter on bootstrapping was not included because of space issues. A good place to start reading is the introductory text by Efron and Tibshirani (1993) and the useful surveys by Horowitz (1997) and Li and Maddala (1996b, 1996a).
- (iii) In Part SIX, in the chapter dealing with modelling the variance of time series, there are important recent developments in stochastic volatility and realized volatility that would be worthy of inclusion. For stochastic volatility, there is an excellent volume of readings edited by Shephard (2005), while the seminal articles in the area of realized volatility are Anderson et al. (2001, 2003).

The fact that these areas have not been covered should not be regarded as a value judgement about their relative importance. Instead the subject matter chosen for inclusion reflects a balance between the interests of the authors and purely operational decisions aimed at preserving the flow and continuity of the book.

Computer Code

Specifically, computer code is available from a companion website to reproduce relevant examples in the text, to reproduce figures in the text that are not part of an example, to reproduce the applications presented in the final section of each chapter, and to complete the exercises. Where applicable, the time series data used in these examples, applications and exercises are also available in a number of different formats.

Presenting numerical results in the examples immediately gives rise to two important issues concerning numerical precision.

- (1) In all of the examples listed in the front of the book where computer code has been used, the numbers appearing in the text are rounded versions of those generated by the code. Accordingly, the rounded numbers should be interpreted as such and **not** be used independently of the computer code to try and reproduce the numbers reported in the text.
- (2) In many of the examples, simulation has been used to demonstrate a concept. Since **GAUSS** and **MATLAB** have different random number generators, the results generated by the different sets of code will not be identical to one another. For consistency we have always used the **GAUSS** output for reporting purposes.

Although **GAUSS** and **MATLAB** are very similar high-level programming languages, there are some important differences that require explanation. Probably the most important difference is one of programming style. **GAUSS** programs are script files that allow calls to both inbuilt **GAUSS** and user-defined *procedures*. **MATLAB**, on the other hand, does not support the use of user-defined *functions* in script files. Furthermore, **MATLAB** programming style favours writing user-defined functions in separate files and then calling them as if they were in-built functions. This style of programming does not suit the learning-by-doing environment that the book tries to create. Consequently, the **MATLAB** programs are written mainly as function files with a main function and all the required user-defined functions required to implement the procedure in the same file. The only exception to this rule is that a few **MATLAB** utility files, which greatly facilitate the conversion and interpretation of code from **GAUSS** to **MATLAB**, which are provided as separate stand-alone **MATLAB** function files. Finally, all the figures in the text were created using **MATLAB** together with a utility file `laprint.m` written by Arno Linnemann of the University of Kassel.²

² A user guide is available at
<http://www.uni-kassel.de/fb16/rat/matlab/laprint/laprintdoc.ps>.

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Vance Martin, Stan Hurn & David Harris
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Code Disclaimer Information

Note that the computer code is provided for illustrative purposes only and although care has been taken to ensure that it works properly, it has not been thoroughly tested under all conditions and on all platforms. The authors and Cambridge University Press cannot guarantee or imply reliability, serviceability, or function of this computer code. All code is therefore provided ‘as is’ without any warranties of any kind.

PART ONE
MAXIMUM LIKELIHOOD

1

The Maximum Likelihood Principle

1.1 Introduction

Maximum likelihood estimation is a general method for estimating the parameters of econometric models from observed data. The principle of maximum likelihood plays a central role in the exposition of this book, since a number of estimators used in econometrics can be derived within this framework. Examples include ordinary least squares, generalized least squares and full-information maximum likelihood. In deriving the maximum likelihood estimator, a key concept is the joint probability density function (pdf) of the observed random variables, y_t . Maximum likelihood estimation requires that the following conditions are satisfied.

- (1) The form of the joint pdf of y_t is known.
- (2) The specification of the moments of the joint pdf are known.
- (3) The joint pdf can be evaluated for all values of the parameters, θ .

Parts ONE and TWO of this book deal with models in which all these conditions are satisfied. Part THREE investigates models in which these conditions are not satisfied and considers four important cases. First, if the distribution of y_t is misspecified, resulting in both conditions 1 and 2 being violated, estimation is by quasi-maximum likelihood (Chapter 9). Second, if condition 1 is not satisfied, a generalized method of moments estimator (Chapter 10) is required. Third, if condition 2 is not satisfied, estimation relies on nonparametric methods (Chapter 11). Fourth, if condition 3 is violated, simulation-based estimation methods are used (Chapter 12).

1.2 Motivating Examples

To highlight the role of probability distributions in maximum likelihood estimation, this section emphasizes the link between observed sample data and

the probability distribution from which they are drawn. This relationship is illustrated with a number of simulation examples where samples of size $T = 5$ are drawn from a range of alternative models. The realizations of these draws for each model are listed in Table 1.1.

Table 1.1
Realisations of y_t from alternative models: $t = 1, 2, \dots, 5$.

Model	t=1	t=2	t=3	t=4	t=5
Time Invariant	-2.720	2.470	0.495	0.597	-0.960
Count	2.000	4.000	3.000	4.000	0.000
Linear Regression	2.850	3.105	5.693	8.101	10.387
Exponential Regression	0.874	8.284	0.507	3.722	5.865
Autoregressive	0.000	-1.031	-0.283	-1.323	-2.195
Bilinear	0.000	-2.721	0.531	1.350	-2.451
ARCH	0.000	3.558	6.989	7.925	8.118
Poisson	3.000	10.000	17.000	20.000	23.000

Example 1.1 Time Invariant Model

Consider the model

$$y_t = \sigma z_t,$$

where z_t is a disturbance term and σ is a parameter. Let z_t be a standardized normal distribution, $N(0, 1)$, defined by

$$f(z) = \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{z^2}{2} \right].$$

The distribution of y_t is obtained from the distribution of z_t using the change of variable technique (see Appendix A for details)

$$f(y; \theta) = f(z) \left| \frac{\partial z}{\partial y} \right|,$$

where $\theta = \{\sigma^2\}$. Applying this rule, and recognising that $z = y/\sigma$, yields

$$f(y; \theta) = \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{(y/\sigma)^2}{2} \right] \left| \frac{1}{\sigma} \right| = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{y^2}{2\sigma^2} \right],$$

or $y_t \sim N(0, \sigma^2)$. In this model, the distribution of y_t is time invariant because neither the mean nor the variance depend on time. This property is highlighted in panel (a) of Figure 1.1 where the parameter is $\sigma = 2$. For comparative purposes the distributions of both y_t and z_t are given. As $y_t = 2z_t$, the distribution of y_t is flatter than the distribution of z_t . $\square$

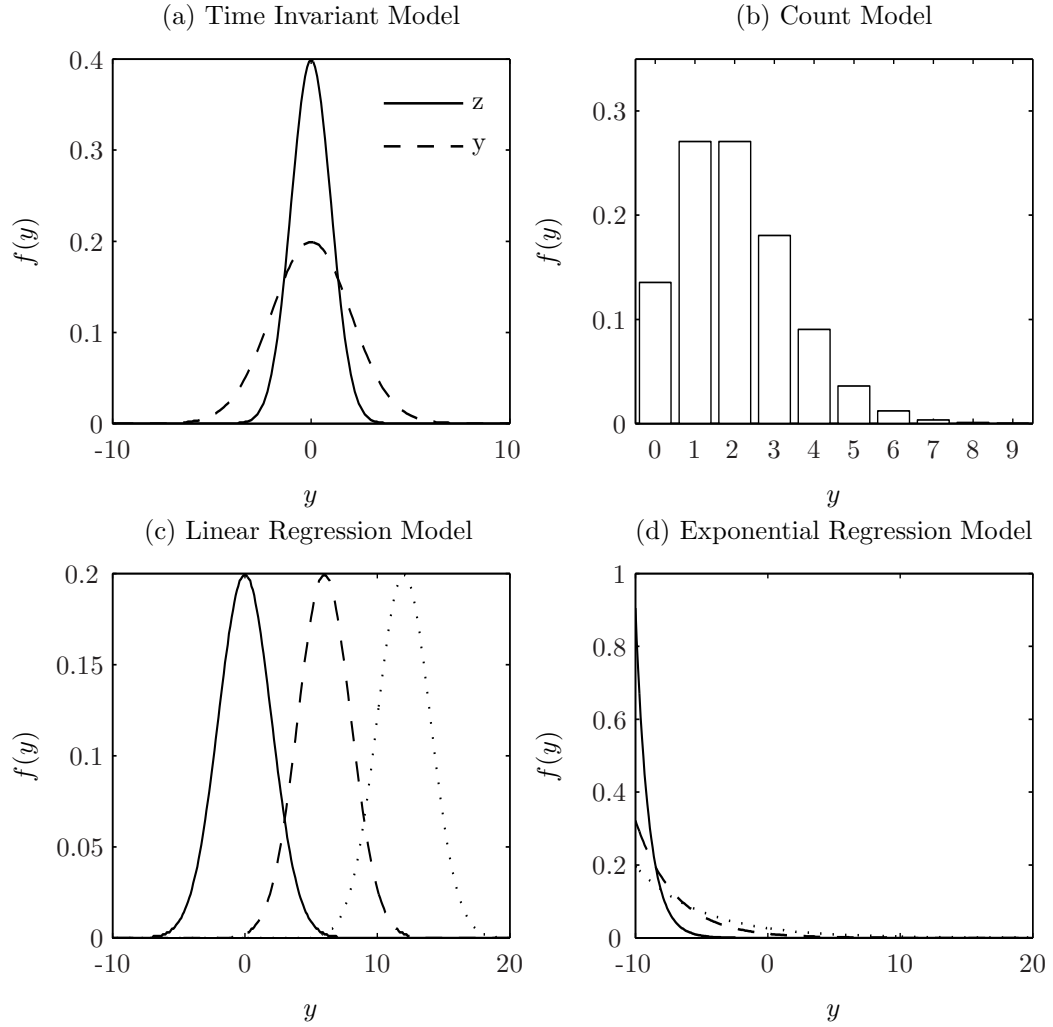


Figure 1.1 Probability distributions of y generated from the time invariant, count, linear regression and exponential regression models. Except for the time invariant and count models, the solid line represents the density at $t = 1$, the dashed line represents the density at $t = 3$ and the dotted line represents the density at $t = 5$.

As the distribution of y_t in Example 1.1 does not depend on lagged values y_{t-i} , y_t is independently distributed. In addition, since the distribution of y_t is the same at each t , y_t is identically distributed. These two properties are abbreviated as *iid*. Conversely, the distribution is dependent if y_t depends on its own lagged values and non-identical if it changes over time.

Example 1.2 Count Model

Consider a time series of counts modelled as a series of draws from a Poisson distribution

$$f(y; \theta) = \frac{\theta^y \exp[-\theta]}{y!}, \quad y = 0, 1, 2, \dots,$$

where $\theta > 0$ is an unknown parameter. A sample of $T = 5$ realizations of y_t , given in Table 1.1, is drawn from the Poisson probability distribution in panel (b) of Figure 1.1 for $\theta = 2$. By assumption, this distribution is the same at each point in time. In contrast to the data in the previous example where the random variable is continuous, the data here are discrete as they are positive integers that measure counts. $\square$

Example 1.3 Linear Regression Model

Consider the regression model

$$y_t = \beta x_t + \sigma z_t, \quad z_t \sim iid N(0, 1),$$

where x_t is an explanatory variable that is independent of z_t and $\theta = \{\beta, \sigma^2\}$. The distribution of y conditional on x_t is

$$f(y | x_t; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y - \beta x_t)^2}{2\sigma^2} \right],$$

which is a normal distribution with conditional mean βx_t and variance σ^2 , or $y_t \sim N(\beta x_t, \sigma^2)$. This distribution is illustrated in panel (c) of Figure 1.1 with $\beta = 3$, $\sigma = 2$ and explanatory variable $x_t = \{0, 1, 2, 3, 4\}$. The effect of x_t is to shift the distribution of y_t over time into the positive region, resulting in the draws of y_t given in Table 1.1 becoming increasingly positive. As the variance at each point in time is constant, the spread of the distributions of y_t is the same for all t . $\square$

Example 1.4 Exponential Regression Model

Consider the exponential regression model

$$f(y | x_t; \theta) = \frac{1}{\mu_t} \exp \left[-\frac{y}{\mu_t} \right],$$

where $\mu_t = \beta_0 + \beta_1 x_t$ is the time-varying conditional mean, x_t is an explanatory variable and $\theta = \{\beta_0, \beta_1\}$. This distribution is highlighted in panel (d) of Figure 1.1 with $\beta_0 = 1$, $\beta_1 = 1$ and $x_t = \{0, 1, 2, 3, 4\}$. As $\beta_1 > 0$, the effect of x_t is to cause the distribution of y_t to become more positively skewed over time. $\square$

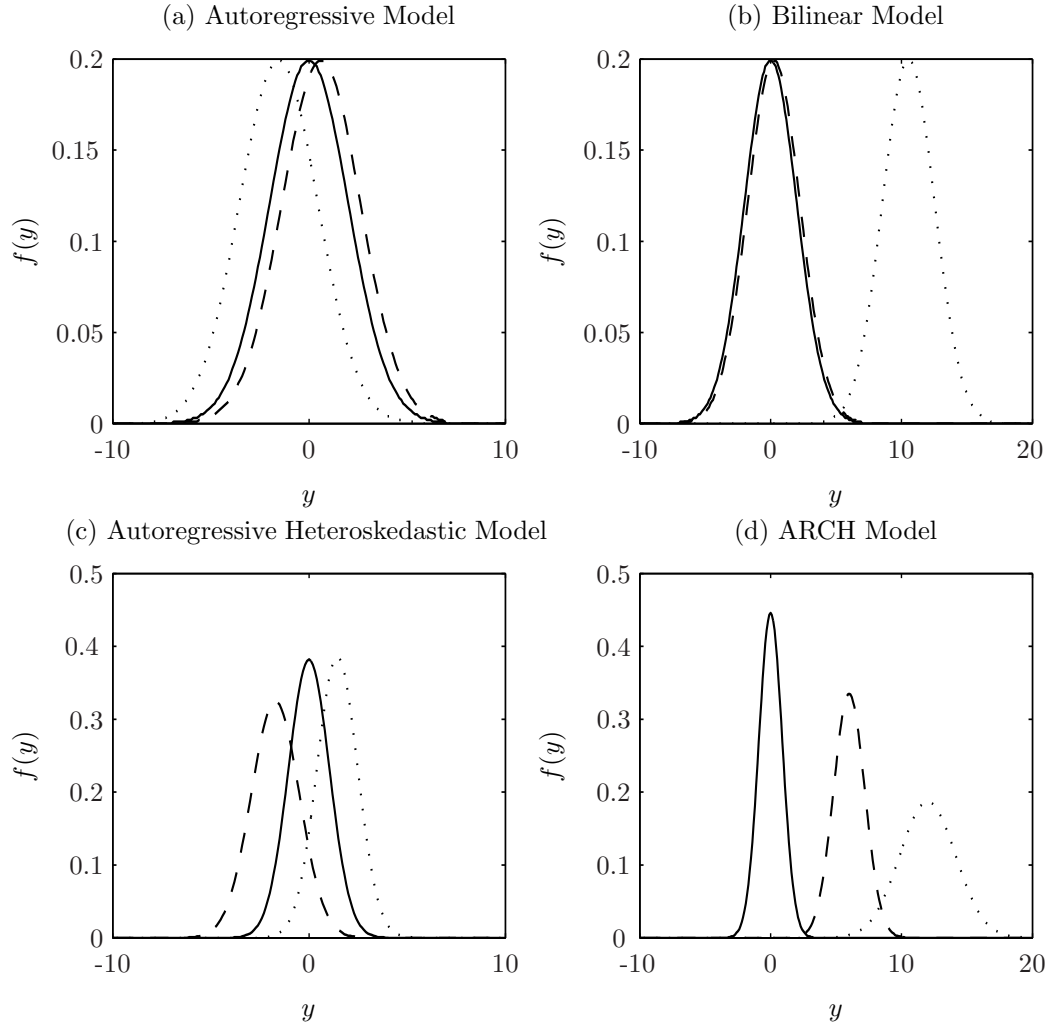


Figure 1.2 Probability distributions of y generated from the autoregressive, bilinear, autoregressive with heteroskedasticity and ARCH models. The solid line represents the density at $t = 1$, the dashed line represents the density at $t = 3$ and the dotted line represents the density at $t = 5$.

Example 1.5 Autoregressive Model

An example of a first-order autoregressive model, denoted $AR(1)$, is

$$y_t = \rho y_{t-1} + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

with $|\rho| < 1$ and $\theta = \{\rho, \sigma^2\}$. The distribution of y , conditional on y_{t-1} , is

$$f(y | y_{t-1}; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y - \rho y_{t-1})^2}{2\sigma^2} \right],$$

which is a normal distribution with conditional mean ρy_{t-1} and variance σ^2 , or $y_t \sim N(\rho y_{t-1}, \sigma^2)$. If $0 < \rho < 1$, then a large positive (negative) value of y_{t-1} shifts the distribution into the positive (negative) region for y_t , raising the probability that the next draw from this distribution is also positive (negative). This property of the autoregressive model is highlighted in panel (a) of Figure 1.2 with $\rho = 0.8$, $\sigma = 2$ and initial value $y_1 = 0$. $\square$

Example 1.6 Bilinear Time Series Model

The autoregressive model discussed above specifies a linear relationship between y_t and y_{t-1} . The following bilinear model is an example of a nonlinear time series model

$$y_t = \rho y_{t-1} + \gamma y_{t-1} u_{t-1} + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

where $y_{t-1} u_{t-1}$ represents the bilinear term and $\theta = \{\rho, \gamma, \sigma^2\}$. The distribution of y_t conditional on y_{t-1} is

$$f(y | y_{t-1}; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y - \mu_t)^2}{2\sigma^2} \right],$$

which is a normal distribution with conditional mean $\mu_t = \rho y_{t-1} + \gamma y_{t-1} u_{t-1}$ and variance σ^2 . To highlight the nonlinear property of the model, substitute out u_{t-1} in the equation for the mean

$$\begin{aligned} \mu_t &= \rho y_{t-1} + \gamma y_{t-1} (y_{t-1} - \rho y_{t-2} - \gamma y_{t-2} u_{t-2}) \\ &= \rho y_{t-1} + \gamma y_{t-1}^2 - \gamma \rho y_{t-1} y_{t-2} - \gamma^2 y_{t-1} y_{t-2} u_{t-2}, \end{aligned}$$

which shows that the mean is a nonlinear function of y_{t-1} . Setting $\gamma = 0$ yields the linear AR(1) model of Example 1.5. The distribution of the bilinear model is illustrated in panel (b) of Figure 1.2 with $\rho = 0.8$, $\gamma = 0.4$, $\sigma = 2$ and initial value $y_1 = 0$. $\square$

Example 1.7 Autoregressive Model with Heteroskedasticity

An example of an AR(1) model with heteroskedasticity is

$$\begin{aligned} y_t &= \rho y_{t-1} + \sigma_t z_t \\ \sigma_t^2 &= \alpha_0 + \alpha_1 w_t \\ z_t &\sim iid N(0, 1), \end{aligned}$$

where $\theta = \{\rho, \alpha_0, \alpha_1\}$ and w_t is an explanatory variable. The distribution

of y_t conditional on y_{t-1} and w_t is

$$f(y | y_{t-1}, w_t; \theta) = \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp \left[-\frac{(y - \rho y_{t-1})^2}{2\sigma_t^2} \right],$$

which is a normal distribution with conditional mean ρy_{t-1} and conditional variance $\alpha_0 + \alpha_1 w_t$. For this model, the distribution shifts because of the dependence on y_{t-1} and the spread of the distribution changes because of w_t . These features are highlighted in panel (c) of Figure 1.2 with $\rho = 0.8$, $\alpha_0 = 0.8$, $\alpha_1 = 0.8$, w_t is defined as a uniform random number on the unit interval and the initial value is $y_1 = 0$. $\square$

Example 1.8 Autoregressive Conditional Heteroskedasticity

The autoregressive conditional heteroskedasticity (ARCH) class of models is a special case of the heteroskedastic regression model where w_t in Example 1.7 is expressed in terms of lagged values of the disturbance term squared. An example of a regression model as in Example 1.3 with ARCH is

$$\begin{aligned} y_t &= \beta x_t + u_t \\ u_t &= \sigma_t z_t \\ \sigma_t^2 &= \alpha_0 + \alpha_1 u_{t-1}^2 \\ z_t &\sim iid N(0, 1), \end{aligned}$$

where x_t is an explanatory variable and $\theta = \{\beta, \alpha_0, \alpha_1\}$. The distribution of y conditional on y_{t-1} , x_t and x_{t-1} is

$$\begin{aligned} f(y | y_{t-1}, x_t, x_{t-1}; \theta) &= \frac{1}{\sqrt{2\pi (\alpha_0 + \alpha_1 (y_{t-1} - \beta x_{t-1})^2)}} \\ &\times \exp \left[-\frac{(y - \beta x_t)^2}{2 (\alpha_0 + \alpha_1 (y_{t-1} - \beta x_{t-1})^2)} \right]. \end{aligned}$$

For this model, a large shock, represented by a large value of u_t , results in an increased variance in the next period if $\alpha_1 > 0$. The distribution from which y_t is drawn in the next period will therefore have a larger variance. The distribution of this model is shown in panel (d) of Figure 1.2 with $\beta = 3$, $\alpha_0 = 0.8$, $\alpha_1 = 0.8$ and $x_t = \{0, 1, 2, 3, 4\}$. $\square$

1.3 Joint Probability Distributions

The motivating examples of the previous section focus on the distribution of y_t at time t which is generally a function of its own lags and the current

and lagged values of explanatory variables x_t . The derivation of the maximum likelihood estimator of the model parameters requires using all of the information $t = 1, 2, \dots, T$ by defining the joint probability density function (pdf). In the case where both y_t and x_t are stochastic, the joint probability pdf for a sample of T observations is

$$f(y_1, y_2, \dots, y_T, x_1, x_2, \dots, x_T; \psi), \quad (1.1)$$

where ψ is a vector of parameters. An important feature of the previous examples is that y_t depends on the explanatory variable x_t . To capture this conditioning, the joint distribution in (1.1) is expressed as

$$\begin{aligned} f(y_1, y_2, \dots, y_T, x_1, x_2, \dots, x_T; \psi) &= f(y_1, y_2, \dots, y_T | x_1, x_2, \dots, x_T; \psi) \\ &\times f(x_1, x_2, \dots, x_T; \psi), \end{aligned} \quad (1.2)$$

where the first term on the right hand side of (1.2) represents the conditional distribution of $\{y_1, y_2, \dots, y_T\}$ on $\{x_1, x_2, \dots, x_T\}$ and the second term is the marginal distribution of $\{x_1, x_2, \dots, x_T\}$. Assuming that the parameter vector ψ can be decomposed into $\{\theta, \theta_x\}$ such that expression (1.2) becomes

$$\begin{aligned} f(y_1, y_2, \dots, y_T, x_1, x_2, \dots, x_T; \psi) &= f(y_1, y_2, \dots, y_T | x_1, x_2, \dots, x_T; \theta) \\ &\times f(x_1, x_2, \dots, x_T; \theta_x). \end{aligned} \quad (1.3)$$

In these circumstances, the maximum likelihood estimation of the parameters θ is based on the conditional distribution without loss of information from the exclusion of the marginal distribution $f(x_1, x_2, \dots, x_T; \theta_x)$.

The conditional distribution on the right hand side of expression (1.3) simplifies further in the presence of additional restrictions.

Independent and identically distributed (*iid*)

In the simplest case, $\{y_1, y_2, \dots, y_T\}$ is independent of $\{x_1, x_2, \dots, x_T\}$ and y_t is *iid* with density function $f(y; \theta)$. The conditional pdf in equation (1.3) is then

$$f(y_1, y_2, \dots, y_T | x_1, x_2, \dots, x_T; \theta) = \prod_{t=1}^T f(y_t; \theta). \quad (1.4)$$

Examples of this case are the time invariant model (Example 1.1) and the count model (Example 1.2).

If both y_t and x_t are *iid* and y_t is dependent on x_t then the decomposition in equation (1.3) implies that inference can be based on

$$f(y_1, y_2, \dots, y_T | x_1, x_2, \dots, x_T; \theta) = \prod_{t=1}^T f(y_t | x_t; \theta). \quad (1.5)$$

Examples include the regression models in Examples 1.3 and 1.4 if sampling is *iid*.

Dependent

Now assume that $\{y_1, y_2, \dots, y_T\}$ depends on its own lags but is independent of the explanatory variable $\{x_1, x_2, \dots, x_T\}$. The joint pdf is expressed as a sequence of conditional distributions where conditioning is based on lags of y_t . By using standard rules of probability the distributions for the first three observations are, respectively,

$$\begin{aligned} f(y_1; \theta) &= f(y_1; \theta) \\ f(y_1, y_2; \theta) &= f(y_2|y_1; \theta)f(y_1; \theta) \\ f(y_1, y_2, y_3; \theta) &= f(y_3|y_2, y_1; \theta)f(y_2|y_1; \theta)f(y_1; \theta), \end{aligned}$$

where y_1 is the initial value with marginal probability density

Extending this sequence to a sample of T observations, yields the joint pdf

$$f(y_1, y_2, \dots, y_T; \theta) = f(y_1; \theta) \prod_{t=2}^T f(y_t|y_{t-1}, y_{t-2}, \dots, y_1; \theta). \quad (1.6)$$

Examples of this general case are the AR model (Example 1.5), the bilinear model (Example 1.6) and the ARCH model (Example 1.8). Extending the model to allow for dependence on explanatory variables, x_t , gives

$$\begin{aligned} f(y_1, y_2, \dots, y_T | x_1, x_2, \dots, x_T; \theta) &= \\ f(y_1 | x_1; \theta) \prod_{t=2}^T f(y_t|y_{t-1}, y_{t-2}, \dots, y_1, x_t, x_{t-1}, \dots, x_1; \theta). \end{aligned} \quad (1.7)$$

An example is the autoregressive model with heteroskedasticity (Example 1.7).

Example 1.9 Autoregressive Model

The joint pdf for the AR(1) model in Example 1.5 is

$$f(y_1, y_2, \dots, y_T; \theta) = f(y_1; \theta) \prod_{t=2}^T f(y_t|y_{t-1}; \theta),$$

where the conditional distribution is

$$f(y_t|y_{t-1}; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y_t - \rho y_{t-1})^2}{2\sigma^2} \right],$$

and the marginal distribution is

$$f(y_1; \theta) = \frac{1}{\sqrt{2\pi\sigma^2/(1-\rho^2)}} \exp \left[-\frac{y_1^2}{2\sigma^2/(1-\rho^2)} \right].$$

□

Non-stochastic explanatory variables

In the case of non-stochastic explanatory variables, because x_t is deterministic its probability mass is degenerate. Explanatory variables of this form are also referred to as fixed in repeated samples. The joint probability in expression (1.3) simplifies to

$$f(y_1, y_2, \dots, y_T, x_1, x_2, \dots, x_T; \psi) = f(y_1, y_2, \dots, y_T | x_1, x_2, \dots, x_T; \theta).$$

Now $\psi = \theta$ and there is no potential loss of information from using the conditional distribution to estimate θ .

1.4 Maximum Likelihood Framework

As emphasized previously, a time series of data represents the observed realization of draws from a joint pdf. The maximum likelihood principle makes use of this result by providing a general framework for estimating the unknown parameters, θ , from the observed time series data, $\{y_1, y_2, \dots, y_T\}$.

1.4.1 The Log-Likelihood Function

The standard interpretation of the joint pdf in (1.7) is that f is a function of y_t for given parameters, θ . In defining the maximum likelihood estimator this interpretation is reversed, so that f is taken as a function of θ for given y_t . The motivation behind this change in the interpretation of the arguments of the pdf is to regard $\{y_1, y_2, \dots, y_T\}$ as a realized data set which is no longer random. The maximum likelihood estimator is then obtained by finding the value of θ which is “most likely” to have generated the observed data. Here the phrase “most likely” is loosely interpreted in a probability sense.

It is important to remember that the likelihood function is simply a re-definition of the joint pdf in equation (1.7). For many problems it is simpler to work with the logarithm of this joint density function. The log-likelihood

function is defined as

$$\begin{aligned} \ln L_T(\theta) &= \frac{1}{T} \ln f(y_1 | x_1; \theta) \\ &\quad + \frac{1}{T} \sum_{t=2}^T \ln f(y_t | y_{t-1}, y_{t-2}, \dots, y_1, x_t, x_{t-1}, \dots, x_1; \theta), \end{aligned} \quad (1.8)$$

where the change of status of the arguments in the joint pdf is highlighted by making θ the sole argument of this function and the T subscript indicates that the log-likelihood is an average over the sample of the logarithm of the density evaluated at y_t . It is worth emphasizing that the term log-likelihood function, used here without any qualification, is also known as the average log-likelihood function. This convention is also used by, among others, Newey and McFadden (1994) and White (1994). This definition of the log-likelihood function is consistent with the theoretical development of the properties of maximum likelihood estimators discussed in Chapter 2, particularly Sections 2.3 and 2.5.1.

For the special case where y_t is *iid*, the log-likelihood function is based on the joint pdf in (1.4) and is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta).$$

In all cases, the log-likelihood function, $\ln L_T(\theta)$, is a scalar that represents a summary measure of the data for given θ .

The maximum likelihood estimator of θ is defined as that value of θ , denoted $\hat{\theta}$, that maximizes the log-likelihood function. In a large number of cases, this may be achieved using standard calculus. Chapter 3 discusses numerical approaches to the problem of finding maximum likelihood estimates when no analytical solutions exist, or are difficult to derive.

Example 1.10 Poisson Distribution

Let $\{y_1, y_2, \dots, y_T\}$ be *iid* observations from a Poisson distribution

$$f(y; \theta) = \frac{\theta^y \exp[-\theta]}{y!},$$

where $\theta > 0$. The log-likelihood function for the sample is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta) = \frac{1}{T} \sum_{t=1}^T y_t \ln \theta - \theta - \frac{\ln(y_1! y_2! \dots y_T!)}{T}.$$

Consider the following $T = 3$ observations, $y_t = \{8, 3, 4\}$. The log-likelihood

function is

$$\ln L_T(\theta) = \frac{15}{3} \ln \theta - \theta - \frac{\ln(8!3!4!)}{3} = 5 \ln \theta - \theta - 5.191.$$

A plot of the log-likelihood function is given in panel (a) of Figure 1.3 for values of θ ranging from 0 to 10. Even though the Poisson distribution is a discrete distribution in terms of the random variable y , the log-likelihood function is continuous in the unknown parameter θ . Inspection shows that a maximum occurs at $\hat{\theta} = 5$ with a log-likelihood value of

$$\ln L_T(5) = 5 \times \ln 5 - 5 - 5.191 = -2.144.$$

The contribution to the log-likelihood function at the first observation $y_1 = 8$, evaluated at $\hat{\theta} = 5$ is

$$\ln f(y_1; 5) = y_1 \ln 5 - 5 - \ln(y_1!) = 8 \times \ln 5 - 5 - \ln(8!) = -2.729.$$

For the other two observations, the contributions are $\ln f(y_2; 5) = -1.963$, $\ln f(y_3; 5) = -1.740$. The probabilities $f(y_t; \theta)$ are between 0 and 1 by definition and therefore all of the contributions are negative because they are computed as the logarithm of $f(y_t; \theta)$. The average of these $T = 3$ contributions is $\ln L_T(5) = -2.144$, which corresponds to the value already given above. A plot of $\ln f(y_t; 5)$ in panel (b) of Figure 1.3 shows that observations closer to $\hat{\theta} = 5$ have a relatively greater contribution to the log-likelihood function than observations further away in the sense that they are smaller negative numbers. $\square$

Example 1.11 Exponential Distribution

Let $\{y_1, y_2, \dots, y_T\}$ be *iid* drawings from an exponential distribution

$$f(y; \theta) = \theta \exp[-\theta y],$$

where $\theta > 0$. The log-likelihood function for the sample is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta) = \frac{1}{T} \sum_{t=1}^T (\ln \theta - \theta y_t) = \ln \theta - \theta \frac{1}{T} \sum_{t=1}^T y_t.$$

Consider the following $T = 6$ observations, $y_t = \{2.1, 2.2, 3.1, 1.6, 2.5, 0.5\}$. The log-likelihood function is

$$\ln L_T(\theta) = \ln \theta - \theta \frac{1}{T} \sum_{t=1}^T y_t = \ln \theta - 2\theta.$$

Plots of the log-likelihood function, $\ln L_T(\theta)$, and the likelihood $L_T(\theta)$ functions are given in Figure 1.4, which show that a maximum occurs at

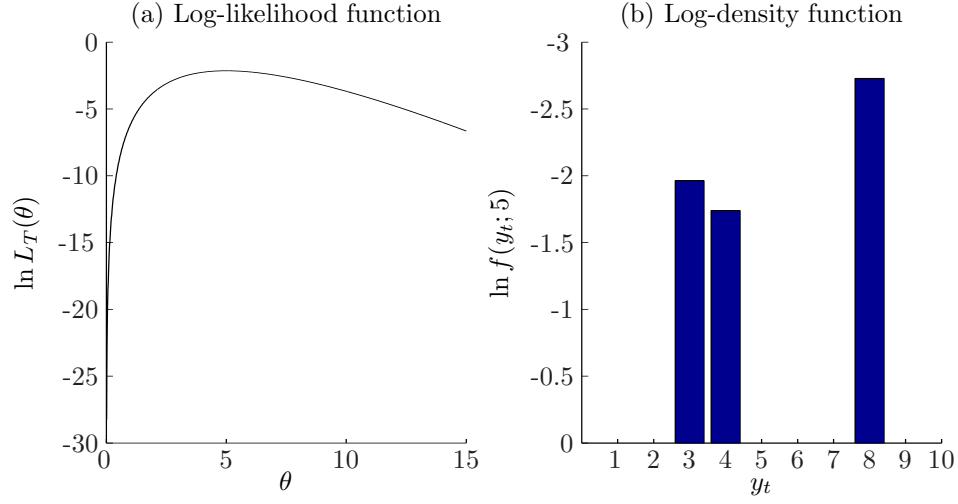


Figure 1.3 Plot of $\ln L_T(\theta)$ and $\ln f(y_t; \hat{\theta} = 5)$ for the Poisson distribution example with a sample size of $T = 3$.

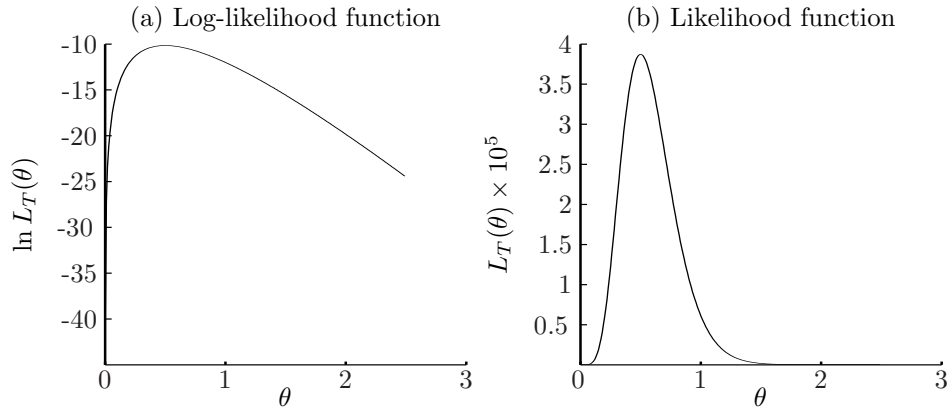


Figure 1.4 Plot of $\ln L_T(\theta)$ for the exponential distribution example.

$\hat{\theta} = 0.5$. Table 1.2 provides details of the calculations. Let the log-likelihood function at each observation evaluated at the maximum likelihood estimate be denoted $\ln l_t(\theta) = \ln f(y_t; \theta)$. The second column shows $\ln l_t(\theta)$ evaluated at $\hat{\theta} = 0.5$

$$\ln l_t(0.5) = \ln(0.5) - 0.5y_t,$$

resulting in a maximum value of the log-likelihood function of

$$\ln L_T(0.5) = \frac{1}{6} \sum_{t=1}^6 \ln l_t(0.5) = \frac{-10.159}{6} = -1.693.$$

Table 1.2

Maximum likelihood calculations for the exponential distribution example. The maximum likelihood estimate is $\hat{\theta}_T = 0.5$.

y_t	$\ln l_t(0.5)$	$g_t(0.5)$	$h_t(0.5)$
2.1	-1.743	-0.100	-4.000
2.2	-1.793	-0.200	-4.000
3.1	-2.243	-1.100	-4.000
1.6	-1.493	0.400	-4.000
2.5	-1.943	-0.500	-4.000
0.5	-0.943	1.500	-4.000
$\ln L_T(0.5) = -1.693$		$G_T(0.5) = 0.000$	$H_T(0.5) = -4.000$

□

Example 1.12 Normal Distribution

Let $\{y_1, y_2, \dots, y_T\}$ be *iid* observations drawn from a normal distribution

$$f(y; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y - \mu)^2}{2\sigma^2} \right],$$

with unknown parameters $\theta = \{\mu, \sigma^2\}$. The log-likelihood function is

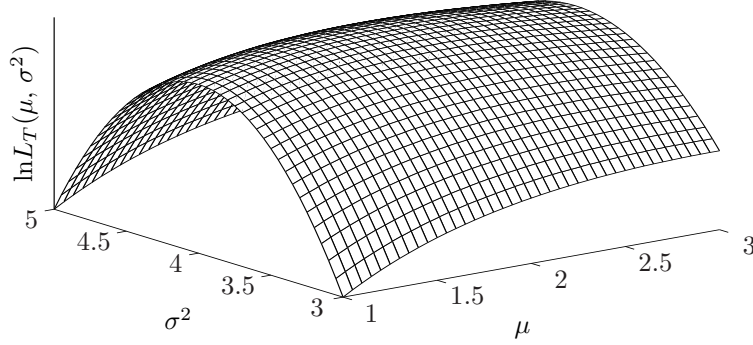
$$\begin{aligned} \ln L_T(\theta) &= \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta) \\ &= \frac{1}{T} \sum_{t=1}^T \left(-\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{(y_t - \mu)^2}{2\sigma^2} \right) \\ &= -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2 T} \sum_{t=1}^T (y_t - \mu)^2. \end{aligned}$$

Consider the following $T = 6$ observations, $y_t = \{5, -1, 3, 0, 2, 3\}$. The log-likelihood function is

$$\ln L_T(\theta) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{1}{12\sigma^2} \sum_{t=1}^6 (y_t - \mu)^2.$$

A plot of this function in Figure 1.5 shows that a maximum occurs at $\hat{\mu} = 2$ and $\hat{\sigma}^2 = 4$. □

Example 1.13 Autoregressive Model

Figure 1.5 Plot of $\ln L_T(\theta)$ for the normal distribution example.

From Example 1.9, the log-likelihood function for the AR(1) model is

$$\begin{aligned} \ln L_T(\theta) = & \frac{1}{T} \left(\frac{1}{2} \ln(1 - \rho^2) - \frac{1}{2\sigma^2} (1 - \rho^2) y_1^2 \right) \\ & - \frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2 T} \sum_{t=2}^T (y_t - \rho y_{t-1})^2. \end{aligned}$$

The first term is commonly excluded from $\ln L_T(\theta)$ as its contribution disappears asymptotically since

$$\lim_{T \rightarrow \infty} \frac{1}{T} \left(\frac{1}{2} \ln(1 - \rho^2) - \frac{1}{2\sigma^2} (1 - \rho^2) y_1^2 \right) = 0.$$

□

As the aim of maximum likelihood estimation is to find the value of θ that maximizes the log-likelihood function, a natural way to do this is to use the rules of calculus. This involves computing the first derivatives and second derivatives of the log-likelihood function with respect to the parameter vector θ .

1.4.2 Gradient

Differentiating $\ln L_T(\theta)$, with respect to a $(K \times 1)$ parameter vector, θ , yields a $(K \times 1)$ gradient vector, also known as the score, given by

$$G_T(\theta) = \frac{\partial \ln L_T(\theta)}{\partial \theta} = \begin{bmatrix} \frac{\partial \ln L_T(\theta)}{\partial \theta_1} \\ \frac{\partial \ln L_T(\theta)}{\partial \theta_2} \\ \vdots \\ \frac{\partial \ln L_T(\theta)}{\partial \theta_K} \end{bmatrix} = \frac{1}{T} \sum_{t=1}^T g_t(\theta), \quad (1.9)$$

where the subscript T emphasizes that the gradient is the sample average of the individual gradients

$$g_t(\theta) = \frac{\partial \ln l_t(\theta)}{\partial \theta}.$$

The maximum likelihood estimator of θ , denoted $\hat{\theta}$, is obtained by setting the gradients equal to zero and solving the resultant K first-order conditions. The maximum likelihood estimator, $\hat{\theta}$, therefore satisfies the condition

$$G_T(\hat{\theta}) = \left. \frac{\partial \ln L_T(\theta)}{\partial \theta} \right|_{\theta=\hat{\theta}} = 0. \quad (1.10)$$

Example 1.14 Poisson Distribution

From Example 1.10, the first derivative of $\ln L_T(\theta)$ with respect to θ is

$$G_T(\theta) = \frac{1}{T\theta} \sum_{t=1}^T y_t - 1.$$

The maximum likelihood estimator is the solution of the first-order condition

$$\frac{1}{T\hat{\theta}} \sum_{t=1}^T y_t - 1 = 0,$$

which yields the sample mean as the maximum likelihood estimator

$$\hat{\theta} = \frac{1}{T} \sum_{t=1}^T y_t = \bar{y}.$$

Using the data for y_t in Example 1.10, the maximum likelihood estimate is $\hat{\theta} = 15/3 = 5$. Evaluating the gradient at $\hat{\theta} = 5$ verifies that it is zero at the

maximum likelihood estimate

$$G_T(\hat{\theta}) = \frac{1}{T\hat{\theta}} \sum_{t=1}^T y_t - 1 = \frac{15}{3 \times 5} - 1 = 0.$$

□

Example 1.15 Exponential Distribution

From Example 1.11, the first derivative of $\ln L_T(\theta)$ with respect to θ is

$$G_T(\theta) = \frac{1}{\theta} - \frac{1}{T} \sum_{t=1}^T y_t.$$

Setting $G_T(\hat{\theta}) = 0$ and solving the resultant first-order condition yields

$$\hat{\theta} = \frac{T}{\sum_{t=1}^T y_t} = \frac{1}{\bar{y}},$$

which is the reciprocal of the sample mean. Using the same observed data for y_t as in Example 1.11, the maximum likelihood estimate is $\hat{\theta} = 6/12 = 0.5$.

The third column of Table 1.2 gives the gradients at each observation evaluated at $\hat{\theta} = 0.5$

$$g_t(0.5) = \frac{1}{0.5} - y_t.$$

The gradient is

$$G_T(0.5) = \frac{1}{6} \sum_{t=1}^6 g_t(0.5) = 0,$$

which follows from the properties of the maximum likelihood estimator. □

Example 1.16 Normal Distribution

From Example 1.12, the first derivatives of the log-likelihood function are

$$\frac{\partial \ln L_T(\theta)}{\partial \mu} = \frac{1}{\sigma^2 T} \sum_{t=1}^T (y_t - \mu), \quad \frac{\partial \ln L_T(\theta)}{\partial (\sigma^2)} = -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4 T} \sum_{t=1}^T (y_t - \mu)^2,$$

yielding the gradient vector

$$G_T(\theta) = \begin{bmatrix} \frac{1}{\sigma^2 T} \sum_{t=1}^T (y_t - \mu) \\ -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4 T} \sum_{t=1}^T (y_t - \mu)^2 \end{bmatrix}.$$

Evaluating the gradient at $\hat{\theta}$ and setting $G_T(\hat{\theta}) = 0$, gives

$$G_T(\hat{\theta}) = \begin{bmatrix} \frac{1}{\hat{\sigma}^2 T} \sum_{t=1}^T (y_t - \hat{\mu}) \\ -\frac{1}{2\hat{\sigma}^2} + \frac{1}{2\hat{\sigma}^4 T} \sum_{t=1}^T (y_t - \hat{\mu})^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Solving for $\hat{\theta} = \{\hat{\mu}, \hat{\sigma}^2\}$, the maximum likelihood estimators are

$$\hat{\mu} = \frac{1}{T} \sum_{t=1}^T y_t = \bar{y}, \quad \hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \bar{y})^2.$$

Using the data from Example 1.12, the maximum likelihood estimates are

$$\begin{aligned} \hat{\mu} &= \frac{5 - 1 + 3 + 0 + 2 + 3}{6} = 2 \\ \hat{\sigma}^2 &= \frac{(5 - 2)^2 + (-1 - 2)^2 + (3 - 2)^2 + (0 - 2)^2 + (2 - 2)^2 + (3 - 2)^2}{6} = 4, \end{aligned}$$

which agree with the values given in Example 1.12. $\square$

1.4.3 Hessian

To establish that $\hat{\theta}$ maximizes the log-likelihood function, it is necessary to determine that the Hessian

$$H_T(\theta) = \frac{\partial^2 \ln L_T(\theta)}{\partial \theta \partial \theta'}, \quad (1.11)$$

associated with the log-likelihood function is negative definite. As θ is a $(K \times 1)$ vector, the Hessian is the $(K \times K)$ symmetric matrix

$$H_T(\theta) = \begin{bmatrix} \frac{\partial^2 \ln L_T(\theta)}{\partial \theta_1 \partial \theta_1} & \frac{\partial^2 \ln L_T(\theta)}{\partial \theta_1 \partial \theta_2} & \cdots & \frac{\partial^2 \ln L_T(\theta)}{\partial \theta_1 \partial \theta_K} \\ \frac{\partial^2 \ln L_T(\theta)}{\partial \theta_2 \partial \theta_1} & \frac{\partial^2 \ln L_T(\theta)}{\partial \theta_2 \partial \theta_2} & \cdots & \frac{\partial^2 \ln L_T(\theta)}{\partial \theta_2 \partial \theta_K} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial^2 \ln L_T(\theta)}{\partial \theta_K \partial \theta_1} & \frac{\partial^2 \ln L_T(\theta)}{\partial \theta_K \partial \theta_2} & \cdots & \frac{\partial^2 \ln L_T(\theta)}{\partial \theta_K \partial \theta_K} \end{bmatrix} = \frac{1}{T} \sum_{t=1}^T h_t(\theta),$$

where the subscript T emphasizes that the Hessian is the sample average of the individual elements

$$h_t(\theta) = \frac{\partial^2 \ln l_t(\theta)}{\partial \theta \partial \theta'}.$$

The second-order condition for a maximum requires that the Hessian matrix evaluated at $\hat{\theta}$,

$$H_T(\hat{\theta}) = \left. \frac{\partial^2 \ln L_T(\theta)}{\partial \theta \partial \theta'} \right|_{\theta=\hat{\theta}}, \quad (1.12)$$

is negative definite. The conditions for negative definiteness are

$$|H_{11}| < 0, \quad \begin{vmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{vmatrix} > 0, \quad \begin{vmatrix} H_{11} & H_{12} & H_{13} \\ H_{21} & H_{22} & H_{23} \\ H_{31} & H_{32} & H_{33} \end{vmatrix} < 0, \quad \dots$$

where H_{ij} is the ij^{th} element of $H_T(\hat{\theta})$. In the case of $K = 1$, the condition is

$$H_{11} < 0. \quad (1.13)$$

For the case of $K = 2$, the condition is

$$H_{11} < 0, \quad H_{11}H_{22} - H_{12}H_{21} > 0. \quad (1.14)$$

Example 1.17 Poisson Distribution

From Examples 1.10 and 1.14, the second derivative of $\ln L_T(\theta)$ with respect to θ is

$$H_T(\theta) = -\frac{1}{\theta^2 T} \sum_{t=1}^T y_t.$$

Evaluating the Hessian at the maximum likelihood estimator, $\hat{\theta} = \bar{y}$, yields

$$H_T(\hat{\theta}) = -\frac{1}{\hat{\theta}^2 T} \sum_{t=1}^T y_t = -\frac{1}{\bar{y}^2 T} \sum_{t=1}^T y_t = -\frac{1}{\bar{y}} < 0.$$

As $\bar{y}$ is always positive because it is the mean of a sample of positive integers, the Hessian is negative and a maximum is achieved. Using the data for y_t in Example 1.10, verifies that the Hessian at $\hat{\theta} = 5$ is negative

$$H_T(\hat{\theta}) = -\frac{1}{\hat{\theta}^2 T} \sum_{t=1}^T y_t = -\frac{15}{5^2 \times 3} = -0.200.$$

□

Example 1.18 Exponential Distribution

From Examples 1.11 and 1.15, the second derivative of $\ln L_T(\theta)$ with respect to θ is

$$H_T(\theta) = -\frac{1}{\theta^2}.$$

Evaluating the Hessian at the maximum likelihood estimator yields

$$H_T(\hat{\theta}) = -\frac{1}{\hat{\theta}^2} < 0.$$

As this term is negative for any $\hat{\theta}$, the condition in equation (1.13) is satisfied and a maximum is achieved. The last column of Table 1.2 shows that the Hessian at each observation evaluated at the maximum likelihood estimate is constant. The value of the Hessian is

$$H_T(0.5) = \frac{1}{6} \sum_{t=1}^6 h_t(0.5) = \frac{-24.000}{6} = -4,$$

which is negative confirming that a maximum has been reached. □

Example 1.19 Normal Distribution

From Examples 1.12 and 1.16, the second derivatives of $\ln L_T(\theta)$ with respect to θ are

$$\begin{aligned} \frac{\partial^2 \ln L_T(\theta)}{\partial \mu^2} &= -\frac{1}{\sigma^2} \\ \frac{\partial^2 \ln L_T(\theta)}{\partial \mu \partial \sigma^2} &= -\frac{1}{\sigma^4 T} \sum_{t=1}^T (y_t - \mu) \\ \frac{\partial^2 \ln L_T(\theta)}{\partial (\sigma^2)^2} &= \frac{1}{2\sigma^4} - \frac{1}{\sigma^6 T} \sum_{t=1}^T (y_t - \mu)^2, \end{aligned}$$

so that the Hessian is

$$H_T(\theta) = \begin{bmatrix} -\frac{1}{\sigma^2} & -\frac{1}{\sigma^4 T} \sum_{t=1}^T (y_t - \mu) \\ -\frac{1}{\sigma^4 T} \sum_{t=1}^T (y_t - \mu) & \frac{1}{2\sigma^4} - \frac{1}{\sigma^6 T} \sum_{t=1}^T (y_t - \mu)^2 \end{bmatrix}.$$

Given that $G_T(\hat{\theta}) = 0$, from Example 1.16 it follows that $\sum_{t=1}^T (y_t - \hat{\mu}) = 0$

and therefore

$$H_T(\hat{\theta}) = \begin{bmatrix} -\frac{1}{\hat{\sigma}^2} & 0 \\ 0 & -\frac{1}{2\hat{\sigma}^4} \end{bmatrix}.$$

From equation (1.14)

$$H_{11} = -\frac{T}{\hat{\sigma}^2} < 0, \quad H_{11}H_{22} - H_{12}H_{21} = -\left(\frac{T}{\hat{\sigma}^2}\right)\left(-\frac{T}{2\hat{\sigma}^4}\right) - 0^2 > 0,$$

establishing that the second-order condition for a maximum is satisfied. Using the maximum likelihood estimates from Example 1.16, the Hessian is

$$H_T(\hat{\mu}, \hat{\sigma}^2) = \begin{bmatrix} -\frac{1}{4} & 0 \\ 0 & -\frac{1}{2 \times 4^2} \end{bmatrix} = \begin{bmatrix} -0.250 & 0.000 \\ 0.000 & -0.031 \end{bmatrix}.$$

□

1.5 Applications

To highlight the features of maximum likelihood estimation discussed thus far, two applications are presented that focus on estimating the discrete time version of the Vasicek (1977) model of interest rates, r_t . The first application is based on the marginal (stationary) distribution while the second focuses on the conditional (transitional) distribution that gives the distribution of r_t conditional on r_{t-1} . The interest rate data used are from Aït-Sahalia (1996). The data, plotted in Figure 1.6, consists of daily 7-day Eurodollar rates (expressed as percentages) for the period 1 June 1973 to the 25 February 1995, a total of $T = 5505$ observations.

The Vasicek model expresses the change in the interest rate, r_t , as a function of a constant and the lagged interest rate

$$\begin{aligned} r_t - r_{t-1} &= \alpha + \beta r_{t-1} + u_t \\ u_t &\sim iid N(0, \sigma^2), \end{aligned} \tag{1.15}$$

where $\theta = \{\alpha, \beta, \sigma^2\}$ are unknown parameters, with the restriction $\beta < 0$.

1.5.1 Stationary Distribution of the Vasicek Model

As a preliminary step to estimating the parameters of the Vasicek model in equation (1.15), consider the alternative model where the level of the interest

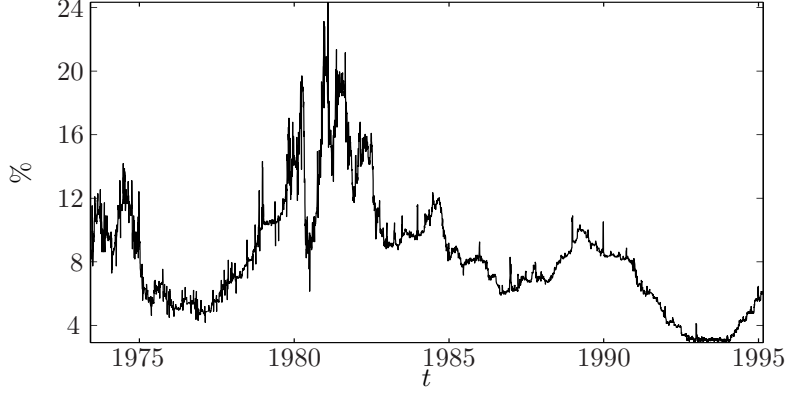


Figure 1.6 Daily 7-day Eurodollar interest rates from the 1 June 1973 to 25 February 1995 expressed as a percentage.

rate is independent of previous interest rates

$$r_t = \mu_s + v_t, \quad v_t \sim iid N(0, \sigma_s^2).$$

The stationary distribution of r_t for this model is

$$f(r; \mu_s, \sigma_s^2) = \frac{1}{\sqrt{2\pi\sigma_s^2}} \exp \left[-\frac{(r - \mu_s)^2}{2\sigma_s^2} \right]. \quad (1.16)$$

The relationship between the parameters of the stationary distribution and the parameters of the model in equation (1.15) is

$$\mu_s = -\frac{\alpha}{\beta}, \quad \sigma_s^2 = -\frac{\sigma^2}{\beta(2 + \beta)}. \quad (1.17)$$

which are obtained as the unconditional mean and variance of (1.15).

The log-likelihood function based on the stationary distribution in equation (1.16) for a sample of T observations is

$$\ln L_T(\theta) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma_s^2 - \frac{1}{2\sigma_s^2 T} \sum_{t=1}^T (r_t - \mu_s)^2,$$

where $\theta = \{\mu_s, \sigma_s^2\}$. Maximizing $\ln L_T(\theta)$ with respect to θ gives

$$\widehat{\mu}_s = \frac{1}{T} \sum_{t=1}^T r_t, \quad \widehat{\sigma}_s^2 = \frac{1}{T} \sum_{t=1}^T (r_t - \widehat{\mu}_s)^2. \quad (1.18)$$

Using the Eurodollar interest rates, the maximum likelihood estimates are

$$\widehat{\mu}_s = 8.362, \quad \widehat{\sigma}_s^2 = 12.893. \quad (1.19)$$

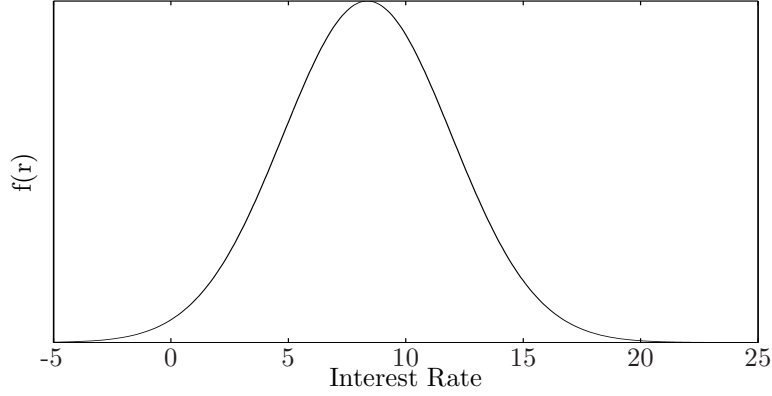


Figure 1.7 Estimated stationary distribution of the Vasicek model based on evaluating (1.16) at the maximum likelihood estimates (1.19), using daily Eurodollar rates from the 1 June 1973 to 25 February 1995.

The stationary distribution is estimated by evaluating equation (1.16) at the maximum likelihood estimates in (1.19) and is given by

$$\begin{aligned} f(r; \hat{\mu}_s, \hat{\sigma}_s^2) &= \frac{1}{\sqrt{2\pi\hat{\sigma}_s^2}} \exp \left[-\frac{(r - \hat{\mu}_s)^2}{2\hat{\sigma}_s^2} \right] \\ &= \frac{1}{\sqrt{2\pi \times 12.893}} \exp \left[-\frac{(r - 8.362)^2}{2 \times 12.893} \right], \end{aligned} \quad (1.20)$$

which is presented in Figure 1.7.

Inspection of the estimated distribution shows a potential problem with the Vasicek stationary distribution, namely that the support of the distribution is not restricted to being positive. The probability of negative values for the interest rate is

$$\Pr(r < 0) = \int_{-\infty}^0 \frac{1}{\sqrt{2\pi \times 12.893}} \exp \left[-\frac{(r - 8.362)^2}{2 \times 12.893} \right] dr = 0.01.$$

To avoid this problem, alternative models of interest rates are specified where the stationary distribution is just defined over the positive region. A well known example is the CIR interest rate model (Cox, Ingersoll and Ross, 1985) which is discussed in Chapters 2, 3 and 12.

1.5.2 Transitional Distribution of the Vasicek Model

In contrast to the stationary model specification of the previous section, the full dynamics of the Vasicek model in equation (1.15) are now used by

specifying the transitional distribution

$$f(r | r_{t-1}; \alpha, \rho, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(r - \alpha - \rho r_{t-1})^2}{2\sigma^2} \right], \quad (1.21)$$

where $\theta = \{\alpha, \rho, \sigma^2\}$ and the substitution $\rho = 1 + \beta$ is made for convenience. This distribution is now of the same form as the conditional distribution of the AR(1) model in Examples 1.5, 1.9 and 1.13.

The log-likelihood function based on the transitional distribution in equation (1.21) is

$$\ln L_T(\theta) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2(T-1)} \sum_{t=2}^T (r_t - \alpha - \rho r_{t-1})^2,$$

where the sample size is reduced by one observation as a result of the lagged term r_{t-1} . This form of the log-likelihood function does not contain the marginal distribution $f(r_1; \theta)$, a point that is made in Example 1.13. The first derivatives of the log-likelihood function are

$$\begin{aligned} \frac{\partial \ln L(\theta)}{\partial \alpha} &= \frac{1}{\sigma^2(T-1)} \sum_{t=2}^T (r_t - \alpha - \rho r_{t-1}) \\ \frac{\partial \ln L(\theta)}{\partial \rho} &= \frac{1}{\sigma^2(T-1)} \sum_{t=2}^T (r_t - \alpha - \rho r_{t-1}) r_{t-1} \\ \frac{\partial \ln L(\theta)}{\partial (\sigma^2)} &= -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4(T-1)} \sum_{t=2}^T (r_t - \alpha - \rho r_{t-1})^2. \end{aligned}$$

Setting these derivatives to zero yields the maximum likelihood estimators

$$\begin{aligned} \hat{\alpha} &= \bar{r}_t - \hat{\rho} \bar{r}_{t-1} \\ \hat{\rho} &= \frac{\sum_{t=2}^T (r_t - \bar{r}_t)(r_{t-1} - \bar{r}_{t-1})}{\sum_{t=2}^T (r_{t-1} - \bar{r}_{t-1})^2} \\ \hat{\sigma}^2 &= \frac{1}{T-1} \sum_{t=2}^T (r_t - \hat{\alpha} - \hat{\rho} r_{t-1})^2, \end{aligned}$$

where

$$\bar{r}_t = \frac{1}{T-1} \sum_{t=2}^T r_t, \quad \bar{r}_{t-1} = \frac{1}{T-1} \sum_{t=2}^T r_{t-1}.$$

The maximum likelihood estimates for the Eurodollar interest rates are

$$\hat{\alpha} = 0.053, \quad \hat{\rho} = 0.994, \quad \hat{\sigma}^2 = 0.165. \quad (1.22)$$

An estimate of β is obtained by using the relationship $\rho = 1 + \beta$. Rearranging for β and evaluating at $\hat{\rho}$ gives $\hat{\beta} = \hat{\rho} - 1 = -0.006$.

The estimated transitional distribution is obtained by evaluating (1.21) at the maximum likelihood estimates in (1.22)

$$f(r | r_{t-1}; \hat{\alpha}, \hat{\rho}, \hat{\sigma}^2) = \frac{1}{\sqrt{2\pi\hat{\sigma}^2}} \exp \left[-\frac{(r - \hat{\alpha} - \hat{\rho}r_{t-1})^2}{2\hat{\sigma}^2} \right]. \quad (1.23)$$

Plots of this distribution are given in Figure 1.8 for three values of the conditioning variable r_{t-1} , corresponding to the minimum (2.9%), median (8.1%) and maximum (24.3%) interest rates in the sample.

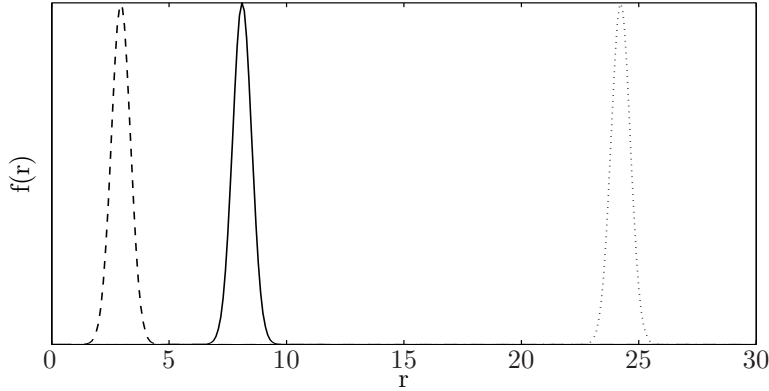


Figure 1.8 Estimated transitional distribution of the Vasicek model, based on evaluating (1.23) at the maximum likelihood estimates in (1.22) using Eurodollar rates from 1 June 1973 to 25 February 1995. The dashed line is the transitional density for the minimum (2.9%), the solid line is the transitional density for the median (8.1%) and the dotted line is the transitional density for the maximum (24.3%) Eurodollar rate.

The location of the three transitional distributions changes over time, while the spread of each distribution remains constant at $\hat{\sigma}^2 = 0.165$. A comparison of the estimates of the variances of the stationary and transitional distributions, in equations (1.19) and (1.22), respectively, shows that $\hat{\sigma}^2 < \hat{\sigma}_s^2$. This result is a reflection of the property that by conditioning on information, in this case r_{t-1} , the transitional distribution is better at tracking the time series behaviour of the interest rate, r_t , than the stationary distribution where there is no conditioning on lagged dependent variables.

Having obtained the estimated transitional distribution using the maximum likelihood estimates in (1.22), it is also possible to use these estimates to reestimate the stationary interest rate distribution in (1.20) by using the expressions in (1.17). The alternative estimates of the mean and variance of the stationary distribution are

$$\begin{aligned}\tilde{\mu}_s &= -\frac{\hat{\alpha}}{\hat{\beta}} = \frac{0.053}{0.006} = 8.308, \\ \tilde{\sigma}_s^2 &= -\frac{\hat{\sigma}^2}{\hat{\beta}(2 + \hat{\beta})} = \frac{0.165}{0.006(2 - 0.006)} = 12.967.\end{aligned}$$

As these estimates are based on the transitional distribution, which incorporates the full dynamic specification of the Vasicek model, they represent the maximum likelihood estimates of the parameters of the stationary distribution. This relationship between the maximum likelihood estimators of the transitional and stationary distributions is based on the invariance property of maximum likelihood estimators which is discussed in Chapter 2. While the parameter estimates of the stationary distribution using the estimates of the transitional distribution are numerically close to estimates obtained in the previous section, the latter estimates are obtained from a misspecified model as the stationary model excludes the dynamic structure in equation (1.15). Issues relating to misspecified models are discussed in Chapter 9.

1.6 Exercises

(1) Sampling Data

Gauss file(s)	<code>basic_sample.g</code>
Matlab file(s)	<code>basic_sample.m</code>

This exercise reproduces the simulation results in Figures 1.1 and 1.2. For each model, simulate $T = 5$ draws of y_t and plot the corresponding distribution at each point in time. Where applicable the explanatory variable in these exercises is $x_t = \{0, 1, 2, 3, 4\}$ and w_t are draws from a uniform distribution on the unit circle.

(a) Time invariant model

$$y_t = 2z_t, \quad z_t \sim iid N(0, 1).$$

(b) Count model

$$f(y; 2) = \frac{2^y \exp[-2]}{y!}, \quad y = 1, 2, \dots$$

- (c) Linear regression model

$$y_t = 3x_t + 2z_t, \quad z_t \sim iid N(0, 1).$$

- (d) Exponential regression model

$$f(y; \theta) = \frac{1}{\mu_t} \exp \left[-\frac{y}{\mu_t} \right], \quad \mu_t = 1 + 2x_t.$$

- (e) Autoregressive model

$$y_t = 0.8y_{t-1} + 2z_t, \quad z_t \sim iid N(0, 1).$$

- (f) Bilinear time series model

$$y_t = 0.8y_{t-1} + 0.4y_{t-1}u_{t-1} + 2z_t, \quad z_t \sim iid N(0, 1).$$

- (g) Autoregressive model with heteroskedasticity

$$y_t = 0.8y_{t-1} + \sigma_t z_t, \quad z_t \sim iid N(0, 1) \\ \sigma_t^2 = 0.8 + 0.8w_t.$$

- (h) The ARCH regression model

$$y_t = 3x_t + u_t \\ u_t = \sigma_t z_t \\ \sigma_t^2 = 4 + 0.9u_{t-1}^2 \\ z_t \sim iid N(0, 1).$$

(2) Poisson Distribution

Gauss file(s)	<code>basic_poisson.g</code>
Matlab file(s)	<code>basic_poisson.m</code>

A sample of $T = 4$ observations, $y_t = \{6, 2, 3, 1\}$, is drawn from the Poisson distribution

$$f(y; \theta) = \frac{\theta^y \exp[-\theta]}{y!}.$$

- (a) Write the log-likelihood function, $\ln L_T(\theta)$.
- (b) Derive and interpret the maximum likelihood estimator, $\hat{\theta}$.
- (c) Compute the maximum likelihood estimate, $\hat{\theta}$.
- (d) Compute the log-likelihood function at $\hat{\theta}$ for each observation.
- (e) Compute the value of the log-likelihood function at $\hat{\theta}$.

(f) Compute

$$g_t(\hat{\theta}) = \left. \frac{d \ln l_t(\theta)}{d\theta} \right|_{\theta=\hat{\theta}} \quad \text{and} \quad h_t(\hat{\theta}) = \left. \frac{d^2 \ln l_t(\theta)}{d\theta^2} \right|_{\theta=\hat{\theta}},$$

for each observation.

(g) Compute

$$G_T(\hat{\theta}) = \frac{1}{T} \sum_{t=1}^T g_t(\hat{\theta}) \quad \text{and} \quad H_T(\hat{\theta}) = \frac{1}{T} \sum_{t=1}^T h_t(\hat{\theta}).$$

(3) Exponential Distribution

Gauss file(s)	<code>basic_exp.g</code>
Matlab file(s)	<code>basic_exp.m</code>

A sample of $T = 4$ observations, $y_t = \{5.5, 2.0, 3.5, 5.0\}$, is drawn from the exponential distribution

$$f(y; \theta) = \theta \exp[-\theta y].$$

- (a) Write the log-likelihood function, $\ln L_T(\theta)$.
- (b) Derive and interpret the maximum likelihood estimator, $\hat{\theta}$.
- (c) Compute the maximum likelihood estimate, $\hat{\theta}$.
- (d) Compute the log-likelihood function at $\hat{\theta}$ for each observation.
- (e) Compute the value of the log-likelihood function at $\hat{\theta}$.
- (f) Compute

$$g_t(\hat{\theta}) = \left. \frac{d \ln l_t(\theta)}{d\theta} \right|_{\theta=\hat{\theta}} \quad \text{and} \quad h_t(\hat{\theta}) = \left. \frac{d^2 \ln l_t(\theta)}{d\theta^2} \right|_{\theta=\hat{\theta}},$$

for each observation.

(g) Compute

$$G_T(\hat{\theta}) = \frac{1}{T} \sum_{t=1}^T g_t(\hat{\theta}) \quad \text{and} \quad H_T(\hat{\theta}) = \frac{1}{T} \sum_{t=1}^T h_t(\hat{\theta}).$$

(4) Alternative Form of Exponential Distribution

Consider a random sample of size T , $\{y_1, y_2, \dots, y_T\}$, of *iid* random variables from the exponential distribution with parameter θ

$$f(y; \theta) = \frac{1}{\theta} \exp \left[-\frac{y}{\theta} \right].$$

- (a) Derive the log-likelihood function, $\ln L_T(\theta)$.
- (b) Derive the first derivative of the log-likelihood function, $G_T(\theta)$.

- (c) Derive the second derivative of the log-likelihood function, $H_T(\theta)$.
- (d) Derive the maximum likelihood estimator of θ . Compare the result with that obtained in Exercise 3.

(5) Normal Distribution

Gauss file(s)	<code>basic_normal.g</code> , <code>basic_normal_like.g</code>
Matlab file(s)	<code>basic_normal.m</code> , <code>basic_normal_like.m</code>

A sample of $T = 5$ observations consisting of the values $\{1, 2, 5, 1, 2\}$ is drawn from the normal distribution

$$f(y; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y - \mu)^2}{2\sigma^2} \right],$$

where $\theta = \{\mu, \sigma^2\}$.

- (a) Assume that $\sigma^2 = 1$.
 - (i) Derive the log-likelihood function, $\ln L_T(\theta)$.
 - (ii) Derive and interpret the maximum likelihood estimator, $\hat{\theta}$.
 - (iii) Compute the maximum likelihood estimate, $\hat{\theta}$.
 - (iv) Compute $\ln l_t(\hat{\theta})$, $g_t(\hat{\theta})$ and $h_t(\hat{\theta})$.
 - (v) Compute $\ln L_T(\hat{\theta})$, $G_T(\hat{\theta})$ and $H_T(\hat{\theta})$.
- (b) Repeat part (a) for the case where both the mean and the variance are unknown, $\theta = \{\mu, \sigma^2\}$.

(6) A Model of the Number of Strikes

Gauss file(s)	<code>basic_count.g</code> , <code>strike.dat</code>
Matlab file(s)	<code>basic_count.m</code> , <code>strike.mat</code>

The data are the number of strikes per annum, y_t , in the U.S. from 1968 to 1976, taken from Kennan (1985). The number of strikes is specified as a Poisson-distributed random variable with unknown parameter θ

$$f(y; \theta) = \frac{\theta^y \exp[-\theta]}{y!}.$$

- (a) Write the log-likelihood function for a sample of T observations.
- (b) Derive and interpret the maximum likelihood estimator of θ .
- (c) Estimate θ and interpret the result.
- (d) Use the estimate from part (c), to plot the distribution of the number of strikes and interpret this plot.

- (e) Compute a histogram of y_t and comment on its consistency with the distribution of strike numbers estimated in part (d).

(7) **A Model of the Duration of Strikes**

Gauss file(s)	<code>basic_strike.g, strike.dat</code>
Matlab file(s)	<code>basic_strike.m, strike.mat</code>

The data are 62 observations, taken from the same source as Exercise 6, of the duration of strikes in the U.S. per annum expressed in days, y_t . Durations are assumed to be drawn from an exponential distribution with unknown parameter θ

$$f(y; \theta) = \frac{1}{\theta} \exp \left[-\frac{y}{\theta} \right].$$

- (a) Write the log-likelihood function for a sample of T observations.
 (b) Derive and interpret the maximum likelihood estimator of θ .
 (c) Use the data on strike durations to estimate θ . Interpret the result.
 (d) Use the estimates from part (c) to plot the distribution of strike durations and interpret this plot.
 (e) Compute a histogram of y_t and comment on its consistency with the distribution of duration times estimated in part (d).

(8) **Asset Prices**

Gauss file(s)	<code>basic_assetprices.g, assetprices.xls</code>
Matlab file(s)	<code>basic_assetprices.m, assetprices.mat</code>

The data consist of the Australian, Singapore and NASDAQ stock market indexes for the period 3 January 1989 to 31 December 2009, a total of $T = 5478$ observations. Consider the following model of asset prices, p_t , that is commonly adopted in the financial econometrics literature

$$\ln p_t - \ln p_{t-1} = \alpha + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

where $\theta = \{\alpha, \sigma^2\}$ are unknown parameters.

- (a) Use the transformation of variable technique to show that the conditional distribution of p is the log-normal distribution

$$f(p | p_{t-1}; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}p} \exp \left[-\frac{\ln p - \ln p_{t-1} - \alpha}{2\sigma^2} \right].$$

- (b) For a sample of size T , construct the log-likelihood function and derive the maximum likelihood estimator of θ based on the conditional distribution of p .

- (c) Use the results in part (b) to compute $\hat{\theta}$ for the three stock indexes.
- (d) Estimate the asset price distribution for each index using the maximum likelihood parameter estimates obtained in part (c).
- (e) Letting $r_t = \ln p_t - \ln p_{t-1}$ represent the return on an asset, derive the maximum likelihood estimator of θ based on the distribution of r_t . Compute $\hat{\theta}$ for the three stock market indexes and compare the estimates to those obtained in part (c).

(9) Stationary Distribution of the Vasicek Model

Gauss file(s)	<code>basic_stationary.g, eurodata.dat</code>
Matlab file(s)	<code>basic_stationary.m, eurodata.mat</code>

The data are daily 7-day Eurodollar rates, expressed as percentages, from 1 June 1973 to the 25 February 1995, a total of $T = 5505$ observations. The Vasicek discrete time model of interest rates, r_t , is

$$r_t - r_{t-1} = \alpha + \beta r_{t-1} + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

where $\theta = \{\alpha, \beta, \sigma^2\}$ are unknown parameters and $\beta < 0$.

- (a) Show that the mean and variance of the stationary distribution are, respectively,

$$\mu_s = -\frac{\alpha}{\beta}, \quad \sigma_s^2 = -\frac{\sigma^2}{\beta(2 + \beta)}.$$

- (b) Derive the maximum likelihood estimators of the parameters of the stationary distribution.
- (c) Compute the maximum likelihood estimates of the parameters of the stationary distribution using the Eurodollar interest rates.
- (d) Use the estimates from part (c) to plot the stationary distribution and interpret its properties.

(10) Transitional Distribution of the Vasicek Model

Gauss file(s)	<code>basic_transitional.g, eurodata.dat</code>
Matlab file(s)	<code>basic_transitional.m, eurodata.mat</code>

The data are the same daily 7-day Eurodollar rates, expressed in percentages, as used in Exercise 9. The Vasicek discrete time model of interest rates, r_t , is

$$r_t - r_{t-1} = \alpha + \beta r_{t-1} + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

where $\theta = \{\alpha, \beta, \sigma^2\}$ are unknown parameters and $\beta < 0$.

- (a) Derive the maximum likelihood estimators of the parameters of the transitional distribution.
- (b) Compute the maximum likelihood estimates of the parameters of the transitional distribution using Eurodollar interest rates.
- (c) Use the estimates from part (b) to plot the transitional distribution where conditioning is based on the minimum, median and maximum interest rates in the sample. Interpret the properties of the three transitional distributions.
- (d) Use the results in part (b) to estimate the mean and the variance of the stationary distribution and compare them to the estimates obtained in part (c) of Exercise 9.

2

Properties of Maximum Likelihood Estimators

2.1 Introduction

Under certain conditions known as regularity conditions, the maximum likelihood estimator introduced in Chapter 1 possesses a number of important statistical properties and the aim of this chapter is to derive these properties. In large samples, this estimator is consistent, efficient and normally distributed. In small samples, it satisfies an invariance property, is a function of sufficient statistics and in some, but not all, cases, is unbiased and unique. As the derivation of analytical expressions for the finite-sample distributions of the maximum likelihood estimator is generally complicated, computationally intensive methods based on Monte Carlo simulations or series expansions are used to examine many of these properties.

The maximum likelihood estimator encompasses many other estimators often used in econometrics, including ordinary least squares and instrumental variables (Chapter 5), nonlinear least squares (Chapter 6), the Cochrane-Orcutt method for the autocorrelated regression model (Chapter 7), weighted least squares estimation of heteroskedastic regression models (Chapter 8) and the Johansen procedure for cointegrated nonstationary time series models (Chapter 18).

2.2 Preliminaries

Before deriving the formal properties of the maximum likelihood estimator, four important preliminary concepts are reviewed. The first presents some stochastic models of time series and briefly discusses their properties. The second is concerned with the convergence of a sample average to its population mean as $T \rightarrow \infty$, known as the weak law of large numbers. The third identifies the scaling factor ensuring convergence of scaled random variables

to non-degenerate distributions. The fourth focuses on the form of the distribution of the sample average around its population mean as $T \rightarrow \infty$, known as the central limit theorem. Four central limit theorems are discussed: the Lindeberg-Levy central limit theorem, the Lindeberg-Feller central limit theorem, the martingale difference sequence central limit theorem and a mixing central limit theorem. These central limit theorems are extended to allow for nonstationary dependence using the functional central limit theorem in Chapter 16.

2.2.1 Stochastic Time Series Models and Their Properties

In this section various classes of time series models and their properties are introduced. These stochastic processes and the behaviour of the moments of their probability distribution functions are particularly important in the establishment of a range of convergence results and central limit theorems that enable the derivation of the properties of maximum likelihood estimators.

Stationarity

A variable y_t is stationary if its distribution, or some important aspect of its distribution, is constant over time. There are two commonly used definitions of stationarity known as weak (or covariance) and strong (or strict) stationarity. A variable that is not stationary is said to be nonstationary, a class of model that is discussed in detail in Part FIVE.

Weak Stationarity

The variable y_t is weakly stationary if the first two unconditional moments of the joint distribution function $F(y_1, y_2, \dots, y_j)$ do not depend on t for all finite j . This definition is summarized by the following three properties

- Property 1 : $E[y_t] = \mu < \infty$
- Property 2 : $\text{var}(y_t) = E[(y_t - \mu)^2] = \sigma^2 < \infty$
- Property 3 : $\text{cov}(y_t y_{t-k}) = E[(y_t - \mu)(y_{t-k} - \mu)] = \gamma_k, \quad k > 0.$

These properties require that the mean, μ , is constant and finite, that the variance, σ^2 , is constant and finite and that the covariance between y_t and y_{t-k} , γ_k , is a function of the time between the two points, k , and is not a function of time, t . Consider two snapshots of a time series which are s

periods apart, a situation which can be represented schematically as follows

$$\begin{array}{ccccccccccccccc}
 y_1, & y_2, & \cdots & y_s, & y_{s+1}, & \cdots & y_j, & y_{j+1}, & \cdots & y_{j+s}, & y_{j+s+1}, & \cdots \\
 \underbrace{\hspace{10em}} & & & & & & & & & & & & \\
 \text{Period 1 (Y}_1\text{)} & & & & & & & & & & & & \\
 & & & & & & & & & & & & \\
 \underbrace{\hspace{10em}} & & & & & & & & & & & & \\
 \text{Period 2 (Y}_2\text{)} & & & & & & & & & & & &
 \end{array}$$

Here Y_1 and Y_2 represent the time series of the two sub-periods. An implication of weak stationarity is that Y_1 and Y_2 are governed by the same parameters μ , σ^2 and γ_k .

Example 2.1 Stationary AR(1) Model

Consider the AR(1) process

$$y_t = \alpha + \rho y_{t-1} + u_t, \quad u_t \sim iid(0, \sigma^2),$$

with $|\rho| < 1$. This process is stationary since

$$\begin{aligned}
 \mu &= E[y_t] = \frac{\alpha}{1 - \rho} \\
 \sigma^2 &= E[(y_t - \mu)^2] = \frac{\sigma^2}{1 - \rho^2} \\
 \gamma_k &= E[(y_t - \mu)(y_{t-k} - \mu)] = \frac{\sigma^2 \rho^k}{1 - \rho^2}.
 \end{aligned}$$

□

Strict Stationarity

The variable y_t is strictly stationary if the joint distribution function $F(y_1, y_2, \dots, y_j)$ do not depend on t for all finite j . Strict stationarity requires that the joint distribution function of two time series s periods apart is invariant with respect to an arbitrary time shift. That is

$$F(y_1, y_2, \dots, y_j) = F(y_{1+s}, y_{2+s}, \dots, y_{j+s}).$$

As strict stationarity requires that all the moments of y_t , if they exist, are independent of t , it follows that higher-order moments such as

$$\begin{aligned}
 E[(y_t - \mu)(y_{t-k} - \mu)] &= E[(y_{t+s} - \mu)(y_{t+s-k} - \mu)] \\
 E[(y_t - \mu)(y_{t-k} - \mu)^2] &= E[(y_{t+s} - \mu)(y_{t+s-k} - \mu)^2] \\
 E[(y_t - \mu)^2(y_{t-k} - \mu)^2] &= E[(y_{t+s} - \mu)^2(y_{t+s-k} - \mu)^2],
 \end{aligned}$$

must be functions of k only.

Strict stationarity does not require the existence of the first two moments

of the joint distribution of y_t . For the special case in which the first two moments do exist and are finite, $\mu, \sigma^2 < \infty$, and the joint distribution function is a normal distribution, weak and strict stationarity are equivalent. In the case where the first two moments of the joint distribution do not exist, y_t can be strictly stationary, but not weakly stationary. An example is where y_t is *iid* with a Cauchy distribution, which is strictly stationary but has no finite moments and is therefore not weakly stationary. Another example is an IGARCH model discussed in Chapter 20, which is strictly stationary but not weakly stationary because the unconditional variance does not exist. An implication of the definition of stationarity is that if y_t is stationary then any function of a stationary process is also stationary, such as higher order terms y_t^2, y_t^3, y_t^4 .

Martingale Difference Sequence

A martingale difference sequence (mds) is defined in terms of its first conditional moment having the property

$$E_{t-1}[y_t] = E[y_t | y_{t-1}, y_{t-2}, \dots] = 0. \quad (2.1)$$

This condition shows that information at time $t-1$ cannot be used to forecast y_t . Two important properties of a mds arising from (2.1) are

$$\begin{aligned} \text{Property 1 :} \quad & E[y_t] = E[E_{t-1}[y_t]] = E[0] = 0 \\ \text{Property 2 :} \quad & E[E_{t-1}[y_t y_{t-k}]] = E[y_{t-k} E_{t-1}[y_t]] = E[y_{t-k} \times 0] = 0. \end{aligned}$$

The first property is that the unconditional mean of a mds is zero which follows by using the law of iterated expectations. The second property shows that a mds is uncorrelated with past values of y_t . The condition in (2.1) does not, however, rule out higher-order moment dependence.

Example 2.2 Nonlinear Time Series

Consider the nonlinear time series model

$$y_t = u_t u_{t-1}, \quad u_t \sim iid(0, \sigma^2).$$

The process y_t is a mds because

$$E_{t-1}[y_t] = E_{t-1}[u_t u_{t-1}] = E_{t-1}[u_t] u_{t-1} = 0,$$

since $E_{t-1}[u_t] = E[u_t] = 0$. The process y_t nonetheless exhibits dependence

in the higher order moments. For example

$$\begin{aligned}
\text{cov}[y_t^2, y_{t-1}^2] &= E[y_t^2 y_{t-1}^2] - E[y_t^2]E[y_{t-1}^2] \\
&= E[u_t^2 u_{t-1}^4 u_{t-2}^2] - E[u_t^2 u_{t-1}^2]E[u_{t-1}^2 u_{t-2}^2] \\
&= E[u_t^2]E[u_{t-1}^4]E[u_{t-2}^2] - E[u_t^2]E[u_{t-1}^2]^2 E[u_{t-2}^2] \\
&= \sigma^4(E[u_{t-1}^4] - \sigma^4) \neq 0.
\end{aligned}$$

□

Example 2.3 Autoregressive Conditional Heteroskedasticity

Consider the ARCH model from Example 1.8 in Chapter 1 given by

$$y_t = z_t \sqrt{\alpha_0 + \alpha_1 y_{t-1}^2}, \quad z_t \sim iid N(0, 1).$$

Now y_t is a mds because

$$E_{t-1}[y_t] = E_{t-1}\left[z_t \sqrt{\alpha_0 + \alpha_1 y_{t-1}^2}\right] = E_{t-1}[z_t] \sqrt{\alpha_0 + \alpha_1 y_{t-1}^2} = 0,$$

since $E_{t-1}[z_t] = 0$. The process y_t nonetheless exhibits dependence in the second moment because

$$E_{t-1}[y_t^2] = E_{t-1}[z_t^2(\alpha_0 + \alpha_1 y_{t-1}^2)] = E_{t-1}[z_t^2](\alpha_0 + \alpha_1 y_{t-1}^2) = \alpha_0 + \alpha_1 y_{t-1}^2,$$

by using the property $E_{t-1}[z_t^2] = E[z_t^2] = 1$. □

In contrast to the properties of stationary time series, a function of a mds is not necessarily a mds.

White Noise

For a process to be white noise its first and second unconditional moments must satisfy the following three properties

$$\begin{aligned}
\text{Property 1 :} & \quad E[y_t] = 0 \\
\text{Property 2 :} & \quad E[y_t^2] = \sigma^2 < \infty \\
\text{Property 3 :} & \quad E[y_t y_{t-k}] = 0, \quad k > 0.
\end{aligned}$$

White noise is a special case of a weakly stationary process with mean zero, constant and finite variance, σ^2 , and zero covariance between y_t and y_{t-k} . A mds with finite and constant variance is also a white noise process since the first two unconditional moments exist and the process is not correlated. If a mds has infinite variance, then it is not white noise. Similarly, a white noise process is not necessarily a mds, as demonstrated by the following example.

Example 2.4 Bilinear Time Series

Consider the bilinear time series model

$$y_t = u_t + \delta u_{t-1} u_{t-2}, \quad u_t \sim iid(0, \sigma^2),$$

where δ is a parameter. The process y_t is white noise since

$$\begin{aligned} E[y_t] &= E[u_t + \delta u_{t-1} u_{t-2}] = E[u_t] + \delta E[u_{t-1}]E[u_{t-2}] = 0, \\ E[y_t^2] &= E[(u_t + \delta u_{t-1} u_{t-2})^2] \\ &= E[u_t^2 + \delta^2 u_{t-1}^2 u_{t-2}^2 + 2\delta u_t u_{t-1} u_{t-2}] = \sigma^2(1 + \delta^2 \sigma^2) < \infty \\ E[y_t y_{t-k}] &= E[(u_t + \delta u_{t-1} u_{t-2})(u_{t-k} + \delta u_{t-1-k} u_{t-2-k})] \\ &= E[u_t u_{t-k} + \delta u_{t-1} u_{t-2} u_{t-k} \\ &\quad + \delta u_t u_{t-1} u_{t-2-k} + \delta^2 u_{t-1} u_{t-2} u_{t-1-k} u_{t-2-k}] = 0, \end{aligned}$$

where the last step follows from the property that every term contains at least two disturbances occurring at different points in time. However, y_t is not a mds because

$$\begin{aligned} E_{t-1}[y_t] &= E_{t-1}[u_t + \delta u_{t-1} u_{t-2}] = E_{t-1}[u_t] + E_{t-1}[\delta u_{t-1} u_{t-2}] \\ &= \delta u_{t-1} u_{t-2} \neq 0. \end{aligned}$$

□

Mixing

As martingale difference sequences are uncorrelated, it is important also to consider alternative processes that exhibit autocorrelation. Consider two sub-periods of a time series s periods apart

$$\begin{array}{ccc} \text{First sub-period} & & \text{Second sub-period} \\ \underbrace{\dots, y_{t-2}, y_{t-1}, y_t}_{Y_{-\infty}^t} & y_{t+1}, y_{t+2}, \dots, y_{t+s-1} & \underbrace{y_{t+s}, y_{t+s+1}, y_{t+s+2}, \dots}_{Y_{t+s}^\infty} \end{array}$$

where $Y_t^s = (y_t, y_{t+1}, \dots, y_s)$. If

$$\text{cov}(g(Y_{-\infty}^t), h(Y_{t+s}^\infty)) \rightarrow 0 \text{ as } s \rightarrow \infty, \quad (2.2)$$

where $g(\cdot)$ and $h(\cdot)$ are arbitrary functions, then as $Y_{-\infty}^t$ and Y_{t+s}^∞ become more widely separated in time, they behave like independent sets of random variables. A process satisfying (2.2) is known as mixing (technically α -mixing or strong mixing). The concepts of strong stationarity and mixing have the convenient property that if they apply to y_t then they also apply to functions of y_t . A more formal treatment of mixing is provided by White (1984)

An *iid* process is mixing because all the covariances are zero and the mixing condition (2.2) is satisfied trivially. As will become apparent from the

results for stationary time series models presented in Chapter 13, a $\text{MA}(q)$ process with *iid* disturbances is mixing because it has finite dependence so that condition (2.2) is satisfied for $k > q$. Provided that the additional assumption is made that u_t in Example 2.1 is normally distributed, the $\text{AR}(1)$ process is mixing since the covariance between y_t and y_{t-k} decays at an exponential rate as k increases, which implies that (2.2) is satisfied. If u_t does not have a continuous distribution then y_t may no longer be mixing (Andrews, 1984).

2.2.2 Weak Law of Large Numbers

The stochastic time series models discussed in the previous section are defined in terms of probability distributions with moments defined in terms of the parameters of these distributions. As maximum likelihood estimators are sample statistics of the data in samples of size T , it is of interest to identify the relationship between the population parameters and the sample statistics as $T \rightarrow \infty$.

Let $\{y_1, y_2, \dots, y_T\}$ represent a set of T *iid* random variables from a distribution with a finite mean μ . Consider the statistic based on the sample mean

$$\bar{y} = \frac{1}{T} \sum_{t=1}^T y_t. \quad (2.3)$$

The weak law of large numbers is about determining what happens to $\bar{y}$ as the sample size T increases without limit, $T \rightarrow \infty$.

Example 2.5 Exponential Distribution

Figure 2.1 gives the results of a simulation experiment from computing sample means of progressively larger samples of size $T = 1, 2, \dots, 500$, comprising *iid* draws from the exponential distribution

$$f(y; \mu) = \frac{1}{\mu} \exp \left[-\frac{y}{\mu} \right], \quad y > 0,$$

with population mean $\mu = 5$. For relatively small sample sizes, $\bar{y}$ is quite volatile, but settles down as T increases. The distance between $\bar{y}$ and μ eventually lies within a ‘small’ band of length $r = 0.2$, that is $|\bar{y} - \mu| < r$, as represented by the dotted lines. $\square$

An important feature of Example 2.5 is that $\bar{y}$ is a random variable, whose value in any single sample need not necessarily equal μ in any deterministic

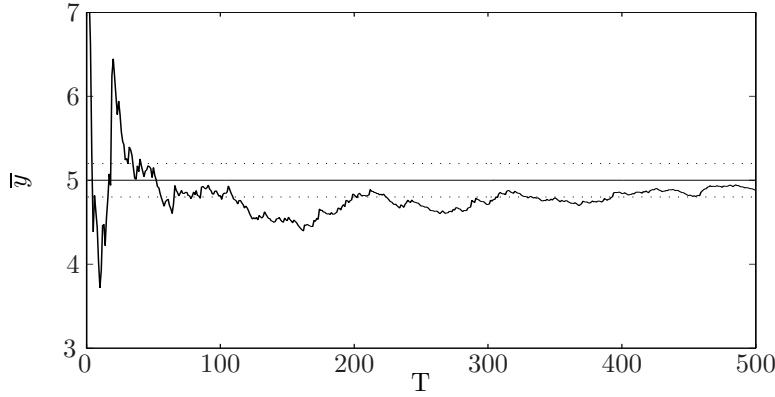


Figure 2.1 The Weak Law of Large Numbers for sample means based on progressively increasing sample sizes drawn from the exponential distribution with mean $\mu = 5$. The dotted lines represent $\mu \pm r$ with $r = 0.2$.

sense, but, rather, $\bar{y}$ is simply ‘close enough’ to the value of μ with probability approaching 1 as $T \rightarrow \infty$. This property is written formally as

$$\lim_{T \rightarrow \infty} \Pr(|\bar{y} - \mu| < r) = 1, \text{ for any } r > 0,$$

or, more compactly, as $\text{plim}(\bar{y}) = \mu$, where the notation plim represents the limit in a probability sense. This is the Weak Law of Large Numbers (WLLN), which states that the sample mean converges in probability to the population mean

$$\frac{1}{T} \sum_{t=1}^T y_t \xrightarrow{p} E[y_t] = \mu, \quad (2.4)$$

where p denotes the convergence in probability or plim . This result also extends to higher order moments

$$\frac{1}{T} \sum_{t=1}^T y_t^i \xrightarrow{p} E[y_t^i], \quad i > 0. \quad (2.5)$$

A necessary condition needed for the weak law of large numbers to be satisfied is that μ is finite (Stuart and Ord, 1994, p.310). A sufficient condition is that $E[\bar{y}] \rightarrow \mu$ and $\text{var}(\bar{y}) \rightarrow 0$ as $T \rightarrow \infty$, so that the sampling distribution of $\bar{y}$ converges to a degenerate distribution with all its probability mass concentrated at the population mean μ .

Example 2.6 Uniform Distribution

Assume that y has a uniform distribution

$$f(y) = 1, \quad -0.5 < y < 0.5.$$

The first four population moments are

$$\int_{-0.5}^{0.5} y f(y) dy = 0, \quad \int_{-0.5}^{0.5} y^2 f(y) dy = \frac{1}{12}, \quad \int_{-0.5}^{0.5} y^3 f(y) dy = 0, \quad \int_{-0.5}^{0.5} y^4 f(y) dy = \frac{1}{80}.$$

Properties of the moments of y simulated from samples of size T drawn from the uniform distribution $(-0.5, 0.5)$. The number of replications is 50000 and the moments have been scaled by 10000.

T	$\frac{1}{T} \sum_{t=1}^T y_t$		$\frac{1}{T} \sum_{t=1}^T y_t^2$		$\frac{1}{T} \sum_{t=1}^T y_t^3$		$\frac{1}{T} \sum_{t=1}^T y_t^4$	
	Mean	Var.	Mean	Var.	Mean	Var.	Mean	Var.
50	-1.380	16.828	833.960	1.115	-0.250	0.450	125.170	0.056
100	0.000	8.384	833.605	0.555	-0.078	0.224	125.091	0.028
200	0.297	4.207	833.499	0.276	0.000	0.112	125.049	0.014
400	-0.167	2.079	833.460	0.139	-0.037	0.056	125.026	0.007
800	0.106	1.045	833.347	0.070	0.000	0.028	125.004	0.003

Table 2.6 gives the mean and the variance of simulated samples of size $T = \{50, 100, 200, 400, 800\}$ for the first four moments given in equation (2.5). The results demonstrate the two key properties of the weak law of large numbers: the means of the sample moments converge to their population means and their variances all converge to zero, with the variance roughly halving as T is doubled. $\square$

Some important properties of plims are as follows. Let $\bar{y}_1$ and $\bar{y}_2$ be the means of two samples of size T , from distributions with respective population means, μ_1 and μ_2 , and let $c(\cdot)$ be a continuous function that is not dependent on T , then

$$\text{Property 1 :} \quad \text{plim}(\bar{y}_1 \pm \bar{y}_2) = \text{plim}(\bar{y}_1) \pm \text{plim}(\bar{y}_2) = \mu_1 \pm \mu_2$$

$$\text{Property 2 :} \quad \text{plim}(\bar{y}_1 \bar{y}_2) = \text{plim}(\bar{y}_1) \text{plim}(\bar{y}_2) = \mu_1 \mu_2$$

$$\text{Property 3 :} \quad \text{plim}\left(\frac{\bar{y}_1}{\bar{y}_2}\right) = \frac{\text{plim}(\bar{y}_1)}{\text{plim}(\bar{y}_2)} = \frac{\mu_1}{\mu_2} \quad (\mu_2 \neq 0)$$

$$\text{Property 4 :} \quad \text{plim } c(\bar{y}) = c(\text{plim}(\bar{y})).$$

Property 4 is known as Slutsky's theorem (see also Exercise 3). These results

generalize to the vector case, where the plim is taken with respect to each element separately.

The WLLN holds under weaker conditions than the assumption of an *iid* process. Assuming only that $\text{var}(y_t) < \infty$ for all t , the variance of $\bar{y}$ can always be written as

$$\text{var}(\bar{y}) = \frac{1}{T^2} \sum_{t=1}^T \sum_{s=1}^T \text{cov}(y_t, y_s) = \frac{1}{T^2} \sum_{t=1}^T \text{var}(y_t) + 2 \frac{1}{T^2} \sum_{s=1}^{T-1} \sum_{t=s+1}^T \text{cov}(y_t, y_{t-s}).$$

If y_t is weakly stationary then this simplifies to

$$\text{var}(\bar{y}) = \frac{1}{T} \gamma_0 + 2 \frac{1}{T} \sum_{s=1}^T \left(1 - \frac{s}{T}\right) \gamma_s, \quad (2.6)$$

where $\gamma_s = \text{cov}(y_t, y_{t-s})$ are the autocovariances of y_t for $s = 0, 1, 2, \dots$.

If y_t is either *iid* or a martingale difference sequence or white noise, then $\gamma_s = 0$ for all $s \geq 1$. In that case (2.6) simplifies to

$$\text{var}(\bar{y}) = \frac{1}{T} \gamma_0 \rightarrow 0$$

as $T \rightarrow \infty$ and the WLLN holds. If y_t is autocorrelated then a sufficient condition for the WLLN is that $|\gamma_s| \rightarrow 0$ as $s \rightarrow \infty$. To show why this works, consider the second term on the right hand side of (2.6). It follows from the triangle inequality that

$$\begin{aligned} \left| \frac{1}{T} \sum_{s=1}^T \left(1 - \frac{s}{T}\right) \gamma_s \right| &\leq \frac{1}{T} \sum_{s=1}^T \left(1 - \frac{s}{T}\right) |\gamma_s| \\ &\leq \frac{1}{T} \sum_{s=1}^T |\gamma_s| \quad \text{since } 1 - s/T < 1 \\ &\rightarrow 0 \text{ as } T \rightarrow \infty, \end{aligned}$$

where the last step uses Cesaro summation.¹ This implies that $\text{var}(\bar{y})$ given in (2.6) disappears as $T \rightarrow \infty$. Thus, any weakly stationary time series whose autocovariances satisfy $|\gamma_s| \rightarrow 0$ as $s \rightarrow \infty$ will obey the WLLN (2.4). If y_t is weakly stationary and strong mixing, then $|\gamma_s| \rightarrow 0$ as $s \rightarrow \infty$ follows by definition, so the WLLN applies to this general class of processes as well.

Example 2.7 WLLN for an AR(1) Model

In the stationary AR(1) model from Example 2.1, since $|\rho| < 1$ it follows

¹ If $a_t \rightarrow a$ as $t \rightarrow \infty$ then $T^{-1} \sum_{t=1}^T a_t \rightarrow a$ as $T \rightarrow \infty$.

that

$$\gamma_s = \frac{\sigma^2 \rho^s}{1 - \rho^2},$$

so that the condition $|\gamma_s| \rightarrow 0$ as $s \rightarrow \infty$ is clearly satisfied. This shows the WLLN applies to a stationary AR(1) process. $\square$

2.2.3 Rates of Convergence

The weak law of large numbers in (2.4) involves computing statistics based on averaging random variables over a sample of size T . Establishing many of the results of the maximum likelihood estimator requires choosing the correct scaling factor to ensure that the relevant statistics have non-degenerate distributions.

Example 2.8 Linear Regression with Stochastic Regressors

Consider the linear regression model

$$y_t = \beta x_t + u_t, \quad u_t \sim iid N(0, \sigma_u^2)$$

where x_t is a *iid* drawing from the uniform distribution on the interval $(-0.5, 0.5)$ with variance σ_x^2 and x_t and u_t are independent. It follows that $E[x_t u_t] = 0$. The maximum likelihood estimator of β is

$$\hat{\beta} = \left[\sum_{t=1}^T x_t^2 \right]^{-1} \sum_{t=1}^T x_t y_t = \beta + \left[\sum_{t=1}^T x_t^2 \right]^{-1} \sum_{t=1}^T x_t u_t,$$

where the last term is obtained by substituting for y_t . This expression shows that the relevant moments to consider are $\sum_{t=1}^T x_t u_t$ and $\sum_{t=1}^T x_t^2$. The appropriate scaling of the first moment to ensure that it has a non-degenerate distribution follows from

$$\begin{aligned} E\left[T^{-k} \sum_{t=1}^T x_t u_t\right] &= 0 \\ \text{var}\left(T^{-k} \sum_{t=1}^T x_t u_t\right) &= T^{-2k} \text{var}\left(\sum_{t=1}^T x_t u_t\right) = T^{1-2k} \sigma_u^2 \sigma_x^2, \end{aligned}$$

which hold for any k . Consequently the appropriate choice of scaling factor is $k = 1/2$ because $T^{-1/2}$ stabilizes the variance and thus prevents it approaching 0 ($k > 1/2$) or ∞ ($k < 1/2$). This property is demonstrated in Table 2.2.3, which gives simulated moments for alternative scale factors, where $\beta = 1$, $\sigma_u^2 = 2$ and $\sigma_x^2 = 1/12$. The variances show that only with the

scale factor $T^{-1/2}$ does $\sum_{t=1}^T x_t u_t$ have a non-degenerate distribution with mean converging to 0 and variance converging to

$$\text{var}(x_t u_t) = \text{var}(u_t) \times \text{var}(x_t) = 2 \times \frac{1}{12} = 0.167, .$$

Since

$$\frac{1}{T} \sum_{t=1}^T x_t^2 \xrightarrow{p} \sigma_x^2,$$

by the WLLN, it follows that the distribution of $\sqrt{T}(\hat{\beta} - \beta)$ is non-degenerate because the variance of both terms on the right hand side of

$$\sqrt{T}(\hat{\beta} - \beta) = \left[\frac{1}{T} \sum_{t=1}^T x_t^2 \right]^{-1} \left[\frac{1}{\sqrt{T}} \sum_{t=1}^T x_t u_t \right],$$

converge to finite non-zero values. $\square$

Simulation properties of the moments of the linear regression model using alternative scale factors. The parameters are $\theta = \{\beta = 1.0, \sigma_u^2 = 2.0\}$, the number of replications is 50000, u_t is drawn from $N(0, 2)$ and the stochastic regressor x_t is drawn from a uniform distribution with support $(-0.5, 0.5)$.

T	$\frac{1}{T^{1/4}} \sum_{t=1}^T x_t u_t$		$\frac{1}{T^{1/2}} \sum_{t=1}^T x_t u_t$		$\frac{1}{T^{3/4}} \sum_{t=1}^T x_t u_t$		$\frac{1}{T} \sum_{t=1}^T x_t u_t$	
	Mean	Var.	Mean	Var.	Mean	Var.	Mean	Var.
50	-0.001	1.177	0.000	0.166	0.000	0.024	0.000	0.003
100	-0.007	1.670	-0.002	0.167	-0.001	0.017	0.000	0.002
200	-0.014	2.378	-0.004	0.168	-0.001	0.012	0.000	0.001
400	-0.001	3.373	0.000	0.169	0.000	0.008	0.000	0.000
800	0.007	4.753	0.001	0.168	0.000	0.006	0.000	0.000

Determining the correct scaling factors for derivatives of the log-likelihood function is important to establishing the asymptotic distribution of the maximum likelihood estimator in Section 2.5.2. The following example highlights this point.

Example 2.9 Higher-Order Derivatives

The log-likelihood function associated with an *iid* sample $\{y_1, y_2, \dots, y_T\}$

from the exponential distribution is

$$\ln L_T(\theta) = \ln \theta - \frac{\theta}{T} \sum_{t=1}^T y_t.$$

The first four derivatives are

$$\begin{aligned} \frac{d \ln L_T(\theta)}{d\theta} &= \theta^{-1} - \frac{1}{T} \sum_{t=1}^T y_t & \frac{d^2 \ln L_T(\theta)}{d\theta^2} &= -\theta^{-2} \\ \frac{d^3 \ln L_T(\theta)}{d\theta^3} &= 2\theta^{-3} & \frac{d^4 \ln L_T(\theta)}{d\theta^4} &= -6\theta^{-4}. \end{aligned}$$

The first derivative $G_T(\theta) = \theta^{-1} - \frac{1}{T} \sum_{t=1}^T y_t$ is an average of *iid* random variables, $g_t(\theta) = \theta^{-1} - y_t$. The scaled first derivative

$$\sqrt{T}G_T(\theta) = \frac{1}{\sqrt{T}} \sum_{t=1}^T g_t(\theta),$$

has zero mean and finite variance because

$$\text{var}(\sqrt{T}G_T(\theta)) = \frac{1}{T} \sum_{t=1}^T \text{var}(\theta^{-1} - y_t) = \frac{1}{T} \sum_{t=1}^T \theta^{-2} = \theta^{-2},$$

by using the *iid* assumption and the fact that $\text{E}[(y_t - \theta^{-1})^2] = \theta^{-2}$ for the exponential distribution. All the other derivatives already have finite limits as they are independent of T . $\square$

2.2.4 Central Limit Theorems

The previous section established the appropriate scaling factor needed to ensure that a statistic has a non-degenerate distribution. The aim of this section is to identify the form of this distribution as $T \rightarrow \infty$, referred to as the asymptotic distribution. The results are established in a series of four central limit theorems.

Lindeberg-Levy Central Limit Theorem

Let $\{y_1, y_2, \dots, y_T\}$ represent a set of T *iid* random variables from a distribution with finite mean μ and finite variance $\sigma^2 > 0$. The Lindeberg-Levy central limit theorem for the scalar case states that

$$\sqrt{T}(\bar{y} - \mu) \xrightarrow{d} N(0, \sigma^2), \quad (2.7)$$

where $\xrightarrow{d}$ represents convergence of the distribution as $T \rightarrow \infty$. In terms of

standardized random variables, the central limit theorem is

$$z = \sqrt{T} \frac{(\bar{y} - \mu)}{\sigma} \xrightarrow{d} N(0, 1). \quad (2.8)$$

Alternatively, the asymptotic distribution is given by rearranging (2.7) as

$$\bar{y} \overset{a}{\sim} N\left(\mu, \frac{\sigma^2}{T}\right), \quad (2.9)$$

where $\overset{a}{\sim}$ signifies convergence to the asymptotic distribution. The fundamental difference between (2.7) and (2.9) is that the former represents a normal distribution with zero mean and constant variance in the limit, whereas the latter represents a normal distribution with mean μ , but with a variance that approaches zero as T grows, resulting in all of its mass concentrated at μ in the limit.

Example 2.10 Uniform Distribution

Let $\{y_1, y_2, \dots, y_T\}$ represent a set of T *iid* random variables from the uniform distribution

$$f(y) = 1, \quad 0 < y < 1.$$

The conditions of the Lindeberg-Levy central limit theorem are satisfied, because the random variables are *iid* with finite mean and variance given by $\mu = 1/2$ and $\sigma^2 = 1/12$, respectively. Based on 5,000 draws, the sampling distribution of

$$z = \sqrt{T} \frac{(\bar{y} - \mu)}{\sigma} = \sqrt{T} \frac{(\bar{y} - 1/2)}{\sqrt{12}},$$

for samples of size $T = 2$ and $T = 10$, are shown in panels (a) and (b) of Figure 2.2 respectively. Despite the population distribution being non-normal, the sampling distributions approach the standardized normal distribution very quickly. Also shown are the corresponding asymptotic distributions of $\bar{y}$ in panels (c) and (d), which become more compact around $\mu = 1/2$ as T increases. $\square$

Example 2.11 Linear Regression with *iid* Regressors

Assume that the joint distribution of y_t and x_t is *iid* and

$$y_t = \beta x_t + u_t, \quad u_t \sim iid(0, \sigma_u^2),$$

where $E[u_t|x_t] = 0$ and $E[u_t^2|x_t] = \sigma_u^2$. From Example 2.8 the least squares of $\hat{\beta}$ is expressed as

$$\sqrt{T}(\hat{\beta} - \beta) = \left[\frac{1}{T} \sum_{t=1}^T x_t^2 \right]^{-1} \left[\frac{1}{\sqrt{T}} \sum_{t=1}^T x_t u_t \right].$$

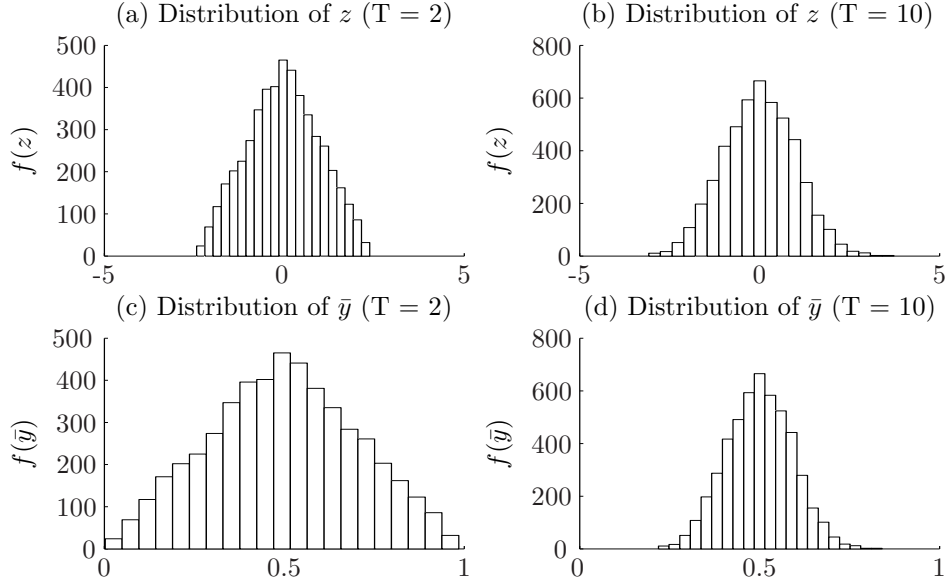


Figure 2.2 Demonstration of the Lindeberg-Levy Central Limit Theorem. Population distribution is the uniform distribution.

To establish the asymptotic distribution of $\hat{\beta}$ the following three results are required

$$\frac{1}{T} \sum_{t=1}^T x_t^2 \xrightarrow{p} \sigma_x^2, \quad \frac{1}{\sqrt{T}} \sum_{t=1}^T x_t u_t \xrightarrow{d} N(0, \sigma_u^2 \sigma_x^2),$$

where the first result follows from the WLLN, and the second result is an application of the Lindeberg-Levy central limit theorem. Combining these three results yields

$$\sqrt{T}(\hat{\beta} - \beta) \xrightarrow{d} N\left(0, \frac{\sigma_u^2}{\sigma_x^2}\right).$$

This is the usual expression for the asymptotic distribution of the maximum likelihood (least squares) estimator. $\square$

The Lindeberg-Levy central limit theorem generalizes to the case where y_t is a vector with mean μ and covariance matrix V

$$\sqrt{T}(\bar{y} - \mu) \xrightarrow{d} N(0, V). \quad (2.10)$$

Lindeberg-Feller Central Limit Theorem

The Lindeberg-Feller central limit theorem is applicable to models based on independent and non-identically distributed random variables, in which

y_t has time-varying mean μ_t and time-varying covariance matrix V_t . For the scalar case, let $\{y_1, y_2, \dots, y_T\}$ represent a set of T independent and non-identically distributed random variables from a distribution with finite time-varying means $E[y_t] = \mu_t < \infty$, finite time-varying variances $\text{var}(y_t) = \sigma_t^2 < \infty$ and finite higher-order moments. The Lindeberg-Feller central limit theorem gives necessary and sufficient conditions for

$$\sqrt{T} \left(\frac{\bar{y} - \bar{\mu}}{\bar{\sigma}} \right) \xrightarrow{d} N(0, 1), \quad (2.11)$$

where

$$\bar{\mu} = \frac{1}{T} \sum_{t=1}^T \mu_t, \quad \bar{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T \sigma_t^2. \quad (2.12)$$

A sufficient condition for the Lindeberg-Feller central limit theorem is given by

$$E[|y_t - \mu_t|^{2+\delta}] < \infty, \quad \delta > 0, \quad (2.13)$$

uniformly in t . This is known as the Lyapunov condition, which operates on moments higher than the second moment. This requirement is in fact a stricter condition than is needed to satisfy this theorem, but it is more intuitive and tends to be an easier condition to demonstrate than the conditions initially proposed by Lindeberg and Feller. Although this condition is applicable to all moments marginally higher than the second, namely $2 + \delta$, considering the first integer moment to which the condition applies, namely the third moment by setting $\delta = 1$ in (2.13), is of practical interest. The condition now becomes

$$E[|y_t - \mu_t|^3] < \infty, \quad (2.14)$$

which represents a restriction on the standardized third moment, or skewness, of y_t .

Example 2.12 Bernoulli Distribution

Let $\{y_1, y_2, \dots, y_T\}$ represent a set of T independent random variables with time-varying probabilities θ_t from a Bernoulli distribution

$$f(y; \theta_t) = \theta_t^y (1 - \theta_t)^{1-y}, \quad 0 < \theta_t < 1.$$

From the properties of the Bernoulli distribution, the mean and the variance are time-varying since $\mu_t = \theta_t$ and $\sigma_t^2 = \theta_t (1 - \theta_t)$. As $0 < \theta_t < 1$ then

$$E[|y_t - \mu_t|^3] = \theta_t (1 - \theta_t)^3 + (1 - \theta_t) \theta_t^3 = \sigma_t^2 \left((1 - \theta_t)^2 - \theta_t^2 \right) \leq \sigma_t^2,$$

so the third moment is bounded. $\square$

Example 2.13 Linear Regression with Bounded Fixed Regressors

Consider the linear regression model

$$y_t = \beta x_t + u_t, \quad u_t \sim iid(0, \sigma_u^2),$$

where u_t has finite third moment $E[u_t^3] = \kappa_3$ and x_t is a uniformly bounded fixed regressor, such as a constant, a level shift dummy variable or seasonal dummy variables.² From Example 2.8 the least squares estimator of $\hat{\beta}$ is

$$\sqrt{T}(\hat{\beta} - \beta) = \left[\frac{1}{T} \sum_{t=1}^T x_t^2 \right]^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^T x_t u_t.$$

The Lindeberg-Feller central limit theorem based on the Lyapunov condition applies to the product $x_t u_t$, because the terms are independent for all t , with mean, variance and uniformly bounded third moment, respectively,

$$\mu = 0, \quad \bar{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T \text{var}(x_t u_t) = \sigma_u^2 \frac{1}{T} \sum_{t=1}^T x_t^2, \quad E[x_t^3 u_t^3] = x_t^3 \kappa_3 < \infty.$$

Substituting into (2.11) gives

$$\frac{(\sum_{t=1}^T x_t^2)^{1/2}}{\sigma_u} (\hat{\beta} - \beta) = \sqrt{T} \frac{T^{-1} \sum_{t=1}^T x_t u_t}{\bar{\sigma}} \xrightarrow{d} N(0, 1).$$

□

As in the case of the Lindeberg-Levy central limit theorem, the Lindeberg-Feller central limit theorem generalizes to independent and non-identically distributed vector random variables with time-varying vector mean μ_t and time-varying positive definite covariance matrix V_t . The theorem states that

$$\sqrt{T} \bar{V}^{-1/2} (\bar{y} - \bar{\mu}) \xrightarrow{d} N(0, I), \quad (2.15)$$

where

$$\bar{\mu} = \frac{1}{T} \sum_{t=1}^T \mu_t, \quad \bar{V} = \frac{1}{T} \sum_{t=1}^T V_t, \quad (2.16)$$

and $\bar{V}^{-1/2}$ represents the square root of the matrix $\bar{V}$.

Martingale Difference Central Limit Theorem

The martingale difference central limit theorem is essentially the Lindberg-Levy central limit theorem, but with the assumption that $y_t = \{y_1, y_2, \dots, y_T\}$ represents a set of T *iid* random variables being replaced with the more

² An example of a fixed regressor that is not uniformly bounded in t is a time trend $x_t = t$.

general assumption that y_t is a martingale difference sequence. If y_t is a martingale difference sequence with mean and variance

$$\bar{y} = \frac{1}{T} \sum_{t=1}^T y_t, \quad \bar{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T \sigma_t^2,$$

and provided that higher order moments are bounded,

$$E[|y_t|^{2+\delta}] < \infty, \quad \delta > 0, \quad (2.17)$$

and

$$\frac{1}{T} \sum_{t=1}^T y_t^2 - \bar{\sigma}^2 \xrightarrow{p} 0, \quad (2.18)$$

then the martingale difference central limit theorem states

$$\sqrt{T} \left(\frac{\bar{y}}{\bar{\sigma}} \right) \xrightarrow{d} N(0, 1). \quad (2.19)$$

The martingale difference property weakens the *iid* assumption, but the assumptions that the sample variance must consistently estimate the average variance and the boundedness of higher moments in (2.17) are stronger than those required for the Lindeberg-Levy central limit theorem.

Example 2.14 Linear AR(1) Model

Consider the autoregressive model from Example 1.5 in Chapter 1, where for convenience the sample contains $T + 1$ observations, $y_t = \{y_0, y_1, \dots, y_T\}$.

$$y_t = \rho y_{t-1} + u_t, \quad u_t \sim iid(0, \sigma^2),$$

with finite fourth moment $E[u_t^4] = \kappa_4 < \infty$ and $|\rho| < 1$. The least squares estimator of $\hat{\rho}$ is

$$\hat{\rho} = \frac{\sum_{t=1}^T y_t y_{t-1}}{\sum_{t=1}^T y_{t-1}^2}.$$

Rearranging and introducing the scale factor $\sqrt{T}$ gives

$$\sqrt{T}(\hat{\rho} - \rho) = \left[\frac{1}{T} \sum_{t=1}^T y_{t-1}^2 \right]^{-1} \left[\frac{1}{\sqrt{T}} \sum_{t=1}^T u_t y_{t-1} \right].$$

To use the mds central limit theorem to find the asymptotic distribution of $\hat{\rho}$, it is necessary to establish that $x_t u_t$ satisfies the conditions of the theorem and also that $T^{-1} \sum_{t=2}^T y_{t-1}^2$ satisfies the WLLN.

The product $u_t y_{t-1}$ is a mds because

$$E_{t-1}[u_t y_{t-1}] = E_{t-1}[u_t] y_{t-1} = 0,$$

since $E_{t-1}[u_t] = 0$. To establish that the conditions of the mds central limit theorem are satisfied, define

$$\begin{aligned}\bar{\mu} &= \frac{1}{T} \sum_{t=1}^T u_t y_{t-1} \\ \bar{\sigma}^2 &= \frac{1}{T} \sum_{t=1}^T \sigma_t^2 = \frac{1}{T} \sum_{t=1}^T \text{var}(u_t y_{t-1}) = \frac{1}{T} \sum_{t=1}^T \frac{\sigma^4}{1 - \rho^2} = \frac{\sigma^4}{1 - \rho^2}.\end{aligned}$$

To establish the boundedness condition in (2.17), choose $\delta = 2$, so that

$$E[|u_t y_{t-1}|^4] = E[u_t^4] E[y_{t-1}^4] < \infty,$$

because $\kappa_4 < \infty$ and it can be shown that $E[y_{t-1}^4] < \infty$ provided that $|\rho| < 1$. To establish (2.18), write

$$\frac{1}{T} \sum_{t=1}^T u_t^2 y_{t-1}^2 = \frac{1}{T} \sum_{t=1}^T (u_t^2 - \sigma^2) y_{t-1}^2 + \sigma^2 \frac{1}{T} \sum_{t=1}^T y_{t-1}^2.$$

The first term is the sample mean of a mds, which has mean zero, so the weak law of large numbers gives

$$\frac{1}{T} \sum_{t=1}^T (u_t^2 - \sigma^2) y_{t-1}^2 \xrightarrow{p} 0.$$

The second term is the sample mean of a stationary process and the weak law of large numbers gives

$$\frac{1}{T} \sum_{t=1}^T y_{t-1}^2 \xrightarrow{p} E[y_{t-1}^2] = \frac{\sigma^2}{1 - \rho^2}.$$

Thus, as required by (2.18),

$$\frac{1}{T} \sum_{t=1}^T u_t^2 y_{t-1}^2 \xrightarrow{p} \frac{\sigma^4}{1 - \rho^2}.$$

Therefore, from the statement of the mds central limit theorem in (2.19) it follows that

$$\sqrt{T} \bar{y} = \frac{1}{\sqrt{T}} \sum_{t=1}^T u_t y_{t-1} \xrightarrow{d} N\left(0, \frac{\sigma^4}{1 - \rho^2}\right).$$

The asymptotic distribution of $\hat{\rho}$ is therefore

$$\begin{aligned}\sqrt{T}(\hat{\rho} - \rho) &\xrightarrow{d} \frac{\sigma^2}{1 - \rho^2} \times N\left(0, \frac{\sigma^4}{1 - \rho^2}\right) \\ &= N(0, 1 - \rho^2).\end{aligned}$$

□

The martingale difference sequence central limit theorem also applies to vector processes with covariance matrix V_t

$$\sqrt{T}\bar{\mu} \xrightarrow{d} N(0, \bar{V}), \quad (2.20)$$

where

$$\bar{V} = \frac{1}{T} \sum_{t=1}^T V_t.$$

Mixing Central Limit Theorem

As will become apparent in Chapter 9, in some situations it is necessary to have a central limit theorem that applies for autocorrelated processes. This is particularly pertinent to situations in which models do not completely specify the dynamics of the dependent variable.

If y_t has zero mean, $E|y_t|^r < \infty$ uniformly in t for some $r > 2$, and y_t is mixing at a sufficiently fast rate then the following central limit theorem applies

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T y_t \xrightarrow{d} N(0, J), \quad (2.21)$$

where

$$J = \lim_{T \rightarrow \infty} \frac{1}{T} E \left[\left(\sum_{t=1}^T y_t \right)^2 \right],$$

assuming this limit exists. If y_t is also weakly stationary, the expression for J simplifies to

$$\begin{aligned}J &= E[y_t^2] + 2 \sum_{j=1}^{\infty} E[y_t y_{t-j}] \\ &= \text{var}(y_t) + 2 \sum_{j=1}^{\infty} \text{cov}(y_t, y_{t-j}),\end{aligned} \quad (2.22)$$

which shows that the asymptotic variance of the sample mean depends on

the variance and all autocovariances of y_t . See Theorem 5.19 of White (1984) for further details of the mixing central limit theorem.

Example 2.15 Sample Moments of an AR(1) Model

Consider the AR(1) model

$$y_t = \rho y_{t-1} + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

where $|\rho| < 1$. The asymptotic distribution of the sample mean and variance of y_t are obtained as follows. Since y_t is stationary, mixing, has mean zero and all moments finite (by normality), the mixing central limit theorem in (2.21) applies to the standardized sample mean $\sqrt{T}\bar{y} = T^{-1/2} \sum_{t=1}^T y_t$ with variance given in (2.22). In the case of the sample variance, since y_t has zero mean, an estimator of its variance $\sigma^2 / (1 - \rho^2)$ is $T^{-1} \sum_{t=1}^T y_t^2$. The function

$$z_t = y_t^2 - \frac{\sigma^2}{1 - \rho^2},$$

has mean zero and inherits stationarity and mixing from y_t , so that

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T \left(y_t^2 - \frac{\sigma^2}{1 - \rho^2} \right) \xrightarrow{d} N(0, J_2),$$

where

$$J_2 = \text{var}(z_t) + 2 \sum_{j=1}^{\infty} \text{cov}(z_t, z_{t-j}),$$

demonstrating that the sample variance is also asymptotically normal. $\square$

2.3 Regularity Conditions

This section sets out a number of assumptions, known as regularity conditions, that are used in the derivation of the properties of the maximum likelihood estimator. Let the true population parameter value be represented by θ_0 and assume that the distribution $f(y; \theta)$ is specified correctly. The following regularity conditions apply to *iid*, stationary, mds and white noise processes as discussed in Section 2.2.1. For simplicity, many of the regularity conditions are presented for the *iid* case.

R1: Existence

The expectation

$$E[\ln f(y_t; \theta)] = \int_{-\infty}^{\infty} \ln f(y_t; \theta) f(y_t; \theta) dy_t, \quad (2.23)$$

exists.

R2: Convergence

The log-likelihood function converges in probability to its expectation

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta) \xrightarrow{P} E[\ln f(y_t; \theta)] , \quad (2.24)$$

uniformly in θ .

R3: Continuity

The log-likelihood function, $\ln L_T(\theta)$, is continuous in θ .

R4: Differentiability

The log-likelihood function, $\ln L_T(\theta)$, is at least twice continuously differentiable in an open interval around θ_0 .

R5: Interchangeability

The order of differentiation and integration of $\ln L_T(\theta)$ is interchangeable.

Condition R1 is a statement of the existence of the population log-likelihood function. Condition R2 is a statement of how the sample log-likelihood function converges to the population value by virtue of the WLLN, provided that this expectation exists in the first place, as given by the existence condition (R1). The continuity condition (R3) is a necessary condition for the differentiability condition (R4). The requirement that the log-likelihood function is at least twice differentiable naturally arises from the discussion in Chapter 1 where the first two derivatives are used to derive the maximum likelihood estimator and establish that a maximum is reached. Even when the likelihood is not differentiable everywhere, the maximum likelihood estimator can, in some instances, still be obtained. An example is given by the Laplace distribution in which the median is the maximum likelihood estimator (see Section 6.6.1 of Chapter 6). Finally, the interchangeability condition (R5) is used in the derivation of many of the properties of the maximum likelihood estimator.

Example 2.16 Likelihood Function of the Normal Distribution

Assume that y has a normal distribution with unknown mean $\theta = \{\mu\}$ and known variance σ_0^2

$$f(y; \theta) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp \left[-\frac{(y - \mu)^2}{2\sigma_0^2} \right] .$$

If the population parameter is defined as $\theta_0 = \{\mu_0\}$, the existence regularity

condition (R1) becomes

$$\begin{aligned}
E[\ln f(y_t; \theta)] &= -\frac{1}{2} \ln(2\pi\sigma_0^2) - \frac{1}{2\sigma_0^2} E[(y_t - \mu)^2] \\
&= -\frac{1}{2} \ln(2\pi\sigma_0^2) - \frac{1}{2\sigma_0^2} E[(y_t - \mu_0)^2 + (\mu_0 - \mu)^2 + 2(y_t - \mu_0)(\mu_0 - \mu)] \\
&= -\frac{1}{2} \ln(2\pi\sigma_0^2) - \frac{1}{2\sigma_0^2} (\sigma_0^2 + (\mu_0 - \mu)^2) \\
&= -\frac{1}{2} \ln(2\pi\sigma_0^2) - \frac{1}{2} - \frac{(\mu_0 - \mu)^2}{2\sigma_0^2},
\end{aligned}$$

which exists because $0 < \sigma_0^2 < \infty$. $\square$

2.4 Properties of the Likelihood Function

This section establishes various features of the log-likelihood function used in the derivation of the properties of the maximum likelihood estimator.

2.4.1 The Population Likelihood Function

Given that the existence condition (R1) is satisfied, an important property of this expectation is

$$\theta_0 = \arg \max_{\theta} E[\ln f(y_t; \theta)]. \quad (2.25)$$

The principle of maximum likelihood requires that the maximum likelihood estimator, $\hat{\theta}$, maximizes the sample log-likelihood function by replacing the expectation in equation (2.25) by the sample average. This property represents the population analogue of the maximum likelihood principle in which θ_0 maximizes $E[\ln f(y_t; \theta)]$. For this reason $E[\ln f(y_t; \theta)]$ is referred to as the population log-likelihood function.

Proof Consider

$$E[\ln f(y_t; \theta)] - E[\ln f(y_t; \theta_0)] = E \left[\ln \frac{f(y_t; \theta)}{f(y_t; \theta_0)} \right] < \ln E \left[\frac{f(y_t; \theta)}{f(y_t; \theta_0)} \right],$$

where $\theta \neq \theta_0$ and the inequality follows from Jensen's inequality.³ Working

³ If $g(y)$ is a concave function in the random variable y , Jensen's inequality states that $E[g(y)] < g(E[y])$. This condition is satisfied here since $g(y) = \ln(y)$ is concave.

with the term on the right-hand side yields

$$\ln E \left[\frac{f(y_t; \theta)}{f(y_t; \theta_0)} \right] = \ln \int_{-\infty}^{\infty} \frac{f(y_t; \theta)}{f(y_t; \theta_0)} f(y_t; \theta_0) dy_t = \ln \int_{-\infty}^{\infty} f(y_t; \theta) dy_t = \ln 1 = 0.$$

It follows immediately that $E[\ln f(y_t; \theta)] < E[\ln f(y_t; \theta_0)]$, for arbitrary θ , which establishes that the maximum occurs just for θ_0 . $\square$

Example 2.17 Population Likelihood of the Normal Distribution

From Example 2.16, the population log-likelihood function based on a normal distribution with unknown mean, μ , and known variance, σ_0^2 , is

$$E[\ln f(y_t; \theta)] = -\frac{1}{2} \ln(2\pi\sigma_0^2) - \frac{1}{2} - \frac{(\mu_0 - \mu)^2}{2\sigma_0^2},$$

which clearly has its maximum at $\mu = \mu_0$. $\square$

2.4.2 Moments of the Gradient

The gradient function at observation t , introduced in Chapter 1, is defined as

$$g_t(\theta) = \frac{\partial \ln f(y_t; \theta)}{\partial \theta}. \quad (2.26)$$

This function has two important properties that are fundamental to maximum likelihood estimation. These properties are also used in Chapter 3 to devise numerical algorithms for computing maximum likelihood estimators, in Chapter 4 to construct test statistics, and in Chapter 9 to derive the quasi-maximum likelihood estimator.

Mean of the Gradient

The first property is

$$E[g_t(\theta_0)] = 0. \quad (2.27)$$

Proof As $f(y_t; \theta)$ is a probability distribution, it has the property

$$\int_{-\infty}^{\infty} f(y_t; \theta) dy_t = 1.$$

Now differentiating both sides with respect to θ gives

$$\frac{\partial}{\partial \theta} \left(\int_{-\infty}^{\infty} f(y_t; \theta) dy_t \right) = 0.$$

Using the interchangeability regularity condition (R5) and the property of natural logarithms

$$\frac{\partial f(y_t; \theta)}{\partial \theta} = \frac{\partial \ln f(y_t; \theta)}{\partial \theta} f(y_t; \theta) = g_t(\theta) f(y_t; \theta),$$

the left-hand side expression is rewritten as

$$\int_{-\infty}^{\infty} g_t(\theta) f(y_t; \theta) dy_t.$$

Evaluating this expression at $\theta = \theta_0$ means the the relevant integral is evaluated using the population density function, $f(y_t; \theta_0)$, thereby enabling it to be interpreted as an expectation. This yields

$$E[g_t(\theta_0)] = 0,$$

which proves the result. $\square$

Variance of the Gradient

The second property is

$$\text{cov}[g_t(\theta_0)] = E[g_t(\theta_0)g_t(\theta_0)'] = -E[h_t(\theta_0)], \quad (2.28)$$

where the first equality uses the result from expression (2.27) that $g_t(\theta_0)$ has zero mean. This expression links the first and second derivatives of the likelihood function and establishes that the expectation of the square of the gradient is equal to the negative of the expectation of the Hessian.

Proof Differentiating

$$\int_{-\infty}^{\infty} f(y_t; \theta) dy_t = 1,$$

twice with respect to θ and using the same regularity conditions to establish the first property of the gradient, gives

$$\begin{aligned} \int_{-\infty}^{\infty} \left[\frac{\partial \ln f(y_t; \theta)}{\partial \theta} \frac{\partial f(y_t; \theta)}{\partial \theta'} + \frac{\partial^2 \ln f(y_t; \theta)}{\partial \theta \partial \theta'} f(y_t; \theta) \right] dy_t &= 0 \\ \int_{-\infty}^{\infty} \left[\frac{\partial \ln f(y_t; \theta)}{\partial \theta} \frac{\partial \ln f(y_t; \theta)}{\partial \theta'} f(y_t; \theta) + \frac{\partial^2 \ln f(y_t; \theta)}{\partial \theta \partial \theta'} f(y_t; \theta) \right] dy_t &= 0 \\ \int_{-\infty}^{\infty} [g_t(\theta)g_t(\theta)' + h_t(\theta)] f(y_t; \theta) dy_t &= 0. \end{aligned}$$

Once again, evaluating this expression at $\theta = \theta_0$ gives

$$E[g_t(\theta_0)g_t(\theta_0)'] + E[h_t(\theta_0)] = 0,$$

which proves the result. $\square$

The properties of the gradient function in equations (2.27) and (2.28) are completely general, because they hold for any arbitrary distribution.

Example 2.18 Gradient Properties and the Poisson Distribution

The first and second derivatives of the log-likelihood function of the Poisson distribution, given in Examples 1.14 and 1.17 in Chapter 1, are, respectively,

$$g_t(\theta) = \frac{y_t}{\theta} - 1, \quad h_t(\theta) = -\frac{y_t}{\theta^2}.$$

To establish the first property of the gradient, take expectations and evaluated at $\theta = \theta_0$

$$E[g_t(\theta_0)] = E\left[\frac{y_t}{\theta_0} - 1\right] = \frac{E[y_t]}{\theta_0} - 1 = \frac{\theta_0}{\theta_0} - 1 = 0,$$

because $E[y_t] = \theta_0$ for the Poisson distribution.

To establish the second property of the gradient, consider

$$E[g_t(\theta_0)g_t(\theta_0)'] = E\left[\left(\frac{y_t}{\theta_0} - 1\right)^2\right] = \frac{1}{\theta_0^2}E[(y_t - \theta_0)^2] = \frac{\theta_0}{\theta_0^2} = \frac{1}{\theta_0},$$

since $E[(y_t - \theta_0)^2] = \theta_0$ for the Poisson distribution. Alternatively

$$E[h_t(\theta_0)] = E\left[-\frac{y_t}{\theta_0^2}\right] = -\frac{E[y_t]}{\theta_0^2} = -\frac{\theta_0}{\theta_0^2} = -\frac{1}{\theta_0},$$

and hence

$$E[g_t(\theta_0)g_t(\theta_0)'] = -E[h_t(\theta_0)] = \frac{1}{\theta_0}.$$

□

The relationship between the gradient and the Hessian is presented more compactly by defining

$$\begin{aligned} J(\theta_0) &= E[g_t(\theta_0)g_t(\theta_0)'] \\ H(\theta_0) &= E[h_t(\theta_0)], \end{aligned}$$

in which case

$$J(\theta_0) = -H(\theta_0). \quad (2.29)$$

The term $J(\theta_0)$ is referred to as the outer product of the gradients. In the more general case where y_t is dependent and g_t is a mds, $J(\theta_0)$ and $H(\theta_0)$

in equation (2.29) become respectively

$$J(\theta_0) = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{E}[g_t(\theta_0)g_t(\theta_0)'] \quad (2.30)$$

$$H(\theta_0) = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{E}[h_t(\theta_0)] . \quad (2.31)$$

2.4.3 The Information Matrix

The definition of the outer product of the gradients in equation (2.29) is commonly referred to as the information matrix

$$I(\theta_0) = J(\theta_0) . \quad (2.32)$$

Given the relationship between $J(\theta_0)$ and $H(\theta_0)$ in equation (2.29) it immediately follows that

$$I(\theta_0) = J(\theta_0) = -H(\theta_0) . \quad (2.33)$$

Equation (2.33) represents the well-known information equality. An important assumption underlying this result is that the distribution used to construct the log-likelihood function is correctly specified. This assumption is relaxed in Chapter 9 on quasi-maximum likelihood estimation.

The information matrix represents a measure of the quality of the information in the sample to locate the population parameter θ_0 . For log-likelihood functions that are relatively flat the information in the sample is dispersed thereby providing imprecise information on the location of θ_0 . For samples that are less diffuse the log-likelihood function is more concentrated providing more precise information on the location of θ_0 . Interpreting information this way follows from the expression of the information matrix in equation (2.33) where the quantity of information in the sample is measured by the curvature of the log-likelihood function, as given by $-H(\theta)$. For relatively flat log-likelihood functions the curvature of $\ln L(\theta)$ means that $-H(\theta)$ is relatively small around θ_0 . For log-likelihood functions exhibiting stronger curvature, the second derivative is correspondingly larger.

If $h_t(\theta)$ represents the information available from the data at time t , if follows from (2.31) that the total information available from a sample of size T is

$$TI(\theta_0) = - \sum_{t=1}^T \mathbb{E}[h_t] . \quad (2.34)$$

Example 2.19 Information Matrix of the Bernoulli Distribution

Let $\{y_1, y_2, \dots, y_T\}$ be *iid* observations from a Bernoulli distribution

$$f(y; \theta) = \theta^y (1 - \theta)^{1-y},$$

where $0 < \theta < 1$. The log-likelihood function at observation t is

$$\ln l_t(\theta) = y_t \ln \theta + (1 - y_t) \ln(1 - \theta).$$

The first and second derivatives are, respectively,

$$g_t(\theta) = \frac{y_t}{\theta} - \frac{1 - y_t}{1 - \theta}, \quad h_t(\theta) = -\frac{y_t}{\theta^2} - \frac{1 - y_t}{(1 - \theta)^2}.$$

The information matrix is

$$I(\theta_0) = -E[h_t(\theta_0)] = \frac{E[y_t]}{\theta_0^2} - \frac{E[1 - y_t]}{(1 - \theta_0)^2} = \frac{\theta_0}{\theta_0^2} + \frac{(1 - \theta_0)}{(1 - \theta_0)^2} = \frac{1}{\theta_0(1 - \theta_0)},$$

because $E[y_t] = \theta_0$ for the Bernoulli distribution. The total amount of information in the sample is

$$TI(\theta_0) = \frac{T}{\theta_0(1 - \theta_0)}.$$

□

Example 2.20 Information Matrix of the Normal Distribution

Let $\{y_1, y_2, \dots, y_T\}$ be *iid* observations drawn from the normal distribution

$$f(y; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y - \mu)^2}{2\sigma^2} \right],$$

where the unknown parameters are $\theta = \{\mu, \sigma^2\}$. From Example 1.12 in Chapter 1, the log-likelihood function at observation t is

$$\ln l_t(\theta) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} (y_t - \mu)^2,$$

and the gradient and Hessian are, respectively

$$g_t(\theta) = \begin{bmatrix} \frac{y_t - \mu}{\sigma^2} \\ -\frac{1}{2\sigma^2} + \frac{(y_t - \mu)^2}{2\sigma^4} \end{bmatrix}, \quad h_t(\theta) = \begin{bmatrix} -\frac{1}{\sigma^2} & -\frac{y_t - \mu}{\sigma^4} \\ -\frac{y_t - \mu}{\sigma^4} & \frac{1}{2\sigma^4} - \frac{(y_t - \mu)^2}{\sigma^6} \end{bmatrix}.$$

Taking expectations of the negative Hessian, evaluating at $\theta = \theta_0$ and scaling

the result by T gives the total information matrix

$$TI(\theta_0) = -T E[h_t(\theta_0)] = \begin{bmatrix} \frac{T}{\sigma_0^2} & 0 \\ 0 & \frac{T}{2\sigma_0^4} \end{bmatrix}.$$

□

2.5 Asymptotic Properties

Assuming that the regularity conditions (R1) to (R5) in Section 2.3 are satisfied, the results in Section 2.4 are now used to study the relationship between the maximum likelihood estimator, $\hat{\theta}$, and the population parameter, θ_0 , as $T \rightarrow \infty$. Three properties are investigated, namely, consistency, asymptotic normality and asymptotic efficiency. The first property focuses on the distance $\hat{\theta} - \theta_0$; the second looks at the distribution of $\hat{\theta} - \theta_0$; and the third examines the variance of this distribution.

2.5.1 Consistency

A desirable property of an estimator $\hat{\theta}$ is that additional information obtained by increasing the sample size, T , yields more reliable estimates of the population parameter, θ_0 . Formally this result is stated as

$$\text{plim}(\hat{\theta}) = \theta_0. \quad (2.35)$$

An estimator satisfying this property is a consistent estimator. Given the regularity conditions in Section 2.3, all maximum likelihood estimators are consistent.

To derive this result, consider a sample of T observations, $\{y_1, y_2, \dots, y_T\}$. By definition the maximum likelihood estimator satisfies the condition

$$\hat{\theta} = \arg \max_{\theta} \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta).$$

From the convergence regularity condition (R2)

$$\frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta) \xrightarrow{P} E[\ln f(y_t; \theta)],$$

which implies that the two functions are converging asymptotically. But,

given the result in equation (2.25), it is possible to write

$$\arg \max_{\theta} \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta) \xrightarrow{p} \arg \max_{\theta} E[\ln f(y_t; \theta)] .$$

So the maxima of these two functions, $\hat{\theta}$ and θ_0 , respectively, must also be converging as $T \rightarrow \infty$, in which case (2.35) holds.

This is a heuristic proof of the consistency property of the maximum likelihood estimator initially given by Wald (1949); see also Newey and McFadden (1994, Theorems 2.1 and 2.5, pp 2111 - 2245). The proof highlights that consistency requires:

- (i) convergence of the sample log-likelihood function to the population log-likelihood function; and
- (ii) convergence of the maximum of the sample log-likelihood function to the maximum of the population log-likelihood function.

These two features of the consistency proof are demonstrated in the following simulation experiment.

Example 2.21 Demonstration of Consistency

Figure 2.3 gives plots of the log-likelihood functions for samples of size $T = \{5, 20, 500\}$ simulated from the population distribution $N(10, 16)$. Also plotted is the population log-likelihood function, $E[\ln f(y_t; \theta)]$, given in Example 2.16. The consistency of the maximum likelihood estimator is first demonstrated with the sample log-likelihood functions approaching the population log-likelihood function $E[\ln f(y_t; \theta)]$ as T increases. The second demonstration of the consistency property is given by the maximum likelihood estimates, in this case the sample means, of the three samples

$$\bar{y}(T = 5) = 7.417, \quad \bar{y}(T = 20) = 10.258, \quad \bar{y}(T = 500) = 9.816,$$

which approach the population mean $\mu_0 = 10$ as $T \rightarrow \infty$. $\square$

A further implication of consistency is that an estimator should exhibit decreasing variability around the population parameter θ_0 as T increases.

Example 2.22 Normal Distribution

Consider the normal distribution

$$f(y; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y - \mu)^2}{2\sigma^2} \right] .$$

From Example 1.16 in Chapter 1, the sample mean, $\bar{y}$, is the maximum likelihood estimator of μ_0 . Figure 2.4 shows that this estimator converges

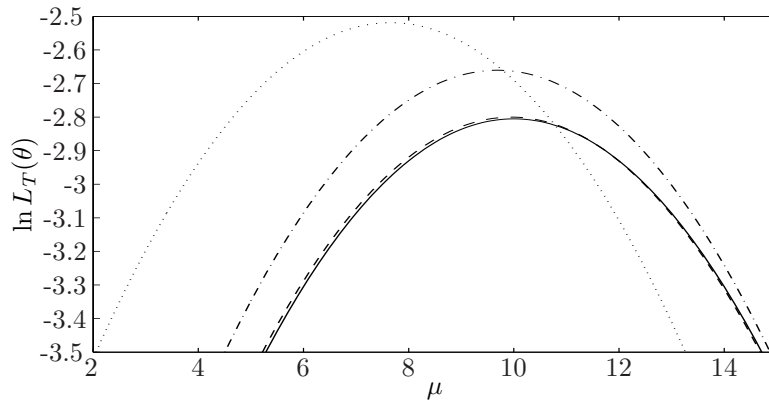


Figure 2.3 Log-likelihood functions for samples of size $T = 5$ (dotted line), $T = 20$ (dot-dashed line) and $T = 500$ (dashed line), simulated from the population distribution $N(10, 16)$. The bold line is the population log-likelihood $E[\ln f(y; \theta)]$ given by Example 2.16.

to $\mu_0 = 1$ for increasing samples of size T while simultaneously exhibiting decreasing variability. $\square$

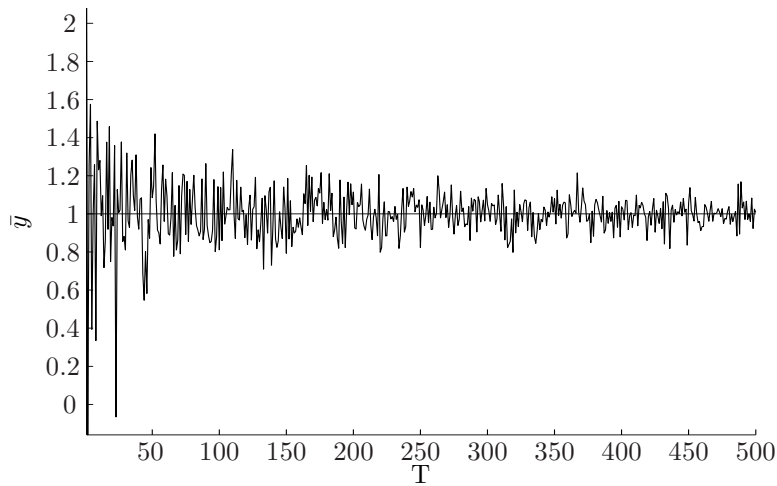


Figure 2.4 Demonstration of the consistency properties of the sample mean when samples of increasing size $T = 1, 2, \dots, 500$ are drawn from a $N(1, 2)$ distribution.

Example 2.23 Cauchy Distribution

The sample mean, $\bar{y}$, and the sample median, $\bar{m}$, are computed from in-

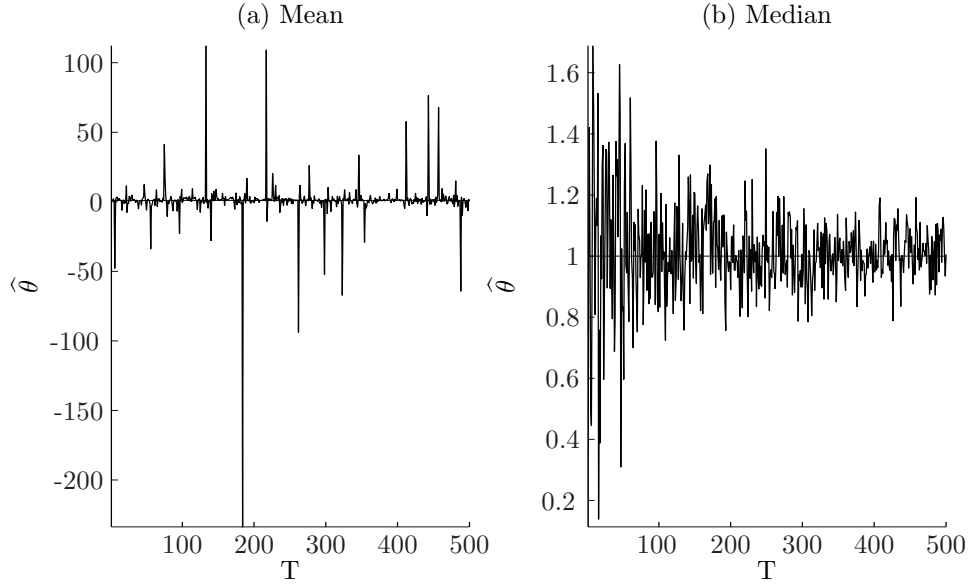


Figure 2.5 Demonstration of the inconsistency of the sample mean and the consistency of the sample median as estimators of the location parameter of a Cauchy distribution with $\theta_0 = 1$, for samples of increasing size $T = 1, 2, \dots, 500$.

creasing samples of size $T = 1, 2, \dots, 500$, drawn from a Cauchy distribution

$$f(y; \theta) = \frac{1}{\pi} \frac{1}{1 + (y - \theta)^2},$$

with location parameter $\theta_0 = 1$. A comparison of panels (a) and (b) in Figure 2.5 suggests that $\bar{y}$ is an inconsistent estimator of θ because its sampling variability does not decrease as T increases. By contrast, the sampling variability of $\bar{m}$ does decrease suggesting that it is a consistent estimator. The failure of $\bar{y}$ to be a consistent estimator stems from the property that the mean of a Cauchy distribution does not exist and therefore represents a violation of the conditions needed for the weak law of large numbers to hold. In this example, neither $\bar{y}$ nor $\bar{m}$ are the maximum likelihood estimators. The maximum likelihood estimator of the location parameter of the Cauchy distribution is investigated further in Chapter 3. $\square$

2.5.2 Normality

To establish the asymptotic distribution of the maximum likelihood estimator, $\hat{\theta}$, consider the first-order condition

$$G_T(\hat{\theta}) = \frac{1}{T} \sum_{t=1}^T g_t(\hat{\theta}) = 0. \quad (2.36)$$

A mean value expansion of this condition around the true value θ_0 , gives

$$0 = \frac{1}{T} \sum_{t=1}^T g_t(\hat{\theta}) = \frac{1}{T} \sum_{t=1}^T g_t(\theta_0) + \left[\frac{1}{T} \sum_{t=1}^T h_t(\theta^*) \right] (\hat{\theta} - \theta_0), \quad (2.37)$$

where θ^* lies between $\hat{\theta}$ and θ_0 , and hence $\theta^* \xrightarrow{p} \theta_0$ if $\hat{\theta} \xrightarrow{p} \theta_0$. Rearranging and multiplying both sides by $\sqrt{T}$ shows that

$$\sqrt{T}(\hat{\theta} - \theta_0) = \left[-\frac{1}{T} \sum_{t=1}^T h_t(\theta^*) \right]^{-1} \left[\frac{1}{\sqrt{T}} \sum_{t=1}^T g_t(\theta_0) \right]. \quad (2.38)$$

Now

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T h_t(\theta^*) &\xrightarrow{p} H(\theta_0) \\ \frac{1}{\sqrt{T}} \sum_{t=1}^T g_t(\theta_0) &\xrightarrow{d} N(0, J(\theta_0)), \end{aligned} \quad (2.39)$$

where

$$\begin{aligned} H(\theta_0) &= \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T E[h_t(\theta_0)] \\ J(\theta_0) &= \lim_{T \rightarrow \infty} E \left[\left(\frac{1}{\sqrt{T}} \sum_{t=1}^T g_t(\theta_0) \right) \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T g_t'(\theta_0) \right) \right]. \end{aligned}$$

The first condition in (2.39) follows from the uniform WLLN and the second condition is based on applying the appropriate central limit theorem based on the time series properties of $g_t(\theta)$. Combining equations (2.38) and (2.39) yields the asymptotic distribution

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, H^{-1}(\theta_0)J(\theta_0)H^{-1}(\theta_0)).$$

Using the information matrix equality in equation (2.33) simplifies the asymptotic distribution to

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, I^{-1}(\theta_0)). \quad (2.40)$$

or

$$\hat{\theta} \stackrel{a}{\sim} N(\theta_0, \frac{1}{T}\Omega), \quad \frac{1}{T}\Omega = \frac{1}{T}I^{-1}(\theta_0). \quad (2.41)$$

This establishes that the maximum likelihood estimator has an asymptotic normal distribution with mean equal to the population parameter, θ_0 , and covariance matrix, $T^{-1}\Omega$, equal to the inverse of the information matrix appropriately scaled to account for the total information in the sample, $T^{-1}I^{-1}(\theta_0)$.

Example 2.24 Asymptotic Normality of the Poisson Parameter

From Example 2.18, equation (2.40) becomes

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \theta_0),$$

because $H(\theta_0) = -1/\theta_0 = -I(\theta_0)$, then $I^{-1}(\theta_0) = \theta_0$. $\square$

Example 2.25 Simulating Asymptotic Normality

Figure 2.6 gives the results of sampling *iid* random variables from an exponential distribution with $\theta_0 = 1$ for samples of size $T = 5$ and $T = 100$, using 5000 replications. The sample means are standardized using the population mean ($\theta_0 = 1$) and the population variance ($\theta_0^2/T = 1^2/T$) as

$$z_i = \frac{\bar{y}_i - 1}{\sqrt{1^2/T}}, \quad i = 1, 2, \dots, 5000.$$

The sampling distribution of z is skewed to the right for samples of size $T = 5$ thus mimicking the positive skewness characteristic of the population distribution. Increasing the sample size to $T = 100$, reduces the skewness in the sampling distribution, which is now approximately normally distributed. $\square$

2.5.3 Efficiency

Asymptotic efficiency concerns the limiting value of the variance of any estimator, say $\tilde{\theta}$, around θ_0 as the sample size increases. The Cramér-Rao lower bound provides a bound on the efficiency of this estimator.

Cramér-Rao Lower Bound: Single Parameter Case

Suppose θ_0 is a single parameter and $\tilde{\theta}$ is any consistent estimator of θ_0 with asymptotic distribution of the form

$$\sqrt{T}(\tilde{\theta} - \theta_0) \xrightarrow{d} N(0, \Omega).$$

(a) Exponential distribution

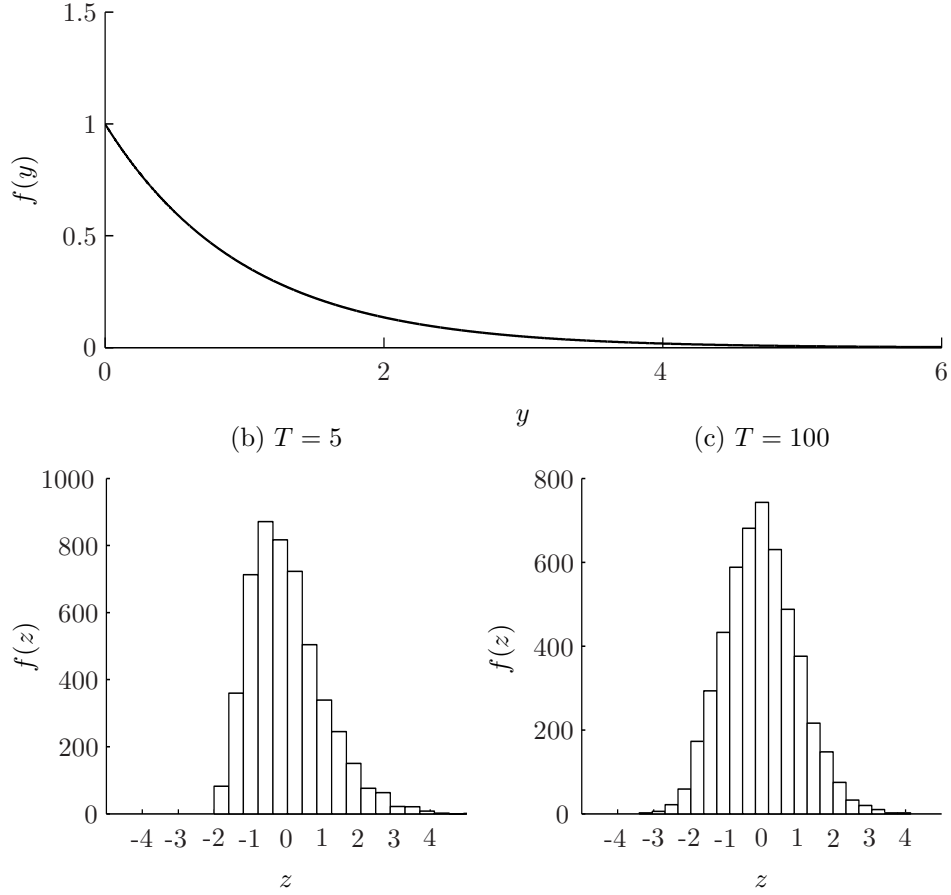


Figure 2.6 Demonstration of asymptotic normality of the maximum likelihood estimator based on samples of size $T = 5$ and $T = 100$ from an exponential distribution, $f(y; \theta_0)$, with mean $\theta_0 = 1$, for 5000 replications.

The Cramér-Rao inequality states that

$$\Omega \geq \frac{1}{I(\theta_0)}. \quad (2.42)$$

Proof An outline of the proof is as follows. A consistent estimator is asymptotically unbiased, so $E[\tilde{\theta} - \theta_0] \rightarrow 0$ as $T \rightarrow 0$, which can be expressed

$$\int \cdots \int (\tilde{\theta} - \theta_0) f(y_1, \dots, y_T; \theta_0) dy_1 \cdots dy_T \rightarrow 0.$$

Differentiating both sides with respect to θ_0 and using the interchangeability

regularity condition (R4) gives

$$- \int \cdots \int f(y_1, \dots, y_T; \theta_0) dy_1 \cdots dy_T \\ + \int \cdots \int (\tilde{\theta} - \theta_0) \frac{\partial f(y_1, \dots, y_T; \theta_0)}{\partial \theta_0} dy_1 \cdots dy_T \rightarrow 0.$$

The first term on the right hand side integrates to 1, since f is a probability density function. Thus

$$\int \cdots \int (\tilde{\theta} - \theta_0) \frac{\partial \ln f(y_1, \dots, y_T; \theta_0)}{\partial \theta_0} f(y_1, \dots, y_T; \theta_0) dy_1 \cdots dy_T \rightarrow 1. \quad (2.43)$$

Using

$$\frac{\partial \ln f(y_1, \dots, y_T; \theta_0)}{\partial \theta_0} = TG_T(\theta_0),$$

equation (2.43) can be expressed

$$\text{cov}(\sqrt{T}(\tilde{\theta} - \theta_0), \sqrt{T}G_T(\theta_0)) \rightarrow 1,$$

since the score $G_T(\theta_0)$ has mean zero.

The squared correlation between $\sqrt{T}(\tilde{\theta} - \theta_0)$ and $G_T(\theta_0)$ satisfies

$$\text{cor}(\sqrt{T}(\tilde{\theta} - \theta_0), \sqrt{T}G_T(\theta_0))^2 = \frac{\text{cov}(\sqrt{T}(\tilde{\theta} - \theta_0), \sqrt{T}G_T(\theta_0))^2}{\text{var}(\sqrt{T}(\tilde{\theta} - \theta_0))\text{var}(\sqrt{T}G_T(\theta_0))} \leq 1$$

and rearranging gives

$$\text{var}(\sqrt{T}(\tilde{\theta} - \theta_0)) \geq \frac{\text{cov}(\sqrt{T}(\tilde{\theta} - \theta_0), \sqrt{T}G_T(\theta_0))^2}{\text{var}(\sqrt{T}G_T(\theta_0))}.$$

Taking limits on both sides of this inequality gives Ω on the left hand side, 1 in the numerator on the right hand side and $I(\theta_0)$ in the denominator, which gives the Cramér-Rao inequality in (2.42) as required. $\square$

Cramér-Rao Lower Bound: Multiple Parameter Case

For a vector parameter the Cramér-Rao inequality (2.42) becomes

$$\Omega \geq I^{-1}(\theta_0), \quad (2.44)$$

where this matrix inequality is understood to mean that $\Omega - I^{-1}(\theta_0)$ is a positive semi-definite matrix. Since equation (2.41) shows that the maximum likelihood estimator, $\hat{\theta}$, has asymptotic variance $I^{-1}(\theta_0)$, the maximum likelihood estimator achieves the Cramér-Rao lower bound and is, therefore, asymptotically efficient. Moreover, since $TI(\theta_0)$ represents the total information available in a sample of size T , the inverse of this quantity provides

a measure of the precision of the information in the sample, as given by the variance of $\hat{\theta}$.

Example 2.26 Lower Bound for the Normal Distribution

From Example 2.20, the log-likelihood function is

$$\ln L_T(\theta) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2 T} \sum_{t=1}^T (y_t - \mu)^2,$$

with information matrix

$$I(\theta) = -E[H_T(\theta)] = \begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{bmatrix}.$$

Evaluating this expression at $\theta = \theta_0$ gives the covariance matrix of the maximum likelihood estimator

$$\frac{1}{T} \Omega = \frac{1}{T} I^{-1}(\theta_0) = \begin{bmatrix} \frac{\sigma_0^2}{T} & 0 \\ 0 & \frac{2\sigma_0^4}{T} \end{bmatrix},$$

so $\text{se}(\hat{\mu}) \approx \sqrt{\sigma_0^2/T}$ and $\text{se}(\hat{\sigma}^2) \approx \sqrt{2\sigma_0^4/T}$. □

Example 2.27 Relative Efficiency of the Mean and Median

The sample mean, $\bar{y}$, and sample median, $\bar{m}$, are both consistent estimators of the population mean, μ , in samples drawn from a normal distribution, with $\bar{y}$ being the maximum likelihood estimator of μ . From Example 2.26 the variance of $\bar{y}$ is $\text{var}(\bar{y}) = \sigma_0^2/T$. The variance of $\bar{m}$ is approximately (Stuart and Ord, 1994, p. 358)

$$\text{var}(\bar{m}) = \frac{1}{4Tf^2},$$

where $f = f(m)$ is the value of the pdf evaluated at the population median (m). In the case of normality with known variance σ_0^2 , $f(m)$ is

$$f(m) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp \left[-\frac{(m - \mu)^2}{2\sigma_0^2} \right] = \frac{1}{\sqrt{2\pi\sigma_0^2}},$$

since $m = \mu$ because of symmetry. The variance of $\bar{m}$ is then

$$\text{var}(\bar{m}) = \frac{\pi\sigma_0^2}{2T} > \text{var}(\bar{y}),$$

because $\pi/2 > 1$, establishing that the maximum likelihood estimator has a smaller variance than another consistent estimator, $\bar{m}$. □

2.6 Finite-Sample Properties

The properties of the maximum likelihood estimator established in the previous section are asymptotic properties. An important application of the asymptotic distribution is to approximate the finite sample distribution of the maximum likelihood estimator, $\hat{\theta}$. There are a number of methods available to approximate the finite sample distribution including simulating the sampling distribution by Monte Carlo methods or using an Edgeworth expansion approach as shown in the following example.

Example 2.28 Edgeworth Expansion Approximations

As illustrated in Example 2.25, the asymptotic distribution of the maximum likelihood estimator of the parameter of an exponential population distribution is

$$z = \sqrt{T} \frac{(\hat{\theta} - \theta_0)}{\theta_0} \xrightarrow{d} N(0, 1),$$

which has asymptotic distribution function

$$F_a(s) = \Phi(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^s e^{-v^2/2} dv.$$

The Edgeworth expansion of the distribution function is

$$F_e(s) = \Phi(s) - \phi(s) \left[\left(1 + \frac{2}{3} H_2(s)\right) \frac{1}{\sqrt{T}} + \left(\frac{5}{2} + \frac{11}{12} H_3(s) + \frac{9}{2} H_5(s)\right) \frac{1}{T} \right],$$

where $H_2(s) = s^2 - 1$, $H_3(s) = s^3 - 3s$ and $H_5(s) = s^5 - 10s^3 + 15s$ are the probabilists' Hermite polynomials and $\phi(s)$ is the standard normal probability density (Severini, 2005, p.144). The finite sample distribution function is available in this case and is given by the complement of the gamma distribution function

$$F(s) = 1 - \frac{1}{\Gamma(s)} \int_0^w e^{-v} v^{s-1} dv, \quad w = T / \left(1 + s/\sqrt{T}\right).$$

Table 2.6 shows that the Edgeworth approximation, $F_e(s)$, improves upon the asymptotic approximation, $F_a(s)$, although the former can yield negative probabilities in the tails of the distribution. $\square$

As the previous example demonstrates, even for simple situations the finite sample distribution approximation of the maximum likelihood estimator is complicated. For this reason asymptotic approximations are commonly employed. However, some other important finite sample properties will now be discussed, namely, unbiasedness, sufficiency, invariance and non-uniqueness.

Comparison of the finite sample, Edgeworth expansion and asymptotic distribution functions of the statistic $\sqrt{T}\theta_0^{-1}(\hat{\theta} - \theta_0)$, for a sample of size $T = 5$ draws from the exponential distribution.

s	Finite	Edgeworth	Asymptotic
-2	0.000	-0.019	0.023
-1	0.053	0.147	0.159
0	0.440	0.441	0.500
1	0.734	0.636	0.841
2	0.872	0.874	0.977

2.6.1 Unbiasedness

Not all maximum likelihood estimators are unbiased. Examples of unbiased maximum likelihood estimators are the sample mean in the normal and Poisson examples. Even in samples known to be normally distributed but with unknown mean, the sample standard deviation is an example of a biased estimator since $E[\hat{\sigma}] \neq \sigma_0$. This result follows from the fact that Slutsky's theorem (see Section 2.2.2) does not hold for the expectations operator. Consequently

$$E[\tau(\hat{\theta})] \neq \tau(E[\hat{\theta}]),$$

where $\tau(\cdot)$ is a monotonic function. This result contrasts with the property of consistency that uses probability limits, because Slutsky's theorem does apply to plims.

Example 2.29 Sample Variance of a Normal Distribution

The maximum likelihood estimator, $\hat{\sigma}^2$, and an unbiased estimator, $\tilde{\sigma}^2$, of the variance of a normal distribution with unknown mean, μ , are, respectively,

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \bar{y})^2, \quad \tilde{\sigma}^2 = \frac{1}{T-1} \sum_{t=1}^T (y_t - \bar{y})^2.$$

As $E[\tilde{\sigma}^2] = \sigma_0^2$, the maximum likelihood estimator underestimates σ_0^2 in finite samples. To highlight the size of this bias, 20000 samples of size $T = 5$ are drawn from a $N(1, 2)$ distribution. The simulated expectations are, respectively,

$$E[\hat{\sigma}^2] \simeq \frac{1}{20000} \sum_{i=1}^{20000} \hat{\sigma}_i^2 = 1.593, \quad E[\tilde{\sigma}^2] \simeq \frac{1}{20000} \sum_{i=1}^{20000} \tilde{\sigma}_i^2 = 1.991,$$

showing a 20.35% underestimation of $\sigma_0^2 = 2$. $\square$

2.6.2 Sufficiency

Let $\{y_1, y_2, \dots, y_T\}$ be *iid* drawings from the joint pdf $f(y_1, y_2, \dots, y_T; \theta)$. Any statistic computed using the observed sample, such as the sample mean or variance, is a way of summarizing the data. Preferably, the statistics should summarize the data in such a way as not to lose any of the information contained by the entire sample. A sufficient statistic for the population parameter, θ_0 , is a statistic that uses all of the information in the sample. Formally, this means that the joint pdf can be factorized into two components

$$f(y_1, y_2, \dots, y_T; \theta) = c(\tilde{\theta}; \theta) d(y_1, \dots, y_T), \quad (2.45)$$

where $\tilde{\theta}$ represents a sufficient statistic for θ .

If a sufficient statistic exists, the maximum likelihood estimator is a function of it. To demonstrate this result, use equation (2.45) to rewrite the log-likelihood function as

$$\ln L_T(\theta) = \frac{1}{T} \ln c(\tilde{\theta}; \theta) + \frac{1}{T} \ln d(y_1, \dots, y_T). \quad (2.46)$$

Differentiating with respect to θ gives

$$\frac{\partial \ln L_T(\theta)}{\partial \theta} = \frac{1}{T} \frac{\partial \ln c(\tilde{\theta}; \theta)}{\partial \theta}. \quad (2.47)$$

The maximum likelihood estimator, $\hat{\theta}$, is given as the solution of

$$\frac{\partial \ln c(\tilde{\theta}; \hat{\theta})}{\partial \theta} = 0. \quad (2.48)$$

Rearranging shows that $\hat{\theta}$ is a function of the sufficient statistic $\tilde{\theta}$.

Example 2.30 Sufficient Statistic of the Geometric Distribution

If $\{y_1, y_2, \dots, y_T\}$ are *iid* observations from a geometric distribution

$$f(y; \theta) = (1 - \theta)^y \theta, \quad 0 < \theta < 1,$$

the joint pdf is

$$\prod_{t=1}^T f(y_t; \theta) = (1 - \theta)^{\tilde{\theta}} \theta^T,$$

where $\tilde{\theta}$ is the sufficient statistic

$$\tilde{\theta} = \sum_{t=1}^T y_t.$$

Defining

$$c(\tilde{\theta}; \theta) = (1 - \theta)^{\tilde{\theta}} \theta^T, \quad d(y_1, \dots, y_T) = 1,$$

equation (2.48) becomes

$$\frac{d \ln c(\hat{\theta}; \hat{\theta})}{d\theta} = -\frac{\tilde{\theta}}{1 - \hat{\theta}} + \frac{T}{\hat{\theta}} = 0,$$

showing that $\hat{\theta} = T/(T + \tilde{\theta})$ is a function of the sufficient statistic $\tilde{\theta}$. $\square$

2.6.3 Invariance

If $\hat{\theta}$ is the maximum likelihood estimator of θ_0 , then for any arbitrary non-linear function, $\tau(\cdot)$, the maximum likelihood estimator of $\tau(\theta_0)$ is given by $\tau(\hat{\theta})$. The invariance property is particularly useful in situations when an analytical expression for the maximum likelihood estimator is not available.

Example 2.31 Invariance Property and the Normal Distribution

Consider the following normal distribution with known mean μ_0

$$f(y; \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y - \mu_0)^2}{2\sigma^2} \right].$$

As shown in Example 1.16, for a sample of size T the maximum likelihood estimator of the variance is $\hat{\sigma}^2 = T^{-1} \sum_{t=1}^T (y_t - \mu_0)^2$. Using the invariance property, the maximum likelihood estimator of σ is

$$\hat{\sigma} = \sqrt{\frac{1}{T} \sum_{t=1}^T (y_t - \mu_0)^2},$$

which immediately follows by defining $\tau(\theta) = \sqrt{\theta}$. $\square$

Example 2.32 Vasicek Interest Rate Model

From the Vasicek model of interest rates in Section 1.5 of Chapter 1, the parameters of the transitional distribution are $\theta = \{\alpha, \beta, \sigma^2\}$. The relationship between the parameters of the transitional distribution and the stationary distribution is

$$\mu_s = -\frac{\alpha}{\beta}, \quad \sigma_s^2 = -\frac{\sigma^2}{\beta(2 + \beta)}.$$

Given the maximum likelihood estimator of the model parameters $\hat{\theta} = \{\hat{\alpha}, \hat{\beta}, \hat{\sigma}^2\}$, the maximum likelihood estimators of the parameters of the stationary distribution are

$$\hat{\mu}_s = -\frac{\hat{\alpha}}{\hat{\beta}}, \quad \hat{\sigma}_s^2 = -\frac{\hat{\sigma}^2}{\hat{\beta}(2 + \hat{\beta})}.$$

□

2.6.4 Non-Uniqueness

The maximum likelihood estimator of θ is obtained by solving

$$G_T(\hat{\theta}) = 0. \quad (2.49)$$

The problems considered so far have a unique and, in most cases, closed-form solution. However, there are examples where there are several solutions to equation (2.49). An example is the bivariate normal distribution, which is explored in Section 2.7.2.

2.7 Applications

Some of the key results from this chapter are now applied to the bivariate normal distribution. The first application is motivated by the portfolio diversification problem in finance. The second application is more theoretical and illustrates the non-uniqueness problem sometimes encountered in the context of maximum likelihood estimation.

Let y_1 and y_2 be jointly *iid* random variables with means $\mu_i = E[y_i]$, variances $\sigma_i^2 = E[(y_i - \mu_i)^2]$, covariance $\sigma_{1,2} = E[(y_1 - \mu_1)(y_2 - \mu_2)]$ and correlation $\rho = \sigma_{1,2}/\sigma_1\sigma_2$. The bivariate normal distribution is

$$f(y_1, y_2; \theta) = \frac{1}{2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}} \exp \left[-\frac{1}{2(1-\rho^2)} \left(\left(\frac{y_1 - \mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{y_1 - \mu_1}{\sigma_1} \right) \left(\frac{y_2 - \mu_2}{\sigma_2} \right) + \left(\frac{y_2 - \mu_2}{\sigma_2} \right)^2 \right) \right], \quad (2.50)$$

where $\theta = \{\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho\}$ are the unknown parameters.

The shape of the bivariate normal distribution is shown in Figure 2.7 for the case of positive correlation $\rho = 0.6$ (left hand column) and zero correlation $\rho = 0$ (right hand column), with $\mu_1 = \mu_2 = 0$ and $\sigma_1^2 = \sigma_2^2 = 1$. The contour plots show that the effect of $\rho > 0$ is to make the contours ellipsoidal, which stretch the mass of the distribution over the quadrants

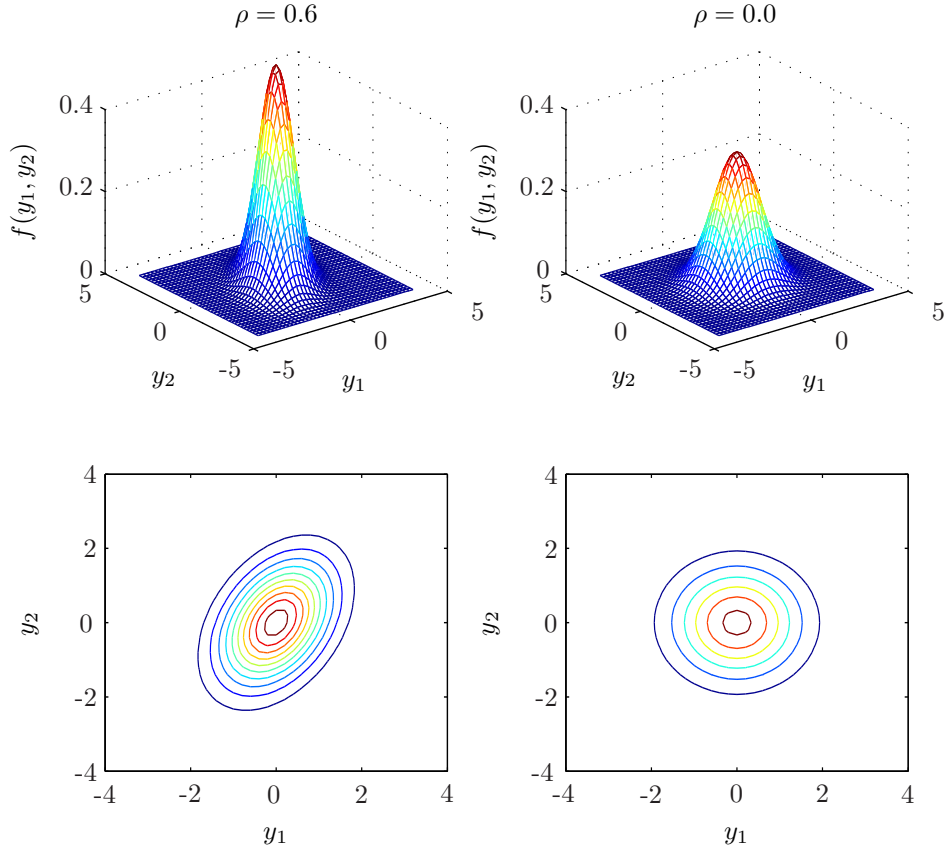


Figure 2.7 Bivariate normal distribution, based on $\mu_1 = \mu_2 = 0$, $\sigma_1^2 = \sigma_2^2 = 1$ and $\rho = 0.6$ (left hand column) and $\rho = 0$ (right hand column).

with y_1 and y_2 having the same signs. The contours are circular for $\rho = 0$, showing that the distribution is evenly spread across all quadrants. In this special case there is no contemporaneous relationship between y_1 and y_2 and the joint distribution reduces to the product of the two marginal distributions

$$f(y_1, y_2; \mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho = 0) = f_1(y_1; \mu_1, \sigma_1^2) f_2(y_2; \mu_2, \sigma_2^2), \quad (2.51)$$

where $f_i(\cdot)$ is a univariate normal distribution.

2.7.1 Portfolio Diversification

A fundamental result in finance is that the risk of a portfolio can be reduced by diversification when the correlation, ρ , between the returns on the assets in the portfolio is not perfect. In the extreme case of $\rho = 1$, all assets move in exactly the same way and there are no gains to diversification.

Figure 2.8 gives a scatter plot of the daily percentage returns on Apple and Ford stocks from 2 January 2001 to 6 August 2010. The cluster of returns exhibits positive, but less than perfect, correlation, suggesting gains to diversification.

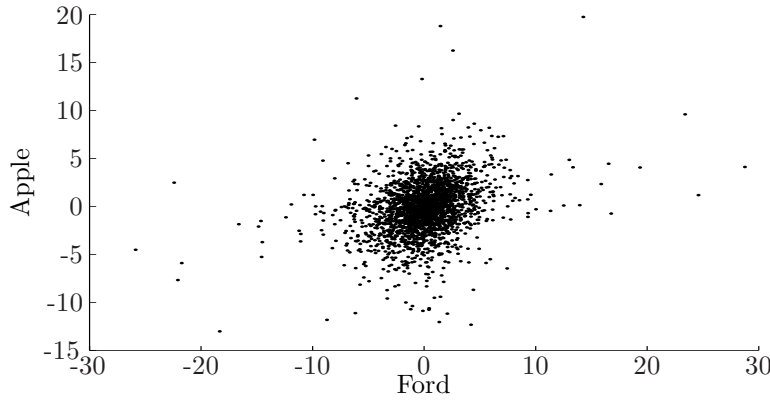


Figure 2.8 Scatter plot of daily percentage returns on Apple and Ford stocks from 2 January 2001 to 6 August 2010.

A common assumption underlying portfolio diversification models is that returns are normally distributed. In the case of two assets, the returns y_1 (Apple) and y_2 (Ford) are assumed to be *iid* with the bivariate normal distribution in (2.50). For $t = 1, 2, \dots, T$ pairs of observations, the log-likelihood function is

$$\begin{aligned} \ln L_T(\theta) = & -\ln 2\pi - \frac{1}{2} (\ln \sigma_1^2 + \ln \sigma_2^2 + \ln(1 - \rho^2)) \\ & - \frac{1}{2(1 - \rho^2)T} \sum_{t=1}^T \left(\left(\frac{y_{1,t} - \mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{y_{1,t} - \mu_1}{\sigma_1} \right) \left(\frac{y_{2,t} - \mu_2}{\sigma_2} \right) + \left(\frac{y_{2,t} - \mu_2}{\sigma_2} \right)^2 \right). \end{aligned} \quad (2.52)$$

To find the maximum likelihood estimator, $\hat{\theta}$, the first-order derivatives

of the log-likelihood function in equation (2.52) are

$$\begin{aligned}\frac{\partial \ln L_T(\theta)}{\partial \mu_i} &= \frac{1}{\sigma_i(1-\rho^2)} \frac{1}{T} \sum_{t=1}^T \left(\left(\frac{y_{i,t} - \mu_i}{\sigma_i} \right) - \rho \left(\frac{y_{j,t} - \mu_j}{\sigma_j} \right) \right) \\ \frac{\partial \ln L_T(\theta)}{\partial \sigma_i^2} &= -\frac{1}{2\sigma_i^2(1-\rho^2)} \left((1-\rho^2) - \frac{1}{T} \sum_{t=1}^T \left(\frac{y_{i,t} - \mu_i}{\sigma_i} \right)^2 \right. \\ &\quad \left. + \frac{\rho}{T} \sum_{t=1}^T \left(\frac{y_{i,t} - \mu_i}{\sigma_i} \right) \left(\frac{y_{j,t} - \mu_j}{\sigma_j} \right) \right) \\ \frac{\partial \ln L_T(\theta)}{\partial \rho} &= \frac{\rho}{1-\rho^2} - \frac{1}{(1-\rho^2)^2} \frac{1}{T} \sum_{t=1}^T \left(\rho \left(\frac{y_{1,t} - \mu_1}{\sigma_1} \right)^2 + \rho \left(\frac{y_{2,t} - \mu_2}{\sigma_2} \right) \right. \\ &\quad \left. + \frac{1+\rho^2}{(1-\rho^2)^2} \left(\frac{y_{1,t} - \mu_1}{\sigma_1} \right) \left(\frac{y_{2,t} - \mu_2}{\sigma_2} \right) \right),\end{aligned}$$

where $i \neq j$. Setting these derivatives to zero and rearranging yields the maximum likelihood estimators

$$\begin{aligned}\hat{\mu}_i &= \frac{1}{T} \sum_{t=1}^T y_{i,t}, \quad \hat{\sigma}_i^2 = \frac{1}{T} \sum_{t=1}^T (y_{i,t} - \hat{\mu}_i)^2, \quad i = 1, 2, \\ \hat{\rho} &= \frac{1}{T\hat{\sigma}_1\hat{\sigma}_2} \sum_{t=1}^T (y_{1,t} - \hat{\mu}_1)(y_{2,t} - \hat{\mu}_2).\end{aligned}$$

Evaluating these expressions using the data in Figure 2.8 gives

$$\hat{\mu}_1 = -0.147, \quad \hat{\mu}_2 = 0.017, \quad \hat{\sigma}_1^2 = 7.764, \quad \hat{\sigma}_2^2 = 10.546, \quad \hat{\rho} = 0.301, \quad (2.53)$$

while the estimate of the covariance is

$$\hat{\sigma}_{1,2} = \hat{\rho}_{1,2}\hat{\sigma}_1\hat{\sigma}_2 = 0.301 \times \sqrt{7.764} \times \sqrt{10.546} = 2.724.$$

The estimate of the correlation $\hat{\rho} = 0.301$ confirms the positive ellipsoidal shape of the scatter plot in Figure 2.8.

To demonstrate the potential advantages of portfolio diversification, define the return on the portfolio of the two assets, Apple and Ford, as

$$r_t = w_1 y_{1,t} + w_2 y_{2,t},$$

where w_1 and w_2 are the respective weights on Apple and Ford in the portfolio, with the property that $w_1 + w_2 = 1$. The risk of this portfolio is

$$\sigma^2 = \text{E}[(r_t - \text{E}[r_t])^2] = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_{1,2}.$$

For the minimum variance portfolio, w_1 and w_2 are the solutions of

$$\arg \min_{w_1, w_2} \sigma^2 \quad \text{s.t.} \quad w_1 + w_2 = 1.$$

The optimal weight on Apple is

$$w_1 = \frac{\sigma_2^2 - \sigma_{1,2}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{1,2}}.$$

Using the sample estimates in (2.53), the estimate of this weight is

$$\hat{w}_1 = \frac{\hat{\sigma}_2^2 - \hat{\sigma}_{1,2}}{\hat{\sigma}_1^2 + \hat{\sigma}_2^2 - 2\hat{\sigma}_{1,2}} = \frac{10.546 - 2.724}{7.764 + 10.546 - 2 \times 2.724} = 0.608.$$

On Ford it is $\hat{w}_2 = 1 - \hat{w}_1 = 0.392$. An estimate of the risk of the optimal portfolio is

$$\hat{\sigma}^2 = 0.608^2 \times 7.764 + 0.392^2 \times 10.546 + 2 \times 0.608 \times 0.392 \times 2.724 = 5.789.$$

From the invariance property $\hat{w}_1$, $\hat{w}_2$ and $\hat{\sigma}^2$ are maximum likelihood estimates of the population parameters. The risk on the optimal portfolio is less than the individual risks on Apple ($\hat{\sigma}_1^2 = 7.764$) and Ford ($\hat{\sigma}_2^2 = 10.546$) stocks, which highlights the advantages of portfolio diversification.

2.7.2 Bimodal Likelihood

Consider the case in (2.50) where $\mu_1 = \mu_2 = 0$ and $\sigma_1^2 = \sigma_2^2 = 1$ and where ρ is the only unknown parameter. The log-likelihood function in (2.52) reduces to

$$\ln L_T(\rho) = -\ln 2\pi - \frac{1}{2} \ln(1-\rho^2) - \frac{1}{2(1-\rho^2)T} \left(\sum_{t=1}^T y_{1,t}^2 - 2\rho \sum_{t=1}^T y_{1,t}y_{2,t} + \sum_{t=1}^T y_{2,t}^2 \right).$$

The gradient is

$$\begin{aligned} \frac{\partial \ln L_T(\rho)}{\partial \rho} &= \frac{\rho}{1-\rho^2} + \frac{1}{(1-\rho^2)T} \sum_{t=1}^T y_{1,t}y_{2,t} \\ &\quad - \frac{\rho}{(1-\rho^2)^2 T} \left(\sum_{t=1}^T y_{1,t}^2 - 2\rho \sum_{t=1}^T y_{1,t}y_{2,t} + \sum_{t=1}^T y_{2,t}^2 \right). \end{aligned}$$

Setting the gradient to zero with $\rho = \hat{\rho}$ and simplifying the resulting expression by multiplying both sides by $(1-\rho^2)^2$, shows that the maximum

likelihood estimator is the solution of the cubic equation

$$\hat{\rho}(1 - \hat{\rho}^2) + (1 + \hat{\rho}^2) \frac{1}{T} \sum_{t=1}^T y_{1,t} y_{2,t} - \hat{\rho} \left(\frac{1}{T} \sum_{t=1}^T y_{1,t}^2 + \frac{1}{T} \sum_{t=1}^T y_{2,t}^2 \right) = 0. \quad (2.54)$$

This equation can have at most three real roots and so the maximum likelihood estimator may not be uniquely defined by the first order conditions in this case.

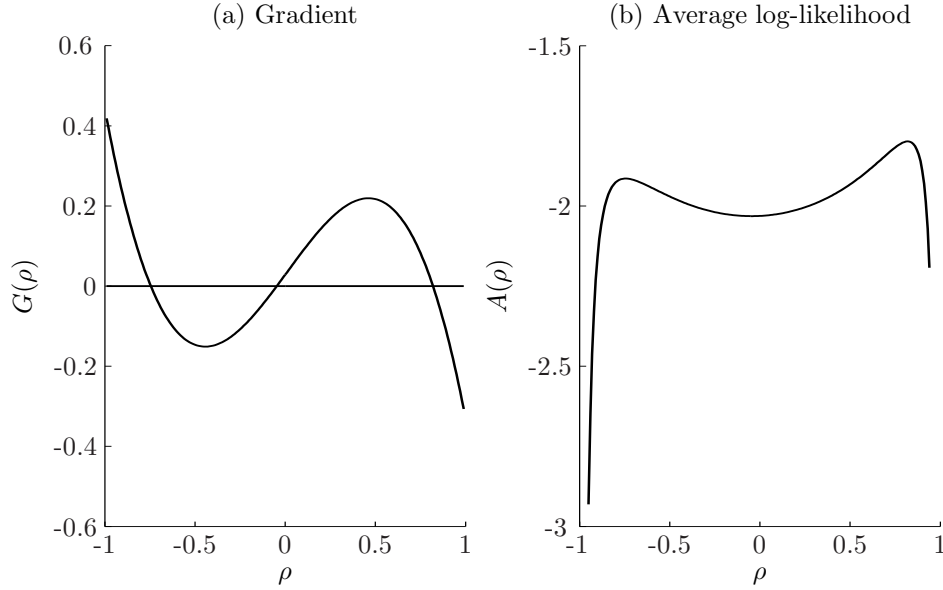


Figure 2.9 Gradient of the bivariate normal model with respect to the parameter ρ for sample size $T = 4$.

An example of multiple roots is given in Figure 2.9. The data are $T = 4$ simulated bivariate normal draws $y_{1,t} = \{-0.6030, -0.0983, -0.1590, -0.6534\}$ and $y_{2,t} = \{0.1537, -0.2297, 0.6682, -0.4433\}$. The population parameters are $\mu_1 = \mu_2 = 0$, $\sigma_1^2 = \sigma_2^2 = 1$ and $\rho = 0.5$. Computing the sample moments yields

$$\frac{1}{T} \sum_{t=1}^T y_{1,t} y_{2,t} = 0.0283, \quad \frac{1}{T} \sum_{t=1}^T y_{1,t}^2 = 0.2064, \quad \frac{1}{T} \sum_{t=1}^T y_{2,t}^2 = 0.1798.$$

From (2.54) define the scaled gradient function as

$$G_T(\rho) = \rho(1 - \rho^2) + (1 + \rho^2)(0.0283) - \rho(0.2064 + 0.1798),$$

which is plotted in panel (a) of Figure 2.9 together with the corresponding

log-likelihood function in panel (b). The function $G_T(\rho)$ has three real roots located at -0.77 , -0.05 and 0.79 , with the middle root corresponding to a minimum. The global maximum occurs at $\rho = 0.79$, so this is the maximum likelihood estimator. It also happens to be the closest root to the true value of $\rho = 0.5$. The solution to the non-uniqueness problem is to evaluate the log-likelihood function at all possible solution values and choose the parameter estimate corresponding to the global maximum.

2.8 Exercises

(1) WLLN (Necessary Condition)

Gauss file(s)	<code>prop_wlln1.g</code>
Matlab file(s)	<code>prop_wlln1.m</code>

- (a) Compute the sample mean of progressively larger samples of size $T = 1, 2, \dots, 500$, comprising *iid* draws from the exponential distribution

$$f(y; \mu) = \frac{1}{\mu} \exp \left[-\frac{y}{\mu} \right], \quad y > 0,$$

with population mean $\mu = 5$. Show that the WLLN holds and hence compare the results with Figure 2.1.

- (b) Repeat part (a) where $f(y; \mu)$ is the Student t distribution with $\mu = 5$ and degrees of freedom parameter $\nu = \{4, 3, 2, 1\}$. Show that the WLLN holds for all cases except $\nu = 1$. Discuss.

(2) WLLN (Sufficient Condition)

Gauss file(s)	<code>prop_wlln2.g</code>
Matlab file(s)	<code>prop_wlln2.m</code>

- (a) A sufficient condition for the WLLN to hold is that $E[\bar{y}] \rightarrow \mu$ and $\text{var}(\bar{y}) \rightarrow 0$ as $T \rightarrow \infty$. Compute the sample moments $m_i = T^{-1} \sum_{t=1}^T y_t^i$, $i = 1, 2, 3, 4$, for $T = \{50, 100, 200, 400, 800\}$ *iid* draws from the uniform distribution

$$f(y) = 1, \quad -0.5 < y < 0.5.$$

Illustrate by simulation that the WLLN holds and compare the results with Table 2.6.

- (b) Repeat part (a) where f is the Student t distribution, with $\mu_0 = 2$, degrees of freedom parameter $\nu_0 = 3$ and where the first two population moments are

$$E[y] = \mu_0, \quad E[y^2] = \frac{\nu_0}{\nu_0 - 2} + \mu_0^2.$$

Confirm that the WLLN holds only for the sample moments m_1 and m_2 , but not m_3 and m_4 .

- (c) Repeat part (b) for $\nu_0 = 4$ and show that the WLLN now holds for m_3 but not for m_4 .
 (d) Repeat part (b) for $\nu_0 = 5$ and show that the WLLN now holds for m_1, m_2, m_3 and m_4 .

(3) Slutsky's Theorem

Gauss file(s)	<code>prop_slutsky.g</code>
Matlab file(s)	<code>prop_slutsky.m</code>

- (a) Consider the sample moment given by the square of the standardized mean

$$m = \left(\frac{\bar{y}}{s} \right)^2,$$

where $\bar{y} = T^{-1} \sum_{t=1}^T y_t$ and $s^2 = T^{-1} \sum_{t=1}^T (y_t - \bar{y})^2$. Simulate this statistic for samples of size $T = \{10, 100, 1000\}$ comprising *iid* draws from the exponential distribution

$$f(y; \mu) = \frac{1}{\mu} \exp \left[-\frac{y}{\mu} \right], \quad y > 0,$$

with mean $\mu = 2$ and variance $\mu^2 = 4$. Given that

$$\text{plim} \left(\frac{\bar{y}}{s} \right)^2 = \frac{(\text{plim } \bar{y})^2}{\text{plim } s^2} = \frac{\mu^2}{\mu^2} = 1,$$

demonstrate Slutsky's theorem where $g(\cdot)$ is the square function.

- (b) Show that Slutsky's theorem does not hold for the statistic

$$m = \left(\sqrt{T} \bar{y} \right)^2$$

by repeating the simulation experiment in part (a). Discuss why the theorem fails in this case?

(4) Normal Distribution

Consider a random sample of size T , $\{y_1, y_2, \dots, y_T\}$, of iid random variables from the normal distribution with unknown mean θ and known variance $\sigma_0^2 = 1$

$$f(y; \theta) = \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{(y - \theta)^2}{2} \right].$$

- (a) Derive expressions for the gradient, Hessian and information matrix.
- (b) Derive the Cramér-Rao lower bound.
- (c) Find the maximum likelihood estimator $\hat{\theta}$ and show that it is unbiased. [Hint: what is $\int_{-\infty}^{\infty} y f(y) dy$?]
- (d) Derive the asymptotic distribution of $\hat{\theta}$.
- (e) Prove that for the normal density

$$\mathbb{E} \left[\frac{d \ln l_t}{d\theta} \right] = 0, \quad \mathbb{E} \left[\left(\frac{d \ln l_t}{d\theta} \right)^2 \right] = -\mathbb{E} \left[\frac{d^2 \ln l_t}{d\theta^2} \right].$$

- (f) Repeat parts (a) to (e) where the random variables are from the exponential distribution

$$f(y; \theta) = \theta \exp[-\theta y].$$

(5) Graphical Demonstration of Consistency

Gauss file(s)	<code>prop_consistency.g</code>
Matlab file(s)	<code>prop_consistency.m</code>

- (a) Simulate samples of size $T = \{5, 20, 500\}$ from the normal distribution with mean $\mu_0 = 10$ and variance $\sigma_0^2 = 16$. For each sample plot the log-likelihood function

$$\ln L_T(\mu, \sigma_0^2) = \frac{1}{T} \sum_{t=1}^T f(y_t; \mu, \sigma^2),$$

for a range of values of μ and compare $\ln L_T(\mu, \sigma_0^2)$ with the population log-likelihood function $\mathbb{E}[\ln f(y_t; \mu, \sigma_0^2)]$. Discuss the consistency property of the maximum likelihood estimator of μ .

- (b) Repeat part (a), except now plot the sample log-likelihood function $\ln L_T(\mu, \sigma^2)$ for different values of σ^2 and compare the result with the population log-likelihood function $\mathbb{E}[\ln f(y_t; \mu_0, \sigma^2)]$. Discuss the consistency property of the maximum likelihood estimator of σ^2 .

(6) Consistency of the Sample Mean Assuming Normality

Gauss file(s)	<code>prop_normal.g</code>
Matlab file(s)	<code>prop_normal.m</code>

This exercise demonstrates the consistency property of the maximum likelihood estimator of the population mean of a normal distribution.

- (a) Generate the sample means for samples of size $T = \{1, 2, \dots, 500\}$, from a $N(1, 2)$ distribution. Plot the sample means for each T and compare the result with Figure 2.4. Interpret the results.
- (b) Repeat part (a) where the distribution is $N(1, 20)$.
- (c) Repeat parts (a) and (b) where the largest sample is now $T = 5000$.

(7) Inconsistency of the Sample Mean of a Cauchy Distribution

Gauss file(s)	<code>prop_cauchy.g</code>
Matlab file(s)	<code>prop_cauchy.m</code>

This exercise shows that the sample mean is an inconsistent estimator of the population mean of a Cauchy distribution, while the median is a consistent estimator.

- (a) Generate the sample mean and median of the Cauchy distribution with parameter $\mu_0 = 1$ for samples of size $T = \{1, 2, \dots, 500\}$. Plot the sample statistics for each T and compare the result with Figure 2.5. Interpret the results.
- (b) Repeat part (a) where the distribution is now Student t with mean $\mu_0 = 1$ and $\nu_0 = 2$ degrees of freedom. Compare the two results.

(8) Efficiency Property of Maximum Likelihood Estimators

Gauss file(s)	<code>prop_efficiency.g</code>
Matlab file(s)	<code>prop_efficiency.m</code>

This exercise demonstrates the efficiency property of the maximum likelihood estimator of the population mean of a normal distribution.

- (a) Generate 10000 samples of size $T = 100$ from a normal distribution with mean $\mu_0 = 1$ and variance $\sigma_0^2 = 2$.
- (b) For each of the 10000 replications compute the sample mean $\bar{y}_i$.
- (c) For each of the 10000 replications compute the sample median $\bar{m}_i$.

- (d) Compute the variance of the sample means around $\mu_0 = 1$ as

$$\text{var}(\bar{y}) = \frac{1}{10000} \sum_{i=1}^{10000} (\bar{y}_i - \mu_0)^2,$$

and compare the result with the theoretical solution $\text{var}(\bar{y}) = \sigma_0^2/T$.

- (e) Compute the variance of the sample medians around $\mu_0 = 1$ as

$$\text{var}(\bar{m}) = \frac{1}{10000} \sum_{i=1}^{10000} (\bar{m}_i - \mu_0)^2,$$

and compare the result with the theoretical solution $\text{var}(\bar{m}) = \pi\sigma_0^2/2T$.

- (f) Use the results in parts (d) and (e) to show that $\text{var}\bar{y} < \text{var}\bar{m}$.

(9) Asymptotic Normality- Exponential Distribution

Gauss file(s)	<code>prop_asymnorm.g</code>
Matlab file(s)	<code>prop_asymnorm.m</code>

This exercise demonstrates the asymptotic normality of the maximum likelihood estimator of the parameter (sample mean) of the exponential distribution.

- (a) Generate 5000 samples of size $T = 5$ from the exponential distribution

$$f(y; \theta) = \frac{1}{\theta} \exp \left[-\frac{y}{\theta} \right], \quad \theta_0 = 1.$$

- (b) For each replication compute the maximum likelihood estimates

$$\hat{\theta}_i = \bar{y}_i, \quad i = 1, 2, \dots, 5000.$$

- (c) Compute the standardized random variables for the sample means using the population mean, θ_0 , and population variance, θ_0^2/T

$$z_i = \sqrt{T} \frac{(\bar{y}_i - 1)}{\sqrt{1^2}}, \quad i = 1, 2, \dots, 5000.$$

- (d) Plot the histogram and interpret its shape.
 (e) Repeat parts (a) to (d) for $T = \{50, 100, 500\}$, and interpret the results.

(10) Asymptotic Normality - Chi Square

Gauss file(s)	<code>prop_chisq.g</code>
Matlab file(s)	<code>prop_chisq.m</code>

This exercise demonstrates the asymptotic normality of the sample mean where the population distribution is a chi-square distribution with one degree of freedom.

- (a) Generate 10000 samples of size $T = 5$ from the chi-square distribution with $\nu_0 = 1$ degrees of freedom.
- (b) For each replication compute the sample mean.
- (c) Compute the standardized random variables for the sample means using $\nu_0 = 1$ and $2\nu_0 = 2$

$$z_i = \sqrt{T} \frac{(\bar{y}_i - 1)}{\sqrt{2}}, \quad i = 1, 2, \dots, 10000.$$

- (d) Plot the histogram and interpret its shape.
- (e) Repeat parts (a) to (d) for $T = \{50, 100, 500\}$, and interpret the results.

(11) **Regression Model with Gamma Disturbances**

Gauss file(s)	<code>prop_gamma.g</code>
Matlab file(s)	<code>prop_gamma.m</code>

Consider the linear regression model

$$y_t = \beta_0 + \beta_1 x_t + (u_t - \rho\alpha),$$

where y_t is the dependent variable, x_t is the explanatory variable and the disturbance term u_t is an *iid* drawing from the gamma distribution

$$f(u; \rho, \alpha) = \frac{1}{\Gamma(\rho)} \left(\frac{1}{\alpha}\right)^\rho u^{\rho-1} \exp\left[-\frac{u}{\alpha}\right],$$

with $\Gamma(\rho)$ representing the gamma function. The term $-\rho\alpha$ in the regression model is included to ensure that $E[u_t - \rho\alpha] = 0$. For samples of size $T = \{50, 100, 250, 500\}$, compute the standardized sampling distributions of the least squares estimators

$$z_{\hat{\beta}_0} = \frac{\hat{\beta}_0 - \beta_0}{\text{se}(\hat{\beta}_0)}, \quad z_{\hat{\beta}_1} = \frac{\hat{\beta}_1 - \beta_1}{\text{se}(\hat{\beta}_1)},$$

based on 5000 draws, parameter values $\beta_0 = 1$, $\beta_1 = 2$, $\rho = 0.25$, $\alpha = 0.1$ and x_t is drawn from a standard normal distribution. Discuss the limiting properties of the sampling distributions.

(12) **Edgeworth Expansions**

Gauss file(s) `prop_edgeworth.g`

Matlab file(s) `prop_edgeworth.m`

Assume that y is *iid* exponential with mean θ_0 and that the maximum likelihood estimator is $\hat{\theta} = \bar{y}$. Define the standardized statistic

$$z = \sqrt{T} \frac{(\hat{\theta} - \theta_0)}{\theta_0}.$$

- (a) For a sample of size $T = 5$ compute the Edgeworth, asymptotic and finite sample distribution functions of z at $s = \{-3, -2, \dots, 3\}$.
- (b) Repeat part (a) for $T = \{10, 100\}$.
- (c) Discuss the ability of the Edgeworth expansion and the asymptotic distribution to approximate the finite sample distribution.

(13) Bias of the Sample Variance

Gauss file(s) `prop_bias.g`

Matlab file(s) `prop_bias.m`

This exercise demonstrates by simulation that the maximum likelihood estimator of the population variance of a normal distribution with unknown mean is biased.

- (a) Generate 20000 samples of size $T = 5$ from a normal distribution with mean $\mu_0 = 1$ and variance $\sigma_0^2 = 2$. For each replication compute the maximum likelihood estimator of σ_0^2 and the unbiased estimator, respectively, as

$$\hat{\sigma}_i^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \bar{y}_i)^2, \quad \tilde{\sigma}_i^2 = \frac{1}{T-1} \sum_{t=1}^T (y_t - \bar{y}_i)^2.$$

- (b) Compute the average of the maximum likelihood estimates and the unbiased estimates, respectively, as

$$E[\hat{\sigma}_T^2] \simeq \frac{1}{20000} \sum_{i=1}^{20000} \hat{\sigma}_i^2, \quad E[\tilde{\sigma}_T^2] \simeq \frac{1}{20000} \sum_{i=1}^{20000} \tilde{\sigma}_i^2.$$

Compare the computed simulated expectations with the population value $\sigma_0^2 = 2$.

- (c) Repeat parts (a) and (b) for $T = \{10, 50, 100, 500\}$. Hence show that the maximum likelihood estimator is asymptotically unbiased.
- (d) Repeat parts (a) and (b) for the case where μ_0 is known. Hence show that the maximum likelihood estimator of the population variance is now unbiased even in finite samples.

(14) **Portfolio Diversification**

Gauss file(s)	<code>prop_diversify.g, apple.csv, ford.csv</code>
Matlab file(s)	<code>prop_diversify.m, diversify.mat</code>

The data files contain daily share prices of Apple and Ford from 2 January 2001 to 6 August 2010, a total of $T = 2413$ observations.

- (a) Compute the daily percentage returns on Apple, $y_{1,t}$, and Ford, $y_{2,t}$. Draw a scatter plot of the returns and interpret the graph.
- (b) Assume that the returns are *iid* from a bivariate normal distribution with means μ_1 and μ_2 , variances σ_1^2 and σ_2^2 , and correlation ρ . Plot the bivariate normal distribution for

$$\rho = \{-0.8, -0.6, -0.4, -0.2, 0.0, 0.2, 0.4, 0.6, 0.8\}.$$

- (c) Derive the maximum likelihood estimators.
- (d) Use the data on returns to compute the maximum likelihood estimates.
- (e) Let the return on a portfolio containing Apple and Ford be

$$p_t = w_1 y_{1,t} + w_2 y_{2,t},$$

where w_1 and w_2 are the respective weights.

- (i) Derive an expression of the risk of the portfolio $\text{var}(p_t)$.
- (ii) Derive expressions of the weights, w_1 and w_2 , that minimize $\text{var}(p_t)$.
- (iii) Use the sample moments in part (d) to estimate the optimal weights and the risk of the portfolio. Compare the estimate of $\text{var}(p_t)$ with the individual sample variances.

(15) **Bimodal Likelihood**

Gauss file(s)	<code>prop_binormal.g</code>
Matlab file(s)	<code>prop_binormal.m</code>

- (a) Simulate a sample of size $T = 4$ from a bivariate normal distribution with zero means, unit variances and correlation $\rho_0 = 0.6$. Plot the log-likelihood function

$$\begin{aligned} \ln L_T(\rho) = & -\ln 2\pi - \frac{1}{2} \ln(1 - \rho^2) \\ & - \frac{1}{2(1 - \rho^2)} \left(\frac{1}{T} \sum_{t=1}^T y_{1,t}^2 - 2\rho \frac{1}{T} \sum_{t=1}^T y_{1,t} y_{2,t} + \frac{1}{T} \sum_{t=1}^T y_{2,t}^2 \right), \end{aligned}$$

and the scaled gradient function

$$G_T(\rho) = \rho(1-\rho^2) + (1+\rho^2) \frac{1}{T} \sum_{t=1}^T y_{1,t} y_{2,t} - \rho \left(\frac{1}{T} \sum_{t=1}^T y_{1,t}^2 + \frac{1}{T} \sum_{t=1}^T y_{2,t}^2 \right),$$

for values of $\rho = \{-0.99, -0.98, \dots, 0.99\}$. Interpret the result and compare the graphs of $\ln L_T(\rho)$ and $G_T(\rho)$ with Figure 2.9.

- (b) Repeat part (a) for $T = \{10, 50, 100\}$, and compare the results with part (a) for the case of $T = 4$. Hence demonstrate that for the case of multiple roots, the likelihood converges to a global maximum resulting in the maximum likelihood estimator being unique (see Stuart, Ord and Arnold, 1999, pp. 50-52, for a more formal treatment of this property).

3

Numerical Estimation Methods

3.1 Introduction

The maximum likelihood estimator is the solution of a set of equations obtained by evaluating the gradient of the log-likelihood function at zero. For many of the examples considered in the previous chapters, a closed-form solution is available. Typical examples consist of the sample mean, or some function of it, the sample variance and the least squares estimator. There are, however, many cases in which the specified model yields a likelihood function that does not admit closed-form solutions for the maximum likelihood estimators.

Example 3.1 Cauchy Distribution

Let $\{y_1, y_2, \dots, y_T\}$ be T *iid* realized values from the Cauchy distribution

$$f(y; \theta) = \frac{1}{\pi} \frac{1}{1 + (y - \theta)^2},$$

where θ is the unknown parameter. The log-likelihood function is

$$\ln L_T(\theta) = -\ln \pi - \frac{1}{T} \sum_{t=1}^T \ln [1 + (y_t - \theta)^2],$$

resulting in the gradient

$$\frac{d \ln L_T(\theta)}{d\theta} = \frac{2}{T} \sum_{t=1}^T \frac{y_t - \theta}{1 + (y_t - \theta)^2}.$$

The maximum likelihood estimator, $\hat{\theta}$, is the solution of

$$\frac{2}{T} \sum_{t=1}^T \frac{y_t - \hat{\theta}}{1 + (y_t - \hat{\theta})^2} = 0.$$

This is a nonlinear function of $\hat{\theta}$ for which no analytical solution exists. $\square$

To obtain the maximum likelihood estimator where no analytical solution is available, numerical optimization algorithms must be used. These algorithms begin by assuming starting values for the unknown parameters and then proceed iteratively until a convergence criterion is satisfied. A general form for the k^{th} iteration is

$$\theta_{(k)} = F(\theta_{(k-1)}),$$

where the form of the function $F(\cdot)$ is governed by the choice of the numerical algorithm. Convergence of the algorithm is achieved when the log-likelihood function cannot be further improved, a situation in which $\theta_{(k)} \simeq \theta_{(k-1)}$, resulting in $\theta_{(k)}$ being the maximum likelihood estimator of θ .

3.2 Newton Methods

From Chapter 1, the gradient and Hessian are defined respectively as

$$G_T(\theta) = \frac{\partial \ln L_T(\theta)}{\partial \theta} = \frac{1}{T} \sum_{t=1}^T g_t, \quad H_T(\theta) = \frac{\partial^2 \ln L_T(\theta)}{\partial \theta \partial \theta'} = \frac{1}{T} \sum_{t=1}^T h_t.$$

A first-order Taylor series expansion of the gradient function around the true parameter vector θ_0 is

$$G_T(\theta) \simeq G_T(\theta_0) + H_T(\theta_0)(\theta - \theta_0), \quad (3.1)$$

where higher-order terms are excluded in the expansion and $G_T(\theta_0)$ and $H_T(\theta_0)$ are, respectively, the gradient and Hessian evaluated at the true parameter value, θ_0 .

As the maximum likelihood estimator, $\hat{\theta}$, is the solution to the equation $G_T(\hat{\theta}) = 0$, the maximum likelihood estimator satisfies

$$G_T(\hat{\theta}) = 0 = G_T(\theta_0) + H_T(\theta_0)(\hat{\theta} - \theta_0), \quad (3.2)$$

where, for convenience, the equation is now written as an equality. This is a linear equation in $\hat{\theta}$ with solution

$$\hat{\theta} = \theta_0 - H_T^{-1}(\theta_0)G_T(\theta_0). \quad (3.3)$$

As it stands, this equation is of little practical use because it expresses the maximum likelihood estimator as a function of the unknown parameter that it seeks to estimate, namely θ_0 . It suggests, however, that a natural way to proceed is to replace θ_0 with a starting value and use (3.3) as an updating scheme. This is indeed the basis of Newton methods. Three algorithms are discussed, differing only in the way that the Hessian, $H_T(\theta)$, is evaluated.

3.2.1 Newton-Raphson

Let $\theta_{(k)}$ be the value of the unknown parameters at the k^{th} iteration. The Newton-Raphson algorithm is given by replacing θ_0 in (3.3) by $\theta_{(k-1)}$ to yield the updated parameter $\theta_{(k)}$

$$\theta_{(k)} = \theta_{(k-1)} - H_{(k-1)}^{-1} G_{(k-1)}, \quad (3.4)$$

where

$$G_{(k)} = \left. \frac{\partial \ln L_T(\theta)}{\partial \theta} \right|_{\theta=\theta_{(k)}}, \quad H_{(k)} = \left. \frac{\partial^2 \ln L_T(\theta)}{\partial \theta \partial \theta'} \right|_{\theta=\theta_{(k)}}.$$

The algorithm proceeds until $\theta_{(k)} \simeq \theta_{(k-1)}$, subject to some tolerance level, which is discussed in more detail later. From (3.4), convergence occurs when

$$\theta_{(k)} - \theta_{(k-1)} = -H_{(k-1)}^{-1} G_{(k-1)} \simeq 0,$$

which can only be satisfied if

$$G_{(k)} \simeq G_{(k-1)} \simeq 0,$$

because both $H_{(k-1)}^{-1}$ and $H_{(k)}^{-1}$ are negative definite. But this is exactly the condition that defines the maximum likelihood estimator, $\hat{\theta}$ so that $\theta_{(k)} \simeq \hat{\theta}$ at the final iteration.

To implement the Newton-Raphson algorithm, both the first and second derivatives of the log-likelihood function, $G(\cdot)$ and $H(\cdot)$, are needed at each iteration. Applying the Newton-Raphson algorithm to estimating the parameter of an exponential distribution numerically highlights the computations required to implement this algorithm. As an analytical solution is available for this example, the accuracy and convergence properties of the numerical procedure can be assessed.

Example 3.2 Exponential Distribution: Newton-Raphson

Let $y_t = \{3.5, 1.0, 1.5\}$ be *iid* drawings from the exponential distribution

$$f(y; \theta) = \frac{1}{\theta} \exp \left[-\frac{y}{\theta} \right],$$

where $\theta > 0$. The log-likelihood function is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta) = -\ln(\theta) - \frac{1}{\theta T} \sum_{t=1}^T y_t = -\ln(\theta) - \frac{2}{\theta}.$$

The first and second derivatives are respectively

$$G_T(\theta) = -\frac{1}{\theta} + \frac{1}{\theta^2 T} \sum_{t=1}^T y_t = -\frac{1}{\theta} + \frac{2}{\theta^2}, \quad H_T(\theta) = \frac{1}{\theta^2} - \frac{2}{\theta^3 T} \sum_{t=1}^T y_t = \frac{1}{\theta^2} - \frac{4}{\theta^3}.$$

Setting $G_T(\hat{\theta}) = 0$ gives the analytical solution

$$\hat{\theta} = \frac{1}{T} \sum_{t=1}^T y_t = \frac{6}{3} = 2.$$

Let the starting value for the Newton-Raphson algorithm be $\theta_{(0)} = 1$. Then the corresponding starting values for the gradient and Hessian are

$$G_{(0)} = -\frac{1}{1} + \frac{2}{1^2} = 1, \quad H_{(0)} = \frac{1}{1^2} - \frac{4}{1^3} = -3.$$

The updated parameter value is computed using (3.4) and is given by

$$\theta_{(1)} = \theta_{(0)} - H_{(0)}^{-1} G_{(0)} = 1 - \left(-\frac{1}{3}\right) \times 1 = 1.333.$$

As $\theta_{(1)} \neq \theta_{(0)}$, the iterations continue. For the next iteration the gradient and Hessian are re-evaluated at $\theta_{(1)} = 1.333$ to give, respectively,

$$G_{(1)} = -\frac{1}{1.333} + \frac{2}{1.333^2} = 0.375, \quad H_{(1)} = \frac{1}{1.333^2} - \frac{4}{1.333^3} = -1.126,$$

yielding the updated value

$$\theta_{(2)} = \theta_{(1)} - H_{(1)}^{-1} G_{(1)} = 1.333 - \left(-\frac{1}{1.126}\right) \times 0.375 = 1.667.$$

As $G_{(1)} = 0.375 < G_{(0)} = 1$, the algorithm is converging to the maximum likelihood estimator where $G_{(k)} \simeq 0$. The calculations for successive iterations are reported in the first block of results in Table 3.1. Using a convergence tolerance of 0.00001, the Newton-Raphson algorithm converges in $k = 7$ iterations to $\hat{\theta} = 2.0$, which is also the analytical solution. $\square$

3.2.2 Method of Scoring

The method of scoring uses the information matrix equality in equation (2.33) of Chapter 2 from which it follows that

$$I(\theta_0) = -E[h_t(\theta_0)].$$

By replacing the expectation by the sample average an estimate of $I(\theta_0)$ is the negative of the Hessian

$$-H_T(\theta_0) = -\frac{1}{T} \sum_{t=1}^T h_t(\theta_0),$$

which is used in the Newton-Raphson algorithm. This suggests that another variation of (3.3) is to replace $-H_T(\theta_0)$ by the information matrix evaluated at $\theta_{(k-1)}$. The iterative scheme of the method of scoring is

$$\theta_{(k)} = \theta_{(k-1)} + I_{(k-1)}^{-1} G_{(k-1)}, \quad (3.5)$$

where $I_{(k)} = E[h_t(\theta_{(k)})]$.

Example 3.3 Exponential Distribution: Method of Scoring

From Example 3.2 the Hessian at time t is

$$h_t(\theta) = \frac{1}{\theta^2} - \frac{2}{\theta^3} y_t.$$

The information matrix is then

$$I(\theta_0) = -E[h_t] = -E\left[\frac{1}{\theta_0^2} - \frac{2}{\theta_0^3} y_t\right] = -\frac{1}{\theta_0^2} + \frac{2}{\theta_0^3} E[y_t] = -\frac{1}{\theta_0^2} + \frac{2\theta_0}{\theta_0^3} = \frac{1}{\theta_0^2},$$

where the result $E[y_t] = \theta_0$ for the exponential distribution is used. Evaluating the gradient and the information matrix at the starting value $\theta_{(0)} = 1$ gives, respectively,

$$G_{(0)} = -\frac{1}{1} + \frac{2}{1^2} = 1, \quad I_{(0)} = \frac{1}{1^2} = 1.$$

The updated parameter value, computed using equation (3.5), is

$$\theta_{(1)} = \theta_{(0)} + I_{(0)}^{-1} G_{(0)} = 1 + \left(\frac{1}{1}\right) \times 1 = 2.$$

The sequence of iterations is in the second block of results in Table 3.1. For this algorithm, convergence is achieved in $k = 1$ iterations since $G_{(1)} = 0$ and $\theta_{(1)} = 2$, which is also the analytical solution. $\square$

As demonstrated by Example 3.3, the method of scoring requires potentially fewer iterations than the Newton-Raphson algorithm to achieve convergence. This is because the scoring algorithm, by replacing the Hessian with the information matrix, uses more information about the structure of the model than does Newton-Raphson. However, for many econometric models the calculation of the information matrix can be difficult, making this algorithm problematic to implement in practice.

3.2.3 BHHH Algorithm

The BHHH algorithm (Berndt, Hall, Hall and Hausman, 1974) uses the information matrix equality in equation (2.33) to express the information

Table 3.1

Demonstration of alternative algorithms to compute the maximum likelihood estimate of the parameter of the exponential distribution.

Iteration	$\theta_{(k-1)}$	$G_{(k-1)}$	$M_{(k-1)}$	$\ln L_{(k-1)}$	$\theta_{(k)}$
Newton-Raphson: $M_{(k-1)} = H_{(k-1)}$					
k = 1	1.0000	1.0000	-3.0000	-2.0000	1.3333
k = 2	1.3333	0.3750	-1.1250	-1.7877	1.6667
k = 3	1.6667	0.1200	-0.5040	-1.7108	1.9048
k = 4	1.9048	0.0262	-0.3032	-1.6944	1.9913
k = 5	1.9913	0.0022	-0.2544	-1.6932	1.9999
k = 6	1.9999	0.0000	-0.2500	-1.6931	2.0000
k = 7	2.0000	0.0000	-0.2500	-1.6931	2.0000
Scoring: $M_{(k-1)} = I_{(k-1)}$					
k = 1	1.0000	1.0000	1.0000	-2.0000	2.0000
k = 2	2.0000	0.0000	0.2500	-1.6931	2.0000
BHHH: $M_{(k-1)} = J_{(k-1)}$					
k = 1	1.0000	1.0000	2.1667	-2.0000	1.4615
k = 2	1.4615	0.2521	0.3192	-1.7479	2.2512
k = 3	2.2512	-0.0496	0.0479	-1.6999	1.2161
k = 4	1.2161	0.5301	0.8145	-1.8403	1.8669
k = 5	1.8669	0.0382	0.0975	-1.6956	2.2586
k = 6	2.2586	-0.0507	0.0474	-1.7002	1.1892
k = 7	1.1892	0.5734	0.9121	-1.8551	1.8178

matrix as

$$I(\theta_0) = J(\theta_0) = E[g_t(\theta_0)g'_t(\theta_0)] . \quad (3.6)$$

Replacing the expectation by the sample average yields an alternative estimate of $I(\theta_0)$ given by

$$J_T(\theta_0) = \frac{1}{T} \sum_{t=1}^T g_t(\theta_0)g'_t(\theta_0) , \quad (3.7)$$

which is the sample analogue of the outer product of gradients matrix. The BHHH algorithm is obtained by replacing $-H_T(\theta_0)$ in equation (3.3) by $J_T(\theta_0)$ evaluated at $\theta_{(k-1)}$.

$$\theta_{(k)} = \theta_{(k-1)} + J_{k-1}^{-1}G_{(k-1)} , \quad (3.8)$$

where

$$J_{(k)} = \frac{1}{T} \sum_{t=1}^T g_t(\theta_{(k)}) g'_t(\theta_{(k)}).$$

Example 3.4 Exponential Distribution: BHHH

To estimate the parameter of the exponential distribution using the BHHH algorithm, the gradient must be evaluated at each observation. From Example 3.2 the gradient at time t is

$$g_t(\theta) = \frac{\partial \ln l_t}{\partial \theta} = -\frac{1}{\theta} + \frac{y_t}{\theta^2}.$$

The outer product of gradients matrix in equation (3.7) is

$$\begin{aligned} J_T(\theta) &= \frac{1}{3} \sum_{t=1}^3 g_t g'_t = \frac{1}{3} \sum_{t=1}^3 g_t^2 \\ &= \frac{1}{3} \left(-\frac{1}{\theta} + \frac{3.5}{\theta^2} \right)^2 + \frac{1}{3} \left(-\frac{1}{\theta} + \frac{1.0}{\theta^2} \right)^2 + \frac{1}{3} \left(-\frac{1}{\theta} + \frac{1.5}{\theta^2} \right)^2. \end{aligned}$$

Using $\theta_{(0)} = 1$ as the starting value gives

$$\begin{aligned} J_{(0)} &= \frac{1}{3} \left(-\frac{1}{1} + \frac{3.5}{1^2} \right)^2 + \frac{1}{3} \left(-\frac{1}{1} + \frac{1.0}{1^2} \right)^2 + \frac{1}{3} \left(-\frac{1}{1} + \frac{1.5}{1^2} \right)^2 \\ &= \frac{2.5^2 + 0.0^2 + 0.5^2}{3} = 2.1667. \end{aligned}$$

The gradient vector evaluated at $\theta_{(0)} = 1$ immediately follows as

$$G_{(0)} = \frac{1}{3} \sum_{t=1}^3 g_t = \frac{2.5 + 0.0 + 0.5}{3} = 1.0.$$

The updated parameter value, computed using equation (3.9), is

$$\theta_{(1)} = \theta_{(0)} + J_{(0)}^{-1} G_{(0)} = 1 + (2.1667)^{-1} \times 1 = 1.4615.$$

The remaining iterations of the BHHH algorithm are contained in the third block of results in Table 3.1. Inspection of these results reveals that the algorithm has still not converged after $k = 7$ iterations with the estimate at this iteration being $\theta_{(7)} = 1.8178$. It is also apparent that successive values of the log-likelihood function at each iteration do not increase monotonically. For iteration $k = 2$, the log-likelihood is $\ln L_{(2)} = -1.6999$, but, for $k = 3$, it decreases to $\ln L_{(3)} = -1.8403$. This problem is addressed in Section 3.4 by using a line-search procedure during the iterations of the algorithm. $\square$

The BHHH algorithm only requires the computation of the gradient of the log-likelihood function and is therefore relatively easy to implement. A potential advantage of this algorithm is that the outer product of the gradients matrix is always guaranteed to be positive semi-definite. The cost of using this algorithm, however, is that it may require more iterations than either the Newton-Raphson or the scoring algorithms do, because information is lost due to the approximation of the information matrix by the outer product of the gradients matrix.

A useful way to think about the structure of the BHHH algorithm is as follows. Let the $(T \times K)$ matrix, X , and the $(T \times 1)$ vector, Y , be given by

$$X = \begin{bmatrix} \frac{\partial \ln l_1(\theta)}{\partial \theta_1} & \frac{\partial \ln l_1(\theta)}{\partial \theta_2} & \cdots & \frac{\partial \ln l_1(\theta)}{\partial \theta_K} \\ \frac{\partial \ln l_2(\theta)}{\partial \theta_1} & \frac{\partial \ln l_2(\theta)}{\partial \theta_2} & \cdots & \frac{\partial \ln l_2(\theta)}{\partial \theta_K} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \ln l_T(\theta)}{\partial \theta_1} & \frac{\partial \ln l_T(\theta)}{\partial \theta_2} & \cdots & \frac{\partial \ln l_T(\theta)}{\partial \theta_K} \end{bmatrix}, \quad Y = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

An iteration of the BHHH algorithm is now written as

$$\theta_{(k)} = \theta_{(k-1)} + (X'_{(k-1)} X_{(k-1)})^{-1} X'_{(k-1)} Y, \quad (3.9)$$

where

$$J_{(k-1)} = \frac{1}{T} X'_{(k-1)} X_{(k-1)}, \quad G_{(k-1)} = \frac{1}{T} X'_{(k-1)} Y.$$

The second term on the right-hand side of equation (3.9) represents an ordinary least squares regression, where the dependent variable Y is regressed on the explanatory variables given by the matrix of gradients, $X_{(k-1)}$, evaluated at $\theta_{(k-1)}$.

3.2.4 Comparative Examples

To highlight the distinguishing features of the Newton-Raphson, scoring and BHHH algorithms, some additional examples are now presented.

Example 3.5 Cauchy Distribution

Let $\{y_1, y_2, \dots, y_T\}$ be T *iid* realized values from the Cauchy distribution.

From Example 3.1, the log-likelihood function is

$$\ln L_T(\theta) = -1 \ln \pi - \frac{1}{T} \sum_{t=1}^T \ln [1 + (y_t - \theta)^2] .$$

Define

$$\begin{aligned} G_T(\theta) &= \frac{2}{T} \sum_{t=1}^T \left[\frac{y_t - \theta}{1 + (y_t - \theta)^2} \right] \\ H_T(\theta) &= \frac{2}{T} \sum_{t=1}^T \frac{(y_t - \theta)^2 - 1}{(1 + (y_t - \theta)^2)^2} \\ J_T(\theta) &= \frac{4}{T} \sum_{t=1}^T \frac{(y_t - \theta)^2}{(1 + (y_t - \theta)^2)^2} \\ I(\theta) &= - \int_{-\infty}^{\infty} \frac{2}{T} \sum_{t=1}^T \frac{(y - \theta)^2 - 1}{(1 + (y - \theta)^2)^2} f(y) dy = \frac{1}{2}, \end{aligned}$$

where the information matrix is as given by Kendall and Stuart (1973, Vol 2). Given the starting value, $\theta_{(0)}$, the first iteration of the Newton-Raphson, scoring and BHHH algorithms are, respectively,

$$\begin{aligned} \theta_{(1)} &= \theta_{(0)} - \left[\frac{2}{T} \sum_{t=1}^T \frac{(y_t - \theta_{(0)})^2 - 1}{(1 + (y_t - \theta_{(0)})^2)^2} \right]^{-1} \left[\frac{2}{T} \sum_{t=1}^T \frac{y_t - \theta_{(0)}}{1 + (y_t - \theta_{(0)})^2} \right] \\ \theta_{(1)} &= \theta_{(0)} + \frac{4}{T} \sum_{t=1}^T \frac{y_t - \theta_{(0)}}{(1 + (y_t - \theta_{(0)})^2)} \\ \theta_{(1)} &= \theta_{(0)} + \frac{1}{2} \left[\frac{1}{T} \sum_{t=1}^T \frac{(y_t - \theta_{(0)})^2}{(1 + (y_t - \theta_{(0)})^2)^2} \right]^{-1} \left[\frac{1}{T} \sum_{t=1}^T \frac{y_t - \theta_{(0)}}{(1 + (y_t - \theta_{(0)})^2)} \right] . \end{aligned}$$

□

Example 3.6 Weibull Distribution

Consider $T = 20$ independent realizations

$$y_t = \{0.293, 0.589, 1.374, 0.954, 0.608, 1.199, 1.464, 0.383, 1.743, 0.022, \\ 0.719, 0.949, 1.888, 0.754, 0.873, 0.515, 1.049, 1.506, 1.090, 1.644\} ,$$

drawn from the Weibull distribution

$$f(y; \theta) = \alpha \beta y^{\beta-1} \exp \left[-\alpha y^\beta \right] ,$$

with unknown parameters $\theta = \{\alpha, \beta\}$. The log-likelihood function is

$$\ln L_T(\alpha, \beta) = \ln \alpha + \ln \beta + (\beta - 1) \frac{1}{T} \sum_{t=1}^T \ln y_t - \alpha \frac{1}{T} \sum_{t=1}^T (y_t)^\beta.$$

Define

$$\begin{aligned} G_T(\theta) &= \begin{bmatrix} \frac{1}{\alpha} - \frac{1}{T} \sum_{t=1}^T y_t^\beta \\ \frac{1}{\beta} + \frac{1}{T} \sum_{t=1}^T \ln y_t - \alpha \frac{1}{T} \sum_{t=1}^T (\ln y_t) y_t^\beta \end{bmatrix} \\ H_T(\theta) &= \begin{bmatrix} -\frac{1}{\alpha^2} & -\frac{1}{T} \sum_{t=1}^T (\ln y_t) y_t^\beta \\ -\frac{1}{T} \sum_{t=1}^T (\ln y_t) y_t^\beta & -\frac{1}{\beta^2} - \alpha \frac{1}{T} \sum_{t=1}^T (\ln y_t)^2 y_t^\beta \end{bmatrix} \\ J_T(\theta) &= \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T \left(\frac{1}{\alpha} - y_t^\beta \right)^2 & \frac{1}{T} \sum_{t=1}^T \left(\frac{1}{\alpha} - y_t^\beta \right) g_{2,t} \\ \frac{1}{T} \sum_{t=1}^T g_{2,t} \left(\frac{1}{\alpha} - y_t^\beta \right) & \frac{1}{T} \sum_{t=1}^T g_{2,t}^2 \end{bmatrix}, \end{aligned}$$

where $g_{2,t} = \beta^{-1} + \ln y_t - \alpha (\ln y_t) y_t^\beta$. Only the iterations of the Newton-Raphson and BHHH algorithms are presented because in this case the information matrix is intractable. Choosing the starting values $\theta_{(0)} = \{0.5, 1.5\}$ yields a log-likelihood function value of $\ln L_{(0)} = -0.959$ and

$$G_{(0)} = \begin{bmatrix} 0.931 \\ 0.280 \end{bmatrix}, \quad H_{(0)} = \begin{bmatrix} -4.000 & -0.228 \\ -0.228 & -0.547 \end{bmatrix}, \quad J_{(0)} = \begin{bmatrix} 1.403 & -0.068 \\ -0.068 & 0.800 \end{bmatrix}.$$

The Newton-Raphson and the BHHH updates are, respectively,

$$\begin{aligned} \begin{bmatrix} \alpha_{(1)} \\ \beta_{(1)} \end{bmatrix} &= \begin{bmatrix} 0.5 \\ 1.5 \end{bmatrix} - \begin{bmatrix} -4.000 & -0.228 \\ -0.228 & -0.547 \end{bmatrix}^{-1} \begin{bmatrix} 0.931 \\ 0.280 \end{bmatrix} = \begin{bmatrix} 0.708 \\ 1.925 \end{bmatrix} \\ \begin{bmatrix} \alpha_{(1)} \\ \beta_{(1)} \end{bmatrix} &= \begin{bmatrix} 0.5 \\ 1.5 \end{bmatrix} + \begin{bmatrix} 1.403 & -0.068 \\ -0.068 & 0.800 \end{bmatrix}^{-1} \begin{bmatrix} 0.931 \\ 0.280 \end{bmatrix} = \begin{bmatrix} 1.183 \\ 1.908 \end{bmatrix}. \end{aligned}$$

Evaluating the log-likelihood function at the updated parameter estimates gives $\ln L_{(1)} = -0.782$ for Newton-Raphson and $\ln L_{(1)} = -0.829$ for BHHH. Both algorithms, therefore, show an improvement in the value of the log-likelihood function after one iteration. $\square$

3.3 Quasi-Newton Methods

The distinguishing feature of the Newton-Raphson algorithm is that it computes the Hessian directly. An alternative approach is to build up an estimate of the Hessian at each iteration, starting from an initial estimate known to be negative definite, usually the negative of the identity matrix. This type of algorithm is known as quasi-Newton. The general form for the updating sequence of the Hessian is

$$H_{(k)} = H_{(k-1)} + U_{(k-1)}, \quad (3.10)$$

where $H_{(k)}$ is the estimate of the Hessian at the k^{th} iteration and $U_{(k)}$ is an update matrix. Quasi-Newton algorithms differ only in their choice of this update matrix. One of the more important variants is the BFGS algorithm (Broyden, 1970; Fletcher, 1970; Goldfarb, 1970; Shanno, 1970) where the updating matrix $U_{(k-1)}$ in equation (3.10) is

$$U_{(k-1)} = -\frac{H_{(k-1)}\Delta_\theta\Delta'_G + \Delta_G\Delta'_\theta H_{(k-1)}}{\Delta'_G\Delta_\theta} + \left(1 + \frac{\Delta'_\theta H_{(k-1)}\Delta_\theta}{\Delta'_G\Delta_\theta}\right)\frac{\Delta_G\Delta'_G}{\Delta'_G\Delta_\theta},$$

where

$$\Delta_\theta = \theta_{(k)} - \theta_{(k-1)}, \quad \Delta_G = G_{(k)} - G_{(k-1)},$$

represent the changes in the parameter values and the gradients between iterations, respectively.

To highlight the properties of the BFGS scheme for updating the Hessian, consider the one parameter case where all terms are scalars. In this situation, the update matrix reduces to

$$U_{(k-1)} = -2H_{(k-1)} + \left(1 + \frac{\Delta_\theta H_{(k-1)}}{\Delta_G}\right)\frac{\Delta_G}{\Delta_\theta},$$

so that the approximation to the Hessian in equation (3.10) is

$$H_{(k)} = \frac{\Delta_G}{\Delta_\theta} = \frac{G_{(k)} - G_{(k-1)}}{\theta_{(k)} - \theta_{(k-1)}}. \quad (3.11)$$

This equation is a numerical approximation to the first derivative of the gradient based on a step length equal to the change in θ across iterations (see Section 3.7.4). For the early iterations of the BFGS algorithm, the numerical approximation is expected to be crude because the size of the step, Δ_θ , is potentially large. As the iterations progress, this step interval diminishes resulting in an improvement in the accuracy of the numerical derivatives as the algorithm approaches the maximum likelihood estimate.

Example 3.7 Exponential Distribution Using BFGS

Continuing the example of the exponential distribution, let the initial value of the Hessian be $H_{(0)} = -1$, and the starting value of the parameter be $\theta_{(0)} = 1.5$. The gradient at $\theta_{(0)}$ is

$$G_{(0)} = -\frac{1}{1.5} + \frac{2}{1.5^2} = 0.2222,$$

and the updated parameter value is

$$\theta_{(1)} = \theta_{(0)} - H_{(0)}^{-1}G_{(0)} = 1.5 - (-1) \times 0.2222 = 1.7222.$$

The gradient evaluated at $\theta_{(1)}$ is

$$G_{(1)} = -\frac{1}{1.7222} + \frac{2}{1.7222^2} = 0.0937,$$

and

$$\begin{aligned}\Delta_{\theta} &= \theta_{(1)} - \theta_{(0)} = 1.7222 - 1.5 = 0.2222 \\ \Delta_G &= G_{(1)} - G_{(0)} = 0.0937 - 0.2222 = -0.1285.\end{aligned}$$

The updated value of the Hessian from equation (3.11) is

$$H_{(1)} = \frac{G_{(1)} - G_{(0)}}{\theta_{(1)} - \theta_{(0)}} = -\frac{0.1285}{0.2222} = -0.5786,$$

so that for iteration $k = 2$

$$\theta_{(2)} = \theta_{(1)} - H_{(1)}^{-1}G_{(1)} = 1.7222 - (-0.5786)^{-1} \times 0.0937 = 1.8841.$$

The remaining iterations are given in Table 3.2. By iteration $k = 6$, the algorithm has converged to the analytical solution $\hat{\theta} = 2$. Moreover, the computed value of the Hessian using the BFGS updating algorithm is equal to its analytical solution of -0.75 . $\square$

3.4 Line Searching

One problem with the simple updating scheme in equation (3.3) is that the updated parameter estimates are not guaranteed to improve the log-likelihood, as in Example 3.4. To ensure that the log-likelihood function increases at each iteration, the algorithm is now augmented by a parameter, λ , that controls the size of updating at each step according to

$$\theta_{(k)} = \theta_{(k-1)} - \lambda H_{(k-1)}^{-1}G_{(k-1)}, \quad 0 \leq \lambda \leq 1. \quad (3.12)$$

Table 3.2

Demonstration of the use of the BFGS algorithm to compute the maximum likelihood estimate of the parameter of the exponential distribution.

Iteration	$\theta_{(k-1)}$	$G_{(k-1)}$	$H_{(k-1)}$	$\ln L_{(k-1)}$	$\theta_{(k)}$
k = 1	1.5000	0.2222	-1.0000	-1.7388	1.7222
k = 2	1.7222	0.0937	-0.5786	-1.7049	1.8841
k = 3	1.8841	0.0327	-0.3768	-1.6950	1.9707
k = 4	1.9707	0.0075	-0.2899	-1.6933	1.9967
k = 5	1.9967	0.0008	-0.2583	-1.6931	1.9999
k = 6	1.9999	0.0000	-0.2508	-1.6931	2.0000
k = 7	2.0000	0.0000	-0.2500	-1.6931	2.0000

For $\lambda = 1$, the full step is taken so updating is as before; for smaller values of λ , updating is not based on the full step. Determining the optimal value of λ at each iteration is a one-dimensional optimization problem known as line searching.

The simplest way to choose λ is to perform a coarse grid search over possible values for λ known as squeezing. Potential choices of λ follow the order

$$\lambda = 1, \lambda = \frac{1}{2}, \lambda = \frac{1}{3}, \lambda = \frac{1}{4}, \dots$$

The strategy is to calculate $\theta_{(k)}$ for $\lambda = 1$ and check to see if $\ln L_{(k)} > \ln L_{(k-1)}$. If this condition is not satisfied, choose $\lambda = 1/2$ and test to see if the log-likelihood function improves. If it does not, then choose $\lambda = 1/3$ and repeat the function evaluation. Once a value of λ is chosen and an updated parameter value is computed, the procedure begins again at the next step with $\lambda = 1$.

Example 3.8 BHHH with Squeezing

In this example, the convergence problems experienced by the BHHH algorithm in Example 3.4 and shown in Table 3.1 are solved by allowing for squeezing. Inspection of Table 3.1 shows that for the simple BHHH algorithm, at iteration $k = 3$, the value of θ changes from $\theta_{(2)} = 2.2512$ to $\theta_{(3)} = 1.2161$ with the value of the log-likelihood function falling from $\ln L_{(2)} = -1.6999$ to $\ln L_{(3)} = -1.8403$. Now squeeze the step interval by $\lambda = 1/2$ so that the updated value of θ at the third iteration is

$$\theta_{(3)} = \theta_{(2)} + \frac{1}{2} J_{(2)}^{-1} G_{(2)} = 2.2512 + \frac{1}{2} \times (0.0479)^{-1} (-0.0496) = 1.7335.$$

Evaluating the log-likelihood function at the new value for $\theta_{(3)}$ gives

$$\ln L_{(3)}(\lambda = 1/2) = -\ln(1.7335) - \frac{2}{1.7335} = -1.7039,$$

which represents an improvement on -1.8403 , but is still lower than $\ln L_{(2)} = -1.6999$.

Table 3.3

Demonstration of the use of the BHHH algorithm with squeezing to compute the maximum likelihood estimate of the parameter of the exponential distribution.

Iteration	$\theta_{(k-1)}$	$G_{(k-1)}$	$J_{(k-1)}$	$\ln L_{(k-1)}$	$\theta_{(k)}$
k=1	1.0000	1.0000	2.1667	-1.7479	1.4615
k=2	1.4615	0.2521	0.3192	-1.6999	2.2512
k=3	2.2512	-0.0496	0.0479	-1.6943	1.9061
k=4	1.9061	0.0258	0.0890	-1.6935	2.0512
k=5	2.0512	-0.0122	0.0661	-1.6934	1.9591
k=6	1.9591	0.0107	0.0793	-1.6932	2.0263
k=7	2.0263	-0.0064	0.0692	-1.6932	1.9801
k=8	1.9801	0.0051	0.0759	-1.6932	2.0136
k=9	2.0136	-0.0033	0.0710	-1.6932	1.9900
k=10	1.9900	0.0025	0.0744	-1.6932	2.0070

By again squeezing the step interval $\lambda = 1/3$, the updated value of θ at the second iteration is now

$$\theta_{(3)} = \theta_{(2)} + \frac{1}{3} J_{(2)}^{-1} G_{(2)} = 2.2512 + \frac{1}{3} \times (0.0479)^{-1}(-0.0496) = 1.9061.$$

Evaluating the log-likelihood function at this value gives

$$\ln L_{(3)}(\lambda = 1/3) = -\ln(1.9061) - \frac{2}{1.9061} = -1.6943.$$

As this value is an improvement on $\ln L_{(2)} = -1.6999$, the value of θ at the second iteration is taken to be $\theta_{(3)} = 1.9061$. Inspection of the log-likelihood function at each iteration in Table 3.3 shows that the improvement in the log-likelihood function is now monotonic. $\square$

3.5 Optimisation Based on Function Evaluation

Practical optimisation problems frequently generate log-likelihood functions with irregular surfaces. In particular, if the gradient is nearly flat in several dimensions, numerical errors can cause a gradient algorithm to misbehave.

Consequently, many iterative algorithms are based solely on function evaluation, including the simplex method of Nelder and Mead (1965) and other more sophisticated schemes such as simulated annealing and genetic search algorithms. These procedures are all fairly robust, but they are more inefficient than gradient-based methods and normally require many more function evaluations to locate the optimum. Because of its popularity in practical work and its simplicity, the simplex algorithm is only briefly described here. For a more detailed account, see Gill, Murray and Wright (1981). This algorithm is usually presented in terms of function minimization rather than the maximising framework adopted in this chapter. This situation is easily accommodated by recognizing that maximizing the log-likelihood function with respect to θ is identical to minimizing the negative log-likelihood function with respect to θ .

The simplex algorithm employs a simple sequence of moves based solely on function evaluations. Consider the negative log-likelihood function $-\ln L_T(\theta)$, which is to be minimized with respect to the parameter vector θ . The algorithm is initialized by evaluating the function for $n + 1$ different starting choices, where $n = \dim(\theta)$, and the function values are ordered so that $-\ln L(\theta_{n+1})$ is the current worst estimate and $-\ln L(\theta_1)$ is the best current estimate, that is $-\ln L(\theta_{n+1}) \geq -\ln L(\theta_n) \geq \dots \geq -\ln L(\theta_1)$. Define

$$\bar{\theta} = \frac{1}{n} \sum_{i=1}^n \theta_i,$$

as the mean (centroid) of the best n vertices. In a two-dimensional problem, $\bar{\theta}$ is the midpoint of the line joining the two best vertices of the current simplex. The basic iteration of the simplex algorithm consists of the following sequence of steps.

Reflect: Reflect the worst vertex through the opposite face of the simplex

$$\theta_r = \bar{\theta} + \alpha(\bar{\theta} - \theta_{n+1}), \quad \alpha > 0.$$

If the reflection is successful, $-\ln L(\theta_r) < -\ln L(\theta_n)$, start the next iteration by replacing θ_{n+1} with θ_r .

Expand: If θ_r is also better than θ_1 , $-\ln L(\theta_r) < -\ln L(\theta_1)$, compute

$$\theta_e = \bar{\theta} + \beta(\theta_r - \bar{\theta}), \quad \beta > 1.$$

If $-\ln L(\theta_e) < -\ln L(\theta_r)$, start the next iteration by replacing θ_{n+1} with θ_e .

Contract: If θ_r is not successful, $-\ln L(\theta_r) > -\ln L(\theta_n)$, contract the sim-

plex as follows:

$$\theta_c = \begin{cases} \bar{\theta} + \gamma(\theta_r - \bar{\theta}) & \text{if } -\ln L(\theta_r) < -\ln L(\theta_{n+1}) \\ \bar{\theta} + \gamma(\theta_{n+1} - \bar{\theta}) & \text{if } -\ln L(\theta_r) \geq -\ln L(\theta_{n+1}), \end{cases}$$

for $0 < \gamma < 1$.

Shrink: If the contraction is not successful, shrink the vertices of the simplex half-way toward the current best point and start the next iteration.

To make the simplex algorithm operational, values for the reflection, α , expansion, β , and contraction, γ , parameters are required. Common choices of these parameters are $\alpha = 1$, $\beta = 2$ and $\gamma = 0.5$ (see Gill, Murray and Wright, 1981; Press, Teukolsky, Vetterling and Flannery, 1992).

3.6 Computing Standard Errors

From Chapter 2, the asymptotic distribution of the maximum likelihood estimator is

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, I^{-1}(\theta_0)).$$

The covariance matrix of the maximum likelihood estimator is estimated by replacing θ_0 by $\hat{\theta}$ and inverting the information matrix

$$\hat{\Omega} = I^{-1}(\hat{\theta}). \quad (3.13)$$

The standard error of each element of $\hat{\theta}$ is given by the square root of the main-diagonal entries of this matrix. In most practical situations, the information matrix is not easily evaluated. A more common approach, therefore, is simply to use the negative of the inverse Hessian evaluated at $\hat{\theta}$:

$$\hat{\Omega} = -H_T^{-1}(\hat{\theta}). \quad (3.14)$$

If the Hessian is not negative-definite at the maximum likelihood estimator, computation of the standard errors from equation (3.14) is not possible. A popular alternative is to use the outer product of gradients matrix, $J_T(\hat{\theta})$ from equation (3.7), instead of the negative of the Hessian

$$\hat{\Omega} = J_T^{-1}(\hat{\theta}). \quad (3.15)$$

Example 3.9 Exponential Distribution Standard Errors

The values of the Hessian and the information matrix, taken from Table 3.1, and the outer product of gradients matrix, taken from Table 3.3, are, respectively,

$$H_T(\hat{\theta}) = -0.250, \quad I(\hat{\theta}) = 0.250, \quad J_T(\hat{\theta}) = 0.074.$$

The standard errors are

$$\begin{aligned}
 \text{Hessian :} \quad \text{se}(\hat{\theta}) &= \sqrt{-\frac{1}{T}H_T^{-1}(\hat{\theta})} = \sqrt{-\frac{1}{3}(-0.250)^{-1}} = 1.547 \\
 \text{Information :} \quad \text{se}(\hat{\theta}) &= \sqrt{\frac{1}{T}I^{-1}(\hat{\theta})} = \sqrt{\frac{1}{3}(0.250)^{-1}} = 1.547 \\
 \text{Outer Product :} \quad \text{se}(\hat{\theta}) &= \sqrt{\frac{1}{T}J_T^{-1}(\hat{\theta})} = \sqrt{\frac{1}{3}(0.074)^{-1}} = 2.122.
 \end{aligned}$$

The standard errors based on the Hessian and information matrices yield the same values, while the estimate based on the outer product of gradients matrix is nearly 40% larger. One reason for this difference is that the outer product of the gradients matrix may not always provide a good approximation to the information matrix. Another reason is that the information and outer product of the gradients matrices may not converge to the same value as T increases. This occurs when the distribution used to construct the log-likelihood function is misspecified (see Chapter 9). $\square$

Estimating the covariance matrix of a nonlinear function of the maximum likelihood estimators, say $C(\theta)$, is a situation that often arises in practice. There are two approaches to dealing with this problem. The first approach, known as the substitution method, simply imposes the nonlinearity and then uses the constrained log-likelihood function to compute standard errors. The second approach, called the delta method, uses a mean value expansion of $C(\hat{\theta})$ around the true parameter θ_0

$$C(\hat{\theta}) = C(\theta_0) + D(\theta^*)(\hat{\theta} - \theta_0),$$

where

$$D(\theta) = \frac{\partial C(\theta)}{\partial \theta'},$$

and θ^* is an intermediate value between $\hat{\theta}$ and θ_0 . As $T \rightarrow \infty$ the mean value expansion gives

$$\begin{aligned}
 \sqrt{T}(C(\hat{\theta}) - C(\theta_0)) &= D(\theta^*)\sqrt{T}(\hat{\theta} - \theta_0) \\
 &\xrightarrow{d} D(\theta_0) \times N(0, I(\theta_0)^{-1}) \\
 &= N(0, D(\theta_0)I(\theta_0)^{-1}D(\theta_0)'),
 \end{aligned}$$

or

$$C(\hat{\theta}) \overset{a}{\sim} N(C(\theta_0), \frac{1}{T}D(\theta_0)I^{-1}(\theta_0)D(\theta_0)').$$

Thus

$$\text{cov}(C(\hat{\theta})) = \frac{1}{T} D(\theta_0) I^{-1}(\theta_0) D(\theta_0)',$$

and this can be estimated by replacing $D(\theta_0)$ with $D(\hat{\theta})$ and $I^{-1}(\theta_0)$ with $\hat{\Omega}$ from any of equations (3.13), (3.14) or (3.15).

Example 3.10 Standard Errors of Nonlinear Functions

Consider the problem of finding the standard error for $\bar{y}^2$ where observations are drawn from a normal distribution with known variance σ_0^2 .

(1) *Substitution Method*

Consider the log-likelihood function for the unconstrained problem

$$\ln L_T(\theta) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma_0^2) - \frac{1}{2\sigma_0^2 T} \sum_{t=1}^T (y_t - \theta)^2.$$

Now define $\psi = \theta^2$ so that the constrained log-likelihood function is

$$\ln L_T(\psi) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma_0^2) - \frac{1}{2\sigma_0^2 T} \sum_{t=1}^T (y_t - \psi^{1/2})^2.$$

The first and second derivatives are

$$\begin{aligned} \frac{d \ln L_T(\psi)}{d\psi} &= \frac{1}{2\sigma_0^2 T} \sum_{t=1}^T (y_t - \psi^{1/2}) \psi^{-1/2} \\ \frac{d^2 \ln L_T(\psi)}{d\psi^2} &= -\frac{1}{2\sigma_0^2 T} \sum_{t=1}^T \left(\frac{1}{2\psi} + (y_t - \psi^{1/2}) \frac{1}{2} \psi^{-3/2} \right). \end{aligned}$$

Recognizing that $E[y_t - \psi_0^{1/2}] = 0$, the information matrix is

$$I(\psi_0) = -E \left[\frac{d^2 \ln l_t}{d\psi^2} \right] = \frac{1}{2\sigma_0^2} \frac{1}{2\psi_0} = \frac{1}{4\sigma_0^4 \psi_0} = \frac{1}{4\sigma_0^2 \theta_0^2}.$$

The standard error is then

$$\text{se}(\hat{\psi}) = \sqrt{\frac{1}{T} I^{-1}(\hat{\theta})} = \sqrt{\frac{4\sigma_0^2 \theta^2}{T}}.$$

(2) *Delta Method*

For a normal distribution, the variance of the maximum likelihood estimator $\hat{\theta} = \bar{y}$ is σ_0^2/T . Define $C(\theta) = \theta^2$ so that

$$\text{se}(\hat{\psi}) = \sqrt{D(\theta_0)^2 \text{var}(\theta_0)} = \sqrt{\frac{(2\theta)^2 \sigma_0^2}{T}} = \sqrt{\frac{4\sigma_0^2 \theta^2}{T}},$$

which agrees with the variance obtained using the substitution method. $\square$

3.7 Hints for Practical Optimization

This section provides an eclectic collection of ideas that may be drawn on to help in many practical situations.

3.7.1 Concentrating the Likelihood

For certain problems, the dimension of the parameter vector to be estimated may be reduced. Such a reduction is known as concentrating the likelihood function and it arises when the gradient can be rearranged to express an unknown parameter as a function of another unknown parameter.

Consider a log-likelihood function that is a function of two unknown parameter vectors $\theta = \{\theta_1, \theta_2\}$, with dimensions $\dim(\theta_1) = K_1$ and $\dim(\theta_2) = K_2$, respectively. The first-order conditions to find the maximum likelihood estimators are

$$\left. \frac{\partial \ln L_T(\theta)}{\partial \theta_1} \right|_{\theta=\hat{\theta}} = 0, \quad \left. \frac{\partial \ln L_T(\theta)}{\partial \theta_2} \right|_{\theta=\hat{\theta}} = 0,$$

which is a nonlinear system of $K_1 + K_2$ equations in $K_1 + K_2$ unknowns. If it is possible to write

$$\hat{\theta}_2 = g(\hat{\theta}_1), \tag{3.16}$$

then the problem is reduced to a K_1 dimensional problem. The log-likelihood function is now maximized with respect to θ_1 to yield $\hat{\theta}_1$. Once the algorithm has converged, $\hat{\theta}_1$ is substituted into (3.16) to yield $\hat{\theta}_2$. The estimator of θ_2 is a maximum likelihood estimator because of the invariance property of maximum likelihood estimators discussed in Chapter 2. Standard errors are obtained from evaluating the full log-likelihood function containing all parameters. An alternative way of reducing the dimension of the problem is to compute the profile log-likelihood function (see Exercise 8).

Example 3.11 Weibull Distribution

Let $y_t = \{y_1, y_2, \dots, y_T\}$ be *iid* observations drawn from the Weibull distribution given by

$$f(y; \alpha, \beta) = \beta \alpha x^{\beta-1} \exp(-\alpha y^\beta).$$

The log-likelihood function is

$$\ln L_T(\theta) = \ln \alpha + \ln \beta + (\beta - 1) \frac{1}{T} \sum_{t=1}^T \ln y_t - \alpha \frac{1}{T} \sum_{t=1}^T y_t^\beta,$$

and the unknown parameters are $\theta = \{\alpha, \beta\}$. The first-order conditions are

$$\begin{aligned} 0 &= \frac{1}{\hat{\alpha}} - \frac{1}{T} \sum_{t=1}^T y_t^{\hat{\beta}} \\ 0 &= \frac{1}{\hat{\beta}} + \frac{1}{T} \sum_{t=1}^T \ln y_t - \hat{\alpha} \frac{1}{T} \sum_{t=1}^T (\ln y_t) y_t^{\hat{\beta}}, \end{aligned}$$

which are two nonlinear equations in $\hat{\theta} = \{\hat{\alpha}, \hat{\beta}\}$. The first equation gives

$$\hat{\alpha} = \frac{T}{\sum_{t=1}^T y_t^{\hat{\beta}}},$$

which is used to substitute for $\hat{\alpha}$ in the equation for $\hat{\beta}$. The maximum likelihood estimate for $\hat{\beta}$ is then found using numerical methods with $\hat{\alpha}$ evaluated at the last step. $\square$

3.7.2 Parameter Constraints

In some econometric applications, the values of the parameters need to be constrained to lie within certain intervals. Some examples are as follows: an estimate of variance is required to be positive ($\theta > 0$); the marginal propensity to consume is constrained to be positive but less than unity ($0 < \theta < 1$); for an MA(1) process to be invertible, the moving average parameter must lie within the unit interval ($-1 < \theta < 1$); and the degrees of freedom parameter in the Student t distribution must be greater than 2, to ensure that the variance of the distribution exists.

Consider the case of estimating a single parameter θ , where $\theta \in (a, b)$. The approach is to transform the parameter θ by means of a nonlinear bijective (one-to-one) mapping, $\phi = c(\theta)$, between the constrained interval (a, b) and the real line. Thus each and every value of ϕ corresponds to a unique value of θ , satisfying the desired constraint, and is obtained by applying the inverse transform $\theta = c^{-1}(\phi)$. When the numerical algorithm returns $\hat{\phi}$ from the invariance property, the associated estimate of θ is given by $\hat{\theta} = c^{-1}(\hat{\phi})$. Some useful one-dimensional transformations, their associated inverse functions and the gradients of the transformations are presented in Table 3.4.

Table 3.4

Some useful transformations for imposing constraints on θ .

Constraint	Transform $\phi = c(\theta)$	Inverse Transform $\theta = c^{-1}(\phi)$	Jacobian $dc(\theta)/d\theta$
$(0, \infty)$	$\phi = \ln \theta$	$\theta = e^\phi$	$\frac{1}{\theta}$
$(-\infty, 0)$	$\phi = \ln(-\theta)$	$\theta = -e^\phi$	$\frac{1}{\theta}$
$(0, 1)$	$\phi = \ln\left(\frac{\theta}{1-\theta}\right)$	$\theta = \frac{1}{1+e^{-\phi}}$	$\frac{1}{\theta(1-\theta)}$
$(0, b)$	$\phi = \ln\left(\frac{\theta}{b-\theta}\right)$	$\theta = \frac{b}{1+e^{-\phi}}$	$\frac{b}{\theta(b-\theta)}$
(a, b)	$\phi = \ln\left(\frac{\theta-a}{b-\theta}\right)$	$\theta = \frac{b+ae^{-\phi}}{1+e^{-\phi}}$	$\frac{b-a}{(\theta-a)(b-\theta)}$
$(-1, 1)$	$\phi = \operatorname{atanh}(\theta)$	$\theta = \tanh(\phi)$	$\frac{1}{1-\theta^2}$
$(-1, 1)$	$\phi = \frac{\theta}{1- \theta }$	$\theta = \frac{\phi}{1+ \phi }$	$\frac{1}{(1- \theta)^2}$
$(-1, 1)$	$\phi = \tan\left(\frac{\pi\theta}{2}\right)$	$\theta = \frac{2}{\pi} \tan^{-1} \phi$	$\frac{\pi}{2} \sec^2\left(\frac{\pi\theta}{2}\right)$

The convenience of using an unconstrained algorithm on what is essentially a constrained problem has a price: the standard errors of the model parameters cannot be obtained simply by taking the square roots of the diagonal elements of the inverse Hessian matrix of the transformed problem. A straightforward way to compute standard errors is the method of substitution discussed in Section 3.6 where the objective function is expressed in terms of the original parameters, θ . The gradient vector and Hessian matrix can then be computed numerically at the maximum of the log-likelihood function using the estimated values of the parameters. Alternatively, the delta method can be used.

3.7.3 Choice of Algorithm

In theory, there is little to choose between the algorithms discussed in this chapter, because in the vicinity of a minimum each should enjoy quadratic

convergence, which means that

$$\|\theta_{(k+1)} - \theta\| < \kappa \|\theta_{(k)} - \theta\|^2, \quad \kappa > 0.$$

If $\theta_{(k)}$ is accurate to 2 decimal places, then it is anticipated that $\theta_{(k+1)}$ will be accurate to 4 decimal places and that $\theta_{(k+2)}$ will be accurate to 8 decimal places and so on. In choosing an algorithm, however, there are a few practical considerations to bear in mind.

- (1) The Newton-Raphson and the method of scoring require the first two derivatives of the log-likelihood function. Because the information matrix is the expected value of the negative Hessian matrix, it is problem specific and typically is not easy to compute. Consequently, the method of scoring is largely of theoretical interest.
- (2) Close to the maximum, Newton-Raphson converges quadratically, but, further away from the maximum, the Hessian matrix may not be negative definite and this may cause the algorithm to become unstable.
- (3) BHHH ensures that the outer product of the gradients matrix is positive semi-definite making it a popular choice of algorithm for econometric problems.
- (4) The current consensus seems to be that quasi-Newton algorithms are the preferred choice. The Hessian update of the BFGS algorithm is particularly robust and is, therefore, the default choice in many practical settings.
- (5) A popular practical strategy is to use the simplex method to start the numerical optimization process. After a few iterations, the BFGS algorithm is employed to speed up convergence.

3.7.4 Numerical Derivatives

For problems where deriving analytical derivatives is difficult, numerical derivatives can be used instead. A first-order numerical derivative is computed simply as

$$\left. \frac{\partial \ln L_T(\theta)}{\partial \theta} \right|_{\theta=\theta_{(k)}} \simeq \frac{\ln L(\theta_{(k)} + s) - \ln L(\theta_{(k)})}{s},$$

where s is a suitably small step size. A second-order derivative is computed as

$$\left. \frac{\partial^2 \ln L_T(\theta)}{\partial \theta^2} \right|_{\theta=\theta_{(k)}} \simeq \frac{\ln L(\theta_{(k)} + s) - 2 \ln L(\theta_{(k)}) + \ln L(\theta_{(k)} - s)}{s^2}.$$

In general, the numerical derivatives are accurate enough to enable the maximum likelihood estimators to be computed with sufficient precision and most good optimization routines will automatically select an appropriate value for the step size, s .

One computational/programming advantage of using numerical derivatives is that it is then necessary to program only the log-likelihood function. A cost of using numerical derivatives is computational time, since the algorithm is slower than if analytical derivatives are used, although the absolute time difference is nonetheless very small given current computer hardware. Gradient algorithms based on numerical derivatives can also be thought of as a form of algorithm based solely on function evaluation, which differs from the simplex algorithm only in the way in which this information is used to update the parameter estimate.

3.7.5 Starting Values

All numerical algorithms require starting values, $\theta_{(0)}$, for the parameter vector. There are a number of strategies to choose starting values.

- (1) **Arbitrary choice:** This method only works well if the log-likelihood function is globally concave. As a word of caution, in some cases $\theta_{(0)} = \{0\}$ is a bad choice of starting value because it can lead to multicollinearity problems causing the algorithm to break down.
- (2) **Consistent estimator:** This approach is only feasible if a consistent estimator of the parameter vector is available. An advantage of this approach is that one iteration of a Newton algorithm yields an asymptotically efficient estimator (Harvey, 1990, pp 142 - 142). An example of a consistent estimator of the location parameter of the Cauchy distribution is given by the median (see Example 2.23 in Chapter 2).
- (3) **Restricted model:** A restricted model is specified in which closed-form expressions are available for the remaining parameters.
- (4) **Historical precedent:** Previous empirical work of a similar nature may provide guidance on the choice of reasonable starting values.

3.7.6 Convergence Criteria

A number of convergence criteria are employed in identifying when the maximum likelihood estimates are reached. Given a convergence tolerance of ε , say equal to 0.00001, some of the more commonly adopted convergence criteria are as follows:

- (1) **Objective function:** $\ln L(\theta_{(k)}) - \ln L(\theta_{(k-1)}) < \varepsilon$.
- (2) **Gradient function:** $G(\theta_{(k)})'G(\theta_{(k)}) < \varepsilon$.
- (3) **Parameter values:** $(\theta_{(k)})'(\theta_{(k)}) < \varepsilon$.
- (4) **Updating function:** $G(\theta_{(k)})H(\theta_{(k)})^{-1}G(\theta_{(k)}) < \varepsilon$.

In specifying the termination rule, there is a tradeoff between the precision of the estimates, which requires a stringent convergence criterion, and the precision with which the objective function and gradients can be computed. Too slack a termination criterion is almost sure to produce convergence, but the maximum likelihood estimator is likely to be imprecisely estimated in these situations.

3.8 Applications

In this section, two applications are presented which focus on estimating the continuous-time model of interest rates, r_t , known as the CIR model (Cox, Ingersoll and Ross, 1985) by maximum likelihood. Estimation of continuous-time models using simulation based estimation are discussed in more detail in Chapter 12. The CIR model is one in which the interest rate evolves over time in steps of dt in accordance with

$$dr = \alpha(\mu - r)dt + \sigma\sqrt{r} dB, \quad (3.17)$$

where $dB \sim N(0, dt)$ is the disturbance term over dt and $\theta = \{\alpha, \mu, \sigma\}$ are model parameters. This model requires the interest rate to revert to its mean, μ , at a speed given by α , with variance $\sigma^2 r$. As long as the condition $2\alpha\mu \geq \sigma^2$ is satisfied, interest rates are never zero.

As in Section 1.5 of Chapter 1, the data for these applications are the daily 7-day Eurodollar interest rates used by Ait-Sahalia (1996) for the period 1 June 1973 to 25 February 1995, $T = 5505$ observations, except that now the data are expressed in raw units rather than percentages. The first application is based on the stationary (unconditional) distribution while the second focuses on the transitional (conditional) distribution.

3.8.1 Stationary Distribution of the CIR Model

The stationary distribution of the interest rate, r_t whose evolution is governed by equation (3.17), is shown by Cox, Ingersoll and Ross (1985) to be a gamma distribution

$$f(r; \nu, \omega) = \frac{\omega^\nu}{\Gamma(\nu)} r^{\nu-1} e^{-\omega r}, \quad (3.18)$$

where $\Gamma(\cdot)$ is the Gamma function with parameters ν and ω . The log-likelihood function is

$$\ln L_T(\nu, \omega) = (\nu - 1) \frac{1}{T} \sum_{t=1}^T \ln(r_t) + \nu \ln \omega - \ln \Gamma(\nu) - \omega \frac{1}{T} \sum_{t=1}^T r_t, \quad (3.19)$$

where $\theta = \{\nu, \omega\}$. The relationship between the parameters of the stationary gamma distribution and the model parameters of the CIR equation (3.17) is

$$\omega = \frac{2\alpha}{\sigma^2}, \quad \nu = \frac{2\alpha\mu}{\sigma^2}. \quad (3.20)$$

As there is no closed-form solution for the maximum likelihood estimator, $\hat{\theta}$, an iterative algorithm is needed. The maximum likelihood estimates obtained by using the BFGS algorithm are

$$\hat{\omega} = \frac{67.634}{(1.310)}, \quad \hat{\nu} = \frac{5.656}{(0.105)}, \quad (3.21)$$

with standard errors based on the inverse Hessian shown in parentheses. An estimate of the mean from equation (3.20) is

$$\hat{\mu} = \frac{\hat{\nu}}{\hat{\omega}} = \frac{5.656}{67.634} = 0.084,$$

or 8.4% per annum.

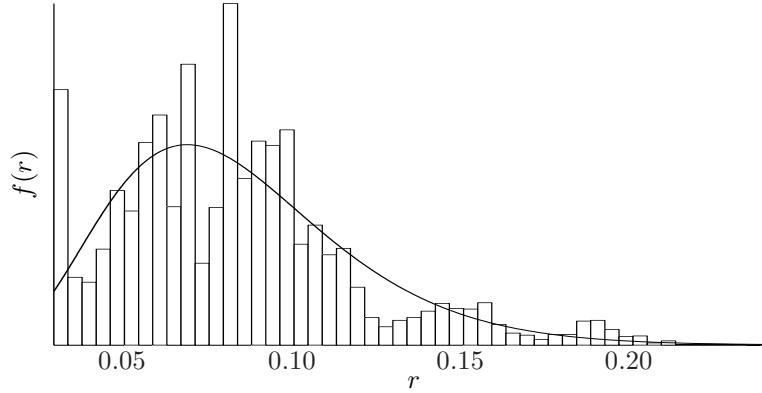


Figure 3.1 Estimated stationary gamma distribution of Eurodollar interest rates from the 1 June 1973 to 25 February 1995.

Figure 3.1 plots the gamma distribution in equation (3.18) evaluated at the maximum likelihood estimates $\hat{\nu}$ and $\hat{\omega}$ given in equation (3.21). The results cast some doubt on the appropriateness of the CIR model for these data, because the gamma density does not capture the bunching effect at

very low interest rates and also underestimates the peak of the distribution. The upper tail of the gamma distribution, however, does provide a reasonable fit to the observed Eurodollar interest rates.

The three parameters of the CIR model cannot all be uniquely identified from the two parameters of the stationary distribution. This distribution can identify only the ratio α/σ^2 and the parameter μ using equation (3.20). Identifying all three parameters of the CIR model requires using the transitional distribution of the process.

3.8.2 Transitional Distribution of the CIR Model

To estimate the parameters of the CIR model in equation (3.17), the transitional distribution must be used to construct the log-likelihood function. The transitional distribution of r_t given r_{t-1} is

$$f(r_t | r_{t-1}; \theta) = ce^{-u-v} \left(\frac{v}{u}\right)^{\frac{q}{2}} I_q(2\sqrt{uv}), \quad (3.22)$$

where $I_q(x)$ is the modified Bessel function of the first kind of order q (see, for example, Abramovitz and Stegun, 1965) and

$$c = \frac{2\alpha}{\sigma^2(1 - e^{-\alpha\Delta})}, \quad u = cr_{t-1}e^{-\alpha\Delta}, \quad v = cr_t, \quad q = \frac{2\alpha\mu}{\sigma^2} - 1,$$

where the parameter Δ is a time step defined to be 1/252 because the data are daily. Cox, Ingersoll and Ross (1985) show that the transformed variable $2cr_t$ is distributed as a non-central chi-square random variable with $2q + 2$ degrees of freedom and non-centrality parameter $2u$.

In constructing the log-likelihood function there are two equivalent approaches. The first is to construct the log-likelihood function for r_t directly from (3.22). In this instance care must be exercised in the computation of the modified Bessel function, $I_q(x)$, because it can be numerically unstable (Hurn, Jeisman and Lindsay, 2007). It is advisable to work with a scaled version of this function

$$I_q^s(2\sqrt{uv}) = e^{-2\sqrt{uv}} I_q(2\sqrt{uv})$$

so that the log-likelihood function at observation t is

$$\ln l_t(\theta) = \log c - u - v + \frac{q}{2} \log \left(\frac{v}{u}\right) + \log(I_q^s(2\sqrt{uv})) + 2\sqrt{uv}, \quad (3.23)$$

where $\theta = \{\alpha, \mu, \sigma\}$. The second approach is to use the non-central chi-square distribution for the variable $2cr_t$ and then use the transformation of variable technique to obtain the density for r_t . These methods are equivalent

and produce identical results. As with the stationary distribution of the CIR model, no closed-form solution for the maximum likelihood estimator, $\hat{\theta}$, exists and an iterative algorithm must be used.

To obtain starting values, a discrete version of equation (3.17)

$$r_t - r_{t-1} = \alpha(\mu - r_{t-1})\Delta + \sigma\sqrt{r_{t-1}}e_t, \quad e_t \sim N(0, \Delta), \quad (3.24)$$

is used. Transforming equation (3.24) into

$$\frac{r_t - r_{t-1}}{\sqrt{r_{t-1}}} = \frac{\alpha\mu\Delta}{\sqrt{r_{t-1}}} - \alpha\sqrt{r_{t-1}}\Delta + \sigma e_t,$$

allows estimates of $\alpha\mu$ and α to be obtained by an ordinary least squares regression of $(r_t - r_{t-1})/\sqrt{r_{t-1}}$ on $\Delta/\sqrt{r_{t-1}}$ and $\sqrt{r_{t-1}}\Delta$. A starting value for σ is obtained as the standard deviation of the ordinary least squares residuals.

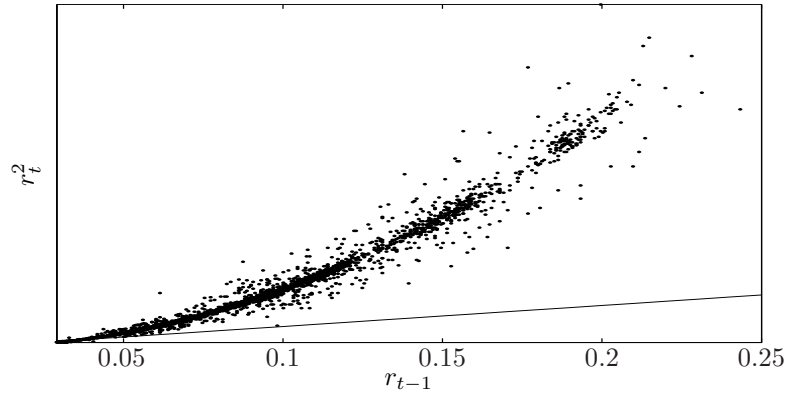


Figure 3.2 Scatter plot of r_t^2 on r_{t-1} together with the model predicted value, $\hat{\sigma}^2 r_{t-1}$ (solid line).

Maximum likelihood estimates, obtained using the BFGS algorithm, are

$$\hat{\alpha} = \frac{1.267}{(0.340)}, \quad \hat{\mu} = \frac{0.083}{(0.009)}, \quad \hat{\sigma} = \frac{0.191}{(0.002)}, \quad (3.25)$$

with standard errors based on the inverse Hessian shown in parentheses. The mean interest rate is 0.083, or 8.3% per annum, and the estimate of variance is $0.192^2 r$. While the estimates of μ and σ appear to be plausible, the estimate of α appears to be somewhat higher than usually found in models of this kind. The solution to this conundrum is to be found in the specification of the variance in this model. Figure 3.2 shows a scatter plot of r_t^2 on r_{t-1} and superimposes on it the predicted value in terms of the

CIR model, $\hat{\sigma}^2 r_{t-1}$. It appears that the variance specification of the CIR model is not dynamic enough to capture the dramatic increases in r_t^2 as r_{t-1} increases. This problem is explored further in Chapter 9 in the context of quasi-maximum likelihood estimation and in Chapter 12 dealing with estimation by simulation.

3.9 Exercises

(1) Maximum Likelihood Estimation using Graphical Methods

Gauss file(s)	<code>max_graph.g</code>
Matlab file(s)	<code>max_graph.m</code>

Consider the regression model

$$y_t = \beta x_t + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

where x_t is an explanatory variable given by $x_t = \{1, 2, 4, 5, 8\}$.

- (a) Simulate the model for $T = 5$ observations using the parameter values $\theta = \{\beta = 1, \sigma^2 = 4\}$.
- (b) Compute the log-likelihood function, $\ln L_T(\theta)$, for:
 - (i) $\beta = \{0.0, 0.1, \dots, 1.9, 2.0\}$ and $\sigma^2 = 4$;
 - (ii) $\beta = \{0.0, 0.1, \dots, 1.9, 2.0\}$ and $\sigma^2 = 3.5$;
 - (iii) plot $\ln L_T(\theta)$ against β for parts (i) and (ii).
- (c) Compute the log-likelihood function, $\ln L_T(\theta)$, for:
 - (i) $\beta = \{1.0\}$ and $\sigma^2 = \{1.0, 1.5, \dots, 10.5, 11\}$;
 - (ii) $\beta = \{0.9\}$ and $\sigma^2 = \{1.0, 1.5, \dots, 10.5, 11\}$;
 - (iii) plot $\ln L_T(\theta)$ against σ^2 for parts (i) and (ii).

(2) Maximum Likelihood Estimation using Grid Searching

Gauss file(s)	<code>max_grid.g</code>
Matlab file(s)	<code>max_grid.m</code>

Consider the regression model set out in Exercise 1.

- (a) Simulate the model for $T = 5$ observations using the parameter values $\theta = \{\beta = 1, \sigma^2 = 4\}$.
- (b) Derive an expression for the gradient with respect to β , $G_T(\beta)$.
- (c) Choosing $\sigma^2 = 4$ perform a grid search of β over $G_T(\beta)$ with $\beta = \{0.5, 0.6, \dots, 1.5\}$ and thus find the maximum likelihood estimator of β conditional on $\sigma^2 = 4$.

- (d) Repeat part (c) except set $\sigma^2 = 3.5$. Find the maximum likelihood estimator of β conditional on $\sigma^2 = 3.5$.

(3) **Maximum Likelihood Estimation using Newton-Raphson**

Gauss file(s)	<code>max_nr.g</code> , <code>max_iter.g</code>
Matlab file(s)	<code>max_nr.m</code> , <code>max_iter.m</code>

Consider the regression model set out in Example 1.

- (a) Simulate the model for $T = 5$ observations using the parameter values $\theta = \{\beta = 1, \sigma^2 = 4\}$.
- (b) Find the log-likelihood function, $\ln L_T(\theta)$, the gradient, $G_T(\theta)$, and the Hessian, $H_T(\theta)$.
- (c) Evaluate $\ln L_T(\theta)$, $G_T(\theta)$ and $H_T(\theta)$ at $\theta_{(0)} = \{1, 4\}$.
- (d) Update the value of the parameter vector using the Newton-Raphson update scheme

$$\theta_{(1)} = \theta_{(0)} - H_{(0)}^{-1} G_{(0)},$$

and recompute $\ln L_T(\theta)$ at $\theta_{(1)}$. Compare this value with that obtained in part (c).

- (e) Continue the iterations in (d) until convergence and compare these values to those obtained from the maximum likelihood estimators

$$\hat{\beta} = \frac{\sum_{t=1}^T x_t y_t}{\sum_{t=1}^T x_t^2}, \quad \hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{\beta} x_t)^2.$$

(4) **Exponential Distribution**

Gauss file(s)	<code>max_exp.g</code>
Matlab file(s)	<code>max_exp.m</code>

The aim of this exercise is to reproduce the convergence properties of the different algorithms in Table 3.1. Suppose that the following observations $\{3.5, 1.0, 1.5\}$ are taken from the exponential distribution

$$f(y; \theta) = \frac{1}{\theta} \exp \left[-\frac{y}{\theta} \right], \quad \theta > 0.$$

- (a) Derive the log-likelihood function $\ln L_T(\theta)$ and also analytical expressions for the gradient, $G_T(\theta)$, the Hessian, $H_T(\theta)$, and the outer product of gradients matrix, $J_T(\theta)$.
- (b) Using $\theta_{(0)} = 1$ as the starting value, compute the first seven iterations of the Newton-Raphson, scoring and BHHH algorithms.

- (c) Redo (b) with $G_T(\theta)$ and $H_T(\theta)$ computed using numerical derivatives.
- (d) Estimate $\text{var}(\hat{\theta})$ based on $H_T(\theta)$, $J_T(\theta)$ and $I(\theta)$.

(5) Cauchy Distribution

Gauss file(s)	<code>max_cauchy.g</code>
Matlab file(s)	<code>max_cauchy.m</code>

An *iid* random sample of size $T = 5$, $y_t = \{2, 5, -2, 3, 3\}$, is drawn from a Cauchy distribution

$$f(y; \theta) = \frac{1}{\pi} \frac{1}{1 + (y - \theta)^2}.$$

- (a) Write the log-likelihood function at the t^{th} observation as well as the log-likelihood function for the sample.
- (b) Choosing the median, m , as a starting value for the parameter θ , update the value of θ with one iteration of the Newton-Raphson, scoring and BHHH algorithms.
- (c) Show that the maximum likelihood estimator converges to $\hat{\theta} = 2.841$ by computing $G_T(\hat{\theta})$. Also show that $\ln L_T(\hat{\theta}) > \ln L_T(m)$.
- (d) Compute an estimate of the standard error of $\hat{\theta}$ based on $H_T(\theta)$, $J_T(\theta)$ and $I(\theta)$.

(6) Weibull Distribution

Gauss file(s)	<code>max_weibull.g</code>
Matlab file(s)	<code>max_weibull.m</code>

- (a) Simulate $T = 20$ observations with $\theta = \{\alpha = 1, \beta = 2\}$ from the Weibull distribution

$$f(y; \theta) = \alpha \beta y^{\beta-1} \exp \left[-\alpha y^\beta \right].$$

- (b) Derive $\ln L_T(\theta)$, $G_T(\theta)$, $H_T(\theta)$, $J_T(\theta)$ and $I(\theta)$.
- (c) Choose as starting values $\theta_{(0)} = \{\alpha_{(0)} = 0.5, \beta_{(0)} = 1.5\}$ and evaluate $G(\theta_{(0)})$, $H(\theta_{(0)})$ and $J(\theta_{(0)})$ for the data generated in part (a). Check the analytical results using numerical derivatives.
- (d) Compute the update $\theta_{(1)}$ using the Newton-Raphson and BHHH algorithms.
- (e) Continue the iterations in part (d) until convergence. Discuss the numerical performances of the two algorithms.

- (f) Compute the covariance matrix, $\hat{\Omega}$, using the Hessian and also the outer product of the gradients matrix.
- (g) Repeat parts (d) and (e) where the log-likelihood function is concentrated with respect to $\hat{\beta}$. Compare the parameter estimates of α and β with the estimates obtained using the full log-likelihood function.
- (h) Suppose that the Weibull distribution is re-expressed as

$$f(y; \theta) = \frac{\beta}{\lambda} \left(\frac{y}{\lambda}\right)^{\beta-1} \exp \left[- \left(\frac{y}{\lambda}\right)^{\beta} \right],$$

where $\lambda = \alpha^{-1/\beta}$. Compute $\hat{\lambda}$ and $\text{se}(\hat{\lambda})$ for $T = 20$ observations by the substitution method and also by the delta method using the maximum likelihood estimates obtained previously.

(7) Simplex Algorithm

Gauss file(s)	<code>max_simplex.g</code>
Matlab file(s)	<code>max_simplex.m</code>

Suppose that the observations $y_t = \{3.5, 1.0, 1.5\}$ are *iid* drawings from the exponential distribution

$$f(y; \theta) = \frac{1}{\theta} \exp \left[-\frac{y}{\theta} \right], \quad \theta > 0.$$

- (a) Based on the negative of the log-likelihood function for this exponential distribution, compute the maximum likelihood estimator, $\hat{\theta}$, using the starting vertices $\theta_1 = 1$ and $\theta_2 = 3$.
- (b) Which move would the first iteration of the simplex algorithm choose?

(8) Profile Log-likelihood Function

Gauss file(s)	<code>max_profile.g</code> , <code>apple.csv</code> , <code>ford.csv</code>
Matlab file(s)	<code>max_profile.m</code> , <code>diversify.mat</code>

The data files contain daily share prices of Apple and Ford from 2 January 2001 to 6 August 2010, a total of $T = 2413$ observations (see also Section 2.7.1 and Exercise 14 in Chapter 2). Let $\theta = \{\theta_1, \theta_2\}$ where θ_1 contains the parameters of interest. The profile log-likelihood function is defined as

$$\ln L_T(\theta_1, \hat{\theta}_2) = \arg \max_{\theta_2} \ln L_T(\theta),$$

where $\hat{\theta}_2$ is the maximum likelihood solution of θ_2 . A plot of $\ln L_T(\theta_1, \hat{\theta}_2)$ over θ_1 provides information on θ_1 .

Assume that the returns on the two assets are *iid* drawings from a bivariate normal distribution with means μ_1 and μ_2 , variances σ_1^2 and σ_2^2 , and correlation ρ . Define $\theta_1 = \{\rho\}$ and $\theta_2 = \{\mu_1, \mu_2, \sigma_1^2, \sigma_2^2\}$.

- (a) Plot $\ln L_T(\theta_1, \hat{\theta}_2)$ over $(-1, 1)$, where $\hat{\theta}_2$ is the maximum likelihood estimate obtained from the returns data.
- (b) Interpret the plot obtained in part (a).

(9) Stationary Distribution of the CIR Model

Gauss file(s)	<code>max_stationary.g, eurodollar.dat</code>
Matlab file(s)	<code>max_stationary.m, eurodollar.mat</code>

The data are daily 7-day Eurodollar rates from 1 June 1973 to 25 February 1995, a total of $T = 5505$ observations. The CIR model of interest rates, r_t , for time steps dt is

$$dr = \alpha(\mu - r)dt + \sigma\sqrt{r} dW,$$

where $dW \sim N(0, dt)$. The stationary distribution of the CIR interest rate is the gamma distribution

$$f(r; \nu, \omega) = \frac{\omega^\nu}{\Gamma(\nu)} r^{\nu-1} e^{-\omega r},$$

where $\Gamma(\cdot)$ is the Gamma function and $\theta = \{\nu, \omega\}$ are unknown parameters.

- (a) Compute the maximum likelihood estimates of ν and ω and their standard errors based on the Hessian.
- (b) Use the results in part (a) to compute the maximum likelihood estimate of μ and its standard error.
- (c) Use the estimates from part (a) to plot the stationary distribution and interpret its properties.
- (d) Suppose that it is known that $\nu = 1$. Using the property of the gamma function that $\Gamma(1) = 1$, estimate ω and recompute the mean interest rate.

(10) Transitional Distribution of the CIR Model

Gauss file(s)	<code>max_transitional.g, eurodollar.dat</code>
Matlab file(s)	<code>max_transitional.m, eurodollar.mat</code>

The data are the same daily 7-day Eurodollar rates used in Exercise 9.

- (a) The transitional distribution of r_t given r_{t-1} for the CIR model in Exercise 9 is

$$f(r_t | r_{t-1}; \theta) = ce^{-u-v} \left(\frac{v}{u}\right)^{\frac{q}{2}} I_q(2\sqrt{uv}),$$

where $I_q(x)$ is the modified Bessel function of the first kind of order q , $\Delta = 1/250$ is the time step and

$$c = \frac{2\alpha}{\sigma^2(1 - e^{-\alpha\Delta})}, \quad u = cr_{t-1}e^{-\alpha\Delta}, \quad v = cr_t, \quad q = \frac{2\alpha\mu}{\sigma^2} - 1.$$

Estimate the CIR model parameters, $\theta = \{\alpha, \mu, \sigma\}$, by maximum likelihood. Compute the standard errors based on the Hessian.

- (b) Use the result that the transformed variable $2cr_t$ is distributed as a non-central chi-square random variable with $2q + 2$ degrees of freedom and non-centrality parameter $2u$ to obtain the maximum likelihood estimates of θ based on the non-central chi-square probability density function. Compute the standard errors based on the Hessian. Compare the results with those obtained in part (a).

4

Hypothesis Testing

4.1 Introduction

The discussion of maximum likelihood estimation has focussed on deriving estimators that maximize the likelihood function. In all of these cases, the potential values that the maximum likelihood estimator, $\hat{\theta}$, can take are unrestricted. Now the discussion is extended to asking if the population parameter has a certain hypothesized value, θ_0 . If this value differs from $\hat{\theta}$, then by definition, it must correspond to a lower value of the log-likelihood function and the crucial question is then how significant this decrease is. Determining the significance of this reduction of the log-likelihood function represents the basis of hypothesis testing. That is, hypothesis testing is concerned about determining if the reduction in the value of the log-likelihood function brought about by imposing the restriction $\theta = \theta_0$ is severe enough to warrant rejecting it. If, however, it is concluded that the decrease in the log-likelihood function is not too severe, the restriction is interpreted as being consistent with the data and it is not rejected. The likelihood ratio test (LR), the Wald test and the Lagrange multiplier test (LM) are three general procedures used in developing statistics to test hypotheses. These tests encompass many of the test statistics used in econometrics, an important feature highlighted in Part TWO of the book. They also offer the advantage of providing a general framework to develop new classes of test statistics that are designed for specific models.

4.2 Overview

Suppose θ is a single parameter and consider the hypotheses

$$H_0 : \theta = \theta_0, \quad H_1 : \theta \neq \theta_0.$$

A natural test based on a comparison of the log-likelihood function evaluated at the maximum likelihood estimator $\hat{\theta}$ and at the null value θ_0 , at both the unrestricted and restricted estimators. A statistic of the form

$$\ln L_T(\hat{\theta}) - \ln L_T(\theta_0) = \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \hat{\theta}) - \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta_0),$$

measures the distance between the maximized log-likelihood $\ln L_T(\hat{\theta})$ and the log-likelihood $\ln L_T(\theta_0)$ restricted by the null hypothesis. This distance is measured on the vertical axis of Figure 4.1 and the test which uses this measure in its construction is known as the likelihood ratio (LR) test.

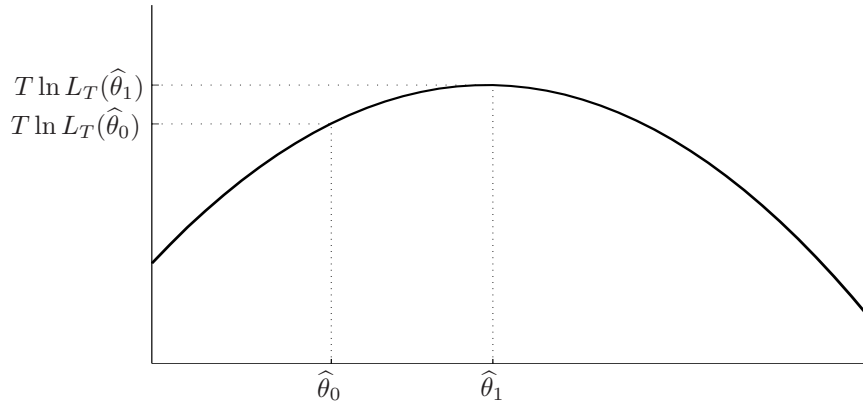


Figure 4.1 Comparison of the value of the log-likelihood function under the null hypothesis, $\hat{\theta}_0$, and under the alternative hypothesis, $\hat{\theta}_1$.

The distance $(\hat{\theta} - \theta_0)$, illustrated on the horizontal axis of Figure 4.1, is an alternative measure of the difference between $\hat{\theta}$ and θ_0 . A test based on this measure is known as a Wald test. The Lagrange multiplier (LM) test is the hypothesis test based on the gradient of the log-likelihood function at the null value θ_0 , $G_T(\theta_0)$. The gradient at the maximum likelihood estimator, $G_T(\hat{\theta})$, is zero by definition (see Chapter 1). The LM statistic is therefore as the distance on the vertical axis in Figure 4.2 between $G_T(\theta_0)$ and $G_T(\hat{\theta}) = 0$.

The intuition behind the construction of these tests for a single parameter can be carried over to provide likelihood-based testing of general hypotheses, which are discussed next.

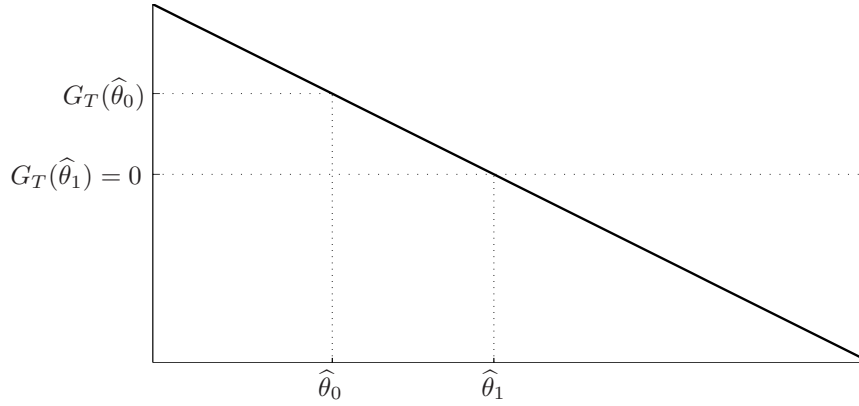


Figure 4.2 Comparison of the value of the gradient of the log-likelihood function under the null hypothesis, $\hat{\theta}_0$, and under the alternative hypothesis, $\hat{\theta}_1$.

4.3 Types of Hypotheses

This section presents detailed examples of types of hypotheses encountered in econometrics, beginning with simple and composite hypotheses and progressing to linear and nonlinear hypotheses.

4.3.1 Simple and Composite Hypotheses

Consider a model based on the distribution $f(y; \theta)$ where θ is an unknown scalar parameter. The simplest form of hypothesis test is based on testing whether or not a parameter takes one of two specific values, θ_0 or θ_1 . The null and alternative hypotheses are, respectively,

$$H_0 : \theta = \theta_0, \quad H_1 : \theta = \theta_1,$$

where θ_0 represents the value of the parameter under the null hypothesis and θ_1 is the value under the alternative. In Chapter 2, θ_0 represents the true parameter value. In hypothesis testing, since the null and alternative hypotheses are distinct, θ_0 still represents the true value, but now interpreted to be under the null hypothesis. Both these hypotheses are simple hypotheses because the parameter value in each case is given and therefore the distribution of the parameter under both the null and alternative hypothesis is fully specified.

If the hypothesis is constructed in such a way that the distribution of the parameter cannot be inferred fully, the hypothesis is referred to as being

composite. An example is

$$H_0 : \theta = \theta_0, \quad H_1 : \theta \neq \theta_0,$$

where the alternative hypothesis is a composite hypothesis because the distribution of the θ under the alternative is not fully specified, whereas the null hypothesis is still a simple hypothesis.

Under the alternative hypothesis, the parameter θ can take any value on either side of θ_0 . This form of hypothesis test is referred to as a two-sided test. Restricting the range under the alternative to be just one side, $\theta > \theta_0$ or $\theta < \theta_0$, would change the test to a one-sided test. The alternative hypothesis would still be a composite hypothesis.

4.3.2 Linear Hypotheses

Suppose that there are K unknown parameters, $\theta = \{\beta_1, \beta_2, \dots, \beta_K\}$, so θ is a $(K \times 1)$ vector, and M linear hypotheses are to be tested simultaneously. The full set of M hypotheses is expressed as

$$H_0 : R\theta = Q, \quad H_1 : R\theta \neq Q,$$

where R and Q are $(M \times K)$ and $(M \times 1)$ matrices, respectively. To highlight the form of R and Q , consider the following cases.

- (1) $K = 1, M = 1, \theta = \{\beta_1\}$:

The null and alternative hypotheses are

$$H_0 : \beta_1 = 0 \quad H_1 : \beta_1 \neq 0,$$

with

$$R = [1], \quad Q = [0].$$

- (2) $K = 2, M = 1, \theta = \{\beta_1, \beta_2\}$:

The null and alternative hypotheses are

$$H_0 : \beta_2 = 0, \quad H_1 : \beta_2 \neq 0,$$

with

$$R = [0 \ 1], \quad Q = [0].$$

This corresponds to the usual example of performing a t-test on the importance of an explanatory variable by testing to see if the pertinent parameter is zero.

- (3) $K = 3, M = 1, \theta = \{\beta_1, \beta_2, \beta_3\}$:

The null and alternative hypotheses are

$$H_0 : \beta_1 + \beta_2 + \beta_3 = 0, \quad H_1 : \beta_1 + \beta_2 + \beta_3 \neq 0,$$

with

$$R = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}, \quad Q = [0].$$

- (4) $K = 4, M = 3, \theta = \{\beta_1, \beta_2, \beta_3, \beta_4\}$:

The null and alternative hypotheses are

$$H_0 : \beta_1 = \beta_2, \beta_2 = \beta_3, \beta_3 = \beta_4$$

$$H_1 : \text{at least one restriction does not hold},$$

with

$$R = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

These restrictions arise in models of the term structure of interest rates.

- (5) $K = 4, M = 3, \theta = \{\beta_1, \beta_2, \beta_3, \beta_4\}$:

The hypotheses are

$$H_0 : \beta_1 = \beta_2, \beta_3 = \beta_4, \beta_1 = 1 + \beta_3 - \beta_4$$

$$H_1 : \text{at least one restriction does not hold},$$

with

$$R = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 1 & 0 & -1 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

4.3.3 Nonlinear Hypotheses

The set of hypotheses entertained is now further extended to allow for nonlinearities. The full set of M nonlinear hypotheses is expressed as

$$H_0 : C(\theta) = Q, \quad H_1 : C(\theta) \neq Q,$$

where $C(\theta)$ is a $(M \times 1)$ matrix of nonlinear restrictions and Q is a $(M \times 1)$ matrix of constants. In the special case where the hypotheses are linear, $C(\theta) = R\theta$. To highlight the construction of these matrices, consider the following cases.

- (1) $K = 2, M = 1, \theta = \{\beta_1, \beta_2\}$:

The null and alternative hypotheses are

$$H_0 : \beta_1\beta_2 = 1, \quad H_1 : \beta_1\beta_2 \neq 1,$$

with

$$C(\theta) = [\beta_1\beta_2], \quad Q = [1].$$

- (2) $K = 2, M = 1, \theta = \{\beta_1, \beta_2\}$:

The null and alternative hypotheses are

$$H_0 : \frac{\beta_1}{1 - \beta_2} = 1, \quad H_1 : \frac{\beta_1}{1 - \beta_2} \neq 1,$$

with

$$C(\theta) = \left[\frac{\beta_1}{1 - \beta_2} \right], \quad Q = [1].$$

This form of restriction often arises in dynamic time series models where restrictions on the value of the long-run multiplier are often imposed.

- (3) $K = 3, M = 2, \theta = \{\beta_1, \beta_2, \beta_3\}$:

The null and alternative hypotheses are

$$H_0 : \beta_1\beta_2 = \beta_3, \quad \frac{\beta_1}{1 - \beta_2} = 1$$

$$H_1 : \text{at least one restriction does not hold,}$$

and

$$C(\theta) = \begin{bmatrix} \beta_1\beta_2 - \beta_3 \\ \beta_1(1 - \beta_2)^{-1} \end{bmatrix}, \quad Q = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

4.4 Likelihood Ratio Test

The LR test requires estimating the model under both the null and alternative hypotheses. The resulting estimators are denoted

$$\hat{\theta}_0 = \text{restricted maximum likelihood estimator,}$$

$$\hat{\theta}_1 = \text{unrestricted maximum likelihood estimator.}$$

The unrestricted estimator $\hat{\theta}_1$ is the usual maximum likelihood estimator. The restricted estimator $\hat{\theta}_0$ is obtained by first imposing the null hypothesis on the model and then estimating any remaining unknown parameters. If the null hypothesis completely specifies the parameter, that is $H_0 : \theta = \theta_0$, then the restricted estimator is simply $\hat{\theta}_0 = \theta_0$. In most cases, however, a null hypothesis will specify only some of the parameters of the model, leaving

the remaining parameters to be estimated in order to find $\hat{\theta}_0$. Examples are given below.

Let

$$T \ln L_T(\hat{\theta}_0) = \sum_{t=1}^T \ln f(y_t; \hat{\theta}_0), T \ln L_T(\hat{\theta}_1) = \sum_{t=1}^T \ln f(y_t; \hat{\theta}_1),$$

be the maximized log-likelihood functions under the null and alternative hypotheses respectively. The general form of the LR statistic is

$$LR = -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)). \quad (4.1)$$

As the maximum likelihood estimator maximizes the log-likelihood function, the term in brackets is non-positive as the restrictions under the null hypothesis in general correspond to a region of lower probability. This loss of probability is illustrated on the vertical axis of Figure 4.1 which gives the term in brackets. The range of LR is $0 \leq LR < \infty$. For values of the statistic near $LR = 0$, the restrictions under the null hypothesis are consistent with the data since there is no serious loss of information from imposing these restrictions. For larger values of LR the restrictions under the null hypothesis are not consistent with the data since a serious loss of information caused by imposing these restrictions now results. In the former case, there is a failure to reject the null, whereas in the latter case the null is rejected in favour of the alternative hypothesis. It is shown in Section 4.7, that LR in equation (4.1) is asymptotically distributed as χ_M^2 under the null hypothesis where M is the number of restrictions.

Example 4.1 Univariate Normal Distribution

The log-likelihood function of a normal distribution with unknown mean and variance, $\theta = \{\mu, \sigma^2\}$, is

$$\ln L_T(\theta) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2 T} \sum_{t=1}^T (y_t - \mu)^2.$$

A test of the mean is based on the null and alternative hypotheses

$$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0.$$

The unrestricted maximum likelihood estimators are

$$\hat{\mu}_1 = \frac{1}{T} \sum_{t=1}^T y_t = \bar{y}, \quad \hat{\sigma}_1^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \bar{y})^2,$$

and the log-likelihood function evaluated at $\hat{\theta}_1 = \{\hat{\mu}_1, \hat{\sigma}_1^2\}$ is

$$\ln L_T(\hat{\theta}_1) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \hat{\sigma}_1^2 - \frac{1}{2\hat{\sigma}_1^2 T} \sum_{t=1}^T (y_t - \hat{\mu}_1)^2 = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \hat{\sigma}_1^2 - \frac{1}{2}.$$

The restricted maximum likelihood estimators are

$$\hat{\mu}_0 = \mu_0, \quad \hat{\sigma}_0^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \mu_0)^2,$$

and the log-likelihood function evaluated at $\hat{\theta}_0 = \{\hat{\mu}_0, \hat{\sigma}_0^2\}$ is

$$\ln L_T(\hat{\theta}_0) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \hat{\sigma}_0^2 - \frac{1}{2\hat{\sigma}_0^2 T} \sum_{t=1}^T (y_t - \hat{\mu}_0)^2 = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \hat{\sigma}_0^2 - \frac{1}{2}.$$

Using equation (4.1), the LR statistic is

$$\begin{aligned} LR &= -2 \left(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1) \right) \\ &= -2 \left[\left(-\frac{T}{2} \ln 2\pi - \frac{T}{2} \ln \hat{\sigma}_0^2 - \frac{T}{2} \right) - \left(-\frac{T}{2} \ln 2\pi - \frac{T}{2} \ln \hat{\sigma}_1^2 - \frac{T}{2} \right) \right] \\ &= T \ln \left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}_1^2} \right). \end{aligned}$$

Under the null hypothesis, the LR statistic is distributed as χ_1^2 . This expression shows that the LR test is equivalent to comparing the variances of the data under the null and alternative hypotheses. If $\hat{\sigma}_0^2$ is close to $\hat{\sigma}_1^2$, the restriction is consistent with the data, resulting in a small value of LR . In the extreme case where no loss of information from imposing the restrictions occurs, $\hat{\sigma}_0^2 = \hat{\sigma}_1^2$ and $LR = 0$. For values of $\hat{\sigma}_0^2$ that are not statistically close to $\hat{\sigma}_1^2$, LR is a large positive value. $\square$

Example 4.2 Multivariate Normal Distribution

The multivariate normal distribution of dimension N at time t is

$$f(y_t; \theta) = \left(\frac{1}{2\pi} \right)^{N/2} |V|^{-1/2} \exp \left[-\frac{1}{2} u_t' V^{-1} u_t \right],$$

where $y_t = \{y_{1,t}, y_{2,t}, \dots, y_{N,t}\}$ is a $(N \times 1)$ vector of dependent variables at time t , $u_t = y_t - \beta x_t$ is a $(N \times 1)$ vector of disturbances with covariance matrix V , and x_t is a $(K \times 1)$ vector of explanatory variables and β is a $(N \times K)$ parameter matrix. The log-likelihood function is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta) = -\frac{N}{2} \ln 2\pi - \frac{1}{2} \ln |V| - \frac{1}{2T} \sum_{t=1}^T u_t' V^{-1} u_t,$$

where $\theta = \{\beta, V\}$. Consider testing M restrictions on β . The unrestricted maximum likelihood estimator of V is

$$\widehat{V}_1 = \frac{1}{T} \sum_{t=1}^T e_t e_t',$$

where $e_t = y_t - \widehat{\beta}_1 x_t$ and $\widehat{\beta}_1$ is the unrestricted estimator of β . The log-likelihood function evaluated at the unrestricted estimator is

$$\begin{aligned} \ln L_T(\widehat{\theta}_1) &= -\frac{N}{2} \ln 2\pi - \frac{1}{2} \ln |\widehat{V}_1| - \frac{1}{2T} \sum_{t=1}^T e_t' \widehat{V}_1^{-1} e_t \\ &= -\frac{N}{2} \ln 2\pi - \frac{1}{2} \ln |\widehat{V}_1| - \frac{N}{2} \\ &= -\frac{N}{2} (1 + \ln 2\pi) - \frac{1}{2} \ln |\widehat{V}_1|, \end{aligned}$$

which uses the result

$$\begin{aligned} \sum_{t=1}^T e_t' \widehat{V}_1^{-1} e_t &= \text{trace} \left(\sum_{t=1}^T e_t' \widehat{V}_1^{-1} e_t \right) = \text{trace} \left(\widehat{V}_1^{-1} \sum_{t=1}^T e_t e_t' \right) \\ &= \text{trace}(\widehat{V}_1^{-1} T \widehat{V}_1) = \text{trace}(T I_N) = TN. \end{aligned}$$

Now consider estimating the model subject to a set of restrictions on β . The restricted maximum likelihood estimator of V is

$$\widehat{V}_0 = \frac{1}{T} \sum_{t=1}^T v_t v_t',$$

where $v_t = y_t - \widehat{\beta}_0 x_t$ and $\widehat{\beta}_0$ is the restricted estimator of β . The log-likelihood function evaluated at the restricted estimator is

$$\begin{aligned} \ln L_T(\widehat{\theta}_0) &= -\frac{N}{2} \ln 2\pi - \frac{1}{2} \ln |\widehat{V}_0| - \frac{1}{2T} \sum_{t=1}^T v_t' \widehat{V}_0^{-1} v_t \\ &= -\frac{N}{2} \ln 2\pi - \frac{1}{2} \ln |\widehat{V}_0| - \frac{N}{2} \\ &= -\frac{N}{2} (1 + \ln 2\pi) - \frac{1}{2} \ln |\widehat{V}_0|. \end{aligned}$$

The LR statistic is

$$LR = -2[T \ln L_T(\widehat{\theta}_0) - T \ln L_T(\widehat{\theta}_1)] = T \ln \left(\frac{|\widehat{V}_0|}{|\widehat{V}_1|} \right),$$

which is distributed asymptotically under the null hypothesis as χ_M^2 . This is the multivariate analogue of Example 4.1 that is commonly adopted when

testing hypotheses within multivariate normal models. It should be stressed that this form of the likelihood ratio test is appropriate only for models based on the assumption of normality. $\square$

Example 4.3 Weibull Distribution

Consider the $T = 20$ independent realizations, given in Example 3.6 in Chapter 3, drawn from the Weibull distribution

$$f(y; \theta) = \alpha \beta y^{\beta-1} \exp \left[-\alpha y^\beta \right],$$

with unknown parameters $\theta = \{\alpha, \beta\}$. A special case of the Weibull distribution is the exponential distribution that occurs when $\beta = 1$. To test that the data are drawn from the exponential distribution, the null and alternative hypotheses are, respectively,

$$H_0 : \beta = 1, \quad H_1 : \beta \neq 1.$$

The unrestricted and restricted log-likelihood functions are

$$\begin{aligned} \ln L_T(\hat{\theta}_1) &= -\hat{\beta}_1 \ln \hat{\alpha}_1 + \ln \hat{\beta}_1 + (\hat{\beta}_1 - 1) \frac{1}{T} \sum_{t=1}^T \ln y_t - \frac{1}{T} \sum_{t=1}^T \left(\frac{y_t}{\hat{\alpha}_1} \right)^{\hat{\beta}_1} \\ \ln L_T(\hat{\theta}_0) &= -\ln \hat{\alpha}_0 - \frac{1}{T} \sum_{t=1}^T \frac{y_t}{\hat{\alpha}_0}, \end{aligned}$$

respectively. Maximizing the two log-likelihood functions yields

$$\begin{array}{lll} \text{Unrestricted} & : & \hat{\alpha}_1 = 0.856 \quad \hat{\beta}_1 = 1.868 \quad T \ln L_T(\hat{\theta}_1) = -15.333, \\ \text{Restricted} & : & \hat{\alpha}_0 = 1.020 \quad \hat{\beta}_0 = 1.000 \quad T \ln L_T(\hat{\theta}_0) = -19.611. \end{array}$$

The likelihood ratio statistic is computed using equation (4.1)

$$LR = -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)) = -2(-19.611 + 15.333) = 8.555.$$

Using the χ_1^2 distribution, the p-value is 0.003 resulting in a rejection of the null hypothesis at the 5% significance level that the data are drawn from an exponential distribution. $\square$

4.5 Wald Test

The LR test requires estimating both the restricted and unrestricted models, whereas the Wald test requires estimation of just the unrestricted model. This property of the Wald test can be very important from a practical point of view, especially in those cases where estimating the model under the null hypothesis is more difficult than under the alternative hypothesis.

The Wald test statistic for the null hypothesis $H_0 : \theta = \theta_0$, a hypothesis which completely specifies the parameter, is

$$W = (\hat{\theta}_1 - \theta_0)' [\text{cov}(\hat{\theta}_1 - \theta_0)]^{-1} (\hat{\theta}_1 - \theta_0),$$

which is distributed asymptotically as χ_1^2 , where $M = 1$ is the number of restrictions under the null hypothesis. The variance of $\hat{\theta}_1$ is given by

$$\text{cov}(\hat{\theta}_1 - \theta_0) = \text{cov}(\hat{\theta}_1) = \frac{1}{T} I^{-1}(\theta_0).$$

This expression is then evaluated at $\theta = \hat{\theta}$, so that the Wald test is

$$W = T(\hat{\theta}_1 - \theta_0)' I(\hat{\theta}_1)(\hat{\theta}_1 - \theta_0). \quad (4.2)$$

The aim of the Wald test is to compare the unrestricted value ($\hat{\theta}_1$) with the value under the null hypothesis (θ_0). If the two values are considered to be close, then W is small. To determine the significance of this difference, the deviation ($\hat{\theta}_1 - \theta_0$) is scaled by the pertinent standard deviation.

4.5.1 Linear Hypotheses

For M linear hypotheses of the form $R\theta = Q$, the Wald statistic is

$$W = [R\hat{\theta}_1 - Q]' [\text{cov}(R\hat{\theta}_1 - Q)]^{-1} [R\hat{\theta}_1 - Q].$$

The covariance matrix is

$$\text{cov}(R\hat{\theta}_1 - Q) = \text{cov}(R\hat{\theta}_1) = R \frac{1}{T} \hat{\Omega} R' \quad (4.3)$$

where $\hat{\Omega}/T$ is the covariance matrix of $\hat{\theta}_1$. The general form of the Wald test of linear restrictions is therefore

$$W = T[R\hat{\theta}_1 - Q]' [R\hat{\Omega}R']^{-1} [R\hat{\theta}_1 - Q]. \quad (4.4)$$

Under the null hypothesis, the Wald statistic is asymptotically distributed as χ_M^2 where M is the number of restrictions.

In practice, the Wald statistic is usually expressed in terms of the relevant method used to compute the covariance matrix $\hat{\Omega}/T$. Given that the maximum likelihood estimator, $\hat{\theta}_1$, satisfies the Information equality in equation (2.33) of Chapter 2, it follows that

$$R \frac{1}{T} \hat{\Omega} R' = R \frac{1}{T} I^{-1}(\hat{\theta}_1) R',$$

where $I(\hat{\theta}_1)$ is the information matrix evaluated at $\hat{\theta}_1$. The information

equality means that the Wald statistic may be written in the following asymptotically equivalent forms

$$W_I = T[R\hat{\theta}_1 - Q]'[R I^{-1}(\hat{\theta}_1) R']^{-1}[R\hat{\theta}_1 - Q], \quad (4.5)$$

$$W_H = T[R\hat{\theta}_1 - Q]'[R (-H_T^{-1}(\hat{\theta}_1)) R']^{-1}[R\hat{\theta}_1 - Q], \quad (4.6)$$

$$W_J = T[R\hat{\theta}_1 - Q]'[R J_T^{-1}(\hat{\theta}_1) R']^{-1}[R\hat{\theta}_1 - Q]. \quad (4.7)$$

All these test statistics have the same asymptotic distribution.

Example 4.4 Normal Distribution

Consider the normal distribution example again where the null and alternative hypotheses are, respectively,

$$H_0 : \mu = \mu_0 \quad H_1 : \mu \neq \mu_0,$$

with $R = [1 \ 0]$ and $Q = [\mu_0]$. The unrestricted maximum likelihood estimators are

$$\hat{\theta}_1 = [\hat{\mu}_1 \ \hat{\sigma}_1^2]' = \left[\bar{y} \quad \frac{1}{T} \sum_{t=1}^T (y_t - \bar{y})^2 \right]'$$

When evaluated at $\hat{\theta}_1$ the information matrix is

$$I(\hat{\theta}_1) = \begin{bmatrix} \frac{1}{\hat{\sigma}_1^2} & 0 \\ 0 & \frac{1}{2\hat{\sigma}_1^4} \end{bmatrix}.$$

Now $[R\hat{\theta}_1 - Q] = [\bar{y} - \mu_0]$ so that

$$[R I^{-1}(\hat{\theta}_1) R'] = \begin{bmatrix} 1 \\ 0 \end{bmatrix}' \begin{bmatrix} \frac{1}{\hat{\sigma}_1^2} & 0 \\ 0 & \frac{1}{2\hat{\sigma}_1^4} \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \hat{\sigma}_1^2.$$

The Wald statistic in equation (4.5) then becomes

$$W = T \frac{(\bar{y} - \mu_0)^2}{\hat{\sigma}_1^2}, \quad (4.8)$$

which is distributed asymptotically as χ_1^2 . This form of the Wald statistic is equivalent to the square of the standard t-test applied to the mean of a normal distribution. $\square$

Example 4.5 Weibull Distribution

Recompute the test of the Weibull distribution in Example 4.3 using a Wald test of the restriction $\beta = 1$ with the covariance matrix computed

using the Hessian. The unrestricted maximum likelihood estimates are $\hat{\theta}_1 = \{\hat{\alpha}_1 = 0.865, \hat{\beta}_1 = 1.868\}$ and the Hessian evaluated at $\hat{\theta}_1$ using numerical derivatives is

$$H_T(\hat{\theta}_1) = \frac{1}{20} \begin{bmatrix} -27.266 & -6.136 \\ -6.136 & -9.573 \end{bmatrix} = \begin{bmatrix} -1.363 & -0.307 \\ -0.307 & -0.479 \end{bmatrix}.$$

Define $R = [0 \ 1]$ and $Q = [1]$ so that

$$R(-H_T^{-1}(\hat{\theta}_1))R' = \begin{bmatrix} 0 \\ 1 \end{bmatrix}' \begin{bmatrix} -1.363 & -0.307 \\ -0.307 & -0.479 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = [2.441].$$

The Wald statistic, given in equation (4.6), is

$$W = 20(1.868 - 1)(2.441)^{-1}(1.868 - 1) = \frac{20(1.868 - 1.000)^2}{2.441} = 6.174.$$

Using the χ_1^2 distribution, the p-value of the Wald statistic is 0.013, resulting in the rejection of the null hypothesis at the 5% significance level that the data come from an exponential distribution. $\square$

4.5.2 Nonlinear Hypotheses

For M nonlinear hypotheses of the form

$$H_0 : C(\theta) = Q, \quad H_1 : C(\theta) \neq Q,$$

the Wald statistic is

$$W = [C(\hat{\theta}_1) - Q]' \text{cov}(C(\hat{\theta}_1))^{-1} [C(\hat{\theta}_1) - Q]. \quad (4.9)$$

To compute the covariance matrix, $\text{cov}(C(\hat{\theta}_1))$ the delta method discussed in Chapter 3 is used. There it is shown that

$$\text{cov}(C(\hat{\theta}_1)) = \frac{1}{T} D(\theta) \Omega(\theta) D(\theta)',$$

where

$$D(\theta) = \frac{\partial C(\theta)}{\partial \theta'}.$$

This expression for the covariance matrix depends on θ , which is estimated by the unrestricted maximum likelihood estimator $\hat{\theta}_1$. The general form of the Wald statistic in the case of nonlinear restrictions is then

$$W = T[C(\hat{\theta}_1) - Q]' [D(\hat{\theta}_1) \hat{\Omega} D(\hat{\theta}_1)']^{-1} [C(\hat{\theta}_1) - Q],$$

which takes the asymptotically equivalent forms

$$W = T[C(\hat{\theta}_1) - Q]'[D(\hat{\theta}_1) I^{-1}(\hat{\theta}_1) D(\hat{\theta}_1)']^{-1}[C(\hat{\theta}_1) - Q] \quad (4.10)$$

$$W = T[C(\hat{\theta}_1) - Q]'[D(\hat{\theta}_1) (-H_T^{-1}(\hat{\theta}_1)) D(\hat{\theta}_1)']^{-1}[C(\hat{\theta}_1) - Q] \quad (4.11)$$

$$W = T[C(\hat{\theta}_1) - Q]'[D(\hat{\theta}_1) J_T^{-1}(\hat{\theta}_1) D(\hat{\theta}_1)']^{-1}[C(\hat{\theta}_1) - Q]. \quad (4.12)$$

Under the null hypothesis, the Wald statistic is asymptotically distributed as χ_M^2 where M is the number of restrictions. If the restrictions are linear, that is $C(\theta) = R\theta$, then

$$\frac{\partial C(\theta)}{\partial \theta'} = R,$$

and equations (4.10), (4.11) and (4.12) reduce to the forms given in equations (4.5), (4.6) and (4.7), respectively.

4.6 Lagrange Multiplier Test

The LM test is based on the property that the gradient, evaluated at the unrestricted maximum likelihood estimator, satisfies $G_T(\hat{\theta}_1) = 0$. Assuming that the log-likelihood function has a unique maximum, evaluating the gradient under the null means that $G_T(\hat{\theta}_0) \neq 0$. This suggests that if the null hypothesis is inconsistent with the data, the value of $G_T(\hat{\theta}_0)$ represents a significant deviation from the unrestricted value of the gradient vector, $G_T(\hat{\theta}_1) = 0$.

The basis of the LM test statistic derives from the properties of the gradient discussed in Chapter 2. The key result is

$$\sqrt{T}(G_T(\hat{\theta}_0) - 0) \xrightarrow{d} N(0, I(\theta_0)). \quad (4.13)$$

This result suggests that a natural test statistic is to compute the squared difference between the sample quantity under the null hypothesis, $G_T(\hat{\theta}_0)$, and the theoretical value under the alternative, $G_T(\hat{\theta}_1) = 0$ and scale the result by the variance, $I(\theta_0)/T$. The test statistic is therefore

$$LM = T[G_T'(\hat{\theta}_0) - 0]'I^{-1}(\hat{\theta}_0)[G_T'(\hat{\theta}_0) - 0] = TG_T'(\hat{\theta}_0)I^{-1}(\hat{\theta}_0)G_T(\hat{\theta}_0). \quad (4.14)$$

It follows immediately from expression (4.13) that this statistic is distributed asymptotically as χ_M^2 where M is the number of restrictions under the null hypothesis. This general form of the LM test is similar to that of the Wald test, where the test statistic is compared to a population value under the null hypothesis and standardized by the appropriate variance.

Example 4.6 Normal Distribution

Consider again the normal distribution in Example 4.1 where the null and alternative hypotheses are, respectively,

$$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0.$$

The restricted maximum likelihood estimators are

$$\hat{\theta}_0 = \begin{bmatrix} \hat{\mu}_0 & \hat{\sigma}_0^2 \end{bmatrix}' = \begin{bmatrix} \mu_0 & \frac{1}{T} \sum_{t=1}^T (y_t - \mu_0)^2 \end{bmatrix}'.$$

The gradient and information matrix evaluated at $\hat{\theta}_0$ are, respectively,

$$G_T(\hat{\theta}_0) = \begin{bmatrix} \frac{1}{\hat{\sigma}_0^2 T} \sum_{t=1}^T (y_t - \mu_0) \\ -\frac{1}{2\hat{\sigma}_0^2} + \frac{1}{2\hat{\sigma}_0^4 T} \sum_{t=1}^T (y_t - \mu_0)^2 \end{bmatrix} = \begin{bmatrix} \frac{1}{\hat{\sigma}_0^2} (\bar{y} - \mu_0) \\ 0 \end{bmatrix},$$

and

$$I(\hat{\theta}_0) = \begin{bmatrix} \frac{1}{\hat{\sigma}_0^2} & 0 \\ 0 & \frac{1}{2\hat{\sigma}_0^4} \end{bmatrix}.$$

From equation (4.14), the LM statistic is

$$LM = T \begin{bmatrix} \frac{1}{\hat{\sigma}_0^2} (\bar{y} - \mu_0) \\ 0 \end{bmatrix}' \begin{bmatrix} \frac{1}{\hat{\sigma}_0^2} & 0 \\ 0 & \frac{1}{2\hat{\sigma}_0^4} \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{\hat{\sigma}_0^2} (\bar{y} - \mu_0) \\ 0 \end{bmatrix} = \frac{T(\bar{y} - \mu_0)^2}{\hat{\sigma}_0^2},$$

which is distributed asymptotically as χ_1^2 . This statistic is of a similar form to the Wald statistic in Example 4.4, except that now the variance in the denominator is based on the restricted estimator, $\hat{\sigma}_0^2$, whereas in the Wald statistic it is based on the unrestricted estimator, $\hat{\sigma}_1^2$. $\square$

As in the computation of the Wald statistic, the information matrix equality, in equation (2.33) of Chapter 2, may be used to replace the information matrix, $I(\theta)$, with the asymptotically equivalent negative Hessian matrix, $-H_T(\theta)$, or the outer product of gradients matrix, $J_T(\theta)$. The asymptotically equivalent versions of the LM statistic are therefore

$$LM_I = T G_T'(\hat{\theta}_0) I^{-1}(\hat{\theta}_0) G_T(\hat{\theta}_0), \quad (4.15)$$

$$LM_H = T G_T'(\hat{\theta}_0) (-H_T^{-1}(\hat{\theta}_0)) G_T(\hat{\theta}_0), \quad (4.16)$$

$$LM_J = T G_T'(\hat{\theta}_0) J_T^{-1}(\hat{\theta}_0) G_T(\hat{\theta}_0). \quad (4.17)$$

Example 4.7 Weibull Distribution

Reconsider the example of the Weibull distribution testing problem in Examples 4.3 and 4.5. The null hypothesis is $\beta = 1$, which is to be tested using a LM test based on the outer product of gradients matrix. The gradient vector evaluated at $\hat{\theta}_0$ using numerical derivatives is

$$G_T(\hat{\theta}_0) = [0.000, 0.599]'.$$

The outer product of gradients matrix using numerical derivatives and evaluated at $\hat{\theta}_0$ is

$$J_T(\hat{\theta}_0) = \begin{bmatrix} 0.248 & -0.176 \\ -0.176 & 1.002 \end{bmatrix}.$$

From equation (4.17), the LM statistic is

$$LM_J = 20 \begin{bmatrix} 0.000 \\ 0.599 \end{bmatrix}' \begin{bmatrix} 0.248 & -0.176 \\ -0.176 & 1.002 \end{bmatrix}^{-1} \begin{bmatrix} 0.000 \\ 0.599 \end{bmatrix} = 8.175.$$

Using the χ_1^2 distribution, the p-value is 0.004, which leads to rejection of the null hypothesis at the 5% significance level that the data are drawn from an exponential distribution. This result is consistent with those obtained using the LR and Wald tests in Examples 4.3 and 4.5, respectively. $\square$

4.7 Distribution Theory

The asymptotic distributions of the LR, Wald and LM tests under the null hypothesis have all been stated to be simply χ_M^2 , where M is the number of restrictions being tested. To show this result formally, the asymptotic distribution of the Wald statistic is derived initially and then used to establish the asymptotic relationships between the three test statistics.

4.7.1 Asymptotic Distribution of the Wald Statistic

To derive the asymptotic distribution of the Wald statistic, the crucial link to be drawn is that between the normal distribution and the chi-square distribution. The chi-square distribution with M degrees of freedom is given by

$$f(y) = \frac{1}{\Gamma(M/2) 2^{M/2}} y^{M/2-1} \exp[-y/2]. \quad (4.18)$$

Consider the simple case of the distribution of $y = z^2$, where $z \sim N(0, 1)$. Note that the standard normal variable z has as its domain the entire real

line, while the transformed variable y is constrained to be positive. This change of domain means that the inverse function is given by $z = \pm\sqrt{y}$. To express the probability distribution of y in terms of the given probability distribution of z , use the change of variable technique (see Appendix A)

$$f(y) = f(z) \left| \frac{dz}{dy} \right| ,$$

where $dz/dy = \pm y^{-1/2}/2$ is the Jacobian of the transformation. The probability of every y therefore has contributions from both $f(-z)$ and $f(z)$

$$f(y) = f(z) \left| \frac{dz}{dy} \right|_{z=-\sqrt{y}} + f(z) \left| \frac{dz}{dy} \right|_{z=\sqrt{y}} . \quad (4.19)$$

Simple substituting of standard normal distribution in equation (4.19) yields

$$\begin{aligned} f(y) &= \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{z^2}{2} \right] \left| \frac{1}{2z} \right|_{z=-\sqrt{y}} + \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{z^2}{2} \right] \left| \frac{1}{2z} \right|_{z=+\sqrt{y}} \\ &= \frac{y^{-1/2}}{\sqrt{2\pi}} \exp \left[-\frac{z^2}{2} \right] \\ &= \frac{y^{-1/2}}{\Gamma(1/2) \sqrt{2}} \exp \left[-\frac{y}{2} \right] , \end{aligned} \quad (4.20)$$

where the last step follows from the property of the Gamma function that $\Gamma(1/2) = \sqrt{\pi}$. This is the chi-square distribution in (4.18) with $M = 1$ degrees of freedom.

Example 4.8 Single Restriction Case

Consider the hypotheses

$$H_0 : \mu = \mu_0 , \quad H_1 : \mu \neq \mu_0 ,$$

to be tested by means of the simple t statistic

$$z = \sqrt{T} \frac{\hat{\mu} - \mu_0}{\hat{\sigma}}$$

where $\hat{\mu}$ is the sample mean and $\hat{\sigma}^2$ is the sample variance. From the Lindberg-Levy central limit theorem in Chapter 2, $z \stackrel{a}{\sim} N(0, 1)$ under H_0 , so that from equation (4.20) it follows that z^2 is distributed as χ_1^2 . But from equation (4.8), the statistic $z^2 = T(\hat{\mu} - \mu_0)^2 / \hat{\sigma}^2$ is the Wald test of the restriction. The Wald statistic is, therefore, asymptotically distributed as a χ_1^2 random variable. $\square$

The relationship between the normal distribution and the chi-square distribution may be generalized to the case of multiple random variables. If

$z_1, z_2, \dots, z_M$ are M independent standard normal random variables, the transformed random variable,

$$y = z_1^2 + z_2^2 + \dots + z_M^2, \quad (4.21)$$

is χ_M^2 , which follows from the additivity property of chi-square random variables.

Example 4.9 Multiple Restriction Case

Consider the Wald statistic given in equation (4.5)

$$W = T[R\hat{\theta}_1 - Q]'[R I^{-1}(\hat{\theta}_1) R']^{-1}[R\hat{\theta}_1 - Q].$$

Using the Choleski decomposition, it is possible to write

$$R I^{-1}(\hat{\theta}_1) R' = S S',$$

where S is a lower triangular matrix. In the special case of a scalar, $M = 1$, S is a standard deviation but in general for $M > 1$, S is interpreted as the standard deviation matrix. It has the property that

$$[R I^{-1}(\hat{\theta}_1) R']^{-1} = (S S')^{-1} = S^{-1'} S^{-1}.$$

It is now possible to write the Wald statistic as

$$W = T[R\hat{\theta}_1 - Q]' S^{-1'} S^{-1} [R\hat{\theta}_1 - Q] = z' z = \sum_{i=1}^M z_i^2,$$

where

$$z = \sqrt{T} S^{-1} [R\hat{\theta}_1 - Q] \sim N(0_M, I_M).$$

Using the additive property of chi-square variables given in (4.21), it follows immediately that $W \sim \chi_M^2$. $\square$

The following simulation experiment highlights the theoretical results concerning the asymptotic distribution of the Wald statistic.

Example 4.10 Simulating the Distribution of the Wald Statistic

The multiple regression model

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \beta_3 x_{3,t} + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

is simulated 10000 times with a sample size of $T = 1000$ with explanatory variables $x_{1,t} \sim U(0, 1)$, $x_{2,t} \sim N(0, 1)$, $x_{3,t} \sim N(0, 1)^2$ and population parameter values $\theta_0 = \{\beta_0 = 0, \beta_1 = 0, \beta_2 = 0, \beta_3 = 0, \sigma^2 = 0.1\}$. The Wald statistic is constructed to test the hypotheses

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0, \quad H_1 : \text{at least one restriction does not hold.}$$

As there are $M = 3$ restrictions, the asymptotic distribution under the null hypothesis of the Wald test is χ_3^2 . Figure 4.3 shows that the simulated distribution (bar chart) of the test statistic matches its asymptotic distribution (continuous line). $\square$

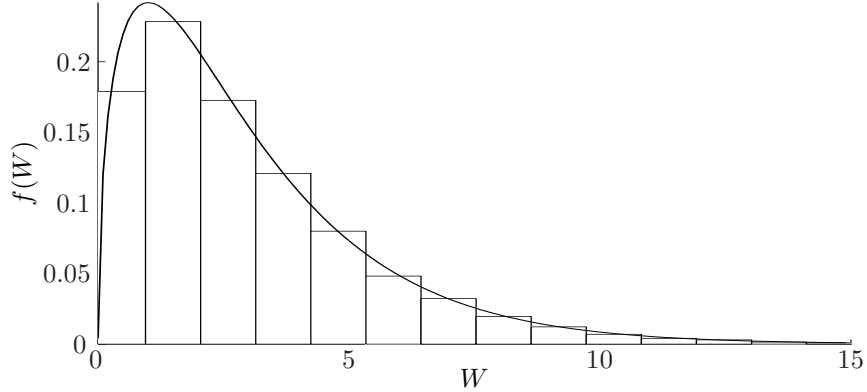


Figure 4.3 Simulated distribution of the Wald statistic (bars) and the asymptotic distribution based on a χ_3^2 distribution.

4.7.2 Asymptotic Relationships Among the Tests

The previous section establishes that the Wald test statistic is asymptotically distributed as χ_M^2 under H_0 , where M is the number of restrictions being tested. The relationships between the LR, Wald and LM tests are now used to demonstrate that all three test statistics have the same asymptotic null distribution.

Suppose the null hypothesis $H_0 : \theta = \theta_0$ is true. Expanding $\ln L_T(\theta)$ in a second-order Taylor series expansion around $\hat{\theta}_1$ and evaluating at $\theta = \theta_0$ gives

$$\ln L_T(\theta_0) \simeq \ln L_T(\hat{\theta}_1) + G_T(\hat{\theta}_1)(\theta_0 - \hat{\theta}_1) + \frac{1}{2}(\theta_0 - \hat{\theta}_1)' H_T(\hat{\theta}_1)(\theta_0 - \hat{\theta}_1), \quad (4.22)$$

where

$$G_T(\hat{\theta}_1) = \left. \frac{\partial \ln L_T(\theta)}{\partial \theta} \right|_{\theta=\hat{\theta}_1}, \quad H_T(\hat{\theta}_1) = \left. \frac{\partial^2 \ln L_T(\theta)}{\partial \theta \partial \theta'} \right|_{\theta=\hat{\theta}_1}. \quad (4.23)$$

The remainder in this Taylor series expansion is asymptotically negligible because $\hat{\theta}_1$ is a $\sqrt{T}$ -consistent estimator of θ_0 . The first order conditions of a maximum likelihood estimator require $G_T(\hat{\theta}_1) = 0$ so that equation (4.22)

reduces to

$$\ln L_T(\theta_0) \simeq \ln L_T(\hat{\theta}_1) + \frac{1}{2}(\theta_0 - \hat{\theta}_1)' H_T(\hat{\theta}_1)(\theta_0 - \hat{\theta}_1).$$

Multiplying both sides by T and rearranging gives

$$-2(T \ln L_T(\theta_0) - T \ln L_T(\hat{\theta}_1)) \simeq -T(\theta_0 - \hat{\theta}_1)' H_T(\hat{\theta}_1)(\theta_0 - \hat{\theta}_1).$$

The left-hand side of this equation is the LR statistic. The right-hand side is the Wald statistic, thereby showing that the LR and Wald tests are asymptotically equivalent under H_0 .

To show the relationship between the LM and Wald tests, expand

$$G_T(\theta) = \frac{\partial \ln L_T(\theta)}{\partial \theta}$$

in terms of a first-order Taylor series expansion around $\hat{\theta}_1$ and evaluate at $\theta = \theta_0$ to get

$$G_T(\theta_0) \simeq G_T(\hat{\theta}_1) + H_T(\hat{\theta}_1)(\theta_0 - \hat{\theta}_1) = H_T(\hat{\theta}_1)(\theta_0 - \hat{\theta}_1),$$

where $G_T(\hat{\theta}_1)$ and $H_T(\hat{\theta}_1)$ are as defined in (4.23). Using the first order conditions of the maximum likelihood estimator yields

$$G_T(\hat{\theta}_0) \simeq \left. \frac{\partial^2 \ln L_T(\theta)}{\partial \theta \partial \theta'} \right|_{\theta=\hat{\theta}_1} (\hat{\theta}_0 - \hat{\theta}_1) = I(\hat{\theta}_1)(\hat{\theta}_1 - \hat{\theta}_0).$$

Substituting this expression into the LM statistic in (4.14) gives

$$LM \simeq T(\hat{\theta}_1 - \theta_0)' I(\hat{\theta}_1)' I^{-1}(\theta_0) I(\hat{\theta}_1)(\hat{\theta}_1 - \theta_0) \simeq T(\hat{\theta}_1 - \theta_0)' I(\hat{\theta}_1)(\hat{\theta}_1 - \theta_0) = W,$$

This demonstrates that the LM and Wald tests are asymptotically equivalent under the null hypothesis.

As the LR, W and LM test statistics have the same asymptotic distribution the choice of which to use is governed by convenience. When it is easier to estimate the unrestricted (restricted) model, the Wald (LM) test is the most convenient to compute. The LM test tends to dominate diagnostic analysis of regression models with normally distributed disturbances because the model under the null hypothesis is often estimated using a least squares estimation procedure. These features of the LM test are developed in Part TWO.

4.7.3 Finite Sample Relationships

The discussion of the LR, Wald and LM test statistics, so far, is based on asymptotic distribution theory. In general, the finite sample distribution of

the test statistics is unknown and is commonly approximated by the asymptotic distribution. In situations where the asymptotic distribution does not provide an accurate approximation of the finite sample distribution, three possible solutions exist.

(1) ***Second-order approximations***

The asymptotic results are based on a first-order Taylor series expansion of the gradient of the log-likelihood function. In some cases, extending the expansions to higher-order terms by using Edgeworth expansions for example (see Example 2.28), will generally provide a more accurate approximation to the sampling distribution of the maximum likelihood estimator. However, this is more easily said than done, because deriving the sampling distribution of nonlinear functions is much more difficult than deriving sampling distributions of linear functions.

(2) ***Monte Carlo methods***

To circumvent the analytical problems associated with deriving the sampling distribution of the maximum likelihood estimator for finite samples using second-order, or even higher-order expansions, a more convenient approach is to use Monte Carlo methods. The approach is to simulate the finite sample distribution of the test statistic for particular values of the sample size, T , by running the simulation for these sample sizes and computing the corresponding critical values from the simulated values.

(3) ***Transformations***

A final approach is to transform the statistic so that the asymptotic distribution provides a better approximation to the finite sample distribution. A well-known example is the distribution of the test of the correlation coefficient, which is asymptotically normally distributed, although convergence is relatively slow as T increases (Stuart and Ord, 1994, p567).

By assuming normality and confining attention to the case of linear restrictions, an important relationship that holds amongst the three test statistics in finite samples is

$$W \geq LR \geq LM.$$

This result implies that the LM test tends to be a more conservative test in finite samples because the Wald statistic tends to reject the null hypothesis more frequently than the LR statistic, which, in turn, tends to reject the null hypothesis more frequently than the LM statistic. This relationship is highlighted by the Wald and LM tests of the normal distribution in Examples

4.4 and 4.6, respectively, because $\hat{\sigma}_1^2 \leq \hat{\sigma}_0^2$, it follows that

$$W = T \frac{(\bar{y} - \theta_0)^2}{\hat{\sigma}_1^2} \geq LM = T \frac{(\bar{y} - \theta_0)^2}{\hat{\sigma}_0^2}.$$

4.8 Size and Power Properties

4.8.1 Size of a Test

The probability of rejecting the null hypothesis when it is true (a Type-1 error) is usually denoted α and called the level of significance or the size of a test. For a test with size $\alpha = 0.05$, therefore, the null is rejected for p-values of less than 0.05. Equivalently, the null is rejected where the test statistic falls within a rejection region, ω , in which case the size of the test is expressed conveniently (in the case of the Wald test) as

$$\text{Size} = P(W \in \omega | H_0). \quad (4.24)$$

In a simulation experiment, the size is computed by simulating the model under the null hypothesis, H_0 , that is when the restrictions are true, and computing the proportion of simulated values of the test statistic that are greater than the critical value obtained from the asymptotic distribution. The asymptotic distribution of the LR , W and LM tests is χ^2 with M degrees of freedom under the null hypothesis; so in this case the critical value is $\chi_M^2(0.05)$.

Subject to some simulation error, the simulated and asymptotic sizes should match. In finite samples, however, this may not be true. In the case where the simulated size is greater than 0.05, the test is oversized with the null hypothesis being rejected more often than predicted by asymptotic theory. In the case where the simulated size is less than 0.05, the test is undersized (conservative) with the null hypothesis being rejected less often than predicted by asymptotic theory.

Example 4.11 Computing the Size of a Test by Simulation

Consider testing the hypotheses

$$H_0 : \beta_1 = 0, \quad H_1 : \beta_1 \neq 0,$$

in the exponential regression model

$$f(y | x_t; \theta) = \mu_t^{-1} \exp[-\mu_t^{-1} y],$$

where $\mu_t = \exp[\beta_0 + \beta_1 x_t]$ and $\theta = \{\beta_0, \beta_1\}$. Computing the size of the test

requires simulating the model 10000 times under the null hypothesis $\beta_1 = 0$ for samples of size $T = 5, 10, 25, 100$ with $x_t \sim iidN(0, 1)$ and intercept $\beta_0 = 1$. For each simulation, the Wald statistic

$$W = \frac{(\hat{\beta}_1 - 0)^2}{\text{var}(\hat{\beta}_1)}.$$

is computed. The size of the Wald test is computed as the proportion of the 10000 statistics with values greater than $\chi_1^2(0.05) = 3.841$. The results are as follows:

T:	5	10	25	100
Size:	0.066	0.053	0.052	0.051
Critical value (Simulated, 5%):	4.288	3.975	3.905	3.873

The test is slightly oversized for $T = 5$ since $0.066 > 0.05$, but the empirical size approaches the asymptotic size of 0.05 very quickly for $T \geq 10$. Also given are the simulated critical values corresponding to the value of the test statistic, which is exceeded by 5% of the simulated values. The fact that the test is oversized results in critical values in excess of the asymptotic critical value of 3.841. $\square$

4.8.2 Power of a Test

The probability of rejecting the null hypothesis when it is false is called the ‘power’ of a test. A second type of error that occurs in hypothesis testing is failing to reject the null hypothesis when it is false (a Type-2 error). The power of a test is expressed formally (in the case of the Wald test) as

$$\text{Power} = P(W \in \omega | H_1), \quad (4.25)$$

so that $1 - \text{Power}$ is the probability of committing a Type-2 error.

In a simulation experiment, the power is computed by simulating the model under the alternative hypothesis, H_1 : that is, when the restrictions stated in the null hypothesis, H_0 , are false. The proportion of simulated values of the test statistic greater than the critical value then gives the power of the test. Here the critical value is not the one obtained from the asymptotic distribution, but rather from simulating the distribution of the statistic under the null hypothesis and then choosing the value that has a fixed size of, say, 0.05. As the size is fixed at a certain level in computing the power of a test, the power is then referred to as a size-adjusted power.

Example 4.12 Computing the Power of a Test by Simulation

Consider again the exponential regression model of Example 4.11 with the null hypothesis given by $\beta_1 = 0$. The power of the Wald test is computed for 10000 samples of size $T = 5$ with $\beta_0 = 1$ and with increasing values for β_1 given by $\beta_1 = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$. For each value of β_1 , the size-adjusted power of the test is computed as the proportion of the 10000 statistics with values greater than 4.288, the critical value from Example 4.11 corresponding to a size of 0.05 for $T = 5$. The results are as follows:

$\beta_1 :$	-4	-3	-2	-1	0	1	2	3	4
Power:	0.99	0.98	0.86	0.38	0.05	0.42	0.89	0.99	0.99

The power of the Wald test at $\beta_1 = 0$ is 0.05 by construction as the powers are size-adjusted. The size-adjusted power of the test increases monotonically as the value of the parameter β_1 moves further and further away from its value under the null hypothesis with a maximum power of 99% attained at $\beta_1 = \pm 4$. $\square$

An important property of any test is that, as the sample increases, the probability of rejecting the null hypothesis when it is false, or the power of the test, approaches unity in the limit

$$\lim_{T \rightarrow \infty} P(W \in \omega | H_1) = 1. \quad (4.26)$$

A test having this property is known as a consistent test.

Example 4.13 Illustrating the Consistency of a Test by Simulation

The testing problem in the exponential regression model introduced in Examples 4.11 and 4.12 is now developed. The power of the Wald test, with respect to testing the null hypothesis $H_0 : \beta_1 = 0$, is computed for 10000 samples using parameter values $\beta_0 = 1$ and $\beta_1 = 1$. The results obtained for increasing sample sizes are as follows:

T:	5	10	25	100
Power:	0.420	0.647	0.993	1.000
Critical value (Simulated, 5%):	4.288	3.975	3.905	3.873

In computing the power for each sample size, a different critical value is used to ensure that the size of the test is 0.05 and, therefore, that the power values reported are size adjusted. The results show that the Wald test is consistent because $\text{Power} \rightarrow 1$ as T is increased. $\square$

4.9 Applications

Two applications that highlight the details of the calculations of the LR, Wald and LM tests are now presented. The first involves performing tests of the parameters of an exponential regression model. The second extends the exponential regression example by generalizing the distribution to a gamma distribution. Further applications of the three testing procedures to regression models are discussed in Part TWO of the book.

4.9.1 Exponential Regression Model

Consider the exponential regression model where y_t is assumed to be independent, but not identically distributed, from an exponential distribution with time-varying mean

$$E[y_t] = \mu_t = \beta_0 + \beta_1 x_t, \quad (4.27)$$

where x_t is the explanatory variable held fixed in repeated samples. The aim is to test the hypotheses

$$H_0 : \beta_1 = 0, \quad H_1 : \beta_1 \neq 0. \quad (4.28)$$

Under the null hypothesis, the mean of y_t is simply β_0 , which implies that y_t is an *iid* random variable. The parameters under the null and alternative hypotheses are, respectively, $\theta_0 = \{\beta_0, 0\}$ and $\theta_1 = \{\beta_0, \beta_1\}$.

As the distribution of y_t is exponential with mean μ_t , the log-likelihood function is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \left(-\ln(\mu_t) - \frac{y_t}{\mu_t} \right) = -\frac{1}{T} \sum_{t=1}^T \ln(\beta_0 + \beta_1 x_t) - \frac{1}{T} \sum_{t=1}^T \frac{y_t}{\beta_0 + \beta_1 x_t}.$$

The gradient vector is

$$G_T(\theta) = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T (-\mu_t^{-1} + \mu_t^{-2} y_t) \\ \frac{1}{T} \sum_{t=1}^T (-\mu_t^{-1} + \mu_t^{-2} y_t) x_t \end{bmatrix},$$

and the Hessian matrix is

$$H_T(\theta) = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T (\mu_t^{-2} - 2\mu_t^{-3} y_t) & \frac{1}{T} \sum_{t=1}^T (\mu_t^{-2} - 2\mu_t^{-3} y_t) x_t \\ \frac{1}{T} \sum_{t=1}^T (\mu_t^{-2} - 2\mu_t^{-3} y_t) x_t & \frac{1}{T} \sum_{t=1}^T (\mu_t^{-2} - 2\mu_t^{-3} y_t) x_t^2 \end{bmatrix}.$$

Taking expectations and changing the sign gives the information matrix

$$I(\theta) = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T \mu_t^{-2} & \frac{1}{T} \sum_{t=1}^T \mu_t^{-2} x_t \\ \frac{1}{T} \sum_{t=1}^T \mu_t^{-2} x_t & \frac{1}{T} \sum_{t=1}^T \mu_t^{-2} x_t^2 \end{bmatrix}.$$

A sample of $T = 2000$ observations on y_t and x_t is generated from the following exponential regression model:

$$f(y; \theta) = \frac{1}{\mu_t} \exp \left[-\frac{y}{\mu_t} \right], \quad \mu_t = \beta_0 + \beta_1 x_t,$$

with parameters $\theta = \{\beta_0, \beta_1\}$. The parameters are set at $\beta_0 = 1$ and $\beta_1 = 2$ and $x_t \sim U(0, 1)$. The unrestricted parameter estimates, the gradient and log-likelihood function value are, respectively,

$$\hat{\theta}_1 = [1.101, 1.760]', \quad G_T(\hat{\theta}_1) = [0.000, 0.000]', \quad \ln L_T(\hat{\theta}_1) = -1.653.$$

Evaluating the Hessian, information and outer product of gradient matrices at the unrestricted parameter estimates gives, respectively,

$$\begin{aligned} H_T(\hat{\theta}_1) &= \begin{bmatrix} -0.315 & -0.110 \\ -0.110 & -0.062 \end{bmatrix} \\ I(\hat{\theta}_1) &= \begin{bmatrix} 0.315 & 0.110 \\ 0.110 & 0.062 \end{bmatrix} \\ J_T(\hat{\theta}_1) &= \begin{bmatrix} 0.313 & 0.103 \\ 0.103 & 0.057 \end{bmatrix}. \end{aligned} \tag{4.29}$$

The restricted parameter estimates, the gradient and log-likelihood function value are, respectively

$$\hat{\theta}_0 = [1.989, 0.000]', \quad G_T(\hat{\theta}_0) = [0.000, 0.037]', \quad \ln L_T(\hat{\theta}_0) = -1.688.$$

Evaluating the Hessian, information and outer product of gradients matrices at the restricted parameter estimates gives, respectively

$$\begin{aligned} H_T(\hat{\theta}_0) &= \begin{bmatrix} -0.377 & -0.092 \\ -0.092 & -0.038 \end{bmatrix} \\ I(\hat{\theta}_0) &= \begin{bmatrix} 0.253 & 0.128 \\ 0.128 & 0.086 \end{bmatrix} \\ J_T(\hat{\theta}_0) &= \begin{bmatrix} 0.265 & 0.165 \\ 0.165 & 0.123 \end{bmatrix}. \end{aligned} \tag{4.30}$$

To test the hypotheses in (4.28), compute the LR statistic as

$$LR = -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)) = -2(-3375.208 + 3305.996) = 138.425.$$

Using the χ_1^2 distribution, the p-value is 0.000 indicating a rejection of the null hypothesis that $\beta_1 = 0$ at conventional significance levels, a result that is consistent with the data-generating process.

To perform the Wald test, define $R = [0 \ 1]$ and $Q = [0]$. Three Wald statistics are computed using the Hessian, information and outer product of gradients matrices in (4.29), with all calculations presented to three decimal points

$$\begin{aligned} W_H &= T[R\hat{\theta}_1 - Q]'[R(-H^{-1}(\hat{\theta}_1))R']^{-1}[R\hat{\theta}_1 - Q] = 145.545 \\ W_I &= T[R\hat{\theta}_1 - Q]'[RI^{-1}(\hat{\theta}_1)R']^{-1}[R\hat{\theta}_1 - Q] = 147.338 \\ W_J &= T[R\hat{\theta}_1 - Q]'[RJ^{-1}(\hat{\theta}_1)R']^{-1}[R\hat{\theta}_1 - Q] = 139.690. \end{aligned}$$

Using the χ_1^2 distribution, all p-values are 0.000, showing that the null hypothesis that $\beta_1 = 0$ is rejected at the 5% significance level for all three Wald tests.

Finally, three Lagrange multiplier statistics are computed using the Hessian, information and outer product of gradients matrices, as in (4.30)

$$\begin{aligned} LM_H &= TG'_T(\hat{\theta}_0)(-H_T^{-1}(\hat{\theta}_0))G_T(\hat{\theta}_0) \\ &= 2000 \begin{bmatrix} 0.000 \\ 0.037 \end{bmatrix}' \begin{bmatrix} 0.377 & 0.092 \\ 0.092 & 0.038 \end{bmatrix}^{-1} \begin{bmatrix} 0.000 \\ 0.037 \end{bmatrix} \\ &= 169.698. \end{aligned}$$

$$\begin{aligned} LM_I &= TG'_T(\hat{\theta}_0)I^{-1}(\hat{\theta}_0)G_T(\hat{\theta}_0) \\ &= 2000 \begin{bmatrix} 0.000 \\ 0.037 \end{bmatrix}' \begin{bmatrix} 0.253 & 0.128 \\ 0.128 & 0.086 \end{bmatrix}^{-1} \begin{bmatrix} 0.000 \\ 0.037 \end{bmatrix} \\ &= 127.482. \end{aligned}$$

$$\begin{aligned} LM_J &= TG'_T(\hat{\theta}_0)J_T^{-1}(\hat{\theta}_0)G_T(\hat{\theta}_0) \\ &= 2000 \begin{bmatrix} 0.000 \\ 0.037 \end{bmatrix}' \begin{bmatrix} 0.265 & 0.165 \\ 0.165 & 0.123 \end{bmatrix}^{-1} \begin{bmatrix} 0.000 \\ 0.037 \end{bmatrix} \\ &= 129.678. \end{aligned}$$

Using the χ_1^2 distribution, all p-values are 0.000, showing that the null hypothesis that $\beta_1 = 0$ is rejected at the 5% significance level for all three LM tests.

4.9.2 Gamma Regression Model

Consider the gamma regression model where y_t is assumed to be independent but not identically distributed from a gamma distribution with time-varying mean

$$E[y_t] = \mu_t = \beta_0 + \beta_1 x_t,$$

where x_t is the explanatory variable. The gamma distribution is given by

$$f(y|x_t; \theta) = \frac{1}{\Gamma(\rho)} \left(\frac{1}{\mu_t}\right)^\rho y^{\rho-1} \exp\left[-\frac{y}{\mu_t}\right], \quad \Gamma(\rho) = \int_0^\infty s^{\rho-1} e^{-s} ds,$$

where $\theta = \{\beta_0, \beta_1\}$. As the gamma distribution nests the exponential distribution when $\rho = 1$, a natural hypothesis to test is

$$H_0 : \rho = 1, \quad H_1 : \rho \neq 1.$$

The log-likelihood function is

$$\ln L_T(\theta) = -\ln \Gamma(\rho) - \frac{\rho}{T} \sum_{t=1}^T \ln(\beta_0 + \beta_1 x_t) + \frac{\rho-1}{T} \sum_{t=1}^T \ln y_t - \frac{1}{T} \sum_{t=1}^T \frac{y_t}{\beta_0 + \beta_1 x_t}.$$

As the gamma function, $\Gamma(\rho)$, appears in the likelihood function, it is convenient to use numerical derivatives to calculate the maximum likelihood estimates and the test statistics.

The following numerical illustration uses the data from the previous application on the exponential regression model. The unrestricted maximum likelihood parameter estimates and log-likelihood function value are, respectively,

$$\hat{\theta}_1 = [1.061, 1.698, 1.037]', \quad \ln L_T(\hat{\theta}_1) = -1.652579.$$

The corresponding restricted values, which are also the unrestricted estimates of the exponential model of the previous application, are

$$\hat{\theta}_0 = [1.101, 1.760, 1.000]', \quad \ln L_T(\hat{\theta}_0) = -1.652998.$$

The LR statistic is

$$LR = -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)) = -2(-3305.996 + 3305.158) = 1.674.$$

Using the χ_1^2 distribution, the p-value is 0.196, which means that the null hypothesis that the distribution is exponential cannot be rejected at the 5% significance level, a result that is consistent with the data generating process in Section 4.9.1.

The Wald statistic is computed with standard errors based on the Hessian evaluated at the unrestricted estimates. The Hessian matrix is

$$H_T(\hat{\theta}_1) = \begin{bmatrix} -0.351 & -0.123 & -0.560 \\ -0.123 & -0.069 & -0.239 \\ -0.560 & -0.239 & -1.560 \end{bmatrix}.$$

Defining $R = [0 \ 0 \ 1]$ and $Q = [1]$, the Wald statistic is

$$W = T[R\hat{\theta}_1 - Q]'[R(-H_T^{-1}(\hat{\theta}_1))R']^{-1}[R\hat{\theta}_1 - Q] = \frac{(1.037 - 1.000)^2}{0.001} = 1.631.$$

Using the χ_1^2 distribution, the p-value is 0.202, which also shows that the null hypothesis that the distribution is exponential cannot be rejected at the 5% significance level.

The LM statistic is based on the outer product of gradients matrix. To calculate the LM statistic, the gradient is evaluated at the restricted parameter estimates

$$G_T(\hat{\theta}_0) = [0.000, 0.000, 0.023]'$$

The outer product of gradients matrix evaluated at $\hat{\theta}_0$ is

$$J_T(\hat{\theta}_0) = \begin{bmatrix} 0.313 & 0.103 & 0.524 \\ 0.103 & 0.057 & 0.220 \\ 0.524 & 0.220 & 1.549 \end{bmatrix},$$

with inverse

$$J_T^{-1}(\hat{\theta}_0) = \begin{bmatrix} 9.755 & -11.109 & -1.728 \\ -11.109 & 51.696 & -3.564 \\ -1.728 & -3.564 & 1.735 \end{bmatrix}.$$

The LM test statistic is

$$\begin{aligned} LM &= TG(\hat{\theta}_0)'J_T^{-1}(\hat{\theta}_0)G(\hat{\theta}_0) \\ &= 20000 \begin{bmatrix} 0.000 \\ 0.000 \\ 0.023 \end{bmatrix}' \begin{bmatrix} 9.755 & -11.109 & -1.728 \\ -11.109 & 51.696 & -3.564 \\ -1.728 & -3.564 & 1.735 \end{bmatrix} \begin{bmatrix} 0.000 \\ 0.000 \\ 0.023 \end{bmatrix} \\ &= 1.853. \end{aligned}$$

Consistent with the results reported for the LR and Wald tests, using the χ_1^2 distribution the p-value of the LM test is 0.173 indicating that the null hypothesis cannot be rejected at the 5% level.

4.10 Exercises

(1) The Linear Regression Model

Gauss file(s)	<code>test_regress.g</code>
Matlab file(s)	<code>test_regress.m</code>

Consider the regression model

$$y_t = \beta x_t + u_t, \quad u_t \sim N(0, \sigma^2)$$

where the independent variable is $x_t = \{1, 2, 4, 5, 8\}$. The aim is to test the hypotheses $H_0 : \beta = 0$, $H_1 : \beta \neq 0$.

- Simulate the model for $T = 5$ observations using the parameter values $\beta = 1, \sigma^2 = 4$.
- Estimate the restricted model and unrestricted models and compute the corresponding values of the log-likelihood function.
- Perform a LR test choosing $\alpha = 0.05$ as the size of the test. Interpret the result.
- Perform a Wald test choosing $\alpha = 0.05$ as the size of the test. Interpret the result.
- Compute the gradient of the unrestricted model, but evaluated at the restricted estimates.
- Compute the Hessian of the unrestricted model, but evaluated at the restricted estimates, $\hat{\theta}_0$, and perform a LM test choosing $\alpha = 0.05$ as the size of the test. Interpret the result.

(2) The Weibull Distribution

Gauss file(s)	<code>test_weibull.g</code>
Matlab file(s)	<code>test_weibull.m</code>

Generate $T = 20$ observations from the Weibull distribution

$$f(y; \theta) = \alpha \beta y^{\beta-1} \exp \left[-\alpha y^\beta \right],$$

where the parameters are $\theta = \{\alpha = 1, \beta = 2\}$.

- Compute the unrestricted maximum likelihood estimates, $\hat{\theta}_1 = \{\hat{\alpha}_1, \hat{\beta}_1\}$ and the value of the log-likelihood function.
- Compute the restricted maximum likelihood estimates, $\hat{\theta}_0 = \{\hat{\alpha}_0, \hat{\beta}_0 = 1\}$ and the value of the log-likelihood function.
- Test the hypotheses $H_0 : \beta = 1$, $H_1 : \beta \neq 1$ using a LR test, a Wald test and a LM test and interpret the results.

- (d) Test the hypotheses $H_0 : \beta = 2$, $H_1 : \beta \neq 2$ using a LR test, a Wald test and a LM test and interpret the results.

(3) Simulating the Distribution of the Wald Statistic

Gauss file(s)	<code>test_asymptotic.g</code>
Matlab file(s)	<code>test_asymptotic.m</code>

Simulate the multiple regression model 10000 times with a sample size of $T = 1000$

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \beta_3 x_{3,t} + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

where the explanatory variables are $x_{1,t} \sim U(0, 1)$, $x_{2,t} \sim N(0, 1)$, $x_{3,t} \sim N(0, 1)^2$ and $\theta = \{\beta_0 = 0, \beta_1 = 0, \beta_2 = 0, \beta_3 = 0, \sigma^2 = 0.1\}$.

- For each simulation, compute the Wald test of the null hypothesis $H_0 : \beta_1 = 0$ and compare the simulated distribution to the asymptotic distribution.
- For each simulation, compute the Wald test of the joint null hypothesis $H_0 : \beta_1 = \beta_2 = 0$ and compare the simulated distribution to the asymptotic distribution.
- For each simulation, compute the Wald test of the joint null hypothesis $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ and compare the simulated distribution to the asymptotic distribution.
- Repeat parts (a) to (c) for $T = 10, 20$ and compare the finite sample distribution of the Wald statistic with the asymptotic distribution as approximated by the simulated distribution based on $T = 1000$.

(4) Simulating the Size and Power of the Wald Statistic

Gauss file(s)	<code>test_size.g, test_power.g</code>
Matlab file(s)	<code>test_size.m, test_power.m</code>

Consider testing the hypotheses

$$H_0 : \beta_1 = 0, \quad H_1 : \beta_1 \neq 0,$$

in the exponential regression model

$$f(y|x_t; \theta) = \mu_t^{-1} \exp[-\mu_t^{-1} x_t],$$

where $\mu_t = \exp[\beta_0 + \beta_1 x_t]$, $x_t \sim N(0, 1)$ and $\theta = \{\beta_0 = 1, \beta_1 = 0\}$.

- (a) Compute the sampling distribution of the Wald test by simulating the model under the null hypothesis 10000 times for a sample of size $T = 5$. Using the 0.05 critical value from the asymptotic distribution of the test statistic, compute the size of the test. Also, compute the critical value from the simulated distribution corresponding to a simulated size of 0.05.
- (b) Repeat part (a) for samples of size $T = 10, 25, 100, 500$. Interpret the results of the simulations.
- (c) Compute the power of the Wald test for a sample of size $T = 5$, $\beta_0 = 1$ and for $\beta_1 = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$.
- (d) Repeat part (c) for samples of size $T = 10, 25, 100, 500$. Interpret the results of the simulations.

(5) **Exponential Regression Model**

Gauss file(s)	<code>test_expreg.g, test_gamareg.g</code>
Matlab file(s)	<code>test_expreg.m, test_gamareg.m</code>

Generate a sample of size $T = 2000$ observations from the following exponential regression model

$$f(y | x_t; \theta) = \frac{1}{\mu_t} \exp \left[-\frac{y}{\mu_t} \right],$$

where $\mu_t = \beta_0 + \beta_1 x_t$, $x_t \sim U(0, 1)$ and the parameter values are $\beta_0 = 1$ and $\beta_1 = 2$.

- (a) Compute the unrestricted maximum likelihood estimates, $\hat{\theta}_1 = \{\hat{\beta}_0, \hat{\beta}_1\}$ and the value of the log-likelihood function, $\ln L_T(\hat{\theta}_1)$.
- (b) Re-estimate the model subject to the restriction that $\beta_1 = 0$ and recompute the value of the log-likelihood function, $\ln L_T(\hat{\theta}_0)$.
- (c) Test the following hypotheses $H_0 : \beta_1 = 0$, $H_1 : \beta_1 \neq 0$, using a LR test; Wald tests based on the Hessian, information and outer product of gradients matrices, respectively, with analytical and numerical derivatives in each case; and LM tests based on the Hessian, information and outer product of gradients matrices, with analytical and numerical derivatives in each case. Interpret the results.
- (d) Now assume that the true distribution is gamma

$$f(y | x_t; \theta) = \frac{1}{\Gamma(\rho)} \left(\frac{1}{\mu_t} \right)^\rho y^{\rho-1} \exp \left(-\frac{y}{\mu_t} \right),$$

where the unknown parameters are $\theta = \{\beta_1, \beta_2, \rho\}$. Compute the

unrestricted maximum likelihood estimates, $\hat{\theta}_1 = \{\hat{\beta}_0, \hat{\beta}_1, \hat{\rho}\}$ and the value of the log-likelihood function, $\ln L(\hat{\theta}_1)$.

(e) Test the following hypotheses

$$H_0 : \rho = 1, \quad H_1 : \rho \neq 1,$$

using a LR test; Wald tests based on the Hessian, information and outer product of gradients matrices, respectively, with numerical derivatives in each case; and LM tests based on the Hessian, information and outer product of gradients matrices, respectively, with numerical derivatives in each case. Interpret the results.

(6) Neyman's Smooth Goodness of Fit Test

Gauss file(s)	<code>test_smooth.g</code>
Matlab file(s)	<code>test_smooth.m</code>

Let $y_{1,t}, y_{2,t}, \dots, y_T$, be *iid* random variables with unknown distribution function F . A test that the distribution function is known and equal to F_0 is given by the respective null and alternative hypotheses

$$H_0 : F = F_0, \quad H_1 : F \neq F_0.$$

The Neyman (1937) smooth goodness of fit test (see also Bera, Ghosh and Xiao (2010) for a recent application) is based on the property that the random variable

$$u = F_0(y) = \int_{-\infty}^y f_0(s) ds,$$

is uniformly distributed under the null hypothesis. The approach is to specify the generalized uniform distribution

$$g(u) = c(\theta) \exp[1 + \theta_1 \phi_1(u) + \theta_2 \phi_2(u) + \theta_3 \phi_3(u) + \theta_4 \phi_4(u)],$$

where $c(\theta)$ is the normalizing constant to ensure that

$$\int_0^1 g(u) du = 1.$$

The terms $\phi_i(u)$ are the Legendre orthogonal polynomials given by

$$\begin{aligned}\phi_1(u) &= \sqrt{32}\left(u - \frac{1}{2}\right) \\ \phi_2(u) &= \sqrt{5}\left(6\left(u - \frac{1}{2}\right)^2 - \frac{1}{2}\right) \\ \phi_3(u) &= \sqrt{7}\left(20\left(u - \frac{1}{2}\right)^3 - 3\left(u - \frac{1}{2}\right)\right) \\ \phi_4(u) &= 3\left(70\left(u - \frac{1}{2}\right)^4 - 15\left(u - \frac{1}{2}\right)^2 + \frac{3}{8}\right),\end{aligned}$$

satisfying the orthogonality property

$$\int_0^1 \phi_i(u)\phi_j(u)du = \begin{cases} 1 & : i = j \\ 0 & : i \neq j. \end{cases}$$

A test of the null and alternative hypotheses is given by the joint restrictions

$$H_0 : \theta_1 = \theta_2 = \theta_3 = \theta_4 = 0, \quad H_1 : \text{at least one restriction fails},$$

as the distribution of u under H_0 is uniform.

- (a) Derive the log-likelihood function, $\ln L_T(\theta)$, in terms of u_t where

$$u_t = F_0(y_t) = \int_{-\infty}^{y_t} f_0(s)ds,$$

as well as the gradient vector $G_T(\theta)$ and the information matrix $I(\theta)$. In writing out the log-likelihood function it is necessary to use the expression of the Legendre polynomials $\phi_i(z)$.

- (b) Derive a LR test.
(c) Derive a Wald test.
(d) Show that a LM test is based on the statistic

$$LM = \sum_{i=1}^4 \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T \phi_i(u_t) \right)^2.$$

In deriving the LM statistic use the result that

$$c(\theta)^{-1} = \int_0^1 \exp[1 + \theta_1\phi_1(u) + \theta_2\phi_2(u) + \theta_3\phi_3(u) + \theta_4\phi_4(u)]du.$$

- (e) Briefly discuss the advantages and disadvantages of the alternative test statistics in parts (b) to (d).

- (f) To examine the performance of the three testing procedures in parts (b) to (d) under the null hypothesis, assume that F_0 is the normal distribution and that the random variables are drawn from $N(0, 1)$.
- (g) To examine the performance of the three testing procedures in parts (b) to (d) under the alternative hypothesis, assume that F_0 is the normal distribution and that the random variables are drawn from χ_1^2 .

PART TWO
REGRESSION MODELS

5

Linear Regression Models

5.1 Introduction

The maximum likelihood framework set out in Part ONE is now applied to estimating and testing regression models. This chapter focuses on linear models, where the conditional mean of a dependent variable is specified to be a linear function of a set of explanatory variables. Both single equation and multiple equations models are discussed. Extensions of the linear class of models are discussed in Chapter 6 (nonlinear regression), Chapter 7 (autocorrelation) and Chapter 8 (heteroskedasticity).

Many of the examples considered in Part ONE specify the distribution of the observable random variable, y_t . Regression models, by contrast, specify the distribution of the unobservable disturbances, u_t . Specifying the distribution in terms u_t means that maximum likelihood estimation cannot be used directly, since this method requires evaluating the log-likelihood function at the observed values of the data. This problem is circumvented by using the transformation of variable technique (see Appendix A), which transforms the distribution of u_t to the distribution of y_t . This technique is used implicitly in the regression examples considered in Part ONE. In Part TWO, however, the form of this transformation must be made explicit. A second important feature of regression models is that the distribution of u_t is usually chosen to be the normal distribution. One of the gains in adopting this assumption is that it can simplify the computation of the maximum likelihood estimators so that they can be obtained simply by least squares regressions.

5.2 Specification

The different types of linear regression models can usefully be illustrated by means of examples which are all similar in the sense that each model includes: one or more endogenous or dependent variables, $y_{i,t}$, that are simultaneously determined by an interrelated series of equations; exogenous variables, $x_{i,t}$, that are assumed to be determined outside the model; and predetermined or lagged dependent variables, $y_{i,t-j}$. Together the exogenous and predetermined variables are referred to as the independent variables.

5.2.1 Model Classification

Example 5.1 Univariate Regression Model

Consider a linear relationship between a single dependent (endogenous) variable, y_t , and a single exogenous variable, x_t , given by

$$y_t = \alpha x_t + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

where u_t is the disturbance term. By definition, x_t is independent of the disturbance term, $E[x_t u_t] = 0$. $\square$

Example 5.2 Seemingly Unrelated Regression Model

An extension of the univariate equation containing two dependent variables, $y_{1,t}$, $y_{2,t}$, and one exogenous variable, x_t , is

$$\begin{aligned} y_{1,t} &= \alpha_1 x_t + u_{1,t} \\ y_{2,t} &= \alpha_2 x_t + u_{2,t}, \end{aligned}$$

where the disturbance term $u_t = (u_{1,t}, u_{2,t})'$ has the properties

$$u_t \sim iid N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix} \right).$$

This system is commonly known as a seemingly unrelated regression model (SUR) and is discussed in greater detail later on. An important feature of the SUR model is that the dependent variables are expressed only in terms of the exogenous variable(s). $\square$

Example 5.3 Simultaneous System of Equations

Systems of equations in which the dependent variables are determinants of other dependent variables, and not just independent variables, are referred to as simultaneous systems of equations. Consider the following system of

equations

$$\begin{aligned} y_{1,t} &= \beta y_{2,t} + u_{1,t} \\ y_{2,t} &= \gamma y_{1,t} + \alpha x_t + u_{2,t}, \end{aligned}$$

where the disturbance term $u_t = (u_{1,t}, u_{2,t})'$ has the properties

$$u_t \sim iid N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{11} & 0 \\ 0 & \sigma_{22} \end{bmatrix} \right).$$

This system is characterized by the dependent variables $y_{1,t}$ and $y_{2,t}$ being functions of each other, with $y_{2,t}$ also being a function of the exogenous variable x_t . $\square$

Example 5.4 Recursive System

A special case of the simultaneous model is the recursive model. An example of a trivariate recursive model is

$$\begin{aligned} y_{1,t} &= \alpha_1 x_t + u_{1,t} \\ y_{2,t} &= \beta_1 y_{1,t} + \alpha_2 x_t + u_{2,t}, \\ y_{3,t} &= \beta_2 y_{1,t} + \beta_3 y_{2,t} + \alpha_3 x_t + u_{3,t}, \end{aligned}$$

where the disturbance term $u_t = (u_{1,t}, u_{2,t}, u_{3,t})'$ has the properties

$$u_t \sim iid N \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{11} & 0 & 0 \\ 0 & \sigma_{22} & 0 \\ 0 & 0 & \sigma_{33} \end{bmatrix} \right).$$

$\square$

5.2.2 Structural and Reduced Forms

Before generalizing the previous examples to many dependent variables and many independent variables, it is helpful to introduce some matrix notation. For example, consider rewriting the simultaneous model of Example 5.3 as

$$\begin{aligned} y_{1,t} - \beta y_{2,t} &= u_{1,t} \\ -\gamma y_{1,t} + y_{2,t} - \alpha x_t &= u_{2,t}, \end{aligned}$$

or more compactly as

$$y_t B + x_t A = u_t, \tag{5.1}$$

where

$$y_t = [y_{1,t} \ y_{2,t}], \ B = \begin{bmatrix} 1 & -\beta \\ -\gamma & 1 \end{bmatrix}, \ A = [0 \ -\alpha], \ u_t = [u_{1,t} \ u_{2,t}].$$

The covariance matrix of the disturbances is

$$V = E[u_t' u_t] = E \begin{bmatrix} u_{1,t}^2 & u_{1,t} u_{2,t} \\ u_{1,t} u_{2,t} & u_{2,t}^2 \end{bmatrix} = \begin{bmatrix} \sigma_{11} & 0 \\ 0 & \sigma_{22} \end{bmatrix}.$$

Equation (5.1) is known as the structural form where y_t represents the endogenous variables and x_t the exogenous variables.

The bivariate system of equations in (5.1) is easily generalized to a system of N equations with K exogenous variables by simply extending the dimensions of the pertinent matrices. For example, the dependent and exogenous variables become

$$\begin{aligned} y_t &= \begin{bmatrix} y_{1,t} & y_{2,t} & \cdots & y_{N,t} \end{bmatrix} \\ x_t &= \begin{bmatrix} x_{1,t} & x_{2,t} & \cdots & x_{K,t} \end{bmatrix}, \end{aligned}$$

and the disturbance terms become

$$u_t = \begin{bmatrix} u_{1,t} & u_{2,t} & \cdots & u_{N,t} \end{bmatrix},$$

so that in equation (5.1) B is now $(N \times N)$, A is $(K \times N)$ and V is a $(N \times N)$ covariance matrix of the disturbances.

An alternative way to write the system of equations in (5.1) is to express the system in terms of y_t ,

$$\begin{aligned} y_t &= -x_t A B^{-1} + u_t B^{-1} \\ &= x_t \Pi + v_t, \end{aligned} \tag{5.2}$$

where

$$\Pi = -A B^{-1}, \quad v_t = u_t B^{-1}, \tag{5.3}$$

and the disturbance term v_t has the properties

$$\begin{aligned} E[v_t] &= E[u_t B^{-1}] = E[u_t] B^{-1} = 0, \\ E[v_t' v_t] &= E[(B^{-1})' u_t' u_t B^{-1}] = (B^{-1})' E[u_t' u_t] B^{-1} = (B^{-1})' V B^{-1}. \end{aligned}$$

Equation (5.2) is known as the reduced form. The reduced form of a set of structural equations serves a number of important purposes.

- (1) It forms the basis for simulating a system of equations.
- (2) It can be used as an alternative way to estimate a structural model. A popular approach is estimating structural vector autoregression models, which is discussed in Chapter 14.
- (3) The reduced form is used to compute forecasts and perform experiments on models.

Example 5.5 Simulating a Simultaneous Model

Consider simulating $T = 500$ observations from the bivariate model

$$\begin{aligned} y_{1,t} &= \beta_1 y_{2,t} + \alpha_1 x_{1,t} + u_{1,t} \\ y_{2,t} &= \beta_2 y_{1,t} + \alpha_2 x_{2,t} + u_{2,t}, \end{aligned}$$

with parameters $\beta_1 = 0.6$, $\alpha_1 = 0.4$, $\beta_2 = 0.2$, $\alpha_2 = -0.5$ and covariance matrix of u_t

$$V = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}.$$

Define the structural parameter matrices

$$\begin{aligned} B &= \begin{bmatrix} 1 & -\beta_2 \\ -\beta_1 & 1 \end{bmatrix} = \begin{bmatrix} 1.000 & -0.200 \\ -0.600 & 1.000 \end{bmatrix} \\ A &= \begin{bmatrix} -\alpha_1 & 0 \\ 0 & -\alpha_2 \end{bmatrix} = \begin{bmatrix} -0.400 & 0.000 \\ 0.000 & 0.500 \end{bmatrix}. \end{aligned}$$

From equation (5.3) the reduced form parameter matrix is

$$\begin{aligned} \Pi &= -AB^{-1} \\ &= - \begin{bmatrix} -0.400 & 0.000 \\ 0.000 & 0.500 \end{bmatrix} \begin{bmatrix} 1.000 & -0.200 \\ -0.600 & 1.000 \end{bmatrix}^{-1} \\ &= - \begin{bmatrix} -0.400 & 0.000 \\ 0.000 & 0.500 \end{bmatrix} \begin{bmatrix} 1.136 & 0.227 \\ 0.681 & 1.136 \end{bmatrix} \\ &= \begin{bmatrix} 0.454 & 0.090 \\ -0.340 & -0.568 \end{bmatrix}. \end{aligned}$$

The reduced form at time t is

$$\begin{bmatrix} y_{1,t} & y_{2,t} \end{bmatrix} = \begin{bmatrix} x_{1,t} & x_{2,t} \end{bmatrix} \begin{bmatrix} 0.454 & 0.090 \\ -0.340 & -0.568 \end{bmatrix} + \begin{bmatrix} v_{1,t} & v_{2,t} \end{bmatrix},$$

where the reduced form disturbances are given by equation (5.3)

$$\begin{bmatrix} v_{1,t} & v_{2,t} \end{bmatrix} = \begin{bmatrix} u_{1,t} & u_{2,t} \end{bmatrix} \begin{bmatrix} 1.136 & 0.227 \\ 0.681 & 1.136 \end{bmatrix}.$$

The simulated series of $y_{1,t}$ and $y_{2,t}$ are given in Figure 5.1, together with scatter plots corresponding to the two equations, where the exogenous variables are chosen as $x_{1,t} \sim N(0, 100)$ and $x_{2,t} \sim N(0, 9)$. $\square$

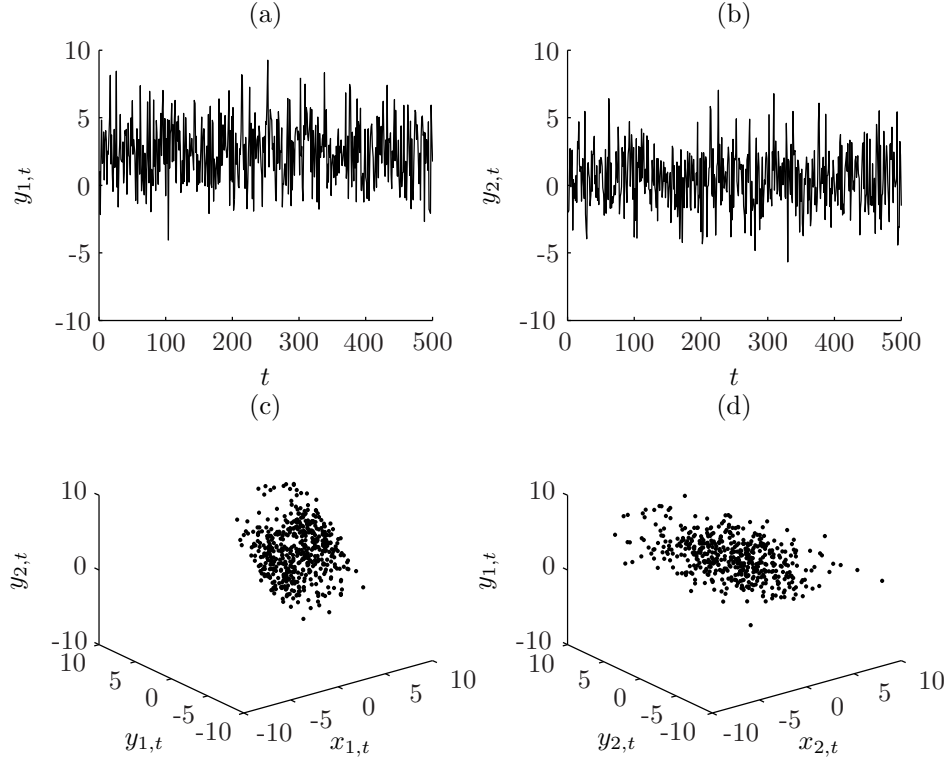


Figure 5.1 Simulating a bivariate regression model.

5.3 Estimation

5.3.1 Single Equation: Ordinary Least Squares

Consider the linear regression model

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + u_t \quad u_t \sim iid N(0, \sigma^2), \quad (5.4)$$

where y_t is the dependent variable, $x_{1,t}$ and $x_{2,t}$ are the independent variables and u_t is the disturbance term. To estimate the parameters $\theta = \{\beta_0, \beta_1, \beta_2, \sigma\}$ by maximum likelihood, it is necessary to use the transformation of variable technique to transform the distribution of the unobservable disturbance, u_t , into the distribution of y_t .

From equation (5.4) the pdf of u_t is

$$f(u_t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{u_t^2}{2\sigma^2} \right].$$

Using the transformation of variable technique, the pdf of y_t is

$$f(y_t) = f(u_t) \left| \frac{\partial u_t}{\partial y_t} \right| = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y_t - \beta_0 - \beta_1 x_{1,t} - \beta_2 x_{2,t})^2}{2\sigma^2} \right], \quad (5.5)$$

where $\partial u_t / \partial y_t$ is

$$\frac{\partial u_t}{\partial y_t} = \frac{\partial}{\partial y_t} [y_t - \beta_0 - \beta_1 x_{1,t} - \beta_2 x_{2,t}] = 1.$$

Given the distribution of y_t in (5.5), the log-likelihood function at time t is

$$\ln l_t(\theta) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} (y_t - \beta_0 - \beta_1 x_{1,t} - \beta_2 x_{2,t})^2.$$

For a sample of $t = 1, 2, \dots, T$ observations the log-likelihood function is

$$\ln L_T(\theta) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2 T} \sum_{t=1}^T (y_t - \beta_0 - \beta_1 x_{1,t} - \beta_2 x_{2,t})^2.$$

Differentiating $\ln L_T(\theta)$ with respect to θ yields

$$\begin{aligned} \frac{\partial \ln L_T(\theta)}{\partial \beta_0} &= \frac{1}{\sigma^2 T} \sum_{t=1}^T (y_t - \beta_0 - \beta_1 x_{1,t} - \beta_2 x_{2,t}) \\ \frac{\partial \ln L_T(\theta)}{\partial \beta_1} &= \frac{1}{\sigma^2 T} \sum_{t=1}^T (y_t - \beta_0 - \beta_1 x_{1,t} - \beta_2 x_{2,t}) x_{1,t} \\ \frac{\partial \ln L_T(\theta)}{\partial \beta_2} &= \frac{1}{\sigma^2 T} \sum_{t=1}^T (y_t - \beta_0 - \beta_1 x_{1,t} - \beta_2 x_{2,t}) x_{2,t} \\ \frac{\partial \ln L_T(\theta)}{\partial \sigma^2} &= -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4 T} \sum_{t=1}^T (y_t - \beta_0 - \beta_1 x_{1,t} - \beta_2 x_{2,t})^2. \end{aligned} \quad (5.6)$$

Setting these derivatives to zero

$$\begin{aligned} \frac{1}{\hat{\sigma}^2 T} \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1,t} - \hat{\beta}_2 x_{2,t}) &= 0 \\ \frac{1}{\hat{\sigma}^2 T} \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1,t} - \hat{\beta}_2 x_{2,t}) x_{1,t} &= 0 \\ \frac{1}{\hat{\sigma}^2 T} \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1,t} - \hat{\beta}_2 x_{2,t}) x_{2,t} &= 0 \\ -\frac{1}{2\hat{\sigma}^2} + \frac{1}{2\hat{\sigma}^4 T} \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1,t} - \hat{\beta}_2 x_{2,t})^2 &= 0, \end{aligned} \quad (5.7)$$

and solving for $\hat{\theta} = \{\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{\sigma}^2\}$ yields the maximum likelihood estimators.

For the system of equations in (5.7) an analytical solution exists. To derive this solution, first notice that the first three equations can be written independently of $\hat{\sigma}^2$ by multiplying both sides by $T\hat{\sigma}^2$ to give

$$\begin{aligned} \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1,t} - \hat{\beta}_2 x_{2,t}) &= 0 \\ \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1,t} - \hat{\beta}_2 x_{2,t}) x_{1,t} &= 0 \\ \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1,t} - \hat{\beta}_2 x_{2,t}) x_{2,t} &= 0, \end{aligned}$$

which is a system of three equations and three unknowns. Writing this system in matrix form,

$$\begin{bmatrix} \sum_{t=1}^T y_t \\ \sum_{t=1}^T y_t x_{1,t} \\ \sum_{t=1}^T y_t x_{2,t} \end{bmatrix} - \begin{bmatrix} T & \sum_{t=1}^T x_{1,t} & \sum_{t=1}^T x_{2,t} \\ \sum_{t=1}^T x_{1,t} & \sum_{t=1}^T x_{1,t}^2 & \sum_{t=1}^T x_{1,t} x_{2,t} \\ \sum_{t=1}^T x_{2,t} & \sum_{t=1}^T x_{1,t} x_{2,t} & \sum_{t=1}^T x_{2,t}^2 \end{bmatrix} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

and solving for $[\hat{\beta}_0 \hat{\beta}_1 \hat{\beta}_2]'$ gives

$$\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} T & \sum_{t=1}^T x_{1,t} & \sum_{t=1}^T x_{2,t} \\ \sum_{t=1}^T x_{1,t} & \sum_{t=1}^T x_{1,t}^2 & \sum_{t=1}^T x_{1,t} x_{2,t} \\ \sum_{t=1}^T x_{2,t} & \sum_{t=1}^T x_{1,t} x_{2,t} & \sum_{t=1}^T x_{2,t}^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T y_t \\ \sum_{t=1}^T x_{1,t} y_t \\ \sum_{t=1}^T x_{2,t} y_t \end{bmatrix},$$

which is the ordinary least squares estimator (OLS) of $[\beta_0 \beta_1 \beta_2]'$. Once $[\hat{\beta}_0 \hat{\beta}_1 \hat{\beta}_2]'$ is computed, the ordinary least squares estimator of the variance, $\hat{\sigma}^2$, is obtained by rearranging the last equation in (5.7) to give

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1,t} - \hat{\beta}_2 x_{2,t})^2. \quad (5.8)$$

This result establishes the relationship between the maximum likelihood estimator and the ordinary least squares estimator in the case of the single equation linear regression model. In computing $\hat{\sigma}^2$, it is common to express the denominator in (5.8) in terms of degrees of freedom, $T - K$, instead of merely T .

Expressing $\hat{\sigma}^2$ analytically in terms of the $\hat{\beta}$ s given in (5.8) means that $\hat{\sigma}^2$ can be concentrated out of the log-likelihood function. Standard errors can

be computed from the negative of the inverse Hessian. If estimation is based on the concentrated log-likelihood function, the estimated variance of $\hat{\sigma}^2$ is

$$\text{var}(\hat{\sigma}^2) = \frac{2\hat{\sigma}^4}{T}.$$

Example 5.6 Estimating a Regression Model

Consider the model

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + u_t, \quad u_t \sim N(0, 4),$$

where $\theta = \{\beta_0 = 1.0, \beta_1 = 0.7, \beta_2 = 0.3, \sigma^2 = 4\}$ and $x_{1,t}$ and $x_{2,t}$ are generated as $N(0, 1)$. For a sample of size $T = 200$, the maximum likelihood parameter estimates without concentrating the log-likelihood function are

$$\hat{\theta} = \{\hat{\beta}_0 = 1.129, \hat{\beta}_1 = 0.719, \hat{\beta}_2 = 0.389, \hat{\sigma}^2 = 3.862\},$$

with covariance matrix based on the Hessian given by

$$\frac{1}{T}\hat{\Omega} = \begin{bmatrix} 0.019 & 0.001 & -0.001 & 0.000 \\ 0.001 & 0.018 & 0.000 & 0.000 \\ -0.001 & 0.000 & 0.023 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.149 \end{bmatrix}.$$

The maximum likelihood parameter estimates obtained by concentrating the log-likelihood function are

$$\hat{\theta}_{\text{conc}} = \{\hat{\beta}_0 = 1.129, \hat{\beta}_1 = 0.719, \hat{\beta}_2 = 0.389\},$$

with covariance matrix based on the Hessian given by

$$\frac{1}{T}\hat{\Omega}_{\text{conc}} = \begin{bmatrix} 0.019 & 0.001 & -0.001 \\ 0.001 & 0.018 & 0.000 \\ -0.001 & 0.000 & 0.023 \end{bmatrix}.$$

The residuals at the second stage are computed as

$$\hat{u}_t = y_t - 1.129 - 0.719x_{1,t} - 0.389x_{2,t}.$$

The residual variance is computed as

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T \hat{u}_t^2 = \frac{1}{200} \sum_{t=1}^{200} \hat{u}_t^2 = 3.862,$$

with variance

$$\text{var}(\hat{\sigma}^2) = \frac{2\hat{\sigma}^4}{T} = \frac{2 \times 3.862^2}{200} = 0.149.$$

□

For the case of K exogenous variables, the linear regression model is

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_K x_{K,t} + u_t.$$

This equation can also be written in matrix form,

$$Y = X\beta + u, \quad E[u] = 0, \quad \text{cov}[u] = E[uu'] = \sigma^2 I_T,$$

where I_T is the $T \times T$ identity matrix and

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_T \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_{1,1} & \cdots & x_{K,1} \\ 1 & x_{1,2} & \cdots & x_{K,2} \\ 1 & x_{1,3} & \cdots & x_{K,3} \\ \vdots & \vdots & \cdots & \vdots \\ 1 & x_{1,T} & \cdots & x_{K,T} \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \vdots \\ \beta_K \end{bmatrix} \quad \text{and} \quad u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_T \end{bmatrix}.$$

Referring to the $K = 2$ case solved previously, the matrix solution is

$$\hat{\beta} = (X'X)^{-1}X'Y. \quad (5.9)$$

Once $\hat{\beta}$ has been computed, an estimate of the variance $\hat{\sigma}^2$ is

$$\hat{\sigma}^2 = \frac{\hat{u}'\hat{u}}{T - K}.$$

5.3.2 Multiple Equations: FIML

The maximum likelihood estimator for systems of equations is commonly referred to as the full-information maximum likelihood estimator (FIML). Consider the system of equations in (5.1). For a system of N equations, the density of u_t is assumed to be the multivariate normal distribution

$$f(u_t) = \left(\frac{1}{\sqrt{2\pi}}\right)^N |V|^{-1/2} \exp \left[-\frac{1}{2} u_t V^{-1} u_t' \right].$$

Using the transformation of variable technique, the density of y_t becomes

$$\begin{aligned} f(y_t) &= f(u_t) \left| \frac{\partial u_t}{\partial y_t} \right| \\ &= \left(\frac{1}{\sqrt{2\pi}}\right)^N |V|^{-1/2} \exp \left[-\frac{1}{2} (y_t B + x_t A) V^{-1} (y_t B + x_t A)' \right] |B|, \end{aligned}$$

because from equation (5.1)

$$u_t = y_t B + x_t A \Rightarrow \frac{\partial u_t}{\partial y_t} = B.$$

The log-likelihood function at time t is

$$\ln l_t(\theta) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |V| + \ln |B| - \frac{1}{2} (y_t B + x_t A) V^{-1} (y_t B + x_t A)',$$

and given $t = 1, 2, \dots, T$ observations, the log-likelihood function is

$$\ln L_T(\theta) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |V| + \ln |B| - \frac{1}{2T} \sum_{t=1}^T (y_t B + x_t A) V^{-1} (y_t B + x_t A)'. \quad (5.10)$$

The FIML estimator of the parameters of the model is obtained by differentiating $\ln L_T(\theta)$ with respect to θ , setting these derivatives to zero and solving to find $\hat{\theta}$. As in the estimation of the single equation model, estimation can be simplified by concentrating the likelihood with respect to the estimated covariance matrix $\hat{V}$. For the N system of equations, the residual covariance matrix is computed as

$$\hat{V} = \frac{1}{T} \begin{bmatrix} \sum_{t=1}^T \hat{u}_{1,t}^2 & \sum_{t=1}^T \hat{u}_{1,t} \hat{u}_{2,t} & \cdots & \sum_{t=1}^T \hat{u}_{1,t} \hat{u}_{N,t} \\ \sum_{t=1}^T \hat{u}_{2,t} \hat{u}_{1,t} & \sum_{t=1}^T \hat{u}_{2,t}^2 & & \sum_{t=1}^T \hat{u}_{2,t} \hat{u}_{N,t} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{t=1}^T \hat{u}_{N,t} \hat{u}_{1,t} & \sum_{t=1}^T \hat{u}_{N,t} \hat{u}_{2,t} & \cdots & \sum_{t=1}^T \hat{u}_{N,t}^2 \end{bmatrix},$$

and $\hat{V}$ can be substituted for V in equation (5.10). This eliminates the need to estimate the variance parameters directly, thus reducing the dimensionality of the estimation problem. Note that this approach is appropriate for simultaneous models based on normality. For other models based on non-normal distributions, all the parameters may need to be estimated jointly. Further, if standard errors of $\hat{V}$ are also required then these can be conveniently obtained by estimating all the parameters.

Example 5.7 FIML Estimation of a Structural Model

Consider the bivariate model introduced in Example 5.3, where the unknown parameters are $\theta = \{\beta, \gamma, \alpha, \sigma_{11}, \sigma_{22}\}$. The log-likelihood function is

$$\begin{aligned} \ln L_T(\theta) = & -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |\sigma_{11} \sigma_{22}| + \ln |1 - \beta \gamma| \\ & - \frac{1}{2 \sigma_{11} T} \sum_{t=1}^T (y_{1,t} - \beta y_{2,t})^2 - \frac{1}{2 \sigma_{22} T} \sum_{t=1}^T (y_{2,t} - \gamma y_{1,t} - \alpha x_t)^2. \end{aligned}$$

The first-order derivatives of $\ln L_T(\theta)$ with respect to θ are

$$\begin{aligned}\frac{\partial \ln L_T(\theta)}{\partial \beta} &= \frac{\gamma}{1 - \beta\gamma} + \frac{1}{\sigma_{11}T} \sum_{t=1}^T (y_{1,t} - \beta y_{2,t}) y_{2,t} \\ \frac{\partial \ln L_T(\theta)}{\partial \gamma} &= -\frac{\beta}{1 - \beta\gamma} + \frac{1}{\sigma_{22}T} \sum_{t=1}^T (y_{2,t} - \gamma y_{1,t} - \alpha x_t) y_{1,t} \\ \frac{\partial \ln L_T(\theta)}{\partial \alpha} &= \frac{1}{\sigma_{22}T} \sum_{t=1}^T (y_{2,t} - \gamma y_{1,t} - \alpha x_t) x_t \\ \frac{\partial \ln L_T(\theta)}{\partial \sigma_{11}} &= -\frac{1}{2\sigma_{11}} + \frac{1}{2\sigma_{11}^2 T} \sum_{t=1}^T (y_{1,t} - \beta y_{2,t})^2 \\ \frac{\partial \ln L_T(\theta)}{\partial \sigma_{22}} &= -\frac{1}{2\sigma_{22}} + \frac{1}{2\sigma_{22}^2 T} \sum_{t=1}^T (y_{2,t} - \gamma y_{1,t} - \alpha x_t)^2.\end{aligned}$$

Setting these derivatives to zero yields

$$\frac{\hat{\gamma}}{1 - \hat{\beta}\hat{\gamma}} + \frac{1}{\hat{\sigma}_{11}T} \sum_{t=1}^T (y_{1,t} - \hat{\beta}y_{2,t}) y_{2,t} = 0 \quad (5.11)$$

$$-\frac{\hat{\beta}}{1 - \hat{\beta}\hat{\gamma}} + \frac{1}{\hat{\sigma}_{22}T} \sum_{t=1}^T (y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t) y_{1,t} = 0 \quad (5.12)$$

$$\frac{1}{\hat{\sigma}_{22}T} \sum_{t=1}^T (y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t) x_t = 0 \quad (5.13)$$

$$-\frac{1}{2\hat{\sigma}_{11}} + \frac{1}{2\hat{\sigma}_{11}^2 T} \sum_{t=1}^T (y_{1,t} - \hat{\beta}y_{2,t})^2 = 0 \quad (5.14)$$

$$-\frac{1}{2\hat{\sigma}_{22}} + \frac{1}{2\hat{\sigma}_{22}^2 T} \sum_{t=1}^T (y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)^2 = 0, \quad (5.15)$$

and solving for $\hat{\theta} = \{\hat{\beta}, \hat{\gamma}, \hat{\alpha}, \hat{\sigma}_{11}, \hat{\sigma}_{22}\}$ gives the maximum likelihood estima-

tors

$$\begin{aligned}
\hat{\beta} &= \frac{\sum_{t=1}^T y_{1,t} x_t}{\sum_{t=1}^T y_{2,t} x_t} \\
\hat{\gamma} &= \frac{\sum_{t=1}^T y_{2,t} \hat{u}_{1,t} \sum_{t=1}^T x_t^2 - \sum_{t=1}^T x_t \hat{u}_{1,t} \sum_{t=1}^T y_{2,t} x_t}{\sum_{t=1}^T y_{1,t} \hat{u}_{1,t} \sum_{t=1}^T x_t^2 - \sum_{t=1}^T x_t \hat{u}_{1,t} \sum_{t=1}^T y_{1,t} x_t} \\
\hat{\alpha} &= \frac{\sum_{t=1}^T y_{1,t} \hat{u}_{1,t} \sum_{t=1}^T y_{2,t} x_t - \sum_{t=1}^T y_{1,t} x_t \sum_{t=1}^T y_{2,t} \hat{u}_{1,t}}{\sum_{t=1}^T y_{1,t} \hat{u}_{1,t} \sum_{t=1}^T x_t^2 - \sum_{t=1}^T x_t \hat{u}_{1,t} \sum_{t=1}^T y_{1,t} x_t} \\
\hat{\sigma}_{11} &= \frac{1}{T} \sum_{t=1}^T (y_{1,t} - \hat{\beta} y_{2,t})^2 \\
\hat{\sigma}_{22} &= \frac{1}{T} \sum_{t=1}^T (y_{2,t} - \hat{\gamma} y_{1,t} - \hat{\alpha} x_t)^2.
\end{aligned}$$

Full details of the derivation of these equations are given in Appendix C. Note that $\hat{\sigma}_{11}$ and $\hat{\sigma}_{22}$ are obtained having already computed the estimators $\hat{\beta}$, $\hat{\gamma}$ and $\hat{\alpha}$. This suggests that a further simplification can be achieved by concentrating the variances and covariances of $\hat{u}_t$ out of the log-likelihood function, by defining

$$\begin{aligned}
\hat{u}_{1,t} &= y_{1,t} - \hat{\beta} y_{2,t} \\
\hat{u}_{2,t} &= y_{2,t} - \hat{\gamma} y_{1,t} - \hat{\alpha} x_t,
\end{aligned}$$

and then maximizing $\ln L_T(\theta)$ with respect to $\hat{\beta}$, $\hat{\gamma}$, and $\hat{\alpha}$ where

$$\hat{V} = \frac{1}{T} \begin{bmatrix} \sum_{t=1}^T \hat{u}_{1,t}^2 & 0 \\ 0 & \sum_{t=1}^T \hat{u}_{2,t}^2 \end{bmatrix}.$$

□

The key result from Section 5.3.1 is that an analytical solution for the maximum likelihood estimator exists for a single linear regression model. It does not necessarily follow, however, that an analytical solution always exists for systems of linear equations. While Example 5.7 is an exception, such exceptions are rare and an iterative algorithm, as discussed in Chapter 3, must usually be used to obtain the maximum likelihood estimates.

Example 5.8 FIML Estimation Based on Iteration

This example uses the simulated data with $T = 500$ given in Figure 5.1

based on the model specified in Example 5.5. The steps to estimate the parameters of this model by FIML are as follows.

Step 1: Starting values are chosen at random to be

$$\theta_{(0)} = \{\beta_1 = 0.712, \alpha_1 = 0.290, \beta_2 = 0.122, \alpha_2 = 0.198\}.$$

Step 2: Evaluate the parameter matrices at the starting values

$$B_{(0)} = \begin{bmatrix} 1 & -\beta_2 \\ -\beta_1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -0.122 \\ -0.712 & 1 \end{bmatrix}$$

$$A_{(0)} = \begin{bmatrix} -\alpha_1 & 0 \\ 0 & -\alpha_2 \end{bmatrix} = \begin{bmatrix} -0.290 & 0.000 \\ 0.000 & -0.198 \end{bmatrix}.$$

Step 3: Compute the residuals at the starting values

$$\hat{u}_{1,t} = y_{1,t} - 0.712 y_{2,t} - 0.290 x_{1,t}$$

$$\hat{u}_{2,t} = y_{2,t} - 0.122 y_{1,t} - 0.198 x_{2,t}.$$

Step 4: Compute the residual covariance matrix at the starting estimates

$$V_{(0)} = \frac{1}{500} \begin{bmatrix} \sum_{t=1}^T \hat{u}_{1,t}^2 & \sum_{t=1}^T \hat{u}_{1,t} \hat{u}_{2,t} \\ \sum_{t=1}^T \hat{u}_{1,t} \hat{u}_{2,t} & \sum_{t=1}^T \hat{u}_{2,t}^2 \end{bmatrix} = \begin{bmatrix} 1.213 & 0.162 \\ 0.162 & 5.572 \end{bmatrix}.$$

Step 5: Compute the log-likelihood function for each observation at the starting values

$$\ln l_t(\theta) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |V_{(0)}| + \ln |B_{(0)}|$$

$$- \frac{1}{2} (y_t B_{(0)} + x_t A_{(0)}) V_{(0)}^{-1} (y_t B_{(0)} + x_t A_{(0)})'.$$

Step 6: Iterate until convergence using a gradient algorithm with the derivatives computed numerically. The residual covariance matrix is computed using the final estimates as follows

$$\hat{V} = \frac{1}{500} \begin{bmatrix} \sum_{t=1}^T \hat{u}_{1,t}^2 & \sum_{t=1}^T \hat{u}_{1,t} \hat{u}_{2,t} \\ \sum_{t=1}^T \hat{u}_{1,t} \hat{u}_{2,t} & \sum_{t=1}^T \hat{u}_{2,t}^2 \end{bmatrix} = \begin{bmatrix} 0.952 & 0.444 \\ 0.444 & 0.967 \end{bmatrix}.$$

Table 5.1

FIML estimates of the bivariate model. Standard errors are based on the Hessian.

Population	Estimate	Std error	t-stat.
$\beta_1 = 0.6$	0.592	0.027	21.920
$\alpha_1 = 0.4$	0.409	0.008	50.889
$\beta_2 = 0.2$	0.209	0.016	12.816
$\alpha_2 = -0.5$	-0.483	0.016	-30.203

The FIML estimates are given in Table 5.1 with standard errors based on the Hessian. The parameter estimates are in good agreement with their population counterparts given in Example 5.5.

□

5.3.3 Identification

The set of first-order conditions given by equations (5.11) - (5.15) is a system of five equations and five unknowns $\hat{\theta} = \{\hat{\beta}, \hat{\gamma}, \hat{\alpha}, \hat{\sigma}_{11}, \hat{\sigma}_{22}\}$. The issue as to whether there is a unique solution is commonly referred to as the identification problem. There exist two conditions for identification:

- (1) A necessary condition for identification is that there are at least as many equations as there are unknowns. This is commonly known as the order condition.
- (2) A necessary and sufficient condition for the system of equations to have a solution is that the Jacobian of this system needs to be nonsingular, which is equivalent to the Hessian or information matrix being nonsingular. This is known as the rank condition for identification.

An alternative way to understand the identification problem is to note that the structural system in (5.1) and the reduced form system in (5.2) are alternative representations of the same system of equations bound by the relationships

$$\begin{aligned} \Pi &= -AB^{-1}, \\ E[v'_t v_t] &= (B^{-1})' V B^{-1}, \end{aligned} \quad (5.16)$$

where the dimensions of the relevant parameter matrices are as follows

$$\begin{array}{lll} \text{Reduced form:} & \Pi \text{ is } (N \times K) & E[v'_t v_t] \text{ is } (N(N+1)/2) \\ \text{Structural form:} & A \text{ is } (N \times K), B \text{ is } (N \times N) & V \text{ is } (N(N+1)/2). \end{array}$$

This equivalence implies that estimation can proceed directly via the structural form to compute A , B and V directly, or indirectly via the reduced form with these parameter matrices being recovered from Π and $E[v'_t v_t]$. For this latter step to be feasible, the system of equations in (5.16) needs to have a solution.

The total number of parameters in the reduced form is $NK + N(N+1)/2$, while the structural system has at most $N^2 + NK + N(N+1)/2$ parameters. This means that there are potentially

$$(NK + N^2 + N(N+1)/2) - (NK + N(N+1)/2) = N^2,$$

more parameters in the structural form than in the reduced form. In order to obtain unique estimates of the structural parameters from the reduced form parameters, it is necessary to reduce the number of unknown structural parameters by at least N^2 . Normalization of the system, by designating $y_{i,t}$ as the dependent variable in the i^{th} equation for $i = 1, \dots, N$, imposes N restrictions leaving a further $N^2 - N$ restrictions yet to be imposed. These additional restrictions can take several forms, including zero restrictions, cross-equation restrictions and restrictions on the covariance matrix of the disturbances, V . Restrictions on the covariance matrix of the disturbances are fundamental to identification in the structural vector autoregression literature (Chapter 14).

Example 5.9 Identification in a Bivariate Simultaneous System

Consider the bivariate simultaneous system introduced in Example 5.3 and developed in Example 5.7 where the structural parameter matrices are

$$B = \begin{bmatrix} 1 & -\gamma \\ -\beta & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 0 & -\alpha \end{bmatrix}, \quad V = \begin{bmatrix} \sigma_{11} & 0 \\ 0 & \sigma_{22} \end{bmatrix}.$$

The system of equations to be solved consists of the two equations

$$\Pi = -AB^{-1} = -\begin{bmatrix} 0 & -\alpha \end{bmatrix} \begin{bmatrix} 1 & -\gamma \\ -\beta & 1 \end{bmatrix}^{-1} = \begin{bmatrix} -\frac{\alpha\beta}{\beta\gamma - 1} & -\frac{\alpha}{\beta\gamma - 1} \end{bmatrix},$$

and three unique equations obtained from the covariance restrictions

$$\begin{aligned} E[v'_t v_t] &= (B^{-1})' V B^{-1} \\ &= \begin{bmatrix} 1 & -\beta \\ -\gamma & 1 \end{bmatrix}^{-1} \begin{bmatrix} \sigma_{1,1} & 0 \\ 0 & \sigma_{2,2} \end{bmatrix} \begin{bmatrix} 1 & -\gamma \\ -\beta & 1 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} \frac{\sigma_{11} + \beta^2 \sigma_{22}}{(\beta\gamma - 1)^2} & \frac{\gamma \sigma_{11} + \beta \sigma_{22}}{(\beta\gamma - 1)^2} \\ \frac{\gamma \sigma_{11} + \beta \sigma_{22}}{(\beta\gamma - 1)^2} & \frac{\sigma_{22} + \gamma^2 \sigma_{11}}{(\beta\gamma - 1)^2} \end{bmatrix}, \end{aligned}$$

representing a system of 5 equations in 5 unknowns $\theta = \{\beta, \gamma, \alpha, \sigma_{11}, \sigma_{22}\}$. $\square$

If the number of parameters in the reduced form and the structural model are equal, the system is just identified resulting in a unique solution. If the reduced form has more parameters in than the structural model, the system is over identified. In this case, the system (5.16) has more equations than unknowns yielding non-unique solutions, unless the restrictions of the model are imposed. The system (5.16) is under identified if the number of reduced form parameters is less than the number of structural parameters. A solution of the system of first-order conditions of the log-likelihood function now does not exist. This means that the Jacobian of this system, which of course is also the Hessian of the log-likelihood function, is singular. Any attempt to estimate an under-identified model using the iterative algorithms from Chapter 3 will be characterised by a lack of convergence and an inability to compute standard errors since it is not possible to invert the Hessian or information matrix.

5.3.4 Instrumental Variables

Instrumental variables estimation is another method that is important in estimating the parameters of simultaneous systems of equations. The ordinary least squares estimator of the structural parameter β in the set of equations

$$\begin{aligned} y_{1,t} &= \beta y_{2,t} + u_{1,t} \\ y_{2,t} &= \gamma y_{1,t} + \alpha x_t + u_{2,t}, \end{aligned} \quad (5.17)$$

is

$$\hat{\beta}_{OLS} = \frac{\sum_{t=1}^T y_{1,t} y_{2,t}}{\sum_{t=1}^T y_{2,t} y_{2,t}}.$$

The ordinary least squares estimator, however, is not a consistent estimator of β because $y_{2,t}$ is not independent of the disturbance term $u_{1,t}$.

From Example 5.7, the FIML estimator of β is

$$\hat{\beta} = \frac{\sum_{t=1}^T y_{1,t} x_t}{\sum_{t=1}^T y_{2,t} x_t}, \quad (5.18)$$

which from the properties of the FIML estimator is a consistent estimator. The estimator in (5.18) is also known as an instrumental variable (IV) estimator. While the variable x_t is not included as an explanatory variable in the first structural equation in (5.17), it nonetheless is used to correct the dependence between $y_{2,t}$ and $u_{1,t}$ by acting as an instrument for $y_{2,t}$. A

quick way to see this is to multiply both sides of the structural equation by x_t and take expectations

$$E[y_{1,t}x_t] = \beta E[y_{2,t}x_t] + E[u_{1,t}x_t].$$

As x_t is exogenous in the system of equations, $E[u_{1,t}x_t] = 0$ and rearranging gives $\beta = E[y_{1,t}x_t] / E[y_{2,t}x_t]$. Replacing the expectations in this expression by the corresponding sample moments gives the instrumental variables estimator in (5.18).

The FIML estimator of all of the structural parameters of the bivariate simultaneous system derived in Example 5.7 can be interpreted in an instrumental variables context. To demonstrate this point, rearrange the first-order conditions from Example 5.7 to be

$$\begin{aligned} \sum_{t=1}^T (y_{1,t} - \hat{\beta}y_{2,t}) x_t &= 0 \\ \sum_{t=1}^T (y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t) \hat{u}_{1,t} &= 0 \\ \sum_{t=1}^T (y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t) x_t &= 0. \end{aligned} \tag{5.19}$$

The first equation shows that β is estimated by using x_t as an instrument for $y_{2,t}$. The second and third equations show that γ and α are estimated jointly by using $\hat{u}_{1,t} = y_{1,t} - \hat{\beta}y_{2,t}$ as an instrument for $y_{1,t}$, and x_t as its own instrument, where $\hat{u}_{1,t}$ is obtained as the residuals from the first instrumental variables regression. Thus, the FIML estimator is equivalent to using an instrumental variables estimator applied to each equation separately. This equivalence is explored in a numerical simulation in Exercise 7.

The discussion of the instrumental variables estimator highlights two key properties that an instrument needs to satisfy, namely, that the instruments are correlated with the variables they are instrumenting and the instruments are uncorrelated with the disturbance term. The choice of the instrument x_t in (5.18) naturally arises from having specified the full model in the first place. Moreover, the construction of the other instrument $\hat{u}_{1,t}$ also naturally arises from the first-order conditions in (5.19) to derive the FIML estimator. In many applications, however, only the single equation is specified leaving the choice of the instrument(s) x_t to the discretion of the researcher. Whilst the properties that a candidate instrument needs to satisfy in theory are transparent, whether a candidate instrument satisfies the two properties in practice is less transparent.

If the instruments are correlated with the variables they are instrumenting, the distribution of the instrumental variables (and FIML) estimators are asymptotically normal. In this example, the focus is on understanding the properties of the sampling distribution of the estimator where this requirement is not satisfied. This is known as the weak instrument problem.

Example 5.10 Weak Instruments

Consider the simple model

$$\begin{aligned} y_{1,t} &= \beta y_{2,t} + u_{1,t} \\ y_{2,t} &= \phi x_t + u_{2,t}, \end{aligned}$$

where

$$u_t \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix} \right).$$

in which $y_{1,t}$ and $y_{2,t}$ are the dependent variables and x_t is an exogenous variable. The parameter σ_{12} controls the strength of the simultaneity bias, where a value of $\sigma_{12} = 0$ would mean that an ordinary least squares regression of $y_{1,t}$ on $y_{2,t}$ results in a consistent estimator of β that is asymptotically normal. The parameter ϕ controls the strength of the instrument. A value of $\phi = 0$ means that there is no correlation between $y_{2,t}$ and x_t , in which case x_t is not a valid instrument. The weak instrument problem occurs when the value of ϕ is ‘small’ relative to σ_{22} .

Let the parameter values be $\beta = 0$, $\phi = 0.25$, $\sigma_{11} = 1$, $\sigma_{22} = 1$ and $\sigma_{12} = 0.99$. Assume further that $x_t \sim N(0, 1)$. The sampling distribution of the instrumental variables estimator, computed by Monte Carlo methods for a sample of size $T = 5$ with 10,000 replications, is given in Figure 5.2. The sampling distribution is far from being normal or centered on the true value of $\beta = 0$. In fact, the sampling distribution is bimodal with neither of the two modes being located near the true value of β . By increasing the value of ϕ , the sampling distribution of the instrumental variables estimator approaches normality with its mean located at the true value of $\beta = 0$. $\square$

A necessary condition for instrumental variable estimation is that there are at least as many instruments, K , as variables requiring to be instrumented, M . From the discussion of the identification problem in Section 5.3.3, the model is just identified when $K = M$, is over identified when $K > M$ and is under identified when $K < M$. Letting X represent a $(T \times K)$ matrix containing the K instruments, Y_1 a $(T \times M)$ matrix of dependent variables and Y_2 represents a $(T \times M)$ matrix containing the M variables to be instrumented. In matrix notation, the instrumental variables estimator

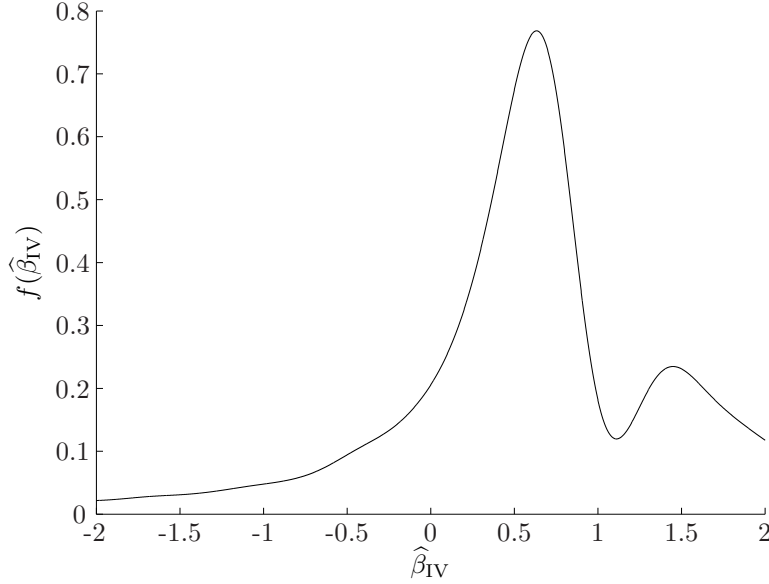


Figure 5.2 Sampling distribution of the instrumental variables estimator in the presence of a weak instrument. The distribution is approximated using a kernel estimate of density based on a Gaussian kernel with bandwidth $h = 0.07$.

of a single equation is

$$\hat{\theta}_{IV} = (Y_2'X(X'X)^{-1}X'Y_2)^{-1}(Y_2'X(X'X)^{-1}X'Y_1). \quad (5.20)$$

The covariance matrix of the instrumental variable estimator is

$$\hat{\Omega}_{IV} = \hat{\sigma}^2(Y_2'X(X'X)^{-1}X'Y_1)^{-1}, \quad (5.21)$$

where $\hat{\sigma}^2$ is the residual variance. For the case of a just identified model, $M = K$, and the instrumental variable estimator reduces to

$$\hat{\theta}_{IV} = (X'Y_2)^{-1}X'Y_1, \quad (5.22)$$

which is the multiple regression version of (5.18) expressed in matrix notation.

Example 5.11 Modelling Contagion

Favero and Giavazzi (2002) propose the following bivariate model to test for contagion

$$\begin{aligned} r_{1,t} &= \alpha_{1,2}r_{2,t} + \theta_1r_{1,t-1} + \gamma_{1,1}d_{1,t} + \gamma_{1,2}d_{2,t} + u_{1,t} \\ r_{2,t} &= \alpha_{2,1}r_{1,t} + \theta_2r_{2,t-1} + \gamma_{2,1}d_{1,t} + \gamma_{2,2}d_{2,t} + u_{2,t}, \end{aligned}$$

where $r_{1,t}$ and $r_{2,t}$ are the returns in two asset markets and $d_{1,t}$ and $d_{2,t}$ are dummy variables representing an outlier in the returns of the i^{th} asset. A test of contagion from asset market 2 to 1 is given by the null hypothesis $\gamma_{1,2} = 0$. As each equation includes an endogenous explanatory variable the model is estimated by FIML. FIML is equivalent to instrumental variables with instruments $r_{1,t-1}$ and $r_{2,t-1}$ because the model is just identified. However, the autocorrelation in returns is likely to be small and potentially zero from an efficient-markets point of view, resulting in weak instrument problems. $\square$

5.3.5 Seemingly Unrelated Regression

An important special case of the simultaneous equations model is the seemingly unrelated regression model (SUR) where each dependent variable only occurs in one equation, so that the structural coefficient matrix B in equation (5.1) is an $(N \times N)$ identity matrix.

Example 5.12 Trivariate SUR Model

An example of a trivariate SUR model is

$$\begin{aligned} y_{1,t} &= \alpha_1 x_{1,t} + u_{1,t} \\ y_{2,t} &= \alpha_2 x_{2,t} + u_{2,t} \\ y_{3,t} &= \alpha_3 x_{3,t} + u_{3,t}, \end{aligned}$$

where the disturbance term $u_t = [u_{1,t} \ u_{2,t} \ u_{3,t}]$ has the properties

$$u_t \sim iid N \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{1,1} & \sigma_{2,1} & \sigma_{3,1} \\ \sigma_{2,1} & \sigma_{2,2} & \sigma_{2,3} \\ \sigma_{3,1} & \sigma_{3,2} & \sigma_{3,3} \end{bmatrix} \right).$$

In matrix notation, this system is written as

$$y_t + x_t A = u_t,$$

where $y_t = [y_{1,t} \ y_{2,t} \ y_{3,t}]$ and $x_t = [x_{1,t} \ x_{2,t} \ x_{3,t}]$ and A is a diagonal matrix

$$A = \begin{bmatrix} -\alpha_1 & 0 & 0 \\ 0 & -\alpha_2 & 0 \\ 0 & 0 & -\alpha_3 \end{bmatrix}.$$

The log-likelihood function is

$$\ln L_T(\theta) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |V| - \frac{1}{2T} \sum_{t=1}^T (y_t + x_t A)' V^{-1} (y_t + x_t A),$$

where $N = 3$. This expression is maximized by differentiating $\ln L_T(\theta)$ with respect to the vector of parameters $\theta = \{\alpha_1, \alpha_2, \alpha_3, \sigma_{1,1}, \sigma_{2,1}, \sigma_{2,2}, \sigma_{3,1}, \sigma_{3,2}, \sigma_{3,3}\}$ and setting these derivatives to zero to find $\hat{\theta}$. $\square$

Example 5.13 Equivalence of SUR and OLS Estimates

Consider the class of SUR models where the independent variables are the same in each equation. An example is

$$y_{i,t} = \alpha_i x_t + u_{i,t},$$

where $u_t = (u_{1,t}, u_{2,t}, \dots, u_{N,t}) \sim N(0, V)$. For this model, $A = [-\alpha_1 - \alpha_2 \dots - \alpha_N]$ and estimation of the model by maximum likelihood yields the same estimates as ordinary least squares applied to each equation individually. $\square$

5.4 Testing

The three tests developed in Chapter 4, namely the likelihood ratio (LR), Wald (W) and Lagrange Multiplier (LM) statistics are now applied to testing the parameters of single and multiple equation linear regression models. Depending on the choice of covariance matrix, various asymptotically equivalent forms of the test statistics are available (see Chapter 4).

Example 5.14 Testing a Single Equation Model

Consider the regression model

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + u_t \quad u_t \sim iid N(0, 4),$$

where $\theta = \{\beta_0 = 1.0, \beta_1 = 0.7, \beta_2 = 0.3, \sigma^2 = 4\}$ and $x_{1,t}$ and $x_{2,t}$ are generated as $N(0, 1)$. The model is simulated with a sample of size $T = 200$ and maximum likelihood estimates of the parameters are reported in Example 5.6. Now consider testing the hypotheses

$$H_0 : \beta_1 + \beta_2 = 1, \quad H_0 : \beta_1 + \beta_2 \neq 1.$$

The unrestricted and restricted maximum likelihood parameter estimates are given in Table 5.2.

The restricted parameter estimates are obtained by imposing the restriction $\beta_1 + \beta_2 = 1$, by writing the model as

$$y_t = \beta_0 + \beta_1 x_{1,t} + (1 - \beta_1) x_{2,t} + u_t.$$

The LR statistic is computed as

$$LR = -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)) = -2 \times (-419.052 + 418.912) = 0.279,$$

Table 5.2

Unrestricted and restricted parameter estimates
of the single equation regression model.

Parameter	Unrestricted	Restricted
β_0	1.129	1.129
β_1	0.719	0.673
β_2	0.389	0.327
σ^2	3.862	3.868
$\ln L_T(\theta)$	-2.0946	-2.0953

which is distributed asymptotically as χ_1^2 under H_0 . The p-value is 0.597 showing that the restriction is not rejected at the 5% level.

Based on the assumption of a normal distribution for the disturbance term, an alternative form for the LR statistic for a single equation model is

$$LR = T(\ln \hat{\sigma}_0^2 - \ln \hat{\sigma}_1^2).$$

The alternative form of this statistic yields the same value:

$$LR = T(\ln \hat{\sigma}_0^2 - \ln \hat{\sigma}_1^2) = 200 \times (\ln 3.876 - \ln 3.8622) = 0.279.$$

To compute the Wald statistic, define

$$R = \begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 \end{bmatrix},$$

and compute the negative Hessian matrix

$$-H_T(\hat{\theta}_1) = \begin{bmatrix} 0.259 & -0.016 & 0.014 & 0.000 \\ -0.016 & 0.285 & -0.007 & 0.000 \\ 0.014 & -0.007 & 0.214 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.034 \end{bmatrix}.$$

The Wald statistic is then

$$W = T[R\hat{\theta}_1 - Q]'[R(-H_T^{-1}(\hat{\theta}_1))R']^{-1}[R\hat{\theta}_1 - Q] = 0.279,$$

which is distributed asymptotically as χ_1^2 under H_0 . The p-value is 0.597 showing that the restriction is not rejected at the 5% level.

The LM statistic requires evaluating the gradients of the unrestricted model at the restricted estimates

$$G'_T(\hat{\theta}_0) = \begin{bmatrix} 0.000 & 0.013 & 0.013 & 0.000 \end{bmatrix},$$

and computing the inverse of the outer product of gradients matrix evaluated at $\hat{\theta}_0$

$$J_T^{-1}(\hat{\theta}_0) = \begin{bmatrix} 3.967 & -0.122 & 0.570 & -0.934 \\ -0.122 & 4.158 & 0.959 & -2.543 \\ 0.570 & 0.959 & 5.963 & -1.260 \\ -0.934 & -2.543 & -1.260 & 28.171 \end{bmatrix}.$$

Using these terms in the LM statistic gives

$$LM = TG'_T(\hat{\theta}_0)J_T^{-1}(\hat{\theta}_0)G_T(\hat{\theta}_0) = 0.399,$$

which is distributed asymptotically as χ_1^2 under H_0 . The p-value is 0.528 showing that the restriction is still not rejected at the 5% level. $\square$

The form of the LR, Wald and LM test statistics in the case of multiple equation regression models is the same as it is for single equation regression models. Once again an alternative form of the LR statistic is available as a result of the assumption of normality. Recall from equation (5.10) that the log-likelihood function for a multiple equation model is

$$\ln L_T(\theta) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |V| + \ln |B| - \frac{1}{2T} \sum_{t=1}^T (y_t B + x_t A) V^{-1} (y_t B + x_t A)'.$$

The unrestricted maximum likelihood estimator of V is

$$\hat{V}_1 = \frac{1}{T} \sum_{t=1}^T \hat{u}_t' \hat{u}_t, \quad \hat{u}_t = y_t \hat{B}_1 + x_t \hat{A}_1.$$

The log-likelihood function evaluated at the unrestricted estimator is

$$\begin{aligned} \ln L_T(\hat{\theta}_1) &= -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |\hat{V}_1| + \ln |\hat{B}_1| \\ &\quad - \frac{1}{2T} \sum_{t=1}^T (y_t \hat{B}_1 + x_t \hat{A}_1) \hat{V}_1^{-1} (y_t \hat{B}_1 + x_t \hat{A}_1)' \\ &= -\frac{N}{2} (1 + \ln 2\pi) - \frac{1}{2} \ln |\hat{V}_1| + \ln |\hat{B}_1|, \end{aligned}$$

which uses the result from Chapter 4 that

$$\frac{1}{T} \sum_{t=1}^T \hat{u}_t \hat{V}_1^{-1} \hat{u}_t' = N.$$

Similarly, the log-likelihood function evaluated at the restricted estimator is

$$\begin{aligned}\ln L_T(\hat{\theta}_0) &= -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |\hat{V}_0| + \ln |\hat{B}_0| \\ &\quad - \frac{1}{2T} \sum_{t=1}^T (y_t \hat{B}_0 + x_t \hat{A}_0) \hat{V}_0^{-1} (y_t \hat{B}_0 + x_t \hat{A}_0)' \\ &= -\frac{N}{2} (1 + \ln 2\pi) - \frac{1}{2} \ln |\hat{V}_0| + \ln |\hat{B}_0|,\end{aligned}$$

where

$$\hat{V}_0 = \frac{1}{T} \sum_{t=1}^T v_t' v_t, \quad v_t = y_t \hat{B}_0 + x_t \hat{A}_0.$$

The LR statistic is

$$LR = -2[\ln L_T(\hat{\theta}_0) - \ln L_T(\hat{\theta}_1)] = T(\ln |\hat{V}_0| - \ln |\hat{V}_1|) - 2T(\ln |\hat{B}_0| - \ln |\hat{B}_1|).$$

In the special case of the SUR model where $B = I_N$, the LR statistic is

$$LR = T(\ln |\hat{V}_0| - \ln |\hat{V}_1|),$$

which is the alternative form given in Chapter 4.

Example 5.15 Testing a Multiple Equation Model

Consider the model

$$\begin{aligned}y_{1,t} &= \beta_1 y_{2,t} + \alpha_1 x_{1,t} + u_{1,t} \\ y_{2,t} &= \beta_2 y_{1,t} + \alpha_2 x_{2,t} + u_{2,t}, \\ u_t &\sim iid N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, V = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix} \right),\end{aligned}$$

in which the hypotheses

$$H_0 : \alpha_1 + \alpha_2 = 0, \quad H_0 : \alpha_1 + \alpha_2 \neq 0,$$

are to be tested. The unrestricted and restricted maximum likelihood parameter estimates are given in Table 5.3. The restricted parameter estimates are obtained by imposing the restriction $\alpha_2 = -\alpha_1$, by writing the model as

$$\begin{aligned}y_{1,t} &= \beta_1 y_{2,t} + \alpha_1 x_{1,t} + u_{1,t} \\ y_{2,t} &= \beta_2 y_{1,t} - \alpha_1 x_{2,t} + u_{2,t}.\end{aligned}$$

The LR statistic is computed as

$$LR = -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)) = -2 \times (-1410.874 + 1403.933) = 13.88,$$

which is distributed asymptotically as χ_1^2 under H_0 . The p-value is 0.000

Table 5.3

Unrestricted and restricted parameter estimates
of the multiple equation regression model.

Parameter	Unrestricted	Restricted
β_1	0.592	0.533
α_1	0.409	0.429
β_2	0.209	0.233
α_2	-0.483	-0.429
$\hat{\sigma}_{11}$	0.952	1.060
$\hat{\sigma}_{12}$	0.444	0.498
$\hat{\sigma}_{22}$	0.967	0.934
$\ln L_T(\theta)$	-2.8079	-2.8217

showing that the restriction is rejected at the 5% level. The alternative form of this statistic gives

$$\begin{aligned}
 LR &= T(\ln |\hat{V}_0| - \ln |\hat{V}_1|) - 2T(\ln |\hat{B}_0| - \ln |\hat{B}_1|) \\
 &= 500 \left(\ln \begin{vmatrix} 1.060 & 0.498 \\ 0.498 & 0.934 \end{vmatrix} - \ln \begin{vmatrix} 0.952 & 0.444 \\ 0.444 & 0.967 \end{vmatrix} \right) \\
 &\quad - 2 \times 500 \left(\ln \begin{vmatrix} 1.000 & -0.233 \\ -0.533 & 1.000 \end{vmatrix} - \ln \begin{vmatrix} 1.000 & -0.209 \\ -0.592 & 1.000 \end{vmatrix} \right) \\
 &= 13.88.
 \end{aligned}$$

To compute the Wald statistic, define

$$R = \begin{bmatrix} 0 & 1 & 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 \end{bmatrix},$$

and compute the negative Hessian matrix

$$-H_T(\hat{\theta}_1) = \begin{bmatrix} 3.944 & 4.513 & -1.921 & 2.921 \\ 4.513 & 44.620 & -9.613 & 0.133 \\ -1.921 & -9.613 & 10.853 & -3.823 \\ 2.921 & 0.133 & -3.823 & 11.305 \end{bmatrix},$$

where $\hat{\theta}_1$ corresponds to the concentrated parameter vector. The Wald statistic is

$$W = T[R\hat{\theta}_1 - Q]'[R(-H_T^{-1}(\hat{\theta}_1))R']^{-1}[R\hat{\theta}_1 - Q] = 13.895,$$

which is distributed asymptotically as χ_1^2 under H_0 . The p-value is 0.000 showing that the restriction is rejected at the 5% level.

The LM statistic requires evaluating the gradients of the unrestricted model at the restricted estimates

$$G'_T(\hat{\theta}_0) = [\ 0.000 \quad -0.370 \quad 0.000 \quad -0.370 \],$$

and computing the inverse of the outer product of gradients matrix evaluated at $\hat{\theta}_0$

$$J_T^{-1}(\hat{\theta}_0) = \begin{bmatrix} 0.493 & -0.071 & -0.007 & -0.133 \\ -0.071 & 0.042 & 0.034 & 0.025 \\ -0.007 & 0.034 & 0.123 & 0.040 \\ -0.133 & 0.025 & 0.040 & 0.131 \end{bmatrix}.$$

Using these terms in the LM statistic gives

$$LM = TG'_T(\hat{\theta}_0)J_T^{-1}(\hat{\theta}_0)G_T(\hat{\theta}_0) = 15.325,$$

which is distributed asymptotically as χ_1^2 under H_0 . The p-value is 0.000 showing that the restriction is rejected at the 5% level. $\square$

5.5 Applications

To highlight the details of estimation and testing in linear regression models two applications are now presented. The first involves estimating a static version of the Taylor rule for the conduct of monetary policy using U.S. macroeconomic data. The second estimates the well-known Klein macroeconomic model for the U.S.

5.5.1 Linear Taylor Rule

In a seminal paper, Taylor (1993) suggests that the monetary authorities follow a simple rule for setting monetary policy. The rule requires policy-makers to adjust the quarterly average of the money market interest rate (Federal Funds Rate), i_t , in response to four-quarter inflation, π_t , and the gap between output and its long-run potential level, y_t , according to

$$i_t = \beta_0 + \beta_1 \pi_t + \beta_2 y_t + u_t, \quad u_t \sim N(0, \sigma^2).$$

Taylor suggested values of $\beta_1 = 1.5$ and $\beta_2 = 0.5$. This static linear version of the so-called Taylor rule is a linear regression model with two independent variables of the form discussed in detail in Section 5.3.

The parameters of the model are estimated using data from the U.S. for the period 1987:Q1 to 1999:Q4, a total of $T = 52$ observations. The variables

are defined in Rudebusch (2002, p1164) in his study of the Taylor rule, with π_t and y_t computed as

$$\pi_t = 400 \times \sum_{j=0}^3 (\log p_{t-j} - \log p_{t-j-1}), \quad y_t = 100 \times ((q_t - q_t^*)/q_t),$$

and where p_t is the U.S. GDP deflator, q_t is real U.S. GDP and q_t^* is real potential GDP as estimated by the Congressional Budget Office. The data are plotted in Figure 5.3.

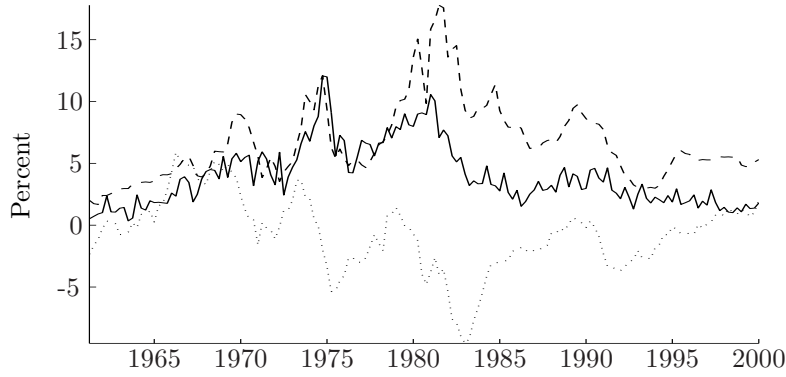


Figure 5.3 U.S. data on the Federal Funds Rate (dashed line), the inflation gap (solid line) and the output gap (dotted line) as defined by Rudebusch (2002, p1164).

The log-likelihood function is

$$\ln L_T(\theta) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2 T} \sum_{t=1}^T (i_t - \beta_0 - \beta_1 \pi_t - \beta_2 y_t)^2,$$

with $\theta = \{\beta_0, \beta_1, \beta_2, \sigma^2\}$. In this particular case, the first-order conditions are solved to yield closed-form solutions for the maximum likelihood estimators that are also the ordinary least squares estimators. The maximum likelihood estimates are

$$\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} 53.000 & 132.92 & -40.790 \\ 132.92 & 386.48 & -123.79 \\ -40.790 & -123.79 & 147.77 \end{bmatrix}^{-1} \begin{bmatrix} 305.84 \\ 822.97 \\ -192.15 \end{bmatrix} = \begin{bmatrix} 2.98 \\ 1.30 \\ 0.61 \end{bmatrix}.$$

Once $[\hat{\beta}_0 \ \hat{\beta}_1 \ \hat{\beta}_2]'$ is computed, the ordinary least squares estimate of the

variance, $\hat{\sigma}^2$, is obtained from

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (i_t - 2.98 - 1.30\pi_t - 0.61y_t)^2 = 1.1136.$$

The covariance matrix of $\hat{\theta} = \{\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2\}$ is

$$\frac{1}{T} \hat{\Omega} = \begin{bmatrix} 0.1535 & -0.0536 & -0.0025 \\ -0.0536 & 0.0227 & 0.0042 \\ -0.0025 & 0.0042 & 0.0103 \end{bmatrix}.$$

The estimated monetary policy response coefficients, namely, $\hat{\beta}_1 = 1.30$ for inflation and $\hat{\beta}_2 = 0.61$ for the response to the output gap, are not dissimilar to the suggested values of 1.5 and 0.5, respectively. A Wald test of the restrictions $\beta_1 = 1.50$ and $\beta_2 = 0.5$ yields a test statistic of 4.062. From the χ^2_2 distribution, the p-value of this statistic is 0.131 showing that the restrictions cannot be rejected at conventional significance levels.

5.5.2 The Klein Model of the U.S. Economy

One of the first macroeconomic models constructed for the U.S. is the Klein (1950) model, which consists of three structural equations and three identities

$$\begin{aligned} C_t &= \alpha_0 + \alpha_1 P_t + \alpha_2 P_{t-1} + \alpha_3 (PW_t + GW_t) + u_{1,t} \\ I_t &= \beta_0 + \beta_1 P_t + \beta_2 P_{t-1} + \beta_3 K_{t-1} + u_{2,t} \\ PW_t &= \gamma_0 + \gamma_1 D_t + \gamma_2 D_{t-1} + \gamma_3 TREND_t + u_{3,t} \\ D_t &= C_t + I_t + G_t \\ P_t &= D_t - TAX_t - PW_t \\ K_t &= K_{t-1} + I_t, \end{aligned}$$

where the key variables are defined as

C_t	=	Consumption
P_t	=	Profits
PW_t	=	Private wages
GW_t	=	Government wages
I_t	=	Investment
K_t	=	Capital stock
D_t	=	Aggregate demand
G_t	=	Government spending
TAX_t	=	Indirect taxes plus net exports
$TREND_t$	=	Time trend, base in 1931 .

The first equation is a consumption function, the second equation is an investment function and the third equation is a labor demand equation. The last three expressions are identities for aggregate demand, private profits and the capital stock, respectively. The variables are classified as

Endogenous	:	$C_t, I_t, PW_t, D_t, P_t, K_t$
Exogenous	:	$CONST, G_t, TAX_t, GW_t, TREND,$
Predetermined	:	$P_{t-1}, D_{t-1}, K_{t-1} .$

To estimate the Klein model by FIML, it is necessary to use the three identities to write the model as a three-equation system just containing the three endogenous variables. Formally, this requires using the identities to substitute P_t and D_t out of the three structural equations. This is done by combining the first two identities to derive an expression for P_t

$$P_t = D_t - TAX_t - PW_t = C_t + I_t + G_t - TAX_t - PW_t ,$$

while an expression for D_t is given directly from the first identity. Notice that the third identity, the capital stock accumulation equation, does not need to be used as K_t does not appear in any of the three structural equations. Substituting the expressions for P_t and D_t into the three structural equations gives

$$\begin{aligned}
C_t &= \alpha_0 + \alpha_1(C_t + I_t + G_t - TAX_t - PW_t) \\
&\quad + \alpha_2 P_{t-1} + \alpha_3(PW_t + GW_t) + u_{1,t} \\
I_t &= \beta_0 + \beta_1(C_t + I_t + G_t - TAX_t - PW_t) \\
&\quad + \beta_2 P_{t-1} + \beta_3 K_{t-1} + u_{2,t} \\
PW_t &= \gamma_0 + \gamma_1(C_t + I_t + G_t) + \gamma_2 D_{t-1} + \gamma_3 TREND_t + u_{3,t} .
\end{aligned}$$

This is now a system of three equations and three endogenous variables (C_t, I_t, PW_t), which can be estimated by FIML. Let

$$\begin{aligned}
 y_t &= [C_t \quad I_t \quad PW_t] \\
 x_t &= [CONST \quad G_t \quad TAX_t \quad GW_t \quad TREND_t \quad P_{t-1} \quad D_{t-1} \quad K_{t-1}] \\
 u_t &= [u_{1,t} \quad u_{2,t} \quad u_{3,t}] \\
 B &= \begin{bmatrix} 1 - \alpha_1 & -\beta_1 & -\gamma_1 \\ -\alpha_1 & 1 - \beta_1 & -\gamma_1 \\ \alpha_1 - \alpha_2 & \beta_1 & 1 \end{bmatrix} \\
 A &= \begin{bmatrix} -\alpha_0 & -\beta_0 & -\gamma_0 \\ -\alpha_1 & -\beta_1 & -\gamma_1 \\ \alpha_1 & \beta_1 & 0 \\ -\alpha_2 & 0 & 0 \\ 0 & 0 & -\gamma_3 \\ -\alpha_3 & -\beta_2 & 0 \\ 0 & 0 & -\gamma_2 \\ 0 & -\beta_3 & 0 \end{bmatrix},
 \end{aligned}$$

then, from (5.1), the system is written as

$$y_t B + x_t A = u_t.$$

The Klein macroeconomic model is estimated over the period 1920 to 1941 using U.S. annual data. As the system contains one lag the effective sample begins in 1921, resulting in a sample of size $T = 21$. The FIML parameter estimates are contained in the last column of Table 5.4. The value of the log-likelihood function is $\ln L_T(\hat{\theta}) = -85.370$. For comparison the ordinary least squares and instrumental variables estimates are also given. The instrumental variables estimates are computed using the 8 variables given in x_t as the instrument set for each equation. Noticeable differences in the magnitudes of the parameter estimates are evident in some cases, particularly in the second equation $\{\beta_0, \beta_1, \beta_2, \beta_3\}$. In this instance, the IV estimates appear to be closer to the FIML estimates than to the ordinary least squares estimates, indicating potential simultaneity problems with the ordinary least squares approach.

5.6 Exercises

(1) Simulating a Simultaneous System

Table 5.4

Parameter estimates of the Klein macroeconomic model for the U.S., 1921 to 1941.

Parameter	OLS	IV	FIML
α_0	16.237	16.555	16.461
α_1	0.193	0.017	0.177
α_2	0.090	0.216	0.210
α_3	0.796	0.810	0.728
β_0	10.126	20.278	24.130
β_1	0.480	0.150	0.007
β_2	0.333	0.616	0.670
β_3	-0.112	-0.158	-0.172
γ_0	1.497	1.500	1.028
γ_1	0.439	0.439	0.317
γ_2	0.146	0.147	0.253
γ_3	0.130	0.130	0.096

Gauss file(s)	<code>linear_simulation.g</code>
Matlab file(s)	<code>linear_simulation.m</code>

Consider the bivariate model

$$y_{1,t} = \beta_1 y_{2,t} + \alpha_1 x_{1,t} + u_{1,t}$$

$$y_{2,t} = \beta_2 y_{1,t} + \alpha_2 x_{2,t} + u_{2,t},$$

where $y_{1,t}$ and $y_{2,t}$ are the dependent variables, $x_{1,t} \sim N(0, 100)$ and $x_{2,t} \sim N(0, 9)$ are the independent variables, $u_{1,t}$ and $u_{2,t}$ are normally distributed disturbance terms with zero means and covariance matrix

$$V = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix},$$

and $\beta_1 = 0.6$, $\alpha_1 = 0.4$, $\beta_2 = 0.2$ and $\alpha_2 = -0.5$.

- Construct A , B and hence compute $\Pi = -AB^{-1}$.
- Simulate the model for $T = 500$ observations and plot the simulated series of $y_{1,t}$ and $y_{2,t}$.

(2) ML Estimation of a Regression Model

Gauss file(s)	<code>linear_estimate.g</code>
Matlab file(s)	<code>linear_estimate.m</code>

Simulate the model for a sample of size $T = 200$

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + u_t$$

$$u_t \sim N(0, 4),$$

where $\beta_0 = 1.0$, $\beta_1 = 0.7$, $\beta_2 = 0.3$, $\sigma^2 = 4$ and $x_{1,t}$ and $x_{2,t}$ are generated as $N(0, 1)$.

- (a) Compute the maximum likelihood parameter estimates using the Newton-Raphson algorithm, without concentrating the log-likelihood function.
- (b) Compute the maximum likelihood parameter estimates using the Newton-Raphson algorithm, by concentrating the log-likelihood function.
- (c) Compute the parameter estimates by ordinary least squares.
- (d) Compare the estimates obtained in parts (a) to (c).
- (e) Compute the covariance matrix of the parameter estimates in parts (a) to (c) and compare the results.

(3) Testing a Single Equation Model

Gauss file(s)	<code>linear_lr.g</code> , <code>linear_w.g</code> , <code>linear_lm.g</code>
Matlab file(s)	<code>linear_lr.m</code> , <code>linear_w.m</code> , <code>linear_lm.m</code>

This exercise is an extension of Exercise 2. Test the hypotheses

$$H_0 : \beta_1 + \beta_2 = 1 \quad H_1 : \beta_1 + \beta_2 \neq 1.$$

- (a) Perform a LR test of the hypotheses.
- (b) Perform a Wald test of the hypotheses.
- (c) Perform a LM test of the hypotheses.

(4) FIML Estimation of a Structural Model

Gauss file(s)	<code>linear_fiml.g</code>
Matlab file(s)	<code>linear_fiml.m</code>

This exercise uses the simulated data generated in Exercise 1.

- (a) Estimate the parameters of the structural model

$$y_{1,t} = \beta_1 y_{2,t} + \alpha_1 x_{1,t} + u_{1,t}$$

$$y_{2,t} = \beta_2 y_{1,t} + \alpha_2 x_{2,t} + u_{2,t},$$

by FIML using an iterative algorithm with the starting estimates taken as draws from a uniform distribution.

- (b) Repeat part (a) by choosing the starting estimates as draws from a normal distribution. Compare the final estimates with the estimates obtained in part (a).
- (c) Re-estimate the model's parameters using an IV estimator and compare these estimates with the FIML estimates obtained in parts (a) and (b).

(5) **Weak Instruments**

Gauss file(s)	<code>linear_weak.g</code>
Matlab file(s)	<code>linear_weak.m</code>

This exercise extends the results on weak instruments in Example 5.10. Consider the model

$$\begin{aligned} y_{1,t} &= \beta y_{2,t} + u_{1,t} \\ y_{2,t} &= \phi x_t + u_{2,t}, \end{aligned} \quad u_t \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1.00 & 0.99 \\ 0.99 & 1.00 \end{bmatrix} \right),$$

where $y_{1,t}$ and $y_{2,t}$ are dependent variables, $x_t \sim U(0, 1)$ is the exogenous variable and the parameter values are $\beta = 0$, $\phi = 0.5$. The sample size is $T = 5$ and 10,000 replications are used to generate the sampling distribution of the estimator.

- (a) Generate the sampling distribution of the IV estimator and discuss its properties.
- (b) Repeat part (a) except choose $\phi = 1$. Compare the sampling distribution of the IV estimator to the distribution obtained in part (a).
- (c) Repeat part (a) except choose $\phi = 10$. Compare the sampling distribution of the IV estimator to the distribution obtained in part (a).
- (d) Repeat part (a) except choose $\phi = 0$. Compare the sampling distribution of the IV estimator to the distribution obtained in part (a). Also compute the sampling distribution of the ordinary least squares estimator for this case. Note that for this model the ordinary least squares estimator has the property (see Stock, Wright and Yogo, 2002)

$$\text{plim}(\hat{\beta}_{OLS}) = \frac{\sigma_{12}}{\sigma_{22}} = 0.99.$$

- (e) Repeat parts (a) to (d) for a larger sample of $T = 50$ and a very large sample of $T = 500$. Are the results in parts (a) to (d) affected by asymptotic arguments?

(6) Testing a Multiple Equation Model

Gauss file(s)	linear_fiml_lr.g, linear_fiml_wd.g, linear_fiml_lm.g
Matlab file(s)	linear_fiml_lr.m, linear_fiml_wd.m, linear_fiml_lm.m

This exercise is an extension of Exercise 4. Test the hypotheses

$$H_0 : \alpha_1 + \alpha_2 = 0 \quad H_1 : \alpha_1 + \alpha_2 \neq 0.$$

- (a) Perform a LR test of the hypotheses.
- (b) Perform a Wald test of the hypotheses.
- (c) Perform a LM test of the hypotheses.

(7) Relationship Between FIML and IV

Gauss file(s)	linear_iv.g
Matlab file(s)	linear_iv.m

Simulate the following structural model for $T = 500$ observations

$$\begin{aligned} y_{1,t} &= \beta y_{2,t} + u_{1,t} \\ y_{2,t} &= \gamma y_{1,t} + \alpha x_t + u_{2,t}, \end{aligned}$$

where $y_{1,t}$ and $y_{2,t}$ are the dependent variables, $x_t \sim N(0, 100)$ is the independent variable, $u_{1,t}$ and $u_{2,t}$ are normally distributed disturbance terms with zero means and covariance matrix

$$V = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix} = \begin{bmatrix} 2.0 & 0.0 \\ 0.0 & 1.0 \end{bmatrix},$$

and the parameters are set at $\beta = 0.6$, $\gamma = 0.4$ and $\alpha = -0.5$.

- (a) Compute the FIML estimates of the model's parameters using an iterative algorithm with the starting estimates taken as draws from a uniform distribution.
- (b) Recompute the FIML estimates using the analytical expressions given in equation (5.16). Compare these estimates with the estimates obtained in part (a).
- (c) Re-estimate the model's parameters using an IV estimator and compare these estimates with the FIML estimates in parts (a) and (b).

(8) Recursive Structural Models

Gauss file(s)	linear_recursive.g
Matlab file(s)	linear_recursive.m

Simulate the trivariate structural model for $T = 200$ observations

$$\begin{aligned} y_{1,t} &= \alpha_1 x_{1,t} + u_{1,t} \\ y_{2,t} &= \beta_1 y_{1,t} + \alpha_2 x_{1,t} + u_{2,t} \\ y_{3,t} &= \beta_2 y_{1,t} + \beta_3 y_{2,t} + \alpha_3 x_{1,t} + u_{3,t}, \end{aligned}$$

where $\{x_{1,t}, x_{2,t}, x_{3,t}\}$ are normal random variables with zero means and respective standard deviations of $\{1, 2, 3\}$. The parameters are $\beta_1 = 0.6$, $\beta_2 = 0.2$, $\beta_3 = 1.0$, $\alpha_1 = 0.4$, $\alpha_2 = -0.5$ and $\alpha_3 = 0.2$. The disturbance vector $u_t = (u_{1,t}, u_{2,t}, u_{3,t})$ is normally distributed with zero means and covariance matrix

$$V = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

- (a) Estimate the model by maximum likelihood and compare the parameter estimates with the population parameter values.
- (b) Estimate each equation by ordinary least squares and compare the parameter estimates to the maximum likelihood estimates. Briefly discuss why the two sets of estimates are the same.

(9) Seemingly Unrelated Regression

Gauss file(s)	<code>linear_sur.g</code>
Matlab file(s)	<code>linear_sur.m</code>

Simulate the following trivariate SUR model for $T = 500$ observations

$$y_{i,t} = \alpha_i x_{i,t} + u_{i,t}, \quad i = 1, 2, 3,$$

where $\{x_{1,t}, x_{2,t}, x_{3,t}\}$ are normal random variables with zero means and respective standard deviations of $\{1, 2, 3\}$. The parameters are $\alpha_1 = 0.4$, $\alpha_2 = -0.5$ and $\alpha_3 = 1.0$. The disturbance vector $u_t = (u_{1,t}, u_{2,t}, u_{3,t})$ is normally distributed with zero means and covariance matrix

$$V = \begin{bmatrix} 1.0 & 0.5 & -0.1 \\ 0.5 & 1.0 & 0.2 \\ -0.1 & 0.2 & 1.0 \end{bmatrix}.$$

- (a) Estimate the model by maximum likelihood and compare the parameter estimates with the population parameter values.
- (b) Estimate each equation by ordinary least squares and compare the parameter estimates to the maximum likelihood estimates.

- (c) Simulate the model using the following covariance matrix

$$V = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

Repeat parts (a) and (b) and comment on the results.

- (d) Simulate the model

$$y_{i,t} = \alpha_i x_{1,t} + u_{i,t}, \quad i = 1, 2, 3,$$

for $T = 500$ observations and using the original covariance matrix. Repeat parts (a) and (b) and comment on the results.

(10) Linear Taylor Rule

Gauss file(s)	<code>linear_taylor.g, taylor.dat</code>
Matlab file(s)	<code>linear_taylor.m, taylor.mat.</code>

The data are $T = 53$ quarterly observations for the U.S. on the Federal Funds Rate, i_t , the inflation gap, π_t , and the output gap, y_t .

- (a) Plot the data and hence reproduce Figure 5.3.
 (b) Estimate the static linear Taylor rule equation

$$i_t = \beta_0 + \beta_1 \pi_t + \beta_2 y_t + u_t, \quad u_t \sim N(0, \sigma^2),$$

by maximum likelihood. Compute the covariance matrix of $\hat{\beta}$.

- (c) Use a Wald test to test the restrictions $\beta_1 = 1.5$ and $\beta_2 = 0.5$.

(11) Klein's Macroeconomic Model of the U.S.

Gauss file(s)	<code>linear_klein.g, klein.dat</code>
Matlab file(s)	<code>linear_klein.m, klein.mat</code>

The data file contains contains 22 annual observations from 1920 to 1941

on the following U.S. macroeconomic variables

C_t	=	Consumption
P_t	=	Profits
PW_t	=	Private wages
GW_t	=	Government wages
I_t	=	Investment
K_t	=	Capital stock
D_t	=	Aggregate demand
G_t	=	Government spending
TAX_t	=	Indirect taxes plus net exports
$TREND_t$	=	Time trend, base in 1931

The Klein (1950) macroeconometric model of the U.S. is

$$\begin{aligned}
 C_t &= \alpha_0 + \alpha_1 P_t + \alpha_2 P_{t-1} + \alpha_3 (PW_t + GW_t) + u_{1,t} \\
 I_t &= \beta_0 + \beta_1 P_t + \beta_2 P_{t-1} + \beta_3 K_{t-1} + u_{2,t} \\
 PW_t &= \gamma_0 + \gamma_1 D_t + \gamma_2 D_{t-1} + \gamma_3 TREND_t + u_{3,t} \\
 D_t &= C_t + I_t + G_t \\
 P_t &= D_t - TAX_t - PW_t \\
 K_t &= K_{t-1} + I_t.
 \end{aligned}$$

- (a) Estimate each of the three structural equations by ordinary least squares. What is the problem with using this estimator to compute the parameter estimates of this model?
- (b) Estimate the model by IV using the following instruments for each equation

$$x_t = [CONST, G_t, TAX_t, GW_t, TREND_t, P_{t-1}, D_{t-1}, K_{t-1}] .$$

What are the advantages over ordinary least squares with using IV to compute the parameter estimates of this model?

- (c) Use the three identities to re-express the three structural equations as a system containing the three endogenous variables, C_t , I_t and PW_t , and estimate this model by FIML. What are the advantages over IV with using FIML to compute the parameter estimates of this model?
- (d) Compare the parameter estimates obtained in parts (a) to (c), and compare your parameter estimates with Table 5.4.

6

Nonlinear Regression Models

6.1 Introduction

The class of linear regression models discussed in Chapter 5 is now extended to allow for nonlinearities in the specification of the conditional mean. Nonlinearity in the specification of the mean of time series models is the subject matter of Chapter 19 while nonlinearity in the specification of the variance is left until Chapter 20. As with the treatment of linear regression models in the previous chapter, nonlinear regression models are examined within the maximum likelihood framework. Establishing this link ensures that methods typically used to estimate nonlinear regression models, including Gauss-Newton, nonlinear least squares and robust estimators, immediately inherit the same asymptotic properties as the maximum likelihood estimator. Moreover, it is also shown that many of the statistics used to test nonlinear regression models are special cases of the LR, Wald or LM tests discussed in Chapter 4. An important example of this property, investigated at the end of the chapter, is that a class of non-nested tests used to discriminate between models is shown to be a LR test.

6.2 Specification

A typical form for the nonlinear regression model is

$$g(y_t; \alpha) = \mu(x_t; \beta) + u_t, \quad u_t \sim iid N(0, \sigma^2), \quad (6.1)$$

where y_t is the dependent variable and x_t is the independent variable. The nonlinear functions $g(\cdot)$ and $\mu(\cdot)$ of y_t and x_t have parameter vectors $\alpha = \{\alpha_1, \alpha_2, \dots, \alpha_m\}$ and $\beta = \{\beta_0, \beta_1, \dots, \beta_k\}$, respectively. The unknown parameters to be estimated are given by the $(m+k+2)$ vector $\theta = \{\alpha, \beta, \sigma^2\}$.

Example 6.1 Zellner-Revankar Production Function

Consider the production function relating output, y_t , to capital, k_t , and labour, l_t , given by

$$\ln y_t + \alpha y_t = \beta_0 + \beta_1 \ln k_t + \beta_2 \ln l_t + u_t ,$$

with

$$g(y_t; \alpha) = \ln y_t + \alpha y_t , \quad \mu(x_t; \beta) = \beta_0 + \beta_1 \ln k_t + \beta_2 \ln l_t .$$

□

Example 6.2 Exponential Regression Model

Consider the nonlinear model

$$y_t = \beta_0 \exp [\beta_1 x_t] + u_t ,$$

where

$$g(y_t; \alpha) = y_t , \quad \mu(x_t; \beta) = \beta_0 \exp [\beta_1 x_t] .$$

□

Examples 6.1 and 6.2 present models that are intrinsically nonlinear in the sense that they cannot be transformed into linear representations of the form of models discussed in Chapter 5. A model that is not intrinsically nonlinear is given by

$$y_t = \beta_0 \exp [\beta_1 x_t + u_t] . \quad (6.2)$$

By contrast with the model in Example 6.2, this model can be transformed into a linear representation using the logarithmic transformation

$$\ln y_t = \ln \beta_0 + \beta_1 x_t + u_t .$$

The properties of these two exponential models are compared in the following example.

Example 6.3 Alternative Exponential Regression Models

Figure 6.1 plots simulated series based on the two exponential models

$$\begin{aligned} y_{1,t} &= \beta_0 \exp [\beta_1 x_t + u_{1,t}] \\ y_{2,t} &= \beta_0 \exp [\beta_1 x_t] + u_{2,t} , \end{aligned}$$

where the sample size is $T = 50$, and the explanatory variable x_t is a linear trend, $u_{1,t}, u_{2,t} \sim iid N(0, \sigma^2)$ and the parameter values are $\beta_0 = 1.0$, $\beta_1 = 0.05$ and $\sigma = 0.5$. Panel (a) of Figure 6.1 shows that both series are increasing exponentially as x_t increases; however, $y_{1,t}$ exhibits increasing volatility for higher levels of x_t whereas $y_{2,t}$ does not. Transforming the series using a

natural log transformation, illustrated in panel (b) of Figure 6.1, renders the volatility of $y_{1,t}$ constant, but this transformation is inappropriate for $y_{2,t}$ where it now exhibits decreasing volatility for higher levels of x_t . $\square$

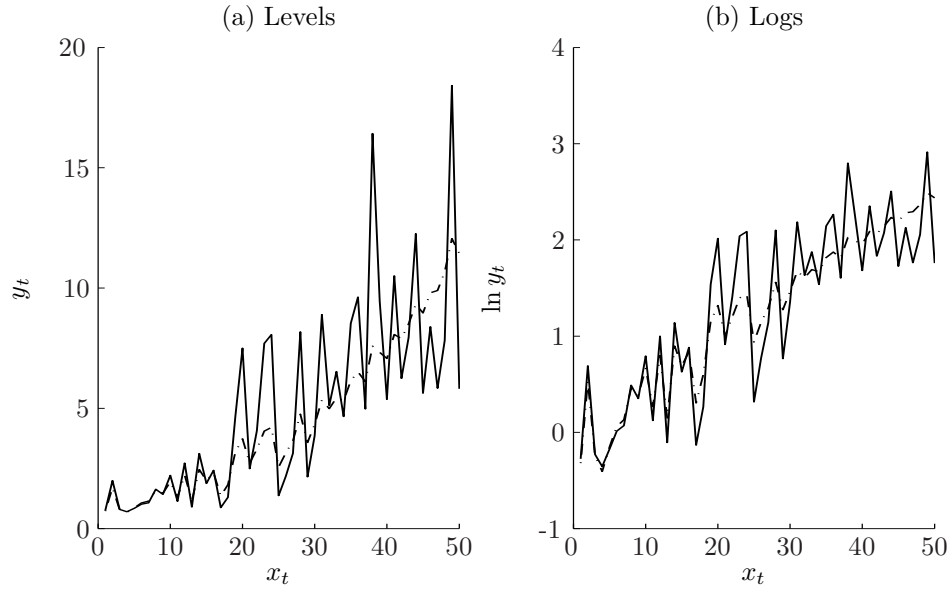


Figure 6.1 Simulated realizations from two exponential models, $y_{1,t}$ (solid line) and $y_{2,t}$ (dot-dashed line), in levels and in logarithms with $T = 50$.

6.3 Maximum Likelihood Estimation

The iterative algorithms discussed in Chapter 3 can be used to find the maximum likelihood estimates of the parameters of the nonlinear regression model in equation (6.1), together with their standard errors. The disturbance term, u , is assumed to be normally distributed given by

$$f(u) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{u^2}{2\sigma^2} \right]. \quad (6.3)$$

The transformation of variable technique (see Appendix A) can be used to derive the corresponding density of y as

$$f(y) = f(u) \left| \frac{du}{dy} \right|. \quad (6.4)$$

Taking the derivative with respect to y_t on both sides of equation (6.1) gives

$$\frac{du_t}{dy_t} = \frac{dg(y_t; \alpha)}{dy_t},$$

so the probability distribution of y_t is

$$f(y_t | x_t; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(g(y_t; \alpha) - \mu(x_t; \beta))^2}{2\sigma^2} \right] \left| \frac{dg(y_t; \alpha)}{dy_t} \right|,$$

where $\theta = \{\alpha, \beta, \sigma^2\}$. The log-likelihood function for $t = 1, 2, \dots, T$ observations, is

$$\begin{aligned} \ln L_T(\theta) = & -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2 T} \sum_{t=1}^T (g(y_t; \alpha) - \mu(x_t; \beta))^2 \\ & + \frac{1}{T} \sum_{t=1}^T \ln \left| \frac{dg(y_t; \alpha)}{dy_t} \right|, \end{aligned}$$

which is maximized with respect to θ .

The elements of the gradient and Hessian at time t are, respectively,

$$\frac{\partial \ln l_t(\theta)}{\partial \alpha} = -\frac{1}{\sigma^2} (g(y_t; \alpha) - \mu(x_t; \beta)) \frac{\partial g(y_t; \alpha)}{\partial \alpha} + \frac{\partial}{\partial \alpha} \ln \left| \frac{dg(y_t; \alpha)}{dy_t} \right|$$

$$\frac{\partial \ln l_t(\theta)}{\partial \beta} = \frac{1}{\sigma^2} (g(y_t; \alpha) - \mu(x_t; \beta)) \frac{\partial \mu(x_t; \beta)}{\partial \beta}$$

$$\frac{\partial \ln l_t(\theta)}{\partial \sigma^2} = -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} (g(y_t; \alpha) - \mu(x_t; \beta))^2,$$

and

$$\begin{aligned}
\frac{\partial^2 \ln l_t(\theta)}{\partial \alpha \partial \alpha'} &= -\frac{1}{\sigma^2} (g(y_t; \alpha) - \mu(x_t; \beta)) \frac{\partial g(y_t; \alpha)}{\partial \alpha \partial \alpha'} - \frac{1}{\sigma^2} \left(\frac{\partial g(y_t; \alpha)}{\partial \alpha \partial \alpha'} \right)^2 \\
&\quad + \frac{\partial^2}{\partial \alpha \partial \alpha'} \ln \left| \frac{dg(y_t; \alpha)}{dy_t} \right| \\
\frac{\partial^2 \ln l_t(\theta)}{\partial \alpha \partial \beta'} &= \frac{1}{\sigma^2} (g(y_t; \alpha) - \mu(x_t; \beta)) \frac{\partial g(y_t; \alpha)}{\partial \alpha} \frac{\partial \mu(x_t; \beta)}{\partial \beta'} \\
\frac{\partial^2 \ln l_t(\theta)}{\partial \beta \partial \beta'} &= \frac{1}{\sigma^2} (g(y_t; \alpha) - \mu(x_t; \beta)) \frac{\partial^2 \mu(x_t; \beta)}{\partial \beta \partial \beta'} - \frac{1}{\sigma^2} \frac{\partial^2 \mu(x_t; \beta)}{\partial \beta \partial \beta'} \\
\frac{\partial^2 \ln l_t}{(\partial \sigma^2)^2} &= -\frac{1}{2\sigma^4} + \frac{1}{\sigma^6} (g(y_t; \alpha) - \mu(x_t; \beta))^2 \\
\frac{\partial^2 \ln l_t(\theta)}{\partial \alpha \partial \sigma^2} &= \frac{1}{\sigma^4} (g(y_t; \alpha) - \mu(x_t; \beta)) \frac{\partial g(y_t; \alpha)}{\partial \alpha} \\
\frac{\partial^2 \ln l_t(\theta)}{\partial \beta \partial \sigma^2} &= -\frac{1}{\sigma^4} (g(y_t; \alpha) - \mu(x_t; \beta)) \frac{\partial \mu(x_t; \beta)}{\partial \beta}.
\end{aligned}$$

The generic parameter updating scheme of the Newton-Raphson algorithm is

$$\theta_{(k)} = \theta_{(k-1)} - H_{(k-1)} G_{(k-1)}, \quad (6.5)$$

which, in the context of the nonlinear regression model may be simplified slightly as follows. Averaging over the $t = 1, 2, \dots, T$ observations, setting the first-order condition for σ^2 equal to zero and solving for $\hat{\sigma}^2$ yields

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (g(y_t; \hat{\alpha}) - \mu(x_t; \hat{\beta}))^2. \quad (6.6)$$

This result is used to concentrate $\hat{\sigma}^2$ out of the log-likelihood function, which is then maximized with respect to $\theta = \{\alpha, \beta\}$. The Newton-Raphson algorithm then simplifies to

$$\theta_{(k)} = \theta_{(k-1)} - H_{1,1}^{-1}(\theta_{(k-1)}) G_1(\theta_{(k-1)}), \quad (6.7)$$

where

$$G_1 = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T \frac{\partial \ln l_t(\theta)}{\partial \alpha} \\ \frac{1}{T} \sum_{t=1}^T \frac{\partial \ln l_t(\theta)}{\partial \beta} \end{bmatrix} \quad (6.8)$$

and

$$H_{1,1} = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \alpha \partial \alpha'} & \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \alpha \partial \beta'} \\ \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \beta \partial \alpha'} & \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \beta \partial \beta'} \end{bmatrix}. \quad (6.9)$$

The method of scoring replaces $-H_{(k-1)}$ in (6.5), by the information matrix $I(\theta)$. The updated parameter vector is calculated as

$$\theta_{(k)} = \theta_{(k-1)} + I_{(k-1)}^{-1} G_{(k-1)}, \quad (6.10)$$

where the information matrix, $I(\theta)$, is given by

$$I(\theta) = -E \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \alpha \partial \alpha'} & \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \alpha \partial \beta'} & \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_T(\theta)}{\partial \alpha \partial \sigma^2} \\ \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \beta \partial \alpha'} & \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \beta \partial \beta'} & \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \beta \partial \sigma^2} \\ \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \sigma^2 \partial \alpha'} & \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \sigma^2 \partial \beta'} & \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial (\sigma^2)^2} \end{bmatrix}.$$

For this class of models $I(\theta)$ is a block-diagonal matrix. To see this, note that from equation (6.1)

$$E[g(y_t; \alpha)] = E[\mu(x_t; \beta) + u_t] = \mu(x_t; \beta),$$

so that

$$\begin{aligned} E \left[\frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_T(\theta)}{\partial \alpha \partial \sigma^2} \right] &= E \left[\frac{1}{\sigma^4 T} \sum_{t=1}^T (g(y_t; \alpha) - \mu(x_t; \beta)) \frac{\partial g(y_t; \alpha)}{\partial \alpha} \right] = 0 \\ E \left[\frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \beta \partial \sigma^2} \right] &= -E \left[\frac{1}{\sigma^4 T} \sum_{t=1}^T (g(y_t; \alpha) - \mu(x_t; \beta)) \frac{\partial \mu(x_t; \beta)}{\partial \beta} \right] = 0. \end{aligned}$$

In this case $I(\theta)$ reduces to

$$I(\theta) = \begin{bmatrix} I_{1,1} & 0 \\ 0 & I_{2,2} \end{bmatrix}, \quad (6.11)$$

where

$$I_{1,1} = -E[H_{1,1}] = -E \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \alpha \partial \alpha'} & \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \alpha \partial \beta'} \\ \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \beta \partial \alpha'} & \frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial \beta \partial \beta'} \end{bmatrix},$$

and

$$I_{2,2} = -E \left[\frac{1}{T} \sum_{t=1}^T \frac{\partial^2 \ln l_t(\theta)}{\partial (\sigma^2)^2} \right].$$

The scoring algorithm now proceeds in two parts

$$\begin{bmatrix} \alpha_{(k)} \\ \beta_{(k)} \end{bmatrix} = \begin{bmatrix} \alpha_{(k-1)} \\ \beta_{(k-1)} \end{bmatrix} + I_{1,1}^{-1}(\theta_{(k-1)}) G_1(\theta_{(k-1)}) \quad (6.12)$$

$$\begin{bmatrix} \sigma_{(k)}^2 \end{bmatrix} = \begin{bmatrix} \sigma_{(k-1)}^2 \end{bmatrix} + I_{2,2}^{-1}(\theta_{(k-1)}) G_2(\theta_{(k-1)}), \quad (6.13)$$

where G_1 is defined in equation (6.8) and

$$G_2 = \left[\frac{1}{T} \sum_{t=1}^T \frac{\partial \ln l_t(\theta)}{\partial \sigma^2} \right].$$

The covariance matrix of the parameter estimators is obtained by inverting the relevant blocks of the information matrix at the last iteration. For example, the variance of $\hat{\sigma}^2$ is simply given by

$$\text{var}(\hat{\sigma}^2) = \frac{2\hat{\sigma}^4}{T}.$$

Example 6.4 Estimation of a Nonlinear Production Function Consider the Zellner-Revankar production function introduced in Example 6.1. The probability density function of u_t is

$$f(u) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{u^2}{2\sigma^2} \right].$$

Using equation (6.4) with

$$\frac{du_t}{dy_t} = \frac{1}{y_t} + \alpha,$$

the density for y_t is

$$f(y_t; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(\ln y_t + \alpha y_t - \beta_0 - \beta_1 \ln k_t - \beta_2 \ln l_t)^2}{2\sigma^2} \right] \left| \frac{1}{y_t} + \alpha \right|.$$

The log-likelihood function for a sample of $t = 1, \dots, T$ observations is

$$\begin{aligned} \ln L_T(\theta) = & -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) + \frac{1}{T} \sum_{t=1}^T \ln \left| \frac{1}{y_t} + \alpha \right| \\ & - \frac{1}{2\sigma^2 T} \sum_{t=1}^T (\ln y_t + \alpha y_t - \beta_0 - \beta_1 \ln k_t - \beta_2 \ln l_t)^2. \end{aligned}$$

This function is then maximized with respect to the unknown parameters $\theta = \{\alpha, \beta_0, \beta_1, \beta_2, \sigma^2\}$. The problem can be simplified by concentrating the log-likelihood function with respect to $\hat{\sigma}^2$ which is given by the variance of the residuals

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (\ln y_t + \hat{\alpha} y_t - \hat{\beta}_0 - \hat{\beta}_1 \ln k_t - \hat{\beta}_2 \ln l_t)^2.$$

□

Example 6.5 Estimation of a Nonlinear Exponential Model

Consider the nonlinear model in Example 6.2. The disturbance term u is assumed to have a normal distribution

$$f(u) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{u^2}{2\sigma^2} \right],$$

so the density of y_t is

$$f(y_t | x_t; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y_t - \beta_0 \exp[\beta_1 x_t])^2}{2\sigma^2} \right].$$

The log-likelihood function for a sample of $t = 1, \dots, T$ observations is

$$\ln L_T(\theta) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2 T} \sum_{t=1}^T (y_t - \beta_0 \exp[\beta_1 x_t])^2.$$

This function is to be maximized with respect to $\theta = \{\beta_0, \beta_1, \sigma^2\}$.

The derivatives of the log-likelihood function with respect θ are

$$\begin{aligned} \frac{\partial \ln L_T(\theta)}{\partial \beta_0} &= \frac{1}{\sigma^2 T} \sum_{t=1}^T (y_t - \beta_0 \exp[\beta_1 x_t]) \exp[\beta_1 x_t] \\ \frac{\partial \ln L_T(\theta)}{\partial \beta_1} &= \frac{1}{\sigma^2 T} \sum_{t=1}^T (y_t - \beta_0 \exp[\beta_1 x_t]) \beta_0 \exp[\beta_1 x_t] x_t \\ \frac{\partial \ln L_T(\theta)}{\partial \sigma^2} &= -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4 T} \sum_{t=1}^T (y_t - \beta_0 \exp[\beta_1 x_t])^2. \end{aligned}$$

The maximum likelihood estimators of the parameters are obtained by set-

ting these derivatives to zero and solving the system of equations

$$\begin{aligned} \frac{1}{\hat{\sigma}^2 T} \sum_{t=1}^T (y_t - \hat{\beta}_0 \exp[\hat{\beta}_1 x_t]) \exp[\hat{\beta}_1 x_t] &= 0 \\ \frac{1}{\hat{\sigma}^2 T} \sum_{t=1}^T (y_t - \hat{\beta}_0 \exp[\hat{\beta}_1 x_t]) \hat{\beta}_0 \exp[\hat{\beta}_1 x_t] x_t &= 0 \\ -\frac{1}{2\hat{\sigma}^2} + \frac{1}{2\hat{\sigma}^4 T} \sum_{t=1}^T (y_t - \hat{\beta}_0 \exp[\hat{\beta}_1 x_t])^2 &= 0. \end{aligned}$$

Estimation of the parameters is simplified by noting that the first two equations can be written independently of $\hat{\sigma}^2$ and that the information matrix is block diagonal. In this case, an iterative algorithm is used to find $\hat{\beta}_0$ and $\hat{\beta}_1$. Once these estimates are computed, $\hat{\sigma}^2$ is obtained immediately from rearranging the last expression as

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{\beta}_0 \exp[\hat{\beta}_1 x_t])^2. \quad (6.14)$$

Using the simulated $y_{2,t}$ data in Panel (a) of Figure 6.1, the maximum likelihood estimates are revealed to be $\hat{\beta}_0 = 1.027$ and $\hat{\beta}_1 = 0.049$. The estimated negative Hessian matrix is

$$-H_T(\hat{\beta}) = \begin{bmatrix} 117.521 & 4913.334 \\ 4913.334 & 215992.398 \end{bmatrix},$$

so that the covariance matrix of $\hat{\beta}$ is

$$\frac{1}{T} \hat{\Omega} = -\frac{1}{T} H_T^{-1}(\hat{\beta}) \begin{bmatrix} 0.003476 & -0.000079 \\ -0.000079 & 0.000002 \end{bmatrix}.$$

The standard errors of the maximum likelihood estimates of β_0 and β_1 are found by taking the square roots of the diagonal terms of $\hat{\Omega}/T$

$$\begin{aligned} \text{se}(\hat{\beta}_0) &= \sqrt{0.003476} = 0.059 \\ \text{se}(\hat{\beta}_1) &= \sqrt{0.000002} = 0.001. \end{aligned}$$

The residual at time t is computed as

$$\hat{u}_t = y_t - \hat{\beta}_0 \exp[\hat{\beta}_1 x_t] = y_t - 1.027 \exp[0.049 x_t],$$

and the residual sum of squares is given by $\sum_{t=1}^T \hat{u}_t^2 = 12.374$. Finally, the

residual variance is computed as

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{\beta}_0 \exp[\hat{\beta}_1 x_t])^2 = \frac{12.374}{50} = 0.247,$$

with standard error

$$\text{se}(\hat{\sigma}^2) = \sqrt{\frac{2\hat{\sigma}^4}{T}} = \sqrt{\frac{2 \times 0.247^2}{50}} = 0.049.$$

□

6.4 Gauss-Newton

For the special case of the nonlinear regression models where $g(y_t; \alpha) = y_t$ in (6.1), the scoring algorithm can be simplified further so that parameter updating can be achieved by means of a least squares regression. This form of the scoring algorithm is known as the Gauss-Newton algorithm.

Consider the model

$$y_t = \mu(x_t; \beta) + u_t, \quad u_t \sim iid N(0, \sigma^2), \quad (6.15)$$

where the unknown parameters are $\theta = \{\beta, \sigma^2\}$. The distribution of y_t is

$$f(y_t | x_t; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{1}{2\sigma^2} \sum_{t=1}^T (y_t - \mu(x_t; \beta))^2 \right], \quad (6.16)$$

and the corresponding log-likelihood function at time t is

$$\ln l_t(\theta) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} (y_t - \mu(x_t; \beta))^2, \quad (6.17)$$

with first derivative

$$g_t(\beta) = \frac{1}{\sigma^2} \frac{\partial(\mu(x_t; \beta))}{\partial \beta} (y_t - \mu(x_t; \beta)) = \frac{1}{\sigma^2} z_t u_t, \quad (6.18)$$

where

$$u_t = y_t - \mu(x_t; \beta), \quad z_t = \frac{\partial(\mu(x_t; \beta))}{\partial \beta}.$$

The gradient with respect to β is

$$G_T(\beta) = \frac{1}{T} \sum_{t=1}^T g_t(\beta) = \frac{1}{\sigma^2 T} \sum_{t=1}^T z_t u_t, \quad (6.19)$$

and the information matrix is, therefore,

$$\begin{aligned} I(\beta) &= \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T g_t(\beta) g_t(\beta)' \right] = \frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[\left(\frac{1}{\sigma^2} z_t u_t \right) \left(\frac{1}{\sigma^2} z_t u_t \right)' \right] \\ &= \frac{1}{\sigma^4 T} \mathbb{E} \left[\sum_{t=1}^T u_t^2 z_t z_t' \right] = \frac{1}{\sigma^2 T} \sum_{t=1}^T z_t z_t', \end{aligned} \quad (6.20)$$

where use has been made of the assumption that u_t *iid* so that $\mathbb{E}[u_t^2] = \sigma^2$.

Because of the block-diagonal property of the information matrix in equation (6.11), the update of β is obtained by using the expressions for $G_T(\beta)$ and $I(\beta)$ in (6.19) and (6.20), respectively,

$$\beta_{(k)} = \beta_{(k-1)} + I^{-1}(\beta_{(k-1)}) G(\beta_{(k-1)}) = \beta_{(k-1)} + \left(\sum_{t=1}^T z_t z_t' \right)^{-1} \sum_{t=1}^T z_t u_t.$$

Let the change in the parameters at iteration k be defined as

$$\hat{\Delta} = \beta_{(k)} - \beta_{(k-1)} = \left(\sum_{t=1}^T z_t z_t' \right)^{-1} \sum_{t=1}^T z_t u_t. \quad (6.21)$$

The Gauss-Newton algorithm, therefore, requires the evaluation of u_t and z_t at $\beta_{(k-1)}$ followed by a simple linear regression of u_t on z_t to obtain $\hat{\Delta}$. The updated parameter vector $\beta_{(k)}$ is simply obtained by adding the parameter estimates from this regression on to $\beta_{(k-1)}$.

Once the Gauss-Newton scheme has converged, the final estimates of $\hat{\beta}$ are the maximum likelihood estimates. In turn, the maximum likelihood estimate of σ^2 is computed as

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \mu(x_t; \hat{\beta}))^2. \quad (6.22)$$

Example 6.6 Nonlinear Exponential Model Revisited

Consider again the nonlinear exponential model in Examples 6.2 and 6.5. Estimating this model using the Gauss-Newton algorithm requires the following steps.

Step 1: Compute the derivatives of $\mu(x_t; \beta)$ with respect to $\beta = \{\beta_0, \beta_1\}$

$$z_{1,t} = \frac{\partial \mu(x_t; \beta)}{\partial \beta_0} = \exp[\beta_1 x_t]$$

$$z_{2,t} = \frac{\partial \mu(x_t; \beta)}{\partial \beta_1} = \beta_0 \exp[\beta_1 x_t] x_t.$$

Step 2: Evaluate u_t , $z_{1,t}$ and $z_{2,t}$ at the starting values of β .

Step 3: Regress u_t on $z_{1,t}$ and $z_{2,t}$ to obtain $\hat{\Delta}_{\beta_0}$ and $\hat{\Delta}_{\beta_1}$.

Step 4: Update the parameter estimates

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}_{(k)} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}_{(k-1)} + \begin{bmatrix} \hat{\Delta}_{\beta_0} \\ \hat{\Delta}_{\beta_1} \end{bmatrix}.$$

Step 5: The iterations continue until convergence is achieved, $|\hat{\Delta}_{\beta_0}|, |\hat{\Delta}_{\beta_1}| < \varepsilon$, where ε is the tolerance level. $\square$

Example 6.7 Estimating a Nonlinear Consumption Function Consider the following nonlinear consumption function

$$c_t = \beta_0 + \beta_1 y_t^{\beta_2} + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

where c_t is real consumption, y_t is real disposable income, u_t is a disturbance term $N(0, \sigma^2)$, and $\theta = \{\beta_0, \beta_1, \beta_2, \sigma^2\}$ are unknown parameters. Estimating this model using the Gauss-Newton algorithm requires the following steps.

Step 1: Compute the derivatives of $\mu(y_t; \beta) = \beta_0 + \beta_1 y_t^{\beta_2}$ with respect to $\beta = \{\beta_0, \beta_1, \beta_2\}$

$$z_{1,t} = \frac{\partial \mu(y_t; \beta)}{\partial \beta_0} = 1$$

$$z_{2,t} = \frac{\partial \mu(y_t; \beta)}{\partial \beta_1} = y_t^{\beta_2}$$

$$z_{3,t} = \frac{\partial \mu(y_t; \beta)}{\partial \beta_2} = \beta_1 y_t^{\beta_2} \ln(y_t).$$

Step 2: Evaluate u_t , $z_{1,t}$, $z_{2,t}$ and $z_{3,t}$ at the starting values for β .

Step 3: Regress u_t on $z_{1,t}$, $z_{2,t}$ and $z_{3,t}$, to get $\hat{\Delta} = \{\hat{\Delta}_{\beta_0}, \hat{\Delta}_{\beta_1}, \hat{\Delta}_{\beta_2}\}$ from this auxiliary regression.

Step 4: Update the parameter estimates

$$\begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}_{(k)} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}_{(k-1)} + \begin{bmatrix} \hat{\Delta}_{\beta_0} \\ \hat{\Delta}_{\beta_1} \\ \hat{\Delta}_{\beta_2} \end{bmatrix}.$$

Step 5: The iterations continue until convergence, $|\hat{\Delta}_{\beta_0}|, |\hat{\Delta}_{\beta_1}|, |\hat{\Delta}_{\beta_2}| < \varepsilon$, where ε is the tolerance level.

U.S. quarterly data for real consumption expenditure and real disposable personal income for the period 1960:Q1 to 2009:Q4, downloaded from the Federal Reserve Bank of St. Louis, are used to estimate the parameters of this nonlinear consumption function. Nonstationary time series The starting values for β_0 and β_1 , obtained from a linear model with $\beta_2 = 1$, are

$$\beta_{(0)} = [-228.540, 0.950, 1.000].$$

After constructing u_t and the derivatives $z_t = \{z_{1,t}, z_{2,t}, z_{3,t}\}$, u_t is regressed on z_t to give the parameter values

$$\hat{\Delta} = [600.699, -1.145, 0.125].$$

The updated parameter estimates are

$$\beta_{(1)} = [-228.540, 0.950, 1.000] + [600.699, -1.145, 0.125] = [372.158, -0.195, 1.125].$$

The final estimates, achieved after five iterations, are

$$\beta_{(5)} = [299.019, 0.289, 1.124].$$

The estimated residual for time t , using the parameter estimates at the final iteration, is computed as

$$\hat{u}_t = c_t - 299.019 - 0.289 y_t^{1.124},$$

yielding the residual variance

$$\hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T \hat{u}_t^2 = \frac{1307348.531}{200} = 6536.743.$$

The estimated information matrix is

$$I(\hat{\beta}) = \frac{1}{\hat{\sigma}^2 T} \sum_{t=1}^T z_t z_t' = \begin{bmatrix} 0.000 & 2.436 & 6.145 \\ 2.436 & 48449.106 & 124488.159 \\ 6.145 & 124488.159 & 320337.624 \end{bmatrix},$$

from which the covariance matrix of $\hat{\beta}$ is computed

$$\frac{1}{T}\hat{\Omega} = \frac{1}{T}I^{-1}(\hat{\beta}) = \begin{bmatrix} 2350.782 & -1.601 & 0.577 \\ -1.601 & 0.001 & -0.0004 \\ 0.577 & -0.0004 & 0.0002 \end{bmatrix}.$$

The standard errors of $\hat{\beta}$ are given as the square roots of the elements on the main diagonal of $\hat{\Omega}/T$

$$\begin{aligned} \text{se}(\hat{\beta}_0) &= \sqrt{2350.782} = 48.485 \\ \text{se}(\hat{\beta}_1) &= \sqrt{0.001} = 0.034 \\ \text{se}(\hat{\beta}_2) &= \sqrt{0.0002} = 0.012. \end{aligned}$$

□

6.4.1 Relationship to Nonlinear Least Squares

A standard procedure used to estimate nonlinear regression models is known as nonlinear least squares. Consider equation (6.15) where for simplicity β is a scalar. By expanding $\mu(x_t; \beta)$ as a Taylor series expansion around $\beta_{(k-1)}$

$$\mu(x_t; \beta) = \mu(x_t; \beta_{(k-1)}) + \frac{d\mu}{d\beta}(\beta - \beta_{k-1}) + \cdots,$$

equation (6.15) is rewritten as

$$y_t - \mu(x_t; \beta_{(k-1)}) = \frac{d\mu}{d\beta}(\beta - \beta_{k-1}) + v_t, \quad (6.23)$$

where v_t is the disturbance which contains u_t and the higher-order terms from the Taylor series expansion. The k^{th} iteration of the nonlinear regression estimation procedure involves regressing $y_t - \mu(x_t; \beta_{(k-1)})$ on the derivative $d\mu/d\beta$, to generate the parameter estimate

$$\hat{\Delta} = \beta_{(k)} - \beta_{k-1}.$$

The updated value of the parameter estimate is then computed as

$$\beta_{(k)} = \beta_{k-1} + \hat{\Delta},$$

which is used to recompute $y_t - \mu(x_t; \beta_{(k-1)})$ and $d\mu/d\beta$. The iterations proceed until convergence.

An alternative way of expressing the linearized regression equation in equation (6.23) is to write it as

$$u_t = z_t(\beta_{(k)} - \beta_{k-1}) + v_t, \quad (6.24)$$

where

$$u_t = y_t - \mu(x_t; \beta_{(k-1)}) , \quad z_t = \frac{d\mu(x_t; \beta_{(k-1)})}{d\beta} .$$

Comparing this equation with the updated Gauss-Newton estimator in (6.21) shows that the two estimation procedures are equivalent.

6.4.2 Relationship to Ordinary Least Squares

For classes of models where the mean function, $\mu(x_t; \beta)$, is linear, the Gauss-Newton algorithm converges in one step regardless of the starting value. Consider the linear regression model where $\mu(x_t; \beta) = \beta x_t$ and the expressions for u_t and z_t are respectively

$$u_t = y_t - \beta x_t , \quad z_t = \frac{\partial \mu(x_t; \beta)}{\partial \beta} = x_t .$$

Substituting these expressions into the Gauss-Newton algorithm (6.21) gives

$$\begin{aligned} \beta_{(k)} &= \beta_{(k-1)} + \left[\sum_{t=1}^T x_t x_t' \right]^{-1} \sum_{t=1}^T x_t (y_t - \beta_{(k-1)} x_t) \\ &= \beta_{(k-1)} + \left[\sum_{t=1}^T x_t x_t' \right]^{-1} \sum_{t=1}^T x_t y_t - \beta_{(k-1)} \left[\sum_{t=1}^T x_t x_t' \right]^{-1} \sum_{t=1}^T x_t x_t \\ &= \beta_{(k-1)} + \left[\sum_{t=1}^T x_t x_t' \right]^{-1} \sum_{t=1}^T x_t y_t - \beta_{(k-1)} \\ &= \left[\sum_{t=1}^T x_t x_t' \right]^{-1} \sum_{t=1}^T x_t y_t, \end{aligned} \tag{6.25}$$

which is just the ordinary least squares estimator obtained when regressing y_t on x_t . The scheme converges in just one step for an arbitrary choice of $\beta_{(k-1)}$ because $\beta_{(k-1)}$ does not appear on the right hand side of equation (6.25).

6.4.3 Asymptotic Distributions

As Chapter 2 shows, maximum likelihood estimators are asymptotically normally distributed. In the context of the nonlinear regression model, this means that

$$\hat{\theta} \overset{a}{\sim} N(\theta_0, \frac{1}{T} I(\theta_0)^{-1}), \tag{6.26}$$

where $\theta_0 = \{\beta_0, \sigma_0^2\}$ is the true parameter vector and $I(\theta_0)$ is the information matrix evaluated at θ_0 . The fact that $I(\theta)$ is block diagonal in the class of models considered here means that the asymptotic distribution of $\hat{\beta}$ can be considered separately from that of $\hat{\sigma}^2$ without any loss of information.

From equation (6.20), the relevant block of the information matrix is

$$I(\beta_0) = \frac{1}{\sigma_0^2 T} \sum_{t=1}^T z_t z_t',$$

so that the asymptotic distribution is

$$\hat{\beta} \stackrel{a}{\sim} N\left(\beta_0, \sigma_0^2 \left(\sum_{t=1}^T z_t z_t'\right)^{-1}\right).$$

In practice σ_0^2 is unknown and is replaced by the maximum likelihood estimator given in equation (6.6). The standard errors of $\hat{\beta}$ are therefore computed by taking the square root of the diagonal elements of the covariance matrix

$$\frac{1}{T} \hat{\Omega} = \hat{\sigma}^2 \left[\sum_{t=1}^T z_t z_t' \right]^{-1}.$$

The asymptotic distribution of $\hat{\sigma}^2$ is

$$\hat{\sigma}^2 \stackrel{a}{\sim} N\left(\sigma_0^2, \frac{1}{T} 2\sigma_0^4\right).$$

As with the standard error of $\hat{\beta}$, σ_0^2 is replaced by the maximum likelihood estimator of σ^2 given in equation (6.6), so that the standard error is

$$\text{se}(\hat{\sigma}^2) = \sqrt{\frac{2\hat{\sigma}^4}{T}}.$$

6.5 Testing

6.5.1 LR, Wald and LM Tests

The LR, Wald and LM tests discussed in Chapter 4 can all be applied to test the parameters of nonlinear regression models. For those cases where the unrestricted model is relatively easier to estimate than the restricted model, the Wald test is particularly convenient. Alternatively, where the restricted model is relatively easier to estimate than the unrestricted model, the LM test is the natural strategy to adopt. Examples of these testing strategies for nonlinear regression models are given below.

Example 6.8 Testing a Nonlinear Consumption Function

A special case of the nonlinear consumption function used in Example 6.7 is the linear version where $\beta_2 = 1$. This suggests that a test of linearity is given by the hypotheses

$$H_0 : \beta_2 = 1 \quad H_1 : \beta_2 \neq 1.$$

This restriction is tested using the same U.S. quarterly data for the period 1960:Q1 - 2009:Q4 on real personal consumption expenditure and real disposable income as in Example 6.7.

To perform the likelihood ratio test, the values of the restricted ($\beta_2 = 1$) and unrestricted ($\beta_2 \neq 1$) log-likelihood functions are respectively

$$\begin{aligned} \ln L_T(\hat{\theta}_0) &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) - \frac{1}{T} \sum_{t=1}^T \frac{(c_t - \beta_0 - \beta_1 y_t)^2}{2\sigma^2} \\ \ln L_T(\hat{\theta}_1) &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) - \frac{1}{T} \sum_{t=1}^T \frac{(c_t - \beta_0 - \beta_1 y_t^{\beta_2})^2}{2\sigma^2}. \end{aligned}$$

The restricted and unrestricted parameter estimates are

$$[-228.540 \quad 0.950 \quad 1.000] \text{ and } [298.739 \quad 0.289 \quad 1.124] \text{ ' .}$$

These estimates produce the respective values of the log-likelihood functions

$$T \ln L_T(\hat{\theta}_0) = -1204.645, \quad T \ln L_T(\hat{\theta}_1) = -1162.307.$$

The value of the LR statistic is

$$LR = -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)) = -2(-1204.645 + 1162.307) = 84.676.$$

From the χ_1^2 distribution, the p-value of the LR test statistic is 0.000 showing that the restriction is rejected at conventional significance levels.

To perform a Wald test define

$$R = [0 \quad 0 \quad 1], \quad Q = [1],$$

and compute the negative Hessian matrix based on numerical derivatives at the unrestricted parameter estimates, $\hat{\beta}_1$,

$$-H_T(\theta) = \begin{bmatrix} 0.000 & 2.435 & 6.145 \\ 2.435 & 48385.997 & 124422.745 \\ 6.145 & 124422.745 & 320409.562 \end{bmatrix}.$$

The Wald statistic is

$$W = T[R\hat{\beta}_1 - Q]'[R(-H_T^{-1}(\hat{\theta}_1))R']^{-1}[R\hat{\beta}_1 - Q] = 64.280.$$

The p-value of the Wald test statistic obtained from the χ_1^2 distribution is 0.000, once again showing that the restriction is strongly rejected at conventional significance levels.

To perform a LM test, the gradient vector of the unrestricted model evaluated at the restricted parameter estimates, $\hat{\beta}_0$, is

$$G_T(\hat{\beta}_0) = [0.000 \quad 0.000 \quad 2.810]',$$

and the outer product of gradients matrix is

$$J_T(\hat{\beta}_0) = \begin{bmatrix} 0.000 & 0.625 & 5.257 \\ 0.625 & 4727.411 & 40412.673 \\ 5.257 & 40412.673 & 345921.880 \end{bmatrix}.$$

The LM statistic is

$$LM = TG_T'(\hat{\beta}_0)J_T^{-1}(\hat{\beta}_0)G_T(\hat{\beta}_0) = 39.908,$$

which, from the χ_1^2 distribution, has a p-value of 0.000 showing that the restriction is still strongly rejected. $\square$

Example 6.9 Constant Marginal Propensity to Consume

The nonlinear consumption function used in Examples 6.7 and 6.8 has a marginal propensity to consume (MPC) given by

$$MPC = \frac{dc_t}{dy_t} = \beta_1\beta_2y_t^{\beta_2-1},$$

whose value depends on the value of income, y_t , at which it is measured. Testing the restriction that the MPC is constant and does not depend on y_t involves testing the hypotheses

$$H_0 : \beta_2 = 1 \quad H_1 : \beta_2 \neq 1.$$

Define $Q = 0$ and

$$\begin{aligned} C(\beta) &= \beta_1\beta_2y_t^{\beta_2} - \beta_1 \\ D(\beta) &= \frac{\partial C(\beta)}{\partial \beta} = [0 \quad \beta_2y_t^{\beta_2-1} - 1 \quad \beta_1y_t^{\beta_2-1}(1 + \beta_2 \ln y_t)]', \end{aligned}$$

then from Chapter 4 the general form of the Wald statistic in the case of nonlinear restrictions is

$$W = T[C(\hat{\beta}) - Q]'[D(\hat{\beta})\hat{\Omega}D(\hat{\beta})']^{-1}[C(\hat{\beta}) - Q],$$

where it is understood that all terms are to be evaluated at the unrestricted maximum likelihood estimates. This statistic is asymptotically distributed as χ_1^2 under the null hypothesis and large values of the test statistic constitute

rejection of the null hypothesis. The test can be performed for each t or it can be calculated for a typical value of y_t , usually the sample mean. $\square$

The LM test has a convenient form for nonlinear regression models because of the assumption of normality. To demonstrate this feature, consider the standard LM statistic, discussed in Chapter 4, which has the form

$$LM = TG'_T(\hat{\beta})I^{-1}(\hat{\beta})G_T(\hat{\beta}), \quad (6.27)$$

where all terms are evaluated at the restricted parameter estimates. Under the null hypothesis, this statistic is distributed asymptotically as χ_M^2 where M is the number of restrictions. From the expression for $G_T(\beta)$ and $I(\beta)$ in (6.19) and (6.20), respectively, the LM statistic is

$$\begin{aligned} LM &= \left[\frac{1}{\hat{\sigma}^2} \sum_{t=1}^T z_t u_t \right]' \left[\frac{1}{\hat{\sigma}^2} \sum_{t=1}^T z_t z_t' \right]^{-1} \left[\frac{1}{\hat{\sigma}^2} \sum_{t=1}^T z_t u_t \right] \\ &= \frac{1}{\hat{\sigma}^2} \left[\sum_{t=1}^T z_t u_t \right]' \left[\sum_{t=1}^T z_t z_t' \right]^{-1} \left[\sum_{t=1}^T z_t u_t \right] \\ &= TR^2, \end{aligned} \quad (6.28)$$

where all quantities are evaluated under H_0 ,

$$\begin{aligned} u_t &= y_t - \mu(x_t; \hat{\beta}) \\ z_t &= - \frac{\partial u_t}{\partial \beta} \bigg|_{\beta=\hat{\beta}} \\ \hat{\sigma}^2 &= \frac{1}{T} \sum_{t=1}^T (y_t - \mu(x_t; \hat{\beta}))^2, \end{aligned}$$

and R^2 is the coefficient of determination obtained by regressing u_t on z_t .

The LM test in (6.28) is implemented by means of two linear regressions. The first regression estimates the constrained model. The second or auxiliary regression requires regressing u_t on z_t , where all of the quantities are evaluated at the constrained estimates. The test statistic is $LM = TR^2$, where R^2 is the coefficient of determination from the auxiliary regression. The implementation of the LM test in terms of two linear regressions is revisited in Chapters 7 and 8.

Example 6.10 Nonlinear Consumption Function

Example 6.9 uses a Wald test to test for a constant marginal propensity to consume in a nonlinear consumption function. To perform an LM test of the same restriction, the following steps are required.

Step 1: Write the model in terms of u_t

$$u_t = c_t - \beta_0 - \beta_1 y_t^{\beta_2}.$$

Step 2: Compute the following derivatives

$$\begin{aligned} z_{1,t} &= -\frac{\partial u_t}{\partial \beta_0} = 1, \\ z_{2,t} &= -\frac{\partial u_t}{\partial \beta_1} = y_t^{\beta_2}, \\ z_{3,t} &= -\frac{\partial u_t}{\partial \beta_2} = \beta_1 y_t^{\beta_2} \ln(y_t). \end{aligned}$$

Step 3: Estimate the restricted model

$$c_t = \beta_0 + \beta_1 y_t + u_t,$$

by regressing c_t on a constant and y_t to generate the restricted estimates $\hat{\beta}_0$ and $\hat{\beta}_1$.

Step 4: Evaluate u_t at the restricted estimates

$$\hat{u}_t = c_t - \hat{\beta}_0 - \hat{\beta}_1 y_t.$$

Step 5: Evaluate the derivatives at the constrained estimates

$$\begin{aligned} z_{1,t} &= 1, \\ z_{2,t} &= y_t, \\ z_{3,t} &= \hat{\beta}_0 y_t \ln(y_t). \end{aligned}$$

Step 6: Regress $\hat{u}_t$ on $\{z_{1,t}, z_{2,t}, z_{3,t}\}$ and compute R^2 from this regression.

Step 7: Evaluate the test statistic, $LM = TR^2$. This statistic is asymptotically distributed as χ_1^2 under the null hypothesis. Large values of the test statistic constitute rejection of the null hypothesis. Notice that the strength of the nonlinearity in the consumption function is determined by the third term in the auxiliary regression in Step 6. If no significant nonlinearity exists, this term should not add to the explanatory power of this regression equation. If the nonlinearity is significant, then it acts as an excluded variable which manifests itself through a non-zero value of R^2 . $\square$

6.5.2 Nonnested Tests

Two models are nonnested if one model cannot be expressed as a subset of the other. While a number of procedures have been developed to test nonnested models, in this application a maximum likelihood approach is

discussed following Vuong (1989). The basic idea is to convert the likelihood functions of the two competing models into a common likelihood function using the transformation of variable technique and perform a variation of a LR test.

Example 6.11 Vuong's Test Applied to U.S. Money Demand

Consider the following two alternative money demand equations

$$\begin{aligned} \text{Model 1: } m_t &= \beta_0 + \beta_1 r_t + \beta_2 y_t + u_{1,t}, & u_{1,t} &\sim iid N(0, \sigma_1^2), \\ \text{Model 2: } \ln m_t &= \alpha_0 + \alpha_1 \ln r_t + \alpha_2 \ln y_t + u_{2,t}, & u_{2,t} &\sim iid N(0, \sigma_2^2), \end{aligned}$$

where m_t is real money, y_t is real income, r_t is the nominal interest rate and $\theta_1 = \{\beta_0, \beta_1, \beta_2, \sigma_1^2\}$ and $\theta_2 = \{\alpha_0, \alpha_1, \alpha_2, \sigma_2^2\}$ are the unknown parameters of the two models, respectively. The models are not nested since one model cannot be expressed as a subset of the other. Another way to view this problem is to observe that Model 1 is based on the distribution of m_t whereas Model 2 is based on the distribution of $\ln m_t$,

$$\begin{aligned} f_1(m_t) &= \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp \left[-\frac{(m_t - \beta_0 - \beta_1 r_t - \beta_2 y_t)^2}{2\sigma_1^2} \right] \\ f_2(\ln m_t) &= \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp \left[-\frac{(\ln m_t - \alpha_0 - \alpha_1 \ln r_t - \alpha_2 \ln y_t)^2}{2\sigma_2^2} \right]. \end{aligned}$$

To enable the comparison of the two models, use the transformation of variable technique to convert the distribution f_2 into a distribution of the level of m_t . Formally this link between the two distributions is given by

$$f_1(m_t) = f_2(\ln m_t) \left| \frac{d \ln m_t}{dm_t} \right| = f_2(\ln m_t) \left| \frac{1}{m_t} \right|,$$

which allows the log-likelihood functions of the two models to be compared. The steps to perform the test are as follows.

Step 1: Estimate Model 1 by regressing m_t on $\{c, r_t, y_t\}$ and construct the log-likelihood function at each observation

$$\ln l_{1,t}(\hat{\theta}_1) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\hat{\sigma}_1^2) - \frac{(m_t - \hat{\beta}_0 - \hat{\beta}_1 r_t - \hat{\beta}_2 y_t)^2}{2\hat{\sigma}_1^2}.$$

Step 2: Estimate Model 2 by regressing $\ln m_t$ on $\{c, \ln r_t, \ln y_t\}$ and construct the log-likelihood function at each observation for m_t by using

$$\begin{aligned} \ln l_{2,t}(\hat{\theta}_2) &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\hat{\sigma}_2^2) - \frac{(\ln m_t - \hat{\alpha}_0 - \hat{\alpha}_1 \ln r_t - \hat{\alpha}_2 \ln y_t)^2}{2\hat{\sigma}_2^2} \\ &\quad - \ln m_t. \end{aligned}$$

Step 3: Compute the difference in the log-likelihood functions of the two models at each observation

$$d_t = \ln l_{1,t}(\hat{\theta}_1) - \ln l_{2,t}(\hat{\theta}_2).$$

Step 4: Construct the test statistic

$$\mathcal{V} = \sqrt{T} \frac{\bar{d}}{s},$$

where

$$\bar{d} = \frac{1}{T} \sum_{t=1}^T d_t, \quad s^2 = \frac{1}{T} \sum_{t=1}^T (d_t - \bar{d})^2,$$

are the mean and the variance of d_t , respectively.

Step 5: Using the result in Vuong (1989), the statistic V is asymptotically normally distributed under the null hypothesis that the two models are equivalent

$$\mathcal{V} \xrightarrow{d} N(0, 1).$$

The nonnested money demand models are estimated using quarterly data for the U.S. on real money, m_t , the nominal interest rate, r_t , and real income, y_t , for the period 1959 to 2005. The estimates of Model 1 are

$$\hat{m}_t = 7.131 + 7.660 r_t + 0.449 y_t.$$

The estimates of Model 2 are

$$\widehat{\ln m}_t = 0.160 + 0.004 \ln r_t + 0.829 \ln y_t.$$

The mean and variance of d_t are, respectively,

$$\begin{aligned} \bar{d} &= -0.159 \\ s^2 &= 0.054, \end{aligned}$$

yielding the value of the test statistic

$$\mathcal{V} = \sqrt{T} \frac{\bar{d}}{s} = \sqrt{188} \frac{-0.159}{\sqrt{0.054}} = -9.380.$$

Since the p-value of the statistic obtained from the standard normal distribution is 0.000, the null hypothesis that the models are equivalent representations of money demand is rejected at conventional significance levels. The statistic being negative suggests that Model 2 is to be preferred because it has the higher value of log-likelihood function at the maximum likelihood estimates. $\square$

6.6 Applications

Two applications are discussed in this section, both focussing on relaxing the assumption of normal disturbances in the nonlinear regression model. The first application is based on the capital asset pricing model (CAPM). A fat-tailed distribution is used to model outliers in the data and thus avoid bias in the parameter estimates of a regression model based on the assumption of normally distributed disturbances. The second application investigates the stochastic frontier model where the disturbance term is specified as a mixture of normal and non-normal terms.

6.6.1 Robust Estimation of the CAPM

One way to ensure that parameter estimates of the nonlinear regression model are robust to the presence of outliers is to use a heavy-tailed distribution such as the Student t distribution. This is a natural approach to modelling outliers since, by definition, an outlier represents an extreme draw from the tails of the distribution. The general idea is that the additional parameters of the heavy-tailed distribution capture the effects of the outliers and thereby help reduce any potential contamination of the parameter estimates that may arise from these outliers.

The approach can be demonstrated by means of the capital asset pricing model

$$r_t = \beta_0 + \beta_1 m_t + u_t, \quad u_t \sim N(0, \sigma^2),$$

where r_t is the return on the i^{th} asset relative to a risk-free rate and m_t is the return on the market portfolio relative to a risk-free rate. The parameter β_1 is of importance in finance because it provides a measure of the risk of the asset. Outliers in the data can properly be accounted for by specifying the model as

$$r_t = \beta_0 + \beta_1 m_t + \sigma \sqrt{\frac{\nu - 2}{\nu}} v_t, \quad (6.29)$$

where the disturbance term v_t now has a Student-t distribution given by

$$f(v_t) = \frac{\Gamma\left(\frac{\nu + 1}{2}\right)}{\sqrt{\pi\nu} \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{v_t^2}{\nu}\right)^{-(\nu+1)/2},$$

where ν is the degrees of freedom parameter and $\Gamma(\cdot)$ is the Gamma function. The term $\sigma\sqrt{(\nu - 2)/\nu}$ in equation (6.29) ensures that the variance of r_t is σ^2 , because the variance of a Student t distribution is $\nu/(\nu - 2)$.

The transformation of variable technique reveals that the distribution of r_t is

$$f(r_t) = f(v_t) \left| \frac{dv_t}{dr_t} \right| = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu} \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{v_t^2}{\nu}\right)^{-(\nu+1)/2} \left| \frac{1}{\sigma} \sqrt{\frac{\nu}{\nu-2}} \right|.$$

The log-likelihood function at observation t is therefore

$$\ln l_t(\theta) = \ln \left(\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu} \Gamma\left(\frac{\nu}{2}\right)} \right) - \frac{\nu+1}{2} \ln \left(1 + \frac{v_t^2}{\nu} \right) - \ln \sigma + \ln \sqrt{\frac{\nu}{\nu-2}}.$$

The parameters $\theta = \{\beta_0, \beta_1, \sigma^2, \nu\}$ are estimated by maximum likelihood using one of the iterative algorithms discussed in Section 6.3.

As an illustration, consider the monthly returns on the company Martin Marietta, over the period January 1982 to December 1986, taken from Butler, McDonald, Nelson and White (1990, pp.321-327). A scatter plot of the data in Figure 6.2 suggests that estimation of the CAPM by least squares may yield an estimate of β_1 that is biased upwards as a result of the outlier in r_t where the monthly excess return of the asset in one month is 0.688.

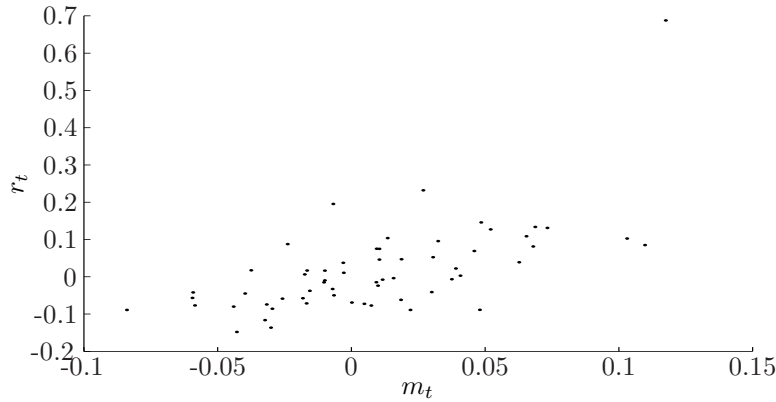


Figure 6.2 Scatter plot of the monthly returns on the company Martin Marietta and return on the market index, both relative to the risk free rate, over the period January 1982 to December 1986.

The results of estimating the CAPM by maximum likelihood assuming normal disturbances, are

$$\hat{r}_t = 0.001 + 1.803 m_t,$$

Table 6.1

Maximum likelihood estimates of the robust capital asset pricing model.
Standard errors based on the inverse of the Hessian.

Parameter	Estimate	Std error	t-stat.
β_0	-0.007	0.008	-0.887
β_1	1.263	0.190	6.665
σ^2	0.008	0.006	1.338
ν	2.837	1.021	2.779

where the estimates are obtained by simply regressing r_t on a constant and m_t . The estimate of 1.803 suggests that this asset is very risky relative to the market portfolio since on average changes in the asset returns amplify the contemporaneous movements in the market excess returns, m_t . A test of the hypothesis that $\beta_1 = 1$, provides a test that movements in the returns on the asset mirror the market one-to-one. The Wald statistic is

$$W = \left(\frac{1.803 - 1}{0.285} \right)^2 = 7.930.$$

The p-value of the statistic obtained from the χ_1^2 distribution is 0.000, showing strong rejection of the null hypothesis.

The maximum likelihood estimates of the robust version of the CAPM model are given in Table 6.1. The estimate of β_1 is now 1.263, which is much lower than the OLS estimate of 1.803. A Wald test of the hypothesis that $\beta_1 = 1$ now yields

$$W = \left(\frac{1.263 - 1}{0.190} \right)^2 = 1.930.$$

The p-value is 0.164 showing that the null hypothesis that the asset tracks the market one-to-one fails to be rejected.

The use of the Student-t distribution to model the outlier has helped to reduce the effect of the outlier on the estimate of β_1 . The degrees of freedom parameter estimate of $\hat{\nu} = 2.837$ shows that the tails of the distribution are indeed very fat, with just the first two moments of the distribution existing.

Another approach to estimate regression models that are robust to outliers is to specify the distribution as the Laplace distribution, also known as the

double exponential distribution

$$f(y_t; \theta) = \frac{1}{2} \exp[-|y_t - \theta|].$$

To estimate the unknown parameter θ , for a sample of size T , the log-likelihood function is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T f(y_t; \theta) = -\ln(2) - \frac{1}{T} \sum_{t=1}^T |y_t - \theta|.$$

In contrast to the log-likelihood functions dealt with thus far, this function is not differentiable everywhere. However, the maximum likelihood estimator can still be derived, which is given as the median of the data (Stuart and Ord, 1999, p. 59)

$$\hat{\theta} = \text{median}(y_t).$$

This result is a reflection of the well-known property that the median is less affected by outliers than is the mean. A generalization of this result forms the basis of the class of estimators known as M-estimators and quantile regression.

6.6.2 Stochastic Frontier Models

In stochastic frontier models the disturbance term u_t of a regression model is specified as a mixture of two random disturbances $u_{1,t}$ and $u_{2,t}$. The most widely used application of this model is in production theory where the production process is assumed to be affected by two types of shocks (Aigner, Lovell and Schmidt, 1977), namely,

- (1) idiosyncratic shocks, $u_{1,t}$, which are either positive or negative; and
- (2) technological shocks, $u_{2,t}$, which are either zero or negative, with a zero (negative) shock representing the production function operates efficiently (inefficiently).

Consider the stochastic frontier model

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t \\ u_t &= u_{1,t} - u_{2,t}, \end{aligned} \tag{6.30}$$

where u_t is a composite disturbance term with independent components, $u_{1,t}$ and $u_{2,t}$, with respective distributions

$$\begin{aligned} f(u_1) &= \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{u_1^2}{2\sigma_1^2}\right], & -\infty < u_1 < \infty, & \text{ [Normal]} \\ f(u_2) &= \frac{1}{\sigma_2} \exp\left[-\frac{u_2}{\sigma_2}\right], & 0 \leq u_2 < \infty, & \text{ [Exponential]} \end{aligned} \quad (6.31)$$

The distribution of u_t has support on the real line $(-\infty, \infty)$, but the effect of $-u_{2,t}$ is to skew the normal distribution to the left as highlighted in Figure 6.3. The strength of the asymmetry is controlled by the parameter σ_2 in the exponential distribution.

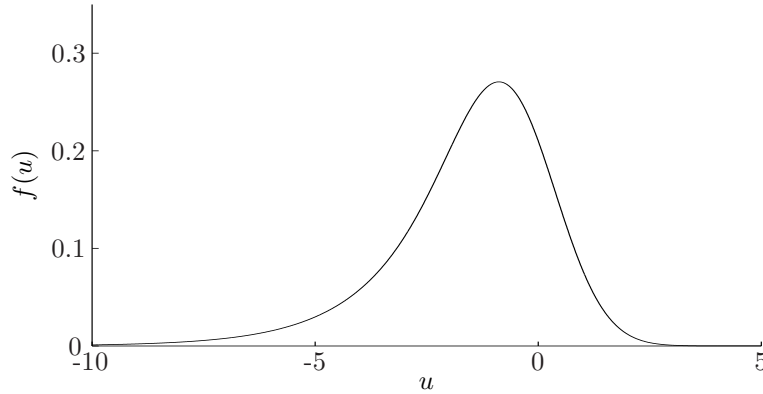


Figure 6.3 Stochastic frontier disturbance distribution as given by expression (6.36) based on a mixture of $N(0, \sigma_1^2)$ with standard deviation $\sigma_1 = 1$ and exponential distribution with standard deviation $\sigma_2 = 1.5$.

To estimate the parameters $\theta = \{\beta_0, \beta_1, \sigma_1, \sigma_2\}$ in (6.30) and (6.31) by maximum likelihood it is necessary to derive the distribution of y_t from u_t . Since u_t is a mixture distribution of two components, its distribution is derived from the joint distribution of $u_{1,t}$ and $u_{2,t}$ using the change of variable technique. However, because the model consists of mapping two random variables, $u_{1,t}$ and $u_{2,t}$, into one random variable u_t , it is necessary to choose an additional variable, v_t , to fill out the mapping for the Jacobian to be nonsingular. Once the joint distribution of (u_t, v_t) is derived, the marginal distribution of u_t is obtained by integrating the joint distribution with respect to v_t .

Let

$$u = u_1 - u_2, \quad v = u_1, \quad (6.32)$$

where the t subscript is excluded for convenience. To derive the Jacobian rearrange these equations as

$$u_1 = v, \quad u_2 = v - u, \quad (6.33)$$

so the Jacobian is

$$|J| = \begin{vmatrix} \frac{\partial u_1}{\partial u} & \frac{\partial u_1}{\partial v} \\ \frac{\partial u_2}{\partial u} & \frac{\partial u_2}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ -1 & 1 \end{vmatrix} = |1| = 1.$$

Using the property that $u_{1,t}$ and $u_{2,t}$ are independent and $|J| = 1$, the joint distribution of (u, v) is

$$\begin{aligned} g(u, v) &= |J| f(u_1) f(u_2) \\ &= |1| \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{u_1^2}{2\sigma_1^2}\right] \times \frac{1}{\sigma_2} \exp\left[-\frac{u_2}{\sigma_2}\right] \\ &= \frac{1}{\sqrt{2\pi\sigma_1^2}} \frac{1}{\sigma_2} \exp\left[-\frac{u_1^2}{2\sigma_1^2} - \frac{u_2}{\sigma_2}\right]. \end{aligned} \quad (6.34)$$

Using the substitution $u_1 = v$ and $u_2 = v - u$, the term in the exponent is

$$-\frac{v^2}{2\sigma_1^2} - \frac{v - u}{\sigma_2} = -\frac{v^2}{2\sigma_1^2} - \frac{v}{\sigma_2} + \frac{u}{\sigma_2} = -\frac{\left(v + \frac{\sigma_1^2}{\sigma_2}\right)^2}{2\sigma_1^2} + \frac{\sigma_1^2}{2\sigma_2^2} + \frac{u}{\sigma_2},$$

where the last step is based on completing the square. Placing this expression into (6.34) and rearranging gives the joint probability density

$$g(u, v) = \frac{1}{\sigma_2} \exp\left[\frac{\sigma_1^2}{2\sigma_2^2} + \frac{u}{\sigma_2}\right] \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left[-\frac{\left(v + \frac{\sigma_1^2}{\sigma_2}\right)^2}{2\sigma_1^2}\right]. \quad (6.35)$$

To derive the marginal distribution of u , as $v = u_1 = u + u_2$ and remembering that u_2 is positive, the range of integration of v is (u, ∞) because

$$\text{Lower: } u_2 = 0 \Rightarrow v = u, \quad \text{Upper: } u_2 > 0 \Rightarrow v > u.$$

The marginal distribution of u is now given by integrating out v in (6.35)

$$\begin{aligned}
 g(u) &= \int_u^\infty g(u, v) dv \\
 &= \frac{1}{\sigma_2} \exp \left[\frac{\sigma_1^2}{2\sigma_2^2} + \frac{u}{\sigma_2} \right] \int_u^\infty \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp \left[-\frac{\left(v + \frac{\sigma_1^2}{\sigma_2}\right)^2}{2\sigma_1^2} \right] dv \\
 &= \frac{1}{\sigma_2} \exp \left[\frac{\sigma_1^2}{2\sigma_2^2} + \frac{u}{\sigma_2} \right] \left(1 - \Phi \left(\frac{u + \frac{\sigma_1^2}{\sigma_2}}{\sigma_1} \right) \right) \\
 &= \frac{1}{\sigma_2} \exp \left[\frac{\sigma_1^2}{2\sigma_2^2} + \frac{u}{\sigma_2} \right] \Phi \left(-\frac{u + \frac{\sigma_1^2}{\sigma_2}}{\sigma_1} \right), \tag{6.36}
 \end{aligned}$$

where $\Phi(\cdot)$ is the cumulative normal distribution function and the last step follows from the symmetry property of the normal distribution. Finally, the distribution in terms of y conditional on x_t is given by using (6.30) to substitute out u in (6.36)

$$g(y|x_t) = \frac{1}{\sigma_2} \exp \left[\frac{\sigma_1^2}{2\sigma_2^2} + \frac{(y - \beta_0 - \beta_1 x_t)}{\sigma_2} \right] \Phi \left(-\frac{y_t - \beta_0 - \beta_1 x_t - \frac{\sigma_1^2}{\sigma_2}}{\sigma_1} \right). \tag{6.37}$$

Using expression (6.37) the log-likelihood function for a sample of T observations is

$$\begin{aligned}
 \ln L_T(\theta) &= \frac{1}{T} \sum_{t=1}^T \ln g(y_t|x_t) \\
 &= -\ln \sigma_2 + \frac{\sigma_1^2}{2\sigma_2^2} + \frac{1}{\sigma_2 T} \sum_{t=1}^T (y_t - \beta_0 - \beta_1 x_t) \\
 &\quad + \frac{1}{T} \sum_{t=1}^T \ln \Phi \left(-\frac{y_t - \beta_0 - \beta_1 x_t - \frac{\sigma_1^2}{\sigma_2}}{\sigma_1} \right).
 \end{aligned}$$

This expression is nonlinear in the parameter θ and can be maximized using an iterative algorithm.

A Monte Carlo experiment is performed to investigate the properties of the maximum likelihood estimator of the stochastic frontier model in (6.30) and (6.31). The parameters are $\theta_0 = \{\beta_0 = 1, \beta_1 = 0.5, \sigma_1 = 1.0, \sigma_2 = 1.5\}$, the explanatory variable is $x_t \sim iid N(0, 1)$, the sample size is $T = 1000$ and the number of replications is 5000. The dependent variable, y_t , is simulated using the inverse cumulative density technique. This involves computing the cumulative density function of u from its marginal distribution in (6.37) for a grid of values of u ranging from -10 to 5 . Uniform random variables are then drawn to obtain draws of u_t which are added to $\beta_0 + \beta_1 x_t$ to obtain a draw of y_t .

Table 6.2

Bias and mean square error (MSE) of the maximum likelihood estimator of the stochastic frontier model in (6.30) and (6.31). Based on samples of size $T = 1000$ and 5000 replications.

Parameter	True	Mean	Bias	MSE
β_0	1.0000	0.9213	-0.0787	0.0133
β_1	0.5000	0.4991	-0.0009	0.0023
σ_1	1.0000	1.0949	0.0949	0.0153
σ_2	1.5000	1.3994	-0.1006	0.0184

The results of the Monte Carlo experiment are given in Table 6.2 which reports the bias and mean square error, respectively, for each parameter. The estimate of β_0 is biased downwards by about 8% while the slope estimate of β_1 exhibits no bias at all. The estimates of the standard deviations exhibit bias in different directions with the estimate of σ_1 biased upwards and the estimate of σ_2 biased downwards.

6.7 Exercises

(1) Simulating Exponential Models

Gauss file(s)	nls_simulate.g
Matlab file(s)	nls_simulate.m

Simulate the following exponential models

$$y_{1,t} = \beta_0 \exp[\beta_1 x_t] + u_t$$

$$y_{2,t} = \beta_0 \exp[\beta_1 x_t + u_t],$$

for a sample size of $T = 50$, where the explanatory variable and the disturbance term are, respectively,

$$u_t \sim iid N(0, \sigma^2), \quad x_t \sim t, \quad t = 0, 1, 2, \dots$$

Set the parameters to be $\beta_0 = 1.0$, $\beta_1 = 0.05$, and $\sigma = 0.5$. Plot the series and compare their time-series properties.

(2) **Estimating the Exponential Model by Maximum Likelihood**

Gauss file(s)	<code>nls_exponential.g</code>
Matlab file(s)	<code>nls_exponential.m</code>

Simulate the model

$$y_t = \beta_0 \exp[\beta_1 x_t] + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

for a sample size of $T = 50$, where the explanatory variable, the disturbance term and the parameters are as defined in Exercise 1.

- Use the Newton-Raphson algorithm to estimate the parameters $\theta = \{\beta_0, \beta_1, \sigma^2\}$, by concentrating out σ^2 . Choose as starting values $\beta_0 = 0.1$ and $\beta_1 = 0.1$.
- Compute the standard errors of $\hat{\beta}_0$ and $\hat{\beta}_1$ based on the Hessian.
- Estimate the parameters of the model without concentrating the log-likelihood function with respect to σ^2 and compute the standard errors of $\hat{\beta}_0$, $\hat{\beta}_1$ and $\hat{\sigma}^2$, based on the Hessian.

(3) **Estimating the Exponential Model by Gauss-Newton**

Gauss file(s)	<code>nls_exponential_gn.g</code>
Matlab file(s)	<code>nls_exponential_gn.m</code>

Simulate the model

$$y_t = \beta_0 \exp[\beta_1 x_t] + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

for a sample size of $T = 50$, where the explanatory variable and the disturbance term and the parameters are as defined in Exercise 1.

- Use the Gauss-Newton algorithm to estimate the parameters $\theta = \{\beta_0, \beta_1, \sigma^2\}$. Choose as starting values $\beta_0 = 0.1$ and $\beta_1 = 0.1$.
- Compute the standard errors of $\hat{\beta}_0$ and $\hat{\beta}_1$ and compare these estimates with those obtained using the Hessian in Exercise 2.

(4) Nonlinear Consumption Function

Gauss file(s)	nls_conest.g, nls_contest.g
Matlab file(s)	nls_conest.m, nls_contest.m

This exercise is based on U.S. quarterly data for real consumption expenditure and real disposable personal income for the period 1960:Q1 to 2009:Q4, downloaded from the Federal Reserve Bank of St. Louis. Consider the nonlinear consumption function

$$c_t = \beta_0 + \beta_1 y_t^{\beta_2} + u_t, \quad u_t \sim iid N(0, \sigma^2).$$

- (a) Estimate a linear consumption function by setting $\beta_2 = 1$.
- (b) Estimate the unrestricted nonlinear consumption function using the Gauss-Newton algorithm. Choose the linear parameter estimates computed in part (a) for β_0 and β_1 and $\beta_2 = 1$ as the starting values.
- (c) Test the hypotheses

$$H_0 : \beta_2 = 1 \quad H_1 : \beta_2 \neq 1,$$

using a LR test, a Wald test and a LM test.

(5) Nonlinear Regression

Consider the nonlinear regression model

$$y_t^{\beta_2} = \beta_0 + \beta_1 x_t + u_t, \quad u_t \sim iid N(0, 1).$$

- (a) Write down the distributions of u_t and y_t .
- (b) Show how you would estimate this model's parameters by maximum likelihood using:
 - (i) the Newton-Raphson algorithm; and
 - (ii) the BHHH algorithm.
- (c) Briefly discuss why the Gauss-Newton algorithm is not appropriate in this case.
- (d) Construct a test of the null hypothesis $\beta_2 = 1$, using:
 - (i) a LR test;
 - (ii) a Wald test;
 - (iii) a LM test with the information matrix based on the outer product of gradients; and
 - (iv) a LM test based on two linear regressions.

(6) **Vuong's Nonnested Test of Money Demand**

Gauss file(s)	nls_money.g
Matlab file(s)	nls_money.m

This exercise is based on quarterly data for the U.S. on real money, m_t , the nominal interest rate, r_t , and real income, y_t , for the period 1959 to 2005.

Consider the following nonnested money demand equations

$$\begin{aligned} \text{Model 1: } m_t &= \beta_0 + \beta_1 r_t + \beta_2 y_t + u_{1,t} \\ u_{1,t} &\sim iid N(0, \sigma_1^2) \end{aligned}$$

$$\begin{aligned} \text{Model 2: } \ln m_t &= \alpha_0 + \alpha_1 \ln r_t + \alpha_2 \ln y_t + u_{2,t} \\ u_{2,t} &\sim iid N(0, \sigma_2^2). \end{aligned}$$

- (a) Estimate Model 1 by regressing m_t on $\{c, r_t, y_t\}$ and construct the log-likelihood at each observation

$$\ln l_{1,t} = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\hat{\sigma}_1^2) - \frac{(m_t - \hat{\beta}_0 - \hat{\beta}_1 r_t - \hat{\beta}_2 y_t)^2}{2\hat{\sigma}_1^2}.$$

- (b) Estimate Model 2 by regressing $\ln m_t$ on $\{c, \ln r_t, \ln y_t\}$ and construct the log-likelihood function of the transformed distribution at each observation

$$\begin{aligned} \ln l_{2,t} = & -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\hat{\sigma}_2^2) - \frac{(\ln m_t - \hat{\alpha}_0 - \hat{\alpha}_1 \ln r_t - \hat{\alpha}_2 \ln y_t)^2}{2\hat{\sigma}_2^2} \\ & - \ln m_t. \end{aligned}$$

- (c) Perform Vuong's nonnested test and interpret the result.

(7) **Robust Estimation of the CAPM**

Gauss file(s)	nls_capm.g
Matlab file(s)	nls_capm.m

This exercise is based on monthly returns data on the company Martin Marietta from January 1982 to December 1986. The data are taken from Butler *et. al.* (1990, pp.321-327).

- (a) Identify any outliers in the data by using a scatter plot of r_t against m_t .

- (b) Estimate the following CAPM model

$$r_t = \beta_0 + \beta_1 m_t + u_t, \quad u_t \sim iid N(0, \sigma^2),$$

and interpret the estimate of β_1 . Test the hypothesis that $\beta_1 = 1$.

- (c) Estimate the following CAPM model

$$r_t = \beta_0 + \beta_1 m_t + \sigma \sqrt{\frac{\nu - 2}{\nu}} v_t, \quad v_t \sim \text{Student } t(0, \nu),$$

and interpret the estimate of β_1 . Test the hypothesis that $\beta_1 = 1$.

- (d) Compare the parameter estimates of $\{\beta_0, \beta_1\}$ in parts (b) and (c) and discuss the robustness properties of these estimates.
- (e) An alternative approach to achieving robustness is to exclude any outliers from the data set and re-estimate the model by OLS using the trimmed data set. A common way to do this is to compute the standardized residual

$$z_t = \frac{\hat{u}_t}{s^2 \text{diag}(I - X(X'X)^{-1}X')},$$

where $\hat{u}_t$ is the least squares residual using all of the data and s^2 is the residual variance. The standardized residual is approximately distributed as $N(0, 1)$, with absolute values in excess of 3 representing extreme observations. Compare the estimates of $\{\beta_0, \beta_1\}$ using the trimmed data approach with those obtained in parts (b) and (c). Hence discuss the role of the degrees of freedom parameter ν in achieving robust parameter estimates to outliers.

- (f) Construct a Wald test of normality based on the CAPM equation assuming Student t errors.

(8) Stochastic Frontier Model

Gauss file(s)	<code>nls_frontier.g</code>
Matlab file(s)	<code>nls_frontier.m</code>

The stochastic frontier model is

$$y_t = \beta_0 + \beta_1 x_t + u_t$$

$$u_t = u_{1,t} - u_{2,t},$$

where $u_{1,t}$ and $u_{2,t}$ are distributed as normal and exponential as defined in (6.31), with standard deviations σ_1 and σ_2 , respectively.

- (a) Use the change of variable technique to show that

$$g(u) = \frac{1}{\sigma_2} \exp \left[\frac{\sigma_1^2}{2\sigma_2^2} + \frac{u}{\sigma_2} \right] \Phi \left(-\frac{u + \frac{\sigma_1^2}{\sigma_2}}{\sigma_1} \right).$$

Plot the distribution and discuss its shape.

- (b) Choose the parameter values $\theta_0 = \{\beta_0 = 1, \beta_1 = 0.5, \sigma_1 = 1.0, \sigma_2 = 1.5\}$. Use the inverse cumulative density technique to simulate u_t , by computing its cumulative density function from its marginal distribution in part (a) for a grid of values of u_t ranging from -10 to 5 and then drawing uniform random numbers to obtain draws of u_t .
- (c) Investigate the sampling properties of the maximum likelihood estimator using a Monte Carlo experiment based on the parameters in part (b), $x_t \sim N(0, 1)$, with $T = 1000$ and 5000 replications.
- (d) Repeat parts (a) to (c) where now the disturbance is $u_t = u_{1,t} + u_{2,t}$ with density function

$$g(u) = \frac{1}{\sigma_2} \exp \left[\frac{\sigma_1^2}{2\sigma_2^2} - \frac{u}{\sigma_2} \right] \Phi \left(\frac{u - \frac{\sigma_1^2}{\sigma_2}}{\sigma_1} \right).$$

- (e) Let $u_t = u_{1,t} - u_{2,t}$, where u_t is normal but now $u_{2,t}$ is half-normal

$$f(u_2) = \frac{2}{\sqrt{2\pi\sigma_2^2}} \exp \left[-\frac{u_2^2}{2\sigma_2^2} \right], \quad 0 \leq u_2 < \infty.$$

Repeat parts (a) to (c) by defining $\sigma^2 = \sigma_1^2 + \sigma_2^2$ and $\lambda = \sigma_2/\sigma_1$, hence show that

$$g(u) = \sqrt{\frac{2}{\pi}} \frac{1}{\sigma} \exp \left[-\frac{u^2}{2\sigma^2} \right] \Phi \left(-\frac{u\lambda}{\sigma} \right).$$

Autocorrelated Regression Models

7.1 Introduction

An important feature of the regression models presented in Chapters 5 and 6 is that the disturbance term is assumed to be independent across time. This assumption is now relaxed and the resultant models are referred to as autocorrelated regression models. The aim of this chapter is to use the maximum likelihood framework set up in Part ONE to estimate and test autocorrelated regression models. The structure of the autocorrelation may be autoregressive, moving average or a combination of the two. Both single equation and multiple equation models are analyzed.

Significantly, the maximum likelihood estimator of the autocorrelated regression model nests a number of other estimators, including conditional maximum likelihood, Gauss-Newton, Zig-zag algorithms and the Cochrane-Orcutt procedure. Tests of autocorrelation are derived in terms of the LR, Wald and LM tests set out in Chapter 4. In the case of LM tests of autocorrelation, the statistics are shown to be equivalent to a number of diagnostic test statistics widely used in econometrics.

7.2 Specification

In Chapter 5, the focus is on estimating and testing linear regression models of the form

$$y_t = \beta_0 + \beta_1 x_t + u_t, \quad (7.1)$$

where y_t is the dependent variable, x_t is the explanatory variable and u_t is the disturbance term assumed to be independently and identically distributed. For a sample of $t = 1, 2, \dots, T$ observations, the joint density function of

this model is

$$f(y_1, y_2, \dots, y_T | x_1, x_2, \dots, x_T; \theta) = \prod_{t=1}^T f(y_t | x_t; \theta), \quad (7.2)$$

where θ is the vector of parameters to be estimated.

The assumption that u_t in (7.1) is independent is now relaxed by augmenting the model to include an equation for u_t that is a function of information at time $t - 1$. Common parametric specifications of the disturbance term are the autoregressive (AR) models and moving average (MA) models

1. AR(1) : $u_t = \rho_1 u_{t-1} + v_t$
2. AR(p) : $u_t = \rho_1 u_{t-1} + \rho_2 u_{t-2} + \dots + \rho_p u_{t-p} + v_t$
3. MA(1) : $u_t = v_t + \delta_1 v_{t-1}$
4. MA(q) : $u_t = v_t + \delta_1 v_{t-1} + \delta_2 v_{t-2} + \dots + \delta_q v_{t-q}$
5. ARMA(p, q) : $u_t = \sum_{i=1}^p \rho_i u_{t-i} + v_t + \sum_{i=1}^q \delta_i v_{t-i}$,

where v_t is independently and identically distributed with zero mean and constant variance σ^2 .

A characteristic of autocorrelated regression models is that a shock at time t , as represented by v_t , has an immediate effect on y_t and continues to have an effect at times $t + 1$, $t + 2$, etc. This suggests that the conditional mean in equation (7.1), $\beta_0 + \beta_1 x_t$, underestimates y for some periods and overestimates it for other periods.

Example 7.1 A Regression Model with Autocorrelation Figure 7.1 panel (a) gives a scatter plot of simulated data for a sample of $T = 200$ observations from the following regression model with an AR(1) disturbance term

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t \\ u_t &= \rho_1 u_{t-1} + v_t \\ v_t &\sim iid N(0, \sigma^2), \end{aligned}$$

with $\beta_0 = 2$, $\beta_1 = 1$, $\rho_1 = 0.95$, $\sigma = 3$ and the explanatory variable is generated as $x_t = 0.5t + N(0, 1)$. For comparative purposes, the conditional mean of y_t , $\beta_0 + \beta_1 x_t$, is also plotted. This figure shows that there are periods when the conditional mean, μ_t , consistently underestimates y_t and other periods when it consistently overestimates y_t . A similar pattern, although less pronounced than that observed in panel (a), occurs in Figure 7.1 panel

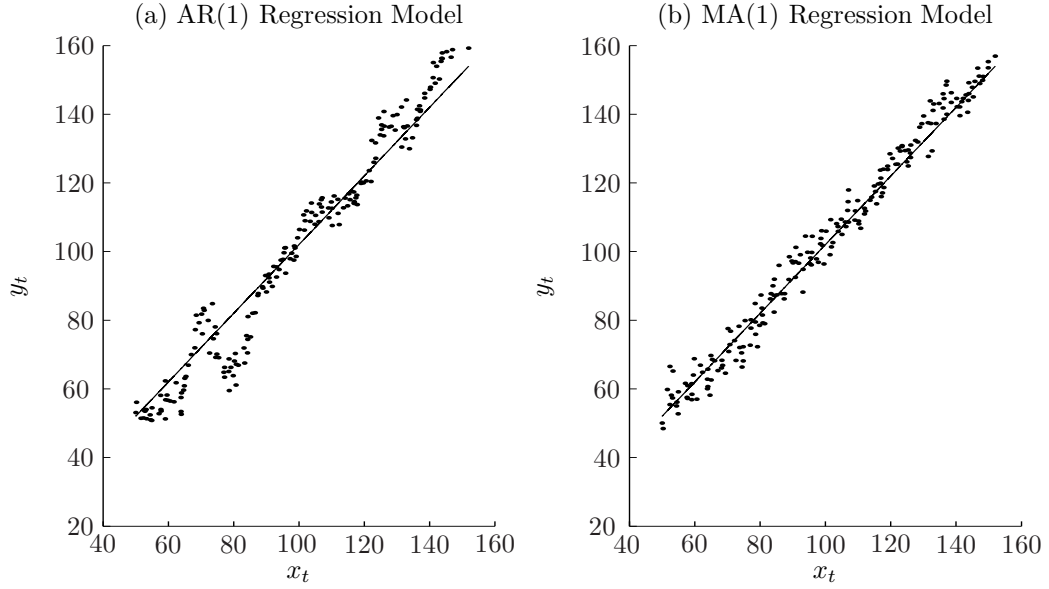


Figure 7.1 Scatter plots of the simulated data from the regression model with an autocorrelated disturbance.

(b), where the disturbance is MA(1)

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t \\ u_t &= v_t + \delta_1 v_{t-1} \\ v_t &\sim iid N(0, \sigma^2), \end{aligned}$$

where x_t is as before and $\beta_0 = 2$, $\beta_1 = 1$, $\delta_1 = 0.95$, $\sigma = 3$. □

7.3 Maximum Likelihood Estimation

From Chapter 1, the joint pdf of $y_1, y_2, \dots, y_T$ dependent observations is

$$\begin{aligned} f(y_1, y_2, \dots, y_T | x_1, x_2, \dots, x_T; \theta) &= f(y_s, y_{s-1}, \dots, y_1 | x_s, x_{s-1}, \dots, x_1; \theta) \\ &\times \prod_{t=s+1}^T f(y_t | y_{t-1}, \dots, x_t, x_{t-1}, \dots; \theta), \end{aligned} \quad (7.3)$$

where $\theta = \{\beta_0, \beta_1, \rho_1, \rho_2, \dots, \rho_p, \delta_1, \delta_2, \dots, \delta_q, \sigma^2\}$ and $s = \max(p, q)$. The first term in equation (7.3) represents the marginal distribution of $y_s, y_{s-1}, \dots, y_1$, while the second term contains the sequence of conditional distributions of y_t . When both terms in the likelihood function in equation (7.3) are used,

the estimator is also known as the exact maximum likelihood estimator. By contrast, when only the second term of equation (7.3) is used the estimator is known as the conditional maximum likelihood estimator. These two estimators are discussed in more detail below.

7.3.1 Exact Maximum Likelihood

From equation (7.3), the log-likelihood function for exact maximum likelihood estimation is

$$\begin{aligned} \ln L_T(\theta) &= \frac{1}{T} \ln f(y_s, y_{s-1}, \dots, y_1 | x_s, x_{s-1}, \dots, x_1; \theta) \\ &\quad + \frac{1}{T} \sum_{t=s+1}^T \ln f(y_t | y_{t-1}, \dots, x_t, x_{t-1}, \dots; \theta), \end{aligned} \quad (7.4)$$

that is to be maximised by choice of the unknown parameters θ . The log-likelihood function is normally nonlinear in θ and must be maximised using one of the algorithms presented in Chapter 3.

Example 7.2 AR(1) Regression Model Consider the model

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t \\ u_t &= \rho_1 u_{t-1} + v_t \\ v_t &\sim iid N(0, \sigma^2), \end{aligned}$$

where $\theta = \{\beta_0, \beta_1, \rho_1, \sigma^2\}$. The distribution of v is

$$f(v) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{v^2}{2\sigma^2} \right].$$

The conditional distribution of u_t for $t > 1$, is

$$f(u_t | u_{t-1}; \theta) = f(v_t) \left| \frac{dv_t}{du_t} \right| = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(u_t - \rho_1 u_{t-1})^2}{2\sigma^2} \right],$$

because $|dv_t/du_t| = 1$ and $v_t = u_t - \rho_1 u_{t-1}$. Consequently, the conditional distribution of y_t for $t > 1$ is

$$f(y_t | x_t, x_{t-1}; \theta) = f(u_t) \left| \frac{du_t}{dy_t} \right| = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(u_t - \rho_1 u_{t-1})^2}{2\sigma^2} \right],$$

because $|du_t/dy_t| = 1$, $u_t = y_t - \beta_0 - \beta_1 x_t$ and $u_{t-1} = y_{t-1} - \beta_0 - \beta_1 x_{t-1}$. To derive the marginal distribution of u_t at $t = 1$, use the result that for the AR(1) model with $u_t = \rho_1 u_{t-1} + v_t$, where $v_t \sim N(0, \sigma^2)$, the marginal

distribution of u_t is $N(0, \sigma^2/(1 - \rho_1^2))$. The marginal distribution of u_1 is, therefore,

$$f(u_1) = \frac{1}{\sqrt{2\pi\sigma^2/(1 - \rho_1^2)}} \exp \left[-\frac{(u_1 - 0)^2}{2\sigma^2/(1 - \rho_1^2)} \right],$$

so that the marginal distribution of y_1 is

$$f(y_1 | x_1; \theta) = f(u_1) \left| \frac{du_1}{dy_1} \right| = \frac{1}{\sqrt{2\pi\sigma^2/(1 - \rho_1^2)}} \exp \left[-\frac{(y_1 - \beta_0 - \beta_1 x_1)^2}{2\sigma^2/(1 - \rho_1^2)} \right],$$

because $|du_1/dy_1| = 1$, and $u_1 = y_1 - \beta_0 - \beta_1 x_1$. It follows, therefore, that the joint probability distribution of y_t is

$$f(y_1, y_2, \dots, y_T | x_1, x_2, \dots, x_T; \theta) = f(y_1 | x_1; \theta) \times \prod_{t=2}^T f(y_t | y_{t-1}, x_t, x_{t-1}; \theta),$$

and the log-likelihood function is

$$\begin{aligned} \ln L_T(\theta) &= \frac{1}{T} \ln f(y_1 | x_1; \theta) + \frac{1}{T} \sum_{t=2}^T \ln f(y_t | y_{t-1}, x_t, x_{t-1}; \theta) \\ &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 + \frac{1}{T} \ln(1 - \rho_1^2) - \frac{1}{2T} \frac{(y_1 - \beta_0 - \beta_1 x_1)^2}{\sigma^2/(1 - \rho_1^2)} \\ &\quad - \frac{1}{2\sigma^2 T} \sum_{t=2}^T (y_t - \rho_1 y_{t-1} - \beta_0(1 - \rho_1) - \beta_1(x_t - \rho_1 x_{t-1}))^2. \end{aligned}$$

This expression shows that the log-likelihood function is a nonlinear function of the parameters. $\square$

7.3.2 Conditional Maximum Likelihood

The maximum likelihood example presented above is for a regression model with an AR(1) disturbance term. Estimation of the regression model with an ARMA(p, q) disturbance term is more difficult, however, since it requires deriving the marginal distribution of $f(y_1, y_2, \dots, y_s)$, where $s = \max(p, q)$. One solution is to ignore this term, in which case the log-likelihood function in (7.4) is taken with respect to an average of the log-likelihoods corresponding to the conditional distributions from $s + 1$ onwards

$$\ln L_T(\theta) = \frac{1}{T - s} \sum_{t=s+1}^T \ln f(y_t | y_{t-1}, \dots, y_{t-s}, x_t, x_{t-1}, \dots, x_{t-s}; \theta). \quad (7.5)$$

As the likelihood is now constructed by treating the first s observations as fixed, estimates based on maximizing this likelihood are referred to as conditional maximum likelihood estimates. Asymptotically the exact and conditional maximum likelihood estimators are equivalent because the contribution of $\ln f(y_s, y_{s-1}, \dots, y_1 | x_s, x_{s-1}, \dots, x_1; \theta)$ to the overall log-likelihood function vanishes for $T \rightarrow \infty$.

Example 7.3 AR(2) Regression Model Consider the model

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t \\ u_t &= \rho_1 u_{t-1} + \rho_2 u_{t-2} + v_t \\ v_t &\sim N(0, \sigma^2). \end{aligned}$$

The conditional log-likelihood function is constructed by computing

$$\begin{aligned} u_t &= y_t - \beta_0 - \beta_1 x_t, & t &= 1, 2, \dots, T \\ v_t &= u_t - \rho_1 u_{t-1} - \rho_2 u_{t-2}, & t &= 3, 4, \dots, T, \end{aligned}$$

where the parameters are replaced by starting values $\theta_{(0)}$. The conditional log-likelihood function is then computed as

$$\ln L_T(\theta) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2(T-2)} \sum_{t=3}^T v_t^2.$$

□

In evaluating the conditional log-likelihood function for ARMA(p, q) models, it is necessary to choose starting values for the first q values of v_t . A common choice is $v_1 = v_2 = \dots v_q = 0$.

Example 7.4 ARMA(1,1) Regression Model Consider the model

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t \\ u_t &= \rho_1 u_{t-1} + v_t + \delta_1 v_{t-1} \\ v_t &\sim iid N(0, \sigma^2). \end{aligned}$$

The conditional log-likelihood is constructed by computing

$$\begin{aligned} u_t &= y_t - \beta_0 - \beta_1 x_t, & t &= 1, 2, \dots, T \\ v_t &= u_t - \rho_1 u_{t-1} - \delta_1 v_{t-1}, & t &= 2, 3, \dots, T, \end{aligned}$$

with $v_1 = 0$ and where the parameters are replaced by starting values $\theta_{(0)}$. The conditional log-likelihood function is then

$$\ln L_T(\theta) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2(T-1)} \sum_{t=2}^T v_t^2.$$

□

Example 7.5 Dynamic Model of U.S. Investment This example uses quarterly data for the U.S. from March 1957 to September 2010 to estimate the following model of investment

$$\begin{aligned}dri_{t-1} &= \beta_0 + \beta_1 dry_t + \beta_2 rint_t + u_t \\ u_t &= \rho_1 u_{t-1} + v_t \\ v_t &\sim iid N(0, \sigma^2),\end{aligned}$$

where dri_t is the quarterly percentage change in real investment, dry_t is the quarterly percentage change in real income, $rint_t$ is the real interest rate expressed as a quarterly percentage, and the parameters are $\theta = \{\beta_0, \beta_1, \beta_2, \rho_1, \sigma^2\}$. The sample begins in June 1957 as one observation is lost from constructing the variables, resulting in a sample of size $T = 214$. The log-likelihood function is constructed by computing

$$\begin{aligned}u_t &= dri_t - \beta_0 - \beta_1 dry_t - \beta_2 rint_t, & t = 1, 2, \dots, T \\ v_t &= u_t - \rho_1 u_{t-1} & t = 2, 3, \dots, T,\end{aligned}$$

where the parameters are replaced by the starting parameter values $\theta_{(0)}$. The log-likelihood function at $t = 1$ is

$$\ln l_1(\theta) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 + \frac{1}{2} \ln(1 - \rho_1^2) - \frac{(u_1 - 0)^2}{2\sigma^2/(1 - \rho_1^2)},$$

while for $t > 1$ it is

$$\ln l_t(\theta) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 - \frac{v_t^2}{2\sigma^2}.$$

The exact maximum likelihood estimates of the investment model are given in Table 7.1 under the heading Exact. The iterations are based on the Newton-Raphson algorithm with all derivatives computed numerically. The standard errors reported are computed using the negative of the inverse of the Hessian. All parameter estimates are statistically significant at the 5% level with the exception of the estimate of ρ_1 . The conditional maximum likelihood estimates which are also given in Table 7.1, yield qualitatively similar results to the exact maximum likelihood estimates. □

7.4 Alternative Estimators

Under certain conditions, the maximum likelihood estimator of the autocorrelated regression model nests a number of other estimation methods as special cases.

Table 7.1

Maximum likelihood estimates of the investment model using the Newton-Raphson algorithm with derivatives computed numerically. Standard errors are based on the Hessian.

Parameter	Exact			Conditional		
	Estimate	SE	t-stat	Estimate	SE	t-stat
β_0	-0.281	0.157	-1.788	-0.275	0.159	-1.733
β_1	1.570	0.130	12.052	1.567	0.131	11.950
β_2	-0.332	0.165	-2.021	-0.334	0.165	-2.023
ρ_1	0.090	0.081	1.114	0.091	0.081	1.125
σ^2	2.219	0.215	10.344	2.229	0.216	10.320
$\ln L_T(\hat{\theta})$	-1.817			-1.811		

7.4.1 Gauss-Newton

The exact and conditional maximum likelihood estimators of the autocorrelated regression model discussed above are presented in terms of the Newton-Raphson algorithm with the derivatives computed numerically. In the case of the conditional likelihood constructing analytical derivatives is straightforward. As the log-likelihood function is based on the normal distribution, the variance of the disturbance, σ^2 , can be concentrated out and the nonlinearities arising from the contribution of the marginal distribution of y_1 are no longer present. Once the Newton-Raphson algorithm is re-expressed in terms of analytical derivatives, it reduces to a sequence of least squares regressions known as the Gauss-Newton algorithm.

To motivate the form of the Gauss-Newton algorithm reconsider the AR(1) regression model in Example 7.2. The first-order derivatives of the log-likelihood function with respect to the parameters $\theta = \{\beta_0, \beta_1, \rho_1, \sigma^2\}$ are

$$\begin{aligned}
 \frac{\partial \ln L_T(\theta)}{\partial \beta_0} &= \frac{1}{\sigma^2(T-1)} \sum_{t=2}^T v_t(1 - \rho_1) \\
 \frac{\partial \ln L_T(\theta)}{\partial \beta_1} &= \frac{1}{\sigma^2(T-1)} \sum_{t=2}^T v_t(x_t - \rho_1 x_{t-1}) \\
 \frac{\partial \ln L_T(\theta)}{\partial \rho_1} &= \frac{1}{\sigma^2(T-1)} \sum_{t=2}^T v_t u_{t-1} \\
 \frac{\partial \ln L_T(\theta)}{\partial \sigma^2} &= -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4(T-1)} \sum_{t=2}^T v_t^2.
 \end{aligned} \tag{7.6}$$

The direct second derivatives are

$$\begin{aligned}
\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_0^2} &= -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T (1 - \rho_1)^2 \\
\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_1^2} &= -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T (x_t - \rho_1 x_{t-1})^2 \\
\frac{\partial^2 \ln L_T(\theta)}{\partial \rho_1^2} &= -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T u_{t-1}^2 \\
\frac{\partial^2 \ln L_T(\theta)}{\partial (\sigma^2)^2} &= \frac{1}{2\sigma^4(T-1)} - \frac{1}{\sigma^6} \sum_{t=2}^T v_t^2,
\end{aligned} \tag{7.7}$$

and cross derivative are

$$\begin{aligned}
\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_0 \partial \beta_1} &= -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T (x_t - \rho_1 x_{t-1})(1 - \rho_1) \\
\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_0 \partial \rho_1} &= -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T (u_{t-1}(1 - \rho_1) + v_t) \\
\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_0 \partial \sigma^2} &= -\frac{1}{\sigma^4(T-1)} \sum_{t=2}^T v_t(1 - \rho_1) \\
\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_1 \partial \rho_1} &= -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T (u_{t-1}(x_t - \rho_1 x_{t-1}) + v_t x_{t-1}) \\
\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_1 \partial \sigma^2} &= -\frac{1}{\sigma^4(T-1)} \sum_{t=2}^T v_t(x_t - \rho_1 x_{t-1}) \\
\frac{\partial^2 \ln L_T(\theta)}{\partial \rho_1 \partial \sigma^2} &= -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T v_t u_{t-1}.
\end{aligned} \tag{7.8}$$

Setting $\partial \ln L_T(\theta) / \partial \sigma^2 = 0$ and rearranging gives the solution

$$\hat{\sigma}^2 = \frac{1}{T-1} \sum_{t=2}^T v_t^2, \tag{7.9}$$

which is used to concentrate σ^2 out of the log-likelihood function. The iterations are now expressed only in terms of the parameters β_0 , β_1 and ρ_1

$$\begin{bmatrix} \beta_0 \\ \beta_1 \\ \rho_1 \end{bmatrix}_{(k)} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \rho_1 \end{bmatrix}_{(k-1)} - H_{(k-1)}^{-1} G_{(k-1)}. \tag{7.10}$$

The gradient, obtained from (7.6), is

$$G_T(\theta) = \frac{1}{\sigma^2 T} \sum_{t=2}^T \begin{bmatrix} 1 - \rho_1 \\ x_t - \rho_1 x_{t-1} \\ u_{t-1} \end{bmatrix} v_t, \tag{7.11}$$

and the Hessian is based on the terms in equations (7.6) - (7.8). The Hessian can be simplified further by using the results $\sum_{t=1}^T v_t = 0$ and $\sum_{t=1}^T v_t x_t = 0$, which follow from the first-order conditions, so that

$$H_T(\theta) = -\frac{1}{\sigma^2 T} \sum_{t=2}^T \begin{bmatrix} 1 - \rho_1 \\ x_t - \rho_1 x_{t-1} \\ u_{t-1} \end{bmatrix} \begin{bmatrix} 1 - \rho_1 & x_t - \rho_1 x_{t-1} & u_{t-1} \end{bmatrix}. \quad (7.12)$$

Substituting expressions (7.11) and (7.12) in equation (7.10) gives

$$\begin{bmatrix} \beta_0 \\ \beta_1 \\ \rho_1 \end{bmatrix}_{(k)} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \rho_1 \end{bmatrix}_{(k-1)} + \left[\sum_{t=2}^T z_t z_t' \right]^{-1} \left[\sum_{t=2}^T z_t v_t \right], \quad (7.13)$$

where $z_t = [1 - \rho_1, x_t - \rho_1 x_{t-1}, u_{t-1}]'$ and all the parameters in z_t and v_t are evaluated at $\theta_{(k-1)}$. This updating algorithm has two important features. First, it does not contain σ^2 because it is canceled out in the calculation of $H_T^{-1}(\theta)G_T(\theta)$. Second, the updating part of the algorithm is equivalent to regressing v_t on the z_t .

A simpler way to motivate the Gauss-Newton algorithm in (7.13) is to express the model in terms of v_t as the dependent variable

$$v_t = u_t - \rho_1 u_{t-1} = (y_t - \beta_0 - \beta_1 x_t) - \rho_1 (y_{t-1} - \beta_0 - \beta_1 x_{t-1}),$$

and construct the derivatives

$$\begin{aligned} z_{1,t} &= -\frac{\partial v_t}{\partial \beta_0} = 1 - \rho_1 \\ z_{2,t} &= -\frac{\partial v_t}{\partial \beta_1} = x_t - \rho_1 x_{t-1} \\ z_{3,t} &= -\frac{\partial v_t}{\partial \rho_1} = u_{t-1}, \end{aligned}$$

which are the same set of variables as those given in z_t .

Example 7.6 Gauss-Newton Estimator of an ARMA(1,1) Model

Re-express the ARMA(1,1) model in Example 7.4 in terms of v_t as

$$v_t = u_t - \rho_1 u_{t-1} - \delta_1 v_{t-1} = (y_t - \beta_0 - \beta_1 x_t) - \rho_1 (y_{t-1} - \beta_0 - \beta_1 x_{t-1}) - \delta_1 v_{t-1}.$$

Construct the derivatives

$$\begin{aligned} z_{1,t} &= -\frac{\partial v_t}{\partial \beta_0} = 1 - \rho_1 - \delta_1 z_{1,t-1} \\ z_{2,t} &= -\frac{\partial v_t}{\partial \beta_1} = x_t - \rho_1 x_{t-1} - \delta_1 z_{2,t-1} \\ z_{3,t} &= -\frac{\partial v_t}{\partial \rho_1} = y_{t-1} - \beta_0 - \beta_1 x_{t-1} - \delta_1 z_{3,t-1} \\ z_{4,t} &= -\frac{\partial v_t}{\partial \delta_1} = v_{t-1} - \delta_1 z_{4,t-1} . \end{aligned}$$

Regress v_t on $z_t = [z_{1,t}, z_{2,t}, z_{3,t}, z_{4,t}]'$, with all terms evaluated at the starting values $[\beta_0, \beta_1, \rho_1, \delta_1]_{(0)}$. Let the parameter estimates be given by $\Delta_{\beta_0}, \Delta_{\beta_1}, \Delta_{\rho_1}, \Delta_{\delta_1}$. Construct new parameter values as

$$[\beta_0, \beta_1, \rho_1, \delta_1]_{(1)} = [\beta_0, \beta_1, \rho_1, \delta_1]_{(0)} + [\Delta_{\beta_0}, \Delta_{\beta_1}, \Delta_{\rho_1}, \Delta_{\delta_1}] ,$$

and repeat the iterations. □

7.4.2 Zig-zag Algorithms

The derivation of the Gauss-Newton algorithm is based on using the first-order conditions to simplify the Hessian. Another way to proceed that leads to even further simplification of the Hessian is to use the scoring algorithm of Chapter 3

$$\theta_{(k)} = \theta_{(k-1)} + I_{(k-1)}^{-1} G_{(k-1)} , \quad (7.14)$$

where the gradient, G , and the information matrix, I , are evaluated at $\theta = \theta_{(k-1)}$. This algorithm is slightly more involved to implement than are the Newton-Raphson and Gauss-Newton algorithms because it requires deriving the information matrix. This is not too difficult for regression models with normal disturbances, as is demonstrated below for the AR(1) regression model set out in Example 7.2.

Taking expectations of the second derivatives given in (7.7) and (7.8)

yields

$$\begin{aligned}
\mathbb{E}\left[\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_0^2}\right] &= \mathbb{E}\left[-\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T (1 - \rho_1)^2\right] = -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T (1 - \rho_1)^2 \\
\mathbb{E}\left[\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_1^2}\right] &= \mathbb{E}\left[-\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T (x_t - \rho_1 x_{t-1})^2\right] = -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T \mathbb{E}[(x_t - \rho_1 x_{t-1})^2] \\
\mathbb{E}\left[\frac{\partial^2 \ln L_T(\theta)}{\partial \rho_1^2}\right] &= \mathbb{E}\left[-\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T u_{t-1}^2\right] = -\frac{1}{\sigma^2(T-1)} \left[\sum_{t=2}^T \frac{\sigma^2}{1 - \rho_1^2}\right] = -\frac{1}{1 - \rho_1^2} \\
\mathbb{E}\left[\frac{\partial^2 \ln L_T(\theta)}{\partial (\sigma^2)^2}\right] &= \mathbb{E}\left[\frac{1}{2\sigma^4} - \frac{1}{\sigma^6(T-1)} \sum_{t=2}^T v_t^2\right] = -\frac{1}{2\sigma^4},
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{E}\left[\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_0 \partial \beta_1}\right] &= -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T \mathbb{E}[x_t - \rho_1 x_{t-1}](1 - \rho_1) \\
\mathbb{E}\left[\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_0 \partial \rho_1}\right] &= -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T (\mathbb{E}[u_{t-1}](1 - \rho_1) + \mathbb{E}[v_t]) = 0 \\
\mathbb{E}\left[\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_0 \partial \sigma^2}\right] &= -\frac{1}{\sigma^4(T-1)} \sum_{t=2}^T \mathbb{E}[v_t](1 - \rho_1) = 0 \\
\mathbb{E}\left[\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_1 \partial \rho_1}\right] &= -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T (\mathbb{E}[u_{t-1}(x_t - \rho_1 x_{t-1})] + \mathbb{E}[v_t x_{t-1}]) = 0 \\
\mathbb{E}\left[\frac{\partial^2 \ln L_T(\theta)}{\partial \beta_1 \partial \sigma^2}\right] &= -\frac{1}{\sigma^4(T-1)} \sum_{t=2}^T \mathbb{E}[v_t(x_t - \rho_1 x_{t-1})] = 0 \\
\mathbb{E}\left[\frac{\partial^2 \ln L_T(\theta)}{\partial \rho_1 \partial \sigma^2}\right] &= -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T \mathbb{E}[v_t] \mathbb{E}[u_{t-1}] = 0.
\end{aligned}$$

Multiplying these terms by -1 , gives the information matrix

$$I(\theta) = \begin{bmatrix} I_{1,1} & 0 & 0 \\ 0 & I_{2,2} & 0 \\ 0 & 0 & I_{3,3} \end{bmatrix}, \quad (7.15)$$

where

$$\begin{aligned} I_{1,1} &= \frac{1}{\sigma^2(T-1)} \sum_{t=2}^T \begin{bmatrix} (1-\rho_1)^2 & (x_t - \rho_1 x_{t-1})(1-\rho_1) \\ (x_t - \rho_1 x_{t-1})(1-\rho_1) & (x_t - \rho_1 x_{t-1})^2 \end{bmatrix} \\ I_{2,2} &= \frac{1}{1-\rho_1^2} \\ I_{3,3} &= \frac{1}{2\sigma^4}. \end{aligned} \quad (7.16)$$

The key feature of the information matrix in equation (7.15) is that it is block diagonal. This suggests that estimation can proceed in three separate blocks, that is in a zig-zag

$$\begin{aligned} \beta_{(k)} &= \beta_{(k-1)} + I_{1,1}^{-1} G_1 \\ \rho_{1(k)} &= \rho_{1(k-1)} + I_{2,2}^{-1} G_2 \\ \sigma_{(k)}^2 &= \sigma_{(k-1)}^2 + I_{3,3}^{-1} G_3, \end{aligned} \quad (7.17)$$

where the gradients from (7.6) are defined as

$$\begin{aligned} G_1 &= \frac{1}{\sigma^2(T-1)} \sum_{t=2}^T \begin{bmatrix} 1-\rho_1 \\ x_t - \rho_1 x_{t-1} \end{bmatrix} v_t \\ G_2 &= \frac{1}{\sigma^2(T-1)} \sum_{t=2}^T \begin{bmatrix} u_{t-1} \end{bmatrix} v_t \\ G_3 &= -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4(T-1)} \sum_{t=2}^T v_t^2, \end{aligned} \quad (7.18)$$

and all parameters are evaluated at $\theta_{(k-1)}$.

Just as in the Gauss-Newton algorithm, where the first-order condition for σ^2 is used to derive an explicit solution of σ^2 , the iterative scheme is only needed for the first two blocks in (7.17). Compared to the Gauss-Newton algorithm, the zig-zag algorithm has the advantage of simplifying the computation since the regression parameters β_0 and β_1 are computed separately to the autocorrelation parameter ρ_1 .

The steps used to update the parameters $\theta = \{\beta_0, \beta_1, \rho_1, \sigma^2\}$ by the zig-zag algorithm are summarized as follows.

- Step 1:** Choose starting values $\theta_{(0)}$.
- Step 2:** Update β_0 and β_1 using the first block in (7.17).
- Step 3:** Update ρ_1 using the second block in (7.17).
- Step 4:** Repeat steps 2 to 3 until convergence.
- Step 5:** Compute σ^2 using (7.9) as the last step.

7.4.3 Cochrane-Orcutt

The zig-zag algorithm is equivalent to the Cochrane-Orcutt algorithm commonly used to estimate regression models with autocorrelated disturbance terms. To see this, consider the updating step of β_0 and β_1 in (7.17) where $I_{1,1}$ and G_1 are given in (7.16) and (7.18), respectively,

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}_{(k)} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}_{(k-1)} + I_{1,1}^{-1} G_1 = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}_{(k-1)} + \left[\sum_{t=2}^T z_t z_t' \right]^{-1} \left[\sum_{t=2}^T z_t v_t \right],$$

where now $z_t = [1 - \rho_1, x_t - \rho_1 x_{t-1}]'$. Given that

$$\begin{aligned} v_t &= y_t - \rho_1 y_{t-1} - \beta_0(1 - \rho_1) - \beta_1(x_t - \rho_1 x_{t-1}) \\ &= y_t - \rho_1 y_{t-1} - z_t' \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \end{aligned}$$

the updating equation reduces to

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}_{(k)} = \left[\sum_{t=2}^T z_t z_t' \right]^{-1} \left[\sum_{t=2}^T z_t (y_t - \rho_1 y_{t-1}) \right].$$

This equation shows that the updated estimates of β_0 and β_1 are immediately obtained by simply regressing $y_t - \rho_1 y_{t-1}$ on z_t .

The updating step for ρ_1 in (7.17) can also be expressed in terms of a least squares regression by replacing $\sigma^2/(1 - \rho_1^2)$ in the $I_{2,2}$ term in (7.16) by u_{t-1}^2

$$\begin{aligned} \rho_{1(k)} &= \rho_{1(k-1)} + I_{2,2}^{-1} G_2 \\ &= \rho_{1(k-1)} + \left[\sum_{t=2}^T u_{t-1}^2 \right]^{-1} \left[\sum_{t=2}^T u_{t-1} v_t \right] \\ &= \rho_{1(k-1)} + \left[\sum_{t=2}^T u_{t-1}^2 \right]^{-1} \left[\sum_{t=2}^T u_{t-1} (u_t - \rho_1 u_{t-1}) \right] \\ &= \left[\sum_{t=2}^T u_{t-1}^2 \right]^{-1} \left[\sum_{t=2}^T u_{t-1} u_t \right]. \end{aligned}$$

This equation shows that the updated estimate of ρ_1 is obtained immediately by simply regressing u_t on u_{t-1} .

The steps used to update the parameters $\theta = \{\beta_0, \beta_1, \rho_1, \sigma^2\}$ by the Cochrane-Orcutt algorithm are summarized as follows.

Step 1: Choose starting values $\theta_{(0)}$.

Step 2: Update β_0 and β_1 by regressing $y_t - \rho_1 y_{t-1}$ on $\{1 - \rho_1, x_t - \rho_1 x_{t-1}\}$.

Step 3: Update ρ_1 by regressing u_t on u_{t-1} .

Step 4: Repeat steps 2 and 3 until convergence.

Step 5: Compute σ^2 using (7.9) as the last step.

Example 7.7 Cochrane-Orcutt Estimator of an AR(p) Model Consider the model

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t \\ u_t &= \rho_1 u_{t-1} + \rho_2 u_{t-2} + \cdots + \rho_p u_{t-p} + v_t \\ v_t &\sim iid N(0, \sigma^2). \end{aligned}$$

The Cochrane-Orcutt algorithm consists of constructing the transformed variables

$$\begin{aligned} y_t^* &= y_t - \rho_1 y_{t-1} - \rho_2 y_{t-2} - \cdots - \rho_p y_{t-p} \\ x_t^* &= x_t - \rho_1 x_{t-1} - \rho_2 x_{t-2} - \cdots - \rho_p x_{t-p}, \end{aligned}$$

given starting values for $\rho_1, \rho_2, \dots, \rho_p$ and regressing y_t^* on a constant (scaled by $1 - \rho_1 - \rho_2 - \cdots - \rho_p$) and x_t^* to get updated values of β_0 and β_1 . Next, regress u_t on $[u_{t-1}, u_{t-2}, \dots, u_{t-p}]$ to get updated estimates of $\rho_1, \rho_2, \dots, \rho_p$. Reconstruct the transformed variables y_t^* and x_t^* using the updated values of ρ_i and repeat the steps until convergence. $\square$

The Cochrane-Orcutt algorithm is very similar to the Gauss-Newton algorithm since both procedures involve performing least squares regressions at each iteration. There are two main difference between these algorithms.

- (1) All parameters are updated jointly in Gauss-Newton whereas in the Cochrane-Orcutt algorithm the regression parameters are updated separately to the autocorrelation parameters.
- (2) Gauss-Newton uses v_t as the dependent variable in the updating of the regression and autocorrelation parameters, whereas Cochrane-Orcutt uses y_t^* as the dependent variable in the updating of the regression parameters and u_t as the dependent variable in the updating of the autocorrelation parameters.

7.5 Distribution Theory

The asymptotic distribution of the maximum likelihood estimator in the case of the autocorrelated regression model is now discussed. Special attention is given to comparing the asymptotic properties of this estimator with the OLS estimator based on misspecifying the presence of autocorrelation in the

disturbance term. Finite sample properties of these estimators are explored in Section 7.9.2.

The asymptotic distributions of the maximum likelihood and ordinary least squares estimators are developed primarily for the AR(1) regression model

$$\begin{aligned} y_t &= \beta x_t + u_t \\ u_t &= \rho u_{t-1} + v_t \\ v_t &\sim iid N(0, \sigma^2), \end{aligned} \quad (7.19)$$

where y_t is the dependent variable, x_t is a (scalar) explanatory variable assumed to be independent of the disturbance term v_t at all lags and the unknown parameters are $\theta = \{\beta, \rho, \sigma^2\}$ with the stationarity restriction $|\rho| < 1$. Both y_t and x_t are assumed to have zero mean. An important feature of the asymptotic distributions is that they depend upon the data-generating process of x_t in (7.19).

7.5.1 Maximum Likelihood Estimator

The maximum likelihood estimator $\hat{\theta}$ of the parameter vector $\theta = \{\beta, \rho, \sigma^2\}$ for the autocorrelated regression model (7.19) has all the properties discussed in Chapter 2. It is consistent, asymptotically efficient and has asymptotic distribution

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1}), \quad (7.20)$$

where $\theta_0 = \{\beta_0, \rho_0, \sigma_0^2\}$ is the true population parameter vector and $I(\theta_0)$ is the information matrix evaluated at the population vector θ_0 . Given that the Gauss-Newton, zig-zag and Cochrane-Orcutt algorithms discussed in Section 7.4 are also maximum likelihood estimators, it immediately follows that these estimators also share the same asymptotic properties.

From equation (7.6) it follows that the gradient $g_t(\theta_0)$ at time t of the parameters of the AR(1) regression model in (7.19) consists of

$$\begin{aligned} g_{1,t} &= \frac{1}{\sigma^2} v_t (x_t - \rho_1 x_{t-1}) \\ g_{2,t} &= \frac{1}{\sigma^2} v_t u_{t-1} \\ g_{3,t} &= -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} v_t^2. \end{aligned}$$

Taking conditional expectations of $g_{1,t}$ yields

$$E_{t-1}[g_{1,t}] = E_{t-1}\left[\frac{1}{\sigma^2} v_t (x_t - \rho_1 x_{t-1})\right] = \frac{1}{\sigma^2} E_{t-1}[v_t] \left[(x_t - \rho_1 x_{t-1})\right] = 0,$$

where the second last step is based on the assumption of x_t being independent of v_t , while the last step also uses the property $E_{t-1}[v_t] = 0$. Similarly, for $g_{2,t}$ it follows that

$$E_{t-1}[g_{2,t}] = \frac{1}{\sigma^2} E_{t-1}[v_t u_{t-1}] = \frac{1}{\sigma^2} E_{t-1}[v_t] u_{t-1} = 0,$$

and finally for $g_{3,t}$

$$E_{t-1}[g_{3,t}] = E_{t-1}\left[-\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} v_t^2\right] = -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} E_{t-1}[v_t^2] = -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} \sigma^2 = 0,$$

since $E_{t-1}[v_t^2] = \sigma^2$. The gradient vector $g_t(\theta_0)$ therefore satisfy the requirements of a mds. The mds central limit theorem from Chapter 2 therefore applies to conclude that

$$\frac{1}{\sqrt{T}} \sum_{t=2}^T g_t(\theta_0) \xrightarrow{d} N(0, \lim_{T \rightarrow \infty} E[g_t(\theta_0) g_t(\theta_0)']) \quad (7.21)$$

where by definition the information matrix is the limiting variance of the score, that is

$$I(\theta_0) = \lim_{T \rightarrow \infty} \text{var}\left(\frac{1}{\sqrt{T}} \sum_{t=2}^T g_t(\theta_0)\right). \quad (7.22)$$

The derivation of the asymptotic normality of the maximum likelihood estimator given in Chapter 2, based on a mean value expansion, then applies to conclude that

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1}), \quad (7.23)$$

as usual.

Because the information matrix in equation (7.15) is block diagonal, the asymptotic distribution of the maximum likelihood estimator separates into three blocks

$$\begin{aligned} \sqrt{T}(\hat{\beta} - \beta_0) &\xrightarrow{d} N(0, I_{1,1}^{-1}) \\ \sqrt{T}(\hat{\rho} - \rho_0) &\xrightarrow{d} N(0, I_{2,2}^{-1}) \\ \sqrt{T}(\hat{\sigma}^2 - \sigma_0^2) &\xrightarrow{d} N(0, I_{3,3}^{-1}). \end{aligned} \quad (7.24)$$

For the conditional maximum likelihood estimator, the relevant information

matrices follow from equation (7.16) and are, respectively,

$$\begin{aligned}
 I_{1,1} &= \lim_{T \rightarrow \infty} -E \left[\frac{\partial^2 \ln L_T(\theta)}{\partial \beta^2} \right] = \lim_{T \rightarrow \infty} E \left[\frac{1}{\sigma_0^2(T-1)} \sum_{t=2}^T E[(x_t - \rho_0 x_{t-1})^2] \right] \\
 I_{2,2} &= \lim_{T \rightarrow \infty} -E \left[\frac{\partial^2 \ln L_T(\theta)}{\partial \rho^2} \right] = \lim_{T \rightarrow \infty} E \left[\frac{1}{\sigma_0^2(T-1)} \sum_{t=2}^T \frac{\sigma_0^2}{1 - \rho_0^2} \right] = \frac{1}{1 - \rho_0^2} \\
 I_{3,3} &= \lim_{T \rightarrow \infty} -E \left[\frac{\partial^2 \ln L_T(\theta)}{\partial (\sigma^2)^2} \right] = \lim_{T \rightarrow \infty} E \left[\frac{1}{\sigma_0^2(T-1)} \sum_{t=2}^T \frac{1}{2\sigma_0^2} \right] = \frac{1}{2\sigma_0^4}.
 \end{aligned} \tag{7.25}$$

Inspection of (7.25) reveals that only in the case of the asymptotic distribution of $\hat{\beta}$ does the form of the information matrix depend upon the assumptions made about the data-generating process of x_t .

Case 1: Deterministic Explanatory Variables

In the case of deterministic x_t , the information matrix of $\hat{\beta}$ in (7.25) is

$$I_{1,1} = \lim_{T \rightarrow \infty} \frac{1}{\sigma_0^2(T-1)} \sum_{t=2}^T (x_t - \rho_0 x_{t-1})^2, \tag{7.26}$$

yielding the approximate variance

$$\text{var}(\hat{\beta}) = \frac{1}{T-1} I_{1,1}^{-1} = \sigma_0^2 \left[\sum_{t=2}^T (x_t - \rho_0 x_{t-1})^2 \right]^{-1}. \tag{7.27}$$

Example 7.8 Asymptotic Distribution Consider the AR(1) regression model

$$y_t = \alpha + \beta x_t + u_t, \quad u_t = \rho u_{t-1} + v_t,$$

where $v_t \sim N(0, \sigma_0^2)$, $x_t \sim U(0, 1) - 0.5$, $\alpha_0 = 1$, $\beta_0 = 1$, $\rho_0 = 0.6$ and $\sigma_0^2 = 10$. For a sample of size $T = 500$,

$$\sum_{t=2}^{500} (x_t - \rho_0 x_{t-1})^2 = 57.4401,$$

and the asymptotic variance of $\hat{\beta}$ using (7.27) is

$$\text{var}(\hat{\beta}) = 10 \left[\sum_{t=2}^T (x_t - 0.6x_{t-1})^2 \right]^{-1} = \frac{10}{57.4401} = 0.1741.$$

The model is simulated 5000 times with the conditional maximum likelihood

estimator computed at each replication. As x_t is deterministic it does not vary over the replications. The mean squared error in the case of $\hat{\beta}$ is

$$\text{MSE} = \frac{1}{5000} \sum_{i=1}^{5000} (\hat{\beta}_i - 1)^2 = 0.1686,$$

which agrees with the theoretical value of 0.1741. A plot of the standardized sampling distribution in Figure 7.2 reveals that it is normally distributed

$$z = \frac{\hat{\beta} - 1}{\sqrt{0.1686}} \xrightarrow{d} N(0, 1).$$

□

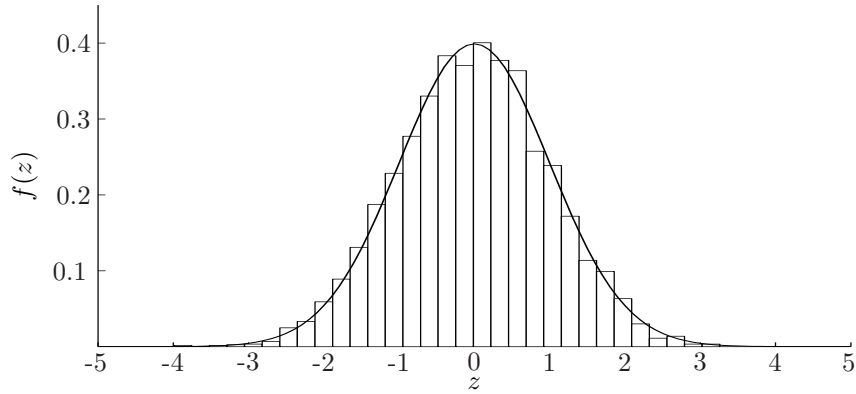


Figure 7.2 Empirical distribution of the maximum likelihood estimator of the slope parameter in the autocorrelated regression model, based on 5000 Monte Carlo replications.

Case 2: Stochastic and Independent Explanatory Variables

In the case where the explanatory variable x_t is stochastic and independent of u_t , it is possible to derive analytical expressions for the asymptotic variance of $\hat{\beta}$ for certain data-generating processes. For example, assume that x_t in (7.19) is the stationary AR(1) process

$$x_t = \phi_0 x_{t-1} + w_t, \quad w_t \sim iid N(0, \eta_0^2), \quad (7.28)$$

where $|\phi_0| < 1$. The information matrix in (7.25) is

$$\begin{aligned} I_{1,1} &= \lim_{T \rightarrow \infty} \frac{1}{\sigma_0^2(T-1)} \sum_{t=2}^T \mathbb{E} [x_t^2 - 2\rho_0 x_t x_{t-1} + \rho_0^2 x_{t-1}^2] \\ &= \lim_{T \rightarrow \infty} \frac{1}{\sigma_0^2(T-1)} \sum_{t=2}^T \left(\frac{\eta_0^2}{1-\phi_0^2} - \frac{2\eta_0^2 \rho_0 \phi_0}{1-\phi_0^2} + \frac{\eta_0^2 \rho_0^2}{1-\phi_0^2} \right) \\ &= \frac{\eta_0^2}{\sigma_0^2} \left(\frac{1-2\rho_0\phi_0+\rho_0^2}{1-\phi_0^2} \right), \end{aligned}$$

where the second line uses a result (which is established in Chapter 13) that for the AR(1) process in (7.28)

$$\mathbb{E} [x_t x_{t+k}] = \eta_0^2 \phi_0^k / (1 - \phi_0^2), \quad k \geq 0.$$

The asymptotic variance of $\hat{\beta}$ in this case is

$$\text{var}(\hat{\beta}) \approx \frac{1}{T-1} I_{1,1}^{-1} = \frac{1}{(T-1)} \frac{\sigma_0^2}{\eta_0^2} \left(\frac{1-\phi_0^2}{1-2\rho_0\phi_0+\rho_0^2} \right). \quad (7.29)$$

7.5.2 Least Squares Estimator

Now compare the maximum likelihood estimator of β in (7.19) to the corresponding least squares estimator obtained by ignoring the autocorrelation in the disturbance term and simply regressing y_t on x_t . Despite misspecifying the dynamics of the model, the OLS estimator of β is still unbiased and asymptotically normally distributed, provided that x_t is independent of the disturbance term. The cost of ignoring the presence of autocorrelation is that the OLS estimator is, in general, asymptotically inefficient relative to the maximum likelihood estimator $\hat{\beta}$ in (7.24). An important special case where no loss of efficiency occurs is demonstrated next.

Case 1: Deterministic Explanatory Variables

The first two derivatives of the log-likelihood function with respect to β in the AR(1) model in (7.19) are, respectively,

$$\begin{aligned} \frac{\partial \ln L_T(\theta)}{\partial \beta} &= \frac{1}{\sigma^2(T-1)} \sum_{t=2}^T v_t (x_t - \rho x_{t-1}) \\ \frac{\partial^2 \ln L_T(\theta)}{\partial \beta^2} &= -\frac{1}{\sigma^2(T-1)} \sum_{t=2}^T (x_t - \rho x_{t-1})^2. \end{aligned} \quad (7.30)$$

The OLS estimator is based on imposing the restriction of no autocorrela-

tion, $\rho = 0$, so that these derivatives simplify to

$$\begin{aligned}\left. \frac{\partial \ln L_T(\theta)}{\partial \beta} \right|_{\rho=0} &= \frac{1}{\sigma^2 T} \sum_{t=1}^T (y_t - \beta x_t) x_t, \\ \left. \frac{\partial^2 \ln L_T(\theta)}{\partial \beta^2} \right|_{\rho=0} &= -\frac{1}{\sigma^2 T} \sum_{t=1}^T x_t^2,\end{aligned}\tag{7.31}$$

since $v_t = u_t = y_t - \beta x_t$ in equation (7.19) when the restriction $\rho = 0$ is imposed and the sample period is now T . The OLS estimator is obtained by setting the first-order derivative in (7.31) to zero

$$\frac{1}{\sigma^2 T} \sum_{t=1}^T (y_t - \hat{\beta}_{LS} x_t) x_t = 0,\tag{7.32}$$

and solving for $\hat{\beta}_{LS}$. Taking conditional expectations of (7.32) and using the result that $y_t = \beta x_t + u_t$ in (7.19) gives

$$\begin{aligned}\mathbb{E} \left[\frac{1}{\sigma^2 T} \sum_{t=1}^T (y_t - \hat{\beta}_{LS} x_t) x_t \right] &= \frac{1}{\sigma^2 T} \left(\sum_{t=1}^T \mathbb{E}[(\beta x_t + u_t) x_t] - \sum_{t=1}^T \mathbb{E}[\hat{\beta}_{LS} x_t^2] \right) \\ &= \frac{1}{\sigma^2 T} \left(\sum_{t=1}^T (\beta_0 x_t^2 + \mathbb{E}[u_t] x_t) - \sum_{t=1}^T \mathbb{E}[\hat{\beta}_{LS}] x_t^2 \right) \\ &= \frac{1}{\sigma^2 T} \sum_{t=1}^T (\beta_0 - \mathbb{E}[\hat{\beta}_{LS}]) x_t^2,\end{aligned}$$

since $\mathbb{E}[u_t] = 0$ by assumption. But from (7.32) this expression should be zero, which can only hold provided that the estimator is unbiased

$$\mathbb{E}[\hat{\beta}_{LS}] = \beta_0.$$

To show that the OLS estimator is asymptotically normal, from (7.32) and again using the result that $y_t = \beta x_t + u_t$ in (7.19), the least squares estimator is

$$\hat{\beta}_{LS} = \left[\sum_{t=1}^T x_t^2 \right]^{-1} \sum_{t=1}^T x_t y_t = \beta_0 + \left[\sum_{t=1}^T x_t^2 \right]^{-1} \sum_{t=1}^T x_t u_t.$$

Rearranging this expression and multiplying both sides by $\sqrt{T}$ gives

$$\sqrt{T}(\hat{\beta}_{LS} - \beta_0) = \left[\frac{1}{T} \sum_{t=1}^T x_t^2 \right]^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^T x_t u_t.$$

By assumption, the term in square brackets on the right hand side converges

to a constant. From equation (7.19), $v_t = (1 - \rho L)u_t$, where L is the lag operator (see Appendix B), so that

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T x_t u_t = \frac{1}{\sqrt{T}} \sum_{t=1}^T x_t \frac{v_t}{1 - \rho L} = \frac{1}{\sqrt{T}} \sum_{t=1}^T x_t \sum_{i=0}^{\infty} \rho^i v_{t-i},$$

where the inverse property of the lag operator has been used. Because the process is stationary, the mean is

$$E\left[\frac{1}{\sqrt{T}} \sum_{t=1}^T x_t u_t\right] = E\left[\frac{1}{\sqrt{T}} \sum_{t=1}^T x_t \sum_{i=0}^{\infty} \rho^i v_{t-i}\right] = \frac{1}{\sqrt{T}} \sum_{t=1}^T x_t \sum_{i=0}^{\infty} \rho^i E[v_{t-i}] = 0,$$

while the variance is

$$\begin{aligned} \text{var}\left(\frac{1}{\sqrt{T}} \sum_{t=1}^T x_t u_t\right) &= E\left[\left(\frac{1}{\sqrt{T}} \sum_{t=1}^T x_t u_t\right)^2\right] = \frac{1}{T} E\left[\left(\sum_{t=1}^T x_t u_t\right)^2\right] \\ &= \frac{1}{\sigma_0^4} E\left[\left(\sum_{t=1}^T x_t^2 u_t^2 + 2 \sum_{i=1}^{T-1} \sum_{t=1}^{T-i} x_t x_{t+i} u_t u_{t+i}\right)^2\right] \\ &= \frac{1}{\sigma_0^4} \left(\sum_{t=1}^T x_t^2 E[u_t^2] + 2 \sum_{i=1}^{T-1} \sum_{t=1}^{T-i} x_t x_{t+i} E[u_t u_{t+i}]\right) \\ &= \frac{1}{\sigma_0^4} \left(\sum_{t=1}^T x_t^2 \frac{\sigma_0^2}{1 - \rho_0^2} + 2 \sum_{i=1}^{T-1} \sum_{t=1}^{T-i} x_t x_{t+i} \frac{\sigma_0^2 \rho_0^i}{1 - \rho_0^2}\right) \\ &= \frac{1}{\sigma_0^2 (1 - \rho_0^2)} \left(\sum_{t=1}^T x_t^2 + 2 \sum_{i=1}^{T-1} \sum_{t=1}^{T-i} \rho_0^i x_t x_{t+i}\right), \quad (7.33) \end{aligned}$$

since $E[u_t u_{t+k}] = \sigma_0^2 \rho_0^k / (1 - \rho_0^2)$, $k \geq 0$, under the true model of first-order autocorrelation. The presence of autocorrelation implies that the gradient vector associated with the OLS estimator in this case is not a martingale difference sequence. However, as the form of the autocorrelation ensures that y_t and hence the gradient vector, operate independently asymptotically, a Central limit theorem nonetheless still applies. This property of asymptotic independence is formally referred to as mixing which is discussed further in Chapter 9.

The efficiency property of the least squares estimator is based on considering the information and outer product of the gradients matrices of this estimator. Using the second derivative in equation (7.31) gives the informa-

tion matrix

$$I(\beta_0) = -E \left[\frac{\partial^2 \ln L_T(\theta)}{\partial \beta^2} \Big|_{\rho=0} \right] = \frac{1}{\sigma_0^2 T} \sum_{t=1}^T x_t^2. \quad (7.34)$$

Using the first derivative in (7.31) gives the OPG matrix

$$\begin{aligned} J_T(\beta_0) &= E \left[\left(\frac{1}{\sigma_0^2 T} \sum_{t=1}^T (y_t - \beta x_t) x_t \right)^2 \right] \\ &= \frac{1}{\sigma_0^2 (1 - \rho_0^2)} \left(\sum_{t=1}^T x_t^2 + 2 \sum_{i=1}^{T-1} \sum_{t=1}^{T-i} \rho_0^i x_t x_{t+i} \right), \end{aligned} \quad (7.35)$$

which follows immediately from (7.33). Comparing expressions (7.34) and (7.35) shows that $I(\beta_0) \neq J_T(\beta_0)$ and so the information equality of Chapter 2 does not hold for the OLS estimator in the case of misspecified autocorrelation. Only for the special case of no autocorrelation, $\rho_0 = 0$, does this equality hold. Chapter 9 shows that the remedy is to combine $I(\beta_0)$ and $J_T(\beta_0)$ by expressing the OLS variance as

$$\text{var}(\hat{\beta}_{LS}) = \frac{1}{T} I^{-1}(\beta_0) J_T(\beta_0) I^{-1}(\beta_0). \quad (7.36)$$

Substituting expressions (7.34) and (7.35) in (7.36) and simplifying yields the variance of the OLS estimator

$$\text{var}(\hat{\beta}_{LS}) = \left[\sum_{t=1}^T x_t^2 \right]^{-2} \left[\frac{\sigma_0^2}{1 - \rho_0^2} \left(\sum_{t=1}^T x_t^2 + 2 \sum_{i=1}^{T-1} \sum_{t=1}^{T-i} \rho_0^i x_t x_{t+i} \right) \right]. \quad (7.37)$$

In Chapter 9, the least squares estimator $\hat{\beta}_{LS}$ with variance given by (7.37) is referred to as a quasi-maximum likelihood estimator.

A measure of relative efficiency is given by the ratio of the OLS variance in (7.37) and the maximum likelihood variance in (7.27)

$$\frac{\text{var}(\hat{\beta}_{LS})}{\text{var}(\hat{\beta})} = \frac{\left[\sum_{t=1}^T x_t^2 \right]^{-2} \left[\sum_{t=1}^T x_t^2 + 2 \sum_{i=1}^{T-1} \sum_{t=1}^{T-i} \rho_0^i x_t x_{t+i} \right]}{(1 - \rho_0^2) \left[\sum_{t=2}^T (x_t - \rho_0 x_{t-1})^2 \right]^{-1}}. \quad (7.38)$$

Values of this ratio greater than unity represents a loss of efficiency from using OLS. The extent of the loss in efficiency depends upon x_t , which is quantified in the Monte Carlo experiments in Section 7.9.2 for alternative assumptions of the explanatory variable. In the special case of $\rho_0 = 0$, expression (7.38) equals unity and so the asymptotic efficiency of the two estimators is the same.

Example 7.9 Efficiency of $\bar{y}$ in the Presence of Autocorrelation

Let $x_t = 1$ in (7.19) so β represents the intercept. The OLS estimator of β is the sample mean and the asymptotic variance from (7.37) is

$$\text{var}(\hat{\beta}_{LS}) = \frac{T^{-2}\sigma_0^2}{1-\rho_0^2} \left(T + 2 \sum_{i=1}^{T-1} \sum_{t=1}^{T-i} \rho_0^i \right) = \frac{T^{-2}\sigma_0^2}{1-\rho_0^2} \left(\frac{T(1-\rho_0^2) - 2\rho_0(1-\rho_0^T)}{(1-\rho_0)^2} \right).$$

Setting $x_t = 1$ in (7.27) gives the asymptotic variance of the conditional maximum likelihood estimator

$$\text{var}(\hat{\beta}) = \sigma_0^2 \left[\sum_{t=2}^T (1-\rho_0)^2 \right]^{-1} = \frac{\sigma_0^2}{(T-1)(1-\rho_0)^2}.$$

The efficiency ratio in (7.38) is now

$$\frac{\text{var}(\hat{\beta}_{LS})}{\text{var}(\hat{\beta})} = \frac{T^{-2}\sigma_0^2}{1-\rho_0^2} \left(\frac{T(1-\rho_0^2) - 2\rho_0(1-\rho_0^T)}{(1-\rho_0)^2} \right) \times \frac{(T-1)(1-\rho_0)^2}{\sigma_0^2}.$$

Upon taking limits

$$\lim_{T \rightarrow \infty} \frac{\text{var}(\hat{\beta}_{LS})}{\text{var}(\hat{\beta})} = \lim_{T \rightarrow \infty} \left(\frac{T(T-1)}{T^2} - 2\rho_0 \frac{(1-\rho_0^T)(T-1)}{1-\rho_0^2} \frac{(T-1)}{T^2} \right) = 1,$$

so that the two estimators are asymptotically efficient. This result is a special case of a more general result established by Anderson (1971, p.581). $\square$

Case 2: Stochastic and Independent Explanatory Variables

In the case where the regressor x_t is stochastic and independent of u_t , it is advantageous to rewrite the OLS variance equation in (7.37) as

$$\text{var}(\hat{\beta}_{LS}) = \frac{1}{T} \left[\frac{1}{T} \sum_{t=1}^T x_t^2 \right]^{-1} \left[\frac{\sigma_0^2}{1-\rho_0^2} \left(1 + 2 \sum_{i=1}^{T-1} \rho_0^i \frac{\frac{1}{T} \sum_{t=1}^{T-i} x_t x_{t+i}}{\frac{1}{T} \sum_{t=1}^T x_t^2} \right) \right].$$

Taking probability limits of both sides of this expression gives

$$\begin{aligned} \text{plim var}(\hat{\beta}_{LS}) &= \frac{1}{T} \left[\text{plim} \left(\frac{1}{T} \sum_{t=1}^T x_t^2 \right) \right]^{-1} \\ &\quad \times \left[\frac{\sigma_0^2}{1-\rho_0^2} \left(1 + 2 \sum_{i=1}^{T-1} \rho_0^i \frac{\text{plim} \left(\frac{1}{T} \sum_{t=1}^{T-i} x_t x_{t+i} \right)}{\text{plim} \left(\frac{1}{T} \sum_{t=1}^T x_t^2 \right)} \right) \right] \\ &= \frac{1}{T} \frac{1}{\text{var}(x_t)} \left[\frac{\sigma_0^2}{1-\rho_0^2} \left(1 + 2 \sum_{i=1}^{T-1} \rho_0^i \frac{\text{cov}(x_t, x_{t+i})}{\text{var}(x_t)} \right) \right]. \end{aligned}$$

Given $E_0 [x_t x_{t+k}] = \eta_0^2 \phi_0^k / (1 - \phi_0^2)$, it follows then that

$$\begin{aligned}
 \text{plim var}(\hat{\beta}_{LS}) &= \frac{1}{T} \frac{1 - \phi_0^2}{\eta_0^2} \left[\frac{\sigma_0^2}{1 - \rho_0^2} \left(1 + 2 \sum_{i=1}^{T-1} \rho_0^i \phi_0^k \right) \right] \\
 &= \frac{1}{T} \frac{1 - \phi_0^2}{1 - \rho_0^2} \frac{\sigma_0^2}{\eta_0^2} \left(1 + 2\rho_0 \phi_0 \left(\sum_{i=0}^{T-2} \rho_0^i \phi_0^k \right) \right) \\
 &= \frac{1}{T} \frac{1 - \phi_0^2}{1 - \rho_0^2} \frac{\sigma_0^2}{\eta_0^2} \left(1 + 2\rho_0 \phi_0 \frac{1 - \rho_0^{T-1} \phi_0^{T-1}}{1 - \rho_0 \phi_0} \right) \\
 &= \frac{1}{T} \frac{1 - \phi_0^2}{1 - \rho_0^2} \frac{\sigma_0^2}{\eta_0^2} \left(\frac{1 + \rho_0 \phi_0 - 2\rho_0^T \phi_0^T}{1 - \rho_0 \phi_0} \right). \tag{7.39}
 \end{aligned}$$

Using (7.39) and (7.29), the relative efficiency of the estimators is

$$\frac{\text{var}(\hat{\beta}_{LS})}{\text{var}(\hat{\beta})} = \frac{(1 - 2\rho_0 \phi_0 + \rho_0^2) (1 + \rho_0 \phi_0 - 2\rho_0^T \phi_0^T)}{(1 - \rho_0^2) (1 - \rho_0 \phi_0)},$$

which, in the limit, becomes

$$\lim_{T \rightarrow \infty} \frac{\text{var}(\hat{\beta}_{LS})}{\text{var}(\hat{\beta})} = \frac{(1 - 2\rho_0 \phi_0 + \rho_0^2) (1 + \rho_0 \phi_0)}{(1 - \rho_0^2) (1 - \rho_0 \phi_0)}. \tag{7.40}$$

Example 7.10 Relative Efficiency of OLS and MLE If $\rho_0 = \phi_0 = 0.6$, then from (7.40) the efficiency ratio is

$$\frac{\text{var}(\hat{\beta}_{LS})}{\text{var}(\hat{\beta})} = \frac{1 + \rho_0^2}{1 - \rho_0^2} = \frac{1 + 0.6^2}{1 - 0.6^2} = 2.125,$$

showing that the asymptotic variance of the OLS estimator is more than twice that of the maximum likelihood estimator. $\square$

7.6 Lagged Dependent Variables

The set of explanatory variables x_t so far are assumed to be either deterministic, or if stochastic are independent of the disturbance term u_t . This restriction of independence is now relaxed by augmenting the set of explanatory variables to include lags of the dependent variable. In the case of the AR(1) regression model in equation (7.19), an extension of this model is

$$\begin{aligned}
 y_t &= \beta x_t + \alpha y_{t-1} + u_t \\
 u_t &= \rho u_{t-1} + v_t \\
 v_t &\sim iid N(0, \sigma^2).
 \end{aligned} \tag{7.41}$$

The parameter α controls the influence of y_{t-1} , which is restricted to the range

$$-1 < \alpha < 1,$$

to ensure that y_t is stationary; see Chapter 13 for a discussion of stationarity. To estimate the parameters $\theta = \{\beta, \alpha, \rho, \sigma^2\}$ by maximum likelihood, the conditional log-likelihood function is given by

$$\begin{aligned} \ln L_T(\theta) &= \frac{1}{T-1} \sum_{t=2}^T \ln f(y_t | y_{t-1}, x_t, x_{t-1}; \theta) \\ &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2(T-1)} \sum_{t=2}^T v_t^2, \end{aligned} \quad (7.42)$$

where

$$\begin{aligned} v_t &= u_t - \rho u_{t-1} \\ &= (y_t - \beta x_t - \alpha y_{t-1}) - \rho(y_{t-1} - \beta x_{t-1} - \alpha y_{t-2}). \end{aligned} \quad (7.43)$$

The log-likelihood function is maximized with respect to θ using any of the algorithms discussed previously.

The maximum likelihood estimator remains consistent even where the regressors include lagged dependent variables. In this respect, it contrasts with the OLS estimator, obtained by regressing y_t on a constant, x_t and y_{t-1} , which is inconsistent because y_{t-1} is correlated with u_t via u_{t-1} . Asymptotic normality follows by invoking the Martingale CLT as the gradient vector g_t , still represents a mds as the inclusion of the explanatory variable y_{t-1} is part of the conditioning set of information resulting in $E_{t-1}[g_t] = 0$.

An alternative efficient estimator for autocorrelated regression models with lagged dependent variables is given by Hatanaka (1974). This estimator is implemented as follows.

Step 1: Obtain consistent estimates of $\{\beta, \alpha\}$ based on an instrumental variables estimator with instruments $\{x_t, x_{t-1}\}$ and construct the residuals $\hat{u}_t$.

Step 2: Regress $\hat{u}_t$ on $\hat{u}_{t-1}$ to obtain $\hat{\rho}_{(1)}$.

Step 3: Run the regression

$$y_t - \hat{\rho} y_{t-1} = \phi_1(x_t - \hat{\rho} x_{t-1}) + \phi_2(y_{t-1} - \hat{\rho} y_{t-2}) + \phi_3 \hat{u}_{t-1} + \nu_t.$$

Efficient estimates of the structural parameters are given by

$$\hat{\beta} = \hat{\phi}_1, \hat{\alpha} = \hat{\phi}_2, \hat{\rho} = \hat{\rho}_{(1)} + \hat{\phi}_3.$$

Step 4: The residual variance σ^2 is obtained by computing the residuals

$$\hat{v}_t = y_t - (\hat{\beta}x_t + \hat{\alpha}y_{t-1}) - \hat{\rho}(y_{t-1} - (\hat{\beta}x_{t-1} + \hat{\alpha}y_{t-2})).$$

The Hatanaka estimator is explored further in Exercise 6.

7.7 Testing

Consider the regression model where the disturbance term u_t is ARMA(p, q)

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t \\ u_t &= \sum_{i=1}^p \rho_i u_{t-i} + v_t + \sum_{i=1}^q \delta_i v_{t-i} \\ v_t &\sim iid N(0, \sigma^2). \end{aligned}$$

A natural test of autocorrelation is a joint test based on the following null and alternative hypotheses

$$\begin{aligned} H_0 &: \rho_1, \rho_2, \dots, \rho_p, \delta_1, \delta_2, \dots, \delta_q = 0 && [\text{No Autocorrelation}] \\ H_1 &: \text{at least one restriction is violated} && [\text{Autocorrelation}] \end{aligned}$$

Some examples of other null hypotheses are

1. Test for AR(1) : $H_0 : \rho_1 = 0$
2. Test for MA(1) : $H_0 : \delta_1 = 0$
3. Test for AR(2) : $H_0 : \rho_1 = \rho_2 = 0$
4. Test for ARMA(1,1) : $H_0 : \rho_1 = \delta_1 = 0$
5. Test for seasonal AR(4) : $H_0 : \rho_4 = 0$.

The last hypothesis is of interest if autocorrelation is thought to be present in quarterly data.

Tests of autocorrelation can be based on the LR, Wald and LM test statistics developed in Chapter 4. The use of the scalar $T-s$, where $s = \max(p, q)$, in the LR and LM tests emphasises that they are constructed by conditioning the log-likelihood function on the first s observations. The LM version of the autocorrelation test has the advantage that it involves estimating the regression model under the null hypothesis only, and this can be achieved by a simple least squares regression. It is for this reason that regression diagnostics often reported in computer packages are LM tests.

Example 7.11 Autocorrelation Test in U.S. Investment Model

The data used in Example 7.5 is revisited in order to test for first-order autocorrelation in the dynamic model of U.S. investment. The model is identical to the one presented in Example 7.5 and all parameter estimates

are based on the conditional log-likelihood function. The hypotheses tested are

$$H_0 : \rho_1 = 0, \quad H_0 : \rho_1 \neq 0.$$

The LR statistic is computed as

$$\begin{aligned} LR &= -2((T-1) \ln L_T(\hat{\theta}_0) - (T-1) \ln L_T(\hat{\theta}_1)) \\ &= -2(-213 \times 1.814 + 213 \times 1.811) = 1.278, \end{aligned}$$

where $\ln L_T(\hat{\theta}_1) = 1.814$ is obtained from Table 7.1, while $\ln L_T(\hat{\theta}_0) = -1.814$ is obtained by re-estimating the model subject to the restriction $\rho_1 = 0$. This statistic is distributed asymptotically as χ_1^2 under H_0 . The p-value is 0.258 resulting in a failure to reject the null hypothesis at the 5% level of significance. To compute the Wald statistic, the unconstrained parameter estimates are given in Table 7.1

$$\hat{\theta}_1' = [-0.275 \quad 1.567 \quad -0.334 \quad 0.091 \quad 2.228],$$

while

$$R = [0 \quad 0 \quad 0 \quad 1 \quad 0], \quad Q = [0],$$

and the covariance matrix of the unconstrained parameter estimates is

$$\frac{1}{T} \hat{\Omega}_1 = -\frac{1}{T} H_T^{-1}(\hat{\theta}_1) = \begin{bmatrix} 0.025 & -0.006 & -0.013 & 0.002 & 0.000 \\ -0.006 & 0.017 & -0.009 & -0.006 & 0.000 \\ -0.013 & -0.009 & 0.027 & 0.002 & 0.000 \\ 0.002 & -0.006 & 0.002 & 0.007 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.047 \end{bmatrix}.$$

The Wald statistic is

$$W = T[R\hat{\theta}_1 - Q]'[R(-H_T^{-1}(\hat{\theta}_1))R']^{-1}[R\hat{\theta}_1 - Q] = 1.266,$$

which is distributed asymptotically as χ_1^2 under H_0 . The p-value is 0.261 yielding the same qualitative results as the LR test at the 5% level of significance. As this test is of a single restriction, an alternative form of the Wald statistic is

$$\sqrt{W} = \sqrt{1.266} = 1.125,$$

which agrees with the t-statistic reported for ρ_1 in Table 7.1 using the conditional maximum likelihood estimator. The LM statistic requires evaluating the gradients of the unconstrained model at the constrained estimates

$$G_T'(\hat{\theta}_0) = [0.000 \quad 0.000 \quad 0.000 \quad 0.067 \quad 0.000],$$

while the pertinent covariance matrix based on the outer product of the gradients matrix is

$$J_T(\hat{\theta}_0) = \begin{bmatrix} 0.446 & 0.235 & 0.309 & -0.079 & -0.103 \\ 0.235 & 0.960 & 0.286 & 0.339 & 0.046 \\ 0.309 & 0.286 & 0.493 & -0.114 & -0.085 \\ -0.079 & 0.339 & -0.114 & 1.216 & 0.144 \\ -0.103 & 0.046 & -0.085 & 0.144 & 0.227 \end{bmatrix}.$$

Using these terms in the LM statistic gives

$$LM = (T - 1)G_T'(\hat{\theta}_0)J_T^{-1}(\hat{\theta}_0)G_T(\hat{\theta}_0) = 0.988,$$

which is distributed asymptotically as χ_1^2 under H_0 . The p-value is 0.320 thereby having the same qualitative outcome as the LR and W tests at the 5% level of significance. $\square$

7.7.1 Alternative LM Test I

An alternative form of the Lagrange multiplier test of autocorrelation that is commonly used is based on replacing $J_T(\hat{\theta}_0)$ by the information matrix $I(\theta)$ evaluated under the null hypothesis

$$LM = (T - 1)G_T'(\hat{\theta}_0)I^{-1}(\hat{\theta}_0)G_T(\hat{\theta}_0). \quad (7.44)$$

This alternative form of the LM test not only has the advantage of simplifying the implementation of the test, but it highlights how other well known tests of autocorrelation can be interpreted as an LM test.

Consider once again the model in Example 7.2. The gradient in equation (7.6) evaluated under the null hypothesis $\rho_1 = 0$, becomes

$$G_T(\hat{\theta}_0) = \frac{1}{\sigma^2(T - 1)} \sum_{t=2}^T \begin{bmatrix} u_t \\ u_t x_t \\ u_t u_{t-1} \\ -\frac{1}{2} + \frac{u_t^2}{2\sigma^2} \end{bmatrix}, \quad (7.45)$$

because under the null hypothesis $v_t = u_t = y_t - \beta_0 - \beta_1 x_t$. The information matrix in (7.15) evaluated under the null hypothesis, $\rho_1 = 0$, is

$$I(\hat{\theta}_0) = \frac{1}{\sigma^2(T - 1)} \sum_{t=2}^T \begin{bmatrix} 1 & x_t & 0 & 0 \\ x_t & x_t^2 & 0 & 0 \\ 0 & 0 & u_{t-1}^2 & 0 \\ 0 & 0 & 0 & \frac{1}{2\sigma^2} \end{bmatrix}, \quad (7.46)$$

where the [3,3] cell is obtained by replacing σ^2 by u_{t-1}^2 . This substitution also implies that the last cell of $G(\theta_0)$ is zero and this, together with the block-diagonal structure of the information matrix in (7.46), means that the LM statistic in (7.44) simplifies to

$$LM = \frac{1}{\sigma_0^2} \begin{bmatrix} \sum_{t=2}^T u_t \\ \sum_{t=2}^T u_t x_t \\ \sum_{t=2}^T u_t u_{t-1} \end{bmatrix}' \begin{bmatrix} T-1 & \sum_{t=2}^T x_t & 0 \\ \sum_{t=2}^T x_t & \sum_{t=2}^T x_t^2 & 0 \\ 0 & 0 & \sum_{t=2}^T u_{t-1}^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=2}^T u_t \\ \sum_{t=2}^T u_t x_t \\ \sum_{t=2}^T u_t u_{t-1} \end{bmatrix}.$$

To understand the structure of this form of the LM test, it is informative to define $z_t = \{1, x_t, u_{t-1}\}$, so that the LM statistic is rewritten as

$$LM = \frac{1}{\sigma^2} \left[\sum_{t=2}^T z_t u_t \right]' \left[\sum_{t=2}^T z_t z_t' \right]^{-1} \left[\sum_{t=2}^T z_t u_t \right] = (T-1)R^2, \quad (7.47)$$

where R^2 is the coefficient of determination from regressing u_t on z_t . The last step arises because $(T-1)\sigma^2 = \sum_{t=2}^T u_t^2$ represents the total sum of squares of the dependent variable, given by u_t , whereas the numerator is the explained sum of squares from the regression of u_t on z_t since $\sum_{t=1}^T u_t = 0$. In the case where the dependent variable does not have zero mean, the R^2 in (7.47) is the uncentered coefficient of determination.

This analysis suggests that the LM test can be constructed as follows.

- Step 1:** Regress y_t on a constant and x_t to get the residuals $\hat{u}_t$.
- Step 2:** Regress $\hat{u}_t$ on a constant, x_t and the lagged residual $\hat{u}_{t-1}$.
- Step 3:** Construct the test statistic $LM = (T-1)R^2$ where T is the sample size and R^2 is the coefficient of determination from the second-stage regression. Under the null hypothesis LM is asymptotically distributed as χ_1^2 .

This version of the LM test is easily extended to test higher-order autocorrelation. For example, to test for autocorrelation of order p , the steps are as follows.

- Step 1:** Regress y_t on a constant and x_t , to get the residuals $\hat{u}_t$.
- Step 2:** Regress $\hat{u}_t$ on a constant, x_t and the lagged residuals $\hat{u}_{t-1}, \dots, \hat{u}_{t-p}$.
- Step 3:** Construct the test statistic $LM = (T-p)R^2$, which is asymptotically distributed as χ_p^2 under the null hypothesis.

7.7.2 Alternative LM Test II

An alternative way to motivate the form of the Lagrange multiplier statistic in (7.47) is in terms of the Gauss-Newton algorithm. In the case of the

regression model with an AR(1) disturbance, the steps are as follows. First, express the model with v_t as the dependent variable

$$v_t = u_t - \rho_1 u_{t-1} = (y_t - \beta_0 - \beta_1 x_t) - \rho_1 (y_{t-1} - \beta_0 - \beta_1 x_{t-1}).$$

Second, construct the derivatives

$$\begin{aligned} z_{1,t} &= -\frac{\partial v_t}{\partial \beta_0} = 1 - \rho_1 \\ z_{2,t} &= -\frac{\partial v_t}{\partial \beta_1} = x_t - \rho_1 x_{t-1} \\ z_{3,t} &= -\frac{\partial v_t}{\partial \rho_1} = y_{t-1} - \beta_0 - \beta_1 x_{t-1}. \end{aligned}$$

Third, evaluate all terms under the null $\rho_1 = 0$

$$\begin{aligned} v_t &= u_t \\ z_{1,t} &= -\frac{\partial v_t}{\partial \beta_0} = 1 \\ z_{2,t} &= -\frac{\partial v_t}{\partial \beta_1} = x_t \\ z_{3,t} &= -\frac{\partial v_t}{\partial \rho_1} = y_{t-1} - \beta_0 - \beta_1 x_{t-1} = u_{t-1}. \end{aligned}$$

The regression of u_t on $\{z_{1,t}, z_{2,t}, z_{3,t}\}$ is the same regression equation used to construct (7.47).

7.7.3 Alternative LM Test III

An even simpler form of the LM test follows from noting that the gradient vector under the null hypothesis is

$$G_T(\hat{\theta}_0) = \frac{1}{\sigma^2(T-1)} \begin{bmatrix} \sum_{t=2}^T u_t \\ \sum_{t=2}^T u_t x_t \\ \sum_{t=2}^T u_t u_{t-1} \end{bmatrix} = \frac{1}{\sigma^2(T-1)} \begin{bmatrix} 0 \\ 0 \\ \sum_{t=2}^T u_t u_{t-1} \end{bmatrix},$$

and that $I_{2,2}$ in expression (7.16) under the null hypothesis is

$$I_{2,20} = \frac{1}{\sigma^2(T-1)} \sum_{t=2}^T \sigma^2 = 1.$$

Using these terms in expression (7.44) yields the LM statistic

$$LM_I = (T-1) \frac{1}{\sigma^4(T-1)^2} \left(\sum_{t=2}^T u_{t-1} u_t \right)^2 = (T-1) \left(\frac{\sum_{t=2}^T u_{t-1} u_t}{\sum_{t=2}^T u_t^2} \right)^2 = (T-1) r_1^2,$$

where r_1 is the first-order autocorrelation coefficient. Because this statistic is distributed as χ_1^2 , another form of the test is

$$\sqrt{T-1} r_1 \sim N(0, 1).$$

7.8 Systems of Equations

Estimation and testing of systems of regression equations with autocorrelated disturbances in theory proceeds as in the case of single equations, although estimation is potentially computationally more demanding. An example of a bivariate system of equations with a vector AR(1) disturbance term is

$$\begin{aligned} y_{1,t} &= \beta_1 y_{2,t} + \alpha_1 x_{1,t} + u_{1,t} \\ y_{2,t} &= \beta_2 y_{1,t} + \alpha_2 x_{2,t} + u_{2,t}, \end{aligned}$$

where the disturbances follow

$$\begin{aligned} u_{1,t} &= \rho_{1,1} u_{1,t-1} + \rho_{1,2} u_{2,t-1} + v_{1,t} \\ u_{2,t} &= \rho_{2,1} u_{1,t-1} + \rho_{2,2} u_{2,t-1} + v_{2,t}, \end{aligned}$$

and

$$\begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{1,1} & \sigma_{1,2} \\ \sigma_{2,1} & \sigma_{2,2} \end{bmatrix} \right),$$

with $\sigma_{1,2} = \sigma_{2,1}$.

More generally, when the specification of systems of regression equations in Chapter 5 is used, a system of N regression equations with first-order vector autocorrelation is

$$\begin{aligned} y_t B + x_t A &= u_t \\ u_t &= u_{t-1} P + v_t \\ v_t &\sim N(0, V), \end{aligned} \tag{7.48}$$

where y_t is a $(1 \times N)$ vector of dependent variables, x_t is a $(1 \times K)$ vector of independent variables, B is a $(N \times N)$, A is a $(K \times N)$, P is a $(N \times N)$ matrix

of autocorrelation parameters and V is the $(N \times N)$ covariance matrix of the disturbances. In the bivariate example

$$P = \begin{bmatrix} \rho_{1,1} & \rho_{1,2} \\ \rho_{2,1} & \rho_{2,2} \end{bmatrix}.$$

Higher-order vector autoregressive systems that include lagged dependent variables as explanatory variables, or even vector ARMA(p,q) models, can also be specified.

7.8.1 Estimation

The starting point for estimating the multivariate model with first-order vector autocorrelation by maximum likelihood is the pdf of v_t , assumed to be the multivariate normality distribution

$$f(v_t) = \left(\frac{1}{\sqrt{2\pi}} \right)^N |V|^{-1/2} \exp \left[-\frac{1}{2} v_t V^{-1} v_t' \right].$$

The transformation of variable technique determines the density of u_t to be

$$\begin{aligned} f(u_t) &= f(v_t) \left| \frac{\partial v_t}{\partial u_t} \right| \\ &= \left(\frac{1}{\sqrt{2\pi}} \right)^N |V|^{-1/2} \exp \left[-\frac{1}{2} (u_t - u_{t-1}P) V^{-1} (u_t - u_{t-1}P)' \right], \end{aligned}$$

because $|\partial v_t / \partial u_t| = 1$. Similarly, the transformation of variable technique determines the density of y_t to be

$$\begin{aligned} f(y_t) &= f(u_t) \left| \frac{\partial u_t}{\partial y_t} \right| \\ &= \left(\frac{1}{\sqrt{2\pi}} \right)^N |V|^{-1/2} \exp \left[-\frac{1}{2} (u_t - u_{t-1}P) V^{-1} (u_t - u_{t-1}P)' \right] |B|, \end{aligned}$$

with $u_t = y_t B + x_t A$ and $|\partial u_t / \partial y_t| = B$. The log-likelihood function for the t^{th} observation is

$$\ln l_t(\theta) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |V| + \ln |B| - \frac{1}{2} v_t V^{-1} v_t',$$

where $v_t = u_t - u_{t-1}P$. The conditional log-likelihood function is

$$\ln L_T(\theta) = \frac{1}{T-1} \sum_{t=2}^T \ln l_t(\theta),$$

which is maximized with respect to $\theta = \{B, A, P, V\}$ in the usual way.

Example 7.12 Estimation of a Vector AR(1) Model This example estimates a bivariate system of equations with a vector first-order autocorrelation disturbances using simulated data. The model in equations (7.48) with parameter matrices

$$\begin{aligned}
 B &= \begin{bmatrix} 1 & -\beta_2 \\ -\beta_1 & 1 \end{bmatrix} = \begin{bmatrix} 1.000 & -0.200 \\ -0.600 & 1.000 \end{bmatrix} \\
 A &= \begin{bmatrix} -\alpha_1 & 0 \\ 0 & -\alpha_2 \end{bmatrix} = \begin{bmatrix} -0.400 & 0.000 \\ 0.000 & 0.500 \end{bmatrix} \\
 P &= \begin{bmatrix} \rho_{1,1} & \rho_{1,2} \\ \rho_{2,1} & \rho_{2,2} \end{bmatrix} = \begin{bmatrix} 0.800 & -0.200 \\ 0.100 & 0.600 \end{bmatrix} \\
 V &= \begin{bmatrix} \sigma_{1,1} & \sigma_{1,2} \\ \sigma_{2,1} & \sigma_{2,2} \end{bmatrix} = \begin{bmatrix} 1.0 & 0.5 \\ 0.5 & 1.0 \end{bmatrix},
 \end{aligned}$$

is simulated for $T = 500$ observations. The maximum likelihood estimates, given in Table 7.2, are statistically indistinguishable from their population parameter counterparts. $\square$

Table 7.2

Maximum likelihood estimates of the bivariate model with a vector AR(1) disturbance, using the BFGS algorithm with derivatives computed numerically: standard errors computed based on the Hessian.

Parameter	Population	Estimate	Std error	t-stat.
β_1	0.6	0.589	0.021	27.851
α_1	0.4	0.396	0.010	38.383
β_2	0.2	0.188	0.024	7.915
α_2	-0.5	-0.500	0.017	-29.764
$\rho_{1,1}$	0.8	0.801	0.027	29.499
$\rho_{1,2}$	0.1	0.078	0.034	2.277
$\rho_{2,1}$	-0.2	-0.189	0.028	-6.726
$\rho_{2,2}$	0.6	0.612	0.035	17.344

7.8.2 Testing

The LR, Wald and LM tests can all be used to test for autocorrelation in systems of equations. An example using the LR test follows.

Example 7.13 Likelihood Ratio Test of a Vector AR(1) Model

To test the null hypothesis of no autocorrelation in the system given by equation (7.48), the restrictions $\rho_{1,1} = \rho_{1,2} = \rho_{2,1} = \rho_{2,2} = 0$ are tested. For the simulated data generated in Example 7.12, the log-likelihood functions of the unconstrained and constrained models are, respectively,

$$(T-1) \ln L_T(\hat{\theta}_1) = -1399.417 \quad (T-1) \ln L_T(\hat{\theta}_0) = -1844.523.$$

The LR statistic is

$$\begin{aligned} LR &= -2((T-1) \ln L_T(\hat{\theta}_0) - (T-1) \ln L_T(\hat{\theta}_1)) \\ &= -2 \times (-1844.523 + 1399.417) = 890.212, \end{aligned}$$

which is distributed asymptotically as χ_4^2 under H_0 . The p-value is 0.000 resulting in the null hypothesis being rejected at the 5% level of significance. An intermediate set of restrictions is given by $\rho_{1,2} = \rho_{2,1} = 0$. The log-likelihood function for this model under the null hypothesis is now $(T-1) \ln L_T(\hat{\theta}_0) = -1419.844$, yielding

$$\begin{aligned} LR &= -2((T-1) \ln L_T(\hat{\theta}_0) - (T-1) \ln L_T(\hat{\theta}_1)) \\ &= -2 \times (-1419.844 + 1399.417) = 40.855, \end{aligned}$$

which is distributed asymptotically as χ_2^2 under H_0 . The p-value is 0.000 resulting in this null hypothesis also being rejected at the 5% level of significance. Both of these test results are consistent with the underlying specification. $\square$

7.9 Applications

7.9.1 Illiquidity and Hedge Funds

Getmansky, Lo and Makarov (2004) argue that as hedge funds tend to contain assets which are not actively traded, the returns they generate are relatively smoother than the returns generated by highly liquid assets such as the S&P500. In the case of the capital asset pricing model (CAPM), where the excess return on a hedge fund is expressed as a function of the excess return on the S&P500, the discrepancy in the autocorrelation properties of the two series manifests itself in autocorrelation in the disturbance term.

The results of estimating the CAPM for Equity hedge fund data are (with standard errors in parentheses based on the Hessian)

$$r_t - f_t = \underset{(0.007)}{0.001} + \underset{(0.005)}{0.226} (m_t - f_t) + \hat{u}_t,$$

where r_t is the return on the hedge fund, m_t is market return and f_t is the risk-free rate of return. The LM test of first-order autocorrelation yields a value of $LM = 20.490$. The number of restrictions is 1, yielding a p-value of 0.000 thereby suggesting significant evidence of serial correlation. Correcting the model for first-order autocorrelation yields the results

$$\begin{aligned} r_t - f_t &= \underset{(0.008)}{0.001} + \underset{(0.006)}{0.217} (m_t - f_t) + \hat{u}_t \\ \hat{u}_t &= \underset{(0.024)}{0.117} \hat{u}_{t-1} + \hat{v}_t. \end{aligned}$$

Applying the LM test of first-order autocorrelation to these residuals results in a value of $LM = 3.356$. The p-value is now 0.067, showing that the adjusted model captures the autocorrelation in the residuals.

When this approach is repeated for the Convertible Arbitrage hedge fund, the estimated CAPM without any adjustment for autocorrelation is (with standard errors in parentheses based on the Hessian)

$$r_t - f_t = \underset{(0.011)}{-0.032} - \underset{(0.008)}{0.030} (m_t - f_t) + \hat{u}_t.$$

The LM test of first-order autocorrelation yields a value of $LM = 118.409$. The number of restrictions is 1, yielding a p-value of 0.000 once again providing significant evidence of serial correlation. Correcting the model for first-order autocorrelation yields the results

$$\begin{aligned} r_t - f_t &= \underset{(0.014)}{-0.032} - \underset{(0.008)}{0.053} (m_t - f_t) + \hat{u}_t \\ \hat{u}_t &= \underset{(0.023)}{0.267} \hat{u}_{t-1} + \hat{v}_t. \end{aligned}$$

Applying the LM test of first-order autocorrelation to these residuals results in a value of $LM = 22.836$. Unlike the results of the Equity Hedge fund, it is now necessary to make allowance for even higher autocorrelation because the residuals for the Convertible Arbitrage hedge fund still display significant autocorrelation.

7.9.2 Beach-Mackinnon Simulation Study

Beach and MacKinnon (1978) investigate the finite sampling properties of the exact and conditional maximum likelihood estimators using Monte Carlo

experiments based on the AR(1) regression model

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t \\ u_t &= \rho_1 u_{t-1} + v_t \\ v_t &\sim N(0, \sigma^2), \end{aligned} \quad (7.49)$$

where y_t is the dependent variable, x_t is the explanatory variable and u_t and v_t are disturbance terms. The true parameter values are $\beta_0 = 1$, $\beta_1 = 1$, $\rho_1 = 0.6$ and $\sigma^2 = 0.0036$. The sample sizes are $T = 20$ and 50 , and the number of replications is 5000 . Finally, the explanatory variable x_t is generated as

$$x_t = \exp(0.04t) + w_t, \quad w_t \sim N(0, 0.0009),$$

where t is a linear trend. The explanatory variable is treated as fixed in repeated samples by drawing random numbers for w_t only once and then holding these values fixed for each of the 5000 replications.

The bias and RMSE of the estimator, computed as

$$\text{Bias} = \frac{1}{5000} \sum_{i=1}^{5000} \hat{\theta}_i - \theta, \quad \text{RMSE} = \frac{1}{5000} \sum_{i=1}^{5000} (\hat{\theta}_i - \theta)^2,$$

are given in Table 7.3.

Table 7.3

Monte Carlo performance of alternative estimators of the autocorrelated regression model parameters in (7.49). The bias and RMSE are expressed as a percentage. The number of replications is 5000 .

Parameter	T	Exact		Conditional		OLS	
		Bias ($\times 100$)	RMSE ($\times 100$)	Bias ($\times 100$)	RMSE ($\times 100$)	Bias ($\times 100$)	RMSE ($\times 100$)
β_0	20	-0.077	11.939	0.297	44.393	-0.045	12.380
	50	-0.069	3.998	-0.101	4.353	-0.084	4.080
β_1	20	0.026	7.315	-0.141	10.302	0.009	7.607
	50	0.027	1.060	0.034	1.114	0.031	1.091
ρ_1	20	-23.985	32.944	-24.189	33.267	n.a.	n.a.
	50	-8.839	15.455	-8.977	15.583	n.a.	n.a.
σ^2	20	-0.061	0.121	-0.068	0.126	n.a.	n.a.
	50	-0.022	0.073	-0.023	0.074	n.a.	n.a.

For comparison, the least squares estimates of β_0 and β_1 obtained by

regressing of y_t on a constant and x_t and based on the assumption of no autocorrelation, are also given. The results show a reduction in bias from using the exact over the conditional maximum likelihood estimators for all four estimators. Both maximum likelihood estimators of ρ_1 and σ^2 are biased downwards. The exact maximum likelihood estimator also provides efficiency gains over the conditional maximum likelihood estimator in small samples. In the case of β_0 , the ratio of the RMSEs of the conditional maximum likelihood estimator to the exact maximum likelihood estimator for $T = 20$ is $44.393/11.939 = 3.718$, indicating a large efficiency differential. The value of the ratio reduces quickly to $4.353/3.998 = 1.089$ for $T = 50$. In the case of β_1 , the ratio for $T = 20$ is $10.302/7.315 = 1.408$ but reduces to $1.114/1.060 = 1.051$ when $T = 50$.

An interesting feature of the finite sample results in Table 7.3 is the performance of the least squares estimator of β_0 and β_1 . For $T = 20$, this estimator has smaller bias than both maximum likelihood estimators while still having lower bias than the conditional maximum likelihood estimator for $T = 50$. Nonetheless, the exact maximum likelihood estimator still exhibits better efficiency than does the least squares estimator for $T = 20$.

7.10 Exercises

(1) Simulating a Regression Model with Autocorrelation

Gauss file(s)	<code>auto_simumlate.g</code>
Matlab file(s)	<code>auto_simulate.m</code>

- (a) Simulate the following regression model with an AR(1) disturbance term for a sample of $T = 200$ observations

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t \\ u_t &= \rho_1 u_{t-1} + v_t \\ v_t &\sim iid N(0, \sigma^2), \end{aligned}$$

with $\beta_0 = 2$, $\beta_1 = 1$, $\rho_1 = 0.95$ and $\sigma = 3$ and with the explanatory variable generated as $x_t = 0.5t + N(0, 1)$. Compare the simulated data to the conditional mean of y_t

$$\mu_t = \beta_0 + \beta_1 x_t.$$

Hence reproduce panel (a) of Figure 7.1.

- (b) Simulate the following regression model with a MA(1) disturbance term for a sample of $T = 200$ observations

$$\begin{aligned}y_t &= \beta_0 + \beta_1 x_t + u_t \\u_t &= v_t + \delta_1 v_{t-1} \\v_t &\sim iid N(0, \sigma^2),\end{aligned}$$

with $\beta_0 = 2$, $\beta_1 = 1$, $\delta_1 = 0.95$ and $\sigma = 3$ and with x_t constructed as above. Compare the simulated data to the conditional mean of y_t . Hence reproduce panel (b) of Figure 7.1.

- (c) Simulate the following regression model with an AR(2) disturbance term for a sample of $T = 200$ observations

$$\begin{aligned}y_t &= \beta_0 + \beta_1 x_t + u_t \\u_t &= \rho_1 u_{t-1} + \rho_2 u_{t-2} + v_t \\v_t &\sim iid N(0, \sigma^2),\end{aligned}$$

with $\beta_0 = 2$, $\beta_1 = 1$, $\rho_1 = 0.1$, $\rho_2 = -0.9$ and $\sigma = 3$ and with x_t constructed as above. Compare the simulated data to the conditional mean of y_t .

- (d) Simulate the following regression model with an ARMA(2,2) disturbance term for a sample of $T = 200$ observations

$$\begin{aligned}y_t &= \beta_0 + \beta_1 x_t + u_t \\u_t &= \rho_1 u_{t-1} + \rho_2 u_{t-2} + v_t + \delta_1 v_{t-1} + \delta_2 v_{t-2} \\v_t &\sim iid N(0, \sigma^2),\end{aligned}$$

with $\beta_0 = 2$, $\beta_1 = 1$, $\rho_1 = 0.1$, $\rho_2 = -0.9$, $\delta_1 = 0.3$, $\delta_2 = 0.2$ and $\sigma = 3$ and with x_t constructed as above. Compare the simulated data to the conditional mean of y_t .

(2) A Dynamic Model of U.S. Investment

Gauss file(s)	auto_invest.g, auto_test.g, usinvest.dat
Matlab file(s)	auto_invest.m, auto_test.g, usinvest.mat

This exercise uses quarterly U.S. data from March 1957 to September 2010. Consider the model

$$\begin{aligned}dri_t &= \beta_0 + \beta_1 dry_t + \beta_2 rint_t + u_t \\u_t &= \rho_1 u_{t-1} + v_t \\v_t &\sim iid N(0, \sigma^2),\end{aligned}$$

where dri_t is the quarterly percentage change in real investment, dry_t is the quarterly percentage change in real income, $rint_t$ is the real interest rate expressed as a quarterly percentage, and the parameters are $\theta = \{\beta_0, \beta_1, \beta_2, \rho_1, \sigma^2\}$.

- (a) Plot the real investment series (dri_t) and interpret its time series properties.
- (b) Estimate the model by exact maximum likelihood using the full log-likelihood function and conditional maximum likelihood. Compare the parameter estimates with the estimates reported in Table 7.1.
- (c) Compute the standard errors using the Hessian matrix and the outer product of the gradients matrix.
- (d) Test the hypotheses

$$H_0 : \rho_1 = 0, \quad H_0 : \rho_1 \neq 0.$$

using a LR test, a Wald test, a LM test, a LM test based on the Gauss-Newton algorithm version and a LM test based on $\sqrt{T-1} r_1$, where r_1 is the first-order autocorrelation coefficient of the residuals obtained by estimating the constrained model.

(3) Asymptotic Distribution

Gauss file(s)	<code>auto_distribution.g</code>
Matlab file(s)	<code>auto_distribution.m</code>

Consider the model

$$\begin{aligned} y_t &= \alpha + \beta x_t + u_t \\ u_t &= \rho u_{t-1} + v_t \\ v_t &\sim iid N(0, \sigma^2), \end{aligned}$$

and let the values of the population parameters be $\alpha_0 = 1$, $\beta_0 = 1$, $\rho_0 = 0.6$ and $\sigma_0^2 = 10$.

- (a) If the explanatory variable is deterministic and generated as $x_t \sim U(0, 1) - 0.5$, use expression (7.27) to compute the asymptotic variance of the conditional maximum likelihood estimator of β for a sample of size $T = 500$. Compare this result by computing the MSE of $\hat{\beta}$ using a Monte Carlo experiment with 5000 replications. Also use the simulation results to compute the empirical distribution of $\hat{\beta}$ and compare the result with the normal distribution.
- (b) Repeat part (a) for the maximum likelihood parameter estimators of α , ρ and σ^2 .

(4) **Efficiency of the Sample Mean**

Gauss file(s)	auto_mean.g
Matlab file(s)	auto_mean.m

Consider the model

$$\begin{aligned}y_t &= \alpha + u_t \\u_t &= \rho u_{t-1} + v_t \\v_t &\sim iid N(0, \sigma^2) .\end{aligned}$$

- (a) Prove that

$$T + 2 \sum_{i=1}^{T-1} \sum_{t=1}^{T-i} \rho_0^i = \frac{T(1 - \rho_0^2) - 2\rho_0(1 - \rho_0^T)}{(1 - \rho_0)^2} .$$

Verify this result by computing the left hand side for various values of T and ρ_0 .

- (b) Derive an expression of the asymptotic variance of α for the conditional maximum likelihood estimator.
- (c) Derive an expression of the asymptotic variance of α for the OLS estimator based on the assumption that $\rho = 0$.
- (d) Compare the asymptotic variances of the two estimators assuming parameter values of $\rho_0 = 0.6$ and $\sigma_0^2 = 10$ and samples of size $T = \{5, 50, 500\}$.

(5) **Relative Efficiency of Maximum Likelihood and Least Squares**

Gauss file(s)	auto_efficiency.g
Matlab file(s)	auto_efficiency.m

In Section 7.5 it is shown that the relative efficiency of the maximum likelihood and least squares estimators in large samples depends upon the generating process of the explanatory variable x_t in the model

$$\begin{aligned}y_t &= \alpha + \beta x_t + u_t \\u_t &= \rho u_{t-1} + v_t \\v_t &\sim iid N(0, \sigma^2) .\end{aligned}$$

The following Monte Carlo experiments, based on the population parameter values $\alpha_0 = 1$, $\beta_0 = 1$, $\rho_0 = 0.6$, $\sigma_0^2 = 0.0036$, with 5000 replications and sample sizes of $T = \{50, 100, 200, 500, 1000, 2000\}$ demonstrate

these properties. For each experiment, compute the MSE

$$MSE = \frac{1}{5000} \sum_{i=1}^{5000} (\hat{\beta}_i - \beta_0)^2,$$

of the maximum likelihood and least squares estimators.

- (a) Simulate the model with $x_t = t$, a time trend, and show that the MSE of the maximum likelihood and least squares estimators of α and β converge to unity.
- (b) Simulate the model with $x_t = \sin(2\pi t/T)$, a sinusoidal trend, and show that the MSE of the maximum likelihood and least squares estimators of α and β converge to unity.
- (c) Simulate the model where the explanatory variable is given by

$$x_t = \phi x_{t-1} + w_t, \quad w_t \sim N(0, 0.0036t),$$

with $\phi = 0.6$. In the simulations, treat x_t as stochastic by redrawing w_t in each replication.

- (i) Show that, as the sample size increases, the ratio of the MSE of the maximum likelihood and least squares estimators for the slope parameter β converges to the asymptotic efficiency ratio given by

$$\frac{\text{var}(\hat{\beta}_{LS})}{\text{var}(\hat{\beta})} = \frac{(1 - 2\rho_0\phi_0 + \rho_0^2)(1 + \rho_0\phi_0)}{(1 - \rho_0^2)(1 - \rho_0\phi_0)}.$$

- (ii) Repeat the exercise for different values of ρ and ϕ .
- (iii) Demonstrate that the asymptotic efficiency ratio is independent of the parameters α and β by choosing different values for these parameters in the experiments.
- (iv) Demonstrate that the asymptotic efficiency ratio is independent of the parameters σ^2 and $\text{var}(w_t)$ by choosing different values for these parameters in the experiments.

(6) Hatanaka Estimator

Gauss file(s)	auto_hatanaka.g
Matlab file(s)	auto_hatanaka.m

Consider the model

$$\begin{aligned}y_t &= \beta_0 + \beta_1 x_t + \alpha y_{t-1} + u_t \\u_t &= \rho u_{t-1} + v_t \\v_t &\sim iid N(0, \sigma_v^2) .\end{aligned}$$

- (a) Assuming that x_t is deterministic, prove that

$$\text{cov}(y_{t-1}, u_t) = \frac{\rho \sigma^2}{(1 - \alpha \rho)(1 - \rho^2)} .$$

- (b) Choosing parameter values of $\beta_0 = 1$, $\beta_1 = 1$, $\alpha_1 = 0.5$, $\rho = 0.6$ and $\sigma_v^2 = 0.1$, verify the result in (a) by simulating the model for $T = 1,000,000$ and computing the sample covariance of y_{t-1} and u_t .
- (c) Show how you would estimate the model using the Gauss-Newton algorithm.
- (d) Compare the sampling properties of the Hatanaka estimator for various sample sizes with the maximum likelihood estimator, as well as with the OLS estimator based on a regression of y_t on a constant, x_t and y_{t-1} .

(7) Systems with Autocorrelation

Gauss file(s)	auto_system.g
Matlab file(s)	auto_system.m

Simulate the following bivariate system of equations with first-order vector autocorrelation disturbances for $T = 500$ observations

$$\begin{aligned}y_t B + x_t A &= u_t \\u_t &= u_{t-1} P + v_t \\v_t &\sim N(0, V) ,\end{aligned}$$

where the parameters are given as

$$\begin{aligned}
 B &= \begin{bmatrix} 1 & -\beta_2 \\ -\beta_1 & 1 \end{bmatrix} = \begin{bmatrix} 1.000 & -0.200 \\ -0.600 & 1.000 \end{bmatrix} \\
 A &= \begin{bmatrix} -\alpha_1 & 0 \\ 0 & -\alpha_2 \end{bmatrix} = \begin{bmatrix} -0.400 & 0.000 \\ 0.000 & 0.500 \end{bmatrix} \\
 P &= \begin{bmatrix} \rho_{1,1} & \rho_{1,2} \\ \rho_{2,1} & \rho_{2,2} \end{bmatrix} = \begin{bmatrix} 0.800 & -0.200 \\ 0.100 & 0.600 \end{bmatrix} \\
 V &= \begin{bmatrix} \sigma_{1,1} & \sigma_{1,2} \\ \sigma_{2,1} & \sigma_{2,2} \end{bmatrix} = \begin{bmatrix} 1.0 & 0.5 \\ 0.5 & 1.0 \end{bmatrix}.
 \end{aligned}$$

- (a) Estimate the model using the conditional maximum likelihood estimator and compare the results with the parameter estimates reported in Table 7.2.
- (b) Use a LR test to test the following restrictions.
 - (i) $\rho_{1,1} = \rho_{1,2} = \rho_{2,1} = \rho_{2,2} = 0$.
 - (ii) $\rho_{1,2} = \rho_{2,1} = 0$.
- (c) Use a Wald test to the following restrictions.
 - (i) $\rho_{1,1} = \rho_{1,2} = \rho_{2,1} = \rho_{2,2} = 0$.
 - (ii) $\rho_{1,1} = \rho_{2,2}, \rho_{1,2} = \rho_{2,1}$.
- (d) Repeat parts (a) to (c), except simulate the model with no autocorrelation

$$P = \begin{bmatrix} \rho_{1,1} & \rho_{1,2} \\ \rho_{2,1} & \rho_{2,2} \end{bmatrix} = \begin{bmatrix} 0.000 & 0.000 \\ 0.000 & 0.000 \end{bmatrix}.$$

(8) Illiquidity and Hedge Funds

Gauss file(s)	auto_hedge.g, hedge.xlsx
Matlab file(s)	auto_hedge.m, hedge.mat

The data consist of $T = 1869$ daily returns for the period 1 April 2003 to the 28 May 2010 on seven hedge funds (Convertible Arbitrage, Distressed Securities, Equity Hedge, Event Driven, Macro, Merger Arbitrage, Equity Market Neutral) and three market indexes (Dow, NASDAQ and S&P500).

- (a) For each of the seven hedge funds, estimate the CAPM

$$r_t - rf_t = \beta_0 + \beta_1(m_t - rf_t) + u_t,$$

where r_t is the return on a hedge fund, m_t is the market return and

rf_t is the risk-free interest rate. Interpret the parameter estimates.
Test for an AR(1) disturbance term.

- (b) For each of the seven hedge funds, estimate a dynamic CAPM where

$$u_t = \rho y_{t-1} + v_t,$$

and reapply the LM test to the estimated residuals.

(9) **Beach-MacKinnon Monte Carlo Study**

Gauss file(s)	<code>auto_beachmack.g</code>
Matlab file(s)	<code>auto_beachmack.m</code>

Simulate the model

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t \\ u_t &= \rho_1 u_{t-1} + v_t \\ v_t &\sim iid N(0, \sigma^2), \end{aligned}$$

with the population parameters given by

$$\theta_0 = \{\beta_0 = 1, \beta_1 = 1, \rho_1 = 0.6, \sigma^2 = 0.0036\},$$

and the independent variable is generated as

$$x_t = \exp(0.04t) + N(0, 0.0009).$$

- (a) For 200 replications and $T = 20$ observations, compute the following statistics on the sampling distributions of the exact and conditional maximum likelihood estimators, as well as the least squares estimators without any autocorrelation adjustment

(i) The mean: $\frac{1}{200} \sum_{i=1}^{200} \hat{\theta}_i.$

(ii) The bias: $\theta_0 - \frac{1}{200} \sum_{i=1}^{200} \hat{\theta}_i.$

(iii) The RMSE: $\sqrt{\frac{1}{200} \sum_{i=1}^{200} (\hat{\theta}_i - \theta_0)^2}.$

- (iv) Efficiency: ratio of least squares RMSE to the exact maximum likelihood RMSE.

- (b) Repeat part (a) using $T = 50$ and comment on the asymptotic properties of the estimators.
(c) Repeat parts (a) and (b) for $\rho_1 = 0.8, 0.99$ and discuss the results.

- (d) Repeat parts (a) to (c) using $x_t = N(0, \sigma_x^2 = 0.0625)$ as the (stationary) explanatory variable, and compare the results with the case of a trending x_t variable.
- (e) For the stationary x_t variable case, compare the square of the RMSE of β_1 to the corresponding asymptotic variances of the maximum likelihood and least squares estimators given in Harvey (1990, pp. 197-198), Harvey, A.C.

$$\text{MLE} \quad : \quad \text{var}(\beta_1) = \frac{1}{T} \frac{\sigma^2}{\sigma_x^2} \frac{1}{1 + \rho_1^2}$$

$$\text{OLS} \quad : \quad \text{var}(\beta_1) = \frac{1}{T} \frac{\sigma^2}{\sigma_x^2} \frac{1}{1 - \rho_1^2},$$

and compare the simulated efficiency results with the asymptotic efficiency ratio given by

$$\frac{\sqrt{1 + \rho_1^2}}{\sqrt{1 - \rho_1^2}}.$$

- (f) Repeat parts (a) to (d) for 10000 and 20000 replications, and discuss the sensitivity of the simulation results to the number of replications.

8

Heteroskedastic Regression Models

8.1 Introduction

The regression models considered in Chapters 5 to 7 have allowed for a time-varying mean and a constant variance. This latter property is known as homoskedasticity. A natural extension of this model is to allow for the variance to be time varying, or heteroskedastic. In this chapter, the maximum likelihood framework is applied to estimating and testing models of heteroskedasticity. More general models are also considered in which both heteroskedasticity and autocorrelation structures are present in systems of equations. In specifying this class of model, the parametric form of the distribution of the disturbances is usually assumed to be normal but this assumption may also be relaxed.

As with autocorrelated regression models dealt with in the previous chapter, estimators and testing procedures commonly applied to heteroskedastic regression models are shown to be special cases of the maximum likelihood framework developed in Part ONE. The estimators include weighted least squares and zig-zag algorithms, while the tests include the Breusch-Pagan and White tests of heteroskedasticity.

8.2 Specification

The classical linear regression model is

$$y_t = \beta_0 + \beta_1 x_t + u_t \quad u_t \sim iid N(0, \sigma^2), \quad (8.1)$$

where the disturbance term, u_t , is independently and identically distributed. To allow for a time-varying variance in equation (8.1), the model becomes

$$y_t = \beta_0 + \beta_1 x_t + u_t \quad u_t \sim N(0, \sigma_t^2). \quad (8.2)$$

The form of the time-varying variance in equation (8.2) is usually specified parametrically. Some common specifications of heteroskedasticity are

1. step heteroskedasticity : $\sigma_t^2 = \gamma_0 + \gamma_1 d_t$
2. linear heteroskedasticity : $\sigma_t^2 = \gamma_0 + \gamma_1 w_t$
3. power heteroskedasticity : $\sigma_t^2 = \gamma_0 + \gamma_1 w_t^2$
4. multiplicative heteroskedasticity : $\sigma_t^2 = \exp(\gamma_0 + \gamma_1 w_t)$,

where w_t is an exogenous explanatory variable and d_t is a suitably defined dummy variable. An important property of σ_t^2 is that it must remain positive for all t . The step function allows for two variances over the sample given by γ_0 and $\gamma_0 + \gamma_1$. Provided that the parameters γ_0 and $\gamma_0 + \gamma_1$ are restricted to be positive, the resultant estimate of σ_t^2 is positive. The linear heteroskedasticity model specifies the time variation in the variance as a linear function of an exogenous variable. Even if $\gamma_0, \gamma_1 > 0$, negative values of the exogenous variable w_t may result in a negative estimate of the variance. The power specification restricts the exogenous variable in the variance equation to be positive so that, if the restrictions $\gamma_0, \gamma_1 > 0$ are enforced, this specification ensures that the variance is positive for all t . The importance and practical appeal of the multiplicative variance specification is that the estimate of the time-varying variance is guaranteed to be positive without the need to restrict the parameters.

Example 8.1 A Regression Model with Heteroskedasticity

Consider the regression model

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t \\ \sigma_t^2 &= \gamma_0 + \gamma_1 w_t \\ u_t &\sim N(0, \sigma_t^2), \end{aligned}$$

where x_t and w_t are exogenous variables and the unknown parameters are $\theta = \{\beta_0, \beta_1, \gamma_0, \gamma_1\}$. The distribution of y_t is

$$f(y_t | x_t, w_t; \theta) = \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp \left[-\frac{(y_t - \beta_0 - \beta_1 x_t)^2}{2\sigma_t^2} \right],$$

where the conditional mean and variance of y_t are, respectively,

$$\begin{aligned} \mu_t &= \beta_0 + \beta_1 x_t \\ \sigma_t^2 &= \gamma_0 + \gamma_1 w_t. \end{aligned}$$

In this example, both the mean and the variance of the distribution of y_t are time varying: the former because of changes in x_t over time and the latter

because of changes in w_t over time. That is, each observation y_t is drawn from a different distribution for each and every t .

Figure 8.1 illustrates a variety of scatter plots of $T = 500$ observations simulated from this model, where $x_t \sim N(0, 1)$, w_t is a linear time trend and $\beta_0 = 1$, $\beta_1 = 2$, $\gamma_0 = 1$ and $\gamma_1 = 0.5$ are the parameters.

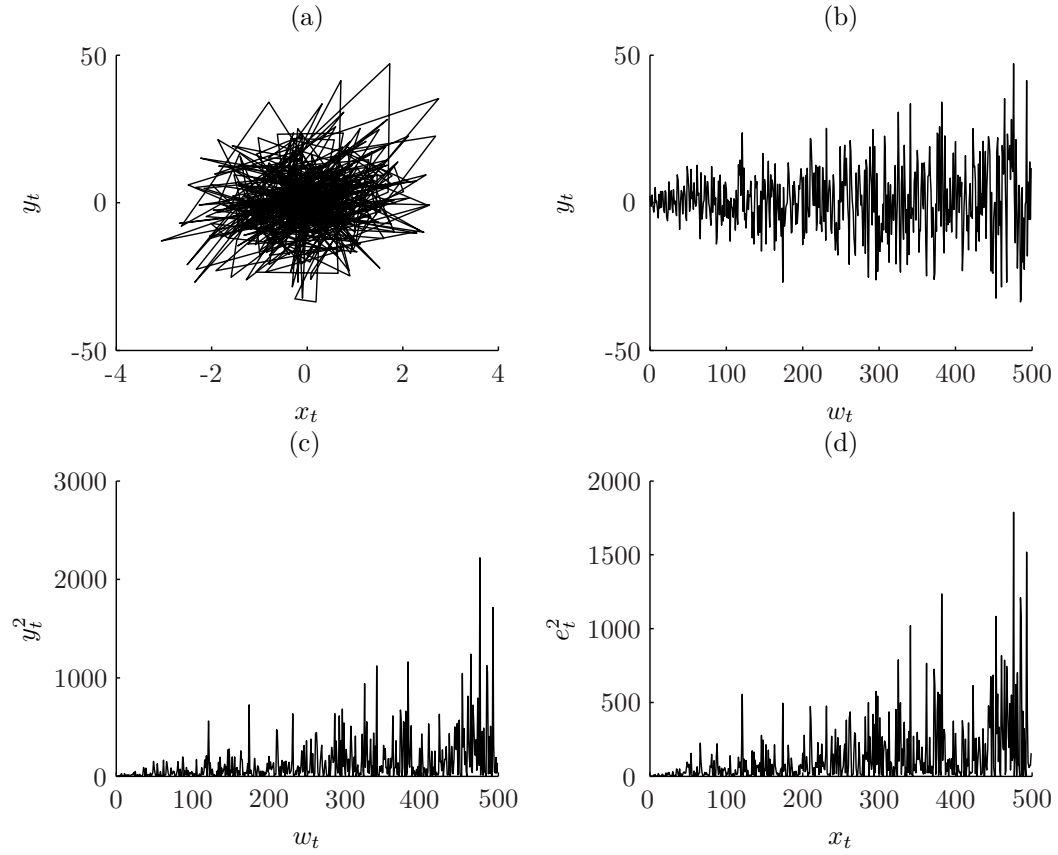


Figure 8.1 Scatter plots of the simulated data from the regression model with heteroskedasticity.

The scatter plot of y_t and x_t in panel (a) of Figure 8.1 does not reveal any evidence of heteroskedasticity. By contrast, the scatter plot between y_t and w_t in panel (b) does provide evidence of a heteroskedastic structure in the variance. The scatter plot in panel (c), between y_t^2 and w_t , provides even stronger evidence of a heteroskedastic structure in the variance, as does panel (d), which replaces y_t^2 with $\hat{u}_t^2$, where $\hat{u}_t$ represents the OLS residual of a the regression of y_t on a constant and x_t . $\square$

For the regression model with heteroskedasticity, let $\beta = \{\beta_0, \beta_1, \dots, \beta_K\}$ be the parameters of the mean equation and let $\gamma = \{\gamma_0, \gamma_1, \dots, \gamma_P\}$ be the parameters of the variance equation. The unknown parameters are, therefore, $\theta = \{\beta_0, \beta_1, \dots, \beta_K, \gamma_0, \gamma_1, \dots, \gamma_P\}$, which are to be estimated by maximum likelihood. A general form for the log-likelihood function of the heteroskedastic regression model is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \ln f(y_t | x_t, w_t; \theta), \quad (8.3)$$

where the form of the pdf $f(y_t | x_t, w_t; \theta)$ is derived from the assumptions of the regression model using the transformation of variable method.

Example 8.2 Multiplicative Heteroskedasticity

Consider the linear regression model with multiplicative heteroskedastic errors

$$\begin{aligned} y_t &= \beta_0 + \beta_1 x_t + u_t, & u_t &\sim N(0, \sigma_t^2), \\ \sigma_t^2 &= \exp(\gamma_0 + \gamma_1 w_t), \end{aligned}$$

where x_t and w_t are exogenous variables and the unknown parameters are $\theta = \{\beta, \gamma_0, \gamma_1\}$. Given the distributional assumptions made for u_t , the distribution of y_t is

$$f(y_t | x_t, w_t; \theta) = \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp \left[-\frac{(y_t - \beta_0 - \beta_1 x_t)^2}{2\sigma_t^2} \right].$$

Using the multiplicative specification for σ_t^2 , the log-likelihood function for a sample of T observations is

$$\ln L_T(\theta) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2T} \sum_{t=1}^T (\gamma_0 + \gamma_1 w_t) - \frac{1}{2T} \sum_{t=1}^T \frac{(y_t - \beta_0 - \beta_1 x_t)^2}{\exp(\gamma_0 + \gamma_1 w_t)}.$$

□

8.3 Estimation

8.3.1 Maximum Likelihood

The log-likelihood function of the heteroskedastic regression model may be maximized with respect to the unknown model parameters, θ , using any of the iterative algorithms described in Chapter 3.

In the case of the Newton-Raphson algorithm, the parameter vector is

updated at iteration k as

$$\theta_{(k)} = \theta_{(k-1)} - H^{-1}(\theta_{(k-1)})G(\theta_{(k-1)}),$$

where the gradient and the Hessian are, respectively,

$$G(\theta_{(k-1)}) = \left. \frac{\partial \ln L_T(\theta)}{\partial \theta} \right|_{\theta=\theta_{(k-1)}} \quad H(\theta_{(k-1)}) = \left. \frac{\partial^2 \ln L_T(\theta)}{\partial \theta \partial \theta'} \right|_{\theta=\theta_{(k-1)}}.$$

Example 8.3 Newton-Raphson Algorithm

The gradient of the multiplicative heteroskedastic model with parameters $\theta = \{\beta_0, \beta_1, \gamma_0, \gamma_1\}$ specified in Example 8.2 is

$$G_T(\theta) = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T \frac{u_t}{\sigma_t^2} \\ \frac{1}{T} \sum_{t=1}^T \frac{x_t u_t}{\sigma_t^2} \\ -\frac{1}{2} + \frac{1}{T} \frac{1}{2} \sum_{t=1}^T \frac{u_t^2}{\sigma_t^2} \\ -\frac{1}{2T} \sum_{t=1}^T w_t + \frac{1}{2T} \sum_{t=1}^T \frac{w_t u_t^2}{\sigma_t^2} \end{bmatrix},$$

and the Hessian, $H_T(\theta)$, is

$$\begin{bmatrix} -\frac{1}{T} \sum_{t=1}^T \frac{1}{\sigma_t^2} & -\frac{1}{T} \sum_{t=1}^T \frac{x_t}{\sigma_t^2} & -\frac{1}{T} \sum_{t=1}^T \frac{u_t}{\sigma_t^2} & -\frac{1}{T} \sum_{t=1}^T \frac{w_t u_t}{\sigma_t^2} \\ -\frac{1}{T} \sum_{t=1}^T \frac{x_t}{\sigma_t^2} & -\frac{1}{T} \sum_{t=1}^T \frac{x_t^2}{\sigma_t^2} & -\frac{1}{T} \sum_{t=1}^T \frac{x_t u_t}{\sigma_t^2} & -\frac{1}{T} \sum_{t=1}^T \frac{x_t w_t u_t}{\sigma_t^2} \\ -\frac{1}{T} \sum_{t=1}^T \frac{u_t}{\sigma_t^2} & -\frac{1}{T} \sum_{t=1}^T \frac{x_t u_t}{\sigma_t^2} & -\frac{1}{2T} \sum_{t=1}^T \frac{u_t^2}{\sigma_t^2} & -\frac{1}{2T} \sum_{t=1}^T \frac{w_t u_t^2}{\sigma_t^2} \\ -\frac{1}{T} \sum_{t=1}^T \frac{w_t u_t}{\sigma_t^2} & -\frac{1}{T} \sum_{t=1}^T \frac{x_t w_t u_t}{\sigma_t^2} & -\frac{1}{2T} \sum_{t=1}^T \frac{w_t u_t^2}{\sigma_t^2} & -\frac{1}{2T} \sum_{t=1}^T \frac{w_t^2 u_t^2}{\sigma_t^2} \end{bmatrix},$$

where $u_t = y_t - \beta_0 - \beta_1 x_t$ and $\sigma_t^2 = \exp(\gamma_0 + \gamma_1 w_t)$.

The data are simulated from the model using the parameter values $\beta_0 = 1$, $\beta_1 = 2$, $\gamma_0 = 0.1$ and $\gamma_1 = 0.1$. The exogenous variables are defined as $x_t \sim N(0, 1)$ and $w_t = 0.1t$, $t = 0, 1, 2, \dots$. The results of estimating the heteroskedastic regression model using the Newton-Raphson algorithm are given in Table 8.1. The point estimates agree satisfactorily with the

population parameters. For comparison, the estimates of the homoskedastic model, with $\gamma_1 = 0$, are also reported. $\square$

Table 8.1

Maximum likelihood estimates of the multiplicative heteroskedasticity model. Standard errors obtained from the inverse of the Hessian matrix are in parentheses.

Parameter	Heteroskedastic Model	Homoskedastic Model
β_0	0.949 (0.105)	0.753 (0.254)
β_1	1.794 (0.103)	2.136 (0.240)
γ_0	0.121 (0.121)	3.475 (0.063)
γ_1	0.098 (0.004)	
$\ln L_T(\theta)$	-2.699	-3.157

An alternative algorithm to Newton-Raphson is the method of scoring, which replaces the Hessian in equation (8.3.1) by the information matrix, $I(\theta)$. The scoring algorithm updating scheme is

$$\theta_{(k)} = \theta_{(k-1)} + I^{-1}(\theta_{(k-1)})G(\theta_{(k-1)}),$$

where the gradient and the information matrix are defined as

$$G(\theta_{(k-1)}) = \left. \frac{\partial \ln L_T(\theta)}{\partial \theta} \right|_{\theta=\theta_{(k-1)}}, \quad I(\theta_{(k-1)}) = -E \left[\left. \frac{\partial^2 \ln L_T(\theta)}{\partial \theta \partial \theta'} \right|_{\theta=\theta_{(k-1)}} \right].$$

Example 8.4 Scoring Algorithm

In the case of the multiplicative heteroskedastic regression model of Example 8.2, the information matrix may be obtained from the Hessian matrix in Example 8.3 by recognising that

$$\begin{aligned} E[y_t - \beta_0 - \beta_1 x_t] &= 0 \\ E \left[\frac{(y_t - \beta_0 - \beta_1 x_t)^2}{\exp(\gamma_0 + \gamma_1 w_t)} \right] &= \frac{E[(y_t - \beta_0 - \beta_1 x_t)^2]}{\sigma_t^2} = \frac{E[u_t^2]}{\sigma_t^2} = 1. \end{aligned}$$

The information matrix is, therefore, the block-diagonal matrix

$$I(\theta) = -E \left[\frac{\partial^2 \ln L_T(\theta)}{\partial \theta \partial \theta'} \right] = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T \frac{1}{\sigma_t^2} & \frac{1}{T} \sum_{t=1}^T \frac{x_t}{\sigma_t^2} & & & \\ \frac{1}{T} \sum_{t=1}^T \frac{x_t}{\sigma_t^2} & \frac{1}{T} \sum_{t=1}^T \frac{x_t^2}{\sigma_t^2} & 0 & 0 & \\ & 0 & \frac{1}{2} & \frac{1}{2T} \sum_{t=1}^T w_t & \\ & 0 & \frac{1}{2T} \sum_{t=1}^T w_t & \frac{1}{2T} \sum_{t=1}^T w_t^2 & \end{bmatrix}.$$

Letting $\beta = \{\beta_0, \beta_1\}$ and $\gamma = \{\gamma_0, \gamma_1\}$, the block-diagonal property of the information matrix allows estimation to proceed in two separate blocks,

$$\begin{aligned} \beta_{(k)} &= \beta_{(k-1)} + I_{1,1}^{-1} G_1 \\ \gamma_{(k)} &= \gamma_{(k-1)} + I_{2,2}^{-1} G_2, \end{aligned} \quad (8.4)$$

where the relevant elements of the gradient are

$$G_1 = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T \frac{u_t}{\sigma_t^2} \\ \frac{1}{T} \sum_{t=1}^T \frac{x_t u_t}{\sigma_t^2} \end{bmatrix}, \quad G_2 = \begin{bmatrix} -\frac{1}{2} + \frac{1}{2T} \sum_{t=1}^T \frac{u_t^2}{\sigma_t^2} \\ -\frac{1}{2T} \sum_{t=1}^T w_t + \frac{1}{2T} \sum_{t=1}^T \frac{w_t u_t^2}{\sigma_t^2} \end{bmatrix},$$

and the relevant blocks of the information matrix are

$$I_{1,1} = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T \frac{1}{\sigma_t^2} & \frac{1}{T} \sum_{t=1}^T \frac{x_t}{\sigma_t^2} \\ \frac{1}{T} \sum_{t=1}^T \frac{x_t}{\sigma_t^2} & \frac{1}{T} \sum_{t=1}^T \frac{x_t^2}{\sigma_t^2} \end{bmatrix}, \quad I_{2,2} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2T} \sum_{t=1}^T w_t \\ \frac{1}{2T} \sum_{t=1}^T w_t & \frac{1}{2T} \sum_{t=1}^T w_t^2 \end{bmatrix},$$

with $u_t = y_t - \beta_0 - \beta_1 x_t$, $\sigma_t^2 = \exp(\gamma_0 + \gamma_1 w_t)$ and where all parameters are evaluated at $\theta_{(k-1)}$. $\square$

8.3.2 Relationship with Weighted Least Squares

The scoring algorithm in Example 8.4 is an example of a zig-zag algorithm where updated estimates of the mean parameters, β , are obtained separately from the updated estimates of the variance parameters, γ . An important

property of the method of scoring for heteroskedastic regression models is that it is equivalent to weighted least squares.

Example 8.5 Weighted Least Squares Estimation

A common way to start the weighted least algorithm is set $\gamma_1 = 0$, that is, to assume that there is no heteroskedasticity. The relevant blocks of the gradient and Hessian of the multiplicative heteroskedastic regression model in Example 8.4 now become, respectively,

$$G_1 = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T \frac{u_t}{\exp(\gamma_0)} \\ \frac{1}{T} \sum_{t=1}^T \frac{x_t u_t}{\exp(\gamma_0)} \end{bmatrix}, \quad G_2 = \begin{bmatrix} -\frac{1}{2} + \frac{1}{2T} \sum_{t=1}^T \frac{u_t^2}{\exp(\gamma_0)} \\ -\frac{1}{2T} \sum_{t=1}^T w_t + \frac{1}{2T} \sum_{t=1}^T \frac{w_t u_t^2}{\exp(\gamma_0)} \end{bmatrix},$$

and

$$I_{1,1} = \begin{bmatrix} \frac{1}{\exp(\gamma_0)} & \frac{1}{T} \sum_{t=1}^T \frac{x_t}{\exp(\gamma_0)} \\ \frac{1}{T} \sum_{t=1}^T \frac{x_t}{\exp(\gamma_0)} & \frac{1}{T} \sum_{t=1}^T \frac{x_t^2}{\exp(\gamma_0)} \end{bmatrix}, \quad I_{2,2} = \begin{bmatrix} \frac{1}{2} & \frac{1}{T} \frac{1}{2} \sum_{t=1}^T w_t \\ \frac{1}{2T} \sum_{t=1}^T w_t & \frac{1}{2T} \sum_{t=1}^T w_t^2 \end{bmatrix},$$

where $u_t = y_t - \beta_0 - \beta_1 x_t$. An important property of this algorithm is that at the first step the update of β is independent of the choice of the starting value. The first step of the update is

$$\begin{aligned} \beta_{(1)} &= \beta_{(0)} + I_{1,1}^{-1}(\beta_{(0)}) G_1(\beta_{(0)}) \\ &= \begin{bmatrix} \beta_{0(0)} \\ \beta_{1(0)} \end{bmatrix} + \begin{bmatrix} T & \sum_{t=1}^T x_t \\ \sum_{t=1}^T x_t & \sum_{t=1}^T x_t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T (y_t - \beta_{0(0)} - \beta_{1(0)} x_t) \\ \sum_{t=1}^T x_t (y_t - \beta_{0(0)} - \beta_{1(0)} x_t) \end{bmatrix} \\ &= \begin{bmatrix} T & \sum_{t=1}^T x_t \\ \sum_{t=1}^T x_t & \sum_{t=1}^T x_t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T y_t \\ \sum_{t=1}^T x_t y_t \end{bmatrix}, \end{aligned}$$

which is equivalent to a linear regression of y_t on $\{1, x_t\}$. The starting value for γ_0 is given by

$$\gamma_{0(0)} = \ln \left(\frac{1}{T} \sum_{t=1}^T (y_t - \beta_{0(0)} - \beta_{1(0)} x_t)^2 \right).$$

From the other component of the scoring algorithm in Example 8.4, the variance parameters are updated as

$$\begin{aligned}\gamma_{(1)} &= \gamma_{(0)} + I_{2,2}^{-1}(\gamma_{(0)})G_2(\gamma_{(0)}) \\ &= \begin{bmatrix} \gamma_{0(0)} \\ \gamma_{1(0)} \end{bmatrix} + \begin{bmatrix} T & \sum_{t=1}^T w_t \\ \sum_{t=1}^T w_t & \sum_{t=1}^T w_t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T v_t \\ \sum_{t=1}^T w_t v_t \end{bmatrix},\end{aligned}$$

where

$$v_t = \frac{(y_t - \beta_{0(0)} - \beta_{1(0)}x_t)^2}{\exp(\gamma_{0(0)})} - 1.$$

The update of the variance parameters is, therefore, obtained from a linear regression of v_t on $\{1, w_t\}$.

These results suggest that estimation by weighted least squares involves the following steps.

Step 1: Estimate $\beta_{(0)}$ by regressing y_t on $\{1, x_t\}$ and compute

$$\gamma_{0(0)} = \ln \left(\frac{1}{T} \sum_{t=1}^T (y_t - \beta_{0(0)} - \beta_{1(0)}x_t)^2 \right).$$

Step 2: Regress v_t on $\{1, w_t\}$, where

$$v_t = \frac{(y_t - \beta_{0(0)} - \beta_{1(0)}x_t)^2}{\exp(\gamma_{0(0)})} - 1.$$

This regression provides estimates of the updates Δ_{γ_0} and Δ_{γ_1} so that the updated parameter estimates of the variance equation are

$$[\gamma_0 \ \gamma_1]_{(1)} = [\gamma_0 \ \gamma_1]_{(0)} + [\Delta_{\gamma_0} \ \Delta_{\gamma_1}].$$

Step 3: Regress the weighted dependent variable, $y_t/\hat{\sigma}_t$, on the weighted independent variable, $\{1/\hat{\sigma}_t, x_t/\hat{\sigma}_t\}$, to obtain $\beta_{(1)} = \{\beta_{0(1)}, \beta_{1(1)}\}$, where

$$\hat{\sigma}_t^2 = \exp(\gamma_{0(1)} + \gamma_{1(1)}w_t).$$

Step 4: Regress v_t on $\{1, w_t\}$ where

$$v_t = \frac{(y_t - \beta_{0(1)} - \beta_{1(1)}x_t)^2}{\exp(\gamma_{0(1)} + \gamma_{1(1)}w_t)} - 1,$$

and update the parameter estimates of the variance equation again.

Step 5: Repeat steps 3 and 4 until convergence is achieved.

□

8.4 Distribution Theory

The asymptotic distribution of the maximum likelihood estimator of $\theta = \{\beta_0, \beta_1, \gamma_0, \gamma_1\}$ of the heteroskedastic regression model is

$$\hat{\theta} \stackrel{a}{\sim} N(\theta_0, \frac{1}{T} I(\theta_0)^{-1}), \quad (8.5)$$

where $I(\theta_0)$ is the information matrix evaluated at the population parameter θ_0 . Because the information matrix is a block-diagonal matrix, the asymptotic distribution of the maximum likelihood estimator of $\beta = \{\beta_0, \beta_1\}$ is

$$\hat{\beta} \stackrel{a}{\sim} N \left(\beta_0, \left[\begin{array}{cc} \sum_{t=1}^T \frac{1}{\sigma_t^2} & \sum_{t=1}^T \frac{x_t}{\sigma_t^2} \\ \sum_{t=1}^T \frac{x_t}{\sigma_t^2} & \sum_{t=1}^T \frac{x_t^2}{\sigma_t^2} \end{array} \right]^{-1} \right). \quad (8.6)$$

From the properties of the maximum likelihood estimator developed in Chapter 2, $\hat{\beta}$ is a consistent and asymptotically efficient estimator of β_0 . This estimator is also the weighted least squares estimator from the discussion in Section 8.3.2.

The ordinary least squares estimator of β_0 is simply a regression of y_t on $\{1, x_t\}$ without any allowance for heteroskedasticity. That is, this estimator is based on specifying the mean of the model correctly but misspecifying the variance. The implication of this result is that the ordinary least squares estimator is consistent, but inefficient compared to $\hat{\beta}$ in (8.6). The details of this situation are explored in more detail in Section 9.5.2.

8.5 Testing

Hypothesis tests on the parameters of heteroskedastic regression models can be based on the LR, Wald and LM test statistics developed in Chapter 4. As in previous chapters, the Wald and LM test statistics can take various asymptotically equivalent forms.

Example 8.6 Testing in a Multiplicative Heteroskedastic Model

Consider again the linear regression model with multiplicative heteroskedasticity specified in Example 8.2. A test of multiplicative heteroskedasticity is

performed by testing the restriction that $\gamma_1 = 0$ because this restriction yields a constant variance, or homoskedasticity. The null and alternative hypotheses are, respectively,

$$\begin{aligned} H_0 : \quad & \gamma_1 = 0 \quad [\text{Homoskedasticity}] \\ H_1 : \quad & \gamma_1 \neq 0 \quad [\text{Heteroskedasticity}]. \end{aligned}$$

Let the parameters of the restricted model be $\hat{\theta}_0 = \{\hat{\beta}_0, \hat{\beta}_1, \hat{\gamma}_0, 0\}$ and the parameters of the unrestricted model be $\hat{\theta}_1 = \{\hat{\beta}_0, \hat{\beta}_1, \hat{\gamma}_0, \hat{\gamma}_1\}$.

The log-likelihood function evaluated at the maximum likelihood estimator under the alternative hypothesis, $\hat{\theta}_1$, is

$$\ln L_T(\hat{\theta}_1) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2T} \sum_{t=1}^T (\hat{\gamma}_0 + \hat{\gamma}_1 w_t) - \frac{1}{2T} \sum_{t=1}^T \frac{(y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t)^2}{\exp(\hat{\gamma}_0 + \hat{\gamma}_1 w_t)}.$$

The log-likelihood function evaluated at the maximum likelihood estimator under the null hypothesis, $\hat{\theta}_0$, is

$$\begin{aligned} \ln L_T(\hat{\theta}_0) &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \hat{\gamma}_0 - \frac{1}{2T} \sum_{t=1}^T \frac{(y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t)^2}{\exp(\hat{\gamma}_0)} \\ &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \hat{\gamma}_0 - \frac{1}{2}, \end{aligned}$$

where

$$\exp(\hat{\gamma}_0) = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t)^2,$$

is the residual variance from regressing y_t on x_t . The LR statistic is

$$\begin{aligned} LR &= -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)) \\ &= T \hat{\gamma}_0|_{H_0} - \sum_{t=1}^T (\hat{\gamma}_0 + \hat{\gamma}_1 w_t)|_{H_1} + T + \sum_{t=1}^T \frac{(y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t)^2}{\exp(\hat{\gamma}_0 + \hat{\gamma}_1 w_t)} \Big|_{H_1}, \end{aligned}$$

which is asymptotically distributed as χ_1^2 under the null hypothesis.

To construct the Wald test statistic define

$$R = [0 \quad 0 \quad 0 \quad 1], \quad Q = [0].$$

From the properties of partitioned matrices

$$I^{-1}(\hat{\theta}_1) = \begin{bmatrix} I_{11}^{-1} & 0 \\ 0 & I_{22}^{-1} \end{bmatrix},$$

where

$$I_{11}^{-1} = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T \frac{1}{\exp(\gamma_0 + \gamma_1 w_t)} & \frac{1}{T} \sum_{t=1}^T \frac{x_t}{\exp(\gamma_0 + \gamma_1 w_t)} \\ \frac{1}{T} \sum_{t=1}^T \frac{x_t}{\exp(\gamma_0 + \gamma_1 w_t)} & \frac{1}{T} \sum_{t=1}^T \frac{x_t^2}{\exp(\gamma_0 + \gamma_1 w_t)} \end{bmatrix}^{-1},$$

and

$$I_{22}^{-1} = \frac{2}{\sum_{t=1}^T (w_t - \bar{w})^2} \begin{bmatrix} \sum_{t=1}^T w_t^2 & -\sum_{t=1}^T w_t \\ -\sum_{t=1}^T w_t & T \end{bmatrix},$$

so that the Wald statistic is

$$\begin{aligned} W &= T[R\hat{\theta}_1 - Q]'[R I^{-1}(\hat{\theta}_1) R']^{-1}[R\hat{\theta}_1 - Q] \\ &= [\hat{\gamma}_1 - 0]' \left[\frac{1}{2} \sum_{t=1}^T (w_t - \bar{w})^2 \right]^{-1} [\hat{\gamma}_1 - 0] \\ &= \frac{(\hat{\gamma}_1 - 0)^2}{\frac{1}{2} \sum_{t=1}^T (w_t - \bar{w})^2}. \end{aligned}$$

An alternative form of the Wald test statistic is

$$W_2 = T[R\hat{\theta}_1 - Q]'[R(-H^{-1}(\hat{\theta}_1)) R']^{-1}[R\hat{\theta}_1 - Q].$$

In this case, the statistic simplifies to

$$W_2 = \frac{(\hat{\gamma}_1 - 0)^2}{\text{var}(\hat{\gamma}_1)},$$

or, when expressed as a t-statistic,

$$\text{t-stat} = \frac{\hat{\gamma}_1 - 0}{\text{se}(\hat{\gamma}_1)}.$$

The elements of the gradient vector under the null hypothesis are

$$G_T(\hat{\theta}) = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t) \exp(-\hat{\gamma}_0) \\ \frac{1}{T} \sum_{t=1}^T x_t (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t) \exp(-\hat{\gamma}_0) \\ -\frac{1}{2} + \frac{1}{2T} \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t)^2 \exp(-\hat{\gamma}_0) \\ -\frac{1}{2T} \sum_{t=1}^T w_t + \frac{1}{2T} \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t)^2 w_t \exp(-\hat{\gamma}_0) \end{bmatrix},$$

while the information matrix under the null hypothesis is

$$I(\hat{\theta}) = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T \exp(-\hat{\gamma}_0) & \frac{1}{T} \sum_{t=1}^T x_t \exp(-\hat{\gamma}_0) & & & \\ \frac{1}{T} \sum_{t=1}^T x_t^1 \exp(-\hat{\gamma}_0) & \frac{1}{T} \sum_{t=1}^T x_t^2 \exp(-\hat{\gamma}_0) & 0 & 0 & \\ & 0 & \frac{1}{2} & \frac{1}{2T} \sum_{t=1}^T w_t & \\ & 0 & \frac{1}{2T} \sum_{t=1}^T w_t & \frac{1}{2T} \sum_{t=1}^T w_t^2 & \end{bmatrix}.$$

As

$$\left. \frac{\partial \ln L_T(\theta)}{\partial \beta} \right|_{H_0} = 0,$$

the information matrix under the null hypothesis is block diagonal, and the LM test statistic simplifies to

$$\begin{aligned} LM &= T G_T'(\hat{\theta}_0) I^{-1}(\hat{\theta}_0) G_T(\hat{\theta}_0) \\ &= \frac{1}{2} \begin{bmatrix} \sum_{t=1}^T v_t \\ \sum_{t=1}^T v_t w_t \end{bmatrix}' \begin{bmatrix} T & \sum_{t=1}^T w_t \\ \sum_{t=1}^T w_t & \sum_{t=1}^T w_t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T v_t \\ \sum_{t=1}^T v_t w_t \end{bmatrix}, \end{aligned}$$

where

$$v_t = \frac{(y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t)^2}{\exp(\hat{\gamma}_0)} - 1.$$

The parameter $\hat{\beta}$ is obtained from a regression of y_t on x_t and

$$\hat{\gamma}_0 = \ln \left(\frac{1}{T} \sum_{t=1}^T (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t)^2 \right).$$

Alternatively, consider the standardized random variable

$$z_t = \frac{y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t}{\sqrt{\exp(\hat{\gamma}_0)}},$$

that is distributed asymptotically as $N(0, 1)$. It follows that

$$\text{plim} \left(\frac{1}{T} \sum v_t^2 \right) = \text{plim} \left(\frac{1}{T} \sum (z_t^2 - 1)^2 \right) = \text{plim} \left(\sum (z_t^4 - 2z_t^2 + 1) \right) = 2,$$

because from the properties of the standard normal distribution $\text{plim}(z_t^2) = 1$ and $\text{plim}(z_t^4) = 3$. Thus, another asymptotic form of the LM test is

$$LM_2 = \frac{T}{\sum_{t=1}^T v_t^2} \begin{bmatrix} \sum_{t=1}^T v_t \\ \sum_{t=1}^T v_t w_t \end{bmatrix}' \begin{bmatrix} T & \sum_{t=1}^T w_t \\ \sum_{t=1}^T w_t & \sum_{t=1}^T w_t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T v_t \\ \sum_{t=1}^T v_t w_t \end{bmatrix} = TR^2,$$

where R^2 is the coefficient of determination from regressing v_t on $\{1, w_t\}$, because under the null hypothesis

$$\bar{v} = \frac{1}{T} \sum_{t=1}^T v_t = 0.$$

Alternatively, v_t can be redefined as $(y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t)^2$, because R^2 is invariant to linear transformations. This suggests that a test of multiplicative heteroskedasticity can be implemented as a two-stage regression procedure as follows.

Step 1: Regress y_t on $\{1, x_t\}$ to obtain the restricted residuals $\hat{u}_t = y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t$.

Step 2: Regress $v_t = \hat{u}_t^2$ on $\{1, w_t\}$.

Step 3: Compute TR^2 from this second-stage regression and compare the computed value of the test statistic to the critical values obtained from the χ_1^2 distribution. $\square$

Example 8.7 Numerical Illustration This example reports LR, Wald and LM tests for heteroskedasticity in the multiplicative heteroskedastic model using the numerical results reported in Table 8.1 and the alternative forms of the test statistics derived in Example 8.6. The LR statistic is

$$LR = -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)) = -2(-1578.275 + 1349.375) = 457.800.$$

Also using the results from the same table, the alternative form of the Wald statistic is

$$W_2 = \frac{(\hat{\gamma}_1 - 0)^2}{\text{var}(\hat{\gamma}_1)} = \left(\frac{0.098 - 0}{0.004} \right)^2 = 578.086.$$

To compute the LM statistic, evaluate the gradient vector and the Hessian

at the constrained estimates $\hat{\theta}_0$ in Table 8.1 as

$$G_T(\hat{\theta}_0) = \begin{bmatrix} 0.000 \\ 0.000 \\ 0.000 \\ 7.978 \end{bmatrix}$$

$$H_T(\hat{\theta}_0) = \begin{bmatrix} -0.031 & 0.000 & 0.000 & 0.125 \\ 0.000 & -0.035 & 0.000 & -0.145 \\ 0.000 & 0.000 & -0.500 & -20.450 \\ 0.125 & -0.145 & -20.450 & -878.768 \end{bmatrix}.$$

The LM statistic is

$$LM = TG'_T(\hat{\theta}_0)[-H_T(\hat{\theta}_0)]^{-1}G_T(\hat{\theta}_0) = 770.154.$$

To compute the regression form of the LM test, in the first stage use the constrained estimates in Table 8.1 to compute the following OLS residuals under the null hypothesis

$$\hat{u}_t = y_t - 0.753 - 2.136x_t.$$

In the second stage regress $v_t = \hat{u}_t^2$ on $\{1, w_t\}$. The results are

$$v_t = -29.424 + 2.474w_t + \hat{e}_t,$$

which yields

$$R^2 = 1 - \frac{\sum \hat{e}_t^2}{\sum (v_t - \bar{v})^2} = 0.157.$$

The regression form of the LM statistic is

$$LM_2 = TR^2 = 500 \times 0.157 = 78.447.$$

The number of restrictions is $M = 1$ and the distribution of all the tests statistics is, therefore, χ_1^2 . For the LR, Wald and two LM tests, the p-values are all 0.000, suggesting that the null hypothesis is rejected at the 5% level. The conclusion that significant evidence of heteroskedasticity exists at the 5% level is consistent with the setup of the model. $\square$

Example 8.8 Breusch-Pagan and White Tests

The regression form of the LM test presented above is perhaps the most commonly adopted test of heteroskedasticity since it is relatively easy to implement, involving just two least squares regressions. This construction of the LM test corresponds to the Breusch-Pagan and White tests of heteroskedasticity, with the difference between these two tests being the set of variables (implicitly) specified in the variance equation.

For the Breusch-Pagan test, the variance equation is specified as

$$\sigma_t^2 = \gamma_0 + \sum_{i=1}^M \gamma_i w_{i,t}.$$

The test involves regressing y_t on x_t to get the least squares residuals $\hat{u}_t$, in the first stage. In the second stage, regress $\hat{u}_t^2$ on $\{1, w_{1,t}, w_{2,t}, \dots, w_{M,t}\}$ and compute $LM = TR^2$ where R^2 is, as before, the coefficient of determination from the second-stage regression. This statistic is asymptotically distributed as χ_M^2 under the null hypothesis.

For the White test, the variance equation is an extension of the Breusch-Pagan specification involving the inclusion of all the relevant cross-product terms

$$\sigma_t^2 = \gamma_0 + \sum_{i=1}^M \gamma_i w_{i,t} + \sum_{i>j}^M \gamma_{i,j} w_{i,t} w_{j,t}.$$

The second-stage regression now consists of regressing $\hat{u}_t^2$ on

$$\{1, w_{1,t}, w_{2,t}, \dots, w_{M,t}, w_{1,t}^2, w_{2,t}^2, \dots, w_{M,t}^2, w_{1,t}w_{2,t}, \dots, w_{M-1,t}w_{M,t}\}.$$

The test statistic is $LM = TR^2$, where R^2 is the coefficient of determination from the second-stage regression, which is asymptotically distributed as χ^2 under the null hypothesis with $(M^2 + 3M)/2$ degrees of freedom.

Implementing the Breusch-Pagan and White tests still requires that the variables in the variance equation, $w_{i,t}, i = 1, 2, \dots, M$, be specified. In practice, it is common to let $w_t = x_t$: that is, the exogenous variables in the regression equation are also used in the variance equation. $\square$

8.6 Heteroskedasticity in Systems of Equations

8.6.1 Specification

Extending the univariate model with heteroskedasticity to a system of equations with heteroskedasticity is relatively straightforward. While the univariate model must ensure that the variance is positive for all t , in the multivariate case a more stringent condition must be satisfied, namely, the estimated covariance matrix must remain positive-definite for all t .

A system of equations where the disturbance vector is heteroskedastic is specified as

$$y_t B + x_t A = u_t, \quad u_t \sim N(0, V_t), \quad (8.7)$$

where y_t is a $(1 \times N)$ vector of dependent variables, x_t is a $(1 \times M)$ vector of

exogenous variables, B is a $(N \times N)$ matrix, A is a $(M \times N)$ matrix and V_t is the $(N \times N)$ covariance matrix of the time-varying disturbances driven by the variable w_t assumed here, for simplicity, to be a scalar. This model can easily be extended to allow for additional variables in the variance equation.

In the specification of V_t , w_t is assumed to be positive for all t . If this restriction is not satisfied, w_t must be transformed to ensure that it is indeed positive over the sample. The approach is to express the variances and covariances in terms of a lower triangular matrix, S , and construct the covariance matrix as

$$V_t = S_t S_t', \quad (8.8)$$

with

$$S_t = C + D w_t, \quad (8.9)$$

where C and D are $(N \times N)$ lower triangular matrices of unknown parameters. For example, consider a bivariate model $N = 2$. The matrix S_t is specified as

$$\begin{aligned} S_t &= \begin{bmatrix} c_{1,1} + d_{1,1}w_t & 0 \\ c_{2,1} + d_{2,1}w_t & c_{2,2} + d_{2,2}w_t \end{bmatrix} \\ &= \begin{bmatrix} c_{1,1} & 0 \\ c_{2,1} & c_{2,2} \end{bmatrix} + \begin{bmatrix} d_{1,1} & 0 \\ d_{2,1} & d_{2,2} \end{bmatrix} w_t \\ &= C + D w_t. \end{aligned}$$

The covariance matrix in (8.8) is then

$$\begin{aligned} V_t &= S_t S_t' \\ &= \begin{bmatrix} c_{1,1} + d_{1,1}w_t & 0 \\ c_{2,1} + d_{2,1}w_t & c_{2,2} + d_{2,2}w_t \end{bmatrix} \begin{bmatrix} c_{1,1} + d_{1,1}w_t & c_{2,1} + d_{2,1}w_t \\ 0 & c_{2,2} + d_{2,2}w_t \end{bmatrix} \\ &= \begin{bmatrix} \sigma_{1,1,t} & \sigma_{1,2,t} \\ \sigma_{2,1,t} & \sigma_{2,2,t} \end{bmatrix}, \end{aligned}$$

where the variances $(\sigma_{1,1,t}, \sigma_{2,2,t})$ and covariance $(\sigma_{1,2,t} = \sigma_{2,1,t})$ are

$$\begin{aligned}\sigma_{1,1,t} &= (c_{1,1} + d_{1,1}w_t)^2 \\ &= c_{1,1}^2 + 2c_{1,1}d_{1,1}w_t + d_{1,1}^2w_t^2\end{aligned}$$

$$\begin{aligned}\sigma_{1,2,t} = \sigma_{2,1,t} &= (c_{1,1} + d_{1,1}w_t)(c_{2,1} + d_{2,1}w_t) \\ &= c_{1,1}c_{2,1} + (c_{1,1}d_{2,1} + c_{2,1}d_{1,1})w_t + d_{1,1}d_{2,1}w_t^2\end{aligned}$$

$$\begin{aligned}\sigma_{2,2,t} &= (c_{2,1} + d_{2,1}w_t)(c_{2,1} + d_{2,1}w_t) + (c_{2,2} + d_{2,2}w_t)(c_{2,2} + d_{2,2}w_t) \\ &= c_{2,1}^2 + c_{2,2}^2 + (2c_{2,1}d_{2,1} + 2c_{2,2}d_{2,2})w_t + (d_{2,1}^2 + d_{2,2}^2)w_t^2,\end{aligned}$$

respectively. This covariance matrix has three features.

- (1) V_t is symmetric since $\sigma_{1,2,t} = \sigma_{2,1,t}$.
- (2) V_t is positive (semi) definite since $\sigma_{1,1,t}\sigma_{2,2,t} \geq \sigma_{1,2,t}^2$.
- (3) The variances, $\sigma_{i,i,t}$, and covariance, $\sigma_{i,j,t}$, are quadratic in w_t .

The matrix S_t in expression (8.8) is the Choleski decomposition of V_t . In the special case of a univariate model, $N = 1$, when the disturbance is homoskedastic $d_{1,1} = 0$, then $\sigma_{1,1,t} = c_{1,1}^2$ so that $c_{1,1}$ represents the standard deviation. For this reason, the matrix S_t is sometimes referred to as the standard-deviation matrix, or, more generally, as the square-root matrix.

8.6.2 Estimation

The multivariate regression model with vector heteroskedasticity is estimated using the full-information maximum likelihood estimator presented in Chapter 5. From the assumption of multivariate normality in (8.7), the distribution of u_t is

$$f(u_t) = \left(\frac{1}{\sqrt{2\pi}}\right)^N |V_t|^{-1/2} \exp \left[-\frac{1}{2} u_t V_t^{-1} u_t' \right]. \quad (8.10)$$

The Jacobian is

$$\frac{\partial u_t}{\partial y_t} = B,$$

so that by the transformation of variable technique, the density of y_t is

$$\begin{aligned}f(y_t | x_t, w_t; \theta) &= f(u_t) \left| \frac{\partial u_t}{\partial y_t} \right| \\ &= \left(\frac{1}{\sqrt{2\pi}}\right)^N |V_t|^{-1/2} \exp \left[-\frac{1}{2} u_t V_t^{-1} u_t' \right] |B|,\end{aligned}$$

where

$$u_t = y_t B + x_t A,$$

and

$$V_t = S_t S_t' = (C + D w_t)(C + D w_t)'.$$

The log-likelihood function at observation t is

$$\ln l_t(\theta) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |V_t| + \ln |B| - \frac{1}{2} u_t V_t^{-1} u_t', \quad (8.11)$$

and for a sample of T observations, the log-likelihood function is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \ln l_t(\theta), \quad (8.12)$$

which is maximized with respect to $\theta = \{B, A, C, D\}$ using one of the iterative algorithms discussed in Chapter 3.

Example 8.9 Estimation of a Vector Heteroskedastic Model

This example simulates and estimates a bivariate system of equations where the covariance matrix of the disturbance vector is time varying. The following model is simulated for $T = 2000$ observations

$$\begin{aligned} y_t B + x_t A &= u_t \\ S_t &= C + D w_t \\ u_t &\sim N(0, V_t = S_t S_t'), \end{aligned}$$

where

$$\begin{aligned} B &= \begin{bmatrix} 1 & -\beta_2 \\ -\beta_1 & 1 \end{bmatrix} = \begin{bmatrix} 1.000 & -0.200 \\ -0.600 & 1.000 \end{bmatrix} \\ A &= \begin{bmatrix} -\alpha_1 & 0 \\ 0 & -\alpha_2 \end{bmatrix} = \begin{bmatrix} -0.400 & 0.000 \\ 0.000 & 0.500 \end{bmatrix} \\ C &= \begin{bmatrix} c_{1,1} & 0 \\ c_{2,1} & c_{2,2} \end{bmatrix} = \begin{bmatrix} 1.000 & 0.000 \\ 0.500 & 2.000 \end{bmatrix} \\ D &= \begin{bmatrix} d_{1,1} & 0 \\ d_{2,1} & d_{2,2} \end{bmatrix} = \begin{bmatrix} 0.500 & 0.000 \\ 0.200 & 0.200 \end{bmatrix}, \end{aligned}$$

and $x_{1,t} \sim U(0, 10)$, $x_{2,t} \sim N(0, 9)$ and $w_t \sim U(0, 1)$.

The log-likelihood function is maximized with respect to the parameters

$$\theta = \{\beta_1, \alpha_1, \beta_2, \alpha_2, c_{1,1}, d_{1,1}, c_{2,1}, d_{2,1}, c_{2,2}, d_{2,2}\}$$

using the BFGS algorithm. Standard errors are computed from the negative of the inverse of the Hessian matrix. The maximum likelihood estimates

Table 8.2

Maximum likelihood estimates of the vector heteroskedastic model using the BFGS algorithm with numerical derivatives and standard errors based on the Hessian.

Parameter	Population	Estimate	Std. error	t-stat.
β_1	0.6	0.591	0.015	38.209
α_1	0.4	0.402	0.005	81.094
β_2	0.2	0.194	0.019	10.349
α_2	-0.5	-0.527	0.018	-29.205
$c_{1,1}$	1.0	1.055	0.037	28.208
$d_{1,1}$	0.5	0.373	0.068	5.491
$c_{2,1}$	0.5	0.502	0.116	4.309
$d_{2,1}$	0.2	0.177	0.170	1.042
$c_{2,2}$	2.0	1.989	0.071	27.960
$d_{2,2}$	0.2	0.304	0.117	2.598

given in Table 8.2 demonstrate good agreement with their population values.

□

8.6.3 Testing

The LR, Wald and LM tests can all be used to test for heteroskedasticity in systems of equations given by (8.7) to (8.9). The null and alternative hypotheses are

$$\begin{aligned} H_0 &: d_{i,j} = 0, \forall i \geq j \\ H_1 &: \text{at least one restriction is not satisfied.} \end{aligned}$$

Example 8.10 Wald Test of Vector Heteroskedasticity

Consider again the vector heteroskedastic model given in Example 8.9. The null and alternative hypothesis are

$$\begin{aligned} H_0 &: d_{1,1} = d_{2,1} = d_{2,2} = 0, \\ H_1 &: \text{at least one restriction is not satisfied,} \end{aligned}$$

so that there are 3 restrictions to be tested. The Wald statistic is

$$W = T[R\hat{\theta}_1 - Q]'[R(-H_T^{-1}(\hat{\theta}_1))R']^{-1}[R\hat{\theta}_1 - Q],$$

where the unconstrained parameter estimates, $\hat{\theta}_1$, are given in Table 8.2,

$\text{var}(\hat{\theta}_1)$ is the covariance matrix of the unconstrained parameter estimates with the square root of the diagonal terms also given in Table 8.2, and

$$R = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Substituting the terms into W and simplifying gives $W = 36.980$, which is distributed asymptotically as χ_3^2 under H_0 . The p-value is 0.000, so the null hypothesis is rejected at the 5% level. $\square$

8.6.4 Heteroskedastic and Autocorrelated Disturbances

A system of equations where the disturbance vector is both heteroskedastic and autocorrelated with one lag is specified as

$$\begin{aligned} y_t B + x_t A &= u_t \\ u_t &= u_{t-1} P + v_t \\ S_t &= C + D w_t \\ v_t &\sim N(0, V_t = S_t S_t'), \end{aligned} \tag{8.13}$$

where y_t is a $(1 \times N)$ vector of dependent variables, x_t is a $(1 \times M)$ vector of independent variables, B is a $(N \times N)$, A is a $(M \times N)$ and P is a $(N \times N)$ matrix of autocorrelation parameters. The covariance matrix of the disturbances, V_t is the $(N \times N)$ matrix with $(N \times N)$ lower triangular parameter matrices C and D and w_t is a positive scalar variable.

The log-likelihood function at observation t is

$$\ln l_t(\theta) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |V_t| + \ln |B| - \frac{1}{2} v_t V_t^{-1} v_t', \tag{8.14}$$

where

$$\begin{aligned} v_t &= u_t - u_{t-1} P \\ u_t &= y_t B + x_t A, \end{aligned}$$

and

$$V_t = S_t S_t' = (C + D w_t)(C + D w_t)'.$$

Conditioning on the first observation gives the log-likelihood function for the entire sample

$$\ln L_T(\theta) = \frac{1}{T-1} \sum_{t=2}^T \ln l_t(\theta), \tag{8.15}$$

which is maximized with respect to $\theta = \{B, A, C, D, P\}$.

The system of equations in (8.13) represents a general and flexible linear model in which to analyze time series and draw inferences. This model can also be used to perform a range of hypothesis tests of heteroskedasticity, autocorrelation or both.

Example 8.11 Testing for Heteroskedasticity and Autocorrelation

This example simulates, estimates and tests a bivariate system of equations (8.13), where the disturbance is a vector AR(1) with a time-varying covariance matrix. The population parameters are

$$\begin{aligned} B &= \begin{bmatrix} 1 & -\beta_2 \\ -\beta_1 & 1 \end{bmatrix} = \begin{bmatrix} 1.000 & -0.200 \\ -0.600 & 1.000 \end{bmatrix} \\ A &= \begin{bmatrix} -\alpha_1 & 0 \\ 0 & -\alpha_2 \end{bmatrix} = \begin{bmatrix} -0.400 & 0.000 \\ 0.000 & 0.500 \end{bmatrix} \\ P &= \begin{bmatrix} \rho_{1,1} & \rho_{1,2} \\ \rho_{2,1} & \rho_{2,2} \end{bmatrix} = \begin{bmatrix} 0.800 & -0.200 \\ 0.100 & 0.600 \end{bmatrix} \\ C &= \begin{bmatrix} c_{1,1} & 0 \\ c_{2,1} & c_{2,2} \end{bmatrix} = \begin{bmatrix} 1.000 & 0.000 \\ 0.500 & 2.000 \end{bmatrix} \\ D &= \begin{bmatrix} d_{1,1} & 0 \\ d_{2,1} & d_{2,2} \end{bmatrix} = \begin{bmatrix} 0.500 & 0.000 \\ 0.200 & 0.200 \end{bmatrix}. \end{aligned}$$

The conditional log-likelihood function in (8.15) is maximized with respect to the parameter vector

$$\theta = \{\beta_1, \alpha_1, \beta_2, \alpha_2, \rho_{1,1}, \rho_{1,2}, \rho_{2,1}, \rho_{2,2}, c_{1,1}, d_{1,1}, c_{2,1}, d_{2,1}, c_{2,2}, d_{2,2}\},$$

using the BFGS algorithm. As before, standard errors are computed using the negative of the inverse of the Hessian matrix. A joint test of heteroskedasticity and autocorrelation is carried out using a Wald test. The hypotheses are

$$\begin{aligned} H_0 &: d_{1,1} = d_{2,1} = d_{2,2} = \rho_{1,1} = \rho_{1,2} = \rho_{2,1} = \rho_{2,2} = 0 \\ H_1 &: \text{at least one restriction is not satisfied,} \end{aligned}$$

representing a total of 7 restrictions. The Wald statistic is

$$W = T[R\hat{\theta}_1 - Q]'[R(-H_T^{-1}(\hat{\theta}_1))R']^{-1}[R\hat{\theta}_1 - Q],$$

where $\hat{\theta}_1$ is the vector of unconstrained parameter estimates, $H_T(\hat{\theta}_1)$ is the

Hessian matrix and

$$R = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Substituting the terms into W gives $W = 5590.061$, which is distributed asymptotically as χ_7^2 under H_0 . The p-value is 0.000 and the null hypothesis is rejected at the 5% level. $\square$

8.7 Applications

8.7.1 The Great Moderation

The Great Moderation refers to the decrease in the volatility of U.S. output growth after the early 1980s by comparison with previous volatility levels. This feature of U.S. output growth is highlighted in Figure 8.2 which gives the annual percentage growth in per capita real GDP for the U.S. from 1947 to 2006. The descriptive statistics presented in Table 8.3 show that the variance decreases from 7.489% in the period 1947 to 1983 to 2.049% in the period 1984 to 2006, a reduction of over 70%.

Table 8.3

Descriptive statistics on the annual percentage growth rate of per capita real GDP in the U.S. for selected sub-periods.

Period	Mean	Variance
1947 to 1983	1.952	7.489
1984 to 2006	2.127	2.049

Consider the following model

$$\begin{aligned} y_t &= \beta_0 + \beta_1 d_t + u_t, & u_t &\sim N(0, \sigma_t^2), \\ \sigma_t^2 &= \exp(\gamma_0 + \gamma_1 d_t) \end{aligned} \quad (8.16)$$

where y_t is the growth rate in real GDP and

$$d_t = \begin{cases} 1 & : \text{1947 to 1983} \\ 0 & : \text{1984 to 2006,} \end{cases} \quad (8.17)$$

is a dummy variable that identifies the structural break in the volatility. This model is an example of the step model of heteroskedasticity where the variances of the two periods are $\sigma_{1947-1983}^2 = \exp(\gamma_0)$ and $\sigma_{1984-2006}^2 = \exp(\gamma_0 + \gamma_1)$. The model also allows for the mean to change over the two sample periods if $\beta_1 \neq 0$ and therefore allows a test of the Great Moderation hypothesis.

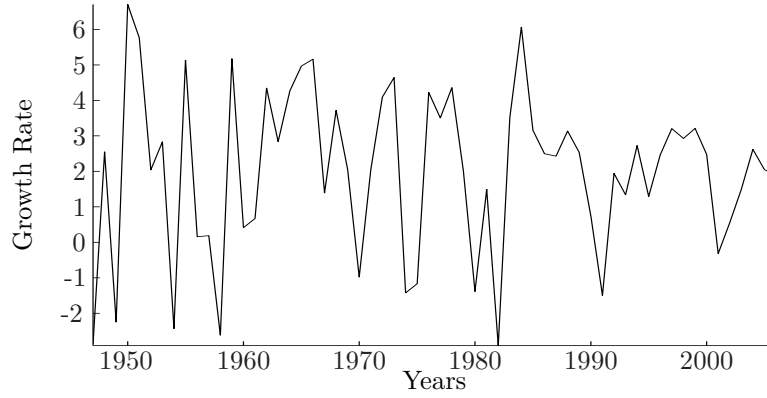


Figure 8.2 Annual percentage growth rate of per capita real GDP in the U.S. for the period 1947 to 2006.

The results from estimating the parameters of this model by maximum likelihood are

$$y_t = \underset{(0.450)}{1.952} + \underset{(0.540)}{0.175}d_t + \hat{u}_t$$

$$\hat{\sigma}_t^2 = \exp(\underset{(0.232)}{2.013} - \underset{(0.375)}{1.296}d_t),$$

where standard errors, computed using the inverse of the Hessian matrix, are given in parentheses. The negative estimate reported for γ_1 shows that there is a fall in the variance in the second sub-period. The estimates of the variances for the two periods are

$$\hat{\sigma}_{1947-1983}^2 = \exp(\hat{\gamma}_0) = \exp(2.013) = 7.489$$

$$\hat{\sigma}_{1984-2006}^2 = \exp(2.013 - 1.296) = 2.049,$$

which equal the variances reported in Table 8.3 for the two sub-periods.

Testing the null and alternative hypotheses

$$\begin{aligned} H_0 : \gamma_1 &= 0 \quad [\text{No Moderation}] \\ H_1 : \gamma_1 &\neq 0 \quad [\text{Moderation}], \end{aligned}$$

using the LR, Wald and LM tests yields

$$LR = 10.226, \quad W = 11.909, \quad LM = 9.279.$$

From the χ_1^2 distribution, the p-values are respectively 0.001, 0.001 and 0.002, providing strong evidence in favour of the Great Moderation hypothesis.

8.7.2 Finite Sample Properties of the Wald Test

The size and power properties of the Wald test for heteroskedasticity in finite samples are now investigated using Monte Carlo methods. Consider the model

$$\begin{aligned} y_t &= 1.0 + 2.0x_t + u_t, \quad u_t \sim N(0, \sigma_t^2), \\ \sigma_t^2 &= \exp(1.0 + \gamma_1 x_t), \end{aligned}$$

where $x_t \sim U[0, 1]$ and $\gamma_1 = \{0.0, 0.5, 1.0, 1.5, \dots, 5.0\}$ controls the strength of heteroskedasticity.

To compute the size of the Wald test, the model is simulated under the null hypothesis of homoskedasticity by setting $\gamma_1 = 0$. The finite sample distribution of the Wald statistic under the null hypothesis for $T = 50$ observations and 10,000 replications, is given in Figure 8.3.

The characteristics of this distribution are similar to the asymptotic distribution, a chi-square distribution with one degree of freedom. The finite sample critical value corresponding to a size of 5%, is computed as that value which 5% of the simulated Wald statistics exceed. The finite sample critical value is in fact 4.435, which is greater than the corresponding asymptotic critical value of 3.841. The size of the test is computed as the average number of times the Wald statistic in the simulations is greater than the asymptotic critical value 3.841

$$\text{SIZE} = \frac{1}{10000} \sum_{i=1}^{10000} D_i(W_i > 3.841) = 0.0653,$$

or 6.530%. This result suggests that the asymptotic distribution is a reasonable approximation to the finite sample distribution for $T = 50$. That the size of the test is slightly greater than the nominal size of 5% suggests that

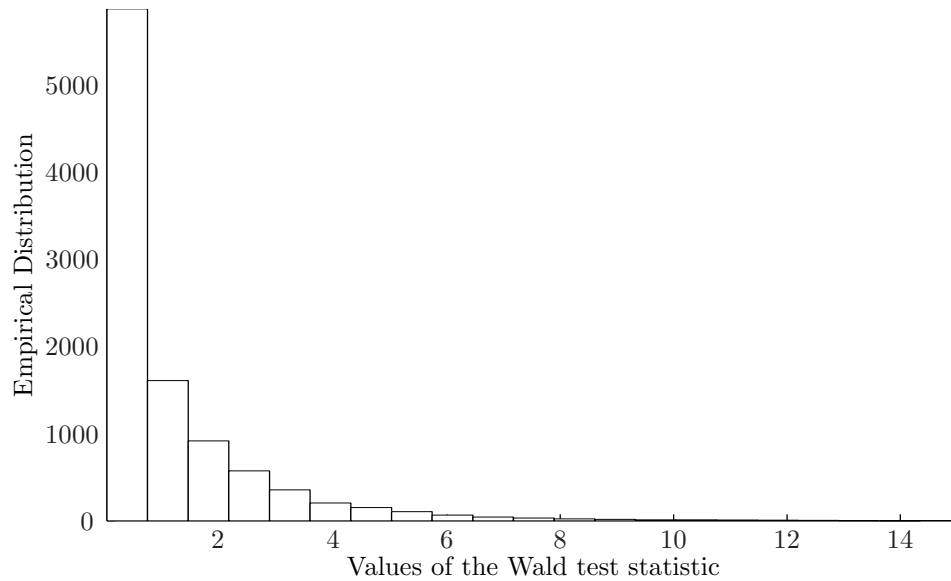


Figure 8.3 Sampling distribution of the Wald test for heteroskedasticity under the null hypothesis. Based on $T = 50$ observations and 10000 replications.

the Wald test rejects the null hypothesis slightly more often than it should when asymptotic critical values are used.

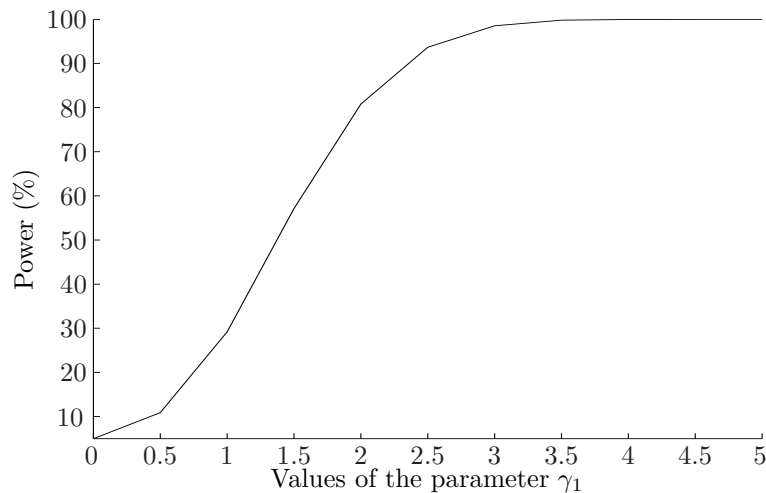


Figure 8.4 Power function of the Wald test of heteroskedasticity, size adjusted. Based on $T = 50$ observations and 10,000 replications.

To compute the power of the Wald test, simulate the model under the

alternative hypothesis of heteroskedasticity by setting $\gamma_1 = 0.5$. Two types of power are computed. The first is the average number of times the simulated Wald statistic is greater than the asymptotic critical value 3.841

$$\text{POWER}_U = \frac{1}{10000} \sum_{i=1}^{10000} D_i(W_i > 3.841) = 0.1285,$$

or 12.850%. The previous results show that under the null hypothesis with a nominal size of 5% the actual empirical size of this test is 6.530%. For this reason POWER_U is referred to as the unadjusted power of the test. The size-adjusted power of the test is computed as

$$\text{POWER}_A = \frac{1}{10000} \sum_{i=1}^{10000} D_i(W_i > 4.435) = 0.1011,$$

or 10.110%. Figure 8.4 gives the size-adjusted power function for values of $\gamma_1 = \{0.5, 1.0, 1.5, \dots, 5.0\}$. The power function is monotonically increasing in γ_1 , with the power reaching unity for $\gamma_1 \geq 4$.

8.8 Exercises

(1) Simulating a Regression Model with Heteroskedasticity

Gauss file(s)	hetero_simulate.g
Matlab file(s)	hetero_simulate.m

Simulate the following model for $T = 500$ observations

$$y_t = \beta_0 + \beta_1 x_t + u_t, \quad u_t \sim N(0, \sigma_t^2),$$

$$\sigma_t^2 = \gamma_0 + \gamma_1 w_t,$$

where $x_t \sim N(0, 1)$, w_t is a time trend and the parameter values are $\beta_0 = 1$, $\beta_1 = 2$, $\gamma_0 = 1$ and $\gamma_1 = 0.5$. Generate and interpret the scatter plots between y_t and x_t , y_t and w_t , y_t^2 and w_t , and $\hat{u}_t^2$ and w_t where $\hat{u}_t$ is the residual from regressing y_t on a constant and x_t .

(2) Estimating a Regression Model with Heteroskedasticity

Gauss file(s)	hetero_estimate.g
Matlab file(s)	hetero_estimate.m

Simulate the following model for $T = 500$ observations

$$y_t = \beta_0 + \beta_1 x_t + u_t, \quad u_t \sim N(0, \sigma_t^2),$$

$$\sigma_t^2 = \exp(\gamma_0 + \gamma_1 w_t),$$

where $x_t \sim N(0, 1)$, w_t is a time trend and the parameter values are $\beta_0 = 1$, $\beta_1 = 2$, $\gamma_0 = 0.1$ and $\gamma_1 = 0.1$.

- (a) Write down the log-likelihood function.
- (b) Compute the maximum likelihood estimates using the Newton-Raphson algorithm and compare the estimates with the population values.

(3) Testing for Heteroskedasticity in the Regression Model

Gauss file(s) hetero_test.g

Matlab file(s) hetero_test.m

Simulate the same model as specified in Exercise 2 for $T = 500$ observations using the same parameter values.

Compute the following:

- (a) a LR test of heteroskedasticity;
- (b) a Wald test of heteroskedasticity; and
- (c) a LM test of heteroskedasticity.

(4) Testing for Vector Heteroskedasticity

Gauss file(s) hetero_system.g

Matlab file(s) hetero_system.m

Simulate the following model for $T = 2000$ observations

$$y_t B + x_t A = u_t$$

$$u_t \sim N(0, V_t = S_t S_t')$$

$$S_t = C + D w_t,$$

where

$$B = \begin{bmatrix} 1 & -\beta_2 \\ -\beta_1 & 1 \end{bmatrix} = \begin{bmatrix} 1.000 & -0.200 \\ -0.600 & 1.000 \end{bmatrix}$$

$$A = \begin{bmatrix} -\alpha_1 & 0 \\ 0 & -\alpha_2 \end{bmatrix} = \begin{bmatrix} -0.400 & 0.000 \\ 0.000 & 0.500 \end{bmatrix}$$

$$C = \begin{bmatrix} c_{1,1} & 0 \\ c_{2,1} & c_{2,2} \end{bmatrix} = \begin{bmatrix} 1.000 & 0.000 \\ 0.500 & 2.000 \end{bmatrix}$$

$$D = \begin{bmatrix} d_{1,1} & 0 \\ d_{2,1} & d_{2,2} \end{bmatrix} = \begin{bmatrix} 0.500 & 0.000 \\ 0.200 & 0.200 \end{bmatrix},$$

and $x_{1,t} \sim U[0, 10]$, $x_{2,t} \sim N(0, 9)$ and $w_t \sim U[0, 1]$.

- (a) Estimate the model by maximum likelihood using the Newton-Raphson algorithm. Compare the parameter estimates with the population values.
- (b) Construct a Wald test for vector heteroskedasticity.

(5) **Testing for Vector Heteroskedasticity and Autocorrelation**

Gauss file(s)	<code>hetero_general.g</code>
Matlab file(s)	<code>hetero_general.m</code>

Simulate the following model for $T = 2000$ observations

$$\begin{aligned}
 y_t B + x_t A &= u_t \\
 u_t &= u_{t-1} P + v_t \\
 v_t &\sim N(0, V_t = S_t S_t') \\
 S_t &= C + D w_t,
 \end{aligned}$$

where

$$\begin{aligned}
 B &= \begin{bmatrix} 1 & -\beta_2 \\ -\beta_1 & 1 \end{bmatrix} = \begin{bmatrix} 1.000 & -0.200 \\ -0.600 & 1.000 \end{bmatrix} \\
 A &= \begin{bmatrix} -\alpha_1 & 0 \\ 0 & -\alpha_2 \end{bmatrix} = \begin{bmatrix} -0.400 & 0.000 \\ 0.000 & 0.500 \end{bmatrix} \\
 P &= \begin{bmatrix} \rho_{1,1} & \rho_{1,2} \\ \rho_{2,1} & \rho_{2,2} \end{bmatrix} = \begin{bmatrix} 0.800 & -0.200 \\ 0.100 & 0.600 \end{bmatrix} \\
 C &= \begin{bmatrix} c_{1,1} & 0 \\ c_{2,1} & c_{2,2} \end{bmatrix} = \begin{bmatrix} 1.000 & 0.000 \\ 0.500 & 2.000 \end{bmatrix} \\
 D &= \begin{bmatrix} d_{1,1} & 0 \\ d_{2,1} & d_{2,2} \end{bmatrix} = \begin{bmatrix} 0.500 & 0.000 \\ 0.200 & 0.200 \end{bmatrix},
 \end{aligned}$$

and $x_{1,t} \sim U[0, 10]$, $x_{2,t} \sim N(0, 9)$ and $w_t \sim U[0, 1]$.

- (a) Estimate the model by maximum likelihood using the Newton-Raphson algorithm. Compare the parameter estimates with the population values.
- (b) Construct a Wald test for vector heteroskedasticity.
- (c) Construct a joint Wald test for vector heteroskedasticity and autocorrelation.

(6) The Great Moderation

Gauss file(s)	hetero_moderation.g
Matlab file(s)	hetero_moderation.m

This exercise is based on annual data on real U.S. GDP per capita for the period 1946 to 2006. The Great Moderation refers to the decrease in the volatility of U.S. output growth after the early 1980s by comparison with previous volatility levels. This proposition is tested by specifying the following model

$$y_t = \beta_0 + \beta_1 d_t + u_t, \quad u_t \sim N(0, \sigma_t^2)$$

$$\sigma_t^2 = \exp(\gamma_0 + \gamma_1 d_t),$$

where y_t is the growth rate in real GDP and d_t is a dummy variable to be defined later.

- (a) Compute the growth rate in real GDP

$$y_t = 100(\ln GDP_t - \ln GDP_{t-1}).$$

- (b) Compute the sample means and sample variances of y_t for the sub-periods 1947 to 1983 and 1984 to 2006.
- (c) Define the dummy variable

$$d_t = \begin{cases} 0 & : \text{1947 to 1983} \\ 1 & : \text{1984 to 2006,} \end{cases}$$

and estimate the parameters of the model, $\theta = \{\beta_0, \beta_1, \gamma_0, \gamma_1\}$, by maximum likelihood. Interpret the parameter estimates by comparing the estimates to the descriptive statistics computed in part (b).

- (d) The Great Moderation suggests that the U.S. GDP growth rate has become less volatile in the post-1983 period. This requires that $\gamma_1 \neq 0$. Perform LR, Wald and LM tests of this restriction and interpret the results.
- (e) The model allows for both the mean and the variance to change. Test the restriction $\beta_1 = 0$. If the null hypothesis is not rejected, then redo part (d) subject to the restriction $\beta_1 = 0$.

(7) Sampling Properties of Heteroskedasticity Estimators

Gauss file(s)	hetero_sampling.g
Matlab file(s)	hetero_sampling.m

Consider the model

$$y_t = \beta_0 + \beta_1 x_t + u_t, \quad u_t \sim N(0, \sigma_t^2),$$

$$\sigma_t^2 = \exp(\gamma_0 + \gamma_1 x_t),$$

where the population parameter values are $\beta_0 = 1.0$, $\beta_1 = 2.0$, $\gamma_0 = 1.0$, $\gamma_1 = 5.0$ and $x_t = U[0, 1]$ are draws from the uniform distribution.

- (a) Simulate the following model for $T = \{20, 50, 100, 200, 500\}$ observations and compute the following statistics on the sampling distribution of the maximum likelihood estimator using 10,000 replications.

(i) The mean: $\frac{1}{10000} \sum_{i=1}^{10000} \hat{\theta}_i.$

(ii) The bias: $\frac{1}{10000} \sum_{i=1}^{10000} \hat{\theta}_i - \theta_0.$

(iii) The RMSE: $\sqrt{\frac{1}{10000} \sum_{i=1}^{10000} (\hat{\theta}_i - \theta_0)^2}.$

- (b) Repeat part (a) for the least squares estimator without adjusting for heteroskedasticity, that is with $\gamma_1 = 0$.
- (c) Using the results in parts (a) and (b), discuss the consistency properties of the maximum likelihood and the OLS estimators of β_0 and β_1 .
- (d) Using the results in parts (a) and (b), discuss the efficiency properties of the maximum likelihood and the OLS estimators of β_0 and β_1 .

(8) Power Properties of the Wald Heteroskedasticity Test

Gauss file(s)	hetero_power.g
Matlab file(s)	hetero_power.m

Consider the model

$$y_t = \beta_0 + \beta_1 x_t + u_t, \quad u_t \sim N(0, \sigma_t^2),$$

$$\sigma_t^2 = \exp(\gamma_0 + \gamma_1 x_t),$$

where the population parameters are $\beta_0 = 1.0$, $\beta_1 = 2.0$, $\gamma_0 = 1.0$, $\gamma_1 = \{0.0, 0.5, 1.0, 1.5, \dots, 5.0\}$ and $x_t = U[0, 1]$ are draws from the uniform distribution.

- (a) Simulate the model under the null hypothesis of no heteroskedasticity by choosing $\gamma_1 = 0.0$ with a sample of $T = 50$ observations and 10,000 replications.

- (i) For each draw, compute the Wald test of heteroskedasticity

$$W = \frac{(\hat{\gamma}_1 - 0)^2}{\text{var}(\hat{\gamma}_1)}.$$

- (ii) Compute the 5% critical value from the sampling distribution.
 (iii) Compute the size of the test based on the 5% critical value from the χ^2 distribution with one degree of freedom.
 (iv) Discuss the size properties of the test.
- (b) Simulate the model under the alternative hypothesis with increasing levels of heteroskedasticity by choosing $\gamma_1 = \{0.5, 1.0, 1.5, \dots, 5.0\}$. Use a sample of $T = 50$ observations and 10,000 replications.
- (i) For each draw compute the Wald test of heteroskedasticity.
 (ii) Compute the power of the test (size unadjusted) based on the 5% critical value from the χ^2 distribution with one degree of freedom.
 (iii) Compute the power of the test (size adjusted) based on the 5% critical value from the sampling distribution of the Wald statistic obtained in part (a).
 (iv) Discuss the power properties of the test.
- (c) Repeat parts (a) and (b) for samples of size $T = 100, 200, 500$. Discuss the consistency properties of the Wald test.

PART THREE

OTHER ESTIMATION METHODS

9

Quasi-Maximum Likelihood Estimation

9.1 Introduction

The class of models discussed in Parts ONE and TWO of the book assume that the specification of the likelihood function, in terms of the joint probability distribution of the variables, is correct and that the regularity conditions set out in Chapter 2 are satisfied. Under these conditions, the maximum likelihood estimator has the desirable properties discussed in Chapter 2, namely that it is consistent, achieves the Cramér-Rao lower bound given by the inverse of the information matrix and is asymptotically normally distributed.

This chapter addresses the problem investigated in the seminal work of White (1982), namely maximum likelihood estimation when the likelihood function is misspecified. In general, the maximum likelihood estimator in the presence of misspecification does not display the usual properties. However, there are a number of important special cases in which the maximum likelihood estimator of a misspecified model still provides a consistent estimator for some of the population parameters in the true model. As the maximum likelihood estimator is based on a misspecified model, this estimator is referred to as the quasi-maximum likelihood estimator. Perhaps the most important case is the estimation of the conditional mean in the linear regression model, discussed in detail in Part TWO, where potential misspecifications arise from assuming either normality, or constant variance, or independence.

One important difference between the maximum likelihood estimator based on the true probability distribution and the quasi-maximum likelihood estimator is that the usual variance formulae, derived in Chapter 2, based on the information matrix equality holding, are in general no longer appropriate for the quasi-maximum likelihood estimator. Nonetheless, an estimate of

the variance is still available, being based on a combination of the Hessian and the outer product of gradient matrices.

9.2 Misspecification

Suppose that the true probability distribution of y_t is $f_0(y_t; \theta, \delta)$ with population parameters $\theta = \theta_0$ and $\delta = \delta_0$, but an incorrect probability distribution given by $f(y_t; \theta, \lambda)$ is specified to construct the likelihood function

$$\begin{aligned} \text{True distribution} & : f_0(y_t; \theta, \delta) \\ \text{Misspecified distribution} & : f(y_t; \theta, \lambda). \end{aligned}$$

The parameter vector of interest is θ , which is common to both models. The additional parameter vector δ is part of the true distribution but not the misspecified model, whilst the reverse holds for the parameter vector λ , which is just part of the misspecified model. Some specific examples of misspecified models are as follows, with the first four examples corresponding to misspecification of the distribution, the next corresponding to misspecification of the mean, and the last representing misspecification of the variance.

Example 9.1 Duration Analysis

To ensure positive durations, the true model of duration times between events follows an exponential distribution with parameter μ , while the misspecified model posits that durations are $N(\mu, 1)$

$$f_0(y_t; \mu) = \frac{1}{\mu} \exp \left[-\frac{y_t}{\mu} \right], \quad f(y_t; \mu) = \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{(y_t - \mu)^2}{2} \right].$$

The parameter $\theta = \{\mu\}$ is common to both models while there are no additional parameters in the true and misspecified models. In this situation, the distribution is misspecified, the mean is not. Panel (a) of Figure 9.1 provides a plot of the true and misspecified log-likelihood functions for a sample of $T = 10$ observations. The data are drawn from the (true) exponential distribution with parameter $\theta_0 = \mu_0 = 1$. $\square$

Example 9.2 Modelling Counts

The true distribution of counts is negative binomial with parameters $\{\mu > 0, 0 < p < 1\}$, whereas the misspecified distribution is Poisson with parameter $\{\mu > 0\}$

$$f_0(y_t; \mu, p) = \frac{\Gamma(y_t + \mu)}{\Gamma(y_t + 1) \Gamma(\mu)} (1 - p)^\mu p^{y_t}, \quad f(y_t; \mu) = \frac{\mu^{y_t} \exp[-\mu]}{y_t!}.$$

The parameter $\theta = \{\mu\}$ is common to both models, while $\delta = \{p\}$ is an

additional parameter in the true model. As the mean and the variance of the negative binomial distribution are, respectively,

$$E[y_t] = \mu \frac{p}{1-p}, \quad \text{var}(y_t) = \mu \frac{p}{(1-p)^2},$$

the true model is characterized by the variance being larger than the mean, commonly known as overdispersion, since

$$\frac{\text{var}(y_t)}{E[y_t]} = \frac{1}{1-p} > 0.$$

The form of misspecification is that the Poisson distribution does not exhibit overdispersion because its mean and variance are equal. Panel (b) of Figure 9.1 illustrates the true and misspecified log-likelihood functions for a sample of $T = 10$ observations. The data are drawn from the (true) negative binomial distribution with parameters $\theta_0 = \mu_0 = 5$ and $p = 0.5$. $\square$

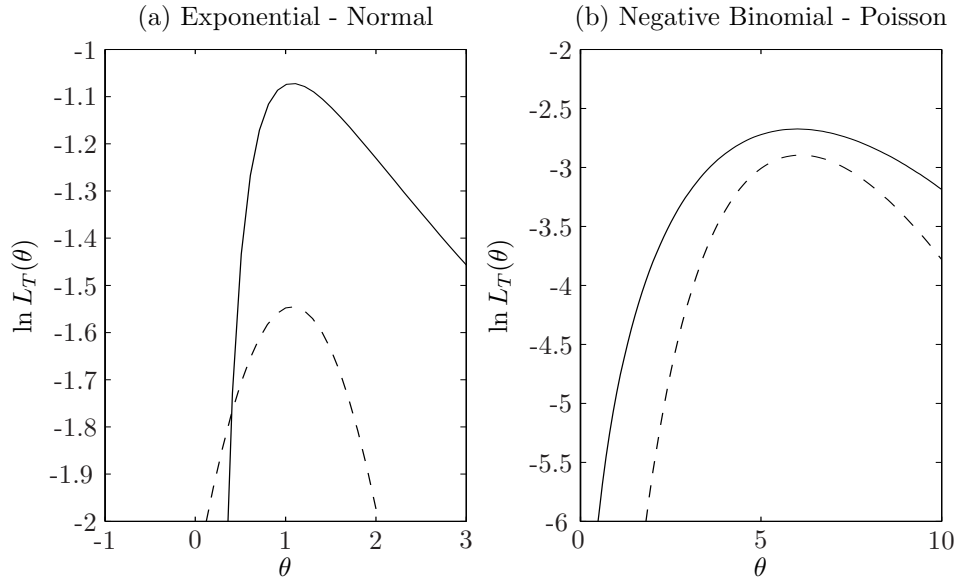


Figure 9.1 Comparison of true and misspecified log-likelihood functions. Panel (a) compares the true, exponential log-likelihood function (solid line) and the misspecified normal log-likelihood function (dashed line) for $T = 10$ observations drawn from an exponential distribution with $\mu_0 = 1$. Panel (b) compares the true, negative binomial distribution (solid line) and the misspecified Poisson distribution for $T = 10$ observations drawn from a negative binomial distribution with $\mu_0 = 5$ and $p = 0.5$.

Example 9.3 Constant Mean Model of Returns

In the constant mean model in finance, returns are assumed to have mean μ and variance σ^2

$$y_t = \mu + \sigma z_t,$$

where z_t is an *iid* $(0, 1)$ disturbance term. In the true model, z_t is a standardized Student *t* distribution with ν degrees of freedom, whereas for the misspecified model it is assumed to be $N(0, 1)$

$$f_0(y_t; \mu, \sigma^2, \nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\sigma^2(\nu-2)}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{(y_t - \mu)^2}{\sigma^2(\nu-2)}\right)^{-(\nu+1)/2}$$

$$f(y_t; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_t - \mu)^2}{2\sigma^2}\right].$$

The common parameters are $\theta = \{\mu, \sigma^2\}$, while the degrees of freedom parameter represents an additional parameter in the true model, $\delta = \{\nu\}$. The degree of misspecification is controlled by the parameter ν , with misspecification disappearing as $\nu \rightarrow \infty$. $\square$

Example 9.4 AR Model with Nonnormal Disturbance

The true model is an AR(1)

$$y_t = \phi y_{t-1} + v_t,$$

where $|\phi| < 1$ and $v_t \sim iid(0, \sigma^2)$ with some non-normal distribution and moment condition $E|v_t|^r$ for $r > 4$, while the misspecified model assumes that v_t is normal

$$f(y_t | y_{t-1}; \gamma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_t - \phi y_{t-1})^2}{2\sigma^2}\right],$$

where $\gamma = \{\phi, \sigma^2\}$. The conditional mean and variance are correctly specified, whereas the distribution is misspecified.

Example 9.5 Time Series Dynamics

The time series y_t follows a MA(1) model whereas the misspecified model is an AR(1)

$$MA(1) : y_t = u_t - \psi u_{t-1}, \quad u_t \sim iid N(0, \sigma^2)$$

$$AR(1) : y_t = \psi y_{t-1} + v_t, \quad v_t \sim iid N(0, \eta^2).$$

The true and misspecified distributions are, respectively,

$$f_0(y_t|y_{t-1}, \dots, y_1; \psi, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y_t - (-\sum_{j=1}^{t-1} \psi^j y_{t-j}))^2}{2\sigma^2} \right]$$

$$f(y_t|y_{t-1}, \dots, y_1; \psi, \eta^2) = \frac{1}{\sqrt{2\pi\eta^2}} \exp \left[-\frac{(y_t - \psi y_{t-1})^2}{2\eta^2} \right],$$

The common parameter is $\theta = \{\psi\}$, whereas $\delta = \{\sigma^2\}$ and $\lambda = \{\eta^2\}$. Inspection of the two distributions shows that the misspecification corresponds to excluding lags greater than the first lag from the model. $\square$

Example 9.6 Heteroskedasticity

The true model is a heteroskedastic regression model with conditional mean βx_t and a normal disturbance with conditional variance αx_t , whereas the misspecified model assumes a constant conditional variance of σ^2

$$f_0(y_t|x_t; \beta, \alpha) = \frac{1}{\sqrt{2\pi(\alpha x_t)}} \exp \left[-\frac{(y_t - \beta x_t)^2}{2(\alpha x_t)} \right]$$

$$f(y_t|x_t; \beta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y_t - \beta x_t)^2}{2\sigma^2} \right].$$

The common parameter vector is $\theta = \{\beta\}$, whereas the additional parameters in the true and misspecified models are, respectively, $\delta = \{\alpha\}$ and $\lambda = \{\sigma^2\}$. $\square$

In Chapter 2, the maximum likelihood estimator based on a correctly specified likelihood is shown to be consistent and asymptotically normal. Also the information equality, equation (2.33), is shown to hold, which implies that the asymptotic variance of the maximum likelihood estimator is given by the inverse of the Fisher information matrix. If the likelihood function is misspecified, however, there are several possible cases to consider. Suppose, a mean or a conditional mean is being estimated using the misspecified likelihood function, then the following possible cases arise.

- (1) The misspecification is mostly harmless because the estimator based on the misspecified log-likelihood function remains consistent, asymptotically normal and has the same asymptotic variance as if it were correctly specified. Examples 9.1, 9.3 and 9.4 fall into this category.
- (2) The estimator is consistent and asymptotically normal, but the information equality no longer holds so that the asymptotic variance of the estimator has a different form. Example 9.6 falls into this category.
- (3) The estimator is inconsistent so that any subsequent inference is likely to be misleading. Example 9.5 is an example of this worst-case scenario.

The details of the model and the type of potential misspecification need to be evaluated on the case-by-case basis to determine which of these three situations may be relevant. While this classification does not, therefore, provide a general rule to analyse misspecification in every case, it does provide some guiding intuition that is often useful when estimating (conditional) means. If the misspecified likelihood function correctly specifies the form of both the mean and the variance, even if other aspects are misspecified, then situation 1 applies. If, on the other hand, the mean is correctly specified but the variance is misspecified then situation 2 applies. Finally, the worst-case scenario arises if the mean is misspecified. The same logic can be applied to estimating a (conditional) variance of y_t , since that is determined by the means of y_t and y_t^2 .

9.3 The Quasi-Maximum Likelihood Estimator

The previous examples provide a number of cases where the parameter θ is common to the true likelihood based on $f_0(y_t; \theta_0, \lambda_0)$ and the misspecified likelihood based on $f(y_t; \theta, \delta_0)$. If $\hat{\theta}$ represents the maximum likelihood estimator of θ_0 using the misspecified model, this estimator is known as the quasi-maximum likelihood estimator since estimation is based on a misspecified distribution and not the true distribution. The pertinent log-likelihood function is referred to as the quasi log-likelihood function

$$\begin{aligned} \text{True log-likelihood function} & : \quad \frac{1}{T} \sum_{t=1}^T \ln f_0(y_t; \theta, \delta) \\ \text{Quasi log-likelihood function} & : \quad \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta, \lambda). \end{aligned}$$

What is of interest is whether or not the maximum likelihood estimator of the misspecified model, $\hat{\theta}$, is a consistent estimator of θ_0 , even though the specified distribution used to construct the likelihood, may not match the true distribution. The conditions that need to be satisfied for this result to occur are now presented.

In Chapter 2, the maximum likelihood estimator is shown to be a consistent estimator of θ_0 , provided that

$$(\theta_0, \delta_0) = \arg \max_{\theta, \delta} E[\ln f_0(y_t; \theta, \delta)], \quad (9.1)$$

and the correctly specified maximum likelihood estimator is the sample

equivalent of this

$$(\tilde{\theta}, \tilde{\delta}) = \arg \max_{\theta, \delta} \frac{1}{T} \sum_{t=1}^T \ln f_0(y_t; \theta, \delta). \quad (9.2)$$

Now consider replacing the true distribution $f_0(y_t; \theta, \delta)$ in (9.1) by the misspecified distribution $f(y_t; \theta, \lambda)$. If the condition

$$(\theta_0, \lambda_0) = \arg \max_{\theta, \lambda} E [\ln f(y_t; \theta, \lambda)], \quad (9.3)$$

is satisfied, then θ_0 also maximises the expectation of the quasi log-likelihood function and the quasi-maximum likelihood estimator is the solution of

$$(\hat{\theta}, \hat{\lambda}) = \arg \max_{\theta, \lambda} \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta, \lambda). \quad (9.4)$$

Provided that (9.3) is shown to hold and that there is a suitable uniform WLLN such that

$$\frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta, \lambda) \xrightarrow{p} E [\ln f(y_t; \theta, \lambda)],$$

it follows by the same reasoning outlined in Chapter 2 that the quasi-maximum likelihood estimator is a consistent estimator of θ_0

$$\hat{\theta} \xrightarrow{p} \theta_0.$$

Example 9.7 Duration Analysis

From Examples 9.1 and 9.10, taking expectations of $\ln f(y_t; \theta)$ in the case of the normal distribution where the true model is exponential gives

$$\begin{aligned} E[\ln f(y_t; \theta)] &= -\frac{1}{2} \ln 2\pi - \frac{1}{2} E[(y_t - \theta)^2] \\ &= -\frac{1}{2} \ln 2\pi - \frac{1}{2} E[(y_t - \theta_0 + \theta_0 - \theta)^2] \\ &= -\frac{1}{2} \ln 2\pi - \frac{1}{2} E[(y_t - \theta_0)^2] - \frac{1}{2} (\theta_0 - \theta)^2 - (\theta_0 - \theta) E[y_t - \theta_0] \\ &= -\frac{1}{2} \ln 2\pi - \frac{1}{2} \theta_0^2 - \frac{1}{2} (\theta_0 - \theta)^2. \end{aligned}$$

Clearly $E[\ln f(y_t; \theta)]$ is maximized at $\theta = \theta_0$. Therefore the normal model satisfies (9.3) for the consistent estimation of the mean of an exponential distribution. $\square$

Example 9.8 Durations with Dependence

The true model of y_t is an exponential distribution conditional on y_{t-1} , with an AR(1) conditional mean

$$E[y_t|y_{t-1}] = \mu_0 + \beta_0(y_{t-1} - \mu_0), \quad t = 2, \dots, T, \quad (9.5)$$

where $|\beta_0| < 1$. The misspecified model is an *iid* exponential distribution with mean μ and quasi log-likelihood function at time t given by

$$\ln f(y_t; \mu) = -\ln \mu - y_t/\mu.$$

Now

$$E[\ln f(y_t; \mu)] = -\ln \mu - \frac{E[y_t]}{\mu} = -\ln \mu - \frac{E[y_t]}{\mu} = -\ln \mu - \frac{\mu_0}{\mu}.$$

The first and second derivatives of $E[\ln f(y_t; \mu)]$ are

$$\frac{dE[\ln f(y_t; \mu)]}{d\mu} = -\frac{1}{\mu} + \frac{\mu_0}{\mu^2}, \quad \frac{d^2E[\ln f(y_t; \mu)]}{d\mu^2} = \frac{1}{\mu^2} - \frac{2\mu_0}{\mu^3}.$$

Setting the derivative to zero shows that $E[\ln f(y_t; \theta)]$ is maximized at $\mu = \mu_0$. The quasi-maximum likelihood estimator provides a consistent estimator of the mean μ_0 , even though it omits the dynamics of the true model.

Example 9.9 Time Series Dynamics

In Example 9.5 the true model is a MA(1) model and the misspecified model is an AR(1) model. The expectation of the misspecified log-likelihood function is

$$E[\ln f(y_t|y_{t-1}, \dots, y_1; \psi, \eta^2)] = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \eta^2 - \frac{1}{2\eta^2} E[(y_t - \psi y_{t-1})^2].$$

Given that the true MA(1) model is $y_t = u_t - \psi_0 u_{t-1}$, then

$$\begin{aligned} E[(y_t - \psi y_{t-1})^2] &= E[((u_t - \psi_0 u_{t-1}) - \psi(u_{t-1} - \psi_0 u_{t-2}))^2] \\ &= \sigma_0^2(1 + (\psi + \psi_0)^2 + \psi^2 \psi_0^2), \end{aligned}$$

as $u_t \sim iid(0, \sigma_0^2)$, resulting in

$$E[\ln f(y_t|y_{t-1}, \dots, y_1; \psi, \eta^2)] = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \eta^2 - \frac{1}{2\eta^2} \sigma_0^2(1 + (\psi + \psi_0)^2 + \psi^2 \psi_0^2).$$

Differentiating with respect to ψ and setting the derivative to zero shows that

$$\psi = -\frac{\psi_0}{1 + \psi_0^2} \neq \psi_0,$$

so the condition in (9.3) is not satisfied. Therefore the estimator of the

coefficient of an AR(1) model is not consistent for the coefficient of an MA(1) model. $\square$

9.4 Asymptotic Distribution

Suppose the quasi-maximum likelihood condition (9.3) holds, so that the quasi-maximum likelihood estimator $\hat{\theta}$ is consistent for θ_0 . The derivation of the asymptotic distribution of $\hat{\theta}$ follows the same steps as those in Chapter 2 except the true likelihood based on $f_0(y_t; \theta, \delta)$ is replaced by the quasi-likelihood based on $f(y_t; \theta, \delta)$. Let $\gamma = \{\theta, \lambda\}$ represent the parameters of the quasi log-likelihood function and $\gamma_0 = \{\theta_0, \lambda_0\}$ correspond to the true parameters, where λ_0 effectively represents a true parameter insofar as it satisfies (9.3) even though it is not actually part of the true model.

The first-order condition of the quasi log-likelihood function corresponding to (9.4) is

$$G_T(\hat{\gamma}) = \frac{1}{T} \sum_{t=1}^T g_t(\hat{\gamma}) = 0, \quad (9.6)$$

where

$$g_t(\gamma) = \frac{\partial \ln f(y_t; \gamma)}{\partial \gamma} \quad (9.7)$$

is the gradient. An important property of $g_t(\gamma)$ is that, provided (9.3) holds,

$$E[g_t(\gamma_0)] = 0. \quad (9.8)$$

This expression can be interpreted as the first-order condition of (9.3) and it shows that, just as in correctly specified maximum likelihood estimation, if the quasi log-likelihood function is well specified for the estimation of θ_0 , the gradient evaluated at the true parameter value has mean zero.

The second order condition for maximisation is that

$$H_T(\hat{\gamma}) = \frac{1}{T} \sum_{t=1}^T h_t(\hat{\gamma})$$

be negative definite, where

$$h_t(\gamma) = \frac{\partial^2 \ln f(y_t; \gamma)}{\partial \gamma \partial \gamma'}.$$

A mean value expansion of this condition around the true value γ_0 , gives

$$0 = \frac{1}{T} \sum_{t=1}^T g_t(\hat{\gamma}) = \frac{1}{T} \sum_{t=1}^T g_t(\gamma_0) + \left[\frac{1}{T} \sum_{t=1}^T h_t(\gamma^*) \right] (\hat{\gamma} - \gamma_0), \quad (9.9)$$

where γ^* lies between $\hat{\gamma}$ and γ_0 , and hence $\gamma^* \xrightarrow{p} \gamma_0$ if $\hat{\gamma} \xrightarrow{p} \gamma_0$. Rearranging and multiplying both sides by $\sqrt{T}$, yields

$$\sqrt{T}(\hat{\gamma} - \gamma_0) = \left[-\frac{1}{T} \sum_{t=1}^T h_t(\gamma^*) \right]^{-1} \left[\frac{1}{\sqrt{T}} \sum_{t=1}^T g_t(\gamma_0) \right]. \quad (9.10)$$

Now the following conditions are used

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T h_t(\gamma^*) &\xrightarrow{p} H(\gamma_0) \\ \frac{1}{\sqrt{T}} \sum_{t=1}^T g_t(\gamma_0) &\xrightarrow{d} N(0, J(\gamma_0)), \end{aligned} \quad (9.11)$$

where

$$H(\gamma_0) = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T E[h_t(\gamma_0)], \quad (9.12)$$

and

$$\begin{aligned} J(\gamma_0) &= \lim_{T \rightarrow \infty} E \left[\left(\frac{1}{\sqrt{T}} \sum_{t=1}^T g_t(\gamma_0) \right) \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T g_t(\gamma_0) \right)' \right] \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T E[g_t(\gamma_0) g_t'(\gamma_0)] \\ &\quad + \lim_{T \rightarrow \infty} \sum_{s=1}^{T-1} \left(\frac{1}{T} \sum_{t=s+1}^T E[g_t(\gamma_0) g_{t-s}'(\gamma_0)] \right) \\ &\quad + \lim_{T \rightarrow \infty} \sum_{s=1}^{T-1} \left(\frac{1}{T} \sum_{t=s+1}^T E[g_{t-s}(\gamma_0) g_t'(\gamma_0)] \right). \end{aligned} \quad (9.13)$$

The first condition in (9.11) follows from a uniform WLLN. The second condition relies on (9.8) to apply a suitable central limit theorem to $g_t(\gamma_0)$. The choice of central limit theorem, as outlined in Chapter 2, is conditioned on the time series properties of $g_t(\theta)$, examples of which are given below. Combining (9.10) and (9.11) yields the asymptotic distribution

$$\sqrt{T}(\hat{\gamma} - \gamma_0) \xrightarrow{d} N(0, \Omega), \quad (9.14)$$

where the asymptotic quasi-maximum likelihood covariance matrix is

$$\Omega = H^{-1}(\gamma_0) J(\gamma_0) H^{-1}(\gamma_0). \quad (9.15)$$

Partitioning $H(\gamma_0)$ and $J(\gamma_0)$ in (9.15) conformably in terms of $\gamma = \{\theta, \lambda\}$

as

$$H(\gamma_0) = \begin{bmatrix} H_{1,1} & H_{1,2} \\ H_{2,1} & H_{2,2} \end{bmatrix}, \quad J(\gamma_0) = \begin{bmatrix} J_{1,1} & J_{1,2} \\ J_{2,1} & J_{2,2} \end{bmatrix},$$

the asymptotic distribution of the quasi-maximum likelihood estimator of θ_0 is

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, H_{1,1}^{-1} J_{1,1} H_{1,1}^{-1}). \quad (9.16)$$

In the derivation of the asymptotic covariance matrix of the maximum likelihood estimator in Chapter 2, the information equality given in equation (2.33), implies that $J_{1,1} = -H_{1,1}$ so that the variance in (9.16) can be simplified to $J_{1,1}^{-1}$ or $-H_{1,1}^{-1}$. However, the information equality does not necessarily hold under misspecification.

9.4.1 Misspecification and the Information Equality

Misspecification will generally imply that

$$J(\gamma_0) \neq -H(\gamma_0). \quad (9.17)$$

To show this, suppose that y_t is *iid* so that the expressions in (9.12) and (9.13) simplify to

$$J(\gamma_0) = E[g_t(\gamma_0)g_t'(\gamma_0)], \quad H(\gamma_0) = E[h_t(\gamma_0)].$$

Now the gradient (9.7) of the misspecified model can be written

$$g_t(\gamma) = \frac{1}{f(y_t; \gamma)} \frac{\partial f(y_t; \gamma)}{\partial \gamma},$$

and differentiating both sides of this representation with respect to γ gives

$$\begin{aligned} h_t(\gamma) &= -\frac{1}{f(y_t; \gamma)^2} \frac{\partial f(y_t; \gamma)}{\partial \gamma} \frac{\partial f(y_t; \gamma)}{\partial \gamma'} + \frac{1}{f(y_t; \gamma)} \frac{\partial^2 f(y_t; \gamma)}{\partial \gamma \partial \gamma'} \\ &= -g_t(\gamma) g_t'(\gamma) + \frac{1}{f(y_t; \gamma)} \frac{\partial^2 f(y_t; \gamma)}{\partial \gamma \partial \gamma'}. \end{aligned}$$

Taking expectations under the true model and rearranging gives

$$E[g_t(\gamma_0) g_t'(\gamma_0)] = -E[h_t(\gamma_0)] + E \left[\frac{1}{f(y_t; \gamma_0)} \frac{\partial^2 f(y_t; \gamma)}{\partial \gamma \partial \gamma'} \bigg|_{\gamma=\gamma_0} \right],$$

which shows that the information equality holds if the second term on the right is zero. If the model is correctly specified, so that $f(y_t; \gamma) = f_0(y_t; \gamma)$,

this term reduces to

$$\begin{aligned} \int \frac{1}{f(y_t; \gamma_0)} \frac{\partial^2 f(y_t; \gamma)}{\partial \gamma \partial \gamma'} \Big|_{\gamma=\gamma_0} f_0(y_t; \gamma_0) dy_t &= \int \frac{\partial^2 f(y_t; \gamma)}{\partial \gamma \partial \gamma'} \Big|_{\gamma=\gamma_0} dy_t \\ &= \frac{\partial^2}{\partial \gamma \partial \gamma'} \int f(y_t; \gamma) dy_t \Big|_{\gamma=\gamma_0} = 0, \end{aligned}$$

since $\int f(y_t; \gamma) dy_t = 1$, resulting in the information equality holding. However, if the model is misspecified then this reasoning does not generally apply and the information equality does not necessarily hold. An implication of the information matrix equality failing to hold is that neither $J^{-1}(\gamma_0)$ nor $H^{-1}(\gamma_0)$ are appropriate variance matrices on which to base standard error calculations. One consequence of using either of these variances as the basis of standard error calculations is that t tests will have sizes that differ from the nominal (5%) level.

Example 9.10 Duration Analysis

In Example 9.1, the misspecified model is $N(\theta, 1)$, yielding a log-likelihood at t of

$$\ln f(y_t; \theta) = -\frac{1}{2} \ln 2\pi - \frac{(y_t - \theta)^2}{2},$$

where $\gamma = \{\theta\}$. The gradient and Hessian are, respectively,

$$g_t(\theta) = \frac{d \ln f(y_t; \theta)}{d\theta} = y_t - \theta, \quad h_t(\theta) = \frac{d^2 \ln f(y_t; \theta)}{d\theta^2} = -1.$$

As the true model is an exponential distribution with mean $E[y_t] = \theta_0$ and variance $\text{var}(y_t) = \theta_0^2$, and given that y_t is *iid*, $H(\theta_0)$ and $J(\theta_0)$ of the misspecified model are, respectively,

$$H(\theta_0) = E[h_t(\theta_0)] = -1, \quad J(\theta_0) = E[g_t(\theta_0)^2] = E[(y_t - \theta_0)^2] = \theta_0^2,$$

verifying that $J(\theta_0) \neq -H(\theta_0)$. By comparison from Chapter 2, the corresponding gradient and Hessian based on the true model are, respectively,

$$g_t(\theta) = \frac{d \ln f(y_t; \theta)}{d\theta} = -\frac{1}{\theta} + \frac{y_t}{\theta^2}, \quad h_t(\theta) = \frac{d^2 \ln f(y_t; \theta)}{d\theta^2} = \frac{1}{\theta^2} - \frac{2y_t}{\theta^3}.$$

As

$$\begin{aligned} H(\theta_0) &= E[h_t(\theta_0)] = E\left[\frac{1}{\theta_0^2} - \frac{2y_t}{\theta_0^3}\right] = \frac{1}{\theta_0^2} - \frac{2E[y_t]}{\theta_0^3} = \frac{1}{\theta_0^2} - \frac{2\theta_0}{\theta_0^3} = -\frac{1}{\theta_0^2}, \\ J(\theta_0) &= E[g_t(\theta_0)^2] = E\left[\left(-\frac{1}{\theta_0} + \frac{y_t}{\theta_0^2}\right)^2\right] = E\left[\frac{(y_t - \theta_0)^2}{\theta_0^4}\right] = E\left[\frac{\theta_0^2}{\theta_0^4}\right] = \frac{1}{\theta_0^2}, \end{aligned}$$

the information equality holds for the true model. $\square$

Example 9.11 AR Model with Nonnormal Disturbance

In Example 9.4, the misspecified model is an AR(1) with normal disturbance yielding a log-likelihood function at t given by

$$\ln f(y_t|y_{t-1}; \gamma) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{(y_t - \phi y_{t-1})^2}{2\sigma^2},$$

with $\gamma = \{\phi, \sigma^2\}$. The gradient vector and Hessian are, respectively,

$$g_t(\gamma) = \begin{bmatrix} \frac{1}{\sigma^2} y_{t-1} (y_t - \phi y_{t-1}) \\ -\frac{1}{2\sigma^2} + \frac{(y_t - \phi y_{t-1})^2}{2\sigma^4} \end{bmatrix},$$

$$h_t(\gamma) = \begin{bmatrix} -\frac{y_{t-1}^2}{\sigma^2} & -\frac{y_{t-1}(y_t - \phi y_{t-1})}{\sigma^4} \\ -\frac{y_{t-1}(y_t - \phi y_{t-1})}{\sigma^4} & \frac{1}{2\sigma^4} - \frac{(y_t - \phi y_{t-1})^2}{\sigma^6} \end{bmatrix},$$

Evaluating at $\gamma = \gamma_0$ and using $v_t = y_t - \phi_0 y_{t-1}$, gives

$$g_t(\gamma_0) = \begin{bmatrix} \frac{1}{\sigma_0^2} y_{t-1} v_t \\ \frac{(v_t^2 - \sigma_0^2)}{2\sigma_0^4} \end{bmatrix}, \quad h_t(\gamma_0) = \begin{bmatrix} -\frac{y_{t-1}^2}{\sigma_0^2} & -\frac{y_{t-1} v_t}{\sigma_0^4} \\ -\frac{y_{t-1} v_t}{\sigma_0^4} & \frac{1}{2\sigma_0^4} - \frac{v_t^2}{\sigma_0^6} \end{bmatrix}.$$

The gradient, $g_t(\gamma_0)$, is a martingale difference sequence because

$$\begin{aligned} E_{t-1} \left[\frac{1}{\sigma_0^2} y_{t-1} v_t \right] &= \frac{1}{\sigma_0^2} y_{t-1} E_{t-1} [v_t] = 0 \\ E_{t-1} \left[\frac{v_t^2 - \sigma_0^2}{2\sigma_0^4} \right] &= \frac{E_{t-1} [v_t^2] - \sigma_0^2}{2\sigma_0^4} = \frac{\sigma_0^2 - \sigma_0^2}{2\sigma_0^4} = 0. \end{aligned}$$

For the AR(1) model, y_t is also stationary so that $J(\gamma_0)$ is given by

$$\begin{aligned} J(\gamma_0) &= E[g_t(\gamma_0)g_t'(\gamma_0)] \\ &= \begin{bmatrix} \frac{E[y_{t-1}^2 v_t^2]}{\sigma_0^4} & \frac{E[y_{t-1} v_t (v_t^2 - \sigma_0^2)]}{2\sigma_0^6} \\ \frac{E[y_{t-1} v_t (v_t^2 - \sigma_0^2)]}{2\sigma_0^6} & \frac{E[(v_t^2 - \sigma_0^2)^2]}{4\sigma_0^8} \end{bmatrix} = \begin{bmatrix} \frac{1}{1 - \phi_0^2} & 0 \\ 0 & \frac{E[v_t^4] - \sigma_0^4}{4\sigma_0^8} \end{bmatrix}. \end{aligned}$$

This expression uses the following results

$$E[y_{t-1}^2 v_t^2] = E[y_{t-1}^2 E_{t-1} [v_t^2]] = E[y_{t-1}^2] \sigma_0^2 = \frac{\sigma_0^4}{1 - \phi_0^2}$$

$$E[y_{t-1} E_{t-1} [v_t (v_t^2 - \sigma_0^2)]] = E[y_{t-1}] E[v_t^3] = 0$$

$$E[(v_t^2 - \sigma_0^2)^2] = E[v_t^4] + \sigma_0^4 - 2E[v_t^2] \sigma_0^2 = E[v_t^4] + \sigma_0^4 - 2\sigma_0^4 = E[v_t^4] - \sigma_0^4,$$

where the first two terms are based on the law of iterated expectations and the property that the first two unconditional moments of the AR(1) model (see also Chapter 13) are $E[y_t] = E[y_{t-1}] = 0$ and $\text{var}(y_t) = \sigma_0^2/(1 - \phi_0^2)$, respectively. Finally, $H(\gamma_0)$ is given by

$$H(\gamma_0) = E[h_t(\gamma_0)] = \begin{bmatrix} -\frac{E[y_{t-1}^2]}{\sigma_0^2} & -\frac{E[y_{t-1}v_t]}{\sigma_0^4} \\ -\frac{E[y_{t-1}v_t]}{\sigma_0^4} & \frac{1}{2\sigma_0^4} - \frac{E[v_t^2]}{\sigma_0^6} \end{bmatrix} = - \begin{bmatrix} \frac{1}{1 - \phi_0^2} & 0 \\ 0 & \frac{1}{2\sigma_0^4} \end{bmatrix},$$

which is based on the same results used to derive $J(\theta_0)$. A comparison of $J(\gamma_0)$ and $H(\theta_0)$ shows that the information equality holds for the autoregressive parameter ϕ , but not for the variance parameter σ^2 . In the special case of no misspecification where the true model is also normal so that $E[v_t^4] = 3\sigma_0^4$, from $J(\gamma_0)$ the pertinent term is $(E[v_t^4] - \sigma_0^4)/4\sigma_0^8 = 1/2\sigma_0^4$ which now equals the corresponding term in $H(\gamma_0)$. $\square$

9.4.2 Independent and Identically Distributed Data

If y_t is *iid* then $g_t(\gamma_0)$ is also *iid* and the expressions for $H(\gamma_0)$ and $J(\gamma_0)$ in (9.12) and (9.13), respectively, reduce to

$$H(\gamma_0) = E[h_t(\gamma_0)]. \quad (9.18)$$

$$J(\gamma_0) = E[g_t(\gamma_0) g_t'(\gamma_0)]. \quad (9.19)$$

as the expectations $E[h_t(\gamma_0)]$ and $E[g_t(\gamma_0) g_t'(\gamma_0)]$ are constant over t because of the identical distribution assumption, and the autocovariance terms in $J(\gamma_0)$ disappear because of independence. As $g_t(\gamma_0)$ is *iid* the asymptotic normality condition in (9.14) follows from the Lindeberg-Levy central limit theorem in Chapter 2.

Example 9.12 Duration Analysis

From Example 9.10, $\gamma = \{\theta\}$ with $H(\gamma_0)$ and $J(\gamma_0)$ given by

$$H(\gamma_0) = -1, \quad J(\gamma_0) = \theta_0^2,$$

respectively. Substituting into (9.15) gives the quasi-maximum likelihood variance as

$$\Omega = H^{-1}(\gamma_0)J(\gamma_0)H^{-1}(\gamma_0) = \theta_0^2,$$

so the asymptotic distribution of the quasi-maximum likelihood estimator from (9.14) is

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \theta_0^2).$$

By comparison, the maximum likelihood estimator, $\tilde{\theta}$, based on the true distribution, has the same variance and hence the same asymptotic distribution. For this example, the erroneous assumption of normality does not result in an inefficient estimator because the quasi-maximum likelihood and maximum likelihood estimators are identical. $\square$

Example 9.13 Duration Analysis Extended

Extending Example 9.10, by choosing a quasi log-likelihood function based on $N(\theta, \sigma^2)$, now yields the gradient and Hessian given by

$$g_t(\theta) = \frac{d \ln f(y_t; \theta)}{d\theta} = \frac{y_t - \theta}{\sigma^2}, \quad h_t(\theta) = \frac{d^2 \ln f(y_t; \theta)}{d\theta^2} = -\frac{1}{\sigma^2}.$$

As the true model is an exponential distribution with mean $E[y_t] = \theta_0$ and variance $\text{var}(y_t) = \theta_0^2 = \sigma_0^2$, now $H(\theta_0)$ and $J(\theta_0)$ of the misspecified model are, respectively,

$$H(\theta_0) = E[h_t(\theta_0)] = -\frac{1}{\sigma_0^2} = -\frac{1}{\theta_0^2},$$

$$J(\theta_0) = E[g_t(\theta_0)^2] = E\left[\frac{(y_t - \theta_0)^2}{\sigma_0^4}\right] = \frac{E[(y_t - \theta_0)^2]}{\theta_0^4} = \frac{\theta_0^2}{\theta_0^4} = \frac{1}{\theta_0^2},$$

so the information equality $J(\theta_0) = -H(\theta_0)$ even holds for this misspecified model. $\square$

Both of these examples are concerned with the estimation of the mean θ_0 . In Example 9.12, the quasi log-likelihood function misspecifies the variance (sets it equal to 1) and, therefore, the information equality does not hold. The difference in Example 9.13 is that the variance is no longer misspecified, so the information equality holds for the estimation of the mean.

9.4.3 Dependent Data: Martingale Difference Score

If the score (or gradient), $g_t(\gamma_0)$, is a martingale difference sequence, the expression for $J(\gamma_0)$ in (9.13) reduces to

$$J(\gamma_0) = \lim_{T \rightarrow \infty} T^{-1} \sum_{t=1}^T E[g_t(\gamma_0) g_t'(\gamma_0)], \quad (9.20)$$

since martingale differences are not autocorrelated. If y_t is also stationary, then (9.20) reduces to (9.19) as the covariances are the same at each t and $H(\gamma_0)$, in expression (9.12), reduces to (9.18). As $g_t(\gamma_0)$ is a martingale difference sequence the asymptotic normality condition in (9.14) follows from the martingale difference central limit theorem in Chapter 2.

Example 9.14 AR Model with Nonnormal Disturbance

From Example 9.11 $\gamma = \{\phi, \sigma^2\}$, while $H(\gamma_0)$ and $J(\gamma_0)$ are, respectively,

$$H(\gamma_0) = - \begin{bmatrix} \frac{1}{1-\phi_0^2} & 0 \\ 0 & \frac{1}{2\sigma_0^4} \end{bmatrix}, \quad J(\gamma_0) = \begin{bmatrix} \frac{1}{1-\phi_0^2} & 0 \\ 0 & \frac{E[v_t^4] - \sigma_0^4}{4\sigma_0^8} \end{bmatrix}.$$

The asymptotic covariance matrix of the quasi-maximum likelihood estimator is

$$\Omega = H^{-1}(\gamma_0) J(\gamma_0) H^{-1}(\gamma_0) = \begin{bmatrix} 1 - \phi_0^2 & 0 \\ 0 & E[v_t^4] - \sigma_0^4 \end{bmatrix}.$$

This matrix is block-diagonal and therefore the asymptotic distributions decompose into

$$\begin{aligned} \sqrt{T}(\hat{\phi} - \phi_0) &\xrightarrow{d} N(0, 1 - \phi_0^2) \\ \sqrt{T}(\hat{\sigma}^2 - \sigma_0^2) &\xrightarrow{d} N(0, E[v_t^4] - \sigma_0^4). \end{aligned}$$

In the special case where the distribution is not misspecified, v_t is normal with fourth moment $E[v_t^4] = 3\sigma_0^4$. Using this result in the asymptotic distribution of $\hat{\sigma}^2$ results in $E[v_t^4] - \sigma_0^4 = 2\sigma_0^4$, which is the asymptotic variance of $\hat{\sigma}^2$ derived in Chapter 2 using the information matrix. $\square$

9.4.4 Dependent Data and Score

If the score (or gradient) $g_t(\gamma_0)$ exhibits autocorrelation and is not, therefore, a martingale difference sequence, it is necessary to use the more general expression of $J(\gamma_0)$ in (9.13) to derive the asymptotic covariance matrix. If y_t is also stationary, then the outer product of the gradients matrix reduces to

$$J(\gamma_0) = E[g_t(\gamma_0) g_t'(\gamma_0)] + \sum_{s=1}^{\infty} \left(E[g_t(\gamma_0) g_{t-s}'(\gamma_0)] + E[g_{t-s}(\gamma_0) g_t'(\gamma_0)] \right). \quad (9.21)$$

Provided that $g_t(\gamma_0)$ is also mixing at a sufficiently fast rate whereby it exhibits independence at least asymptotically, the asymptotic normality result in (9.11) follows from the mixing central limit theorem of Chapter 2. Finally, as y_t is also stationary the Hessian is given by (9.18).

Example 9.15 Durations with Dependence

From Example 9.8, $\gamma = \{\mu\}$, with expected Hessian

$$H(\mu_0) = E[h_t(\mu_0)] = \frac{1}{\mu_0^2} - \frac{2\mu_0}{\mu_0^3} = -\frac{1}{\mu_0^2}.$$

To find $J(\gamma_0)$, as the gradient is $g_t(\mu) = (y_t - \mu) / \mu^2$, the first term of (9.21) is

$$E[g_t(\gamma_0) g'_t(\gamma_0)] = \frac{1}{\mu_0^4} E[(y_t - \mu_0)^2] = \frac{1}{\mu_0^4} \frac{\mu_0^2}{1 - 2\beta_0^2},$$

since $\text{var}(y_t) = \mu_0^2 / (1 - 2\beta_0^2)$.¹ For the second term of (9.21), the AR(1) structure of (9.5) implies

$$E[g_t(\gamma_0) g'_{t-s}(\gamma_0)] = \beta_0^s \frac{1}{\mu_0^4} \frac{\mu_0^2}{1 - 2\beta_0^2}.$$

Thus (9.21) is

$$J(\gamma_0) = \frac{1}{\mu_0^4} \frac{\mu_0^2}{1 - 2\beta_0^2} \left(1 + 2 \sum_{s=1}^{\infty} \beta_0^s \right) = \frac{1}{\mu_0^4} \frac{\mu_0^2}{1 - 2\beta_0^2} \frac{1 + \beta_0}{1 - \beta_0}.$$

The quasi-maximum likelihood estimator $\hat{\mu}$, has variance

$$\Omega = H^{-1}(\gamma_0) J(\gamma_0) H^{-1}(\gamma_0) = \frac{\mu_0^2}{1 - 2\beta_0^2} \frac{1 + \beta_0}{1 - \beta_0},$$

and from (9.14) has asymptotic distribution

$$\sqrt{T}(\hat{\mu} - \mu_0) \xrightarrow{d} N\left(0, \frac{\mu_0^2}{1 - 2\beta_0^2} \frac{1 + \beta_0}{1 - \beta_0}\right).$$

In the special case where $\beta_0 = 0$, the asymptotic distribution simplifies to $\sqrt{T}(\hat{\mu} - \mu_0) \xrightarrow{d} N(0, \mu_0^2)$, which is the *iid* case given in Example 9.12. $\square$

9.4.5 Variance Estimation

If a quasi-maximum likelihood estimator is consistent and has asymptotic distribution given by (9.14), in practice it is necessary to estimate the asymptotic covariance matrix

$$\Omega = H^{-1}(\gamma_0) J(\gamma_0) H^{-1}(\gamma_0), \quad (9.22)$$

with a consistent estimator. A covariance estimator of this form is also known as a sandwich estimator.

¹ To see this, use $\text{var}(y_t) = E(\text{var}(y_t|y_{t-1})) + \text{var}(E(y_t|y_{t-1}))$ and then $E(\text{var}(y_t|y_{t-1})) = \mu_0^2 + \beta_0^2 \text{var}(y_{t-1})$, which follows from the exponential property that the variance is the square of the mean, and also $\text{var}(E(y_t|y_{t-1})) = \beta_0^2 \text{var}(y_{t-1})$.

In the case where y_t is stationary, a consistent estimator of $H(\gamma_0)$ in (9.12) is obtained by using a uniform WLLN to replace the expectations operator by the sample average and evaluating γ_0 at the quasi-maximum likelihood estimator $\hat{\gamma}$

$$H_T(\hat{\gamma}) = \frac{1}{T} \sum_{t=1}^T h_t(\hat{\gamma}). \quad (9.23)$$

The appropriate form of estimator for $J(\gamma_0)$ depends on the autocorrelation properties of the gradient $g_t(\gamma_0)$. If the gradient is not autocorrelated the autocovariance terms in (9.13) are ignored. In this case an estimator of $J(\gamma_0)$ is obtained by using a uniform WLLN to replace the expectations operator by the sample average and evaluating γ_0 at the quasi-maximum likelihood estimator $\hat{\gamma}$

$$J_T(\hat{\gamma}) = \frac{1}{T} \sum_{t=1}^T g_t(\hat{\gamma}) g_t'(\hat{\gamma}). \quad (9.24)$$

which is the outer product of gradients estimator of $J(\gamma_0)$ discussed in Chapter 3. Using (9.23) and (9.24) in (9.22) yields the covariance matrix

$$\hat{\Omega} = \left[\frac{1}{T} \sum_{t=1}^T h_t(\hat{\gamma}) \right]^{-1} \left[\frac{1}{T} \sum_{t=1}^T g_t(\hat{\gamma}) g_t'(\hat{\gamma}) \right] \left[\frac{1}{T} \sum_{t=1}^T h_t(\hat{\gamma}) \right]^{-1}, \quad (9.25)$$

If there is autocorrelation in $g_t(\gamma_0)$ it is necessary to allow for the autocovariance terms in (9.13) by defining

$$J_T(\hat{\gamma}) = \hat{\Gamma}_0 + \sum_{i=1}^P w_i (\hat{\Gamma}_i + \hat{\Gamma}_i'), \quad (9.26)$$

where

$$\hat{\Gamma}_i = \frac{1}{T} \sum_{t=i+1}^T g_t(\hat{\gamma}) g_{t-i}'(\hat{\gamma}), \quad i = 0, 1, 2, \dots \quad (9.27)$$

with P representing the maximum lag length and w_i the weights which ensure that $\hat{\Omega}$ in (9.26) is positive definite and consistent. Substituting (9.23) and (9.26) in (9.22) yields the asymptotic covariance matrix

$$\hat{\Omega} = \left[\frac{1}{T} \sum_{t=1}^T h_t(\hat{\gamma}) \right]^{-1} \left[\hat{\Gamma}_0 + \sum_{i=1}^P w_i (\hat{\Gamma}_i + \hat{\Gamma}_i') \right] \left[\frac{1}{T} \sum_{t=1}^T h_t(\hat{\gamma}) \right]^{-1}, \quad (9.28)$$

where $\hat{\Gamma}_i$ are defined in (9.27).

The choices of P and w_i are summarized in Table 9.1 for three weighting

schemes, namely Bartlett, Parzen and quadratic spectral. The determination of the maximum lag length P follows Newey and West (1994) which is based on minimizing the asymptotic mean squared error of $J_T(\gamma_0)$. The steps are as follows.

Step 1: Choose a weighting scheme and hence a preliminary value of the maximum lag length P , given by the first column of Table 9.1.

Step 2: Compute the quantities

$$\hat{J}_0 = \hat{\Gamma}_0 + \sum_{i=1}^P (\hat{\Gamma}_i + \hat{\Gamma}'_i), \quad \hat{J}_1 = 2 \sum_{i=1}^P i \hat{\Gamma}_i, \quad \hat{J}_2 = 2 \sum_{i=1}^P i^2 \hat{\Gamma}_i, \quad (9.29)$$

and update the maximum lag length according to the second column of Table 9.1. In the case of Bartlett weights the updated value of P is

$$\hat{P} = \text{int} \left[1.1447 \left(\frac{\hat{\nu}_1^2}{\hat{\nu}_0^2} T \right)^{1/3} \right], \quad (9.30)$$

where $\hat{\nu}_i = \iota' \hat{J}_i \iota$ for $i = 0, 1, 2$ and ι is a conformable column vector of ones.

Step 3 Compute the weights using the third column of Table 9.1 using the updated maximum lag length P from Step 2.

9.5 Quasi-Maximum Likelihood and Linear Regression

Let the true relationship between y_t and x_t be represented by the linear equation

$$y_t = x_t' \beta + u_t, \quad (9.31)$$

with population parameter $\beta = \beta_0$ and where u_t is a disturbance term. The aim of this section is to investigate the properties of the quasi-maximum likelihood estimator of β_0 based on a linear regression model that assumes

$$u_t | x_t \sim iid N(0, \sigma^2). \quad (9.32)$$

This assumption may not be true in a variety of ways, introducing misspecification into the quasi log-likelihood function. The common parameter vector is $\theta = \{\beta\}$ and the additional parameter in the quasi log-likelihood function is $\lambda = \{\sigma^2\}$. An important result derived below is that under certain conditions the quasi-maximum likelihood estimator of β_0 is consistent with an asymptotic normal distribution. As these properties are shown to still hold

Table 9.1

Alternative choices of lag length, P , and weights, w_i , $i \geq 1$, to compute the quasi-maximum likelihood covariance matrix in equation (9.28). Updated lag lengths computed using equations (9.29) and (9.30).

Preliminary P	Updated P	Weighting Scheme, w_i
Bartlett:		
$\text{int} \left[4 \left(\frac{T}{100} \right)^{\frac{2}{9}} \right]$	$\text{int} \left[1.1447 \left(\frac{\hat{\nu}_0^2}{\nu_0^2} T \right)^{1/3} \right]$	$w_i = 1 - \frac{i}{P+1}$
Parzen:		
$\text{int} \left[4 \left(\frac{T}{100} \right)^{\frac{4}{25}} \right]$	$\text{int} \left[2.6614 \left(\frac{\hat{\nu}_0^2}{\nu_0^2} T \right)^{1/5} \right]$	$w_i = \begin{cases} 1 - 6 \left(\frac{i}{P+1} \right)^2 - 6 \left(\frac{i}{P+1} \right)^3 & 0 \leq i \leq \frac{P+1}{2} \\ 2 \left(1 - \frac{i}{P+1} \right)^3 & \text{otherwise} \end{cases}$
Quadratic Spectral:		
$\text{int} \left[4 \left(\frac{T}{100} \right)^{\frac{2}{25}} \right]$	$\text{int} \left[1.3221 \left(\frac{\hat{\nu}_0^2}{\nu_0^2} T \right)^{1/5} \right]$	$w_i = \frac{25}{12\pi^2 \left(\frac{i}{P} \right)^2} \left(\frac{1}{\frac{6\pi i}{5P}} \sin \frac{6\pi i}{5P} - \cos \frac{6\pi i}{5P} \right)$

in the presence of various types of misspecifications, the quasi-maximum likelihood estimator is referred to as a robust estimator.

The quasi log-likelihood function is

$$\ln f(y_t|x_t; \beta, \sigma^2) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} (y_t - x'_t \beta)^2, \quad (9.33)$$

which has respective gradient and Hessian

$$g_t(\gamma) = \begin{bmatrix} \frac{1}{\sigma^2} x_t (y_t - \beta x_t) \\ -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} (y_t - x'_t \beta)^2 \end{bmatrix}, \quad (9.34)$$

$$h_t(\gamma) = \begin{bmatrix} -\frac{1}{\sigma^2} x_t x'_t & -\frac{1}{\sigma^4} x_t (y_t - x'_t \beta) \\ -\frac{1}{\sigma^4} (y_t - \beta x_t) x'_t & \frac{1}{2\sigma^4} - \frac{1}{\sigma^6} (y_t - x'_t \beta)^2 \end{bmatrix}. \quad (9.35)$$

The quasi-maximum likelihood estimator of β_0 is the least squares estimator

$$\hat{\beta} = \left(\sum_{t=1}^T x_t x'_t \right)^{-1} \sum_{t=1}^T x_t y_t, \quad (9.36)$$

where $\hat{\sigma}^2 = T^{-1} \sum_{t=1}^T (y_t - x'_t \hat{\beta})^2$ is the residual variance.

Using (9.33) in the quasi-maximum likelihood condition in (9.3), it is easily verified that

$$\beta_0 = \arg \max_{\beta} E [\ln f(y_t | x_t; \beta, \sigma^2)].$$

so $\hat{\beta}$ in (9.36) is a consistent estimator of β_0 ,

$$\hat{\beta} \xrightarrow{P} \beta_0.$$

This result, which is a general one, is not surprising given the well-known properties of the least squares estimator. An important assumption underlying the result is that the conditional mean of the quasi log-likelihood function is specified correctly as

$$E[y_t | x_t] = x_t' \beta_0$$

Situations where this assumption may not hold are discussed below.

To derive the asymptotic distribution of $\hat{\beta}$, partition $H(\gamma_0)$ and $J(\gamma_0)$ in (9.15) conformably in terms of $\gamma = \{\beta, \sigma^2\}$ as

$$H(\gamma_0) = \begin{bmatrix} H_{1,1} & H_{1,2} \\ H_{2,1} & H_{2,2} \end{bmatrix}, \quad J(\gamma_0) = \begin{bmatrix} J_{1,1} & J_{1,2} \\ J_{2,1} & J_{2,2} \end{bmatrix}.$$

The Hessian is block diagonal since

$$H_{1,2} = -\frac{1}{\sigma_0^4} E[x_t(y_t - x_t' \beta_0)] = -\frac{1}{\sigma_0^4} E[x_t(E[y_t | x_t] - x_t' \beta_0)] = 0.$$

Simplifying notation by defining

$$H(\beta_0) = H_{1,1}, \quad J(\beta_0) = J_{1,1},$$

from (9.14) the asymptotic distribution of $\hat{\beta}$ can therefore be written

$$\sqrt{T}(\hat{\beta} - \beta_0) \xrightarrow{d} N(0, \Omega), \quad (9.37)$$

with asymptotic covariance matrix

$$\Omega = H_{1,1}^{-1} J_{1,1} H_{1,1}^{-1} = H^{-1}(\beta_0) J(\beta_0) H^{-1}(\beta_0). \quad (9.38)$$

For the quasi log-likelihood function based on the linear regression model in (9.32), $H(\beta_0)$ and $J(\beta_0)$ have well-known forms. In the case of $H(\beta_0)$ substituting $h_t(\gamma)$ from (9.35) into (9.12) gives

$$H(\beta_0) = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T E[h_t(\beta_0)] = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \left[-\frac{1}{\sigma_0^2} E[x_t^2] \right]. \quad (9.39)$$

For $J(\beta_0)$, the gradient in (9.34) is substituted into (9.13) to give

$$\begin{aligned} J(\beta_0) &= \lim_{T \rightarrow \infty} \frac{1}{T} E \left[\left(\sum_{t=1}^T g_t(\beta_0) \right) \left(\sum_{t=1}^T g_t(\beta_0) \right)' \right] \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} E \left[\left(\sum_{t=1}^T \frac{1}{\sigma_0^2} x_t u_t \right) \left(\sum_{t=1}^T \frac{1}{\sigma_0^2} x_t u_t \right)' \right], \end{aligned}$$

or

$$\begin{aligned} J(\beta_0) &= \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \frac{1}{\sigma_0^4} E[u_t^2 x_t x_t'] + \lim_{T \rightarrow \infty} \sum_{s=1}^{T-1} \left(\frac{1}{T} \sum_{t=s+1}^T \frac{1}{\sigma_0^4} E[u_t u_{t-s} x_t x_{t-s}'] \right) \\ &\quad + \lim_{T \rightarrow \infty} \sum_{s=1}^{T-1} \left(\frac{1}{T} \sum_{t=s+1}^T \frac{1}{\sigma_0^4} E[u_t u_{t-s} x_{t-s} x_t'] \right). \end{aligned} \quad (9.40)$$

As in the general discussion of the quasi-maximum likelihood estimator in Section 9.4, the form of the asymptotic distribution of $\hat{\beta}$ in (9.37) varies depending upon the assumptions underlying the data. Three special cases, namely nonnormality, heteroskedasticity and autocorrelation, are now discussed in detail.

9.5.1 Nonnormality

Suppose that $\{y_t, x_t\}$ is *iid* in the true model and the assumption that $\text{var}(u_t|x_t) = \sigma_0^2$ is correct, but the assumption of normality is incorrect. The *iid* assumption means that $H(\beta_0)$ and $J(\beta_0)$ in (9.39) and (9.40) are, respectively,

$$H(\beta_0) = -\frac{1}{\sigma_0^2} E[x_t x_t'] \quad (9.41)$$

$$J(\beta_0) = \frac{1}{\sigma_0^4} E[u_t^2 x_t x_t'] = \frac{1}{\sigma_0^4} E[E[u_t^2|x_t] x_t x_t'] = \frac{1}{\sigma_0^4} E[\sigma_0^2 x_t x_t'] = \frac{1}{\sigma_0^2} E[x_t x_t'].$$

Using these expressions in (9.38) shows that the quasi-maximum likelihood estimator has asymptotic covariance matrix

$$\Omega = H^{-1}(\beta_0) J(\beta_0) H^{-1}(\beta_0) = \sigma_0^2 (E[x_t x_t'])^{-1}.$$

Other than the conditional variance σ_0^2 , the asymptotic distribution of $\hat{\beta}$ does not depend on the form of the conditional distribution of y_t . The asymptotic variance of $\hat{\beta}$ can be calculated as if y_t were truly conditionally normal, even

if this is not the case. So far as estimating β_0 is concerned, the misspecification of the conditional distribution has no effect on the consistency or the asymptotic distribution of the estimator, although now $\hat{\beta}$ is asymptotically inefficient relative to the maximum likelihood estimator based on the true conditional distribution.

9.5.2 Heteroskedasticity

Suppose that $\{y_t, x_t\}$ is assumed *iid*, but the assumption of homoskedasticity is incorrect, so that

$$\sigma_t^2 = \text{var}(y_t|x_t) = E[u_t^2|x_t].$$

The *iid* assumption means that $H(\beta_0)$ and $J(\beta_0)$ in (9.39) and (9.40) simplify to

$$H(\beta_0) = -\frac{1}{\sigma_0^2} E[x_t x_t'] \quad (9.42)$$

$$J(\beta_0) = \frac{1}{\sigma_0^4} E[u_t^2 x_t x_t'] = \frac{1}{\sigma_0^4} E[E[u_t^2|x_t] x_t x_t'] = \frac{1}{\sigma_0^4} E[\sigma_t^2 x_t x_t'].$$

The quasi-maximum likelihood estimator has the asymptotic covariance matrix

$$\Omega = H^{-1}(\beta_0) J(\beta_0) H^{-1}(\beta_0) = (E[x_t x_t'])^{-1} E[u_t^2 x_t x_t'] (E[x_t x_t'])^{-1}. \quad (9.43)$$

Example 9.16 Exponential Regression Model

The true model is that (y_t, x_t) are *iid* with conditional exponential distribution with conditional mean $\mu_t = x_t' \beta_0$. From the properties of the exponential distribution the conditional variance is $\sigma_t^2 = (x_t' \beta_0)^2$. For this model the quasi log-likelihood function based on the normal distribution is misspecified in both its distribution and its assumption of homoskedasticity. From (9.42)

$$H(\beta_0) = -\frac{1}{\sigma_0^2} E[x_t x_t'], \quad J(\beta_0) = \frac{1}{\sigma_0^4} E[(x_t' \beta_0)^2 x_t x_t'],$$

so the asymptotic covariance matrix of $\hat{\beta}$ in (9.43) becomes

$$\Omega = (E[x_t x_t'])^{-1} E[(x_t' \beta_0)^2 x_t x_t'] (E[x_t x_t'])^{-1}.$$

The asymptotic covariance matrix of $\hat{\beta}$ for this model is a function of the second and fourth moments of x_t . For example if x_t is a single regressor the asymptotic variance is $\beta_0^2 E[x_t^4] / E[x_t^2]^2$. $\square$

Even where the quasi-maximum likelihood estimator is consistent, the cost of misspecifying the true model in general manifests itself into inefficient parameter estimates relative to the true maximum likelihood estimator. The following example illustrates this using the exponential regression model from the previous example.

Example 9.17 Relative Efficiency

The example explores the relative efficiency of maximum likelihood estimation and quasi-maximum likelihood estimation. From Example 9.16 the log-likelihood function of the true model is

$$\sum_{t=1}^T \ln f_0(y_t|x_t; \beta) = \sum_{t=1}^T \left(-\ln \mu_t - \frac{y_t}{\mu_t} \right),$$

with $\mu_t = \beta x_t$, and x_t is a single regressor for simplicity. The respective gradient and Hessian are

$$\frac{\partial \ln f_0(y_t|x_t; \beta)}{\partial \beta} = -\frac{1}{\mu_t} x_t + \frac{y_t}{\mu_t^2} x_t, \quad \frac{\partial^2 \ln f_0(y_t|x_t; \beta)}{\partial \beta \partial \beta'} = \frac{1}{\mu_t^2} x_t x_t' - 2 \frac{y_t}{\mu_t^3} x_t x_t'.$$

Taking expectations of the Hessian

$$H(\beta) = E \left[\frac{1}{\mu_t^2} x_t x_t' - 2 \frac{y_t}{\mu_t^3} x_t x_t' \right] = \frac{1}{\mu_t^2} x_t x_t' - 2 \frac{E[y_t]}{\mu_t^3} x_t x_t' = -\frac{x_t x_t'}{\mu_t^2},$$

the asymptotic variance of the true model for the single regressor case is

$$\Omega_{MLE} = -H^{-1}(\beta_0) = \left(E \left[\frac{x_t x_t'}{(\beta_0 x_t)^2} \right] \right)^{-1} = \beta_0^2.$$

By comparison from Example 9.16, the asymptotic variance of the quasi-maximum likelihood estimator is $\Omega_{QMLE} = \beta_0^2 E(x_t^4) / E(x_t^2)^2$, which is larger than β_0^2 from the Cauchy-Schwarz inequality. If $x_t \sim U[0, a]$ then $E(x_t^4) / E(x_t^2)^2 = 1.8$,² and the asymptotic variance of the quasi-maximum likelihood estimator is 80% larger than the asymptotic variance of the maximum likelihood estimator. $\square$

9.5.3 Autocorrelation

Now suppose the *iid* assumption is relaxed to allow dependence amongst (y_t, x_t) in the true model, subject to the mixing condition of Section 9.4.4, while retaining the identical distribution assumption so that (y_t, x_t) is a

² This result uses the property of the uniform distribution,

$$\int x^j dx = a^{-1} \int_0^a x^j dx = a^{-1} \left[\frac{x^{j+1}}{j+1} \right]_0^a = \frac{a^j}{j+1}.$$

stationary process. Assuming that the conditional mean is specified correctly then from (9.12) and (9.13)

$$H(\beta_0) = -\frac{1}{\sigma_0^2} E[x_t x_t']$$

$$J(\beta_0) = \frac{1}{\sigma_0^4} \left(E[u_t^2 x_t x_t'] + \sum_{j=1}^{\infty} (E[u_t u_{t-j} x_t x_{t-j}'] + E[u_{t-j} u_t x_{t-j} x_t']) \right),$$

with the simplification because the stationarity of (y_t, x_t) means that the expectations in this expression are constant over t . The quasi-maximum likelihood estimator has asymptotic covariance matrix

$$\begin{aligned} \Omega &= H^{-1}(\beta_0) J(\beta_0) H^{-1}(\beta_0) \\ &= (E[x_t x_t'])^{-1} \left(E[u_t^2 x_t x_t'] + \sum_{j=1}^{\infty} (E[u_t u_{t-j} x_t x_{t-j}'] + E[u_{t-j} u_t x_{t-j} x_t']) \right) (E[x_t x_t'])^{-1}. \end{aligned} \quad (9.44)$$

The autocorrelation in $u_t x_t$ is used to adjust the variance.

Example 9.18 Overlapping Data

Under forward market efficiency the h -period forward rate at time t is an unbiased predictor of the spot rate at $t + h$

$$E_t[s_{t+h}] = f_{t,h}.$$

This unbiasedness property implies that the spread $y_t = s_t - f_{t-h,h}$ satisfies

$$E_{t-h}[y_t] = E_{t-h}[s_t] - f_{t-h,h} = 0,$$

which in turn, by the law of iterated expectations, implies that

$$E[y_t] = E[E_{t-h}[y_t]] = 0,$$

and hence for any lag $j \geq h$

$$\text{cov}[y_t, y_{t-j}] = E[y_t y_{t-j}] = E[E_{t-h}[y_t] y_{t-j}] = 0.$$

This derivation does not apply for $j = 1, \dots, h-1$, so y_t may be auto-correlated at lags $1, \dots, h-1$, but not at lags $h, h+1, \dots$. Therefore one simple way to test whether forward market efficiency holds, empirically, is to specify a regression model of the form

$$y_t = \beta_1 + \beta_2 y_{t-h} + u_t,$$

in which the theory implies the testable restrictions $\beta_{1,0} = 0$ and $\beta_{2,0} = 0$.

If the theory is true then $y_t = u_t$ and the disturbances in this regression may be autocorrelated at lags $1, \dots, h-1$. In this case $x_t = (1, y_{t-h})'$ so the matrices in (9.44) are

$$H(\beta_0) = \begin{pmatrix} 1 & 0 \\ 0 & E[y_{t-h}^2] \end{pmatrix},$$

and

$$\begin{aligned} J(\beta_0) &= E \left[u_t^2 \begin{pmatrix} 1 & y_{t-h} \\ y_{t-h} & y_{t-h}^2 \end{pmatrix} \right] \\ &+ \sum_{j=1}^{h-1} E \left[u_t u_{t-j} \begin{pmatrix} 1 & y_{t-h-j} \\ y_{t-h} & y_{t-h} y_{t-h-j} \end{pmatrix} \right] \\ &+ E \left[u_t u_{t-j} \begin{pmatrix} 1 & y_{t-h} \\ y_{t-h-j} & y_{t-h} y_{t-h-j} \end{pmatrix} \right]. \end{aligned}$$

□

Not every regression where (y_t, x_t) is autocorrelated will result in the product $x_t u_t$ being autocorrelated. In fact, the specification of many time series regressions precludes $x_t u_t$ being autocorrelated, in which case (9.44) can be simplified to (9.43). For example, consider an AR(1) regression specification

$$E_{t-1}[y_t] = \beta_0 y_{t-1}, \quad (9.45)$$

where E_{t-1} means to take expectations conditional on the set $\{y_{t-1}, y_{t-2}, \dots\}$. The representation of this model in terms of disturbances is

$$y_t = \beta_0 y_{t-1} + u_t,$$

in which (9.45) can be seen to imply that u_t is a mds

$$E_{t-1}[u_t] = E_{t-1}[y_t] - \beta_0 y_{t-1} = 0.$$

This in turn implies that u_t is not autocorrelated since the law of iterated expectations gives $E[u_t] = E[E_{t-1}[u_t]] = 0$ and hence for any $j \geq 1$

$$\text{cov}(u_t, u_{t-j}) = E[u_t u_{t-j}] = E[E_{t-1}[u_t u_{t-j}]] = E[E_{t-1}[u_t] u_{t-j}] = 0. \quad (9.46)$$

Moreover the product $u_t y_{t-1}$ is not autocorrelated since

$$\begin{aligned} E[u_t u_{t-j} y_{t-1} y_{t-1-j}] &= E[E_{t-1}[u_t u_{t-j} y_{t-1} y_{t-1-j}]] \\ &= E[E_{t-1}[u_t] u_{t-j} y_{t-1} y_{t-1-j}] \\ &= 0, \end{aligned} \quad (9.47)$$

which means that all of the autocovariances in the expression for $J(\beta_0)$ in

(9.44) are zero³. Therefore the AR(1) specification (9.45) implies that autocorrelation cannot be an issue. Conversely, if autocorrelation is discovered in the residuals of an AR(1) model in practice, the implication is that the conditional mean (9.45) has been misspecified. A model with a misspecified conditional mean does not give a quasi log-likelihood function that can consistently estimate that mean, and the adoption of the quasi-maximum likelihood covariance in (9.44) will not correct this misspecification. Instead additional lags or alternative functional form must be tried to correct the conditional mean specification.

Not every regression implies the absence of autocorrelation. The crucial step in both (9.46) and (9.47) is that the conditioning set $\{y_{t-1}, y_{t-2}, \dots\}$ in E_{t-1} is rich enough to contain all lags of u_t and $u_t y_{t-1}$. For example, in (9.46) this allows the step $E_{t-1}[u_t u_{t-j}] = E_{t-1}[u_t] u_{t-j}$. The disturbances of a static regression such as

$$E[y_t|x_t] = x_t' \beta_0 \quad (9.48)$$

will always satisfy $E[u_t|x_t] = 0$ by definition. However, $\text{cov}(u_t, u_{t-j})$ cannot be shown to be zero in this case since, unlike in the AR(1) model, $E[u_t u_{t-j}|x_t] \neq E[u_t|x_t] u_{t-j}$. The conditioning set of this regression is not rich enough to include the lagged disturbances. Therefore, the absence of autocorrelation is not an implication of such a model. Conversely, a practical finding of autocorrelated residuals would not automatically imply the conditional mean is misspecified. If the conditional mean is correct then autocorrelation simply suggests that (9.44) is the appropriate asymptotic variance.

Note that if the regression (9.48) were instead specified as

$$E[y_t|x_t, y_{t-1}, x_{t-1}, \dots] = x_t' \beta_0$$

then the conditioning set is sufficient to conclude that autocorrelation cannot be present.

To summarise, if a regression conditions on the full history of the dependent variable and regressors then the disturbance term is an mds with respect to this conditioning, residual autocorrelation must not be present and (9.43) is the relevant variance matrix. Practically, if residual autocorrelation is present, the regression itself is misspecified. On the other hand if a regression does not condition on this full history then residual autocorrelation does not necessarily imply regression misspecification and (9.44) is the relevant variance.

³ Specifically the gradient $g_t(\beta_0)$ of this model is an mds.

9.5.4 Variance Estimation

The White Estimator

In the case of neglected heteroskedasticity, consistent estimator of the asymptotic covariance matrix, Ω , in (9.43) is given by

$$\hat{\Omega} = H_T^{-1}(\hat{\gamma}) J_T(\hat{\gamma}) H_T^{-1}(\hat{\gamma}). \quad (9.49)$$

$H_T(\hat{\gamma})$ and $J_T(\hat{\gamma})$ are consistent estimators of $H(\gamma_0)$ and $J(\gamma_0)$, respectively, and are obtained by replacing the expectations operator by the sample average and evaluating the parameters at the quasi-maximum likelihood estimators, $\hat{\gamma} = \{\hat{\beta}, \hat{\sigma}^2\}$, to give

$$H_T(\hat{\gamma}) = -\frac{1}{\hat{\sigma}^2} \frac{1}{T} \sum_{t=1}^T x_t x_t', \quad J_T(\hat{\gamma}) = \frac{1}{\hat{\sigma}^2} \frac{1}{T} \sum_{t=1}^T \hat{u}_t^2 x_t x_t', \quad (9.50)$$

where $\hat{\sigma}^2 = T^{-1} \sum_{t=1}^T \hat{u}_t^2$ and $\hat{u}_t = y_t - x_t' \hat{\beta}$. Substituting (9.50) in (9.49) and simplifying gives

$$\hat{\Omega} = \left[\frac{1}{T} \sum_{t=1}^T x_t x_t' \right]^{-1} \left[\frac{1}{T} \sum_{t=1}^T \hat{u}_t^2 x_t x_t' \right] \left[\frac{1}{T} \sum_{t=1}^T x_t x_t' \right]^{-1}. \quad (9.51)$$

This covariance matrix estimator is also known as the White estimator.

Standard errors for individual ordinary least squares coefficients are computed as the square roots of the diagonal elements of the normalised covariance matrix

$$\frac{1}{T} \hat{\Omega} = \left[\sum_{t=1}^T x_t x_t' \right]^{-1} \left[\sum_{t=1}^T \hat{u}_t^2 x_t x_t' \right] \left[\sum_{t=1}^T x_t x_t' \right]^{-1}. \quad (9.52)$$

These standard errors are different from the usual ordinary least squares standard errors given by the square roots of the diagonal elements of the matrix

$$\frac{1}{T} \hat{\Omega}_{OLS} = \hat{\sigma}^2 \left[\sum_{t=1}^T x_t x_t' \right]^{-1}, \quad (9.53)$$

which are inconsistent in the presence of neglected heteroskedasticity.

For computational purposes, it is useful to define the matrices

$$X = \begin{pmatrix} x_1' \\ \vdots \\ x_T' \end{pmatrix}, \quad Z = \begin{pmatrix} \hat{u}_1 x_1' \\ \vdots \\ \hat{u}_T x_T' \end{pmatrix}, \quad (9.54)$$

so that (9.52) is computed as

$$\frac{1}{T}\hat{\Omega} = (X'X)^{-1} (Z'Z) (X'X)^{-1}.$$

Note that to compute $\sum_{t=1}^T \hat{u}_t^2 x_t x_t'$ the form $Z'Z$ is much faster than than the more commonly cited $X'WX$, where $W = \text{diag}(\hat{u}_1^2, \dots, \hat{u}_T^2)$ is a $T \times T$ matrix. The $Z'Z$ computations make substantial differences in execution times in Monte Carlo simulations for example.

Example 9.19 Sampling Properties of the White Estimator

The true model is the heteroskedastic AR(1) model

$$y_t = \beta_0 y_{t-1} + (\omega_0 + \alpha_0 y_{t-1}^2)^{1/2} v_t, \quad v_t \sim iid N(0, 1),$$

with conditional mean $E[y_t|y_{t-1}] = \beta_0 y_{t-1}$ and conditional variance $\text{var}(y_t|y_{t-1}) = \omega_0 + \alpha_0 y_{t-1}^2$. The appropriate asymptotic variance for estimates of the conditional mean in a heteroskedastic model is given by (9.51), so the White estimator is the appropriate variance estimator. The true values of the parameters are set to $\beta_0 = 0.5$, $\omega_0 = 1$ and $\alpha_0 = 0.5$ in the simulation. Table 9.2 gives statistics on the sampling distribution of the quasi-maximum likelihood estimator $\hat{\beta}$ based on a homoskedastic AR(1) model, computed by regressing y_t on y_{t-1} . The mean of $\hat{\beta}$ shows the well known negative finite sample bias in the estimator of the AR(1) model, but also shows this bias disappearing as the sample size increases. Also the variance is decreasing towards zero as the sample size increases. These findings demonstrate the consistency of the quasi-maximum likelihood estimator of β_0 . The t-statistic, $t = (\hat{\beta} - 0.5)/\text{se}(\hat{\beta})$, exhibits substantial over-rejections based on ordinary least squares standard errors, with the problem worsening as the sample size increases. The White standard errors largely correct this problem, especially for the larger sample sizes. $\square$

The Newey-West Estimator

Provided that the conditional mean is misspecified correctly, a consistent estimator of the asymptotic covariance matrix, Ω , in (9.44) in the presence of neglected autocorrelation is given by

$$\hat{\Omega} = H_T^{-1}(\hat{\gamma}) J_T(\hat{\gamma}) H_T^{-1}(\hat{\gamma}), \quad (9.55)$$

The terms $H_T(\hat{\gamma})$ and $J_T(\hat{\gamma})$ are consistent estimators of $H(\gamma_0)$ and $J(\gamma_0)$, respectively, and are obtained by replacing the expectations operator by the sample average and evaluating the parameters at the quasi-maximum

Table 9.2

Finite sample properties of the quasi-maximum likelihood estimator where the true model is an heteroskedastic autoregression with population parameter $\beta_0 = 0.5$. Based on 100000 Monte Carlo replications.

T	Statistics		$\Pr(t > 1.96)$	
	Mean $\hat{\beta}$	Var $\hat{\beta}$	OLS	White
25	0.406	0.053	0.145	0.100
50	0.431	0.032	0.183	0.082
100	0.450	0.020	0.231	0.069
200	0.464	0.013	0.283	0.061
400	0.474	0.008	0.342	0.059
800	0.482	0.006	0.398	0.053
1600	0.487	0.004	0.454	0.050

likelihood estimators $\hat{\gamma} = \{\hat{\beta}, \hat{\sigma}^2\}$

$$H_T(\hat{\gamma}) = -\frac{1}{\hat{\sigma}^2} \frac{1}{T} \sum_{t=1}^T x_t x_t', \quad J_T(\hat{\gamma}) = \frac{1}{\hat{\sigma}^4} \left(\hat{\Gamma}_0 + \sum_{i=1}^P w_i (\hat{\Gamma}_i + \hat{\Gamma}_i') \right), \quad (9.56)$$

where

$$\hat{\Gamma}_i = \frac{1}{T} \sum_{t=i+1}^T \hat{u}_t \hat{u}_{t-i} x_t x_{t-i}', \quad i = 0, 1, 2, \dots, \quad (9.57)$$

are sample autocovariance matrices, $\hat{u}_t = y_t - x_t' \hat{\beta}$ represents the least squares residuals with variance $\hat{\sigma}^2 = T^{-1} \sum_{t=1}^T \hat{u}_t^2$. Substituting (9.56) in (9.55) gives the estimator of the asymptotic covariance matrix

$$\hat{\Omega} = \left[\frac{1}{T} \sum_{t=1}^T x_t x_t' \right]^{-1} \left[\hat{\Gamma}_0 + \sum_{i=1}^P w_i (\hat{\Gamma}_i + \hat{\Gamma}_i') \right] \left[\frac{1}{T} \sum_{t=1}^T x_t x_t' \right]^{-1}, \quad (9.58)$$

commonly known as the Newey-West estimator. The value of the truncation parameter P and the choice of weights w_t , in (9.58) are determined using Table 9.1.

Standard errors for individual ordinary least squares coefficients are computed as the square roots of the diagonal elements of the normalised covariance matrix

$$\frac{1}{T} \hat{\Omega} = \frac{1}{T} H_T^{-1}(\hat{\gamma}) J_T(\hat{\gamma}) H_T^{-1}(\hat{\gamma}). \quad (9.59)$$

Example 9.20 Foreign Exchange Market Efficiency

From the discussion of Example 9.18, a model of forward market efficiency is based on the linear regression equation

$$y_t = \beta_1 + \beta_2 y_{t-h} + u_t,$$

where $y_{t+h} = s_{t+h} - f_{t,h}$ is the spread between the spot rate at $t+h$, s_{t+h} , and the h -period forward contract at t , $f_{t,h}$. The quasi-maximum likelihood estimates of the regression are computed using monthly data on the spot and 3-month forward exchange rates ($h = 3$) between the U.S. dollar and British pound from April 1979 to August 2011. The data are plotted in Figure 9.2 where 3 observations are lost from computing the spread which results in a sample of size $T = 386$.

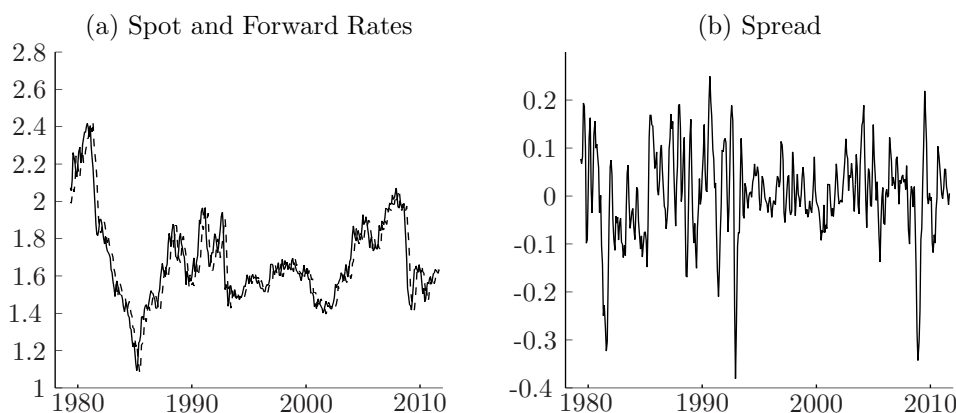


Figure 9.2 Monthly data on the spot (solid line) and 3-month forward (dashed line) exchange rates between the U.S. dollar and British pound, April 1979 to August 2011.

The least squares estimates are $\hat{\beta}_1 = 0.0026$ and $\hat{\beta}_2 = 0.1850$. The first 6 autocorrelations of the residuals $\hat{u}_t = y_t - \hat{\beta}_1 - \hat{\beta}_2 y_{t-3}$ and the product $\hat{u}_t y_{t-3}$ reveal evidence of autocorrelation at lags 1 and 2. Using the critical value $2/\sqrt{T} = 0.102$, these autocorrelations are also statistically significant while autocorrelations at higher lags are not.

Lag	1	2	3	4	5	6
$\hat{u}_t$	0.747	0.337	0.006	-0.061	-0.070	-0.077
$\hat{u}_t y_{t-3}$	0.600	0.138	-0.074	-0.087	-0.089	-0.061

To calculate the Newey-West standard errors for $\hat{\beta}_1$ and $\hat{\beta}_2$ using Bartlett

weights, from Table 9.1 the preliminary value of the maximum lag in (9.58) is $P = \text{int} \left[4(T/100)^{2/9} \right] = 5$. Now

$$\begin{aligned}\hat{J}_0 &= \hat{\Gamma}_0 + \sum_{i=1}^2 (\hat{\Gamma}_i + \hat{\Gamma}'_i) = \begin{bmatrix} 9.377 & -0.339 \\ -0.339 & 0.080 \end{bmatrix}, \\ \hat{J}_1 &= 2 \sum_{i=1}^2 i \hat{\Gamma}_i = \begin{bmatrix} 5.436 & 0.400 \\ -1.542 & -0.011 \end{bmatrix},\end{aligned}$$

and

$$\begin{aligned}\hat{\nu}_0 &= \begin{bmatrix} 1 \\ 1 \end{bmatrix}' \begin{bmatrix} 9.377 & -0.339 \\ -0.339 & 0.080 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 0.080 \\ \hat{\nu}_1 &= \begin{bmatrix} 1 \\ 1 \end{bmatrix}' \begin{bmatrix} 5.436 & 0.400 \\ -1.542 & -0.011 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = -0.011.\end{aligned}$$

result in an updated value of the lag length given by

$$P = \text{int} \left[1.1447 \left(\frac{\hat{\nu}_1^2}{\hat{\nu}_0^2} T \right)^{1/3} \right] = \text{int} \left[1.1447 \left(\frac{0.0111^2}{0.0802^2} \times 386 \right)^{1/3} \right] = 2.$$

The Newey-West standard errors based on $P = 2$ are $\text{se}(\hat{\beta}_1) = 0.007$ and $\text{se}(\hat{\beta}_2) = 0.084$. Given the least squares point estimates, it is concluded that $\hat{\beta}_2$ is significant at the 5% level but $\hat{\beta}_1$ is not. The conclusion that $\hat{\beta}_2$ is significant represents a contradiction of the efficient market hypothesis. $\square$

9.6 Testing

If the quasi-maximum likelihood estimator is consistent, then the Wald and LM test statistics discussed in Chapter 4 are made robust to model misspecification by using the quasi-maximum likelihood covariance matrix $\hat{\Omega}$ in (9.22). In regression models, the choice of (9.25) will correct the test for heteroskedasticity, while using (9.28) will correct the test for both heteroskedasticity and autocorrelation.

Consider testing the hypotheses

$$H_0 : \theta = \theta_0, \quad H_1 : \theta \neq \theta_0,$$

where θ_0 represents the population parameter under the null hypothesis. A robust version of the Wald test is

$$W = T[\hat{\theta} - \theta_0]' \hat{\Omega}_{1,1}^{-1} [\hat{\theta} - \theta_0], \quad (9.60)$$

where $\hat{\theta}$ is the quasi-maximum likelihood estimator without imposing the

constraints and $\widehat{\Omega}_{1,1}$ is the sub-matrix of $\widehat{\Omega}$ corresponding to θ , which is evaluated at $\widehat{\theta}$.

A robust LM test is found by first defining the restricted version of the quasi-maximum likelihood estimator in (9.4) to be $\widehat{\gamma}_0 = \{\theta_0, \widehat{\lambda}_0\}$ where

$$\widehat{\lambda}_0 = \arg \max_{\lambda} \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta_0, \lambda),$$

represents the restricted estimator of λ under the null hypothesis. The LM statistic is computed as

$$LM = TG'_T(\widehat{\gamma}_0) \widehat{\Omega}_0 G_T(\widehat{\gamma}_0), \quad (9.61)$$

where

$$G_T(\widehat{\gamma}_0) = \frac{1}{T} \sum_{t=1}^T g_t(\widehat{\gamma}_0),$$

and $\widehat{\Omega}_0$ is computed as in (9.22) but with $g_t(\widehat{\gamma})$ and $h_t(\widehat{\gamma})$ replaced by $g_t(\widehat{\gamma}_0)$ and $h_t(\widehat{\gamma}_0)$. The asymptotic null distribution of both statistics is the same as for their likelihood versions, that is $W, LM \xrightarrow{d} \chi_p^2$ where p is the dimension of θ , and both tests reject for large values of the statistic.

No similar robust LR statistic is available. The robust analogues of the Wald and LM tests have correct size under misspecification, whereas the LR test does not.

Example 9.21 Test of Foreign Exchange Market Efficiency

From Example 9.20 a joint test of the market efficiency is based on the null hypothesis

$$H_0 : \alpha = \beta = 0.$$

Using the calculated value of $\widehat{\Omega}$ in Example 9.20 which is based on Newey-West covariance matrix with $P = 2$ lags, the Wald test statistic is

$$\begin{aligned} W &= T \begin{bmatrix} \widehat{\beta}_1 \\ \widehat{\beta}_2 \end{bmatrix}' \begin{bmatrix} \widehat{\Omega}_{1,1} & \widehat{\Omega}_{1,2} \\ \widehat{\Omega}_{2,1} & \widehat{\Omega}_{2,2} \end{bmatrix}^{-1} \begin{bmatrix} \widehat{\beta}_1 \\ \widehat{\beta}_2 \end{bmatrix} \\ &= 386 \begin{bmatrix} 0.0026 \\ 0.1850 \end{bmatrix}' \begin{bmatrix} 0.0189 & -0.0612 \\ -0.0612 & 2.6970 \end{bmatrix}^{-1} \begin{bmatrix} 0.0026 \\ 0.1850 \end{bmatrix} \\ &= 5.927. \end{aligned}$$

The p value is 0.052, computed from a χ_2^2 distribution, shows that the null hypothesis of foreign exchange market efficiency is not rejected at the 5%

level of significance. As the test is based on the Newey-West covariance matrix, this joint test is robust to both autocorrelation and heteroskedasticity.

9.7 Applications

The first application is a simulation study that evaluates the properties of two quasi-maximum likelihood estimators when estimating a dynamic model for count data. Count data are also discussed in Chapter 21. The second application continues the theme of exploring estimation issues in time series models of the interest rate, previously discussed in Chapters 1 and 3.

9.7.1 Autoregressive Models for Count Data

Let $y_1, y_2, \dots, y_T$ represent a time series of count data where the true conditional distribution is negative binomial

$$f_0(y_t | y_{t-1}; \alpha, \beta) = \frac{\Gamma(y + \mu_t)}{\Gamma(y + 1) \Gamma(\mu_t)} (1 - p)^{\mu_t} p^y, \quad (9.62)$$

with μ_t and p specified as

$$\mu_t = \alpha + \beta y_{t-1}, \quad p = 0.5. \quad (9.63)$$

From the properties of the negative binomial distribution the conditional mean and variance are, respectively,

$$E[y_t | y_{t-1}] = \mu_t \frac{p}{1 - p} = \mu_t = \alpha + \beta y_{t-1} \quad (9.64)$$

$$\text{var}(y_t | y_{t-1}) = \mu_t \frac{p}{(1 - p)^2} = 2\mu_t = 2(\alpha + \beta y_{t-1}). \quad (9.65)$$

This model is characterized by the conditional variance being twice the size of the conditional mean as

$$\frac{\text{var}(y_t | y_{t-1})}{E[y_t | y_{t-1}]} = \frac{1}{1 - p} = 2.$$

Suppose that the model specified is the Poisson distribution

$$f(y_t | y_{t-1}; \alpha, \beta) = \frac{\mu_t^{y_t} \exp[-\mu_t]}{y_t!}, \quad (9.66)$$

where μ_t is defined in (9.63). The parameters of the true and misspecified models are common with $\theta = \{\alpha, \beta\}$. The Poisson distribution represents a misspecification of the true distribution as the mean and the variance are

restricted to be equal, compared to the true process which exhibits overdispersion. The quasi-maximum likelihood estimator is the solution of

$$\hat{\theta} = \arg \max_{\theta} \frac{1}{T} \sum_{t=1}^T \ln f(y_t | y_{t-1}; \theta). \quad (9.67)$$

The first and second derivatives at observation t are

$$g_t(\theta) = \frac{y_t - \mu_t}{\mu_t} \frac{\partial \mu_t}{\partial \theta} = \frac{y_t - \mu_t}{\mu_t} \begin{bmatrix} 1 \\ y_{t-1} \end{bmatrix} \quad (9.68)$$

$$h_t(\theta) = -\frac{y_t}{\mu_t^2} \frac{\partial \mu_t}{\partial \theta} \frac{\partial \mu_t}{\partial \theta'} = -\frac{y_t}{\mu_t^2} \begin{bmatrix} 1 & y_{t-1} \\ y_{t-1} & y_{t-1}^2 \end{bmatrix}. \quad (9.69)$$

Recognizing that $E_{t-1}[y_t] = \mu_t$ in (9.64), it follows that the gradient in (9.68) is a martingale difference sequence since

$$E_{t-1}[g_t] = E_{t-1} \left[\frac{y_t - \mu_t}{\mu_t} \frac{\partial \mu_t}{\partial \theta} \right] = \frac{E_{t-1}[y_t] - \mu_t}{\mu_t} \frac{\partial \mu_t}{\partial \theta} = 0.$$

By using the law of iterated expectations, the unconditional expectation under the true model is given by

$$E[g_t] = E[E_{t-1}[g_t]] = 0,$$

which is required for the quasi-maximum likelihood estimator to be consistent.

Taking conditional expectations of the Hessian in (9.69) gives

$$E_{t-1}[h_t] = E_{t-1} \left[-\frac{y_t}{\mu_t^2} \frac{\partial \mu_t}{\partial \theta} \frac{\partial \mu_t}{\partial \theta'} \right] = -\frac{E_{t-1}[y_t]}{\mu_t^2} \frac{\partial \mu_t}{\partial \theta} \frac{\partial \mu_t}{\partial \theta'} = -\frac{1}{\mu_t} \frac{\partial \mu_t}{\partial \theta} \frac{\partial \mu_t}{\partial \theta'} \quad (9.70)$$

since $E_{t-1}[y_t] = \mu_t$. Similarly, taking conditional expectations of the outer product of gradients matrix yields

$$E_{t-1}[g_t g_t'] = E_{t-1} \left[\left(\frac{y_t - \mu_t}{\mu_t} \right)^2 \frac{\partial \mu_t}{\partial \theta} \frac{\partial \mu_t}{\partial \theta'} \right] = \left[\frac{2}{\mu_t} \frac{\partial \mu_t}{\partial \theta} \frac{\partial \mu_t}{\partial \theta'} \right] \quad (9.71)$$

since $E_{t-1}[(y_t - \mu_t)^2] = \text{var}(y_t | y_{t-1}) = 2\mu_t$ from (9.65). By using the law of iterated expectations, the unconditional expectations under the true model of (9.70) and (9.71) are, respectively,

$$\begin{aligned} H(\theta_0) &= E[E_{t-1}[h_t]] = E \left[-\frac{1}{\mu_t} \frac{\partial \mu_t}{\partial \theta} \frac{\partial \mu_t}{\partial \theta'} \right] = E \left[-\frac{1}{\mu_t} \begin{bmatrix} 1 & y_{t-1} \\ y_{t-1} & y_{t-1}^2 \end{bmatrix} \right] \\ J(\theta_0) &= E[E_{t-1}[g_t g_t']] = E \left[\frac{2}{\mu_t} \frac{\partial \mu_t}{\partial \theta} \frac{\partial \mu_t}{\partial \theta'} \right] = E \left[\frac{2}{\mu_t} \begin{bmatrix} 1 & y_{t-1} \\ y_{t-1} & y_{t-1}^2 \end{bmatrix} \right]. \end{aligned}$$

These results verify that the information equality does not hold as $J(\theta_0) \neq -H(\theta_0)$. The asymptotic distribution of $\hat{\theta}$ is, therefore,

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \Omega), \quad (9.72)$$

where

$$\Omega = H^{-1}(\theta_0)J(\theta_0)H^{-1}(\theta_0) = 2\left[E\left[\frac{1}{\mu_t}\frac{\partial\mu_t}{\partial\theta}\frac{\partial\mu_t}{\partial\theta'}\right]\right]^{-1}.$$

Consistent estimators of $H(\theta_0)$ and $J(\theta_0)$ are obtained by approximating the expectations operator with the sample average and evaluating these terms at the quasi-maximum likelihood estimator $\hat{\theta}$

$$H_T(\hat{\theta}) = -\frac{1}{T}\sum_{t=1}^T \frac{1}{(\hat{\alpha} + \hat{\beta}y_{t-1})} \begin{bmatrix} 1 & y_{t-1} \\ y_{t-1} & y_{t-1}^2 \end{bmatrix}, \quad (9.73)$$

$$J_T(\hat{\theta}) = \frac{2}{T}\sum_{t=1}^T \frac{1}{(\hat{\alpha} + \hat{\beta}y_{t-1})} \begin{bmatrix} 1 & y_{t-1} \\ y_{t-1} & y_{t-1}^2 \end{bmatrix},$$

where the form of the outer product of gradients matrix in (9.24) is applicable for $J_T(\hat{\theta})$ because g_t is a martingale difference sequence and there is no need to adjust for autocorrelation. The quasi-maximum likelihood covariance matrix is, therefore, consistently estimated by

$$\hat{\Omega} = H_T^{-1}(\hat{\theta})J_T(\hat{\theta})H_T^{-1}(\hat{\theta}). \quad (9.74)$$

The square roots of the diagonal elements of $\hat{\Omega}_T/T$ provide standard errors for the individual coefficients.

To investigate the sampling properties of the quasi-maximum likelihood estimator in the case where the true model is a negative binomial distribution and the misspecified model is a Poisson distribution, Table 9.3 provides statistics on the sampling distribution of the quasi-maximum likelihood estimator of the autoregressive parameter β from a Monte Carlo experiment using 10000 replications. The true parameter values are $\theta_0 = \{\alpha_0 = 1, \beta_0 = 0.5\}$. For comparison the results of using a quasi log-likelihood function based on the assumption of a normal distribution are also presented. Both quasi-maximum likelihood estimators behave like consistent estimators. The means of the two sampling distributions approach the true value of $\beta_0 = 0.5$ and the variances both approach zero, roughly halving as the sample size doubles.

Also reported in Table 9.3 is the tail probability from the simulations of

Table 9.3

Sampling properties of the quasi-maximum likelihood estimator of β , where the true distribution is negative binomial and the misspecified distribution is Poisson. Based on 10,000 Monte Carlo replications where the true parameters in (9.63) are $\alpha_0 = 1$ and $\beta_0 = 0.5$. The estimate of the standard error is based on the covariance matrix in (9.74) for the Poisson quasi log-likelihood function and (9.52) for the normal quasi log-likelihood function.

T	mean $\hat{\beta}$		var $\hat{\beta}$		$\Pr \left(\left (\hat{\beta} - \beta_0) / se(\hat{\beta}) \right > 1.96 \right)$	
	Poisson	Normal	Poisson	Normal	Poisson	Normal
25	0.389	0.357	0.046	0.046	0.159	0.197
50	0.441	0.421	0.024	0.024	0.105	0.142
100	0.471	0.459	0.012	0.013	0.078	0.113
200	0.486	0.478	0.006	0.007	0.069	0.090
400	0.493	0.489	0.003	0.004	0.061	0.075
800	0.496	0.494	0.002	0.002	0.055	0.065
1600	0.498	0.497	0.001	0.001	0.051	0.056

the t-statistic $(\hat{\beta} - \beta_0) / se(\hat{\beta})$. The standard error $se(\hat{\beta})$ for the quasi log-likelihood function based on the Poisson distribution is based on the square root of the diagonal element of (9.74). For the quasi log-likelihood function based on the assumption of a normal distribution, the standard error is based on the White variance estimator in (9.52). The results demonstrate that the quasi-maximum likelihood variance estimators work well for both quasi-maximum likelihood estimators, with the asymptotic normal approximation resulting in tail probabilities approaching 0.05 as the sample size increases.

9.7.2 Estimating the Parameters of the CKLS Model

The CKLS model of Chan, Karolyi, Longstaff and Saunders (1992) specify the interest rate r_t , to evolve as

$$r_t - r_{t-1} = \alpha(\mu - r_{t-1})\Delta + \Delta^{1/2}\sigma r_{t-1}^\gamma z_t, \quad (9.75)$$

where z_t is an *iid* disturbance distributed as $N(0, 1)$. The unknown parameters are $\theta = \{\alpha, \mu, \sigma, \gamma\}$ where α is the adjustment parameter, μ is the mean interest rate, σ is a volatility parameter and γ represents the levels-effect parameter. The scale factor Δ corresponds to the frequency of the data which is used to annualize the parameters of the model. Imposing the constraint

$\gamma = 0.5$ corresponds to the square-root diffusion or CIR model (Cox, Ingersoll and Ross, 1985) in the case of the continuous time version of (9.75) discussed in Section 3.8 of Chapter 3. An important feature of the CIR model is that the conditional distribution of the interest rate is nonnormal, having a non-central chi-square distribution. This observation suggests that even for the discrete version of the model in (9.75), the normality assumption of z_t is a misspecification of the log-likelihood function.

The quasi log-likelihood function is

$$\ln L_T(\theta) = \frac{1}{T-1} \sum_{t=2}^T \ln f(r_t|r_{t-1}; \theta), \quad (9.76)$$

where the conditional distribution is

$$f(r_t|r_{t-1}; \theta) = \frac{1}{\Delta^{1/2} \sigma r_{t-1}^\gamma \sqrt{2\pi}} \exp \left[-\frac{(r_t - r_{t-1} - \alpha(\mu - r_{t-1})\Delta)^2}{2\Delta\sigma^2 r_{t-1}^{2\gamma}} \right]. \quad (9.77)$$

The gradients at time t are

$$\begin{aligned} g_{1,t} &= \frac{\partial \ln f(r_t|r_{t-1}; \theta)}{\partial \alpha} = \frac{u_t(\mu - r_{t-1})}{\sigma^2 r_{t-1}^{2\gamma}} \\ g_{2,t} &= \frac{\partial \ln f(r_t|r_{t-1}; \theta)}{\partial \mu} = \frac{u_t \alpha}{\sigma^2 r_{t-1}^{2\gamma}} \\ g_{3,t} &= \frac{\partial \ln f(r_t|r_{t-1}; \theta)}{\partial \sigma^2} = \left(\frac{u_t^2}{\Delta \sigma^2 r_{t-1}^{2\gamma}} - 1 \right) \frac{1}{2\sigma^2} \\ g_{4,t} &= \frac{\partial \ln f(r_t|r_{t-1}; \theta)}{\partial \gamma} = \left(\frac{u_t^2}{\Delta \sigma^2 r_{t-1}^{2\gamma}} - 1 \right) \ln r_{t-1}, \end{aligned} \quad (9.78)$$

where the disturbance term u_t , is given by

$$u_t = r_t - r_{t-1} - \alpha(\mu - r_{t-1})\Delta.$$

The quasi-maximum likelihood estimator is obtained by maximizing (9.76). As the gradients in (9.78) are nonlinear in the parameters, the BFGS algorithm is adopted.

As in Sections 1.5 and 3.8, the data for this application is the daily annualized 7-day Eurodollar interest rate used by Ait-Sahalia (1996) for the period 1 June 1973 to 25 February 1995, $T = 5505$ observations. The scale parameter in (9.75) is set at $\Delta = 1/250$ as the frequency of the data is daily and the interest rates are annualized. The method described in Section 3.8.2 is used to obtain starting values for α , μ and σ , and the starting value of γ

is set equal to 1. The quasi-maximum likelihood estimates are

$$\hat{\alpha} = \begin{matrix} 0.300 \\ (0.303) \end{matrix}, \quad \hat{\mu} = \begin{matrix} 0.115 \\ (0.069) \end{matrix}, \quad \hat{\sigma} = \begin{matrix} 1.344 \\ (0.194) \end{matrix}, \quad \hat{\gamma} = \begin{matrix} 1.311 \\ (0.059) \end{matrix},$$

with quasi-maximum likelihood standard errors based on White's estimator given in parentheses. In this model, the gradients are martingale difference sequences so the White estimator rather than the Newey-West estimator is the appropriate one.

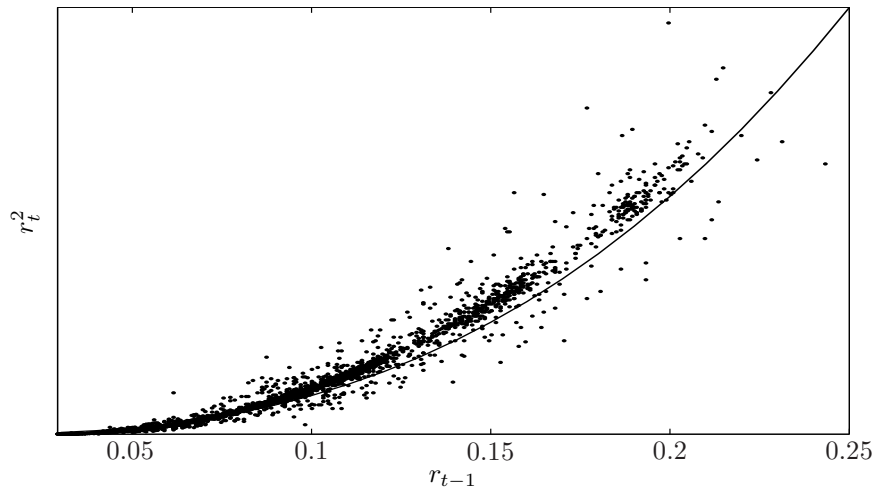


Figure 9.3 Scatter plot of r_t^2 on r_{t-1} together with the model predicted value, $\hat{\sigma}^2 r_t^{2\hat{\gamma}}$ (solid line).

Interestingly, the estimate of α , the speed of adjustment parameter, is now broadly as expected, unlike the estimate returned by the CIR model in Section 3.8.2. The reason for this is that diffusion function of the CKLS model is more dynamic than that of the CIR model and thus more capable of capturing the volatility of the short-term interest rate. The estimates of the drift parameters are, therefore, not contaminated by trying to compensate for a specification of diffusion that is far too restrictive. Note that the estimate of γ is 1.311, which is much larger than the value of 0.5 imposed by the CIR model. The estimated standard error of γ suggests that its value is significantly different from 0.5, thereby providing support for the contention that this parameter should be estimated rather than imposed.

Figure 9.3 shows a scatter plot of r_t^2 on r_{t-1} and superimposes on it the predicted value in terms of the model $\hat{\sigma}^2 r_{t-1}$. Comparison of this figure with Figure 3.2 in Chapter 3 demonstrates that the CKLS model is much better

able to capture the behaviour of the actual scatter of interest rates as the level of the interest rate increases. This added flexibility is largely due to the influence of γ .

9.8 Exercises

(1) Graphical Analysis

Gauss file(s)	qmle_graph.g
Matlab file(s)	qmle_graph.m

- (a) Simulate $T = 10$ observations from an exponential distribution with parameter $\theta_0 = \mu_0 = 1$. for $\theta = \{0.1, 0.2, \dots, 3\}$, plot the following log-likelihood functions

$$\begin{aligned} \text{Exponential: } \ln L_T(\theta) &= -\ln \theta - \frac{1}{\theta} \frac{1}{T} \sum_{t=1}^T y_t \\ \text{Normal: } \ln L_T(\theta) &= -\frac{1}{2} \ln 2\pi - \frac{1}{2} \sum_{t=1}^T (y_t - \theta)^2. \end{aligned}$$

Comment on the results.

- (b) Simulate $T = 10$ observations from a negative binomial distribution with parameters $\theta_0 = \mu_0 = 5$ and $p = 0.5$. For $\theta = \{0.1, 0.2, \dots, 10\}$, plot the following log-likelihood functions

$$\begin{aligned} \text{Neg. bin: } \ln L_T(\theta) &= \frac{1}{T} \sum_{t=1}^T \frac{\ln \Gamma(y_t + \theta)}{\Gamma(y_t + 1) \Gamma(\theta)} + \theta \ln(1 - p) + \frac{1}{T} \sum_{t=1}^T y_t \ln p \\ \text{Poisson: } \ln L_T(\theta) &= \frac{1}{T} \sum_{t=1}^T y_t \ln \theta - \theta - \sum_{t=1}^T \ln y_t!. \end{aligned}$$

Comment on the results.

- (c) Simulate $T = 10$ observations from a Standardized Student distribution with parameters $\theta_0 = \mu_0 = 5$, $\sigma^2 = 1$ and degrees of freedom $\nu = 10$. for $\theta = \{0.1, 0.2, \dots, 3\}$, plot the following log-likelihood functions

$$\begin{aligned} \text{Stud. t: } \ln L_T(\theta) &= \ln \frac{\Gamma((\nu + 1)/2)}{\sqrt{\pi(\nu - 2)} \Gamma(\nu/2)} - \frac{1}{2} \ln \sigma^2 \\ &\quad - \frac{\nu + 1}{2} \sum_{t=1}^T \ln \left(1 + \frac{(y_t - \theta)^2}{\sigma^2(\nu + 2)} \right) \\ \text{Normal: } \ln L_T(\theta) &= -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{1}{2} \sum_{t=1}^T \frac{(y_t - \theta)^2}{\sigma^2}. \end{aligned}$$

Comment on the results.

- (d) Simulate $T = 10$ observations from a Poisson distribution with parameter $\theta_0 = \mu_0 = 5$. For $\theta = \{0.1, 0.2, \dots, 10\}$, plot the following log-likelihood functions

$$\begin{aligned} \text{Poisson:} \quad \ln L_T(\theta) &= \frac{1}{T} \sum_{t=1}^T y_t \ln \theta - \theta - \sum_{t=1}^T \ln y_t! \\ \text{Normal:} \quad & \ln L_T(\theta) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \theta - \frac{1}{2} \sum_{t=1}^T \frac{(y_t - \theta)^2}{\theta}. \end{aligned}$$

Comment on the results.

(2) Bernoulli Distribution

Gauss file(s)	<code>qmle_bernoulli.g</code>
Matlab file(s)	<code>qmle_bernoulli.m</code>

Suppose that the true and misspecified models are, respectively,

$$\begin{aligned} f_0(y_t; \theta_0) &= \theta_0^{y_t} (1 - \theta_0)^{1-y_t} \\ f(y_t; \theta) &= \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{(y_t - \theta)^2}{2} \right]. \end{aligned}$$

- (a) Using the true model find:
- (i) the maximum likelihood estimator $\hat{\theta}_0$;
 - (ii) the variance based on the information matrix; and
 - (iii) the variance based on the outer product of the gradients matrix.
- (b) Using the misspecified model find:
- (i) the maximum likelihood estimator $\hat{\theta}$;
 - (ii) the variance based on the information matrix;
 - (iii) the variance based on the outer product of gradients matrix; and
 - (iv) the quasi-maximum likelihood variance based on the assumption of independence.
- (c) Compare the variance expressions based on the true model in part (a) with the variance expressions obtained using the misspecified model.
- (d) Generate a sample of size $T = 100$ from the Bernoulli distribution. Repeat parts (a) to (c) using the simulated data.

(3) **Poisson Distribution**

Gauss file(s)	qmle_poisson.g
Matlab file(s)	qmle_poisson.m

Suppose that the true and misspecified models are, respectively,

$$f_0(y_t; \theta_0) = \frac{\theta_0^{y_t} \exp[-\theta_0]}{y_t!}$$

$$f(y_t; \theta) = \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{(y_t - \theta)^2}{2} \right].$$

- (a) Using the true model find:
 - (i) the maximum likelihood estimator $\hat{\theta}_0$;
 - (ii) the variance based on the information matrix; and
 - (iii) the variance based on the outer product of the gradients matrix.
- (b) Using the misspecified model find:
 - (i) the maximum likelihood estimator $\hat{\theta}$;
 - (ii) the variance based on the information matrix;
 - (iii) the variance based on the outer product of gradients matrix; and
 - (iv) the quasi-maximum likelihood variance based on the assumption of independence.
- (c) Compare the variance expressions based on the true model in part (a) with the variance expressions obtained using the misspecified model.
- (d) Generate a sample of size $T = 100$ from the Poisson distribution. Repeat parts (a) to (c) using the simulated data.

(4) **Duration Analysis**

Gauss file(s)	qmle_exponential.g
Matlab file(s)	qmle_exponential.m

The true model of duration times between significant events follows an exponential distribution with parameter θ_0 . The misspecified model posits that the durations are normally distributed with mean θ and unit variance

$$f_0(y_t; \theta_0) = \frac{1}{\theta_0} \exp \left[-\frac{y_t}{\theta_0} \right]$$

$$f(y_t; \theta) = \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{(y_t - \theta)^2}{2} \right].$$

- (a) Simulate $T = 500000$ observations on y_t drawn from the exponential distribution.
- (b) Compute the maximum likelihood estimate $\hat{\theta}$ based on the misspecified model.
- (c) Derive an expression for the Hessian, $H(\theta)$, and use the data on y_t to compute $H(\hat{\theta})$.
- (d) Derive an expression for the outer product of gradients matrix, $J(\theta)$, assuming independence and use the data on y_t to compute $J(\hat{\theta})$.

(5) **Effect of Misspecifying the Log-likelihood Function**

Gauss file(s)	qmle_student1.g
Matlab file(s)	qmle_student1.m

Suppose the true and misspecified models are

$$f_0(y_t; \theta_0) = \frac{\Gamma\left(\frac{\gamma_0 + 1}{2}\right)}{\sqrt{\pi\sigma_0^2(\gamma_0 - 2)}\Gamma\left(\frac{\gamma_0}{2}\right)} \left(1 + \frac{(y_t - \mu_0)^2}{\sigma_0^2(\gamma_0 - 2)}\right)^{-(\gamma_0 + 1)/2}$$

$$f(y_t; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_t - \mu)^2}{2\sigma^2}\right],$$

with parameters $\theta_0 = \{\mu_0, \sigma_0^2, \gamma_0\}$ and $\theta = \{\mu, \sigma^2\}$, respectively.

- (a) Simulate $T = 500000$ observations of y_t drawn from the Student t distribution with parameters $\{\mu_0 = 1, \sigma_0^2 = 1\}$ for values of $\gamma_0 = \{5, 10, 20, 30, 100, 500\}$.
- (b) In each case, compute the maximum likelihood estimators of $\theta = \{\mu, \sigma^2\}$ using the misspecified log-likelihood function.
- (c) For each choice of γ_0 and associated estimators $\hat{\theta} = \{\hat{\mu}, \hat{\sigma}^2\}$, tabulate the elements of the information matrix, the elements of the outer product of the gradients matrix and the differences between them. Comment on your results.

(6) **Student t and Normal Distributions**

Gauss file(s)	qmle_student2.g
Matlab file(s)	qmle_student2.m

Let the true distribution be the standardized Student t distribution with parameters $\theta_0 = \{\mu_0 = 10, \sigma_0^2 = 1, \gamma_0 = 5\}$ and the misspecified model be normal with parameters $\theta = \{\mu, \sigma^2\}$.

- (a) Simulate a sample of size $T = 500000$ and compute the covariance matrix of the parameters of the true model based on the Hessian, outer product of gradients matrix and quasi-maximum likelihood estimator. Comment on your results.
- (b) Compute the covariance matrix of the parameters of the misspecified model. Comment on your results.

(7) Linear Regression Model

Gauss file(s)	qmle_ols.g
Matlab file(s)	qmle_ols.m

Consider the linear regression model

$$y_t = \beta_1 + \beta_2 x_t + u_t, \quad u_t \sim N(0, \sigma^2),$$

with parameters $\theta = \{\beta_1, \beta_2, \sigma^2\}$. Data on y_t and x_t are $y_t = \{3, 2, 4, 6, 10\}$ and $x_t = \{1, 2, 3, 4, 5\}$.

- (a) Compute the maximum likelihood parameter estimates $\hat{\theta}$.
- (b) Compute the covariance matrix of the mean parameters $\hat{\beta}_1$ and $\hat{\beta}_2$ using the Hessian, the outer product of gradients matrix assuming independence and the outer product of gradients matrix assuming dependence. Compare these estimates of the covariance matrix.
- (c) Now suppose the data on y_t and x_t are $y_t = \{3, 2, 4, 6, 5\}$ and $x_t = \{2, 4, 4, 2, 4\}$. Repeat parts (a) and (b).

(8) Simulation Experiment

Gauss file(s)	qmle_ols_simul.g
Matlab file(s)	qmle_ols_simul.m

Simulate $T = 200$ observations from the heteroskedastic and autocorrelated linear regression model

$$\begin{aligned} y_t &= \beta_1 + \beta_2 x_t + u_t \\ u_t &= \rho_1 u_{t-1} + v_t \\ \sigma_t^2 &= \exp(\gamma_1 + \gamma_2 x_t) \\ v_t &\sim N(0, \sigma_t^2), \end{aligned}$$

using the parameter values $\beta_1 = 1$, $\beta_2 = 0.2$, $\rho_1 = 0.2$, $\gamma_1 = 1$ and $\gamma_2 = 0.2$.

- (a) Compute the maximum likelihood parameter estimates $\hat{\theta} = \{\hat{\beta}_1, \hat{\beta}_2, \hat{\sigma}^2\}$ of the misspecified model

$$y_t = \beta_1 + \beta_2 x_t + u_t, \quad u_t \sim N(0, \sigma^2).$$

- (b) Compute the covariance matrix of the mean parameters $\hat{\beta}_1$ and $\hat{\beta}_2$ using the Hessian, the outer product of gradients matrix assuming independence and the outer product of gradients matrix assuming dependence. Compare these estimates of the covariance matrix.
- (c) Adopt a nonparametric solution to correct for misspecification by computing a quasi-maximum likelihood covariance matrix based on either White or Newey-West.
- (d) Estimate the correctly specified model and recompute the covariance matrices in parts (b) and (c). Comment on your results.

(9) **U.S. Investment**

Gauss file(s)	qmle_invest.g, investment.xls
Matlab file(s)	qmle_invest.m, investment.mat

Consider the dynamic investment model

$$\ln ri_t = \beta_0 + \beta_1 \ln ry_t + \beta_2 rint_t + u_t, \quad u_t \sim N(0, \sigma^2),$$

where ri_t is real investment, ry_t is real income and $rint_t$ is the real interest rate expressed as a percentage. The data are U.S. quarterly data beginning in June 1957 and ending just prior to the start of the global financial crisis in June 2007, $T = 201$.

- (a) Compute the maximum likelihood parameter $\hat{\theta} = \{\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{\sigma}^2\}$.
- (b) Compute the covariance matrix of $\hat{\theta}$ using the Hessian, the outer product of gradients matrix assuming independence, the outer product of gradients matrix assuming dependence, the White estimator and the Newey-West estimator. Compare these estimates of the covariance matrix and interpret the results.

(10) **Estimating Parameters of the CKLS Model**

Gauss file(s)	qmle_ckls.g, eurod.dat
Matlab file(s)	qmle_ckls.m, eurod.mat

The data are the daily 7-day Eurodollar interest rates used by Aït-Sahalia (1996) for the period 1 June 1973 to 25 February 1995, $T = 5505$ observations, expressed in raw units.

- (a) The CKLS model of the short-term interest rate, r_t , is given by

$$dr = \alpha(\mu - r)dt + \sigma r^\gamma dW ,$$

where dW is the differential of the Wiener process $W(t)$. Estimate the parameters $\theta = \{\alpha, \mu, \sigma, \gamma\}$ of the model by quasi-maximum likelihood based on the assumption that the unknown transitional pdf required to construct the log-likelihood function is approximated in discrete time by the normal distribution

$$f(r_t|r_{t-1};\theta) = \frac{1}{\sigma r_{t-1}^\gamma \sqrt{2\pi} \Delta} \exp \left[- \frac{(r_t - r_{t-1} - \alpha(\mu - r_{t-1})\Delta)^2}{\sigma^2 r_{t-1}^{2\gamma}} \right] .$$

- (b) Compute the quasi-maximum likelihood standard errors.

10

Generalized Method of Moments

10.1 Introduction

The maximum likelihood method of estimation is based on specifying a likelihood function which, in turn, requires specifying a particular form for the joint distribution of the underlying random variables. This requirement is relaxed in this Chapter so that the model is based upon the specification of moments of certain functions of the random variables. This approach is known as generalized method of moments (GMM). In the case where the moments used in the GMM procedure correspond to the distribution specified in the maximum likelihood procedure, the two estimators are equivalent. In essence the choice between maximum likelihood and GMM then boils down to a trade-off between the statistical efficiency of a maximum likelihood estimator based on the full distribution against the ease of specification and robustness of a GMM estimator based only on certain moments.

The use of the GMM estimator is often a natural estimation framework in economics and finance as the moments of a model often correspond to the first-order conditions of a dynamic optimization problem. Moreover, as theory tends to provide little or even no guidance on the specification of the distribution, this means that to compute maximum likelihood estimators it is necessary to make potentially *ad hoc* assumptions about the underlying stochastic processes. This is not the case with using GMM. On the other hand, GMM estimation requires the construction of a sufficient number of moment conditions by choosing instruments that may not be directly related to the theoretical model.

Under general regularity conditions, GMM estimators are consistent and asymptotically normal. This class of estimators also has the additional advantage that the potential problems of misspecifying the likelihood function inherent with the maximum likelihood estimator discussed in the context

of quasi-maximum likelihood estimation in Chapter 9, are circumvented. As will be shown, however, the covariance matrix of the GMM and quasi-maximum likelihood estimators discussed in Chapter 9, are identical under certain conditions. The GMM family of estimators also nests a number of well known estimators, including the maximum likelihood estimator dealt with in Part ONE and the OLS and instrumental variable estimators discussed in Part TWO. A further property of GMM is that it forms the basis of many recently developed simulation estimation procedures, the subject matter of Chapter 12.

10.2 Motivating Examples

In this section various statistical and economic examples are used to motivate the GMM methodology. The GMM approach is based on specifying certain moments of the random variables, not necessarily their full distribution as the maximum likelihood estimator requires.

10.2.1 Population Moments

Given some random variables $y_1, y_2, \dots, y_T$ and a K dimensional parameter vector θ , a GMM model is specified in terms of an N dimensional vector of functions $m(y_t; \theta)$ such that the true value θ_0 satisfies

$$E[m(y_t; \theta_0)] = 0. \quad (10.1)$$

These N equations are called the population moment conditions, or population moments. These population moments define the GMM model.

The usual moments of a random variable can be written in the form of (10.1). Some examples are as follows.

- (1) If $E[y_t] = \theta_0$ for all t then

$$m(y_t; \theta) = y_t - \theta, \quad (10.2)$$

so that (10.1) is satisfied. In this case $K = N = 1$.

- (2) If $E[y_t] = \mu_0$ and $\text{var}[y_t] = \sigma_0^2$ for all t then $\theta = (\mu, \sigma^2)'$ and

$$m(y_t; \theta) = \begin{bmatrix} y_t - \mu \\ (y_t - \mu)^2 - \sigma^2 \end{bmatrix}, \quad (10.3)$$

so the (10.1) with $\theta_0 = (\mu_0, \sigma_0^2)'$ implies $E[y_t] = \mu_0$ and $E[(y_t - \mu_0)^2] = \sigma_0^2$, as required. In this case $K = N = 2$.

- (3) A GMM model may specify any number of uncentered moments, say $E[y_t^i] = \theta_{i,0}$ for $i = 1, 2, \dots, K$. Then $\theta = (\theta_1, \dots, \theta_K)'$ and

$$m(y_t; \theta) = \begin{bmatrix} y_t - \theta_1 \\ y_t^2 - \theta_2 \\ \vdots \\ y_t^K - \theta_K \end{bmatrix}. \quad (10.4)$$

- (4) Centered moments may be specified in a similar manner. In this case $E[y_t] = \theta_{1,0}$ as before, and then $E[(y_t - \theta_{1,0})^i] = \theta_{i,0}$ so that $\theta = (\theta_1, \dots, \theta_K)'$ and

$$m(y_t; \theta) = \begin{bmatrix} y_t - \theta_1 \\ (y_t - \theta_1)^2 - \theta_2 \\ \vdots \\ (y_t - \theta_1)^K - \theta_K \end{bmatrix}. \quad (10.5)$$

- (5) Other functions besides polynomials can also be used. For example, if $E[\ln y_t] = \theta_0$ then

$$m(y_t; \theta) = \ln y_t - \theta, \quad (10.6)$$

or if $E[1/y_t] = \theta_0$ then

$$m(y_t; \theta) = \frac{1}{y_t} - \theta. \quad (10.7)$$

10.2.2 Empirical Moments

The idea of GMM is to replace the population moments with their sample counterparts, the empirical moments. For any θ , the empirical moments have the general form

$$M_T(\theta) = \frac{1}{T} \sum_{t=1}^T m(y_t; \theta).$$

If $K = N$, as in all of the examples above, then the model is called exactly identified and the GMM estimator $\hat{\theta}$ is defined as the sample counterpart of (10.1), that is

$$M_T(\hat{\theta}) = 0. \quad (10.8)$$

If $N > K$ then the model is called over identified and an exact solution to (10.8) is not available because there are more equations than unknowns. GMM estimation for this case is discussed in section 10.3.

- (1) In order to estimate the mean $\theta_0 = E[y_t]$, the GMM model is specified by (10.2), so that

$$M_T(\theta) = \frac{1}{T} \sum_{t=1}^T (y_t - \theta) = \frac{1}{T} \sum_{t=1}^T y_t - \theta.$$

Solving (10.8) for $\hat{\theta}$ gives

$$\hat{\theta} = \frac{1}{T} \sum_{t=1}^T y_t.$$

That is, the application of the method of moments provides the sample mean as the estimator of the population mean.

- (2) The GMM model to estimate both mean and variance is specified by (10.3), so that

$$M_T(\theta) = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T y_t - \mu \\ \frac{1}{T} \sum_{t=1}^T (y_t - \mu)^2 - \sigma^2 \end{bmatrix},$$

and solving (10.8) gives the sample mean and sample variance respectively

$$\hat{\mu} = \frac{1}{T} \sum_{t=1}^T y_t, \quad \hat{\sigma}^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{\mu})^2. \quad (10.9)$$

- (3) The GMM model to estimate the first K uncentered moments is specified by (10.4), giving

$$M_T(\theta) = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T y_t - \theta_1 & \frac{1}{T} \sum_{t=1}^T y_t^2 - \theta_2 & \cdots & \frac{1}{T} \sum_{t=1}^T y_t^K - \theta_K \end{bmatrix}',$$

so solving (10.8) gives the usual sample uncentered moments of y_t

$$\hat{\theta}_i = \frac{1}{T} \sum_{t=1}^T y_t^i,$$

for $i = 1, 2, \dots, K$.

- (4) The GMM model to estimate the first K centered moments is specified by (10.5), giving

$$M_T(\theta) = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T y_t - \theta_1 & \frac{1}{T} \sum_{t=1}^T (y_t - \theta_1)^2 - \theta_2 & \cdots & \frac{1}{T} \sum_{t=1}^T (y_t - \theta_1)^K - \theta_K \end{bmatrix}',$$

and solving (10.8) gives

$$\hat{\theta}_1 = \frac{1}{T} \sum_{t=1}^T y_t, \quad \hat{\theta}_i = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{\theta}_1)^i, \quad i = 2, 3, \dots, K.$$

(5) If $E[\ln y_t] = \theta_0$ then (10.6) and (10.8) give

$$\hat{\theta} = \frac{1}{T} \sum_{t=1}^T \ln y_t.$$

If $E[1/y_t] = \theta_0$ then (10.7) and (10.8) give

$$\hat{\theta} = \frac{1}{T} \sum_{t=1}^T \frac{1}{y_t}.$$

Table 10.1 gives some examples of empirical moments for a data set consisting of $T = 10$ observations. It shows the computation of the first four sample uncentered moments of y_t , the first two centered moments and the means of $\ln y_t$ and $1/y_t$.

Table 10.1

Calculation of various empirical moments for the $T = 10$ observations on the variable y_t .

t	y_t	y_t^2	y_t^3	y_t^4	$y_t - m_1$	$(y_t - m_1)^2$	$\ln(y_t)$	$1/y_t$
1	2.000	4.000	8.000	16.000	-2.900	8.410	0.693	0.500
2	7.000	49.000	343.000	2401.000	2.100	4.410	1.946	0.143
3	5.000	25.000	125.000	625.000	0.100	0.010	1.609	0.200
4	6.000	36.000	216.000	1296.000	1.100	1.210	1.792	0.167
5	4.000	16.000	64.000	256.000	-0.900	0.810	1.386	0.250
6	8.000	64.000	512.000	4096.000	3.100	9.610	2.079	0.125
7	5.000	25.000	125.000	625.000	0.100	0.010	1.609	0.200
8	5.000	25.000	125.000	625.000	0.100	0.010	1.609	0.200
9	4.000	16.000	64.000	256.000	-0.900	0.810	1.386	0.250
10	3.000	9.000	27.000	81.000	-1.900	3.610	1.099	0.333
Moment:	4.900	26.900	160.900	1027.700	0.000	2.890	1.521	0.237

The approach to GMM model specification is different from that of maximum likelihood. A GMM model specifies only those aspects of the distribution of the random variables that are of interest. A maximum likelihood model specifies the entire joint distribution of the random variables, even if interest is confined only to a small selection of moments. Nevertheless,

as shown in the following examples, the GMM approach can be used to estimate the parameters of distributions.

Example 10.1 Normal Distribution

Suppose that the distribution of y_t is normal with pdf

$$f(y; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{1}{2} \left(\frac{y - \mu}{\sigma} \right)^2 \right],$$

where μ and σ^2 are the mean and variance respectively. By definition the mean and variance are

$$E[y_t] = \mu_0, \quad \text{var}[y_t^2] = \sigma_0^2,$$

and so GMM model (10.3) applies. The GMM estimators are therefore the sample mean and variance given in (10.9). Using the data reported in Table 10.1, the GMM estimates are

$$\hat{\mu} = 4.900, \quad \hat{\sigma}^2 = 2.890.$$

□

Example 10.2 Student t distribution

Suppose that the distribution of y_t is Student t

$$f(y; \theta) = \frac{\Gamma[(\nu + 1)/2]}{\sqrt{\pi\nu}\Gamma[\nu/2]} \left[1 + \frac{(y - \mu)^2}{\nu} \right]^{-(\nu+1)/2},$$

where μ is the mean and ν is the degrees of freedom parameter. Consider the population moments

$$E[y_t] = \mu_0, \quad E[(y_t - \mu_0)^2] = \frac{\nu_0}{\nu_0 - 2}.$$

The GMM model for $\theta = (\mu, \nu)'$ is

$$m(y_t; \theta) = \begin{bmatrix} y_t - \mu \\ (y_t - \mu)^2 - \frac{\nu}{\nu - 2} \end{bmatrix}.$$

Solving (10.8) in this case gives

$$\hat{\mu} = \frac{1}{T} \sum_{t=1}^T y_t,$$

and

$$\frac{\hat{\nu}}{\hat{\nu} - 2} = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{\mu})^2,$$

which rearranges to give

$$\hat{\nu} = \frac{2 \frac{1}{T} \sum_{t=1}^T (y_t - \hat{\mu})^2}{\frac{1}{T} \sum_{t=1}^T (y_t - \hat{\mu})^2 - 1}.$$

Using the data reported in Table 10.1, the GMM estimates are

$$\hat{\mu} = 4.900, \quad \hat{\nu} = \frac{2 \times 2.890}{2.890 - 1} = 3.058.$$

These estimators are different from the maximum likelihood estimators, but having such estimators in closed form may be useful to provide starting values for numerical computation of the maximum likelihood estimators. $\square$

Example 10.3 Gamma Distribution

Suppose that the distribution of y_t is gamma

$$f(y; \theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \exp[-\beta y] y^{\alpha-1}, \quad y \geq 0, \alpha > 0, \beta > 0, \quad (10.10)$$

where $\theta = (\alpha, \beta)'$ are parameters. Consider the population moments

$$E[y_t] = \frac{\alpha_0}{\beta_0}, \quad E[y_t^2] = \frac{\alpha_0(\alpha_0 + 1)}{\beta_0^2}, \quad E\left[\frac{1}{y_t}\right] = \frac{\beta_0}{\alpha_0 - 1},$$

which result in the model

$$m(y_t; \theta) = \left[y_t - \frac{\alpha}{\beta} \quad y_t^2 - \frac{\alpha(\alpha + 1)}{\beta^2} \quad \frac{1}{y_t} - \frac{\beta}{\alpha - 1} \right]'$$

This model is over-identified, since it has more moments than parameters. Estimation for over-identified models is discussed in section 10.3. Exactly identified models are specified by selecting two of the three moments. In the case of the first and third moments, the model is

$$m(y_t; \theta) = \begin{bmatrix} y_t - \frac{\alpha}{\beta} \\ \frac{1}{y_t} - \frac{\beta}{\alpha - 1} \end{bmatrix}.$$

From (10.8) the equations to solve are

$$\frac{1}{T} \sum_{t=1}^T y_t = \frac{\hat{\alpha}}{\hat{\beta}}, \quad \frac{1}{T} \sum_{t=1}^T \frac{1}{y_t} = \frac{\hat{\beta}}{\hat{\alpha} - 1},$$

which have solutions

$$\hat{\alpha} = \frac{T^{-1} \sum_{t=1}^T y_t}{T^{-1} \sum_{t=1}^T y_t - T^{-1} \sum_{t=1}^T 1/y_t}, \quad \hat{\beta} = \frac{\hat{\alpha}}{T^{-1} \sum_{t=1}^T y_t}.$$

Using the data in Table 10.1, the estimates are

$$\hat{\alpha} = \frac{4.900}{4.900 - 0.237} = 1.051, \quad \hat{\beta} = \frac{1.051}{4.900} = 0.214.$$

Using another combination of moments yields a different model and hence a different set of estimates. This illustrates that in GMM, unlike maximum likelihood, the specification of the model is often not unique. However, a method is given below to combine the available moment conditions optimally in over-identified situations. $\square$

10.2.3 GMM Models from Conditional Expectations

Many models are expressed in terms of conditional expectations, whether derived from statistical or economic principles. For example, a regression is a model of a conditional mean. In general terms, suppose the model is expressed as

$$E[u(y_t; \theta_0)|w_t] = 0, \quad (10.11)$$

where $u(y_t; \theta)$ is a function of the random variables and parameters that define the functional form of the model and w_t are conditioning variables. The law of iterated expectations gives

$$E[w_t u(y_t; \theta_0)] = E[w_t E[u(y_t; \theta_0)|w_t]] = 0, \quad (10.12)$$

implying a GMM model of the form

$$m(y_t; \theta) = w_t u(y_t; \theta). \quad (10.13)$$

This logic applies for any subset of the variables contained in w_t , so there is a different GMM model for each subset. This construction is now illustrated in some common modelling situations.

Example 10.4 Linear Regression

Consider the regression equation

$$E[y_t|x_t] = x_t' \beta_0.$$

This can be written as

$$E[y_t - x_t' \beta_0 | x_t] = 0,$$

implying $u(y_t, x_t; \beta) = y_t - x_t' \beta$ and $w_t = x_t$ in (10.11). Following (10.12), the Law of Iterated Expectations immediately gives

$$E[x_t(y_t - x_t' \beta_0)] = E[x_t E[y_t - x_t' \beta_0 | x_t]] = 0,$$

so the GMM model is

$$m(y_t, x_t; \beta) = x_t (y_t - x_t' \beta).$$

Applying (10.8) gives

$$M_T(\hat{\beta}) = \frac{1}{T} \sum_{t=1}^T x_t (y_t - x_t' \hat{\beta}) = 0,$$

which is solved to give the OLS estimator

$$\hat{\beta} = \left[\sum_{t=1}^T x_t x_t' \right]^{-1} \sum_{t=1}^T x_t y_t.$$

□

Example 10.5 Instrumental Variables

Consider a model of the form

$$y_t = x_t' \beta_0 + u_t, \quad (10.14)$$

where u_t does not satisfy $E[u_t|x_t] = 0$, so that u_t and x_t may be correlated. It follows that the regression model $E[y_t|x_t] = x_t' \beta_0$ is misspecified for the estimation of β_0 since $E[y_t|x_t] = x_t' \beta_0 + E[u_t|x_t] \neq x_t' \beta_0$. For example, if $E[u_t|x_t] = x_t' \gamma_0$ then $E[y_t|x_t] = x_t' (\beta_0 + \gamma_0)$, which shows the well-known “endogeneity bias” in this regression. Instead, suppose there is a vector of instrumental variables w_t that satisfy $E[u_t|w_t] = 0$ and $E[x_t|w_t] \neq E[x_t]$. Taking expectations of (10.14) conditional on w_t gives

$$E[y_t|w_t] = E[x_t|w_t]' \beta_0,$$

or

$$E[y_t - x_t' \beta_0 | w_t] = 0,$$

so this conditional expectation is correctly specified for the estimation of β_0 . It has the form (10.11) with $u(y_t, x_t; \beta) = y_t - x_t' \beta$ and $z_t = w_t$. By (10.12) it follows that

$$E[w_t (y_t - x_t' \beta_0)] = 0,$$

and the GMM model is

$$m(y_t, x_t, w_t; \beta) = w_t (y_t - x_t' \beta).$$

If w_t and x_t have the same dimension the model is exactly identified, and

(10.8) becomes

$$M_T(\hat{\beta}) = \frac{1}{T} \sum_{t=1}^T w_t (y_t - x_t' \hat{\beta}) = 0,$$

with solution

$$\hat{\beta} = \left[\sum_{t=1}^T w_t x_t' \right]^{-1} \sum_{t=1}^T w_t y_t,$$

which is the IV estimator given in Chapter 5. If there are more instruments than regressors then the model is over-identified. $\square$

Example 10.6 C-CAPM

The consumption capital asset pricing model (C-CAPM) is based on the assumption that a representative agent maximizes the inter-temporal utility function

$$\sum_{i=0}^{\infty} \beta_0^i E_t \left[\frac{c_{t+i}^{1-\gamma_0} - 1}{1 - \gamma_0} \right],$$

subject to a budget constraint, where c_t is real consumption and E_t represents expectations conditional on all relevant variables available at time t . The parameters are the discount rate, β , and the relative risk aversion parameter, γ . The first-order condition is

$$E_t \left[\beta_0 \left(\frac{c_{t+1}}{c_t} \right)^{-\gamma_0} (1 + r_{t+1}) - 1 \right] = 0,$$

where r_t is the interest rate. This has the form (10.11) where

$$u \left(\frac{c_{t+1}}{c_t}, r_{t+1}; \beta, \gamma \right) = \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\gamma} (1 + r_{t+1}) - 1.$$

Letting $w_t = \{1, e_t\}$ represent the relevant conditioning variables, where e_t represents real returns on equities, this yields an exactly identified model of the form (10.13). The GMM estimator defined by (10.8) satisfies

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T \left(\hat{\beta} \left(\frac{c_{t+1}}{c_t} \right)^{-\hat{\gamma}} (1 + r_{t+1}) - 1 \right) &= 0 \\ \frac{1}{T} \sum_{t=1}^T e_t \left(\hat{\beta} \left(\frac{c_{t+1}}{c_t} \right)^{-\hat{\gamma}} (1 + r_{t+1}) - 1 \right) &= 0. \end{aligned}$$

This is a nonlinear system of two equations which is solved using one of the iterative methods discussed in Chapter 3. Expanding the information set to

$w_t = \{1, c_t/c_{t-1}, r_t, e_t\}$ results in an over-identified model where the number of over-identifying restrictions is 2. $\square$

10.2.4 GMM and Maximum Likelihood

Consider a maximum likelihood model based on a density function $f(y_t; \theta)$ for *iid* random variables $y_1, y_2, \dots, y_T$. From Chapter 2 the expected score at the true value $\theta = \theta_0$ is zero

$$\mathbb{E} \left[\frac{\partial \ln f(y_t; \theta)}{\partial \theta} \Big|_{\theta=\theta_0} \right] = 0. \quad (10.15)$$

This suggests that a GMM model is defined as

$$m(y_t; \theta) = \frac{\partial \ln f(y_t; \theta)}{\partial \theta}. \quad (10.16)$$

This model is exactly identified because there is an element of the gradient for each parameter in θ , so the GMM estimator from (10.8) satisfies

$$\frac{1}{T} \sum_{t=1}^T \frac{\partial \ln f(y_t; \theta)}{\partial \theta} \Big|_{\theta=\hat{\theta}} = 0, \quad (10.17)$$

which also defines the first order conditions for the maximum likelihood estimator.

Note that the GMM model (10.16) does not fully encapsulate the maximum likelihood model, because it only represents the first order conditions. The expectations in (10.15) are the first order conditions of the likelihood inequality

$$\theta_0 = \arg \max_{\theta} \mathbb{E} [\ln f(y_t; \theta)], \quad (10.18)$$

and this maximization aspect of the maximum likelihood model is not captured by the GMM formulation. For example, if there are multiple solutions to the equations

$$\mathbb{E} \left[\frac{\partial \ln f(y_t; \theta)}{\partial \theta} \right] = 0,$$

only one of which is $\theta = \theta_0$, the GMM model considered in isolation provides no guidance on which solution is the desired one. In a practical situation solving (10.17), the GMM principle alone is not informative about how to proceed when there are multiple solutions to (10.17). The additional structure provided by the maximum likelihood model suggests that the desired solution of (10.17) is the one that also maximizes $T^{-1} \sum_{t=1}^T \ln f(y_t; \theta)$.

10.3 Estimation

In the analysis that follows, a GMM estimator is defined that covers over-identified models, and it is shown to include the exactly identified estimator (10.8) as a special case.

10.3.1 The GMM Objective Function

To define the GMM criterion function, it is convenient to define the simplified notation

$$m_t(\theta) = m(y_t; \theta),$$

for the GMM model, where $\theta = (\theta_1, \dots, \theta_K)'$ is the parameter vector and $m_t(\theta)$ is the $N \times 1$ vector of moments

$$m_t(\theta) = \begin{bmatrix} m_{1,t}(\theta) \\ m_{2,t}(\theta) \\ \vdots \\ m_{N,t}(\theta) \end{bmatrix}.$$

Following (10.1), the true value θ_0 satisfies

$$E[m_t(\theta_0)] = 0. \quad (10.19)$$

The sample mean of these moment conditions is

$$M_T(\theta) = \frac{1}{T} \sum_{t=1}^T m_t(\theta),$$

but in the over identified case the GMM estimator cannot be defined to satisfy (10.8), which only works for exactly identified models. Instead the GMM estimator is defined as

$$\hat{\theta} = \arg \min_{\theta} Q_T(\theta), \quad (10.20)$$

where

$$Q_T(\theta) = \frac{1}{2} M_T'(\theta) W_T^{-1} M_T(\theta), \quad (10.21)$$

and W_T is an $N \times N$ positive definite weighting matrix that determines how each moment condition is weighted in the estimation. The first-order condition for a minimum of this function is

$$\left. \frac{\partial Q_T(\theta)}{\partial \theta} \right|_{\theta=\hat{\theta}} = D_T(\hat{\theta}) W_T^{-1} M_T(\hat{\theta}) = 0, \quad (10.22)$$

where

$$D_T(\theta) = \frac{\partial M_T(\theta)}{\partial \theta'},$$

so (10.22) must be solved to find the GMM estimator $\hat{\theta}$.

The definition (10.20) is valid in the exactly identified case. The exactly identified GMM estimator was initially defined as (10.8) but the two definitions do not conflict. In an exactly identified model it is possible to choose $\hat{\theta}$ to solve $M_T(\hat{\theta}) = 0$ exactly, so that $Q_T(\hat{\theta}) = 0$. For all other $\theta \neq \hat{\theta}$, the positive definiteness of W_T^{-1} implies that $Q_T(\theta) > 0$. This shows that the solutions to (10.8) and (10.20) coincide under exact identification. This also implies there is no role for the weighting matrix W_T in exactly identified estimation. This can also be seen in the first order conditions (10.22), where exact identification means that the matrix $D_T(\hat{\theta})$ is square and hence can be inverted, so that (10.22) simplifies to $M_T(\hat{\theta}) = 0$, which is (10.8) again.

10.3.2 Asymptotic Properties

Under some general assumptions the GMM estimator defined in (10.20) is consistent and asymptotically normal.

Consistency

Define the population moments

$$M(\theta) = E[m_t(\theta)],$$

for any θ , so that the population moment conditions (10.19) can be represented

$$M(\theta_0) = 0. \quad (10.23)$$

The true value θ_0 is identified by the model if

$$M(\theta) = 0, \text{ if and only if } \theta = \theta_0. \quad (10.24)$$

The “if” part of this identification condition is (10.23). The “only if” part is an extra assumption required to ensure there is only one candidate value for θ_0 identified by the model. In some particular models this “only if” part of the identification assumption might be replaced with some other identification rule, such as (10.18) in a quasi-maximum likelihood model, which would allow for multiple solutions to the first order conditions.

Example 10.7 Gamma Distribution

From Example 10.3, the gamma distribution of y_t for the special case $\beta = 1$ is given by

$$f(y; \alpha) = \frac{1}{\Gamma(\alpha)} \exp[-y] y^{\alpha-1}, \quad y \geq 0, \alpha > 0,$$

with $\theta = \{\alpha\}$ and the first two uncentered population moments are $E[y_t] = \alpha_0$ and $E[y_t^2] = \alpha_0(\alpha_0 + 1)$. The GMM model based on these two moments is

$$m_t(\alpha) = \begin{bmatrix} y_t - \alpha \\ y_t^2 - \alpha(\alpha + 1) \end{bmatrix}, \quad (10.25)$$

so

$$M(\alpha) = \begin{bmatrix} \alpha_0 - \alpha \\ \alpha_0(\alpha_0 + 1) - \alpha(\alpha + 1) \end{bmatrix} = \begin{bmatrix} \alpha_0 - \alpha \\ (\alpha_0 - \alpha)(\alpha + \alpha_0 + 1) \end{bmatrix},$$

which shows that $M(\alpha) = 0$ if and only if $\alpha = \alpha_0$, so α_0 is identified by (10.25). $\square$

The WLLN provides the convergence in probability of the sample moments to the population moments for any θ

$$M_T(\theta) \xrightarrow{p} M(\theta), \quad (10.26)$$

and we assume that the model is sufficiently regular for this convergence to be uniform in θ (see Theorem 2.6 of Newey and McFadden (1994) for example). It is also assumed that

$$W_T \xrightarrow{p} W, \quad (10.27)$$

for some positive definite matrix W . Combining (10.26) and (10.27) in the definition of $Q_T(\theta)$ in (10.21) gives

$$Q_T(\theta) \xrightarrow{p} Q(\theta), \quad (10.28)$$

uniformly in θ , where

$$Q(\theta) = \frac{1}{2} M'(\theta) W^{-1} M(\theta). \quad (10.29)$$

The population moment conditions (10.23) imply that $Q(\theta_0) = 0$, while the identification condition (10.24) and the positive definiteness of W together imply that $Q(\theta) > 0$ for all $\theta \neq \theta_0$. Thus

$$\theta_0 = \arg \min_{\theta} Q(\theta), \quad (10.30)$$

which shows that minimizing $Q(\theta)$ identifies θ_0 . The definition of the GMM

estimator in (10.20), the convergence requirement in (10.28) and the identification property in (10.30), when combined imply the consistency of the GMM estimator

$$\hat{\theta} \xrightarrow{p} \theta_0.$$

This result does not depend on the choice of the weighting matrix W_T , provided that it is positive definite in the limit.

Normality

The mean value theorem for $M_T(\theta)$ gives

$$M_T(\hat{\theta}) = M_T(\theta_0) + D_T(\theta^*)(\hat{\theta} - \theta_0),$$

for some intermediate value θ^* between $\hat{\theta}$ and θ_0 . Pre-multiplying this expression by $D_T(\hat{\theta})'W_T^{-1}$ gives

$$D_T'(\hat{\theta})W_T^{-1}M_T(\hat{\theta}) = D_T'(\hat{\theta})W_T^{-1}M_T(\theta_0) + D_T'(\hat{\theta})W_T^{-1}D_T(\theta^*)(\hat{\theta} - \theta_0).$$

The left hand side of this expression is zero by the first order conditions (10.22), so this can be re-arranged to give

$$\sqrt{T}(\hat{\theta} - \theta_0) = - \left(D_T'(\hat{\theta})W_T^{-1}D_T(\theta^*) \right)^{-1} D_T'(\hat{\theta})W_T^{-1}\sqrt{T}M_T(\theta_0). \quad (10.31)$$

It is now assumed that there is a uniform WLLN such that

$$D_T(\theta) = \frac{1}{T} \sum_{t=1}^T \frac{\partial m_t(\theta)}{\partial \theta'} \xrightarrow{p} E \left[\frac{\partial m_t(\theta)}{\partial \theta'} \right],$$

uniformly in θ , and that the expected derivative is continuous so that

$$D_T(\hat{\theta}), D_T(\theta^*) \xrightarrow{p} D, \quad (10.32)$$

where

$$D = E \left[\frac{\partial m_t(\theta)}{\partial \theta'} \Big|_{\theta=\theta_0} \right].$$

It is also assumed that there is a central limit theorem such that

$$\sqrt{T}M_T(\theta_0) = \frac{1}{\sqrt{T}} \sum_{t=1}^T m_t(\theta_0) \xrightarrow{d} N(0, J), \quad (10.33)$$

for some variance matrix J . This central limit theorem in (10.33) requires some conditions be imposed on the model, in particular that $m_t(\theta_0)$ have more than two finite moments and that it have at most weak dependence (as in Chapter 9). The form of the variance matrix J depends on the dependence properties of $m_t(\theta_0)$. For example if $m_t(\theta_0)$ is a stationary martingale

difference sequence, including *iid* as a special case, then from Chapter 9 it immediately follows that

$$J = E[m_t(\theta_0) m_t'(\theta_0)], \quad (10.34)$$

while if $m_t(\theta_0)$ is stationary and autocorrelated then

$$J = E[m_t(\theta_0) m_t'(\theta_0)] + \sum_{s=1}^{\infty} (E[m_t(\theta_0) m_{t-s}'(\theta_0)] + E[m_{t-s}(\theta_0) m_t'(\theta_0)]). \quad (10.35)$$

The covariance matrix J corresponds to the outer product of gradients matrix discussed in Chapter 9 where $m_t(\theta_0)$ corresponds to the gradient vector.

Example 10.8 Gamma Distribution

From Example 10.7, the derivative of the model is

$$\frac{\partial m_t(\alpha)}{\partial \alpha} = - \begin{bmatrix} 1 \\ 2\alpha + 1 \end{bmatrix},$$

so

$$D = - \begin{bmatrix} 1 \\ 2\alpha_0 + 1 \end{bmatrix}.$$

To derive J it is necessary to make some assumption about the dynamics of $m_t(\alpha_0)$. If y_t is *iid* so $m_t(\alpha_0)$ is also *iid*, then $J = \text{var}[m_t(\alpha_0)]$. From the properties of the gamma distribution with $\beta = 1$, the first four moments are $E[y_t] = \alpha_0$, $E[(y_t - \alpha_0)^2] = \alpha_0$, $E[(y_t - \alpha_0)^3] = 2\alpha_0$ and $E[(y_t - \alpha_0)^4] = 3\alpha_0^2 + 6\alpha_0$, whereby

$$\begin{aligned} J &= \text{var}[m_t(\alpha_0)] \\ &= \begin{bmatrix} \text{var}[y_t - \alpha_0] & \text{cov}[(y_t - \alpha_0), (y_t^2 - \alpha_0(\alpha_0 + 1))] \\ \text{cov}[(y_t - \alpha_0), (y_t^2 - \alpha_0(\alpha_0 + 1))] & \text{var}[y_t^2 - \alpha_0(\alpha_0 + 1)] \end{bmatrix} \\ &= \begin{bmatrix} \alpha_0 & 2\alpha_0(\alpha_0 + 1) \\ 2\alpha_0(\alpha_0 + 1) & \alpha_0(\alpha_0 + 2)(4\alpha_0 + 3) \end{bmatrix}. \end{aligned}$$

□

Combining (10.27), (10.32) and (10.33) in (10.31) gives

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \Omega_W), \quad (10.36)$$

where

$$\Omega_W = (D'W^{-1}D)^{-1} D'W^{-1}JW^{-1}D (D'W^{-1}D)^{-1}.$$

This is the general form of the asymptotic distribution of the GMM estimator in the over-identified case.

The result in (10.36) also applies to an exactly identified GMM model, by noting that in this case the matrix D is square, so that $(D'W^{-1}D)^{-1}$ becomes $D^{-1}WD^{-1'}$ and hence Ω_W simplifies to

$$\Omega_W = D^{-1}JD^{-1'},$$

for any W . The invariance of Ω_W to W under exact identification is consistent with the fact that the GMM estimator itself does not depend on W_T in this case.

Efficiency

The weighting matrix W_T in (10.21) can be any positive definite matrix that satisfies (10.27). The dependence of the asymptotic variance Ω_W on W in (10.36) raises the question of how best to choose W_T . It turns out that if $W = J$ then the expression for Ω_W simplifies to

$$\Omega_J = [D'J^{-1}D]^{-1}, \quad (10.37)$$

and, most importantly, it can be shown that $\Omega_W - \Omega_J$ is positive semi-definite for any W (Hansen, 1982), so that Ω_J represents a lower bound on the asymptotic variance of a GMM estimator for θ_0 based on the model $m_t(\theta)$. It is therefore desirable to implement the GMM estimator defined in terms of (10.21) using a weighting matrix W_T such that $W_T \xrightarrow{p} J$. This will produce a GMM estimator with asymptotic variance Ω_J .

It is important to distinguish between this concept of asymptotic efficiency for GMM and that associated with the maximum likelihood estimator. The GMM estimator computed with weighting matrix satisfying $W_T \xrightarrow{p} J$ is asymptotically efficient among those estimators based on the particular model, $m_t(\theta)$. A different form for the moments will result in a different efficient GMM estimator. This is a more restricted concept of asymptotic efficiency than maximum likelihood, which is asymptotically efficient among all regular consistent estimators. A GMM estimator can match the asymptotic variance of the maximum likelihood estimator if $m_t(\theta)$ happens to be the gradient of the true likelihood function, but otherwise in general a GMM estimator, even with efficient choice of W_T , is less efficient than the maximum likelihood estimator based on a correctly specified model. The trade-off with GMM is that it is not necessary to be able to specify correctly the joint distribution of the random variables, but the price paid for this flexibility is a loss of asymptotic efficiency relative to the maximum likelihood estimator.

Example 10.9 Gamma Distribution

From Example 10.8, the asymptotic variance of the efficient GMM estimator in (10.37) is

$$\begin{aligned}\Omega_J &= \begin{bmatrix} 1 \\ 2\alpha_0 + 1 \end{bmatrix}' \begin{bmatrix} \alpha_0 & 2\alpha_0(\alpha_0 + 1) \\ 2\alpha_0(\alpha_0 + 1) & \alpha_0(\alpha_0 + 2)(4\alpha_0 + 3) \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 2\alpha_0 + 1 \end{bmatrix} \\ &= \frac{\alpha_0(3\alpha_0 + 2)}{3(\alpha_0 + 1)},\end{aligned}$$

resulting in the asymptotic distribution

$$\sqrt{T}(\hat{\alpha} - \alpha_0) \xrightarrow{d} N\left(0, \frac{\alpha_0(3\alpha_0 + 2)}{3(\alpha_0 + 1)}\right).$$

□

10.3.3 Estimation Strategies

Variance Estimation

In order to implement a GMM estimator with a weighting matrix that satisfies $W_T \xrightarrow{p} J$, it is necessary to consider the consistent estimation of J . From Chapter 9, if $m_t(\theta_0)$ is not autocorrelated, so J has the form (10.34), an estimator of $W_T(\theta)$ is

$$W_T(\theta) = \frac{1}{T} \sum_{t=1}^T m_t(\theta) m_t'(\theta), \quad (10.38)$$

which is defined for any θ . If autocorrelation in $m_t(\theta_0)$ is suspected then J has the general form (10.35) and an estimator of the form

$$W_T(\theta) = \hat{\Gamma}_0(\theta) + \sum_{i=1}^P w_i \left(\hat{\Gamma}_i(\theta) + \hat{\Gamma}_i'(\theta) \right), \quad (10.39)$$

is used, where

$$\hat{\Gamma}_i(\theta) = \frac{1}{T} \sum_{t=i+1}^T m_t(\theta) m_{t-i}'(\theta), \quad i = 0, 1, 2, \dots$$

Efficient GMM Estimators

The estimators $W_T(\theta)$ in (10.38) and (10.39) are consistent for J if evaluated at some consistent estimator of θ_0 . One approach is to compute an initial GMM estimator $\hat{\theta}_{(1)}$, based on a known weighting matrix W_T in (10.21), with $W_T = I_N$ being one possibility. This initial estimator is generally not

asymptotically efficient, but it is consistent, since consistency does not depend on the choice of W_T . Thus $W_T(\hat{\theta}_{(1)})$ provides a consistent estimator for J and

$$\hat{\theta}_{(2)} = \arg \min_{\theta} \frac{1}{2} M_T'(\theta) W_T^{-1}(\hat{\theta}_{(1)}) M_T(\theta), \quad (10.40)$$

is then an asymptotically efficient GMM estimator for θ_0 . This is referred to as the two-step estimator since two estimators of θ_0 are computed.

The two step approach can be extended by re-estimating J using the second asymptotically efficient estimator $\hat{\theta}_{(2)}$ in $W_T(\theta)$, on the grounds that $W_T(\hat{\theta}_{(2)})$ might give a better estimator of J . This updated estimator of J , in turn, is used to update the estimator of θ_0 . This process is repeated, giving a sequence of estimators of the form

$$\hat{\theta}_{(j+1)} = \arg \min_{\theta} \frac{1}{2} M_T'(\theta) W_T(\hat{\theta}_{(j)}^{-1}) M_T(\theta), \quad (10.41)$$

for $j = 1, 2, \dots$ until the estimators and the weighting matrix converge to within some tolerance. The iterations for $j > 2$ provide no further improvement in asymptotic properties, but it is hoped they provide some improvement in the finite sample properties of the estimator. This is referred to as the iterative estimator.

Rather than switching between the separate estimation of θ_0 and J , an alternative strategy is to estimate them jointly using

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{2} M_T'(\theta) W_T^{-1}(\theta) M_T(\theta). \quad (10.42)$$

This is referred to as the continuous updating estimator. The dependence of $W_T(\theta)$ on θ makes the first order conditions more complicated because there is an extra term arising from the derivative of $W_T^{-1}(\theta)$ with respect to θ . Also the derivations of the asymptotic properties outlined above do not allow for the dependence of $W_T(\theta)$ on θ . Nevertheless, as described in Hansen, Heaton and Yaron (1996) and references therein, the asymptotic properties of this continuous updating estimator are the same as those for the other two estimators. They also point out the continuous updating estimator has some nice invariance properties not shared by the other two estimators and present simulation evidence on the three estimators.

Computations

For models where $m_t(\theta)$ is nonlinear in the unknown parameters θ , implementation of the estimation strategies in (10.40) to (10.42) requires the minimization to be performed using an iterative algorithm such as the Newton-Raphson method discussed in Chapter 3. In the case of the continuous updat-

ing estimator in (10.42), implementation of a gradient algorithm requiring first and second derivatives is even more involved as a result of the effect of θ on the weighting matrix. To see this, given (10.42), consider the criterion function

$$Q_T(\theta) = \frac{1}{2} M_T'(\theta) W_T(\theta)^{-1} M_T(\theta). \quad (10.43)$$

Letting $M_T(\theta) = (M_{i,T}(\theta))_{i=1}^N$ and $W_T(\theta)^{-1} = (w_{i,j}(\theta))_{i,j=1}^N$, the first derivative of $Q_T(\theta)$ is

$$\sum_{i=1}^N \sum_{j=1}^N M_{i,T}(\theta) w_{i,j}(\theta) \frac{\partial M_{j,T}(\theta)}{\partial \theta} + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N M_{i,T}(\theta) \frac{\partial w_{i,j}(\theta)}{\partial \theta} M_{j,T}(\theta), \quad (10.44)$$

and the second derivative is

$$\begin{aligned} & \sum_{i=1}^N \sum_{j=1}^N \frac{\partial M_{i,T}(\theta)}{\partial \theta} w_{i,j}(\theta) \frac{\partial M_{j,T}(\theta)}{\partial \theta'} + \sum_{i=1}^N \sum_{j=1}^N M_{i,T}(\theta) w_{i,j}(\theta) \frac{\partial^2 M_{j,T}(\theta)}{\partial \theta \partial \theta'} \\ & + \sum_{i=1}^N \sum_{j=1}^N M_{i,T}(\theta) \frac{\partial w_{i,j}(\theta)}{\partial \theta} \frac{\partial M_{j,T}(\theta)}{\partial \theta'} + \sum_{i=1}^N \sum_{j=1}^N \frac{\partial M_{i,T}(\theta)}{\partial \theta} \frac{\partial w_{i,j}(\theta)}{\partial \theta'} M_{j,T}(\theta) \\ & + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N M_{i,T}(\theta) \frac{\partial^2 w_{i,j}(\theta)}{\partial \theta \partial \theta'} M_{j,T}(\theta). \end{aligned} \quad (10.45)$$

This suggests that a more convenient approach is to use a gradient algorithm where the derivatives are computed numerically (see Chapter 3).

The leading term in the expression for the second derivative of $Q_T(\theta)$ in (10.45) is

$$\left(\frac{\partial M_T(\theta)}{\partial \theta'} \right)' W_T(\theta)^{-1} \left(\frac{\partial M_T(\theta)}{\partial \theta'} \right) = D_T'(\theta) W_T^{-1}(\theta) D_T(\theta).$$

When evaluated at the estimator $\hat{\theta}$, the inverse of this expression provides a consistent estimator of the asymptotic variance $\Omega_J = (D'J^{-1}D)^{-1}$. The remaining terms in the second derivative of $Q_T(\theta)$ in (10.45) all involve $M_{i,T}(\theta)$, and $M_{i,T}(\hat{\theta}) \xrightarrow{P} 0$, as $M_T(\hat{\theta}) \xrightarrow{P} E[m_t(\theta_0)] = 0$. Therefore the inverse of the second derivative of $Q_T(\theta)$ is a consistent estimator of the asymptotic variance of the efficient GMM estimator. It follows that standard errors can be obtained as the square roots of the diagonal elements of the matrix

$$\frac{1}{T} \left[\frac{\partial^2 Q_T(\theta)}{\partial \theta \partial \theta'} \Big|_{\theta=\hat{\theta}} \right]^{-1},$$

where $Q_T(\theta)$ is as defined in (10.43).

Alternatively, the usual variance estimator suggested in the GMM literature is

$$\hat{\Omega}_J = \left[D'_T(\hat{\theta}) W_T^{-1}(\hat{\theta}) D_T(\hat{\theta}) \right]^{-1},$$

which is also consistent and standard errors can be obtained as the square roots of the diagonal elements of $T^{-1}\hat{\Omega}_J$.

Example 10.10 Gamma Distribution

Consider estimating α in the Gamma distribution using the data in Table 10.1 based on the over-identified model in Example 10.7, for which

$$M_T(\alpha) = \frac{1}{T} \sum_{t=1}^T \begin{bmatrix} y_t - \alpha \\ y_t^2 - \alpha(\alpha + 1) \end{bmatrix}$$

$$W_T(\alpha) = \frac{1}{T} \sum_{t=1}^T \begin{bmatrix} y_t - \alpha \\ y_t^2 - \alpha(\alpha + 1) \end{bmatrix} \begin{bmatrix} y_t - \alpha \\ y_t^2 - \alpha(\alpha + 1) \end{bmatrix}'.$$

A convenient starting value is available for the numerical search by using just the first moment

$$\alpha_{(0)} = \frac{1}{T} \sum_{t=1}^T y_t = 4.9.$$

The numerical first and second derivatives of (10.43) at this value are computed as

$$\left. \frac{dQ_T(\alpha)}{d\alpha} \right|_{\alpha=4.9} = 0.0709, \quad \left. \frac{d^2Q_T(\alpha)}{d\alpha^2} \right|_{\alpha=4.9} = 0.3794,$$

and hence the first Newton-Raphson iteration is

$$\alpha_{(1)} = \alpha_{(0)} - \left[\left. \frac{d^2Q_T(\alpha)}{d\alpha^2} \right|_{\alpha_{(0)}} \right]^{-1} \left. \frac{dQ_T(\alpha)}{d\alpha} \right|_{\alpha_{(0)}} = 4.9 - \frac{0.0709}{0.3794} = 4.7130.$$

The second iteration is

$$\alpha_{(2)} = \alpha_{(1)} - \left[\left. \frac{d^2Q_T(\alpha)}{d\alpha^2} \right|_{\alpha_{(1)}} \right]^{-1} \left. \frac{dQ_T(\alpha)}{d\alpha} \right|_{\alpha_{(1)}} = 4.7130 - \frac{-0.0021}{0.3905} = 4.7184.$$

The first derivative of $Q_T(\alpha)$ at $\alpha_{(2)}$ is 1.6×10^{-6} , so the search is deemed to have converged at this point. As the second derivative evaluated at $\alpha_{(2)}$ is 0.3912, and given that the sample size is $T = 10$, the estimated variance is $1/(10 \times 0.3912) = 0.2556$ and the standard error is $\text{se}(\hat{\alpha}) = \sqrt{0.2556} = 0.5056$. $\square$

Example 10.11 Estimating the C-CAPM

The C-CAPM introduced in Example 10.6

$$m\left(\frac{c_{t+1}}{c_t}, r_{t+1}, w_t; \beta, \gamma\right) = \left(\beta\left(\frac{c_{t+1}}{c_t}\right)^{-\gamma} (1 + r_{t+1}) - 1\right) w_t,$$

is now estimated by GMM using a data set consisting of $T = 238$ observations on the ratio of inter-temporal real U.S. consumption, c_t/c_{t-1} , the real return on Treasury bills, r_t , and the real return on the value weighted index e_t . The data are a revised version of the original data used by Hansen and Singleton (1982). The full set of instruments is chosen as $w_t = \{1, c_t/c_{t-1}, r_t, e_t\}$ resulting in the GMM model having four elements

$$\begin{aligned} &\left(\beta\left(\frac{c_{t+1}}{c_t}\right)^{-\gamma} (1 + r_{t+1}) - 1\right) \\ &\left(\beta\left(\frac{c_{t+1}}{c_t}\right)^{-\gamma} (1 + r_{t+1}) - 1\right) \left(\frac{c_t}{c_{t-1}}\right) \\ &\left(\beta\left(\frac{c_{t+1}}{c_t}\right)^{-\gamma} (1 + r_{t+1}) - 1\right) r_t \\ &\left(\beta\left(\frac{c_{t+1}}{c_t}\right)^{-\gamma} (1 + r_{t+1}) - 1\right) e_t, \end{aligned}$$

which is a set of four nonlinear moments and two unknown parameters. The GMM parameter estimates of $\theta = \{\beta, \gamma\}$ are reported in Table 10.2 for various instrument sets based on the continuous updating estimator with all derivatives computed numerically. The estimate of β for all instrument sets is 0.998. This implies a discount rate of

$$d = \frac{1}{\hat{\beta}} - 1 = \frac{1}{0.998} - 1 = 0.002,$$

or 0.2%, which appears to be quite low. The estimates of the relative risk aversion parameter, γ , range from 0.554 to 0.955. These estimates suggest that relative risk aversion is also relatively low over the sample period considered.

□

10.4 Over-Identification Testing

An over-identified GMM model for K parameters has $N > K$ population moment conditions of the form

$$M(\theta_0) = 0, \tag{10.46}$$

Table 10.2

GMM estimates of the consumption capital asset pricing model for alternative instrument sets. Estimation is based on continuous updating with standard errors computed using the heteroskedasticity consistent weighting matrix in equation (10.38).

Parameter	Instrument Set		
	$w_t = \{1, c_t/c_{t-1}\}$	$w_t = \{1, c_t/c_{t-1}, r_t\}$	$w_t = \{1, c_t/c_{t-1}, r_t, e_t\}$
β	0.998 (0.004)	0.998 (0.004)	0.998 (0.004)
γ	0.554 (1.939)	0.955 (1.863)	0.660 (1.761)
J_{HS} p-value	0.000 (n.a.)	1.067 (0.785)	1.278 (0.865)

where $M(\theta) = E[m_t(\theta)]$ for any θ . However, if some of the moments are misspecified then some or all of the elements of $M(\theta_0)$ will be non-zero. In the presence of such misspecification, a uniform WLLN can still apply to the sample moments so that the GMM estimator (10.20) converges in probability to

$$\theta^* = \arg \min_{\theta} \frac{1}{2} M'(\theta) W^{-1} M(\theta). \quad (10.47)$$

The effect of the misspecification is that $\theta^* \neq \theta_0$, so that the GMM estimator is no longer consistent for θ_0 . In general this implies that

$$M(\theta^*) \neq 0. \quad (10.48)$$

An implication of this result is that the probability limit of the GMM estimator now depends on the choice of weighting matrix W . This is in contrast to the GMM estimator of a correctly specified model in which (10.46) holds.

Example 10.12 Gamma Distribution

In Example 10.7, the estimation of the shape parameter of a Gamma distribution with $\beta = 1$ is based on the moments $E[y_t] = \alpha_0$ and $E[y_t^2] = \alpha_0(\alpha_0 + 1)$. Now suppose that the true distribution of y_t is exponential with parameter α_0 , for which the moments are $E[y_t] = \alpha_0$ and $E[y_t^2] = 2\alpha_0^2$. This means that the first moment based on the gamma distribution is correctly specified, but the second moment is not, except in the special case of $\alpha_0 = 1$.

The moment conditions of the misspecified model (10.25) are

$$M(\alpha) = E[m_t(\alpha)] = \begin{bmatrix} E[y_t] - \alpha \\ E[y_t^2] - \alpha(\alpha + 1) \end{bmatrix} = \begin{bmatrix} \alpha_0 - \alpha \\ 2\alpha_0^2 - \alpha(\alpha + 1) \end{bmatrix},$$

showing that there is no value of α where $M(\alpha) = 0$, as the first population moment is zero for $\alpha = \alpha_0$, while the second population moment is zero for $\alpha = \sqrt{2\alpha_0^2 + \frac{1}{4}} - \frac{1}{2}$. Following (10.47), the probability limit of the GMM estimator in this model is the value of α that minimises $\frac{1}{2}\mu(\alpha)'W\mu(\alpha)$, denoted α^* . The value of α^* will depend on W ; for example if $W = I_2$ then α^* minimises $\left((\alpha_0 - \alpha)^2 + (2\alpha_0^2 - \alpha(\alpha + 1))^2\right)/2$. So, if the GMM model (10.25) based on the Gamma distribution is used to estimate α_0 , when the truth is an exponential distribution with parameter α_0 , the GMM estimator will converge to this implied value of α^* , not the correct value α_0 . Also α^* will generally be neither α_0 nor $\sqrt{2\alpha_0^2 + \frac{1}{4}} - \frac{1}{2}$, so neither of the population moment conditions will be equal to zero when evaluated at α^* . The practical implication of this is that even though only one of the two moment conditions is misspecified, both of them turn out to be non-zero when evaluated at the probability limit of the GMM estimator. Therefore knowing that both elements of $\mu(\alpha^*)$ are non-zero is not informative about which moments are misspecified. $\square$

A general misspecification test is based on the property that (10.46) holds under correct specification, but (10.48) holds under misspecification. In particular, under correct specification $\hat{\theta} \xrightarrow{p} \theta_0$ and

$$Q_T(\hat{\theta}) \xrightarrow{p} \frac{1}{2}M'(\theta_0)W^{-1}M(\theta_0) = 0,$$

by (10.46), whereas under misspecification

$$Q_T(\hat{\theta}) \xrightarrow{p} \frac{1}{2}M'(\theta^*)W^{-1}M(\theta^*) > 0,$$

by (10.48) and the positive-definiteness of W . This suggests that the value of the minimized criterion function $Q_T(\hat{\theta})$ provides a test of misspecification.

In general terms the hypotheses are

H_0 : model correctly specified

H_1 : model not correctly specified.

The Hansen-Sargan J test statistic is defined as

$$J_{HS} = 2T Q_T(\hat{\theta}), \quad (10.49)$$

This statistic must be computed from a criterion function in which the weighting matrix is chosen optimally so $W_T \xrightarrow{p} J$. Under the null J_{HS} is distributed asymptotically as χ^2_{N-K} , where the number of over-identifying restrictions, represented by $N - K$, is the number of degrees of freedom. Under the alternative, J_{HS} diverges to $+\infty$ as $T \rightarrow \infty$, since $Q_T(\hat{\theta})$ itself converges to a strictly positive limit. The null hypothesis H_0 is therefore rejected for values of J_{HS} larger than the critical value from the χ^2_{N-K} distribution. As J_{HS} represents a general test of misspecification, it is not informative about which moments are misspecified, as illustrated in Example 10.12.

The J test exploits the presence of over-identification to detect any misspecification, so this approach to specification testing does not apply for exactly identified models. In that case misspecification will still generally result in inconsistent estimation, as $\hat{\theta} \xrightarrow{p} \theta^* \neq \theta_0$, where $M(\theta^*) = 0$. In the sample, the exactly identified estimator is obtained to solve $M_T(\hat{\theta}) = 0$, so by construction $Q_T(\hat{\theta}) = 0$, regardless of any misspecification.

Example 10.13 Gamma Distribution

From Example 10.12, suppose that only the second moment of the Gamma distribution is used to define the GMM model

$$m_t(\alpha) = y_t^2 - \alpha(\alpha + 1),$$

while the true distribution of y_t is exponential with parameter α_0 . This is an exactly identified model with

$$M(\alpha) = 2\alpha_0^2 - \alpha(\alpha + 1),$$

As $M(\alpha^*) = 0$ yields as the solution $\alpha^* = \sqrt{2\alpha_0^2 + \frac{1}{4}} - \frac{1}{2} \neq \alpha_0$, the GMM estimator is inconsistent for α_0 and the J_{HS} test cannot be used to detect the misspecification. $\square$

The majority of applications of GMM in econometrics arise from models specified in terms of conditional expectations, such as (10.11). When a sufficient number of instruments is chosen in (10.12) to over-identify the model, the J_{HS} test has the potential to detect whether any of the instruments are invalid or if the functional form of $u(y_t; \theta)$ is misspecified. This is illustrated in the following example.

Example 10.14 Regression with an Invalid Instrument

Consider the model in Example 10.5 where

$$y_t = x_t\beta_0 + u_t,$$

and x_t is a single regressor for which $E(u_t|x_t) \neq 0$. Suppose there are two potential instruments, $z_{1,t}$ and $z_{2,t}$, that are used to specify the GMM model

$$m(y_t, x_t, z_{1,t}, z_{2,t}; \beta) = \begin{bmatrix} z_{1,t}(y_t - x_t\beta) \\ z_{2,t}(y_t - x_t\beta) \end{bmatrix}. \quad (10.50)$$

This model is correctly specified if $E[z_{i,t}u_t] = 0$ and $E[z_{i,t}x_t] \neq 0$ for $i = 1, 2$. Suppose however that the condition $E[z_{2,t}u_t] \neq 0$ does not hold, so that $z_{2,t}$ is not a valid instrument. The model (10.50) is then misspecified for the estimation of β_0 . This is shown by using $y_t - x_t\beta = u_t - x_t(\beta - \beta_0)$ to rewrite the moment conditions as

$$M(\beta) = E[m(y_t, x_t, z_{1,t}, z_{2,t}; \beta)] = \begin{bmatrix} -E[z_{1,t}x_t](\beta - \beta_0) \\ E[z_{2,t}u_t] - E[z_{2,t}x_t](\beta - \beta_0) \end{bmatrix}.$$

The first of these population moments is zero if $\beta = \beta_0$ (using $E[z_{1,t}x_t] \neq 0$), but the second population moment is zero for

$$\beta = \beta_0 + \frac{E[z_{2,t}u_t]}{E[z_{2,t}x_t]},$$

showing there is no value of β such that $M(\beta) = 0$. Hence the J test will have power against this misspecification. Note that the J test cannot distinguish which instrument is not valid. $\square$

Example 10.15 Regression with Incorrect Functional Form

Suppose the true functional relationship between y_t and x_t is

$$y_t = x_t^2\beta_0 + u_t,$$

but the GMM model (10.50) is specified assuming linearity. Assume that both instruments are now valid. The population moments are

$$E[z_{i,t}(y_t - x_t\beta)] = E[z_{i,t}(u_t + x_t^2\beta_0 - x_t\beta)] = E[z_{i,t}x_t^2]\beta_0 - E[z_{i,t}x_t]\beta, .$$

for $i = 1, 2$. The first population moment condition is zero at

$$\beta = \frac{E[z_{1,t}x_t^2]}{E[z_{1,t}x_t]}\beta_0,$$

and the second is zero at a different value

$$\beta = \frac{E[z_{2,t}x_t^2]}{E[z_{2,t}x_t]}\beta_0.$$

Therefore there is no single value of β such that $\mu(\beta) = 0$ and the J test will have power against this misspecification. Taken together, these two examples show that a rejection by the J test might be because of a misspecified

functional form or an invalid instrument, but it is not informative about the type of misspecification. $\square$

Example 10.16 Testing the C-CAPM

Tests of the C-CAPM in Example 10.11 are based on the J_{HS} tests reported in Table 10.2. For the first instrument set $w_t = \{1, c_t/c_{t-1}\}$, the model is exactly identified, so $J_{HS} = 0$ and there is nothing to test. For the instrument set $w_t = \{1, c_t/c_{t-1}, r_t\}$, there is one over-identifying restriction resulting in a value of the test statistic of $J_{HS} = 1.067$. The statistic is distributed asymptotically as χ_1^2 under the null yielding a p-value of 0.785. For the instrument set $w_t = \{1, c_t/c_{t-1}, r_t, e_t\}$, there are two over-identifying restrictions resulting in a value of the test statistic of $J_{HS} = 1.278$. This statistic is distributed asymptotically as χ_2^2 under the null yielding a p-value 0.865. For the last two instrument sets the model specification cannot be rejected at conventional significance levels. $\square$

10.5 Applications

Two applications of the GMM estimation framework are presented. The first application focusses on the finite sample properties of the GMM estimator and its associated test statistics, using a range of Monte Carlo experiments. The second application is empirical which uses GMM to estimate an interest rate model with level effects.

10.5.1 Monte Carlo Evidence

Estimation

The data generating process y_t is *iid* with Gamma distribution for $t = 1, 2, \dots, T$ with shape parameter α_0 and known scale parameter $\beta_0 = 1$, resulting in the moments $E[y_t] = \alpha_0$ and $E[y_t^2] = \alpha_0(1 + \alpha_0)$. Two GMM estimators are computed. The first is based only on the first of these moments, for which the GMM model $m_t(\alpha) = y_t - \alpha$ is exactly identified and gives the sample mean as the estimator of α_0

$$\hat{\alpha}_1 = \frac{1}{T} \sum_{t=1}^T y_t.$$

The second estimator is based on both moments, for which the GMM model is (10.25). This model is over-identified, and the continuous updating estimator is computed using

$$\hat{\alpha}_2 = \arg \min_{\alpha} Q_T(\alpha),$$

where

$$Q_T(\alpha) = \frac{1}{2} M_T'(\alpha) W_T^{-1}(\alpha) M_T(\alpha),$$

and where $W_T(\alpha)$ is given in (10.38). The maximum likelihood estimator is also computed by solving

$$\hat{\alpha}_{ML} = \arg \max_{\alpha} \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \alpha),$$

where the pdf is given in (10.10). A gradient algorithm is used to compute both $\hat{\alpha}_2$ and $\hat{\alpha}_{ML}$, with $\hat{\alpha}_1$ used as a starting value in both cases

Results for the bias and standard deviation of the sampling distributions of these three estimators are shown in Table 10.3 for $\alpha_0 = \{1, 2, 3, 4, 5\}$ and $T = \{50, 100, 200\}$. The sample mean $\hat{\alpha}_1$ is exactly unbiased, so any non-zero bias in the table only reflects simulation error. The GMM estimator $\hat{\alpha}_2$ is not exactly unbiased and shows a small negative bias that tends to increase in magnitude as α_0 increases and decrease in magnitude as T increases, the latter being a reflection of the consistency of the estimator. The MLE $\hat{\alpha}_{ML}$ shows a very small positive bias that decreases as T increases. The slightly smaller bias for $\hat{\alpha}_{ML}$ relative to $\hat{\alpha}_2$ is not predicted from the asymptotic theory.

The variances of all three estimators increase as α_0 increases, which increases dispersion of y_t . For example, the variances of $\hat{\alpha}_2$ are $0.1374^2 = 0.0189$ (for $T = 50$), 0.0086 ($T = 100$) and 0.0041 ($T = 200$), so the variances approximately halves as the sample size doubles, as is expected from an estimator whose approximate variance has the form $T^{-1}\Omega_J$.

Comparing the estimators, $\hat{\alpha}_2$ is slightly more efficient than $\hat{\alpha}_1$ across the range of α_0 and T considered. This efficiency gain reflects the extra information available to $\hat{\alpha}_2$ in the form of the second moment condition. The maximum likelihood estimator $\hat{\alpha}_{ML}$ is more efficient again than $\hat{\alpha}_2$, which illustrates the asymptotic efficiency property of the maximum likelihood estimator working in finite samples in this case. The maximum likelihood estimator imposes more information as it uses knowledge of the entire distribution, compared to the GMM estimator which imposes only the first two moments. One way to interpret the magnitude of efficiency differences is to compute the relative efficiency $\text{var}(\hat{\alpha}_{ML})/\text{var}(\hat{\alpha}_2)$. For example, for $T = 100$ and $\alpha_0 = 1$, $\text{var}(\hat{\alpha}_{ML})/\text{var}(\hat{\alpha}_2) = 0.0790^2/0.0925^2 = 0.7294$, or relative efficiency of 73%. Intuitively, using $\hat{\alpha}_2$ with 100 observations is as efficient as using $\hat{\alpha}_{ML}$ with only 73 observations.

Hypothesis Testing

Table 10.3

Simulation results for maximum likelihood ($\hat{\alpha}_{ML}$) and GMM ($\hat{\alpha}_1, \hat{\alpha}_2$) estimators.
Based on a gamma distribution with shape parameter α_0 and scale parameter $\beta_0 = 1$.

α_0	$\hat{\alpha}_{ML}$		$\hat{\alpha}_1$		$\hat{\alpha}_2$	
	Bias	S.D.	Bias	S.D.	Bias	S.D.
$T = 50$						
1	0.0105	0.1125	0.0016	0.1413	-0.0292	0.1374
2	0.0099	0.1748	-0.0020	0.1974	-0.0352	0.1925
3	0.0151	0.2283	0.0050	0.2474	-0.0338	0.2427
4	0.0074	0.2706	-0.0020	0.2863	-0.0431	0.2843
5	0.0100	0.3001	0.0013	0.3156	-0.0426	0.3136
$T = 100$						
1	0.0050	0.0790	0.0006	0.0997	-0.0110	0.0925
2	0.0066	0.1251	0.0018	0.1406	-0.0136	0.1338
3	0.0044	0.1590	-0.0008	0.1734	-0.0189	0.1663
4	0.0056	0.1877	0.0008	0.1996	-0.0193	0.1940
5	0.0066	0.2117	0.0010	0.2228	-0.0203	0.2180
$T = 200$						
1	0.0030	0.0557	0.0006	0.0709	-0.0031	0.0637
2	0.0029	0.0876	0.0010	0.0996	-0.0054	0.0932
3	0.0015	0.1116	-0.0005	0.1227	-0.0091	0.1159
4	0.0015	0.1347	-0.0008	0.1438	-0.0102	0.1382
5	0.0021	0.1492	0.0004	0.1575	-0.0106	0.1521

The finite sample properties of hypothesis tests based on the three estimators are now investigated. The hypotheses tested are

$$H_0 : \alpha_0 = \alpha \quad H_1 : \alpha_0 \neq \alpha,$$

using the t -statistic

$$t = \frac{\hat{\alpha} - \alpha}{\text{se}(\hat{\alpha})}.$$

The decision rule is to reject H_0 if $|t| > 1.96$, giving a test with an asymptotic significance level of 5%.

The t -statistic based on $\hat{\alpha}_{ML}$, denoted t_{ML} , uses a standard error from

the square root of the inverse of the scaled Hessian

$$\text{se}(\hat{\alpha}_{ML}) = \left[\sum_{t=1}^T \frac{\partial^2 \ln f(y_t; \alpha)}{\partial \alpha \partial \alpha} \Big|_{\alpha=\hat{\alpha}_{ML}} \right]^{-1/2},$$

which is available immediately once the numerical maximization algorithm has converged.

The t -statistic based on the GMM estimator $\hat{\alpha}_1$ uses as the standard error

$$\text{se}(\hat{\alpha}_1) = \frac{1}{T} \left[\sum_{t=1}^T (y_t - \hat{\alpha}_1)^2 \right]^{1/2},$$

which is the usual standard error for the sample mean.

The t -statistic based on the GMM estimator $\hat{\alpha}_2$ has two possible standard errors. The first is

$$\text{se}(\hat{\alpha}_2) = \left[T \frac{\partial^2 Q_T(\alpha)}{\partial \alpha \partial \alpha'} \Big|_{\alpha=\hat{\alpha}_2} \right]^{-1/2},$$

which is immediately available as the final Hessian on the convergence of the numerical search for $\hat{\alpha}_2$. The second standard error is

$$\text{se}(\hat{\alpha}_2) = [TD'(\hat{\alpha}_2) W_T^{-1}(\hat{\alpha}_2) D(\hat{\alpha}_2)]^{-1/2},$$

where $D(\alpha) = \partial m_t(\alpha) / \partial \alpha = (1, 2\alpha + 1)'$. The t -statistics using these standard errors are denoted as t_2^Q and t_2 respectively.

The simulated finite sample sizes of the four t -tests are shown in Table 10.4 for $T = 100$ and $T = 200$. The test based on t_{ML} has finite sample size closest to the nominal size of 0.05 for each T and α_0 . In the case of the GMM based tests, all tests are marginally over-sized with the t_1 test yielding sizes closest to the nominal size of 0.05.

Some of the simulated finite sample powers of the t tests for $H_0 : \alpha_0 = 1$ are shown in Table 10.5 for $T = 200$. The larger sample size used because the sizes, reproduced in the first row of the table, are closer to the nominal level, making powers more comparable. The t_{ML} test has the highest power, by a substantial margin in most cases. This is a direct result of the greater efficiency of the maximum likelihood estimator revealed in Table 10.3. The t_2 and t_2^Q tests have higher power than the t_1 test. Some of this power advantage is due to the t_2 and t_2^Q tests having slightly higher size, and the rest of the power difference is attributed to the greater efficiency of the estimator based on two moments instead of one. The power results illustrate that more efficient estimators translate to more powerful hypothesis tests.

Table 10.4

Empirical sizes for t-tests of $\alpha = 1$. Based on a gamma distribution with shape parameter α_0 and scale parameter $\beta_0 = 1$.

α_0	t_{ML}	t_1	t_2^Q	t_2
$T = 100$				
1	0.0524	0.0588	0.0820	0.0749
2	0.0496	0.0575	0.0737	0.0680
3	0.0473	0.0530	0.0712	0.0651
4	0.0500	0.0592	0.0732	0.0661
5	0.0538	0.0575	0.0735	0.0670
$T = 200$				
1	0.0536	0.0537	0.0638	0.0600
2	0.0512	0.0546	0.0620	0.0584
3	0.0480	0.0528	0.0613	0.0570
4	0.0509	0.0540	0.0618	0.0589
5	0.0525	0.0565	0.0644	0.0602

Table 10.5

Empirical powers for t-tests with $T = 200$. Based on a gamma distribution with shape parameter α_0 and scale parameter $\beta_0 = 1$.

α_0	t_{ML}	t_1	t_2^Q	t_2
1.00	0.0539	0.0537	0.0638	0.0600
1.05	0.1343	0.0847	0.1057	0.1016
1.10	0.3993	0.2496	0.3074	0.2995
1.15	0.7173	0.5052	0.5937	0.5857
1.20	0.9176	0.7626	0.8353	0.8271
1.25	0.9859	0.9163	0.9535	0.9506
1.30	0.9981	0.9806	0.9914	0.9905

Misspecification

The properties of the various estimators and tests are now evaluated under some misspecification of the model. In this case, the data generating process of y_t is *iid* from an exponential distribution with parameter α_0 , and moments $E[y_t] = \alpha_0$ and $E[y_t^2] = 2\alpha_0^2$. The maximum likelihood and the GMM models continue to assume that y_t is gamma distributed. In the special case that $\alpha_0 = 1$ there is no misspecification as the exponential and gamma

distributions coincide. For all other values of α_0 , there is some misspecification. (The Gamma likelihood can not be considered a quasi-likelihood for an Exponential model.)

The GMM model $m_t(\alpha) = y_t - \alpha$ based only on the mean is correctly specified for every α . The GMM model (10.25) based on the first two moments of the Gamma distribution is misspecified for all $\alpha_0 \neq 1$. The maximum likelihood model is also misspecified for all $\alpha_0 \neq 1$.

Finite sample properties of the three estimators are shown Table 10.6 for $T = 200$ and $T = 400$. As expected, all three estimators have small bias for $\alpha_0 = 1$ when the gamma model is correctly specified. As α_0 increases away from 1, both $\hat{\alpha}_{ML}$ and $\hat{\alpha}_2$ become increasingly negatively biased. For $\alpha_0 = 2$ the maximum likelihood estimator appears to be converging to a value of about 1.6, while $\hat{\alpha}_2$ appears to be converging to an α^* value of about 1.7. These biases do not disappear in large samples, as illustrated by doubling the sample size from $T = 200$ to $T = 400$. Regardless of the value of α_0 , the sample mean $\hat{\alpha}_1$ remains unbiased and is a consistent estimator of α_0 .

The implications of the misspecification for the size properties of the t tests are shown in Table 10.7. The table shows the empirical sizes of the t tests of $H_0 : \alpha_0 = \alpha$ against $H_1 : \alpha_0 \neq \alpha$ for $\alpha = 1.0, 1.2, \dots, 2.0$ and for $T = 200$. For $\alpha_0 = 1$, where there is no misspecification, the finite sample sizes are close to the nominal level of 0.05 as expected. As α_0 increases away from 1, the sizes of all tests except t_1 increase dramatically above 0.05. If $\alpha_0 = 2$, the t_{ML} test wrongly rejects $H_0 : \alpha_0 = 2$ about 98% of the time. These poor size properties are a direct result of the large biases in $\hat{\alpha}_{ML}$ and $\hat{\alpha}_2$ under misspecification. The size properties of t_1 are completely unaffected by α_0 because the sample mean and its associated t -statistic are exactly invariant to the true value of the mean.

To summarize, these results illustrate the trade-offs involved in model specification. Imposing correct restrictions on a model, with maximum likelihood as the ultimate example, can produce more efficient estimators and more powerful hypothesis tests, but imposing incorrect restrictions on a model can produce inconsistent estimators and highly distorted hypothesis tests.

As discussed in Example 10.12, the J_{HS} test for over-identifying restrictions can be applied in the GMM model (10.25) to detect a distributional misspecification. The finite sample properties of the J_{HS} test are shown in Table 10.8 where the data generating process for y_t is *iid* from an exponential distribution with parameter α_0 , and the sample sizes are $T = 200, 400, 800$. When $\alpha_0 = 1$ the gamma model is correctly specified so the rejection frequencies in that case show the empirical size of the J_{HS} test. Its finite sample

Table 10.6

Simulation results for GMM and maximum likelihood estimators under misspecification. Based on a gamma distribution with shape parameter α_0 and scale parameter $\beta_0 = 1$.

α_0	$\hat{\alpha}_{ML}$		$\hat{\alpha}_1$		$\hat{\alpha}_2$	
	Bias	S.D.	Bias	S.D.	Bias	S.D.
$T = 200$						
1.0	0.0030	0.0557	0.0006	0.0709	-0.0031	0.0637
1.2	-0.0763	0.0646	0.0019	0.0842	-0.0485	0.0730
1.4	-0.1578	0.0745	0.0011	0.0987	-0.1071	0.0833
1.6	-0.2407	0.0848	-0.0001	0.1127	-0.1704	0.0947
1.8	-0.3236	0.0947	-0.0008	0.1286	-0.2347	0.1070
2.0	-0.4095	0.1054	-0.0018	0.1410	-0.3012	0.1192
$T = 400$						
1.0	0.0018	0.0389	0.0011	0.0500	-0.0001	0.0444
1.2	-0.0784	0.0460	0.0004	0.0598	-0.0480	0.0517
1.4	-0.1604	0.0530	-0.0010	0.0700	-0.1052	0.0598
1.6	-0.2426	0.0595	-0.0007	0.0799	-0.1648	0.0679
1.8	-0.3248	0.0664	0.0003	0.0896	-0.2240	0.0777
2.0	-0.4082	0.0743	0.0019	0.0994	-0.2838	0.0879

size is clearly above the nominal size for the smaller sample sizes and takes at least 800 observations to approach the nominal size of 0.05. There is also evidence that the test is biased for small samples, with power less than size for $T = 200$ and $\alpha_0 = 1.2$. The test does show reasonable power as α_0 increases further away from 1 and also as T increases.

10.5.2 Level Effect in Interest Rates

Consider again the discretized version of the CKLS model of the short-term interest rate that was estimated by quasi-maximum likelihood in Section 9.7

$$r_{t+1} - r_t = \alpha_0 + \beta_0 r_t + \sigma_0 r_t^{\gamma_0} z_{t+1}, \quad (10.51)$$

where r_t is the interest rate, $z_t \sim iid N(0, 1)$, and $\theta = \{\alpha, \beta, \sigma, \gamma\}$ are parameters. This model (10.51) implies the conditional first moment.

$$E_t[r_{t+1} - r_t] = \alpha_0 + \beta_0 r_t, \quad (10.52)$$

Table 10.7

Empirical sizes for t-tests with $T = 200$ under misspecification. Based on a gamma distribution with shape parameter α_0 and scale parameter $\beta_0 = 1$.

α_0	t_{ML}	t_1	t_2^Q	t_2
1.0	0.0539	0.0537	0.0638	0.0600
1.2	0.2827	0.0537	0.1237	0.1311
1.4	0.6601	0.0537	0.2775	0.3117
1.6	0.8737	0.0537	0.4602	0.5146
1.8	0.9567	0.0537	0.6241	0.6781
2.0	0.9840	0.0537	0.7343	0.7828

Table 10.8

Rejection frequencies for the J_{HS} test. Based on a gamma distribution with shape parameter α_0 and scale parameter $\beta_0 = 1$.

α_0	$T = 200$	$T = 400$	$T = 800$
		Size	
1.0	0.1298	0.0979	0.0755
		Power	
1.2	0.0761	0.2264	0.5410
1.4	0.3717	0.7903	0.9889
1.6	0.7279	0.9829	0.9998
1.8	0.9159	0.9987	1.0000
2.0	0.9743	0.9990	1.0000

since $E_t[\sigma r_t^\gamma z_{t+1}] = \sigma r_t^\gamma E_t[z_{t+1}] = 0$. It also exhibits heteroskedasticity because

$$\begin{aligned}
 \text{var}_t[r_{t+1} - r_t] &= E_t[(r_{t+1} - r_t - E_t[r_{t+1} - r_t])^2] \\
 &= E_t[(\sigma_0 r_t^{\gamma_0} z_{t+1})^2] \\
 &= \sigma_0^2 r_t^{2\gamma_0} E_t[z_{t+1}^2] \\
 &= \sigma_0^2 r_t^{2\gamma_0}.
 \end{aligned} \tag{10.53}$$

That is, the variance of the change in the interest rate varies with the level of the interest rate. The strength of this relationship is determined by the parameter γ_0 .

Defining

$$u(r_{t+1}, r_t; \theta) = \begin{bmatrix} r_{t+1} - r_t - \alpha - \beta r_t \\ (r_{t+1} - r_t - \alpha - \beta r_t)^2 - \sigma^2 r_t^{2\gamma} \end{bmatrix},$$

allows (10.52) and (10.53) to be expressed as

$$E_t[u(r_{t+1}, r_t; \theta_0)] = E[u(r_{t+1}, r_t; \theta_0)|w_t] = 0, \quad (10.54)$$

with w_t representing the information set of all relevant variables available up to and including time t . This conditional expectation has the general form (10.11). Using the law of iterated expectations gives

$$E[u(r_{t+1}, r_t; \theta_0)w_t'] = E[E[u(r_{t+1}, r_t; \theta_0)|w_t]w_t'] = 0,$$

implying a GMM model of the form

$$m(r_{t+1}, r_t; \theta) = u(r_{t+1}, r_t; \theta)w_t'.$$

Letting the instrument set be $w_t = \{1, r_t\}$ and writing the model in vector form gives

$$m(r_t, r_{t-1}; \theta) = \begin{bmatrix} r_t - r_{t-1} - \alpha - \beta r_{t-1} \\ (r_t - r_{t-1} - \alpha - \beta r_{t-1})r_{t-1} \\ (r_t - r_{t-1} - \alpha - \beta r_{t-1})^2 - \sigma^2 r_{t-1}^{2\gamma} \\ ((r_t - r_{t-1} - \alpha - \beta r_{t-1})^2 - \sigma^2 r_{t-1}^{2\gamma})r_{t-1} \end{bmatrix}.$$

As the number of moment conditions equals the number of parameters, the parameters in θ are just identified.

The results of estimating the interest rate model by GMM are given in Table 10.9. The interest rates are monthly U.S. zero coupon yields beginning December 1946 and ending February 1991. Equation (10.54) shows that $u(r_{t+1}, r_t; \theta_0)$ is a martingale difference, and hence not autocorrelated, so $P = 0$ is chosen in the weighting matrix estimator. The results show that there is a strong level effect in U.S. interest rates which changes over the maturity of the asset. The parameter estimates of γ increase in magnitude as maturity increases from 0 to 6 months, reaching a peak at 6 months, and then tapering off thereafter.

Table 10.9

GMM estimates of interest rate models with level effects. Standard errors in parentheses are based on the heteroskedasticity and autocorrelation consistent weighting matrix with $P = 0$.

Maturity	α	β	σ	γ
0 months	0.184 (0.048)	-0.038 (0.013)	0.204 (0.033)	0.852 (0.079)
1 month	0.106 (0.048)	-0.020 (0.013)	0.049 (0.020)	1.352 (0.201)
3 months	0.090 (0.049)	-0.015 (0.012)	0.035 (0.019)	1.425 (0.254)
6 months	0.089 (0.050)	-0.015 (0.012)	0.026 (0.016)	1.533 (0.307)
9 months	0.091 (0.049)	-0.015 (0.011)	0.027 (0.016)	1.517 (0.296)
10 years	0.046 (0.026)	-0.006 (0.006)	0.029 (0.007)	1.178 (0.120)

Maturity	α	β	σ	γ
0 months	0.184 (0.064)	-0.038 (0.018)	0.204 (0.039)	0.852 (0.106)
1 month	0.106 (0.059)	-0.020 (0.016)	0.049 (0.018)	1.352 (0.188)
3 months	0.090 (0.054)	-0.015 (0.014)	0.035 (0.017)	1.425 (0.231)
6 months	0.089 (0.057)	-0.015 (0.014)	0.026 (0.015)	1.533 (0.283)
9 months	0.091 (0.056)	-0.015 (0.013)	0.027 (0.015)	1.517 (0.278)
10 years	0.046 (0.027)	-0.006 (0.006)	0.029 (0.007)	1.178 (0.114)

10.6 Exercises

(1) Method of Moments Estimation

Gauss file(s) `gmm_table.g`

Matlab file(s) `gmm_table.m`

This exercise is based on the data in Table 10.1.

- (a) Estimate μ and σ^2 using the moment conditions

$$m(y_t; \mu, \sigma^2) = \begin{bmatrix} y_t - \mu \\ y_t^2 - \sigma^2 - \mu^2 \end{bmatrix}.$$

- (b) Suppose that the moment conditions come from a Student t distribution.

- (i) Estimate μ and ν using the moment conditions

$$m(y_t; \mu, \nu) = \begin{bmatrix} y_t - \mu \\ (y_t - \mu)^2 - \frac{\nu}{\nu - 2} \end{bmatrix}.$$

- (ii) Estimate μ and ν using the moment conditions

$$m(y_t; \mu, \nu) = \begin{bmatrix} y_t - \mu \\ (y_t - \mu)^4 - \frac{3\nu^2}{(\nu - 2)(\nu - 4)} \end{bmatrix}.$$

Compare the point estimates of ν .

- (c) Suppose that the moment conditions come from a gamma distribution.

- (i) Estimate α and β using the moment conditions

$$m(y_t; \alpha, \beta) = \begin{bmatrix} y_t - \frac{\alpha}{\beta} \\ \frac{1}{y_t} - \frac{\beta}{\alpha - 1} \end{bmatrix}.$$

- (ii) Estimate α and β using the moment conditions

$$m(y_t; \alpha, \beta) = \begin{pmatrix} y_t - \frac{\alpha}{\beta} \\ y_t^2 - \frac{\alpha(\alpha + 1)}{\beta^2} \end{pmatrix}.$$

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T d_{1,t} &= m_1 - E[y_t] = \frac{1}{T} \sum_{t=1}^T y_t - \frac{\alpha}{\beta} \\ \frac{1}{T} \sum_{t=1}^T d_{2,t} &= m_2 - E[y_t^2] = \frac{1}{T} \sum_{t=1}^T y_t^2 - \frac{\alpha(\alpha + 1)}{\beta^2}. \end{aligned}$$

Compare the two sets of point estimates for α and β .

(2) The Relationship Between GMM and OLS

Gauss file(s)	<code>gmm.g, gmm_opt.g</code>
Matlab file(s)	<code>gmm.m, gmm_opt.m</code>

Consider the linear model

$$y_t = \beta_0 + \beta_1 x_t + u_t.$$

- (a) Estimate the model by OLS.
- (b) Estimate the model by GMM based on the instruments $w_t = \{1, x_t\}$.
- (c) Compare the GMM point estimates of the two procedures with the point estimates obtained from estimating the model by *OLS*.

(3) Graphical Demonstration of Consistency

Gauss file(s)	<code>gmm_consistency.g</code>
Matlab file(s)	<code>gmm_consistency.m</code>

- (a) Simulate $T = 100000$ observations from the gamma distribution with parameter $\alpha_0 = 10$ and compute the GMM objective function

$$Q_T(\alpha) = M_T'(\alpha) W_T^{-1}(\alpha) M_T(\alpha)$$

for values of $\alpha = \{4, \dots, 16\}$ using the following moments

$$E[y_t] = \alpha_0, \quad E[y_t^2] = \alpha_0(\alpha_0 + 1), \quad E\left[\frac{1}{y_t}\right] = \frac{1}{\alpha_0 - 1}.$$

- (b) Repeat part (a) for $T = 10, 25, 50, 100, 200, 400$ and discuss the consistency property of the GMM estimator.
- (c) Repeat parts (a) and (b) just using $E[y_t] = \alpha_0$ and $E[y_t^2] = \alpha_0(\alpha_0 + 1)$.
- (d) Repeat parts (a) and (b) just using $E[y_t] = \alpha_0$.

(4) Estimating a Gamma Distribution

Gauss file(s)	<code>gmm_gamma.g</code>
Matlab file(s)	<code>gmm_gamma.m</code>

The first two uncentered moments of the gamma distribution are

$$E[y_t] = \frac{\alpha_0}{\beta_0}, \quad E[y_t^2] = \frac{\alpha_0(\alpha_0 + 1)}{\beta_0^2}.$$

- (a) Using the data in Table 10.1 and $\theta_{(0)} = \{\alpha_{(0)} = 8, \beta_{(0)} = 2\}$ compute the following

$$M_T(\theta_{(0)}), W_T(\theta_{(0)}), Q_T(\theta_{(0)}), \frac{\partial M_T(\theta_{(0)})}{\partial \theta'}, \frac{\partial Q_T(\theta_{(0)})}{\partial \theta}, \frac{\partial^2 Q_T(\theta_{(0)})}{\partial \theta \partial \theta'}$$

where the derivatives are computed numerically. Hence compute the Newton-Raphson update

$$\theta_{(1)} = \theta_{(0)} - \left[\frac{\partial^2 Q_T(\theta_{(0)})}{\partial \theta \partial \theta'} \right]^{-1} \left[\frac{\partial Q_T(\theta_{(0)})}{\partial \theta} \right]$$

- (b) Iterate until convergence to find the GMM parameter estimates and the corresponding standard errors.
 (c) Repeat parts (a) and (b) for the Student t distribution

$$f(y) = \frac{\Gamma[(\nu+1)/2]}{\sqrt{\pi\nu}\Gamma[\nu/2]} \left[1 + \frac{(y-\mu)^2}{\nu} \right]^{-(\nu+1)/2},$$

using the following two moments

$$E[y_t] = \mu_0, \quad E[y_t^2] = \frac{\nu_0}{\nu_0 - 2}.$$

(5) **The Consumption Based Capital Asset Pricing Model (C-CAPM)**

Gauss file(s)	<code>gmm_ccapm.g</code> , <code>ccapm.dat</code>
Matlab file(s)	<code>gmm_ccapm.m</code> , <code>ccapm.mat</code>

The data are 238 observations on the real U.S. consumption ratio, *CRATIO*, the real Treasury bill rate, *R*, and the real value weighted returns *E*. This is the adjusted Hansen and Singleton (1982) data set used in their original paper.

- (a) Consider the first order condition of the C-CAPM

$$E_t[\beta_0(c_{t+1}/c_t)^{-\gamma_0}(1+r_{t+1})-1] = 0, \quad (10.55)$$

where c_t is real consumption and r_t is the real interest rate. The parameters are the discount factor, β , and the relative risk aversion coefficient γ . Estimate the parameters by GMM using $w_t = \{1, c_t/c_{t-1}\}$ as instruments. Let the starting values be $\beta = 0.9$ and $\gamma = 2$.

- (i) Interpret the parameter estimates.
 (ii) Compute the value of the GMM objective function $Q_T(\hat{\theta})$ and interpret the result. Can you test the number of over-identifying restrictions in this case?

- (b) Re-estimate the model by GMM using $w_t = \{1, c_t/c_{t-1}, r_t\}$ as instruments.
 - (i) Interpret the parameter estimates.
 - (ii) Test the number of over-identifying restrictions and interpret the result.
- (c) Repeat part (b) using the instrument set $w_t = \{1, c_t/c_{t-1}, r_t, e_t\}$, where e_t is the real value weighted returns on equities.

(6) Modelling Contagion in the Asian Crisis

Gauss file(s)	<code>gmm_contagion.g</code> , <code>contagion.dat</code>
Matlab file(s)	<code>gmm_contagion.m</code> , <code>contagion.mat</code>

The data file contains daily data on the exchange rates of the following seven countries: South Korea; Indonesia; Malaysia; Japan; Australia; New Zealand; and Thailand. The sample period is 2 June 1997 to 31 August 1998, a total of 319 observations. Let $s_{i,t}$ represent the exchange rate of the i^{th} country at time t . For each country, compute the exchange rate return $r_{i,t} = \ln(s_{i,t}) - \ln(s_{i,t-1})$ and ensure that the returns have zero mean by computing $e_{i,t} = r_{i,t} - \bar{r}_i$.

- (a) Plot $e_{i,t}$ for each country. Describe the time-series properties of the exchange rates.
- (b) Compute descriptive statistics of $e_{i,t}$ for each country. Describe the statistical properties of the exchange rates.
- (c) Estimate the model

$$e_{i,t} = \lambda_i F_{1,t} + \theta F_{2,t} + \phi_i u_{i,t} + \gamma_i u_{7,t}, \quad i = 1, 2, \dots, 7,$$

by GMM with $\gamma_7 = 0$ and where the factors $\{F_{1,t}, F_{2,t}, u_{1,t}, \dots, u_{7,t}\}$ are all *iid*.

- (d) For each country, estimate the proportion of volatility arising from
 - (i) The variable common shock, V_t ,

$$\frac{\hat{\lambda}_i^2}{\hat{\lambda}_i^2 + \hat{\theta}^2 + \hat{\phi}_i^2 + \hat{\gamma}_i^2} \quad i = 1, 2, \dots, 7$$

- (ii) The fixed common shock, F_t ,

$$\frac{\hat{\theta}^2}{\hat{\lambda}_i^2 + \hat{\theta}^2 + \hat{\phi}_i^2 + \hat{\gamma}_i^2}, \quad i = 1, 2, \dots, 7$$

- (iii) The idiosyncratic shock, $u_{i,t}$,

$$\frac{\widehat{\phi}_i^2}{\widehat{\lambda}_i^2 + \widehat{\theta}^2 + \widehat{\phi}_i^2 + \widehat{\gamma}_i^2}, \quad i = 1, 2, \dots, 7$$

- (iv) Contagion, $u_{7,t}$,

$$\frac{\widehat{\gamma}_i^2}{\widehat{\lambda}_i^2 + \widehat{\theta}^2 + \widehat{\phi}_i^2 + \widehat{\gamma}_i^2}, \quad i = 1, 2, \dots, 6.$$

- (e) Discuss ways how the model can be improved. *Hint: review the descriptive statistics computed above.*

(7) Level Effects in U.S. Interest Rates

Gauss file(s)	<code>gmm_level.g</code> , <code>level.dat</code>
Matlab file(s)	<code>gmm_level.m</code> , <code>level.mat</code>

The data are monthly and cover the period December 1946 to February 1991. The zero coupon bonds have maturities of 0, 1, 3, 6, 9 months and 10 years.

- (a) For each yield estimate the following interest rate equation by GMM

$$r_{t+1} - r_t = \alpha + \beta r_t + \sigma r_t^\gamma z_{t+1},$$

where z_t is *iid* $(0, 1)$, and the instrument set is $w_t = \{1, r_t\}$;

- (b) For each yield test the following restrictions: $\gamma = 0.0$, $\gamma = 0.5$ and $\gamma = 1.0$.
 (c) If the level effect model of the interest rate captures time-varying volatility, and $\alpha, \beta \simeq 0$, then

$$E \left[\left(\frac{r_{t+1} - r_t}{r_t^\gamma} \right)^2 \right] \simeq \sigma^2.$$

Plot the series

$$\frac{r_{t+1} - r_t}{r_t^\gamma},$$

for $\gamma = 0.0, 0.5, 1.0, 1.5$, and discuss the properties of the series.

(8) Monte Carlo Evidence for the Gamma model

Gauss file(s)	<code>gammasim.g</code>
Matlab file(s)	<code>gammasim.m</code>

Carry out Monte Carlo simulations of the GMM estimator of the shape parameter of a gamma distribution as discussed in section 10.5.1.

- (a) Let y_t have an *iid* gamma distribution for $t = 1, \dots, T$ with shape parameter α_0 and scale parameter $\beta_0 = 1$. Use this as the data generating process with $\alpha_0 = 1, 2, 3, 4, 5$ and $T = 50, 100, 200$ to investigate the finite sample bias and variance properties of
 - (i) the maximum likelihood estimator of α_0 with $\beta_0 = 1$ known,
 - (ii) the GMM estimator of α_0 based on the first moment of the gamma distribution,
 - (iii) the GMM estimator of α_0 based on the first two moments of the gamma distribution.
- (b) Using the same data generating process, investigate the finite sample size properties of the t tests based on these three estimators.
- (c) For $T = 200$, use $\alpha_0 \in \{1.05, 1.10, \dots, 1.30\}$ as the data generating process to investigate the power properties of the three tests.
- (d) Using an appropriate range of α_0 values, extend the power analysis to $T = 400$.
- (e) Let the data generating process be an exponential distribution with parameter α_0 . Using $\alpha_0 \in \{1.0, 1.2, \dots, 2.0\}$ investigate the finite sample bias and variance properties of the three estimators based on the now misspecified maximum likelihood and GMM models based on the gamma distribution for $T = 100, 200, 400$.
- (f) For the same exponential data generating process, investigate the finite sample size properties of the associated t tests for $T = 100, 200, 400$.
- (g) For the same exponential data generating process, investigate the finite sample size and power properties of the J test for misspecification for $T = 100, 200, 400$.
- (h) Extend the analysis to also include a GMM estimator based on the first two moments of the gamma distribution as well as the additional moment $E[1/y_t] = 1/(\alpha_0 - 1)$ introduced in Example 10.3.

(9) **Risk Aversion and the Equity Premium Puzzle**

Gauss file(s)	<code>gmm_risk_aversion.g</code> , <code>equity_mp.dat</code>
Matlab file(s)	<code>gmm_risk_aversion.m</code> , <code>equity_mp.mat</code>

In this exercise the risk aversion parameter is estimated by GMM using the data originally used by Mehra and Prescott (1985) in their work on the equity premium puzzle. The data are annual for the period 1889 to 1978, a total of 91 observations on the following U.S. variables: the real stock price, S_t ; real dividends, D_t ; real per capita consumption, C_t ; the

nominal risk free rate on bonds, expressed as a per annum percentage, R_t ; and the price of consumption goods, P_t .

- (a) Compute the following returns series for equities, bonds and consumption respectively

$$R_{s,t+1} = \frac{S_{t+1} + D_t - S_t}{S_t}$$

$$R_{b,t+1} = (1 + R_t) \left(\frac{P_t}{P_{t+1}} \right) - 1$$

$$R_{c,t+1} = \frac{C_{t+1} - C_t}{C_t}.$$

- (b) Consider the first order conditions of the C-CAPM model

$$E_t[\beta(1 + R_{c,t+1})^{-\gamma}(1 + R_{b,t+1}) - 1] = 0$$

$$E_t[\beta(1 + R_{c,t+1})^{-\gamma}(1 + R_{s,t+1}) - 1] = 0$$

where the parameters are the discount factor (β) and the relative risk aversion coefficient (γ). Estimate the parameters by *GMM* using the non-optimal weighting matrix $W_T = I$, with the following set of instruments, $w_t = \{1, R_{b,t}, R_{s,t}\}$.

- (i) Interpret the parameter estimates.
 - (ii) Compute the value of the objective function $Q_T(\hat{\theta})$ and test the over-identifying restrictions.
- (c) Re-estimate the model using the optimal weighting matrix with a lag window of $P = 0$, and the same instrument set as in part (b).
- (i) Interpret the parameter estimates.
 - (ii) Compute the value of the objective function $Q_T(\hat{\theta})$ and test the over-identifying restrictions.
- (d) Re-estimate the model using the optimal weighting matrix with a lag window of $P = 0$, but with an extended instrument set given by $w_t = \{1, R_{b,t}, R_{s,t}, R_{c,t}\}$. Compare the parameter estimates across the different models and discuss the robustness properties of these estimates.

11

Nonparametric Estimation

11.1 Introduction

Earlier chapters in Part THREE of the book explore circumstances in which the likelihood function is misspecified. The case of incorrectly specifying the entire distribution is considered in Chapter 9, while Chapter 10 deals with the situation in which the form of the distribution is unknown but the functional form of the moments of the distribution are known. In this chapter, the assumptions underlying the model are relaxed even further to the extent that neither the distribution of the dependent variable nor the functional form of the moments of the distribution are specified. Estimation is nonparametric in the sense that the conditional moments of the dependent variable, y_t , at each point in time, are estimated conditional on a set of explanatory variables, x_t , without specifying the functional form of this relationship. Instead a set of smoothness conditions linking the dependent variable y_t and the explanatory variables x_t are imposed. This estimator is known as the Nadaraya-Watson kernel estimator of the relevant conditional moment.

The cost of using a nonparametric estimator is that the rate of convergence of the estimator to the true model is slower (less than $\sqrt{T}$) than the case where a parametric model is correctly specified ($\sqrt{T}$). Intuitively, this slower rate of convergence is the cost of providing less information about the model's structure than would be the case when using parametric estimation methods. The gain however, is that if the parametric form of the model is misspecified then the maximum likelihood estimator is likely to be biased and inconsistent while the nonparametric estimator is still consistent.

Prior to dealing with the nonparametric estimation of regression models, however, it is necessary to establish a number of preliminary results by discussing briefly the problem of estimating an unknown probability density function of a random variable using only sample information. The formal

results given in this chapter concerning both density estimation and non-parametric regression are initially presented for observations that are independently and identically distributed, which are then generalized to non *iid* observations.

11.2 The Kernel Density Estimator

The problem is to estimate the unknown probability density function, $f(y)$, using only the information available in a sample of T observations $y_t = \{y_1, y_2, \dots, y_T\}$. A simple, but useful first approximation to $f(y)$ is the histogram. A simple algorithm for constructing a histogram is as follows.

- Step 1:** Order the y_t data from lowest to highest.
Step 2: Construct a grid of values of y starting at the lowest value of y_t and ending at the highest value of y_t . Let the distance between the y values be h , where h is known as the bandwidth. This interval is also commonly referred to as the bin.
Step 3: Choose the first value of y and count the number of ordered y_t values falling in the interval $(y - h/2, y + h/2)$.
Step 4: Repeat the exercise for the next value of y in the grid and continue until the last value in the grid.

Formally, the histogram estimator of $f(y)$, for a sample of size T , is

$$\begin{aligned}\hat{f}(y) &= \frac{1}{Th} [\text{number of observations in the interval } (y - h/2, y + h/2)] \\ &= \frac{1}{Th} \sum_{t=1}^T \mathcal{I}\left(y - \frac{h}{2} \leq y_t \leq y + \frac{h}{2}\right) = \frac{1}{Th} \sum_{t=1}^T \mathcal{I}\left(\left|\frac{y - y_t}{h}\right| \leq \frac{1}{2}\right),\end{aligned}$$

where

$$\mathcal{I}(\cdot) = \begin{cases} 1 : & \left|\frac{y - y_t}{h}\right| \leq \frac{1}{2} \\ 0 : & \left|\frac{y - y_t}{h}\right| > \frac{1}{2}, \end{cases} \quad (11.1)$$

is the indicator function. This estimator can be written as

$$\hat{f}(y) = \frac{1}{T} \sum_{t=1}^T w_t, \quad w_t = \frac{1}{h} \mathcal{I}\left(\left|\frac{y - y_t}{h}\right| \leq \frac{1}{2}\right),$$

which emphasises that the histogram is the sample mean of the weights when evaluated at the value, y . This estimate of the density is in essence a local histogram where the width of the bin, h , controls the amount of smoothing.

The main disadvantage with this approach is that the weights, w_t , are discontinuous, switching from 1 to 0 immediately wherever $|(y_t - y)/h| > 1/2$. This in turn, leads to an estimator of $f(y)$ that is jagged. A natural solution is to replace the indicator function with a smooth function. This is the kernel density solution because the chosen smoothing function is known as the kernel. In selecting the kernel function, the approach is to choose kernels that are non-negative functions and which enclose unit probability mass, thereby satisfying automatically the basic prerequisites of a probability distribution. The kernel estimate of a density therefore has the generic form

$$\hat{f}(y) = \frac{1}{Th} \sum_{t=1}^T K\left(\frac{y - y_t}{h}\right) = \frac{1}{T} \sum_{t=1}^T w_t, \quad w_t = \frac{1}{h} K\left(\frac{y - y_t}{h}\right), \quad (11.2)$$

where h is a window length controlling the averaging, known as the kernel bandwidth and $K(\cdot)$ which is a suitable non-negative function defined over the required interval and encloses unit mass. As emphasized by the expression in (11.2), the density estimator is interpreted as a kernel with bandwidth h placed on all the data points with the resultant density mass at y being averaged to obtain $\hat{f}(y)$. It follows that the greater is the number of observations near y , the greater is the probability mass computed at this point.

Table 11.1

Commonly-used kernel functions, $z = (y - y_t)/h$.

Kernel	$K(z)$	Range
Uniform	1	$I(z \leq 1/2)$
Triangle	$(1 - z)$	$I(z \leq 1)$
Epanechnikov	$\frac{3}{4}(1 - z^2)$	$I(z \leq 1)$
Biweight	$\frac{15}{16}(1 - z^2)^2$	$I(z \leq 1)$
Triweight	$(1 - z^2)^3$	$I(z \leq 1)$
Gaussian	$\frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}z^2)$	$-\infty < z < \infty$

Table 11.1 presents some commonly used kernel functions, of which the Gaussian kernel is the most commonly used. All of the tabulated kernel

functions have the following properties:

$$\begin{aligned} \int_{-\infty}^{\infty} K(z_t) dz_t &= 1 & \int_{-\infty}^{\infty} z_t K(z_t) dz_t &= 0 \\ \int_{-\infty}^{\infty} z_t^2 K(z_t) dz_t &= \mu_2 & \int_{-\infty}^{\infty} z_t^3 K(z_t) dz_t &= 0, \end{aligned} \quad (11.3)$$

where the variance of the kernel, μ_2 , is a positive constant whose actual value depends on the choice of the kernel function. Further, all the higher odd-order moments are zero by virtue of the fact that the kernel is a symmetric function. As will become clear, the choice of kernel function is not particularly crucial to the quality of the density estimate. The most important factor is the choice of the bandwidth, h .

A simple algorithm for computing a kernel estimate of the density, $f(y)$, is as follows:

- Step 1:** Order the y_t data from lowest to highest.
- Step 2:** Construct a grid of values of y starting at the lowest value of y_t and ending at the highest value of y_t .
- Step 3:** Choose the first value of y and compute $f(y)$ as in equation (11.2).
- Step 4:** Repeat for the next value of y in the grid and continue until the last value in the grid.

Example 11.1 Numerical Illustration

Consider the following $T = 20$ ordered observations, given in the first column of Table 11.2, that are drawn from an unknown distribution $f(y)$. The density is estimated at three points $y = \{5, 10, 15\}$ using a Gaussian kernel and two bandwidths $h = \{1, 2\}$. The weights w_t associated with each observation for $y = 5$ and $h = 1$, are given in the second column. For example, for the first ordered observation $y_1 = 2$ the weight is computed as

$$\begin{aligned} w_1 &= \frac{1}{h} K\left(\frac{y - y_1}{h}\right) = \frac{1}{h} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{y - y_1}{h}\right)^2\right) \\ &= \frac{1}{1} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{5 - 2}{1}\right)^2\right) = 0.004. \end{aligned}$$

For the second ordered observation

$$\begin{aligned} w_2 &= \frac{1}{h} K\left(\frac{y - y_2}{h}\right) = \frac{1}{h} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{y - y_2}{h}\right)^2\right) \\ &= \frac{1}{1} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{5 - 3}{1}\right)^2\right) = 0.054. \end{aligned}$$

This procedure is repeated for all $T = 20$ observations, with the kernel

Table 11.2

Kernel density estimation example based on a Gaussian kernel with bandwidths $h = 1, 2$, and a sample of size $T = 20$. The grid points selected are $y = \{5, 10, 15\}$.

y_t (ordered)	w_t ($h = 1$)			w_t ($h = 2$)		
	$y = 5$	$y = 10$	$y = 15$	$y = 5$	$y = 10$	$y = 15$
2.000	0.004	0.000	0.000	0.065	0.000	0.000
3.000	0.054	0.000	0.000	0.121	0.000	0.000
5.000	0.399	0.000	0.000	0.199	0.009	0.000
7.000	0.054	0.004	0.000	0.121	0.065	0.000
8.000	0.004	0.054	0.000	0.065	0.121	0.000
8.000	0.000	0.054	0.000	0.065	0.121	0.000
9.000	0.000	0.242	0.000	0.027	0.176	0.002
9.000	0.000	0.242	0.000	0.027	0.176	0.002
10.000	0.000	0.399	0.000	0.009	0.199	0.009
10.000	0.000	0.399	0.000	0.009	0.199	0.009
11.000	0.000	0.242	0.000	0.002	0.176	0.027
11.000	0.000	0.242	0.000	0.002	0.176	0.027
11.000	0.000	0.242	0.000	0.002	0.176	0.027
12.000	0.000	0.054	0.004	0.000	0.121	0.065
12.000	0.000	0.054	0.004	0.000	0.121	0.065
14.000	0.000	0.000	0.242	0.000	0.027	0.176
15.000	0.000	0.000	0.399	0.000	0.009	0.199
17.000	0.000	0.000	0.054	0.000	0.000	0.121
18.000	0.000	0.000	0.004	0.000	0.000	0.065
20.000	0.000	0.000	0.000	0.000	0.000	0.009
$\hat{f}(y) = \frac{1}{T} \sum_{t=1}^T w_t :$	0.026	0.111	0.035	0.036	0.094	0.040

estimate at $y = 5$, given by the sample average of the weights

$$\hat{f}(5) = \frac{1}{20} \sum_{t=1}^{20} w_t = \frac{0.004 + 0.054 + 0.399 + 0.054 \cdots 0.000}{20} = 0.026,$$

which is given in the last row of the second column in Table 11.2. Columns 3 and 4 repeat the calculations for $y = 10$ and $y = 15$ with $h = 1$, yielding the respective estimates of 0.111 and 0.035. Of course, the kernel density estimator can be evaluated at a finer number of grid points.

The last three columns of Table 11.2 repeat the calculations for a wider bandwidth $h = 2$, but using the same values of y . Comparing the two sets of estimates shows that the distribution based on $h = 1$ yields a more peaked distribution than is the case when $h = 2$ is used. Intuitively this is due to the fact that larger bandwidths assign relatively greater weight to observations

further away from the grid points $y = \{5, 10, 15\}$, thus creating smoother estimates of $f(y)$ than those obtained using smaller bandwidths. $\square$

The kernel estimator generalizes naturally to the deal with the multi-variate case. The problem in multiple dimensions is to approximate the N -dimensional probability density function evaluated at $f(y_1, \dots, y_N)$ given the matrix of observations

$$y_{t,n} = \begin{bmatrix} y_{1,1} & y_{1,2} & \cdots & y_{1,n} \\ y_{2,1} & y_{2,2} & \cdots & y_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ y_{T,1} & y_{T,2} & \cdots & y_{T,N} \end{bmatrix}.$$

The multivariate kernel density estimator in the N -dimensional case is defined as

$$\hat{f}(\mathbf{y}) = \frac{1}{T} \sum_{t=1}^T \frac{1}{h_1, \dots, h_N} K\left(\frac{y_1 - y_{t,1}}{h_1}, \dots, \frac{y_N - y_{t,N}}{h_N}\right),$$

where K denotes a multivariate kernel and where it is assumed that the bandwidth h is in fact a vector of bandwidths $h_1, \dots, h_N$. In practice the form of this multivariate kernel function $K(\mathbf{y})$ is simplified to the so-called product kernel estimator of the multivariate density

$$\hat{f}(y_1, \dots, y_N) = \frac{1}{Th_1, \dots, h_N} \sum_{t=1}^T \left\{ \prod_{n=1}^N K\left(\frac{y_n - y_{t,n}}{h_n}\right) \right\},$$

which amounts to using the same univariate kernel in each dimension but with a different smoothing parameter for each dimension. A major difficulty with multidimensional estimation however, is that very large sample sizes are needed to obtain accurate estimates of the density. Yatchew (2003) provides a good discussion of the curse of dimensionality .

11.3 Properties of the Kernel Density Estimator

The properties of the kernel estimator, $f(y)$, outlined in this section are based on the assumption of *iid* sample data. Generalizations to dependent data are discussed at the end of the section. The derivations of some of the properties are presented in Appendix D along with some other useful results.

11.3.1 Finite Sample Properties

Bias

The bias of the kernel estimator, $\hat{f}(y)$, is given by

$$\text{bias}(\hat{f}(y)) = E[\hat{f}(y)] - f(y) = \frac{h^2}{2} f^{(2)}(y) \mu_2 + O(h^4), \quad (11.4)$$

where

$$f^{(k)}(y) = \frac{d^k f(y)}{dy^k}.$$

This expression shows that the bias is *positively* related to the bandwidth h and disappears as $h \rightarrow 0$.

Variance

The variance of the kernel estimator, $\hat{f}(y)$, is given by

$$\text{var}(\hat{f}(y)) = \frac{1}{Th} f(y) \int_{-\infty}^{\infty} K^2(z) dz + o\left(\frac{1}{Th}\right). \quad (11.5)$$

By contrast with the expression for the bias, the variance is *negatively* related to the bandwidth h with the variance disappearing as $Th \rightarrow \infty$.

Equations (11.4) and (11.5) show that there is an inverse relationship between bias and variance for different values of h , as illustrated in the following example.

Example 11.2 Bias, Variance and Bandwidth

The effect on bias of a ‘large’ bandwidth, $h = 1$, is demonstrated in panel (a) of Figure 11.1 for the case where the true distribution is standard normal. The kernel estimator underestimates the peak (biased downwards) and overestimates the tails (biased upwards) because of over-smoothing. Panel (b) of Figure 11.1 shows that for a ‘smaller’ bandwidth, $h = 0.1$, the bias is considerably smaller in the peak and the tails of the distribution. The cost of the smaller bandwidth is to generate greater variance, as represented by the jagged behaviour of the kernel estimate. This contrasts with panel (a) where the variance of the kernel estimator is lower. $\square$

11.3.2 Optimal Bandwidth Selection

The aim of data-driven bandwidth selection is to choose h to resolve the trade-off between bias and variance in an optimal way. The approach is to choose h to minimize the asymptotic mean integrated squared error of the

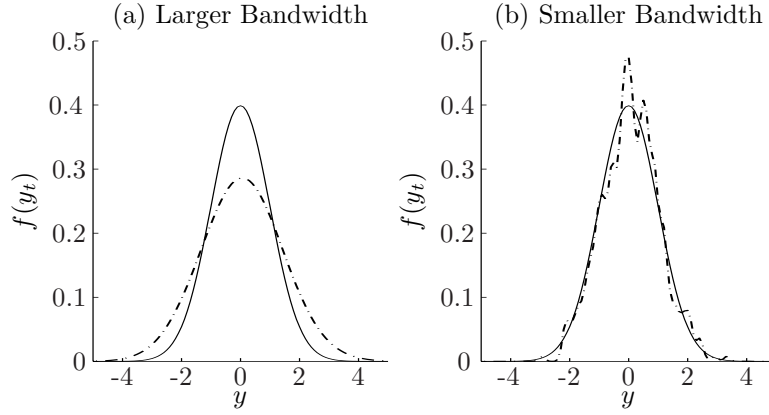


Figure 11.1 Bias and variance of the kernel density estimator (broken line) for different choices of bandwidth. Data are independent drawings from the standard normal distribution (continuous line)

density estimator, hereafter *AMISE*, obtained by combining expressions (11.4) and (11.5) as

$$AMISE = \int_{-\infty}^{\infty} (\text{bias}^2 + \text{var}(\hat{f}(y))) dy = \frac{h^4}{4} \mu_2^2 R(f^{(2)}(y)) + \frac{1}{Th} R(K),$$

where terms up to order h are retained and

$$R(K) = \int_{-\infty}^{\infty} K^2(y) dy,$$

represents the roughness of the kernel density $K(y)$ (see Appendix D).

The optimal bandwidth is obtained by minimizing the AMISE by differentiating this expression with respect to h

$$\frac{d}{dh} AMISE = h^3 \mu_2^2 R(f^{(2)}(y)) - \frac{1}{Th^2} R(K).$$

Setting this derivative to zero

$$0 = h_{opt}^3 \mu_2^2 R(f^{(2)}(y)) - \frac{1}{Th_{opt}^2} R(K),$$

and rearranging gives the optimal bandwidth

$$h_{opt} = \left[\frac{R(K)}{T \mu_2^2 R(f^{(2)}(y))} \right]^{1/5}. \quad (11.6)$$

The bandwidth depends positively on the roughness of the kernel function,

$R(K)$, and inversely on the sample size, T , the variance of the kernel function, μ_2 , and the roughness of the true distribution, $R(f^{(2)}(y))$. The optimal bandwidth can also be written as

$$h_{opt} = cT^{-1/5}, \quad c = \left[\frac{R(K)}{\mu_2^2 R(f^{(2)}(y))} \right]^{1/5}, \quad (11.7)$$

a useful form for examining the asymptotic properties of the kernel.

The expression for c in (11.7) involves the unknown function $R(f^{(2)}(y))$, the roughness of the second derivative of the true population distribution. The simplest solution is to assume that the population distribution, $f(y)$, is normal with unknown mean μ and variance σ^2 . From the properties of the Gaussian kernel,

$$R(f^{(2)}) = \frac{3}{8\sigma^5\sqrt{\pi}}, \quad \mu_2 = 1, \quad R(K) = \frac{1}{2\sqrt{\pi}},$$

and substituting these expressions into equation (11.7) gives

$$c = \left(\frac{1}{2\sqrt{\pi}} \frac{8\sigma^5\sqrt{\pi}}{3} \right)^{1/5} \simeq 1.06\sigma.$$

From equation (11.6) the optimal bandwidth is

$$h_{opt} = 1.06 \sigma T^{-1/5}. \quad (11.8)$$

Replacing σ by an unbiased estimator, s , the sample standard deviation, is an adequate strategy. Silverman (1986), however, suggests that the choice

$$\hat{\sigma} = \min \left[s, \frac{3IQR}{4} \right], \quad (11.9)$$

where IQR is the interquartile range of the data, is a more robust choice. Consequently, the normal-reference-rule or rule-of-thumb (ROT) bandwidth is

$$\tilde{h}_{rot} = 0.9 \hat{\sigma} T^{-1/5}. \quad (11.10)$$

This ROT bandwidth is based on a Gaussian kernel. Choosing an alternative kernel function will change the value of c in (11.7).

Example 11.3 FTSE All Ordinaries Index

A plot of the continuously compounded returns of the FTSE All Ordinaries share index is given in panel (a) of Figure 11.2. The returns are daily beginning on 20 November 1973 and ending 23 July 2001, a total of $T = 7000$ observations. The sample mean and standard deviation are respectively $\bar{y} = 0.000$ and $s = 0.011$, with a minimum daily return of -0.130

and a maximum daily return of 0.089. The kernel estimate of the distribution is presented in panel (b) of Figure 11.2 using a Gaussian kernel with bandwidth

$$h_{opt} = 1.06 s T^{-1/5} = 1.06 \times 0.011 \times 7000^{-1/5} = 0.002.$$

As all of the observations, apart from one, fall in the range -0.1 to 0.1 , this is chosen as the range of the grid points which increase in steps of 0.001 . For comparative purposes, a normal distribution with mean $\mu = 0$ and variance $\sigma^2 = 0.011^2$ is also plotted. The empirical distribution of y_t based on the kernel estimator exhibits leptokurtosis as the peak is higher than the normal distribution and the tails are relatively fatter. This is a standard empirical result for financial returns distributions. $\square$

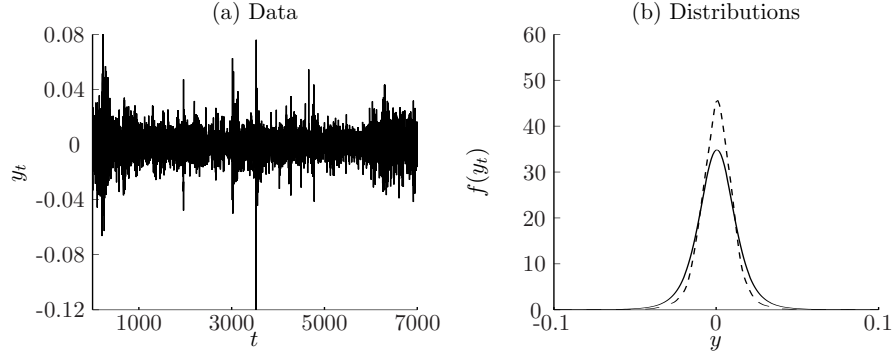


Figure 11.2 Daily returns on the FTSE, 1973 to 2001.in (a), and a comparison of the normal distribution (continuous line) and the kernel estimate of returns (broken line) in (b).

The choice of bandwidths for the N -dimensional multivariate product kernels are based on the same principles as in the univariate case. If a normal kernel is used, the bandwidth in each dimension that minimizes the multivariate AMISE is given by

$$h_{n,\text{rot}} = \left(\frac{4}{N+2} \right)^{1/(N+4)} \hat{\sigma}_n T^{-1/(N+4)}, \quad n = 1, 2, \dots, N, \quad (11.11)$$

where $\hat{\sigma}_n$ is the standard deviation of the n^{th} series $\{y_{1,n}, y_{2,n}, \dots, y_{T,n}\}$. The univariate bandwidth in (11.8) is recovered by setting $N = 1$ since

$$\left(\frac{4}{1+2} \right)^{1/5} = 1.0592 \approx 1.06.$$

Furthermore, when $N = 2$ the constant in equation (11.11) is exactly equal

to 1. As a consequence, Scott (1992, p152) suggests that an easy-to-remember rule for bandwidth computation in a multivariate setting is

$$h_{n,\text{rot}} = \hat{\sigma}_n T^{-1/(N+4)} . \quad (11.12)$$

Example 11.4 Bivariate Normal Distribution

Figure 11.3 contains a bivariate Gaussian kernel density estimated using a product kernel with bandwidth given by equation (11.12). The sample data are drawn from $N = 2$ independent standardized normal distributions and the sample size is $T = 20000$ observations.

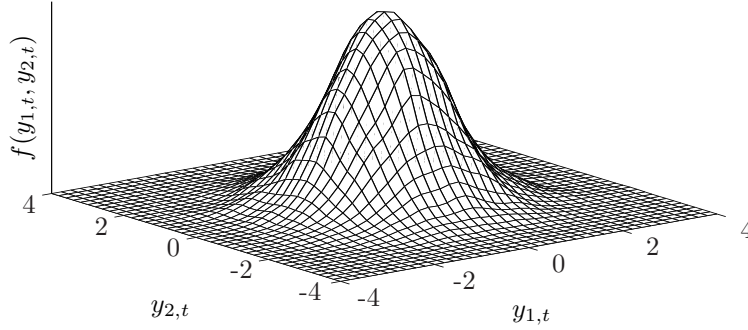


Figure 11.3 Product kernel estimate of the bivariate normal distribution.

□

11.3.3 Asymptotic Properties

To develop the properties of consistency and asymptotic normality of the kernel estimator, it is useful to write equation (11.7) in the form

$$h = cT^{-k}, \quad \text{where} \quad \begin{cases} k > 1/5 & \rightarrow \text{under : smoothing} \\ k = 1/5 & \rightarrow \text{optimal : smoothing} \\ k < 1/5 & \rightarrow \text{over : smoothing} . \end{cases} \quad (11.13)$$

Consistency

Consistency requires that both the bias and the variance approach zero as T increases so that

$$\text{plim}(\hat{f}(y)) = f(y) .$$

Recall from equations (11.4) and (11.5), that the bias decreases as $h \rightarrow 0$ and the variance decreases as $Th \rightarrow \infty$. Both of these conditions must be satisfied simultaneously for consistency to hold. This is clearly the case for the optimal bandwidth $h = cT^{-1/5}$ given in equation (11.7), but it is also true for other values of k in equation (11.13). If $k = 1$ in equation (11.13), then $h = cT^{-1}$ and $Th = TcT^{-1} = c \nrightarrow \infty$ as T grows so that the variance does not disappear. On the other hand, if $k = 0$ in equation (11.13), then h is the constant cT^0 and $h \nrightarrow 0$ as T grows so that the bias will not disappear. In other words, consistency requires that k satisfies $0 < k < 1$.

Asymptotic normality

The asymptotic distribution of the kernel estimator is

$$\sqrt{Th}(\hat{f}(y) - E[\hat{f}(y)]) \xrightarrow{d} N\left(0, f(y) \int_{-\infty}^{\infty} K^2(z_t) dz_t\right). \quad (11.14)$$

For this expression to be of practical use for undertaking inference, it is necessary for $E[\hat{f}(y)]$ to converge in the limit to the true distribution $f(y)$. It turns out that this convergence is only assured if the bandwidth h satisfies an even tighter constraint than that required for the consistency property. To see why, consider the left hand side of equation (11.14)

$$\begin{aligned} \sqrt{Th}(\hat{f}(y) - E[\hat{f}(y)]) &= \sqrt{Th}(\hat{f}(y) - f(y) + f(y) - E[\hat{f}(y)]) \\ &= \sqrt{Th}(\hat{f}(y) - f(y) - \text{bias}(\hat{f}(y))) \\ &= \sqrt{Th}\left(\hat{f}(y) - f(y) - \frac{h^2}{2}f^{(2)}(y)\mu_2\right) \\ &= \sqrt{Th}(\hat{f}(y) - f(y)) - \sqrt{Th^5}\left(\frac{1}{2}f^{(2)}(y)\mu_2\right), \end{aligned}$$

where terms of $O(h^2)$ are just retained from (11.4). The expectation $E[\hat{f}(y)]$ converges to the true distribution $f(y)$ only if the last term in this expression disappears as $T \rightarrow \infty$. Interestingly enough, this condition is not satisfied by the optimal bandwidth $h = cT^{-1/5}$ because

$$\sqrt{Th^5} = \sqrt{Tc^5(T^{-1/5})^5} = \sqrt{c^5T^0},$$

is constant as $T \rightarrow \infty$. It turns out that by under-smoothing, that is setting $k > 1/5$, the bias term disappears faster than the variance and $E[\hat{f}(y)]$ converges to $f(y)$. To ensure the asymptotic normality of the kernel estimator, therefore, the relevant condition on k is $1/5 < k < 1$.

Provided that this bandwidth condition is satisfied, the nonparametric

estimator of $f(y)$ has an asymptotic normal distribution

$$\sqrt{Th}(\hat{f}(y) - f(y)) \xrightarrow{d} N\left(0, f(y) \int_{-\infty}^{\infty} K^2(z_t) dz_t\right), \quad (11.15)$$

which is similar to the expression in (11.14), but with $E[\hat{f}(y)]$ replaced by $f(y)$. Equation (11.15) can be used to construct the 95% confidence interval as

$$\hat{f}(y) \pm 1.96(Th)^{-1/2} \left[f(y) \int_{-\infty}^{\infty} K^2(z) dz \right]^{1/2}.$$

In the case of the Gaussian kernel

$$\int_{-\infty}^{\infty} K^2(z) dz = \frac{3}{8\sqrt{\pi}} = 0.2821,$$

so that the 95% confidence interval simplifies to

$$\hat{f}(y) \pm 1.96(Th)^{-1/2} [f(y) \times 0.2821]^{1/2},$$

or

$$\hat{f}(y) \pm 1.041(Th)^{-1/2} f(y)^{1/2}.$$

Finally, to be able to implement this interval in practice it is necessary to replace $f(y)$ by a consistent estimator, $\hat{f}(y)$, in which case the 95% confidence interval that is used in practice is

$$\hat{f}(y) \pm 1.041(Th)^{-1/2} \hat{f}(y)^{1/2}.$$

11.3.4 Dependent Data

The key theoretical results of the nonparametric density estimator presented above are based on the assumption that the time series y_t , is *iid*. For the more general case where y_t is not independent of y_{t-k} as is generally the case in time series data, these theoretical results still hold under weak conditions (Wand and Jones (1995) and Pagan and Ullah (1999)). Intuitively this result arises as the kernel density estimator is based on the spatial dependence structure of the data given that $f(y)$ is estimated as a weighted average of the observations around y , the support of the density. This weighting procedure effectively reduces the role of the time dependence structure of y_t in estimating $f(y)$, while emphasizing instead the spatial dependence structure of y_t .

A more formal justification is to assume that y_t behaves as an *iid* process, at least asymptotically, in the sense that y_t and y_{t-k} , are independent of each

other as $k \rightarrow \infty$ (Robinson, 1983). Consider two sets of time series k periods apart, $Y_t = \{\dots, y_1, y_2, \dots, y_t\}$ and $Y_{t+k} = \{y_{t+k}, y_{t+k+1}, \dots\}$. The formal condition for asymptotic independence is based on comparing the absolute difference of the joint probability of the two sets of time series and the product of their marginal probabilities

$$\alpha(k) = \sup_{k \rightarrow \infty} |P(Y_t \cap Y_{t+k}) - P(Y_t)P(Y_{t+k})|. \quad (11.16)$$

If $\lim_{k \rightarrow \infty} \alpha(k) = 0$, the joint probability equals the product of the two marginal probabilities, which is the standard condition of independence. This condition is known as strong mixing, or α -mixing.

11.4 Semi-Parametric Density Estimation

Density estimators which also use parametric information are referred to as semi-parametric estimators. An example is given by Stachurski and Martin (2008) where parametric information on the transitional density of an autoregressive time series model, $f(y|y_{t-1})$ is used to estimate the marginal density $f(y)$. In this context $f(y)$ represents the stationary density of the time series model as discussed in Chapters 1 and 3. An important advantage of the approach is that the convergence rate is now $\sqrt{T}$ compared to the slower rate of convergence arising from the nonparametric kernel estimator.

Let $f(y|y_{t-1})$ represent the transitional density of a parametric model. Under certain conditions, the stationary density is given by

$$\lim_{T \rightarrow \infty} \hat{f}(y) = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T f(y|y_{t-1}) = \int f(y|y_{t-1}) f(y_{t-1}) dy_{t-1}, \quad (11.17)$$

suggesting that a suitable estimator of $f(y)$ is given by the sample average of the heights of the transitional densities at y

$$\hat{f}(y) = \frac{1}{T} \sum_{t=1}^T f(y|y_{t-1}). \quad (11.18)$$

This estimator is also known as the look-ahead estimator (LAE). As the transitional density approaches regions of higher probability more frequently than regions of lower probability, by averaging across the transitional densities over the sample an estimator of the stationary density is constructed.

The relationship between the LAE and the nonparametric kernel density estimator is highlighted for the case of an AR(1) model with normal disturbances

$$y_t = \phi_0 + \theta_1 y_{t-1} + \sigma z_t, \quad z_t \sim N(0, 1). \quad (11.19)$$

The LAE is

$$\hat{f}(y) = \frac{1}{T} \sum_{t=1}^T f(y|y_{t-1}) = \frac{1}{T} \sum_{t=1}^T \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{1}{2} \left(\frac{y - \phi_0 - \phi_1\theta y_{t-1}}{\sigma} \right)^2 \right],$$

By redefining y_{t-1} as y_t , this expression is rewritten as

$$\hat{f}(y) = \frac{1}{T} \sum_{t=1}^T \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{1}{2} \left(\frac{y - \phi_0 - \phi_1\theta y_t}{\sigma} \right)^2 \right]. \quad (11.20)$$

Comparing the LAE in (11.20) and the nonparametric kernel density estimator using a Gaussian kernel with bandwidth h , shows that the two density estimators are equivalent where $\phi_0 = 0$, $\phi_1\theta = 1$ and $\sigma = h$. The last condition is important as it suggests that if the AR(1) model is derived from economic theory, the bandwidth has an economic interpretation. Moreover, this relationship also suggests that the choice of the kernel can be motivated from the assumptions underlying the economic model.

Example 11.5 LAE of a Threshold AR(1) Model

Consider the Theshold AR(1) model

$$y_t = \theta |y_{t-1}| + \sqrt{1 - \theta^2} z_t, \quad z_t \sim N(0, 1),$$

where z_t is an *iid* disturbance. The stationary and transitional densities are respectively

$$f(y) = 2\phi(y)\Phi(\delta y), \quad \delta = \theta/\sqrt{1 - \theta^2}$$

$$f(y|y_{t-1}) = \frac{1}{\sqrt{2\pi(1 - \theta^2)}} \exp \left[-\frac{1}{2} \left(\frac{y - \theta |y_{t-1}|}{\sqrt{1 - \theta^2}} \right)^2 \right].$$

The LAE is computed as

$$\hat{f}(y) = \frac{1}{T} \sum_{t=1}^T f(y|y_{t-1}) = \frac{1}{T} \sum_{t=1}^T \frac{1}{\sqrt{2\pi(1 - \theta^2)}} \exp \left[-\frac{1}{2} \left(\frac{y - \theta |y_{t-1}|}{\sqrt{1 - \theta^2}} \right)^2 \right].$$

Panel (a) of Figure 11.4 gives the LAE for parameter $\theta = 0.5$ and representative transitional densities used to construct it based on the three observations $y_t = \{0.600, -0.279, -1.334, 0.847, 0.894\}$. It is apparent from panel (b) of Figure 11.4 that the LAE based on $T = 5$ provides a better approximation of the true density than that of a fully nonparametric kernel estimate based on a Gaussian kernel using a rule-of-thumb bandwidth. $\square$

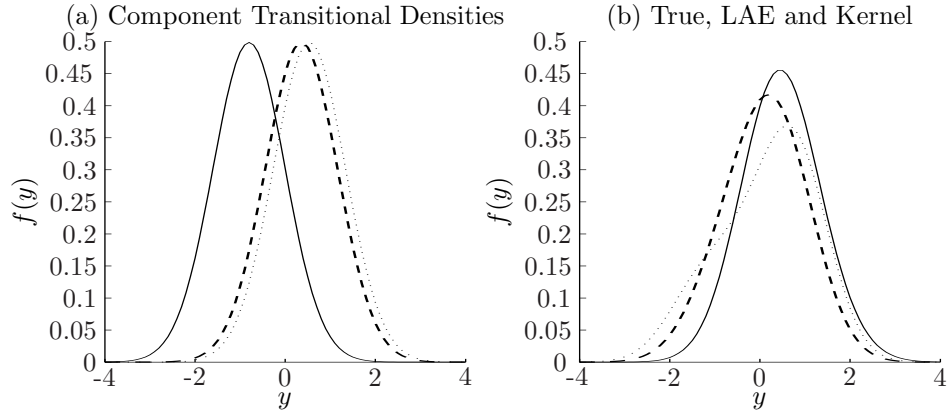


Figure 11.4 Comparison of the LAE and the nonparametric density estimator of the stationary density of a threshold AR(1) model with $\theta = 0.5$. Panel (a) illustrates the component transitional densities of the LAE for the observations $y_t = 0.6$ (dashed line), $y_t = -1.334$ (solid line) and $y_t = 0.894$ (dotted line). Panel (b) compares the true density (solid line), the LAE (dashed line) and the nonparametric kernel estimator (dotted line).

11.5 The Nadaraya-Watson Kernel Regression Estimator

Consider the simple regression problem of estimating the conditional mean

$$y_t = m(x_t) + u_t, \quad (11.21)$$

where $m(x_t)$ is the conditional mean and u_t is a disturbance term. The problem with adopting a parametric specification for $m(x_t)$ is that if either the functional form or the choice of the distribution of the disturbance term is incorrect, then the estimator of the conditional mean is unlikely to be consistent. This is demonstrated in the following example, where the parametric specification of the conditional mean is misspecified.

Example 11.6 Parametric Estimates of the Conditional Mean

A sample of $T = 500$ observations are simulated from (11.21) with conditional mean

$$m(x_t) = 0.3 \exp(-4(x_t + 1)^2) + 0.7 \exp(-16(x_t - 1)^2),$$

where $x_t \sim U[-2, 2]$ and $u_t \sim N(0, 0.01)$ is an *iid* disturbance. Least squares estimates of a linear and a nonlinear parametric model of $m(x)$, are respectively

$$\begin{aligned} \hat{m}(x_t)_{\text{linear}} &= 0.155 + 0.008 x_t \\ \hat{m}(x_t)_{\text{nonlinear}} &= 0.058 + 0.244 x_t - 0.035 x_t^2 - 0.021 x_t^3 - 0.080 x_t^4. \end{aligned}$$

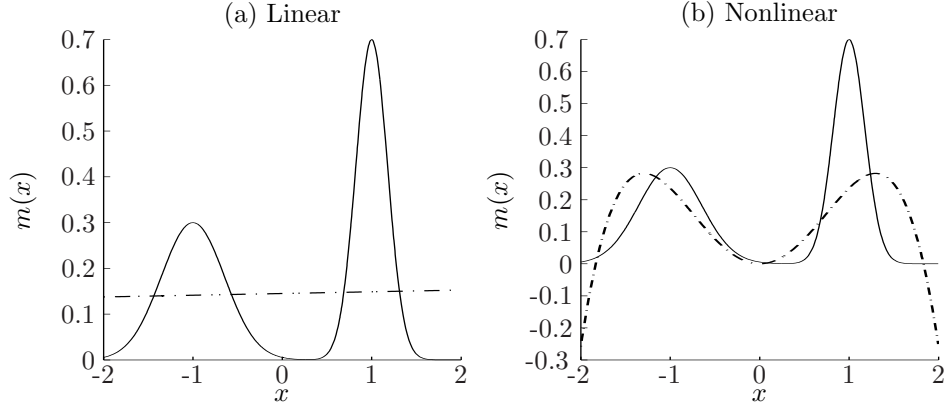


Figure 11.5 Comparison of the true conditional mean (continuous line) with estimated parametric models based on (a) linear, and (b) nonlinear parametric specifications (broken lines).

Figure 11.5 provides a comparison of the linear and nonlinear estimated parametric conditional means and the true conditional mean. The linear model predicts a slight upward drift in y_t , but fails to capture any of the turning points. The nonlinear model identifies some of the nonlinear features of the data, but misses the amplitudes of the peaks and troughs, and performs poorly in the tails. $\square$

To avoid misspecifying the conditional moments, an alternative approach, known as nonparametric regression is adopted. The aim of nonparametric regression is to recover (estimate) the conditional distribution of the dependent variable, y_t , at each point in time given a set of conditioning variables, x_t , but without specifying the functional form of the relationship between y_t and x_t . The most widely used nonparametric estimator adopted in practice is the Nadaraya-Watson estimator, where the conditional mean is estimated by smoothing over y_t using an appropriate weighting function that is a function of x_t . In other words, it is essentially the nonparametric kernel regression analogue to nonparametric density estimation.

The Nadaraya-Watson nonparametric regression estimator, for the case where y_t depends only on one exogenous variable x_t , is constructed as follows. From the definition of conditional expectations, the conditional mean $m(x)$ is defined as

$$m(x) = E[y|x_t = x] = \int y f_{y|x} dy = \frac{\int y f_{yx}(y, x) dy}{f_x(x)}, \quad (11.22)$$

where $f_{y|x}$ is the conditional distribution of y given x , $f_{yx}(y, x)$ is the joint distribution of y and x and $f_x(x)$ is the marginal distribution of x . By replacing $f_{yx}(y, x)$ and $f_x(x)$ by nonparametric density estimators, a nonparametric estimator of $m(x)$ is

$$\hat{m}(x) = \frac{\int y \hat{f}_{yx}(y, x) dy}{\hat{f}_x(x)}, \quad (11.23)$$

where

$$\hat{f}_{yx}(y, x) = \frac{1}{Th^2} \sum_{t=1}^T K\left(\frac{y - y_t}{h}\right) K\left(\frac{x - x_t}{h}\right) \quad (11.24)$$

$$\hat{f}_x(x) = \frac{1}{Th} \sum_{t=1}^T K\left(\frac{x - x_t}{h}\right), \quad (11.25)$$

and all bandwidths are assumed to be equal to h . The choice of a product kernel in (11.24) for the bivariate distribution enables the numerator of (11.23) to be written as

$$\begin{aligned} \int y \hat{f}_{yx}(x) dy &= \int y \left[\frac{1}{Th^2} \sum_{t=1}^T K\left(\frac{y - y_t}{h}\right) K\left(\frac{x - x_t}{h}\right) \right] dy \\ &= \frac{1}{Th^2} \sum_{t=1}^T K\left(\frac{x - x_t}{h}\right) \int y K\left(\frac{y - y_t}{h}\right) dy \\ &= \frac{1}{Th} \sum_{t=1}^T y_t K\left(\frac{x - x_t}{h}\right). \end{aligned} \quad (11.26)$$

The last step follows by using a change of variable

$$s = \frac{y - y_t}{h},$$

and writing

$$\begin{aligned} \int y K\left(\frac{y - y_t}{h}\right) dy &= h \int (sh + y_t) K(s) ds = h(h \int s K(s) ds + y_t \int K(s) ds) \\ &= h(h \times 0 + y_t) = hy_t, \end{aligned}$$

which recognizes that for a symmetric kernel $\int K(s) ds = 1$ and $\int s K(s) ds = 0$. Substituting (11.26) and (11.25) into (11.23) gives

$$\hat{m}(x) = \frac{\frac{1}{Th} \sum_{t=1}^T y_t K\left(\frac{x - x_t}{h}\right)}{\frac{1}{Th} \sum_{t=1}^T K\left(\frac{x - x_t}{h}\right)} = \sum_{t=1}^T y_t w_t, \quad (11.27)$$

where w_t represents the weight given by

$$w_t = \frac{\frac{1}{Th} K\left(\frac{x - x_t}{h}\right)}{\frac{1}{Th} \sum_{t=1}^T K\left(\frac{x - x_t}{h}\right)}. \quad (11.28)$$

The kernel regression estimator therefore has the intuitive interpretation of a weighted average of the dependent variable where the weights are a function of the explanatory variable x_t .

The steps to compute the kernel regression estimator are as follows.

- Step 1:** Choose a grid of values of x starting at the lowest value of x_t and ending at the highest value of x_t .
- Step 2:** Choose the first value of x in the grid and weight the observations according to (11.28) with the largest weight given to the x_t values closest to x . This involves choosing a kernel function K and a bandwidth h .
- Step 3:** Compute (11.27), the estimate of the conditional mean, $E[y_t|x]$.
- Step 4:** Repeat the computation for all the values of x in the grid. Notice that changing x changes the weights and hence the estimate of the conditional mean.

Example 11.7 Computing the Nadaraya-Watson Estimator

Table 11.3 provides a breakdown the computation of the nonparametric kernel regression estimates for the data on y_t and x_t given in columns 2 and 3 of the table. A Gaussian kernel with a bandwidth of $h = 0.5$ is chosen with seventeen grid points, $x = \{-2.0, -1.75, -0.5, \dots, 1.75, 2.0\}$.

The calculations are presented for the first grid point in the table. For $x = -2.0$

$$\begin{aligned} \frac{1}{Th} \sum_{t=1}^T K\left(\frac{-2 - x_t}{h}\right) &= \frac{2.164}{20 \times 0.5} = 0.216 \\ \frac{1}{Th} \sum_{t=1}^T y_t K\left(\frac{-2 - x_t}{h}\right) &= \frac{0.157}{20 \times 0.5} = 0.016, \end{aligned}$$

and (allowing for rounding error)

$$\hat{m}(-2.0) = \frac{0.016}{0.216} = 0.073.$$

Figure 11.6 illustrates the Nadaraya-Watson conditional mean estimates for all seventeen grid points of x together with the realized values of y_t .

Table 11.3

Nadaraya-Watson kernel regression estimates of the conditional mean, $m(x)$, for selected values of x . The bandwidth is $h = 0.5$ and $K(\cdot)$ is the Gaussian kernel.

t	y_t	x_t	$x = -2.0$		$x = -1.75$	
			$K\left(\frac{x - x_t}{h}\right)$	$y_t K\left(\frac{x - x_t}{h}\right)$	$K\left(\frac{x - x_t}{h}\right)$	$y_t K\left(\frac{x - x_t}{h}\right)$
1	0.433	0.886	0.000	0.000	0.000	0.000
2	0.236	-1.495	0.239	0.057	0.350	0.083
3	0.299	-1.149	0.094	0.028	0.194	0.058
4	0.030	1.921	0.000	0.000	0.000	0.000
5	-0.041	-1.970	0.398	-0.016	0.362	-0.015
6	0.185	-1.697	0.332	0.062	0.397	0.074
7	0.053	-0.324	0.001	0.000	0.007	0.000
8	0.459	0.844	0.000	0.000	0.000	0.000
9	-0.038	-1.968	0.398	-0.015	0.363	-0.014
10	-0.061	-1.829	0.376	-0.023	0.394	-0.024
11	0.147	0.510	0.000	0.000	0.000	0.000
12	-0.041	-0.380	0.002	0.000	0.009	0.000
13	0.042	0.230	0.000	0.000	0.000	0.000
14	0.224	-0.678	0.012	0.003	0.040	0.009
15	-0.043	1.954	0.000	0.000	0.000	0.000
16	0.046	1.477	0.000	0.000	0.000	0.000
17	-0.055	1.925	0.000	0.000	0.000	0.000
18	-0.054	0.099	0.000	0.000	0.000	0.000
19	0.205	-1.641	0.308	0.063	0.390	0.080
20	-0.084	-0.339	0.002	0.000	0.007	-0.001
Sum			2.164	0.157	2.514	0.250

The kernel estimates of the conditional mean tend to be overly smooth with $h = 0.5$, but are more jagged for $h = 0.1$.

□

11.6 Properties of Kernel Regression Estimators

This section provides a discussion of some of the finite sample and asymptotic properties of the kernel regression estimator. The results are presented for the *iid* case, with generalizations allowing for data exhibiting limited temporal dependence discussed at the end of the section.

Bias and Variance

Just as in the case of the kernel estimate of density, the bias of the kernel regression is positively related to the size of the bandwidth, h , and the

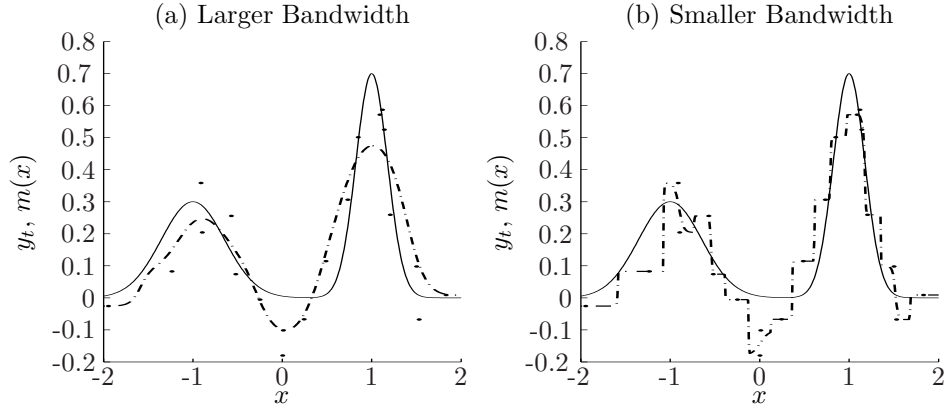


Figure 11.6 Actual values (dots), true conditional mean (line) and the nonparametric kernel conditional mean estimates (dashed) for different bandwidths.

variance is negatively related to h . These properties are highlighted in the following example.

Example 11.8 Bias and Variance in a Kernel Regression

A sample of $T = 500$ observations are simulated from the model

$$y_t = 0.3 \exp(-4(x_t + 1)^2) + 0.7 \exp(-16(x_t - 1)^2) + u_t,$$

where $u_t \sim N(0, 0.01)$ and $x_t \sim U[-2, 2]$. The kernel regression estimates of the conditional mean for different choices of the bandwidth are illustrated in Figure 11.7.

The nonparametric regression estimator exhibits some bias for $h = 0.2$ in panel (a) of Figure 11.7. In particular, the estimator underestimates the two peaks and overestimates the troughs, suggesting that the estimate is over-smoothed. Panel (b) of Figure 11.7 shows that reducing the bandwidth to $h = 0.02$ decreases this bias, but at the expense of introducing greater variance. The jagged nature of the estimator in panel (b) is indicative of under-smoothing. $\square$

Consistency

Consistency requires that both the respective bias and variance expressions for $\hat{f}_{yx}(x)$ and $\hat{f}_x(x)$ approach zero as T increases so that

$$\text{plim}(\hat{f}_{yx}(x)) = f_{yx}(x) \quad \text{and} \quad \text{plim}(\hat{f}_x(x)) = f_x(x).$$

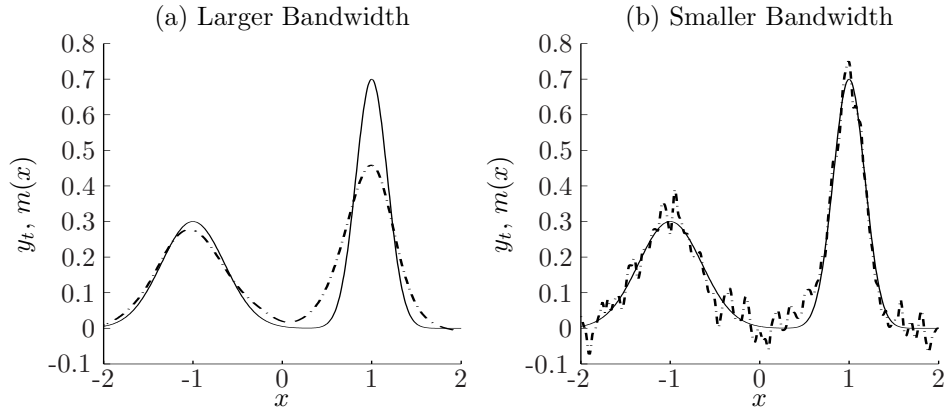


Figure 11.7 True conditional mean (solid line) and nonparametric estimates (broken lines) demonstrate the effect of the bandwidth h on the kernel regression estimator. Larger bandwidth given by $h = 0.2$, and smaller bandwidth given by $h = 0.02$.

Using Slutsky's theorem

$$\begin{aligned} \text{plim}(\hat{m}(x)) &= \text{plim}\left(\frac{\int y \hat{f}_{yx}(y, x) dy}{\hat{f}_x(x)}\right) = \frac{\text{plim}(\int y \hat{f}_{yx}(y, x) dy)}{\text{plim}(\hat{f}_x(x))} = \frac{\int y f_{yx}(y, x) dy}{f_x(x)} \\ &= m(x), \end{aligned}$$

which establishes the consistency of the nonparametric kernel regression estimate of the conditional mean. Pagan and Ullah (1999) establish the formal conditions underlying this result. Just as in the case of the kernel density estimator, the required condition on the bandwidth used in the kernel regression is

$$h = cT^{-k}, \quad 0 < k < 1,$$

where, as before, c is a constant whose value depends upon the kernel function chosen to compute the estimator.

Asymptotic Normality

The asymptotic distribution of $\hat{m}(x)$ is

$$\sqrt{Th}(\hat{m}(x) - E[\hat{m}(x)]) \xrightarrow{d} N\left(0, \frac{\sigma_{y|x}^2}{f_x(x)} \int K^2(z) dz\right),$$

where

$$\sigma_{y|x}^2 = \int (y_t - m(x))^2 f_{y|x} dy,$$

is the conditional variance of y given x . Following the argument in Section 11.3.3, to make this result useful for statistical inference it is necessary for $E[\hat{m}(x)]$ to converge to $m(x)$ in which case $\hat{m}(x)$ has an asymptotic normal distribution

$$\sqrt{Th}(\hat{m}(x) - m(x)) \xrightarrow{d} N\left(0, \frac{\sigma_{y|x}^2}{f_x(x)} \int K^2(z) dz\right), \quad (11.29)$$

which is the same as the previous expression, but with $E[\hat{m}(x)]$ replaced by $m(x)$. The bandwidth condition needed for the convergence of $E[\hat{m}(x)]$ to $m(x)$ is

$$h_T = cT^{-k}, \quad 1/5 < k < 1,$$

where c is a constant. This is the same condition used to establish the asymptotic normality property of the kernel density estimator.

The 95% confidence interval for the kernel regression estimator is

$$\hat{m}(x) \pm 1.96 (Th)^{-1/2} \left[\frac{\sigma_{y|x}^2}{f_x(x)} \int K^2(z_t) dz_t \right]^{1/2}.$$

In the case of a Gaussian kernel

$$\int_{-\infty}^{\infty} K^2(z_t) dz_t = \frac{3}{8\sqrt{\pi}} = 0.2821,$$

the 95% confidence interval simplifies to

$$\hat{m}(x) \pm 1.96 (Th)^{-1/2} \left[\frac{\sigma_{y|x}^2}{f_x(x)} \times 0.2821 \right]^{1/2},$$

or

$$\hat{m}(x) \pm 1.041 (Th)^{-1/2} \frac{\sigma_{y|x}}{f_x(x)^{1/2}}.$$

Finally, to be able to implement this interval in practice $\sigma_{y|x}$ and $f_x(x)$ are replaced by consistent estimators, giving the 95% confidence interval used in practice as

$$\hat{m}(x) \pm 1.041 (Th)^{-1/2} \frac{\hat{\sigma}_{y|x}}{\hat{f}_x(x)^{1/2}},$$

where

$$\hat{u}_t = y_t - \hat{m}(x_t), \quad \hat{\sigma}_{y|x} = \sum_{t=1}^T w_t \hat{u}_t^2,$$

and the weights w_t are given in equation (11.28).

Dependent Data

As in the case of the nonparametric density estimator where the theoretical results based on *iid* also hold at least asymptotically for dependent data, the theoretical results of the kernel regression estimator derived above are still satisfied where y_t and y_{t-k} for $k > 0$ are correlated (Robinson, 1983; Pagan and Ullah, 1999).

11.7 Bandwidth Selection for Kernel Regression

The problem of bandwidth selection is now one of ensuring the best possible estimator of the mean of y conditional on x . A natural approach is to treat the bandwidth h , as a parameter and choose the value that minimizes the distance between y_t and its conditional mean $m(x_t)$, using the criterion

$$\arg \min_h \mathcal{S} = \sum_{t=1}^T (y_t - \hat{m}(x_t))^2, \quad (11.30)$$

where $\hat{m}(x_t)$ is an estimator of $m(x_t)$. The problem with this approach is that it leads to the degenerate solution of $h = 0$, as then $\hat{m}(x_t) = y_t$ resulting in a perfect fit with zero error. To see why this is the case, consider the nonparametric estimator of $m(x_t)$ in equation (11.27), evaluated at the first observation of x

$$\hat{m}(x_1) = \frac{\frac{1}{Th} \sum_{t=1}^T y_t K\left(\frac{x_1 - x_t}{h}\right)}{\frac{1}{Th} \sum_{t=1}^T K\left(\frac{x_1 - x_t}{h}\right)}.$$

In the case of a Gaussian kernel

$$\begin{aligned} \hat{m}(x_1) &= \frac{\frac{1}{Th} \left[\frac{y_1}{\sqrt{2\pi}} + \sum_{t=2}^T \frac{y_t}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x_1 - x_t}{h}\right)^2\right) \right]}{\frac{1}{Th} \left[\frac{1}{\sqrt{2\pi}} + \sum_{t=2}^T \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x_1 - x_t}{h}\right)^2\right) \right]} \\ &= \frac{\frac{y_1}{\sqrt{2\pi}} + \sum_{t=2}^T \frac{y_t}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x_1 - x_t}{h}\right)^2\right)}{\frac{1}{\sqrt{2\pi}} + \sum_{t=2}^T \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x_1 - x_t}{h}\right)^2\right)}. \end{aligned}$$

Allowing $h \rightarrow 0$, the exponential terms all approach zero, leaving

$$\lim_{h \rightarrow 0} \hat{m}(x_1) = \frac{\frac{y_1}{\sqrt{2\pi}} + 0}{\frac{1}{\sqrt{2\pi}} + 0} = y_1.$$

That is, the best estimate of y_1 is indeed y_1 without any error! Repeating these calculations for $t \geq 2$, the sum of squares function in equation (11.30) is minimized where

$$\lim_{h \rightarrow 0} \mathcal{S} = 0. \quad (11.31)$$

The degenerate solution of the bandwidth given in (11.31) is circumvented by removing the j^{th} observation when estimating $m(x_j)$. This results in the so-called leave-one-out kernel estimator of $m(x_t)$ given by

$$\tilde{m}(x_j) = \frac{\frac{1}{Th} \sum_{\substack{t=1 \\ t \neq j}}^T y_j K\left(\frac{x_j - x_t}{h}\right)}{\frac{1}{Th} \sum_{\substack{t=1 \\ t \neq j}}^T K\left(\frac{x_j - x_t}{h}\right)}, \quad (11.32)$$

and the associated criterion for the choice of bandwidth

$$\arg \min_h \tilde{\mathcal{S}} = \sum_{t=1}^T (y_t - \tilde{m}(x_t))^2. \quad (11.33)$$

This method of bandwidth selection is also known as cross validation.

The leave-one-out approach to choosing the bandwidth is particularly sensitive to outliers. There are two common methods to address this problem.

- (1) Eliminate the values of x_t that fall in the upper and lower 5th percentiles of the data by using a non-negative weight function

$$\omega(x_t) = \mathcal{H}(x \in [P_{0.05}, P_{0.95}]), \quad (11.34)$$

where $\mathcal{H}$ is the Heavyside operator. The objective function is then

$$\arg \min_h \tilde{\mathcal{S}} = \sum_{t=1}^T (y_t - \tilde{m}(x_t))^2 \omega(x_t). \quad (11.35)$$

- (2) Penalize the sum of squares function in (11.30) for ‘small’ choices of h .

Here the bandwidth is the solution of the constrained problem

$$\arg \min_h \hat{\mathcal{S}} = \sum_{t=1}^T (y_t - \hat{m}(x_t))^2 \omega(x_t) \Xi(h), \quad (11.36)$$

where $\hat{m}(x_t)$ is constructed using all the available observations, $\omega(x_t)$ is as defined in (11.34) and $\Xi(h)$ is a suitable penalty function that penalizes small bandwidths. Rice (1984) suggests using one of the following penalty functions

$$\Xi(h) = 1 + 2u, \quad \Xi(h) = \exp(2u), \quad \Xi(h) = (1 - 2u)^{-1},$$

where $u = K(0)/Th$.

Example 11.9 Cross Validation

A sample of $T = 500$ observations are simulated from the model

$$y_t = 0.3 \exp(-4(x_t + 1)^2) + 0.7 \exp(-16(x_t - 1)^2) + u_t,$$

where $u_t \sim N(0, 0.01)$ and $x_t \sim U[-2, 2]$. Figure 11.8 plots the sum of squares function in (11.36) based on the leave-one-out kernel estimator of $m(x_t)$ in (11.32) using a grid of bandwidths given by $h = \{0.001, 0.002, \dots, 0.5\}$. The optimal bandwidth based on cross validation is $h = 0.07$ where the objective function is $\hat{\mathcal{S}} = 5.378$. For comparison the value of (11.36) is also computed without cross validation by using (11.27) as the estimator of $m(x_t)$. $\square$

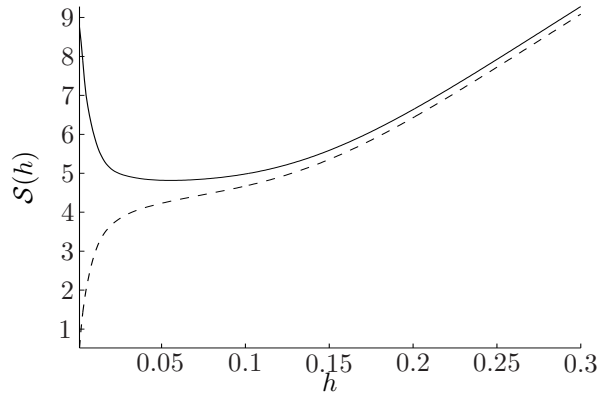


Figure 11.8 Illustrating the importance of the leave-one-out principle in kernel regression bandwidth selection. The sum of squares function for the cross validation procedure (solid line) shows a well-defined minimum at $h > 0$, whereas the sum of squares function for the bandwidth with all observations included (dashed line) reaches its minimum at $h = 0$.

Ultimately, however, there are no convenient rules-of-thumb for bandwidth selection in nonparametric regression. Yatchew (2003, p46) argues that trying different bandwidths and examining the resultant estimate of the regression function is often a useful way of obtaining a general indication of possible over- or under-smoothing.

11.8 Multivariate Kernel Regression

The analysis of kernel regression so far has concentrated on a single explanatory variable, x_t . In theory the kernel regression estimator can be expanded to allow for N regressors. Consider

$$y_t = m(x_t) + u_t,$$

where $x_t = \{x_{1,t}, x_{2,t}, \dots, x_{N,t}\}$ and u_t is a disturbance term. Using the results of multivariate kernel density estimation discussed in Section 11.2, the multivariate analogue of the Nadaraya-Watson kernel regression estimator, based on product kernels, is

$$\hat{m}(x) = \frac{\int y \hat{f}_{yx}(y, x) dy}{\hat{f}_x(x)} \quad (11.37)$$

where

$$\begin{aligned} \int y \hat{f}_{yx}(y, x) dy &= \frac{1}{T \times h_1 \times \dots \times h_N} \sum_{t=1}^T y_t \prod_{n=1}^N K\left(\frac{x_n - x_{n,t}}{h_n}\right) \\ \hat{f}_x(x) &= \frac{1}{T \times h_1 \times \dots \times h_N} \sum_{t=1}^T \prod_{n=1}^N K\left(\frac{x_n - x_{n,t}}{h_n}\right). \end{aligned}$$

Note that in implementing the multivariate kernel regression, a different bandwidth is needed for each $x_{i,t}$ to accommodate differences in scaling. The difficulty of selecting $h_1, \dots, h_N$ bandwidths means that a commonly-adopted approach is simply to use the Scott (1992) rule-of thumb approach, equation (11.12), for bandwidth selection in a multivariate kernel density setting.

Example 11.10 Two-dimensional Product Kernel Regression

Consider the following nonlinear model

$$y_t = x_{1,t} x_{2,t}^2 + u_t,$$

where y_t is the dependent variable, $x_{1,t}$ and $x_{2,t}$ are the explanatory variables and $u_t \sim N(0, 0.01)$ is a disturbance term. The $N = 2$ explanatory variables

are taken as realizations from independent $N(0, 1)$ distributions. The model is simulated for $T = 20,000$ observations. To perform a multivariate non-parametric kernel regression the bandwidths for each explanatory variable are chosen as

$$h_1 = \hat{\sigma}_1 T^{-1/(N+4)}, \quad h_2 = \hat{\sigma}_2 T^{-1/(N+4)},$$

where $\hat{\sigma}_i$ is the sample standard deviation of $x_{i,t}$. The Nadaraya-Watson multivariate kernel regression estimator in equation (11.37) requires computation of

$$\begin{aligned} \hat{f}_x(x) &= \frac{1}{Th_1h_2} \sum_{t=1}^T K\left(\frac{x_1 - x_{1,t}}{h_1}\right) K\left(\frac{x_2 - x_{2,t}}{h_2}\right) \\ \int y \hat{f}_{yx}(y, x) dy &= \frac{1}{Th_1h_2} \sum_{t=1}^T y_t K\left(\frac{x_1 - x_{1,t}}{h_1}\right) K\left(\frac{x_2 - x_{2,t}}{h_2}\right). \end{aligned}$$

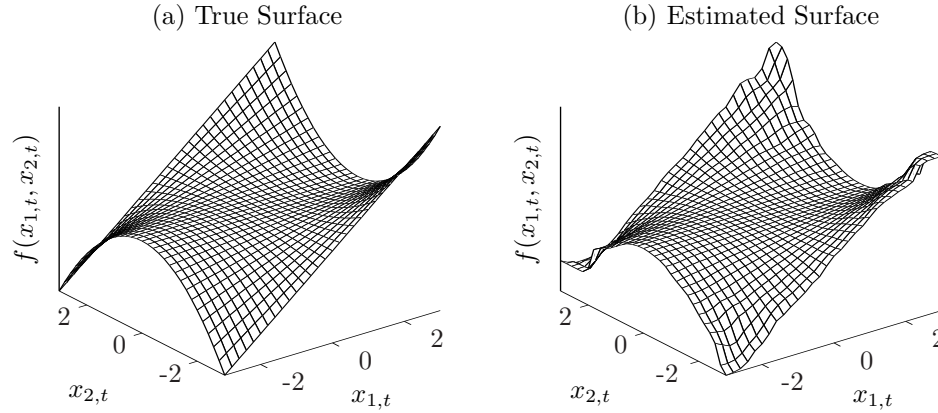


Figure 11.9 True and two-dimensional product kernel estimates of the conditional mean where the regression equation is $y_t = x_{1,t}x_{2,t}^2 + u_t$.

The true surface in panel (a) of Figure 11.9 is compared with the bivariate kernel estimates in panel (b) based on a Gaussian kernel. The kernel estimates are very accurate at the center of the surface, but become less accurate near the edges where data are sparse. $\square$

The previous example highlights a common problem in multi-dimensional nonparametric estimation, already mentioned in Section 11.2, namely the ‘curse of dimensionality’. This feature arises because the accuracy of the estimates of a one-dimensional problem with sample size of T can only be reproduced in two-dimensions with T^2 observations. As a result applications of

Multivariate kernel regressions are less common and alternative approaches that impose more structure on the problem, such as semi-parametric regression estimators are developed.

11.9 Semi-parametric Regression of the Partial Linear Model

Consider the partial linear model where the dependent variable y_t is a function of the explanatory variables $\{x_{1,t}, x_{2,t}, x_{3,t}\}$

$$y_t = \beta_1 x_{1,t} + \beta_2 x_{2,t} + g(x_{3,t}) + u_t, \quad (11.38)$$

where $g(x_{3,t})$ is an unknown function of $x_{3,t}$ and u_t is a disturbance term. Part of the model has a well defined parametric structure, but the functional form of the remaining part is unknown. Using a multivariate nonparametric kernel to estimate the conditional mean of y in this instance ignores the parametric information $\beta_1 x_{1,t} + \beta_2 x_{2,t}$. One approach to estimating the parameters of this partial linear model, is the difference procedure described in detail by Yatchew (2003). The approach presented here is the semi-parametric method developed by Robinson (1988).

Take expectations of equation (11.38) conditioning on $x_{3,t}$

$$E[y_t|x_{3,t}] = \beta_1 E[x_{1,t}|x_{3,t}] + \beta_2 E[x_{2,t}|x_{3,t}] + g(x_{3,t}), \quad (11.39)$$

where the results $E[g(x_{3,t})|x_{3,t}] = g(x_{3,t})$ and $E[u_t|x_{3,t}] = 0$ are used. Subtracting equation (11.39) from equation (11.38) yields

$$y_t - E[y_t|x_{3,t}] = \beta_1 (x_{1,t} - E[x_{1,t}|x_{3,t}]) + \beta_2 (x_{2,t} - E[x_{2,t}|x_{3,t}]) + u_t. \quad (11.40)$$

The estimation proceeds as follows.

Step 1: Estimate $E[y_t|x_{3,t}]$, $E[x_{1,t}|x_{3,t}]$ and $E[x_{2,t}|x_{3,t}]$ by three separate bivariate nonparametric regressions.

Step 2: Substitute the nonparametric estimates from Step 1 into equation (11.40), and estimate β_1 and β_2 by an OLS regression of $y_t - E[y_t|x_{3,t}]$ on $x_{1,t} - E[x_{1,t}|x_{3,t}]$ and $x_{2,t} - E[x_{2,t}|x_{3,t}]$.

Step 3: An estimate of $g(x_{3,t})$ is now obtained as

$$\hat{g}(x_{3,t}) = E[y_t|x_{3,t}] - \hat{\beta}_1 E[x_{1,t}|x_{3,t}] - \hat{\beta}_2 E[x_{2,t}|x_{3,t}],$$

where all quantities are replaced by their sample estimates.

Example 11.11 The Partial Linear Model

DiNardo and Tobias (2001), consider the model

$$y_t = 2x_{1,t} + x_{2,t} + g(x_{3,t}) + u_t,$$

where $x_{1,t} \sim N(0.5x_{3,t}, 1)$, $x_{2,t}$ is distributed as a Student-t distribution with 4 degrees of freedom, $x_{3,t}$ is distributed as $U[-2, 2]$ and $u_t \sim N(0, 0.1)$. In this example, the true functional form of $g(x_{3,t})$ is

$$g(x_{3,t}) = 0.3 \exp(-4(x_{3,t} + 1)^2) + 0.7 \exp(-16(x_{3,t} - 1)^2),$$

which is treated as unknown when using the semi-parametric estimator. The simulated data of sample size $T = 500$, is shown in panel (a) of Figure 11.10.

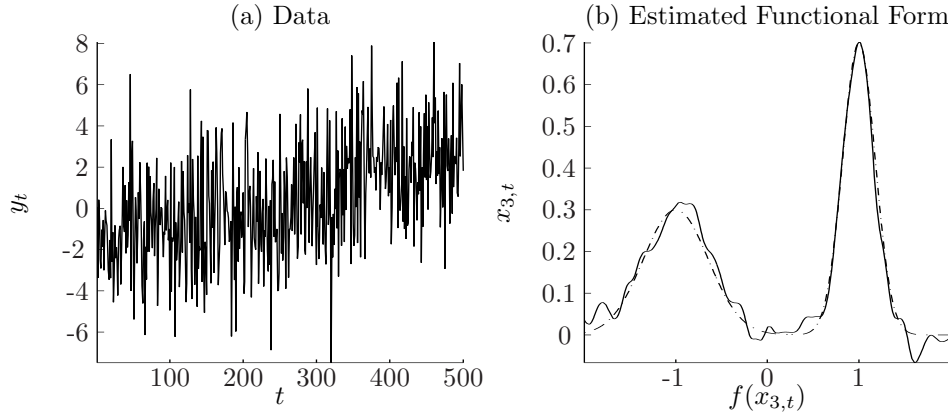


Figure 11.10 Semiparametric estimation of the partial linear model. Panel (b) compares the true model (dashed line) with the semiparametric estimate (solid line).

Three kernel regressions are computed with bandwidth $h = 0.05$, to estimate $E[y_t|x_{3,t}]$, $E[x_{1,t}|x_{3,t}]$ and $E[x_{2,t}|x_{3,t}]$. An OLS regression is then estimated yielding parameter estimates

$$\hat{\beta}_1 = 1.999, \hat{\beta}_2 = 1.001,$$

which are in good agreement with the true parameter values of 2 and 1 respectively. The nonparametric estimate of $g(x_{3,t})$ is plotted in panel (b) of Figure 11.10 and compares favourably with the true function. $\square$

11.10 Applications

The first application requires estimating the derivatives of a nonlinear production function that relates output to two inputs and is adapted from Pagan and Ullah (1999). The second application is based on Chapman and Pearson (2000).

11.10.1 Derivatives of a Nonlinear Production Function

Consider the problem of estimating the derivatives of a nonlinear production function that relates y_t (the natural logarithm of output) and $x_{1,t}$ and $x_{2,t}$ (the natural logarithms of the inputs). As the variables are in logarithms, the derivatives represent elasticities. The true model is given by

$$\begin{aligned} y_t &= m(x_{1,t}, x_{2,t}) + u_t \\ &= -0.2 \ln(\exp(-5x_{1,t}) + 2 \exp(-5x_{2,t})) + u_t, \end{aligned}$$

where $u_t \sim N(0, 0.01)$ and the logarithms of the inputs are $x_{1,t}, x_{2,t} \sim U(0, 1)$. The sample size is $T = 200$.

In computing the first-order derivatives, it is necessary to use a finite-difference approximation with step length h_i . The finite difference approximations are

$$\begin{aligned} D_1 &\approx \frac{m(x_1 + h_1, x_2) - m(x_1 - h_1, x_2)}{2h_1} \\ D_2 &\approx \frac{m(x_1, x_2 + h_2) - m(x_1, x_2 - h_2)}{2h_2}. \end{aligned}$$

These expressions suggest that the nonparametric estimates of the derivatives are obtained by replacing $m(x)$ by $\hat{m}(x)$. In which case

$$\hat{D}_1 = \frac{\hat{m}(x_1 + h_1, x_2) - \hat{m}(x_1 - h_1, x_2)}{2h_1} \quad (11.41)$$

$$\hat{D}_2 = \frac{\hat{m}(x_1, x_2 + h_2) - \hat{m}(x_1, x_2 - h_2)}{2h_2}. \quad (11.42)$$

In computing these estimates, the optimal bandwidth used is adjusted by the order of the derivative being approximated. Pagan and Ullah (1999, p189) give the optimal bandwidth for the S^{th} derivative for the case of N explanatory variables as

$$h_i = \hat{\sigma}_i T^{-1/(4+N+2S)}. \quad (11.43)$$

This choice of the optimal bandwidth shows that to estimate the derivatives, the nonparametric estimator needs to be based on a larger bandwidth than would be used in the computation of the conditional mean.

Figure 11.11 plots the nonparametric kernel regression estimator of the production function using a Gaussian product kernel, with the bandwidths based on $S = 0$ and $N = 2$

$$\begin{aligned} h_1 &= \hat{\sigma}_1 T^{-1/(4+N+2S)} = 0.306 \times 200^{-1/(4+2+0)} = 0.126 \\ h_2 &= \hat{\sigma}_2 T^{-1/(4+N+2S)} = 0.288 \times 200^{-1/(4+2+0)} = 0.119, \end{aligned}$$

where $\hat{\sigma}_i$ are the sample standard deviations of $x_{i,t}$. The grid points are chosen as

$$x_1, x_2 = \{0.00, 0.02, 0.04, \dots, 0.98, 1.00\}.$$

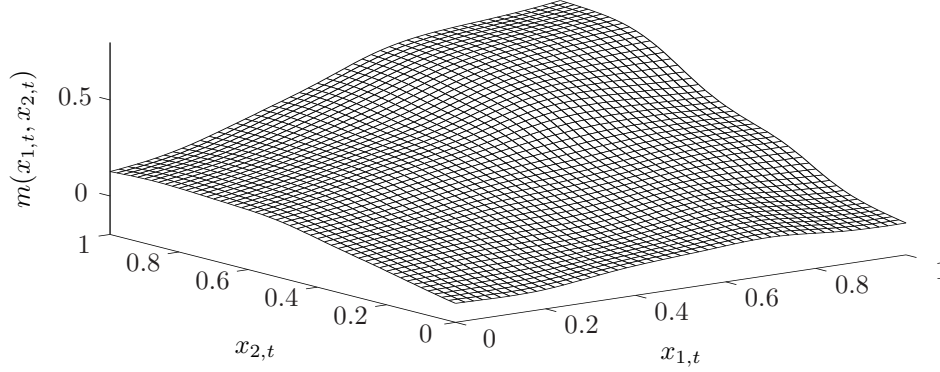


Figure 11.11 Bivariate nonparametric kernel regression estimator of a production function.

To compute the first derivatives, the bandwidths with $S = 1$, are

$$\begin{aligned} h_1 &= \hat{\sigma}_1 T^{-1/(4+N+2S)} = 0.306 \times 200^{-1/(4+2+2)} = 0.158 \\ h_2 &= \hat{\sigma}_2 T^{-1/(4+N+2S)} = 0.288 \times 200^{-1/(4+2+2)} = 0.149, \end{aligned}$$

The nonparametric estimates of the two elasticities evaluated at $x_1 = x_2 = 0.5$, are computed using equations (11.41) and (11.42) as

$$\begin{aligned} \hat{D}_1 &= \frac{\hat{m}(0.5 + 0.158, 0.5) - \hat{m}(0.5 - 0.158, 0.5)}{2 \times 0.158} = \frac{0.306 - 0.187}{2 \times 0.158} = 0.378 \\ \hat{D}_2 &= \frac{\hat{m}(0.5, 0.5 + 0.149) - \hat{m}(0.5, 0.5 - 0.149)}{2 \times 0.149} = \frac{0.336 - 0.155}{2 \times 0.149} = 0.609, \end{aligned}$$

where the pertinent conditional mean estimates $\hat{m}(x_1, x_2)$, are given in Table 11.4.

For comparative purposes, the population elasticities, evaluated at $x_1 = x_2 = 0.5$, are

$$\begin{aligned} D_1 &= \frac{\partial m(x_1, x_2)}{\partial x_1} = \frac{\exp(-5x_1)}{\exp(-5x_1) + 2\exp(-5x_2)} = 0.333, \\ D_2 &= \frac{\partial m(x_1, x_2)}{\partial x_2} = \frac{2\exp(-5x_2)}{\exp(-5x_1) + 2\exp(-5x_2)} = 0.667, \end{aligned}$$

showing that the nonparametric estimates of the derivatives are accurate to at least one decimal place.

Table 11.4

Nonparametric estimates of the elasticities of the nonlinear production function example. Derivatives are evaluated at $x_1 = x_2 = 0.5$.

$\widehat{m}(x_1, x_2)$				Nonpara.		True	
				$\frac{\partial m(x)}{\partial x_1}$	$\frac{\partial m(x)}{\partial x_2}$	$\frac{\partial m(x)}{\partial x_1}$	$\frac{\partial m(x)}{\partial x_2}$
				</			

man and Pearson (2000, pp.361-362),

$$\alpha = 0.21459, \mu = 0.08571, \sigma = 0.07830,$$

where $\Delta = 1/250$ represents an approximate daily frequency. In simulating the model the total length of the sample is chosen as $T = 7,500$ with the initial value of the interest rate set at μ .

Defining $E_{t-1}[r_t - r_{t-1}]$ as the conditional expectations operator using information up to time $t - 1$, the conditional mean is then

$$E_{t-1}[r_t - r_{t-1}] = E_{t-1}[\alpha(\mu - r_{t-1})\Delta + \sigma r_{t-1}^{1/2} \Delta^{1/2} e_t] = \alpha(\mu - r_{t-1})\Delta,$$

as $E_{t-1}[e_t] = 0$. Alternatively

$$\frac{E_{t-1}[r_t - r_{t-1}]}{\Delta} = \alpha(\mu - r_{t-1}),$$

which is linear in the interest rate. A plot of $(r_t - r_{t-1})/\Delta$ is given in panel (a) of Figure 11.12.

Further, the conditional variance is given by

$$\begin{aligned} E_{t-1}[(r_t - r_{t-1})^2] &= E_{t-1}[(\alpha(\mu - r_{t-1})\Delta + \sigma r_{t-1}^{1/2} \Delta^{1/2} e_t)^2] \\ &= E_{t-1}[(\alpha(\mu - r_{t-1})\Delta)^2 + (\sigma r_{t-1}^{1/2} \Delta^{1/2} e_t)^2 + 2\alpha(\mu - r_{t-1})\sigma r_{t-1}^{1/2} \Delta^{3/2} e_t] \\ &= \alpha^2(\mu - r_{t-1})^2 \Delta^2 + \sigma^2 r_{t-1} \Delta E_{t-1}[e_t^2] + 2\alpha(\mu - r_{t-1})\sigma r_{t-1}^{1/2} \Delta^{3/2} E_{t-1}[e_t] \\ &= \alpha^2(\mu - r_{t-1})^2 \Delta^2 + \sigma^2 r_{t-1} \Delta \\ &= \sigma^2 r_{t-1} \Delta + O(\Delta^2). \end{aligned}$$

Ignoring terms of Δ^2 , which are of a smaller magnitude than Δ , an alternative expression is

$$\frac{E_{t-1}[(r_t - r_{t-1})^2]}{\Delta} = \sigma^2 r_{t-1},$$

which shows that the conditional variance is also linear in r_t . A plot of $(r_t - r_{t-1})^2/\Delta$ is given in panel (c) of Figure 11.12.

To estimate the conditional mean using the nonparametric regression estimator

$$m(x) = \frac{\sum_{t=1}^T y_t K\left(\frac{x - x_t}{h}\right)}{\sum_{t=1}^T K\left(\frac{x - x_t}{h}\right)},$$

define the variables

$$y_t = \frac{r_t - r_{t-1}}{\Delta}, \quad x_t = r_{t-1}.$$

Using a Gaussian kernel with a bandwidth of $h = 0.023$, the nonparametric

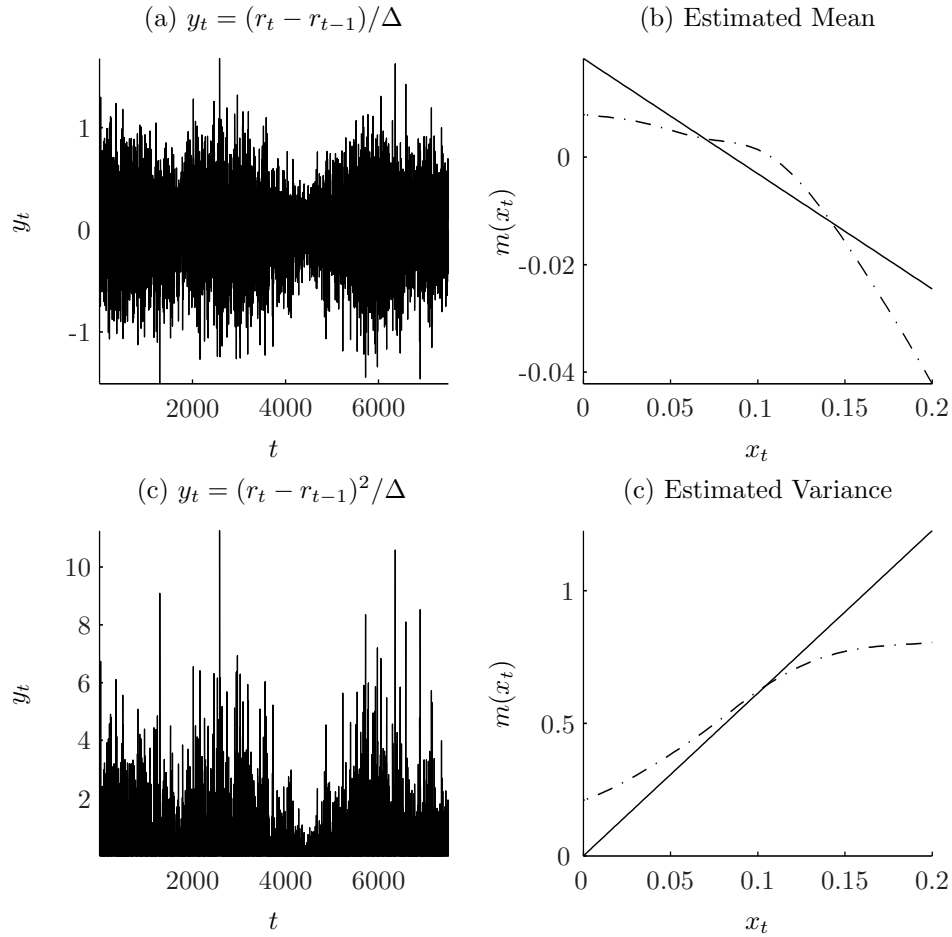


Figure 11.12 Nonparametric estimates of the conditional mean and the conditional variance (dashed lines) of the stochastic differential equation of the interest rate.

conditional mean estimates are given in panel (b) of Figure 11.12 based on the grid $x = r = \{0.000, 0.002, 0.004, \dots, 0.200\}$. For comparison, the true conditional mean, $\alpha(\mu - r)$, is also shown. The nonparametric estimates of the conditional mean suggest a nonlinear structure when in fact the true conditional mean is linear. This feature of the simulation experiment highlights a more general problem in using nonparametric kernel regression procedures; namely, the estimates can become imprecise in the tails of the data.

To repeat the exercise for the conditional variance, define the variables

$$y_t = \frac{(r_t - r_{t-1})^2}{\Delta} \quad x_t = r_{t-1}.$$

The nonparametric estimates of the conditional variance are given in panel (d) of Figure 11.12, using a Gaussian kernel with a bandwidth of $h = 0.023$ over the same grid of values for $x = r$ as used in computing the conditional mean. For comparison, the true conditional variance, $\sigma^2 r$, is also given. As with the conditional mean, the nonparametric estimates of the conditional variance exhibit a spurious nonlinearity when compared with the true linear conditional variance.

11.11 Exercises

(1) Parametric Approximations

Gauss file(s)	<code>npd_parametric.g</code>
Matlab file(s)	<code>npd_parametric.m</code>

Simulate $T = 200$ random numbers from a normal distribution

$$f(y; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - \mu)^2}{2\sigma^2}\right),$$

with parameters $\mu = 0$ and $\sigma^2 = 9$.

- (a) Estimate the population distribution using a normal distribution parametric density estimator where the mean and the variance are estimated as

$$\bar{y} = \frac{1}{T} \sum_{t=1}^{200} y_t \quad s^2 = \frac{1}{T-1} \sum_{t=1}^{200} (y_t - \bar{y})^2.$$

- (b) Estimate the population distribution using a Student t parametric density estimator where the mean and variance are computed as in (b) and the degrees of freedom parameter is computed as

$$\nu = \frac{2s^2}{s^2 - 1}.$$

- (c) Plot the parametric density estimates and compare the results with the normal distribution, $N(0, 9)$.

(2) Numerical Illustration of the Kernel Density Estimator

Gauss file(s)	<code>npd_kernel.g</code>
Matlab file(s)	<code>npd_kernel.m</code>

This exercise uses the $T = 20$ observations on y_t that are tabulated in Table 11.2 and which are drawn from an unknown distribution, $f(y)$.

- Using the Gaussian kernel with a bandwidth of $h = 1$, estimate the nonparametric density at the points $y = \{5, 10, 15\}$. Repeat for $h = 2$.
- Plot the nonparametric density estimates for the two bandwidths and comment on the differences between the two estimates. Compare your results with the estimates reported in Table 11.2.

(3) Normal Distribution

Gauss file(s)	<code>npd_normal.g</code>
Matlab file(s)	<code>npd_normal.m</code>

Simulate $T = 200$ random numbers from a normal distribution

$$f(y; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - \mu)^2}{2\sigma^2}\right),$$

with parameters $\mu = 0$ and $\sigma^2 = 9$. Define a grid of points $y = \{-10, -0.99, \dots, 10\}$.

- Use a nonparametric kernel estimator to estimate the density with the Gaussian kernel and bandwidths of $h = \{1.0, 2.0\}$.
- Use a nonparametric kernel estimator to estimate the density with the Gaussian kernel and a bandwidth of $h_{\text{opt}} = 1.06 \hat{\sigma} T^{-1/5}$.
- Estimate the density using the normal distribution where the mean and the variance are the corresponding sample statistics.
- Compare the parametric and nonparametric results.

(4) Sampling Properties of Kernel Density Estimators

Gauss file(s)	<code>npd_property.g</code>
Matlab file(s)	<code>npd_property.m</code>

Generate a sample of size $T = 500$ from a standardized normal distribution.

- (a) Compute Gaussian kernel density estimators with bandwidths $h = \{0.1, 1.0\}$ and compare your estimate with the population distribution. Compare the results with Figure 11.1.
- (b) Discuss the relationship between bias and bandwidth, and variance and bandwidth.

(5) **Kernel Density of the FTSE Share Index**

Gauss file(s)	<code>npd_ftse.g, ftse.dat</code>
Matlab file(s)	<code>npd_ftse.m, ftse.mat</code>

This exercise is based on $T = 7,000$ daily returns on the FTSE index for the period 20 November 1973 to 23 July 2001.

- (a) Estimate the kernel density using a Gaussian kernel with bandwidth $h_{\text{opt}} = 1.06 \hat{\sigma} T^{-1/5}$.
- (b) Re-estimate the kernel density using bandwidths $h = \{0.0001, 0.01, 0.05\}$.
- (c) Compare the kernel estimates and comment on the size of the bias and variance of each of the estimates.
- (d) Compare the kernel estimates with the normal distribution where the mean and the variance are the corresponding sample statistics.

(6) **Kernel Density of the S&P Share Index**

Gauss file(s)	<code>npd_sp500.g, s&p500.dat</code>
Matlab file(s)	<code>npd_sp500.m, s&p500.mat</code>

This exercise is based on $T = 12,000$ daily returns on the S&P 500 index for the period 9 February 1954 to 23 July 2001.

Repeat parts (a) to (d) of Question 5.

(7) **Sampling Distribution of the t-Statistic**

Gauss file(s)	<code>npd_cauchy.g</code>
Matlab file(s)	<code>npd_cauchy.m</code>

Simulate 10000 samples of size $T = 100$ observations from the Cauchy distribution

$$f(y_t) = \frac{1}{\pi} \frac{1}{1 + y_t^2}.$$

You can use the inverse cumulative density technique to draw from a Student t distribution with 1 degree of freedom.

- (a) For each replication compute the t-statistic

$$t = \frac{\sqrt{T}\bar{y}}{s_y},$$

where

$$\bar{y} = \frac{1}{T} \sum_{t=1}^T y_t, \quad s_y^2 = \frac{1}{T-1} \sum_{t=1}^T (y_t - \bar{y})^2,$$

are respectively the sample mean and the sample variance.

- (b) Plot the 10000 values of the t-statistics.
 (c) Compute the histogram of the t-statistics with 21 bins. Discuss the shape of the sampling distribution.
 (d) Compute kernel density estimates of the t-statistics using the following bandwidths $h = \{0.1, 0.2, 0.5, 1.0, 1.06 \hat{\sigma} T^{-1/5}\}$.
 (e) Compare the estimated sampling distributions with the asymptotic (normal) distribution. Why are the two distributions different in this case? *Hint: Redo the experiment with the degrees of freedom equal to 2.*

(8) Multivariate Kernel Density Estimation

Gauss file(s)	<code>npd_bivariate.g</code>
Matlab file(s)	<code>npd_bivariate.m</code>

Estimate a bivariate Gaussian product kernel based on $T = 20000$ observations from two independent normal distributions. Compare your answer with the results in Figure 11.3.

(9) Semi-parametric Density Estimation

Gauss file(s)	<code>npd_semilin.g</code> , <code>npd_seminonlin.g</code>
Matlab file(s)	<code>npd_semilin.m</code> , <code>npd_seminonlin.m</code>

- (a) Consider the AR(1) linear model

$$y_t = \phi_0 + \phi_1 y_{t-1} + \sigma z_t, \quad z_t \sim N(0, 1).$$

Using the observations $y_t = \{1.000, 0.701, -0.527, 0.794, 1.880\}$ and the parameter values $\theta = \{\mu = 1.0, \phi = 0.5, \sigma^2 = 1.0\}$ estimate the stationary density using the LAE and the nonparametric kernel density estimator based on a Gaussian kernel with a rule-of-thumb bandwidth. Compare these estimates with the true stationary density.

- (b) Repeat part (a) for the nonlinear model

$$y_t = \theta |y_{t-1}| + \sqrt{1 - \theta^2} z_t, \quad z_t \sim N(0, 1).$$

where the observations are $y_t = \{0.600, -0.279, -1.334, 0.847, 0.894\}$ and the parameter is $\theta = 0.5$.

(10) **Parametric Conditional Mean Estimator**

Gauss file(s)	<code>npr_parametric.g</code>
Matlab file(s)	<code>npr_parametric.m</code>

Simulate the following model for $T = 500$ observations

$$y_t = 0.3 \exp(-4(x_t + 1)^2) + 0.7 \exp(-16(x_t - 1)^2) + u_t,$$

where $u_t \sim N(0, 0.1)$ and $x_t \sim U[-2, 2]$.

- (a) Estimate the conditional mean of the linear model

$$y_t = \beta_0 + \beta_1 x_t + v_t.$$

- (b) Estimate the conditional mean of the quartic polynomial model

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_t^2 + \beta_3 x_t^3 + \beta_4 x_t^4 + v_t.$$

- (c) Compare the conditional mean estimates in parts (a) and (b) with the true conditional mean

$$m(x) = 0.3 \exp(-4(x + 1)^2) + 0.7 \exp(-16(x - 1)^2).$$

Also compare your results with Figure 11.5.

(11) **Nadaraya-Watson Conditional Mean Estimator**

Gauss file(s)	<code>npr_nadwatson.g</code>
Matlab file(s)	<code>npr_nadwatson.m</code>

Simulate $T = 20$ observations from the model used in Exercise 10.

- (a) Using a Gaussian kernel with a bandwidth of $h = 0.5$ for $x = -2.0$, and compute $\hat{m}(-2.0)$.

$$\hat{f}_x(-2.0), \quad \hat{f}_{yx}(-2.0), \quad .$$

Compare these results with Table 11.3.

- (b) Using a Gaussian kernel with a bandwidth of $h = 0.5$ for $x = -1.75$, compute

$$\hat{f}_{xx}(-1.75), \quad \hat{f}_{yx}(-1.75), \quad \hat{m}(-1.75).$$

Compare these results with Table 11.3.

- (c) Compute the Nadaraya-Watson estimator using a Gaussian kernel with a bandwidth of $h = 0.5$ for the $N = 17$ grid points

$$x = \{-2.0, -1.75, -0.5, \dots, 1.75, 2.0\}.$$

Compare the conditional mean estimates with the actual values y_t , and the true conditional mean

$$m(x) = 0.3 \exp(-4(x+1)^2) + 0.7 \exp(-16(x-1)^2).$$

Also compare your results with Figure 11.6 panel (a).

- (d) Repeat part (c) with $h = 0.1$. Compare your results with Figure 11.6 panel (b). Discuss the effects on the nonparametric estimates of $m(x)$ by changing the bandwidth from $h = 0.5$ to $h = 0.1$.

(12) Properties of the Nadaraya-Watson Estimator

Gauss file(s)	npr_property.g
Matlab file(s)	npr_property.m

Simulate $T = 500$ observations from the model used in Exercises 10 and 11.

- (a) Estimate the conditional mean using the Nadaraya-Watson estimator with a Gaussian kernel and bandwidths $h = \{0.02, 0.2\}$.
- (b) Compare the nonparametric estimates with the true conditional mean and comment on the bias and variance of the estimator as a function of h .

(13) Cross Validation Bandwidth Selection

Gauss file(s)	npr_crossvalid.g
Matlab file(s)	npr_crossvalid.m

Simulate $T = 100$ observations from the model used in Exercises 10, 11 and 12.

- (a) Find the optimal bandwidth by cross validation using a grid of bandwidths, $h = \{0.001, 0.002, \dots, 0.5\}$.

- (b) Repeat the exercise for samples of size $T = 200, 500, 1000$ and comment on the relationship between the optimal bandwidth based on cross validation and the sample size.
- (c) Show that by not using cross validation, whereby the conditional mean is computed using (11.27), this leads to the degenerate solution of $h = 0$.

(14) **Bivariate Nonparametric Regression**

Gauss file(s)	<code>npr_bivariate.g</code>
Matlab file(s)	<code>npr_bivariate.m</code>

Simulate the model

$$y_t = x_{1,t}x_{2,t}^2 + u_t,$$

where $u_t \sim N(0, 0.01)$ and $x_{i,t} \sim N(0, 1), i = 1, 2$, for $T = 20,000$ observations. Compute the Nadaraya-Watson estimator with bandwidths given by equation (11.12). Compare the result with Figure 11.9.

(15) **Nonparametric Estimates of Derivatives**

Gauss file(s)	<code>npr_production.g</code>
Matlab file(s)	<code>npr_production.m</code>

Consider $T = 200$ observations from the nonlinear production function relating output y_t and the inputs $x_{1,t}$ and $x_{2,t}$

$$y_t = -0.2 \ln(\exp(-5x_{1,t}) + 2 \exp(-5x_{2,t})) + u_t,$$

where $u_t \sim N(0, 0.01)$ and $x_{1,t}, x_{2,t} \sim U(0, 1)$.

- (a) Compute the nonparametric estimates of the first derivatives

$$D_1 = \frac{\partial m(x_1, x_2)}{\partial x_1}, \quad D_2 = \frac{\partial m(x_1, x_2)}{\partial x_2},$$

with the derivatives evaluated at $x_1 = x_2 = 0.5$. Compare these estimates and the corresponding true values of the derivatives.

- (b) Repeat part (a) for the second derivatives

$$\frac{\partial^2 m(x_1, x_2)}{\partial x_1^2}, \frac{\partial^2 m(x_1, x_2)}{\partial x_2^2}, \frac{\partial^2 m(x_1, x_2)}{\partial x_1 \partial x_2}.$$

(16) Estimating Drift and Diffusion Functions

Gauss file(s)	<code>npr_chapman.g</code>
Matlab file(s)	<code>npr_chapman.m</code>

The true data generating process of the interest rate is given by the SDE

$$dr = \alpha(\mu - r)dt + \sigma r^{1/2}dW,$$

where $dW \sim N(0, dt)$ and $\{\alpha, \mu, \sigma\}$ are parameters.

- (a) Generate a sample of daily observations of size $T = 2500$, by simulating the following model of the interest rate r_t

$$r_t - r_{t-1} = \alpha(\mu - r_{t-1})\Delta + \sigma r_{t-1}^{1/2}\Delta^{1/2}e_t,$$

where $\Delta = 1/250$, $e_t \sim N(0, 1)$ and the parameters are

$$\alpha = 0.21459 \quad \mu = 0.08571 \quad \sigma = 0.07830.$$

- (b) Plot the following following series and discuss their properties

$$y_{1,t} = \frac{r_t - r_{t-1}}{\Delta}, \quad y_{2,t} = \frac{(r_t - r_{t-1})^2}{\Delta}.$$

- (c) Let the dependent and independent variables be respectively

$$y_t = \frac{r_t - r_{t-1}}{\Delta} \quad x_t = r_{t-1}.$$

Estimate the conditional mean using the Nadaraya-Watson non-parametric estimator with a Gaussian kernel and a bandwidth of $h = 0.023$ over the grid of values $x = r = \{0.000, 0.002, 0.004, \dots, 0.200\}$.

Compare the estimated values with the true value, $\alpha(\mu - r)$.

- (d) Let the dependent and independent variables be respectively

$$y_t = \frac{(r_t - r_{t-1})^2}{\Delta} \quad x_t = r_{t-1}.$$

Estimate the conditional mean using the Nadaraya-Watson non-parametric estimator with the same specification as for (c). Compare the estimated values with the true value, $\sigma^2 r$.

12

Estimation by Simulation

12.1 Introduction

As emphasized in Chapter 1, maximum likelihood estimation requires that the likelihood function, defined in terms of the probability distribution of the observed data, y_t , is tractable. Some important examples in economics and finance where the likelihood cannot be evaluated easily are continuous-time models where data are observed discretely, factor GARCH models and ARMA models where the random variable is discrete.

A popular solution to the problem of an intractable likelihood function is to use an estimation procedure based on simulation. There are two important requirements for the successful implementation of this approach.

- (1) The actual model with the intractable likelihood function whose parameters, θ , are to be estimated, can be simulated.
- (2) There exists an alternative (auxiliary) model with parameters β that provides a good approximation to the true model and can be estimated by conventional methods.

The underlying idea behind the simulation estimator is that the estimates $\hat{\theta}$ are chosen in such a way as to ensure that the data simulated using the true model yield the same estimates of the auxiliary model, $\hat{\beta}_s(\theta)$, as does the observed data, $\hat{\beta}$.

Under certain conditions the simulation estimator is consistent, asymptotically normally distributed and achieves the same asymptotic efficiency as the maximum likelihood estimator. The algorithm for computing the simulation estimator is based on the GMM estimator discussed in Chapter 10 as it involves choosing the unknown parameters of the true model so that the empirical moments (based on the data) are equated to the theoretical moments (based on the simulated data), with the form of the moments de-

terminated by the auxiliary model. Three simulation estimators are discussed here, namely, Indirect Inference (Gouriéroux, Monfort and Renault, 1993); Efficient Method of Moments (Gallant and Tauchen, 1996); and Simulated Method of Moments (Duffie and Singleton, 1993; Smith, 1993). Although not discussed in this chapter, Bayesian numerical methods are frequently used in estimation by simulation. For a general introduction to these methods, see Chib and Greenberg (1996), Geweke (1999) and Chib (2001, 2008).

12.2 Motivating Example

To motivate the simulation procedures and to establish notation, consider the problem of estimating the parameters of the MA(1)

$$y_t = u_t - \theta u_{t-1}, \quad u_t \sim iid N(0, 1), \quad (12.1)$$

where $|\theta| < 1$ to ensure that the process is invertible. The aim is to estimate the unknown parameter θ . An appropriate estimation strategy is to use maximum likelihood. The log-likelihood function is

$$\begin{aligned} \ln L_T(\theta) &= \frac{1}{T-1} \sum_{t=2}^T \ln f_0(y_t | y_{t-1}, y_{t-2}, \dots; \theta) \\ &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2(T-1)} \sum_{t=2}^T (y_t + \theta u_{t-1})^2, \end{aligned}$$

which is maximized using a nonlinear iterative algorithm. Of course this is not a difficult numerical problem, but it does highlight the key estimation issues that underlie more difficult problems which motivate the need for a simulation based estimation procedure.

Suppose that instead of specifying the MA(1) model, an AR(1) model is incorrectly specified instead

$$y_t = \rho y_{t-1} + v_t, \quad v_t \sim iid N(0, 1), \quad (12.2)$$

and ρ is an unknown parameter. Estimation of the AR(1) model is certainly more simple as ρ can be estimated by maximum likelihood in one iteration. The log-likelihood function for this model is

$$\ln L_T(\theta) = \frac{1}{T-1} \sum_{t=2}^T \ln f_1(y_t | y_{t-1}; \rho) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2(T-1)} \sum_{t=2}^T (y_t - \rho y_{t-1})^2,$$

where the log-likelihood function is conditioned on the first observation y_0 ,

which is assumed known. Solving the first-order condition

$$\frac{1}{T-1} \sum_{t=2}^T y_{t-1}(y_t - \hat{\rho}y_{t-1}) = 0, \quad (12.3)$$

yields the maximum likelihood estimator

$$\hat{\rho} = \frac{\sum_{t=2}^T y_{t-1}y_t}{\sum_{t=2}^T y_{t-1}^2}, \quad (12.4)$$

which is simply the ordinary least squares estimator obtained by regressing y_t on y_{t-1} .

To show the relationship between the true model and the misspecified model, take expectations of the first-order condition in (12.3) with respect to the true MA(1) model

$$E\left[\frac{1}{T-1} \sum_{t=2}^T y_{t-1}(y_t - \hat{\rho}y_{t-1})\right] = 0.$$

Using the properties of the MA(1) model, the expectations are

$$E[y_t^2] = (1 + \theta_0^2), \quad E[y_{t-1}y_t] = -\theta_0,$$

the first-order condition becomes

$$-\theta_0 - \rho_0(1 + \theta_0^2) = 0,$$

which uses the result $\text{plim } \hat{\rho} = \rho_0$. It follows that

$$\rho_0 = -\frac{\theta_0}{(1 + \theta_0^2)},$$

which provides an analytical expression relating the parameter of the auxiliary model, ρ_0 , to the parameter of the true model, θ_0 . A natural way to estimate θ_0 is to replace ρ_0 with its estimate, $\hat{\rho}$, obtained from the auxiliary model

$$\hat{\rho} = -\frac{\hat{\theta}}{(1 + \hat{\theta}^2)},$$

and solve for $\hat{\theta}$. The solution is a quadratic equation in $\hat{\theta}$

$$\hat{\rho}\hat{\theta}^2 + \hat{\theta} + \hat{\rho} = 0.$$

This equation has two roots, with the root falling within the unit circle (see Chapter 13) taken as the estimator of θ

$$\hat{\theta} = \frac{-1 + \sqrt{1 - 4\hat{\rho}^2}}{2\hat{\rho}}.$$

Example 12.1 AR(1) Model

The MA(1) model in equation (12.1) is simulated with $\theta_0 = 0.5$ and $u_t \sim iid N(0, 1)$ to obtain $T = 250$ observations. The simulated data are presented in Figure 12.1.

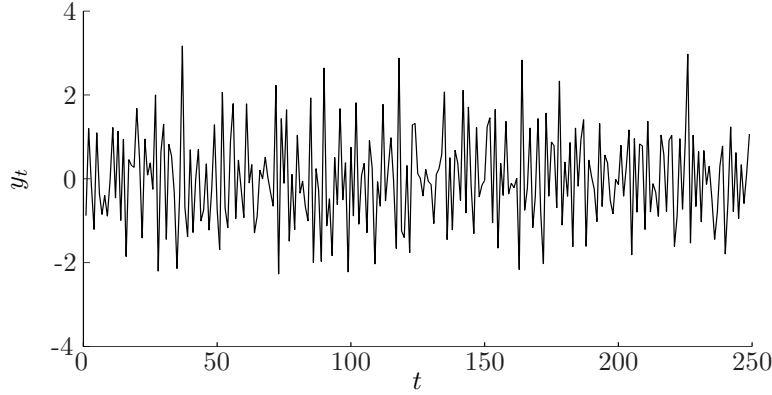


Figure 12.1 Simulated data from a AR(1) model.

The first-order autocorrelation coefficient, estimated on the zero mean data, is $\hat{\rho} = -0.411$. The estimate of $\hat{\theta}$ is therefore

$$\hat{\theta} = \frac{-1 + \sqrt{1 - 4\hat{\rho}^2}}{2\hat{\rho}} = \frac{-1 + \sqrt{1 - 4(-0.411)^2}}{-20.411} = 0.525,$$

which is in good agreement with the true population parameter $\theta_0 = 0.5$. $\square$

The analysis of the MA(1) model highlights an important point that is exploited by estimation methods using simulation, namely, that the empirical AR(1) coefficient estimated from the sample data, $\hat{\rho}$, is equated with the theoretical AR(1) coefficient, $-\theta/(1 + \theta^2)$ and used to obtain an estimate of the MA(1) parameter. Simulation estimation methods use this approach but with the theoretical parameter replaced by coefficient estimates obtained from data simulated using the true model.

12.3 Indirect Inference

This estimator was first proposed by Gouriéroux, Monfort and Renault (1993). In this approach, parameters of the true model are not estimated directly by maximizing the log-likelihood function, but indirectly through another model, known as the auxiliary model. There is also a developing

literature on approximate Bayesian computation which is linked to indirect inference (Beaumont, Cornuet, Marin and Robert, 2009; Drovandi, Pettitt and Faddy, 2011).

12.3.1 Estimation

As usual, let θ denote the parameter vector of the true model to be estimated and let β represent the parameters of an auxiliary model which is easily estimated. Indirect estimation extracts information about θ indirectly through the estimates of β , by ensuring that the estimates of the parameters of the auxiliary model are linked to the parameters of the true model by a binding function $\beta = g(\theta)$. If the dimension of β is identical to that of θ , then the binding function is invertible and θ is estimated indirectly using $\theta = g^{-1}(\beta)$. In practice, the binding function between the β and θ is recovered by simulation. Observations are simulated from the true model for given θ and then used to find the corresponding estimate for β thus creating a realization of the binding function.

A central feature of estimation is the simulation of the true model for given values of θ . The length of the simulated series is $S = T \times K$ where K is a constant. When simulating the model it is common practice to simulate more observations than strictly required and then to discard the supernumerary initial observations. This practice, known as burn in, has the effect of reducing dependence of the simulated series on the choice of initial conditions. Another advantage of using this practice is that if p lagged values of y_t are present in the auxiliary model the last p discarded observations from the burn-in sample may be treated as the initial conditions for these lags. This means that none of the S observations are lost when estimating the auxiliary model using the simulated data.

The estimation procedure is now as follows. Estimate the parameters of the auxiliary model using the observed sample data, $\hat{\beta}$. Simulate the model for given θ and use the S simulated observations to estimate the parameters of the auxiliary model from the simulated data, $\hat{\beta}_s$. The indirect estimator is the solution of

$$\hat{\theta} = \arg \min_{\theta} (\hat{\beta} - \hat{\beta}_s(\theta))' W_T^{-1} (\hat{\beta} - \hat{\beta}_s(\theta)), \quad (12.5)$$

where W_T is the optimal weighting matrix discussed in in Chapters 9 and 10. The need for a weighting matrix stems from the fact that the auxiliary model with parameters β represents a misspecification of the true model

with parameters θ . If independence is assumed then

$$W = E[(\hat{\beta} - \hat{\beta}_s(\theta))(\hat{\beta} - \hat{\beta}_s(\theta))'] . \quad (12.6)$$

The model is just identified if the dimension of β equals the dimension of θ . The model is over-identified if the dimension of β exceeds the dimension of θ . If the dimension of β is less than the dimension of θ , then the model is under-identified in which case it is not possible to compute the indirect estimator. For the over-identified models where the weighting matrix is chosen to be the identity matrix, the indirect estimator is still consistent but not asymptotically efficient.

Example 12.2 AR(1) Model

Consider the MA(1) model in equation (12.1) which is simulated in Example 12.1. The MA(1) parameter θ may be estimated by indirect inference using an AR(1) model as the auxiliary model by implementing the following steps.

Step 1: Estimate the parameter of the auxiliary model, ρ , using the observed data y_t to give

$$\hat{\rho} = \frac{\sum_{t=2}^T y_t y_{t-1}}{\sum_{t=2}^T y_{t-1}^2} .$$

Step 2: Choose a value of θ initially, say $\theta_{(0)}$.

Step 3: Let $y_{s,t}$ represent a time series of simulated data of length $S = T \times K$ from the true model, in this case a AR(1) model, given $\theta = \theta_{(0)}$.

Step 4: Compute the first-order autocorrelation coefficient using the simulated data

$$\hat{\rho}_s = \frac{\sum_{t=1}^S y_{s,t-1} y_{s,t}}{\sum_{t=1}^S y_{s,t-1}^2} ,$$

where the limits of the summation reflect the fact that the starting value y_0 is available from the discarded burn-in observations.

Step 5: The indirect estimator is to choose θ , so that

$$\hat{\rho} = \hat{\rho}_s ,$$

in which case $-\hat{\theta}/(1 + \hat{\theta}^2)$ is replaced by $\hat{\rho}_s$ in (12.2).

Setting $K = 1$ results in the simulated series having the same length as the actual series. As K increases, the simulated AR(1) coefficient, $\hat{\rho}_s$, approaches the theoretical AR(1) coefficient, ρ_0 . In the limit as $K \rightarrow \infty$, the simulated and theoretical AR(1) coefficients are equal. $\square$

Example 12.3 Accuracy of Simulated Moments

The true model is a MA(1)

$$y_t = u_t - \theta u_{t-1}, \quad u_t \sim iid N(0, 1),$$

with $\theta_0 = 0.5$. The sample size is $T = 250$. Table 12.1 gives the AR(1) coefficient estimates for various values of K . The true AR(1) parameter is

$$\rho_0 = -\frac{\theta_0}{(1 + \theta_0^2)} = -\frac{0.5}{(1 + 0.5^2)} = -0.4.$$

The results reported in Table 12.1 demonstrate that as K increases, the simulated AR(1) coefficient estimates approach the true population autocorrelation parameter value of -0.4 . Note that the sequence of AR(1) estimates do not monotonically approach $\rho = -0.4$. This is simply a reflection of simulation error, where for different simulation runs, different seeds for the random number generator will yield different numerical results. The overall qualitative behaviour of the results, however, remains the same. $\square$

Table 12.1

Estimated AR(1) coefficient estimates obtained from simulating a MA(1) model with $\theta_0 = 0.5$, for various simulation sample sizes. The sample size of the observed data is $T = 250$.

K	$S = T \times K$	AR(1)
1	250	-0.411
2	500	-0.375
3	750	-0.360
4	1000	-0.385
5	1250	-0.399
10	2500	-0.403
100	5000	-0.399
True:		-0.400

Rewrite the MA(1) model from Examples 12.1 and 12.2 as an infinite AR model as follows

$$\begin{aligned}
 y_t &= u_t - \theta u_{t-1} \\
 y_t &= (1 - \theta L)u_t \\
 (1 - \theta L)^{-1}y_t &= u_t \\
 (1 + \theta L + \theta^2 L^2 + \cdots)y_t &= u_t \\
 y_t &= -\theta y_{t-1} - \theta^2 y_{t-2} - \cdots + u_t, \quad (12.7)
 \end{aligned}$$

Table 12.2

Sampling properties of the indirect estimator of θ for the MA(1) example. Based on a Monte Carlo experiment with 1000 replications. The sample size is $T = 250$.

Statistics	Auxiliary model		
	AR(1)	AR(2)	AR(3)
True	0.500	0.500	0.500
Mean	0.534	0.508	0.501
St. dev.	0.184	0.108	0.088
RMSE	0.187	0.108	0.088

where L is the lag operator (see Appendix B). The AR(1) model is therefore a first order approximation of the more correct auxiliary model, given by an infinite AR model. This suggests that a better approximation to the true model is obtained by choosing as the auxiliary model an AR model with P lags. In the limit the auxiliary model approaches the true model as $P \rightarrow \infty$. In this case the efficiency of the indirect estimator approaches the maximum likelihood estimator as given by the Cramér-Rao lower bound. In practice, the indirect estimator achieves a comparable efficiency level as that of the maximum likelihood estimator for a finite lag structure.

Example 12.4 Over-Identified AR(1) Model

Table 12.2 gives statistics on the sampling properties of the indirect estimator for a sample of size $T = 250$ using a Monte Carlo experiment based on 1000 replications. The indirect estimator is computed by setting the length of the simulated series to be $S = T = 250$ observations, that is $K = 1$. Three auxiliary models are used, beginning with an AR(1) model, then an AR(2) model and finally an AR(3) model. For convenience, the weighting matrix is taken to be the identity matrix in this example.

The mean of the simulations shows that the bias of the indirect estimator decreases as the dimension of the auxiliary model increases. There is also a decrease in the RMSE showing that the indirect estimator based on the AR(3) model is statistically more efficient than the indirect estimators based on lower AR lags. $\square$

12.3.2 Relationship with Indirect Least Squares

The goal of indirect inference is to obtain a consistent estimator of the true model even when the estimated model is misspecified. This objective is not new to econometrics and has been around for a long time as a method for obtaining consistent estimates of the parameters of simultaneous equation models (see Chapter 5).

Consider the simple Keynesian two-sector model consisting of a consumption function and the income identity

$$\begin{aligned} c_t &= \alpha + \phi y_t + u_t \\ y_t &= c_t + i_t \\ u_t &\sim iid N(0, \sigma^2), \end{aligned} \tag{12.8}$$

where c_t is consumption, y_t is income, i_t is investment which is assumed to be exogenous so that $E[i_t u_t] = 0$, u_t and $\theta = \{\alpha, \phi\}$ are the parameters to be estimated. Estimation of the first equation by least squares is inappropriate as a result of simultaneity bias. The equation for y_t can be written as

$$y_t = \frac{\alpha}{1 - \phi} + \frac{1}{1 - \phi} i_t + \frac{1}{1 - \phi} u_t,$$

so that

$$E[y_t u_t] = \frac{\sigma^2}{1 - \phi} \neq 0.$$

The earliest method to obtain a consistent parameter estimator of θ is known as indirect least squares. This is obtained by first solving (12.8) for the reduced form. In the case of the consumption equation the pertinent equation is

$$c_t = \frac{\alpha}{1 - \phi} + \frac{\phi}{1 - \phi} i_t + \frac{1}{1 - \phi} u_t = \delta + \gamma i_t + v_t, \tag{12.9}$$

where $v_t = (1 - \phi)^{-1} u_t$ and

$$\delta = \frac{\alpha}{1 - \phi}, \quad \gamma = \frac{\phi}{1 - \phi}, \tag{12.10}$$

are the reduced form parameters. Estimation of (12.9) by least squares yields consistent estimators of $\hat{\delta}$ and $\hat{\gamma}$. Consistent estimators of the consumption function parameters in (12.8) are obtained by rearranging (12.10) and evaluating at $\hat{\delta}$ and $\hat{\gamma}$ to give the indirect least squares estimators

$$\hat{\alpha} = \frac{\hat{\delta}}{1 + \hat{\gamma}}, \quad \hat{\phi} = \frac{\hat{\gamma}}{1 + \hat{\gamma}}. \tag{12.11}$$

The indirect least squares solutions in (12.11) equivalent to the indirect inference solution where the simulation length is $K \rightarrow \infty$. The reduced form equation in (12.9) is nothing other than the auxiliary model and (12.10) corresponds to the binding function that links the true parameters $\theta = \{\alpha, \phi\}$ to the auxiliary parameters $\beta = \{\delta, \gamma\}$. Thus to estimate the consumption function parameters in (12.8) by indirect inference, the strategy would be to estimate auxiliary model (reduced form) in (12.9) by least squares using the actual data to yield $\hat{\beta} = \{\hat{\delta}, \hat{\gamma}\}$. The consumption function in (12.8) is then simulated using some starting values of $\{\alpha_{(0)}, \phi_{(0)}\}$ to generate simulated consumption data $c_{s,t}$. The auxiliary model (reduced form) in (12.9) is then re-estimated using the simulated consumption data, $c_{s,t}$, and the actual investment data, i_t , to generate $\hat{\beta}_s$. As the model is just identified, the indirect inference estimators are obtained when $\theta = \{\alpha, \phi\}$ is iterated until $\hat{\beta} = \hat{\beta}_s$.

12.4 Efficient Method of Moments (EMM)

Gallant and Tauchen (1996) introduce a variant of the indirect estimator, commonly known as the efficient method of moments (EMM).

12.4.1 Estimation

Let β represent the $M \times 1$ vector of parameters of the auxiliary model and $\hat{\beta}$ the corresponding parameter estimates using the actual data. The EMM estimator is the solution of

$$\hat{\theta} = \arg \min_{\theta} G'_S(\theta) W_T^{-1} G_S(\theta), \quad (12.12)$$

where $G_S(\theta)$ is a $M \times 1$ vector of gradients of the auxiliary model evaluated at the maximum likelihood estimates of the auxiliary model, $\hat{\beta}$, but using the simulated data. The use of the simulated data in computing the gradients establishes the required binding function between θ and β . The optimal weighting matrix W_T is given by equation (10.39) in Chapter 10.

The intuition behind the EMM estimator is as follows. The gradient of the auxiliary model, $G_T(\beta)$, computed using the actual data and evaluated at the maximum likelihood estimator, $\beta = \hat{\beta}$, is zero by definition, that is $G_T(\hat{\beta}) = 0$. EMM chooses that θ which generates simulated data such that the gradients of the auxiliary model, $G_S(\hat{\theta})$ match the gradients obtained using the actual data, namely $G_T(\hat{\beta}) = 0$. This idea is similar to the indirect inference approach where θ is chosen to generate simulated data which yields the same estimates of the parameters of the auxiliary model as is obtained

using the actual data. An important computational advantage of the EMM estimator is that it is just necessary to estimate the auxiliary model once when solving (12.12). This is in contrast to the indirect estimator in (12.5) where it is necessary to re-estimate the auxiliary model at each iteration of the algorithm.

Example 12.5 EMM Estimation of a AR(1) Model

Consider again the MA(1) model in equation (12.1) with $\theta_0 = 0.5$. Three auxiliary models are used to estimate the parameter θ , an AR(1) model, an AR(2) model and an AR(3) model. If the model is simulated $S = T \times K$ times, in the case of the AR(1) model, the gradient is given by

$$G_S(\theta) = \frac{1}{S} \sum_{t=1}^S y_{s,t-1}(y_{s,t} - \hat{\rho}_1 y_{s,t-1}),$$

where the starting value y_0 is available from the burn-in sample. The approach is to choose θ so that (12.12) is satisfied.

Table 12.3 gives statistics on the sampling properties of the EMM estimator for a sample of size $T = 250$ using a Monte Carlo experiment. The number of replications is 1000. The EMM estimator is based on $S = T = 250$ observations, that is $K = 1$. The optimal weighting matrix is based on $P = 0$ lags in the expression for W_T given in (10.39).

Table 12.3

Sampling properties of the EMM estimator for the AR(1) example. Based on a Monte Carlo experiment with 1000 replications. The sample size is $T = 250$. The optimal weighting matrix is based on $P = 0$ lags.

Statistics	Auxiliary model		
	AR(1)	AR(2)	AR(3)
True	0.500	0.500	0.500
Mean	0.531	0.492	0.482
St. dev.	0.178	0.106	0.089
RMSE	0.180	0.106	0.091

The sampling distribution of the EMM estimator is similar to the sampling distribution of the indirect estimator given in Table 12.3. In particular, the fall in the RMSE as the lag length of the auxiliary model increases shows

the improvement in the efficiency of this estimator as the approximation of the auxiliary model improves. $\square$

12.4.2 Relationship with Instrumental Variables

The relationship between simulation estimation and simultaneous equations estimators can be generalized by showing that the instrumental variables estimator is equivalent to the EMM estimator. Consider the model

$$y_t = \theta x_t + u_t, \quad u_t \sim iid N(0, 1), \quad (12.13)$$

where $E[x_t u_t] \neq 0$, in which case least squares is inappropriate. The EMM estimator is based on estimating an auxiliary model specified as

$$x_t = \phi z_t + v_t, \quad v_t \sim iid N(0, 1), \quad (12.14)$$

where the variable z_t has the property that $E[z_t v_t] = 0$. In the simultaneous equations literature, (12.13) is the structural equation and (12.14) is the reduced form. The restriction $E[z_t u_t] = 0$, means that z_t represents a valid instrument.

The reduced form in the case of y_t is obtained by substituting x_t from (12.14) into (12.13)

$$y_t = \beta z_t + w_t, \quad (12.15)$$

where $\beta = \phi\theta$ and $w_t = u_t + \theta v_t$. The population moment condition is

$$E[(y_t - \beta_0 z_t)z_t] = 0. \quad (12.16)$$

Evaluating the expectation by the sample average yields the first order condition

$$\frac{1}{T} \sum_{t=1}^T (y_t - \hat{\beta} z_t) z_t = 0, \quad (12.17)$$

where $\hat{\beta}$ is obtained as a least squares regression of y_t on z_t . The EMM estimator is obtained by simulating (12.13) for a particular value of θ according to $y_{t,s} = \theta x_t + u_{t,s}$, with x_t treated as given, and evaluating y_t in equation (12.17) at $y_{t,s}$ according to

$$\frac{1}{S} \sum_{t=1}^S (y_{t,s} - \hat{\beta} z_t) z_t = \frac{1}{S} \sum_{t=1}^S (\hat{\theta} x_t + u_{t,s} - \hat{\beta} z_t) z_t = 0.$$

Letting $S \rightarrow \infty$ amounts to replacing the summation by the expectations

operator

$$\lim_{S \rightarrow \infty} \frac{1}{S} \sum_{t=1}^S (\hat{\theta} x_t + u_{t,s} - \hat{\beta} z_t) z_t = E[(\theta_0 x_t - \beta_0 z_t) z_t] = \theta_0 E[x_t z_t] - E[y_t z_t] = 0.$$

where the second equality uses $E[u_{t,s} z_t] = 0$, and the third equality uses $\beta_0 E[z_t^2] = E[y_t z_t]$ from (12.16). Solving for θ_0 gives

$$\theta_0 = \frac{E[y_t z_t]}{E[x_t z_t]}.$$

Replacing the expectations by the sample moments gives the instrumental variables estimator discussed in Chapter 5

$$\hat{\theta} = \frac{T^{-1} \sum_{t=1}^T x_t z_t}{T^{-1} \sum_{t=1}^T z_t^2},$$

which by construction is also the EMM estimator.

The establishment of the link between the EMM estimator and the instrumental variables estimator means that statements concerning the quality of the instrument are equivalent to statements about the quality of the auxiliary model to approximate the true model. If the auxiliary model represents a good approximation of the true model, the sampling distribution of the EMM estimator under the regularity conditions in Chapter 2 is asymptotically normal. However, if the approximation is poor, the sampling distribution will not be asymptotically normal, but will inherit the sampling distribution of the instrumental variables estimator when the instruments are weak, as investigated in Chapter 5.

12.5 Simulated Generalized Method of Moments (SMM)

An important feature of the indirect estimator defined by (12.5) is that it involves estimating the parameter vector of the true model, θ , by equating the parameter estimates of the auxiliary model using the observed data, $\hat{\beta}$, with the parameter vector of the auxiliary model computed using the simulated data, $\hat{\beta}_s$. In the case of the MA(1) example where the auxiliary model is chosen to be an AR(1), $\hat{\beta}$ is the first-order autocorrelation coefficient based on the actual data, whereas $\hat{\beta}_s$ is the first-order autocorrelation coefficient based on the simulated data. More succinctly, the indirect estimator amounts to equating the empirical moment, $\hat{\beta}$, with the theoretical moments, $\hat{\beta}_s$,

$$\begin{array}{ccc} \hat{\beta} & = & \hat{\beta}_s \\ \text{(Empirical)} & & \text{(Theoretical)} \end{array}$$

This interpretation of the indirect estimator is also equivalent to the simulated generalized method of moments estimator proposed by Duffie and Singleton (1993). Perhaps the relationship between SMM and EMM estimators is even more transparent since the EMM estimator is based on the scores, which is equivalent to the moments used in SMM.

In general, the auxiliary model can represent a (misspecified) likelihood function that yields first-order conditions corresponding to a range of moments. These moments and hence the auxiliary model, should be chosen to capture the empirical characteristics of the data. In turn, these empirical moments are used to identify the parameters of the true model being estimated by a simulation estimation procedure. In fact, Gallant and Tauchen (1996) suggest a semi-nonparametric model that provides sufficient flexibility to approximate many models. This class of models captures empirical characteristics such as autocorrelation in the mean, autocorrelation in the variance and non-normality.

A further relationship between the three simulation procedures discussed are the diagnostic tests of the models proposed. For example, the Hansen-Sargan J test of the number of over-identifying restrictions, discussed in Chapter 10, can also be applied in the indirect and EMM estimation frameworks. To implement the test, the value of the objective function is

$$Q(\hat{\theta}) = G'_S(\theta)W_T^{-1}G_S(\theta), \quad (12.18)$$

where all quantities are evaluated at the EMM estimator, $(\hat{\theta})$. Under the null hypothesis that the model is correctly specified, the test statistic is

$$J_{HS} = TQ_T \xrightarrow{d} \chi^2_{N-M},$$

where N is the number of moment conditions and M is the number of parameters.

Example 12.6 Over-Identification Test of a AR(1) Model

The true model is a MA(1)

$$y_t = u_t - \theta u_{t-1}, \quad u_t \sim iid N(0, 1),$$

with $\theta_0 = 0.5$. Table 12.4 gives the results of the over-identification test using the EMM estimator for a sample of size $T = 250$. The EMM estimator is based on $S = T = 250$ observations, that is $K = 1$. The optimal weighting matrix based on $P = 0$ lags. Three auxiliary models are used, namely an AR(1) model, an AR(2) model and finally an AR(3) model.

For the AR(1) auxiliary model, the system is just-identified in which case there is nothing to test. For the AR(2) and AR(3) auxiliary models the

computed values of the J_{HS} test statistic are 0.039 and 1.952, respectively, with associated p-values of 0.844 and 0.377. In both cases the null hypothesis that the true model is specified correctly is not rejected at conventional significance levels.

Table 12.4

Over-identification test of the AR(1) model based on the EMM estimator. The sample size is $T = 250$ and the optimal weighting matrix is based on $P = 0$ lags.

Statistics	Auxiliary model		
	AR(1)	AR(2)	AR(3)
Q	0.000	0.001	0.008
TQ	0.000	0.039	1.952
$N - M$	0.000	1.000	2.000
p-val	n.a.	0.844	0.377

12.6 Estimating Continuous-Time Models

An important area of the use of simulation estimation in economics and finance is the estimation of continuous-time models as in the application presented in Section 3.8 of Chapter 3. This class of models is summarized by the following stochastic differential equation

$$dy_t = \mu(y_t; \alpha)dt + \sigma(y_t; \beta)dB, \quad (12.19)$$

where $B(t)$ is a Wiener process such that $dB \sim N(0, dt)$. The function $\mu(y_t; \alpha)$ is the mean of dy_t , with parameter vector α , and the function $\sigma^2(y_t; \beta)$ is the variance of dy_t , with parameter vector β .

Panel (a) of Figure 12.2 gives an example of a continuous time series with $\mu(y_t; \alpha) = 0$ and $\sigma^2(y_t; \beta) = 1$, known as a Brownian motion, by simulating (12.19) using a time interval of $\Delta t = 1/60$. This choice of Δt is chosen to correspond to minute data. In practice continuous data are not available, but are measured at discrete points in time, $y_1, y_2, \dots, y_T$. For example, in Figure 12.2 panel (b) the data correspond to measuring y every 10 minutes. In Figure 12.2 panel (c) the data are hourly as y is observed every 60 minutes. In Figure 12.2 panel (d) the data are daily as y is observed every $60 \times 24 = 1440$ minutes.

In general, maximum likelihood estimation for continuous-time processes

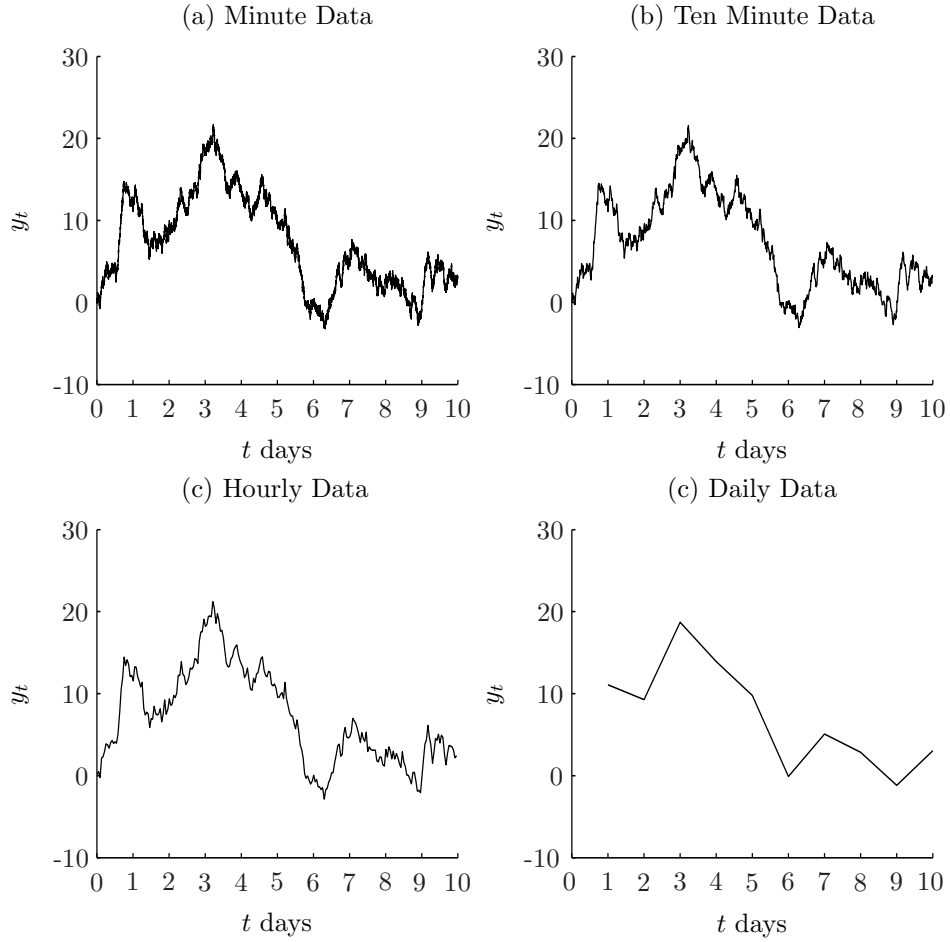


Figure 12.2 Continuous, weekly, monthly and annual data, sampled by simulating equation (12.19) with $\mu(y_t; \alpha) = 0$, $\sigma^2(y_t; \beta) = 1$ (Brownian motion) and $\Delta t = 1/60$.

is intractable as the following simple illustration demonstrates. Consider two discrete observations y_{t+1} and y_t . The records of continuous data between these two observations may be regarded as missing observations. Based on the assumption that there are $n - 1$ missing observations between y_{t+1} and y_t the joint pdf may be represented schematically as

$$f(\underbrace{y_{t+1}}_{\text{known}}, \underbrace{\cdots, y_{t+(n-1)/n}, \cdots}_{\text{missing}}, \underbrace{y_t}_{\text{known}}) \quad (12.20)$$

To derive the conditional distribution $f(y_{t+1}|y_t)$ all the missing observations

in equation (12.20) must be integrated out an operation that requires the computation of an $(n - 1)$ -dimensional integral

$$f(y_{t+1}|y_t) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(y_{t+1}, \dots, y_t) dy_{t+1/n} dy_{t+2/n} \cdots dy_{t+(n-1)/n}.$$

The construction of the likelihood function requires that this integration be computed for each transition in the data, effectively involving the evaluation of a $T \times (n - 1)$ -dimensional integral.

To estimate the parameters α and β of (12.19), using standard estimation methods including maximum likelihood, it is necessary to have continuous data. The earliest approach adopted to resolve the difference in frequency between the model and the data, was to discretize the model thereby matching the frequency of the model to the data. A problem with this strategy is that the parameters of the discrete model are in general not the parameters of the continuous time model, thereby resulting in biased estimators of α and β .

To highlight the problem of converting a continuous time model into a discrete time, consider the simplest of all models containing no parameters

$$dy_t = y dB. \quad (12.21)$$

An exact discretisation of this model is given by

$$y_t = y_{t-1} \exp[u_t - 0.5], \quad (12.22)$$

where $u_t \sim iid N(0, 1)$. An inexact discretisation is given by choosing the time interval as $\Delta t = 1$ in (12.21)

$$y_t - y_{t-1} = y_{t-1} v_t \iff \frac{y_t - y_{t-1}}{y_{t-1}} = v_t, \quad (12.23)$$

where $v_t \sim iid N(0, 1)$. In both (12.22) and (12.23) y_t is growing at a certain rate, but in (12.22) the growth is continuous whereas in (12.23) the growth is discrete. As the functional forms of the two models differ, the adoption of (12.23) represents a misspecification of the true functional form as given in (12.22).

Simulation estimation procedures circumvent this problem by simulating the continuous model, and constructing a discrete time series of simulated data which are calibrated with the observed discrete data through a set of moment conditions based on a chosen auxiliary model. Even though the incorrect discretized model is estimated, it is important to remember that the actual parameters being estimated are the parameters of the continuous-time model which are used to simulate a continuous-time process. To highlight

the approach, a simulation estimation framework is developed to estimate models of Brownian motion, geometric Brownian motion and a continuous time model of stochastic volatility.

12.6.1 Brownian Motion

Consider estimating the parameters $\theta = \{\mu, \sigma\}$ of the stochastic differential equation

$$dy_t = \mu dt + \sigma dB, \quad (12.24)$$

where $B(t)$ is a Wiener process such that $dB \sim N(0, dt)$. Equation (12.24) is known as Brownian motion.

The auxiliary model is chosen as a discretization of this model, obtained by setting $\Delta t = 1$

$$y_t - y_{t-1} = \mu_\Delta + \sigma_\Delta v_t, \quad v_t \sim iid N(0, 1), \quad (12.25)$$

where $\beta = \{\mu_\Delta, \sigma_\Delta\}$ are the parameters of the auxiliary model. The log-likelihood function of the auxiliary model is

$$\ln L_T(\beta) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma_\Delta^2 - \frac{1}{2\sigma_\Delta^2(T-1)} \sum_{t=2}^T (y_t - y_{t-1} - \mu_\Delta)^2, \quad (12.26)$$

with gradient vector

$$G_T(\beta) = \begin{bmatrix} \frac{1}{\sigma_\Delta^2(T-1)} \sum_{t=2}^T (y_t - y_{t-1} - \mu_\Delta) \\ -\frac{1}{2\sigma_\Delta^2} + \frac{1}{2\sigma_\Delta^4(T-1)} \sum_{t=2}^T (y_t - y_{t-1} - \mu_\Delta)^2 \end{bmatrix}. \quad (12.27)$$

Setting $G_T(\hat{\beta}) = 0$, yields the maximum likelihood estimators

$$\begin{aligned} \hat{\mu}_\Delta &= \frac{1}{T-1} \sum_{t=2}^T (y_t - y_{t-1}) \\ \hat{\sigma}_\Delta^2 &= \frac{1}{T-1} \sum_{t=2}^T (y_t - y_{t-1} - \hat{\mu}_\Delta)^2. \end{aligned} \quad (12.28)$$

Indirect Inference

The indirect estimator of the parameters of the model in (12.24) uses (12.28) to form the moment conditions. To estimate (12.24) by indirect inference requires the following steps.

Step 1: Estimate $\hat{\beta} = \{\hat{\mu}_\Delta, \hat{\sigma}_\Delta^2\}$ from the auxiliary model using (12.28) based on the observed data.

Step 2: Choose starting values for $\theta = \{\mu, \sigma^2\}$, say $\mu_{(0)}$ and $\sigma_{(0)}^2$, and generate continuous time data using a small time step, say $\Delta t = 0.1$

$$y_{s,t+\Delta t} - y_{s,t} = \mu_{(0)}\Delta t + \sigma_{(0)}v_{t+\Delta t},$$

where $v_t \sim iid N(0, \Delta t)$. Let the length of the simulated series be $S = T \times K$.

Step 3: Generate discrete time data $y_{s,1}, y_{s,2}, \dots, y_{s,S}$, by choosing every $1/\Delta t$ observations from the simulated continuous time series.

Step 4: Compute the parameter estimates of the auxiliary model, $\hat{\beta}_s = \{\hat{\mu}_{\Delta,s}, \hat{\sigma}_{\Delta,s}^2\}$, based on the simulated data

$$\begin{aligned}\hat{\mu}_{\Delta,s} &= \frac{1}{S} \sum_{t=1}^S (y_{s,t} - y_{s,t-1}) \\ \hat{\sigma}_{\Delta,s}^2 &= \frac{1}{S} \sum_{t=1}^S (y_{s,t} - y_{s,t-1} - \hat{\mu}_{\Delta,s})^2.\end{aligned}$$

Step 5: Define the moment conditions

$$G_S(\theta) = \begin{bmatrix} \hat{\mu}_{\Delta} - \hat{\mu}_{\Delta,s} \\ \hat{\sigma}_{\Delta}^2 - \hat{\sigma}_{\Delta,s}^2 \end{bmatrix}.$$

Step 6: Choose $\theta = \{\mu, \sigma\}$ to satisfy

$$\hat{\theta} = \arg \min_{\theta} G'_S(\theta) G_S(\theta).$$

EMM Estimation

The EMM estimator of (12.24) uses (12.27) to form the moment conditions. To estimate the parameters of the model in equation (12.24) by EMM, the steps are as follows:

Step 1: Estimate $\hat{\beta} = \{\hat{\mu}_{\Delta}, \hat{\sigma}_{\Delta}^2\}$ from the auxiliary model using (12.28) based on the observed data.

Step 2: Using equation (12.27) evaluate $G_T(\hat{\beta})$ in equation (12.27) at the maximum likelihood estimators of the auxiliary model, $\hat{\mu}_{\Delta}$ and $\hat{\sigma}_{\Delta}^2$, obtained in Step 1 and compute the weighting matrix W for a given lag length P .

Step 3: Choose starting values for $\theta = \{\mu, \sigma^2\}$, say $\mu_{(0)}$ and $\sigma_{(0)}^2$, and generate continuous time data using a small time step, say $\Delta t = 0.1$

$$y_{s,t+\Delta t} - y_{s,t} = \mu_{(0)}\Delta t + \sigma_{(0)}v_{t+\Delta t},$$

where $v_t \sim iid N(0, \Delta t)$.

Step 4: Generate discrete time data $y_{s,1}, y_{s,2}, \dots, y_{s,S}$, by choosing every $1/\Delta t$ observations from the simulated continuous time series.

Step 5: Evaluate the gradients in (12.27) using the simulated data and the maximum likelihood parameter estimates in (12.28)

$$G_S(\hat{\theta}) = \begin{bmatrix} \frac{1}{\hat{\sigma}_\Delta^2 S} \sum_{t=1}^S (y_{s,t} - y_{s,t-1} - \hat{\mu}_\Delta) \\ -\frac{1}{2\hat{\sigma}_\Delta^2} + \frac{1}{2\hat{\sigma}_\Delta^4 S} \sum_{t=1}^S (y_{s,t} - y_{s,t-1} - \hat{\mu}_\Delta)^2 \end{bmatrix}.$$

Step 6: Choose $\theta = \{\mu, \sigma\}$ to satisfy

$$\hat{\theta} = \arg \min_{\theta} G'_S(\theta) W^{-1} G_S(\theta).$$

Example 12.7 Simulation Estimation of Brownian Motion

Table 12.5 gives the sampling properties of the indirect and EMM estimators from estimating the parameters of the model in (12.24) with true parameter values $\mu_0 = 0.5$ and $\sigma_0^2 = 0.25$. The sample size is $T = 500$, the time interval is set at $\Delta t = 0.1$ and $K = 1/\Delta t = 10$. The optimal weighting matrix used to compute the EMM estimates is $P = 0$. The sampling distributions are based on 1000 Monte Carlo replications.

Table 12.5

Sampling properties of the indirect and EMM estimators of the Brownian motion parameters μ_0 and σ_0^2 . The sample size is $T = 500$, $\Delta t = 0.1$, $K = 1/\Delta t$, and the number of Monte Carlo replications is 1000.

Estimator	Parameter	Mean	Std dev.	RMSE
Indirect	μ	0.501	0.031	0.031
	σ^2	0.252	0.023	0.023
EMM	μ	0.501	0.031	0.031
	σ^2	0.252	0.028	0.028
True	μ	0.500	n.a.	n.a.
	σ^2	0.250	n.a.	n.a.

The results in Table 12.5 show that the sampling distributions of the two estimators are nearly equivalent. Apart from asymptotic reasons why the two estimators should be similar for large samples at least, this is to be

expected in this example as the moment conditions used in both estimation procedures are equivalent.

As this model is just identified, the EMM solution is where the first moment condition of the EMM estimator is satisfied, namely,

$$\frac{1}{S\hat{\sigma}_\Delta^2} \sum_{t=1}^S (y_{s,t} - y_{s,t-1} - \hat{\mu}_\Delta) = 0.$$

This is equivalent to

$$\frac{1}{S} \sum_{t=12}^S (y_{s,t} - y_{s,t-1}) - \hat{\mu}_\Delta = \hat{\mu}_{\Delta,s} - \hat{\mu}_\Delta,$$

which is the first moment used in the indirect inference procedure. The same result applies for the second moment condition. Any differences, therefore, between the numerical values of the two estimators in this example are purely due to numerical precision. $\square$

12.6.2 Geometric Brownian Motion

Consider estimating the parameters $\theta = \{\mu, \sigma\}$, of the stochastic differential equation

$$dy_t = \mu y_t dt + \sigma y_t dB, \quad (12.29)$$

where $B(t)$ is a Wiener process such that $dB \sim N(0, dt)$. The stochastic process satisfying equation (12.29) is known as geometric Brownian motion. An exact discretisation of this model is given by

$$\ln y_t - \ln y_{t-1} = (\mu - 0.5\sigma^2)\Delta t + \sigma u_t, \quad u_t \sim iid N(0, 1), \quad (12.30)$$

and an Euler discretization of this model is obtained by setting $\Delta t = 1$ in equation (12.29)

$$y_t - y_{t-1} = \mu_\Delta y_{t-1} + \sigma_\Delta y_{t-1} v_t, \quad v_t \sim iid N(0, 1). \quad (12.31)$$

Alternatively the auxiliary model is expressed as

$$z_t = \mu_\Delta + \sigma_\Delta v_t, \quad (12.32)$$

where

$$z_t = \frac{y_t - y_{t-1}}{y_{t-1}},$$

is the (discrete) growth rate. The log-likelihood function of the auxiliary model is

$$\ln L_T(\beta) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma_\Delta^2 - \frac{1}{2\sigma_\Delta^2(T-1)} \sum_{t=2}^T (z_t - \mu_\Delta)^2. \quad (12.33)$$

The gradient is given by

$$G_T(\beta) = \begin{bmatrix} \frac{1}{\sigma_\Delta^2(T-1)} \sum_{t=2}^T (z_t - \mu_\Delta) \\ -\frac{1}{2\sigma_\Delta^2} + \frac{1}{2\sigma_\Delta^4(T-1)} \sum_{t=2}^T (z_t - \mu_\Delta)^2 \end{bmatrix}, \quad (12.34)$$

and setting $G_T(\hat{\beta}) = 0$ yields the maximum likelihood estimators

$$\begin{aligned} \hat{\mu}_\Delta &= \frac{1}{T-1} \sum_{t=2}^T z_t \\ \hat{\sigma}^2 &= \frac{1}{T-1} \sum_{t=2}^T (z_t - \hat{\mu}_\Delta)^2. \end{aligned} \quad (12.35)$$

Indirect Inference

The indirect estimator of (12.29) uses (12.35) to form the moment conditions. To estimate (12.29) by indirect inference, the steps are as follows.

Step 1: Estimate $\hat{\mu}_\Delta$ and $\hat{\sigma}_\Delta^2$ from the auxiliary model using (12.35) based on the observed data.

Step 2: Choose starting values for $\theta = \{\mu, \sigma^2\}$, say $\mu_{(0)}$ and $\sigma_{(0)}^2$, and generate continuous time data using a small time step, say $\Delta t = 0.1$

$$y_{s,t+\Delta t} - y_{s,t} = \mu_{(0)} y_{s,t} \Delta t + \sigma_{(0)} y_{s,t} v_{t+\Delta t}, \quad v_t \sim iid N(0, \Delta t).$$

Let the length of the simulated series be $S = T \times K$. Alternatively, simulate

$$z_{s,t+\Delta t} = \mu \Delta t + \sigma v_{t+\Delta t},$$

and construct $y_{s,t+\Delta t}$ using the recursive product

$$y_{s,t+\Delta t} = y_{s,t}(1 + z_{s,t+\Delta t}).$$

Step 3: Generate discrete time data $y_{s,1}, y_{s,2}, \dots, y_{s,S}$, by choosing every $1/\Delta t$ observations from the simulated continuous time series.

Step 4: Compute the parameter estimates of the auxiliary model, $\hat{\beta}_s = \{\hat{\mu}_{\Delta,s}, \hat{\sigma}_{\Delta,s}^2\}$ based on the simulated data

$$\hat{\mu}_{\Delta,s} = \frac{1}{S} \sum_{t=1}^S z_{s,t}$$

$$\hat{\sigma}_{\Delta,s}^2 = \frac{1}{S} \sum_{t=1}^S (z_{s,t} - \hat{\mu}_{\Delta,s})^2.$$

Step 5: Define the moment conditions

$$G_S(\theta) = \begin{bmatrix} \hat{\mu} - \hat{\mu}_{\Delta,s} \\ \hat{\sigma}^2 - \hat{\sigma}_{\Delta,s}^2 \end{bmatrix}.$$

Step 6: Choose $\theta = \{\mu, \sigma\}$ to satisfy

$$\hat{\theta} = \arg \min_{\theta} G'_S(\theta) G_S(\theta).$$

EMM Estimation

The EMM estimator is computed as follows.

Step 1: Estimate $\hat{\mu}_{\Delta}$ and $\hat{\sigma}_{\Delta}^2$ from the auxiliary model using (12.35) based on the observed data.

Step 2: Compute the gradient vector in equation (12.34) using the maximum likelihood estimators $\hat{\mu}_{\Delta}$ and $\hat{\sigma}_{\Delta}^2$ and compute the weighting matrix W_T using a given lag length, P .

Step 3: Choose starting values for $\theta = \{\mu, \sigma^2\}$, say $\mu_{(0)}$ and $\sigma_{(0)}^2$, and generate continuous time data using a small time step, say $\Delta t = 0.1$

$$y_{s,t+\Delta t} - y_{s,t} = \mu_{(0)} \Delta t + \sigma_{(0)} v_{t+\Delta t}, \quad v_t \sim iid N(0, \Delta t).$$

Step 4: Generate discrete time data $y_{s,1}, y_{s,2}, \dots, y_{s,S}$, by choosing every $1/\Delta t$ observations from the simulated continuous time series.

Step 5: Evaluate the gradients in (12.34) using the simulated data and the maximum likelihood parameter estimates in (12.35)

$$G_S(\theta) = \begin{bmatrix} \frac{1}{\hat{\sigma}_{\Delta}^2 S} \sum_{t=1}^S (z_t - \hat{\mu}_{\Delta}) \\ -\frac{1}{2\hat{\sigma}_{\Delta}^2} + \frac{1}{2\hat{\sigma}_{\Delta}^4 S} \sum_{t=1}^S (z_t - \hat{\mu}_{\Delta})^2 \end{bmatrix}.$$

Step 6: Choose $\theta = \{\mu, \sigma\}$ to satisfy

$$\hat{\theta} = \arg \min_{\theta} G'_S(\theta) W_T^{-1} G_S(\theta).$$

12.6.3 Stochastic Volatility

Consider estimating the parameters $\theta = \{\mu, \alpha, \beta, \gamma\}$, of the stochastic volatility model

$$\begin{aligned} dy_t &= \mu y_t dt + \sigma_t y_t dB_1 \\ d \log \sigma_t^2 &= \alpha(\beta - \log \sigma_t^2) dt + \gamma dB_2, \end{aligned} \quad (12.36)$$

where B_1 and B_2 are Wiener processes which are assumed to be uncorrelated for simplicity, although this assumption can be easily relaxed. The stochastic volatility model is used here because it is specified in continuous time and therefore provides a useful example of estimation by simulation for intractable log-likelihood functions.

In the first equation y_t follows geometric Brownian motion whilst in the second equation $\ln \sigma_t$ follows a Ornstein-Uhlenbeck process. Estimation of this problem is more involved than the previous examples because not only are the data observed discretely but the volatility series σ_t is latent.

To estimate the parameters of (12.36), it is necessary to have at least four moments, the dimension of the parameter space. Consider the auxiliary model obtained by discretizing (12.36) with the time interval $\Delta t = 1$

$$\begin{aligned} \ln y_t - \ln y_{t-1} &= \mu_\Delta + \sigma_t v_{1,t} \\ \log \sigma_t^2 - \log \sigma_{t-1}^2 &= \alpha_\Delta(\beta_\Delta - \log \sigma_{t-1}^2) + \gamma_\Delta v_{2,t}, \end{aligned} \quad (12.37)$$

where $v_{i,t} \sim N(0, 1)$. The parameter μ_Δ is estimated as

$$\hat{\mu}_\Delta = \frac{1}{T-1} \sum_{t=2}^T (\ln y_t - \ln y_{t-1}). \quad (12.38)$$

To estimate the volatility parameters, consider transforming the first equation in (12.37) as

$$r_t^2 = (\ln y_t - \ln y_{t-1} - \mu_\Delta)^2 = \ln \sigma_t^2 + \ln v_{1,t}^2,$$

which represents the mean-adjusted squared return. Use this expression to substitute out $\ln \sigma_t^2$ and $\ln \sigma_{t-1}^2$ in the second equation in (12.37)

$$r_t^2 - \ln v_{1,t}^2 - r_{t-1}^2 + \ln v_{1,t-1}^2 = \alpha_\Delta(\beta_\Delta - r_{t-1}^2 + \ln v_{1,t-1}^2) + \gamma_\Delta v_{2,t},$$

which is simplified as

$$r_t^2 = \delta_0 + \delta_1 r_{t-1}^2 + \delta_2 \eta_t. \quad (12.39)$$

The parameters $\{\delta_0, \delta_1, \delta_2\}$ are functions of the parameters $\{\alpha_\Delta, \beta_\Delta, \gamma_\Delta\}$, and η_t is a disturbance term. Estimates of $\{\delta_0, \delta_1, \delta_2\}$ can be obtained by regressing r_t^2 on a constant and r_{t-1}^2 , with the estimate of δ_2 obtained from the standard deviation of the ordinary least squares residuals.

The quality of the approximation of the auxiliary model can be improved by estimating (12.39) with a first order moving average disturbance as η_t contains both $\ln v_{1,t}^2$ and $\ln v_{1,t-1}^2$. This modification to computing the empirical moments does not change the consistency properties of the indirect estimator, but it can improve its efficiency in small samples.

Indirect Inference

The indirect estimator of (12.36) is based on the following steps:

Step 1: Estimate the parameters of the auxiliary model using the observed data. The parameter $\hat{\mu}_\Delta$ is estimated using equation (12.38) and $\hat{\delta}_0$, $\hat{\delta}_1$ and $\hat{\delta}_2$ are estimated using equation (12.39)

Step 2: Choose starting values for $\theta = \{\mu, \alpha, \beta, \gamma\}$ and generate continuous time data using a small time step, say $\Delta t = 0.1$

$$\begin{aligned}\ln y_{s,t+\Delta t} - \ln y_{s,t} &= \mu_{(0)}\Delta t + \sigma_{s,t+\Delta t}v_{1,t+\Delta t} \\ \log \sigma_{s,t+\Delta t}^2 - \log \sigma_{s,t}^2 &= \alpha_{(0)}(\beta_{(0)} - \log \sigma_{s,t}^2)\Delta t + \gamma_{(0)}v_{2,t+\Delta t},\end{aligned}$$

where $v_{i,t} \sim N(0, \Delta t)$. Let the length of the simulated series be $S = T \times K$.

Step 3: Generate discrete time data $y_{s,1}, y_{s,2}, \dots, y_{s,N}$, by choosing every $1/\Delta t$ observations from the simulated continuous time series.

Step 4: Compute the parameter estimates of the auxiliary model based on the simulated data $\hat{\beta}_s = \{\hat{\mu}_{\Delta,s}, \hat{\delta}_{0,s}, \hat{\delta}_{1,s}, \hat{\delta}_{2,s}\}$, where

$$\hat{\mu}_{\Delta,s} = \frac{1}{S} \sum_{t=1}^S (\ln y_{s,t} - \ln y_{s,t-1}),$$

and $\hat{\delta}_{0,s}, \hat{\delta}_{1,s}, \hat{\delta}_{2,s}$ are the ordinary least squares estimates obtained by estimating the regression equation

$$r_{s,t}^2 = \delta_0 + \delta_1 r_{s,t-1}^2 + \delta_2 \eta_{s,t},$$

where

$$r_{s,t} = \ln y_{s,t} - \ln y_{s,t-1} - \hat{\mu}_{\Delta,s}.$$

Step 5: Define the moment conditions

$$G'_S(\theta) = \begin{bmatrix} \hat{\mu}_s - \hat{\mu} & \hat{\delta}_{0,s} - \hat{\delta}_0 & \hat{\delta}_{1,s} - \hat{\delta}_1 & \hat{\delta}_{2,s}^2 - \hat{\delta}_2^2 \end{bmatrix}.$$

Step 6: Choose $\theta = \{\mu, \sigma, \alpha, \beta, \gamma\}$ to satisfy

$$\hat{\theta} = \arg \min_{\theta} G'_S(\theta) G_S(\theta).$$

An alternative auxiliary model is not to transform the auxiliary model into the regression equation (12.39), but treat

$$\begin{aligned} (\ln y_t - \ln y_{t-1} - \mu)^2 &= \mu + \ln \sigma_t^2 + \ln v_{1,t}^2, \\ \log \sigma_t^2 - \log \sigma_{t-1}^2 &= \alpha(\beta - \log \sigma_{t-1}^2) + \gamma v_{2,t}, \end{aligned}$$

as a state-space model where the first equation is the measurement equation and the second equation is the state equation. This model can be estimated by maximum likelihood using a Kalman filter (see Chapter 15), first with the actual data ($\hat{\beta}$) and second with the simulated data ($\hat{\beta}_s$), with the indirect estimates of the stochastic volatility model (θ) obtained by equating the two sets of parameter estimates of the auxiliary (state-space) model. However, this would involve a nonlinear estimation problem at each iteration of the indirect algorithm to compute ($\hat{\beta}_s$). An alternative approach which circumvents the need to re-estimate the state-space model using the simulated data is to use the EMM estimator. This requires estimating the state-space model using the actual data ($\hat{\beta}$). The gradients of the likelihood of the Kalman filter then act as the moments in the EMM algorithm which are simply evaluated at the parameter estimates of the Kalman filter ($\hat{\beta}$) but using the simulated data. Bayesian methods have also featured prominently in the estimation of stochastic volatility models (Jacquier, Polson and Rossi, 2002, 2004; Chib, Nardari and Shephard, 2002).

12.7 Applications

Consider the following one factor model of the business cycle proposed by Stock and Watson (2005)

$$\begin{aligned} y_{i,t} &= \lambda_i s_t + \sigma_i z_{i,t}, & i &= 1, 2, \dots, N \\ s_t &= \phi s_{t-1} + \eta_t \\ z_{i,t}, \eta_t &\sim iid N(0, 1), \end{aligned} \tag{12.40}$$

where $y_{i,t}$ is an indicator, s_t is the business cycle factor and the unknown parameters are

$$\theta = \{\lambda_1, \lambda_2, \dots, \lambda_N, \sigma_1, \sigma_2, \dots, \sigma_N, \phi\}.$$

The distinguishing feature of the model is that data are available on the indicators, but not on the business cycle so s_t is treated as a latent process.

In this application an EMM estimator is devised to estimate the parameters in (12.7). To investigate the properties of the estimator, a Monte Carlo experiment is presented initially, followed by an empirical application where

the model is applied to Australian business cycle data. For the class of latent factor models in (12.40) it is shown in Chapter 15 that the parameters can be estimated directly using maximum likelihood methods using a Kalman filter. However, this is not the case for generalizations of the latent process in (12.40) that allow for either conditional volatility (Diebold and Nerlove, 1989), or Markov switching (Kim, 1994), or both (Diebold and Rudebusch, 1996). For these extensions it is necessary to use simulation methods such as EMM.

12.7.1 Simulation Properties

To motivate the choice of the auxiliary model to estimate the latent factor business cycle model in (12.40) by EMM, rewrite the first equation of (12.40) as

$$\begin{aligned} (1 - \phi L) y_{i,t} &= \lambda_i (1 - \phi L) s_t + \sigma_i (1 - \phi L) z_{i,t} \\ (1 - \phi L) y_{i,t} &= \lambda_i \eta_t + \sigma_i (1 - \phi L) z_{i,t}, \quad i = 1, 2, \dots, N, \end{aligned} \quad (12.41)$$

where the second step uses the second equation of (12.40). The latent factor model is now expressed as a system of N univariate ARMA(1,1) equations.

Following the discussion above of the choice of the auxiliary model to estimate a AR(1) model, an AR(2) model represents a suitable specification for the auxiliary model

$$y_{i,t} = \rho_{i,0} + \rho_{i,1}y_{i,t-1} + \rho_{i,2}y_{i,t-2} + \sigma_{v,i}v_{i,t}, \quad i = 1, 2, \dots, N, \quad (12.42)$$

where the parameters of the auxiliary model are $\beta = \{\rho_{i,0}, \rho_{i,1}, \rho_{i,2}, \sigma_{v,i}\}$ and $v_{i,t} \sim iid N(0, 1)$. The auxiliary log-likelihood function at time t for the i^{th} indicator is

$$\ln l_{t,i}(\beta) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma_{v,i}^2 - \frac{1}{2} \frac{1}{\sigma_{v,i}^2} (y_{i,t} - \rho_{i,0} - \rho_{i,1}y_{i,t-1} - \rho_{i,2}y_{i,t-2})^2. \quad (12.43)$$

Differentiating with respect to the auxiliary parameters gives the gradients

for $i = 1, 2, \dots, N$,

$$\begin{aligned}
\frac{\partial \ln l_{t,i}(\beta)}{\partial \rho_{i,1}} &= \frac{1}{\sigma_{v,i}^2} (y_{i,t} - \rho_{i,0} - \rho_{i,1}y_{i,t-1} - \rho_{i,2}y_{i,t-2}) \\
\frac{\partial \ln l_{t,i}(\beta)}{\partial \rho_{i,1}} &= \frac{1}{\sigma_{v,i}^2} (y_{i,t} - \rho_{i,0} - \rho_{i,1}y_{i,t-1} - \rho_{i,2}y_{i,t-2}) y_{i,t-1} \\
\frac{\partial \ln l_{t,i}(\beta)}{\partial \rho_{i,2}} &= \frac{1}{\sigma_{v,i}^2} (y_{i,t} - \rho_{i,0} - \rho_{i,1}y_{i,t-1} - \rho_{i,2}y_{i,t-2}) y_{i,t-2} \\
\frac{\partial \ln l_{t,i}(\beta)}{\partial \sigma_{v,i}^2} &= -\frac{1}{2\sigma_{v,i}^2} + \frac{1}{2\sigma_{v,i}^4} (y_{i,t} - \rho_{i,0} - \rho_{i,1}y_{i,t-1} - \rho_{i,2}y_{i,t-2})^2.
\end{aligned} \tag{12.44}$$

The log-likelihood function of the auxiliary model is

$$\ln L_T(\beta) = \frac{1}{T-1} \sum_{t=2}^T \sum_{i=1}^N \ln l_{t,i}.$$

The log-likelihood function separates into N components because the information matrix is block-diagonal and therefore each of the N AR(2) models are estimated by ordinary least squares separately.

To investigate the ability of the EMM estimator to recover the population parameters, the following Monte Carlo experiment is performed. The DGP has $N = 6$ indicators with parameter values θ given by $\lambda = \{1, 0.8, 0.7, 0.5, -0.5, -1\}$, $\sigma = \{0.2, 0.4, 0.6, 0.8, 1.0, 1.2\}$ and $\phi = 0.8$, a total of 13 parameters. The model in (12.40) is simulated for a sample of size $T = 600$, which corresponds to the length of data used in the empirical application below.

The first step of the EMM algorithm is to estimate the auxiliary model based on the $N = 6$ univariate AR(2) models in (12.42). The estimated auxiliary models are

$$\begin{aligned}
y_{1,t} &= 0.019 + 0.801y_{1,t-1} + 0.012y_{1,t-2} + 0.988\hat{v}_{1,t} \\
y_{2,t} &= 0.023 + 0.660y_{2,t-1} + 0.101y_{2,t-2} + 0.838\hat{v}_{2,t} \\
y_{3,t} &= 0.021 + 0.533y_{3,t-1} + 0.180y_{3,t-2} + 1.009\hat{v}_{3,t} \\
y_{4,t} &= 0.036 + 0.345y_{4,t-1} + 0.209y_{4,t-2} + 1.117\hat{v}_{4,t} \\
y_{5,t} &= -0.017 + 0.256y_{5,t-1} + 0.208y_{5,t-2} + 1.457\hat{v}_{5,t} \\
y_{6,t} &= -0.029 + 0.423y_{6,t-1} + 0.210y_{6,t-2} + 2.787\hat{v}_{6,t},
\end{aligned} \tag{12.45}$$

where the total number of estimated parameters is 24. Also computed is the (24×24) weighting matrix W which is the average of the outer product of the gradients in (12.44) evaluated using the auxiliary parameter estimates in (12.45).

The first iteration of the EMM estimator involves choosing starting values of the parameters in (12.7) and simulating (12.40) for a simulated sample of $K \times T = 10 \times 600 = 6000$ observations. The iterations continue by solving

$$\hat{\theta} = \arg \min_{\theta} G'_S(\theta) W_T^{-1} G_S(\theta), \quad (12.46)$$

where $G_S(\theta)$ is evaluated using the estimates of the parameters of the auxiliary models given in (12.45) by replacing $y_{i,t}$ in (12.44) with the simulated data and averaging the gradients in (12.44) over the 6000 simulated observations. The algorithm converges in 16 iterations. The estimated model with standard errors in parentheses is

$$\begin{aligned} y_{1,t} &= \underset{(0.051)}{0.958} s_t + \underset{(0.127)}{0.224} \hat{z}_{1,t} & y_{2,t} &= \underset{(0.049)}{0.741} s_t + \underset{(0.056)}{0.418} \hat{z}_{2,t} \\ y_{3,t} &= \underset{(0.052)}{0.689} s_t + \underset{(0.050)}{0.620} \hat{z}_{3,t} & y_{4,t} &= \underset{(0.047)}{0.503} s_t + \underset{(0.038)}{0.822} \hat{z}_{4,t} \\ y_{5,t} &= \underset{(0.046)}{-0.478} s_t + \underset{(0.038)}{0.973} \hat{z}_{5,t} & y_{6,t} &= \underset{(0.075)}{-0.937} s_t + \underset{(0.063)}{1.205} \hat{z}_{6,t} \\ \hat{s}_t &= \underset{(0.025)}{0.828} \hat{s}_{t-1} + \hat{\eta}_t. \end{aligned} \quad (12.47)$$

The EMM parameter estimates in (12.47) compare favourably with the true population parameter values. The value of the objective function at the minimum is $Q(\hat{\theta}) = 0.0047$, which produces a Hansen-Sargan J test statistic of the over-identifying restrictions given by

$$H_{JS} = TQ_T(\hat{\theta}) = 600 \times 0.0047 = 2.8180.$$

As the number of over-identifying restrictions is $24 - 13 = 11$, the p-value of the test statistic using the χ^2_{11} distribution is 0.9929, a result that provides strong support for the estimated model.

12.7.2 Empirical Properties

The EMM estimator of the latent factor business cycle model in (12.40) is applied to Australian business cycle data using the auxiliary model developed in the previous section. The data are monthly beginning September 1959 and ending September 2009. Six indicators are used corresponding to the indicators used to construct the coincident index: employment, $y_{1,t}$, GDP, $y_{2,t}$, household income, $y_{3,t}$, industrial production, $y_{4,t}$, retail sales, $y_{5,t}$, and the unemployment rate, $y_{6,t}$. Quarterly data are converted into

monthly data by linear interpolation over the quarter. All indicators are expressed as quarterly percentage growth rates by computing the annual span of the natural logarithm of the raw series. In the case of the unemployment rate, this series is inverted to make it move pro-cyclically over the business cycle. Finally, each growth rates is standardized by subtracting the sample mean and dividing by the sample standard deviation. The total sample after computing annual growth rates is $T = 598$.

The correlations between the six indicators and the quarterly percentage growth rate of the coincident index are

Coincident index	Employ. 0.833	GDP 0.830	Income 0.081	Prod. 0.611	Sales 0.199	Unemp. 0.795
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All series move positively with the coincident index, although the correlation with household income is relatively low at 0.081.

The auxiliary model is an AR(2) without a constant, resulting in a total of 18 parameters. The starting values of θ to solve (12.46) are chosen from a uniform distribution with the exception of ϕ , which is constrained using the hyperbolic tangent function to ensure that it is within the unit circle. Once the algorithm has coveredged, this restriction is relaxed and the estimation repeated in order to compute the standard error on $\hat{\phi}$. The length of the simulated series in the EMM algorithm is chosen as $K \times T = 30 \times 598 = 17670$ observations. The estimated latent factor model, with standard errors in parentheses, is

$$\begin{aligned}
 \text{Employment} & : y_{1,t} = 0.470 s_t + 0.353 \hat{z}_{1,t} \\
 & \quad (0.048) \quad (0.053) \\
 \text{GDP} & : y_{2,t} = 0.338 s_t + 0.010 \hat{z}_{2,t} \\
 & \quad (0.015) \quad (0.049) \\
 \text{Household Income} & : y_{3,t} = 0.376 s_t + 0.009 \hat{z}_{3,t} \\
 & \quad (0.015) \quad (0.002) \\
 \text{Industrial Production} & : y_{4,t} = 0.281 s_t + 0.147 \hat{z}_{4,t} \\
 & \quad (0.150) \quad (0.012) \\
 \text{Retail Sales} & : y_{5,t} = 0.548 s_t + 0.503 \hat{z}_{5,t} \\
 & \quad (0.076) \quad (0.093) \\
 \text{Unemployment Rate} & : y_{6,t} = 0.586 s_t + 0.002 \hat{z}_{6,t} \\
 & \quad (0.031) \quad (0.091) \\
 \text{Factor} & : \hat{s}_t = 0.775 \hat{s}_{t-1} + \hat{\eta}_t \\
 & \quad (0.028)
 \end{aligned}$$

The loadings on each indicator are similar with values ranging from 0.281 for industrial production to 0.586 for the unemployment rate. The business cycle exhibits a strong positive correlation structure with $\hat{\phi} = 0.775$.

The value of the objective function at the minimum is $Q(\hat{\theta}) = 0.1607$, resulting in the Hansen-Sargan J test of the $18 - 13 = 5$ over-identifying

restrictions given by

$$J_{HS} = TQ(\hat{\theta}) = 598 \times 0.1607 = 96.085.$$

This value of J_{HS} produces a p-value of 0.000 using the χ_5^2 distribution, suggesting evidence of misspecification. Four possible sources of misspecification are: (i) the number of latent factors is greater than one; (ii) the dynamics of the latent factor is of a higher order; (iii) the disturbance terms $z_{i,t}$ are not white noise but also exhibit some autocorrelation; (iv) additional dynamic structures including conditional volatility and Markov Switching affect the latent factor. The first three types of misspecification are investigated in Chapter 15. Conditional volatility is discussed in Chapter 20 and Markov switching is discussed in Chapter 21.

12.8 Exercises

(1) Method of Moments Estimation of a MA Model

Gauss file(s)	<code>sim_mom.g</code>
Matlab file(s)	<code>sim_mom.m</code>

Simulate a AR(1) model

$$y_t = u_t - \theta u_{t-1}, \quad u_t \sim iid N(0, 1),$$

for a sample size of $T = 250$ observations with $\theta_0 = 0.5$.

- Estimate the model using the method of moments estimator in equation (12.2) and compare the parameter estimate with the true parameter value.
- Repeat part (a) for samples of size $T = \{500, 1000, 2000\}$. Compare the parameter estimate with the true parameter value in each case and discuss the statistical properties of the method of moments estimator.

(2) Computing Autocorrelation Coefficients by Simulation

Gauss file(s)	<code>sim_accuracy.g</code>
Matlab file(s)	<code>sim_accuracy.m</code>

Consider the AR(1) model

$$y_t = u_t - \theta u_{t-1}, \quad u_t \sim iid N(0, 1),$$

where $\theta_0 = 0.5$.

- (a) Compute the true AR(1) parameter ρ_0 .
- (b) Simulate the AR(1) model and compute the first-order autocorrelation coefficient using the simulated data for simulated series of length $S = T \times K$, with $T = 250$ and $K = \{1, 2, 3, 4, 5, 10, 100\}$. Compare the results with the theoretical autocorrelation parameter and hence reproduce Table 12.1.
- (c) Repeat part (b) for $\theta = 0.8$ and $\theta = -0.5$.

(3) Indirect Estimation of the AR(1) Model

Gauss file(s)	<code>sim_ma1indirect.g</code>
Matlab file(s)	<code>sim_ma1indirect.m</code>

This exercise is concerned with reproducing the Monte Carlo results presented in Gouriéroux, Monfort and Renault (1993, Table I, p.S98).

- (a) Simulate a AR(1) model

$$y_t = u_t - \theta u_{t-1}, \quad u_t \sim iid N(0, 1),$$

for a sample size of $T = 250$ observations with $\theta_0 = 0.5$. Let the number of replications be 1000.

- (b) Estimate the model using the indirect estimator and choosing as the auxiliary models an AR(1) model, an AR(2) model and an AR(3) model. In each case, compare the parameter estimate with the true parameter value.
- (c) Compare the sampling properties of the indirect estimator for each auxiliary model. Compare the results with Table 12.2.

(4) EMM Estimation of the AR(1) Model

Gauss file(s)	<code>sim_ma1emm.g</code>
Matlab file(s)	<code>sim_ma1emm.m</code>

- (a) Simulate a AR(1) model

$$y_t = u_t - \theta u_{t-1}, \quad u_t \sim iid N(0, 1),$$

for a sample size of $T = 250$ observations with $\theta_0 = 0.5$. Let the number of replications be 1000.

- (b) Estimate the model using the EMM estimator and choosing as auxiliary models an AR(1) model, an AR(2) model and an AR(3) model. In each case use an optimal weighting matrix based on $P = 0$. Compare the resultant estimates with the true parameter value.

- (c) Compare the sampling properties of the EMM estimator for each auxiliary model. Compare the results with Table 12.3.
- (d) Repeat parts (b) and (c) where the optimal weighting matrix is replaced by the identity matrix. Discuss the properties of the resultant estimator.

(5) **Over-identification Test of a Moving Average Model**

Gauss file(s)	<code>sim_ma1overid.g</code>
Matlab file(s)	<code>sim_ma1overid.m</code>

- (a) Simulate a AR(1) model

$$y_t = u_t - \theta u_{t-1}, \quad u_t \sim iid N(0, 1),$$

for a sample size of $T = 250$ observations with $\theta_0 = 0.5$.

- (b) Perform the over-identification test of the true model based on the EMM estimator using the AR(1), AR(2) and AR(3) auxiliary models with the optimal weighting matrix based on $P = 0$.
- (c) Compare the results with the results reported in Table 12.4.

(6) **Brownian Motion**

Gauss file(s)	<code>sim_brown.g</code>
Matlab file(s)	<code>sim_brown.m</code>

Consider the stochastic differential equation

$$dt_t = \mu dt + \sigma dB, \quad dB \sim N(0, dt),$$

where $B(t)$ is a Wiener process and $\theta = \{\mu = 0, \sigma^2 = 1\}$ are parameters.

- (a) Generate a minute time series of length $T = 14400$ (10 days), by simulating the model for $\Delta t = 1/60$. Compare the results with panel (a) of Figure 12.2.
- (b) Generate a 10 minute series by extracting every 10th observation, an ‘hourly’ series by extracting every 60th observation and a ‘daily’ time series by extracting every 1440th observation. Compare the results with panels (b), (c) and (d) of Figure 12.2.
- (c) Repeat parts (a) and (b) using as parameters $\mu = 0.05$ and $\sigma^2 = 2.0$.

(7) **Brownian Motion**

Gauss file(s)	<code>sim_brownind.g, sim_brownemm.g</code>
Matlab file(s)	<code>sim_brownind.m, sim_brownemm.m</code>

Consider estimating the parameters $\theta = \{\mu, \sigma^2\}$ of the stochastic differential equation

$$dy_t = \mu dt + \sigma dB, \quad dB \sim N(0, dt),$$

where $B(t)$ is a Wiener process. Let the sample size be $T = 500$ and true parameter values be $\theta = \{\mu = 0.5, \sigma^2 = 0.25\}$.

- (a) Compute the indirect estimates of the parameters for $K = \{1/\Delta t, 2/\Delta t\}$ simulation paths where $\Delta t = 0.1$. Compare the properties of the estimators.
- (b) Repeat part (a) using the EMM estimator.

(8) Geometric Brownian Motion

Gauss file(s)	<code>sim_geobrind.g</code>
Matlab file(s)	<code>sim_geobrind.m</code>

This exercise reproduces the Monte Carlo results presented in Gouriéroux, Monfort and Renault (1993, Table II, p.S101). Consider estimating the parameters $\theta = \{\mu, \sigma^2\}$ of

$$dy_t = \mu y_t dt + \sigma y_t dB, \quad dB \sim N(0, dt),$$

where $B(t)$ is a Wiener process. Let the sample size be $T = 500$ and true parameter values be $\theta = \{\mu = 0.5, \sigma^2 = 0.25\}$.

- (a) Compute the indirect estimates of the parameters for $K = \{1/\Delta t, 2/\Delta t\}$ simulation paths where $\Delta t = 0.1$.
- (b) Compare the properties of the resultant estimators in part (a).

(9) Ornstein-Uhlenbeck Process

Gauss file(s)	<code>sim_ouind.g</code>
Matlab file(s)	<code>sim_ouind.m</code>

This exercise reproduces the Monte Carlo results presented in Gouriéroux, Monfort and Renault (1993, Table III, p.S103). The Ornstein-Uhlenbeck process is given by

$$dy_t = \alpha(\kappa - y_t)dt + \sigma dB, \quad dB \sim N(0, dt),$$

where $B(t)$ is a Wiener process. Let the sample size be $T = 250$ and the

true parameters values be $\theta = \{\alpha = 0.1, \kappa = 0.8, \sigma = 0.06^2\}$. An exact discretization is

$$y_{t+\Delta t} = \alpha(1 - \exp[-\kappa\Delta t]) + \exp[-\kappa\Delta t]y_t + \sigma \left[\frac{1 - \exp[-2\kappa\Delta t]}{2\kappa} \right] u_t,$$

where $u_t \sim N(0, \Delta t)$. Choose as the auxiliary model

$$y_t = (1 - \kappa\Delta t)y_{t-1} + \kappa\alpha\Delta t + \sigma v_t, \quad v_t \sim N(0, \Delta t).$$

- (a) Compute the indirect estimates of the parameters for $K = \{10/\Delta t, 20/\Delta t\}$ simulation paths where $\Delta t = 0.1$.
- (b) Compare the properties of the resultant estimators.

(10) **A Level Effects Model of UK Gilts**

Gauss file(s)	<code>sim_ukgilts.g, ukgilts.dat</code>
Matlab file(s)	<code>sim_ukgilts.m, ukgilts.mat</code>

For the data on UK gilts, r_t , consider the model

$$dr_t = \kappa(\beta - r_t)dt + \sigma r_t^\alpha dB, \quad dB \sim N(0, dt),$$

where $B(t)$ is a Wiener process.

- (a) Compute the EMM estimates of the parameters using $\Delta t = 0.1$ and $\Delta t = 0.01$ and compare the properties of the resultant estimators.
- (b) Compare the properties of the resultant estimators in part (a).

(11) **Business Cycles**

Gauss file(s)	<code>sim_bcycle.g, sim_stockwatson.g,</code> <code>bcycle_monthly.xlsx</code>
Matlab file(s)	<code>sim_bcycle.m, sim_stockwatson.m,</code> <code>bcycle_monthly.xlsx</code>

- (a) Simulate the latent factor business cycle model

$$\begin{aligned} y_{i,t} &= \lambda_i s_t + \sigma_i z_{i,t}, & i &= 1, 2, \dots, 6 \\ s_t &= \phi s_{t-1} + \eta_t \\ z_{i,t}, \eta_t &\sim iid N(0, 1), \end{aligned}$$

for a sample of size $T = 600$ and parameter values

$$\begin{aligned} \lambda &= \{1, 0.8, 0.7, 0.5, -0.5, -1\} \\ \sigma &= \{0.2, 0.4, 0.6, 0.8, 1.0, 1.2\}, \end{aligned}$$

and $\phi = 0.8$. Choosing the auxiliary model as a set of $N = 6$ AR(2) equations, estimate the model by EMM and compare the parameter estimates with the true values. Discuss other choices of auxiliary models.

- (b) Estimate the latent factor business cycle model in part (a) using data on employment, GDP, household income, industrial production, retail sales and the unemployment rate from September 1959 to September 2009 for Australia. Comment on the estimated model.

PART FOUR
STATIONARY TIME SERIES

13

Linear Time Series Models

13.1 Introduction

The maximum likelihood framework presented in Part ONE is now applied to estimating and testing a general class of dynamic models known as stationary time series models. Both univariate and multivariate models are discussed. The dynamics enter the model in one of two ways. The first is through lags of the variables, referred to as the autoregressive part, and the second is through lags of the disturbance term, referred to as the moving average part. In the case where the dynamics of a single variable are being modelled this class of models is referred to as the autoregressive moving average model, or ARMA model. In the multivariate case where the dynamics of multiple variables are modelled, this class of models is referred to as the vector autoregressive moving average (VARMA) model. Jointly, these models are called stationary time series models where stationarity refers to the types of dynamics allowed for. The case of nonstationary dynamics is discussed in Part FIVE.

The specification of dynamics through the inclusion of lagged variables and lagged disturbances is not new. It was discussed in Part TWO in the context of the linear regression model in Chapter 5 and more directly in Chapter 7 where autoregressive and moving-average dynamics were specified in the context of the autoregressive regression model. In fact, a one-to-one relationship exists between the VARMA class of models investigated in this chapter and the structural class of regression models of Part TWO, where the VARMA model is interpreted as the reduced form of a structural model. However, as the VARMA class of models is widely used in applied econometric modelling, it is appropriate to discuss the properties of these models in detail.

The VARMA model includes an important special case distinguished by

dynamics that are just driven by the lags of the variables themselves. This special case is known as a vector autoregression, or VAR model, which was first investigated by Sims (1980) using U.S. data on the nominal interest rate, money, prices and output. Given the widespread use of VARs in empirical work, a large part of this chapter is devoted to understanding their properties.

Despite the attraction of VARs in modelling economic processes, behind this so-called generality are implicit assumptions concerning the identification of the model that impose very strict relationships amongst the variables. These restrictions are identified here and are relaxed in Chapter 14 where the VAR class of models is extended to another class of models known as structural vector autoregressive, or SVAR, models. Another potential problem of VARs is the dimension of the specified system. In practice, VARs are specified for relatively small systems as estimation involves many unknown parameters. One solution to this problem is to use economic theory to impose restrictions on the dynamics of a VAR. Some examples of this strategy are given at the end of the chapter. Another approach is discussed in Chapter 15 where the dimensionality problem is reduced by specifying the model in terms of latent factors. The latent factors are identified and the model parameters are estimated using a Kalman filter.

13.2 Time Series Properties of Data

Figure 13.1 gives plots from January 1960 to December 1998 of key U.S. macroeconomic variables consisting of the interest rate, the logarithm of money, the logarithm of prices and the logarithm of output: $y_t = [r_t, lm_t, lp_t, lo_t]'$. Money, price and output all display smooth upward trends, whereas the interest rate tends to display relatively greater volatility over the sample with a positive trend in the first part of the period followed by a negative trend in the second part. Taking first differences of the variables (column 2), $\Delta y_t = y_t - y_{t-1}$, expresses the variables in quarterly changes or quarterly growth rates. The times series of the differenced variables do not have trends and exhibit noisy behaviour relative to their levels. Money in particular displays very strong seasonality behaviour. By taking 12th differences (column 3), $\Delta_{12}y_t = y_t - y_{t-12}$, the variables are expressed in annual changes or annual growth rates. These transformed variables still do not exhibit trends but now reveal stronger cyclical behaviour over the sample period.

To understand the dynamic properties of the variables in Figure 13.1, refer to their autocorrelation functions (ACF) and partial autocorrelation functions (PACF) given in Table 13.1. The ACF at lag k is the estimated

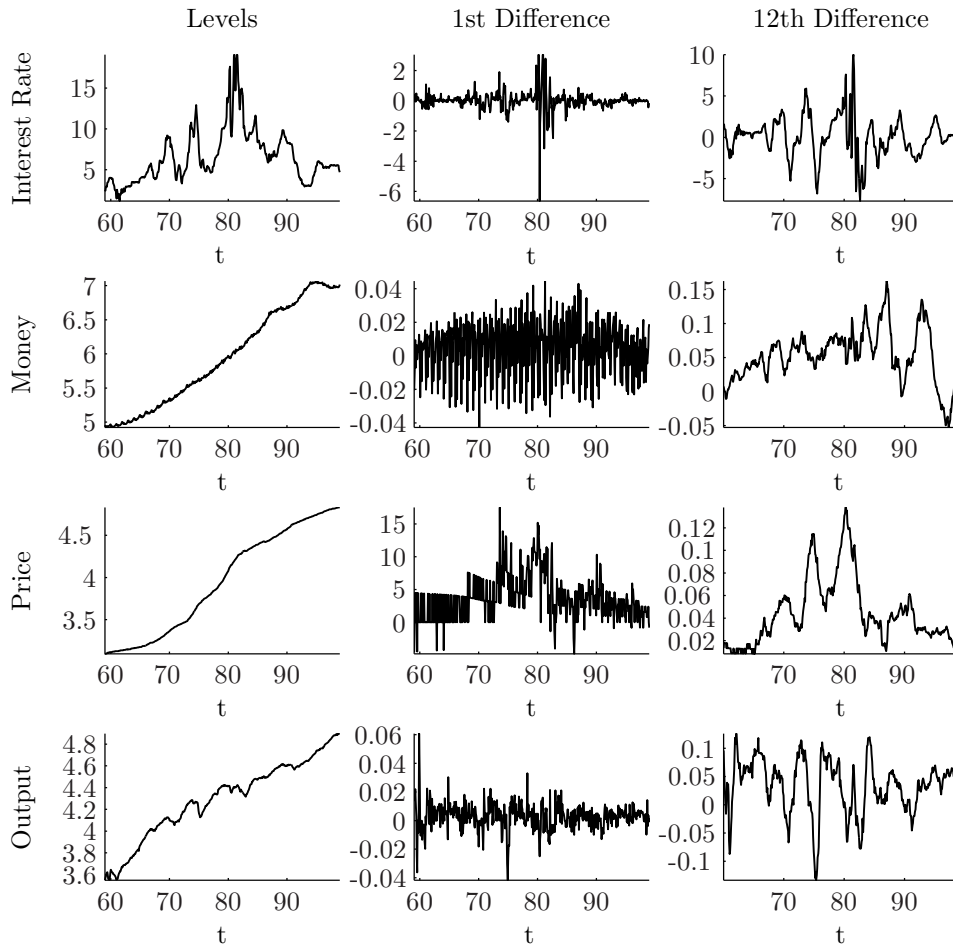


Figure 13.1 Plots of U.S. monthly macroeconomic data, January 1960 to December 1998. The interest rate is the nominal interest rate expressed as an annual percentage, money is the annual percentage growth in nominal money, price is the annual percentage growth in the CPI, and real output is the annual percentage growth in industrial production.

coefficient on y_{t-k} in the linear regression of y_t on a constant and y_{t-k} , while the PACF is the estimated coefficient on y_{t-k} in the linear regression of y_t on a constant and $y_{t-1}, y_{t-2}, \dots, y_{t-k}$.

The ACFs of the levels of the variables in Table 13.1 show very slow decay, which is representative of the strong trends exhibited by these variables. The PACFs reveal that it is the first lag that is the most important in explaining y_t . The ACF and PACF of the first-differenced variables reveal

Table 13.1

ACF and PACF of U.S. macroeconomic data, January 1959 to December 1998: interest rate, r_t , the logarithm of the money stock, lm_t , the logarithm of price, lp_t and logarithm of real output, lo_t .

Lag	ACF				PACF			
	r_t	lm_t	lp_t	lo_t	r_t	lm_t	lp_t	lo_t
<i>Level (y_t)</i>								
1	0.98	1.00	1.00	1.00	0.98	1.00	1.00	1.00
2	0.95	1.00	1.00	1.00	-0.39	0.08	-0.56	-0.37
3	0.92	1.00	1.00	0.99	0.16	0.37	-0.28	-0.07
<i>1st Difference ($y_t - y_{t-1}$)</i>								
1	0.37	-0.08	0.56	0.37	0.37	-0.08	0.56	0.37
2	-0.01	-0.36	0.51	0.20	-0.18	-0.37	0.28	0.07
3	-0.10	0.25	0.48	0.13	-0.04	0.20	0.18	0.05
<i>12th Difference ($y_t - y_{t-12}$)</i>								
1	0.92	0.98	0.99	0.96	0.92	0.98	0.99	0.96
2	0.79	0.96	0.98	0.90	-0.41	-0.32	-0.25	-0.37
3	0.67	0.93	0.97	0.83	0.17	0.01	-0.16	-0.17

quite different dynamics with output dynamics dominated by the first lag, whereas for the other variables higher-order lags are important. The ACF of the 12th-differenced variables decay slowly, which is consistent with the relatively smooth time series properties of these variables given in Figure 13.1. The corresponding PACFs show the importance of higher-order lags in all of the variables.

13.3 Specification

From Chapter 5, the class of single equation linear regression models with an ARMA(p, q) disturbance is specified as

$$\begin{aligned}
 y_t &= \beta_0 + \beta_1 x_t + u_t \\
 u_t &= \sum_{i=1}^p \rho_i u_{t-i} + v_t + \sum_{i=1}^q \delta_i v_{t-i} \\
 v_t &\sim iid N(0, \sigma_v^2),
 \end{aligned} \tag{13.1}$$

where y_t is the dependent variable, x_t is the independent variable and v_t is the disturbance term. Consider a special case of this model by setting $\beta_1 = 0$ so that y_t is fully determined by its own dynamics

$$\begin{aligned} y_t &= \beta_0 + u_t \\ &= \beta_0 + \sum_{i=1}^p \rho_i u_{t-i} + v_t + \sum_{i=1}^q \delta_i v_{t-i} \\ &= \mu + \sum_{i=1}^p \phi_i y_{t-i} + v_t + \sum_{i=1}^q \psi_i v_{t-i}, \end{aligned} \quad (13.2)$$

where $\mu = \beta_0 - \beta_0 \sum_{i=1}^p \rho_i$, $\phi_i = \rho_i$ and $\psi_i = \delta_i$. This model represents the univariate class of linear autoregressive moving average models with p autoregressive lags and q moving average lags, ARMA(p, q). The multivariate analogue of this model is known as the VARMA(p, q) model.

13.3.1 Univariate Model Classification

Some common specifications of the ARMA(p, q) model in (13.2) are

1. ARMA(0, 0) = White noise : $y_t = \mu + v_t$
2. ARMA(1, 0) = AR(1) : $y_t = \mu + \phi_1 y_{t-1} + v_t$
3. ARMA(2, 0) = AR(2) : $y_t = \mu + \phi_1 y_{t-1} + \phi_2 y_{t-2} + v_t$
4. ARMA(0, 1) = MA(1) : $y_t = \mu + v_t + \psi_1 v_{t-1}$
5. ARMA(0, 2) = MA(2) : $y_t = \mu + v_t + \psi_1 v_{t-1} + \psi_2 v_{t-2}$
6. ARMA(1, 1) : $y_t = \mu + \phi_1 y_{t-1} + v_t + \psi_1 v_{t-1}$.

The ARMA(p, q) model in (13.2) is rewritten more conveniently in terms of lag polynomials

$$\begin{aligned} \phi_p(L)y_t &= \mu + \psi_q(L)v_t \\ v_t &\sim iid N(0, \sigma_v^2), \end{aligned} \quad (13.3)$$

where

$$\begin{aligned} \phi_p(L) &= (1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p) \\ \psi_q(L) &= (1 + \psi_1 L + \psi_2 L^2 + \cdots + \psi_q L^q), \end{aligned} \quad (13.4)$$

are polynomials in the lag operator L and $\{\mu, \phi_1, \phi_2, \dots, \phi_p, \psi_1, \psi_2, \dots, \psi_q, \sigma_v^2\}$ are unknown parameters. Appendix B contains further details on lag operators. This model provides a general framework to capture the dynamics of a univariate time series y_t .

Example 13.1 Simulation Properties of an ARMA(2,2) Model

Consider the ARMA(2,2) model

$$y_t = \mu + \phi_1 y_{t-1} + \phi_2 y_{t-2} + v_t + \psi_1 v_{t-1} + \psi_2 v_{t-2}$$

$$v_t \sim iid N(0, \sigma_v^2).$$

The first column of Figure 13.2 gives plots of y_t obtained by simulating the model for alternative parameterizations. The relevant starting values for y_t are taken to be 0 and $\sigma_v^2 = 0.1^2$. The first 100 simulated values are discarded to reduce the dependence on the choice of starting values and the remaining $T = 200$ observations are used.

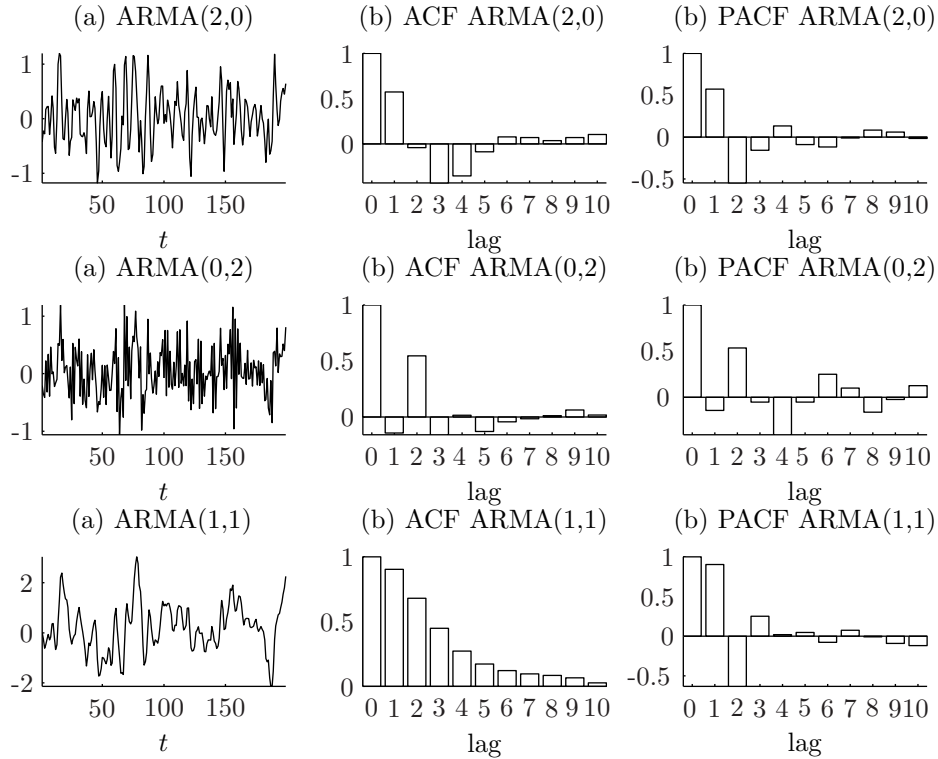


Figure 13.2 Plots of simulated series for AR(2), MA(2) and ARMA(1,1) models together with their autocorrelation (ACF) and partial autocorrelation (PACF) functions.

Inspection of the time series tends not to reveal any clear patterns in the data, whereas inspection of the ACF and the PACF in the second and third columns, respectively, identify strong dynamical behaviour. The ACF for the AR(2) model exhibits damped oscillatory behaviour, while the PACF

is characterized by spikes equal to the number of significant lags of y_t . The MA(2) model displays qualitatively the opposite pattern to the AR model, with the ACF now exhibiting spikes equal to the number of moving average lags and the PACF characterized by damped oscillatory behaviour. The ARMA(1,1) model is a combination of AR and MA models and exhibits both a damped ACF and PACF. $\square$

13.3.2 Multivariate Model Classification

A natural extension of the univariate ARMA class of models is where y_t represents a vector of N time series. Let Φ_i and Ψ_i be $(N \times N)$ matrices

$$\Phi_i = \begin{bmatrix} \phi_{i,1,1} & \cdots & \phi_{i,1,N} \\ \vdots & \ddots & \vdots \\ \phi_{i,N,1} & \cdots & \phi_{i,N,N} \end{bmatrix}, \quad \Psi_i = \begin{bmatrix} \psi_{i,1,1} & \cdots & \psi_{i,1,N} \\ \vdots & \ddots & \vdots \\ \psi_{i,N,1} & \cdots & \psi_{i,N,N} \end{bmatrix}, \quad (13.5)$$

where $\phi_{i,j,k}$ is the autoregressive parameter at lag i in equation j on variable k and $\psi_{i,j,k}$ is the corresponding moving-average parameter. The multivariate analogue of the ARMA(p,q) model in (13.2) is the vector autoregressive model VARMA(p,q)

$$\begin{aligned} y_t &= \mu + \sum_{i=1}^p \Phi_i y_{t-i} + v_t + \sum_{i=1}^q \Psi_i v_{t-i} \\ v_t &\sim N(0, V), \end{aligned} \quad (13.6)$$

where v_t is an N dimensional disturbance vector with zero mean vector and $(N \times N)$ covariance matrix V and $\{\mu, \Phi_1, \Phi_2, \dots, \Phi_p, \Psi_1, \Psi_2, \dots, \Psi_q, V\}$ are unknown parameters. Using lag operators, the VARMA(p,q) class of model is represented as

$$\Phi_p(L)y_t = \mu + \Psi_q(L)v_t, \quad v_t \sim iid N(0, V), \quad (13.7)$$

where

$$\begin{aligned} \Phi_p(L) &= I - \Phi_1 L - \Phi_2 L^2 - \cdots - \Phi_p L^p \\ \Psi_q(L) &= I + \Psi_1 L + \Psi_2 L^2 + \cdots + \Psi_q L^q, \end{aligned} \quad (13.8)$$

are matrix polynomials in the lag operator L .

An important special case of the VARMA model is one in which there are p autoregressive lags and no moving-average lags, $q = 0$. This special case is known as a VAR(p) model

$$y_t = \mu + \Phi_1 y_{t-1} + \cdots + \Phi_p y_{t-p} + v_t. \quad (13.9)$$

A VAR has the property that each variable is expressed as a function of its

own lags and the lags of all of the other variables in the system, with the result that the lag structure on all variables in all equations are the same. Thus, the right hand side variables in each equation in a VAR are identical.

Example 13.2 VAR(1) Model

A trivariate ($N = 3$) VAR with one lag is

$$\begin{aligned} y_{1,t} &= \mu_1 + \phi_{1,1,1}y_{1,t-1} + \phi_{1,1,2}y_{2,t-1} + \phi_{1,1,3}y_{3,t-1} + v_{1,t} \\ y_{2,t} &= \mu_2 + \phi_{1,2,1}y_{1,t-1} + \phi_{1,2,2}y_{2,t-1} + \phi_{1,2,3}y_{3,t-1} + v_{2,t} \\ y_{3,t} &= \mu_3 + \phi_{1,3,1}y_{1,t-1} + \phi_{1,3,2}y_{2,t-1} + \phi_{1,3,3}y_{3,t-1} + v_{3,t}, \end{aligned}$$

In matrix notation, the model becomes

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \\ y_{3,t} \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} + \begin{bmatrix} \phi_{1,1,1} & \phi_{1,1,2} & \phi_{1,1,3} \\ \phi_{1,2,1} & \phi_{1,2,2} & \phi_{1,2,3} \\ \phi_{1,3,1} & \phi_{1,3,2} & \phi_{1,3,3} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \\ y_{3,t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \\ v_{3,t} \end{bmatrix}$$

$$y_t = \mu + \Phi_1 y_{t-1} + v_t.$$

□

A second special case involves a VARMA model with q moving average lags and no autoregressive lags, $p = 0$. This special case is known as a VMA(q) model

$$y_t = \mu + v_t + \Psi_1 v_{t-1} + \Psi_2 v_{t-2} + \cdots + \Psi_q v_{t-q}. \quad (13.10)$$

A VMA(q) has the property that each variable is expressed as a function of its own disturbance and the lags of all of the other disturbances in the system, with the lags structure on all lagged disturbances in all equations being the same.

Example 13.3 VMA(2) Model

A trivariate ($N = 3$) VMA with two lags is

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \\ y_{3,t} \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \\ v_{3,t} \end{bmatrix} + \begin{bmatrix} \psi_{1,1,1} & \psi_{1,1,2} & \psi_{1,1,3} \\ \psi_{1,2,1} & \psi_{1,2,2} & \psi_{1,2,3} \\ \psi_{1,3,1} & \psi_{1,3,2} & \psi_{1,3,3} \end{bmatrix} \begin{bmatrix} v_{1,t-1} \\ v_{2,t-1} \\ v_{3,t-1} \end{bmatrix}$$

$$+ \begin{bmatrix} \psi_{2,1,1} & \psi_{2,1,2} & \psi_{2,1,3} \\ \psi_{2,2,1} & \psi_{2,2,2} & \psi_{2,2,3} \\ \psi_{2,3,1} & \psi_{2,3,2} & \psi_{2,3,3} \end{bmatrix} \begin{bmatrix} v_{1,t-2} \\ v_{2,t-2} \\ v_{3,t-2} \end{bmatrix}$$

$$y_t = \mu + v_t + \Psi_1 v_{t-1} + \Psi_2 v_{t-2}.$$

□

13.3.3 Likelihood

Specifying the likelihood function for a VARMA(p, q) model, requires deriving the joint probability density function of y_t . As the process y_t is dependent, from Chapter 1, the joint pdf factorizes as

$$f(y_1, y_2, \dots, y_T; \theta) = f(y_s, y_{s-1}, \dots, y_1; \theta) \times \prod_{t=s+1}^T f(y_t | y_{t-1}, \dots, y_1; \theta), \quad (13.11)$$

where $s = \max(p, q)$ represents the maximum lag in the model and θ contains the unknown parameters $\theta = \{\Phi_1, \Phi_2, \dots, \Phi_p, \Psi_1, \Psi_2, \dots, \Psi_q, V\}$. The log-likelihood function is

$$\ln L_T(\theta) = \frac{1}{T} \left(\ln f(y_s, y_{s-1}, \dots, y_1; \theta) + \sum_{t=s+1}^T \ln f(y_t | y_{t-1}, \dots, y_1; \theta) \right). \quad (13.12)$$

The specification of the log-likelihood function is complicated by the presence of $f(y_s, y_{s-1}, \dots, y_1; \theta)$, the joint distribution of the initial observations to allow for the s lags. The solution adopted in Chapter 7, in which these s observations are treated as fixed, is also adopted here. In this case, $f(y_s, y_{s-1}, \dots, y_1; \theta) = 1$ in (13.12), which produces the simpler conditional log-likelihood function

$$\ln L_T(\theta) = \frac{1}{T-s} \sum_{t=s+1}^T \ln f(y_t | y_{t-1}, \dots, y_1; \theta). \quad (13.13)$$

13.4 Stationarity

An important feature of the changes and the growth rates of the macroeconomic variables in Figure 13.1 is that they do not exhibit trends. This characteristic in time series is formally referred to as stationarity, which is formally defined in Chapter 2. The simplest approach to identify the weak stationarity properties of y_t is to look at the first two moments of the distribution, which are determined directly from the specification of the ARMA and VARMA models given in the previous section.

13.4.1 Univariate Examples

The following examples identify the stationarity properties of the variable y_t using the moments of its distribution. A useful concept in helping to establishing stationarity is the autocovariance function defined as

$$\gamma_k = E[(y_t - E[y_t])(y_{t-k} - E[y_{t-k}])]. \quad (13.14)$$

Setting $k = 0$ gives the variance. The related autocorrelation function is

$$\rho_k = \frac{\gamma_k}{\gamma_0}. \quad (13.15)$$

In all of the following examples, v_t is *iid* with zero mean and constant variance σ_v^2 .

Example 13.4 Moments of Stationary ARMA Models

Consider the ARMA(p, q) model

$$y_t = \mu + \sum_{i=1}^p \phi_i y_{t-i} + v_t + \sum_{i=1}^q \psi_i v_{t-i},$$

where v_t is *iid* with zero mean and constant variance σ_v^2 . The first two moments for alternative parameterizations are as follows.

White Noise	Mean:	$E[y_t] = \mu + E[v_t] = \mu$
	Variance:	$\gamma_0 = E[v_t^2] = \sigma_v^2 > 0$
	Covariance:	$\gamma_k = E[v_t v_{t-k}] = 0, k \neq 0.$
MA(1)	Mean:	$E[y_t] = E[\mu + v_t + \psi_1 v_{t-1}] = \mu$
	Variance:	$\gamma_0 = \sigma_v^2(1 + \psi_1^2)$
	Covariance:	$\gamma_k = \begin{cases} \sigma_v^2 \psi_1 & \text{if } k = \pm 1 \\ 0 & \text{otherwise.} \end{cases}$
MA(q)	Mean:	$E[y_t] = \mu$
	Variance:	$\gamma_0 = \sigma_v^2(1 + \psi_1^2 + \psi_2^2 + \cdots + \psi_q^2)$
	Covariance:	$\gamma_k = \begin{cases} \sigma_v^2 \sum_{i=0}^{q-k} \psi_i \psi_{i+k} & \text{if } k = 1, 2, \dots, q \\ 0 & k > q. \end{cases}$
AR(1)	Mean:	$E[y_t] = E[\mu + \phi_1 y_{t-1}] = \frac{\mu}{1 - \phi_1}, \quad \phi_1 < 1$
	Variance:	$\gamma_0 = \frac{\sigma_v^2}{1 - \phi_1^2}$
	Covariance:	$\gamma_k = \frac{\sigma_v^2 \phi_1^k}{1 - \phi_1^2}, \quad k \geq 1.$

As all of the moments are not a function of time, y_t is weakly stationary in each model. If v_t is distributed as $N(0, \sigma_v^2)$, then y_t is also strongly stationary. $\square$

The fact that the moving average examples yield moments that are finite sums of the moving average parameters, implies that all (finite) moving average models are stationary. In the case of the AR(1) example, the restriction $|\phi_1| < 1$ is derived formally below. Without this restriction, the moments of y_t are no longer finite, as highlighted in the next example where the moments of y_t are shown to be a function of t .

Example 13.5 Random Walk with Drift

Consider the model

$$y_t = \mu + y_{t-1} + v_t ,$$

which is a special case of the AR(1) model with $\phi_1 = 1$. The first two moments are

$$\begin{aligned} \text{Mean:} & \quad E[y_t] = t\mu \\ \text{Variance:} & \quad \gamma_0 = t\sigma_v^2 \\ \text{Covariance:} & \quad \gamma_k = (t-k)\sigma_v^2. \end{aligned}$$

As the first two moments are a function of t , y_t is not stationary. The time series properties of y_t are characterized by an increasing ($\mu > 0$) trend and increasing volatility. In the special case of a random walk without drift $\mu = 0$, y_t is still not stationary because its variance is a function of t even though its mean is zero. $\square$

13.4.2 Multivariate Examples

The multivariate analogue of the autocovariance function in (13.14) is defined as

$$\Gamma_k = E[(y_t - E[y_t])(y_{t-k} - E[y_{t-k}])'] , \quad (13.16)$$

where y_t is now a $(N \times 1)$ vector. Setting $k = 0$ gives the $(N \times N)$ covariance matrix. A property of the autocovariance matrix is

$$\Gamma_k = \Gamma_{-k}' . \quad (13.17)$$

The autocorrelation matrix is

$$P_k = D^{-1/2} \Gamma_k D^{-1/2} , \quad (13.18)$$

where D is a diagonal matrix with the variances down the main diagonal.

Example 13.6 Moments of Stationary VARMA Models

Consider the VARMA(p, q) model

$$y_t = \mu + \sum_{i=1}^p \Phi_i y_{t-i} + v_t + \sum_{i=1}^q \Psi_i v_{t-i},$$

where v_t is an N dimensional *iid* vector with zero mean covariance matrix V . The first two moments for alternative parameterizations are as follows.

VMA(1)	Mean:	$E[y_t] = E[\mu + v_t + \Psi_1 v_{t-1}] = \mu$
	Variance:	$\Gamma_0 = V + \Psi_1 V \Psi_1'$
	Covariance:	$\gamma_k = \begin{cases} \Psi_1 V & \text{if } k = 1 \\ V \Psi_1' & \text{if } k = -1 \\ 0 & \text{otherwise.} \end{cases}$

VAR(1)	Mean:	$E[y_t] = E[\mu + \Phi_1 y_{t-1}] = [I - \Phi_1]^{-1} \mu, \quad \phi_1 < 1$
	Variance:	$\text{vec}(\Gamma_0) = [I - \Phi_1 \otimes \Phi_1]^{-1} \text{vec}(V)$
	Covariance:	$\Gamma_k = \Phi_1 \Gamma_{k-1}, \quad k \geq 1.$

As all of the moments are not a function of time, y_t is weakly stationary in each model. If v_t is distributed as $N(0, V)$, then y_t is also strongly stationary. $\square$

The stationarity condition for the VAR(1) is based on the N dimensional matrix Φ_1 , which is now slightly more involved than it is for the univariate AR(1) model, which simply requires $|\phi_1| < 1$.

13.4.3 The Stationarity Condition

The N dimensional variable y_t is stationary provided that the roots of the polynomial $|\Phi_p(z)| = 0$ lie outside the unit circle. The notation $|\cdot|$ represents the determinant of the matrix argument.

Example 13.7 Stationarity of an AR(1) Model

The polynomial is $\phi_p(L) = 1 - \phi_1 L$ so the equation to be solved is

$$|1 - \phi_1 z| = 1 - \phi_1 z = 0.$$

This equation has only one root, which is given by $z_1 = \phi_1^{-1}$. For stationarity $|z_1| > 1$, which is satisfied if $|\phi_1^{-1}| > 1$, or $|\phi_1| < 1$, a result which that was stated earlier. $\square$

Example 13.8 Stationarity of the Interest Rate

Consider the following AR(4) model of the interest rate, $y_t = r_t$,

$$r_t = 0.141 + 1.418 r_{t-1} - 0.588 r_{t-2} + 0.125 r_{t-3} - 0.024 r_{t-4} + v_t.$$

The polynomial $1 - 1.418 z + 0.588 z^2 - 0.125 z^3 + 0.024 z^4 = 0$ has two real roots and two complex roots given by

$$z_1 = -8.772, \quad z_2 = 1.285 - 1.713i, \quad z_3 = 1.285 + 1.713i, \quad z_4 = 1.030.$$

Because $|z_1| = 8.772 > 1$, $|z_2| = |z_3| = 2.141 > 1$ and $|z_4| = 1.030 > 1$, the nominal interest rate is stationary. $\square$

Example 13.9 Stationarity of Money Growth and Inflation

Consider a VAR(2) model with $N = 2$ variables containing annual percentage money growth and inflation, $y_t = [100\Delta_{12}lm_t, 100\Delta_{12}lp_t]'$, with autoregressive parameter matrices

$$\Phi_1 = \begin{bmatrix} 1.279 & -0.355 \\ 0.002 & 1.234 \end{bmatrix}, \quad \Phi_2 = \begin{bmatrix} -0.296 & 0.353 \\ 0.007 & -0.244 \end{bmatrix}.$$

The polynomial $|I - \Phi_1 z - \Phi_2 z^2| = 0$, has four real roots

$$z_1 = 4.757, \quad z_2 = 2.874, \quad z_3 = 1.036, \quad z_4 = 1.011.$$

Because $|z_i| > 0, \forall i$ both variables are jointly stationary. $\square$

13.4.4 Wold's Representation Theorem

Suppose that y_t is purely stochastic so that it contains no mean, trends, structural breaks or other deterministic terms. This assumption implies that $E[y_t] = 0$. If y_t is also weakly stationary, then Wold's theorem gives a useful representation for understanding its second moment properties.

Wold Representation Theorem

If y_t is weakly stationary, then it can be represented as

$$y_t = \sum_{j=0}^{\infty} C_j v_{t-j}, \quad (13.19)$$

where $\sum_{j=0}^{\infty} C_j^2 < \infty$ and v_t is white noise. The infinite sum is understood as the mean square limit of $\sum_{j=0}^n C_j v_{t-j}$ as $n \rightarrow \infty$.

■

The intuition behind this representation is that the left and right hand sides of (13.19) have the same first and second moments. Note that Wold's theorem does not state that all weakly stationary time series must be generated according to (13.19), only that they can be represented as (13.19).

Example 13.10 Wold Representation

Consider the random sequence, y_t , generated by

$$y_t = u_t u_{t-1}, \quad u_t \sim iid(0, \sigma^2).$$

The variable y_t is white noise and has a Wold representation of the form

$$y_t = v_t,$$

where v_t has variance σ^4 . This Wold representation captures the second order properties of y_t . But other properties may be overlooked, such as, in this example, the fact that

$$\text{cov}(y_t^2, y_{t-1}^2) = \sigma^4 (m_4 - \sigma^4),$$

where $m_4 = E[v_t^4]$. □

It is common and convenient to assume that a weakly stationary process is generated by (13.19). In this case, y_t is referred to as a linear process. In view of the preceding comments, Wold's theorem can not be invoked to state that every zero mean weakly stationary process is generated by a linear process, only that for every weakly stationary process there exists a linear process with the same first and second moments. Moreover, for the development of asymptotic distribution theory, it is necessary to strengthen the condition on v_t in the linear process so that it is *iid* or at least a martingale difference sequence, since first and second moment conditions alone are not sufficient for asymptotics. This is just a cautionary note that all results for linear processes need not extend automatically to all weakly stationary processes by appeal to Wold's theorem.

13.4.5 Transforming a VAR to a VMA

An important feature of the AR and MA examples given in Figure 13.2 is that they mirror each other, with the AR model generating a decaying ACF and a PACF that cuts-off at the lag of the AR, whereas the MA model gives the opposite result with the PACF exhibiting a decaying pattern and the ACF cutting-off at the lag of the MA. The relationship between autoregressive and moving-average models is now formalized. To highlight

the relationship between AR and MA models, consider initially the simplest example of converting an AR(1) model to an infinite MA model.

Example 13.11 Transforming an AR(1) to a MA(∞)

The AR(1) model expressed in terms of lag operators is

$$(1 - \phi_1 L)y_t = v_t.$$

The polynomial $(1 - \phi_1 L)$ is inverted as (see Appendix B)

$$\begin{aligned} y_t &= (1 - \phi_1 L)^{-1} v_t = (1 + \phi_1 L + \phi_1^2 L^2 + \cdots) v_t \\ &= v_t + \phi_1 v_{t-1} + \phi_1^2 v_{t-2} + \cdots = \sum_{i=0}^{\infty} \psi_i v_{t-i}, \end{aligned}$$

which is now an infinite moving average model with $\psi_i = \phi_1^i$. Provided that $|\phi_1| < 1$, which is the condition required for stationarity given previously, the observed time path of y_t is determined by the complete history of shocks $\{v_t, v_{t-1}, \dots\}$, with the weights ϕ_1^i in the summation decaying at an exponential rate. $\square$

The conversion of an autoregressive model into a moving average model where there are several autoregressive lags and the dimension is $N > 1$ requires a recursive algorithm. To express a VAR(p) model as an infinite VMA model, consider

$$(I - \Phi_1 L - \Phi_2 L^2 - \cdots - \Phi_p L^p) y_t = v_t, \quad (13.20)$$

where, for simplicity, the vector of intercepts is set to zero. Invert the polynomial $\Phi_p(L)$ to generate an infinite moving average process

$$y_t = (I - \Phi_1 L - \Phi_2 L^2 - \cdots - \Phi_p L^p)^{-1} v_t = (I + \Psi_1 L + \Psi_2 L^2 + \cdots) v_t, \quad (13.21)$$

where by definition

$$(I - \Phi_1 L - \Phi_2 L^2 - \cdots - \Phi_p L^p)^{-1} = (I + \Psi_1 L + \Psi_2 L^2 + \cdots). \quad (13.22)$$

It follows from expression (13.22) that

$$(I - \Phi_1 L - \Phi_2 L^2 - \cdots - \Phi_p L^p)(I + \Psi_1 L + \Psi_2 L^2 + \cdots) = I \quad (13.23)$$

$$I + (\Psi_1 - \Phi_1)L + (\Psi_2 - \Phi_1\Psi_1 - \Phi_2)L^2 + \cdots = I. \quad (13.24)$$

Equating the parameters on the powers on L using expressions (13.23) and (13.24) gives a recursion for deriving the moving average parameter matrices

from the autoregressive parameter matrices

$$\begin{aligned}
 \Psi_1 &= \Phi_1 \\
 \Psi_2 &= \Phi_1 \Psi_1 + \Phi_2 \\
 \Psi_3 &= \Phi_1 \Psi_2 + \Phi_2 \Psi_1 + \Phi_3 \\
 &\vdots \\
 \Psi_i &= \sum_{j=1}^i \Phi_j \Psi_{i-j} \quad \text{where } \Phi_j = 0, j > p.
 \end{aligned} \tag{13.25}$$

Example 13.12 Transforming a VAR(1) to a VMA(∞)

As $\Phi_i = 0$ for $i > 1$, the VMA parameter matrices associated with the first three lags using the recursion in (13.25) are

$$\begin{aligned}
 \Psi_1 &= \Phi_1 \\
 \Psi_2 &= \Phi_1 \Psi_1 + \Phi_2 = \Phi_1 \Psi_1 = \Phi_1 \Phi_1 \\
 \Psi_3 &= \Phi_1 \Psi_2 + \Phi_2 \Psi_1 + \Phi_3 = \Phi_1 \Psi_2 = \Phi_1 \Phi_1 \Phi_1.
 \end{aligned}$$

This result is a multivariate version of the univariate AR(1) example shown previously in which $\psi_i = \phi_1^i$. $\square$

Example 13.13 Transforming a VAR(2) to a VMA(∞)

Recall the VAR(2) model of money growth and inflation from Example 13.9, where the VAR parameter matrices are

$$\Phi_1 = \begin{bmatrix} 1.28 & -0.36 \\ 0.00 & 1.23 \end{bmatrix}, \quad \Phi_2 = \begin{bmatrix} -0.30 & 0.35 \\ 0.01 & -0.24 \end{bmatrix}.$$

As $\Phi_i = 0$ for $i > 2$, the first three lagged VMA parameter matrices are

$$\begin{aligned}
 \Psi_1 &= \Phi_1 = \begin{bmatrix} 1.28 & -0.36 \\ 0.00 & 1.23 \end{bmatrix} \\
 \Psi_2 &= \Phi_1 \Psi_1 + \Phi_2 \\
 &= \begin{bmatrix} 1.28 & -0.36 \\ 0.00 & 1.23 \end{bmatrix} \begin{bmatrix} 1.28 & -0.36 \\ 0.00 & 1.23 \end{bmatrix} + \begin{bmatrix} -0.30 & 0.35 \\ 0.01 & -0.24 \end{bmatrix} \\
 &= \begin{bmatrix} 1.34 & -0.54 \\ 0.01 & 1.28 \end{bmatrix} \\
 \Psi_3 &= \Phi_1 \Psi_2 + \Phi_2 \Psi_1 \\
 &= \begin{bmatrix} 1.28 & -0.36 \\ 0.00 & 1.23 \end{bmatrix} \begin{bmatrix} 1.34 & -0.54 \\ 0.01 & 1.28 \end{bmatrix} \\
 &\quad + \begin{bmatrix} -0.30 & 0.35 \\ 0.01 & -0.24 \end{bmatrix} \begin{bmatrix} 1.28 & -0.36 \\ 0.00 & 1.23 \end{bmatrix} \\
 &= \begin{bmatrix} 1.33 & -0.60 \\ 0.03 & 1.27 \end{bmatrix}.
 \end{aligned}$$

□

13.5 Invertibility

The results of the previous section on stationarity relate to the conditions needed for transforming a VAR to a VMA. Reversing this transformation and working from a VMA to a VAR requires in the same type of condition which is known as the invertibility property. As will be seen in the section on estimation, invertibility is required for estimation of the parameters.

13.5.1 The Invertibility Condition

The N dimensional variable y_t is invertible provided that the roots of the polynomial $|\Psi_p(z)| = 0$ lie outside the unit circle, where as before $|\cdot|$ represents the determinant of the matrix.

Example 13.14 MA(1)

The polynomial is $\psi_p(L) = 1 + \psi_1 L$ so the equation to be solved is $1 + \psi_1 z = 0$. There is only one root, which is given by $z_1 = -\psi_1^{-1}$. For invertibility $|z_1| > 1$, which is satisfied if $|\psi_1^{-1}| > 1$, or $|\psi_1| < 1$. □

13.5.2 Transforming a VMA to a VAR

Now consider the problem of transforming a moving average model into an autoregressive model.

Example 13.15 Transforming a MA(1) to an AR(∞)

Consider the MA(1) model

$$y_t = (1 + \psi_1 L)v_t,$$

where $|\psi_1| < 1$. Inverting the lag polynomial $(1 + \psi_1 L)$ gives

$$\begin{aligned} (1 + \psi_1 L)^{-1} y_t &= v_t \\ (1 - \psi_1 L + \psi_1^2 L^2 - \cdots) y_t &= v_t \\ y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2} - \cdots &= v_t \\ y_t &= \sum_{i=1}^{\infty} \phi_i y_{t-i} + v_t, \end{aligned}$$

which is now an infinite autoregressive model with $\phi_i = \psi_1^i (-1)^{i+1}$. Provided that $|\psi_1| < 1$, the weights ϕ_i in the summation decay at an exponential rate for longer lags. $\square$

This last example suggests that if the autoregressive lag order of y_t is relatively long, then a more parsimonious way of modelling the dynamics is to specify a MA model with a relatively small number of lags.

An important result is that all finite autoregressive processes are invertible, which is the mirror of the stationarity result found for moving average processes. The invertibility results for the univariate MA model generalize to an N dimensional process. Once again a recursion as in expression (13.13) is required, with the roles of the parameters being reversed.

13.6 Estimation

The parameters of the ARMA and VARMA models are estimated by maximum likelihood methods by choosing the parameter vector θ to maximize the conditional log-likelihood function in (13.13). One of the iterative algorithms presented in Chapter 3 is needed because the inclusion of moving average terms in the model results in the likelihood being nonlinear in the parameters. As the assumption of normality is commonly adopted in specifying linear time series models, the Gauss-Newton algorithm, as discussed in Chapter 6, can also be used. For the important special case where there

are no moving average terms, it is shown that the maximum likelihood estimates are obtained by ordinary least squares and so convergence is achieved in one iteration.

Example 13.16 Estimating an ARMA(1,1) Model

Consider the model

$$y_t = \mu + \phi_1 y_{t-1} + v_t + \psi_1 v_{t-1}, \quad v_t \sim iid N(0, \sigma_v^2),$$

with unknown parameters $\theta = \{\mu, \phi_1, \psi_1, \sigma_v^2\}$. The conditional log-likelihood function with $s = \max(p, q) = 1$ in (13.13) is

$$\begin{aligned} \ln L_T(\theta) &= \frac{1}{T-1} \sum_{t=2}^T \ln f(y_t | y_{t-1}, \dots, y_1; \theta) \\ &= \frac{1}{T-1} \sum_{t=2}^T \left(-\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma_v^2 - \frac{1}{2\sigma_v^2} (y_t - \mu - \phi_1 y_{t-1} - \psi_1 v_{t-1})^2 \right) \\ &= -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma_v^2 - \frac{1}{2\sigma_v^2(T-1)} \sum_{t=2}^T (y_t - \mu - \phi_1 y_{t-1} - \psi_1 v_{t-1})^2, \end{aligned}$$

with gradients

$$\begin{aligned} \frac{\partial \ln L_T(\theta)}{\partial \mu} &= \frac{1}{\sigma_v^2(T-1)} \sum_{t=2}^T v_t \left(1 + \psi_1 \frac{\partial v_{t-1}}{\partial \mu} \right) \\ \frac{\partial \ln L_T(\theta)}{\partial \phi_1} &= \frac{1}{\sigma_v^2(T-1)} \sum_{t=2}^T v_t \left(y_{t-1} + \psi_1 \frac{\partial v_{t-1}}{\partial \phi_1} \right) \\ \frac{\partial \ln L_T(\theta)}{\partial \psi_1} &= \frac{1}{\sigma_v^2(T-1)} \sum_{t=2}^T v_t \left(v_{t-1} + \psi_1 \frac{\partial v_{t-1}}{\partial \psi_1} \right) \\ \frac{\partial \ln L_T(\theta)}{\partial \sigma_v^2} &= -\frac{T-1}{2\sigma_v^2(T-1)} + \frac{1}{2\sigma_v^4} \sum_{t=2}^T v_t^2. \end{aligned}$$

An iterative algorithm is needed to maximize the likelihood since the gradients are nonlinear functions of the parameters because of the presence of $\partial v_{t-1} / \partial \theta$. To circumvent the need to derive $\partial v_{t-1} / \partial \theta$, it is convenient to use numerical derivatives in the optimization algorithm. This algorithm is simplified by concentrating out σ_v^2 using the maximum likelihood estimator

$$\hat{\sigma}_v^2 = (T-1)^{-1} \sum_{t=2}^T \hat{v}_t^2, \quad \hat{v}_t = y_t - \hat{\mu} - \hat{\phi}_1 y_{t-1} - \hat{\psi}_1 \hat{v}_{t-1},$$

computed at each iteration of the algorithm. □

This example highlights the importance of the invertibility property of ARMA and VARMA models in estimation. Consider the gradient expression in the previous example corresponding to ϕ_1 that contains the term

$$\frac{\partial v_t}{\partial \phi_1} = y_{t-1} + \psi_1 \frac{\partial v_{t-1}}{\partial \phi_1}.$$

This is a first-order difference equation in $\partial v_{t-1}/\partial \phi_1$ with parameter ψ_1 . If the MA term is not invertible, $|\psi_1| > 1$, then this equation may eventually explode resulting in the algorithm failing.

Example 13.17 Estimating a VARMA(1,1) Model

Consider the model

$$y_t = \mu + \Phi_1 y_{t-1} + v_t + \Psi_1 v_{t-1}, \quad v_t \sim N(0, V),$$

where y_t is of dimension N . The conditional log-likelihood function with $s = 1$ in (13.13) is

$$\begin{aligned} \ln L_T(\theta) &= \frac{1}{T-1} \sum_{t=2}^T \ln f(y_t | y_{t-1}, \dots, y_1; \theta) \\ &= -\frac{N}{2} \ln 2\pi - \frac{1}{2} \ln |V| - \frac{1}{2(T-1)} \sum_{t=2}^T v_t V^{-1} v_t', \end{aligned}$$

where the $(1 \times N)$ disturbance vector at time t is $v_t = y_t - \mu - \Phi_1 y_{t-1} - \Psi_1 v_{t-1}$. Table 13.2 gives the maximum likelihood estimates from estimating the VARMA(1,1)

$$\begin{aligned} y_{1,t} &= \mu_1 + \phi_{1,1,1} y_{1,t-1} + v_{1,t} + \psi_{1,1,1} v_{1,t-1} + \psi_{1,1,2} v_{2,t-1} \\ y_{2,t} &= \mu_2 + \phi_{1,2,2} y_{2,t-1} + v_{2,t} + \psi_{1,2,1} v_{1,t-1} + \psi_{1,2,2} v_{2,t-1}. \end{aligned}$$

$T = 500$ observations are generated by simulating the model with the true parameters given in the table. The disturbance term, v_t , has zero mean and covariance matrix given by the identity matrix. There is good agreement between the parameter estimates and the true population parameters. Also reported are the quasi-maximum likelihood standard errors and t-statistics that allow for heteroscedasticity. The residual covariance matrix is estimated as

$$\hat{V} = \frac{1}{499} \sum_{t=2}^{500} \hat{v}_t' \hat{v}_t = \begin{bmatrix} 0.956 & 0.040 \\ 0.040 & 0.966 \end{bmatrix},$$

which is obtained at the final iteration. □

Table 13.2

Maximum likelihood estimates of the VARMA(1, 1) model: standard errors and t-statistics are based on quasi-maximum likelihood estimates of the covariance matrix allowing for heteroscedasticity.

Parameter	Population	Estimate	Std error	t-stat.
μ_1	0.0	0.028	0.059	0.471
$\phi_{1,1,1}$	0.6	0.600	0.045	13.354
$\psi_{1,1,1}$	0.2	0.244	0.063	3.871
$\psi_{1,1,2}$	-0.5	-0.485	0.046	-10.619
μ_2	0.0	0.083	0.072	1.164
$\phi_{2,2,2}$	0.4	0.415	0.053	7.756
$\psi_{1,2,1}$	0.2	0.255	0.029	8.717
$\psi_{1,2,2}$	0.6	0.576	0.051	11.400
$\ln L_T(\theta) = -2.789$				

Example 13.18 Estimating a VAR(p) Model

The conditional log-likelihood function of the N dimensional VAR(p) model in (13.13) is

$$\begin{aligned}\ln L_T(\theta) &= \frac{1}{T-p} \sum_{t=p+1}^T \ln f(y_t | y_{t-1}, \dots, y_1; \theta) \\ &= -\frac{N}{2} \ln 2\pi - \frac{1}{2} \ln |V| - \frac{1}{2(T-p)} \sum_{t=p+1}^T v_t V^{-1} v_t',\end{aligned}$$

where the $(1 \times N)$ disturbance vector at time t is

$$v_t = y_t - \mu - \Phi_1 y_{t-1} - \Phi_2 y_{t-2} - \dots - \Phi_p y_{t-p},$$

and $\theta = \{\mu, \Phi_1, \Phi_2, \dots, \Phi_p, V\}$, is the set of unknown parameters. As all of the explanatory variables in each of the equations are identical, from the results in Chapter 5, the maximum likelihood estimates are obtained by ordinary least squares applied to each equation separately. There are no efficiency gains from estimating the system jointly. Once these estimates are obtained, V is computed as

$$\hat{V} = \frac{1}{T-p} \sum_{t=p+1}^T \hat{v}_t \hat{v}_t'.$$

In the special case where $N = 1$, the maximum likelihood estimates of the $AR(p)$ model are obtained by regressing y_t on a constant and its lags. $\square$

13.7 Optimal Choice of Lag Order

An important practical consideration in estimating stationary time series models is the optimal choice of lag order. A common data-driven way of selecting the lag order is to use information criteria. An information criterion is a scalar metric that is a simple but effective way of balancing the improvement in the value of the log-likelihood function with the loss of degrees of freedom which results from increasing the lag order of a time series model.

The three most commonly used information criteria for selecting a parsimonious time series model are the Akaike information criterion (AIC) (Akaike, 1974, 1976), the Hannan information criterion (HIC) (Hannan and Quinn, 1979; Hannon, 1980) and the Schwarz information criterion (SIC) (Schwarz, 1978). If k is the number of parameters estimated in the current model, these information criteria are given by

$$\begin{aligned} AIC &= -2 \ln L_T(\hat{\theta}) + \frac{2k}{T} & [AIC] \\ HIC &= -2 \ln L_T(\hat{\theta}) + \frac{2k \ln(\ln(T))}{T} & [HIC] \\ SIC &= -2 \ln L_T(\hat{\theta}) + \frac{k \ln(T)}{T}, & [SIC] \end{aligned} \quad (13.26)$$

where it is understood that the number of observations, T , is suitably adjusted for the maximum lag order.

For the general VARMA(p, q) class of time series model given in equation (13.6) in which the disturbances, v_t , are assumed to be normally distributed, the information criteria in equation (13.26) take on a simple form. Using the results established in Chapter 4, the log-likelihood function evaluated at the maximum likelihood estimates is

$$\ln L_T(\hat{\theta}) = -\frac{N}{2}(1 + \ln 2\pi) - \frac{1}{2} \ln |\hat{V}|. \quad (13.27)$$

Recognising that the first two terms on the right hand side of (13.27) are constants and do not depend on the choice of lag order, the three information

criteria in equation (13.26) become

$$\begin{aligned}
 AIC &= \ln |\widehat{V}| + \frac{2k}{T} & [AIC] \\
 HIC &= \ln |\widehat{V}| + \frac{2k \ln(\ln(T))}{T} & [HIC] \\
 SIC &= \ln |\widehat{V}_v| + \frac{k \ln(T)}{T} & [SIC].
 \end{aligned} \tag{13.28}$$

In the scalar case, $\ln |\widehat{V}|$ is replaced by $\ln \widehat{\sigma}_v^2$.

Choosing an optimal lag order using information criteria requires the following steps.

- Step 1:** Choose a maximum number of lags for the AR and MA components of the time series model, respectively, $p_{\max} > p$ and $q_{\max} > q$, where p and q are the true but unknown lag lengths. The choice of $p_{\max}$ and $q_{\max}$ is informed by the ACFs and PACFs of the data and also the frequency with which the data are observed.
- Step 2:** Systematically estimate the model for all combinations of lag orders. For VAR(p) models this involves estimating the model sequentially for $p = 1, 2, \dots, p_{\max}$. For a VMA(q) specification, the model is estimated sequentially for $q = 1, 2, \dots, q_{\max}$. For VARMA(p, q) specifications, the model is systematically estimated for all combinations of $p = 1, 2, \dots, p_{\max}$ and $q = 1, 2, \dots, q_{\max}$. For each regression the relevant information criteria are computed.
- Step 3:** Choose the specification of the model corresponding to the minimum values of the information criteria. In some cases there will be disagreement between different information criteria and the final choice is then an issue of judgement.

Example 13.19 Lag Length Selection in an AR(3) Model

Consider the AR(3) model

$$y_t = 0.0 + 0.2y_{t-1} - 0.15y_{t-2} + 0.05y_{t-3} + u_t, \quad u_t \sim iid N(0, 0.5).$$

The model is simulated using starting values of 0.0 for y_{t-i} , $i = 1, 2, 3$ for $T = 200$ observations. Note that the first 100 simulated values are discarded to reduce the dependence on initial conditions. The AIC, HIC and SIC are used to select the optimal lag order in 2000 replications and the results are summarized in Figure 13.3.

In this example, the AIC and HIC select 2 lags as the optimal lag length most often, while the SIC favours the choice of only 1 lag. The SIC penalizes additional variables more harshly than the AIC because $k \ln(T)/T > 2k/T$

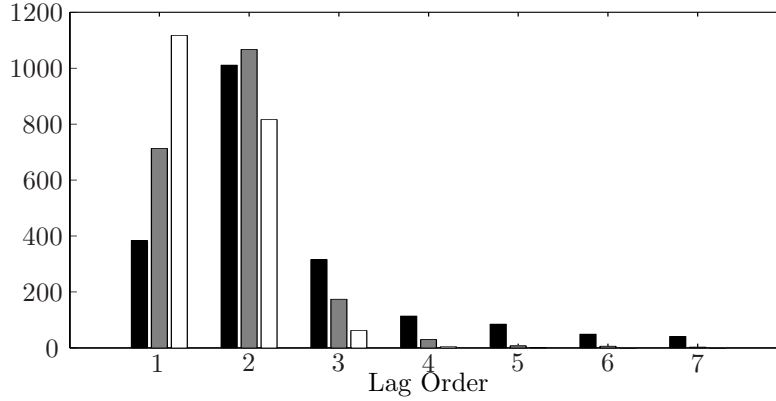


Figure 13.3 Bar graph of the optimal choice of lag order returned by the AIC (black bar), HIC (gray bar) and SIC (white bar).

for $T \geq 8$, so this result is not surprising. It is also apparent from Figure 13.3 that only the AIC chooses a lag length greater than 3 a significant number of times. $\square$

Finally, it is worth noting that the use of information criteria as a method of selecting parsimonious models extends beyond stationary time series models. In particular, the use of information criteria in tests for unit roots will be dealt with in Chapter 17.

13.8 Distribution Theory

Let θ_0 be the true population parameter. Assuming that y_t is stationary and that the conditions of the martingale difference sequence central limit theorem from Chapter 2 are satisfied, the asymptotic distribution of the quasi-maximum likelihood estimator, $\hat{\theta}$, is

$$\hat{\theta} \stackrel{a}{\sim} N\left(\theta_0, \frac{1}{T} I^{-1}(\theta_0) J(\theta_0) I^{-1}(\theta_0)\right), \quad (13.29)$$

where $I(\theta_0)$ and $J(\theta_0)$ are, respectively, the information matrix and the outer product of the gradients evaluated at θ_0 . The expression for the covariance matrix is the same expression derived for the covariance matrix of the quasi-maximum likelihood estimator in Chapter 9.

Example 13.20 Distribution of the Autocorrelation Coefficient

Assuming that y_t has zero mean, consider the AR(1) model

$$y_t = \phi_1 y_{t-1} + v_t, \quad v_t \sim N(0, \sigma_v^2).$$

For a sample of size T , maximizing the log-likelihood function in the case of ϕ_1 gives the first and second derivatives

$$\begin{aligned}\frac{\partial \ln L_T(\theta)}{\partial \phi_1} &= \frac{1}{\sigma_v^2(T-1)} \sum_{t=2}^T v_t y_{t-1} = \frac{1}{\sigma_v^2(T-1)} \sum_{t=2}^T (y_t - \phi_1 y_{t-1}) y_{t-1} \\ \frac{\partial^2 \ln L_T(\theta)}{\partial \phi_1^2} &= -\frac{1}{\sigma_v^2(T-1)} \sum_{t=2}^T y_{t-1}^2.\end{aligned}$$

The maximum likelihood estimator is the sample correlation coefficient

$$\hat{\phi}_1 = \frac{\sum_{t=2}^T y_t y_{t-1}}{\sum_{t=2}^T y_{t-1}^2}.$$

Using the definition of the variance of an AR(1) model gives the information matrix

$$\begin{aligned}I(\theta) &= -E \left[\frac{\partial^2 \ln L_T(\theta)}{\partial \phi_1^2} \right] = \frac{1}{\sigma_v^2(T-1)} \sum_{t=2}^T E[y_{t-1}^2] \\ &= \frac{1}{\sigma_v^2(T-1)} \sum_{t=2}^T \frac{\sigma_v^2}{1 - \phi_1^2} = \frac{1}{1 - \phi_1^2}.\end{aligned}$$

As v_t is independent, $I(\theta) = J(\theta)$ and the asymptotic distribution of r_1 is

$$\hat{\phi}_1 \stackrel{a}{\sim} N\left(0, \frac{1}{T} (1 - \phi_1^2)\right).$$

See also Example 2.14 in Chapter 2. □

To explore the asymptotic distribution in (13.29) in the context of estimating stationary time series models, consider a situation in which the true model is the AR(1) model

$$y_t = \mu + \phi_1 y_{t-1} + v_t, \quad v_t \sim N(0, \sigma_v^2),$$

but a misspecified model given by

$$y_t = \alpha + w_t, \quad w_t \sim N(0, \sigma_w^2),$$

is estimated instead. The gradient and Hessian of the log-likelihood function for the misspecified model are, respectively,

$$G_T(\theta) = \frac{1}{\sigma_w^2} \sum_{t=1}^T w_t, \quad H_T(\theta) = -\frac{T}{\sigma_w^2},$$

and the maximum likelihood estimator is simply the sample mean

$$\hat{\theta} = \hat{\alpha} = \frac{1}{T} \sum_{t=1}^T y_t = \bar{y}.$$

The information matrix is

$$I(\theta) = -E[H_T(\theta)] = \frac{1}{\sigma_w^2}.$$

Since

$$g_t = \frac{w_t}{\sigma_w^2} = \frac{y_t - \alpha}{\sigma_w^2},$$

then

$$E[g_t g_t'] = \frac{1}{\sigma_w^4} E[w_t^2] = \frac{\gamma_0}{\sigma_w^4},$$

where γ_0 is the variance of y_t given in (13.14) for $k = 0$ and the expectation is taken with respect to the true model. Also

$$E[g_t g_{t-j}'] = E\left[\left(\frac{w_t}{\sigma_w^2}\right)\left(\frac{w_{t-j}}{\sigma_w^2}\right)\right] = \frac{1}{\sigma_w^4} E[w_t w_{t-j}] = \frac{\gamma_j}{\sigma_w^4},$$

where γ_j from (13.14) is the j th autocovariance given by

$$\gamma_j = E[w_t w_{t-j}] = E_0[(y_t - \alpha)(y_{t-j} - \alpha)].$$

Using these results in the general expression for the outer product of the gradients matrix gives

$$E[g_t(\theta) g_t'(\theta)] = \frac{1}{\sigma_w^4} (\gamma_0 + 2\gamma_1 + 2\gamma_2 + \cdots) = \frac{1}{\sigma_w^4} \sum_{j=-\infty}^{\infty} \gamma_j.$$

Substituting the relevant terms into the covariance matrix in (13.29) yields

$$I^{-1}(\theta) J(\theta) I^{-1}(\theta) = \left[\frac{1}{\sigma_w^2}\right]^{-1} \left[\frac{1}{\sigma_w^4} \sum_{j=-\infty}^{\infty} \gamma_j\right] \left[\frac{1}{\sigma_w^2}\right]^{-1} = \left[\sum_{j=-\infty}^{\infty} \gamma_j\right],$$

so that the asymptotic distribution is

$$\bar{y} \sim N\left(\frac{\mu}{1 - \phi_1}, \frac{1}{T} \sum_{j=-\infty}^{\infty} \gamma_j\right).$$

Alternatively,

$$\sqrt{T}\left(\bar{y} - \frac{\mu}{1 - \phi_1}\right) \xrightarrow{d} N\left(0, \sum_{j=-\infty}^{\infty} \gamma_j\right). \quad (13.30)$$

This is known as the Anderson central limit theorem for the case of dependent data. The importance of stationarity to this result derives from the fact that the term $\sum_{j=-\infty}^{\infty} \gamma_j$ only converges if stationarity is satisfied. If this is not the case, the variance does not exist and the asymptotic distribution is not defined.

Example 13.21 Simulating Anderson's CLT

Let the true AR(1) model have parameter values $\mu = 0.1$, $\phi_1 = 0.9$ and $\sigma_v^2 = 0.1$. The distribution of $\bar{y}$ has mean

$$E[\bar{y}] = \frac{\mu}{1 - \phi_1} = \frac{0.1}{1 - 0.9} = 1.0,$$

and variance

$$\begin{aligned} E[(\bar{y} - E[\bar{y}])^2] &= \frac{1}{T} \sum_{j=-\infty}^{\infty} \gamma_j = \frac{1}{T} \left(\frac{\sigma_v^2}{1 - \phi_1^2} \right) (1 + 2\phi_1 + 2\phi_1^2 + \cdots) \\ &= \frac{1}{T} \left(\frac{\sigma_v^2}{1 - \phi_1^2} \right) \left(\frac{2}{1 - \phi_1} - 1 \right) \\ &= \frac{1}{T} \left(\frac{0.1}{1 - 0.9^2} \right) \left(\frac{2}{1 - 0.9} - 1 \right) \\ &= \frac{10}{T}. \end{aligned}$$

Simulating the AR(1) model 10000 times for a sample of size $T = 200$ yields the summary statistics

$$\frac{1}{10000} \sum_{i=1}^{10000} \bar{y}_i = 0.996, \quad \frac{1}{10000} \sum_{i=1}^{10000} (\bar{y}_i - 1)^2 = 0.0473,$$

which are close to their respective population values of 1.0 and $10/200 = 0.05$. $\square$

If the true model is a VAR(1), the multivariate version of (13.30) is

$$\sqrt{T}(\bar{y} - [I - \Phi_1]^{-1} \mu) \xrightarrow{d} N\left(0, \sum_{j=-\infty}^{\infty} \Gamma_j\right), \quad (13.31)$$

where all terms are now of dimension N and Γ_j is the $(N \times N)$ autocovariance matrix at lag j defined in (13.16).

13.9 Testing

The LR, Wald and LM test statistics, presented in Chapter 4, can be used to test hypotheses on the parameters of ARMA and VARMA models. Many

of the statistics used to test hypotheses in stationary time series models are LM tests. This is because the models are relatively easy to estimate under the null hypothesis.

Example 13.22 Testing the Autocorrelation Coefficient

Consider testing for the independence of y_t using the AR(1) model

$$y_t = \phi_1 y_{t-1} + v_t, \quad v_t \sim iid N(0, \sigma_v^2),$$

which amounts to testing the restriction $\phi_1 = 0$. As demonstrated previously, the asymptotic distribution of $\hat{\phi}_1$ corresponds to the asymptotic distribution of the sample autocorrelation coefficient r_1 . Evaluating this distribution under the null gives $r_1 \sim N(0, T^{-1})$, suggesting that a LM test of independence is based on $\sqrt{T}r_1 \sim N(0, 1)$. $\square$

The general form of the LM test statistic is

$$LM = TG'_T(\hat{\theta}_0)J_T^{-1}(\hat{\theta}_0)G_T(\hat{\theta}_0), \quad (13.32)$$

where $G_T(\hat{\theta}_0)$ is the gradient and $J_T(\hat{\theta}_0)$ is the sample outer product of the gradients, both evaluated at the restricted estimator $\hat{\theta}_0$. From Chapter 4, the Lagrange multiplier statistic in the case of normality becomes

$$\begin{aligned} LM &= T \left[\frac{1}{\sigma_v^2 T} \sum_{t=1}^T z_t v_t \right]' \left[\frac{1}{\sigma_v^2 T} \sum_{t=1}^T z_t z_t' \right]^{-1} \left[\frac{1}{\sigma_v^2 T} \sum_{t=1}^T z_t v_t \right] \\ &= \frac{1}{\sigma_v^2} \left[\sum_{t=1}^T z_t v_t \right]' \left[\sum_{t=1}^T z_t z_t' \right]^{-1} \left[\sum_{t=1}^T z_t v_t \right] \\ &= TR^2, \end{aligned} \quad (13.33)$$

where R^2 is the coefficient of determination from a regression of v_t on $z_t = -\partial v_t / \partial \theta$ with all terms evaluated under the null hypothesis. The LM statistic is distributed asymptotically under the null hypothesis as χ_K^2 where K is the number of restrictions.

Example 13.23 LM Test of a MA(1)

Consider testing the ARMA(1, 1) model

$$y_t = \phi_1 y_{t-1} + v_t + \psi_1 v_{t-1},$$

for no moving average term, $\psi_1 = 0$. Write the model as

$$v_t = y_t - \phi_1 y_{t-1} - \psi_1 v_{t-1},$$

and construct the derivatives

$$z_{1,t} = -\frac{\partial v_t}{\partial \phi_1} = y_{t-1} - \psi_1 z_{1,t-1}$$

$$z_{2,t} = -\frac{\partial v_t}{\partial \psi_1} = -\psi_1 z_{2,t-1} + v_{t-1}.$$

Evaluating these expressions under the null hypothesis at $\psi_1 = 0$ gives

$$v_t = y_t - \phi_1 y_{t-1}$$

$$z_{1,t} = y_{t-1}$$

$$z_{2,t} = v_{t-1}.$$

To perform the LM test the following steps are required.

- Step 1:** Estimate the restricted model by regressing y_t on y_{t-1} and get the restricted residuals $\hat{v}_t$.
- Step 2:** Estimate the second-stage regression by regressing $\hat{v}_t$ on $\{y_{t-1}, \hat{v}_{t-1}\}$.
- Step 3:** Compute the test statistic $LM = TR^2$, where T is the sample size and R^2 is the coefficient of determination from the second-stage regression. Under the null hypothesis, LM is asymptotically distributed as χ_1^2 .

□

13.10 Analyzing Vector Autoregressions

This section focuses on some of the tools used to analyze the dynamic interrelationships between the variables of VARs. As outlined previously, a VAR(p) model is specified as

$$y_t = \mu + \Phi_1 y_{t-1} + \cdots + \Phi_p y_{t-p} + v_t, \quad v_t \sim N(0, V), \quad (13.34)$$

with all of the dynamics captured by the lagged variables and no additional dynamics arising from any moving average terms. In specifying a VAR, the key determinants are the number of variables, N , and the number of lags, p . It is this simplicity of the specification of a VAR that is an important reason for its popularity in modelling economic systems.

Example 13.24 VAR Estimates of a U.S. Macroeconometric Model

Consider the following VAR(2) model of the U.S. consisting of an interest rate, money growth, inflation and output growth, $N = 4$. Define $y_t = [r_t, \Delta_{12}lm_t, \Delta_{12}lp_t, \Delta_{12}lo_t]'$ so that the VAR is

$$y_t = \mu + \Phi_1 y_{t-1} + \Phi_2 y_{t-2} + v_t.$$

The maximum likelihood estimates are

$$\begin{aligned}\hat{\Phi}_1 &= \begin{bmatrix} 1.31 & 0.29 & 0.12 & 0.04 \\ -0.21 & 1.25 & -0.24 & 0.04 \\ 0.07 & 0.03 & 1.16 & 0.01 \\ 0.08 & 0.27 & -0.07 & 1.25 \end{bmatrix}, \\ \hat{\Phi}_2 &= \begin{bmatrix} -0.35 & -0.28 & -0.07 & -0.02 \\ 0.19 & -0.26 & 0.24 & -0.05 \\ -0.07 & -0.02 & -0.16 & 0.01 \\ -0.13 & -0.23 & 0.03 & -0.31 \end{bmatrix}, \\ \hat{\mu} &= \begin{bmatrix} -0.04 \\ 0.22 \\ -0.11 \\ 0.49 \end{bmatrix}, \quad \hat{V} = \begin{bmatrix} 0.29 & -0.02 & 0.01 & 0.11 \\ -0.02 & 0.38 & -0.01 & -0.05 \\ 0.01 & -0.01 & 0.10 & 0.00 \\ 0.11 & -0.05 & 0.00 & 1.27 \end{bmatrix}.\end{aligned}$$

The log-likelihood function evaluated at $\hat{\theta}$ is $\ln L_T(\hat{\theta}) = -3.471$. $\square$

The VAR represents the reduced form of a dynamic structural model of the type discussed in Chapter 14. To highlight this relationship consider the following structural model

$$B_0 y_t = B_1 y_{t-1} + u_t, \quad (13.35)$$

where B_0 and B_1 are matrices of parameters. The $(N \times 1)$ structural disturbance vector u_t has the properties

$$E[u_t] = 0, \quad E[u_t u_t'] = D, \quad (13.36)$$

where D is a diagonal matrix containing the variances on the main diagonal. The fact that D is a diagonal matrix means that the disturbances are independent, a restriction implied by the structural model. Expressing this system in terms of its reduced form gives

$$y_t = B_0^{-1} B_1 y_{t-1} + B_0^{-1} u_t, \quad = \Phi_1 y_{t-1} + v_t, \quad (13.37)$$

which is a VAR(1) where the parameters of the VAR and the structural model are related by

$$\Phi_1 = B_0^{-1} B_1, \quad (13.38)$$

while the disturbance vector of the VAR is related to the structural disturbances by

$$v_t = B_0^{-1} u_t. \quad (13.39)$$

One of the costs of a VAR is that it generates a large number of parameters, even for relatively small models. For example, a VAR with $N = 4$ variables and $p = 2$ lags, yields $N^2 \times p = 32$ autoregressive parameters and $N = 4$ constants. The fact that VARs produce large numbers of parameters creates difficulties in understanding the dynamic interrelationships amongst the variables in the system. There are three common methods used in empirical work to circumvent this problem: namely, Granger causality, impulse response analysis and variance decomposition.

13.10.1 Granger Causality Testing

A natural approach to understanding the dynamic structure of a VAR is to determine if a particular variable is explained by the lags of any other variable in the VAR other than its own lags. This suggests testing whether the parameters on blocks of lags are jointly zero. As a VAR is characterized by each variable being expressed in terms of lags of all variables in the system, this test is equivalent to identifying which variables in the VAR are important in predicting any particular dependent variable. This test is commonly referred to as the Granger causality test.

Reconsider the VAR(2) macroeconomic model of the U.S., estimated in Example 13.24, containing the nominal interest rate, money growth, inflation and output growth. The Granger causality tests for the interest rate equation consist of the following hypotheses:

$$H_0 : \phi_{1,1,2} = \phi_{2,1,2} = 0 \quad (\text{money} \nrightarrow \text{interest})$$

$$H_1 : \text{at least one restriction fails} \quad (\text{money} \rightarrow \text{interest})$$

$$H_0 : \phi_{1,1,3} = \phi_{2,1,3} = 0 \quad (\text{price} \nrightarrow \text{interest})$$

$$H_1 : \text{at least one restriction fails} \quad (\text{price} \rightarrow \text{interest})$$

$$H_0 : \phi_{1,1,4} = \phi_{2,1,4} = 0 \quad (\text{output} \nrightarrow \text{interest})$$

$$H_1 : \text{at least one restriction fails} \quad (\text{output} \rightarrow \text{interest})$$

Under the null hypothesis, money growth, inflation and output growth fail to Granger cause (denoted $\nrightarrow$) the interest rate, while under the alternative hypothesis there is a causal link (denoted $\rightarrow$). A similar set of hypotheses applies to the other three variables in the system resulting in a total of 12 Granger causality tests that can be performed for this model.

To implement the Granger causality test, it is convenient to use the Wald statistic from Chapter 4. The Wald test just uses the unrestricted maximum likelihood estimates of the VAR, which are simply computed by ap-

plying ordinary least squares to each equation. The Wald statistic is

$$W = T[R\hat{\theta}_1 - Q]'[R\hat{\Omega}R']^{-1}[R\hat{\theta}_1 - Q], \quad (13.40)$$

where $\hat{\theta}_1$ is the vector of unrestricted parameter estimates, $\hat{\Omega}$ is the asymptotic covariance matrix of $\hat{\theta}_1$ and R and Q are matrices based on the restrictions. Under the null hypothesis, the Wald statistic is distributed asymptotically as χ^2 where the degrees of freedom equals the number of zero restrictions being tested.

Example 13.25 Granger Causality Properties of the U.S. VAR

In the macroeconometric model of the U.S., θ is (36×1) with the parameters ordered as

$$\theta = [\mu_1, \phi_{1,1,1}, \phi_{1,1,2}, \phi_{1,1,3}, \phi_{1,1,4}, \phi_{2,1,1}, \phi_{2,1,2}, \phi_{2,1,3}, \phi_{2,1,4}, \mu_2, \dots]'$$

To test the hypothesis that money fails to Granger cause the interest rate (r), R is (2×36) and Q is (2×1)

$$R = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & \cdots & 0 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The other Granger causality tests for this model are obtained by simply changing R . All the results use the parameter estimates from Example 13.24.

Table 13.3

Granger causality tests of the VAR(2) model of the U.S. economy containing the interest rate, r_t , and annual percentage changes in the money stock, $\Delta_{12}lm_t$, prices, $\Delta_{12}lp_t$, and output, $\Delta_{12}lo_t$, $N = 4$. Results are presented for the interest rate and output equations.

Null hypothesis			Wald statistic	DOF	p-value
money	$\nrightarrow$	interest rate	57.366	2	0.000
price	$\nrightarrow$	interest rate	15.839	2	0.000
output	$\nrightarrow$	interest rate	12.602	2	0.002
interest rate	$\nrightarrow$	output	5.036	2	0.081
money	$\nrightarrow$	output	16.958	2	0.000
price	$\nrightarrow$	output	1.927	2	0.382

The Granger causality tests of the interest rate equation are given in the first block of Table 13.3. All restrictions are statistically significant showing that money growth, inflation and output growth all Granger cause the interest rate. The second block gives Granger causality tests for output. There

is Granger causality from money to output, while the causal link from the interest rate to output is significant, but only at the 10% level. The results show that price fails to Granger cause output with a p-value of 0.382. $\square$

13.10.2 Impulse Response Functions

The Granger causality test of the previous section focusses on the lags of the VAR. An alternative approach is to transform the VAR into a vector moving average and identify the dynamic properties of the VAR from the parameters of the VMA. Consider the VAR(p) in (13.34). Assuming stationarity, the VMA is infinite dimensional

$$y_t = v_t + \Psi_1 v_{t-1} + \Psi_2 v_{t-2} + \Psi_3 v_{t-3} + \cdots \quad (13.41)$$

Taking conditional expectations of (13.41) based on information at time $t-1$ and subtracting from (13.41) gives the one-step-ahead forecast error

$$y_t - E_{t-1}[y_t] = y_t - (\Psi_1 v_{t-1} + \Psi_2 v_{t-2} + \Psi_3 v_{t-3} + \cdots) = v_t.$$

Writing the model at $t+1$

$$y_{t+1} = v_{t+1} + \Psi_1 v_t + \Psi_2 v_{t-1} + \Psi_3 v_{t-2} + \cdots \quad (13.42)$$

reveals that v_t also affects y_{t+1} through Ψ_1 . By deduction, v_t affects y_{t+h} by Ψ_h , which suggests that a natural way to understand the dynamics of a VAR is to analyze the VMA parameters Ψ_h .

Presenting the impulse responses in terms of v_t in (13.42) is problematic because V is, in general, a non-diagonal matrix and therefore the disturbances are correlated with each other. This implies that a change in any component of v_t represents a weighted average of the same set of structural shocks in the model, with only the weights differing. To circumvent this problem, v_t in (13.42) is replaced by the structural shocks, u_t , in (13.39)

$$y_t = B_0^{-1} u_t + \Psi_1 B_0^{-1} u_{t-1} + \Psi_2 B_0^{-1} u_{t-2} + \Psi_3 B_0^{-1} u_{t-3} + \cdots \quad (13.43)$$

But this is just shifting the problem, since the structural parameters in B_0 now need to be identified. The most common way to solve this problem dates back to Sims (1980) where B_0 is specified as a triangular matrix, so the contemporaneous relationships amongst the variables in y_t is recursive. This is the implicit assumption in performing VAR analysis and reporting the results of impulse responses. A more general identification approach that does not rely on a strict recursive structure is given in Chapter 14.

In the case of VAR(2) model estimated in Example 13.24, for example,

the structural model (not the VAR) is specified as the following recursive model

$$\begin{aligned}
 y_{1t} &= & + \{lags\} & + u_{1,t} \\
 y_{2,t} &= \beta_{2,1}y_{1,t} & + \{lags\} & + u_{2,t} \\
 y_{3,t} &= \beta_{3,1}y_{1,t} + \beta_{3,2}y_{2,t} & + \{lags\} & + u_{3,t} \\
 y_{4,t} &= \beta_{4,1}y_{1,t} + \beta_{4,2}y_{2,t} + \beta_{4,3}y_{3,t} & + \{lags\} & + u_{4,t},
 \end{aligned} \tag{13.44}$$

where B_0 in (13.35) and D in (13.36) are, respectively,

$$B_0 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -\beta_{2,1} & 1 & 0 & 0 \\ -\beta_{3,1} & -\beta_{3,2} & 1 & 0 \\ -\beta_{4,1} & -\beta_{4,2} & -\beta_{4,3} & 1 \end{bmatrix}, \quad D = \text{diag} \begin{pmatrix} d_{1,1} \\ d_{2,2} \\ d_{3,3} \\ d_{4,4} \end{pmatrix}. \tag{13.45}$$

A natural way to compute B_0 and D is to estimate the full structural model in (13.44) by maximum likelihood and extract estimates of B_0 and D , which are substituted into (13.43) to obtain the impulse responses. In fact, given the recursive structure of the specified model, there are four possible ways to estimate the structural model which are numerically equivalent.

Method 1: Structural equation approach

Each structural equation in (13.44) is estimated by ordinary least squares

$$\begin{aligned}
 y_{1,t} &= -0.036 \\
 &\quad + 1.309y_{1,t-1} + 0.287y_{2,t-1} + 0.043y_{3,t-1} + 0.044y_{4,t-1} \\
 &\quad - 0.353y_{1,t-2} - 0.280y_{2,t-2} - 0.071y_{3,t-2} - 0.023y_{4,t-2} + \hat{u}_{1,t}, \\
 y_{2,t} &= 0.219 - 0.053y_{1,t} \\
 &\quad - 0.137y_{1,t-1} + 1.263y_{2,t-1} - 0.235y_{3,t-1} + 0.040y_{4,t-1} \\
 &\quad + 0.172y_{1,t-2} - 0.278y_{2,t-2} + 0.241y_{3,t-2} - 0.053y_{4,t-2} + \hat{u}_{2,t}, \\
 y_{3,t} &= -0.109 + 0.027y_{1,t} - 0.013y_{2,t} \\
 &\quad + 0.032y_{1,t-1} + 0.035y_{2,t-1} + 1.149y_{3,t-1} + 0.007y_{4,t-1} \\
 &\quad - 0.054y_{1,t-2} - 0.016y_{2,t-2} - 0.153y_{3,t-2} + 0.010y_{4,t-2} + \hat{u}_{3,t}, \\
 y_{4,t} &= 0.519 + 0.387y_{1,t} - 0.112y_{2,t} - 0.042y_{3,t} \\
 &\quad - 0.444y_{1,t-1} + 0.296y_{2,t-1} - 0.091y_{3,t-1} + 1.239y_{4,t-1} \\
 &\quad + 0.026y_{1,t-2} - 0.152y_{2,t-2} + 0.076y_{3,t-2} - 0.308y_{4,t-2} + \hat{u}_{4,t},
 \end{aligned}$$

yielding the matrices

$$\hat{B}_0 = \begin{bmatrix} 1.000 & 0.000 & 0.000 & 0.000 \\ 0.053 & 1.000 & 0.000 & 0.000 \\ -0.027 & 0.013 & 1.000 & 0.000 \\ -0.387 & 0.112 & 0.042 & 1.000 \end{bmatrix}, \quad \hat{D} = \text{diag} \begin{pmatrix} 0.285 \\ 0.370 \\ 0.097 \\ 1.192 \end{pmatrix},$$

where the i^{th} diagonal element of $\hat{D}$ is computed as $T^{-1} \sum_{t=1}^T \hat{u}_{i,t}^2$.

Method 2: Reduced form approach

Each equation of the VAR (reduced form) is estimated by ordinary least squares to yield estimates of $\hat{v}_t$. Now rewrite (13.39) as $B_0 v_t = u_t$, which corresponds to a recursive structure in the VAR residuals. This suggests that this system of equations can also be estimated by ordinary least squares with $v_t = [v_{1,t}, v_{2,t}, v_{3,t}, v_{4,t}]$ replaced by $\hat{v}_t$, the VAR residuals from above. The results are

$$\begin{aligned} \hat{v}_{1,t} &= \hat{u}_{1,t} \\ \hat{v}_{2,t} &= -0.053 \hat{v}_{1,t} + \hat{u}_{2,t} \\ \hat{v}_{3,t} &= 0.027 \hat{v}_{1,t} - 0.013 \hat{v}_{2,t} + \hat{u}_{3,t} \\ \hat{v}_{4,t} &= 0.387 \hat{v}_{1,t} - 0.112 \hat{v}_{2,t} - 0.042 \hat{v}_{3,t} + \hat{u}_{4,t}, \end{aligned}$$

where the $\hat{u}_{i,t}$ s are the residuals from the estimated equations, which also correspond to the structural residuals. Notice that the parameter estimates of this system, apart from a change in sign, do indeed correspond to the $\hat{B}_0$ matrix using Method 1. The estimates of the structural variances are then computed from these residuals to compute $\hat{D}$.

Method 3: Choleski decomposition approach

The first stage is as in Method 2 where each equation of the VAR is estimated by ordinary least squares and the VAR residuals are used to compute $\hat{V}$. At the second stage, a Choleski decomposition of $\hat{V}$ is performed by defining

$$\hat{V} = \hat{S}\hat{S}'.$$

Performing this decomposition yields

$$S = \begin{bmatrix} 0.534 & 0.000 & 0.000 & 0.000 \\ -0.028 & 0.608 & 0.000 & 0.000 \\ 0.015 & -0.008 & 0.311 & 0.000 \\ 0.209 & -0.068 & -0.013 & 1.092 \end{bmatrix}.$$

To recover B_0 from S , define the structural variances as the square of the

diagonal elements of S

$$\hat{D} = \text{diag} \begin{pmatrix} 0.534^2 \\ 0.608^2 \\ 0.311^2 \\ 1.092^2 \end{pmatrix} = \text{diag} \begin{pmatrix} 0.285 \\ 0.370 \\ 0.097 \\ 1.192 \end{pmatrix},$$

which agrees with $\hat{D}$ given by Method 1. Now from (13.39)

$$V = E[v_t v_t'] = E[B_0^{-1} u_t u_t' B_0^{-1}] = B_0^{-1} E[u_t u_t'] B_0^{-1} = B_0^{-1} D B_0^{-1},$$

which implies that $S = B_0^{-1} D^{1/2}$ and hence

$$B_0 = (S D^{-1/2})^{-1}.$$

Dividing each column by the respective diagonal element and taking the inverse gives

$$\begin{aligned} \hat{B}_0 &= \begin{bmatrix} 0.534/0.534 & 0.000/0.608 & 0.000/0.311 & 0.000/1.092 \\ -0.028/0.534 & 0.608/0.608 & 0.000/0.311 & 0.000/1.092 \\ 0.015/0.534 & -0.008/0.608 & 0.311/0.311 & 0.000/1.092 \\ 0.209/0.534 & -0.068/0.608 & -0.013/0.311 & 1.092/1.092 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} 1.000 & 0.000 & 0.000 & 0.000 \\ -0.053 & 1.000 & 0.000 & 0.000 \\ 0.028 & -0.013 & 1.000 & 0.000 \\ 0.391 & -0.112 & -0.042 & 1.000 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} 1.000 & 0.000 & 0.000 & 0.000 \\ 0.053 & 1.000 & 0.000 & 0.000 \\ -0.027 & 0.013 & 1.000 & 0.000 \\ -0.386 & 0.112 & 0.042 & 1.000 \end{bmatrix}, \end{aligned}$$

which is the same $\hat{B}_0$ matrix as obtained using Method 1.

Method 4: Maximum likelihood approach

The VAR is estimated by ordinary least squares, as in Methods 2 to 3, to yield the VAR residuals $\hat{v}_t$. The log-likelihood function given by

$$\ln L_T(\theta) = -\frac{N}{2} \ln 2\pi - \frac{1}{2} \ln |V| - \frac{1}{2(T-2)} \sum_{t=3}^T \hat{v}_t' V^{-1} \hat{v}_t,$$

which is maximized with respect to $\theta = \{B_0, D\}$ subject to the restriction

$V = B_0^{-1}DB_0^{-1}$, with

$$B_0 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -b_{2,1} & 1 & 0 & 0 \\ -b_{3,1} & -b_{3,2} & 1 & 0 \\ -b_{4,1} & -b_{4,2} & -b_{4,3} & 1 \end{bmatrix}, \quad D = \begin{bmatrix} d_{1,1} & 0 & 0 & 0 \\ 0 & d_{2,2} & 0 & 0 \\ 0 & 0 & d_{3,3} & 0 \\ 0 & 0 & 0 & d_{4,4} \end{bmatrix}.$$

The maximum likelihood estimates with standard errors based on the Hessian in parentheses, are given by

$$\hat{B}_0 = \begin{bmatrix} 1.000 & 0.000 & 0.000 & 0.000 \\ (-) & (-) & (-) & (-) \\ 0.053 & 1.000 & 0.000 & 0.000 \\ (0.053) & (-) & (-) & (-) \\ -0.027 & 0.013 & 1.000 & 0.000 \\ (0.027) & (0.024) & (-) & (-) \\ -0.386 & 0.112 & 0.042 & 1.000 \\ (0.095) & (0.083) & (0.165) & (-) \end{bmatrix}, \quad \hat{D} = \text{diag} \begin{pmatrix} 0.285 \\ (0.019) \\ 0.370 \\ (0.024) \\ 0.097 \\ (0.006) \\ 1.192 \\ (0.078) \end{pmatrix},$$

where the value of the log-likelihood function is $\ln L_T(\hat{\theta}) = -3.471$. These estimates are the same as the estimates obtained using the other three methods.

The last method, in particular, demonstrates that for recursive VAR models, asymptotically efficient parameter estimates can be obtained using a sequence of ordinary least squares regressions. It also demonstrates that there is more than one way to do this depending on whether the structural form or the reduced form is estimated. Additionally, the results also show that, for a recursive model, a Choleski decomposition can also be performed using a sequence of ordinary least squares regressions and that these estimates are also the maximum likelihood estimates. An important advantage of the maximum likelihood approach given by Method 4 is that it yields standard errors thereby facilitating hypothesis testing. By contrast, the Choleski method only delivers point estimates.

The structural shocks u_t can also be expressed in terms of one-standard deviation impulse responses by defining the standardized random variable

$$z_t = D^{-1/2}u_t, \quad (13.46)$$

where $D^{1/2}$ is given above and represents a diagonal matrix with the structural standard deviations down the main diagonal. As the standardized random variable z_t has covariance matrix I_N , it is referred to as the orthogonalized shock. The impulse responses of the one-standard deviation orthogonalized impulse response functions are obtained by using (13.46) to substitute out u_t in (13.43)

$$\begin{aligned}
y_t &= B_0^{-1}D^{1/2}z_t + \Psi_1 B_0^{-1}D^{1/2}z_{t-1} + \Psi_2 B_0^{-1}D^{1/2}z_{t-2} + \cdots \\
&= Sz_t + \Psi_1 Sz_{t-1} + \Psi_2 Sz_{t-2} + \Psi_3 Sz_{t-3} + \cdots,
\end{aligned} \tag{13.47}$$

where, as before, $S = B_0^{-1}D^{1/2}$. The one-standard deviation orthogonalized impulse responses are then

$$IR_h = \frac{\partial y_{t+h-1}}{\partial z'_t} = \Psi_{h-1}S, \quad h = 1, 2, \dots \tag{13.48}$$

Example 13.26 Impulse Responses of the U.S. VAR

Consider again the VAR(2) estimated using the U.S. macroeconomic data in Example 13.24 with $N = 4$. At $h = 1$, the one-standard deviation orthogonalized impulse responses are

$$\begin{aligned}
IR_1 &= \Psi_0 S \\
&= \begin{bmatrix} 1.00 & 0.00 & 0.00 & 0.00 \\ 0.00 & 1.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & 1.00 & 0.00 \\ 0.00 & 0.00 & 0.00 & 1.00 \end{bmatrix} \begin{bmatrix} 0.53 & 0.00 & 0.00 & 0.00 \\ -0.03 & 0.61 & 0.00 & 0.00 \\ 0.02 & -0.01 & 0.31 & 0.00 \\ 0.21 & -0.07 & -0.01 & 1.09 \end{bmatrix} \\
&= \begin{bmatrix} 0.53 & 0.00 & 0.00 & 0.00 \\ -0.03 & 0.61 & 0.00 & 0.00 \\ 0.02 & -0.01 & 0.31 & 0.00 \\ 0.21 & -0.07 & -0.01 & 1.09 \end{bmatrix}.
\end{aligned}$$

At $h = 2$ the impulse responses are

$$\begin{aligned}
IR_2 &= \Psi_1 S \\
&= \begin{bmatrix} 1.31 & 0.29 & 0.12 & 0.04 \\ -0.21 & 1.25 & -0.24 & 0.04 \\ 0.07 & 0.03 & 1.16 & 0.01 \\ 0.08 & 0.27 & -0.07 & 1.25 \end{bmatrix} \begin{bmatrix} 0.53 & 0.00 & 0.00 & 0.00 \\ -0.03 & 0.61 & 0.00 & 0.00 \\ 0.02 & -0.01 & 0.31 & 0.00 \\ 0.21 & -0.07 & -0.01 & 1.09 \end{bmatrix} \\
&= \begin{bmatrix} 0.70 & 0.17 & 0.04 & 0.05 \\ -0.14 & 0.76 & -0.08 & 0.04 \\ 0.06 & 0.01 & 0.36 & 0.01 \\ 0.30 & 0.08 & -0.04 & 1.37 \end{bmatrix}.
\end{aligned}$$

The impulse responses for longer lags are computed as $IR_h = \Psi_{h-1}S$. For this example, there are $N^2 = 4^2 = 16$ impulse response functions for each h , with the effects of an interest rate shock of the $N = 4$ variables given in the first column, the effects of a money shock in the second column etc. $\square$

13.10.3 Variance Decompositions

The one-standard deviation impulse response function in (13.47) gives the conditional mean of the distribution of y_t . The variance decomposition is concerned with the conditional variance of the impulse responses. Write the VAR in equation (13.47) at time $t + h$ as

$$y_{t+h} = Sz_{t+h} + \Psi_1 Sz_{t+h-1} + \Psi_2 Sz_{t+h-2} + \Psi_3 Sz_{t+h-3} + \cdots .$$

From the properties of z_t , the conditional mean is

$$\begin{aligned} E_t[y_{t+h}] &= E_t[Sz_{t+h} + \Psi_1 Sz_{t+h-1} + \Psi_2 Sz_{t+h-2} + \Psi_3 Sz_{t+h-3} + \cdots] \\ &= \Psi_h Sz_t + \Psi_{h+1} Sz_{t-1} + \cdots . \end{aligned}$$

The conditional variance is decomposed as

$$\begin{aligned} VD_h &= E_t[(y_{t+h} - E_t[y_{t+h}])(y_{t+h} - E_t[y_{t+h}])'] \\ &= E_t[(Sz_{t+h} + \cdots + \Psi_{h-1} Sz_{t+1})(Sz_{t+h} + \cdots + \Psi_{h-1} Sz_{t+1})'] \\ &= SE_t[z_{t+h} z_{t+h}'] S' + \Psi_1 SE_t[z_{t+h-1} z_{t+h-1}'] S' \Psi_1' + \cdots \\ &\quad + \Psi_{h-1} SE_t[z_{t+1} z_{t+1}'] S' \Psi_{h-1}' \\ &= SS' + \Psi_1 SS' \Psi_1' + \Psi_2 SS' \Psi_2' + \cdots + \Psi_{h-1} SS' \Psi_{h-1}' \\ &= \sum_{i=0}^{h-1} \Psi_i SS' \Psi_i', \quad h = 1, 2, \dots, \end{aligned} \tag{13.49}$$

with $\Psi_0 = I$. The last step follows from the fact that z_t is orthogonalized, which means that there are no cross products when conditional expectations are taken and $E[z_t z_t'] = I$ by definition.

Alternatively, (13.49) is written in terms of the separate contributions of each of the N shocks in the system to the h -step ahead variances

$$VD_h = \sum_{i=0}^{h-1} \Psi_i S_1 S_1' \Psi_i' + \sum_{i=0}^{h-1} \Psi_i S_2 S_2' \Psi_i' + \cdots + \sum_{i=0}^{h-1} \Psi_i S_N S_N' \Psi_i', \quad h = 1, 2, \dots,$$

where S_i is the i^{th} column of S . An even simpler expression of the variance decomposition is just

$$VD_h = \sum_{i=0}^{h-1} \begin{bmatrix} \Psi_i S \end{bmatrix} \odot \begin{bmatrix} \Psi_i S \end{bmatrix} = \sum_{i=1}^h IR_i \odot IR_i,$$

where $\odot$ is the Hadamard product, which is computed by squaring each element of the matrix IR_i defined in (13.48).

Example 13.27 Variance Decomposition of the U.S. VAR

Consider again the VAR(2) model estimated in Example 13.24 and used in for the illustration of impulse response analysis in Example 13.26. The computation of the first two variance decompositions requires

$$\begin{aligned} IR_1 \odot IR_1 &= \begin{bmatrix} 0.53^2 & 0.00^2 & 0.00^2 & 0.00^2 \\ -0.03^2 & 0.61^2 & 0.00^2 & 0.00^2 \\ 0.02^2 & -0.01^2 & 0.31^2 & 0.00^2 \\ 0.21^2 & -0.07^2 & -0.01^2 & 1.09^2 \end{bmatrix} \\ &= \begin{bmatrix} 0.29 & 0.00 & 0.00 & 0.00 \\ 0.00 & 0.37 & 0.00 & 0.00 \\ 0.00 & 0.00 & 0.10 & 0.00 \\ 0.04 & 0.01 & 0.00 & 1.19 \end{bmatrix}, \end{aligned}$$

and

$$\begin{aligned} IR_2 \odot IR_2 &= \begin{bmatrix} 0.70^2 & 0.17^2 & 0.04^2 & 0.05^2 \\ -0.14^2 & 0.76^2 & -0.08^2 & 0.04^2 \\ 0.06^2 & 0.01^2 & 0.36^2 & 0.01^2 \\ 0.30^2 & 0.08^2 & -0.04^2 & 1.37^2 \end{bmatrix} \\ &= \begin{bmatrix} 0.49 & 0.03 & 0.00 & 0.00 \\ 0.02 & 0.58 & 0.01 & 0.00 \\ 0.00 & 0.00 & 0.13 & 0.00 \\ 0.09 & 0.01 & 0.00 & 1.87 \end{bmatrix}, \end{aligned}$$

The variance decomposition at $h = 1$ is

$$\begin{aligned} VD_1 &= IR_1 \odot IR_1 \\ &= \begin{bmatrix} 0.29 & 0.00 & 0.00 & 0.00 \\ 0.00 & 0.37 & 0.00 & 0.00 \\ 0.00 & 0.00 & 0.10 & 0.00 \\ 0.04 & 0.01 & 0.00 & 1.19 \end{bmatrix}. \end{aligned}$$

The variance decomposition at $h = 2$ is

$$\begin{aligned} VD_2 &= IR_1 \odot IR_1 + IR_2 \odot IR_2 \\ &= \begin{bmatrix} 0.29 & 0.00 & 0.00 & 0.00 \\ 0.00 & 0.37 & 0.00 & 0.00 \\ 0.00 & 0.00 & 0.10 & 0.00 \\ 0.04 & 0.01 & 0.00 & 1.19 \end{bmatrix} + \begin{bmatrix} 0.49 & 0.03 & 0.00 & 0.00 \\ 0.02 & 0.58 & 0.01 & 0.00 \\ 0.00 & 0.00 & 0.13 & 0.00 \\ 0.09 & 0.01 & 0.00 & 1.87 \end{bmatrix} \\ &= \begin{bmatrix} 0.78 & 0.03 & 0.00 & 0.00 \\ 0.02 & 0.95 & 0.01 & 0.00 \\ 0.00 & 0.00 & 0.23 & 0.00 \\ 0.13 & 0.01 & 0.00 & 3.06 \end{bmatrix}. \end{aligned}$$

The total variance of each variable is given by the rows sums of each VD , with the elements representing the contribution of each of the $N = 4$ variables to the total variance. For example, the total variance of output at $h = 2$ is given by the last column of VD_2

$$\text{var}(\text{output}) = 0.13 + 0.01 + 0.00 + 3.06 = 3.20,$$

where 0.13 is the contribution of the interest rate, 0.01 the contribution of money, 0.00 the contribution of price and 3.06 the contribution of output. $\square$

13.11 Applications

This section provides examples of how economic theory can be used to impose cross-equation restrictions on VARs to reduce the number of unknown parameters and thus simplify the estimation.

13.11.1 Barro's Rational Expectations Model

Consider the following rational expectations model of money, m_t , and output, o_t ,

$$\Delta \ln m_t = \alpha_0 + \alpha_1 \Delta \ln m_{t-1} + \alpha_2 \Delta \ln m_{t-2} + v_{1,t} \quad (13.50)$$

$$\begin{aligned} \Delta \ln o_t = & \beta_0 + \beta_1 (\Delta \ln m_{t-1} - E_{t-2} [\Delta \ln m_{t-1}]) \\ & + \beta_2 (\Delta \ln m_{t-2} - E_{t-3} [\Delta \ln m_{t-2}]) + v_{2,t}. \end{aligned} \quad (13.51)$$

This is an abbreviated version of the model originally specified by Barro (1978). The first equation is an AR(2) model of nominal money, while the second equation shows that the growth rate in real output responds to lags in the unanticipated growth rate in nominal money, m_t .

Using (13.50) to form the conditional expectations of m_t , the unanticipated money shock is

$$v_{1,t} = \Delta \ln m_t - E_{t-1} [\Delta \ln m_t] = \Delta \ln m_t - \alpha_0 - \alpha_1 \Delta \ln m_{t-1} - \alpha_2 \Delta \ln m_{t-2}.$$

Using this expression in equation (13.51) gives

$$\begin{aligned} \Delta \ln o_t = & (\beta_0 - \beta_1 \alpha_0 - \beta_2 \alpha_0) + \beta_1 \Delta \ln m_{t-1} + (\beta_2 - \beta_1 \alpha_1) \Delta \ln m_{t-2} \\ & - (\beta_1 \alpha_2 + \beta_2 \alpha_1) \Delta \ln m_{t-3} - \beta_2 \alpha_2 \Delta \ln m_{t-4} + v_{2,t}, \\ = & \delta_0 + \delta_1 m_{t-1} + \delta_2 m_{t-2} + \delta_3 m_{t-3} + \delta_4 m_{t-4} + v_{2,t}. \end{aligned} \quad (13.52)$$

Equations (13.50) and (13.52) form a restricted bivariate VAR in the sense that lags in $\Delta \ln o_t$ are excluded from the model, the lags of $\Delta \ln m_t$ in

the two equations differ and, most important from a rational expectations perspective, there are two cross-equation restrictions given by

$$\delta_3 = -\delta_1\alpha_2 - (\delta_2 + \delta_1\alpha_1)\alpha_1, \quad \delta_4 = -(\delta_2 + \delta_1\alpha_1)\alpha_2.$$

The restricted VAR can be estimated by maximum likelihood methods using an iterative algorithm to yield consistent and asymptotically efficient parameters estimates. This contrasts with the earlier methods used to estimate (13.50) and (13.51), in which equation (13.50) would be estimated by ordinary least squares to get estimates of the α_i s and, in particular, the residuals $\hat{v}_{1,t}$. The output equation would then be estimated by simply regressing o_t on a constant and the lagged residuals $\hat{v}_{1,t-1}$ and $\hat{v}_{1,t-2}$. The problem with this method is that the standard errors are incorrect because the regressors based on lags of $\hat{v}_{1,t}$ do not recognize that these variables contain the estimated parameters $\hat{\alpha}_i$ s, which are estimated subject to error. The maximum likelihood method avoids this problem as the system is estimated jointly with the restrictions imposed.

A test of rational expectations is a test of the cross-equation restrictions. One approach is to estimate equations (13.50) and (13.52) jointly by maximum likelihood methods and use the nonlinear form of the Wald test to determine if the restrictions are consistent with the data. From Chapter 4, the nonlinear version of the Wald statistic is

$$W = T[C(\hat{\theta}_1) - Q]'[D(\hat{\theta}_1)(-H_T^{-1}(\hat{\theta}_1))D(\hat{\theta}_1)'] [C(\hat{\theta}_1) - Q],$$

where $\theta = \{\alpha_0, \alpha_1, \alpha_2, \delta_0, \delta_1, \delta_2, \delta_3, \delta_4\}$, and

$$C(\theta) = \begin{bmatrix} \delta_3 + \delta_1\alpha_2 + (\delta_2 + \delta_1\alpha_1)\alpha_1 \\ \delta_4 + (\delta_2 + \delta_1\alpha_1)\alpha_2 \end{bmatrix}, \quad Q = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

$$D(\hat{\theta}_1) = \frac{\partial C}{\partial \theta'} = \begin{bmatrix} 0 & 2\delta_1\alpha_1 & \delta_1 & 0 & \alpha_2 & \alpha_1 & 1 & 0 \\ 0 & \delta_1\alpha_2 & \delta_1\alpha_1 & 0 & \alpha_1\alpha_2 & \alpha_2 & 0 & 1 \end{bmatrix}.$$

Under the null hypothesis that the restrictions are valid, W is distributed asymptotically as χ^2 with two degrees of freedom. A failure to reject the null hypothesis is evidence in favour of rational expectations.

13.11.2 The Campbell-Shiller Present Value Model

The present value model relates the stock price, s_t , to the discounted stream of dividends, d_t . Campbell and Shiller (1987) show that the present value

of the stock price is given by

$$s_t = \alpha(1 - \delta)E_t \sum_{i=0}^{\infty} \delta^i d_{t+i}, \quad (13.53)$$

where δ is the discount factor and $\alpha = 27.342$ is a coefficient of proportionality which is estimated from the least squares equation

$$s_t = \beta + \alpha d_t + \eta_t \quad (13.54)$$

where β is an intercept and η_t is a disturbance term. Although α is estimated, which again raises the potential problem of a generated regressor, Part FIVE shows that this estimator is superconsistent under certain conditions because (13.54) represents what is known as a cointegrating equation. Campbell and Shiller's argument is that the superconsistency of α circumvents the generated-regressors problem. Let the dynamics of $y_t = [\Delta d_t, \eta_t]'$ be represented by a VAR(1)

$$\begin{bmatrix} \Delta d_t \\ \eta_t \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \phi_{1,1,1} & \phi_{1,1,2} \\ \phi_{1,2,1} & \phi_{1,2,2} \end{bmatrix} \begin{bmatrix} \Delta d_{t-1} \\ \eta_{t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix}. \quad (13.55)$$

Campbell and Shiller show that

$$\phi_{1,2,1} = -\alpha\phi_{1,1,1}, \quad \phi_{1,2,2} = \delta^{-1} - \alpha\phi_{1,1,2}. \quad (13.56)$$

Substituting these parameters into (13.55) gives the restricted VAR

$$\begin{bmatrix} \Delta d_t \\ \eta_t \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \phi_{1,1,1} & \phi_{1,1,2} \\ -\alpha\phi_{1,1,1} & \delta^{-1} - \alpha\phi_{1,1,2} \end{bmatrix} \begin{bmatrix} \Delta d_{t-1} \\ \eta_{t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix}. \quad (13.57)$$

There are five parameters to be estimated $\theta = \{\mu_1, \mu_2, \phi_{1,1,1}, \phi_{1,1,2}, \delta\}$, compared to the unrestricted VAR in equation (13.55) where there are six parameters. There is, thus, one cross-equation restriction imposed by theory.

Table 13.4 gives the maximum likelihood estimates of (13.57) using data on monthly real equity prices, s_t , and real dividends, d_t , from January 1933 to December 1990 for the U.S. The parameter estimate on the discount parameter is $\hat{\delta} = 0.806$, which implies a discount rate of 24.07% as

$$\hat{r} = \hat{\delta}^{-1} - 1 = \frac{1}{0.806} - 1 = 0.2407.$$

To test the cross-equation restriction, a LR statistic is used. The value of the unconstrained log-likelihood function obtained by estimating a bivariate VAR(1) is $\ln L_T(\hat{\theta}_1) = -4.100$. Using the value of the constrained log-likelihood function reported in Table 13.4 gives the LR statistic

$$LR = -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)) = -2(-694 \times 5.172 + 694 \times 4.100) = 1489.171.$$

Table 13.4

Maximum likelihood estimates of the present value model. Standard errors and t-statistics are based on quasi-maximum likelihood estimates allowing for heteroskedasticity.

Parameter	Estimate	Std error	t-stat.
μ_1	1.630	0.094	17.323
$\phi_{1,1,1}$	0.026	0.009	2.850
$\phi_{1,1,2}$	0.009	0.006	1.460
μ_2	1.177	0.395	2.982
δ	0.806	0.116	6.960
$\ln L_T(\hat{\theta}) = -5.172$			

The p-value is 0.000, which results in a clear rejection of the restriction at the 5% level.

13.12 Exercises

(1) Time Series Properties of U.S. Macroeconomic Variables

Gauss file(s)	stsm_properties.g, sims_data.dat
Matlab file(s)	stsm_properties.m, sims_data.mat

The data file contains monthly macroeconomic data for the U.S. for $y_t = [r_t, lm_t, lp_t, lo_t]'$, where r_t is the interest rate, lm_t is the logarithm of the money stock, lp_t is logarithm of prices and lo_t is logarithm of output.

- Plot the variables and interpret their time series properties.
- Repeat part (a) for the 1st differenced variables, Δy_t .
- Repeat part (a) for the 12th differenced variables, $\Delta_{12}y_t$.
- Compute the ACF and PACF for y_t , Δy_t and $\Delta_{12}y_t$ and interpret the results.

(2) Simulating ARMA Models

Gauss file(s)	stsm_simulate.g
Matlab file(s)	stsm_simulate.m

Consider the ARMA(2, 2) model

$$y_t = \mu + \phi_1 y_{t-1} + \phi_2 y_{t-2} + v_t + \psi_1 v_{t-1} + \psi_2 v_{t-2},$$

where $v_t \sim iid N(0, \sigma_v^2)$.

- (a) Simulate the following models for $T = 200$, with the following parameter values

Model	μ	ϕ_1	ϕ_2	ψ_1	ψ_2	σ_v^2
ARMA(0, 0) :	0.0	0.0	0.0	0.0	0.0	0.1
ARMA(1, 0) :	0.0	0.7	0.0	0.0	0.0	0.1
ARMA(2, 0) :	0.0	0.7	-0.5	0.0	0.0	0.1
ARMA(0, 1) :	0.0	0.0	0.0	0.9	0.0	0.1
ARMA(0, 2) :	0.0	0.0	0.0	-0.2	0.7	0.1
ARMA(1, 1) :	0.0	0.8	0.0	0.7	0.0	0.1

(Hint: You should simulate $T + 100$ observations and then exclude the first 100 observations to avoid the initialization problem).

- (b) Estimate the ACF for each model and interpret the results. Compare the estimated ACF with the theoretical ACF in (13.15).
 (c) Estimate the PACF for each model and interpret the results.

(3) Stationarity Properties of U.S. Macroeconomic Variables

Gauss file(s)	<code>stsm_roots.g</code> , <code>sims_data.dat</code>
Matlab file(s)	<code>stsm_roots.m</code> , <code>sims_data.mat</code>

The data file contains monthly macroeconomic data for the U.S. from January 1959 to December 1998.

- (a) Estimate an AR(2) model for the annual growth rate in output and determine if the process is stationary by computing the roots of the polynomial $1 - \phi_1 z - \phi_2 z^2 = 0$.
 (b) Estimate an AR(4) model for the interest rate and determine if the process is stationary by computing the roots of the polynomial $1 - \phi_1 z - \phi_2 z^2 - \phi_3 z^3 - \phi_4 z^4 = 0$.
 (c) Repeat part (a) but now for the level of logarithm of output.
 (d) Estimate a VAR(2) containing a constant, the annual growth rate of money and the annual inflation rate, and determine if the processes are stationary by computing the roots of the polynomial $|1 - \Phi_1 z - \Phi_2 z^2| = 0$.

- (e) Estimate a VAR(2) containing a constant, the interest rate, the annual growth rate of money, the annual inflation rate and the annual growth rate of output, and determine if the processes are stationary by computing the roots of the polynomial $|1 - \Phi_1 z - \Phi_2 z^2| = 0$.
- (f) Repeat part (e) for a VAR(4).

(4) **ARMA Models of U.S. Macroeconomic Variables**

Gauss file(s)	<code>stsm_arma.g</code> , <code>sims_data.dat</code>
Matlab file(s)	<code>stsm_arma.m</code> , <code>sims_data.mat</code>

The data file contains monthly macroeconomic data for the U.S. from January 1959 to December 1998.

- (a) Estimate an ARMA(1,1) model for the interest rate, r_t ,

$$y_t = \mu + \phi_1 y_{t-1} + v_t + \psi_1 v_{t-1}, \quad v_t \sim N(0, 1).$$

- (b) Perform Wald tests of the restrictions (i) $\phi_1 = 0$; (ii) $\psi_1 = 0$; (iii) $\phi_1 = \psi_1 = 0$.
- (c) Perform a LR test of the restrictions $\phi_1 = \psi_1 = 0$.
- (d) Repeat parts (a) to (c) for annual inflation, $100\Delta_{12}lp_t$.
- (e) Repeat parts (a) to (c) for the annual growth rate in output, $100\Delta_{12}lo_t$.

(5) **The Gauss-Newton Algorithm**

Gauss file(s)	<code>stsm_gaussn.g</code>
Matlab file(s)	<code>stsm_gaussn.m</code>

Consider the ARMA(1,1) model

$$y_t = \mu + \phi_1 y_{t-1} + v_t + \psi_1 v_{t-1}, \quad v_t \sim N(0, 1).$$

- (a) Show how the parameters of the model would be estimated using the Gauss-Newton algorithm. Show all of the steps.
- (b) Simulate the ARMA(1,1) model for $T = 200$ observations using the population parameters $\theta_0 = \{\mu = 0.0, \phi_1 = 0.8, \psi_1 = 0.3\}$.
- (c) Using the simulated data in part (b), estimate the model using the Gauss-Newton algorithm where the starting values are chosen as $\theta_{(0)} = \{\mu = 0.2, \phi_1 = 0.2, \psi_1 = 0.2\}$. In addition, estimate the asymptotic covariance matrix as well as the asymptotic t-statistics. Compare the maximum likelihood estimates $\hat{\theta}$ with θ_0 .
- (d) Perform a Wald test of the hypothesis $\psi_1 = 0$.
- (e) Perform a LM test of the hypothesis $\psi_1 = 0$. What advantage does the LM test have over the Wald test in this example?

- (f) Repeat part (c) with the starting values $\theta_{(0)} = \{\mu = 0.0, \phi_1 = 0.0, \psi_1 = 0.0\}$. Why does the algorithm fail in this case?
- (g) Repeat part (c) with the starting values $\theta_{(0)} = \{\mu = 0.0, \phi_1 = 0.3, \psi_1 = 1.1\}$. Estimate the model for a larger sample size of $T = 500$. Why does the algorithm fail for the larger sample of $T = 500$, but not for the smaller sample size of $T = 200$? Should not more data provide greater precision?
- (h) Repeat part (c) with the starting values $\theta_{(0)} = \{\mu = 0.0, \phi_1 = 1.1, \psi_1 = 0.3\}$. Comparing the results to part (g), why does the algorithm not fail for this model? In addition, suggest an alternative and perhaps better way of estimating this model still using the same algorithm.

(6) Properties of LM Tests of ARMA Models

Gauss file(s)	stsm_lm.g
Matlab file(s)	stsm_lm.m

Simulate $T = 200$ observations using the ARMA(1, 1) model of Exercise 5.

- (a) (*Equivalence Result*) Use the simulated data to perform the following LM regression-based tests.
 - (i) A test of $\phi_1 = 0$ in the AR(1) $y_t = \mu + \phi_1 y_{t-1} + v_t$.
 - (ii) A test of $\psi_1 = 0$ in the MA(1) model $y_t = \mu + v_t + \psi_1 v_{t-1}$. Hence show the equivalence of the two LM tests (Poskitt and Tremayne, 1980).
- (b) (*Singularity Result*) Use the simulated data to perform a LM regression-based test of $\phi_1 = \psi_1 = 0$ in the ARMA(1, 1) model $y_t = \mu + \phi_1 y_{t-1} + v_t + \psi_1 v_{t-1}$. Hence show that the test breaks down because of a singularity in the second-stage regression equation.

(7) Estimating and Testing VARMA Models

Gauss file(s)	stsm_varma.g
Matlab file(s)	stsm_varma.m

Simulate $T = 500$ observations from the VARMA(1, 1) model

$$y_{1,t} = \mu_1 + \phi_{1,1,1}y_{1,t-1} + v_{1,t} + \psi_{1,1,1}v_{1,t-1} + \psi_{1,1,2}v_{2,t-1}$$

$$y_{2,t} = \mu_2 + \phi_{1,2,2}y_{2,t-1} + v_{2,t} + \psi_{1,2,1}v_{1,t-1} + \psi_{1,2,2}v_{2,t-1},$$

where v_t is bivariate normal with zero mean and covariance matrix given

by the identity matrix, and the population parameters are given in Table 13.2.

- (a) Compute the maximum likelihood estimates using the Newton-Raphson algorithm.
- (b) Repeat part (a) using $T = 1000$ observations. Discuss the asymptotic properties of the maximum likelihood estimator.
- (c) Perform a Wald test of the hypothesis

$$H_0 : \psi_{1,1,1} = \psi_{1,1,2} = \psi_{1,2,1} = \psi_{1,2,2} = 0.$$

Choose $T = 500, 1000$ and discuss the consistency properties of the Wald statistic.

- (d) Perform a LR test of the multivariate moving average part

$$H_0 : \psi_{1,1,1} = \psi_{1,1,2} = \psi_{1,2,1} = \psi_{1,2,2} = 0.$$

Choose $T = 500, 1000$ and discuss the consistency properties of the LR statistic.

(8) Testing VARMA Models for Nonlinearities

Gauss file(s)	<code>stsm_varmab.g</code>
Matlab file(s)	<code>stsm_varmab.m</code>

Simulate $T = 500$ observations of the VARMA(1,1) model set out in Exercise 7, using the population parameters given in Table 13.2.

- (a) Use the simulated data to estimate the following nonlinear time series model

$$\begin{aligned}
 y_{1,t} &= \mu_1 + \phi_{1,1,1}y_{1,t-1} + v_{1,t} + \psi_{1,1,1}v_{1,t-1} + \psi_{1,1,2}v_{2,t-1} \\
 &\quad + \delta_1 y_{1,t-1}v_{1,t-1} \\
 y_{2,t} &= \mu_2 + \phi_{1,2,2}y_{2,t-1} + v_{2,t} + \psi_{1,2,1}v_{1,t-1} + \psi_{1,2,2}v_{2,t-1} \\
 &\quad + \delta_2 y_{2,t-1}v_{2,t-1}.
 \end{aligned}$$

The inclusion of the nonlinearities yields a multivariate bilinear model that is studied in greater detail in Chapter 19. Perform a test of multivariate bilinearity

$$H_0 : \delta_1 = \delta_2 = 0,$$

using a LR test and a Wald test.

- (b) Briefly discuss how you could make the multivariate test of bilinearity robust to nonnormality.

(9) **Finite Sample Distribution**

Gauss file(s)	<code>stsm_finite.g</code>
Matlab file(s)	<code>stsm_finite.m</code>

Simulate the AR(1) model $y_t = \phi_1 y_{t-1} + v_t$ where $v_t \sim N(0, 1)$ and $\phi_1 = \{0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 0.99\}$. Choose 20,000 replications and a sample size of $T = 50$.

- Compute the finite sample distribution of the maximum likelihood estimator of ϕ_1 and show that this estimator is biased downwards with the size of the bias approximately equal to $2\phi_1 T^{-1}$ (Shenton and Johnson, 1975).
- Show that the bias decreases as the sample size increases, by repeating part (a) for $T = 100, 500$.

(10) **Anderson's Central Limit Theorem**

Gauss file(s)	<code>stsm_anderson.g</code>
Matlab file(s)	<code>stsm_anderson.m</code>

- Simulate an AR(1) model 10,000 times for a sample of size $T = 200$, with parameters $\theta = \{\mu = 0.1, \Phi_1 = 0.9, \sigma_v^2 = 0.1\}$. Compute the summary statistics

$$\frac{1}{10000} \sum_{i=1}^{10000} \bar{y}_i, \quad \frac{1}{10000} \sum_{i=1}^{10000} \left(\bar{y}_i - \frac{\mu}{1 - \Phi_1} \right)^2,$$

and compare these values with the population values based on Anderson's central limit theorem.

- Repeat part (a) for $T = 500$ and discuss the asymptotic properties of the sample mean.
- Repeat parts (a) and (b) for $\theta = \{\mu = 0.1, \phi_1 = 0.95, \sigma_v^2 = 0.1\}$.
- Repeat parts (a) and (b) for $\theta = \{\mu = 0.0, \phi_1 = 1.1, \sigma_v^2 = 0.1\}$. Why does the simulation exercise fail in this case?

(11) **Granger Causality**

Gauss file(s)	<code>stsm_recursive.g, sims_data.dat</code>
Matlab file(s)	<code>stsm_recursive.m, sims_data.mat</code>

The data file contains monthly macroeconomic data for the U.S. from January 1959 to December 1998.

- (a) Estimate a VAR(2) containing a constant, the interest rate, the annual growth rate of money, the annual inflation rate and the annual growth of output.
- (b) Use a Wald test to construct Granger causality tests between the 4 variables.
- (c) Use a Wald test to construct a joint test that a variable fails to be Granger caused by all variables. That is, test the exogeneity of each of the 4 variables.

(12) Impulse Responses and Variance Decompositions

Gauss file(s)	<code>stsm_recursive.g, sims_data.dat</code>
Matlab file(s)	<code>stsm_recursive.m, sims_data.mat</code>

The data file contains monthly macroeconomic data for the U.S. from January 1959 to December 1998.

- (a) Estimate a VAR(2) containing a constant, the interest rate, the annual growth rate of money, the annual inflation rate and the annual growth rate of output.
- (b) Transform the VAR(2) into a VMA(5) and, hence, compute $\Psi_0, \Psi_1, \Psi_2, \Psi_3, \Psi_4$ and Ψ_5 .
- (c) Compute B_0 and D and hence $S = B_0^{-1}D^{1/2}$, using the structural equation approach, the reduced form approach, the Choleski decomposition approach and the maximum likelihood approach.
- (d) Compute the orthogonalized one-standard deviation impulse responses

$$IR_h = \Psi_{h-1}S, \quad h = 1, 2, \dots, 5,$$

and interpret the results.

- (e) Compute the variance decompositions based on the orthogonalized one-standard deviation impulse responses

$$VD_h = \sum_{i=1}^h IR_i \odot IR_i, \quad h = 1, 2, \dots, 5,$$

and interpret the results.

(13) Diebold-Yilmaz Spillover Index

Gauss file(s)	<code>stsm_spillover.g, diebold_yilmaz.xls</code>
Matlab file(s)	<code>stsm_spillover.m, diebold_yilmaz.xls</code>

Diebold and Yilmaz (2009) construct spillover indexes of international

real asset returns and volatility based on the variance decomposition of a VAR. The data file contains weekly data on real asset returns, *RETURNS*, and volatility *VOLATILITY* of 7 developed countries and 12 emerging countries from the first week of January 1992 to the fourth week of November 2007.

- (a) Compute descriptive statistics of the 19 real asset market returns in *RETURNS*. Compare the estimates to the results reported in Table 1 of Diebold and Yilmaz.
- (b) Estimate a VAR(2) containing a constant and the 19 real asset market returns.
- (c) Compute S using the Choleski decomposition of the residual covariance matrix of the VAR.
- (d) Compute the orthogonalized impulse response function for horizon 10.
- (e) Compute the variance decomposition for horizon 10, VD_{10} . Compare the estimates to the results reported in Table 3 of Diebold and Yilmaz.
- (f) Compute the 'Contribution from Others' by summing each row of VD_{10} excluding the diagonal elements. Interpret the results.
- (g) Compute the 'Contribution to Others' by summing each column of VD_{10} excluding the diagonal elements. Interpret the results.
- (h) Repeat parts (a) to (g) with the 19 series in *RETURNS* replaced by *VOLATILITY*, and the comparisons now based on Tables 2 and 4 in Diebold and Yilmaz.

(14) **Campbell-Shiller Present Value Model**

Gauss file(s)	<code>stsm_camshiller.g</code>
Matlab file(s)	<code>stsm_camshiller.m</code>

The data file contains monthly data on real equity prices, s_t , and real dividends, d_t , for the U.S. from January 1933 to December 1990. Consider the present value model of Campbell and Shiller (1987)

$$\begin{bmatrix} \Delta d_t \\ \eta_t \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \begin{bmatrix} \phi_{1,1,1} & \phi_{1,1,2} \\ -\alpha\phi_{1,1,1} & \delta^{-1} - \alpha\phi_{1,1,2} \end{bmatrix} \begin{bmatrix} \Delta d_{t-1} \\ \eta_{t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix}.$$

where η_t is the disturbance of the regression equation

$$s_t = \beta + \alpha d_t + \eta_t.$$

- (a) Estimate the parameter α by regressing s_t on a constant and d_t , and setting $\hat{\alpha}$ equal to the slope of this regression equation.

- (b) Given an estimate of α in part (a), compute the maximum likelihood estimates of $\theta = \{\mu_1, \mu_2, \phi_{1,1,1}, \phi_{1,1,2}, \delta\}$.
- (c) Perform a LR test of the restriction of the model and interpret the result.
- (d) Repeat parts (b) and (c) by increasing the number of lags to two.

14

Structural Vector Autoregressions

14.1 Introduction

The vector autoregression model (VAR) discussed in Chapter 13 provides a convenient framework for modelling dynamic systems of equations. Maximum likelihood estimation of the model is performed one equation at a time using ordinary least squares, while the dynamics of the system are analyzed using impulse response analysis and variance decompositions. Although the VAR framework is widely applied in econometrics, it requires the imposition of additional structure on the model in order to give the impulse responses and variance decompositions structural interpretations. For example, in macroeconomic applications, the key focus is often on understanding the effects of a monetary shock on the economy, but this requires the ability to identify precisely what the monetary shock is. In Chapter 13, a recursive structure known as a triangular ordering is adopted to identify shocks. This is a purely statistical approach to identification that imposes a very strict and rigid structure on the dynamics of the model that may not necessarily be consistent with the true structure of the underlying processes. This approach becomes even more problematic when alternative orderings of variables are tried, since the number of combinations of orderings increases dramatically as the number of variables in the model increases.

Structural vector autoregressive (SVAR) models alleviate the problems of imposing a strict recursive structure on the model by specifying restrictions that, in general, are motivated by economic theory. Four common sets of restrictions are used to identify SVARS: namely, short-run restrictions, long-run restrictions, a combination of the two and sign restrictions. Despite the additional acronyms associated with the SVAR literature and the fact that nature of the applications may seem different at first glance, SVARS

simply represent a subset of the class of linear simultaneous equations models discussed in Chapter 5.

14.2 Specification

A SVAR is a dynamic structural model of the type studied in Chapter 13. In its most general representation, all variables in the system are endogenous, as highlighted by the following model

$$B_0 y_t = \sum_{i=1}^p B_i y_{t-i} + u_t, \quad u_t \sim N(0, D), \quad (14.1)$$

where y_t is a $(N \times 1)$ vector of the endogenous variables and B_i is a $(N \times N)$ matrix containing the parameters on the i^{th} lag, with B_0 representing the contemporaneous interactions between the variables. The $(N \times 1)$ vector of disturbances, u_t , represents the structural shocks and has covariance matrix D , which is a diagonal matrix containing the variances. It is the fact that the covariances of u_t are all zero that gives u_t its structural interpretation, since each shock is, by definition, unique. As will be seen, the matrices B_0 and D are fundamental to the specification and estimation of SVAR models.

In macroeconomic applications of SVARs, the structural shocks, u_t , may consist of policy shocks, monetary or fiscal, shocks in the goods market arising from either aggregate demand or supply, nominal shocks and asset market shocks. The following example helps to clarify the relationship between y_t and u_t .

Example 14.1 A Prototype Macroeconomic Model

Consider the four-variable macroeconomic model consisting of the logarithm of output, the interest rate, the logarithm of prices and the logarithm of money: $y_t = [\ln o_t \ r_t \ \ln p_t \ \ln m_t]$. The model is

$$\begin{aligned} \ln \left(\frac{p_t}{p_{t-1}} \right) &= b_1 (\ln o_t - u_{as,t}) && \text{Aggregate supply} \\ \ln o_t &= -b_2 (r_t - \ln \left(\frac{p_t}{p_{t-1}} \right) - u_{is,t}) && \text{IS equation} \\ \ln \left(\frac{m_t}{p_t} \right) &= b_3 \ln o_t - b_4 r_t - u_{md,t} && \text{Money demand} \\ \ln m_t &= u_{ms,t}, && \text{Money supply} \end{aligned}$$

where all parameters b_i are positive. The structural shocks are

$$u_t = [u_{as,t} \ u_{is,t} \ u_{md,t} \ u_{ms,t}],$$

which correspond to shocks in aggregate supply, the IS schedule, money

demand and money supply. The parameter matrices for the specification in equation (14.1) with $p = 1$ are

$$B_0 = \begin{bmatrix} 1 & 0 & -b_1^{-1} & 0 \\ b_2^{-1} & 1 & -1 & 0 \\ b_3 & -b_4 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B_1 = \begin{bmatrix} 0 & 0 & -b_1^{-1} & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

where B_0 represents the contemporaneous effects of shocks of y_t , and B_1 captures the dynamics of the model. $\square$

An alternative representation of equation (14.1), commonly adopted in modelling SVARs, is to define the standardized random variable

$$z_t = D^{-1/2}u_t, \quad (14.2)$$

where $D^{1/2}$ contains the standard deviations on the diagonal and

$$\begin{aligned} E[z_t] &= D^{-1/2}E[u_t] = 0 \\ E[z_t z_t'] &= D^{-1/2}E[u_t u_t'](D^{-1/2})' = D^{-1/2}DD^{-1/2} = I. \end{aligned} \quad (14.3)$$

It follows from equations (14.1) and (14.3) that $z_t \sim N(0, I)$, in which case equation (14.1) is rewritten in terms of standard deviation shocks z_t

$$B_0 y_t = \sum_{i=1}^p B_i y_{t-i} + D^{1/2} z_t. \quad (14.4)$$

Shocks presented in this form are usually referred to as one-standard deviation shocks since an increase in $z_{i,t}$ is multiplied by $D_{i,i}^{1/2}$, which is by definition the standard deviation of the structural shock $u_{i,t}$.

The dynamics of the structural model in equation (14.4) are summarized by its reduced form representation (see Chapter 5)

$$y_t = \sum_{i=1}^p B_0^{-1} B_i y_{t-i} + B_0^{-1} D^{1/2} z_t = \sum_{i=1}^p \Phi_i y_{t-i} + v_t, \quad (14.5)$$

where

$$\Phi_i = B_0^{-1} B_i, \quad (14.6)$$

is the matrix of autoregressive parameters at lag i and

$$v_t = B_0^{-1} D^{1/2} z_t = S z_t, \quad (14.7)$$

is the reduced form disturbance vector with

$$S = B_0^{-1} D^{1/2}. \quad (14.8)$$

From the properties of z_t in equation (14.3), v_t satisfies

$$\begin{aligned} E[v_t] &= S E[z_t] = 0 \\ E[v_t v_t'] &= S E[z_t z_t'] S' = S I S' = S S' = V, \end{aligned} \quad (14.9)$$

and it follows from the distribution of z_t that $v_t \sim N(0, V)$. Unlike the structural shocks z_t , which are orthogonal, v_t in (14.5) is not orthogonal since V is not necessarily a diagonal matrix.

The dynamics of the reduced form in equation (14.5) are highlighted by writing this equation in terms of the lag operator (see Appendix B)

$$\begin{aligned} y_t - \sum_{i=1}^p \Phi_i y_{t-i} &= v_t \\ (I - \Phi_1 L - \Phi_2 L^2 - \dots - \Phi_p L^p) y_t &= v_t \\ \Phi(L) y_t &= v_t, \end{aligned} \quad (14.10)$$

where $L^k y_t = y_{t-k}$ defines the lag operator and

$$\Phi(L) = I - \Phi_1 L - \Phi_2 L^2 - \dots - \Phi_p L^p. \quad (14.11)$$

From Chapter 13, inverting equation (14.11) gives the vector moving average representation

$$y_t = \Phi(L)^{-1} v_t = \Psi(L) v_t, \quad (14.12)$$

where

$$\Psi(L) = \Psi_0 + \Psi_1 L + \Psi_2 L^2 + \dots + \Psi_q L^q + \dots, \quad (14.13)$$

and $\Psi_0 = I_N$. The effect of a shock in v_t on the future time path of the dependent variables, $y_t, y_{t+1}, y_{t+2}, \dots$, are, respectively, the $(N \times N)$ parameter matrices $\Psi_0, \Psi_1, \Psi_2, \dots$. These are the impulse responses of the non-orthogonalized shocks v_t , as the following example demonstrates.

Example 14.2 A Simple Bivariate Model

Consider a model consisting of $N = 2$ variables expressed in first differences $y_t = [\Delta y_{1,t} \ \Delta y_{2,t}]'$ and the structural shocks given by $u_t = [u_{1,t} \ u_{2,t}]'$. The impulse responses of the non-orthogonalized shocks $v_t = [v_{1,t} \ v_{2,t}]'$ are given in Figure 14.1 and are based on specifying (14.5) with $p = 2$. The shocks are cumulated to express the impulses in terms of the level of the variables. The contemporaneous effects are given by

$$\Psi_0 = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

which directly translate into the 2×2 window of impulse responses at $h = 0$.

The intermediate impulses are obtained from the 2×2 matrices $\Psi_1, \Psi_2, \dots$, with the long-run effects computed as

$$\sum_{i=0}^{40} \Psi_i = \begin{bmatrix} 1.560 & -3.585 \\ 0.338 & 7.245 \end{bmatrix},$$

which are the last elements of the impulses in Figure 14.1. $\square$

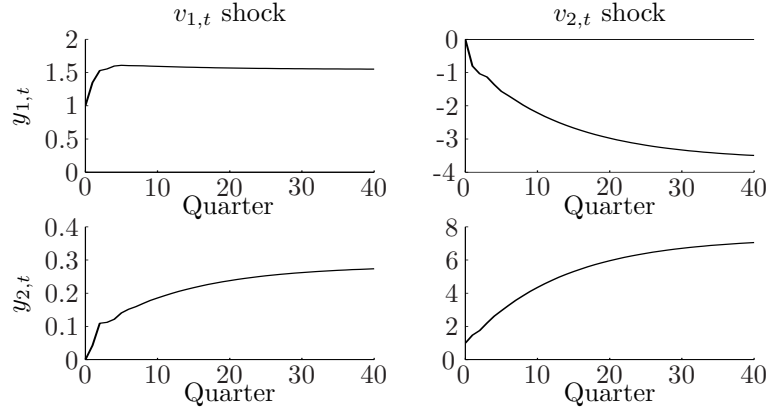


Figure 14.1 Non-orthogonalized impulse responses of a bivariate SVAR model.

From the discussion in Chapter 13, the reduced form in equation (14.5) also represents a VAR where each variable in the system is expressed as a function of the lags of all of the variables in the system with parameters given by equation (14.6). This link between a VAR and the structural model in equation (14.4) via the reduced form in equation (14.5) is important in practice, because it is a VAR that is commonly estimated at the first stage of SVAR modelling. But this initial estimation causes a problem since the VAR generates non-orthogonalized shocks that are not the structural shocks. This problem is addressed at the second stage of SVAR modelling where identification of the structural shocks, z_t , from the VAR shocks, v_t , is achieved through the imposition of restrictions on the model. What this means in practice is that estimates of the structural parameters $B_0, B_1, \dots, B_p$ and D in equation (14.7) need to be obtained from the estimates of the VAR in equation (14.5), because it is these structural parameters that represent the effects of z_t on y_t . Four sets of restrictions are used to achieve identification.

- (1) **Short-run:** Restrictions are imposed on the impulse responses at $h = 0$.

In Figure 14.1, all off-diagonal terms of the non-orthogonalized shocks

are set to zero by definition as $\Psi_0 = I_N$. A less extreme approach is to set some of the initial shocks to zero. For example, monetary shocks have no short-run effect on output whereas aggregate demand shocks do (Bernanke and Blinder, 1992).

- (2) **Long-run:** Restrictions are imposed on the impulse responses at $h \rightarrow \infty$. Inspection of Figure 14.1 reveals that the long-run impulses of the non-orthogonalized shocks are $\{1.560, -3.585, 0.338, 7.245\}$. A long-run restriction could take the form of a zero restriction, which would be appropriate in the case of nominal shocks having no long-run effect on real variables.
- (3) **Combination of Short-run and Long-run:** Restrictions are imposed on the impulse responses at $h = 0$ and $h \rightarrow \infty$. An example would be that monetary shocks have no contemporaneous or long-run effects on output (Peersman, 2005).
- (4) **Sign:** Intermediate impulses, as well as contemporaneous and possibly long-run impulses, are used to identify shocks. For example, if $y_{1,t}$ and $y_{2,t}$ in Figure 14.1 represent output and price, respectively, the shock $v_{2,t}$ does not represent aggregate demand shocks since it has a negative effect on output and a positive effect on price. If anything, the impulse response patterns of $v_{1,t}$ look more like the effects of aggregate demand shocks, but now the impulse responses of $v_{2,t}$ do not look like (negative) aggregate supply shocks because the initial effect on $y_{1,t}$ is zero, which does not make sense. This method involves looking at a range of alternative orthogonalizations, which is a refinement of the earlier practice of just looking at alternative recursive orderings of the variables in the VAR.

14.2.1 Short-Run Restrictions

The earliest method of identifying the shocks in SVAR models is based on short-run restrictions. This is formally obtained by imposing zero restrictions on the matrix B_0 in (14.4) since it represents the contemporaneous relationships between the variables y_t . The following two examples highlight the nature of these restrictions.

Example 14.3 The Sims Recursive Model

The Sims (1980) model of the U.S. economy, discussed in Chapter 13, can be interpreted as representing the first SVAR model that imposes a recursive structure on the short-run parameters. The model contains $N = 4$ variables: namely, the interest rate, the logarithm of money, the logarithm of prices

and the logarithm of output, $y_t = [r_t \ln m_t \ln p_t \ln o_t]$. The model has a recursive structure with the contemporaneous matrix B_0 restricted to be lower triangular

$$B_0 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -b_{2,1} & 1 & 0 & 0 \\ -b_{3,1} & -b_{3,2} & 1 & 0 \\ -b_{4,1} & -b_{4,2} & -b_{4,3} & 1 \end{bmatrix}.$$

This means that the model imposes the short-run ordering

$$r_t \longrightarrow \ln m_t \longrightarrow \ln p_t \longrightarrow \ln o_t$$

on the model variables. $\square$

Example 14.4 The Kim and Roubini Nonrecursive Model

Kim and Roubini (2000) specify a seven-variable macroeconomic model with non-recursive restrictions imposed on the short-run parameters. The model includes the variables used in the Sims model but also adds three other variables: namely, the logarithm of the oil price, the Federal funds rate and the logarithm of the exchange rate, $y_t = [r_t \ln m_t \ln p_t \ln o_t \ln oil_t ffr_t \ln e_t]'$. The standardized structural shocks comprise shocks to the money supply, money demand, prices, output, the oil price, monetary policy and the exchange rate, $z_t = [z_{ms,t} z_{md,t} z_{p,t} z_{o,t} z_{oil,t} z_{ffr,t} z_{ue,t}]'$. The contemporaneous matrix, B_0 , is specified as

$$B_0 = \begin{bmatrix} 1 & -b_{1,2} & 0 & 0 & -b_{1,5} & 0 & -b_{1,7} \\ -b_{2,1} & 1 & -b_{2,2} & -b_{2,3} & 0 & 0 & 0 \\ 0 & 0 & 1 & -b_{3,4} & -b_{3,5} & 0 & 0 \\ 0 & 0 & 0 & 1 & -b_{4,5} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -b_{6,5} & 1 & 0 \\ -b_{7,1} & -b_{7,2} & -b_{7,3} & -b_{7,4} & -b_{7,5} & -b_{7,6} & 1 \end{bmatrix}.$$

As B_0 is non-triangular, the model has a non-recursive structure. $\square$

To translate the short-run restrictions given in B_0 to the orthogonalized impulse responses, rewrite the moving average representation in equation (14.10) in terms of the orthogonalized shocks z_t by using $v_t = B_0^{-1} D^{1/2} z_t = S z_t$ in (14.7) to substitute v_t as follows

$$\begin{aligned} y_t &= v_t + \Psi_1 v_{t-1} + \Psi_2 v_{t-2} + \cdots \\ &= B_0^{-1} D^{1/2} z_t + \Psi_1 B_0^{-1} D^{1/2} z_{t-1} + \Psi_2 B_0^{-1} D^{1/2} z_{t-2} + \cdots \\ &= S z_t + \Psi_1 S z_{t-1} + \Psi_2 S z_{t-2} + \Psi_3 S z_{t-3} + \cdots \end{aligned} \quad (14.14)$$

Inspection of equation (14.14) reveals that converting the impulse responses from non-orthogonalized shocks to orthogonalized shocks with identification based on imposing short-run restrictions on the matrix B_0 , it is simply a matter of rescaling the vector moving average parameter matrices Ψ_i , by $S = B_0^{-1}D^{1/2}$. Thus the orthogonalized one-standard impulse responses are computed as the $(N \times N)$ parameter matrices $S, \Psi_1 S, \Psi_2 S, \dots$.

Once B_0 and D are specified, S is determined from $S = B_0^{-1}D^{1/2}$. This suggests that short-run restrictions can be imposed either on B_0 or on S directly. Restrictions on B_0 control contemporaneous relationships between the y_t variables, whereas restrictions on S control the relationships between the y_t variables and the structural shocks. Both methods are commonly adopted in SVAR models. If B_0 is triangular, then B_0^{-1} and S are both triangular, resulting in the equivalence of the two methods for imposing short-run restrictions. This is not the case for non-recursive models.

Example 14.5 A Model with Short-Run Restrictions

Consider the bivariate SVAR model of Example 14.2, but with the restriction that $u_{2,t}$ does not contemporaneously affect $y_{1,t}$. Let the matrix S be

$$S = \begin{bmatrix} 3 & 0 \\ -1 & 2 \end{bmatrix},$$

where the zero element corresponds to the short-run restriction. The cumulated impulse responses of the orthogonalized shocks in Figure 14.2 show that, by construction, the initial effects are represented by S , with $u_{2,t}$ having no contemporaneous effect on $y_{1,t}$. In the long-run, both shocks have non-zero effects on the variables. $\square$

14.2.2 Long-Run Restrictions

The next way to impose restrictions is to constrain the impulse responses in the long run to have values that are motivated by economic theory. The form of these long-run restrictions is highlighted by investigating the long-run solution of the prototype theoretical model presented earlier.

Example 14.6 A Prototype Macroeconomic Model Revisited

Consider the prototype macroeconomic model of Example 14.1 where

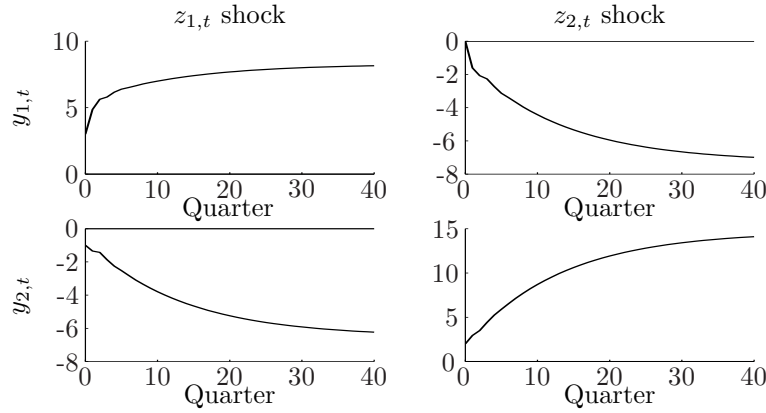


Figure 14.2 Orthogonalized impulse responses of a bivariate SVAR model with short-run restrictions.

long-run equilibrium requires $\ln p_t - \ln p_{t-1} = 0$ and the model reduces to

$$\begin{aligned}\ln o_t &= u_{as,t} \\ r_t &= u_{is,t} - b_2^{-1} u_{as,t} \\ \ln p_t &= u_{ms,t} + b_4 u_{is,t} - (b_3 + b_4 b_2^{-1}) u_{as,t} + u_{md,t} \\ m_t &= u_{ms,t}.\end{aligned}$$

These equations impose a number of restrictions on the long-run relationship between the model variables $y_t = [\ln o_t \ r_t \ \ln p_t \ \ln m_t]'$ and the structural shocks $u_t = [u_{as,t} \ u_{is,t} \ u_{md,t} \ u_{ms,t}]'$. $\square$

Example 14.7 The Blanchard and Quah Model

Blanchard and Quah (1989) specify a bivariate macroeconomic model consisting of the growth rate in output and the unemployment rate, $y_t = [\Delta \ln o_t \ ue_t]'$. The one-standard deviation structural shocks correspond to aggregate supply and aggregate demand, $z_t = [z_{as,t} \ z_{ad,t}]'$. The model has a recursive long-run structure with the long-run parameter matrix F specified as

$$F = \begin{bmatrix} f_{1,1} & 0 \\ f_{2,1} & f_{2,2} \end{bmatrix},$$

where the zero shows that aggregate demand shocks have no long-run effect on output ($f_{1,2} = 0$). $\square$

The previous two examples emphasize the role of zero restrictions in the long run. The next example uses zero long-run restrictions in conjunction with cross-equation restrictions.

Example 14.8 The Rapach Model

Rapach (2001) specifies a four-variable model consisting of the first differences of the logarithm of prices, the logarithm of real stock prices, the interest rate and the logarithm of output, $y_t = [\Delta \ln p_t \ \Delta \ln s_t \ \Delta r_t \ \Delta \ln o_t]'$. The one-standard deviation structural shocks comprise nominal shocks, portfolio shocks, aggregate demand shocks and aggregate supply shocks, $z_t = [z_{n,t} \ z_{s,t} \ z_{ad,t} \ z_{as,t}]'$. The model has a non-recursive long-run structure with long-run parameter matrix, F , given by

$$F = \begin{bmatrix} f_{1,1} & f_{1,2} & f_{1,3} & f_{1,4} \\ 0 & f_{2,2} & f_{2,3} & f_{2,4} \\ 0 & 0.025f_{2,2} & f_{3,3} & f_{3,4} \\ 0 & 0 & 0 & f_{4,4} \end{bmatrix}.$$

The first row of the F matrix shows that all shocks affect nominal prices in the long run. The zeros in the second and third rows show that nominal shocks have no long-run effect on the real stock price and the interest rate. The last row represents the natural rate hypothesis whereby only supply shocks affect output in the long run. The restriction $0.025f_{2,2}$ imposes a long-run relationship between real stock prices and the interest rate, so that a $f_{22} = 10\%$ permanent increase in stock prices requires a $0.025 \times 10 = 2.5\%$ permanent increase in the interest rate to maintain long-run portfolio equilibrium. An alternative identification mechanism is proposed by Fry, Hocking and Martin (2008) discussed later in Section 14.6. $\square$

To impose long-run restrictions on the orthogonalized impulse responses, assume that all variables in y_t are expressed in first differences as in the Rapach model. The long-run effects of a one-standard deviation shock in z_t on the levels of these variables are computed by summing the impulse responses $\Psi_\ell S$. Letting F represent the long-run matrix, from equation (14.14) the long-run impulse responses are obtained as

$$F = S + \Psi_1 S + \Psi_2 S + \Psi_3 S + \cdots = \Psi(1)S, \quad (14.15)$$

with

$$\Psi(1) = I + \Psi_1 + \Psi_2 + \Psi_3 + \cdots.$$

It follows from equation (14.15) that the way to restrict impulse responses in the long-run is to impose restrictions on F . Re-arranging equation (14.15) as

$$S = \Psi(1)^{-1}F = \Phi(1)F, \quad (14.16)$$

shows that S is now equal to the long-run matrix F weighted by the sum of

the VAR parameters in equation (14.10)

$$\Phi(1) = I - \Phi_1 - \Phi_2 - \cdots - \Phi_p. \quad (14.17)$$

Not surprisingly, in the special case of no dynamics, there is no distinction between the short-run and long-run because $\Phi(1) = I$ since $\Phi_i = 0 \forall i$.

Example 14.9 A Model with Long-Run Restrictions

Consider the bivariate SVAR model of Example 14.2, but with the restriction that $u_{2,t}$ does not affect $y_{1,t}$ in the long run. Suppose that the sum of the VAR parameters is

$$\Phi(1) = \begin{bmatrix} 0.579 & 0.287 \\ -0.027 & 0.123 \end{bmatrix},$$

and the long-run matrix F is given by

$$F = \begin{bmatrix} 3 & 0 \\ -1 & 2 \end{bmatrix},$$

where the zero element corresponds to the long-run restriction. From equation (14.16), the S matrix is computed as

$$S = \Phi(1) F = \begin{bmatrix} 0.579 & 0.287 \\ -0.027 & 0.123 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1.451 & 0.573 \\ -0.204 & 0.246 \end{bmatrix}.$$

The cumulated impulse responses in Figure 14.3 show that both shocks have short-run effects given by S , while the long-run effects are given by F with the effect of $u_{2,t}$ on $y_{1,t}$ approaching its long-run value of zero. $\square$

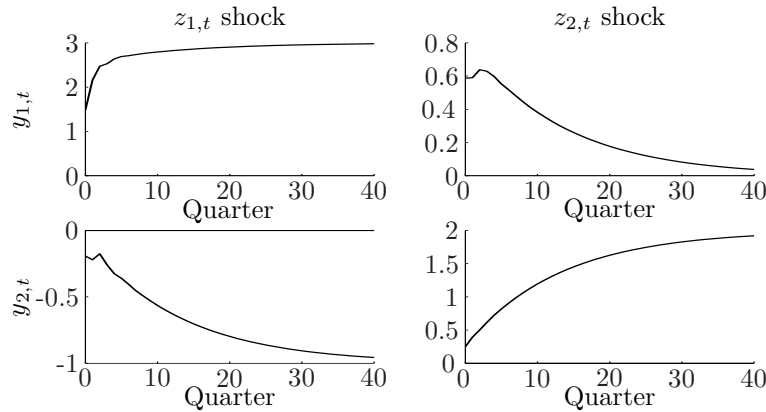


Figure 14.3 Orthogonalized impulse responses of a bivariate SVAR model with long-run restrictions.

14.2.3 Short-Run and Long-Run Restrictions

Another way to impose structure on the model is to use a combination of short-run and long-run restrictions. This requires that constraints are imposed on the contemporaneous impulse responses and on their long-run behaviour. In so doing, the aim is still to specify the structure of the matrix S in terms of these restrictions.

Example 14.10 Combining Short-Run and Long-Run Restrictions

Consider once again the bivariate SVAR model introduced in Example 14.2, but with the restriction that $u_{1,t}$ does not affect $y_{2,t}$ in the short-run and $u_{2,t}$ does not affect $y_{1,t}$ in the long run. The short-run restriction is $s_{2,1} = 0$. The long-run restriction can be derived from

$$\Phi(1)^{-1} = \begin{bmatrix} 0.579 & 0.287 \\ -0.027 & 0.123 \end{bmatrix}^{-1} = \begin{bmatrix} 1.557 & -3.630 \\ 0.342 & 7.336 \end{bmatrix}.$$

Using $F = \Phi(1)^{-1}S$ makes the long-run restriction

$$0 = f_{1,2} = a^{1,1}s_{1,2} + a^{1,2}s_{2,2},$$

which is solved for $s_{2,2}$ to yield

$$S = \begin{bmatrix} s_{1,1} & s_{1,2} \\ 0 & s_{2,2} = -\frac{a^{1,1}}{a^{1,2}}s_{1,2} \end{bmatrix}.$$

The cumulated impulse responses are given in Figure 14.4 for $s_{1,1} = 2$ and $s_{1,2} = 1$. The impulses capture the short-run and long-run restrictions imposed on the model. $\square$

The following example by Gali (1992) is slightly more complex, involving as it does the use of short-run and long-run restrictions in a four-variable macroeconomic model.

Example 14.11 The Gali Model

The Gali model consists of the growth rate of output, the change in the nominal interest rate, the real interest rate and the growth rate of real money, $y_t = [\Delta \ln o_t \quad \Delta r_t \quad (r_t - \Delta \ln p_t) \quad (\Delta \ln m_t - \Delta \ln p_t)]'$. The one-standard deviation structural shocks, $z_t = [z_{as,t} \ z_{ms,t} \ z_{md,t} \ z_{ad,t}]'$, consist of shocks to aggregate supply, the money supply, money demand and aggregate demand. The restrictions are as follows.

- (1) **Short-run:** The money market shocks, $z_{ms,t}$ and $z_{md,t}$, do not have a contemporaneous effect on output, $s_{1,2} = s_{1,3} = 0$, where $s_{i,j}$ is the relevant entry of S in equation (14.16).

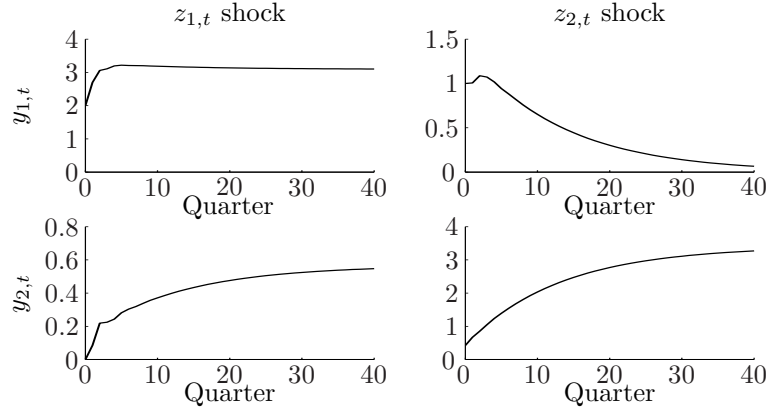


Figure 14.4 Orthogonal impulse responses of a bivariate SVAR model with short-run and long-run restrictions.

- (2) **Short-run:** Output does not enter the monetary policy rule contemporaneously, $b_{2,1} = 0$, where $b_{i,j}$ is a representative element of B_0 . To derive the effect of this restriction on S , rewrite (14.8) as $B_0 = D^{1/2} S^{-1}$ so that the restriction becomes

$$0 = d_{1,1}^{1/2} s^{2,1},$$

where $s^{2,1}$ is now the pertinent element of S^{-1} . For $d_{1,1} \neq 0$ it follows that $s^{2,1} = 0$. This equation represents a nonlinear restriction on all of the N elements of S which in theory can be solved for any $s_{i,j}$. However, because $s^{2,1} = 0$ it is also true that $s_{2,1} = 0$, resulting in S having the structure

$$S = \begin{bmatrix} s_{1,1} & 0 & 0 & s_{1,4} \\ 0 & s_{2,2} & s_{2,3} & s_{2,4} \\ s_{3,1} & s_{3,2} & s_{3,3} & s_{3,4} \\ s_{4,1} & s_{4,2} & s_{4,3} & s_{4,4} \end{bmatrix},$$

after all three short-run restrictions have been imposed.

- (3) **Long-run:** Output is assumed to operate at its natural rate so that nominal shocks, $z_{ms,t}$, $z_{md,t}$ and $z_{ad,t}$, have no long-run effect. These long-run restrictions imply that $f_{1,2} = f_{1,3} = f_{1,4} = 0$. To derive the effect of these restriction on S , rewrite equation (14.16) as $F = \Phi(1)^{-1} S$, which implies that

$$\begin{aligned} 0 &= f_{1,2} = \phi^{1,1} s_{1,2} + \phi^{1,2} s_{2,2} + \phi^{1,3} s_{3,2} + \phi^{1,4} s_{4,2} \\ 0 &= f_{1,3} = \phi^{1,1} s_{1,3} + \phi^{1,2} s_{2,3} + \phi^{1,3} s_{3,3} + \phi^{1,4} s_{4,3} \\ 0 &= f_{1,4} = \phi^{1,1} s_{1,4} + \phi^{1,2} s_{2,4} + \phi^{1,3} s_{3,4} + \phi^{1,4} s_{4,4}, \end{aligned}$$

where, as before, $\phi^{i,j}$ is an element of $\Phi(1)^{-1}$. Recalling that $s_{1,2} = s_{1,3} = 0$ from the short-run restrictions, it follows that

$$\begin{aligned} s_{2,2} &= -\frac{\phi^{1,3}}{\phi^{1,2}} s_{3,2} - \frac{\phi^{1,4}}{\phi^{1,2}} s_{4,2} \\ s_{2,3} &= -\frac{\phi^{1,3}}{\phi^{1,2}} s_{3,3} - \frac{\phi^{1,4}}{\phi^{1,2}} s_{4,3} \\ s_{1,4} &= -\frac{\phi^{1,2}}{\phi^{1,1}} s_{2,4} - \frac{\phi^{1,3}}{\phi^{1,1}} s_{3,4} - \frac{\phi^{1,4}}{\phi^{1,1}} s_{4,4}. \end{aligned}$$

Once these expressions are substituted into S , there are 10 remaining unknown parameters, $\{s_{1,1}, s_{2,4}, s_{3,1}, s_{3,2}, s_{3,3}, s_{3,4}, s_{4,1}, s_{4,2}, s_{4,3}, s_{4,4}\}$.

□

14.2.4 Sign Restrictions

The earliest method of imposing restrictions is to specify recursive SVAR models for different variable orderings as discussed in Chapter 13. In a recursive bivariate model consisting of $y_{1,t}$ and $y_{2,t}$, the two orderings are identified by the respective S matrices

$$S_{y_1 \rightarrow y_2} = \begin{bmatrix} s_{1,1} & 0 \\ s_{2,1} & s_{2,2} \end{bmatrix} \quad S_{y_2 \rightarrow y_1} = \begin{bmatrix} s_{1,1} & s_{1,2} \\ 0 & s_{2,2} \end{bmatrix}. \quad (14.18)$$

In the first case, a shock in $y_{2,t}$ does not contemporaneously affect $y_{1,t}$, whereas, in the second case, a shock in $y_{1,t}$ does not contemporaneously affect $y_{2,t}$. The choice of variable ordering, and hence SVAR model, is based on the model that yields impulse responses with signs consistent with economic theory. The sign restriction methodology to specifying SVAR models represents a generalization of this approach.

The strategy for choosing different variable orderings can be understood by defining the matrix

$$Q = \begin{bmatrix} \cos \vartheta & -\sin \vartheta \\ \sin \vartheta & \cos \vartheta \end{bmatrix}, \quad 0 \leq \vartheta \leq \pi, \quad (14.19)$$

which is orthonormal as $QQ' = Q'Q = I$. Now rewrite (14.7) as

$$v_t = Sz_t = SQ'Qz_t = SQ'w_t, \quad (14.20)$$

where $w_t = Qz_t$ represents a new set of standardized structural shocks that are also orthogonal because

$$E[w_t] = QE[z_t] = 0, \quad E[w_t w_t'] = QE[z_t z_t'] Q' = QQ' = I.$$

These new structural shocks are simply a weighted average of the original structural shocks, z_t ,

$$w_t = Qz_t = \begin{bmatrix} \cos \vartheta & -\sin \vartheta \\ \sin \vartheta & \cos \vartheta \end{bmatrix} \begin{bmatrix} z_{1,t} \\ z_{2,t} \end{bmatrix} = \begin{bmatrix} z_{1,t} \cos \vartheta - z_{2,t} \sin \vartheta \\ z_{1,t} \sin \vartheta + z_{2,t} \cos \vartheta \end{bmatrix}.$$

The two variable orderings given for the bivariate model in (14.18) are special cases of (14.20). For a lower triangular matrix S , as in the first ordering in equation (14.18), and for Q , as defined in equation (14.19),

$$\begin{aligned} SQ' &= \begin{bmatrix} s_{1,1} & 0 \\ s_{2,1} & s_{2,2} \end{bmatrix} \begin{bmatrix} \cos \vartheta & -\sin \vartheta \\ \sin \vartheta & \cos \vartheta \end{bmatrix}' \\ &= \begin{bmatrix} s_{1,1} \cos \vartheta & s_{1,1} \sin \vartheta \\ s_{2,1} \cos \vartheta - s_{2,2} \sin \vartheta & s_{2,1} \sin \vartheta + s_{2,2} \cos \vartheta \end{bmatrix}. \end{aligned}$$

Setting $\vartheta = 0$ yields the first ordering in equation (14.18). Setting $\vartheta = \vartheta^*$ where ϑ^* is chosen so that $s_{2,1} \cos \vartheta^* - s_{2,2} \sin \vartheta^* = 0$ gives the second ordering. More generally, other structural shocks are identified by choosing alternative values of ϑ within the range $0 \leq \vartheta \leq \pi$.

The approach of generating alternative models using the orthonormal rotation SQ' and selecting those models that generate impulse responses consistent with economic theory is known as the sign restrictions approach. Canova and de Nicolo (2002) and Peersman (2005) provide recent examples of this method. For models with dimensions greater than two, Q in (14.20) is based on either the Givens transformation (Uhlig, 2005) or the Householder transformation (Fry and Pagan, 2007). In practice the sign restriction approach to specifying a SVAR consists of the following steps.

- Step 1:** Estimate a VAR and compute S , where $SS' = V$.
- Step 2:** Draw a value of ϑ from $[0, \pi]$ and compute Q . For $N = 2$, the matrix Q is given by equation (14.19).
- Step 3:** Compute a finite number of impulse responses. For quarterly data, it is common to choose the first four impulses: $SQ', \Psi_1 SQ', \Psi_2 SQ', \Psi_3 SQ'$, where the Ψ_i matrices are defined in (14.12).
- Step 4:** If all impulses have the correct sign, select the model, otherwise discard it.
- Step 5:** Repeat steps 2 to 4 and generate other models that satisfy the restrictions.
- Step 6:** Select a definitive model from the set of candidate models according to some rule. See Fry and Pagan (2007) for a discussion of alternative rules.

Example 14.12 A Model with Sign Restrictions

Consider again the bivariate SVAR model introduced in Example 14.2, but imposing the sign restrictions that $u_{1,t}$ has a positive effect on $y_{1,t}$ and a negative effect on $y_{2,t}$, whereas $u_{2,t}$ has positive effects on both $y_{1,t}$ and $y_{2,t}$. For comparative purposes, let the matrix S be

$$S = \begin{bmatrix} 3 & 0 \\ -1 & 2 \end{bmatrix},$$

which was used earlier to generate the impulse responses of the short-run restricted version of the model. If $\vartheta = 0.4\pi$, the orthonormal matrix Q is

$$Q = \begin{bmatrix} \cos 0.4\pi & -\sin 0.4\pi \\ \sin 0.4\pi & \cos 0.4\pi \end{bmatrix} = \begin{bmatrix} 0.309 & -0.951 \\ 0.951 & 0.309 \end{bmatrix},$$

and hence the relevant matrix used to generate the orthogonal impulse responses is

$$SQ' = \begin{bmatrix} 3 & 0 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 0.309 & 0.951 \\ -0.951 & 0.309 \end{bmatrix} = \begin{bmatrix} 0.927 & 2.853 \\ -2.211 & -0.333 \end{bmatrix}.$$

The cumulated impulse responses given in Figure 14.5 reveal a different pattern to the impulses given in Figure 14.2 where the latter are equivalent to setting $\vartheta = 0.0$ despite both sets of impulses being based on orthogonalized shocks. Imposing the sign restrictions results in choosing the former set of impulses as representing the pertinent orthogonal shocks and discarding the latter set. $\square$

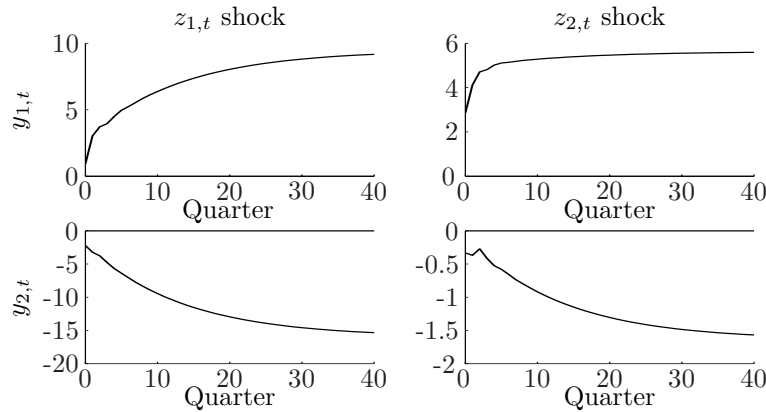


Figure 14.5 Orthogonal impulse responses of a bivariate SVAR model with sign restrictions.

14.3 Estimation

Let the parameters of the SVAR model be given by $\theta = \{S, B_1, \dots, B_p\}$, where

$$S = \begin{cases} B_0^{-1} D^{1/2} & : \text{Short-run restrictions} \\ \Phi(1)F & : \text{Long-run restrictions,} \end{cases} \quad (14.21)$$

and $\Phi(1)$ is obtained from the VAR in equation (14.5). The parameters can be estimated by maximum likelihood using the standard gradient algorithms discussed in Chapter 3. Conditioning on the first p observations, the log-likelihood function is

$$\ln L_T(\theta) = \frac{1}{T-p} \sum_{t=p+1}^T \ln l_t(\theta).$$

There are two strategies in performing maximum likelihood estimation, depending on whether the log-likelihood function is expressed in terms of the structural model or the reduced form.

Structural Form

$$\ln L_T(\theta) = -\frac{N}{2} \ln 2\pi - \frac{1}{2} \ln |D| - \frac{1}{2(T-p)} \sum_{t=p+1}^T u_t' D^{-1} u_t, \quad (14.22)$$

where $u_t \sim N(0, D)$ as given in equation (14.1).

Reduced Form

$$\ln L_T(\theta) = -\frac{N}{2} \ln 2\pi - \frac{1}{2} \ln |V| - \frac{1}{2(T-p)} \sum_{t=p+1}^T v_t' V^{-1} v_t, \quad (14.23)$$

where v_t is given in equation (14.5) and $V = SS'$.

Maximizing either log-likelihood function produces the same parameter estimates since both procedures are full information maximum likelihood (FIML) procedures, as discussed in Chapter 5. In practice, however, estimation of SVARs tends to be based on the reduced form log-likelihood function because historically a VAR is usually estimated first. As the log-likelihood function is based on the assumption of normality, estimation of the mean and the variance parameters can be performed separately as the information matrix is block diagonal. Thus estimation can proceed in two stages.

Stage 1: Mean Parameters

The maximum likelihood estimates of the VAR are obtained by maximizing the log-likelihood function with respect to

$$\Phi_1, \Phi_2, \dots, \Phi_p,$$

with V concentrated out of the log-likelihood function. This means that each equation of the VAR (reduced form) is estimated individually by ordinary least squares to yield

$$\hat{\Phi}_1, \hat{\Phi}_2, \dots, \hat{\Phi}_p.$$

The VAR residuals are given by

$$\hat{v}_t = y_t - \sum_{i=1}^p \hat{\Phi}_i y_{t-i},$$

which are used to compute the $(N \times N)$ covariance matrix

$$\hat{V} = \frac{1}{T-p} \sum_{t=p+1}^T \hat{v}_t \hat{v}_t'.$$

Stage 2: Variance Parameters

The maximum likelihood estimates of B_0 and D , in the case of the short-run parameters, and F , in the case of the long-run parameters, are obtained by maximizing

$$\ln L_T(\theta) = -\frac{N}{2} \ln 2\pi - \frac{1}{2} \ln |V| - \frac{1}{2(T-p)} \sum_{t=p+1}^T \hat{v}_t' V^{-1} \hat{v}_t,$$

with $V = SS'$ and S defined in equation (14.21). This maximization problem is equivalent to solving the nonlinear system of equations

$$\hat{V} = \hat{S} \hat{S}',$$

with $\hat{V}$ determined from the first stage of the estimation. This expression defines the first-order conditions of the maximum likelihood estimator with respect to these parameters. From equation (14.8) $S = B_0^{-1} D^{1/2}$, in which case estimates of B_0 and D are obtained from $\hat{S}$. One $\hat{B}_0$ is obtained the $\hat{B}_i$ s are then obtained from $\hat{B}_i = \hat{B}_0 \hat{\Phi}_i$, where the $\hat{\Phi}_i$ s are computed from the VAR in the first stage.

Example 14.13 Estimating a SVAR with Short-Run Restrictions

As an example of estimating a SVAR with short-run restrictions, consider a dynamic system of interest rates $r_t = [r_{1,t} \ r_{2,t} \ r_{3,t}]'$, where $r_{1,t}$ is the 0-month yield, $r_{2,t}$ is the 1-month yield and $r_{3,t}$ is the 3-month yield. The structural model is given by

$$B_0 r_t = B_1 r_{t-1} + u_t,$$

where the unknown parameters are

$$B_0 = \begin{bmatrix} 1 & 0 & 0 \\ -b_{2,1} & 1 & 0 \\ -b_{3,1} & 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} d_{1,1} & 0 & 0 \\ 0 & d_{2,2} & 0 \\ 0 & 0 & d_{3,3} \end{bmatrix},$$

plus the 9 unknown parameters in B_1 . The short-run restrictions show that the only contemporaneous relationships are $r_{1,t}$ to $r_{2,t}$ and $r_{3,t}$. The parameters to be estimated are $\theta = \{b_{2,1}, b_{3,1}, d_{1,1}, d_{2,2}, d_{3,3}, B_1\}$. The data are monthly U.S. zero coupon yields from December 1946 to February 1991. Estimation is performed in two stages. In the first stage, a VAR(1) is estimated with the results given by

$$\begin{aligned} r_{1,t} &= -0.124 + 0.060r_{1,t-1} + 0.321r_{2,t-1} + 0.559r_{3,t-1} + \hat{v}_{1,t} \\ r_{2,t} &= -0.032 - 0.146r_{1,t-1} + 0.472r_{2,t-1} + 0.635r_{3,t-1} + \hat{v}_{2,t} \\ r_{3,t} &= 0.068 + 0.074r_{1,t-1} - 0.279r_{2,t-1} + 1.185r_{3,t-1} + \hat{v}_{3,t}, \end{aligned}$$

with residual covariance matrix

$$\hat{V} = \frac{1}{T-1} \sum_{t=1}^{T-1} \hat{v}_t' \hat{v}_t = \begin{bmatrix} 0.491 & 0.345 & 0.224 \\ 0.345 & 0.314 & 0.268 \\ 0.224 & 0.268 & 0.289 \end{bmatrix}.$$

In the second stage, the estimates of $\{b_{2,1}, b_{3,1}, d_{1,1}, d_{2,2}, d_{3,3}\}$ are computed by maximizing the log-likelihood function in equation (14.23) with $T = 530$, $p = 1$ and

$$\begin{aligned} V &= SS' = B_0^{-1} D (B_0^{-1})' \\ &= \begin{bmatrix} 1 & 0 & 0 \\ -b_{2,1} & 1 & 0 \\ -b_{3,1} & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} d_{1,1} & 0 & 0 \\ 0 & d_{2,2} & 0 \\ 0 & 0 & d_{3,3} \end{bmatrix} \begin{bmatrix} 1 & -b_{2,1} & -b_{3,1} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1}. \end{aligned}$$

The value of the log-likelihood function evaluated at the maximum likelihood estimates is $\ln L_T(\hat{\theta}) = -1.744$. The parameter estimates, with standard errors based on the Hessian matrix given in parentheses, are

$$\hat{B}_0 = \begin{bmatrix} \underline{1.000} & \underline{0.000} & \underline{0.000} \\ -0.703 & \underline{1.000} & \underline{0.000} \\ -0.457 & \underline{0.000} & \underline{1.000} \end{bmatrix} \quad \hat{D} = \begin{bmatrix} \underline{0.491} & \underline{0.000} & \underline{0.000} \\ (0.030) & \underline{0.072} & \underline{0.000} \\ \underline{0.000} & (0.004) & \underline{0.187} \end{bmatrix},$$

and hence the estimates of the contemporaneous impact of the three shocks

on the three interest rates are

$$\hat{S} = \hat{B}_0^{-1} \hat{D}^{1/2} = \begin{bmatrix} 0.700 & 0.000 & 0.000 \\ 0.492 & 0.268 & 0.000 \\ 0.320 & 0.000 & 0.432 \end{bmatrix}.$$

□

Example 14.14 Estimating a SVAR with Long-Run Restrictions

Consider again the SVAR in Example 14.13, except that identification is now based on imposing the long-run restrictions that only shocks in $r_{1,t}$ have a long-run effect on all three interest rates. This implies that the long-run parameter matrix is

$$F = \begin{bmatrix} f_{1,1} & 0 & 0 \\ f_{2,1} & f_{2,2} & 0 \\ f_{3,1} & 0 & f_{3,3} \end{bmatrix}.$$

The parameters to be estimated are $\theta = \{f_{1,1}, f_{2,1}, f_{3,1}, f_{2,2}, f_{3,3}, B_1\}$. The results are based on the same data as given in Example 14.13, namely monthly U.S. zero coupon yields from December 1946 to February 1991. Estimation is performed in two stages with the first stage the same as the first stage in the previous example. In the second stage, estimates of $\{f_{1,1}, f_{2,1}, f_{3,1}, f_{2,2}, f_{3,3}\}$ are computed by maximizing the log-likelihood function in equation (14.23) with $T = 530$, $p = 1$ and $V = SS'$. From the first stage,

$$\begin{aligned} \hat{\Phi}(1) &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.060 & 0.321 & 0.559 \\ -0.146 & 0.472 & 0.635 \\ 0.074 & -0.279 & 1.185 \end{bmatrix} \\ &= \begin{bmatrix} 0.940 & -0.321 & -0.559 \\ 0.146 & 0.528 & -0.635 \\ -0.074 & 0.279 & -0.185 \end{bmatrix}, \end{aligned}$$

so that, in the second stage,

$$\begin{aligned} S &= \hat{\Phi}(1)F \\ &= \begin{bmatrix} 0.940 & -0.321 & -0.559 \\ 0.146 & 0.528 & -0.635 \\ -0.074 & 0.279 & -0.185 \end{bmatrix} \begin{bmatrix} f_{1,1} & 0 & 0 \\ f_{2,1} & f_{2,2} & 0 \\ f_{3,1} & 0 & f_{3,3} \end{bmatrix}. \end{aligned}$$

The value of the log-likelihood function evaluated at the maximum likelihood estimates is $\ln L_T(\hat{\theta}) = -1.593$. The long-run parameter estimates, with

standard errors based on the Hessian matrix given in parentheses, are

$$\hat{F} = \begin{bmatrix} 32.941 & 0.000 & 0.000 \\ (1.063) & - & - \\ 34.018 & 0.451 & 0.000 \\ (1.098) & (0.014) & - \\ 35.131 & 0.000 & 0.792 \\ (1.134) & - & (0.024) \end{bmatrix}.$$

The estimates of the contemporaneous impacts of the three shocks on the three interest rates are

$$\hat{S} = (I - \hat{\Phi}_1)\hat{F} = \begin{bmatrix} 0.391 & -0.145 & -0.443 \\ 0.479 & 0.238 & -0.503 \\ 0.535 & 0.126 & -0.147 \end{bmatrix}.$$

□

An important special case of SVAR models occurs when B_0 is recursive. Under these conditions, all of the parameters θ can be estimated exactly without any need for an iterative algorithm. In fact, Chapter 13 shows four ways of estimating θ that are all numerically equivalent. Here we focus on the maximum likelihood method.

Example 14.15 Estimating a Recursive SVAR

Consider again the SVAR in Example 14.13 and the monthly U.S. zero coupon yields from December 1946 to February 1991. The maximum likelihood approach consists of estimating the VAR by OLS to obtain the VAR residuals $\hat{v}_t$. The log-likelihood function in equation (14.23) is then maximized with respect to $B_0 = \{b_{2,1}, b_{3,1}, b_{3,2}\}$ and $D = \{d_{1,1}, d_{2,2}, d_{3,3}\}$, with $T = 530$, $p = 1$ and

$$\begin{aligned} V &= B_0^{-1}D(B_0^{-1})' \\ &= \begin{bmatrix} 1 & 0 & 0 \\ -b_{2,1} & 1 & 0 \\ -b_{3,1} & -b_{3,2} & 1 \end{bmatrix}^{-1} \begin{bmatrix} d_{1,1} & 0 & 0 \\ 0 & d_{2,2} & 0 \\ 0 & 0 & d_{3,3} \end{bmatrix} \begin{bmatrix} 1 & -b_{2,1} & -b_{3,1} \\ 0 & 1 & -b_{3,2} \\ 0 & 0 & 1 \end{bmatrix}^{-1}. \end{aligned}$$

The value of the log-likelihood function evaluated at the maximum likelihood estimates is $\ln L_T(\hat{\theta}) = -0.535$. The parameter estimates, with standard errors based on the Hessian matrix given in parentheses, are

$$\hat{B}_0 = \begin{bmatrix} 1.000 & 0.000 & 0.000 \\ -0.703 & 1.000 & 0.000 \\ 0.627 & -1.542 & 1.000 \\ (0.017) & - & (0.017) & 0.021 & - \end{bmatrix} \quad \hat{D} = \begin{bmatrix} 0.491 & 0.000 & 0.000 \\ (0.030) & - & - \\ 0.000 & 0.072 & 0.000 \\ - & (0.004) & - \\ 0.000 & 0.000 & 0.017 \\ - & - & (0.001) \end{bmatrix}.$$

□

In the case where the model is partially recursive in the sense that the lower triangular part of B_0 contains some zero elements, ordinary least squares estimates of each equation are still consistent but not asymptotically efficient. To achieve asymptotic efficiency in this case it would be necessary to use maximum likelihood methods. If the SVAR is non-recursive, ordinary least squares will not be consistent because there will be simultaneity bias arising from the presence of endogenous variables in each equation. In this situation maximum likelihood methods are needed. Alternatively, instrumental variable methods can be used to achieve consistent estimates following the approach of Pagan and Robertson (1989).

14.4 Identification

The discussion of the estimation of the SVAR parameters is based on the assumption that the unknown parameters in θ are identified, where identification is based on the imposition of either short-run or long-run restrictions, or restrictions based on the signs of the impulse responses. The issue of identification is now formalized.

As pointed out in Section 14.3, in the second estimation stage the maximum likelihood estimator is found as the solution of the expression $\hat{V} = \hat{S}\hat{S}'$. As V contains N variances and $N(N-1)/2$ covariances, the maximum number of parameters that can be identified from this set of equations is $N(N+1)/2$. This restriction on the number of identifiable parameters is the order condition, which is a necessary condition for identification. If $\theta = \{B_0, D\}$ is the vector of parameters to be estimated at the second stage, there are three cases to consider:

$$\begin{array}{lll} \text{Just-identified} & : & \dim(\theta) = N(N+1)/2 \\ \text{Over-identified} & : & \dim(\theta) < N(N+1)/2 \\ \text{Under-identified} & : & \dim(\theta) > N(N+1)/2. \end{array}$$

In the just-identified case, $\hat{V}$ is exactly equal to $\hat{S}\hat{S}'$. In the over-identified case, this need not be true, but, if the model is correctly specified, then the difference between $\hat{V}$ and $\hat{S}\hat{S}'$ should not be statistically significant. These over-identifying restrictions can be verified by testing, which is the subject matter of Section 14.5. In the under-identified case, insufficient information exists to be able to identify all of the parameters, implying the need for further restrictions on the model.

Example 14.16 Order Conditions of Empirical SVAR Models

The order conditions for some of the SVAR models used in previous ex-

amples and are given below where in each case $\theta = \{B_0, D\}$.

Model	N	$\frac{N(N+1)}{2}$	$\dim(\theta)$	Order condition
Sims	4	10	10	Just-identified model
Kim and Roubini	7	28	23	Over-identified model
Blanchard and Quah	3	3	3	Just-identified model
Rapach	4	10	10	Just-identified model
Gali	4	10	10	Just-identified model

□

The previous example is based on the order condition, which is a necessary condition for identification. As discussed in Chapter 5, a necessary and sufficient condition for identification is the rank condition, which requires that the Hessian of the log-likelihood function in equation (14.23) is negative definite. As $\hat{V} = \hat{S}\hat{S}'$ is the first-order condition satisfied by the maximum likelihood estimator, the rank condition is based on the matrix

$$\frac{\partial \text{vech}(V)}{\partial \theta'} = \frac{\partial \text{vech}(B_0^{-1}D(B_0^{-1})')}{\partial \theta'},$$

which has $N(N+1)/2$ rows and $\dim(\theta)$ columns and $\text{vech}(V)$ represents stacking the $N(N+1)/2$ unique elements of the system of equations. The rank condition is that the columns of the matrix are non-singular, while the order condition is that the $N(N+1)/2$ rows are at least as great as the number of columns.

14.5 Testing

Hypothesis tests on SVAR models can be performed using the LR, Wald and LM test statistics discussed in Chapter 4. Of particular interest in the context of SVARs is a test of the number of over-identifying restrictions. In the second stage of the estimation, an important question is whether or not the covariance matrix of the VAR residuals from the first stage, $\hat{V}$, is equal to the covariance matrix from the second stage, $\hat{S}\hat{S}'$. As stated previously, this condition does not hold for over-identified models, but, if the restrictions are valid, the difference should not be statistically significant. Consequently, a test of the over-identifying restrictions is based on comparing each (unique) element of V with each element of SS' . The hypotheses are

$$H_0 : \text{vech}(V - SS') = 0 \quad H_1 : \text{vech}(V - SS') \neq 0.$$

A straightforward way to perform the test is to use the LR statistic

$$LR = -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)), \quad (14.24)$$

where $\ln L_T(\hat{\theta}_0)$ is the value of the log-likelihood function subject to the over-identifying restrictions (constrained), and $\ln L_T(\hat{\theta}_1)$ is the value of the log-likelihood function corresponding to the just-identified model (unconstrained). Under the null hypothesis that the over-identifying restrictions are valid, LR is asymptotically distributed as χ^2 with degrees of freedom equal to the number of over-identifying restrictions

$$\frac{N(N+1)}{2} - \dim(\theta_0).$$

Example 14.17 Over-identifying Test of the Interest Rate Model

Consider again the estimation of an SVAR model for the system of U.S. interest rates with $N = 3$ introduced in Example 14.13. The interest rate model based on a recursive structure corresponds to the just-identified case where the value of the (unconstrained) log-likelihood function is $\ln L_T(\hat{\theta}_1) = -0.535$. Using the results of the over-identified interest rate model based on short-run restrictions, yields the value of the (constrained) log-likelihood function $\ln L_T(\hat{\theta}_0) = -1.744$. The value of the LR statistic is

$$\begin{aligned} LR &= -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)) \\ &= -2 \times (-530 \times 0.535 + 530 \times 1.744) = 1281.872, \end{aligned}$$

where the number of over-identifying restrictions is

$$\frac{N(N+1)}{2} - \dim(\theta_0) = 6 - 5 = 1.$$

From the chi-square distribution with one degree of freedom, the p-value is 0.000 giving a clear rejection of the over-identifying restriction. $\square$

The LR form of the over-identifying test requires estimating the model under the null and alternative hypotheses. A more convenient form of this test is to recognize that the covariance matrix of the VAR residuals $\hat{V}$ represents the unconstrained covariance matrix and $\hat{S}\hat{S}'$ represents the constrained covariance matrix obtained from maximizing the log-likelihood function. As the disturbance term is assumed to be normal, from Chapter 4 an alternative form of the LR statistic in equation (14.24) is

$$LR = T(\ln |\hat{S}\hat{S}'| - \ln |\hat{V}|).$$

Example 14.18 Over-identifying Test Revisited

Using the parameter estimates from Example 14.17, the unconstrained VAR residual covariance matrix is

$$\hat{V} = \begin{bmatrix} 0.491 & 0.345 & 0.224 \\ 0.345 & 0.314 & 0.268 \\ 0.224 & 0.268 & 0.289 \end{bmatrix}.$$

The constrained covariance based of the SVAR with the over-identifying restrictions is

$$\begin{aligned} \hat{S}\hat{S}' &= \begin{bmatrix} 0.700 & 0.000 & 0.000 \\ 0.492 & 0.268 & 0.000 \\ 0.320 & 0.000 & 0.432 \end{bmatrix} \begin{bmatrix} 0.700 & 0.492 & 0.320 \\ 0.000 & 0.268 & 0.000 \\ 0.000 & 0.000 & 0.432 \end{bmatrix} \\ &= \begin{bmatrix} 0.491 & 0.345 & 0.224 \\ 0.345 & 0.314 & 0.158 \\ 0.224 & 0.158 & 0.289 \end{bmatrix}. \end{aligned}$$

Computing the determinants

$$|\hat{V}| = 0.000585, \quad |\hat{S}\hat{S}'| = 0.006571,$$

yields the value of the LR statistic

$$\begin{aligned} LR &= T(\ln |\hat{S}\hat{S}'| - \ln |\hat{V}|) \\ &= 530 \times (\ln(0.006571) - \ln(0.000585)) = 1281.969, \end{aligned}$$

which differs from the previous value only because of rounding error. $\square$

14.6 Applications**14.6.1 Peersman's Model of Oil Price Shocks**

Peersman (2005) specifies and estimates a four-variable SVAR model using U.S. data from June 1979 to June 2002, to identify the effects of oil price shocks on the U.S. economy. The variables in the model are the growth rate in oil prices, the growth rate of output, the inflation rate and the interest rate, $y_t = [\Delta \ln oil_t \quad \Delta \ln o_t \quad \Delta \ln p_t \quad r_t]'$.

The VAR is estimated with $p = 3$ lags, a constant and a time trend. From the first stage of estimation, the VAR residual covariance matrix is

$$\hat{V} = \begin{bmatrix} 168.123 & 0.173 & 1.190 & 1.935 \\ 0.173 & 0.281 & 0.013 & 0.173 \\ 1.190 & 0.013 & 0.047 & 0.026 \\ 1.935 & 0.173 & 0.026 & 0.763 \end{bmatrix}, \quad (14.25)$$

and

$$\hat{\Phi}(1) = I - \hat{\Phi}_1 - \hat{\Phi}_2 - \hat{\Phi}_3 = \begin{bmatrix} 1.053 & -4.035 & -5.155 & -0.414 \\ -0.007 & 0.342 & -0.039 & 0.099 \\ -0.004 & 0.066 & 0.256 & 0.012 \\ -0.003 & -1.158 & -2.040 & 0.233 \end{bmatrix}, \quad (14.26)$$

so that

$$\hat{\Phi}(1)^{-1} = \begin{bmatrix} 1.028 & 5.736 & 11.902 & -1.211 \\ -0.006 & 1.802 & -4.258 & -0.555 \\ 0.013 & -0.566 & 4.377 & 0.038 \\ 0.097 & 4.075 & 17.286 & 1.851 \end{bmatrix}. \quad (14.27)$$

The one-standard deviation structural shocks comprise shocks to oil prices, aggregate supply, aggregate demand and money, $z_t = [z_{oil,t} \ z_{as,t} \ z_{ad,t} \ z_{m,t}]'$. Identification of these shocks is based on a combination of four short-run and two long-run restrictions on the SVAR, which results in a just-identified system.

- (1) **Short-run:** The non-oil price shocks have no contemporaneous effects on oil prices, $s_{1,2} = s_{1,3} = s_{1,4} = 0$.
- (2) **Short-run:** Money shocks have no contemporaneous effect on output, $s_{2,4} = 0$.
- (3) **Long-run:** Aggregate demand shocks and money shocks have no long-run effects on output, $f_{2,3} = f_{2,4} = 0$.

The effect of the two long-run restrictions on S are derived from the second and third rows of $F = \Phi(1)^{-1}S$, which are

$$\begin{aligned} f_{2,3} &= \phi^{2,1}s_{1,3} + \phi^{2,2}s_{2,3} + \phi^{2,3}s_{3,3} + \phi^{2,4}s_{4,3} \\ f_{2,4} &= \phi^{2,1}s_{1,4} + \phi^{2,2}s_{2,4} + \phi^{2,3}s_{3,4} + \phi^{2,4}s_{4,4}, \end{aligned}$$

where $\phi^{i,j}$ is an element of $\Phi(1)^{-1}$ given in (14.27). As $f_{2,3} = f_{2,4} = 0$ and $s_{1,3} = s_{1,4} = s_{2,4} = 0$, these equations are rearranged to give

$$\begin{aligned} s_{4,3} &= -\frac{\phi^{2,2}}{\phi^{2,4}}s_{2,3} - \frac{\phi^{2,3}}{\phi^{2,4}}s_{3,3} = \frac{1.802}{0.555}s_{2,3} - \frac{4.258}{0.555}s_{3,3} \\ s_{3,4} &= -\frac{\phi^{2,4}}{\phi^{2,3}}s_{4,4} = -\frac{0.555}{4.258}s_{4,4}. \end{aligned}$$

Once all short-run and long-run restrictions are imposed, the matrix S is

$$S = \begin{bmatrix} s_{1,1} & 0 & 0 & 0 \\ s_{2,1} & s_{2,2} & s_{2,3} & 0 \\ s_{3,1} & s_{3,2} & s_{3,3} & -\frac{0.555}{4.258}s_{4,4} \\ s_{4,1} & s_{4,2} & \frac{1.802}{0.555}s_{2,3} - \frac{4.258}{0.555}s_{3,3} & s_{4,4} \end{bmatrix}. \quad (14.28)$$

The parameters of the SVAR,

$$\theta = \{s_{1,1}, s_{2,1}, s_{2,2}, s_{2,3}, s_{3,1}, s_{3,2}, s_{3,3}, s_{4,1}, s_{4,2}, s_{4,4}\},$$

are estimated by maximizing the reduced form log-likelihood function in (14.23) subject to the restriction $V = SS'$, where v_t is replaced by $\hat{v}_t$, V is replaced by $\hat{V}$ in (14.25) and S is given by (14.28). The short-run and long-run estimates are, respectively,

$$\hat{S} = \begin{bmatrix} 12.966 & 0.000 & 0.000 & 0.000 \\ 0.013 & 0.309 & 0.431 & 0.000 \\ 0.092 & -0.128 & 0.118 & -0.090 \\ 0.149 & -0.131 & 0.491 & 0.695 \end{bmatrix} \quad (14.29)$$

$$\hat{F} = \hat{\Phi}(1)^{-1}\hat{S} = \begin{bmatrix} 14.317 & 0.412 & 3.284 & -1.918 \\ -0.525 & 1.174 & 0.000 & 0.000 \\ 0.566 & -0.739 & 0.293 & -0.369 \\ 3.171 & -1.191 & 4.709 & -0.278 \end{bmatrix}. \quad (14.30)$$

The full set of impulse responses are presented in Figure 14.6 with the initial values given by $\hat{S}$ in equation (14.29). The impulse responses on the logarithms of oil, output and prices are cumulated, because these variables are expressed in first differences in the model. This means that the final impulse values of these variables correspond to the last three columns of $\hat{F}$ in equation (14.30). The impulses on the interest rate are not cumulated because this variable is already expressed in levels in the model. However, cumulating these impulses would indeed result in estimates given in the first column of $\hat{F}$.

The oil price shock in Figure 14.6 represents a negative supply shock with output falling and prices increasing. There is a positive effect on the interest rate, which dissipates in the long run.

14.6.2 A Portfolio SVAR Model of Australia

Fry, Hocking and Martin (2008) specify a five-variable SVAR model of the Australian economy that focuses on the transmission of wealth effects

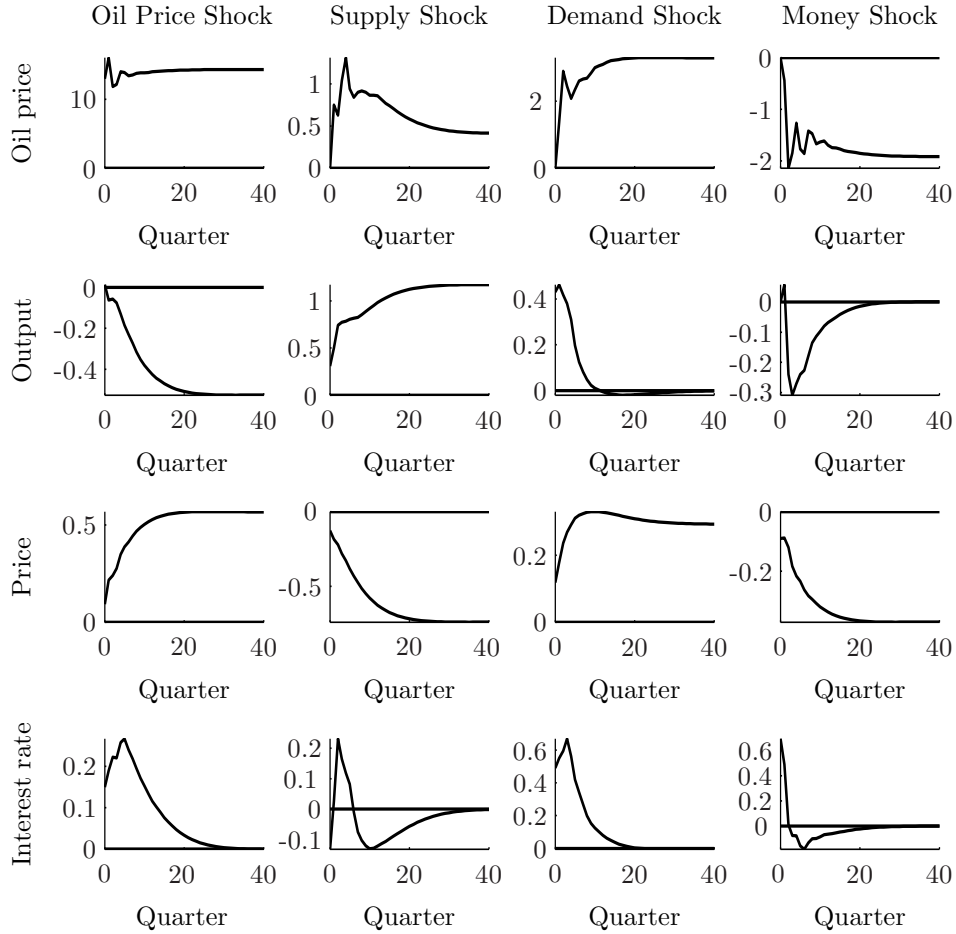


Figure 14.6 Impulse responses of the Peersman model of the U.S., June 1979 to June 2002. All impulses are expressed in terms of the levels of the variables.

arising from portfolio shocks, with identification based on the imposition of long-run restrictions. The variables are the growth rate in output, the change in the logarithm of the interest rate, real stock returns on Australian equities, goods market inflation and real stock returns on U.S. equities, $y_t = [\Delta \ln o_t \ \Delta \ln r_t \ \Delta \ln s_t \ \Delta \ln p_t \ \Delta \ln f_t]'$. The one-standard deviation structural shocks comprise aggregate supply shocks, aggregate demand shocks, Australian portfolio shocks, nominal shocks and foreign portfolio shocks, $z_t = [z_{as,t} \ z_{ad,t} \ z_{au,t} \ z_{p,t} \ z_{us,t}]'$.

Identification is based on imposing the following long-run restrictions

$$F = \begin{bmatrix} f_{1,1} & 0 & 0 & 0 & 0 \\ f_{2,1} & f_{2,2} & f_{2,3} & 0 & 0 \\ f_{3,1} & -f_{2,2} & f_{3,3} & 0 & f_{3,5} \\ f_{4,1} & f_{4,2} & f_{4,3} & f_{4,4} & 0 \\ 0 & 0 & 0 & 0 & f_{5,5} \end{bmatrix}. \quad (14.31)$$

The long-run restriction

$$\frac{\partial \ln r_t}{\partial z_{ad,t}} = -\frac{\partial \ln s_t}{\partial z_{ad,t}} = -f_{2,2},$$

is obtained from assuming that financial assets are priced in the long run at present value levels. This specification contrasts with the long-run restrictions imposed by Rapach (2001) given in Example 14.8.

The SVAR is estimated using quarterly data from March 1980 to June 2005, with the lag length set at $p = 2$ lags. The results of estimating the long-run parameters by maximum likelihood are given in Table 14.1.

Table 14.1

Maximum likelihood estimates of the long-run parameters of the Australian SVAR model. Quasi-maximum likelihood standard errors (se) and p-values (pv) are reported.

Equation	Shock	Parameter	Estimate	se	pv
Output	Aggregate supply	$f_{1,1}$	1.039	0.075	0.000
Interest	Aggregate supply	$f_{2,1}$	4.301	1.158	0.000
	Aggregate demand	$f_{2,2}$	5.181	0.376	0.000
	Australian portfolio	$f_{2,3}$	10.109	0.927	0.000
Aust. equity	Aggregate supply	$f_{3,1}$	2.708	0.535	0.000
	Aggregate demand	$-f_{2,2}$	-5.181	0.376	0.000
	Australian portfolio	$f_{3,3}$	1.429	0.751	0.057
	U.S. portfolio	$f_{3,5}$	3.513	0.525	0.000
Price	Aggregate supply	$f_{4,1}$	-0.712	0.274	0.009
	Aggregate demand	$f_{4,2}$	0.777	0.153	0.000
	Australian portfolio	$f_{4,3}$	1.621	0.223	0.000
	Nominal	$f_{4,4}$	1.349	0.099	0.000
U.S. equity	U.S. portfolio	$f_{5,5}$	10.752	0.809	0.000
$T \ln L_T(\hat{\theta}) = -1132.578$					

Fry, Hocking and Martin (2008) show that the aggregate supply (demand) shock results in a fall (increase) in the price level. The Australian and U.S. portfolio shocks have a wealth effect causing an increase in aggregate demand that results in a permanent increase in the interest rate and a temporary increase in output. The nominal shock causes a fall in the interest rate, known as the liquidity effect, and an increase in the price level through its positive effect on aggregate demand.

14.7 Exercises

(1) Different Ways to Impose Restrictions

Gauss file(s)	<code>svar_bivariate.g</code> , <code>peersman_data.dat</code>
Matlab file(s)	<code>svar_bivariate.m</code> , <code>peersman_data.mat</code>

Consider a model consisting of two variables expressed in first differences, $y_t = [\Delta y_{1,t} \quad \Delta y_{2,t}]'$ and with structural shocks $u_t = [u_{1,t} \quad u_{2,t}]'$. Compute impulse responses of the orthogonalized shocks for the following models where the VAR is estimated with two lags and a constant.

- (a) $S = [3 \ 0, -1 \ 2]$.
- (b) $S = [3 \ 1, 0 \ 2]$.
- (c) $s_{1,1} = 3, s_{2,2} = 2, s_{1,2} = s_{2,1}$.
- (d) $S = \Phi(1)F$, where $F = [3 \ 0, -1 \ 2]$.
- (e) $S = \Phi(1)F$, where $F = [3 \ -1, 0 \ 2]$.
- (f) $S = \Phi(1)F$, where $F = [3 \ 0, 0 \ 2]$.
- (g) $s_{1,1} = 3, s_{2,2} = 2$, $u_{1,t}$ does not affect $y_{2,t}$ in the short run and $u_{2,t}$ does not affect $y_{1,t}$ in the long run.
- (h) $s_{1,1} = 3, s_{2,2} = 2$, $u_{1,t}$ does not affect $y_{2,t}$ in the long run and $u_{2,t}$ does not affect $y_{1,t}$ in the short run.
- (i) SQ' where $S = [3 \ 0, -1 \ 2]$ and Q is given by (14.19) with $\vartheta = 0.4\pi$.
- (j) SQ' where $S = [3 \ 0, -1 \ 2]$ and Q is given by (14.19) with $\vartheta = 0.5\pi$.
- (k) SQ' where $S = [3 \ 0, -1 \ 2]$ and Q is given by (14.19) with $\vartheta = 0.6\pi$.

(2) A Model of Interest Rates with Short-Run Restrictions

Gauss file(s)	<code>svar_shortrun.g</code> , <code>mcnew.dat</code>
Matlab file(s)	<code>svar_shortrun.m</code>

Consider a dynamic system of three interest rates, $r_t = [r_{1,t} \quad r_{2,t} \quad r_{3,t}]'$,

where $r_{1,t}$ is the 0-month yield, $r_{2,t}$ is the 1-month yield and $r_{3,t}$ is the 3-month yield. The structural model is

$$B_0 r_t = B_1 r_{t-1} + D^{1/2} z_t,$$

where the short-run restrictions are

$$B_0 = \begin{bmatrix} 1 & 0 & 0 \\ -b_{2,1} & 1 & 0 \\ 0 & -b_{3,1} & 1 \end{bmatrix}, \quad D = \begin{bmatrix} d_{1,1} & 0 & 0 \\ 0 & d_{2,2} & 0 \\ 0 & 0 & d_{3,3} \end{bmatrix}.$$

Using monthly data on U.S. zero coupon yields from December 1946 to February 1991, estimate the parameters of the model by maximum likelihood and test the number of over-identifying restrictions.

(3) **A Model of Interest Rates with Long-Run Restrictions**

Gauss file(s)	<code>svar_longrun.g, mcnew.dat</code>
Matlab file(s)	<code>svar_longrun.m</code>

Consider a dynamic system of three interest rates $r_t = [r_{1,t} \ r_{2,t} \ r_{3,t}]'$, where $r_{1,t}$ is the 0-month yield, $r_{2,t}$ is the 1-month yield and $r_{3,t}$ is the 3-month yield. The structural model is

$$B_0 r_t = B_1 r_{t-1} + D^{1/2} z_t,$$

where the long-run restrictions are

$$F = \begin{bmatrix} f_{1,1} & 0 & 0 \\ f_{1,1} & f_{2,2} & 0 \\ f_{1,1} & 0 & f_{3,3} \end{bmatrix}.$$

Using monthly data on U.S. zero coupon yields from December 1946 to February 1991, estimate the parameters of the model by maximum likelihood and test the number of over-identifying restrictions.

(4) **The Extended Sims Model**

Gauss file(s)	<code>svar_sims.g</code>
Matlab file(s)	<code>svar_sims.m, simsdata.mat</code>

This is an extension of the four-variable Sims model, consisting of the interest rate, r_t , the logarithm of the exchange rate, $\ln e_t$, the logarithm of commodity prices, $\ln cp_t$, the logarithm of money, $\ln m_t$, the logarithm of prices, $\ln p_t$, and the logarithm of output, $\ln o_t$. The data are monthly beginning January 1959 and ending December 1998.

- (a) Estimate a six-variable VAR with 14 lags, a constant and 11 seasonal dummy variables. Compute the VAR residuals $\hat{v}_t$ and the covariance matrix $\hat{V}$.
- (b) Choosing the ordering of variables as

$$r_t \rightarrow \ln e_t \rightarrow \ln cp_t \rightarrow \ln m_t \rightarrow \ln p_t \rightarrow \ln o_t ,$$

compute B_0 , D and hence $S = B_0^{-1}D$, using maximum likelihood.

(5) The Peersman SVAR Model of Oil Price Shocks

Gauss file(s)	<code>svar_peersman.g, peersman_data.dat</code>
Matlab file(s)	<code>svar_peersman.m, peersman_data.mat</code>

Consider a four-variable model of the U.S. containing the growth rate in the oil price, the growth rate in output, the inflation rate and the interest rate, $y_t = [\Delta \ln oil_t \quad \Delta \ln o_t \quad \Delta \ln p_t \quad r_t]'$. The structural shocks comprise oil price shocks, aggregate supply shocks, aggregate demand shocks and money shocks, $z_t = [z_{oil,t} \quad z_{as,t} \quad z_{ad,t} \quad z_{m,t}]'$.

- (a) Estimate the VAR with $p = 3$ lags, a constant and a time trend using data from June 1979 to June 2002.
- (b) Estimate the SVAR subject to the restrictions:
 - (i) the three non-oil price shocks, $z_{as,t}$, $z_{ad,t}$ and $z_{m,t}$ do not have a contemporaneous effect on oil prices;
 - (ii) the money shock, $z_{m,t}$ has no contemporaneous effect on output;
 - (iii) the aggregate demand shock, $z_{ad,t}$, and the money shock, $z_{m,t}$, have no long-run effect on output.
- (c) Compute the impulse responses and interpret the results.

(6) A Portfolio SVAR Model

Gauss file(s)	<code>svar_port.g, svar_port_u.g, port_data.dat</code>
Matlab file(s)	<code>svar_port.m, svar_port_u.m, port_data.mat</code>

- (a) Estimate a five-variable portfolio SVAR model of the Australian economy where identification is based on the long-run restrictions in (14.31). Compute the value of the constrained log-likelihood function.
- (b) The constrained model is based on the restriction that an aggregate demand shock that raises interest rates results in a matching fall in real equity values in Australia, $f_{3,2} = -f_{2,2}$. Use a LR statistic to test this restriction.

(7) The Blanchard-Quah Model and Okun's Law

Gauss file(s)	svar_bq.g, bq_us.dat, bq_uk.dat, bq_jp.dat
Matlab file(s)	svar_bq.m, bq_us.mat, bq_uk.mat, bq_jp.mat

The Blanchard and Quah (1989) model is a bivariate SVAR consisting of the growth rate in output and the unemployment rate, $y_t = [\Delta \ln o_t \text{ } ue_t]'$. The one-standard deviation structural shocks correspond to aggregate supply and aggregate demand, $z_t = [z_{as,t} \text{ } z_{ad,t}]'$, which are identified by the long-run restriction that aggregate demand shocks have no long-run effect on output

$$F = \begin{bmatrix} f_{1,1} & 0 \\ f_{2,1} & f_{2,2} \end{bmatrix}.$$

- Estimate an SVAR for the U.S. using quarterly data over the period March 1950 to September 2009, where the VAR is specified to have $p = 8$ lags and a constant. Interpret the parameter estimates of F .
- Compute impulse responses for positive shocks to aggregate supply and demand. Interpret the patterns of the impulse responses and compare the results to the impulses reported by Blanchard and Quah in Figures 1 to 6, which are based on the smaller sample of 1950 to 1987. *In computing the impulses for positive shocks, it may be necessary to change the sign on some of the diagonal elements of the long-run restriction matrix, F , since a positive demand shock, for example, should result in a decrease in unemployment and an increase in output.*
- Okun's Law predicts that a 1% decrease in the unemployment rate corresponds to a 3% increase in output (see also Blanchard and Quah (1989, p.663), who choose 2.5%). Interpreting this as a long-run relationship, use the estimates of the long-run matrix, F , to construct a Wald test of Okun's Law based first on supply shocks and then also on aggregate demand shocks.
- Re-estimate the SVAR with the additional restriction that Okun's Law holds.
- Repeat parts (a) to (d) for the U.K. using quarterly data over the period June 1971 to September 2009.
- Repeat parts (a) to (d) for Japan using quarterly data over the period March 1980 to September 2009.

(8) Sign Restrictions

Gauss file(s)	svar_sign.g, sign.dat
Matlab file(s)	svar_sign.m, sign.mat

Consider a bivariate SVAR consisting of the annualized quarterly growth rates in output and prices, $y_t = [\Delta \ln o_t \ \Delta \ln p_t]'$. Sign restrictions are used to identify aggregate supply and aggregate demand shocks, $z_t = [z_{as,t} \ z_{ad,t}]'$, where an aggregate supply shock has a positive effect on output and a negative effect on prices, whereas an aggregate demand shock has a positive effect on both output and prices.

- (a) Estimate a VAR for the U.S. using quarterly data over the period March 1950 to December 2009, where the VAR is specified to have $p = 2$ lags and a constant and a time trend. Show that the residual covariance matrix is

$$\hat{V} = \begin{bmatrix} 12.172 & -0.309 \\ -0.309 & 5.923 \end{bmatrix}.$$

- (b) Use the estimate of $\hat{V}$ in part (a) to compute S based on a Choleski decomposition. Hence, show that the impulse responses do not conform with shocks corresponding to aggregate supply and aggregate demand.
- (c) Redo the impulse responses by redefining S as SQ' , where Q is defined in (14.19) with $\vartheta = 0.2\pi$. Show that the impulse responses now correspond to aggregate supply and aggregate demand shocks.
- (d) Generate 10000 simulated values of ϑ from the interval $[0, \pi]$ and select the set of impulse responses for each draw that satisfy the sign restrictions. Compute the median impulse responses from this set. In computing the median it is important to ensure that the four impulses chosen come from the same model, otherwise the shocks are not orthogonalized. *The approach adopted is to extract the contemporaneous impulse values from the selected impulses and group these values into the 4 impulse sets. These are each standardized by subtracting the respective median and dividing by the respective standard deviation. The median set of impulses is taken as the model that has the smallest absolute deviation from summing across the four impulses.*

15

Latent Factor Models

15.1 Introduction

The simplest representation of the models specified and estimated so far is that of an observed dependent variable, y_t , expressed as a function of an observable explanatory variable, x_t , and an unobservable disturbance term, u_t . For a linear specification this relationship is

$$y_t = \beta x_t + u_t . \quad (15.1)$$

In this chapter, an important class of model is introduced that relaxes the assumption that the explanatory variable, x_t , is observed. To highlight this change in the status of the explanatory variable, the model is rewritten as

$$y_t = \lambda s_t + u_t , \quad (15.2)$$

where s_t represents an unobserved or latent factor. This model is easily extended to include N dependent variables, $y_t = \{y_{1,t}, \dots, y_{N,t}\}$, and K latent factors, $s_t = \{s_{1,t}, \dots, s_{K,t}\}$, where $K < N$.

The latent factor class of model is encountered frequently in economics and finance and prominent examples include multi-factor models of the term structure of interest rates, dating of business cycles, the identification of *ex ante* real interest rates, real business cycle models with technology shocks, the capital asset pricing model and models of stochastic volatility. A number of advantages stem from being able to identify the existence of a latent factor structure underlying the behaviour of economic and financial time series. First, it provides a parsimonious way of modelling dynamic multivariate systems. As seen in Chapter 13, the curse of dimensionality arises even in relatively small systems of unrestricted VARs. Second, it avoids the need to use *ad hoc* proxy variables for the unobservable variables which can result in biased and inconsistent parameters estimates. Examples of this include

the capital asset pricing model and the identification of *ex ante* real interest rates.

Despite the fact that it is latent, the factor s_t in equation (15.2) can nonetheless be characterized via the time series properties of the observed dependent variables. The system of equations capturing this structure is commonly referred to as state-space form and the technique that enables this identification and extraction of latent factors is known as the Kalman filter. An important assumption underlying the Kalman filter is that of normality of the disturbance terms impacting on the system. The assumption of normality makes it possible to summarize the entire distribution of the latent factors using conditional means and variances alone. Consequently, these moments feature prominently in the derivations of the recursions that define the Kalman filter. For the case in which the assumption of normality of the disturbance terms is inappropriate, a quasi-maximum likelihood estimator may be available. The details of this estimator are discussed in Chapter 9. An alternative approach to estimating this class of models is the generalized method of moments discussed in Chapter 10. Bayesian approaches to estimating state-space models are dealt with by Carter and Kohn (1994) and Stroud, Muller and Polson (2003).

15.2 Motivating Examples

To motivate the Kalman filter, two sets of examples are explored.

15.2.1 Empirical

The first set of examples is essentially empirical and highlights the usefulness of the filter in overcoming the curse of dimensionality.

Example 15.1 Term Structure of Interest Rates

Figure 15.1 plots U.S. daily zero coupon rates from 4 October 1988 to 28 December 2001. The maturities are, respectively, three months, $r_{1,t}$, one year, $r_{2,t}$, three years, $r_{3,t}$, five years, $r_{4,t}$, seven years, $r_{5,t}$, and ten years, $r_{6,t}$. Inspection of the figure shows that all the yields exhibit very similar dynamics, behaviour that suggest that they are all driven by one or two unobservable factors.

□

In the case of a single latent factor, $K = 1$, the term structure model for interest rates is

$$r_{i,t} = \lambda_i s_t + u_{i,t} \quad i = 1, \dots, 6, \quad (15.3)$$

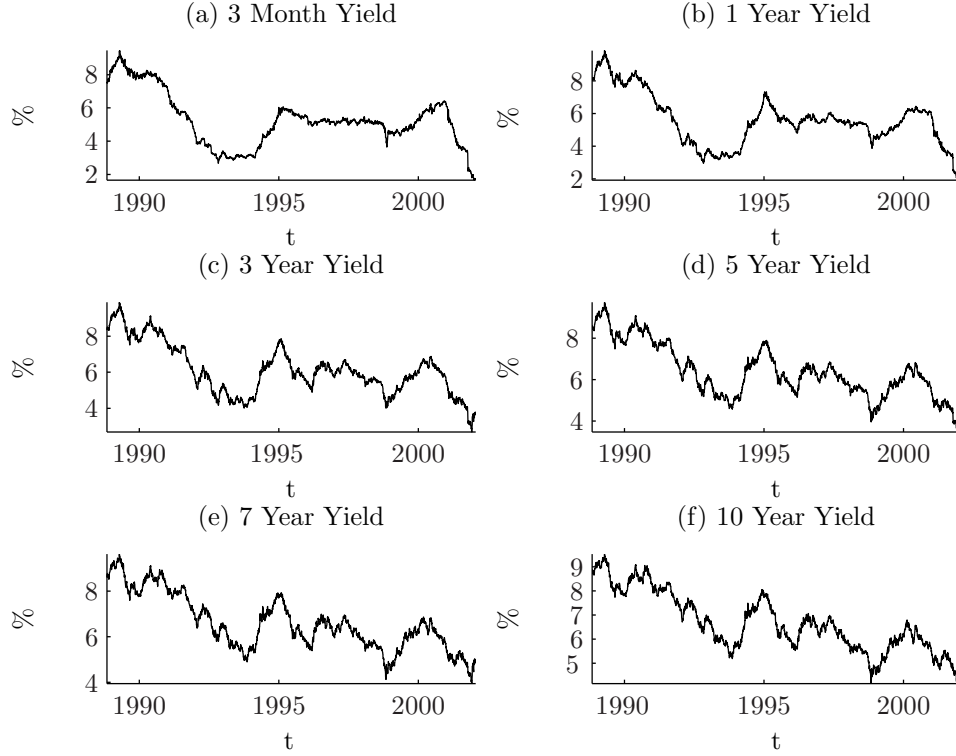


Figure 15.1 Daily U.S. zero coupon rates from 4 October 1988 to 28 December 2001.

where s_t is a latent factor and $u_{i,t} \sim iid N(0, \sigma_i^2)$. The parameter λ_i controls the strength of the relationship between the i^{th} yield and the factor. These parameters are commonly referred to as the factor loadings and this equation is commonly referred to as the measurement equation. The disturbance term, $u_{i,t}$, allows for idiosyncratic movements in the i^{th} yield which are not explained by movements in the factor. The strength of these idiosyncratic movements are controlled by the parameters σ_i . This set of equations can be written as a single equation in matrix form as

$$r_t = \Lambda s_t + u_t, \quad (15.4)$$

where r_t is the column vector of yields, u_t is the column vector of idiosyncratic terms and

$$\Lambda = \begin{bmatrix} \lambda_1 & \lambda_2 & \lambda_3 & \lambda_4 & \lambda_5 & \lambda_6 \end{bmatrix}'. \quad (15.5)$$

To complete the specification of the model in equation (15.3), it remains

only to specify the dynamics of the latent factor, s_t . As the disturbance terms are independent, any dynamics displayed by the yields must be captured by the dynamics of the factor. The simplest representation is to let the factor follow a first-order autoregressive process

$$s_t = \phi s_{t-1} + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2), \quad (15.6)$$

where ϕ is the autoregressive parameter and η_t is a disturbance term. This equation is commonly known as the state equation and together with the measurement equation (15.3) the system is known as a state-space system.

Example 15.2 Business Cycle Dating

A common approach to identifying the peaks and troughs of a business cycle is to look at the turning points of a range of indicators of economic activity. Typical examples of indicators include GDP growth rates, $I_{1,t}$, changes in unemployment, $I_{2,t}$, movements in employment, $I_{3,t}$, and retail sales, $I_{4,t}$. The business cycle is unobservable and is to be inferred from these observable indicators. $\square$

The business cycle example suggests the following model for the indicators, $I_{i,t}$,

$$I_{i,t} = \lambda_i s_t + u_{i,t}, \quad i = 1, \dots, 4, \quad (15.7)$$

where s_t is the latent business cycle and $u_{i,t}$ is a disturbance term. The dynamics of the business cycle are represented by the second-order autoregressive model

$$s_t = \phi_1 s_{t-1} + \phi_2 s_{t-2} + \eta_t, \quad \eta_t \sim iid N(0, \sigma_\eta^2). \quad (15.8)$$

In addition, the assumption that the $u_{i,t}$'s are independent can be relaxed to allow additional dynamics over and above the dynamics induced by the business cycle s_t , by specifying

$$u_{i,t} = \delta_i u_{i,t-1} + w_{i,t}, \quad w_{i,t} \sim iid N(0, \sigma_{w,i}^2). \quad (15.9)$$

15.2.2 Theoretical

The second set of examples is more theoretical in nature and illustrates the importance of being able to deal effectively with latent factors in econometrics.

Example 15.3 Capital Asset Pricing Model

The capital asset pricing model (CAPM) regression equation is

$$e_{i,t} = \alpha_i + \beta_i e_{m,t} + u_{i,t},$$

where $e_{i,t}$ is the observed excess return on an asset, $e_{m,t}$ is the observed excess return on the market portfolio, β_i is known as the beta of the i^{th} asset and $u_{i,t}$ is the disturbance term with zero mean and variance σ_i^2 . In theory, the CAPM equation should be specified in terms of the excess return on all invested wealth. As this an unobservable variable, the excess return on the market portfolio, $e_{m,t}$, is essentially serving as a proxy. $\square$

An alternative approach in the CAPM that avoids the use of a proxy variable is to treat the return on wealth as a latent factor, s_t , and re-specify the model as

$$e_{i,t} = \alpha_i + \beta_i s_t + u_{i,t}, \quad i = 1, \dots, N. \quad (15.10)$$

An important feature of this model is that aggregate wealth is identified by the dynamics of the excess returns on all of the assets in the system. The model is completed by specifying the dynamics of the excess return on wealth, s_t .

Example 15.4 *Ex Ante* Real Interest Rates

In economic models, decisions are made on *ex ante* real interest rates that are unobserved because they are based on the expected inflation rate. The relationship between the *ex post* and *ex ante* real interest rates is as follows. The *ex post* real interest rate is

$$r_t = i_t - \pi_t,$$

where i_t is the nominal interest rate and π_t is the observed inflation rate. Expanding this expression to allow for a constant α and expected inflation, π_t^e , gives

$$\begin{aligned} r_t &= \alpha + i_t - \pi_t^e - \alpha + \pi_t^e - \pi_t \\ r_t &= \alpha + s_t + u_t, \end{aligned}$$

where $s_t = i_t - \pi_t^e - \alpha$ is now the *ex ante* real interest rate and $u_t = \pi_t^e - \pi_t$ is the inflation expectations error. $\square$

15.3 The Recursions of the Kalman Filter

The Kalman filter is an algorithm to form predictions of the latent factors based on their conditional means and then to update these predictions in a systematic fashion as more measurements of the observed variables become available. Presenting both univariate ($N = K = 1$) and multivariate ($N > K \geq 1$) versions of the Kalman filter highlights the recursive structure and other properties of this algorithm. In deriving the algorithm, it is assumed

that the parameters of the model are known. This assumption is relaxed in Section 15.6.

15.3.1 Univariate

Consider the univariate model relating a single observed dependent variable, y_t , to a single latent factor, s_t , given by

$$y_t = \lambda s_t + u_t \quad (15.11)$$

$$s_t = \phi s_{t-1} + \eta_t, \quad (15.12)$$

where the disturbances are distributed as

$$u_t \sim N(0, \sigma^2) \quad (15.13)$$

$$\eta_t \sim N(0, \sigma_\eta^2), \quad (15.14)$$

where $E[u_t \eta_t] = 0$ and λ , ϕ , σ^2 and σ_η^2 are known parameters.

Prediction

Consider forming a prediction of s_t using information at time $t-1$, defined as $s_{t|t-1} = E_{t-1}[s_t]$. This can be achieved by taking conditional expectations of equation (15.12)

$$\begin{aligned} s_{t|t-1} &= E_{t-1}[s_t] \\ &= E_{t-1}[\phi s_{t-1} + \eta_t] \\ &= E_{t-1}[\phi s_{t-1}] + E_{t-1}[\eta_t] \\ &= \phi E_{t-1}[s_{t-1}] \\ &= \phi s_{t-1|t-1}, \end{aligned} \quad (15.15)$$

where $E_{t-1}[\eta_t] = 0$ by assumption and by definition

$$s_{t-1|t-1} = E_{t-1}[s_{t-1}]. \quad (15.16)$$

The conditional variance of s_t given information at time $t-1$ is by definition

$$P_{t|t-1} = E_{t-1}[(s_t - s_{t|t-1})^2]. \quad (15.17)$$

This expression can be obtained by subtracting $s_{t|t-1}$, given in equation (15.15), from both sides of (15.12) to give

$$s_t - s_{t|t-1} = \phi(s_{t-1} - s_{t-1|t-1}) + \eta_t. \quad (15.18)$$

Squaring and taking expectations conditional on information at time $t-1$

gives

$$\begin{aligned}
 P_{t|t-1} &= E_{t-1} [(s_t - s_{t|t-1})^2] \\
 &= \phi^2 E_{t-1} [(s_{t-1} - s_{t-1|t-1})^2] + E_{t-1} [\eta_t^2] \\
 &\quad + 2\phi E_{t-1} [(s_{t-1} - s_{t-1|t-1})\eta_t] \\
 &= \phi^2 P_{t-1|t-1} + \sigma_\eta^2,
 \end{aligned}$$

where

$$\begin{aligned}
 P_{t-1|t-1} &= E_{t-1} [(s_{t-1} - s_{t-1|t-1})^2] \\
 E_{t-1} [(s_{t-1} - s_{t-1|t-1})\eta_t] &= (s_{t-1} - s_{t-1|t-1})E_{t-1} [\eta_t] = 0 \\
 E_{t-1} [\eta_t^2] &= \sigma_\eta^2.
 \end{aligned}$$

The expressions for the conditional mean and variance of s_t given information at $t-1$ require values for $s_{t-1|t-1}$ and $P_{t-1|t-1}$, respectively. This highlights the iterative nature of the Kalman filter algorithm as $s_{t-1|t-1}$ and $P_{t-1|t-1}$ will be available from the updating phase of the previous iteration of the filter as outlined below.

Observation

From (15.11), the conditional expectation of y_t based on information at time $t-1$ is

$$\begin{aligned}
 \mu_{t|t-1} &= E_{t-1} [y_t] \\
 &= E_{t-1} [\lambda s_t + u_t] \\
 &= \lambda E_{t-1} [s_t] + E_{t-1} [u_t] \\
 &= \lambda E_{t-1} [s_t] \\
 &= \lambda s_{t|t-1},
 \end{aligned} \tag{15.19}$$

where $E_{t-1} [u_t] = 0$, by assumption, and $s_{t|t-1}$ is available from the prediction phase.

The conditional variance of y_t is derived by defining the one-step ahead conditional prediction error by subtracting (15.19) from (15.11)

$$u_{t|t-1} = y_t - \mu_{t|t-1} \tag{15.20}$$

$$\begin{aligned}
 &= \lambda s_t + u_t - \lambda s_{t|t-1} \\
 &= \lambda(s_t - s_{t|t-1}) + u_t.
 \end{aligned} \tag{15.21}$$

The conditional variance of y_t is then

$$\begin{aligned}
 V_{t|t-1} &= E_{t-1} [(y_t - \mu_{t|t-1})^2] \\
 &= \lambda^2 E_{t-1} [(s_t - s_{t|t-1})^2] + E_{t-1} [u_t^2] \\
 &= \lambda^2 P_{t|t-1} + \sigma^2,
 \end{aligned} \tag{15.22}$$

where $E_{t-1}[u_t^2] = \sigma^2$, by assumption, and $P_{t|t-1}$ is available from the prediction phase.

Updating

The conditional forecast of the factor s_t in (15.15) is based on information at time $t - 1$. However, y_t is available at time t so an improved forecast of s_t can be derived using this information. To achieve this objective, consider the regression equation that relates the errors in forecasting s_t and y_t using information at time $t - 1$, namely

$$s_t - s_{t|t-1} = \kappa(y_t - \mu_{t|t-1}) + \zeta_t, \quad (15.23)$$

where ζ_t is a disturbance term and κ is a parameter to be identified below. This equation can be rearranged as

$$\begin{aligned} s_t &= s_{t|t-1} + \kappa(y_t - \mu_{t|t-1}) + \zeta_t \\ &= s_{t|t} + \zeta_t, \end{aligned}$$

where

$$s_{t|t} = s_{t|t-1} + \kappa(y_t - \mu_{t|t-1}), \quad (15.24)$$

represents the conditional expectation of s_t at time t using all of the information up to and including time t .

Equation (15.23) represents a least squares regression where $s_t - s_{t|t-1}$ is the dependent variable, $y_t - \mu_{t|t-1}$ is the explanatory variable and κ is the unknown parameter. The fit of the regression line is given by (15.24). It follows immediately that κ is given by

$$\kappa = \frac{E_{t-1}[(s_t - s_{t|t-1})(y_t - \mu_{t|t-1})]}{E_{t-1}[(y_t - \mu_{t|t-1})^2]}. \quad (15.25)$$

The numerator of equation (15.25) is simplified by substituting for $y_t - \mu_{t|t-1}$ using equation (15.21) to give

$$\begin{aligned} E_{t-1}[(s_t - s_{t|t-1})(y_t - \mu_{t|t-1})] &= E_{t-1}[(s_t - s_{t|t-1})(\lambda(s_t - s_{t|t-1}) + u_t)] \\ &= \lambda E_{t-1}[(s_t - s_{t|t-1})^2] \\ &\quad + E_{t-1}[(s_t - s_{t|t-1})u_t] \\ &= \lambda P_{t|t-1}. \end{aligned} \quad (15.26)$$

The simplification in equation (15.26) uses the definition of $P_{t|t-1}$ given in equation (15.17) and also equation (15.18) to deduce that

$$\begin{aligned} E_{t-1}[(s_t - s_{t|t-1})u_t] &= E_{t-1}[(\phi(s_{t-1} - s_{t-1|t-1}) + \eta_t)u_t] \\ &= \phi(s_{t-1} - s_{t-1|t-1})E_{t-1}[u_t] + E_{t-1}[\eta_t u_t] = 0. \end{aligned}$$

Now using equation (15.26) as the numerator in equation (15.25) and equation (15.22) as the denominator gives

$$\kappa = \frac{\lambda P_{t|t-1}}{V_{t|t-1}}. \quad (15.27)$$

This term is commonly referred to as the Kalman gain. Thus (15.24) can be rewritten as

$$\begin{aligned} s_{t|t} &= s_{t|t-1} + \kappa(y_t - E_{t-1}[y_t]) \\ &= s_{t|t-1} + \frac{\lambda P_{t|t-1}}{V_{t|t-1}}(y_t - \mu_{t|t-1}) \\ &= s_{t|t-1} + \frac{\lambda P_{t|t-1}}{V_{t|t-1}}(y_t - \lambda s_{t|t-1}), \end{aligned} \quad (15.28)$$

where the last step is based on using $\mu_{t|t-1} = \lambda s_{t|t-1}$ in (15.19).

The conditional variance of s_t based on information at time t is

$$P_{t|t} = E_{t-1}[(s_t - s_{t|t})^2].$$

Using equation (15.28) for $s_{t|t}$ and expanding gives

$$\begin{aligned} P_{t|t} &= E_{t-1} \left[\left(s_t - s_{t|t-1} - \frac{\lambda P_{t|t-1}}{V_{t|t-1}} (y_t - \lambda s_{t|t-1}) \right)^2 \right] \\ &= E_{t-1} \left[(s_t - s_{t|t-1})^2 + \left(\frac{\lambda P_{t|t-1}}{V_{t|t-1}} (y_t - \lambda s_{t|t-1}) \right)^2 \right. \\ &\quad \left. - 2 \frac{\lambda P_{t|t-1}}{V_{t|t-1}} (s_t - s_{t|t-1}) (y_t - \lambda s_{t|t-1}) \right] \\ &= E_{t-1} [(s_t - s_{t|t-1})^2] + \left(\frac{\lambda P_{t|t-1}}{V_{t|t-1}} \right)^2 E_{t-1} [(y_t - \lambda s_{t|t-1})^2] \\ &\quad - 2 \frac{\lambda P_{t|t-1}}{V_{t|t-1}} E_{t-1} [(s_t - s_{t|t-1}) (y_t - \lambda s_{t|t-1})]. \end{aligned}$$

Given the conditional expectations

$$\begin{aligned} E_{t-1} [(s_t - s_{t|t-1})^2] &= P_{t|t-1} \\ E_{t-1} [(y_t - \lambda s_{t|t-1})^2] &= V_{t|t-1} \\ E_{t-1} [(s_t - s_{t|t-1}) (y_t - \lambda s_{t|t-1})] &= \lambda P_{t|t-1}, \end{aligned}$$

where the last result uses equation (15.26) with $\mu_{t|t-1} = \lambda s_{t|t-1}$, the updated

conditional variance of the factor is simply

$$P_{t|t} = P_{t|t-1} - \frac{\lambda^2 P_{t|t-1}^2}{V_{t|t-1}}.$$

Iterating

Now consider predicting y_{t+1} using information at time t . From (15.19) this prediction requires $s_{t+1|t}$ which from (15.15) is based on $s_{t|t}$ where the latter is computed from the previous step given in (15.28). This means that the entire sequence of one-step-ahead predictions can be constructed for both the observable variable, y_t , and the unobservable factor, s_t .

Initialization

For the first observation, $t = 1$, the algorithm requires some starting values for $s_{1|0}$ and $P_{1|0}$. The simplest approach is to assume stationarity and set the starting values equal to the unconditional moments of s_t obtained from equation (15.12). These are, respectively,

$$s_{1|0} = 0$$

$$P_{1|0} = \frac{1}{1 - \phi^2}.$$

Example 15.5 Recursions of the Univariate Kalman Filter

The recursive features of the Kalman filter algorithm for the model in equations (15.11) to (15.14) are highlighted with a simple numerical example in Table 15.1. The parameters are specified as $\lambda = 1.0$, $\phi = 0.8$, $\sigma^2 = 0.5^2$ and $\sigma_\eta^2 = 1^2$. The sample size is $T = 5$, with the actual values of y_t given in column (2) of Table 15.1.

The numerical calculations of the recursions of the Kalman filter proceed as follows.

(1) Prediction: t=1 (Initialization)

The starting values for the conditional mean and variance of the factor at $t = 1$ are given in columns (3) and (4),

$$s_{1|0} = 0.0,$$

$$P_{1|0} = \frac{1}{1 - \phi^2} = \frac{1}{1 - 0.8^2} = 2.778.$$

(2) Observation: t=1

The conditional mean and variance of y_1 are given in columns (5) and

(6), respectively,

$$\begin{aligned}\mu_{1|0} &= \lambda s_{1|0} = 1.0 \times 0.0 = 0.0, \\ V_{1|0} &= \lambda^2 P_{1|0} + \sigma^2 = 1.0^2 \times 2.778 + 0.5^2 = 3.028,\end{aligned}$$

while the forecast error of y_1 is given in column (7),

$$u_{1|0} = y_1 - \mu_{1|0} = -0.680 - 0.0 = -0.680.$$

(3) Updating: t=1

The update of the mean and the variance of the factor are given in columns (8) and (9). These are

$$\begin{aligned}s_{1|1} &= s_{1|0} + \frac{\lambda P_{1|0}}{V_{1|0}}(y_1 - \mu_{1|0}) \\ &= 0.0 + \frac{1.0 \times 2.778}{3.028}(-0.680 - 0.0) = -0.624, \\ P_{1|1} &= P_{1|0} - \frac{\lambda^2 P_{1|0}^2}{V_{1|0}} \\ &= 2.778 - \frac{1.0^2 \times 2.778^2}{3.028} = 0.229.\end{aligned}$$

(4) Prediction: t=2

From columns (3) and (4),

$$\begin{aligned}s_{2|1} &= \phi s_{1|1} = 0.8 \times (-0.624) = -0.499, \\ P_{2|1} &= \phi^2 P_{1|1} + 1 = 0.8^2 \times 0.229 + 1.0 = 1.147.\end{aligned}$$

(5) Repeat calculations for $\mu_{2|1}, V_{2|1}, s_{2|2}, P_{2|2}$, etc.

(6) Repeat calculations for $t = 3, 4, 5$.

□

15.3.2 Multivariate

Consider the case of N variables $\{y_{1,t}, \dots, y_{N,t}\}$ and K factors $\{s_{1,t}, \dots, s_{K,t}\}$. The multivariate version of the state-space system is

$$\begin{aligned}y_t &= \Lambda s_t + u_t \\ s_t &= \Phi s_{t-1} + \eta_t,\end{aligned}\tag{15.29}$$

Table 15.1

Numerical illustration of the recursions of the Kalman filter.
Parameters are $\lambda = 1.0$, $\sigma = 0.5$ and $\phi = 0.8$.

Data		Prediction		Observation			Updating		Like.
t	y_t	$s_{t t-1}$	$P_{t t-1}$	$\mu_{t t-1}$	$V_{t t-1}$	$u_{t t-1}$	$s_{t t}$	$P_{t t}$	$\ln l_t$
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
1	-0.680	0.000	2.778	0.000	3.028	-0.680	-0.624	0.229	-1.549
2	0.670	-0.499	1.147	-0.499	1.397	1.169	0.461	0.205	-1.576
3	0.012	0.369	1.131	0.369	1.381	-0.356	0.077	0.205	-1.126
4	-0.390	0.062	1.131	0.062	1.381	-0.451	-0.308	0.205	-1.154
5	-1.477	-0.246	1.131	-0.246	1.381	-1.231	-1.255	0.205	-1.629

where the disturbances are distributed as

$$\begin{aligned} u_t &\sim N(0, R) \\ \eta_t &\sim N(0, Q), \end{aligned} \quad (15.30)$$

where

$$E[u_t u_t'] = R, \quad E[\eta_t \eta_t'] = Q. \quad (15.31)$$

The dimensions of the parameter matrices are as follows: Λ is $(N \times K)$, Φ is $(K \times K)$, R is $(N \times N)$ and Q is $(K \times K)$.

The recursions of the multivariate Kalman filter are as follows.

(1) **Prediction:**

$$\begin{aligned} s_{t|t-1} &= \Phi s_{t-1|t-1} \\ P_{t|t-1} &= \Phi P_{t-1|t-1} \Phi' + Q. \end{aligned}$$

(2) **Observation:**

$$\begin{aligned} \mu_{t|t-1} &= \Lambda s_{t|t-1} \\ V_{t|t-1} &= \Lambda P_{t|t-1} \Lambda' + R \\ u_{t|t-1} &= y_t - \mu_{t|t-1}. \end{aligned}$$

(3) Updating:

$$s_{t|t} = s_{t|t-1} + P_{t|t-1} \Lambda' V_{t|t-1}^{-1} (y_t - \mu_{t|t-1})$$

$$P_{t|t} = P_{t|t-1} - P_{t|t-1} \Lambda' V_{t|t-1}^{-1} \Lambda P_{t|t-1},$$

where the Kalman gain is given by

$$\mathcal{K}_t = P_{t|t-1} \Lambda' V_{t|t-1}^{-1}.$$

To start the recursion, the initial values $s_{1|0}$ and $P_{1|0}$ for the multivariate K factor model, assuming stationarity, are

$$s_{1|0} = 0, \quad (15.32)$$

$$\text{vec}(P_{1|0}) = (I_{K \times K} - (\Phi \otimes \Phi))^{-1} \text{vec}(Q). \quad (15.33)$$

Example 15.6 Recursions of the Multivariate Kalman Filter

Consider a factor model as in equations (15.29) to (15.31), with three observable variables ($N = 3$), two factors ($K = 2$) and parameter matrices

$$\Lambda = \begin{bmatrix} 1.00 & 0.50 \\ 1.00 & 0.00 \\ 1.00 & -0.50 \end{bmatrix}, \quad \Phi = \begin{bmatrix} 0.80 & 0.00 \\ 0.00 & 0.50 \end{bmatrix},$$

$$R = \begin{bmatrix} 0.25 & 0.00 & 0.00 \\ 0.00 & 0.16 & 0.00 \\ 0.00 & 0.00 & 0.09 \end{bmatrix}, \quad Q = I_2.$$

For $T = 5$ observations, the variables are

$$y_1 = \{1.140, 2.315, -0.054, -1.545, -0.576\}$$

$$y_2 = \{3.235, 0.552, -0.689, 1.382, 0.718\}$$

$$y_3 = \{1.748, 1.472, -1.413, -0.199, 1.481\}.$$

The Kalman filter iterations proceed as follows.

(1) Prediction: t=1 (Initialization)

$$s_{1|0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{vec}(P_{1|0}) = \left(I_4 - \left(\begin{bmatrix} 0.80 & 0.00 \\ 0.00 & 0.50 \end{bmatrix} \otimes \begin{bmatrix} 0.80 & 0.00 \\ 0.00 & 0.50 \end{bmatrix} \right) \right)^{-1} \text{vec}(I_2),$$

so that

$$P_{1|0} = \begin{bmatrix} 2.778 & 0.000 \\ 0.000 & 1.333 \end{bmatrix}.$$

(2) **Observation: t=1**

$$\begin{aligned} \mu_{t|t-1} &= \Lambda s_{t|t-1} \\ &= \begin{bmatrix} 1.00 & 0.50 \\ 1.00 & 0.00 \\ 1.00 & -0.50 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} V_{t|t-1} &= \Lambda P_{t|t-1} \Lambda' + R \\ &= \begin{bmatrix} 1.00 & 0.50 \\ 1.00 & 0.00 \\ 1.00 & -0.50 \end{bmatrix} \begin{bmatrix} 2.778 & 0.000 \\ 0.000 & 1.333 \end{bmatrix} \begin{bmatrix} 1.00 & 0.50 \\ 1.00 & 0.00 \\ 1.00 & -0.50 \end{bmatrix}' + \text{diag} \begin{pmatrix} 0.25 \\ 0.16 \\ 0.09 \end{pmatrix} \\ &= \begin{bmatrix} 3.361 & 2.778 & 2.444 \\ 2.778 & 2.938 & 2.778 \\ 2.444 & 2.778 & 3.201 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} u_{t|t-1} &= y_t - \mu_{t|t-1} \\ &= \begin{bmatrix} 2.5 \\ 2.0 \\ 1.5 \end{bmatrix} - \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 2.0 \\ 1.5 \end{bmatrix}. \end{aligned}$$

(3) **Updating: t=1**

$$\begin{aligned} s_{t|t} &= s_{t|t-1} + P_{t|t-1} \Lambda' V_{t|t-1}^{-1} (y_t - \mu_{t|t-1}) \\ &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 2.778 & 0.000 \\ 0.000 & 1.333 \end{bmatrix} \begin{bmatrix} 1.00 & 0.50 \\ 1.00 & 0.00 \\ 1.00 & -0.50 \end{bmatrix}' \\ &\quad \times \begin{bmatrix} 3.361 & 2.778 & 2.444 \\ 2.778 & 2.938 & 2.778 \\ 2.444 & 2.778 & 3.201 \end{bmatrix}^{-1} \begin{bmatrix} 1.140 \\ 3.235 \\ 1.748 \end{bmatrix} \\ &= \begin{bmatrix} 2.027 \\ -0.049 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
P_{t|t} &= P_{t|t-1} - P_{t|t-1} \Lambda' V_{t|t-1}^{-1} \Lambda P_{t|t-1} \\
&= \begin{bmatrix} 2.778 & 0.000 \\ 0.000 & 1.333 \end{bmatrix} - \begin{bmatrix} 2.778 & 0.000 \\ 0.000 & 1.333 \end{bmatrix} \begin{bmatrix} 1.00 & 0.50 \\ 1.00 & 0.00 \\ 1.00 & -0.50 \end{bmatrix}' \\
&\times \begin{bmatrix} 3.361 & 2.778 & 2.444 \\ 2.778 & 2.938 & 2.778 \\ 2.444 & 2.778 & 3.201 \end{bmatrix}^{-1} \begin{bmatrix} 1.00 & 0.50 \\ 1.00 & 0.00 \\ 1.00 & -0.50 \end{bmatrix} \begin{bmatrix} 2.778 & 0.000 \\ 0.000 & 1.333 \end{bmatrix} \\
&= \begin{bmatrix} 0.053 & 0.041 \\ 0.041 & 0.253 \end{bmatrix}.
\end{aligned}$$

(4) Repeat calculations for $t = 2, 3, 4, 5$.

□

15.4 Extensions

The derivation of the Kalman filter in Section 15.3 is based on a state-space model in which the latent factors have zero mean, AR(1) dynamics which are stationary and no exogenous or predetermined variables. These assumptions are now relaxed.

15.4.1 Intercepts

The state-space representation can be extended to allow for intercepts as in the case of the expected real interest rate example discussed in Section 15.2. For example, in the case of the univariate model, the model is now rewritten as

$$\begin{aligned}
y_t &= \lambda_0 + \lambda s_t + u_t \\
s_t &= \phi_0 + \phi s_{t-1} + \eta_t.
\end{aligned}$$

If the factor is specified to have a zero mean, $\phi_0 = 0$, then λ_0 is the sample mean of y_t . This implies that the number of unknown parameters in the model can be reduced by subtracting the sample mean from y_t .

15.4.2 Dynamics

The focus of the dynamics so far has been a first-order autoregressive representation for s_t with the idiosyncratic disturbances u_t being white noise. An exception is the business cycle example where the factor is specified as

an AR(2) process, and the disturbance, u_t , is AR(1). To allow for greater generality in the dynamics, consider two extensions of the state-space equation.

(1) AR(2) State Dynamics

Consider the business cycle model in equations (15.7) and (15.8), where the latent factor follows the AR(2) process

$$s_t = \phi_1 s_{t-1} + \phi_2 s_{t-2} + \eta_t.$$

This equation is rewritten as a vector AR(1) model

$$\begin{bmatrix} s_t \\ s_{t-1} \end{bmatrix} = \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} s_{t-1} \\ s_{t-2} \end{bmatrix} + \begin{bmatrix} \eta_t \\ 0 \end{bmatrix}.$$

The Kalman filter proceeds, as before, except now there are two factors, s_t and s_{t-1} , with

$$\Phi = \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{bmatrix}.$$

Further, the loading parameter is expanded to accommodate the additional factor as

$$y_t = \begin{bmatrix} \lambda & 0 \end{bmatrix} \begin{bmatrix} s_t \\ s_{t-1} \end{bmatrix} + u_t.$$

(2) AR(p) State Dynamics

For an AR(p) model

$$s_t = \phi_1 s_{t-1} + \phi_2 s_{t-2} + \cdots + \phi_p s_{t-p} + \eta_t,$$

the state equation is written as

$$\begin{bmatrix} s_t \\ s_{t-1} \\ s_{t-2} \\ \vdots \\ s_{t-p+1} \end{bmatrix} = \begin{bmatrix} \phi_1 & \phi_2 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix} \begin{bmatrix} s_{t-1} \\ s_{t-2} \\ s_{t-3} \\ \vdots \\ s_{t-p} \end{bmatrix} + \begin{bmatrix} \eta_t \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

The model can be viewed as having p factors $\{s_t, s_{t-1}, \dots, s_{t-p+1}\}$, although it is really just the first element of this set of factors that is of interest. The measurement equation now becomes

$$y_t = \begin{bmatrix} \lambda & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} s_t \\ s_{t-1} \\ \vdots \\ s_{t-p} \end{bmatrix} + \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

(3) Idiosyncratic Dynamics

Consider the extended business cycle model in equations (15.7) to (15.9)

$$\begin{aligned} y_{i,t} &= \lambda_i s_t + \sigma_i u_{i,t}, \quad i = 1, \dots, 4 \\ s_t &= \phi_1 s_{t-1} + \phi_2 s_{t-2} + \eta_t \\ u_{i,t} &= \delta_i u_{i,t-1} + w_{i,t}, \end{aligned}$$

where $u_{i,t} \sim N(0, I)$ and $w_{i,t} \sim N(0, I)$. The state equation is now augmented to accommodate the dynamics in the idiosyncratic terms as follows

$$\begin{bmatrix} s_t \\ s_{t-1} \\ u_{1,t} \\ u_{2,t} \\ u_{3,t} \\ u_{4,t} \end{bmatrix} = \begin{bmatrix} \phi_1 & \phi_2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \delta_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \delta_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \delta_3 & 0 \\ 0 & 0 & 0 & 0 & 0 & \delta_4 \end{bmatrix} \begin{bmatrix} s_{t-1} \\ s_{t-2} \\ u_{1,t-1} \\ u_{2,t-1} \\ u_{3,t-1} \\ u_{4,t-1} \end{bmatrix} + \begin{bmatrix} \eta_t \\ 0 \\ w_{1,t} \\ w_{2,t} \\ w_{3,t} \\ w_{4,t} \end{bmatrix}.$$

The model can be viewed as having six factors $\{s_t, s_{t-1}, u_{1,t}, u_{2,t}, u_{3,t}, u_{4,t}\}$. In this scenario, the idiosyncratic terms are redefined as factors. From equation (15.31), $E[u_t u_t'] = R$, which now becomes $R = 0_N$ and the full model is

$$\begin{aligned} y_t &= \Lambda s_t \\ s_t &= \Phi s_{t-1} + \eta_t, \quad \eta_t \sim N(0, I), \end{aligned} \tag{15.34}$$

where s_t represents the vector of six factors and

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & \sigma_1 & 0 & 0 & 0 \\ \lambda_2 & 0 & 0 & \sigma_2 & 0 & 0 \\ \lambda_3 & 0 & 0 & 0 & \sigma_3 & 0 \\ \lambda_4 & 0 & 0 & 0 & 0 & \sigma_4 \end{bmatrix}.$$

15.4.3 Nonstationary Factors

The assumption that s_t is stationary as defined in Chapter 13 may be relaxed. In the case of nonstationarity, the roots of Φ lie on the unit circle resulting in the variance in equation (15.33) being undefined. To circumvent this problem, starting values may be taken to be

$$s_{1|0} = \psi \tag{15.35}$$

$$P_{1|0} = \omega \text{vec}(Q), \tag{15.36}$$

where ψ represents the best guess of starting value for the conditional mean and ω is a positive constant (also, see Harvey, 1989, pp 120 – 121). For big values of ω , the distribution has large variance and is thus diffuse. This can be viewed as a device for controlling numerical precision when the factors are nonstationary. Alternatively, setting $\omega = 0$ yields a degenerate initial distribution for s_t where the mass of the distribution falls on ψ . Figure 15.2 plots the distribution of the factor s_t for various choices of ω .

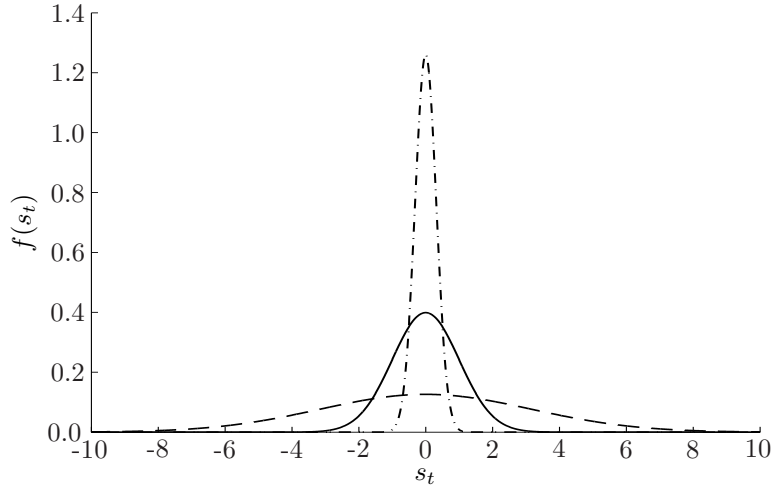


Figure 15.2 The initial distribution of the factor s_t with standard normal prior (solid line), diffuse prior (dashed line) and tight prior (dot-dashed line).

Example 15.7 The Bai-Ng PANIC Model

Bai and Ng (2004) propose the following factor model to capture nonstationarity in panel data

$$\begin{aligned} y_{i,t} &= \lambda_{0,i} + \lambda_i s_t + u_{i,t}, & i &= 1, 2, \dots, N, \\ s_t &= s_{t-1} + \eta_t, \\ u_{i,t} &\sim N(0, \sigma_i^2), \quad \eta_t \sim N(0, 1) \end{aligned}$$

known as Panel Analysis of Nonstationarity in Idiosyncratic and Common components (PANIC). The intercept $\lambda_{0,i}$ represents a fixed effect, while the dynamics in the dependent variables, $y_{i,t}$, are captured by the nonstationary factor, s_t . $\square$

15.4.4 Exogenous and Predetermined Variables

A set of M exogenous or lagged dependent variables, x_t , can be included in the state-space model in one of two different ways. The first is in the measurement equation

$$y_t = \Lambda f_t + \Gamma x_t + u_t,$$

where Γ is $(N \times M)$ and x_t is $(M \times 1)$. A special case of this model is the factor VAR model (F-VAR), where $x_t = y_{t-1}$ and Γ is a $(N \times N)$ diagonal matrix. The second approach is to include the exogenous or predetermined variables in the state equation

$$s_t = \Phi s_{t-1} + \Gamma x_t + u_t,$$

where Γ is now a $(K \times M)$ matrix of parameters.

15.5 Factor Extraction

In many applications, an important objective is to extract estimates of the latent factors, s_t , and interpret their time series properties. By assumption at each t , s_t is normally distributed with a conditional mean and conditional variance. From the discussion in Section 15.3, estimates of these conditional moments are automatically obtained as a by-product of the recursions of the Kalman filter.

Three possible estimates of the factor are available depending on the form of the conditioning information set. These are as follows.

(1) **Predicted**

$$s_{t|t-1} = E_{t-1} [s_t] .$$

(2) **Updated**

$$s_{t|t} = E_t [s_t] .$$

(3) **Smoothed**

$$s_{t|T} = E_T [s_t] .$$

The predicted and updated estimates are as defined in Section 15.3. The smoothed estimate takes account of all the data in the observed sample represented by conditioning on T . This approach, also known as fixed-interval smoothing has the effect of generating smoother estimates than either of the

two previous estimates. Formally, the equations for the smoothed conditional mean and variance are, respectively,

$$s_{t|T} = s_{t|t} + J_t(s_{t+1|T} - s_{t+1|t}) \quad (15.37)$$

$$P_{t|T} = P_{t|t} + J_t(P_{t+1|T} - P_{t+1|t})J'_t, \quad (15.38)$$

where

$$J_t = P_{t|t}\Phi'P_{t+1|t}^{-1}. \quad (15.39)$$

The practical process of computing $s_{t|T}$ and $P_{t|T}$ essentially requires that the Kalman filter algorithm be run backwards starting at T .

An important application of the smooth-factor estimate is in dating business cycles. Since it uses all the information in the data, it provides more precise estimates of the timing of peaks and troughs from a historical perspective than the un-smoothed estimates.

Example 15.8 Computing Smooth-Factor Estimates

Using the results in Table 15.1 where the sample size is $T = 5$, at $t = 5$ the smoothed estimates equal the updated factor estimates

$$s_{5|5} = -1.255$$

$$P_{5|5} = 0.205.$$

Using equation (15.39) at $t = 4$

$$J_4 = P_{4|4}\Phi'P_{5|4}^{-1} = 0.205 \times 0.8 \times 1.131^{-1} = 0.145,$$

and from (15.37) and (15.38) yields the respective smoothed estimates

$$\begin{aligned} s_{4|5} &= s_{4|4} + J_4(s_{5|5} - s_{5|4}) \\ &= -0.308 + 0.145 \times (-1.255 + 0.246) \\ &= -0.454, \end{aligned}$$

and

$$\begin{aligned} P_{4|5} &= P_{4|4} + J_4(P_{5|5} - P_{5|4})J'_4 \\ &= 0.205 + 0.145 \times (0.205 - 1.131) \times 0.145 \\ &= 0.185. \end{aligned}$$

Similarly, the remaining smooth-factor estimates conditional on information at time $T = 5$ are

$$\begin{aligned} s_{3|5} &= 0.002, s_{2|5} = 0.408, s_{1|5} = -0.479 \\ P_{3|5} &= 0.185, P_{2|5} = 0.185, P_{1|5} = 0.205. \end{aligned}$$

□

15.6 Estimation

The discussion so far has assumed given values for the population parameters Λ , Φ , R and Q in equations (15.29) to (15.31). In general, however, it is necessary to estimate these parameters. If the factors are observable, then the parameters are estimated by simply regressing y_t on s_t and regressing s_t on s_{t-1} . But as s_t is unobservable, an alternative estimation strategy is needed. Two estimators are discussed. The first is the maximum likelihood estimator that constructs the log-likelihood function from the prediction errors of the Kalman filter. The second approach is a sequence of least squares regressions proposed by Stock and Watson (2005) that circumvents potential numerical problems associated with high-dimensional systems. Before dealing with the mechanics of estimation, the issue of identification is addressed.

15.6.1 Identification

The state-space model in (15.29) to (15.31) is under identified and therefore all of the parameters cannot be estimated unless some further restrictions are imposed. The difficulty is seen by noting that the volatility in the factor is controlled by Q , but the impact of the factor on y_t is given by Λ . An infinite number of combinations of Q and Λ are consistent with the volatility of y_t . Thus, it is necessary to fix one of these quantities. A common approach is to set

$$Q = I_K. \quad (15.40)$$

This is the case in the term structure example in Section 15.2. Another approach is to place restrictions on Λ and allow Q to be estimated as is the case in the expected real interest rate example in the same section.

15.6.2 Maximum Likelihood

The maximum likelihood approach is to choose values of the parameters that generate latent factors that maximize the likelihood of the observed variables. The conditional distribution of the $(N \times 1)$ vector y_t is multivariate normal

$$y_t \sim N(\mu_{t|t-1}, V_{t|t-1}),$$

where the conditional mean and variance are, respectively,

$$\begin{aligned} \mu_{t|t-1} &= \Lambda s_{t|t-1} \\ V_{t|t-1} &= \Lambda P_{t|t-1} \Lambda' + R. \end{aligned}$$

The log-likelihood function for the t^{th} observation is given by

$$\ln l_t = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |V_{t|t-1}| - \frac{1}{2} (y_t - \mu_{t|t-1})' V_{t|t-1}^{-1} (y_t - \mu_{t|t-1}).$$

Example 15.9 Log-Likelihood Function of the Kalman Filter

Continuing the univariate Kalman filter example in Table 15.1 for $N = 1$, the log-likelihood function at $t = 1$ reported in column (10) for the starting value $\mu_{1|0} = 0$ is computed as

$$\begin{aligned} \ln l_1 &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln |V_{1|0}| - \frac{1}{2} (y_1 - \mu_{1|0})' V_{1|0}^{-1} (y_1 - \mu_{1|0}) \\ &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln |3.028| - \frac{1}{2} (-0.680)' 3.028^{-1} (-0.680) \\ &= -1.549. \end{aligned}$$

The values for $\ln l_2, \ln l_3, \dots, \ln l_5$ are also given in Table 15.1. $\square$

For a sample of $t = 1, 2, \dots, T$ observations, the log-likelihood function is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \ln l_t(\theta). \quad (15.41)$$

This expression is maximized using an iterative algorithm from Chapter 3 because the parameters Λ, Φ, R and Q enter the expressions for $\mu_{t|t-1}$ and $V_{t|t-1}$ nonlinearly.

Example 15.10 A One-Factor Model of the Term Structure

Consider the yields $r_{i,t}, i = 1, 2, \dots, 6$ on U.S. zero coupon bonds given in Figure 15.1. From equations (15.4) to (15.6), a one-factor model is given by the state-space representation

$$\begin{aligned} r_t &= \Lambda s_t + u_t, & u_t &\sim N(0, R) \\ s_t &= \phi s_{t-1} + \eta_t, & \eta_t &\sim N(0, Q), \end{aligned}$$

where s_t is the factor, $R = \text{diag}(\sigma_i^2)$ and $Q = \sigma_\eta^2 = I$. The maximum likelihood estimates are given in Table 15.2, with standard errors based on the Hessian matrix. A diffuse prior is used to allow for a possible nonstationary factor by specifying the starting values to be $\psi = 0$ and $\omega = 0.1$ as in equations (15.35) and (15.36). The estimates of λ_i are similar in magnitude suggesting that s_t captures the level of the yield curve. The estimates of the idiosyncratic parameter, σ_i , exhibit a U-shape with the shortest and longest maturities having the greatest volatility and with the 3-year yield having the smallest. As the estimate of ϕ is close to unity, it appears that the factor is nonstationary. $\square$

Table 15.2

Maximum likelihood estimates of the one-factor term structure model, yields expressed in basis points. Standard errors are based on the Hessian matrix.

Variable	Parameter	Estimate	Std. Error	pv
3 month	λ_1	7.354	0.157	0.000
1 year	λ_2	7.636	0.157	0.000
3 year	λ_3	7.004	0.139	0.000
5 year	λ_4	6.293	0.127	0.000
7 year	λ_5	5.848	0.120	0.000
10 year	λ_6	5.536	0.118	0.000
3 month	σ_1	65.743	0.818	0.000
1 year	σ_2	43.667	0.554	0.000
3 year	σ_3	6.287	0.130	0.000
5 year	σ_4	26.591	0.345	0.000
7 year	σ_5	36.680	0.463	0.000
10 year	σ_6	50.466	0.629	0.000
Factor	ϕ	0.999	0.001	0.000
$T \ln L_T(\hat{\theta}) = -98107.146$				

15.6.3 Principal Components Estimator

When the number of measurement and state equations is large, estimation by maximum likelihood can be problematic. To circumvent this situation Stock and Watson (2005) suggest an iterative least squares approach. To highlight the steps involved in this algorithm, consider the model

$$\begin{aligned}
 y_t &= \Lambda s_t + \Gamma y_{t-1} + u_t \\
 s_t &= \Phi s_{t-1} + \eta_t \\
 u_t &\sim N(0, R) \\
 v_t &\sim N(0, Q),
 \end{aligned} \tag{15.42}$$

where y_t is $(N \times 1)$, s_t is $(K \times 1)$, Λ is $(N \times K)$, Γ is $(N \times N)$, Φ is $(K \times K)$, R is $(N \times N)$ and Q is $(K \times K)$. The algorithm proceeds as follows.

Step 1: Standardize the y_t variables to have zero mean and unit variance.

Step 2: Perform a principal components decomposition on the standard-

ized y_t variables to obtain an initial estimate of the K factors, $\hat{s}_t$, where K is based on the magnitude of the eigenvalues.¹

Step 3: Regress $y_{i,t}$ on $\{\hat{s}_t, y_{i,t-1}\}$, $i = 1, 2, \dots, N$ and compute $\hat{\Lambda}$ and $\hat{\Gamma}$.

Step 4: Redefine y_t as $y_t - \Gamma y_{t-1}$.

Step 5: Repeat steps 2 to 4 until there is no change in the parameter estimates across iteration to some desired tolerance level.

Step 6: Regress $\hat{s}_t$ on $\hat{s}_{t-1}$ and compute $\hat{\Phi}$.

Example 15.11 A Multi-Factor Model of the Term Structure

A three-factor model ($K = 3$) of U.S. yields ($N = 30$) with maturities ranging from 1 year to 30 years, inclusive, is estimated using the Stock-Watson estimator. The data are daily, beginning on 2nd of January 1990 and ending on 21st of August 2006. The estimates of the factor loadings on the three factors are given in Figure 15.3 and demonstrate that the factors represent the level, slope and curvature of the yield curve. The estimates of the factor dynamics are

$$\hat{\Phi} = \begin{bmatrix} 0.853 & 0.451 & -0.039 \\ 0.034 & 0.835 & 0.056 \\ -0.020 & 0.131 & 0.316 \end{bmatrix}.$$

These estimates reveal an interesting causal relationship between the three factors, with the estimates of 0.451 and 0.131 showing unidirectional Granger causality, as defined in Chapter 13, from the slope factor to both the level and curvature factors, respectively.

□

Assuming stationary s_t and u_t , the principal components estimator based on y_t yields consistent estimators of s_t (Stock and Watson, 2002), and Λ (Bai and Ng, 2004) for $N, T \rightarrow \infty$. Consistency of this estimator also holds for nonstationary s_t and stationary u_t , but not for nonstationary s_t and u_t (Bai and Ng, 2004). An alternative estimator proposed by Bai and Ng is to perform the principal components decomposition on Δy_t to generate $\Delta \hat{s}_t$ and estimate Λ by regressing Δy_t on a constant and $\Delta \hat{s}_t$. The estimate of the level of the factor is computed as $\hat{s}_t = \sum_{i=1}^t \Delta \hat{s}_i$. The advantage of these estimators of s_t and Λ is that they are always consistent.

Example 15.12 Properties of the Bai-Ng Estimator

The properties of the Bai-Ng estimator are demonstrated in Table 15.3 using some Monte Carlo experiments conducted on the Bai-Ng PANIC model outlined previously. The efficiency of the Stock-Watson approach over the

¹ Bai and Ng (2002) provide a more formal testing procedure to choose the number of factors.

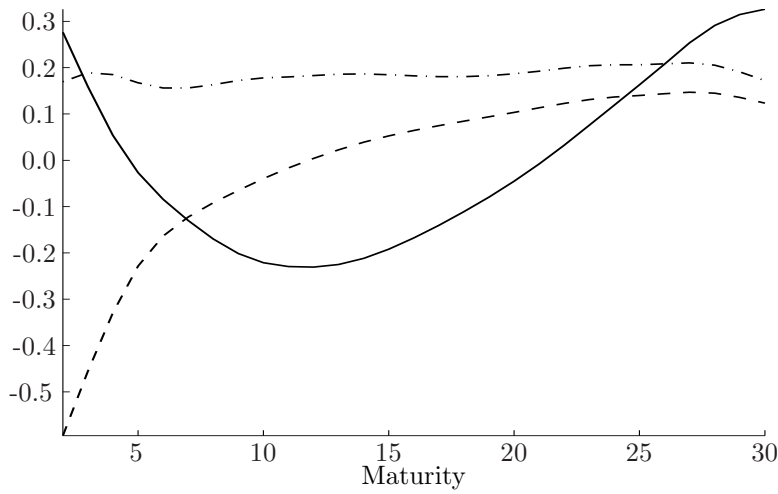


Figure 15.3 Stock-Watson factor loading estimates of standardized U.S. yields for yearly maturities from 1 year to 30 years. The three factors are the level (dot-dashed line), slope (dashed line) and curvature (solid line).

Table 15.3

Sampling properties of the principal components estimator of λ in the PANIC model of Bai and Ng (2004). The Bias and RMSE are averages across the N cross-sections. The sample size is $T = 100$, with the Monte Carlo experiments based on 5,000 replications.

Type		$N = 10$		$N = 30$		$N = 50$	
		Bias	RMSE	Bias	RMSE	Bias	RMSE
Δy_t	$\rho = 0.0$	-0.894	1.180	-0.633	0.975	-0.731	0.939
	$\rho = 0.5$	-0.872	1.083	-0.594	0.877	-0.716	0.891
	$\rho = 1.0$	-0.847	1.021	-0.581	0.838	-0.714	0.880
y_t	$\rho = 0.0$	-0.031	0.395	0.008	0.102	0.005	0.101
	$\rho = 0.5$	-0.290	0.907	0.009	0.197	0.006	0.191
	$\rho = 1.0$	-0.672	3.620	-0.681	3.532	-0.787	3.529

Bai-Ng approach for the stationary case, $\rho < 1$, is evident by the lower RMSEs. For the nonstationary case of $\rho = 1$ this is no longer true and the Bai-Ng estimator now yields smaller RMSEs. $\square$

15.7 Relationship to VARMA Models

Consider the $N = 3$ variable state-space model

$$\begin{aligned} y_{i,t} &= \lambda_i f_t + u_{i,t}, & i = 1, 2, \dots, N \\ s_t &= \phi s_{t-1} + \eta_t. \end{aligned}$$

where $u_{i,t} \sim N(0, \sigma_i^2)$ and $\eta_t \sim N(0, 1)$. Rewrite the state equation using the lag operator

$$(1 - \phi L) s_t = \eta_t.$$

Multiplying the measurement equation on both sides by $1 - \phi L$ gives

$$\begin{aligned} (1 - \phi L) y_{i,t} &= \lambda_i (1 - \phi L) s_t + (1 - \phi L) u_{i,t} \\ &= \lambda_i \eta_t + (1 - \phi L) u_{i,t}, \end{aligned}$$

or

$$y_{i,t} = \phi y_{i,t-1} + v_{i,t}, \quad i = 1, 2, \dots, N, \quad (15.43)$$

with

$$v_{i,t} = \lambda_i \eta_t + u_{i,t} - \phi u_{i,t-1}. \quad (15.44)$$

This is a restricted VARMA(1, 1) model as the autoregressive and moving-average lags have the same parameter, namely ϕ , which is of course due to the fact that the dynamics are driven by the single factor s_t . This model is of a similar form to the factor models of contagion proposed by Dungey and Martin (2007) where a VAR is estimated to extract the dynamics and a factor structure of the form $\lambda_i \eta_t + u_{i,t}$ is used to model the contemporaneous linkages. Estimation by maximum likelihood is complicated by the composite disturbance term containing η_t and $u_{i,t}$.

One way to proceed is to use a GMM estimator as discussed in Chapter 10. The theoretical moments all have zero means by assumption. The pertinent moments of v_t are

$$\begin{aligned} E[v_{i,t}^2] &= \lambda_i^2 + \sigma_i^2 (1 + \phi^2), \\ E[v_{i,t} v_{j,t}] &= \lambda_i \lambda_j, \\ E[v_{i,t} v_{i,t-1}] &= -\phi \sigma_i^2, \\ E[v_{i,t} v_{j,t-1}] &= 0. \end{aligned} \quad (15.45)$$

A simple, just-identified GMM estimator consists of estimating the scalar ϕ from a stacked (pooled) regression of $\text{vec}(y_t)$ on $\text{vec}(y_{t-1})$. Having estimated ϕ , the estimates of σ_i^2 are computed using the second-last moment condition, with estimates of λ_i then coming from the first moment condition. The second moment condition is also used in (15.45), so that a cross-equation

restriction is imposed, which causes the system of equations to become overidentified.

15.8 Applications

15.8.1 The Hodrick-Prescott Filter

Separating out trends and cycles is fundamental to much of macroeconomic analysis and represents an important example of factor extraction. A commonly used technique for estimating the trend in economic and financial time series is the Hodrick-Prescott filter (Hodrick and Prescott, 1997). In this application, this filter is recast in a state-space representation following Harvey and Jaeger (1993). This alternative formulation allows the estimation of an important parameter that the conventional Hodrick-Prescott filter simply imposes. Consequently, this parameter may be tested against the value commonly adopted in the conventional approach.

Let the series, y_t , be decomposed in terms of a trend, τ_t , and a cycle, c_t , so that

$$y_t = \tau_t + c_t, \quad t = 1, 2, \dots, T. \quad (15.46)$$

The Hodrick-Prescott filter defines the trend in terms of the following minimization problem

$$\min_{\{\tau_t\}} Q = \left[\sum_{t=1}^T (y_t - \tau_t)^2 + \gamma \sum_{t=1}^T [(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})]^2 \right], \quad (15.47)$$

for some appropriate choice of the smoothing parameter γ . The first term in equation (15.47) penalizes the lack of fit of the trend component while the second term penalizes the lack of smoothness in τ_t .

A convenient analytical solution to the minimization problem posed in equation (15.47) is available, which allows the Hodrick-Prescott filter to be implemented easily. Rewriting equation (15.47) using lag operators (see

Appendix B for details) results in

$$\begin{aligned}
 \min_{\{\tau_t\}} Q &= \sum_{t=1}^T (y_t - \tau_t)^2 + \gamma \sum_{t=1}^T ((L^{-1} - I)\tau_t - (I - L)\tau_t)^2 \\
 &= \sum_{t=1}^T (y_t - \tau_t)^2 + \gamma \sum_{t=1}^T ((L^{-1} - 2I + L)\tau_t)^2 \\
 &= \sum_{t=1}^T (y_t - \tau_t)^2 + \gamma \sum_{t=1}^T (L^{-1} - 2I + L)^2 \tau_t^2 \\
 &= \sum_{t=1}^T (y_t - \tau_t)^2 + \gamma \sum_{t=1}^T (L^{-2} - 4L^{-1} + 6I - 4L + L^2) \tau_t^2,
 \end{aligned}$$

where the last term represents a fifth-order moving average in the squared trend. Differentiating with respect to τ_t and setting the derivative to zero yields the first-order condition

$$-2(y_t - \tau_t) + 2\gamma(L^{-2} - 4L^{-1} + 6I - 4L + L^2)\tau_t = 0.$$

Solving for y_t gives

$$\gamma(L^{-2} - 4L^{-1} + 6I - 4L + L^2)\tau_t + \tau_t = y_t. \quad (15.48)$$

This expression is solved for all t by defining τ as a $(T \times 1)$ vector containing the trend component and y as a $(T \times 1)$ vector of the observed data. Now equation (15.48) becomes

$$(\gamma F + I_T)\tau = y,$$

where F is a $(T \times T)$ matrix given by

$$\begin{bmatrix}
 1 & -2 & 1 & 0 & \cdots & & & \cdots & 0 \\
 -2 & 5 & -4 & 1 & 0 & \cdots & & & 0 \\
 1 & -4 & 6 & -4 & 1 & 0 & \cdots & & 0 \\
 0 & 1 & -4 & 6 & -4 & 1 & 0 & \cdots & \vdots \\
 \vdots & \ddots & & & & & \ddots & & \vdots \\
 & & & & & & & \ddots & \vdots \\
 \vdots & & & & & & & & \vdots \\
 0 & & & & & 0 & 1 & -4 & 6 & -4 & 1 & 0 \\
 0 & & & & & \cdots & 0 & 1 & -4 & 6 & -4 & 1 \\
 0 & & & & & & \cdots & 0 & 1 & -4 & 5 & -2 \\
 0 & & & & & & & \cdots & 0 & 1 & -2 & 1
 \end{bmatrix},$$

and where suitable adjustments are made at the beginning and at the end

of the sample. Upon solving this expression for τ , the Hodrick-Prescott estimate of the trend becomes

$$\tau = (\gamma F + I_T)^{-1} y. \quad (15.49)$$

By construction, the trend τ at each time t is a weighted average of all of the sample observations y .

The smoothness parameter, γ , is central to estimating the trend: as $\gamma \rightarrow 0$, the trend component approaches the actual series and, as $\gamma \rightarrow \infty$, τ_t becomes the linear trend. Conventional choices for γ are 100 for annual data, 1600 for quarterly data and 14400 for monthly data. An alternative approach is to estimate the smoothing parameter from the data. One way to do this is to recast the Hodrick-Prescott filter in state-space form as

$$\begin{aligned} y_t &= \tau_t + c_t & c_t &\sim N(0, \sigma_c^2) \\ \tau_t &= \tau_{t-1} + \beta_{t-1} \\ \beta_t &= \beta_{t-1} + \zeta_t & \zeta_t &\sim N(0, \sigma_\zeta^2), \end{aligned}$$

where, as before, τ_t is the stochastic trend component and c_t is the cyclical or transitory component of y_t . Note that the transitory component, c_t , and ζ_t are mutually independent normally distributed variables. This model is commonly known as an unobserved-component model.

Define

$$c_t = u_t, \quad s_t = \begin{bmatrix} \tau_t \\ \beta_t \end{bmatrix},$$

and parameter matrices

$$\Lambda = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad \Phi = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad R = \sigma_c^2, \quad Q = \begin{bmatrix} 0 & 0 \\ 0 & \sigma_\zeta^2 \end{bmatrix}.$$

The model now has exactly the representation of the multivariate Kalman filter as in equations (15.29) to (15.31). This system contains two unknown parameters, namely the variances σ_c^2 and σ_ζ^2 , which together define the signal-to-noise ratio $q = \sigma_c^2 / \sigma_\zeta^2$.

The results of estimating this model using the logarithm of quarterly real U.S. GDP for the period March 1940 to December 2000 ($T = 244$ observations) by maximum likelihood for different restrictions on the parameters σ_c^2 and σ_ζ^2 are reported in Table 15.4.

The maximum likelihood estimates of the parameters for the unconstrained model yield a value of $T \ln L_T(\hat{\theta}_2) = -367.847$. Imposing the normalization $\sigma_c = 1$ yields a value for the partially constrained model of $T \ln L_T(\hat{\theta}_1) =$

Table 15.4

Maximum likelihood estimates of the Hodrick-Prescott factor model.
Standard errors based on the Hessian matrix in brackets.

Parameter	Unconstrained	Partially Constrained	Fully Constrained
σ_c	0.363 (0.051)	1.000	1.000
σ_ζ	0.716 (0.073)	0.465 (0.056)	1/40
$T \ln L_T(\hat{\theta})$	-367.847	-424.866	-696.845

-424.866. Harvey and Jaeger (1993) show that when the further restriction

$$\sigma_\zeta^2 = \frac{1}{\gamma} = \frac{1}{1600},$$

is imposed on the partially constrained model, the trend component obtained by the Kalman filter with this restricted signal-to-noise ratio is identical to that obtained from a conventional Hodrick-Prescott filter. In this case, there are no parameters to estimate resulting in a value of $T \ln L_T(\hat{\theta}_0) = -696.845$. Performing a LR test of this restriction yields a value of

$$LR = -2(-696.845 + 424.866) = 543.958.$$

The p-value from the chi-square distribution with one degree of freedom is 0.000, which indicates that the Hodrick-Prescott choice of γ is strongly rejected by the data.

To ensure that the estimated trend obtained from the filter is conditioned on all the information in the sample, compute the smoothed version of the latent factor using equations (15.37) and (15.38). This makes sense because the Hodrick-Prescott trend τ at each t in equation (15.49) is a weighted average of all the sample observations. Figure 15.4 shows the trend component extracted from the U.S. GDP data using the Kalman filter implementation of the Hodrick-Prescott filter for the sub-period 1940 to 1952, which is numerically identical to the conventional Hodrick-Prescott filter in equation (15.49).

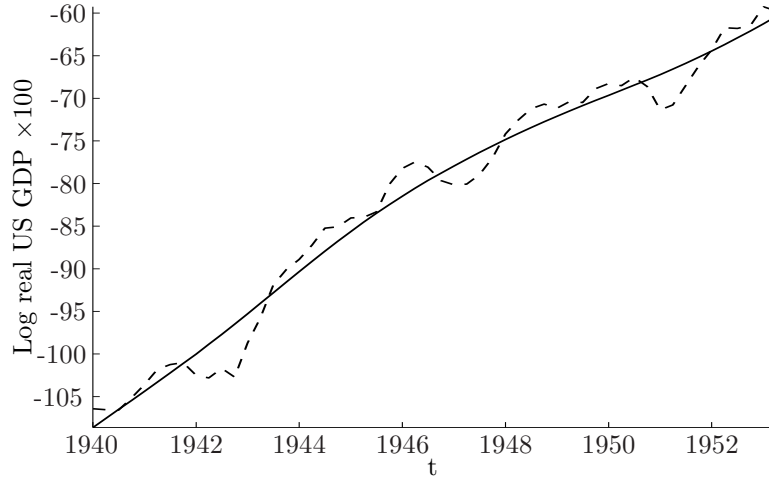


Figure 15.4 The logarithm of real U.S. GDP for the sub-period 1940 to 1952 (dashed line) with the smooth trend component extracted using the Kalman filter (solid line). The data are centered before being scaled by a factor of 100.

15.8.2 A Factor Model of Spreads with Money Shocks

The earlier empirical work on the term structure of interest rates focussed on identifying statistical factors from sets of interest rates of differing maturities. These factors are commonly classified in terms of level, slope and curvature encountered previously in Figure 15.3 in Section 15.5. More recently, understanding the economic processes underlying these statistical factors has generated interest (Gurkaynak, Sack, and Swanson, 2005; Cochrane and Piazzesi, 2009; Craine and Martin 2008, 2009). In this application, the focus is on identifying the role of money shocks in determining the factors and hence the movements in the term structure over time. Extending the analysis to include macroeconomic shocks is discussed in an exercise at the end of this chapter.

The spread at time t between a one-year forward rate maturing at time $t + n$, $f_{n,t}$, and the one-year interest rate at time t , $r_{1,t}$ is given by

$$sp_{t,n} = \ln \left(1 + \frac{f_{n,t}}{100} \right) - \ln \left(1 + \frac{r_{1,t}}{100} \right), \quad (15.50)$$

where both rates are expressed as a percentage. Let y_t be a (5×1) vector containing the $n = 2, 4, 6, 8, 10$ year spreads, which are expressed in basis points and whose mean has been removed. To identify the role of money shocks, x_t , in determining the statistical factors, consider the following two-

The maximum likelihood estimates from estimating the factor model are given in Table 15.5 based on daily U.S. spreads beginning on 2nd of January 1990 and ending on 21st of August 2006. Money shocks are defined as the change in the Eurodollar 1-month rate on Federal Reserve Board meeting dates. The estimates of the loadings on the first factor, $\lambda_{i,1}$, suggest a slope factor with a positive shock to the factor widening all spreads, with longer maturities being most affected. The loading estimates on the second factor, $\lambda_{i,2}$, suggest a curvature factor with a shock to the factor increasing (decreasing) spreads on shorter (longer) maturities, although only the parameter estimates on the 2 and 4 year spreads are statistically significant.

The parameter estimates on the money shock show that money has a significant effect on the slope factor, but not the curvature factor, a result which suggests that the slope factor could be relabelled as a money factor. The negative sign on γ_1 also suggests that positive money shocks have the effect of narrowing spreads, with the longer-term spreads being the most affected.

15.9 Exercises

(1) Recursions of the Univariate Kalman Filter

Gauss file(s)	kal_uni.g, kal_smooth.g
Matlab file(s)	kal_uni.m, kal_smooth.m

The model is given by

$$\begin{aligned} y_t &= \lambda s_t + u_t, & u_t &\sim N(0, \sigma^2) \\ s_t &= \phi s_{t-1} + \eta_t, & \eta_t &\sim N(0, \sigma_\eta^2) \end{aligned}$$

with parameters $\lambda = 1.0$, $\sigma = 0.5$, $\phi = 0.8$ and $\sigma_\eta = 1$. The sample size is $T = 5$ with the realized values of y_t given by

$$y_t = \{-0.680, 0.670, 0.012, -0.390, -1.477\}.$$

(a) At time $t = 1$ compute the following.

(i) The prediction (initial) equations

$$\begin{aligned} s_{1|0} &= 0.0 \\ P_{1|0} &= \frac{1}{1 - \phi^2}. \end{aligned}$$

(ii) The observation equations

$$\begin{aligned}\mu_{1|0} &= \lambda s_{1|0} \\ V_{1|0} &= \lambda^2 P_{1|0} + \sigma^2 \\ u_{1|0} &= y_1 - \mu_{1|0} .\end{aligned}$$

(iii) The updating equations

$$\begin{aligned}s_{1|1} &= s_{1|0} + \frac{\lambda P_{1|0}}{V_{1|0}}(y_1 - \mu_{1|0}) \\ P_{1|1} &= P_{1|0} - \frac{\lambda^2 P_{1|0}^2}{V_{1|0}} .\end{aligned}$$

(b) At time $t = 2$ compute the following.

(i) The prediction equations

$$\begin{aligned}s_{2|1} &= \phi s_{1|1} \\ P_{2|1} &= \phi^2 P_{1|1} + 1 .\end{aligned}$$

(ii) The observation equations

$$\begin{aligned}\mu_{2|1} &= \lambda s_{2|1} \\ V_{2|1} &= \lambda^2 P_{2|1} + \sigma^2 \\ u_{2|1} &= y_2 - \mu_{2|1} .\end{aligned}$$

(iii) The updating equations

$$\begin{aligned}s_{2|2} &= s_{2|1} + \frac{\lambda P_{2|1}}{V_{2|1}}(y_2 - \mu_{2|1}) \\ P_{2|2} &= P_{2|1} - \frac{\lambda^2 P_{2|1}^2}{V_{2|1}} .\end{aligned}$$

(c) Continue for $t = 3, 4, 5$ and compare the results with those reported in Table 15.1.

(d) Using the estimates of the conditional factor means and variances obtained previously, compute the smoothed factor estimates $s_{t|T}$ and $P_{t|T}$ for $t = 5, 4, 3, 2, 1, .$

(2) Recursions of the Multivariate Kalman Filter

Gauss file(s)	<code>kal_multi.g</code>
Matlab file(s)	<code>kal_multi.m</code>

Consider a factor model containing $N = 3$ variables and $K = 2$ factors with parameter matrices

$$\Lambda = \begin{bmatrix} 1.00 & 0.50 \\ 1.00 & 0.00 \\ 1.00 & -0.50 \end{bmatrix}, \Phi = \begin{bmatrix} 0.80 & 0.00 \\ 0.00 & 0.50 \end{bmatrix},$$

$$R = \begin{bmatrix} 0.25 & 0.00 & 0.00 \\ 0.00 & 0.16 & 0.00 \\ 0.00 & 0.00 & 0.09 \end{bmatrix}, Q = I_2.$$

with variables for $T = 5$ observations given by

$$y_1 = \{2.500, 2.017, -0.107, -0.739, -0.992\}$$

$$y_2 = \{2.000, 1.032, -0.535, 0.061, 0.459\}$$

$$y_3 = \{1.500, 0.047, -0.964, 0.862, 1.910\}.$$

For $t = 1, 2, 3, 4, 5$, compute the prediction, observation and updating equations.

(3) Term Structure of Interest Rates

Gauss file(s)	kal_term.g, kal_termfig.g, usdata.dat
Matlab file(s)	kal_term.m, kal_termfig.m, usdata.mat

The data are daily U.S. zero coupon bond yields beginning 4 October 1988 and ending 28 December 2001. The maturities are 3 months, 1 year, 3 years, 5 years, 7 years and 10 years.

- (a) Plot the yields and discuss the time series properties of the series.
- (b) Transform the data by scaling the yields by a factor of 100 and removing the mean from the scaled data.

- (i) Estimate the following one-factor model of the yields, $r_{i,t}$,

$$r_{i,t} = \lambda_i s_t + u_{i,t}, \quad u_t \sim N(0, R)$$

$$s_t = \phi s_{t-1} + \eta_t, \quad \eta_t \sim N(0, Q)$$

where $r_{i,t}$, $i = 1, 2, \dots, 6$, $R = \text{diag}(\sigma_i^2)$ and $Q = 1$.

- (ii) Interpret the parameter estimates obtained.
 - (iii) Test the restrictions $\lambda_1 = \lambda_2 = \dots = \lambda_6 = \lambda$ and interpret the results.

(4) Alternative Formulation of the Kalman Filter

Gauss file(s)	kal_term_adj.g, usdata.dat
Matlab file(s)	kal_term_adj.m, usdata.dat

An alternative way to express the state-space model of the term structure is to treat the disturbance term $u_{i,t}$ in the measurement equation as a latent (idiosyncratic) factor with loading σ_i . The factor model is now expressed as

$$\begin{bmatrix} r_{1,t} \\ r_{2,t} \\ r_{3,t} \\ r_{4,t} \\ r_{5,t} \\ r_{6,t} \end{bmatrix} = \begin{bmatrix} \lambda_1 & \sigma_1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_2 & 0 & \sigma_2 & 0 & 0 & 0 & 0 \\ \lambda_3 & 0 & 0 & \sigma_3 & 0 & 0 & 0 \\ \lambda_2 & 0 & 0 & 0 & \sigma_4 & 0 & 0 \\ \lambda_1 & 0 & 0 & 0 & 0 & \sigma_5 & 0 \\ \lambda_2 & 0 & 0 & 0 & 0 & 0 & \sigma_6 \end{bmatrix} \begin{bmatrix} s_{1,t} \\ s_{2,t} \\ s_{3,t} \\ s_{4,t} \\ s_{5,t} \\ s_{6,t} \\ s_{7,t} \end{bmatrix}$$

and

$$\begin{bmatrix} s_{1,t} \\ s_{2,t} \\ s_{3,t} \\ s_{4,t} \\ s_{5,t} \\ s_{6,t} \\ s_{7,t} \end{bmatrix} = \text{diag} \begin{pmatrix} \phi \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \\ v_{3,t} \\ v_{4,t} \\ v_{5,t} \\ v_{6,t} \\ v_{7,t} \end{bmatrix}$$

where now

$$R = E[u_t u_t'] = 0, \quad Q = E[\eta_t \eta_t'] = I.$$

Re-estimate the term structure model by maximum likelihood using the adjusted state-space representation and show that the parameter estimates are the same as the estimates obtained in the previous exercise.

(5) An F-VAR Model of the Term Structure

Gauss file(s)	kal_fvar.g, daily_finance.xlsx
Matlab file(s)	kal_fvar.m, daily_finance.xlsx

The F-VAR model in the case of one lag is specified as

$$\begin{aligned} y_t &= \Lambda f_t + \Gamma y_{t-1} + u_t, \\ f_t &= \Phi f_{t-1} + \eta_t, \\ u_t &\sim N(0, R), \\ v_t &\sim N(0, Q), \end{aligned}$$

where Λ is a $(N \times K)$ matrix, Γ is a $(N \times N)$ diagonal matrix, Φ is a $(K \times K)$ matrix, R is a $(N \times N)$ matrix, Q is a $(K \times K)$ matrix, Γ is a $(N \times M)$ matrix and x_t is a $(M \times 1)$.

- (a) Estimate the latent factors and the parameters of the model using the Stock-Watson algorithm.
- (b) Show that the factor loadings can be interpreted as level, slope and curvature consistent with the analysis of the term structure by Knez, Litterman and Scheinkman (1994).

(6) Sampling Properties of the Principal Components Estimator

Gauss file(s)	<code>kal_panic.g</code>
Matlab file(s)	<code>kal_panic.m</code>

Consider the nonstationary $K = 1$ factor model

$$\begin{aligned} y_{it} &= \lambda_i f_t + u_{i,t}, & i &= 1, 2, \dots, N \\ f_t &= f_{t-1} + \eta_t \\ u_{i,t} &= \rho u_{i,t-1} + w_{i,t}, \end{aligned}$$

where $w_{i,t} \sim N(0, 1)$, $\eta_t \sim N(0, 1)$ and $\lambda_i \sim N(0, 1)$.

- (a) The Stock-Watson estimator of λ is based on estimating the factor from a principal components decomposition of y_t and regressing y_t on a constant and the estimated factor. Compute the mean, bias and RMSE of the sampling distribution of λ , for cross-sections of $N = 10, 30, 50$, idiosyncratic dynamics of $\rho = 0, 0.5, 1.0$ and a sample size of $T = 100$. Choose 5000 replications.
- (b) The Bai-Ng estimator of λ is based on estimating the factor from a principal components decomposition of Δy_t and regressing Δy_t on a constant and the estimated factor. Repeat part (a) for this estimator.
- (c) Compare the results in parts (a) and (b) and show that (i) for $\rho < 1$ both estimators are consistent with the Stock-Watson estimator being more efficient; and (ii) the Bai-Ng estimator is consistent whereas the Stock-Watson estimator is not.

(7) Hodrick-Prescott Filter

Gauss file(s)	<code>kal_hp.g</code> , <code>usgdp.dat</code>
Matlab file(s)	<code>kal_hp.m</code> , <code>usgdp.mat</code>

The data are quarterly real U.S. GDP for the period 1940:Q1 to 2000:Q4.

- (a) Verify that the traditional implementation of the Hodrick-Prescott filter and the Kalman filter version given by

$$\begin{aligned} y_t &= \tau_t + c_t & c_t &\sim N(0, \sigma_c^2) \\ \tau_t &= \tau_{t-1} + \beta_{t-1} \\ \beta_t &= \beta_{t-1} + \zeta_t & \zeta_t &\sim N(0, \sigma_\zeta^2), \end{aligned}$$

where

$$\Lambda = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad s_t = \begin{bmatrix} \tau_t \\ \beta_t \end{bmatrix}, \quad \Phi = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$$

$$R = \sigma_c^2, \quad Q = \begin{bmatrix} 0 & 0 \\ 0 & \sigma_\zeta^2 \end{bmatrix}$$

produce identical results for the trend component of the logarithm of real U.S. GDP when the smoothing parameter γ is set to 1600.

- (b) Test the validity of the assumption that the smoothing parameter is 1600 by means of a LR test. That is, test the hypotheses

$$H_0 : \sigma_\zeta^2 = 1/\gamma \quad H_1 : \sigma_\zeta^2 \neq 1/\gamma.$$

(8) A Multi-Factor Model of Spreads with Money Shocks

Gauss file(s)	<code>kal_spreads.g, daily_finance.xlsx</code>
Matlab file(s)	<code>kal_spreads.m, daily_finance.xlsx</code>

Let y_t be a (5×1) vector containing the spreads, expressed in basis points, between the one year forwards maturing at $n = 2, 4, 6, 8, 10$ years and the one year interest rate.

- (a) Estimate the two-factor model by maximum likelihood methods

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \\ y_{3,t} \\ y_{4,t} \\ y_{5,t} \end{bmatrix} = \begin{bmatrix} \lambda_{1,1} & \lambda_{1,2} \\ \lambda_{2,1} & \lambda_{2,2} \\ \lambda_{3,1} & \lambda_{3,2} \\ \lambda_{4,1} & \lambda_{4,2} \\ \lambda_{5,1} & \lambda_{5,2} \end{bmatrix} \begin{bmatrix} s_{1,t} \\ s_{2,t} \end{bmatrix} + \begin{bmatrix} u_{1,t} \\ u_{2,t} \\ u_{3,t} \\ u_{4,t} \\ u_{5,t} \end{bmatrix}$$

$$\begin{bmatrix} s_{1,t} \\ s_{2,t} \end{bmatrix} = \begin{bmatrix} \phi_1 & 0 \\ 0 & \phi_2 \end{bmatrix} \begin{bmatrix} s_{1,t-1} \\ s_{2,t-1} \end{bmatrix} + \begin{bmatrix} \gamma_1 \\ \gamma_2 \end{bmatrix} \begin{bmatrix} x_t \end{bmatrix} + \begin{bmatrix} \eta_{1,t} \\ \eta_{2,t} \end{bmatrix},$$

where x_t is the money factor, $R = E[u_t u_t'] = \text{diag}(\sigma_i^2)$ and $Q = E[\eta_t \eta_t'] = I$. Interpret the parameter vector $\Gamma = [\gamma_1 \ \gamma_2]'$, which controls the effect of money shocks on the factors.

- (b) Test the restriction

$$\gamma_1 = \gamma_2 = 0,$$

and interpret the result.

- (c) Extend the analysis to include a set of macroeconomic shocks given by capacity utilization, consumer confidence, the CPI, advance GDP, the index of business activity, nonfarm payroll, new home sales and retail sales.

(9) ***Ex Ante* Real Interest Rates**

Gauss file(s)	kal_exante.g, exante.xls
Matlab file(s)	kal_exante.m, exante.xls

The data are monthly consisting of the U.S. Consumer Price Index, p_t , and the 1-month (% p.a.) Eurodollar rate, r_t , from January 1971 to December 2009, a total of $T = 468$ observations. A state-space model of the *ex ante* real interest rate is

$$\begin{aligned} y_t &= \alpha + s_t + u_t \\ s_t &= \phi s_{t-1} + \eta_t \\ u_t &\sim N(0, \sigma^2) \\ \eta_t &\sim N(0, \sigma_\eta^2), \end{aligned}$$

where y_t is the *ex post* real interest rate, $s_t + \alpha$ is the *ex ante* real interest rate and u_t is the expectations error in measuring inflation.

- (a) Compute the 1-month *ex post* real interest rate (% p.a.)

$$y_t = r_t - 1200 \times (\ln p_t - \ln p_{t-12}).$$

- (b) Estimate the unknown parameters $\{\alpha, \phi, \sigma^2, \sigma_\eta^2\}$ by maximum likelihood.
- (c) Estimate the unconditional mean and variance of the *ex post* real interest rate and compare these values with the sample mean and sample variance of the *ex post* real interest rate.
- (d) Compute a time series on the *ex ante* real interest rate using $s_{t|t-1} + \alpha$ and compare the result with the *ex post* real interest rate y_t .
- (e) Compute a time series on the expectations error in measuring inflation and interpret its time series properties.

(10) **Capital Asset Pricing Model**

Gauss file(s) `kal_capm.g, capm.xls`
Matlab file(s) `kal_capm.m, capm.mat`

The data are quarterly from March 1990 to March 2010, a total of $T = 81$ observations, on 10 U.S. stock prices (Microsoft, Intel, Pfizer, Exxon, Procter & Gamble, AT&T, General Electric, Chevron, Bank of America), the S&P500 index and the U.S. 3-month Treasury rate (p.a.). The state-space representation of the CAPM is

$$\begin{aligned} y_{i,t} &= \lambda_{0,i} + \lambda_i s_t + u_{i,t}, & i = 1, 2, \dots, 10 \\ s_t &= \phi s_{t-1} + \eta_t \\ u_{i,t} &\sim N(0, \sigma_i^2), & i = 1, 2, \dots, 10 \\ \eta_t &\sim N(0, 1), \end{aligned}$$

where y_t is the excess return on asset i and $\theta = \{\lambda_0, \lambda, \phi, \sigma^2\}$ are parameters.

- (a) Compute the annual excess return of each asset

$$y_{i,t} = 4 \times (\ln p_{i,t} - \ln p_{i,t-1}) - r_t/100, \quad i = 1, 2, \dots, 10,$$

where $p_{i,t}$ is the asset price on asset i and r_t is the risk-free rate of interest.

- (b) Estimate the unknown parameters θ by maximum likelihood.
(c) Compute the excess return on all invested wealth using the updated conditional mean $s_{t|t}$. Now compute the covariance matrix and correlation matrix of $s_{t|t}$ and the annual excess market return based on the S&P500 index, m_t , where

$$m_t = 4 \times (\ln sp500_t - \ln sp500_{t-4}) - r_t/100.$$

Compare the two different measures of the excess market return.

- (d) Compare the estimate of $\lambda_{0,i}$ with the mean of y_i , for each of the 10 assets.
(e) Compare the estimate of λ_i with the estimate of β_i for each of the 10 assets, where the latter is obtained as the least squares estimate from the proxy CAPM regression equation

$$y_{i,t} = \alpha_i + \beta_i m_t + v_{i,t},$$

where m_t is the market excess return based on the S&P500 defined above.

- (f) From part (c), the variances of $s_{t|t}$ and m_t are very different in magnitude, a difference which makes the comparison of λ_i and β_i in

part (e) inappropriate. One way to proceed is to rescale λ by the ratio of the standard deviation of $s_{t|t}$ to the standard deviation of m_t . Compare the rescaled estimate of λ_i with the estimate of β_i for each of the 10 assets.

(11) Business Cycles

Gauss file(s)	<code>kal_bcycle.g, bcycle.dat</code>
Matlab file(s)	<code>kal_bcycle.m, bcycle.mat</code>

The data are monthly consisting of 6 Australian indicators on the business cycle (GDP, unemployment rate, employment, retail sales, household income and industrial production), and the coincident index, beginning in January 1980 and ending in September 2009, a total of $T = 357$ observations. The state-space business cycle model is

$$\begin{aligned} y_{i,t} &= \lambda_i s_t + u_{i,t}, & i &= 1, 2, \dots, 6 \\ s_t &= \phi_1 s_{t-1} + \phi_2 s_{t-2} + \eta_t \\ u_{i,t} &\sim N(0, \sigma_i^2), & i &= 1, 2, \dots, 6 \\ \eta_t &\sim N(0, 1), \end{aligned}$$

where y_t is the zero-mean percentage growth on indicator i and $\theta = \{\lambda, \phi, \sigma^2\}$ are parameters.

- (a) Compute the annual percentage growth rate of each indicator with the exception of the unemployment rate

$$y_{i,t} = 100 \times (\ln I_{i,t} - \ln I_{i,t-12}), \quad i = 1, 3, 4, 5, 6,$$

where $I_{i,t}$ is the i^{th} indicator. For the unemployment rate, compute

$$y_{2,t} = I_{2,t} - I_{2,t-12}.$$

Ensure that all the indicators are centered to have zero mean.

- (b) Estimate the unknown parameters θ by maximum likelihood and interpret the parameter estimates.
- (c) Estimate the business cycle using the smoothed conditional mean $s_{t|T}$ and using the annual percentage growth rate on the coincident index

$$bc_t = 100 \times (\ln CI_t - \ln CI_{t-12}).$$

Rescale $s_{t|T}$ to have the same sample variance as $bcycle_t$ by multiplying $s_{t|T}$ by the ratio of the standard deviation of bc_t to the standard deviation of $s_{t|T}$. Compare the turning points of these two estimates of the business cycle.

PART FIVE

NON-STATIONARY TIME SERIES

16

Nonstationary Distribution Theory

16.1 Introduction

A common feature of many economic and financial variables is that they exhibit trending behaviour. Typical examples in economics are output and consumption, while, in finance, examples consist of asset prices and dividends. The earliest approach to capture trends involves augmenting the specification of the model with a deterministic time trend. Nelson and Plosser (1982), however, show that this strategy can represent a misspecification of the dynamics of the model and argue that a stochastic trend modelled by a random walk is the more appropriate specification to capture trends in the data. Not only does this observation have important implications for the interpretation of the model's parameters, it also has implications for the distribution theory of the maximum likelihood estimator and associated test statistics used to perform inference.

The aim of this chapter is to develop the distribution theory of nonstationary processes with stochastic trends. Formally, the move from a stationary world to a nonstationary world based on stochastic trends involves increasing the absolute value of the parameter ϕ of the AR(1) model investigated in Chapter 13

$$y_t = \phi y_{t-1} + v_t, \quad v_t \sim iid(0, \sigma^2),$$

from the stationary region, $|\phi| < 1$, to the nonstationary region, $|\phi| \geq 1$. This seemingly innocuous change in ϕ , however, leads to fundamental changes to the distribution theory in three key ways.

- (1) Sample moments no longer have finite limits, as they do for stationary processes, but converge (weakly) to random variables.
- (2) The least squares estimator of ϕ is super consistent with convergence rates greater than the usual $\sqrt{T}$ rate that occurs for stationary processes.

- (3) The asymptotic distribution of the least squares estimator is non-standard in contrast to the asymptotic normality result for stationary processes.

The non-standard behaviour of the asymptotic distribution of nonstationary processes with stochastic trends carries over into hypothesis testing for unit roots (Chapter 17) and to estimation and testing using maximum likelihood methods in multivariate nonstationary models (Chapter 18), where the focus is on identifying linear combinations of random walk processes that are stationary, known as cointegration.

16.2 Specification

In this section, two competing models of trends are proposed. The first is a deterministic trend and the second is a stochastic trend modelled using a random walk. To highlight the widespread nature of trends in economic and financial data, the following example revisits the original data set used by Nelson and Plosser (1982).

Example 16.1 Nelson and Plosser Study

Figure 16.1 provides plots of 14 U.S. annual macroeconomic variables from 1860 to 1970. The variables are expressed in natural logarithms with the exception of the bond yield. All variables exhibit an upward trend with the exception of velocity, which shows a strong downward trend, and the unemployment rate, which tends to fluctuate around a constant level. $\square$

16.2.1 Models of Trends

Two possible specifications to model trends in the data are

$$\begin{aligned} y_t &= \beta_0 + \beta_1 t + e_t && \text{[Deterministic trend]} \\ y_t &= \delta + y_{t-1} + v_t, && \text{[Stochastic trend]} \end{aligned} \quad (16.1)$$

where e_t and v_t are *iid* disturbance terms. Either specification represents a nonstationary model. The deterministic trend specification assumes transitory (stationary) deviations (e_t) around a deterministic trend (t). The stochastic trend model is a random walk with drift ($\delta \neq 0$) in which y_t drifts upwards if $\delta > 0$ without following a deterministic path. For this model, it is the changes in the variable ($y_t - y_{t-1}$) that are transitory (v_t).

Example 16.2 Simulating Trends

Figure 16.1 also gives plots simulated data from the two nonstationary models in (16.1) for a sample size of $T = 111$. The simulated time paths of the two series are very different. The deterministic trend has a clear positive

linear trajectory whereas the stochastic trend displays a more circuitous positive path. A visual comparison of the two simulated trend series with the actual data in Figure 16.1 suggests that the stochastic trend model is the more appropriate specification for characterizing the trends observed in the data. $\square$

Differences in the two trend models in (16.1) can be highlighted more formally by solving the stochastic trend model backwards recursively to give

$$y_t = y_0 + \delta t + v_t + v_{t-1} + v_{t-2} + \cdots + v_1, \quad (16.2)$$

where y_0 represents the initial value of y_t which for exposition is assumed to be fixed. This equation is of a form similar to the deterministic trend model in (16.1) with y_0 representing the intercept and δ now representing the slope parameter on the time trend, t . However, the fundamental difference between the deterministic trend model in (16.1) and (16.2) is that the disturbance term in the latter is the cumulative sum of all shocks to the system

$$e_t = \sum_{i=1}^t v_i. \quad (16.3)$$

As the weight on each shock is the same, the effect of a shock on y_t does not decay over time. This means that if the data generating process is based on a stochastic trend, the specification of a deterministic trend model corresponds to a disturbance term having permanent (nonstationary) shocks rather than transitory (stationary) shocks.

Example 16.3 Nelson and Plosser Data Revisited

Table 16.1 provides estimates of the two trend models in (16.1) applied to the Nelson-Plosser data, and also of a combined model given by

$$y_t = \delta + \gamma t + \phi y_{t-1} + w_t,$$

where w_t is an *iid* disturbance term. Estimates of ϕ in the stochastic trend model are around unity, with the exception of the unemployment rate where the estimate is 0.754. Including a deterministic time trend results in slightly smaller estimates of ϕ but does not qualitatively change the conclusion that the trends in the data appear to be best characterized as random walks and not as deterministic trends. $\square$

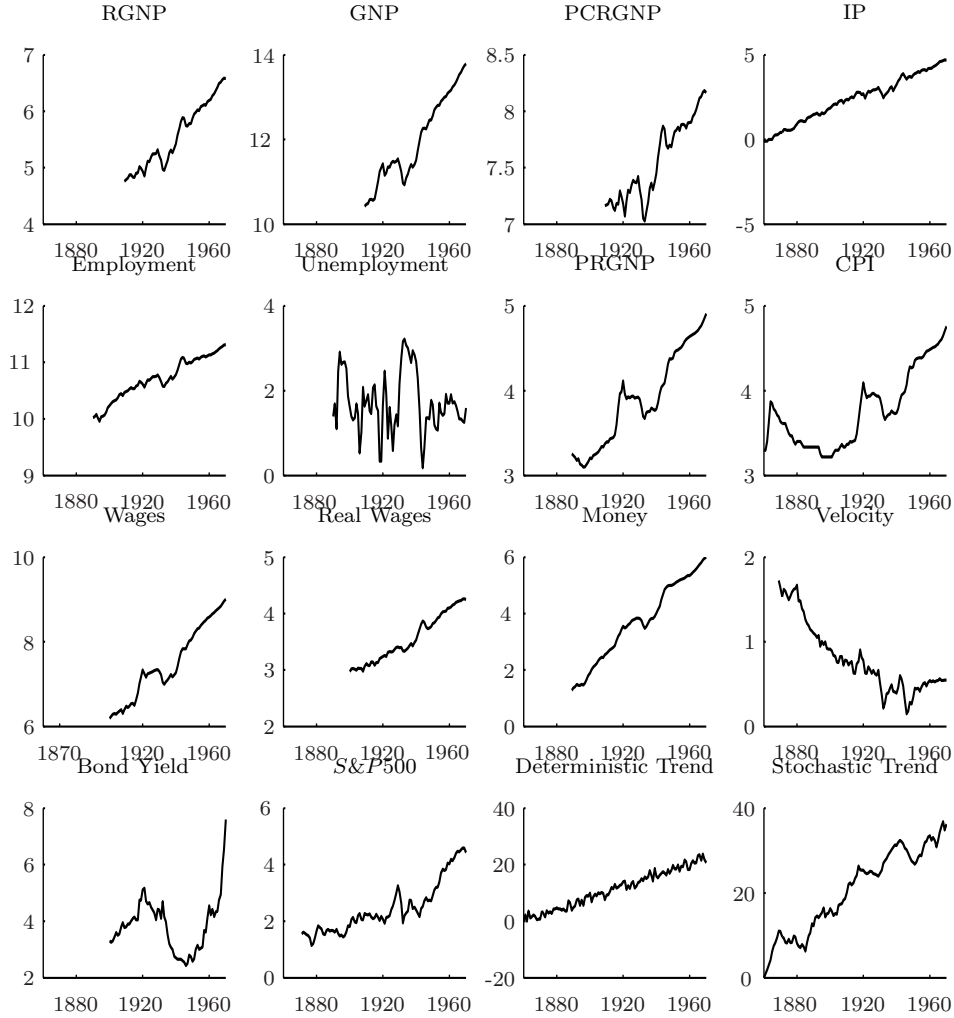


Figure 16.1 US annual macroeconomic variables from 1860 to 1970. All variables are expressed in natural logarithms with the exception of the bond yield. Also given are simulated series based on a deterministic trend and a stochastic trend, as given in (16.1). The parameter values of these two models are $\beta_0 = 0.1$, $\beta_1 = 0.2$ and $\delta_0 = 0.3$, with $e_t, v_t \sim iid N(0, 1)$.

16.2.2 Integration

Another way to write the stochastic trend model in (16.1) is

$$\Delta y_t = y_t - y_{t-1} = \delta + v_t. \quad (16.4)$$

Table 16.1

Ordinary least squares estimates of the deterministic and stochastic trend models in (16.1) and a combined trend model applied to the Nelson-Plosser data with standard errors in parentheses. The point estimates and standard errors of the time deterministic time trend are scaled up by 100 by defining the time trend as $t/100$ in the regression equations.

Variable	Deterministic		Stochastic		Combined		
	β_0	β_1	δ	ϕ	δ	γ	ϕ
Real GNP	4.614 (0.033)	3.099 (0.093)	0.007 (0.082)	1.004 (0.015)	0.591 (0.276)	0.418 (0.189)	0.876 (0.060)
Nominal GNP	10.341 (0.066)	5.343 (0.186)	0.021 (0.154)	1.003 (0.013)	0.710 (0.483)	0.386 (0.257)	0.935 (0.047)
Real GNP (per capita)	7.002 (0.033)	1.810 (0.093)	0.034 (0.182)	0.998 (0.024)	0.928 (0.424)	0.273 (0.118)	0.868 (0.060)
Industrial Prod.	0.050 (0.034)	4.212 (0.053)	0.055 (0.019)	0.995 (0.007)	0.056 (0.018)	0.662 (0.217)	0.841 (0.051)
Employment	10.095 (0.018)	1.532 (0.039)	0.127 (0.125)	0.990 (0.012)	1.136 (0.504)	0.162 (0.079)	0.889 (0.050)
Unemployment	1.871 (0.151)	-0.391 (0.325)	0.424 (0.136)	0.754 (0.073)	0.491 (0.171)	-0.142 (0.222)	0.748 (0.074)
GNP deflator	3.003 (0.034)	2.188 (0.072)	-0.015 (0.042)	1.009 (0.011)	0.208 (0.109)	0.182 (0.082)	0.933 (0.036)
CPI	3.161 (0.050)	1.090 (0.079)	-0.026 (0.048)	1.010 (0.013)	0.041 (0.065)	0.041 (0.028)	0.987 (0.020)
Wages	6.083 (0.044)	4.011 (0.110)	0.016 (0.073)	1.003 (0.010)	0.410 (0.254)	0.276 (0.171)	0.938 (0.042)
Real wages	2.848 (0.018)	1.996 (0.045)	0.008 (0.038)	1.003 (0.011)	0.380 (0.154)	0.272 (0.110)	0.871 (0.054)
Money stock	1.344 (0.042)	5.796 (0.089)	0.068 (0.019)	0.997 (0.005)	0.131 (0.049)	0.287 (0.205)	0.949 (0.035)
Velocity	1.408 (0.038)	-1.209 (0.064)	0.019 (0.015)	0.962 (0.016)	0.052 (0.051)	-0.032 (0.049)	0.941 (0.035)
Bond yield	3.685 (0.223)	0.486 (0.550)	-0.227 (0.160)	1.076 (0.041)	-0.359 (0.164)	0.393 (0.166)	1.075 (0.040)
Stock price	1.064 (0.078)	2.819 (0.137)	0.021 (0.047)	1.003 (0.018)	0.082 (0.053)	0.285 (0.124)	0.921 (0.040)
Det. trend	0.017 (0.180)	20.122 (0.283)	0.480 (0.293)	0.975 (0.023)	0.216 (0.182)	20.426 (1.949)	-0.015 (0.096)
Random walk	-1.524 (0.723)	30.696 (1.136)	0.500 (0.211)	0.980 (0.013)	0.242 (0.187)	0.935 (0.794)	0.969 (0.024)

This expression shows that the first difference of y_t is stationary provided that v_t is stationary. In this case, y_t is commonly referred to as being difference stationary, since by differencing the series once it is rendered stationary.

Similarly, in the case of the deterministic trend model, y_t is interpreted as being trend stationary because the subtraction of a deterministic trend from y_t renders the variable stationary. If differencing a series once achieves stationarity, the series is identified as integrated of order one, or $I(1)$. This definition follows from the fact that y_t is a partial sum as in (16.3), the integral, of lagged disturbances.

In some cases, it is necessary to difference a series twice before rendering it stationary

$$\Delta^2 y_t = (1 - L)^2 y_t = (1 + 2L - L^2) y_t = y_t + 2y_{t-1} - y_{t-2} = \delta + v_t,$$

where L is the lag operator (see Appendix B). Now y_t is integrated of order two, or $I(2)$. Variables that are $I(2)$ tend to be smoothly evolving over time, in contrast to $I(1)$ variables, which are also nonstationary but tend to show more jagged movements. This characteristic of an $I(2)$ process is not too surprising since y_t is a function of both y_{t-1} and y_{t-2} , which means that movements in y_t are averaged across two periods. This is not the case for an $I(1)$ process where no such averaging takes place since y_t is simply a function of y_{t-1} .

Extending the definition of integration to the case where y_t is $I(d)$, then

$$\Delta^d y_t = (1 - L)^d y_t = \delta + v_t,$$

is stationary. In the special case where y_t is $I(0)$, the series does not need to be differenced ($d = 0$) to achieve stationarity because it already is stationary.

16.3 Estimation

The parameter estimates of the deterministic and stochastic trend models in (16.1) for the Nelson and Plosser data are reported in Table 16.1. These estimates are based on the least squares estimator, which also corresponds to the maximum likelihood estimator in the case where the disturbance term is normally distributed. For this estimator to have the same asymptotic properties as the maximum likelihood estimator discussed in Chapters 2 and 13, it is necessary that certain sample moments have finite limits for the weak law of large numbers (WLLN) to be satisfied.

Given that the deterministic and stochastic trend models discussed in Section 16.2 yield nonstationary processes, it is not immediately obvious that a WLLN holds for the sample moments of these models. This section shows that, provided different scale factors are adopted to those used for stationary processes, the moments of both the deterministic and the stochastic trend models converge. For the deterministic trend model the sample moments

converge to finite limits, but, in the case of the stochastic trend model, the sample moments converge to random variables.

In order to provide a benchmark in terms of the properties of the least squares estimator, against which the asymptotic properties of these nonstationary models may be highlighted, the stationary AR(1) model is discussed initially.

16.3.1 Stationary Case

Consider the stationary AR(1) model

$$y_t = \delta + \phi y_{t-1} + v_t, \quad v_t \sim iid(0, \sigma^2), \quad (16.5)$$

where $|\phi| < 1$ is required for the process to be stationary. The ordinary least squares estimator of this equation is

$$\begin{bmatrix} \hat{\delta} \\ \hat{\phi} \end{bmatrix} = \begin{bmatrix} \sum_{t=2}^T 1 & \sum_{t=2}^T y_{t-1} \\ \sum_{t=2}^T y_{t-1} & \sum_{t=2}^T y_{t-1}^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=2}^T y_t \\ \sum_{t=2}^T y_{t-1} y_t \end{bmatrix}. \quad (16.6)$$

Using (16.5) to substitute out v_t , and rearranging gives

$$\begin{bmatrix} \hat{\delta} \\ \hat{\phi} \end{bmatrix} - \begin{bmatrix} \delta \\ \phi \end{bmatrix} = \begin{bmatrix} \sum_{t=2}^T 1 & \sum_{t=2}^T y_{t-1} \\ \sum_{t=2}^T y_{t-1} & \sum_{t=2}^T y_{t-1}^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=2}^T v_t \\ \sum_{t=2}^T y_{t-1} v_t \end{bmatrix}.$$

Now scale both sides by $\sqrt{T}$

$$\begin{aligned} \sqrt{T} \begin{bmatrix} \hat{\delta} - \delta \\ \hat{\phi} - \phi \end{bmatrix} &= \sqrt{T} \begin{bmatrix} \sum_{t=2}^T 1 & \sum_{t=2}^T y_{t-1} \\ \sum_{t=2}^T y_{t-1} & \sum_{t=2}^T y_{t-1}^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=2}^T v_t \\ \sum_{t=2}^T y_{t-1} v_t \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{T} \sum_{t=2}^T 1 & \frac{1}{T} \sum_{t=2}^T y_{t-1} \\ \frac{1}{T} \sum_{t=2}^T y_{t-1} & \frac{1}{T} \sum_{t=2}^T y_{t-1}^2 \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{\sqrt{T}} \sum_{t=2}^T v_t \\ \frac{1}{\sqrt{T}} \sum_{t=2}^T y_{t-1} v_t \end{bmatrix}. \quad (16.7) \end{aligned}$$

To understand the asymptotic distribution of the least squares estimator in expression (16.7), it is necessary to understand the convergence properties of the moments on the right-hand side of this equation. From Chapter 13,

the AR(1) model in (16.5) is rewritten as

$$y_t = \delta \sum_{j=0}^{t-1} \phi^j + \sum_{j=0}^{t-1} \phi^j v_{t-j} + \phi^t y_0, \quad (16.8)$$

where the initial observation, y_0 , is assumed to be fixed. The fixed initial value means that y_t is not stationary, but it is asymptotically stationary in the sense that the first two moments have fixed and finite limits as $t \rightarrow \infty$. The mean is

$$E[y_t] = \delta \sum_{j=0}^{t-1} \phi^j + \sum_{j=0}^{t-1} \phi^j E[v_{t-j}] + \phi^t y_0 = \phi^t y_0 + \delta \sum_{j=0}^{t-1} \phi^j, \quad (16.9)$$

and the variance is

$$\text{var}(y_t) = E[(y_t - E[y_t])^2] = E \left[\left(\sum_{j=0}^{t-1} \phi^j v_{t-j} \right)^2 \right] = \sigma^2 \sum_{j=0}^{t-1} \phi^{2j}, \quad (16.10)$$

where the last step follows from the *iid* assumption about v_t and $E[v_t^2] = \sigma^2$. Taking the limits of (16.9) and (16.10) gives, respectively,

$$\begin{aligned} \lim_{t \rightarrow \infty} E[y_t] &= \lim_{t \rightarrow \infty} \phi^t y_0 + \lim_{t \rightarrow \infty} \sum_{j=0}^{t-1} \phi^j \delta = \frac{\delta}{1 - \phi} \\ \lim_{t \rightarrow \infty} \text{var}(y_t) &= \lim_{t \rightarrow \infty} \sigma^2 \sum_{j=0}^{t-1} \phi^{2j} = \sigma^2 (1 + \phi^2 + \cdots) = \frac{\sigma^2}{1 - \phi^2}, \end{aligned} \quad (16.11)$$

and hence

$$\lim_{t \rightarrow \infty} E[y_t^2] = \frac{\sigma^2}{1 - \phi^2} + \frac{\delta^2}{(1 - \phi)^2}.$$

See also the analysis of the moments of stationary processes in Chapter 13.

Given that the first two moments have fixed and finite limits, the WLLN applies, so that

$$\frac{1}{T} \sum_{t=2}^T y_{t-1} \xrightarrow{p} \lim_{t \rightarrow \infty} E[y_t], \quad \frac{1}{T} \sum_{t=2}^T y_{t-1}^2 \xrightarrow{p} \lim_{t \rightarrow \infty} E[y_t^2].$$

Example 16.4 Simulating a Stationary AR(1) Model

The AR(1) model in (16.5) is simulated for $T = \{50, 100, 200, 400, 800, 1600\}$, using 50000 draws, with parameters $\theta = \{\delta = 0.0, \phi = 0.8, \sigma^2 = 1.0\}$ and

starting value of $y_0 = 0.0$. For each draw, the sample moments

$$m_1 = \frac{1}{T} \sum_{t=2}^T y_{t-1}, \quad m_2 = \frac{1}{T} \sum_{t=2}^T y_{t-1}^2,$$

are computed, with the means and variances of these quantities reported in Table 16.2. The two means converge to their respective theoretical values given by (16.11),

$$\begin{aligned} \lim_{t \rightarrow \infty} E[y_t] &= \frac{\delta}{1 - \phi} = \frac{0.0}{1 - 0.8} = 0.0 \\ \lim_{t \rightarrow \infty} \text{var}(y_t) &= \frac{\sigma^2}{1 - \phi^2} = \frac{1.0}{1 - 0.8^2} = 2.778, \end{aligned}$$

as T increases. The limits of these moments are finite with both variances approaching zero with the variances roughly halving as T is doubled. $\square$

Table 16.2

Simulation properties of the stationary AR(1) model in (16.5). The parameters are $\theta = \{\delta = 0.0, \phi = 0.8, \sigma^2 = 1.0\}$ with a starting value of $y_0 = 0.0$. The number of replications is 50000.

T	$m_1 = \frac{1}{T} \sum_{t=2}^T y_{t-1}$		$m_2 = \frac{1}{T} \sum_{t=2}^T y_{t-1}^2$	
	Mean	Variance	Mean	Variance
50	-0.001	0.428	2.620	1.240
100	-0.003	0.231	2.701	0.661
200	-0.002	0.120	2.738	0.340
400	-0.002	0.061	2.756	0.173
800	0.000	0.031	2.767	0.087
1600	0.000	0.016	2.772	0.044

From Chapter 2, the least squares estimator in (16.6) has an asymptotic normal distribution because $y_{t-1}v_t$ in (16.7) is a martingale difference sequence. This property is illustrated in panel (a) of Figure 16.2 which gives simulated sampling distributions of the standardized least squares estimator of ϕ where the true parameter is 0.8. The normal approximation for the smaller sample sizes is reasonable except in the tails, but the quality of the approximation quickly improves as the sample size increases.

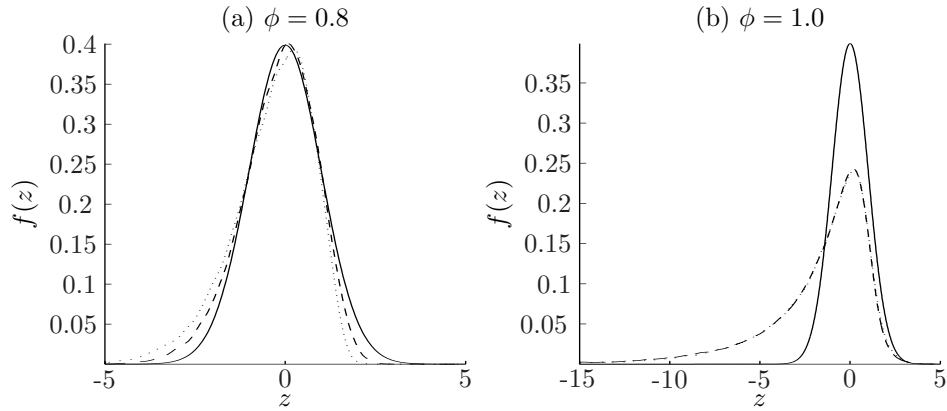


Figure 16.2 Finite sample distributions of the standardized least squares estimator of the AR(1) model for sample size $T = 50$ (dotted line) and $T = 200$ (dashed line) compared to the standard normal distribution (solid line). Simulated distributions are based on 50000 replications. Choice of standardization is $z = \sqrt{T}(\hat{\phi} - \phi)/\sqrt{1 - \phi^2}$ for $\phi < 1$ and $z = T(\hat{\phi} - \phi)$ for $\phi = 1$.

16.3.2 Nonstationary Case: Stochastic Trends

Consider the stochastic trend model in (16.1), which is obtained by setting $\phi = 1$ in equation (16.5). Now using $\phi = 1$ in (16.9) and (16.10) to find the moments of y_t reveals a very different result than the one obtained for the stationary model. The moment in the case of the mean is

$$E[y_t] = \phi^t y_0 + \sum_{j=0}^{t-1} \phi^j \delta = y_0 + \delta t,$$

whereas the variance is

$$\text{var}(y_t) = \sigma^2 \sum_{j=0}^{t-1} \phi^{2j} = \sigma^2 (1 + \phi^2 + \phi^4 + \cdots) = \sigma^2 t.$$

Both moments are nonstationary because they are functions of t . This suggests that the scaling of the moments $\sum_{t=2}^T y_{t-1}$ and $\sum_{t=2}^T y_{t-1}^2$ by T^{-1} in (16.7), as adopted in the stationary case, does not result in a convergent sequence in the nonstationary case. In fact, the appropriate scaling factors for these moments are $T^{-3/2}$ and T^{-2} , respectively, as the following example demonstrates.

Example 16.5 Sample Moments of a Stochastic Trend Model

The stochastic trend model

$$y_t = \delta + \phi y_{t-1} + w_t, \quad w_t \sim iid N(0, \sigma^2),$$

is simulated for samples of size $T = \{50, 100, 200, 400, 800, 1600\}$, using 50000 replications, with parameters $\theta = \{\delta = 0.0, \phi = 1.0, \sigma^2 = 1.0\}$ and starting value $y_0 = 0.0$. For each replication the moments

$$m_1 = \frac{1}{T^{3/2}} \sum_{t=2}^T y_{t-1}, \quad m_2 = \frac{1}{T^2} \sum_{t=2}^T y_{t-1}^2,$$

are computed with the means and variances of these quantities reported in Table 16.3. The means of the two quantities converge respectively to 0.0 and 0.5. In contrast to the stationary case, the variances of both quantities do not approach zero, but tend to converge to 1/3. This property is a reflection that m_1 and m_2 now converge to random variables and not to a finite limit as is the case for stationary processes. $\square$

Table 16.3

Simulation properties of the first two moments of the nonstationary AR(1) model. The number of replications is 50000.

T	$m_1 = \frac{1}{T^{3/2}} \sum_{t=2}^T y_{t-1}$		$m_2 = \frac{1}{T^2} \sum_{t=2}^T y_{t-1}^2$	
	Mean	Variance	Mean	Variance
50	-0.001	0.323	0.490	0.317
100	-0.002	0.329	0.496	0.330
200	-0.003	0.331	0.497	0.334
400	-0.002	0.328	0.494	0.324
800	-0.001	0.335	0.501	0.339
1600	0.003	0.336	0.503	0.339

This example demonstrates a fundamental difference between the moments of a stationary process and the moments of a nonstationary process. Whilst the moments of a stationary process converge to a finite limit using a suitable scaling, the moments of the nonstationary process converge, after suitable scaling, to a random variable, not a finite limit. Moreover, the scaling factor needed for the moments of nonstationary processes is not only higher than it is for the moments of stationary processes, but it also varies across moments according to the power of the pertinent moment.

The fact that sample moments converge to random variables and not finite

limits, suggests that the distribution of the least squares estimator in (16.6) is affected by this result. This is indeed the case, as demonstrated in panel (b) of Figure 16.2, which shows the distribution of the least squares estimator of ϕ corresponding to the AR(1) model in (16.5) where $\phi = 1$. The distribution of $\hat{\phi}$ is nonnormal exhibiting negative skewness with a relatively long left tail. Increasing the sample size reveals no sign of convergence to normality. The nonnormality feature of the distribution is also highlighted in Table 16.4. In fact, there is very little difference in the sampling distributions of $\hat{\phi}$ for the various sample sizes, suggesting that convergence to the asymptotic distribution is indeed super fast. These properties are formally derived in Section 16.6.

Table 16.4

Distributional properties of the sampling distribution of the least squares estimator of the AR(1) model in equation (16.5). The parameters are $\delta = 0.0$, $\phi = \{0.0, 0.8, 1.0\}$ and $\sigma^2 = 1.0$, with a starting value of $y_0 = 0.0$. The sample sizes are $T = \{50, 100, 200, 400, 800\}$ and the number of replications is 50000.

Statistics given are skewness (Skew), kurtosis (Kurt) and the proportion of simulated values of $\hat{\phi}$ less than ϕ (Prop).

T	$\phi = 0.0$			$\phi = 0.8$			$\phi = 1.0$		
	Skew.	Kurt.	Prop.	Skew.	Kurt.	Prop.	Skew.	Kurt.	Prop.
50	-0.005	2.871	0.496	-1.947	6.286	0.568	-3.882	16.134	0.677
100	-0.007	2.946	0.498	-1.461	4.854	0.550	-3.991	17.121	0.682
200	0.007	2.927	0.503	-1.094	4.028	0.538	-4.129	18.345	0.684
400	0.013	2.960	0.500	-0.757	3.496	0.527	-4.151	18.896	0.680
800	-0.023	2.984	0.502	-0.576	3.270	0.516	-4.097	18.022	0.681

16.3.3 Nonstationary Case: Deterministic Trends

Consider the nonstationary deterministic trend model in (16.1). The ordinary least squares estimator of the parameters of model is

$$\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} = \begin{bmatrix} \sum_{t=1}^T 1 & \sum_{t=1}^T t \\ \sum_{t=1}^T t & \sum_{t=1}^T t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T y_t \\ \sum_{t=1}^T ty_t \end{bmatrix}. \quad (16.12)$$

Substituting for y_t and rearranging gives

$$\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} - \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} \sum_{t=1}^T 1 & \sum_{t=1}^T t \\ \sum_{t=1}^T t & \sum_{t=1}^T t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T e_t \\ \sum_{t=1}^T te_t \end{bmatrix}. \quad (16.13)$$

In the stationary case both sides are multiplied by $\sqrt{T}$ since all elements in the matrix on the right-hand side of the equation converge to a stationary value in the limit. As with the nonstationary stochastic trend model, this result does not hold here because the rates of convergence of the estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ differ.

From the properties of time trends, the following results hold

$$\begin{aligned} \sum_{t=1}^T 1 &= T = O(T) \\ \sum_{t=1}^T t &= \frac{1}{2}T(T+1) = O(T^2) \\ \sum_{t=1}^T t^2 &= \frac{1}{6}T(T+1)(2T+1) = O(T^3), \end{aligned}$$

and consequently

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T 1 &= \lim_{T \rightarrow \infty} \frac{T}{T} = 1 \\ \lim_{T \rightarrow \infty} \frac{1}{T^2} \sum_{t=1}^T t &= \frac{1}{2} \lim_{T \rightarrow \infty} \frac{T(T+1)}{T^2} = \frac{1}{2} \lim_{T \rightarrow \infty} \left(1 + \frac{1}{T}\right) = \frac{1}{2} \\ \lim_{T \rightarrow \infty} \frac{1}{T^3} \sum_{t=1}^T t^2 &= \frac{1}{6} \lim_{T \rightarrow \infty} \frac{T(T+1)(2T+1)}{T^3} = \frac{1}{6} \lim_{T \rightarrow \infty} \left(2 + \frac{3}{T} + \frac{1}{T^2}\right) = \frac{1}{3}. \end{aligned} \quad (16.14)$$

The results in (16.14) suggest that (16.13) be scaled as follows

$$\begin{aligned} \begin{bmatrix} T^{1/2} & 0 \\ 0 & T^{3/2} \end{bmatrix} \begin{bmatrix} \hat{\beta}_0 - \beta_0 \\ \hat{\beta}_1 - \beta_1 \end{bmatrix} &= \begin{bmatrix} T^{1/2} & 0 \\ 0 & T^{3/2} \end{bmatrix} \begin{bmatrix} \sum_{t=1}^T 1 & \sum_{t=1}^T t \\ \sum_{t=1}^T t & \sum_{t=1}^T t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T e_t \\ \sum_{t=1}^T te_t \end{bmatrix} \\ &= \left[\begin{bmatrix} T^{1/2} & 0 \\ 0 & T^{3/2} \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T 1 & \sum_{t=1}^T t \\ \sum_{t=1}^T t & \sum_{t=1}^T t^2 \end{bmatrix} \begin{bmatrix} T^{1/2} & 0 \\ 0 & T^{3/2} \end{bmatrix} \right]^{-1} \\ &\quad \times \begin{bmatrix} T^{1/2} & 0 \\ 0 & T^{3/2} \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T e_t \\ \sum_{t=1}^T te_t \end{bmatrix}, \end{aligned}$$

or

$$\begin{bmatrix} T^{1/2}(\hat{\beta}_0 - \beta_0) \\ T^{3/2}(\hat{\beta}_1 - \beta_1) \end{bmatrix} = \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T 1 & \frac{1}{T^2} \sum_{t=1}^T t \\ \frac{1}{T^2} \sum_{t=1}^T t & \frac{1}{T^3} \sum_{t=1}^T t^2 \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{T^{1/2}} \sum_{t=1}^T e_t \\ \frac{1}{T^{1/2}} \sum_{t=1}^T te_t \end{bmatrix}. \quad (16.15)$$

From (16.14), the first term on the right-hand side has the limit

$$\lim_{T \rightarrow \infty} \begin{bmatrix} \frac{1}{T} \sum_{t=1}^T 1 & \frac{1}{T^2} \sum_{t=1}^T t \\ \frac{1}{T^2} \sum_{t=1}^T t & \frac{1}{T^3} \sum_{t=1}^T t^2 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{3} \end{bmatrix}. \quad (16.16)$$

As with the stationary AR(1) model, this is a finite limit, in contrast to the nonstationary stochastic trend model where the limit is a random variable.

Example 16.6 Sample Moments of a Deterministic Trend Model

The deterministic trend model in (16.1) is simulated for samples of size $T = \{50, 100, 200, 400, 800, 1600\}$ using 50000 replications, with parameters $\theta = \{\beta_0 = 0.1, \beta_1 = 0.2, \sigma^2 = 1.0\}$. For each replication the moments

$$m_1 = \frac{1}{T^2} \sum_{t=2}^T y_{t-1}, \quad m_2 = \frac{1}{T^3} \sum_{t=2}^T y_{t-1}^2,$$

are computed with the means and variances of these quantities reported in Table 16.5. The two means converge to their respective theoretical values of $1/2$ and $1/3$ in (16.16) as T increases. The sample variances are zero for all sample sizes verifying that m_1 and m_2 have finite limits. $\square$

Table 16.5

Simulation properties of the nonstationary deterministic trend model in (16.1) with parameters $\theta = \{\beta_0 = 0.1, \beta_1 = 0.2, \sigma^2 = 1.0\}$. The number of draws is 50000.

T	$\frac{1}{T^2} \sum_{t=1}^T y_{t-1}$		$\frac{1}{T^3} \sum_{t=1}^T y_{t-1}^2$	
	Mean	Variance	Mean	Variance
50	0.510	0.000	0.343	0.000
100	0.505	0.000	0.338	0.000
200	0.502	0.000	0.336	0.000
400	0.501	0.000	0.335	0.000
800	0.501	0.000	0.334	0.000
1600	0.500	0.000	0.334	0.000

Finally, since the second term on the right hand side of equation (16.15) satisfies the Lindberg-Feller central limit theorem of Chapter 2, the asymp-

otic distribution of the least squares estimator is

$$\begin{bmatrix} T^{1/2}(\hat{\beta}_0 - \beta_0) \\ T^{3/2}(\hat{\beta}_1 - \beta_1) \end{bmatrix} \xrightarrow{d} N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \sigma^2 \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{3} \end{bmatrix} \right). \quad (16.17)$$

These results show that the least squares estimator of the nonstationary deterministic trend model is consistent, although the rates of convergence differ: the intercept rate of convergence is $\sqrt{T}$ which occurs for the stationary model, whereas for $\hat{\beta}_1$ it is much faster occurring at the rate $T^{3/2}$. This higher rate of convergence is referred to as super-consistency. Furthermore, unlike the nonstationary model, but like the stationary model, the asymptotic distribution is normal provided that the true form of nonstationarity is a deterministic trend.

16.4 Asymptotics for Integrated Processes

The distinction between alternative trend model specifications is fundamental to time series analysis of nonstationary processes. The distribution theory associated with estimation and hypothesis testing in the case of the deterministic trend model is standard since it is based on asymptotic normality. However, in the case of the stochastic trend model, the distribution theory is non-standard. Given the importance of the stochastic trend model in characterizing economic and financial variables, the tools for understanding the non-standard nature of this distribution theory are now investigated. This section aims to outline the asymptotic distribution theory for integrated processes. Three fundamental tools are developed following the approach of Phillips (1987). The first is to express the discrete time random walk as a continuous-time stochastic process. The second is to introduce a new central limit theorem known as the functional central limit theorem (FCLT), which shows that this continuous-time process converges to a normally distributed stochastic process called Brownian motion. The third is to use a result known as the continuous mapping theorem which provides a generalization of the Slutsky theorem discussed in Chapter 2. The main theoretical result is that statistics based on $I(1)$ processes converge (weakly) to random variables that are functionals of Brownian motion.

16.4.1 Brownian Motion

As outlined in Chapter 12, Brownian motion is the continuous-time analogue of the discrete time random walk model

$$y_t = y_{t-1} + v_t, \quad v_t \sim iid N(0, \sigma^2), \quad (16.18)$$

or

$$y_t = y_{t-1} + \sigma z_t, \quad z_t \sim iid N(0, 1). \quad (16.19)$$

The movement from $t - 1$ to t , $\Delta y_t = y_t - y_{t-1}$ involves a discrete step of size σz_t .

To motivate the continuous-time feature of Brownian motion, consider breaking the step in (16.19) into smaller intervals. For example, suppose that the full step z_t represents a working week, broken down into $n = 5$ independent steps,

$$z_t = z_{1,t} + z_{2,t} + z_{3,t} + z_{4,t} + z_{5,t},$$

corresponding to each day of the working week, with each daily step distributed as

$$z_{i,t} \sim N\left(0, \frac{1}{5}\right).$$

As the $z_{i,t}$ are independent, the moments of the full step, z_t , still have zero mean

$$\begin{aligned} E[z_t] &= E[z_{1,t} + z_{2,t} + z_{3,t} + z_{4,t} + z_{5,t}] \\ &= E[z_{1,t}] + E[z_{2,t}] + E[z_{3,t}] + E[z_{4,t}] + E[z_{5,t}] \\ &= 0, \end{aligned}$$

and unit variance

$$\begin{aligned} E[z_t^2] &= E[(z_{1,t} + z_{2,t} + z_{3,t} + z_{4,t} + z_{5,t})^2] \\ &= E[z_{1,t}^2] + E[z_{2,t}^2] + E[z_{3,t}^2] + E[z_{4,t}^2] + E[z_{5,t}^2] \\ &= \frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \frac{1}{5} \\ &= 1. \end{aligned}$$

For n steps, the random walk in (16.19) becomes

$$y_t = y_{t-1} + \sigma \sum_{i=1}^n z_{i,t},$$

where now each step is

$$z_{i,t} \sim N\left(0, \frac{1}{n}\right).$$

Allowing for infinitely smaller time intervals, $n \rightarrow \infty$, the limit is a continuous-time process. This suggests the following continuous-time representation of a random walk where the continuous movement in $y(t)$ over a small time interval, $s = dt$, is given by

$$dy(t) = \sigma dB(t),$$

where $B(t)$ is known as standard Brownian motion with the property that it has independent increments distributed as

$$dB(t) \sim N(0, s).$$

Note that the t subscript is now placed in parentheses to emphasize that the function is continuous in time.

Formally, standard Brownian motion, denoted $B(s)$, $s \in [0, 1]$, is the stochastic process satisfying

- (i) $B(0) = 0$
- (ii) $B(s) \sim N(0, s)$
- (iii) $B(s) - B(r)$, is independent of $B(r)$ for all $0 \leq r < s \leq 1$.

Brownian motion is, thus, a normally distributed stochastic process whose increments are independent. The intuition behind the independence property is that future changes in a Brownian motion cannot be predicted by past Brownian motion. This result follows from the fact that $B(s)$ is the sum of independent random variables that also contain the independent random variables in $B(r)$. It follows, therefore, that $B(s) - B(r)$ contains just random variables that are in $B(s)$ but not in $B(r)$ and hence is independent of $B(r)$. The sample paths of Brownian motion are continuous but nowhere differentiable. Figure 12.2 in Chapter 12 provides an illustration of simulated Brownian motion for various time intervals.

16.4.2 Functional Central Limit Theorem

To derive asymptotic theory for a random walk, consider again the partial summation representation of (16.18)

$$y_t = y_0 + \sum_{j=1}^t v_j. \quad v_j \sim iid(0, \sigma^2), \quad (16.20)$$

It is assumed that $T^{-1/2}y_0 \xrightarrow{P} 0$ so y_0 can be treated as either fixed or random. It is useful to re-express (16.20) using alternative notation as

$$y_{[Ts]} = y_0 + \sum_{j=1}^{[Ts]} v_j, \quad 0 < s \leq 1, \quad (16.21)$$

where $[x]$ represents the largest integer less than or equal to x . The function $y_{[Ts]}$ is a right-continuous step function on $[0, 1]$. In this representation, a particular value of s measures the s^{th} proportion of the sample used in the summation. For example, if $T = 200$ and $s = 1/4$, $y_{[Ts]} = y_{50}$ is the observation a quarter way through the sample (first quartile), for $s = 1/2$, $y_{[Ts]} = y_{100}$ is the observation half way through the sample (median) and similarly if $s = 1$, $y_{[Ts]} = y_{200}$ which is the last observation in the sample. As $T \rightarrow \infty$, $y_{[Ts]}$ in (16.21) becomes continuously observed on the unit interval $0 \leq s \leq 1$, in contrast to y_t , which is still defined discretely on the integers $t = 1, \dots, T$, as in (16.20).

Example 16.7 Constructing a Continuous-Time Process

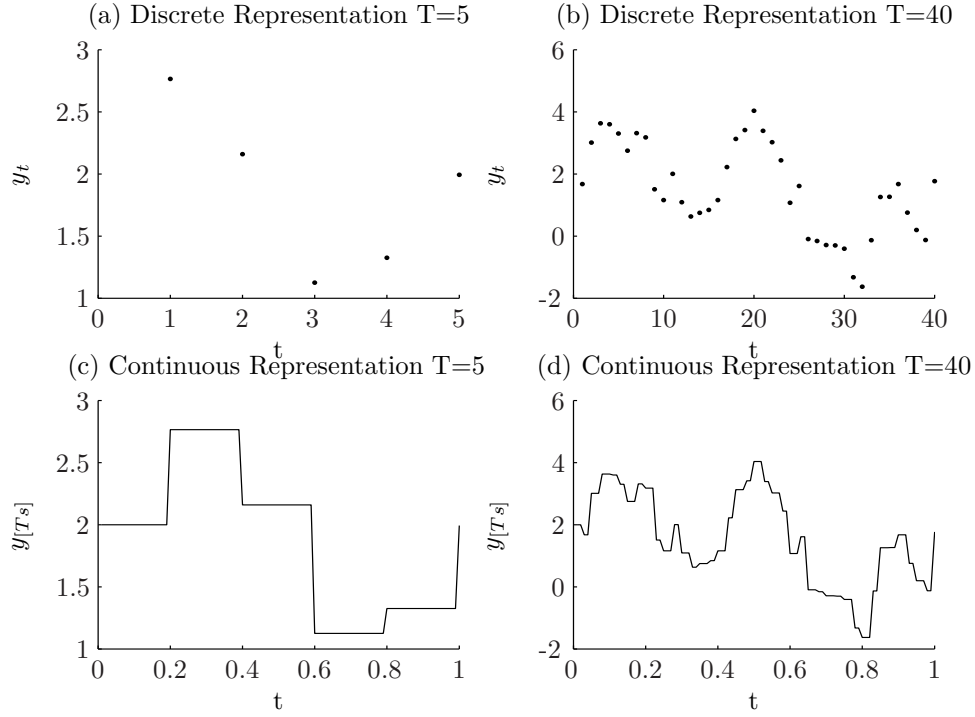
Panel (a) of Figure 16.3 gives a plot of $T = 5$ observations simulated from a random walk, $y_t = \{2.3, 3.3, 2.7, 4.7, 5.3\}$, with the construction of the right-continuous step function $y_{[Ts]}$ over $s \in [0, 1]$ given immediately below in panel (c). Increasing the sample size to $T = 40$, as in panel (b) of Figure 16.3, shows that the picture becomes more dense with the horizontal axis increasing in t , while for the continuous representation in Figure 16.3 panel (d) the horizontal axis still remains in the interval $[0, 1]$. $\square$

The previous example shows that the relationship between the discretely observed time series y_t in (16.20) and the continuous function $y_{[Ts]}$ in (16.21) is conveniently represented as

$$y_{[Ts]} = \begin{cases} 0 & : 0/T \leq s < 1/T \\ y_1 = v_1 & : 1/T \leq s < 2/T \\ y_2 = v_1 + v_2 & : 2/T \leq s < 3/T \\ \vdots & \vdots \\ y_T = v_1 + v_2 + \dots + v_T & : s = 1. \end{cases} \quad (16.22)$$

Inspection of Figure 16.3 shows that, by construction, y_t is a constant within the t^{th} interval, so the height and the width of the t^{th} bar of the histogram are respectively

$$\text{HEIGHT}_t = y_t, \quad \text{WIDTH}_t = \frac{t+1}{T} - \frac{t}{T} = \frac{1}{T},$$

Figure 16.3 Construction of $y_{[Ts]}$ for sample sizes of $T = 5$ and $T = 40$.

and the associated area for observation $s = t/T$ is

$$\text{AREA}_t = \int_{t/T}^{(t+1)/T} y_{[Ts]} ds = \int_{t/T}^{(t+1)/T} y_{[T \times \frac{t}{T}]} ds = \text{HEIGHT}_t \times \text{WIDTH}_t = \frac{y_t}{T}.$$

The total area corresponding to the $y_{[Ts]}$ function is

$$\begin{aligned} \text{AREA} &= \int_0^1 y_{[Ts]} ds = \sum_{t=1}^T \int_{t/T}^{(t+1)/T} y_{[T \times \frac{t}{T}]} ds \\ &= \frac{1}{T} \int_{1/T}^{2/T} y_{[T \times \frac{1}{T}]} ds + \frac{1}{T} \int_{2/T}^{3/T} y_{[T \times \frac{2}{T}]} ds + \dots \\ &= \frac{y_1}{T} + \frac{y_2}{T} + \dots + \frac{y_T}{T} \\ &= \frac{1}{T} \sum_{t=1}^T y_t, \end{aligned} \tag{16.23}$$

which is the sample mean, so that in effect each bar of the histogram in

panels (c) and (d) of Figure 16.3 represent the contribution of each respective observation to the sample mean of y_t .

To construct asymptotic theory for $y_{[Ts]}$ defined in (16.22) for a given $s \in [0, 1]$, note that $[Ts] \rightarrow \infty$ as $T \rightarrow \infty$. Equation (16.21) is then rescaled to yield

$$[Ts]^{-1/2} y_{[Ts]} = [Ts]^{-1/2} y_0 + [Ts]^{-1/2} \sum_{j=1}^{[Ts]} v_j \xrightarrow{d} N(0, \sigma^2),$$

or, by using the result $[Ts]/T \rightarrow s$,

$$T^{-1/2} y_{[Ts]} = \left(\frac{[Ts]}{T} \right)^{1/2} [Ts]^{-1/2} y_{[Ts]} \xrightarrow{d} s^{1/2} N(0, \sigma^2) = N(0, \sigma^2 s).$$

Alternatively, the standardized function

$$Y_T(s) = \sigma^{-1} T^{-1/2} y_{[Ts]}, \quad 0 \leq s \leq 1, \quad (16.24)$$

satisfies

$$Y_T(s) \xrightarrow{d} N(0, s), \quad (16.25)$$

for a given s . As an important special case, the standard central limit theorem arises from (16.25) where $s = 1$.

Now consider the entire random function $Y_T(\cdot)$ on $[0, 1]$, not just for a single s . Here $Y_T(\cdot)$ is referred to as a functional. The convergence properties of $Y_T(\cdot)$ is based on the following theorem.

Functional Central Limit Theorem

If y_t is the random walk $y_t = y_{t-1} + v_t$ where $T^{-1/2} y_0 \xrightarrow{p} 0$ and v_t is iid $(0, \sigma^2)$, the standardized function Y_T in (16.24) satisfies

$$Y_T(\cdot) \xrightarrow{d} B(\cdot), \quad (16.26)$$

where $B(s)$ is standard Brownian motion on $[0, 1]$. ■

Despite their apparent similarity, the FCLT result (16.26) and the asymptotic normality result (16.25) show important mathematical differences. The simple manipulations leading to (16.25) do not constitute a proof of the FCLT because convergence in distribution of $Y_T(s)$ to $N(0, s)$ for each s is not sufficient to conclude that $Y_T(\cdot)$ converges as a function to Brownian motion $B(\cdot)$ (Billingsley, 1968; Davidson, 1994).

Example 16.8 Order Statistics of a Random Walk

The random walk

$$y_t = y_{t-1} + v_t, \quad v_t \sim iid N(0, \sigma^2)$$

is simulated 10000 times with a starting value of $y_0 = 0$, $\sigma^2 = 1$ and a sample size of $T = 500$. The distributions of the standardized functions $Y_T(s) = \sigma^{-1}T^{-1/2}y_{[Ts]}$, for $s = 1/4$ (first quartile), $s = 1/2$ (median) and $s = 1$ (last observation) are given in Figure 16.4. All three sampling distributions are normally distributed with zero means but with differing variances. The variances of the simulated standardized functions are computed, respectively, as 0.2478, 0.4976, 1.0004 which agree with the theoretical values of $s = \{1/4, 1/2, 1.0\}$. $\square$

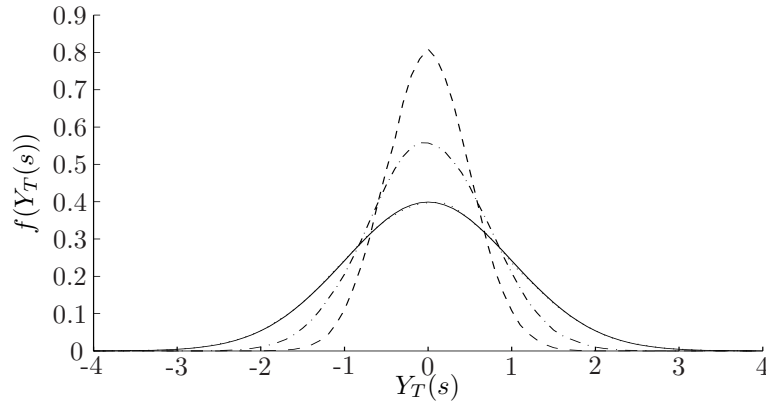


Figure 16.4 Sampling distributions of the standardized first quartile (dashed line), the median (dot-dashed line) and the last observation (dotted line) of a random walk compared to the standard normal distribution (solid line). The sampling distributions are based on a sample size of $T = 500$ and 10000 replications.

16.4.3 Continuous Mapping Theorem

An important additional tool to derive the distribution of statistics based on stochastic trends is the continuous mapping theorem. This theorem represents an extension of Slutsky's theorem encountered in Chapter 2, which is used in standard asymptotic distribution theory to derive the properties of continuous functions of random variables. Now the approach is to derive the properties of continuous functionals of random functions.

Continuous Mapping Theorem

Let $f(\cdot)$ be a continuous function. If $Y_T(s)$ is a sequence of functions such that $Y_T(s) \xrightarrow{d} B(s)$ then $f(Y_T(s)) \xrightarrow{d} f(B(s))$.

■

The continuous mapping theorem allows further results to be derived from the FCLT such as

$$\begin{aligned} (Y_T(s))^2 &\xrightarrow{d} B(s)^2, \\ \int_0^1 Y_T(s) ds &\xrightarrow{d} \int_0^1 B(s) ds, \\ \left[\int_0^1 Y_T(s) ds \right]^{-1} &\xrightarrow{d} \left[\int_0^1 B(s) ds \right]^{-1}. \end{aligned} \quad (16.27)$$

Example 16.9 Sample Mean of a Random Walk

From equation (16.23)

$$\frac{1}{T} \sum_{t=1}^T y_t = \int_0^1 y_{[Ts]} ds.$$

Multiplying both sides by $\sigma^{-1}T^{-1/2}$ and using the definition of $Y_T(s)$ in (16.24) shows that

$$\frac{1}{\sigma T^{3/2}} \sum_{t=1}^T y_t = \int_0^1 \sigma^{-1}T^{-1/2} y_{[Ts]} ds = \int_0^1 Y_T(s) ds \xrightarrow{d} \int_0^1 B(s) ds,$$

where the last step uses the continuous mapping theorem result in (16.27). Thus, a suitably standardized sample mean of a random walk converges weakly in distribution to an integral of a Brownian motion. □

From property (ii) of the definition of Brownian motion, $\int_0^1 B(s) ds$ is a continuous sum of normal random variables over s and is therefore also normally distributed. From the properties of the normal distribution, the mean of this term is zero since it is the continuous sum of normal random variables each having zero mean. In the case of the random walk, the variance of the standardized mean is $1/3$ (Banerjee, Dolado, Galbraith and Hendry, 1993), in which case

$$\frac{1}{\sigma\sqrt{T}} \bar{y} = \frac{1}{\sigma T^{3/2}} \sum_{t=1}^T y_t \xrightarrow{d} N(0, 1/3), \quad (16.28)$$

so the variance of $\int_0^1 B(s) ds$ is $\frac{1}{3}$. This result is developed in Exercise 6.

16.4.4 Stochastic Integrals

The discussion, so far, of the asymptotic properties of the least squares estimator in (16.7) in the presence of a random walk has focussed on the moments just involving y_t such as $\sum_{t=2}^T y_{t-1}$. Now the focus is on the sample moment

$$\sum_{t=2}^T y_{t-1} v_t,$$

which is important in establishing the asymptotic distribution of the least squares estimator discussed in Section 16.3. The asymptotic properties of this expression are given by the following theorem.

Stochastic Integral Theorem

If y_t is a random walk $y_t = y_{t-1} + v_t$ and v_t is iid $(0, \sigma^2)$, then

$$\frac{1}{\sigma^2 T} \sum_{t=2}^T y_{t-1} v_t \xrightarrow{d} \int_0^1 B(s) dB(s), \quad (16.29)$$

where $B(s)$ is standard Brownian motion. ■

An integral taken with respect to a Brownian motion is called a stochastic integral. This differs from an integral such as $\int_0^1 B(s) ds$, which is a random variable, but is not termed a stochastic integral since it is taken with respect to s . This functional can be expressed in terms of a centred chi-squared distribution as the following example demonstrates.

Example 16.10 Distribution of a Stochastic Integral

Consider the random walk

$$y_t = y_{t-1} + v_t, \quad v_t \sim iid(0, \sigma^2).$$

Square both sides, sum from $t = 2, \dots, T$ and rearrange to give

$$\sum_{t=2}^T y_{t-1} v_t = \frac{1}{2} (y_T^2 - y_1^2 - \sum_{t=2}^T v_t^2).$$

Scaling by $(\sigma^2 T)^{-1}$ results in the sample moment

$$\frac{1}{\sigma^2 T} \sum_{t=2}^T y_{t-1} v_t = \frac{1}{2} \left(\frac{1}{\sigma^2 T} y_T^2 - \frac{1}{\sigma^2 T} y_1^2 - \frac{1}{\sigma^2 T} \sum_{t=2}^T v_t^2 \right) \xrightarrow{d} \frac{1}{2} (B(1)^2 - 1),$$

by using (16.25) with $s = 1$, $\text{plim } T^{-1} y_1^2 = 0$ and $\text{plim } T^{-1} \sum_{t=2}^T v_t^2 = \sigma^2$. As $B(1)$ is $N(0, 1)$ and $B(1)^2$ is χ^2 with one degree of freedom with mean 1 and

variance 2, the distribution is a centred chi-square distribution with mean 0 and variance of $1/2$. This result is verified in Figure 16.5, which compares the analytical distribution with the sampling distribution obtained by 10000 simulations of a random walk and computing the standardized statistic m for each draw. The mean is 0.0002 and the variance is 0.4914, which agree with the theoretical moments, respectively. $\square$

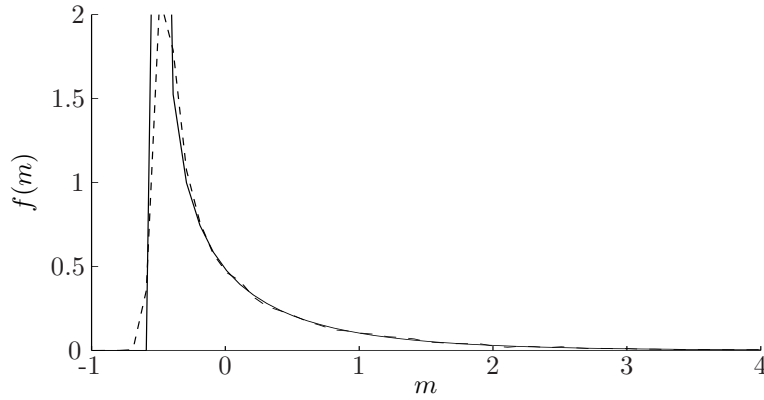


Figure 16.5 Sampling distribution of the standardized statistic $m = \sigma^{-2}T^{-1} \sum_{t=2}^T y_{t-1}v_t$ of a random walk. Distribution based on a sample size of $T = 500$ (dashed line) and 10,000 replications, compared to the centred χ_1^2 distribution (solid line).

The stochastic integral theorem is generalized by using the continuous mapping theorem whereby, if

$$f\left(\frac{1}{\sigma\sqrt{T}}y_{[Ts]}\right) = f(Y_T(s)) \xrightarrow{d} f(B(s)),$$

for some continuous functional $f(\cdot)$, then

$$\frac{1}{T} \sum_{t=2}^T f(y_{t-1})v_t \xrightarrow{d} \int_0^1 f(B(s))dB(s).$$

See also Exercise 7.

16.5 Multivariate Analysis

The derivation of the asymptotic distribution of the previous section is based on a single integrated process. Extending the scalar case to N integrated processes is relatively straightforward and provides a natural generalization

of the results from the previous section. Consider an N dimensional random walk

$$y_{i,t} = y_{i,t-1} + v_{i,t}, \quad i = 1, 2, \dots, N, \quad (16.30)$$

and $v_t = [v_{1,t}, v_{2,t}, \dots, v_{N,t}]'$ is a $(N \times 1)$ iid vector of disturbances with zero mean and covariance matrix

$$V = E[v_t v_t'] = \begin{bmatrix} \sigma_{1,1} & \sigma_{1,2} & \cdots & \sigma_{1,N} \\ \sigma_{2,1} & \sigma_{2,2} & \cdots & \sigma_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{N,1} & \sigma_{N,2} & \cdots & \sigma_{N,N} \end{bmatrix}.$$

The key multivariate results are summarized as follows.

Multivariate Functional Central Limit Theorem

Let $B(s)$ be an N dimensional Brownian motion with covariance matrix V with properties

- (i) $B(0) = 0$
- (ii) $B(s) \sim N(0, V)$
- (iii) $B(s) - B(r)$, is independent of $B(r)$ for all $0 \leq r < s \leq 1$.

It follows that

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{[Ts]} v_t \xrightarrow{d} B(s) \text{ on } s \in [0, 1].$$

■

Stochastic Integral Convergence

Defining the N dimensional cumulative sum as $y_t = \sum_{j=1}^t v_j$ yields the $(N \times N)$ matrix

$$\frac{1}{\sigma^2 T} \sum_{t=2}^T y_{t-1} v_t' \xrightarrow{d} \int_0^1 B(s) dB(s)'. \quad \blacksquare$$

Example 16.11 Bivariate and Mixed Normality

Suppose $N = 2$, in which case

$$\int_0^1 B(s) dB(s)' = \begin{bmatrix} \int_0^1 B_1(s) dB_1(s) & \int_0^1 B_1(s) dB_2(s) \\ \int_0^1 B_2(s) dB_1(s) & \int_0^1 B_2(s) dB_2(s) \end{bmatrix}.$$

The diagonal elements are centred chi-squared distributions from Example

16.10. For the off-diagonal elements, suppose B_1 and B_2 are independent, $\sigma_{1,2} = 0$. The distribution of $\int_0^1 B_2(s)dB_1(s)$ conditional on B_2 is

$$\int_0^1 B_2(s)dB_1(s) \Big| B_2 \sim N(0, \sigma_{1,1} \int_0^1 B_2(s)^2 ds).$$

Thus, $\int_0^1 B_2(s)dB_1(s)$ is conditionally normal given B_2 and hence unconditionally mixed normal. Figure 16.6 compares a mixed normal to a standard normal distribution. If B_1 and B_2 are dependent, $\sigma_{1,2} \neq 0$, then $\int_0^1 B_2(s)dB_1(s)$ is no longer unconditionally mixed normal, but a combination of mixed normal and χ_1^2 components. To see this, express B_1 as $B_1 = (\sigma_{1,2}/\sigma_{2,2})B_2 + B_{1|2}$, which is essentially a population regression of B_1 on B_2 with residual $B_{1|2}$, which is independent of B_2 by construction. It follows that

$$\int_0^1 B_2(s)dB_1(s) = \int_0^1 B_2(s)dB_{1|2}(s) + \frac{\sigma_{1,2}}{\sigma_{2,2}} \int_0^1 B_2(s)dB_2(s),$$

which is the sum of mixed normal and χ_1^2 components. $\square$

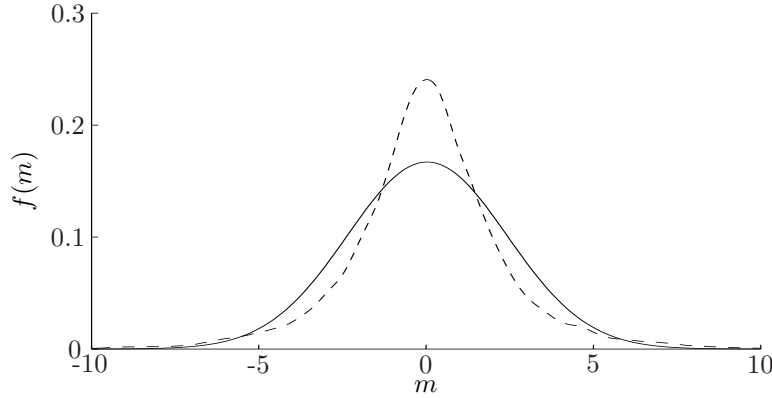


Figure 16.6 Comparison of a mixed normal (dashed line) and a normal distribution (solid line). The mixed normal distribution is obtained by simulating $V_{1,t}, V_{2,t} \sim N(0, 1)$ with $T = 4000$ and regressing $V_{2,t}$ on $y_{1,t}$ where $y_{1,t} = y_{1,t-1} + v_{1,t}$ to obtain an estimate of the regression parameter $\hat{\beta}$. The statistic is $m = T\hat{\beta}$. The number of replications is 20000.

16.6 Applications

The following two applications use the FCLT and the continuous mapping theorem to derive the asymptotic distribution of the least squares estimator

in the presence of integrated processes for two types of models. The first application focusses on estimating an AR(1) model where the true model is $I(1)$. The shape of the derived asymptotic distribution is nonstandard, which has important implications for understanding the properties of unit root tests discussed in Chapter 17. The second application looks at the problem of misspecifying the trend for the case where a deterministic trend is adopted while the true model is a random walk.

16.6.1 Least Squares Estimator of the AR(1) Model

Consider an AR(1) model

$$y_t = \phi y_{t-1} + v_t, v_t \sim iid(0, \sigma^2), \quad (16.31)$$

Suppose the true value is $\phi = 1$, so that y_t is a random walk without drift. The least squares estimator is

$$\hat{\phi} = \frac{\sum_{t=2}^T y_{t-1} y_t}{\sum_{t=2}^T y_{t-1}^2}. \quad (16.32)$$

As y_t is generated by the random walk in (16.31), this expression is used to rewrite (16.32) as

$$\hat{\phi} = \frac{\sum_{t=2}^T y_{t-1} (y_{t-1} + v_t)}{\sum_{t=2}^T y_{t-1}^2} = 1 + \frac{\sum_{t=2}^T y_{t-1} v_t}{\sum_{t=2}^T y_{t-1}^2}.$$

Rearranging and scaling both sides by T gives

$$T(\hat{\phi} - 1) = \frac{T^{-1} \sum_{t=2}^T y_{t-1} v_t}{T^{-2} \sum_{t=2}^T y_{t-1}^2}, \quad (16.33)$$

where the choice of scaling is motivated by Example (16.5) and (16.29). Thus,

$$\begin{aligned} \frac{1}{T^2} \sum_{t=2}^T y_{t-1}^2 &\xrightarrow{d} \sigma^2 \int_0^1 B(s)^2 ds \\ \frac{1}{T} \sum_{t=2}^T y_{t-1} v_t &\xrightarrow{d} \sigma^2 \int_0^1 B(s) dB(s) = \frac{\sigma^2}{2} (B(1)^2 - 1). \end{aligned}$$

Using these expressions in (16.33) shows that the least squares estimator, $\hat{\phi}$, converges weakly to the following random variable

$$T(\hat{\phi} - 1) \xrightarrow{d} \frac{\frac{\sigma^2}{2}(B(1)^2 - 1)}{\sigma^2 \int_0^1 B(s)^2 ds} = \frac{(B(1)^2 - 1)}{2 \int_0^1 B(s)^2 ds}. \quad (16.34)$$

There are two main implications of (16.34).

(1) **Super Consistency**

Since the right-hand side of (16.34) is a random variable independent of T , it follows that

$$T(\hat{\phi} - 1) = O_p(1),$$

where the subscript p denotes probability. Dividing both sides by T yields

$$\hat{\phi} - 1 = O_p(T^{-1}),$$

and taking the limit as $T \rightarrow \infty$, the right-hand side goes to zero as the random variable is eventually dominated by T^{-1} . That is,

$$\hat{\phi} - 1 \xrightarrow{p} 0,$$

and $\hat{\phi}$ is a consistent estimator of $\phi = 1$. As the rate of convergence is T , which is faster than the usual convergence rate of $\sqrt{T}$, the least squares estimator in (16.32) is super consistent.

(2) **Non-standard Distribution**

The distribution of the random variable in (16.34) is non-standard because it is the ratio of a centred chi-square random variable with one degree of freedom, $0.5(B(1)^2 - 1)$, and a non-normal random variable, $\int_0^1 B(s)^2 ds$. The distribution of the least squares estimator in (16.32) was given in panel (b) of Figure 16.2 based on Monte Carlo simulation. The strong negative skewness property of the sampling distribution arises from $B(1)^2$ having a chi-square distribution with one degree of freedom. In particular, the probability that a random variable (x) is less than unity is

$$p(x < 1) = \int_0^1 \frac{1}{\Gamma(0.5)2^{0.5}} x^{-0.5} \exp(-x/2) dx = 0.683, \quad (16.35)$$

which corresponds to the proportion of simulated draws of $\hat{\phi} < 1$ reported in Table 16.4. For a sample of size $T = 50$ this proportion is 0.677, which quickly approaches the asymptotic value of 0.683 in (16.35) for larger sample sizes.

16.6.2 Trend Misspecification

Consider the case where the true form of nonstationarity is stochastic given by a random walk with drift but the data are modelled using a deterministic linear trend

$$\begin{aligned} \text{True:} \quad y_t &= \delta + y_{t-1} + v_t \\ \text{Specified:} \quad y_t &= \beta_0 + \beta_1 t + e_t, \end{aligned} \quad (16.36)$$

where v_t and e_t are disturbances, with v_t assumed to be *iid* $(0, \sigma^2)$. Rewrite the true model as

$$y_t = \delta t + u_t, \quad (16.37)$$

where, for simplicity, $y_0 = 0$ and u_t is also a disturbance term given by the accumulation of the true disturbance, v_t in (16.36),

$$u_t = \sum_{i=1}^t v_i. \quad (16.38)$$

The least squares slope estimator of the specified deterministic trend model in (16.36) is

$$\hat{\beta}_1 = \frac{\sum_{t=1}^T (t - \bar{t})(y_t - \bar{y})}{\sum_{t=1}^T (t - \bar{t})^2} = \frac{T \sum_{t=1}^T t y_t - (\sum_{t=1}^T t)(\sum_{t=1}^T y_t)}{T \sum_{t=1}^T t^2 - (\sum_{t=1}^T t)^2}. \quad (16.39)$$

As y_t is generated by (16.37), equation (16.39) is rewritten as

$$\hat{\beta}_1 - \delta = \frac{T \sum_{t=1}^T t u_t - \sum_{t=1}^T t \sum_{t=1}^T u_t}{T \sum_{t=1}^T t^2 - (\sum_{t=1}^T t)^2},$$

or

$$\sqrt{T}(\hat{\beta}_1 - \delta) = \frac{T^{-5/2} \sum_{t=1}^T t u_t - (T^{-2} \sum_{t=1}^T t)(T^{-3/2} \sum_{t=1}^T u_t)}{T^{-3} \sum_{t=1}^T t^2 - (T^{-2} \sum_{t=1}^T t)^2}. \quad (16.40)$$

Now consider

$$\frac{1}{T^{5/2}} \sum_{t=1}^T t u_t \xrightarrow{d} \sigma \int_0^1 s B(s) ds, \quad \frac{1}{T^{3/2}} \sum_{t=1}^T u_t \xrightarrow{d} \sigma \int_0^1 B(s) ds,$$

and from (16.14)

$$\lim_{T \rightarrow \infty} \frac{1}{T^2} \sum_{t=1}^T t = \frac{1}{2}, \quad \lim_{T \rightarrow \infty} \frac{1}{T^3} \sum_{t=1}^T t^2 = \frac{1}{3}.$$

Using these expressions in (16.40) shows that $\hat{\beta}_1$ converges weakly to the following random variable

$$\begin{aligned}\sqrt{T}(\hat{\beta}_1 - \delta) &\xrightarrow{d} \frac{\sigma \int_0^1 sB(s)ds - \left(\frac{1}{2}\right)\left(\sigma \int_0^1 B(s)ds\right)}{\frac{1}{3} - \left(\frac{1}{2}\right)^2} \\ &= 12\sigma \left(\int_0^1 sB(s)ds - \frac{1}{2} \int_0^1 B(s)ds \right) \\ &= 12\sigma \left(\int_0^1 \left(s - \frac{1}{2}\right) B(s)ds \right).\end{aligned}$$

The distribution of this random variable is known and given by Durlauf and Phillips (1988)

$$\sqrt{T}(\hat{\beta}_1 - \delta) \xrightarrow{d} N\left(0, \frac{6\sigma^2}{5}\right). \quad (16.41)$$

Equation (16.41) shows that the least squares estimator $\hat{\beta}_1$ in (16.39) is consistent as it converges to δ at the rate $\sqrt{T}$. Durlauf and Phillips (1988) show that the pertinent t-statistic diverges, which leads to the incorrect conclusion that y_t has a deterministic linear trend. To demonstrate this last point, the following random walk with drift of $\delta = 0.5$, is simulated for a sample of size $T = 100$

$$y_t = 0.5 + y_{t-1} + v_t,$$

where $v_t \sim N(0, 1)$ and $y_0 = 0.0$. The least squares estimates of the deterministic trend model, with t-statistics in parentheses, are

$$y_t = \underset{(-1.739)}{-0.683} + \underset{(81.189)}{0.557t} + \hat{v}_t.$$

The estimate of the slope on the time trend provides a good estimate of the drift parameter $\delta = 0.5$. The t-statistic shows that this parameter estimate is highly significant, a finding which is misleading because it suggests the deterministic time trend is an important explanatory variable when the true model is a random walk with drift.

16.7 Exercises

(1) Nelson-Plosser Data

Gauss file(s)	nts_nelplos.g, nelson_plosser.xls
Matlab file(s)	nts_nelplos.m, nelson_plosser.mat

The data consist of 14 annual U.S. macroeconomic time series with varying starting dates spanning the period 1860 to 1970. All the variables are expressed in natural logarithms with exception of the bond yield.

- (a) Plot the 14 times series and discuss their time series properties.
- (b) Compute and interpret the ACF of each series. Compare the results with those reported in Nelson and Plosser (1982, Table 2, p.147) as well as in Table 16.1.
- (c) Consider the regression equation

$$y_t = \delta + \delta_1 t + \phi y_{t-1} + v_t, \quad v_t \sim iid(0, \sigma^2).$$

For each series, estimate the following models and interpret the results

- (i) Deterministic trend model ($\phi = 0$).
 - (ii) AR(1) model with drift ($\delta_1 = 0$).
 - (iii) AR(1) model with drift and a deterministic trend.
- (d) Simulate the deterministic trend model

$$y_t = 0.1 + 0.2t + v_t, \quad v_t \sim iid(0, \sigma^2),$$

with $\sigma^2 = 1$ and $T = 11$. Repeat parts (a) to (c).

- (e) Simulate the stochastic trend model (random walk with drift)

$$y_t = 0.3 + y_{t-1} + v_t, \quad v_t \sim iid(0, \sigma^2),$$

where $\sigma^2 = 1$ and $T = 111$. Repeat parts (a) to (c).

(2) Integration

Gauss file(s)	<code>nts_integration.g</code>
Matlab file(s)	<code>nts_integration.m</code>

Consider the following four models:

- | | | |
|-------|---|------------------------|
| (i) | $y_t = \delta + v_t$ | [I(0), White noise] |
| (ii) | $y_t = \delta + 0.5y_{t-1} + v_t$ | [I(0), Stationary] |
| (iii) | $y_t = \delta + y_{t-1} + v_t$ | [I(1), Nonstationary] |
| (iv) | $y_t = \delta + 2y_{t-1} - y_{t-2} + v_t$ | [I(2), Nonstationary], |

where $v_t \sim N(0, \sigma^2)$ with $\sigma^2 = 1$.

- (a) For each model, simulate $T = 200$ observations with the initial condition $y_0 = 0.0$ and with the drift parameter set at $\delta = 0$. Plot the series and compare their time series properties.
- (b) Repeat part (a) with a drift parameter of $\delta = 0.1$.

- (c) Repeat part (a) with a drift parameter of $\delta = 0.5$.

(3) Properties of Moments for Alternative Models

Gauss file(s)	<code>nts_moment.g</code>
Matlab file(s)	<code>nts_moment.m</code>

- (a) Simulate the AR(1) model

$$y_t = 0.0 + 0.8y_{t-1} + v_t, \quad v_t \sim N(0, \sigma^2),$$

with $\sigma^2 = 1$ for $T = \{50, 100, 200, 400, 800, 1600\}$ using 50000 draws and a starting value of $y_0 = 0.0$. For each draw, compute the moments

$$m_i = \frac{1}{T} \sum_{t=2}^T y_{t-1}^i, \quad i = 1, 2, 3, 4,$$

and the means and variances of these quantities.

- (b) Simulate the stochastic trend model (random walk without drift)

$$y_t = y_{t-1} + v_t, \quad v_t \sim N(0, \sigma^2),$$

with $\sigma^2 = 1$ for $T = \{50, 100, 200, 400, 800, 1600\}$ using 50000 draws and a starting value of $y_0 = 0.0$. For each draw, compute the moments

$$m_i = \frac{1}{T^{(1+i/2)}} \sum_{t=2}^T y_{t-1}^i, \quad i = 1, 2, 3, 4,$$

and the means and variances of these quantities.

- (c) Simulate the deterministic trend model

$$y_t = 0.1 + 0.2t + v_t, \quad v_t \sim N(0, \sigma^2),$$

with $\sigma^2 = 1$ for $T = \{50, 100, 200, 400, 800, 1600\}$ using 50000 draws and a starting value of $y_0 = 0.0$. For each draw, compute the moments

$$m_i = \frac{1}{T^{(1+i)}} \sum_{t=2}^T t^i, \quad i = 1, 2, 3, 4,$$

and the means and variances of these quantities.

- (d) Discuss the properties of the moments computed in (a) to (c).

(4) Sampling Distribution of the AR(1) Least Squares Estimator

Gauss file(s)	<code>nts_distribution.g</code>
Matlab file(s)	<code>nts_distribution.m</code>

Consider the AR(1) model

$$y_t = \phi y_{t-1} + v_t, \quad v_t \sim N(0, \sigma^2).$$

- (a) Simulate the model with $\phi = 0$ and $\sigma^2 = 1$ for samples of size $T = \{50, 100, 200, 400, 800, 1600\}$ using 50000 draws. For each draw, compute the standardized statistic

$$z = \frac{\sqrt{T}(\hat{\phi} - \phi)}{\sqrt{1 - \phi^2}},$$

where $\hat{\phi}$ is the least squares estimator from regressing y_t on y_{t-1} . Use a nonparametric kernel to compute the sampling distributions and compare these distributions with the standardized normal distribution. Compute the skewness and kurtosis of the sampling distributions as well as the proportion of $\hat{\phi}$ draws less than ϕ .

- (b) Repeat part (a) with $\phi = 0.8$.
 (c) Repeat part (a) with $\phi = 1.0$, except compute the statistic

$$z = T(\hat{\phi} - \phi).$$

(5) Demonstrating the Functional Central Limit Theorem

Gauss file(s)	<code>nts_yts.g</code> , <code>nts_fcflt.g</code>
Matlab file(s)	<code>nts_yts.m</code> , <code>nts_fcflt.m</code>

- (a) Simulate the random walk

$$y_t = y_{t-1} + v_t, \quad v_t \sim N(0, \sigma^2),$$

with $\sigma^2 = 1$ and $y_0 = 0$ for a sample size of $T = 5$. Construct the variable

$$y_{[Ts]} = \sum_{j=1}^{[Ts]} v_j,$$

for $s = \{0.2, 0.4, 0.6, 0.8, 1.0\}$.

- (b) Repeat part (a) for $T = 10$ and $s = \{0.1, 0.2, 0.3, \dots, 1.0\}$.
 (c) Repeat part (a) for $T = 40$ and $s = \{1/40, 2/40, 3/40, \dots, 1.0\}$.

- (d) Now simulate the random walk in part (a) 50000 times with a sample of size $T = 500$. For each draw, compute the standardized statistic

$$Y_T(s) = \sigma^{-1} T^{-1/2} y_{[Ts]}, \quad s = 1/40 \leq s \leq 1,$$

with $s = \{0.25, 0.5, 1.0\}$ and show that the simulated distributions are $Y_T(s) \xrightarrow{d} N(0, s)$.

(6) Demonstrating the Continuous Mapping Theorem

Gauss file(s)	<code>nts_cmt.g</code>
Matlab file(s)	<code>nts_cmt.m</code>

Simulate the random walk

$$y_t = y_{t-1} + v_t, \quad v_t \sim N(0, \sigma^2),$$

with $\sigma^2 = 1$ and $y_0 = 0$ for a sample size of $T = 500$. The number of draws is 10000.

- (a) Given that $m = \sigma^{-1} T^{-1/2} \bar{y} \xrightarrow{d} \int_0^1 B(s) ds$, simulate the sampling distribution of m and verify that the functional can be expressed as $N(0, 1/3)$.
- (b) Given that $m = \sigma^{-1} T^{-5/2} \sum_{t=1}^T t y_t \xrightarrow{d} \int_0^1 s B(s) ds$, simulate the sampling distribution of m and verify that the functional can be expressed as $N(0, 2/15)$.

(7) Simulating the Distribution of Stochastic Integrals

Gauss file(s)	<code>nts_stochint.g</code>
Matlab file(s)	<code>nts_stochint.m</code>

Simulate the random walk

$$y_t = y_{t-1} + v_t, \quad v_t \sim N(0, \sigma^2),$$

with $\sigma^2 = 1$ and $y_0 = 0$ for a sample size of $T = 500$. The number of draws is 10000.

- (a) Given that $m = \sigma^{-2} T^{-1} \sum_{t=2}^T y_{t-1} v_t \xrightarrow{d} \int_0^1 B(s) dB(s)$, simulate the sampling distribution of m and verify that the functional can be expressed as a centred chi-squared distribution with zero mean and variance $1/2$.

- (b) Given that $m = \sigma^{-3}T^{-3/2} \sum_{t=2}^T y_{t-1}^2 v_t \xrightarrow{d} \int_0^1 B(s)^2 dB(s)$ from applying the continuous mapping theorem to a stochastic integral, simulate the distribution of the functional and compare its shape with the standardized normal distribution.

(8) Simulating a Mixed Normal Distribution

Gauss file(s)	<code>nts_mixednormal.g</code>
Matlab file(s)	<code>nts_mixednormal.m</code>

For a sample of size $T = 4000$, simulate two independent normal random variables $V_{1,t}, V_{2,t} \sim N(0, 1)$ and compute the slope coefficient, $\hat{\beta}$, from a regression of $V_{2,t}$ on $y_{1,t}$, where

$$y_{1,t} = y_{1,t-1} + V_{1,t}.$$

Compute the scaled statistic $T\hat{\beta}$ and repeat 20000 times. Plot the distribution of $T\hat{\beta}$ and compare the shape of this distribution with the standard normal distribution based on the mean and the variance of $\hat{\beta}$.

(9) Spurious Regression Problem

Gauss file(s)	<code>nts_spurious1.g, nts_spurious2.g</code>
Matlab file(s)	<code>nts_spurious1.m, nts_spurious2.m</code>

A spurious relationship occurs when two independent variables are incorrectly identified as being related. A simple test of independence is based on the estimated correlation coefficient, $\hat{\rho}$.

- (a) Consider the following bivariate models

- | | | |
|-------|---|--|
| (i) | $y_{1,t} = v_{1,t},$ | $y_{2,t} = v_{2,t}$ |
| (ii) | $y_{1,t} = y_{1,t-1} + v_{1,t},$ | $y_{2,t} = y_{2,t-1} + v_{2,t}$ |
| (iii) | $y_{1,t} = y_{1,t-1} + v_{1,t},$ | $y_{2,t} = 2y_{2,t-1} - y_{2,t-2} + v_{2,t}$ |
| (iv) | $y_{1,t} = 2y_{1,t-1} - y_{1,t-2} + v_{1,t},$ | $y_{2,t} = 2y_{2,t-1} - y_{2,t-2} + v_{2,t}$ |

where $v_{1,t}, v_{2,t}$ are *iid* $N(0, \sigma^2)$ with $\sigma^2 = 1$. Simulate each bivariate model 10000 times for a sample of size $T = 100$ and compute the correlation coefficient, $\hat{\rho}$, of each draw. Compute the sampling distributions of $\hat{\rho}$ for the four sets of bivariate models and discuss the properties of these distributions in the context of the spurious regression problem.

- (b) Repeat part (a) with $T = 500$. What do you conclude?

- (c) Repeat part (a), except for each draw estimate the regression model

$$y_{2,t} = \beta_0 + \beta_1 y_{1,t} + u_t, \quad u_t \sim iid(0, \sigma^2).$$

Compute the sampling distributions of the least squares estimator $\hat{\beta}_1$ and its t-statistic for the four sets of bivariate models and discuss the properties of these distributions in the context of the spurious regression problem.

(10) **Trend Misspecification**

Gauss file(s)	<code>nts_trend.g</code>
Matlab file(s)	<code>nts_trend.m</code>

- (a) Simulate the random walk with drift

$$y_t = \delta + y_{t-1} + v_t, \quad v_t \sim N(0, \sigma^2),$$

where $\delta = 0.5$, $\sigma^2 = 1$ and $y_0 = 0$ for a sample size of $T = 100$. Using the simulated data, estimate the deterministic trend model

$$y_t = \beta_0 + \beta_1 t + e_t,$$

and comment on the parameter estimates $\hat{\beta}_0$ and $\hat{\beta}_1$.

- (b) Repeat part (a) with drift parameter $\delta = 1.0$.
 (c) Repeat parts (a) and (b) for $T = 1000$.
 (d) Simulate the random walk with drift model in part (a) 50000 times for a sample of size $T = 5000$. Compute the mean and the variance of the sampling distribution of $\hat{\beta}_1$ in the deterministic trend model of part (a) and compare these statistics with the moments of the asymptotic distribution $N(0, 6\sigma^2/5)$ derived in (16.41).

17

Unit Root Testing

17.1 Introduction

An important conclusion to emerge from Chapter 16 is that many economic processes may exhibit stochastic trends as described by a random walk. This observation suggests that a natural test of a random walk is to estimate the AR(1) model

$$y_t = \phi y_{t-1} + v_t, \quad v_t \sim iid(0, \sigma^2), \quad (17.1)$$

and test whether the slope parameter ϕ is equal to unity using a t-statistic. As the test on the slope is equivalent to testing that the root of the polynomial $(1 - \phi L)$ is unity, these tests are commonly referred to as unit root tests.

When a unit root test is performed, the distribution of the t-statistic is nonstandard because the process is nonstationary under the null hypothesis of $\phi = 1$. This situation is different to hypothesis tests performed on stationary processes under the null conducted in Chapter 13, where a standard asymptotic distribution arises, based on $\sqrt{T}$ consistency and normality. The results of Chapter 16 show that the least squares estimator $\hat{\phi}$ in (17.1) has a nonstandard distribution expressed in terms of functionals of Brownian motion. It is not too surprising, therefore, that this nonstandard distribution of $\hat{\phi}$ also manifests itself in the distribution of the t-statistic.

17.2 Specification

Since the original unit root test proposed by Dickey and Fuller (1979, 1981), there has been an explosion of proposed testing procedures, a survey of which is to be found in Maddala and Kim (1998). The major themes of unit root testing may be summarized as follows:

- (1) choice of detrending method;
- (2) procedures to test $\phi = 1$;
- (3) size and power properties of the test statistics;
- (4) autocorrelation dynamics;
- (5) structural breaks; and
- (6) the importance of initial conditions.

To motivate some of these themes, consider Figure 17.1, which shows a plot of the log of real quarterly U.S. GDP from March 1947 to March 2007, a sample size of $T = 241$. The series exhibits fluctuations around a strong positive trend, beginning with an initial value of 7.359 in March 1947. The durations and magnitudes of the fluctuations around the trend vary over time reflecting the presence of the business cycle. There also appears to be a break in the trend in the early 1970s corresponding to the slow down in economic growth following the OPEC oil crisis of 1973.

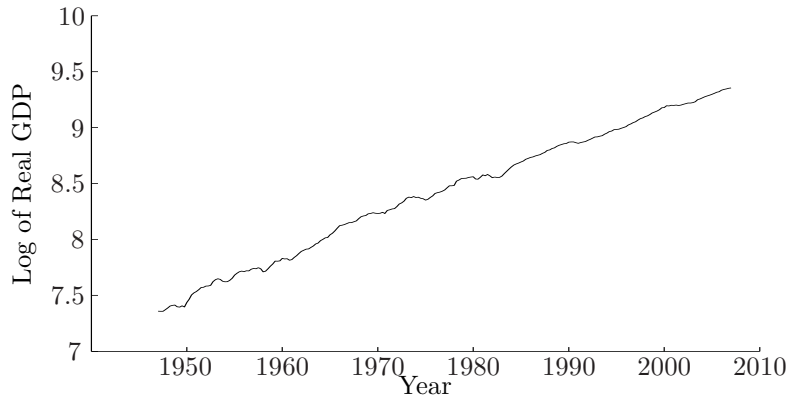


Figure 17.1 The logarithm of real U.S. GDP for the period March 1947 to March 2007, $T = 241$.

The time series characteristics of U.S. GDP in Figure 17.1 suggest the following specification to model its dynamics

$$y_t = \delta + \gamma t + u_t \quad (17.2)$$

$$u_t = \phi u_{t-1} + v_t, \quad v_t \sim iid(0, \sigma^2). \quad (17.3)$$

The first equation captures the deterministic trend features of y_t through the linear trend $\delta + \gamma t$. The second equation captures the stochastic trend features of the y_t where a unit root corresponds to $\phi = 1$. Further extensions allowing for richer dynamics are investigated where v_t in (17.3) is specified

to have an ARMA representation (Section 17.7), the inclusion of structural breaks (Section 17.8) and issues related to initial conditions (Section 17.9).

The aim is to test for a unit root in y_t , which, from (17.3) is achieved by testing for a unit root in u_t . The null and alternative hypotheses are

$$\begin{aligned} H_0 &: \phi = 1 && [\text{Nonstationary}] \\ H_1 &: \phi < 1. && [\text{Stationary}] \end{aligned} \quad (17.4)$$

To implement the test, a two-step approach is adopted.

Step 1: Detrending

Equation (17.2) is estimated to yield the least squares residuals $\hat{u}_t$, which represent the (deterministic) detrended data

$$\hat{u}_t = y_t - \hat{\delta} - \hat{\gamma}t. \quad (17.5)$$

Step 2: Testing

Using the residuals $\hat{u}_t$ from the first step, the second step tests for a unit root in (17.3) with u_t replaced by $\hat{u}_t$.

An advantage of this two-step approach is that within each step a range of options occur, each designed to improve the performance of the unit root test. A further advantage of the two-step approach is that it provides a broad conceptual framework in which to compare alternative strategies to test for unit roots.

17.3 Detrending

Equations (17.2) and (17.3) represent an autocorrelated regression model as discussed in Chapter 7. As the disturbance term has an AR(1) structure, the log-likelihood function for a sample of $t = 1, 2, \dots, T$, is

$$\ln L_T(\theta) = \frac{1}{T} \left(\ln f(y_1; \theta) + \sum_{t=2}^T \ln f(y_t | y_{t-1}; \theta) \right) \quad (17.6)$$

where $\theta = \{\delta, \gamma, \phi\}$ are the unknown parameters. The conditional distribution $\ln f(y_t | y_{t-1}; \theta)$ in (17.6) is obtained by combining equations (17.2) and (17.3) to derive an expression for y_t , applying the lag operator $(1 - \phi L)$ to both sides of (17.2) and using (17.3) to substitute out u_t . The expression for y_t is

$$(1 - \phi L) y_t = \delta (1 - \phi) + \gamma (1 - \phi L) t + v_t,$$

or

$$y_t - \phi y_{t-1} = \delta (1 - \phi) + \gamma (t - \phi(t-1)) + v_t. \quad (17.7)$$

Assuming that $v_t \sim N(0, \sigma^2)$, the conditional distribution is

$$\begin{aligned} \ln f(y_t|y_{t-1}; \theta) &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 \\ &\quad - \frac{1}{2\sigma^2} (y_t - \phi y_{t-1} - \delta(1 - \phi) - \gamma(t - \phi(t - 1)))^2. \end{aligned} \quad (17.8)$$

The marginal distribution, $\ln f(y_1; \theta)$, is obtained by writing equations (17.2) and (17.3) at $t = 1$ as

$$\begin{aligned} y_1 &= \delta + \gamma \times 1 + u_1 \\ u_1 &= \phi u_0 + v_1. \end{aligned}$$

By assuming the initial condition $u_0 = 0$, it follows that $u_1 = v_1$. An alternative approach is to specify $u_0 \sim iid(0, \sigma^2/(1 - \rho^2))$ following Elliot (1999). If the assumption of normality for v_t is invoked, the marginal distribution is

$$\ln f(y_1; \theta) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} (y_1 - \delta - \gamma)^2. \quad (17.9)$$

Substituting (17.8) and (17.9) into (17.6) gives the full log-likelihood function

$$\begin{aligned} \ln L_T(\theta) &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 - \frac{1}{2T} \frac{(y_1 - \delta - \gamma)^2}{\sigma^2} \\ &\quad - \frac{1}{2T\sigma^2} \sum_{t=2}^T (y_t - \phi y_{t-1} - \delta(1 - \phi) - \gamma(t - \phi(t - 1)))^2. \end{aligned} \quad (17.10)$$

The approach to estimate θ adopted in Chapter 7 is to use an iterative algorithm to compute $\hat{\theta}$ because of the nonlinear parameter structure of the log-likelihood function. An alternative approach that is commonly followed to test for unit roots is to set $\phi = \phi^*$, where ϕ^* is a constant, and estimate the remaining parameters using the full sample of $t = 1, 2, \dots, T$ by the regression

$$y_t^* = x_t^* \beta^* + u_t^*, \quad (17.11)$$

where $\beta^* = \{\delta, \gamma\}$ and y_t^* and x_t^* are defined, respectively, as

$$y^* = \begin{bmatrix} y_1 \\ y_2 - \phi^* y_1 \\ y_3 - \phi^* y_2 \\ \vdots \\ y_T - \phi^* y_{T-1} \end{bmatrix}, \quad x^* = \begin{bmatrix} 1 & 1 \\ 1 - \phi^* & 2 - \phi^* \\ 1 - \phi^* & 3 - 2\phi^* \\ \vdots & \vdots \\ 1 - \phi^* & T - (T - 1)\phi^* \end{bmatrix}. \quad (17.12)$$

Least squares estimation of equation (17.11) for a particular choice of ϕ^*

gives the least squares estimates $\hat{\beta}^* = \{\hat{\delta}, \hat{\gamma}\}$ which are used in (17.5) to compute $\hat{u}_t$.

Three choices of ϕ^* are as follows:

1. *Ordinary Least Squares* (Dickey and Fuller, 1979, 1981)

$$\phi^* = 0$$

2. *Differencing* (Schmidt and Phillips, 1992)

$$\phi^* = 1$$

3. *Generalized Least Squares* (Elliott, Rothenberg and Stock, 1996)

$$\phi^* = 1 + \frac{\bar{c}}{T}, \text{ where } \begin{cases} \bar{c} = -7 & [\text{Constant } (\delta \neq 0, \gamma = 0)] \\ \bar{c} = -13.5 & [\text{Trend } (\delta \neq 0, \gamma \neq 0)] \end{cases}.$$

17.3.1 Ordinary Least Squares: Dickey and Fuller

The ordinary least squares detrending approach of setting $\phi^* = 0$ in (17.12) means that the transformed variables y^* and x^* are the original variables

$$y^* = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_T \end{bmatrix}, \quad x^* = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ \vdots & \vdots \\ 1 & T \end{bmatrix}. \quad (17.13)$$

In this case (17.2) is estimated directly by ordinary least squares and the residuals from this regression are then the detrended data as in (17.5).

This suggestion is very closely related the original Dickey-Fuller regressions. Suppose that (17.2) just includes a constant ($\gamma = 0$), in which case (17.7) is

$$y_t = \delta(1 - \phi) + \phi y_{t-1} + v_t, \quad t = 2, \dots, T. \quad (17.14)$$

The ordinary least squares estimator of ϕ in (17.14) is

$$\hat{\phi}_{DF} = \frac{\sum_{t=2}^T (y_t - \bar{y}_{(0)}) (y_{t-1} - \bar{y}_{(1)})}{\sum_{t=2}^T (y_{t-1} - \bar{y}_{(1)})^2}, \quad (17.15)$$

where $\bar{y}_{(0)} = (T-1)^{-1} \sum_{t=2}^T y_t$ and $\bar{y}_{(1)} = (T-1)^{-1} \sum_{t=2}^T y_{t-1}$. By comparison, (17.5) in this case is $\hat{u}_t = y_t - \bar{y}$, where $\bar{y} = T^{-1} \sum_{t=1}^T y_t$, and so the ordinary least squares estimator of ϕ obtained by substituting $\hat{u}_t$ into

(17.3) is

$$\hat{\phi} = \frac{\sum_{t=2}^T (y_t - \bar{y})(y_{t-1} - \bar{y})}{\sum_{t=2}^T (y_{t-1} - \bar{y})^2}. \quad (17.16)$$

The terms $\bar{y}$, $\bar{y}_{(0)}$ and $\bar{y}_{(1)}$ only differ in whether or not they include the observations y_1 and y_T . Comparing (17.15) and (17.16) shows that the estimation of (17.14) by ordinary least squares is very closely related to the two-step estimation of (17.2) and (17.3). Indeed, $\hat{\phi}_{DF}$ and $\hat{\phi}$ are asymptotically equivalent.

17.3.2 First Differences: Schmidt and Phillips

This approach involves evaluating the model under the null hypothesis of a unit root by imposing the restriction $\phi^* = 1$ on (17.3) so that the transformed variables y^* and x^* (17.12) become

$$y^* = \begin{bmatrix} y_1 \\ y_2 - y_1 \\ y_3 - y_2 \\ \vdots \\ y_T - y_{T-1} \end{bmatrix}, \quad x^* = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 1 \\ \vdots & \vdots \\ 0 & 1 \end{bmatrix}. \quad (17.17)$$

Once the parameter estimates $\hat{\delta}$ and $\hat{\gamma}$ in equation (17.2) have been estimated using the transformed variables y^* and x^* , the detrended data are computed using (17.5).

The first differencing detrending method is conveniently summarized by estimating (17.2) in first differences to obtain the parameter estimates but using the levels equation in (17.2) to compute the residuals. The ability of the first-differenced form of (17.2) to estimate all parameters, including the intercept δ stems from the treatment of the initial condition in (17.17). In the case where (17.2) includes a constant ($\gamma = 0$), the least squares estimator of δ is

$$\hat{\delta} = \frac{\sum_{t=1}^T x_t^* y_t^*}{\sum_{t=1}^T (x_t^*)^2} = y_1.$$

The mean of y_t is now estimated by the single observation y_1 , which is an unbiased but inconsistent estimator. If (17.2) includes a constant and time trend, the least squares estimators of δ and γ are

$$\begin{pmatrix} \hat{\delta} \\ \hat{\gamma} \end{pmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & T \end{bmatrix}^{-1} \begin{bmatrix} y_1 \\ y_1 + \sum_{t=2}^T \Delta y_t \end{bmatrix} = \frac{1}{T-1} \begin{bmatrix} T y_1 - y_T \\ y_T - y_1 \end{bmatrix},$$

which have the property

$$\hat{\delta} + \hat{\gamma} = \frac{T y_1 - y_T}{T - 1} + \frac{y_T - y_1}{T - 1} = \frac{(T - 1) y_1}{T - 1} = y_1.$$

17.3.3 Generalized Least Squares: Elliott, Rothenberg and Stock

Consider the estimation of (17.2) using a value of ϕ local to the null, rather than at the null itself, by defining some fixed value $\phi^* < 1$ as

$$\phi^* = 1 + \frac{\bar{c}}{T},$$

where $\bar{c} < 0$ is itself a fixed value that is a function of the choice of the deterministic variables in (17.2). The choices of $\bar{c} = -7$ (constant) and $\bar{c} = -13.5$ (linear trend) are justified in Section 17.6. The transformed variables y^* and x^* (17.12) for the case of a linear time trend become

$$y^* = \begin{bmatrix} y_1 \\ y_2 - (1 + \frac{\bar{c}}{T})y_1 \\ y_3 - (1 + \frac{\bar{c}}{T})y_2 \\ \vdots \\ y_T - (1 + \frac{\bar{c}}{T})y_{T-1} \end{bmatrix}, \quad x^* = \begin{bmatrix} 1 & 1 \\ 1 - (1 + \frac{\bar{c}}{T}) & 2 - (1 + \frac{\bar{c}}{T}) \\ 1 - (1 + \frac{\bar{c}}{T}) & 3 - 2(1 + \frac{\bar{c}}{T}) \\ \vdots & \vdots \\ 1 - (1 + \frac{\bar{c}}{T}) & T - (T - 1)(1 + \frac{\bar{c}}{T}) \end{bmatrix}. \quad (17.18)$$

which depend on the sample size, T . Once the parameter estimates $\hat{\delta}$ and $\hat{\gamma}$ in equation (17.2) have been estimated using the transformed variables y^* and x^* , the detrended data are computed using (17.5).

The ordinary least squares ($\bar{c} = -T$) and first-difference detrending methods ($\bar{c} = 0$) are both special cases of the generalized least squares approach.

Example 17.1 Effects of Different Detrending Methods on GDP

The effects of the three detrending methods on U.S. GDP are given in Figure 17.2, which contains plots of the residuals $\hat{u}_t$ from estimating (17.5). As a time trend is included in the detrending, $\bar{c} = -13.5$, and the GLS filter is based on setting

$$\phi^* = 1 - \frac{13.5}{241} = 0.9440.$$

The estimated trend equations are

$$\begin{array}{lll} \text{OLS} & (\phi^* = 0) & : y_t = 7.410223 + 0.008281 t + \hat{u}_t \\ \text{Differencing} & (\phi^* = 1) & : y_t = 7.350836 + 0.008313 t + \hat{u}_t \\ \text{GLS} & (\phi^* = 0.9440) & : y_t = 7.362268 + 0.008521 t + \hat{u}_t. \end{array}$$

All three methods yield very similar patterns. The generalized least squares detrended GDP initially tracks the first-differenced detrended GDP, eventually drifting towards the ordinary least squares detrended series. $\square$

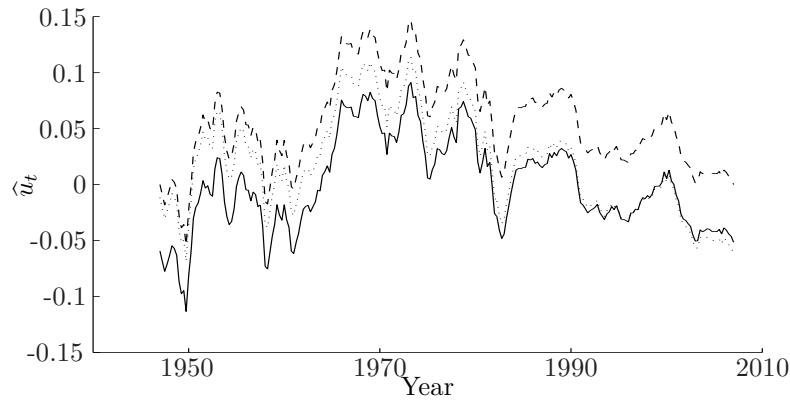


Figure 17.2 Different detrending methods applied to the logarithm of real U.S. GDP for the period March 1947 to March 2007. The detrending methods used are ordinary least squares detrending (solid line), first-difference detrending (dashed line) and generalized least squares detrending (dotted line).

17.4 Testing

Given the residuals $\hat{u}_t$ in (17.5) computed using the estimates β^* from the regression in (17.11) based on one of the detrending methods described in Section 17.3, tests of (17.4) using (17.3) are now considered. Two broad classes of tests are discussed: the Dickey-Fuller test proposed by Dickey and Fuller (1979, 1981) and the class of M tests proposed by Stock (1999) and Perron and Ng (1996). For an alternative Bayesian perspective on testing for unit roots, see Phillips (1991b).

17.4.1 Dickey-Fuller Tests

As v_t is *iid*, the ordinary least squares estimator of ϕ in (17.3) is obtained by regressing $\hat{u}_t$ on $\hat{u}_{t-1}$ to obtain

$$\hat{\phi} = \frac{\sum_{t=2}^T \hat{u}_t \hat{u}_{t-1}}{\sum_{t=2}^T \hat{u}_{t-1}^2}. \quad (17.19)$$

Two versions of the Dickey-Fuller test are given based on (17.19). The first is

$$DF_\alpha = T(\hat{\phi} - 1), \quad (17.20)$$

while the second is given by the t-statistic evaluated under the null of a unit root, $\phi = 1$,

$$DF_t = \frac{\hat{\phi} - 1}{\text{se}(\hat{\phi})}. \quad (17.21)$$

An alternative approach, equivalent to (17.21), is to rearrange (17.3) by subtracting u_{t-1} from both sides to obtain

$$\begin{aligned} \Delta u_t &= (\phi - 1)u_{t-1} + v_t \\ &= \alpha u_{t-1} + v_t, \end{aligned} \quad (17.22)$$

where $\alpha = \phi - 1$. A test of $\phi = 1$ is now a test of $\alpha = 0$ so that the hypotheses in terms of α are

$$\begin{array}{ll} H_0 & : \alpha = 0 \quad \text{[Nonstationary]} \\ H_1 & : \alpha < 0. \quad \text{[Stationary]} \end{array} \quad (17.23)$$

The Dickey-Fuller test now requires regressing $\Delta \hat{u}_t$ on $\hat{u}_{t-1}$ to obtain

$$\hat{\alpha} = \frac{\sum_{t=2}^T \Delta \hat{u}_t \hat{u}_{t-1}}{\sum_{t=2}^T \hat{u}_{t-1}^2}, \quad (17.24)$$

and using the usual regression t-statistic

$$DF_t = \frac{\hat{\alpha}}{\text{se}(\hat{\alpha})}. \quad (17.25)$$

The test statistics (17.21) and (17.25) give identical results since $\hat{\alpha} = \hat{\phi} - 1$ and $\text{se}(\hat{\alpha}) = \text{se}(\hat{\phi})$, although (17.21) is often computed for convenience.

17.4.2 M Tests

The M tests are given by

$$MZ_\alpha = \frac{\frac{1}{2} (T^{-1} \hat{u}_T^2 - \hat{\sigma}^2)}{\left(T^{-2} \sum_{t=2}^T \hat{u}_{t-1}^2 \right)} \quad (17.26)$$

$$MSB = \left(\frac{T^{-2} \sum_{t=2}^T \hat{u}_{t-1}^2}{\hat{\sigma}^2} \right)^{1/2} \quad (17.27)$$

$$MZ_t = MZ_\alpha \times MSB = \frac{\frac{1}{2} (T^{-1} \hat{u}_T^2 - \hat{\sigma}^2)}{\hat{\sigma} \left(T^{-2} \sum_{t=2}^T \hat{u}_{t-1}^2 \right)^{1/2}}, \quad (17.28)$$

where

$$\hat{\sigma}^2 = \frac{1}{T-1} \sum_{t=2}^T \hat{v}_t^2, \quad (17.29)$$

and $\hat{v}_t$ are the residuals from the regression of $\Delta \hat{u}_t$ on $\hat{u}_{t-1}$ used to compute the Dickey-Fuller test above.

The two Dickey-Fuller statistics in (17.21) and (17.25) and the three M tests in (17.26) to (17.28) are quite closely related. To understand this relationship, consider the identity $\hat{u}_t = \hat{u}_{t-1} + \Delta \hat{u}_t$. Squaring both sides of this identity and summing over t gives

$$\sum_{t=2}^T \hat{u}_t^2 = \sum_{t=2}^T \hat{u}_{t-1}^2 + 2 \sum_{t=2}^T \Delta \hat{u}_t \hat{u}_{t-1} + \sum_{t=2}^T (\Delta \hat{u}_t)^2,$$

which, upon rearranging, yields

$$\sum_{t=2}^T \Delta \hat{u}_t \hat{u}_{t-1} = \frac{1}{2} \left(\sum_{t=2}^T \hat{u}_t^2 - \sum_{t=2}^T \hat{u}_{t-1}^2 - \sum_{t=2}^T (\Delta \hat{u}_t)^2 \right) = \frac{1}{2} \left(\hat{u}_T^2 - \hat{u}_1^2 - \sum_{t=2}^T \Delta \hat{u}_t^2 \right).$$

From the Dickey-Fuller statistic in (17.20) with $\hat{\alpha}$ defined as (17.24), it follows that

$$\begin{aligned} DF_\alpha &= T\hat{\alpha} = \frac{T \sum_{t=2}^T \Delta \hat{u}_t \hat{u}_{t-1}}{\sum_{t=2}^T \hat{u}_{t-1}^2} \\ &= \frac{\frac{1}{2} \left(T^{-1} \hat{u}_T^2 - T^{-1} \hat{u}_1^2 - T^{-1} \sum_{t=2}^T \Delta \hat{u}_t^2 \right)}{T^{-2} \sum_{t=2}^T \hat{u}_{t-1}^2}. \end{aligned} \quad (17.30)$$

Using (17.30), the Dickey-Fuller version of the t -test in (17.25) is

$$DF_t = \frac{\frac{1}{2} \left(T^{-1} \hat{u}_T^2 - T^{-1} \hat{u}_1^2 - T^{-1} \sum_{t=2}^T \Delta \hat{u}_t^2 \right)}{\hat{\sigma} \left(T^{-2} \sum_{t=2}^T \hat{u}_{t-1}^2 \right)^{1/2}}. \quad (17.31)$$

A comparison of MZ_α in (17.26) and DF_α in (17.30) shows that the two tests are closely related, with two differences. The first is that the initial residual term $T^{-1} \hat{u}_1^2$ appears in DF_α but not MZ_α . This initial term disappears asymptotically in some cases. The second difference is that MZ_α has the residual variance $\hat{\sigma}^2$ while DF_α uses the dependent variable variance $T^{-1} \sum_{t=2}^T \Delta \hat{u}_t^2$. These variances are asymptotically equivalent under the null hypothesis. Similarly, the MZ_t test is closely related to the Dickey-Fuller t -test.

Example 17.2 Unit Root Test Calculations

The ordinary least squares detrended residuals from Example 17.1 have the following summary statistics

$$\sum_{t=2}^T \hat{u}_{t-1}^2 = 0.4103, \quad \sum_{t=2}^T (\hat{u}_t - \hat{u}_{t-1}) \hat{u}_{t-1} = -0.0120, \quad \hat{u}_T = -0.0515,$$

where $T = 241$. The estimated regression equation to perform the unit root tests is

$$\hat{u}_t - \hat{u}_{t-1} = -0.0291 \hat{u}_{t-1} + \hat{v}_t,$$

where the slope, $\hat{\alpha}$ is computed from the summary statistics $-0.0120/0.4103 = -0.0291$ and the residual variance is

$$\hat{\sigma}^2 = \frac{1}{T-1} \sum_{t=2}^T \hat{v}_t^2 = \frac{0.0227}{241-1} = 0.9457 \times 10^{-4}.$$

The Dickey-Fuller statistics based on ordinary least squares detrending are

$$DF_\alpha^{OLS} = T\hat{\alpha} = -241 \times 0.029 = -7.0238$$

$$DF_t^{OLS} = \frac{\hat{\alpha}}{\text{s.e.}(\hat{\alpha})} = \frac{-0.0291}{\sqrt{\frac{0.9457 \times 10^{-4}}{0.4103}}} = -1.9197.$$

The M statistics based on ordinary least squares detrending are calculated

as

$$\begin{aligned}
 MZ_{\alpha}^{OLS} &= \frac{\frac{1}{2}(T^{-1}\hat{u}_T^2 - \hat{\sigma}^2)}{T^{-2}\sum_{t=2}^T \hat{u}_{t-1}^2} = \frac{\frac{1}{2}\left(241^{-1}(-0.0515)^2 - 0.9457 \times 10^{-4}\right)}{241^{-2} \times 0.4103} \\
 &= -5.9146 \\
 MSB^{OLS} &= \left(\frac{T^{-2}\sum_{t=2}^T \hat{u}_{t-1}^2}{\hat{\sigma}^2}\right)^{1/2} = \left(\frac{241^{-2} \times 0.4103}{0.9457 \times 10^{-4}}\right)^{1/2} = 0.2733 \\
 MZ_t^{OLS} &= MZ_{\alpha}^{OLS} \times MSB^{OLS} = -5.9146 \times 0.2733 = -1.6166.
 \end{aligned}$$

Replacing the ordinary least squares detrended residuals with the generalized least squares detrended residuals from Example 17.1 yields the following test statistics

$$\begin{aligned}
 DF_{\alpha}^{GLS} &= -4.1587, & DF_t^{GLS} &= -1.3249 \\
 MZ_{\alpha}^{GLS} &= -4.1142, & MSB^{GLS} &= 0.3186, & MZ_t^{GLS} &= -1.3108.
 \end{aligned}$$

□

17.5 Distribution Theory

This section presents the asymptotic null distributions of five possible test statistics, namely DF_{α} , DF_t , MZ_{α} , MSB and MZ_t statistics. Their distributions depend only on that of the residual process $\hat{u}_t$. With an appropriate FCLT from Chapter 16 for $\hat{u}_t$, the asymptotic distributions of all of the statistics follow reasonably straightforwardly and are all functionals of Brownian motion. The specific form of the Brownian motion appearing in these distributions depends on the type of detrending used (ordinary least squares or generalized least squares) and also on the choice of deterministic variables in the detrending regression (constant or constant and time trend). For ease of exposition, the basic distributional results are given below and the forms of these Brownian motions are derived in Sections 17.5.1 and 17.5.2.

Distribution of the DF_{α} test

The asymptotic distribution of the DF_{α} test in equation (17.30) is

$$DF_{\alpha} = \frac{\frac{1}{2}\left(\frac{1}{T}\hat{u}_T^2 - \frac{1}{T}\hat{u}_1^2 - \frac{1}{T}\sum_{t=2}^T \Delta \hat{u}_t^2\right)}{\frac{1}{T^2}\sum_{t=2}^T \hat{u}_{t-1}^2} \xrightarrow{d} \frac{\frac{1}{2}(B_X(1)^2 - B_X(0)^2 - 1)}{\int_0^1 B_X(s)^2 ds}, \quad (17.32)$$

which uses the results

$$\frac{1}{T}\hat{u}_T^2 \xrightarrow{d} \sigma^2 B_X(1)^2, \quad \frac{1}{T}\hat{u}_0^2 \xrightarrow{d} \sigma^2 B_X(0)^2, \quad \frac{1}{T^2} \sum_{t=2}^T \hat{u}_{t-1}^2 \xrightarrow{d} \sigma^2 \int_0^1 B_X(s)^2 ds,$$

and

$$\begin{aligned} \frac{1}{T} \sum_{t=2}^T \hat{v}_t^2 &= \frac{1}{T} \sum_{t=2}^T (\Delta \hat{u}_t - \alpha \hat{u}_{t-1})^2 \\ &= \frac{1}{T} \sum_{t=2}^T \Delta \hat{u}_t^2 - \frac{1}{T} \times (T\hat{\alpha})^2 \times \frac{1}{T^2} \sum_{t=2}^T \hat{u}_{t-1}^2 \\ &= \frac{1}{T} \sum_{t=2}^T \Delta \hat{u}_t^2 + o_p(1), \end{aligned}$$

so that $\hat{\sigma}^2 \xrightarrow{p} \sigma^2$.

Distribution of the DF_t test

The asymptotic distribution of the DF_t test in equation (17.31) is

$$DF_t = \frac{\frac{1}{2} \left(\frac{1}{T} \hat{u}_T^2 - \frac{1}{T} \hat{u}_1^2 - \frac{1}{T} \sum_{t=2}^T \Delta \hat{u}_t^2 \right)}{\hat{\sigma} \left(\frac{1}{T^2} \sum_{t=2}^T \hat{u}_{t-1}^2 \right)^{1/2}} \xrightarrow{d} \frac{\frac{1}{2} (B_X(1)^2 - B_X(0)^2 - 1)}{\left(\int_0^1 B_X(s)^2 ds \right)^{1/2}}. \quad (17.33)$$

Distribution of the MZ_α test

The asymptotic distribution of the MZ_α test in equation (17.26) is

$$MZ_\alpha = \frac{\frac{1}{2} \left(\frac{1}{T} \hat{u}_T^2 - \hat{\sigma}^2 \right)}{\left(\frac{1}{T^2} \sum_{t=2}^T \hat{u}_{t-1}^2 \right)} \xrightarrow{d} \frac{\frac{1}{2} (B_X(1)^2 - 1)}{\int_0^1 B_X(s)^2 ds}. \quad (17.34)$$

Distribution of the MSB test

The asymptotic distribution of the MSB test (17.27) is

$$MSB = \left(\frac{\frac{1}{T^2} \sum_{t=2}^T \hat{u}_{t-1}^2}{\hat{\sigma}^2} \right)^{1/2} \xrightarrow{d} \left(\int_0^1 B_X(s)^2 ds \right)^{1/2}. \quad (17.35)$$

Distribution of the MZ_t test

The asymptotic distribution of the MZ_t test (17.28) is

$$MZ_t = \frac{\frac{1}{2} \left(\frac{1}{T} \hat{u}_T^2 - \hat{\sigma}^2 \right)}{\hat{\sigma} \left(\frac{1}{T^2} \sum_{t=2}^T \hat{u}_{t-1}^2 \right)^{1/2}} \xrightarrow{d} \frac{\frac{1}{2} (B_X(1)^2 - 1)}{\left(\int_0^1 B_X(s)^2 ds \right)^{1/2}}. \quad (17.36)$$

The relevant forms of the Brownian motions in the asymptotic distributions in equations (17.32) to (17.36) are now derived for ordinary least squares and generalized least squares detrending. To derive these results it is useful to express the detrended residuals from (17.5) in the form

$$\hat{u}_t = u_t - x'_t \left[\sum_{t=1}^T x_t^* x_t^{*'} \right]^{-1} \sum_{t=1}^T x_t^* u_t^*, \quad (17.37)$$

where u^* is defined in (17.11). Under the null hypothesis of a unit root in $\hat{u}_t$, the basic FCLT states

$$\frac{1}{\sqrt{T}} u_{[Ts]} = \frac{1}{\sqrt{T}} \sum_{t=1}^{[Ts]} u_t \xrightarrow{d} \sigma B(s),$$

which is used the derivations to follow.

17.5.1 Ordinary Least Squares Detrending

The detrended ordinary least squares residuals are given by equation (17.37) where y^* and x^* are defined in equation (17.13) with $\phi^* = 0$.

Constant

In the case of detrending based on a constant, $x_t^* = 1$ and

$$\hat{u}_t = u_t - \frac{1}{T} \sum_{t=1}^T u_t.$$

since now $u_t^* = u_t$. It follows, therefore, that

$$\frac{1}{\sqrt{T}} \hat{u}_{[Ts]} = \frac{1}{\sqrt{T}} u_{[Ts]} - \frac{1}{T^{3/2}} \sum_{t=1}^T u_{[Ts]} \xrightarrow{d} \sigma \left(B(s) - \int_0^1 B(s) ds \right) = \sigma B_X(s). \quad (17.38)$$

which is de-meaned Brownian motion.

Constant and Time Trend

In the case of detrending based on a constant and a time trend, $x_t^* = [1 \ t]'$ and

$$\hat{u}_t = u_t - \begin{bmatrix} 1 \\ t \end{bmatrix}' \left[\sum_{t=1}^T \begin{bmatrix} 1 & t \\ t & t^2 \end{bmatrix} \right]^{-1} \sum_{t=1}^T \begin{bmatrix} 1 \\ t \end{bmatrix} u_t.$$

Let $D_T = \text{diag}(1, T^{-1})$ so that

$$D_T x_{[Ts]}^* = \begin{bmatrix} 1 & 0 \\ 0 & T^{-1} \end{bmatrix} \begin{bmatrix} 1 \\ [Ts] \end{bmatrix} \xrightarrow{d} \begin{bmatrix} 1 \\ s \end{bmatrix}.$$

Now

$$\begin{aligned} \frac{1}{\sqrt{T}} \hat{u}_{[Ts]} &= \frac{1}{\sqrt{T}} u_{[Ts]} - \begin{bmatrix} 1 \\ t \end{bmatrix}' D_T \left[D_T \sum_{t=1}^T \begin{bmatrix} 1 & t \\ t & t^2 \end{bmatrix} D_T \right]^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^T D_T \begin{bmatrix} 1 \\ t \end{bmatrix} u_t \\ &\xrightarrow{d} \sigma \left(B(s) - \begin{bmatrix} 1 \\ s \end{bmatrix}' \left[\int_0^1 \begin{bmatrix} 1 & t \\ t & t^2 \end{bmatrix} ds \right]^{-1} \int_0^1 \begin{bmatrix} 1 \\ s \end{bmatrix} B(s) ds \right) \\ &= \sigma \left(B(s) - \begin{bmatrix} 1 \\ s \end{bmatrix}' \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}^{-1} \int_0^1 \begin{bmatrix} 1 \\ s \end{bmatrix} B(s) ds \right) \\ &= \sigma \left(B(s) - (4 - 6s) \int_0^1 B(s) ds + (6 - 12s) \int_0^1 s B(s) ds \right) \\ &= \sigma B_X(s). \end{aligned} \tag{17.39}$$

which represents detrended Brownian motion.

It follows, therefore, that for detrending by ordinary least squares, the appropriate Brownian motions for the purposes of deriving the asymptotic distributions of the unit root tests in expressions (17.32) to (17.36) are the expressions for $\sigma B_X(s)$ in equations (17.38) and (17.38), respectively.

17.5.2 Generalized Least Squares Detrending

The detrended least squares residuals are given by equation (17.37) where y^* and x^* are defined in equation (17.18) with $\phi^* = 1 + \frac{\bar{c}}{T}$.

Constant

In the case of detrending based on a constant, $x_1^* = 1$ and $x_t^* = -\bar{c}/T$ for $t = 2, \dots, T$ so that

$$\begin{aligned} \sum_{t=1}^T x_t^* x_t^{*'} &= x_1^* x_1^{*'} + \sum_{t=2}^T x_t^* x_t^{*'} = 1^2 + \sum_{t=2}^T \left(\frac{\bar{c}}{T} \right)^2 = 1 + \bar{c}^2 \frac{T-1}{T^2} \\ &= 1 + o(1). \end{aligned} \tag{17.40}$$

Since $\phi^* = 1 + \bar{c}T^{-1}$ it follows that

$$u_t^* = u_t - (1 + \bar{c}T^{-1})u_{t-1} = \Delta u_t - \bar{c}T^{-1}u_{t-1},$$

and

$$\begin{aligned} \sum_{t=1}^T x_t^* u_t^* &= x_1^* u_1^* + \sum_{t=2}^T x_t^* u_t^* \\ &= u_1 - \frac{\bar{c}}{T} \sum_{t=2}^T u_t^* \\ &= u_1 - \frac{\bar{c}}{T} \sum_{t=2}^T (\Delta u_t - \frac{\bar{c}}{T} u_{t-1}) \\ &= u_1 - \frac{\bar{c}}{T} \sum_{t=2}^T \Delta u_t + \left(\frac{\bar{c}}{T}\right)^2 \sum_{t=2}^T u_{t-1} \\ &= u_1 - \frac{\bar{c}}{T} (u_T - u_1) + \left(\frac{\bar{c}}{T}\right)^2 \sum_{t=2}^T u_{t-1} \\ &= \left(1 + \frac{\bar{c}}{T}\right) u_1 - \frac{\bar{c}}{T} u_T + \left(\frac{\bar{c}}{T}\right)^2 \sum_{t=2}^T u_{t-1} \\ &= u_1 + o_p(1). \end{aligned} \tag{17.41}$$

The results in (17.40) and (17.41) may now be substituted in equation (17.37) to give

$$\frac{1}{\sqrt{T}} \hat{u}_{[Ts]} = \frac{1}{\sqrt{T}} u_{[Ts]} - \frac{1}{\sqrt{T}} (1 + o(1))^{-1} (u_1 + o_p(1)) \xrightarrow{d} \sigma B(s). \tag{17.42}$$

The unit root tests have asymptotic distributions as given in equations (17.32) to (17.36), except that $B_X(s)$ is replaced by $B(s)$. That is, unlike ordinary least squares detrending when only a constant is used, generalized least squares has no asymptotic effect on the asymptotic distribution of the residual process $\hat{u}_{[Ts]}$. Furthermore, since $B(0) = 0$, it follows that DF_α^{GLS} and DF_t^{GLS} are asymptotically equivalent to MZ_α^{GLS} and MZ_t^{GLS} , respectively, under the null hypothesis..

Constant and Time Trend

In the case of detrending based on a constant and linear trend, $x_1^* = [1 \ 1]'$ and $x_t^* = [-\bar{c}T^{-1}, 1 - (t-1)\bar{c}T^{-1}]'$ for $t > 1$ which uses the result that the quasi-differenced trend variable is given by

$$t - (t-1)\phi^* = t - (t-1)(1 + \bar{c}T^{-1}) = 1 - (t-1)\bar{c}T^{-1}.$$

The procedure is now similar to that adopted for generalised least squares detrending with only a constant term present where expressions are sought for the appropriately scaled sums $\sum_{t=1}^T D_T x_t^* x_t'^* D_T$ and $D_T \sum_{t=1}^T x_t^* u_t^*$, where the scaling is given by the diagonal matrix $D_T = \text{diag}(1, T^{-1})$. In this case it can be shown that

$$\begin{aligned}
T^{-1/2} \hat{u}_{[Ts]} &= T^{-1/2} u_{[Ts]} - T^{-1/2} x'_{[Ts]} D_T \left[D_T \sum_{t=1}^T x_t^* x_t'^* D_T \right]^{-1} \left[D_T \sum_{t=1}^T x_t^* u_t^* \right] \\
&= T^{-1/2} u_{[Ts]} - \begin{bmatrix} T^{-1/2} \\ [Ts]/T \end{bmatrix}' \begin{bmatrix} 1 & 0 \\ 0 & 1 - \bar{c} + \bar{c}^2/3 \end{bmatrix}^{-1} \\
&\quad \times \begin{bmatrix} (1 + \frac{\bar{c}}{T}) u_1 - \bar{c} T^{-1} u_T + \bar{c}^2 T^{-2} \sum_{t=2}^T u_{t-1} \\ (1 - \bar{c} \frac{T-1}{T}) T^{-1/2} u_T + \frac{\bar{c}^2}{T^{5/2}} \sum_{t=2}^{T-1} t u_t - (1 - \frac{\bar{c}^2}{T^2}) T^{-1/2} u_1 \end{bmatrix} + o_p(1) \\
&\xrightarrow{d} \sigma(B(s) - s \left(\frac{1 - \bar{c}}{1 - \bar{c} + \bar{c}^2/3} B(1) + \frac{\bar{c}^2}{1 - \bar{c} + \bar{c}^2/3} \int_0^1 s B(s) ds \right)) \\
&= \sigma B_{\bar{c}}(s), \tag{17.43}
\end{aligned}$$

where $B_{\bar{c}}(s)$ represents generalized least squares detrended Brownian motion.

For generalized least squares detrending with both a constant and linear trend, therefore, the asymptotic distributions given in equations (17.32) to (17.36) require that $B_X(s)$ is replaced by $B_{\bar{c}}(s)$ from (17.43). Since $B_{\bar{c}}(0) = 0$, it again follows that DF_{α}^{GLS} and DF_t^{GLS} are asymptotically equivalent to MZ_{α}^{GLS} and MZ_t^{GLS} , respectively, under the null hypothesis.

17.5.3 Simulating Critical Values

The simplest method of obtaining critical values from the non-standard distributions involving Brownian motion is to simulate the distributions of the relevant test statistics. To obtain the critical values for unit root tests by simulation requires the following steps.

- Step 1:** Generate a sample of size T from $y_t = y_{t-1} + v_t$, $v_t \sim iid N(0, \sigma^2)$ with $\sigma^2 = 1$ and $y_0 = 0$.
- Step 2:** Compute each of the statistics using (a) no deterministic regression, (b) ordinary and generalized least squares de-meaning and (c) ordinary and generalized least squares linear de-trending.
- Step 3:** Repeat Steps 1 and 2 a large number of times, say 1000000 replications.

Step 4: Compute the quantiles of the various statistics.

The simulated 5% critical values are reported in Table 17.1 for the DF_t and the MZ_t tests for sample sizes $T = \{25, 50, 100, 250, 500, 1000\}$. Values of the test statistic less than the critical value leads to a rejection of the null hypothesis of a unit root at the 5% level.

Table 17.1

Simulated 5% critical values for the Dickey-Fuller and M tests for alternative sample sizes.

T	25	50	100	250	500	1000
Constant						
DF_t^{OLS}	-3.089	-2.972	-2.913	-2.885	-2.872	-2.866
MZ_t^{OLS}	-2.036	-2.250	-2.357	-2.432	-2.455	-2.467
DF_t/MZ_t^{GLS}	-2.559	-2.297	-2.138	-2.031	-1.988	-1.966
Constant and linear trend						
DF_t^{OLS}	-3.812	-3.601	-3.500	-3.446	-3.428	-3.420
MZ_t^{OLS}	-2.327	-2.710	-2.921	-3.059	-3.107	-3.130
DF_t/MZ_t^{GLS}	-3.580	-3.219	-3.035	-2.925	-2.886	-2.867

These simulated critical values should not be considered as exact finite sample critical values but rather as asymptotic approximations. They would only be exact if the disturbances in the observed data are actually normally distributed. Whether asymptotic critical values are superior to finite sample critical values is a question for Monte Carlo experimentation in each different case.

17.6 Power

The critical values of the unit root tests presented in Table 17.1 are constructed to generate a test with size of 5% under the null hypothesis of a unit root. In this section the focus is on examining the power of unit root tests to reject the null hypothesis of $\phi = 1$ when in fact the process is stationary $-1 < \phi < 1$. For values of ϕ just less than unity, say 0.99, even though the process is stationary, the behaviour of the process in finite samples is nearer to a unit root process than a stationary process as illustrated in Figure 17.3. This behaviour makes the detection of stationarity difficult and consequently the test statistics have low power. As ϕ moves further into the stationary region the process will exhibit more of the characteristics of a stationary process resulting in an increase in power.

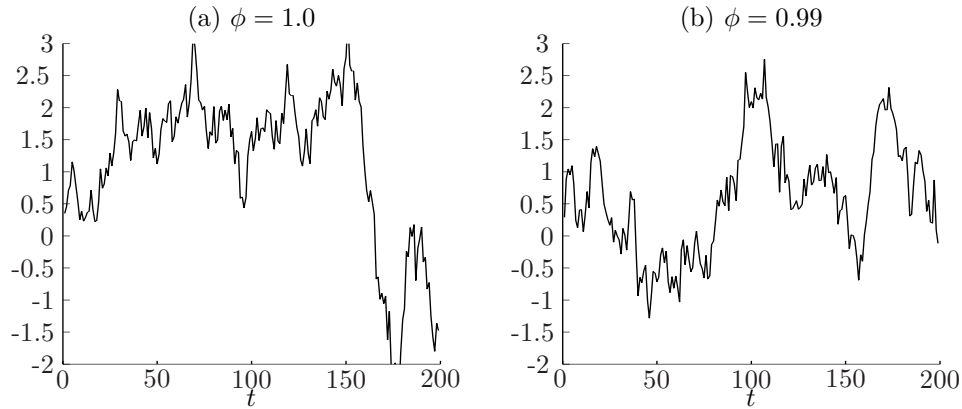


Figure 17.3 Illustrating the similarity of a unit root process ($\phi = 1$) and a near unit root process ($\phi = 0.99$) with $\sigma^2 = 0.1$ and sample size $T = 200$.

To model the sequence of stationary alternatives near the null of a unit root, ϕ in (17.3) is given by

$$\phi = \phi_c = 1 + \frac{c}{T}, \quad c < 0. \quad (17.44)$$

A process y_t having this specification for ϕ is referred to as a near unit root process. The choice of the sign of the constant c ensures that values of ϕ occur in the stationary region. As T increases, ϕ approaches 1 and the alternative hypothesis approaches the null hypothesis. The asymptotic power of a test against such a sequence is called the asymptotic local power and is given as a function of c . In standard hypothesis tests, as opposed to unit root tests, the appropriate sequence of local alternatives uses an exponent of $-\frac{1}{2}$ for T . In this instance, the exponent of T in (17.44), namely -1 , is chosen so that the asymptotic local power function is not degenerate. If this exponent was -2 ($-\frac{1}{2}$), then the asymptotic local power function would be one (zero) for all c . Phillips (1988) uses $\phi = \exp[c/T]$, which is asymptotically equivalent to (17.44).

17.6.1 Near Integration and the Ornstein-Uhlenbeck Processes

The asymptotic theory for near integrated processes is more like that for integrated processes than it is for stationary processes, a result that is not too surprising given the observation made concerning the time series patterns of the near unit processes given in Figure 17.3. The asymptotic distribution theory for near integrated processes is an extension of the FCLT. If u_t is

generated by (17.3) with ϕ given by (17.44), then

$$\frac{1}{\sigma\sqrt{T}}u_{[Ts]} \xrightarrow{d} B_c(s), \quad 0 \leq s \leq 1, \quad (17.45)$$

where $B_c(s)$ is an Ornstein–Uhlenbeck (OU) process given by

$$B_c(s) = \int_0^s \exp[c(s-r)]dB(r), \quad (17.46)$$

and $B(s)$ is standard Brownian motion. The process $B_c(s)$ is similar to Brownian motion in that it is a normally distributed continuous stochastic process, but its variance for a fixed s and $c \neq 0$ is

$$\int_0^s \exp[2c(s-r)]dr = \frac{e^{2cs} - 1}{2c}.$$

In the special case of the null of a unit root, $c = 0$ and (17.46) reduces to $B_0(s) = \int_0^s dB(r) = B(s)$, which is the basic FCLT result for a unit root process.

The asymptotic distributions of the unit root statistics under (17.44) have the same form as the null distributions given above, except with the standard Brownian $B(s)$ is replaced by the OU process $B_c(s)$ in (17.46). For example, for ordinary least squares detrending, the asymptotic distribution of DF_t^{OLS} when ϕ is given by (17.44) is

$$DF_t^{OLS} \xrightarrow{d} \frac{\frac{1}{2}(B_{X,c}(1)^2 - B_{X,c}(0)^2 - 1)}{\left(\int_0^1 B_{X,c}(s)^2 ds\right)^{1/2}}. \quad (17.47)$$

where $B_X(s)$ in (17.33) is replaced by

$$B_{X,c}(s) = B_c(s) - (4 - 6s) \int_0^1 B_c(s)ds + (6 - 12s) \int_0^1 sB_c(s)ds.$$

For generalized least squares detrending, the asymptotic distribution is

$$DF_t^{GLS} \xrightarrow{d} \frac{\frac{1}{2}(B_{\bar{c},c}(1)^2 - 1)}{\left(\int_0^1 B_{\bar{c},c}(s)^2 ds\right)^{1/2}} \quad (17.48)$$

where

$$B_{\bar{c},c}(s) = B_c(s) - s \left(\frac{1 - \bar{c}}{1 - \bar{c} + \bar{c}^2/3} B_c(1) + \frac{\bar{c}^2}{1 - \bar{c} + \bar{c}^2/3} \int_0^1 sB_c(s)ds \right),$$

noting that $B_{\bar{c},c}(0) = 0$.

17.6.2 Asymptotic Local Power

The asymptotic local power of a unit root test is its asymptotic power against local alternatives of the form (17.44). To illustrate the ideas, consider a comparison of the asymptotic local power of the DF_t test under ordinary and generalized least squares detrending. The asymptotic local power function for the DF_t test is

$$POWER(c) = \lim_{T \rightarrow \infty} \Pr(DF_t < cv_\infty),$$

where cv_∞ is the asymptotic critical value from Table 17.1 corresponding to a size of 5% and the limit is taken to reflect the fact it is the asymptotic power function that is required. For OLS detrending based on a linear time trend cv_∞ is -3.420 and for GLS detrending it is -2.867 which corresponds to $\bar{c} = -13.5$.

Example 17.3 Asymptotic Local Power Curves

Consider the near unit root model

$$y_t = \left(1 + \frac{c}{T}\right)y_{t-1} + v_t, \quad v_t \sim iid N(0, \sigma^2),$$

with $\sigma^2 = 1$, $y_0 = 0$, $T = 1000$ and $c = -30, -29, \dots, -1, 0$ controlling the size of the departure from a unit root. The powers of the DF_t^{OLS} and DF_t^{GLS} test statistics where detrending is based on a linear time trend, are computed for each c by simulating the test statistics 100000 times and computing the proportion of values less than their respective critical values, -3.420 and -2.867 respectively. The results in Figure 17.4 show a clear power advantage for using generalized least squares over ordinary least squares detrending. However, it is important to note that this power advantage relies on the assumption that $y_0 = 0$, a point developed in Section 17.8. $\square$

The asymptotic local power can be used to approximate the power of a test in finite samples. In Example 17.3, when $c = -20$ the asymptotic local powers of the OLS and GLS DF_t tests are respectively 0.61 and 0.85. In a finite sample of size $T = 100$ these powers would approximate the finite sample power corresponding to $\phi = 1 - 20/100 = 0.8$. If $T = 200$ then those powers would apply to $\phi = 1 - 20/200 = 0.9$.

17.6.3 Point Optimal Tests

It is possible to place an upper bound on the asymptotic local power that can be obtained by unit root tests, assuming normality. Consider the log-

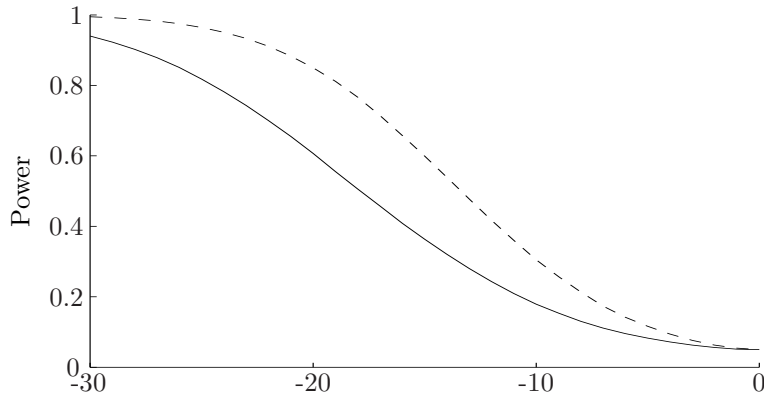


Figure 17.4 Approximate size-adjusted asymptotic power curves of the DF_t^{OLS} test (solid line) and the DF_t^{GLS} test (dashed line) using alternative detrending methods. Computed using Monte Carlo methods with 100000 replications and a sample size of 1000.

likelihood in (17.10) with ϕ defined as (17.44)

$$\begin{aligned} \ln L_T(c, \delta, \gamma) = & -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 \\ & - \frac{1}{2\sigma^2} (y_t - \phi y_{t-1} - \delta(1 - \phi) - \gamma(t - \phi(t-1)))^2, \end{aligned}$$

so c represents an additional parameter in the likelihood together with δ and γ . This suggests that a test of a unit root under the null and a near unit root under the alternative can be based on the parameter c according to

$$\begin{aligned} H_0 : \quad \phi &= 1 & [\text{Unit root, } c = 0] \\ H_1 : \quad \phi &= 1 + \frac{\bar{c}}{T} & [\text{Near unit root, } c < 0] \end{aligned}$$

Elliott, Rothenberg and Stock (1996) show that the most powerful invariant test has the form of the LR test (see Chapter 4) given by

$$LR = -2(T \ln L_T(0, \hat{\delta}_0, \hat{\gamma}_0) - T \ln L_T(\bar{c}, \hat{\delta}_{\bar{c}}, \hat{\gamma}_{\bar{c}})),$$

where $\hat{\delta}_c$ and $\hat{\gamma}_c$ are the generalized least squares de-trending estimators using either $c = \bar{c}$ or $c = 0$. The null hypothesis of a unit root is rejected for large values of this statistic. An alternative form of the LR statistic is

$$\frac{\sum_{t=1}^T \hat{v}_{\bar{c},t}^2 - \sum_{t=1}^T \hat{v}_{0,t}^2}{\sigma^2},$$

where the null hypothesis is rejected for small values. This statistic is infeasible in practice since it depends on the unknown parameter σ^2 . An asymp-

totically equivalent version of this LR test is

$$P_{\bar{c}} = \frac{\sum_{t=1}^T \hat{v}_{\bar{c},t}^2 - (1 + \bar{c}/T) \sum_{t=1}^T \hat{v}_{0,t}^2}{\hat{\sigma}^2}, \quad (17.49)$$

where $\hat{\sigma}^2$ is an estimator such as $T^{-1} \sum_{t=1}^T v_{\bar{c},t}^2$. The inclusion of the additional $(1 + \bar{c}/T)$ term in the numerator only changes the statistic by a constant asymptotically but allows the statistic to be easily modified to allow for autocorrelation in v_t by replacing $\hat{\sigma}^2$ with a different estimator, a point taken up in Section 17.7.

For a particular value of $\bar{c}$, the $P_{\bar{c}}$ test can be used as a benchmark against which to compare the power of the other unit root tests. Since $P_{\bar{c}}$ is constructed to be optimal against a particular point of the alternative hypothesis, it is referred to as a point optimal test. The asymptotic distribution of $P_{\bar{c}}$ is derived using the FCLT result used previously and is given by

$$\begin{aligned} P_{\bar{c}} &\xrightarrow{d} \bar{c}^2 \int_0^1 B_c(s)^2 ds - \bar{c} B_c(1)^2, & [\text{Constant}] \\ P_{\bar{c}} &\xrightarrow{d} \bar{c}^2 \int_0^1 B_{\bar{c},c}(s)^2 ds + (1 - \bar{c}) B_{\bar{c},c}(1)^2, & [\text{Constant and Trend}] \end{aligned}$$

17.6.4 Asymptotic Power Envelope

The $P_{\bar{c}}$ test in (17.49) can be used as a practical unit root test. Although asymptotically optimal only at the fixed value $\bar{c}$, it does have good local power properties against other values of c . However, it is used in the construction of the asymptotic local power envelope, which defines the maximum local power attainable by any unit root test, under the joint assumptions of normality, of asymptotically negligible initial value and of invariance to the deterministic regressors.

The asymptotic local power function for the point optimal test $P_{\bar{c}}$ in (17.6.3) is

$$POWER(c) = \lim_{T \rightarrow \infty} \Pr(P_{\bar{c}} < cv_{\infty}), \quad (17.50)$$

where the critical value cv_{∞} corresponds to a size of 0.05 ($c = 0$). By construction, this is the maximum attainable power for a unit root test when $c = \bar{c}$. The asymptotic local power envelope is defined by evaluating the point optimal test in (17.49) at all values of c and not just $\bar{c}$

$$POWER^{\text{env}}(c) = \lim_{T \rightarrow \infty} \Pr(P_c < cv_{\infty}),$$

No single unit root test can attain this envelope, so the task is instead to find tests that are as close as possible to the bound.

Elliott, Rothenberg and Stock (1996) show there is no uniformly most powerful unit root test and therefore no single optimal choice for $\bar{c}$. However, choosing $\bar{c}$ to be the value at which the asymptotic local power envelope has power of 0.5 produces tests with good power properties across a wide range of values of c . Values $\bar{c} = -7$ (for a constant) and $\bar{c} = -13.5$ (for a constant and linear trend) are appropriate practical choices for these constants, which correspond to the choices adopted for generalized least squares detrending in Section 17.3.3.

Example 17.4 Asymptotic Local Power Envelope

The simulation design in Example 17.3 is extended to produce the asymptotic power envelope and the asymptotic local power of the DF_t^{OLS} and DF_t^{GLS} tests as well as the point optimal test $P_{\bar{c}}$. Detrending is based on a linear time trend so $\bar{c} = -13.5$ for the point optimal test. Figure 17.5 shows that the asymptotic local power of the DF_t^{GLS} test and the point optimal test are barely distinguishable from the power envelope, which shows that there is no point in searching for asymptotic power improvements in this particular model. The DF_t^{OLS} test has inferior asymptotic local power. $\square$

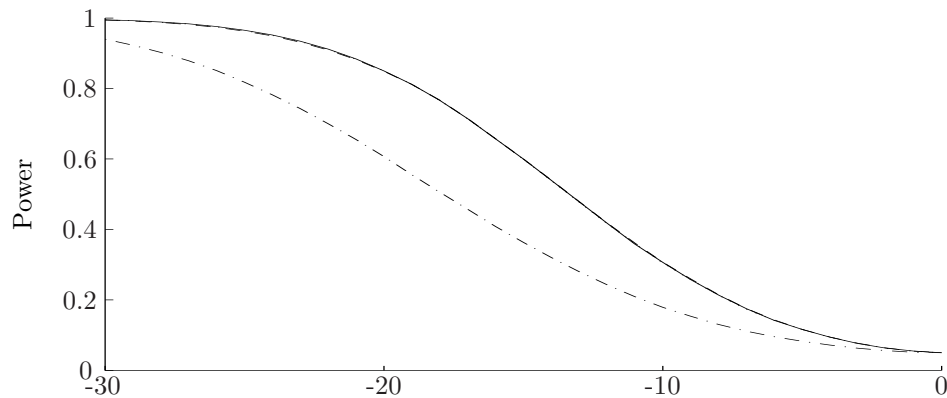


Figure 17.5 Asymptotic power envelope (solid line) and approximate size-adjusted asymptotic power curves of the point optimal test (dashed line), the DF_t^{GLS} test (dot dashed line) and DF_t^{OLS} test (dotted line). Computed using Monte Carlo methods with 100000 replications and a sample size of 1000.

17.7 Autocorrelation

In practice, an AR(1) model for u_t , as in equation (17.3), is unlikely to be sufficient to ensure that v_t is not autocorrelated. Consequently, this section explores unit root testing in the presence of autocorrelation.

17.7.1 Dickey-Fuller Test with Autocorrelation

The Dickey-Fuller regression (17.22) requires augmenting with lagged differences of u_t to correct for autocorrelation. The augmented test regression is

$$\Delta u_t = \alpha u_{t-1} + \sum_{j=1}^p \delta_j \Delta u_{t-j} + v_t, \quad (17.51)$$

where p is a lag length chosen to ensure that v_t is not autocorrelated, at least approximately. It is not necessary that the true data generating process be an finite order AR model; only that the data generating process can be approximated by an AR model. Chang and Park (2002) formalise the sense of this approximation, showing that a general class of linear processes can be approximated by (17.51) by letting $p \rightarrow \infty$ as $T \rightarrow \infty$. This allows unit root inference to be carried out semi-parametrically with respect to the form of the autocorrelation in Δu_t . The selection of p affects both the size and power properties of a unit root test. If p is chosen to be too small, then substantial autocorrelation will remain in the error term of equation (17.51) and this will result in size distortions since the asymptotic null distribution no longer applies in the presence of autocorrelation. However, including an excessive number of lags will have an adverse effect on the power of the test.

A lag-length selection procedure that has good properties in testing for unit roots is the modified Akaike information criterion (MAIC) method proposed by Ng and Perron (2001). The lag length is chosen to satisfy

$$\hat{p} = \arg \min_p \text{MAIC}(p) = \log(\hat{\sigma}^2) + \frac{2(\tau_p + p)}{T - p_{\max}}, \quad (17.52)$$

where

$$\tau_p = \frac{\hat{\alpha}^2}{\hat{\sigma}^2} \sum_{t=p_{\max}+1}^T \hat{u}_{t-1}^2,$$

and the maximum lag length is chosen as $p_{\max} = \text{int}[12(T/100)^{1/4}]$. In estimating $\hat{p}$, it is important that the sample over which the computations are performed is held constant. The inclusion of the stochastic penalty term,

τ_p , in the AIC allows for the possible presence of MA terms in the true data generating process for (17.51). The autoregression (17.51) can be used to approximate moving average autocorrelation provided p is allowed to be sufficiently large. This is especially important for MA processes with roots close to -1 . The stochastic penalty, τ_p , will be small in the presence of such MA processes, hence allowing the selection of longer lags. The MAIC will work much like the standard AIC in other situations, such as when the true data generating process is a finite order AR.

17.7.2 *M Tests with Autocorrelation*

Consider the MZ_t statistic

$$MZ_t = \frac{\frac{1}{2} \left(\frac{1}{T} \hat{u}_T^2 - \hat{\sigma}^2 \right)}{\hat{\sigma} \left(\frac{1}{T^2} \sum_{t=2}^T \hat{u}_{t-1}^2 \right)^{1/2}}$$

with

$$\hat{\sigma}^2 = \frac{1}{T-1} \sum_{t=2}^T \hat{v}_t^2,$$

but where v_t is now autocorrelated. If the long-run variance of v_t is given by

$$\omega^2 = \sum_{j=-\infty}^{\infty} E[v_t v_{t-j}],$$

then Phillips (1987) and Phillips and Perron (1988), suggest replacing $\hat{\sigma}^2$ in the definition of MZ_t with a consistent estimator of ω^2 . Perron and Ng (1998) and Stock (1999) propose an estimator of ω^2 given by

$$\hat{\omega}^2 = \frac{\hat{\sigma}_p^2}{(1 - \sum_{j=1}^p \hat{\delta}_j)^2},$$

where $\hat{\delta}_j$ and $\hat{\sigma}_p^2$ are the estimators from the Dickey-Fuller equation (17.51). This is a consistent estimator for ω^2 under the null and local alternatives. The resultant MZ_t test

$$MZ_t = \frac{\frac{1}{2} \left(\frac{1}{T} \hat{u}_T^2 - \hat{\omega}^2 \right)}{\hat{\omega} \left(\frac{1}{T^2} \sum_{t=2}^T \hat{u}_{t-1}^2 \right)^{1/2}}$$

has the same asymptotic distribution under the null hypothesis given in (17.36). This form of the test, therefore, has the advantage that the asymptotic critical values reported earlier for MZ_t apply in the presence of autocorrelation in v_t . The MZ_α and MSB statistics are similarly modified to allow for autocorrelation in v_t . If p in (17.51) is selected using the MAIC, then the resulting unit root tests are found to have good finite sample properties.

Example 17.5 Unit Root Tests and Autocorrelation

Monte Carlo simulations are used to derive the finite sample properties of the DF_t test and the MZ_t test in the presence of autocorrelation in v_t . The data generating process is given by expressions (17.2), (17.3) and

$$(1 - \rho L)v_t = (1 + \theta L)e_t, \quad e_t \sim iid N(0, \sigma_e^2),$$

with $\sigma_e^2 = 1$, $\rho = \{0, 0.3, 0.6\}$ and $\theta = \{-0.8, -0.4, 0.0, 0.4, 0.8\}$. The sample size is $T = 200$ and $\phi = 1.0$ to compute sizes of the tests and $\phi = \{0.90, 0.95\}$ to compute the powers of the test. For the MZ_t test, two methods of choosing p are considered, namely, the MAIC and the AIC, where the latter is obtained by setting $\tau_p = 1$ in (17.52). All three tests are computed using generalized least squares detrending.

Table 17.2

Finite sample sizes ($\phi = 1.0$) of the DF_t^{GLS} and MZ_t^{GLS} tests with v_t autocorrelated. Lag length selection based on the AIC and MAIC criteria for $T = 200$ and using 10000 repetitions.

Test	ρ	θ				
		-0.8	-0.4	0.0	0.4	0.8
MZ_t^{GLS} , MAIC	0.0	0.058	0.042	0.030	0.046	0.032
	0.3	0.060	0.042	0.043	0.049	0.084
	0.6	0.068	0.041	0.049	0.014	0.144
MZ_t^{GLS} , AIC	0.0	0.198	0.122	0.085	0.126	0.318
	0.3	0.200	0.128	0.107	0.116	0.426
	0.6	0.220	0.134	0.115	0.098	0.405
DF_t^{GLS} , MAIC	0.0	0.026	0.037	0.040	0.058	0.105
	0.3	0.026	0.033	0.043	0.060	0.130
	0.6	0.024	0.028	0.043	0.013	0.153

The results in Table 17.2 show that the MZ_t test with lag length selected by MAIC generates the best sizes with values close to 0.05. Basing the lag

Table 17.3

Finite sample powers ($\phi = \{0.90, 0.95\}$) of the DF_t^{GLS} and MZ_t^{GLS} tests with v_t autocorrelated. Lag length selection based on the AIC and MAIC criteria for $T = 200$ and using 10000 repetitions.

Test	ϕ	ρ	θ				
	0.90		-0.8	-0.4	0.0	0.4	0.8
MZ_t , MAIC		0.0	0.478	0.596	0.641	0.498	0.236
		0.3	0.473	0.551	0.528	0.628	0.410
		0.6	0.434	0.477	0.591	0.262	0.644
MZ_t , AIC		0.0	0.499	0.425	0.348	0.446	0.629
		0.3	0.502	0.411	0.384	0.474	0.763
		0.6	0.500	0.408	0.385	0.334	0.828
DF_t , MAIC		0.0	0.373	0.571	0.682	0.561	0.433
		0.3	0.346	0.514	0.541	0.641	0.535
		0.6	0.291	0.435	0.557	0.305	0.677
Test	ϕ	ρ	θ				
	0.95		-0.8	-0.4	0.0	0.4	0.8
MZ_t , MAIC		0.0	0.215	0.225	0.204	0.216	0.102
		0.3	0.220	0.205	0.214	0.274	0.233
		0.6	0.221	0.196	0.236	0.078	0.421
MZ_t , AIC		0.0	0.198	0.122	0.085	0.126	0.318
		0.3	0.200	0.128	0.107	0.116	0.426
		0.6	0.220	0.134	0.115	0.098	0.405
DF_t , MAIC		0.0	0.130	0.206	0.250	0.261	0.257
		0.3	0.126	0.177	0.213	0.307	0.337
		0.6	0.119	0.160	0.217	0.082	0.445

length on the AIC generally results in oversized tests, while the DF_t test tends to be undersized for negative moving average parameters and more over-sized than the MZ_t test for positive moving average parameters. The finite sample powers of the tests given in Table 17.3 suggest that the MZ_t test, with the lag length chosen by MAIC, has the highest power, rejecting the null 20% of the time for $\phi = 0.95$ and 50% of the time for $\phi = 0.90$. The DF_t test has similar power, except where it is undersized in which case the power is correspondingly reduced. $\square$

17.8 Structural Breaks

The analysis is now extended to allow for structural breaks in testing for a unit root in y_t . Both known break points and unknown break points will be dealt with. To allow for structural in the trend, equation (17.2) is augmented

as follows

$$y_t = \delta + \gamma t + \beta DT_t + u_t, \quad (17.53)$$

where

$$DT_t = \begin{cases} 0, & t \leq T_B, \\ t - T_B, & t > T_B, \end{cases}$$

The following example highlights the potential structural break in U.S. real GDP following the oil crisis of 1973 (Perron, 1989).

Example 17.6 U.S. Post-war GDP

Panel (a) of Figure 17.6 gives a plot of the log of quarterly U.S. real GDP, y_t , from March 1947 to March 2007 with two different trend functions superimposed corresponding to a structural break in March 1973. The trend lines are computed by estimating equation (17.53) with $T_B = 105$, which corresponds to the timing of the structural break in the sample. The dashed trend line shows a break in the trend compared to the dotted line based on a linear trend ($\beta = 0$). Panel (b) shows that the residuals from the structural break appear to be more stationary. This may reflect a general result that an omitted structural break can make a time series appear to be more like a unit root process than it actually is. $\square$

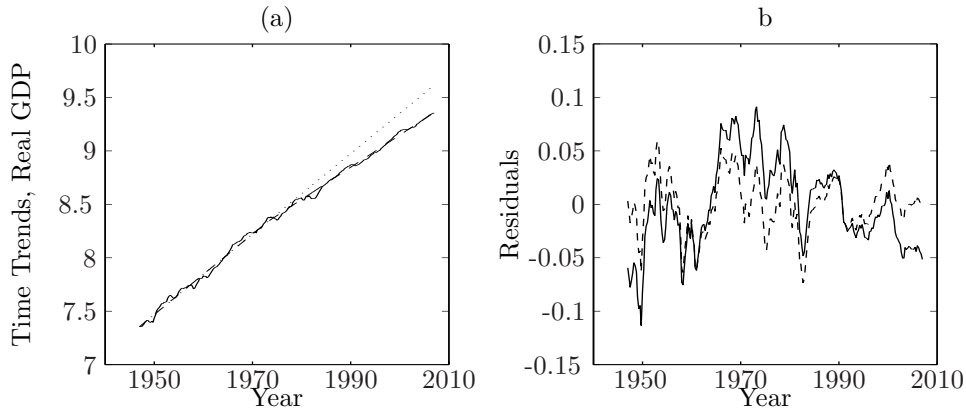


Figure 17.6 Comparison of residuals from excluding and including a structural break in the logarithm of real U.S. GDP in March 1973. Panel (a) shows the series (solid line) together with the fitted linear trend (dotted line) and the trend allowing for the break (dashed line). Panel (b) compares the residuals from the linear trend (solid line) to the residuals from the trend incorporating the break (dashed line).

Example 17.7 The Effect of Omitted Breaks

The data generating process is as specified in Example 17.6, with $\delta = \gamma = 0$ and $\beta = \{0, 0.2, 0.4\}$. The disturbances, u_t , are generated using equation (17.3) with parameter $\phi = \{1, 0.95, 0.90\}$. The sample size is $T = 200$ and the time of the break is parameterised as $T_B = [\lambda T]$, where $\lambda = \{0.25, 0.50, 0.75\}$ represents the fraction of the sample at which the break occurs.

Table 17.4

The effects of omitted breaks on the performance of unit root tests. Detrending is based on a linear time trend.

β	λ	ϕ	DF_t^{OLS}	DF_t^{GLS}	MZ_t^{OLS}	MZ_t^{GLS}
0	n.a.	1.00	0.060	0.065	0.041	0.049
		0.95	0.197	0.345	0.211	0.279
		0.90	0.647	0.880	0.706	0.819
0.2	0.25	1.00	0.052	0.032	0.023	0.025
		0.95	0.125	0.034	0.045	0.025
		0.90	0.259	0.028	0.098	0.018
0.2	0.5	1.00	0.026	0.027	0.015	0.020
		0.95	0.025	0.010	0.009	0.007
		0.90	0.012	0.002	0.004	0.001
0.2	0.75	1.00	0.032	0.036	0.022	0.026
		0.95	0.045	0.069	0.042	0.053
		0.90	0.062	0.115	0.064	0.086
0.4	0.25	1.00	0.050	0.005	0.003	0.003
		0.95	0.053	0.000	0.000	0.000
		0.90	0.046	0.000	0.000	0.000
0.4	0.5	1.00	0.006	0.003	0.002	0.002
		0.95	0.000	0.000	0.000	0.000
		0.90	0.000	0.000	0.000	0.000
0.4	0.75	1.00	0.007	0.006	0.004	0.004
		0.95	0.001	0.001	0.000	0.000
		0.90	0.000	0.000	0.000	0.000

Table 17.4 shows that by neglecting a structural break is a strong tendency not to reject the null of a unit root. The extent to which this occurs varies with the type of test and the timing of the break, but the overall finding is very clear. Therefore, it is necessary to be able to allow for structural breaks

in the deterministic component to be able to carry out an effective unit root test. $\square$

17.8.1 Known Break Point

To derive the asymptotic distribution in the presence of a known structural break, the dummy variable DT_t in equation (17.53) is re-defined as

$$DT_t = \begin{cases} 0, & t \leq [\lambda T], \\ t - [\lambda T], & t > [\lambda T], \end{cases},$$

where λ is the known break fraction. The distribution theory for ordinary least squares and generalized least squares detrending is now derived.

Ordinary Least Squares Detrending

Define $D_T = \text{diag}(1, T^{-1}, T^{-1})$, so that

$$D_T^{-1} x_{[Ts]} = \begin{bmatrix} 1 \\ T^{-1} [Ts] \\ T^{-1} (\max(0, [Ts] - [\lambda T])) \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ s \\ \max(0, s - \lambda) \end{bmatrix} = X(s).$$

Hence

$$\frac{1}{\sqrt{T}} \hat{u}_{[Ts]} \xrightarrow{d} \sigma B_X(s), \quad (17.54)$$

where

$$\begin{aligned} B_X(s) &= B(s) - X(s)' \left[\int_0^1 X(s) X(s)' \right]^{-1} \int_0^1 X(s) B(s) ds \\ &= B(s) - \begin{bmatrix} 1 \\ s \\ s - \lambda \end{bmatrix}' \begin{bmatrix} 1 & \frac{1}{2} & \frac{(1-\lambda)^2}{2} \\ \frac{1}{2} & \frac{1}{3} & \frac{(\lambda+2)(1-\lambda)^2}{6} \\ \frac{(1-\lambda)^2}{2} & \frac{(\lambda+2)(1-\lambda)^2}{6} & \frac{(1-\lambda)^3}{3} \end{bmatrix}^{-1} \\ &\quad \times \begin{bmatrix} \int_0^1 B(s) ds \\ \int_0^1 s B(s) ds \\ \int_\lambda^1 (s - \lambda) B(s) ds \end{bmatrix} \\ &= B(s) - \frac{3s - \lambda - 2}{1 - \lambda} \int_0^1 B(s) ds + \frac{3(3s - \lambda - 2)}{\lambda(1 - \lambda)} \int_0^1 s B(s) ds \\ &\quad + \frac{3(2s\lambda - \lambda - 3s + 2)}{\lambda(1 - \lambda)^3} \int_\lambda^1 (s - \lambda) B(s) ds. \end{aligned}$$

The distributions of the unit root tests under ordinary least squares detrending in the presence of a known structural break are given by equations (17.32) to (17.36) with $B_X(s)$ now given by (17.54).

Generalized Least Squares Detrending

For generalized least squares detrending, similar derivations show that

$$\frac{1}{\sqrt{T}}\widehat{u}_{[Ts]} \xrightarrow{d} \sigma B_{\bar{c}}(s), \quad (17.55)$$

where

$$\begin{aligned} B_{\bar{c}}(s) = B(s) - \left[\begin{array}{c} s \\ \max(0, s - \lambda) \end{array} \right]' \times \\ \left[\begin{array}{cc} \int_0^1 (1 - \bar{c}s)^2 ds & \int_\lambda^1 (1 - \bar{c}s)(1 - \bar{c}(s - \lambda)) ds \\ \int_\lambda^1 (1 - \bar{c}s)(1 - \bar{c}(s - \lambda)) ds & \int_\lambda^1 (1 - \bar{c}(s - \lambda))^2 ds \end{array} \right]^{-1} \\ \times \left[\begin{array}{c} (1 - \bar{c})B(1) + \bar{c}^2 \int_0^1 sB(s) ds \\ (1 - \bar{c}(1 - \lambda))B(1) + \bar{c}^2 \int_\lambda^1 (s - \lambda)B(s) ds - B(\lambda) \end{array} \right]. \end{aligned}$$

The asymptotic distributions of the generalized least squares-based tests are all functions of $B_{\bar{c}}(s)$ (see Theorem 1 of Perron and Rodriguez (2003) for further details). The distributions of the unit root tests under generalized least squares detrending in the presence of a known structural break are given by equations (17.32) to (17.36) with $B_X(s)$ now replaced by $B_{\bar{c}}(s)$ given by expression (17.55).

The important thing to note from equations (17.54) and (17.55) is that the Brownian motions are functions of the break fraction λ . Consequently, the asymptotic distributions of the unit root tests also depend on the timing of the break, although, by construction, they are invariant to the size of the break. Two important implications follow from this result.

- (1) The choice of $\bar{c}$ for generalized least squares detrending depends on λ . Appropriate choices of $\bar{c}$ for different values of the break fraction parameter, λ , are given in Table 17.5.
- (2) Critical values of unit root tests depend on λ . Approximate asymptotic critical values are given in Table 17.6 for various unit root tests with detrending based on a linear time trend.

Table 17.5

Values of $\bar{c}$ for GLS detrending based on a linear trend in the presence of a known trend break with break fraction λ .

λ	0.15	0.20	0.25	0.30	0.35
$\bar{c}$	-17.6	-18.1	-18.3	-18.4	-18.4
λ	0.40	0.45	0.50	0.55	0.60
$\bar{c}$	-18.4	-18.4	-18.2	-18.1	-17.8
λ	0.65	0.70	0.75	0.80	0.85
$\bar{c}$	-17.8	-17.5	-17.0	-16.6	-16.0

Table 17.6

Asymptotic 5% critical values with a trend break with break fraction λ .
Detrending is based on a linear time trend.

λ	DF_t^{OLS}	DF_t/MZ_t^{GLS}	MZ_t^{OLS}
0.15	-3.58	-3.38	-3.42
0.20	-3.67	-3.42	-3.48
0.25	-3.73	-3.43	-3.53
0.30	-3.77	-3.44	-3.56
0.35	-3.81	-3.45	-3.58
0.40	-3.84	-3.45	-3.60
0.45	-3.86	-3.45	-3.61
0.50	-3.87	-3.44	-3.62
0.55	-3.87	-3.43	-3.62
0.60	-3.88	-3.42	-3.61
0.65	-3.87	-3.40	-3.60
0.70	-3.85	-3.37	-3.58
0.75	-3.83	-3.33	-3.55
0.80	-3.79	-3.28	-3.51
0.85	-3.74	-3.23	-3.46

Example 17.8 Unit Root Test of U.S. GDP (Known Break)

Consider testing for a unit root in U.S. GDP while allowing for a trend break in March 1973. The break in March 1973 corresponds to $\lambda = 0.44$ so that the MZ_t^{OLS} and MZ_t^{GLS} tests have 5% asymptotic critical values of -3.61 and -3.45 , respectively. Computation of the test statistics gives $MZ_t^{OLS} = -4.111$ and $MZ_t^{GLS} = -4.106$, so each individual test rejects the unit root null. The conclusion from these tests is that real GDP is stationary

around a broken trend, which is not a surprising finding given the graphical evidence in Figure 17.6. $\square$

17.8.2 Unknown Break Point

The preceding analysis is based on the assumption that the timing of the break in trend is known. A straightforward way to proceed in the more likely event of the break point being unknown is to estimate the break point initially, use that break point to estimate the rest of the detrending regression and then perform the unit root test. The critical requirement for this approach is that λ be consistently estimated with a rate of convergence faster than the standard $O_p(T^{-1/2})$ rate. In particular, if an estimator $\hat{\lambda}$ is obtained such that $\hat{\lambda} - \lambda = o_p(T^{-1/2})$, then $\hat{\lambda}$ can be used in the detrending regression and unit root test as if it were the true value without changing the asymptotic properties of the test.

To make the role of λ explicit, consider

$$y_t = \delta + \gamma t + \beta_2 DT_t(\lambda) + u_t, \quad (17.56)$$

where

$$DT_t(\lambda) = \begin{cases} 0, & t \leq [\lambda T], \\ t - [\lambda T], & t > [\lambda T]. \end{cases} \quad (17.57)$$

A two-step approach suggested by Carrion-i-Silverstre, Kim and Perron (2009) satisfies this condition.

Step 1 Choose λ to satisfy

$$\hat{\lambda} = \arg \min_{\lambda} \sum_{t=1}^T \hat{u}_t(\lambda)^2, \quad (17.58)$$

where $\hat{u}_t(\lambda)$ denote the ordinary least squares residuals in (17.56) for an arbitrary choice of λ . The range of values for λ is typically restricted to some subset of $[0, 1]$, say $[0.15, 0.85]$.

Step 2 Using Table 17.5, look up the appropriate value for $\bar{\epsilon}$ corresponding to $\hat{\lambda}$ and then re-estimate λ from the generalized least squares regression in (17.56).

Setting $\bar{\epsilon} = 0$ in the two-step approach results in the generalized least squares detrending approach becoming a simple first-differencing procedure, which is the method suggested by Harris, Harvey, Leybourne and Taylor (2009).

Example 17.9 Unit Root Test of U.S. GDP (Unknown Break)

The two-step procedure of testing for a unit root when the timing of the structural break is unknown is now applied to the U.S. GDP data. The estimator (17.58) from the levels regression (17.56) is $\hat{\lambda} = 0.39$, which corresponds to a break date of March 1970, somewhat earlier than the assumed date of March 1973. However, this is only a preliminary estimate used to obtain a value of $\bar{c}$ that is then used to compute the generalized least squares estimate. Since $\hat{\lambda} = 0.39$ is closest to $\lambda = 0.4$, the appropriate choice is $\bar{c} = -18.5$. This value is used to construct the generalized least squares transformation of (17.56) and then to re-estimate λ , giving $\hat{\lambda} = 0.44$ or a break date of June 1973. Recall that the convergence properties of these estimates of λ mean that $\bar{c}$ can be obtained from Table 17.5. Based on $\hat{\lambda} = 0.44$, the appropriate choice is $\bar{c} = -18.4$. The critical values for the ordinary least squares and generalized least squares tests are taken from Table 17.6 and are -3.61 and -3.45 , respectively. The test statistics are found to be $MZ_t^{OLS} = -4.095$ and $MZ_t^{GLS} = -4.091$ so each test rejects the unit root null. It appears, therefore, that irrespective of a break being imposed in March 1973 based on historical knowledge, or estimated to be in June 1973, post-war U.S. real GDP is stationary around a broken trend. $\square$

17.9 Applications**17.9.1 Power and the Initial Value**

The preceding analysis suggests that tests based on generalized least squares rather than ordinary least squares detrending provide a substantial power advantage. This conclusion stems from the seemingly innocuous assumption that the extent to which the time series, y_t , differs from its underlying trend is asymptotically negligible, that is $u_0 = 0$ at least asymptotically. It turns out, however, that the power of unit root tests does depend critically on u_0 .

The effects of the initial condition on the properties of unit root tests is highlighted in Figure 17.7, which gives the results of simulating the MZ_t test for a sample of $T = 200$ using ordinary least squares and generalized least squares detrending and initial condition values of $u_0 = \{0, 2.5, 5.0\}$. The results demonstrate that the ordering of the power functions of the ordinary least squares and generalized least squares tests changes as u_0 is varies. For $u_0 = 0$ in panel (a), the generalized least squares test is superior to the ordinary least squares test for all values of C . As u_0 increases from $u_0 = 2.5$ (panel (b)) to $u_0 = 5.0$ (panel (c)) the power advantage associated

with generalized least squares detrending diminishes while the performance of ordinary least squares is largely unchanged.

Muller and Elliott (2001) construct an asymptotic theory for unit root tests that captures this dependence on the initial value by allowing u_0 to grow with the sample size, specifically $u_0 = O(T^{1/2})$. In that case, u_0 is no longer asymptotically negligible and its role in determining power can be analysed theoretically. A practical approach to this problem, proposed by Harvey, Leybourne and Taylor (2009), is to do both the MZ_t^{OLS} and MZ_t^{GLS} tests and reject the null hypothesis if either or both tests reject it. If u_0 is small, then rejections will tend to come from the more powerful MZ_t^{GLS} test. If u_0 is large, then rejections will tend to come from the MZ_t^{OLS} test. This is known as a union of rejections strategy. To control the overall size of this joint procedure, the decision rule is to reject H_0 if $MZ_t^{OLS} < \tau \cdot cv^{OLS}$ and/or $MZ_t^{GLS} < \tau \cdot cv^{GLS}$, where τ is a constant chosen to ensure a size of 5%.

Figure 17.7 shows the performance of the union of rejections strategy for a linear trend, a sample size of $T = 200$ and $\tau = 1.038$. Doing two tests instead of one means that the critical values have been increased by a factor of τ and hence the union of rejections approach can never achieve the power of the individual MZ_t^{OLS} or MZ_t^{GLS} tests. Nevertheless, it does not fall far short of the best test in each case and avoids the worst of the power losses that can arise.

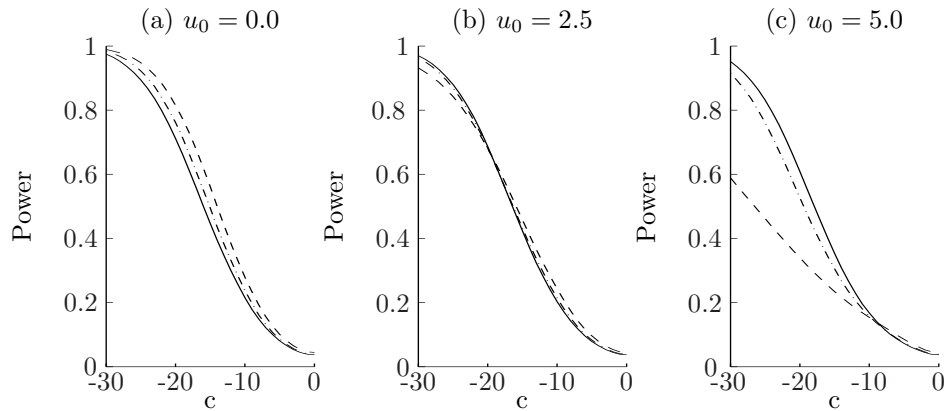


Figure 17.7 The effects of alternative initial conditions on the power of the MZ_t^{OLS} test (solid line) and MZ_t^{GLS} test (dashed line) and union of rejections method (dot-dashed line). Computed using Monte Carlo methods with 50000 replications and a sample size of 200.

17.9.2 Nelson-Plosser Data Revisited

Unit root tests are applied to the 14 U.S. annual macroeconomic variables of the Nelson-Plosser dataset analyzed in Chapter 16. The results are presented for the MZ_t unit root test with the lag length chosen by MAIC and a union of rejections decision rule used to combine the ordinary least squares and generalized least squares detrending methods assuming a linear trend and no structural break. Three decision rules are adopted, namely,

- (1) reject H_0 if $MZ_t^{OLS} < -3.130$, [M test OLS detrending]
- (2) reject H_0 if $MZ_t^{GLS} < -2.867$, [M test GLS detrending]
- (3) reject H_0 if $MZ_t^{OLS} < (\tau \times -3.130)$ and/or $MZ_t^{GLS} < (\tau \times -2.867)$, where $\tau = 1.038$. [Union of Rejections test]

where the critical values for the first two decision rules are the asymptotic values ($T = 1000$) from Table 17.1.

The results are shown in Table 17.7. All series are found to be $I(1)$ using the MZ_t^{OLS} test. For the MZ_t^{GLS} version of the test, the unemployment rate and the money stock are found to be $I(0)$, a finding supported by the union of rejections rule. This result is reinforced by inspection of Figure 16.1 in Chapter 16, which suggests that the money stock is trend stationary. Moreover, the ability of the generalized least squares version of the test to reject the null may reflect the extra power of this test established previously.

In the case of the unemployment rate, inspection of Figure 16.1 in Chapter 16 suggests that a linear trend is not required for this series. Repeating the calculations where detrending includes only a constant term for the unemployment rate yields values of -3.062 (MZ_t^{OLS}) and -2.876 (MZ_t^{GLS}) for the two tests. The asymptotic critical values from Table 17.1 are -2.467 (OLS) and -1.966 (GLS), so the unit root null hypothesis is rejected by both tests. This is confirmed by the union of rejections approach with $\tau = 1.084$, which also finds the unemployment rate to be $I(0)$. The change in the classification of the unemployment rate for the MZ_t^{OLS} test suggests that the failure to reject the null of a unit where a time trend is included indicates that the time trend is redundant and its inclusion results in a loss of power. The simple remedy is to perform the test based on just a constant.

17.10 Exercises

(1) U.S. GDP

Gauss file(s)	unit_qusgdp.g, usgdp.txt
Matlab file(s)	unit_qusgdp.m, usgdp.txt

Table 17.7

Unit root tests for the Nelson-Plosser data using the MZ_t^{OLS} and MZ_t^{GLS} tests, together with the union of rejections (UR) test.

Variable	Statistics		Classifications		UR
	MZ_t^{OLS}	MZ_t^{GLS}	MZ_t^{OLS}	MZ_t^{GLS}	
Real GNP	-1.813	-1.740	$I(1)$	$I(1)$	$I(1)$
Nominal GNP	-2.455	-2.450	$I(1)$	$I(1)$	$I(1)$
Real per cap. GNP	-1.858	-1.769	$I(1)$	$I(1)$	$I(1)$
Ind. production	-2.830	-2.725	$I(1)$	$I(1)$	$I(1)$
Employment	-2.829	-2.717	$I(1)$	$I(1)$	$I(1)$
Unemployment rate	-3.109	-2.996	$I(1)$	$I(0)$	$I(0)$
GNP deflator	-2.410	-2.169	$I(1)$	$I(1)$	$I(1)$
Consumer prices	-1.427	-1.442	$I(1)$	$I(1)$	$I(1)$
Wages	-2.413	-2.378	$I(1)$	$I(1)$	$I(1)$
Real wages	-1.928	-1.776	$I(1)$	$I(1)$	$I(1)$
Money stock	-3.097	-3.077	$I(1)$	$I(0)$	$I(0)$
Velocity	-1.292	-1.102	$I(1)$	$I(1)$	$I(1)$
Bond Yield	-1.997	-2.326	$I(1)$	$I(1)$	$I(1)$
SP500	-1.365	-1.193	$I(1)$	$I(1)$	$I(1)$

The data file contains real quarterly U.S. GDP from March 1947 to March 2007. Let y_t be the natural logarithm of the data.

- Test for a unit root test in y_t without making any allowance for autocorrelation in the residuals of the test regression.
- Repeat part (a) making any allowance for autocorrelation in the residuals of the test regression.
- Repeat part (a) based on the assumption of a structural break in the trend in March 1973.
- Test for a unit root test in y_t based on the assumption of a structural break whose location is unknown.
- Does an answer to the question of whether or not U.S. GDP has a unit root emerge?

(2) Near Unit Root Processes

Gauss file(s)	unit_nearplot.g
Matlab file(s)	unit_nearplot.m

Consider the AR(1) model

$$y_t = \phi y_{t-1} + v_t, \quad v_t \sim iid N(0, \sigma^2).$$

- (a) Simulate a unit root process with parameters $\phi = 1$ and $\sigma^2 = 0.1$ for a sample of size of $T = 200$.
- (b) Repeat part (a) for a near unit root process with $\phi = 0.99$.
- (c) Repeat part (a) for a near unit root process with $\phi = 0.95$.
- (d) Plot the simulated series in parts (a) to (c) and compare their time series properties.
- (e) For parts (a) to (c) compute the ACF and the PACF and compare their values.

(3) Asymptotic Local Power

Gauss file(s)	<code>unit_asympower1.g</code>
Matlab file(s)	<code>unit_asympower1.m</code>

Consider the data generating process

$$y_t = \phi y_{t-1} + v_t, \quad t = 2, \dots, T,$$

$$y_1 = v_1,$$

with $v_t \sim iid N(0, 1)$, $T = 1000$ and $\phi = 1 + c/T$ with $c \in \{-30, -29, \dots, 0\}$.

- (a) For $c = 0$, carry out a simulation of 100000 replications to obtain the 5% critical values of the DF_t^{OLS} and DF_t^{GLS} tests where the initial regression includes only a constant term. Use $\bar{c} = -7$ for the DF_t^{GLS} test.
- (b) For each $c = -1, -2, \dots, -30$, simulate the power of the DF_t^{OLS} and DF_t^{GLS} tests and plot the resulting asymptotic local power curves.
- (c) Repeat part (a) using an initial regression on a constant and time trend and using $\bar{c} = -13.5$ for the DF_t^{GLS} test.
- (d) In the case of a constant and time trend, generate asymptotic local power curves for the DF_t^{GLS} test using $\bar{c} = -5$ and $\bar{c} = -20$ to get some idea of the effect of changing this constant.
- (e) Repeat the previous two parts using the Schmidt-Phillips de-meaning and detrending procedure.

(4) Asymptotic Power Envelope

Gauss file(s)	<code>unit_asympowerenv.g</code>
Matlab file(s)	<code>unit_asympowerenv.m</code>

Consider the problem of obtaining the asymptotic local power envelope for unit root testing in the model

$$\begin{aligned}y_t &= x_t' \beta + u_t, \\u_t &= \phi u_{t-1} + v_t, \\u_0 &= \xi, \\v_t &\sim iid(0, \sigma^2),\end{aligned}$$

for $x_t = 1$, $x_t = (1, t)'$ and ξ is a constant.

- (a) Simulate the power envelope for $c = \{-1, -2, \dots, -30\}$, $T = 1000$ and using 100000 repetitions.
- (b) Repeat for quadratic detrending. Observe the effect of increasing the degree of the detrending regression on the power available to a unit root test.

(5) Values of $\bar{c}$ and Critical Values in the Presence of a Trend Break.

Gauss file(s)	<code>unit_cbar.g, unit_breakcv.g</code>
Matlab file(s)	<code>unit_cbar.m, unit_breakcv.m</code>

- (a) The asymptotic distribution of unit root tests in the presence of a trend break depends on the timing of the break. Simulate the asymptotic local power envelope in the presence of a trend break at a given fraction λ , and hence obtain the values of GLS detrending constants for each of $\lambda = \{0.15, 0.20, \dots, 0.85\}$.
- (b) Using these values of $\bar{c}$, carry out a simulation to obtain approximate asymptotic critical values for MZ_t^{GLS} for each λ . Also obtain critical values for MZ_t^{GLS} .
- (c) Carry out a simulation to investigate the effect of the initial condition on the power of MZ_t^{GLS} and MZ_t^{OLS} for $\lambda = \{0.25, 0.50, 0.75\}$.

(6) Phillips-Perron Test

Gauss file(s)	<code>unit_ppcv.g, unit_ppsim.g</code>
Matlab file(s)	<code>unit_ppcv.m, unit_ppsim.m</code>

Consider the AR(1) model

$$y_t = \phi y_{t-1} + v_t,$$

where v_t is a stationary autocorrelated disturbance with mean zero, variance σ^2 and satisfying the FCLT for autocorrelated processes

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{\lfloor Ts \rfloor} v_t \xrightarrow{d} \omega B(s),$$

where B is standard Brownian motion and ω^2 is the long-run variance $\omega^2 = \sum_{j=-\infty}^{\infty} E(v_t v_{t-j})$.

- (a) Derive the asymptotic distribution of the OLS estimator of ϕ in the AR(1) model where $\phi = 1$ and show that it depends on the autocorrelation properties of v_t .
- (b) Suppose that consistent estimators $\hat{\omega}^2$ and $\hat{\sigma}^2$ of ω^2 and σ^2 , respectively, are available. Define the transformed estimator

$$\tilde{\phi} = \hat{\phi} - \frac{1}{2} \frac{\hat{\omega}^2 - \hat{\sigma}^2}{T^{-1} \sum_{t=2}^T y_{t-1}^2}$$

and derive its asymptotic distribution when $\phi = 1$, hence showing that the asymptotic distribution of $\tilde{\phi}$ does not depend on the autocorrelation properties of v_t .

- (c) Define a unit root test based on $\tilde{\phi}$ and simulate approximate asymptotic critical values. This test is the Z_α test suggested in Phillips (1986) and extended to allow for constant and time trend in Phillips and Perron (1988).
- (d) Carry out a Monte Carlo experiment to investigate the finite sample size properties of the test in part (c). Use $T = \{100, 200\}$ with v_t given by

$$(1 - \phi L)v_t = (1 + \theta L)\varepsilon_t, \quad \varepsilon_t \sim iidN(0, 1)$$

where $\phi = \{0, 0.3, 0.6, 0.9\}$ and $\theta = \{-0.8, -0.4, 0, 0.4, 0.8\}$. Use $\hat{\sigma}^2 = (T-1)^{-1} \sum_{t=2}^T \hat{v}_t^2$ and let $\hat{\omega}^2$ be the Newey-West long-run variance estimator with quadratic spectral lag weights (see Chapter 9) and pre-whitening as suggested by Andrews and Monahan (1992).

(7) Unit Root Test without Lags or Long-Run Variance

Gauss file(s)	<code>unit_breitung_size.g</code> , <code>unit_breitung_power.g</code>
Matlab file(s)	<code>unit_breitung_size.m</code> , <code>unit_breitung_power.m</code>

Consider the AR(1) model $y_t = \phi y_{t-1} + v_t$ where v_t is a disturbance

satisfying the same conditions as set out in Exercise 6. Consider the unit root test statistic

$$\rho = \frac{T^{-4} \sum_{t=1}^T S_t^2}{T^{-2} \sum_{t=1}^T y_t^2},$$

where $S_t = \sum_{j=1}^t y_j$, as proposed by Breitung (2002).

- (a) Show that the asymptotic distribution of this statistic under $H_0 : \phi = 1$ is

$$\rho \xrightarrow{d} \frac{\int_0^1 B_S(r)^2 dr}{\int_0^1 B(r)^2 dr},$$

where $B_S(r) = \int_0^r B(s) ds$, regardless of the autocorrelation properties of v_t (no autoregression or long-run variance is required to make this test operational).

- (b) Carry out a Monte Carlo experiment to investigate the finite sample size properties of this test. Use the design from part d of the previous exercise.
- (c) Use a simulation to compare the asymptotic local power of the test to that of the DF_t test.

(8) Testing the Null Hypothesis of Stationarity

Gauss file(s) unit_kpss_cv.g, unit_kpssmc.g

Matlab file(s) unit_kpss_cv.m, unit_kpssmc.m

Consider the regression model

$$y_t = \beta x_t + z_t,$$

where x_t is a $(k \times 1)$ vector of fixed regressors (constant, time trend), β is $(1 \times k)$ vector of coefficients and z_t has the unobserved components representation

$$z_t = w_t + u_t,$$

$$w_t = w_{t-1} + v_t,$$

where $w_0 = 0$ and

$$\begin{bmatrix} u_t \\ v_t \end{bmatrix} \sim iid N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_u^2 & 0 \\ 0 & \sigma_v^2 \end{bmatrix} \right).$$

Define the standardized test statistic

$$s = \frac{T^{-2} \sum_{t=1}^T S_t^2}{\hat{\omega}^2},$$

where $S_t = \sum_{j=1}^t \hat{z}_j$, $\hat{z} = (\hat{z}_1, \dots, \hat{z}_T)'$ is the vector of ordinary least squares residuals from a regression of y on x and $\hat{\omega}^2$ is a consistent Newey-West form of estimator of ω^2 based on $\hat{z}_t$. This test statistic s can be interpreted as a test of over-differencing and/or a moving average unit root and is most commonly known as the KPSS test, after Kwiatkowski, Phillips, Schmidt and Shin (1992).

- (a) Show that z_t is $I(1)$ if $\sigma_v^2 > 0$ and z_t is $I(0)$ if $\sigma_v^2 = 0$ and hence that a test of $\sigma_v^2 = 0$ against $\sigma_v^2 > 0$ is a test of $I(0)$ against $I(1)$.
- (b) Using the Newey-West form of long-run variance estimator, but without pre-whitening, to carry out a simulation of the size properties of the test when $x_t = 1$. Use $T = \{100, 200\}$ with u_t given by

$$(1 - \phi L)u_t = (1 + \theta L)\varepsilon_t, \quad \varepsilon_t \sim iidN(0, 1)$$

where $\phi = \{0, 0.3, 0.6, 0.9\}$ and $\theta = \{-0.8, -0.4, 0, 0.4, 0.8\}$.

(9) Union of Rejections tests

Gauss file(s)	<code>unit_urtau.g</code> , <code>unit_urbreak.g</code> , <code>unit_pow0.g</code>
Matlab file(s)	<code>unit_urtau.m</code> , <code>unit_urbreak.m</code> , <code>unit_pow0.m</code>

- (a) Carry out a simulation to obtain the value of τ for the MZ_t^{OLS} and MZ_t^{GLS} tests such that the decision rule reject H_0 if $MZ_t^{OLS} < \tau \cdot cv^{OLS}$ and/or $MZ_t^{GLS} < \tau \cdot cv^{GLS}$ has asymptotic size of 5%, where cv^{OLS} and cv^{GLS} are the usual 5% asymptotic critical values for the tests taken from Table 17.1. To do this, choose $T = 1000$, generate data under the null (y_t will be a random walk) and obtain a large number of replications of both MZ_t^{OLS} and MZ_t^{GLS} . Once these replications are available, search for the appropriate value of τ such that $MZ_t^{OLS} < \tau \cdot cv^{OLS}$ and/or $MZ_t^{GLS} < \tau \cdot cv^{GLS}$ occurs in just 5% of the replications. Carry this out for $x_t = [1]$ and $x_t = [1 \ t]'$.
- (b) Repeat the simulation for $x_t = [1 \ t \ DT_t]'$, where DT_t is the trend break variable for $\lambda = \{0.25, 0.50, 0.75\}$ and observe the insensitivity of τ to λ .
- (c) Carry out a simulation to compare the power properties of the union of rejections test based on MZ_t^{OLS} and MZ_t^{GLS} with the individual tests. Obtain results for $x_t = 1$ and $x_t = [1 \ t]'$. Obtain results for $u_0 = \{0, 2.5, 5.0\}$ to evaluate the robustness of the union of rejections test to the initial value.

(10) **Role of Initial Conditions and Union of Rejections**

Gauss file(s)	unit_poweru0.g, unit_powerour.g
Matlab file(s)	unit_poweru0.m, unit_powerour.m

To investigate the effect of the initial condition on unit root tests, consider the model

$$\begin{aligned}
 y_t &= x_t' \beta + u_t, \\
 u_t &= \phi u_{t-1} + v_t, \\
 u_0 &= \xi, \\
 v_t &\sim iid(0, \sigma^2),
 \end{aligned}$$

where ξ is a constant.

- (a) Show that any unit root test that includes de-meaning (x_t includes 1 so that the test is invariant to adding a constant to y_t) is invariant to ξ under the null that $\phi = 1$. Show that this does not apply when $\phi < 1$.
- (b) Carry out a simulation to explore the effect of ξ on the finite sample power of the MZ_t^{OLS} and MZ_t^{GLS} tests. Use $T = 200$ and consider the values $\xi = \{0, 2.5, 5\}$.
- (c) Repeat the simulation with $\xi = -2.5$ and $\xi = -5$ to see if the effects of ξ on the power of the tests are symmetric.
- (d) Using the same simulation design, include the union of rejections test in the comparison. That is, the null is rejected if $MZ_t^{OLS} < \tau \cdot cv_{OLS}$ and/or $MZ_t^{GLS} < \tau \cdot cv_{GLS}$ where $cv_{OLS} = -2.467$, $cv_{GLS} = -1.966$, $\tau = 1.084$ when $x_t = 1$, and $cv_{OLS} = -3.130$, $cv_{GLS} = -2.867$, $\tau = 1.038$ when $x_t = (1, t)'$.

(11) **Nelson and Plosser Data**

Gauss file(s)	unit_nelplos.g
Matlab file(s)	unit_nelplos.m

- (a) Test for a unit root in each of the Nelson and Plosser series using the MZ_t^{OLS} and MZ_t^{GLS} tests and the union of rejections of the two tests. Use a constant and linear trend for each series except unemployment, for which a constant is sufficient. Use the MAIC for lag length selection. The results should replicate those in Table 17.7.
- (b) Test whether nominal GNP can be considered to be $I(2)$ (test for a unit root in the first difference of nominal GNP).

18

Cointegration

18.1 Introduction

An important implication of the analysis of stochastic trends presented in Chapter 16 and of the unit root tests discussed in Chapter 17 is that nonstationary time series can be rendered stationary through differencing the series. This use of the differencing operator represents a univariate approach to achieving stationarity since the discussion of nonstationary processes so far has concentrated on a single time series. In the case of $N > 1$ nonstationary time series $y_t = \{y_{1,t}, y_{2,t}, \dots, y_{N,t}\}$, an alternative method of achieving stationarity is to form linear combinations of the series. The ability to find stationary linear combinations of nonstationary time series is known as cointegration (Engle and Granger, 1987).

The existence of cointegration amongst sets of nonstationary time series has three important implications.

- (1) Cointegration implies a set of dynamic long-run equilibria where the weights used to achieve stationarity represent the parameters of the equilibrium relationship.
- (2) The estimates of the weights to achieve stationarity (the long-run parameter estimates) converge to their population values at a super-consistent rate of T compared to the usual $\sqrt{T}$ rate of convergence.
- (3) Modelling a system of cointegrated variables allows for specification of both long-run and short-run dynamics. The resultant model is known as a vector error correction model (VECM).

Maximum likelihood methods are used to estimate the parameters of and test restrictions on VECMs. As cointegration yields a set of nonlinear restrictions on the VECM, estimation proceeds using either the iterative gradient algorithms of Chapter 3, or the Johansen algorithm (1988,1991,1995b)

that decomposes the log-likelihood function in terms of its eigenvalues. Hypothesis tests of the VECM are computed using the statistics discussed in Chapter 4. A widely adopted test is the LR test of cointegration, commonly known as the trace test, which represents a multivariate generalization of the Dickey-Fuller unit root test discussed in Chapter 17.

18.2 Long-Run Economic Models

To highlight the dynamic interrelationships between economic processes with stochastic trends and how linear combinations of nonstationary variables can result in stationary series, three examples are presented.

Example 18.1 Permanent Income Hypothesis

The permanent income hypothesis represents a long-run relationship between log real consumption, lrc_t , and log real income, lry_t ,

$$lrc_t = \beta_c + \beta_y lry_t + u_t,$$

where u_t is a disturbance term and β_c and β_y are parameters. Panel (a) of Figure 18.1 shows that even though lrc_t and lry_t are nonstationary, the 2-dimensional scatter plot in panel (b) of Figure 18.1 reveals a 1-dimensional relationship between suggesting that lrc_t does not drift too far from its permanent income level of $\beta_c + \beta_y lry_t$. $\square$

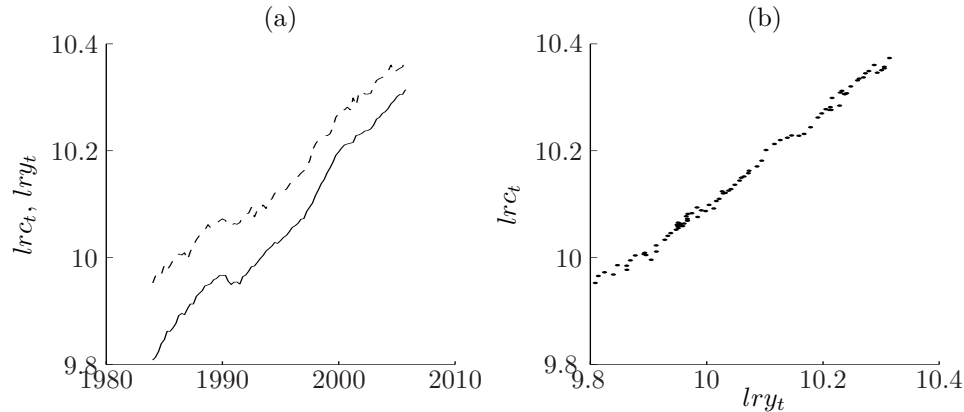


Figure 18.1 U.S. data on the logarithm of real consumption per capita, lrc_t , and the logarithm of real income per capita, lry_t , for the period March 1984 to December 2005.

Example 18.2 Demand for Money

The long-run theory of the demand for money predicts that log real money,

lrm_t , is jointly determined by log real income, lry_t , and the spread between the yields on bonds and money, $spread_t$,

$$lrm_t = \beta_c + \beta_y lry_t + \beta_s spread_t + u_t,$$

where u_t is a disturbance term and β_c, β_y and β_s are parameters. A 3-dimensional scatter plot of the data in panel (a) of Figure 18.2 reveals a 2-dimensional surface suggesting that lrm_t does not persistently deviate from its long-run level of $\beta_c + \beta_y lry_t + \beta_s spread_t$. This feature is supported in panel (b) of Figure 18.2, which shows that the residuals from estimating the long-run money demand equation by ordinary least squares do not exhibit a trend. $\square$

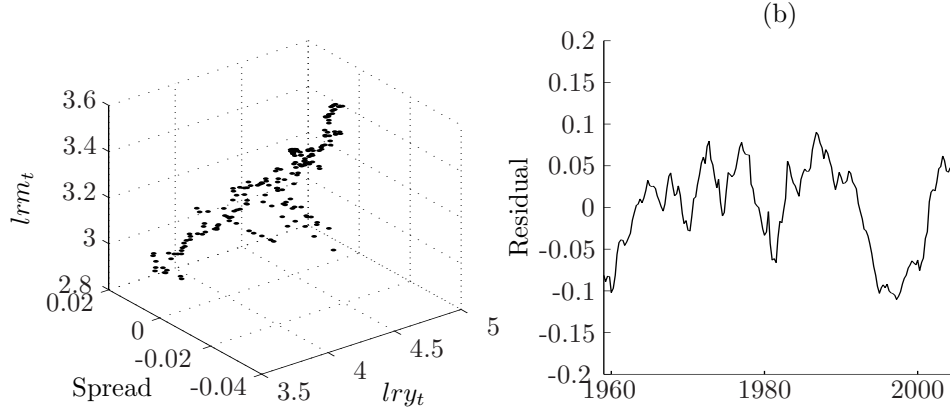


Figure 18.2 U.S. data on log real M2, lrm_t , log real GDP, lry_t , and the spread between the Treasury bill rate and Federal funds rate, $spread_t$, for the period March 1959 to December 2005. The residuals are computed from the least squares estimated equation $lrm_t = 0.149 + 0.828 lry_t - 0.785 spread_t + \hat{u}_t$.

Example 18.3 Term Structure of Interest Rates

The term structure is the relationship between the yields on bonds of differing maturities. Figure 18.3 shows that while the 1-year, 5-year and 10-year yields are nonstationary, a 3-dimensional scatter plot shows that all yields fall on a 1-dimensional line. This suggests two long-run relationships between the three yields

$$\begin{aligned} r_{3,t} &= \beta_{c,1} + \beta_{1,1} r_{1,t} + u_{1,t} \\ r_{2,t} &= \beta_{c,2} + \beta_{2,1} r_{1,t} + u_{2,t}, \end{aligned}$$

where $u_{1,t}$ and $u_{2,t}$ are stationary disturbances. By deduction, N yields result

in $N - 1$ long-run relationships and hence $N - 1$ spreads. In the special case where $\beta_{1,1} = \beta_{2,1} = 1$ and $\beta_{c,1} = \beta_{c,2} = 0$, the disturbance terms are equal to the spreads, which are stationary. $\square$

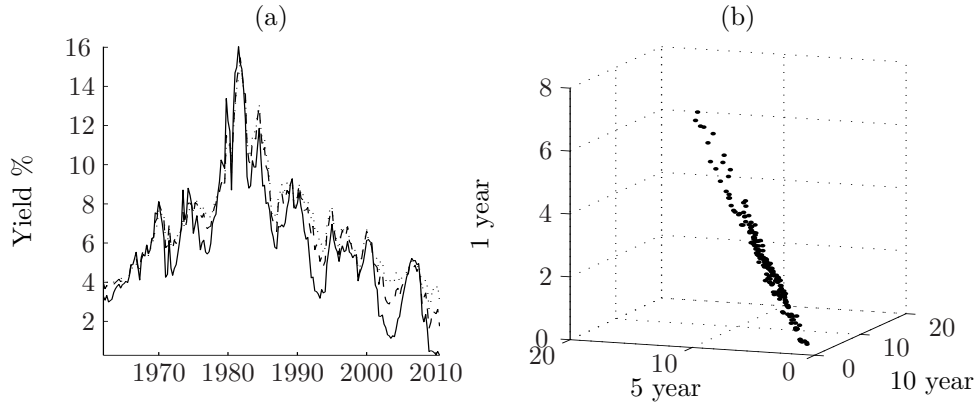


Figure 18.3 Yields (% p.a.) on 1-year, 5-year and 10-year U.S. bonds, for the period March 1962 to September 2010.

18.3 Specification: VECM

The examples in Section 18.2 demonstrate the existence of long-run equilibrium relationships between nonstationary variables where short-run deviations from equilibrium are stationary.

18.3.1 Bivariate Models

Consider a bivariate model containing two $I(1)$ variables $y_{1,t}$ and $y_{2,t}$, with the long-run relationship given by

$$y_{1,t} = \beta_c + \beta_y y_{2,t} + u_t, \quad (18.1)$$

where $\beta_c + \beta_y y_{2,t}$ represents the long-run equilibrium, and u_t represents the short-run deviations from equilibrium, which by assumption is stationary. The long run is represented in Figure 18.4 by the straight line assuming $\beta_y > 0$. Suppose that the two variables are in equilibrium at point A. From (18.1), the effect of a positive shock in the previous period ($u_{t-1} > 0$) immediately raises $y_{1,t}$ to point B while leaving $y_{2,t-1}$ unaffected. For the process to converge back to its long-run equilibrium, there are three possible trajectories.

(1) *Adjustments are made by $y_{1,t}$*

Equilibrium is restored by $y_{1,t}$ decreasing toward point A while $y_{2,t}$ remains unchanged at its initial position. Assuming that the short-run movements in $y_{1,t}$ are a linear function of the size of the shock, u_t , the adjustment in $y_{1,t}$ is given by

$$y_{1,t} - y_{1,t-1} = \alpha_1 u_{t-1} + v_{1,t} = \alpha_1 (y_{1,t-1} - \beta_c - \beta_y y_{2,t-1}) + v_{1,t}, \quad (18.2)$$

where $\alpha_1 < 0$, is a parameter and $v_{1,t}$ is a disturbance term.

(2) *Adjustments are made by $y_{2,t}$*

Equilibrium is restored by $y_{2,t}$ increasing toward point C, with $y_{1,t}$ remaining unchanged after the initial shock. Assuming that the short-run movements in $y_{2,t}$ are a linear function of the size of the shock, u_t , the adjustment in $y_{2,t}$ is given by

$$y_{2,t} - y_{2,t-1} = \alpha_2 u_{t-1} + v_{2,t} = \alpha_2 (y_{1,t-1} - \beta_c - \beta_y y_{2,t-1}) + v_{2,t}, \quad (18.3)$$

where $\alpha_2 > 0$ is a parameter and $v_{2,t}$ is a disturbance term.

(3) *Adjustments are made by both $y_{1,t}$ and $y_{2,t}$*

Both equations (18.2) and (18.3) are now in operation with $y_{1,t}$ and $y_{2,t}$ converging to a point on the long-run equilibrium such as D. The relative strengths of the two adjustment paths depend upon the relative magnitudes of the adjustment parameters α_1 and α_2 .

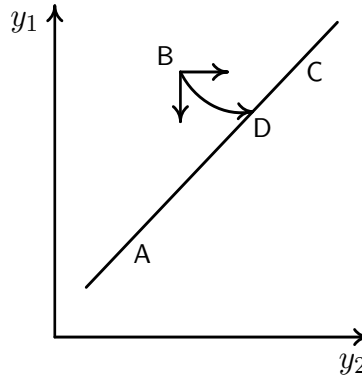


Figure 18.4 Phase diagram demonstrating a vector error correction model.

Equations (18.2) and (18.3) represent a VECM as the two variables correct themselves in the next period according to the error from being out of equilibrium. For this reason, the parameters α_1 and α_2 are known as the error correction parameters. An important characteristic of the VECM is

that it is a special case of a VAR where the parameters are subject to a set of cross-equation restrictions because all the variables are governed by the same long-run equation(s). To highlight this, rewrite the VECM in (18.2) and (18.3) as

$$\begin{bmatrix} \Delta y_{1,t} \\ \Delta y_{2,t} \end{bmatrix} = \begin{bmatrix} -\alpha_1 \beta_c \\ -\alpha_2 \beta_c \end{bmatrix} + \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} \begin{bmatrix} 1 & -\beta_y \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix},$$

or, in terms of a VAR

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} -\alpha_1 \beta_c \\ -\alpha_2 \beta_c \end{bmatrix} + \begin{bmatrix} 1 + \alpha_1 & -\alpha_1 \beta_y \\ \alpha_2 & 1 - \alpha_2 \beta_y \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix}$$

$$y_t = \mu + \Phi_1 y_{t-1} + v_t,$$

where

$$\mu = \begin{bmatrix} -\alpha_1 \beta_c \\ -\alpha_2 \beta_c \end{bmatrix}, \quad \Phi_1 = \begin{bmatrix} 1 + \alpha_1 & -\alpha_1 \beta_y \\ \alpha_2 & 1 - \alpha_2 \beta_y \end{bmatrix}. \quad (18.4)$$

This is a first-order VAR with two restrictions on the parameters. In an unconstrained VAR, the vector of intercepts μ contains 2 parameters, and the matrix of autocorrelation parameters Φ_1 contains 4 parameters, for a total of 6 parameters. However, from (18.4), the model consists of just 4 parameters $\{\beta_c, \beta_y, \alpha_1, \alpha_2\}$, resulting in a total of $6 - 4 = 2$ restrictions.

18.3.2 Multivariate Models

The relationship between a VECM and a VAR generalizes to N variables $y_t = (y_{1,t}, \dots, y_{N,t})'$ and p lags. First consider the case of an N dimensional model with $p = 1$ lags

$$y_t = \mu + \Phi_1 y_{t-1} + v_t. \quad (18.5)$$

Subtracting y_{t-1} from both sides and rearranging gives

$$\Delta y_t = \mu - (I_N - \Phi_1) y_{t-1} + v_t = \mu - \Phi(1) y_{t-1} + v_t, \quad (18.6)$$

where $\Phi(1) = I_N - \Phi_1$. This is a VECM, but with $p - 1 = 0$ lags. The operation to transform the VAR in (18.5) to the VECM in (18.6) is exactly the same operation used in Chapter 17 to derive the Dickey-Fuller regression equation from an AR(1) model of y_t to test for a unit root.

Expanding the VAR in (18.5) to include p lags gives

$$\Phi(L) y_t = \mu + v_t, \quad (18.7)$$

where v_t is an N dimensional vector of iid disturbances and $\Phi(L) = I_N -$

$\Phi_1 L - \dots - \Phi_p L^p$ is a polynomial in the lag operator (see Appendix B). The resulting VECM has $p - 1$ lags given by

$$\Delta y_t = \mu - \Phi(1)y_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t. \quad (18.8)$$

In the special case $N = 1$, (18.8) reduces to the augmented Dickey-Fuller equation to test for a unit root discussed in Chapter 17.

The formal operation linking (18.7) and (18.8) is the Beveridge-Nelson decomposition, (Beveridge and Nelson, 1981) which re-expresses the polynomial $\Phi(L)$ in (18.7) as

$$\Phi(L) = \Phi(1)L + \Gamma(L)(1 - L), \quad (18.9)$$

where

$$\Phi(1) = I_N - \Phi_1 - \dots - \Phi_p, \quad \Gamma(L) = I_N - \sum_{j=1}^{p-1} \Gamma_j L^j, \quad \Gamma_j = - \sum_{i=j+1}^p \Phi_i. \quad (18.10)$$

Substituting (18.9) into (18.7) gives the VECM in (18.8),

$$(\Phi(1)L + \Gamma(L)(1 - L))y_t = \Phi(1)y_{t-1} + \Delta y_t - \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} = \mu + v_t.$$

18.3.3 Cointegration

If the vector time series y_t is assumed to be $I(1)$, then y_t is cointegrated if there exists a $N \times r$ full column rank matrix, β , with $1 \leq r < N$, such that the r linear combinations

$$\beta' y_t = u_t, \quad (18.11)$$

are $I(0)$. The dimension r is called the cointegrating rank and the columns of β are called the cointegrating vectors. In matrix notation the cointegrating system is given by

$$\begin{bmatrix} \beta_{1,1} & \beta_{1,2} & \cdots & \beta_{1,r} \\ \beta_{2,1} & \beta_{2,2} & \cdots & \beta_{2,r} \\ \beta_{3,1} & \beta_{3,2} & \cdots & \beta_{3,r} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{N,1} & \beta_{N,2} & \cdots & \beta_{N,r} \end{bmatrix}' \begin{bmatrix} y_{1,t} \\ y_{2,t} \\ y_{3,t} \\ \vdots \\ y_{N,t} \end{bmatrix} = \begin{bmatrix} u_{1,t} \\ u_{2,t} \\ u_{3,t} \\ \vdots \\ u_{N,t} \end{bmatrix}.$$

This result suggests that $N - r$ common trends exist that are $I(1)$, which are the driving factors behind the $I(1)$ features of y_t .

Example 18.4 Graphical Detection of the Rank

In the $N = 2$ dimensional area of the scatter plot of the permanent income example, panel (b) of in Figure 18.1, a single line suggests a rank of $r = 1$ and $N - r = 2 - 1 = 1$ common trend. In the $N = 3$ dimensional area of the scatter plot of the money demand example, panel (b) of Figure 18.2, a two-dimensional surface suggests a rank of $r = 1$ and $N - r = 3 - 1 = 2$ common trends. Finally, in the $N = 3$ dimensional area of the scatter plot of the term structure example, panel (b) of Figure 18.3, a single line suggests a rank of $r = 2$ and $N - r = 3 - 2 = 1$ common trend. $\square$

The Granger representation theorem (Engle and Granger, 1987) is the fundamental result showing the formal connection between cointegration in equation (18.11) and the vector error correction model in (18.8). The main result of the theorem is that all of the information needed to analyze cointegration is contained in the $\Phi(1)$ matrix in (18.8).

Granger Representation Theorem

Suppose y_t , generated by (18.8), is either $I(1)$ or $I(0)$.

- (a) If $\Phi(1)$ has full rank, $r = N$, then y_t is $I(0)$.
- (b) If $\Phi(1)$ has reduced rank, r for $0 < r < N$, $\Phi(1) = -\alpha\beta'$ where α and β are each $(N \times r)$ matrices with full column rank, then y_t is $I(1)$ and $\beta'y_t$ is $I(0)$ with cointegrating vectors(s) given by the columns of β .
- (c) If $\Phi(1)$ has zero rank, $r = 0$, $\Phi(1) = 0$ and y_t is $I(1)$ and not cointegrated.

■

Example 18.5 Cointegrating Rank of Long-Run Economic Models

The form of $\Phi(1)$ for the long-run economic models in Section 18.2 is

$$\begin{array}{ll}
 \text{Permanent income} & : \quad \Phi(1) = -\alpha\beta' = - \begin{bmatrix} \alpha_{1,1} \\ \alpha_{2,1} \end{bmatrix} \begin{bmatrix} 1 \\ -\beta_y \end{bmatrix}' \\
 (N = 2, r = 1) & \\
 \text{Money demand} & : \quad \Phi(1) = -\alpha\beta' = - \begin{bmatrix} \alpha_{1,1} \\ \alpha_{2,1} \\ \alpha_{3,1} \end{bmatrix} \begin{bmatrix} 1 \\ -\beta_y \\ -\beta_s \end{bmatrix}' \\
 (N = 3, r = 1) & \\
 \text{Term structure} & : \quad \Phi(1) = -\alpha\beta' = - \begin{bmatrix} \alpha_{1,1} & \alpha_{1,2} \\ \alpha_{2,1} & \alpha_{2,2} \\ \alpha_{3,1} & \alpha_{3,2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{bmatrix}' \\
 (N = 3, r = 2) &
 \end{array}$$

□

The Granger representation theorem shows how cointegration arises from the rearranged VAR in (18.7) and suggests the form of the model that should be estimated. If $\Phi(1)$ has full rank, N , then all of the time series in y_t are stationary and the original VAR in (18.7) is specified in levels. If $\Phi(1) = 0$, then equation (18.8) shows that the appropriate model is a VAR($p - 1$) in first differences

$$\Delta y_t = \mu + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t. \quad (18.12)$$

If $\Phi(1)$ has reduced rank, r with $0 < r < N$, then the VECM in (18.8) is subject to the cointegrating restrictions $\Phi(1) = -\alpha\beta'$ and is given by

$$\Delta y_t = \mu + \alpha\beta' y_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t. \quad (18.13)$$

18.3.4 Deterministic Components

A natural extension of the VECM in (18.13) is to include a deterministic time trend t as follows

$$\Delta y_t = \mu_0 + \mu_1 t + \alpha\beta' y_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t, \quad (18.14)$$

where now μ_0 and μ_1 are $(N \times 1)$ vectors of parameters associated with the intercept and time trend, respectively. A property that the long-run economic models presented in Section 18.2 have in common is the presence of an intercept. This suggests that, in general, the deterministic components contribute both to the short-run and long-run properties of y_t . To capture this feature of these models, the deterministic elements in (18.14) are decomposed into their short-run and long-run contributions by defining

$$\underbrace{\mu_j}_{\text{Total}} = \underbrace{\delta_j}_{\text{Short-run}} + \underbrace{\alpha\beta'_j}_{\text{Long-run}}, \quad j = 0, 1, \quad (18.15)$$

where δ_j is $(N \times 1)$ and β_j is $(1 \times r)$. Now rewrite (18.14) as

$$\Delta y_t = \delta_0 + \delta_1 t + \alpha(\beta'_0 + \beta'_1 t + \beta' y_{t-1}) + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t. \quad (18.16)$$

The term $\beta'_0 + \beta'_1 t + \beta' y_{t-1}$ in (18.16) represents the long-run relationship among the variables. The parameter δ_0 provides a drift component in the

equation of Δy_t that contributes a linear trend to y_t . Similarly $\delta_1 t$ allows for a linear time trend in Δy_t that contributes a quadratic trend to y_t . By contrast, β_0 contributes just a constant to y_t and $\beta_1' t$ contributes a linear trend. Thus the unrestricted term μ_0 in (18.14) has potentially two effects on y_t , contributing both a constant and a linear trend depending on its linear relationship with α in (18.16).

The decomposition of μ_j in (18.15) into β_j and δ_j is given by

$$\delta_j = \alpha_{\perp} [\alpha'_{\perp} \alpha_{\perp}]^{-1} \alpha'_{\perp} \mu_j, \quad \beta_j = [\alpha' \alpha]^{-1} \alpha' \mu_j, \quad (18.17)$$

where $\alpha_{\perp}$ is the $(N \times (N-r))$ orthogonal complement matrix of the $(N \times r)$ matrix of error correction parameters α , such that $\alpha'_{\perp} \alpha = 0_{(N-r) \times r}$. To derive these expressions, multiply both sides of the identity $I_N = \alpha(\alpha' \alpha)^{-1} \alpha' + \alpha_{\perp}(\alpha'_{\perp} \alpha_{\perp})^{-1} \alpha'_{\perp}$, by μ_j to give

$$\mu_j = \alpha_{\perp}(\alpha'_{\perp} \alpha_{\perp})^{-1} \alpha'_{\perp} \mu_j + \alpha(\alpha' \alpha)^{-1} \alpha' \mu_j = \delta_j + \alpha \beta_j.$$

An important implication of this decomposition is that even though δ_j has N elements in (18.16), there are at most only $N-r$ linear independent elements, which is the difference between the dimension of the total system μ_j and the dimension of the number of cointegrating vectors β_j . Alternatively, the VECM in (18.16) contains $N+r$ intercepts with r cross-equation restrictions on the parameters given by (18.17).

Equation (18.16) is a general model containing five important special cases summarized in Table 18.1. Model 1 is the simplest and most restricted version of the VECM as it contains no deterministic components. Model 2 allows for r intercepts corresponding to the r long-run equations, whereas Model 3 allows for contributions of the intercepts in the short run and the long run for a total of N intercepts. Model 4 extends Model 3 by including r time trends just in the long-run equation, whereas Model 5 allows for N time trends that contribute to y_t in both the short run and the long run. The time series properties of these models are investigated in Exercise 2.

Additional types of deterministic components allowing for dummy variables as a result of seasonality or structural breaks can also be included. Seasonal dummy variables are included in the model by appending them to the constant term. Provided that the dummy variables are standardized to sum to zero, they do not have any effect on the asymptotic properties of estimated parameters of the model (Johansen, 1995b, p166).

Structural breaks can arise from

- (1) a change in the number of cointegrating equations; and/or
- (2) a change in the parameters of the VECM.

Table 18.1

Summary of alternative VECM specifications with $\mu_j = \delta_j + \alpha\beta'_j, j = 0, 1$.

Model	Specification
1	$\Delta y_t = \alpha\beta' y_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t$ Restrictions: $\{\delta_0 = 0, \delta_1 = 0, \beta_0 = 0, \beta_1 = 0\}$
2	$\Delta y_t = \alpha(\beta'_0 + \beta' y_{t-1}) + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t$ Restrictions: $\{\delta_0 = 0, \delta_1 = 0, \beta_1 = 0\}$
3	$\Delta y_t = \delta_0 + \alpha(\beta'_0 + \beta' y_{t-1}) + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t$ $= \mu_0 + \alpha\beta' y_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t$ Restrictions: $\{\delta_1 = 0, \beta_1 = 0\}$
4	$\Delta y_t = \delta_0 + \alpha(\beta'_0 + \beta'_1 t + \beta' y_{t-1}) + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t$ $= \mu_0 + \alpha(\beta'_1 t + \beta' y_{t-1}) + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t$ Restrictions: $\{\delta_1 = 0\}$
5	$\Delta y_t = \delta_0 + \delta_1 t + \alpha(\beta'_0 + \beta'_1 t + \beta' y_{t-1}) + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t$ $= \mu_0 + \mu_1 t + \alpha\beta' y_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t$ Restrictions: None

The statistical properties of the estimators depend upon whether the timing of the break is known or unknown as well as on where the break occurs in the sample.

18.4 Estimation

Assuming a parametric specification for the distribution of the disturbance vector v_t in (18.14), the parameters of a VECM are estimated by maximum likelihood by estimating a VAR subject to the cross-equation restrictions arising from cointegration. Two maximum likelihood estimators are presented. The first is an iterative estimator based on the algorithms of Chapter 3 as a result of the cross-equation restrictions on the parameters of the

VAR. The second is the estimator proposed by Johansen (1988, 1991) based on an eigen decomposition of the log-likelihood function. Both estimators yield identical point estimates. The second estimator has a computational advantage since it does not require an iterative solution but the former estimator has the advantage of being appropriate for more general classes of cointegrating models.

If v_t is assumed to be distributed as $v_t \sim iid N(0, V)$, the conditional log-likelihood function of the VECM based on p lags is

$$\begin{aligned} \ln L_T(\theta) &= \frac{1}{T-p} \sum_{t=p+1}^T \ln f(y_t | y_{t-1}, y_{t-2}, \dots, y_{t-p}; \theta) \\ &= -\frac{N}{2} \ln 2\pi - \frac{1}{2} \ln |V| - \frac{1}{2(T-p)} \sum_{t=p+1}^T v_t V^{-1} v_t', \end{aligned} \quad (18.18)$$

where θ are the unknown parameters. The maximum likelihood estimator is obtained by choosing θ to maximize $\ln L_T(\theta)$, by solving the following first-order conditions for $\hat{\theta}$,

$$G_T(\hat{\theta}) = \left. \frac{\partial \ln L_T(\theta)}{\partial \theta} \right|_{\theta=\hat{\theta}} = 0. \quad (18.19)$$

The solution of (18.19) depends upon the rank of the matrix $\Phi(1)$ in (18.8). From the Granger representation theorem, three cases need to be considered: full rank ($r = N$), reduced rank ($0 < r < N$), zero rank ($r = 0$).

18.4.1 Full-Rank Case

In the case where $\Phi(1)$ in (18.8) has full rank, the VECM is equivalent to the unconstrained VAR in (18.7) with unknown parameters

$$\theta = \{\mu_0, \mu_1, \Phi_1, \Phi_2, \dots, \Phi_p, V\} = \{\mu_0, \mu_1, \Phi(1), \Gamma_1, \Gamma_2, \dots, \Gamma_{p-1}, V\},$$

where a time trend is included. When the results of Chapter 13 are used, the maximum likelihood estimator of θ is simply obtained by applying ordinary least squares to each equation separately to obtain $\{\hat{\mu}_0, \hat{\mu}_1, \hat{\Phi}_1, \hat{\Phi}_2, \dots, \hat{\Phi}_p\}$, with V estimated from the least squares residuals $\{\hat{v}_{1,t}, \hat{v}_{2,t}, \dots, \hat{v}_{N,t}\}$ at the last step since

$$\hat{V} = \frac{1}{T-p} \sum_{t=p+1}^T \hat{v}_t \hat{v}_t'. \quad (18.20)$$

Evaluating (18.18) at $\hat{\theta}$, by using $\hat{V}$ in (18.20), gives the maximised log-likelihood function

$$\ln L_T(\hat{\theta}) = -\frac{N}{2}(\ln 2\pi + 1) - \frac{1}{2} \ln |\hat{V}|. \quad (18.21)$$

Example 18.6 Full-Rank Estimates of the Term Structure Model

Consider the bivariate, $N = 2$, model of the term structure of interest rates where $y_{1,t}$ and $y_{2,t}$ are the 10-year and 1-year interest rates respectively given in Figure 18.3 and the sample size is $T = 195$ observations. Using $p = 1$ lags in (18.7) with a constant and no time trend yields the estimated model (with standard errors in parentheses)

$$\begin{aligned} y_{1,t} &= \underset{(0.120)}{0.257} + \underset{(0.039)}{0.888}y_{1,t-1} + \underset{(0.034)}{0.085}y_{2,t-1} + \hat{v}_{1,t} \\ y_{2,t} &= \underset{(0.189)}{0.221} - \underset{(0.062)}{0.026}y_{1,t-1} + \underset{(0.053)}{0.990}y_{2,t-1} + \hat{v}_{2,t}. \end{aligned}$$

The residual covariance matrix is estimated as

$$\hat{V} = \frac{1}{194} \sum_{t=2}^{195} \hat{v}_t \hat{v}_t' = \begin{bmatrix} 0.274 & 0.341 \\ 0.341 & 0.675 \end{bmatrix},$$

and $|\hat{V}| = 0.0690$. From (18.21), the value of the log-likelihood function evaluated at $\hat{\theta}$ is

$$\ln L_T(\hat{\theta}) = -\frac{2}{2}(\ln 2\pi + 1) - \frac{1}{2} \ln(0.068981) = -1.5009.$$

□

18.4.2 Reduced-Rank Case: Iterative Estimator

If $\Phi(1)$ in (18.8) has reduced rank, then the cross-equation restrictions in (18.11) are imposed on the model and it is the constrained VECM in (18.16) that is estimated. The unknown parameters

$$\theta = \{\delta_0, \delta_1, \alpha, \beta_0, \beta_1, \beta, \Gamma_1, \Gamma_2, \dots, \Gamma_{p-1}, V\},$$

are estimated by means of an iterative algorithm from Chapter 3.

Consider the bivariate VECM,

$$\Delta y_t = \alpha u_{t-1} + v_t, \quad u_{t-1} = y_{1,t-1} - \beta y_{2,t-1}, \quad (18.22)$$

where $\Delta y_t = [\Delta y_{1,t}, \Delta y_{2,t}]'$, $\alpha = [\alpha_1, \alpha_2]'$, $v_t = [v_{1,t}, v_{2,t}]'$ and the parameters

are $\theta = \{\alpha_1, \alpha_2, \beta\}$. The gradient and Hessian of (18.18) are, respectively,

$$G_T(\theta) = \frac{1}{T-1} \sum_{t=2}^T \begin{bmatrix} V^{-1}v_t u_{t-1} \\ -\alpha' V^{-1}v_t y_{2,t-1} \end{bmatrix},$$

$$H_T(\theta) = \frac{1}{T-1} \sum_{t=2}^T \begin{bmatrix} -V^{-1}u_{t-1}^2 & -V^{-1}(v_t - \alpha u_{t-1})y_{2,t-1} \\ -V^{-1}(v_t - \alpha u_{t-1})y_{2,t-1} & -\alpha' V^{-1}\alpha y_{2,t-1}^2 \end{bmatrix}. \quad (18.23)$$

The Newton-Raphson algorithm requires evaluating $G_T(\theta)$ and $H_T(\theta)$ at the starting values $\theta_{(0)}$ and updating according to $\theta_{(1)} = \theta_{(0)} - H_{(0)}^{-1}G_{(0)}$, until convergence.

A convenient way of providing starting estimates for the gradient algorithm is to use the Engle-Granger two-step estimator. In the case of two variables, $y_{1,t}$ and $y_{2,t}$, the first step involves estimating the cointegrating equation by ordinary least squares by regressing $y_{1,t}$ on a constant and $y_{2,t}$ to get $\hat{u}_t$ and an estimate of the cointegrating vector $\hat{\beta}$. The second step involves regressing $\Delta y_{1,t}$ on $\hat{u}_{t-1}$ to get $\hat{\alpha}_1$, and $\Delta y_{2,t}$ on $\hat{u}_{t-1}$ to get $\hat{\alpha}_2$. Any lags of $\Delta y_{1,t}$ and $\Delta y_{2,t}$ in the VECM are included in the second-stage regressions.

Example 18.7 Iterative Estimates of the Term Structure Model

A bivariate, $N = 2$, VECM containing the 10-year, $y_{1,t}$, and 1-year yields, $y_{2,t}$, using Model 2 in Table 18.1, with $p = 1$ lags and rank of $r = 1$, is specified as

$$y_{1,t} - y_{1,t-1} = \alpha_1(y_{1,t-1} - \beta_c - \beta_s y_{2,t-1}) + v_{1,t}$$

$$y_{2,t} - y_{2,t-1} = \alpha_2(y_{1,t-1} - \beta_c - \beta_s y_{2,t-1}) + v_{2,t},$$

where $v_t = [v_{1,t}, v_{2,t}]$ is distributed as $N(0, V)$ and the unknown parameters are $\theta = \{\beta_c, \beta_s, \alpha_1, \alpha_2\}$. The conditional log-likelihood in (18.18) is maximized in 5 iterations using the Newton-Raphson algorithm with starting estimates based on the Engle-Granger two-step estimator. The parameter estimates are

$$\hat{\theta} = [\hat{\beta}_0, \hat{\beta}_s, \hat{\alpha}_1, \hat{\alpha}_2]' = [1.434, 0.921, -0.092, 0.012]',$$

with estimated covariance matrix based on the Hessian given by

$$-\frac{1}{T-1} H_T^{-1}(\hat{\theta}) = \begin{bmatrix} 0.332418 & -0.047472 & -0.010395 & -0.010112 \\ -0.047472 & 0.008112 & 0.001861 & 0.001849 \\ -0.010395 & 0.001861 & 0.001734 & 0.002042 \\ -0.010112 & 0.001849 & 0.002042 & 0.003619 \end{bmatrix}.$$

The estimated VECM, with standard errors in parentheses, is

$$\begin{aligned}\Delta y_{1,t} &= \begin{matrix} -0.092 \\ (0.042) \end{matrix} (y_{1,t-1} - \begin{matrix} 1.434 \\ (0.576) \end{matrix} - \begin{matrix} 0.921 \\ (0.090) \end{matrix} y_{2,t-1}) + \hat{v}_{1,t} \\ \Delta y_{2,t} &= \begin{matrix} 0.012 \\ (0.060) \end{matrix} (y_{1,t-1} - \begin{matrix} 1.434 \\ (0.576) \end{matrix} - \begin{matrix} 0.921 \\ (0.090) \end{matrix} y_{2,t-1}) + \hat{v}_{2,t},\end{aligned}$$

and the residual covariance matrix is estimated as

$$\hat{V} = \frac{1}{194} \sum_{t=2}^{195} \hat{v}_t \hat{v}_t' = \begin{bmatrix} 0.684 & 0.346 \\ 0.346 & 0.277 \end{bmatrix}.$$

From (18.21), the value of the log-likelihood function evaluated at $\hat{\theta}$ is

$$\ln L_T(\hat{\theta}) = -\frac{2}{2}(\ln 2\pi + 1) - \frac{1}{2} \ln(0.069914) = -1.5076.$$

□

18.4.3 Reduced Rank Case: Johansen Estimator

An alternative maximum likelihood estimator proposed by Johansen (1988, 1991) has the advantage that the restricted VECM in (18.16) is estimated without the need for an iterative algorithm. The fundamental difference in approach from the iterative estimator is that the Johansen approach does not first identify the individual coefficients cointegrating vector(s) before estimation. The Johansen approach estimates a basis for the vector space spanned by the cointegrating vectors and then imposes identification on the coefficients afterwards.

Each of the VECM specifications in Table 18.1 can be written in the general form

$$\Delta y_t = \alpha \gamma' z_{1,t} + \Psi z_{2,t} + v_t, \quad (18.24)$$

where for model 1:

$$\begin{aligned}z_{1,t} &= y_{t-1}, & \gamma &= \beta, \\ z_{2,t} &= (\Delta y_{t-1}', \dots, \Delta y_{t-p+1}')', & \Psi &= (\Gamma_1, \dots, \Gamma_{p-1}),\end{aligned}$$

for model 2:

$$\begin{aligned}z_{1,t} &= (1, y_{t-1})', & \gamma &= (\beta_0', \beta')', \\ z_{2,t} &= (\Delta y_{t-1}', \dots, \Delta y_{t-p+1}')', & \Psi &= (\Gamma_1, \dots, \Gamma_{p-1}),\end{aligned}$$

for model 3:

$$\begin{aligned}z_{1,t} &= y_{t-1}, & \gamma &= \beta, \\ z_{2,t} &= (1, \Delta y_{t-1}', \dots, \Delta y_{t-p+1}')', & \Psi &= (\mu_0, \Gamma_1, \dots, \Gamma_{p-1}),\end{aligned}$$

for model 4:

$$\begin{aligned} z_{1,t} &= (1, t, y_{t-1})', & \gamma &= (\beta'_0, \beta'_1, \beta')', \\ z_{2,t} &= (1, \Delta y'_{t-1}, \dots, \Delta y'_{t-p+1})', & \Psi &= (\delta_0, \Gamma_1, \dots, \Gamma_{p-1}), \end{aligned}$$

and for model 5:

$$\begin{aligned} z_{1,t} &= y_{t-1}, & \gamma &= \beta, \\ z_{2,t} &= (1, t, \Delta y'_{t-1}, \dots, \Delta y'_{t-p+1})', & \Psi &= (\mu_0, \mu_1, \Gamma_1, \dots, \Gamma_{p-1}), \end{aligned}$$

Estimation is achieved by concentrating the log-likelihood function (see Chapter 3) on γ to yield

$$\ln L_T(\gamma) = \ln L_T(\hat{\alpha}, \gamma, \hat{\Psi}, \hat{V}).$$

The idea behind this strategy is based on the recognition that if γ is known, estimation of the remaining parameters of the VECM in (18.24) no longer involves a nonlinear algorithm and simply requires an N dimensional multivariate regression of Δy_t on $\gamma' z_{1,t}$ and $z_{2,t}$. This second step also constitutes the second step of the Engle-Granger estimator used to generate starting estimates for the iterative estimator in Subsection 18.4.2. As each equation contains the same set of variables, the maximum likelihood estimator is ordinary least squares applied to each equation separately.

Concentrating the log-likelihood function involves three stages. The first stage uses (18.20) to estimate V to yield the log-likelihood function in (18.21), concentrated on α , γ and Ψ . The second stage involves estimating Ψ for fixed α and γ using two multivariate regressions of Δy_t on $z_{2,t}$ and $z_{1,t}$ on $z_{2,t}$ to obtain residuals $R_{0,t}$ and $R_{1,t}$ respectively. The details of these regressions are summarized in Table 18.2. For example, in the case of Model 2, first regress Δy_t on $\{\Delta y_{t-1}, \Delta y_{t-2}, \dots, \Delta y_{t-p+1}\}$ to obtain the $(N \times 1)$ vector of residuals $R_{0,t}$ and then regress $\{1, y_{t-1}\}$ on $\{\Delta y_{t-1}, \Delta y_{t-2}, \dots, \Delta y_{t-p+1}\}$ to obtain the $((N+1) \times 1)$ vector of residuals $R_{1,t}$. The estimated covariance matrix $\hat{V}$ in (18.20) is now defined as

$$\begin{aligned} \hat{V} &= \frac{1}{T-p} \sum_{t=p+1}^T (R_{0,t} - \alpha \gamma' R_{1,t})(R_{0,t} - \alpha \gamma' R_{1,t})' \\ &= S_{00} - \alpha \gamma' S_{10} - S_{01} \gamma \alpha' + \alpha \gamma' S_{11} \gamma \alpha', \end{aligned}$$

where

$$S_{ij} = \frac{1}{T-p} \sum_{t=p+1}^T R_{i,t} R'_{j,t}, \quad (18.25)$$

and the concentrated log-likelihood function is

$$\begin{aligned} \ln L_T(\alpha, \gamma) &= -\frac{N}{2}(\ln 2\pi + 1) \\ &\quad - \frac{1}{2} \ln |S_{00} - \alpha\gamma'S_{10} - S_{01}\gamma\alpha' + \alpha\gamma'S_{11}\gamma\alpha'|. \end{aligned} \quad (18.26)$$

Table 18.2

Summary of intermediate ordinary least squares regressions to concentrate out the deterministic and lagged variable parameters from the log-likelihood function for alternative model specifications of the VECM.

Model	Dep. Variables	Independent Variables	Residuals
1	$\{\Delta y_t\}$	$\{\Delta y_{t-1}, \Delta y_{t-2}, \dots, \Delta y_{t-p+1}\}$	$R_{0,t}$
	$\{y_{t-1}\}$	$\{\Delta y_{t-1}, \Delta y_{t-2}, \dots, \Delta y_{t-p+1}\}$	$R_{1,t}$
2	$\{\Delta y_t\}$	$\{\Delta y_{t-1}, \Delta y_{t-2}, \dots, \Delta y_{t-p+1}\}$	$R_{0,t}$
	$\{1, y_{t-1}\}$	$\{\Delta y_{t-1}, \Delta y_{t-2}, \dots, \Delta y_{t-p+1}\}$	$R_{1,t}$
3	$\{\Delta y_t\}$	$\{1, \Delta y_{t-1}, \Delta y_{t-2}, \dots, \Delta y_{t-p+1}\}$	$R_{0,t}$
	$\{y_{t-1}\}$	$\{1, \Delta y_{t-1}, \Delta y_{t-2}, \dots, \Delta y_{t-p+1}\}$	$R_{1,t}$
4	$\{\Delta y_t\}$	$\{1, \Delta y_{t-1}, \Delta y_{t-2}, \dots, \Delta y_{t-p+1}\}$	$R_{0,t}$
	$\{t, y_{t-1}\}$	$\{1, \Delta y_{t-1}, \Delta y_{t-2}, \dots, \Delta y_{t-p+1}\}$	$R_{1,t}$
5	$\{\Delta y_t\}$	$\{1, t, \Delta y_{t-1}, \Delta y_{t-2}, \dots, \Delta y_{t-p+1}\}$	$R_{0,t}$
	$\{y_{t-1}\}$	$\{1, t, \Delta y_{t-1}, \Delta y_{t-2}, \dots, \Delta y_{t-p+1}\}$	$R_{1,t}$

The third stage is to maximize (18.26) with respect to α . The first-order condition is

$$S_{01}\gamma - \hat{\alpha}\gamma'S_{11}\gamma = 0,$$

and solving for $\hat{\alpha}$ gives

$$\hat{\alpha} = S_{01}\gamma [\gamma'S_{11}\gamma]^{-1}.$$

This is the ordinary least squares estimator of α for a known γ . That is, $\hat{\alpha}$ is found from a regression of $R_{0,t}$ on $\gamma'R_{1,t}$. Substituting into (18.26) gives the concentrated log-likelihood function

$$\begin{aligned} \ln L_T(\gamma) &= -\frac{N}{2}(\ln 2\pi + 1) - \frac{1}{2} \ln |S_{00} - S_{01}\gamma [\gamma'S_{11}\gamma]^{-1} \gamma'S_{10}| \\ &= -\frac{N}{2}(\ln 2\pi + 1) - \frac{1}{2} \ln |S_{00}| \\ &\quad - \frac{1}{2} \ln |\gamma'(S_{11} - S_{10}S_{00}^{-1}S_{01})\gamma|, \end{aligned}$$

which uses the property of determinants that

$$\begin{aligned} |S_{00} - S_{01}\gamma [\gamma' S_{11}\gamma]^{-1} \gamma' S_{10}| &= |\gamma' S_{11}\gamma|^{-1} |S_{00}| |\gamma' S_{11}\gamma - \gamma' S_{10} S_{00}^{-1} S_{01}\gamma| \\ &= |S_{00}| |\gamma' (S_{11} - S_{10} S_{00}^{-1} S_{01}) \gamma|, \end{aligned}$$

where the normalization $\gamma' S_{11}\gamma = I_r$ is imposed. The maximum likelihood estimator, $\hat{\gamma}$, is the value of γ that maximizes $\ln L_T(\gamma)$, which is equivalent to solving the following eigenvalue problem (Anderson, 1984)

$$|\hat{\lambda} S_{11} - S_{10} S_{00}^{-1} S_{01}| = 0, \quad (18.27)$$

where $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_r \geq \dots \geq \hat{\lambda}_N \geq 0$ and choosing $\hat{\gamma}$ to be the eigenvectors corresponding to the largest r eigenvalues.

To solve (18.27), use the Choleski decomposition $S_{11} = LL'$ to rewrite equation (18.27) as

$$|\hat{\lambda} I - L^{-1} S_{10} S_{00}^{-1} S_{01} L'^{-1}| = 0, \quad (18.28)$$

which uses the result $LS_{11}L' = I_N$. Solving this eigenvalue problem yields $\hat{\lambda}_1, \dots, \hat{\lambda}_N$ and the corresponding matrix of eigenvectors E . The estimates of the eigenvalues have the property $0 < \hat{\lambda}_i < 1$. The estimated eigenvectors of (18.27) are normalised as EL'^{-1} in order to satisfy $\hat{\gamma}' S_{11} \hat{\gamma} = I_r$. Finally, the maximized log-likelihood is

$$\ln L_T(\hat{\gamma}) = -\frac{N}{2}(1 + \ln 2\pi) - \frac{1}{2} \ln |S_{00}| - \frac{1}{2} \sum_{i=1}^r \ln(1 - \hat{\lambda}_i), \quad (18.29)$$

which represents a decomposition of the log-likelihood function in terms of the r largest eigenvalues.

A summary of the steps to implement the Johansen estimator are as follows.

Step 1: Compute the residual vectors $R_{0,t}$ and $R_{1,t}$ as given in Table 18.2.

Step 2: Compute the sums of squares matrices S_{ij} in (18.25).

Step 3: Compute the Choleski decomposition $S_{11} = LL'$.

Step 4: Perform an eigen decomposition on $L^{-1} S_{10} S_{00}^{-1} S_{01} L'^{-1}$, which yields the estimated eigenvalues $\hat{\lambda}$ and matrix of eigenvectors E .

Step 5: Normalize the eigenvector matrix as $L'^{-1}E$ and obtain $\hat{\gamma}$ as the r columns of $L'^{-1}E$ corresponding to the largest r eigenvalues. From $\hat{\gamma}$, obtain $\hat{\beta}$ and, if necessary, $\hat{\beta}_0$ and $\hat{\beta}_1$.

Example 18.8 Johansen Estimates of the Term Structure Model

To re-estimate the bivariate term structure model in Example 18.7 by the Johansen estimator, the sums of squares matrices are

$$\begin{aligned} S_{00} &= \frac{1}{T-1} \sum_{t=2}^T R_{0,t} R'_{0,t} = \begin{bmatrix} 0.286 & 0.344 \\ 0.344 & 0.684 \end{bmatrix} \\ S_{11} &= \frac{1}{T-1} \sum_{t=2}^T R_{1,t} R'_{1,t} = \begin{bmatrix} 53.971 & 48.103 & 6.882 \\ 48.103 & 44.444 & 5.942 \\ 6.882 & 5.942 & 1.000 \end{bmatrix} \\ S_{01} &= \frac{1}{T-1} \sum_{t=2}^T R_{0,t} R'_{1,t} = \begin{bmatrix} -0.163 & -0.060 & -0.006 \\ -0.349 & -0.369 & -0.015 \end{bmatrix}, \end{aligned}$$

where $R_{0,t} = [\Delta y_{1,t}, \Delta y_{2,t}]'$ and $R_{1,t} = [y_{1,t-1}, y_{2,t-1}, 1]'$. The Choleski decomposition of S_{11} is

$$L = \begin{bmatrix} 7.346 & 0.000 & 0.000 \\ 6.548 & 1.253 & 0.000 \\ 0.937 & -0.153 & 0.315 \end{bmatrix}.$$

Also compute

$$L^{-1} S_{10} S_{00}^{-1} S_{01} L^{-1'} = \begin{bmatrix} 0.003327 & 0.004630 & -0.004132 \\ 0.004630 & 0.076978 & 0.032473 \\ -0.004132 & 0.032473 & 0.025845 \end{bmatrix}.$$

The estimated eigenvalues and eigenvectors of this matrix are, respectively,

$$\hat{\lambda} = \begin{bmatrix} 0.092804 \\ 0.013346 \\ 0.000000 \end{bmatrix}, \quad E = \begin{bmatrix} 0.026 & -0.494 & 0.869 \\ 0.900 & -0.366 & -0.236 \\ 0.435 & 0.788 & 0.435 \end{bmatrix},$$

and the rescaled eigenvectors are

$$\begin{aligned} L^{-1'} E &= \begin{bmatrix} 7.346 & 0.000 & 0.000 \\ 6.548 & 1.253 & 0.000 \\ 0.937 & -0.153 & 0.315 \end{bmatrix}^{-1'} \begin{bmatrix} 0.026 & -0.494 & 0.869 \\ 0.900 & -0.366 & -0.236 \\ 0.435 & 0.788 & 0.435 \end{bmatrix} \\ &= \begin{bmatrix} -0.962494 & -0.397778 & -0.040307 \\ 0.886447 & 0.012790 & -0.019751 \\ 1.380589 & 2.502564 & 1.381752 \end{bmatrix}. \end{aligned}$$

In this application there is a single cointegrating vector, so just the first vector is taken from $L^{-1'} E$ to give

$$\hat{\gamma} = [-0.962494, 0.886447, 1.380589]'$$

These coefficients correspond to $y_{1,t}$, $y_{2,t}$ and a constant respectively. It

is convenient for the interpretation of a single cointegrating vector to re-normalize this vector so that the coefficient corresponding to one of the variables is 1. For example, normalizing on $y_{1,t}$ gives

$$\begin{aligned}\hat{\gamma} &= \left[1, \frac{0.886447}{-0.962494}, \frac{1.380589}{-0.962494} \right]' \\ &= [1, -0.921, -1.434]'. \end{aligned}$$

Thus

$$\hat{\beta}_0 = -1.434, \quad \hat{\beta} = [1, -0.921]',$$

which agree with the estimates obtained using the iterative estimator in Example 18.7. From (18.29) with $p = 1$ and $r = 1$,

$$\begin{aligned}\ln L_T(\hat{\theta}) &= -\frac{N}{2}(\ln 2\pi + 1) - \frac{1}{2} \ln |S_{00}| - \frac{1}{2} \sum_{i=1}^r \ln(1 - \hat{\lambda}_i) \\ &= -\frac{2}{2}(\ln 2\pi + 1) - \frac{1}{2} \ln 0.077066 - \frac{1}{2} \ln(1 - 0.092804) \\ &= -1.5076, \end{aligned}$$

which matches the value of the log-likelihood reported in Example 18.7. $\square$

Example 18.9 Permanent Income Hypothesis

A bivariate VECM consisting of log real consumption per capita, rc_t , and the logarithm real income per capita, ry_t , is specified using Model 3 in Table 18.1 with $r = 1$ long-run equation and $p = 4$ lags. Based on the data in Figure 18.1, the eigenvalues and rescaled eigenvectors are, respectively,

$$\hat{\lambda} = \begin{bmatrix} 0.200582 \\ 0.000010 \end{bmatrix}, \quad L^{-1}E = \begin{bmatrix} -134.915366 & -18.086849 \\ 155.253208 & 29.423442 \end{bmatrix},$$

so that

$$\hat{\gamma} = [-134.915366, 155.253208]'$$

Normalizing $\hat{\gamma}$ in terms of real consumption ($y_{1,t}$) gives

$$\hat{\beta} = \left[\frac{134.915}{134.915}, -\frac{155.253}{134.915} \right]' = [1, -1.151]'$$

The estimates of μ_0 and α from the Johansen estimator are

$$\hat{\mu}_0 = [-0.208390, 0.504582]', \quad \hat{\alpha} = [-0.130, 0.309]'$$

The estimates of β_0 and δ_0 are recovered using (18.17) by first computing

the orthogonal complement matrix $\hat{\alpha}_\perp = \begin{bmatrix} 0.922 & 0.387 \end{bmatrix}'$. The estimates are

$$\begin{aligned}\hat{\beta}_0 &= [\hat{\alpha}'\hat{\alpha}]^{-1} \hat{\alpha}'\hat{\mu} \\ &= \begin{bmatrix} 0.130^2 + 0.309^2 \end{bmatrix}^{-1} [-0.130, 0.309] \begin{bmatrix} -0.208390 \\ 0.504582 \end{bmatrix} = 1.628 \\ \hat{\delta}_0 &= \hat{\alpha}_\perp [\hat{\alpha}'_\perp \hat{\alpha}_\perp]^{-1} \hat{\alpha}'_\perp \hat{\mu} \\ &= \begin{bmatrix} 0.922 \\ 0.387 \end{bmatrix} \begin{bmatrix} 0.922^2 + 0.387^2 \end{bmatrix}^{-1} \begin{bmatrix} 0.922 \\ 0.387 \end{bmatrix}' \begin{bmatrix} -0.208390 \\ 0.504582 \end{bmatrix} = \begin{bmatrix} 0.003 \\ 0.001 \end{bmatrix}.\end{aligned}$$

The estimated long-run permanent income equation is

$$\ln rc_t = -1.628 + 1.151 \ln ry_t + \hat{u}_t.$$

The estimated long-run marginal propensity to consume is 1.151, which is similar to the least squares estimate given by

$$lrc_t = -1.771 + 1.165 lry_t + \hat{u}_t.$$

□

18.4.4 Zero-Rank Case

The final case is where $\Phi(1)$ in (18.8) has zero rank, resulting in the VECM reducing to a VAR in first differences, as in (18.12), with unknown parameters $\theta = \{\mu_0, \mu_1, \Gamma_1, \Gamma_2, \dots, \Gamma_{p-1}, V\}$. As with the full-rank model, the maximum likelihood estimator of (18.12) is the ordinary least squares estimator applied to each equation separately to estimate $\mu_0, \mu_1, \Gamma_1, \Gamma_2, \dots, \Gamma_{p-1}$, with V estimated at the last step using (18.20).

Example 18.10 Bivariate Term Structure Model with Zero Rank

In the case of $r = 0$ rank, the VECM in Example 18.8 reduces to

$$y_{1,t} - y_{1,t-1} = v_{1,t}, \quad y_{2,t} - y_{2,t-1} = v_{2,t}.$$

The residual covariance matrix is computed immediately using $\hat{v}_t = v_t$ as

$$\hat{V} = \frac{1}{194} \sum_{t=2}^{195} \hat{v}_t \hat{v}_t' = \begin{bmatrix} 0.286 & 0.344 \\ 0.344 & 0.684 \end{bmatrix}.$$

From (18.21), the value of the log-likelihood function evaluated at $\hat{\theta}$ is

$$\ln L_T(\hat{\theta}) = -\frac{2}{2}(\ln 2\pi + 1) - \frac{1}{2} \ln(0.077066) = -1.5563.$$

□

18.5 Identification

An important feature of the Johansen estimator is the need to normalize the rescaled matrix of eigenvectors to estimate the cointegrating vectors $\hat{\beta}$. In the bivariate term structure and permanent income examples, the normalization takes the form of designating one of the variables in the system as the dependent variable. When this normalization is adopted, the estimates from the Johansen and iterative estimators match, as is demonstrated in the bivariate term structure example. Two general approaches to normalization are now discussed.

18.5.1 Triangular Restrictions

Triangular restrictions on a model with r long-run relationships, consists of transforming the top $(r \times r)$ block of $\hat{\beta}$ to be the identity matrix (Ahn and Reinsel, 1990; Phillips, 1991a). Formally this is achieved by deriving the row echelon form of $\hat{\beta}$. If $r = 1$ then this corresponds to normalizing one of the coefficients to one as in the previous two examples. If there are $N = 3$ variables and $r = 2$ cointegrating equations, $\hat{\beta}$ is normalized as

$$\hat{\beta} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ \hat{\beta}_{3,1} & \hat{\beta}_{3,2} \end{bmatrix}.$$

This form of the normalized estimated cointegrated vector is appropriate for the trivariate term structure model in Example 18.3. This structure is also appropriate for term structure models of dimension N as the model predicts $r = N - 1$ cointegrating vectors.

Example 18.11 Normalizing a Trivariate Term Structure Model

A trivariate VECM consisting of the 10-year, $y_{1,t}$, 5-year, $y_{2,t}$ and 1-year, $y_{3,t}$, yields is specified using Model 2 with $p = 1$ lags. The estimates of the eigenvalues and the matrix of rescaled eigenvectors using the Johansen reduced-rank estimator are, respectively,

$$\hat{\lambda} = \begin{bmatrix} 0.159 \\ 0.087 \\ 0.015 \\ 0.000 \end{bmatrix} \quad L^{-1}E = \begin{bmatrix} 4.358 & 2.513 & -0.370 & -0.339 \\ -6.306 & -2.152 & -0.013 & 0.441 \\ 2.064 & -0.231 & -0.004 & -0.081 \\ -0.542 & -1.733 & 2.561 & -1.098 \end{bmatrix},$$

where the variables are ordered as $\{y_{1,t}, y_{2,t}, y_{3,t}, 1\}$. Choosing a rank of $r = 2$, inspection of the first two columns of $L^{-1}E$ does not reveal a standard term structure relationship amongst the yields. However, adopting a

triangular normalization gives

$$\hat{\beta} = \begin{bmatrix} 1.000 & 0.000 & -0.912 & -1.510 \\ 0.000 & 1.000 & -0.958 & -0.957 \end{bmatrix}',$$

resulting in the following two long-run equations

$$\begin{aligned} y_{1,t} &= 1.510 + 0.912y_{3,t} + \hat{u}_{1,t} \\ y_{2,t} &= 0.957 + 0.958y_{3,t} + \hat{u}_{2,t}. \end{aligned}$$

These equations now have a term structure interpretation with the 10-year, $y_{1,t}$, and 5-year, $y_{2,t}$, yields expressed as a function of the 1-year yield, $y_{3,t}$, with slope estimates numerically close to unity. $\square$

18.5.2 Structural Restrictions

Along the line of the discussion of identification in Chapter 5, alternative restrictions can be imposed for identification, including exclusion restrictions, cross-equation restrictions, and restrictions on the disturbance covariance matrix (also see Chapter 14). In the context of cointegration and VECMs, this topic is discussed from different perspectives by Johansen (1995a), Boswijk (1995), Hsiao (1997), Davidson (1998) and Pesaran and Shin (2002).

Example 18.12 Open Economy Model

Johansen and Juselius (1992) propose an open economy model in which $y_t = \{s_t, p_t, p_t^*, i_t, i_t^*\}$ represents, respectively, the spot exchange rate, the domestic price, the foreign price, the domestic interest rate and the foreign interest rate, $N = 5$. Assuming $r = 2$ long-run equations, the following restrictions consisting of normalization, exclusion and cross-equation restrictions on γ yield the normalized long-run parameter matrix

$$\beta' = \begin{bmatrix} 1 & -\beta_{2,1} & \beta_{2,1} & 0 & 0 \\ 0 & 0 & 0 & 1 & -\beta_{5,1} \end{bmatrix}.$$

The long-run equations now represent PPP and UIP

$$\begin{aligned} s_t &= \beta_{2,1}(p_t - p_t^*) + u_{1,t} && \text{[Purchasing power parity]} \\ i_t &= \beta_{5,1}i_t^* + u_{2,t} && \text{[Uncovered interest parity]} \end{aligned}$$

Stricter forms of PPP and UIP are given by also imposing $\beta_{2,1} = \beta_{5,1} = 1$. $\square$

Alternative identification strategies are investigated in Exercise 11. One implication of this analysis is that the Johansen estimator is inappropriate when imposing over-identifying restrictions on the cointegrating vector. In

these cases, the iterative estimator needs to be used to obtain maximum likelihood estimators.

18.6 Distribution Theory

This section provides a summary of the key results regarding the asymptotic properties of the estimated VECM parameters. Understandably, the asymptotics are a function of the type of identifying restrictions discussed in Section 18.5 that are imposed on the model. For full derivations of the results, see Johansen (1995b) for diagonal restrictions, Ahn and Reinsel (1990) and Phillips (1991a) for systems based on triangular identification and Pesaran and Shin (2002) for systems with general nonlinear identifying restrictions.

18.6.1 Asymptotic Distribution of the Eigenvalues

An important feature of the Johansen maximum likelihood estimator in Section 18.4 is that the unrestricted log-likelihood function, for a system of N equations in which the disturbances are assumed to normally distributed, evaluated at $\hat{\theta}$ is

$$\ln L_T(\hat{\theta}) = -\frac{N}{2}(1 + \ln 2\pi) - \frac{1}{2} \ln |S_{00}| - \frac{1}{2} \sum_{i=1}^N \ln(1 - \hat{\lambda}_i), \quad (18.30)$$

where $\hat{\lambda}_1 > \hat{\lambda}_2 > \dots, \hat{\lambda}_N$ are the estimated eigenvalues. For a model with reduced rank r , the first r estimated eigenvalues converge in probability to constants, with the remaining $N - r$ estimated eigenvalues converging to zero. These properties may be summarized as follows.

Let the notation $\lambda_j(A)$ denote the j^{th} largest eigenvalue of A . Under the conditions of the Granger representation theorem, the estimated eigenvalues $\hat{\lambda}_1 > \dots > \hat{\lambda}_N$ satisfy

$$\begin{aligned} \text{Largest eigenvalues} & : \quad \hat{\lambda}_j \xrightarrow{p} \lambda_j(\Omega_{uu}^{-1}\Omega_{u0}\Omega_{00}^{-1}\Omega_{0u}), \quad j = 1, 2, \dots, r \\ \text{Smallest eigenvalues} & : \quad T(\hat{\lambda}_j - 0) \xrightarrow{d} \lambda_{j-r}(M), \quad j = r + 1, \dots, N, \end{aligned}$$

where

$$\Omega_{00} = E(\Delta y_t \Delta y_t'), \quad \Omega_{u0} = E(u_{t-1} \Delta y_t'), \quad \Omega_{uu} = E(u_{t-1} u_{t-1}'),$$

and $u_t = \beta' y_t$ is the error correction term. The stochastic matrix, M , is given by

$$M = \int_0^1 dB_{N-r}(s) F(s)' \left[\int_0^1 F(s) F(s)' ds \right]^{-1} \int_0^1 F(s) dB_{N-r}(s)', \quad (18.31)$$

where $B_{N-r}(s)$ is an $N - r$ dimensional Brownian motion with increments $dB_{N-r}(s)$ and $F(s)$ is a function of $B(s)$ and t , which depends upon the model type in Table 18.1. The derivation is given in Theorem 11.1 of Johansen (1995b). In the univariate case ($N = 1$), the distribution of M is the same as that of the asymptotic distribution of the square of the DF_t statistic given in Chapter 17.

The r largest eigenvalues are $O_p(1)$ thereby converging to a constant as is required for a positive rank r . The $N - r$ smallest eigenvalues are $O_p(T^{-1})$. As this rate of convergence is T compared to the usual $\sqrt{T}$ rate, $\hat{\lambda}_{r+1} > \cdots > \hat{\lambda}_N$ are super-consistent estimators of $\lambda_{r+1} = \cdots = \lambda_N = 0$. This contrasting order of the eigenvalues provides the basis for the asymptotic properties of the various methods for selecting r . The properties of the asymptotic distribution of $\hat{\lambda}$ are highlighted in the following example.

Example 18.13 Asymptotic Properties of Estimated Eigenvalues

The data generating process is a triangular bivariate VECM based on Model 1 in Table 18.1 with rank $r = 1$, given by

$$y_{1,t} = y_{2,t} + u_t, \quad y_{2,t} = y_{2,t-1} + v_t, \quad \begin{bmatrix} u_t \\ v_t \end{bmatrix} \sim iid N(0, I_2).$$

The model is simulated 10000 times with samples of size $T = \{100, 200, 400, 800\}$. As

$$\Delta y_{1,t} = \Delta y_{2,t} + \Delta u_t = v_t + \Delta u_t,$$

the population quantities are

$$\begin{aligned} \Omega_{00} &= E \begin{bmatrix} \Delta y_{1,t}^2 & \Delta y_{1,t} \Delta y_{2,t} \\ \Delta y_{1,t} \Delta y_{2,t} & \Delta y_{2,t}^2 \end{bmatrix} \\ &= E \begin{bmatrix} (\Delta u_t + v_t)^2 & (\Delta u_t + v_t) v_t \\ (\Delta u_t + v_t) v_t & v_t^2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix} \\ \Omega_{0u} &= E \begin{bmatrix} \Delta y_{1,t} u_{t-1} \\ \Delta y_{2,t} u_{t-1} \end{bmatrix} = E \begin{bmatrix} (\Delta u_t + v_t) u_t \\ v_t u_t \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \\ \Omega_{uu} &= E [u_{t-1}^2] = E [v_{1,t-1}^2] = 1, \end{aligned}$$

in which case the population eigenvalue is simply

$$\Omega_{uu}^{-1} \Omega_{u0} \Omega_{00}^{-1} \Omega_{0u} = 1^{-1} \begin{bmatrix} 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ -1 \end{bmatrix} = 0.5.$$

For each simulation, the eigenvalues are computed using $L^{-1} S_{10} S_{00}^{-1} S_{01} L^{-1'}$, where $S_{i,j}$ are the sums of squares matrices in (18.25) with $R_{0,t} = \Delta y_t$ and $R_{1,t} = y_{t-1}$ and $LL' = S_{1,1}$. Summary descriptive statistics of the

sampling distribution of $\hat{\lambda}_1$ and $\hat{\lambda}_2$ are given in Table 18.3. The results demonstrate that as T increases $\hat{\lambda}_1$ converges to a constant equal to the population eigenvalue of $\lambda_1 = 0.5$, while $\hat{\lambda}_2$ approaches $\lambda_2 = 0$. The rate of convergence of $\hat{\lambda}_2$ to λ_2 is the super-consistent rate of T because the sample standard deviation approximately halves as T doubles. $\square$

Table 18.3

Summary statistics of the sampling distribution of $\hat{\lambda}_1$ and $\hat{\lambda}_2$ from a bivariate VECM with rank $r = 1$. Based on 10000 replications.

Statistic	$T =$	$\hat{\lambda}_1$				$\hat{\lambda}_2$			
		100	200	300	400	100	200	300	400
Mean		0.510	0.505	0.502	0.501	0.050	0.035	0.025	0.018
Std. Dev.		0.011	0.006	0.003	0.001	0.015	0.007	0.004	0.002

18.6.2 Asymptotic Distribution of the Parameters

The asymptotic distribution of the maximum likelihood estimator $\hat{\theta}$ is derived using the approach of Chapter 2 by expanding the gradient $G_T(\hat{\theta})$ in (18.19) around the true population parameter θ_0 . Upon rearranging,

$$\hat{\theta} - \theta_0 = -H_T^{-1}(\theta_0)G_T(\theta_0) + o_p(\hat{\theta} - \theta_0), \quad (18.32)$$

where $H_T(\theta_0)$ is the Hessian and the remainder term contains the higher-order derivatives of the Taylor series, which is small provided that $\hat{\theta}$ is a consistent estimator of θ_0 . In Chapters 2 and 13, the scale factor used to derive the asymptotic distribution is $T^{1/2}$ as the processes there are assumed to be stationary. This is not the case for nonstationary processes as demonstrated in Chapter 16, where the scale factor in general not only has to vary across parameters but also has to be of an order higher than $T^{1/2}$ when sums of nonstationary variables are involved.

To highlight the choice of scale parameters needed to derive the asymptotic distribution of $\hat{\theta}$, reconsider the bivariate VECM in (18.22), which involves the error correction parameters $\alpha = [\alpha_1, \alpha_2]'$ and the cointegrating parameter β . This model can be viewed as a concentrated log-likelihood function where the remaining parameters of the VECM are concentrated out. As α is associated with the error correction term u_{t-1} , which is stationary, and β is associated with the variable $y_{2,t-1}$, which is nonstationary, a suitable

choice of the scale factors corresponding to the elements of $\theta = \{\alpha, \beta\}$ is

$$D_T = \begin{bmatrix} \sqrt{T} & 0 \\ 0 & T \end{bmatrix}. \quad (18.33)$$

Applying this expression to (18.32) gives

$$D_T \begin{bmatrix} \hat{\alpha} - \alpha_0 \\ \hat{\beta} - \beta_0 \end{bmatrix} = [(T-1)D_T^{-1}H_T(\theta_0)D_T^{-1}]^{-1} (T-1)D_T^{-1}G_T(\theta_0) + o_p(1). \quad (18.34)$$

When the expressions of the gradient and the Hessian in (18.23) are used, then

$$(T-1)D_T^{-1}G_T(\theta_0) = \sum_{t=2}^T \begin{bmatrix} \frac{1}{\sqrt{T}}V_0^{-1}v_t u_{t-1} \\ -\frac{1}{T}\alpha'_0 V_0^{-1}v_t y_{2,t-1} \end{bmatrix}, \quad (18.35)$$

and $(T-1)D_T^{-1}H_T(\theta_0)D_T^{-1}$ is given by the expression

$$\sum_{t=2}^T \begin{bmatrix} -\frac{1}{T}V_0^{-1}u_{t-1}^2 & -\frac{1}{T^{3/2}}V_0^{-1}(v_t - \alpha_0 u_{t-1})y_{2,t-1} \\ -\frac{1}{T^{3/2}}V_0^{-1}(v_t - \alpha_0 u_{t-1})'y_{2,t-1} & -\frac{1}{T^2}\alpha'_0 V_0^{-1}\alpha_0 y_{2,t-1}^2 \end{bmatrix}, \quad (18.36)$$

where V_0 is the true value of the disturbance covariance matrix because $G_T(\theta)$ and $H_T(\theta)$ are evaluated at the population parameter.

From Chapter 16, the diagonal terms of the Hessian in (18.36) are

$$\frac{1}{T} \sum_{t=2}^T u_{t-1}^2 = O_p(1), \quad \frac{1}{T^2} \sum_{t=2}^T y_{2,t-1}^2 = O_p(1),$$

where the first term converges to a fixed limit and the second term converges to a random variable that is a functional of Brownian motion. In the case of the off-diagonal term in (18.36), from the stochastic integral theorem of Chapter 16

$$\frac{1}{T} \sum_{t=2}^T (v_t - \alpha_0 u_{t-1})y_{2,t-1} = O_p(1), \quad (18.37)$$

which also converges to a functional of Brownian motion. But, as the scale factor in the off-diagonal term in (18.36) is $T^{-3/2}$, then multiplying both sides of (18.37) by $T^{-1/2}$ shows that

$$\frac{1}{T^{3/2}} \sum_{t=2}^T \Delta(v_t - \alpha_0 u_{t-1})y_{2,t-1} = O_p(T^{-1/2}), \quad (18.38)$$

which disappears in the limit. The implication of (18.38) is that the scaled Hessian in (18.36) becomes block diagonal asymptotically, allowing the expressions in (18.34) to separate as

$$\sqrt{T}(\hat{\alpha} - \alpha_0) = \left[\frac{1}{T} \sum_{t=2}^T u_{t-1}^2 \right]^{-1} \left[\frac{1}{T^{1/2}} \sum_{t=2}^T v_t u_{t-1} \right] + o_p(1), \quad (18.39)$$

and

$$T(\hat{\beta} - \beta_0) = \left[\frac{1}{T^2} \sum_{t=2}^T y_{2,t-1}^2 \right]^{-1} \left[\frac{1}{T} \sum_{t=2}^T z_t y_{2,t-1} \right] + o_p(1), \quad (18.40)$$

respectively, where

$$z_t = -[\alpha_0' V_0^{-1} \alpha_0]^{-1} \alpha_0' V_0^{-1} v_t. \quad (18.41)$$

Asymptotic Theory for $\hat{\alpha}$

The asymptotic theory for $\hat{\alpha}$ is standard being based on the results of Chapter 2. When the WLLN is used for the first term on the right hand side of (18.39) and the martingale difference central limit theorem is used for the second term

$$\text{plim} \frac{1}{T} \sum_{t=2}^T u_{t-1}^2 = \sigma_u^2, \quad \frac{1}{\sqrt{T}} \sum_{t=2}^T v_t u_{t-1} \xrightarrow{d} N(0, \sigma_u^2 V_0),$$

it immediately follows that

$$\sqrt{T}(\hat{\alpha} - \alpha_0) \xrightarrow{d} N(0, \sigma_u^{-2} V_0). \quad (18.42)$$

This result shows that the error correction parameter estimator $\hat{\alpha}$ is consistent at the standard rate of $T^{1/2}$ and is asymptotically normally distributed. The asymptotic properties of $\hat{\alpha}$ are the same as if β is known, which is a result of the super-consistency of $\hat{\beta}$, see (18.45) below. This means that α_0 is estimated by regressing Δy_t on u_{t-1} and that the distribution theory for $\hat{\alpha}$ is standard.

Asymptotic Theory for $\hat{\beta}$

The asymptotic theory for $\hat{\beta}$ requires the functional central limit theorem of Chapter 16. The basic FCLT using the *iid* disturbances of the VECM is

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{[Ts]} v_t \xrightarrow{d} V(s),$$

where $V(s) = \text{BM}(V_0)$, which is used to derive the asymptotic theory for

the terms involving $y_{2,t}$ and z_t in (18.40). From the Granger representation theorem (see Theorem 4.2 of Johansen (1995b) omitting deterministic terms and initial values) y_t in (18.22) is rewritten as

$$y_t = \beta_{\perp}(\alpha'_{\perp}\beta_{\perp})^{-1}\alpha'_{\perp} \sum_{j=1}^t v_j + \eta_t,$$

where $\eta_t \sim I(0)$, which gives (using $e_2 = \begin{bmatrix} 0 & 1 \end{bmatrix}$)

$$\begin{aligned} \frac{1}{\sqrt{T}} y_{2,[Ts]-1} &= e'_2 \beta_{\perp}(\alpha'_{\perp}\beta_{\perp})^{-1}\alpha'_{\perp} \frac{1}{\sqrt{T}} \sum_{t=1}^{[Ts]-1} v_t + O_p(T^{-1/2}) \\ &\xrightarrow{d} e'_2 \beta_{\perp}(\alpha'_{\perp}\beta_{\perp})^{-1}\alpha'_{\perp} V(s) = Y_2(s). \end{aligned} \quad (18.43)$$

In addition, from (18.41)

$$\begin{aligned} \frac{1}{\sqrt{T}} \sum_{t=1}^{[Ts]} z_t &= -(\alpha'_0 V_0^{-1} \alpha_0)^{-1} \alpha'_0 V_0^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^{[Ts]} v_t \\ &\xrightarrow{d} -(\alpha'_0 V_0^{-1} \alpha_0)^{-1} \alpha'_0 V_0^{-1} V(s) = Z(s). \end{aligned} \quad (18.44)$$

These two Brownian motions are independent because

$$\begin{aligned} E[Y_2(s)Z(s)] &= -(\alpha'_0 V_0^{-1} \alpha_0)^{-1} \alpha'_0 V_0^{-1} E[V(s)V(s)'] \alpha_{\perp}(\beta'_{\perp} \alpha_{\perp})^{-1} \beta'_{\perp} e_2 \\ &= -s(\alpha'_0 V_0^{-1} \alpha_0)^{-1} \alpha'_0 V_0^{-1} \Omega_0 \alpha_{\perp}(\beta'_{\perp} \alpha_{\perp})^{-1} \beta'_{\perp} e_2 \\ &= -s(\alpha'_0 V_0^{-1} \alpha_0)^{-1} \alpha'_0 \alpha_{\perp}(\beta'_{\perp} \alpha_{\perp})^{-1} \beta'_{\perp} e_2 \\ &= 0, \end{aligned}$$

using $E(V(s)V(s)') = sV_0$ and $\alpha'_0 \alpha_{\perp} = 0$. Combining (18.43) and (18.44) in (18.40) gives

$$T(\hat{\beta} - \beta_0) \xrightarrow{d} \left[\int_0^1 Y_2(s)^2 ds \right]^{-1} \int_0^1 Y_2(s) dZ(s). \quad (18.45)$$

Since Y_2 and Z are independent, it follows from Chapter 16 that the second term in (18.45) is mixed normal, in which case $T(\hat{\beta} - \beta_0)$ is asymptotically mixed normal. In contrast to the error correction parameter estimator, $\hat{\alpha}$, the estimator of the cointegrating parameter $\hat{\beta}$ is superconsistent, converging at the rate T , whilst the asymptotic distribution of $T(\hat{\beta} - \beta_0)$ has fatter tails than a normal distribution. Nevertheless, the results of Phillips (1991a) and Saikkonen (1991) show that $\hat{\beta}$ is asymptotically efficient.

Remaining Parameters

The asymptotic distribution of the remaining parameters, including the

lagged dependent variables in the VECM, Γ_i , the intercepts, μ , and the disturbance covariance matrix, V , all have standard asymptotic distributions following the line of reasoning given for (18.42). In particular, when it is assumed that β is known, the VECM is characterized by all of its variables being stationary: (i) since y_t is $I(1)$ the dependent variable Δy_t is $I(0)$ as are the lags of Δy_t ; (ii) since there is cointegration, the error correction term, u_{t-1} , is also $I(0)$.

18.7 Testing

This section uses the distribution theory results of the previous section to undertake a range of tests on the parameters of the VECM.

18.7.1 Cointegrating Rank

In estimating models of the term structure of interest rates, three models have been estimated without making any judgement as to whether the model has full ($r = N$), reduced ($0 < r < N$) or zero rank ($r = 0$). A test of these alternative models amounts to testing the cointegrating rank r , with the aim of identifying the true cointegrating rank r_0 by determining the number of estimated eigenvalues $\hat{\lambda} = \{\hat{\lambda}_1, \hat{\lambda}_2, \dots, \hat{\lambda}_N\}$ that are significantly greater than zero. The null and alternative hypotheses are

$$\begin{aligned} H_0 : r &= r_0 && \text{[Reduced Rank]} \\ H_1 : r &= N && \text{[Full Rank]} \end{aligned}$$

Since the three term structure models are nested within each other, with the full rank model being the most unrestricted and the zero rank model being the most restricted, the LR test discussed in Chapter 4 can be used to choose among the models and hence r . The LR statistic is

$$LR = -2((T - p) \ln L_T(\hat{\theta}_0) - (T - p) \ln L_T(\hat{\theta}_1)), \quad (18.46)$$

where $\hat{\theta}_0$ and $\hat{\theta}_1$ are, respectively, the restricted and unrestricted parameter estimates corresponding to models with different ranks. Another way of constructing this statistic is to use the eigen decomposition form of the log-likelihood function in (18.30). The full-rank ($r = N$) and reduced-rank

($r < N$) log-likelihood functions are respectively

$$\begin{aligned}\ln L_T(\hat{\theta}_1) &= -\frac{N}{2}(1 + \ln 2\pi) - \frac{1}{2} \ln |S_{00}| - \frac{1}{2} \sum_{i=1}^N \ln(1 - \hat{\lambda}_i) \\ \ln L_T(\hat{\theta}_0) &= -\frac{N}{2}(1 + \ln 2\pi) - \frac{1}{2} \ln |S_{00}| - \frac{1}{2} \sum_{i=1}^r \ln(1 - \hat{\lambda}_i).\end{aligned}$$

Using these expressions in (18.46) gives an alternative but numerically equivalent form for the LR statistic, known as the trace statistic,

$$LR = -(T - p) \sum_{i=r+1}^N \ln(1 - \hat{\lambda}_i). \quad (18.47)$$

Table 18.4

Quantiles of $\text{tr}M$ based on a sample of size $T = 1000$ and 100000 replications.

		Number of common trends ($N - r$)					
		1	2	3	4	5	6
Model 1	0.90	2.983	10.460	21.677	36.877	56.041	78.991
	0.95	4.173	12.285	24.102	39.921	59.829	83.428
	0.99	6.967	16.380	29.406	46.267	67.174	92.221
Model 2	0.90	7.540	17.869	32.058	50.206	72.448	98.338
	0.95	9.142	20.205	34.938	53.734	76.559	103.022
	0.99	12.733	25.256	41.023	60.943	84.780	112.655
Model 3	0.90	2.691	13.347	26.948	44.181	65.419	90.412
	0.95	3.822	15.430	29.616	47.502	69.293	95.105
	0.99	6.695	19.810	35.130	54.307	77.291	103.980
Model 4	0.90	10.624	23.224	39.482	59.532	83.681	111.651
	0.95	12.501	25.726	42.585	63.336	88.089	116.781
	0.99	16.500	30.855	49.047	70.842	96.726	126.510
Model 5	0.90	2.706	16.090	31.874	51.136	74.462	101.484
	0.95	3.839	18.293	34.788	54.680	78.588	106.265
	0.99	6.648	22.978	40.776	61.744	86.952	115.570

The quantiles of LR are given in Table 18.4 and vary for each model type in Table 18.1. The asymptotic distribution of LR in (18.47) is nonstandard being based on the trace of the stochastic matrix M

$$LR \xrightarrow{d} \text{tr}M, \quad (18.48)$$

where M is defined in expression (18.31). This result contrasts with the standard asymptotic distribution given by the chi-square distribution.

Example 18.14 Bivariate Term Structure Model

To test the hypothesis $H_0 : r = 0$ against $H_1 : r = 2$ in a model of the term structure given by Model 2 in Table 18.1, the LR statistic in (18.46) using the log-likelihood function values from Examples 18.10 and 18.6, is

$$LR = -2 \times (195 - 1) \times (-1.5563 + 1.5009) = 21.502.$$

Alternatively, using the eigenvalues in Example 18.8 produces the same value of LR as using (18.47) since

$$\begin{aligned} LR &= -(T - p) \sum_{i=1}^2 \ln(1 - \hat{\lambda}_i) \\ &= -(195 - 1)(\ln(1 - 0.092804) + \ln(1 - 0.013346)) = 21.502. \end{aligned}$$

From Table 18.4, under the null hypothesis the number of common trends is $N - r = 2 - 0 = 2$, which corresponds to a critical value of 20.205 for a test with size 5%. As $21.502 > 20.205$, the null of zero rank is rejected at the 5% level. To test the hypothesis $H_0 : r = 1$ against $H_1 : r = N$, the value of the LR statistic, based on the log-likelihood function values from Examples 18.7 and 18.6, is

$$LR = -2 \times (195 - 1) \times (-1.5076 + 1.5009) = 2.607,$$

which also agrees with the LR statistic based on (18.47) as

$$LR = -(T - p) \sum_{i=2}^2 \ln(1 - \hat{\lambda}_i) = -(195 - 1) \ln(1 - 0.013346) = 2.607.$$

The number of common trends under the null hypothesis is now $N - 1 = 2 - 1 = 1$, which corresponds to a critical value of 9.142 for a test with size 5%. As $2.607 > 9.142$, the null cannot be rejected at the 5% level, leading to the conclusion that there is only one cointegrating vector. This result is consistent with the bivariate term structure of interest rates model. $\square$

Example 18.15 Trivariate Term Structure Model

The results of the cointegration test for a trivariate VECM consisting of the 10-year, $y_{1,t}$, 5-year, $y_{2,t}$, and 1-year, $y_{3,t}$, yields estimated in Example 18.11, are given in Table 18.5.

The critical values are based on Table 18.4 for Model 2. The first null hypothesis is rejected at the 5% level ($54.120 > 34.938$) as is the second null hypothesis ($20.557 > 20.205$). In contrast, the third null is not rejected

Table 18.5

Results of the cointegration test for a trivariate VECM consisting of quarterly observations on the 10-year, $y_{1,t}$, 5-year, $y_{2,t}$, and 1-year, $y_{3,t}$, yields on U.S. bonds for the period March 1962 to September 2010.

H_0	H_1	$\hat{\lambda}$	LR	$CV(5\%)$
$r = 0$	$r = 3$	0.159	54.120	34.938
$r = 1$	$r = 3$	0.087	20.557	20.205
$r = 2$	$r = 3$	0.015	2.900	9.142

($2.900 < 9.142$), thereby providing evidence of $r = 2$ cointegrating equations at the 5% level. $\square$

As $\hat{\lambda}$ is a random variable, the estimated cointegrating rank, $\hat{r}$, is also a random variable, albeit a discrete random variable. A property of the trace test (see Johansen, 1995b, Theorem 12.3) with asymptotic size of δ at each stage as $T \rightarrow \infty$, is

$$\Pr(\hat{r} = r) \rightarrow \begin{cases} 0, & r < r_0 \\ 1 - \delta, & r = r_0 \\ \leq \delta, & r > r_0. \end{cases} \quad (18.49)$$

This result shows that if the test is computed using an asymptotic size of $\delta = 0.05$, the probability of selecting the true cointegrating rank of r_0 converges to $1 - \delta = 0.95$. Even though this result suggests that $\hat{r}$ is not consistent as $\Pr(\hat{r} = r_0) \not\rightarrow 1$, the degree of inconsistency of $\hat{r}$ arises only from a small probability (δ) of overspecifying ($r > r_0$) the cointegrating rank. The finite sample properties of this test are investigated and compared with another cointegration test based on the information criteria in Section 18.9.

18.7.2 Cointegrating Vector

Hypothesis tests on the cointegrating vector using the procedures of Chapter 4, constitute tests of long-run economic theories. In the term structure example, a test that the cointegrating vector is $(1, -1)$ is a test that yields move one to one in the long run, so the spreads between yields are stationary. In contrast to the cointegration test discussed immediately above, where the distribution is nonstandard, the Wald, LR and LM statistics are distributed asymptotically as χ^2 under the null hypothesis that the restrictions are valid. The Wald test has the advantage that only the unrestricted

model needs to be estimated. If there are M linear restrictions of the form $R\theta_1 - Q$, where R is $(M \times K)$ and Q is $(M \times 1)$, the Wald statistic is

$$W = (T - p)[R\hat{\theta}_1 - Q]'[R\hat{\Omega}R']^{-1}[R\hat{\theta}_1 - Q],$$

where $\hat{\Omega}/(T - p)$ is the estimated covariance matrix of $\hat{\theta}_1$. Alternatively, a LR test can be adopted that involves estimating both the restricted and unrestricted models. As with the cointegration test, the LR statistic can be computed using the log-likelihood values evaluated at the null and alternative hypotheses or expressed in terms of its estimated eigenvalues.

Example 18.16 Testing for the Spread in the Term Structure

From Example 18.7, the null and alternative hypotheses are

$$H_0 : \beta_s = 1, \quad H_1 : \beta_s \neq 1,$$

so the VECM under the null becomes

$$\begin{aligned} y_{1,t} - y_{1,t-1} &= \alpha_1(y_{2,t-1} - \beta_c - y_{1,t-1}) + v_{1,t} \\ y_{2,t} - y_{2,t-1} &= \alpha_2(y_{2,t-1} - \beta_c - y_{1,t-1}) + v_{2,t}. \end{aligned}$$

Given that $\theta = \{\beta_c, \beta_s, \alpha_1, \alpha_2\}$, define $R = [0 \ 1 \ 0 \ 0]$ and $Q = [1]$, with $\hat{\theta}_1$ and relevant covariance matrix given in Example 18.7, the Wald statistic is

$$W = (T - 1)[R\hat{\theta}_1 - Q]'[R(-H_T^{-1}(\hat{\theta}_1)R')^{-1}[R\hat{\theta}_1 - Q] = \frac{(0.921 - 1.000)^2}{0.008104} = 0.766,$$

which is distributed under the null hypothesis as chi-square with one degree of freedom. The p-value is 0.382 resulting in a failure to reject the null hypothesis that $\beta_s = 1$ at the 5% level. $\square$

The practical usefulness of the asymptotic mixed normality of $\hat{\beta}$ can be demonstrated by considering a t -test of $H_0 : \beta = \beta_0$ with statistic

$$t = \frac{\hat{\beta} - \beta_0}{\text{se}(\hat{\beta})}.$$

The standard error can be found from the inverse of the Hessian

$$\text{se}(\hat{\beta}) = \frac{1}{T} [H_{\beta\beta}(\hat{\theta}) - H_{\beta\alpha}(\hat{\theta})H_{\alpha\alpha}^{-1}(\hat{\theta})H_{\alpha\beta}(\hat{\theta})]^{-1/2},$$

where the Hessian is partitioned as

$$H_T(\theta) = \begin{bmatrix} H_{\alpha\alpha}(\theta) & H_{\alpha\beta}(\theta) \\ H_{\beta\alpha}(\theta) & H_{\beta\beta}(\theta) \end{bmatrix}.$$

Following the same steps used to deal with (18.36) shows that

$$\begin{aligned} T\text{se}(\hat{\beta}) &= [T^{-2}H_{\beta\beta}(\hat{\theta}) - T^{-3/2}H_{\beta\alpha}(\hat{\theta}) \cdot [T^{-1}H_{\alpha\alpha}(\hat{\theta})]^{-1} \cdot T^{-3/2}H_{\alpha\beta}(\hat{\theta})]^{-1/2} \\ &= [\hat{\alpha}'\hat{V}^{-1}\hat{\alpha} \cdot T^{-2} \sum_{t=2}^T y_{2,t-1}^2]^{-1/2} + o_p(1) \\ &= [\alpha_0'V_0^{-1}\alpha_0 \cdot T^{-2} \sum_{t=2}^T y_{2,t-1}^2]^{-1/2} + o_p(1). \end{aligned}$$

The second line shows that $H_{\beta\beta}(\hat{\theta})^{-1/2}$ can be used as an alternative formula for $\text{se}(\hat{\beta})$. Combining this expression with (18.40) gives

$$\begin{aligned} t &= \frac{T(\hat{\beta} - \beta_0)}{T\text{se}(\hat{\beta})} \\ &= [\alpha_0'V_0^{-1}\alpha_0]^{1/2} \left[\frac{1}{T^2} \sum_{t=2}^T y_{2,t-1}^2 \right]^{-1/2} \frac{1}{T} \sum_{t=2}^T z_t y_{2,t-1} + o_p(1) \\ &\xrightarrow{d} \left[\int_0^1 Y_2(s)^2 ds \right]^{-1/2} \int_0^1 Y_2(s) dZ^*(s), \end{aligned}$$

where $Z^* = (\alpha_0'V_0^{-1}\alpha_0)^{1/2}Z$. Note that $\text{var}(Z) = (\alpha_0'V_0^{-1}\alpha_0)^{-1}$ implies that $\text{var}(Z^*) = 1$. Thus, following Example 16.11 of Chapter 16, the conditional distribution of

$$\left[\int_0^1 Y_2(s)^2 ds \right]^{-1/2} \int_0^1 Y_2(s) dZ^*(s),$$

is

$$\left[\int_0^1 Y_2(s)^2 ds \right]^{-1/2} \int_0^1 Y_2(s) dZ^*(s) \Big| Y_2 \sim N(0, 1).$$

As this conditional distribution does not depend on Y_2 , it is also the unconditional distribution. In this case, the asymptotic of the t-statistic under the null hypothesis is

$$t \stackrel{a}{\sim} N(0, 1).$$

The same reasoning can be generalized to show that the usual tests (LR, Wald and LM) for β have asymptotic χ^2 distributions, even though $\hat{\beta}$ (and its gradient) are asymptotically mixed normal rather than asymptotically normal.

18.7.3 Exogeneity

An important feature of a VECM is that all of the variables in the system are endogenous. When the system is out of long-run equilibrium, all of the variables interact with each other to move the system back into equilibrium, as is demonstrated in Figure 18.4. In a VECM, this interaction formally occurs through the impact of lagged variables so that variable $y_{i,t}$ is affected by the lags of the other variables either through the error correction term, u_{t-1} , with parameter, α_i , or through the lags of $\Delta y_{j,t}$, $j \neq i$, with parameters given by the rows of $\Gamma_1, \Gamma_2, \dots, \Gamma_{p-1}$.

If the first channel does not exist so that the error correction parameter satisfies the restriction $\alpha_i = 0$, the variable $y_{i,t}$ is weakly exogenous. If both channels do not exist, $y_{i,t}$ is strongly exogenous as only lagged values of $y_{i,t}$ are important in explaining movements in itself. This definition of strong exogeneity is also equivalent to the other variables failing to Granger cause $y_{i,t}$, as discussed in Chapter 13.

Tests of weak and strong exogeneity are conveniently based on the Wald test if the unrestricted model is estimated. From Section 18.6, this statistic has a standard distribution asymptotically. If there is a single cointegrating equation ($r = 1$), weak exogeneity tests reduce to a t -test since only one error correction parameter associated with each variable in the system exists. An alternative approach based on the LR test is explored in Exercise 12.

Example 18.17 Exogeneity Tests of the Term Structure Model

A t -test of the restriction the error correction parameter in the 10-year yield VECM equation in Example 18.7 is zero is $(-0.092 - 0.000)/0.042 = -2.161$. The p -value is 0.031 showing that the 10-year yield is not weakly exogenous. In contrast, performing the same test on the error correction parameter in the 1-year yield VECM equation produces a t -statistic of $(0.012 - 0.000)/0.060 = 0.198$, which is now statistically insignificant with a p -value of 0.843, providing evidence that this yield is weakly exogenous. As $p = 1$ in this example, there are no lags of Δy_t in the VECM, in which case the 1-year yield is also strongly exogenous. $\square$

An implication of weak exogeneity is that, as the cointegrating vector β does not appear in the equations corresponding to the weakly exogenous variables, inference on β can be performed without loss of generality using a partial model of dimension r , which excludes the weakly exogenous variables. For example, consider the bivariate VECM

$$\begin{bmatrix} \Delta y_{1,t} \\ \Delta y_{2,t} \end{bmatrix} = \begin{bmatrix} \alpha_{1,1} \\ \alpha_{2,1} \end{bmatrix} \begin{bmatrix} 1 \\ \beta_{2,1} \end{bmatrix}' \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} \Gamma_{1,1} & \Gamma_{1,2} \\ \Gamma_{2,1} & \Gamma_{2,2} \end{bmatrix} \begin{bmatrix} \Delta y_{1,t-1} \\ \Delta y_{2,t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix},$$

where

$$\begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right),$$

and $\alpha_{2,1} = 0$ so $y_{2,t}$ is weakly exogenous. Now define the regression equation

$$v_{1,t} = \rho v_{2,t} + w_t,$$

where the disturbance term w_t has the property $E[v_{2,t}w_t] = 0$, and use the VECM to substitute out $v_{1,t}$ and $v_{2,t}$ to derive the partial model for $y_{1,t}$ as

$$\begin{aligned} \Delta y_{1,t} &= \alpha_{1,1}(y_{1,t-1} - \beta_{2,1}y_{2,t-1}) + \rho\Delta y_{2,t} + (\Gamma_{1,1} - \rho\Gamma_{1,2})\Delta y_{1,t-1} \\ &\quad + (\Gamma_{2,1} - \rho\Gamma_{2,2})\Delta y_{2,t-1} + w_t. \end{aligned}$$

This equation can be used to estimate $\beta_{2,1}$ without the need for the equation of $y_{2,t}$ by simply regressing $\Delta y_{1,t}$ on $\{c, y_{1,t-1}, y_{2,t-1}, \Delta y_{2,t}, \Delta y_{1,t-1}, \Delta y_{2,t-1}\}$. This estimator is also equivalent to the maximum likelihood estimator of $\beta_{2,1}$ from estimating the bivariate VECM subject to the weak exogeneity restriction $\alpha_{2,1} = 0$. This result immediately generalizes to the case where $y_{1,t}$ and $y_{2,t}$ consist of r and $N - r$ variables respectively, with the partial VECM of $y_{1,t}$ now representing a system of r equations.

18.8 Dynamics

18.8.1 Impulse responses

The dynamics of a VECM can be investigated using impulse response functions as discussed in Chapter 13. As the impulse response functions of Chapter 13 are derived within the context of a VAR, the approach is to re-express a VECM as a VAR by using the property that a VECM is a VAR subject to cross-equation restrictions. For example, the VECM

$$\Delta y_t = \mu + \alpha\beta'y_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j} + v_t, \quad (18.50)$$

is a VAR in levels

$$y_t = \mu + \sum_{j=1}^p \Phi_j y_{t-j} + v_t, \quad (18.51)$$

subject to the restrictions $\Phi_1 = \alpha\beta' + \Gamma_1 - I_N$, and $\Phi_j = \Gamma_j - \Gamma_{j-1}$, $j = 2, 3, \dots, p$.

The impulses are computed as

$$IR_i = \Psi_i S, \quad (18.52)$$

where Ψ_i are obtained from the VMA and S is obtained as $V = SS'$, the Choleski decomposition of V , if a triangular ordering is adopted (see Chapter 14). Shocks are characterized as having restricted contemporaneous channels determined by the triangular ordering, but unrestricted channels thereafter. For an alternative way of identifying structural shocks in VECMs, see Pagan and Pesaran (2008).

Phillips (1998) shows that restricted and unrestricted estimation of (18.51) gives consistent estimators of the short run impulse responses, but only the VECM gives consistent estimators of the long-run impulse responses.

18.8.2 Cointegrating Vector Interpretation

Interpretation of the cointegrating vector itself can be clarified using impulse responses. Suppose there is a cointegrating equation of the form

$$y_{1,t} = \beta_2 y_{2,t} + \beta_3 y_{3,t} + u_t, \quad (18.53)$$

where u_t is $I(0)$. The coefficient β_2 is interpreted as the expected long-run change in $y_{1,t}$ given a one unit long-run change in $y_{2,t}$, while holding $y_{3,t}$ constant in the long run. This is very similar to the standard interpretation of a regression coefficient, but the emphasis on long-run changes is important. For example, this cointegrating equation does not necessarily imply that a single period one unit shock to $y_{2,t}$, holding $y_{3,t}$ constant, is expected to change $y_{1,t}$ by β_2 in either the short or long run.

In terms of impulse responses, Johansen (2005) shows that an impulse of the form $S = \Gamma(1)F$, where $\Gamma(1)$ is defined in (18.10) and $F = [\beta_2, 1, 0]'$, produces the desired long-run changes of a one unit change in $y_{2,t}$, a zero unit change in $y_{3,t}$ and the resulting β_2 change in $y_{1,t}$. That is, there exists some shock that can be applied to (18.50) to produce the long run changes implied by (18.53).

The shock $S = \Gamma(1)F$ is different from those implied by other identifications, such as the Choleski decomposition of V . These latter shocks will, therefore, have different effects in both the short run and the long run. In some cases the long-run effects of the Choleski shocks can even differ in sign from what might be expected from (18.53). This is because the long-run dynamics of the model are determined by $\Gamma(1)$ as well as α and β .

18.9 Applications

Two applications are presented that focus on extensions of the trace LR test of cointegration. The first investigates the performance of information

criteria to select the rank, while the second focusses on the size properties of the trace test when the *iid* assumption is relaxed.

18.9.1 Rank Selection Based on Information Criteria

Identifying the cointegrating rank by means of information criteria (see also Chapter 13) requires that the log-likelihood in (18.29) be augmented by a penalty factor, c_T ,

$$IC(r) = -2(T - p) \ln L_T(r) + c_T m_r,$$

where $c_T = \ln T$ for the Schwarz information criterion (*SIC*) and $m_r = \dim(\alpha, \beta, V) = Nr + (N - r)r + N(N + 1)/2$, is the number of free parameters corresponding to rank r having normalized r of the N variables in the system. The estimated rank is chosen to satisfy

$$\hat{r} = \arg \min_{0 \leq r \leq N} IC(r). \quad (18.54)$$

Asymptotic Properties

If r_0 represents the true cointegrating rank, then from Section 18.6.1, $\{\hat{\lambda}_1, \dots, \hat{\lambda}_{r_0}\}$ are $O_p(1)$ and $\{\hat{\lambda}_{r_0+1}, \dots, \hat{\lambda}_N\}$ are $O_p(T^{-1})$. If the model is underspecified, ($r < r_0$), the asymptotic difference between the information criteria based on r and the true rank r_0 , is

$$\lim_{T \rightarrow \infty} (IC(r) - IC(r_0)) = \lim_{T \rightarrow \infty} (T - p) \sum_{j=r+1}^{r_0} \ln(1 - \hat{\lambda}_j) + \lim_{T \rightarrow \infty} c_T (m_r - m_{r_0}).$$

The first term diverges to $+\infty$ at the rate T ,

$$\lim_{T \rightarrow \infty} (T - p) \sum_{j=r+1}^{r_0} \ln(1 - \hat{\lambda}_j) = O(T)O_p(1) = O_p(T).$$

The second term

$$\lim_{T \rightarrow \infty} c_T (m_r - m_{r_0}),$$

diverges to $-\infty$ since $m_r - m_{r_0} < 0$ and $\lim_{T \rightarrow \infty} c_T = \ln T \rightarrow \infty$. However, because this is a lower rate of convergence than T , the first term dominates, resulting in $\Pr(IC(r) > IC(r_0)) = 1$ as $T \rightarrow \infty$, and hence $\Pr(\hat{r} < r_0) \rightarrow 0$.

If the model is overspecified ($r > r_0$), the asymptotic difference between the information criteria based on r and the true rank r_0 is now

$$\lim_{T \rightarrow \infty} (IC(r) - IC(r_0)) = \lim_{T \rightarrow \infty} (T - p) \sum_{j=r_0+1}^r \ln(1 - \hat{\lambda}_j) + \lim_{T \rightarrow \infty} c_T (m_r - m_{r_0}).$$

The first term converges to a constant as

$$\lim_{T \rightarrow \infty} (T-1) \sum_{j=r_0+1}^r \ln(1 - \hat{\lambda}_j) = O(T)O_p(T^{-1}) = O_p(1).$$

The second term converges to $+\infty$ as $m_r - m_{r_0} > 0$ and $\lim_{T \rightarrow \infty} c_T = \ln T \rightarrow \infty$. In this case, $IC(r) - IC(r_0) \rightarrow \infty$ for $r > r_0$ and hence $\Pr(\hat{r} > r_0) \rightarrow 0$. The combination of $\Pr(\hat{r} < r_0) \rightarrow 0$ and $\Pr(\hat{r} > r_0) \rightarrow 0$ for the SIC imply that $\Pr(\hat{r} = r_0) \rightarrow 1$, so $\hat{r}$ is a consistent estimator of r_0 .

Finite Sample Properties

To identify the finite sample properties of the SIC in determining the true cointegrating rank, r_0 , the following Monte Carlo experiments are conducted using the data generating process ($N = 2$)

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} \phi_1 & 0 \\ 0 & \phi_2 \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix}, \quad (18.55)$$

where $v_t = [v_{1,t}, v_{2,t}]'$ and $v_t \sim iid N(0, I_2)$. Three experiments are conducted with the true cointegrating rank of $r_0 = \{0, 1, 2\}$, respectively, with the results for a sample of size $T = 100$ and 10000 replications reported in Table 18.6. For comparison the results are also presented for the trace LR test based on Model 1 with the selection of the rank determined using the asymptotic 5% critical value.

Table 18.6

Estimates of $\Pr(\hat{r} = r)$ for alternative tests of cointegration where the true rank is r_0 , with the value of r_0 in each case indicated with an asterisk. The simulation is based on (18.55) with a sample of size $T = 100$ and 10000 replications.

Stat.	Experiment I ($\phi_1 = 1, \phi_2 = 1$)			Experiment II ($\phi_1 = 1, \phi_2 = 0.8$)			Experiment III ($\phi_1 = \phi_2 = 0.8$)		
	$r = 0^*$	$r = 1$	$r = 2$	$r = 0$	$r = 1^*$	$r = 2$	$r = 0$	$r = 1$	$r = 2^*$
SIC	0.984	0.016	0.000	0.590	0.391	0.019	0.136	0.010	0.854
LR	0.949	0.046	0.005	0.329	0.626	0.045	0.003	0.013	0.984

The SIC tends to be a conservative test in finite samples with a tendency to under-select the cointegrating rank. In Experiment II ($r_0 = 1$), the probability of choosing $r = 0$ is 0.590, whereas for Experiment II ($r_0 = 2$), the probability of choosing a lower rank is 0.136+0.010. The trace LR test yields empirical sizes close to the nominal size of 5% in Experiment I (0.046+0.005) and Experiment III (0.003 + 0.013). In Experiment II where the true rank

is $r_0 = 1$, the probability of choosing this rank is just 0.626. For this experiment the estimated probability of 0.329 shows that the power of the trace test of testing $H_0: r = 0$ against $H_1: r > 0$, is $1 - 0.329 = 0.671$.

18.9.2 Effects of Heteroskedasticity on the Trace Test

An important assumption underlying the computation of the critical values of the trace test in Table 18.4 is that the disturbance term v_t is *iid*. This assumption is now relaxed by using a range of Monte Carlo experiments to investigate the effects of heteroskedasticity on the size of the trace test. Two forms of heteroskedasticity are considered with the first based on a generalized autoregressive conditional heteroskedastic (or GARCH) variance (see Chapter 20 for a discussion of GARCH models) and the second based on a one-off structural break in the variance. A sufficient condition for the asymptotic null distribution theory of the trace test to remain valid in the first case is for the GARCH process to have at least finite fourth moments (Cavaliere, Rahbek and Taylor, 2010). In the case of structural breaks in the variance, the asymptotic distribution does change being affected by the timing of the structural break.

The data generating process under the null of no cointegration ($r = 0$) is an $N = 2$ dimensional VECM based on Model 1 with $P = 1$ lags

$$\Delta y_t = v_t,$$

where $v_t = [v_{1,t}, v_{2,t}]'$ is now a heteroskedastic martingale difference disturbance term, with zero mean and variance $\text{var}(v_{j,t}) = \sigma_{j,t}^2$, given by

$$\begin{aligned} \text{GARCH} & : \sigma_{j,t}^2 = (1 - \phi_1 - \phi_2) + \phi_1 v_{i,t-1}^2 + \phi_2 \sigma_{j,t-1}^2, \quad \forall j, \\ \text{Structural Breaks} & : \sigma_{j,t}^2 = \begin{cases} 1 & : t \leq \lfloor \tau T \rfloor \\ 2 & : t > \lfloor \tau T \rfloor \end{cases}, \quad \forall j. \end{aligned}$$

The results of the Monte Carlo experiments are given in Table 18.7 for samples ranging from $T = 100$ to $T = 3200$. The size of the trace test is computed by simulating the DGP 10000 times and computing the proportion of calculated values of the trace statistic in excess of the pertinent critical values given in Table 18.4. As a benchmark, the results of the *iid* case are also presented. Two sets of parameters are used for the GARCH model. In the first set, $\{\phi_1 = 0.3, \phi_2 = 0.6\}$ yields a finite fourth moment for v_t (Lee and Tse, 1996) because

$$3\phi_1^2 + 2\phi_1\phi_2 + \phi_2^2 < 1, \quad (18.56)$$

and the trace test has its usual asymptotic distribution. For the second set

of parameters $\{\phi_1 = 0.3, \phi_2 = 0.69\}$, the fourth moment condition in (18.56) is no longer satisfied.

Table 18.7

Size of the trace test in the presence of heteroskedasticity from GARCH and structural breaks. Results are based on $N = 2$ with 10000 replications and the critical values of Model 1 in Table 18.4.

T	iid	GARCH ($\phi_1 = 0.3$)		Structural Break		
		$\phi_2 = 0.6$	$\phi_2 = 0.69$	$(\tau = 0.1)$	$(\tau = 0.5)$	$(\tau = 0.9)$
100	0.0538	0.0704	0.0912	0.0645	0.1081	0.0915
200	0.0506	0.0698	0.0901	0.0671	0.1027	0.0888
400	0.0501	0.0676	0.0932	0.0632	0.1051	0.0964
800	0.0528	0.0638	0.1009	0.0690	0.1054	0.0943
1600	0.0511	0.0571	0.0971	0.0679	0.1057	0.0923
3200	0.0513	0.0560	0.0994	0.0658	0.1057	0.0926

The Monte Carlo results in Table 18.7 show that the size of the trace test approaches the nominal size of 0.05 as T increases for the first parameter set which results in the GARCH model having a finite fourth moment. In contrast, for the GARCH model which does not satisfy the fourth moment condition, the usual asymptotic distribution is affected since there is now no evidence of the size approaching 0.05 as T increases.

The effects of structural breaks in the variance results in size distortions that are largest when the break is in the middle of the sample ($\tau = 0.5$), although the other choices of τ also result in significant size distortions. Furthermore, there is no sign of amelioration in the size distortion as T increases.

18.10 Exercises

(1) Graphical Analysis of Long-Run Economic Theories

Gauss file(s)	coint_lrgraphs.g, permincome.xlsx, moneydemand.xlsx, usmacro.xlsx
Matlab file(s)	lrgraphs_usmacro.m, permincome.mat, moneydemand.mat, usmacro.mat

The data files contain quarterly macroeconomic data for the U.S. for various sample periods.

- (a) In the context of the permanent income hypothesis, discuss the time series properties of and relationships between the logarithms of real consumption per capita and real income per capita from March 1984 to December 2005.
- (b) In the context of the long-run demand for money, discuss the time series properties of and relationships between the logarithms of real money and real income, and the relative interest rate from March 1959 to December 2005.
- (c) In the context of the term structure of interest rates, discuss the time series properties of and relationships between the yields on 10-year, 5-year and 1-year bonds from March 1962 to September 2010.

(2) Time Series Properties of a VECM

Gauss file(s)	coint_ecmsim.g
Matlab file(s)	coint_ecmsim.m

The following question explores the time series properties of a VECM by simulating the model under various parameterizations. Consider the bivariate VECM with rank $r = 1$

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} \delta_{1,0} \\ \delta_{2,0} \end{bmatrix} + \begin{bmatrix} \alpha_{1,1} \\ \alpha_{2,1} \end{bmatrix} (2 + \beta_1 t + y_{1,t-1} - y_{2,t-1}) + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix},$$

where $v_t = [v_{1,t}, v_{2,t}]'$ is $N(0, V)$ with $V = I_2$. For a sample of size $T = 200$, simulate the model for the parameterizations given in the

following table and interpret the time series properties of $y_t = [y_{1,t}, y_{2,t}]'$.

Experiment	Model	$\delta_{0,1}$	$\delta_{0,2}$	β_1	$\alpha_{1,1}$	$\alpha_{2,1}$
I	2	0.0	0.0	0.0	-0.1	0.1
II	2	0.0	0.0	0.0	-0.01	0.01
III	3	0.1414	0.1414	0.0	-0.1	0.1
IV	4	0.0	0.0	0.1	-0.1	0.1

(3) Term Structure of Interest Rates

Gauss file(s)	<code>coint_bivterm.g, coint_triterm.g, usmacro.xlsx</code>
Matlab file(s)	<code>coint_bivterm.m, coint_triterm.m, usmacro.mat</code>

The data file contains quarterly data on 1-year, 5-year and 10-year yields for the U.S. from March 1962 to September 2010.

- Estimate a bivariate VAR containing the 10-year and 1-year yields, with a constant and $p = 1$ lags. Interpret the parameter estimates.
- Using the results of part (a), estimate a bivariate VECM with a constant in the cointegrating equation only (Model 2). Interpret the parameter estimates. Use both the iterative and Johansen estimators.
- Estimate a bivariate VAR containing the first differences of the 10-year and 1-year yields, no constant and the lag length corresponding to the lag length of the VECM in part (b). Interpret the parameter estimates.
- Use the log-likelihood values of the estimated models in parts (a) to (c) to test for cointegration.
- Test for weak exogeneity and strong exogeneity.
- Repeat parts (a) to (e) for the 10-year and 5-year yields.
- Repeat parts (a) to (e) for the 5-year and 1-year yields.
- Repeat parts (a) to (e) for the 10-year, 5-year and 1-year yields.

(4) Permanent Income Hypothesis

Gauss file(s)	<code>coint_permincome.g, permincome.xlsx</code>
Matlab file(s)	<code>coint_permincome.m, permincome.mat</code>

The data file contains quarterly data on real consumption per capita, rc_t , and real income per capita, ry_t , for the U.S. from March 1984 to December 2005.

- (a) Test for cointegration between $\ln rc_t$ and $\ln ry_t$ using a VECM based on Model 3 in Table 18.1 with $p = 4$ lags.
- (b) Given the results of the cointegration test in part (a), estimate a VECM and interpret the long-run parameter estimates, β , and the error correction parameter estimates, α .
- (c) Using the results of parts (a) and (b), discuss whether the permanent income hypothesis is satisfied.

(5) **Simulating the Eigenvalue Distribution**

Gauss file(s)	<code>coint_simevals.g</code>
Matlab file(s)	<code>coint_simevals.m</code>

- (a) Consider the bivariate VECM based on Model 1 in Table 18.1 with rank $r = 1$

$$y_{1,t} = y_{2,t} + v_{1,t}, \quad y_{2,t} = y_{2,t-1} + v_{2,t},$$

where $v_t = [v_{1,t}, v_{2,t}]' \sim iid N(0, V)$ with $V = I_2$. Simulate the model 10000 times with samples of size $T = \{100, 200, 400, 800\}$ and discuss the asymptotic properties of the sampling distributions of the estimated eigenvalues obtained from

$$L^{-1} S_{10} S_{00}^{-1} S_{01} L^{-1'},$$

where $S_{i,j}$ are defined in (18.25) with $R_{0,t} = \Delta y_t$ and $R_{1,t} = y_{t-1}$ and $LL' = S_{1,1}$.

- (b) Repeat part (a) for the trivariate VECM

$$\begin{aligned} y_{1,t} &= 0.5y_{2,t} + 0.5y_{3,t} + v_{1,t} \\ y_{2,t} &= y_{2,t-1} + v_{2,t} \\ y_{3,t} &= y_{3,t-1} + v_{3,t}. \end{aligned}$$

- (c) Repeat part (a) for the trivariate VECM

$$\begin{aligned} y_{1,t} &= y_{3,t} + v_{1,t} \\ y_{2,t} &= y_{3,t} + v_{1,t} \\ y_{3,t} &= y_{3,t-1} + v_{3,t}. \end{aligned}$$

(6) **Computing the Quantiles of the Trace Statistic by Simulation**

Gauss file(s)	<code>coint_tracecv.g</code>
Matlab file(s)	<code>coint_tracecv.m</code>

The quantiles of the trace statistic reported in Table 18.4 for models 1 to 5 are computed by simulating the model under the null hypothesis of $N - r$ common trends.

- (a) For Model 1, simulate the following K dimensional process under the null hypothesis

$$\Delta y_t = v_t, \quad v_t \sim iid N(0, V),$$

where $V = I_K$ and v_t is $(T \times K)$ matrix which approximates the Brownian increments $dB(s)$ and y_t approximates $B(s)$. In the computation of B and $dB(s)$, the latter is treated as a forward difference relative to B . Compute the trace statistic

$$LR = \text{tr} \left(\left[T^{-1} \sum_{t=1}^T v_t y'_{t-1} \right] \left[T^{-2} \sum_{t=1}^T y_{t-1} y'_{t-1} \right]^{-1} \left[T^{-1} \sum_{t=1}^T y_{t-1} v'_t \right] \right),$$

100000 times and find the 90%, 95% and 99% quantiles of LR for $K = 1, 2, \dots, 6$ common trends.

- (b) For Model 2, repeat part (a), except that B is augmented by including a constant.
- (c) For Model 3, repeat part (a), except replace one of the common trends in B by a deterministic time trend and then demean B .
- (d) For Model 4, repeat part (a), except augment B by including a deterministic time trend and then demean B .
- (e) For Model 5, repeat part (a), except replace one of the common trends in B by a squared deterministic time trend and then detrend B by regressing B on a constant and a deterministic time trend.

(7) Demand for Money

Gauss file(s)	<code>coint_impulse.g, moneydemand.xlsx</code>
Matlab file(s)	<code>coint_impulse.m, moneydemand.mat</code>

The data file contains quarterly data on nominal M2, m_t , nominal GDP, gdp_t , the consumer price level, cpi_t , the Treasury bill yield, $tbill_t$ and the Federal funds rate, $ffunds_t$, for the U.S. from March 1959 to December 2005.

- (a) Test for cointegration among $\ln(m_t/p_t)$, $\ln(gdp_t/p_t)$, $tbill_t$ and $ffunds_t$, where the VECM is based on Model 3 from Table 18.1 with $p = 4$ lags.

- (b) Given the results of the cointegration test in part (a), estimate a VECM and interpret the normalized long-run parameter estimates, β , and the error-correction parameter estimates, α .
- (c) Test the following restrictions on the long-run parameters and interpret the results.
 - (i) The income elasticity is unity.
 - (ii) The two interest rate semi-elasticities sum to zero.
- (d) Given the results of part (c), compute the impulse responses of the VECM based on short-run restrictions given by a triangular ordering and interpret the results.
- (e) Compute the impulse responses for an income shock based on the long-run restrictions between money and income while holding the two yields fixed.

(8) Information Criteria Test of Cointegration

Gauss file(s)	<code>coint_ic.g</code>
Matlab file(s)	<code>coint_ic.m</code>

The information criteria approach to choosing the cointegrating rank is based on solving

$$\hat{r} = \arg \min_{0 \leq r \leq N} (-2(T-p) \ln L_T(r) + c_T m_r),$$

where $c_T = 2(AIC)$, $\ln T(SIC)$, $2 \ln \ln T(HIC)$ and $m_r = \dim(\alpha, \beta, V) = Nr + (N-r)r + N(N+1)/2$, the number of free parameters corresponding to rank r having normalized r of the N variables in the system.

- (a) Show that the BIC and the HIC yield consistent estimators of the true cointegrating rank r_0 , whereas AIC does not.
- (b) The following data generating process is used to investigate the small sample properties of the information criteria to select the cointegrating rank

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} \phi_1 & 0 \\ 0 & \phi_2 \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix},$$

$$\begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix} \sim iid N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}\right),$$

where the sample size is $T = 100$, the number of replications is 10000 and the parameter values are

- (i) $r_0 = 0 : \phi_1 = 1.0, \phi_2 = 1.0, \rho = 0.0$

- (ii) $r_0 = 1 : \phi_1 = 1.0, \phi_2 = 0.8, \rho = 0.0$
 - (iii) $r_0 = 1 : \phi_1 = 1.0, \phi_2 = 0.8, \rho = 0.9$
 - (iv) $r_0 = 2 : \phi_1 = 0.8, \phi_2 = 0.8, \rho = 0.0$.
- (c) Redo the Monte Carlo experiment in part (b) for the trace test using the asymptotic 5% critical value reported in Table 18.4. Compare the finite sample performance of the trace and information criteria approaches to select the cointegrating rank.

(9) Heteroskedasticity and the Distribution of the Trace Statistic

Gauss file(s)	<code>coint_hetero.g</code>
Matlab file(s)	<code>coint_hetero.m</code>

Consider the following VECM with heteroskedasticity

$$\Delta y_{i,t} = v_{i,t}, \quad v_{i,t} = \sqrt{h_{i,t}} z_{i,t}, \quad z_{i,t} \sim N(0, 1), \quad i = 1, 2, \dots, N.$$

- (a) Assume that the heteroskedasticity is of the form

$$\sigma_{i,t}^2 = (1 - \phi_1 - \phi_2) + \phi_1 v_{i,t-1}^2 + \phi_2 \sigma_{j,t-1}^2.$$

Simulate the size of the trace test for $N = 2$, a sample of $T = 100$ and using 10000 replications. Choose two sets of parameters given by $\{\phi_1 = 0.3, \phi_2 = 0.6\}$ and $\{\phi_1 = 0.3, \phi_2 = 0.69\}$. Discuss the results.

- (b) Repeat part (a) for dimensions of $N = 4, 6, 8$. Discuss the results.
- (c) Assume that the heteroskedasticity is a structural break

$$h_{j,t} = \begin{cases} 1 & : t \leq \lfloor \tau T \rfloor \\ 2 & : t > \lfloor \tau T \rfloor \end{cases}.$$

Simulate the size of the trace test for a sample of $T = 100$ using 10000 replications and break points of $\tau = \{0.1, 0.25, 0.5, 0.75, 0.9\}$. Discuss the results.

- (d) Repeat part (c) for dimensions of $N = 4, 6, 8$. Discuss the results.

(10) Likelihood Ratio Test of a Deterministic Time Trends

Gauss file(s)	<code>coint_trend.g</code>
Matlab file(s)	<code>coint_trend.m</code>

Simulate the following bivariate VECM based on Model 3 for a sample of size $T = 200$

$$\begin{bmatrix} \Delta y_{1,t} \\ \Delta y_{2,t} \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.2 \end{bmatrix} + \begin{bmatrix} -0.2 \\ 0.2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}' \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \\ 1 \end{bmatrix} + \begin{bmatrix} -0.2 & 0 \\ 0 & 0.4 \end{bmatrix} \begin{bmatrix} \Delta y_{1,t-1} \\ \Delta y_{2,t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix}.$$

where $v_t = [v_{1,t} \ v_{2,t}]'$ and $v_t \sim iid N(0, V)$ with $V = I_2$.

- (a) Estimate two VECMs using the Johansen estimator with the first based on Model 5 in Table 18.1 and the second based on Model 4. Perform a LR test and interpret the result given that the true DGP is based on Model 3. Use the property that under the null hypothesis the LR test is distributed asymptotically as χ^2 with $\dim(\mu_1) - \dim(\beta_1) = N - r$ degrees of freedom since the total number of deterministic time trends in the unrestricted model is N , while in the restricted model the number of trends is r .
- (b) Estimate two VECMs using the Johansen estimator with the first based on Model 4 in Table 18.1 and the second based on Model 3. Perform a LR test and interpret the result given that the true DGP is based on Model 3. Use the property that under the null hypothesis the LR test is distributed asymptotically as χ^2 with $\dim(\beta_1) = r$ restrictions.

(11) Identification of VECMs

Gauss file(s)	<code>coint_ident.g</code>
Matlab file(s)	<code>coint_ident.m</code>

Simulate the following trivariate VECM based on Model 3 for a sample of size $T = 200$

$$\begin{bmatrix} \Delta y_{1,t} \\ \Delta y_{2,t} \\ \Delta y_{3,t} \end{bmatrix} = \begin{bmatrix} 0.04286 \\ 0.02857 \\ 0.08571 \end{bmatrix} + \begin{bmatrix} -0.2 & 0.0 \\ 0.0 & -0.3 \\ 0.1 & 0.1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -0.8 & -0.8 \\ 4 & 1 \end{bmatrix}' \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \\ y_{3,t-1} \\ 1 \end{bmatrix} + \begin{bmatrix} -0.2 & 0 & 0 \\ 0 & 0.4 & 0 \\ 0 & 0 & 0.1 \end{bmatrix} \begin{bmatrix} \Delta y_{1,t-1} \\ \Delta y_{2,t-1} \\ \Delta y_{3,t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \\ v_{3,t} \end{bmatrix},$$

where $v_t = [v_{1,t} \ v_{2,t} \ v_{3,t}]' \sim iid N(0, V)$ with $V = I_3$ and $y_0 = y_{-1} = 0$.

- (a) Use the Johansen algorithm to estimate the VECM based on model 3 with rank $r = 2$.
- (b) Estimate the nonnormalized cointegrating vectors using the first two columns of $\hat{\gamma} = L'^{-1}E$ and compare the estimates with the population parameter values.
- (c) Estimate β using the triangular normalization and compare the estimates with the population parameter values.
- (d) Use the iterative algorithm to estimate the VECM based on model 3 with rank $r = 2$, where the triangular normalization is subject to the cross-equation restriction that the long-run parameter estimates on $y_{3,t}$ in the two cointegrating equations are equal.

(12) Exogeneity Testing

Gauss file(s)	<code>coint_exogeneity.g</code>
Matlab file(s)	<code>coint_exogeneity.m</code>

Simulate the following bivariate VECM based on Model 1 for a sample of size $T = 200$

$$\begin{bmatrix} \Delta y_{1,t} \\ \Delta y_{2,t} \end{bmatrix} = \begin{bmatrix} -0.4 \\ 0.0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}' \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} -0.2 & 0.0 \\ 0.0 & 0.4 \end{bmatrix} \begin{bmatrix} \Delta y_{1,t-1} \\ \Delta y_{2,t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix}$$

$$\begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix} \sim iid N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1.0 & 0.5 \\ 0.5 & 1.0 \end{bmatrix}\right).$$

In this model $y_{2,t}$ is both weakly and strongly exogenous.

- (a) Estimate the following VECM based on Model 1 with $p = 2$ lags

$$\begin{bmatrix} \Delta y_{1,t} \\ \Delta y_{2,t} \end{bmatrix} = \begin{bmatrix} \alpha_{1,1} \\ \alpha_{2,1} \end{bmatrix} \begin{bmatrix} 1.0 \\ \beta_{2,1} \end{bmatrix}' \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} \Gamma_{1,1} & \Gamma_{1,2} \\ \Gamma_{2,1} & \Gamma_{2,2} \end{bmatrix} \begin{bmatrix} \Delta y_{1,t-1} \\ \Delta y_{2,t-1} \end{bmatrix} + \begin{bmatrix} v_{1,t} \\ v_{2,t} \end{bmatrix},$$

using the iterative estimator based on Newton-Raphson. Test the following hypotheses

$$\begin{array}{ll} H_0 : \alpha_{2,1} = 0 & [\text{Weak exogeneity}] \\ H_0 : \alpha_{2,1} = 0, \Gamma_{2,1} = 0 & [\text{Strong exogeneity}], \end{array}$$

using a Wald statistic.

- (b) Estimate the VECM in part (a) subject to the restriction $\alpha_{2,1} = 0$. Test for weak exogeneity using a LR statistic.
- (c) Estimate the VECM in part (a) subject to the restrictions $\alpha_{2,1} = 0$ and $\Gamma_{2,1} = 0$. Test for strong exogeneity using a LR statistic.
- (d) Show that, under weak exogeneity, the estimate of the cointegrating parameter $\beta_{2,1}$, obtained in part (b), can also be recovered from estimating the regression equation

$$\Delta y_{1,t} = \nu_1 y_{1,t-1} + \nu_2 y_{2,t-1} + \nu_3 \Delta y_{2,t} + \nu_4 \Delta y_{1,t-1} + \nu_5 \Delta y_{2,t-1} + w_t,$$

by ordinary least squares and defining $\beta_{2,1} = -\nu_2/\nu_1$. Compare the estimate of $\hat{\beta}_{2,1}$ obtained using the partial model with the estimate using the full system obtained in part (b) subject to the restriction $\alpha_{2,1} = 0$.

- (e) Estimate the VECM in part (a) based on Model 1 with $p = 2$ lags using the Johansen estimator and compare the results obtained using the Newton-Raphson algorithm.

(13) Term Structure of Interest Rates with Level Effects

Gauss file(s)	coint_level.g, usmacro.xlsx
Matlab file(s)	coint_level.m, usmacro.mat

The data file contains quarterly data on 1-year, 5-year and 10-year yields for the US from March 1962 to September 2010.

- (a) Consider the bivariate VECM of y_t containing the 10-year and 1-year yields, using Model 2 with $p = 1$ lags,

$$\begin{bmatrix} \Delta y_{1,t} \\ \Delta y_{2,t} \end{bmatrix} = \begin{bmatrix} \alpha_{1,1} \\ \alpha_{2,1} \end{bmatrix} (y_{1,t-1} - \beta_0 - \beta_s y_{2,t-1}) + v_t$$

$$L = \begin{bmatrix} L_{1,1} & 0 \\ L_{1,2} & L_{2,2} \end{bmatrix}$$

$$v_t \sim iid N(0, V = LL').$$

Estimate the parameters $\theta = \{\alpha_1, \alpha_2, \beta_c, \beta_s, L_{1,1}, L_{1,2}, L_{2,2}\}$ using the iterative estimator based on the Newton-Raphson algorithm. Compare the estimates with those obtained in part (b) of Exercise 3.

- (b) Now estimate the VECM with level effects parameters κ_1 and κ_2 by redefining L and V in part(a) as

$$L_t = \begin{bmatrix} L_{1,1} & 0 \\ L_{1,2} & L_{2,2} \end{bmatrix} \begin{bmatrix} y_{1,t-1}^{\kappa_1} \\ y_{2,t-1}^{\kappa_2} \end{bmatrix}, \quad V_t = L_t L_t'.$$

- (c) Compare the estimates obtained in parts (a) and (b). Test for weak and strong exogeneity. Perform a joint test of level effects in the term structure of interest rates by testing the restrictions $\kappa_1 = \kappa_2 = 0$.

PART SIX
NONLINEAR TIME SERIES

19

Nonlinearities in Mean

19.1 Introduction

The stationary time series models developed in Part FOUR and the nonstationary time series models developed in Part FIVE are characterized by the mean being a linear function of the lagged dependent variables (autoregressive) and/or the lagged disturbances (moving average). These models are able to capture many of the characteristics observed in time series data, including randomness, cycles and stochastic trends. Where these models come up short, however, is in the modelling of more extreme events such as jumps and asymmetric adjustments across cycles that cannot be captured adequately by a linear representation. This chapter deals with models in which the linear mean specification is augmented by the inclusion of nonlinear terms so that conditional mean becomes nonlinear in the lagged dependent variables and lagged disturbances.

Examples of nonlinear models investigated are thresholds time series models (TAR), artificial neural networks (ANN), bilinear models and Markov switching models. Nonparametric methods are also investigated where a parametric specification of the nonlinearity is not imposed on the structure of the model. Further nonlinear specifications are investigated in Chapters 20 and 21. In Chapter 20, nonlinearities in variance are introduced and developed in the context of GARCH and MGARCH models. In Chapter 21, nonlinearities arise from the specification of time series models of discrete random variables.

19.2 Motivating Examples

The class of stationary linear time series models presented in Chapter 13 yield solutions that are characterized by convergence to a single equilibrium

point, with the trajectory path exhibiting either monotonic or oscillatory behaviour.

Example 19.1 AR(2) Model

Consider the linear AR(2) process

$$y_t = 0.9 + 0.7y_{t-1} - 0.6y_{t-2}. \quad (19.1)$$

Panel (a) of Figure 19.1 shows that the dynamics exhibit damped oscillatory behaviour, starting from the initial value $y_0 = 0.5$ and converging to the equilibrium point $y = 0.9/(1 - 0.7 + 0.6) = 1.0$. The phase diagram in Panel (b) of Figure 19.1 highlights how the process spirals towards this equilibrium point. $\square$

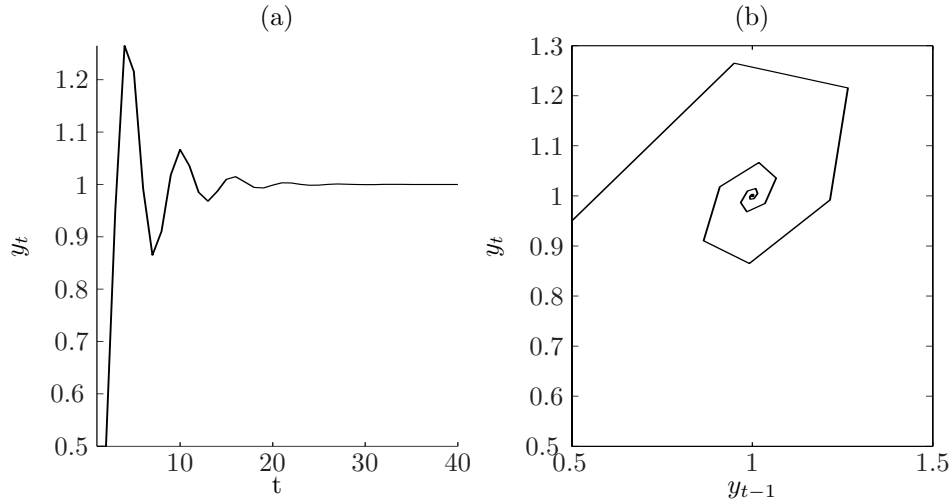


Figure 19.1 Properties of the AR(2) process, $y_t = 0.9 + 0.7y_{t-1} - 0.6y_{t-2}$, with $y_0 = 0.5$.

While this linear class of models can capture a wide range of patterns commonly observed in time series data, there are some empirical characteristics that these models cannot explain. Examples of this nonlinear behaviour include limit cycles, strange attractors, multiple equilibria and asymmetries.

Example 19.2 Limit Cycle

Consider the following model, proposed by Tong (1983, p85), where y_t is governed by separate AR processes depending upon whether it is above or

below the threshold value of 3.05

$$y_t = \begin{cases} 0.8023 + 1.0676y_{t-1} - 0.2099y_{t-2} + 0.1712y_{t-3} - 0.4528y_{t-4} + 0.2237y_{t-5} - 0.0331y_{t-6} & : y_{t-2} \leq 3.05 \\ 2.2964 + 1.4246y_{t-1} - 1.0795y_{t-2} - 0.0900y_{t-3} & : y_{t-2} > 3.05. \end{cases}$$

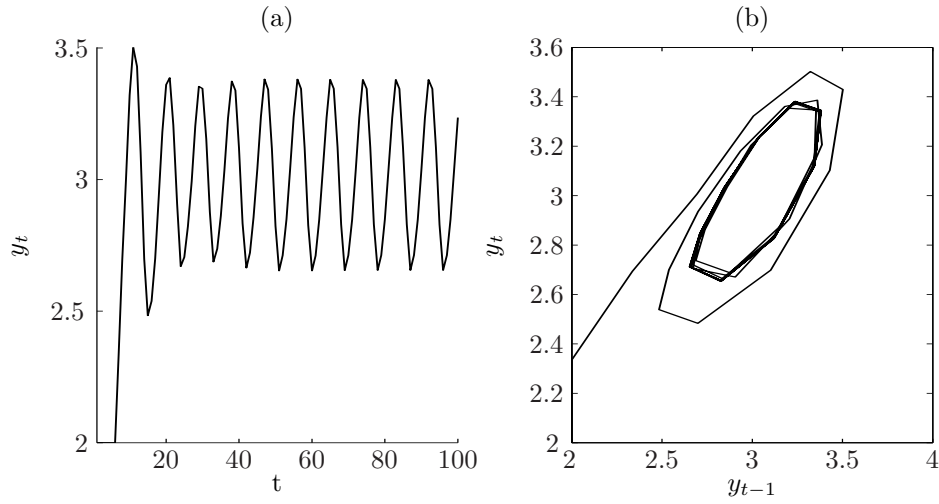


Figure 19.2 Demonstration of a limit cycle based on the example given by Tong (1983).

Panel (a) of Figure 19.2 shows that the process converges to a 9-period cycle. The phase diagram in panel (b) of Figure 19.2 illustrates how the process eventually settles down into a cycle, known as a limit cycle. This behaviour is in contrast to the pattern observed for the linear time series model in Figure 19.1 where the limit of the process is a single point. $\square$

Example 19.3 Strange Attractors

Strange attractors are regions to which a process converges, but once in the region the process exhibits aperiodic behaviour, commonly known as chaos. This is in contrast to the limit cycle depicted in Figure 19.2 where the process settles upon the outer perimeter of the region. An example of a strange attractor is the nonlinear business cycle model of Kaldor (see Lorenz, 1989, p130; Creedy and Martin, 1994) given by

$$\begin{aligned} y_{t+1} - y_t &= 20(i_t - s_t) && \text{[Output]} \\ k_{t+1} - k_t &= i_t - 0.05k_t && \text{[Capital]} \\ i_t &= 20x2^{-(0.01y_t+0.00001)^{-2}} + 0.05y_t + 5(280k_t^{-1})^{4.5} && \text{[Investment]} \\ s_t &= 0.21y_t, && \text{[Savings]} \end{aligned}$$

where y_t is output, k_t is capital, i_t is investment and s_t is savings.

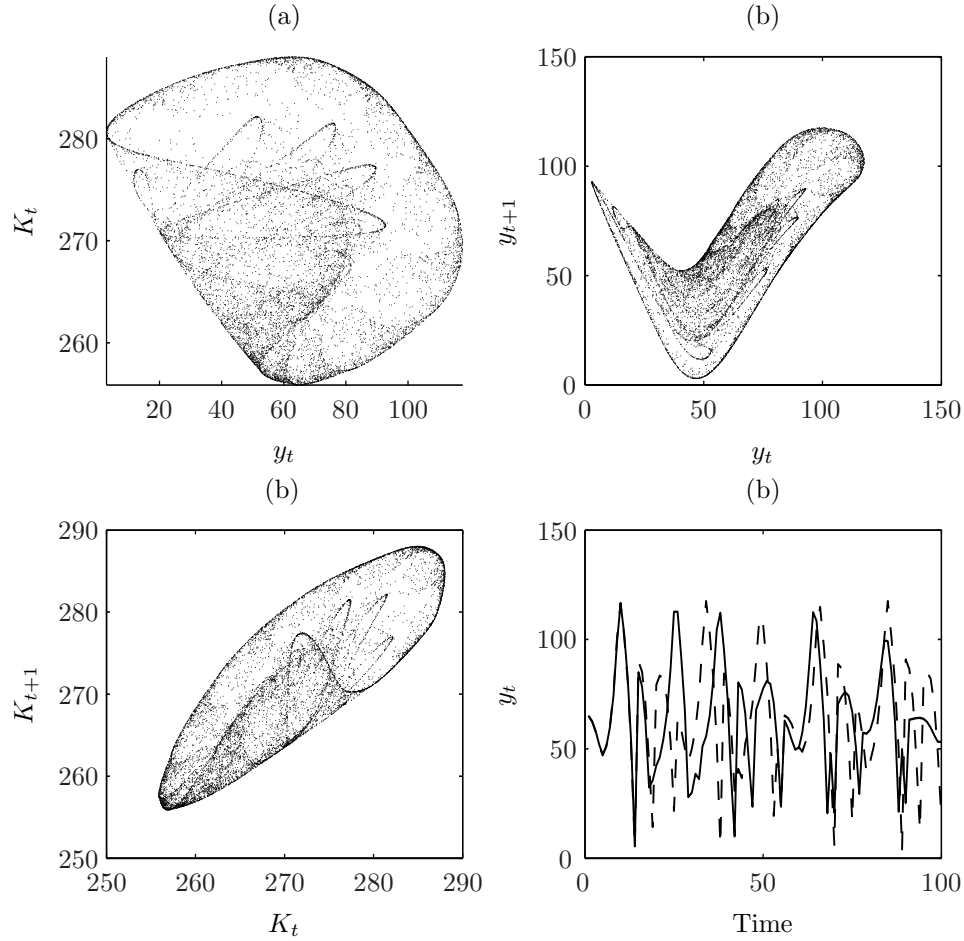


Figure 19.3 Strange attractor properties of the Kaldorian nonlinear business cycle model.

The dynamic properties of the model are given in Figure 19.3. The phase diagrams, in panels (a) to (c) of Figure 19.3, highlight the strange attraction properties of the model with regions of attraction characterized by aperiodic behaviour. Panel (d) of Figure 19.3 demonstrates the sensitive dependence to initial conditions property of chaotic attractors whereby the time paths of two processes with very similar initial conditions eventually diverge and follow completely different time trajectories. $\square$

Example 19.4 Multiple Equilibria

The stationary linear times series model exhibits a unique equilibrium. However, more than one equilibrium arises in many situations in economics. The nonlinear business cycle model of Kaldor presented in Figure 19.3 provides an example of an equilibrium region, referred to as an attractor, where the process never remains at any single point because of chaotic behaviour. As another example of multiple equilibria, consider the nonlinear error correction model of the logarithm of real money, m_t , which is a simplification of the model estimated in Ericsson, Hendry and Prestwich (1998, p.296)

$$\Delta m_t = 0.45\Delta m_{t-1} - 0.10\Delta^2 m_{t-2} - 2.54(m_{t-1} - 0.1)m_{t-1}^2 + v_t,$$

where v_t is an $iid N(0, 0.005^2)$ disturbance, $\Delta m_t = m_t - m_{t-1}$, $\Delta^2 m_{t-2} = m_{t-2} - 2m_{t-3} + m_{t-4}$ and v_t . The nonlinearity is given by the error correction term $(m_{t-1} - 0.1)m_{t-1}^2$, which is a cubic with three roots: one at 0.1 (stable) and two at 0.0 (point of inflection). Simulating this model for $T = 50000$ periods generates the histogram given in Figure 19.4. The majority of the mass is around the stable equilibrium value of 0.1, with smaller mass at the inflection point of 0.0. $\square$

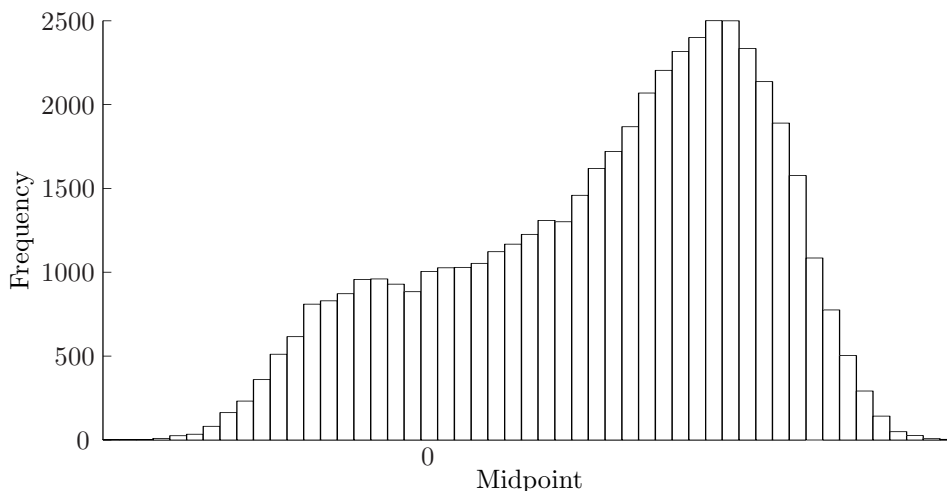


Figure 19.4 Histogram obtained from simulating the nonlinear error correction model of Ericsson, Hendry and Prestwich (1998).

Example 19.5 Asymmetries

A property of linearity is that positive and negative shocks have the same effect on the absolute value of y_t , so that a positive shock followed by a negative shock of equal magnitude would return the process to its original position. This degree of symmetry is not always evident in the data. An

example where adjustment is asymmetric arises in the analysis of business cycles where strong empirical evidence exists that the duration of expansions is longer on average than the duration of contractions (Harding and Pagan, 2002, p372)..

Table 19.1

U.S. business cycle turning points (month/year) for the period 1945 to 2007.

Peak	Trough	Duration Contraction (Peak to Trough)	Expansion (Trough to Peak)
02/1945	10/1945	8	80
11/1948	10/1949	11	37
07/1953	05/1954	10	45
08/1957	04/1958	8	39
04/1960	02/1961	10	24
12/1969	11/1970	11	106
11/1973	03/1975	16	36
01/1980	07/1980	6	58
07/1981	11/1982	16	12
07/1990	03/1991	8	92
03/2001	11/2001	8	120
12/2007	—	—	73
Average		10	57

Source: <http://www.nber.org/cycles.html>

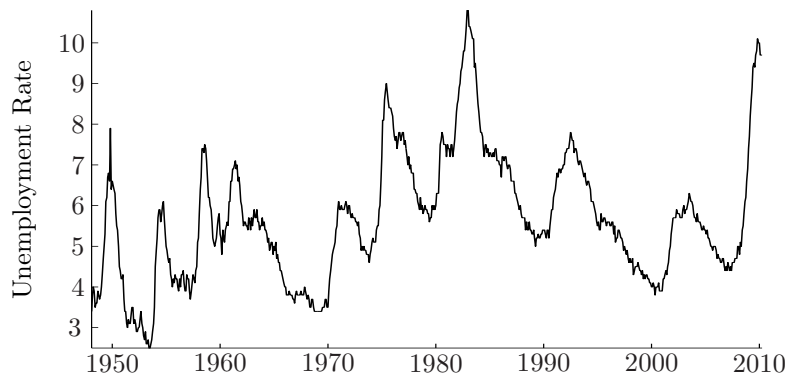


Figure 19.5 U.S. monthly aggregate unemployment rate measured for the period January 1947 to March 2010.

Table 19.1 gives the business cycle turning points for the U.S. for the pe-

riod 1945 to 2007. The average duration of contractions is about 10 months, compared to 57 months for expansions. The asymmetric behaviour of the business cycle is highlighted in Figure 19.5 which gives the monthly unemployment rate for the U.S. from 1947 to 2010. Large, sharp upward movements in unemployment (contraction phase) are followed by slow, downward drifts (expansion phase). This suggests that separate time series models may be needed to capture differences in the expansion and contraction phases of the business cycle. $\square$

19.3 Threshold Models

19.3.1 Specification

Threshold time series models assume that the dependent variable y_t , is governed by more than a single regime at any point in time. In the case of two regimes where each sub-model has an autoregressive process, the model is expressed as

$$y_t = E_{t-1}[y_t] + v_t, \quad (19.2)$$

where v_t is an *iid* disturbance term and $E_{t-1}[y_t]$ is the conditional model given by

$$E_{t-1}[y_t] = \begin{cases} \phi_0 + \sum_{i=1}^p \phi_i y_{t-i} & : \text{ with probability } 1 - w_t \\ \alpha_0 + \sum_{i=1}^p \alpha_i y_{t-i} & : \text{ with probability } w_t. \end{cases} \quad (19.3)$$

The variable w_t represents a time-varying weighting function with the property

$$0 \leq w_t \leq 1. \quad (19.4)$$

which, in general, is function of lagged values of y_t and a set of explanatory variables x_t . For values of w_t close to unity (zero) the conditional mean of y_t is dominated by the first (second) regime. Combining equations (19.2) and (19.3), the model is written more compactly as

$$\begin{aligned} y_t &= (\phi_0 + \sum_{i=1}^p \phi_i y_{t-i})(1 - w_t) + (\alpha_0 + \sum_{i=1}^p \alpha_i y_{t-i})w_t + v_t \\ &= \phi_0 + \sum_{i=1}^p \phi_i y_{t-i} + (\beta_0 + \sum_{i=1}^p \beta_i y_{t-i})w_t + v_t, \end{aligned} \quad (19.5)$$

where $\beta_0 = \alpha_0 - \phi_0$ and $\beta_i = \alpha_i - \phi_i$. Equation (19.5) forms the basis of the class of threshold time series models, which, in turn, nests a number of well-known nonlinear time series models.

The key distinguishing feature of the nonlinear class of time series models in (19.5) is the way the weighting function, w_t , is specified. Four popular choices of this function are

$$\begin{aligned}
 \text{STAR} & : w_t = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\gamma(y_{t-d}-c)} e^{-0.5s^2} ds, \\
 \text{LSTAR} & : w_t = \frac{1}{1 + \exp(-\gamma(y_{t-d} - c))}, \quad \gamma > 0, \\
 \text{SETAR} & : w_t = \begin{cases} 1 & y_{t-d} \geq c \\ 0 & y_{t-d} < c, \end{cases} \\
 \text{ESTAR} & : w_t = 1 - \exp(-\gamma(y_{t-d} - c)^2), \quad \gamma > 0,
 \end{aligned} \tag{19.6}$$

which represent, respectively, the smooth transition autoregressive model (STAR), logistic smooth transition autoregressive (LSTAR) model, self-exciting threshold autoregressive model (SETAR) and exponential smooth transition autoregressive model (ESTAR).

The properties of w_t in (19.6) are highlighted in Figure 19.3.1 for the four models with $\gamma = 1$, $c = 0$ and $d = 1$. The parameter d controls the delay in moving between regimes, the parameter c represents the threshold where switching occurs and the parameter γ determines the speed of adjustment in switching between regimes. At $\gamma = 0$ there is no switching with the STAR, LSTAR and ESTAR models reducing to the linear model. For $\gamma > 0$, there is smooth adjustment between the two regimes for the STAR, LSTAR and ESTAR models. As $\gamma \rightarrow \infty$, there is infinitely fast switching between regimes with the STAR and LSTAR models reducing to the SETAR model, whilst the ESTAR model approaches the linear model again. For the STAR, LSTAR and SETAR models, switching occurs from regime 1 to regime 2 in (19.3) since y_{t-d} increases from values less than c to values greater c . In the case of the ESTAR model, the dynamics of y_t are governed by regime 1 for values of y_{t-d} near $c = 0$ and by a weighted average of the two regimes for high and low values of y_{t-d} .

19.3.2 Estimation

Threshold models can be estimated by maximum likelihood using standard iterative algorithms. Assuming a normal disturbance term, $v_t \sim N(0, \sigma^2)$,

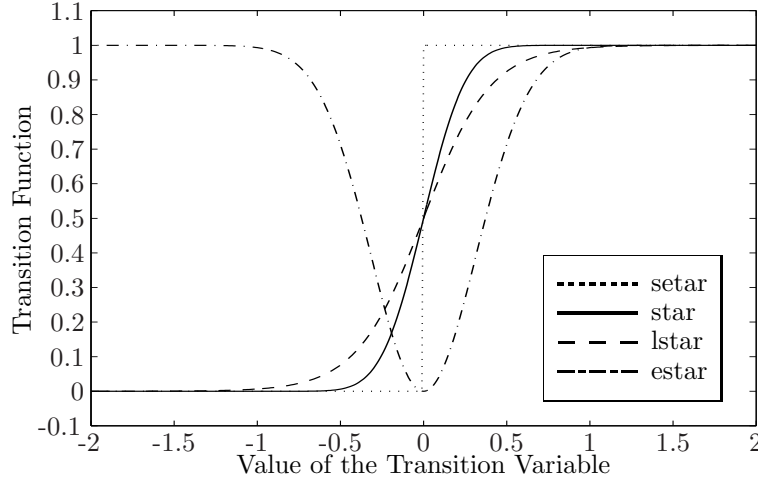


Figure 19.6 Alternative threshold functions used in threshold autoregressive models: $\gamma = 1$, $c = 0$ and $d = 1$.

the conditional probability distribution of y_t for the threshold model in equation (19.5) with weighting function w_t in (19.6) is

$$f(y_t|y_{t-1}, \dots, y_{t-s}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{v_t^2}{2\sigma^2}\right], \quad (19.7)$$

where $s = \max(p, d)$.

For a sample of $t = 0, 1, 2, \dots, T$, observations, the conditional log-likelihood function from Chapter 7 for the case of known delay parameter, d , is

$$\begin{aligned} \ln L_T(\theta) &= \frac{1}{T-s} \sum_{t=s+1}^T \ln f(y_t|y_{t-1}, \dots, y_{t-s}). \\ &= -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 \\ &\quad - \frac{1}{2\sigma^2(T-s)} \sum_{t=s+1}^T \left(y_t - \phi_0 - \sum_{i=1}^p \phi_i y_{t-i} - (\beta_0 - \sum_{i=1}^p \beta_i y_{t-i}) w_t \right)^2, \end{aligned}$$

where $s = \max(p, d)$ and $\theta = \{\phi_0, \phi_1, \dots, \phi_p, \beta_0, \beta_1, \dots, \beta_p, \gamma, c, \sigma^2\}$ are the unknown parameters. The log-likelihood function is maximized with respect to θ using one of the iterative algorithms discussed in Chapter 3. This algorithm is simplified by using the maximum likelihood estimator $\hat{\sigma}^2 = (T-s)^{-1} \sum_{t=s+1}^T \hat{v}_t^2$ to concentrate out σ^2 from $\ln L_T(\theta)$, where $\hat{v}_t$ is the residual. For the case of unknown d , a grid search is performed by

estimating the model for values of $d = \{1, 2, 3, \dots\}$ and choosing $\hat{d}$ that maximizes $\ln L_T(\theta)$.

An alternative method to compute the maximum likelihood estimator is to perform a grid search on both d and c . The advantage of this method is that for each choice of d and c , w_t is now an observable time series so the maximum likelihood estimator of the remaining parameters is computed by regressing y_t on $\{\text{constant}, y_{t-1}, y_{t-2}, \dots, y_{t-p}, w_t, y_{t-1}w_t, \dots, y_{t-p}w_t\}$.

19.3.3 Testing

A natural test of the threshold model in (19.5) is to perform a test of linearity by seeing if the model reduces to the linear autoregressive model

$$y_t = \phi_0 + \sum_{i=1}^p \phi_i y_{t-i} + v_t. \quad (19.8)$$

A comparison of (19.5) and (19.8) suggests that a natural test statistic to determine whether or not the parameters in the second regime are zero, $\beta_i = 0, \forall i$, can be based on either the Wald or LM principle. The distribution of any test statistic that tests this hypothesis, however, is nonstandard because the parameters γ and c are not defined under the null hypothesis and conventional maximum likelihood theory is, therefore, not directly applicable. An alternative approach is to test $\gamma = 0$ as w_t in (19.6) becomes a constant resulting in the STAR, LSTAR and ESTAR models reducing to a linear AR model. However, in this case it is the parameters c and β_i that are now not identified under the null hypothesis. There are two possible ways to address this problem. Luukkonen, Saikkonen and Teräsvirta (1988) suggest focussing on local asymptotics at $\gamma = 0$. This approach has the advantage of yielding a test statistic with a standard distribution under the null hypothesis. Alternatively, Hansen (1996) proposes a solution based on local asymptotics at $\beta = 0$, which yields a test statistic whose distribution must be approximated by bootstrapping. The discussion that follows focusses on the first approach.

Let $z_t = \gamma(y_{t-d} - c)$ so that under the null hypothesis $\gamma = 0$ and thus $z_t = 0$. For the LSTAR weighting function $w_t = 1/(1 + \exp[-z_t])$ where for simplicity the t subscript is omitted. The first three derivatives of the LSTAR

weighting function with respect to z_t , evaluated at $z_t = 0$, are as follows

$$\begin{aligned} w^{(1)}(0) &= \left. \frac{\partial w_t}{\partial z_t} \right|_{z_t=0} = \left. \frac{\exp(-z_t)}{(1 + \exp(-z_t))^{-2}} \right|_{z_t=0} = \frac{1}{4} \\ w^{(2)}(0) &= \left. \frac{\partial^2 w_t}{\partial z_t^2} \right|_{z_t=0} = \left. -\frac{\exp(-z_t) - \exp(-2z_t)}{(1 + \exp(-z_t))^{-3}} \right|_{z_t=0} = 0 \\ w^{(3)}(0) &= \left. \frac{\partial^3 w_t}{\partial z_t^3} \right|_{z_t=0} = \left. \frac{\exp(3z_t) - 4\exp(2z_t) + \exp(z_t)}{6\exp(2z_t) + 4\exp(3z_t) + \exp(4z_t) + 4\exp(z_t) + 1} \right|_{z_t=0} \\ &= -\frac{1}{8}. \end{aligned}$$

Using these derivatives in a third-order Taylor expansion of $w(z_t)$ around $z_t = 0$ gives

$$\begin{aligned} w_t &\approx w(0) + w^{(1)}(0)(z_t - 0) + \frac{1}{2}w^{(2)}(0)(z_t - 0)^2 + \frac{1}{6}w^{(3)}(0)(z_t - 0)^3 \\ &= \frac{1}{2} + \frac{1}{4}z_t + 0 - \frac{1}{48}z_t^3. \end{aligned}$$

The expansion of $z_t^3 = (y_{t-d} - c)^3$ has terms in y_{t-d}^3 , y_{t-d}^2 and y_{t-d} , so that the weighting function is approximated by the cubic

$$w_t \approx \delta_0 + \delta_1 y_{t-d} + \delta_2 y_{t-d}^2 + \delta_3 y_{t-d}^3. \quad (19.9)$$

This Taylor series approximation represents the local behaviour of the function in the vicinity of $\gamma = 0$ and therefore provides a basis of a test for linearity.

Substituting (19.9) in (19.5) for w_t gives the regression equation

$$y_t = \pi_0 + \sum_{i=1}^p \pi_{0,i} y_{t-i} + \sum_{i=1}^p \pi_{1,i} y_{t-i} y_{t-d} + \sum_{i=1}^p \pi_{2,i} y_{t-i} y_{t-d}^2 + \sum_{i=1}^p \pi_{3,i} y_{t-i} y_{t-d}^3 + u_t, \quad (19.10)$$

where the disturbance term u_t is a function of v_t in (19.5) and the approximation error from truncating the Taylor series expansion in (19.9). Where the delay parameter is known, this regression equation forms the basis of the LST linearity test of Luukkonen, Saikkonen and Teräsvirta (1988). Under the null hypothesis, there are no interaction terms, $\pi_{j,i} = 0$ with $j = 1, 2, 3$ and $i = 1, 2, \dots, p$. The steps to perform the linearity test are as follows.

Step 1: Regress y_t on $\{1, y_{t-1}, \dots, y_{t-p}\}$ to get the restricted residuals $\hat{u}_t$.

Step 2: Regress $\hat{u}_t$ on

$$\{1, y_{t-1}, \dots, y_{t-p}, y_{t-1}y_{t-d}, \dots, y_{t-p}y_{t-d}, y_{t-1}y_{t-d}^2, \dots, y_{t-p}y_{t-d}^2, y_{t-1}y_{t-d}^3, \dots, y_{t-p}y_{t-d}^3\}.$$

Step 3: Compute the test statistic $LM = TR^2$, where T is the sample size and R^2 is the coefficient of determination from the second-stage regression. Under the null hypothesis, LM is asymptotically distributed as χ^2 with $3p$ degrees of freedom.

Three other variants of a test based on equation (19.10) are obtained by imposing the following restrictions.

Variant 1: $\pi_{2,i} = \pi_{3,i} = 0, \forall i$.

Variant 2: $\pi_{3,i} = 0, \forall i$.

Variant 3: $\pi_{2,i} = 0, \forall i$.

These alternative versions of the test may improve the power properties of the test if the lower-order terms in the Taylor series expansion in (19.9) are sufficient to capture the nonlinearities in the model. This may occur, for example, if the weighting function in (19.6) is of the ESTAR type, a property which is developed in the exercises.

Example 19.6 Sampling Properties of the LST Linearity Test

Table 19.2 gives the finite sample performance of the linearity test based on (19.10) using a small Monte Carlo experiment. The data generating process is identical to that used by Luukkonen, Saikkonen and Teräsvirta (1988) and is given by

$$y_t = -0.5y_{t-1} + \beta y_{t-1} w_t + v_t$$

$$w_t = \begin{cases} \Phi(\gamma(y_{t-1} - 0)) \\ (1 + \exp(-\gamma(y_{t-1} - 0)))^{-1}, \end{cases}$$

where v_t is *iid* $N(0, 25)$. The size of the test is 4.36%, given in the column headed $\beta = 0$, which is close to the nominal size of 5%. The size-adjusted power function increases monotonically as β becomes more positive. This pattern holds for all values of γ , although the power tends to taper off for larger values of γ . $\square$

In the case where the delay parameter d is unknown, the regression equation in (19.10) includes the intermediate lags of y_{t-d}^2 as well. Assuming $d = p$ for simplicity, the augmented regression equation is

$$y_t = \pi + \sum_{i=1}^p \pi_{0,i} y_{t-i} + \sum_{i=1}^p \sum_{j=1}^p \pi_{1,i,j} y_{t-i} y_{t-j} + \sum_{i=1}^p \sum_{j=1}^p \pi_{2,i,j} y_{t-i} y_{t-j}^2$$

$$+ \sum_{i=1}^p \sum_{j=1}^p \pi_{3,i,j} y_{t-i} y_{t-j}^3 + u_t. \quad (19.11)$$

Table 19.2

Sampling properties of the linearity test in (19.10), for a sample of $T = 100$ and 10000 replications. The null hypothesis is $\beta = 0$, which gives the size of the test (%) based on a nominal size of 5%. The alternative hypothesis is $\beta > 0$, which gives the size-adjusted power of the test (%).

	w_{t-1} is STAR			w_{t-1} is LSTAR		
	$\beta = 0.0$	$\beta = 0.5$	$\beta = 1.0$	$\beta = 0.0$	$\beta = 0.5$	$\beta = 1.0$
$\gamma = 0.5$	4.360	24.480	67.930	4.360	25.230	70.550
$\gamma = 1.0$	4.360	22.880	63.320	4.360	24.080	66.560
$\gamma = 5.0$	4.360	22.200	61.170	4.360	22.290	61.310

A potential problem with this auxiliary regression is that the presence of a large number of regressors may result in a loss in power of the test statistic. One solution is to exclude the second-order terms by setting $\pi_{2,i,j} = 0$ so that, under the null hypothesis, the test is based on the restrictions $\pi_{1,i,j} = \pi_{3,i,j} = 0$ with $i, j = 1, 2, \dots, p$. This version of the test statistic has a χ^2 distribution with $p(p+1)/2 + p$ degrees of freedom.

19.4 Artificial Neural Networks

Artificial neural networks (ANNs) provide a flexible framework in which to approximate nonlinear processes. In their simplest form, they represent a threshold time series model in which the weighting function in (19.6) is now referred to as a squasher. This colourful choice of terminology stems from engineering where ANNs were initially developed. The aim of this section, following Creedy and Martin (1994), is to couch a discussion of ANNs in an econometric framework of model specification, estimation and testing. Further discussion is given in Kuan and White (1994) and Lee, White and Granger (1993).

19.4.1 Specification

Consider the nonlinear time series model

$$y_t = \phi y_{t-1} + \beta \left[\frac{1}{1 + \exp(-(\delta_0 + \delta_1 y_{t-1}))} \right] + v_t, \quad (19.12)$$

where v_t is an *iid* disturbance term. This is a special case of the TAR model in equation (19.5) with lag $p = 1$ and weighting function in (19.6) given

by the logistic function. In artificial neural network parlance, the nonlinear time series model in (19.12) is known as a hidden-layer, feed-forward artificial neural network. The hidden layer refers to the term $\delta_0 + \delta_1 y_{t-1}$, because it provides for an additional linkage between y_t and y_{t-1} that is nested/hidden within the logistic function. Feed forward indicates that the model uses information at time $t - 1$ to make predictions of the process at t . The artificial neural network component is given by the parameters δ_0 and δ_1 which attempt to model the brain. The logistic function, or squasher, dampens the impact on large movements in the lagged values of y_t . The logistic function is the most commonly used squasher. Some alternatives are the cumulative normal distribution and the hyperbolic tangent.

Example 19.7 Decomposition of an Artificial Neural Network

Figure 19.7 decomposes the conditional mean of the ANN in equation (19.12) into a linear and a nonlinear component

$$\begin{aligned} \text{Linear} & : \phi y_{t-1} \\ \text{Nonlinear} & : \beta \frac{1}{1 + \exp(-(\delta_0 + \delta_1 y_{t-1}))} \\ \text{Total} & : \phi y_{t-1} + \beta \frac{1}{1 + \exp(-(\delta_0 + \delta_1 y_{t-1}))} . \end{aligned}$$

The nonlinear prediction ranges from 0 to $\beta = 2$ since the parameter β has the effect of stretching the unit-interval domain of the logistic function. The parameter δ_0 causes the nonlinear function to shift to the right, while the parameter δ_1 controls the steepness of the nonlinear function around its point of inflection. $\square$

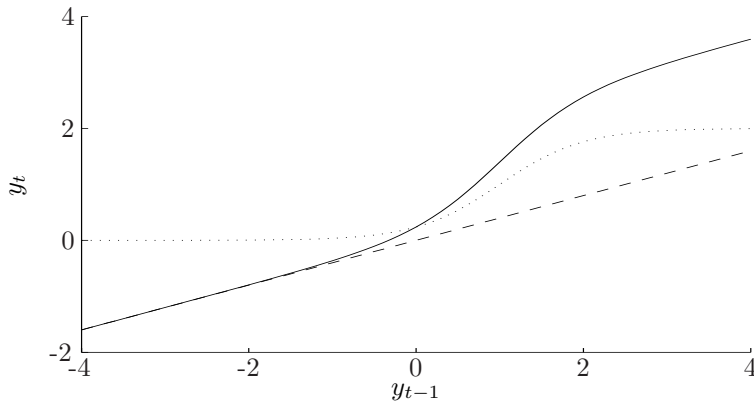


Figure 19.7 Decomposition of the predictions of an ANN (solid line) into linear (dashed line) and nonlinear (dotted line) components, based on equation (19.12). Parameter values are $\phi = 0.4$, $\beta = 2.0$, $\delta_0 = -2$ and $\delta_1 = 2$.

The specification of the model in (19.12) is commonly referred to as architecture. Other forms of architecture (model specifications) consist of including additional lags, additional logistic functions, different types of nonlinear functions and additional variables. For example, the following hidden-layer feed-forward artificial neural network has four lags and three logistic functions, with two of the logistic functions feeding into a third logistic function

$$y_t = \sum_{i=1}^4 \phi_i y_{t-i} + \beta \left[\frac{1}{1 + \exp(-(\delta_{0,0} + \delta_{0,1} w_{1,t} + \delta_{0,2} w_{2,t}))} \right] + v_t, \quad (19.13)$$

where

$$w_{j,t} = \left[\frac{1}{1 + \exp(-(\delta_{j,0} + \sum_{i=1}^4 \delta_{j,i} y_{t-i}))} \right], \quad j = 1, 2.$$

Table 19.3

Artificial neural network terminology.

Terminology	Translation
Architecture	Specification
Target	Dependent variable, y_t
Input	Explanatory variable, y_{t-1}
Output	Predictor, $E_{t-1}[y]$
Squasher	Logistic function
Hidden layer	Equation inside the squasher, $\delta_0 + \delta_1 y_{t-1}$
Neurodes	The unknown parameters, $\theta = \{\phi, \beta, \delta_0, \delta_1\}$
Bias	The parameter δ_0 in the logistic function
Connection strengths	The parameter δ_1 in the logistic function
Training, Learning	Estimation
Fitness function	Objective function used in estimation
Tolerance	Convergence criteria
Epochs	Number of iterations used in estimation
Learning rate	Line search parameter used in estimation
Momentum	Backstep parameter used in estimation

The econometrics equivalents of other commonly-used engineering terms used in the context of ANNs is given in Table 19.3. The unknown parameters are called nodes (or neurodes). The parameter δ_0 in (19.12) is called the bias as it pushes the point of inflection of the logistic function away from zero. The parameter δ_1 in (19.12) is called the connection strength since it controls the influence of the lag y_{t-1} on the squasher and hence the output (predictor).

19.4.2 Estimation

Maximum Likelihood

For a sample of $t = 1, 2, \dots, T$ observations, the unknown parameters of the ANN given in equation (19.12), $\theta = \{\phi, \beta, \delta_0, \delta_1, \sigma^2\}$ can be estimated by maximum likelihood methods. Assuming that the disturbance v_t in (19.13) is $N(0, \sigma^2)$ and the maximum lag length is $p = 1$, the conditional log-likelihood function is

$$\ln L_T(\theta) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2(T-1)} \sum_{t=2}^T (y_t - \phi y_{t-1} - \beta w_t)^2, \quad (19.14)$$

where

$$w_t = \frac{1}{1 + \exp(-(\delta_0 + \delta_1 y_{t-1}))}.$$

This function is maximized using one of the iterative algorithms discussed in Chapter 3.

In most applications of ANNs, there is no explicit discussion of estimation issues. Instead the terminology refers to presenting the data to the neural network and the ANN being trained to learn the unknown structure underlying the dynamic processes connecting the target to the inputs. Learning is based on a fitness function as well as additional controls for tolerance, momentum and the number of epochs. Table 19.3 shows that training and leaning are just synonyms for estimation and the fitness function is simply the log-likelihood function. Learning arises because starting with some initial parameter values, $\theta_{(0)}$, does not generally yield good outputs (predictors) of the target (dependent) variable. As the algorithm is an iterative procedure, at the next step the parameters are updated, thereby yielding better predictors. The number of iterations required to achieve convergence is referred to as the number of epochs and the convergence criteria used is called the tolerance.

Back Propagation

The gradient algorithms discussed in Chapter 3 to estimate the unknown parameters of an econometric model are based on first and second derivatives. Although these algorithms are well-known and widely used in econometrics, they tend not to be widely adopted in ANN applications. Instead the so-called back-propagation algorithm is most commonly used. The representative iteration of this algorithm is given by

$$\theta_{(k)} = \theta_{(k-1)} - \eta \left. \frac{\partial \ln l_t(\theta)}{\partial \theta} \right|_{\theta=\theta_{(k-1)}} + v(\theta_{(k-1)} - \theta_{(k-2)}), \quad (19.15)$$

where $\theta_{(k)}$ is the parameter vector at iteration k , $\ln l_t(\theta)$ is the log-likelihood function at observation t , η is the learning rate and v is the momentum. The latter two parameters are used to aid convergence and correspond to line searching and back stepping respectively (see Table 19.3). White (1989) and Kuan and White (1994) show that by choosing the learning parameter as

$$\eta \propto T^{-k} \quad 0.5 < k \leq 1, \quad (19.16)$$

$\theta_{(k)}$ converges almost surely to θ_0 .

The most notable features of the back-propagation algorithm are that it uses first derivatives only and that its iterations correspond to scrolling through the sample evaluating the log-likelihood function at time t only and not summing over all t as would be the case with standard gradient algorithms. This suggests that the algorithm is computationally less efficient than gradient algorithms that use additional information in the form of second derivatives.

Neural Methods

For certain ANNs, the number of neurodes (unknown parameters) can be extremely large. In fact, it is not uncommon in some applications for the number of neurodes to exceed the number of observations, T . Apart from the potential problems of imprecision in the parameter estimates in trying to train an ANN with this number of neurodes, a gradient algorithm used to compute the maximum likelihood estimates can be expected to breakdown as a result of multicollinearity causing the information matrix to be singular. Even for the ANN given by equation (19.13), a gradient algorithm can still be expected to have problems in achieving convergence.

One approach that is used in training the ANN, is to choose values of the δ_i neurodes (parameters) from a random number generator. The motivation for doing this is that squashes with many neurodes can begin to mimic the brain which has millions of neurons. One advantage of this approach is that the ANN in equation (19.12), for example, now becomes linear in the remaining parameters

$$y_t = \phi y_{t-1} + \beta w_t^{(1)} + v_t,$$

where

$$w_t^{(1)} = \left[\frac{1}{1 + \exp(-(\delta_0^{(1)} + \delta_1^{(1)} y_{t-1}))} \right],$$

and $\delta_0^{(1)}$ and $\delta_1^{(1)}$ are random draws from the first step. The parameters ϕ and β are then estimated by regressing y_t on $\{y_{t-1}, w_t^{(1)}\}$. The procedure is

repeated for an another draw of δ_0 and δ_1 , with ϕ and β once again estimated by OLS. After performing the algorithm N times, the parameter estimates are chosen as the values that minimize the fitness function.

19.4.3 Testing

Under the regularity conditions discussed in Chapter 2, the maximum likelihood estimator $\hat{\theta}$ has an asymptotic normal distribution

$$\hat{\theta} \stackrel{a}{\sim} N(\theta_0, I^{-1}(\theta_0)) .$$

In general, this asymptotic distribution can be used as the basis of undertaking tests on the parameters of the ANN. However, as already observed in the case of the threshold time series class of models, care needs to be exercised when performing hypothesis tests when some parameters are not identified under the null hypothesis. This problem arises in testing for a hidden layer based on the hypotheses

$$H_0 : \beta = 0 \quad H_1 : \beta \neq 0.$$

Following from the discussion of the threshold time series models this also constitutes a natural test of nonlinearity as the ANN in equation (19.12) reduces to a linear model under the null hypothesis.

To circumvent the issue of some parameters not being identified under the null hypothesis, one solution is to condition the model on values of δ_0 and δ_1 , which are drawn randomly from a specified distribution. Given values for these parameters, the first-order conditions for ϕ and β under H_0 are used to construct a LM test similar to the approach adopted to test for linearity in the threshold time series model. This approach is suggested by Lee, Granger and White (1993), who choose a rectangular distribution with range $[-2, 2]$ to draw the random numbers for δ_0 and δ_1 . The steps of the LM test are as follows.

Step 1: Estimate the (linear) model under the null by regressing y_t on y_{t-1} and extract the OLS residuals $\hat{v}_t$.

Step 2: Choose δ_0, δ_1 randomly from a rectangular distribution in the range $[-2, 2]$, say $\hat{\delta}_0, \hat{\delta}_1$.

Step 3: Construct the activation variable(s)

$$\hat{w}_t = \left[\frac{1}{1 + \exp(-(\hat{\delta}_0 + \hat{\delta}_1 y_{t-1}))} \right] .$$

Step 4: Regress $\hat{v}_t$ on $\{y_{t-1}, \hat{w}_{t-1}\}$ and extract the coefficient of determination R^2 , from this regression.

Step 5: Construct the test statistic $LM = TR^2$, where T is the sample size.

Step 6: Reject the null hypothesis for $LM > \chi_1^2$.

The intuition behind this test is that any important nonlinearity in the data excluded at the first stage (Step 1), will show up in the second-stage regression (Step 4) in the form of a high value for R^2 .

19.5 Bilinear Time Series Models

Bilinear times series models constitute a class of nonlinear models where the linear ARMA model is augmented by the product of the AR and MA terms (Granger and Anderson, 1978; Subba Rao and Gabr, 1984). These models are able to generate a wide range of time series patterns including bursting behaviour as observed in stock market bubbles.

19.5.1 Specification

Consider the ARMA(1, 1) model

$$y_t = \phi y_{t-1} + v_t + \psi v_{t-1}, \quad (19.17)$$

where v_t is an *iid* disturbance. From Chapter 13, values of $|\phi| < 1$ result in the model exhibiting uniform variation around its unconditional mean $E[y_t] = 0$. Now consider the introduction of a nonlinear term $y_{t-1}v_{t-1}$, which is simply the product of the AR(1) and MA(1) components of the model

$$y_t = \phi y_{t-1} + v_t + \psi v_{t-1} + \gamma y_{t-1}v_{t-1}. \quad (19.18)$$

This additional component is known as the bilinear term, which has the effect of generating a range of interesting nonlinear features that cannot be generated from the linear ARMA(1, 1) model in (19.17).

Example 19.8 Properties of the Bilinear Time Series Model

Simulated time series of size $T = 200$ from the bilinear model

$$y_t = 0.4y_{t-1} + v_t + 0.2v_{t-1} + \gamma y_{t-1}v_{t-1}, \quad v_t \text{ iid} \sim N(0, 1),$$

are given in Figure 19.8 for alternative values of the bilinear parameter γ . For values of $\gamma = 0$ and $\gamma = 0.4$, y_t exhibits the typical pattern of a linear ARMA(1,1) model. Increasing γ further causes the model to exhibit bursting ($\gamma = 0.8$) and extreme bursting ($\gamma = 1.2$) behaviour. $\square$

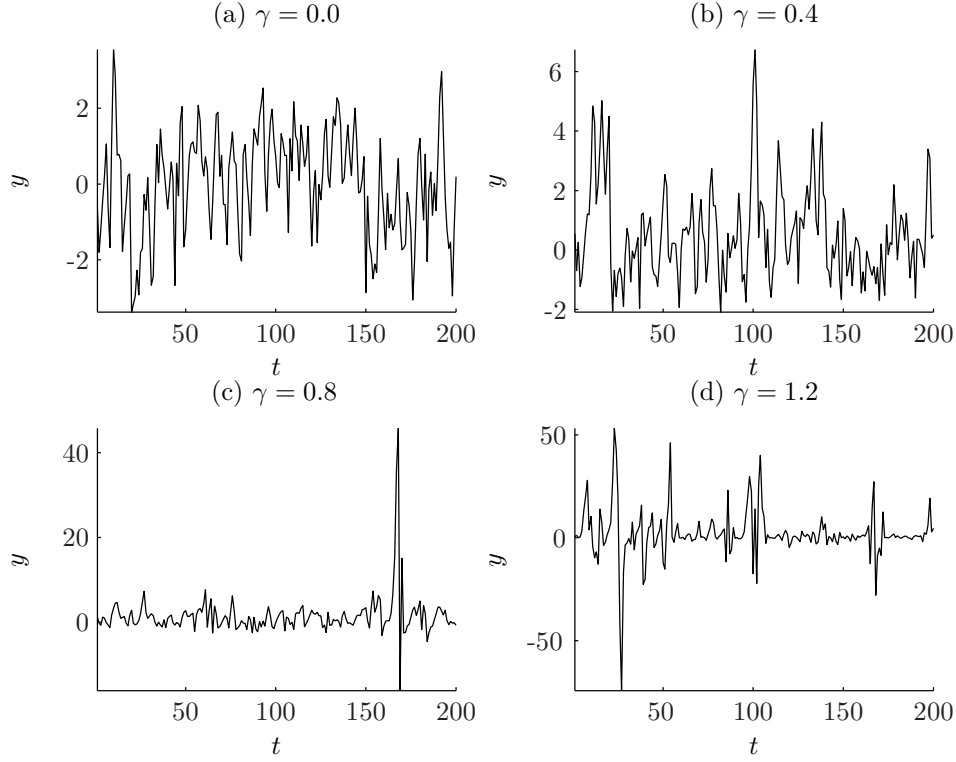


Figure 19.8 Simulated bilinear time series models for different values of γ with $T = 200$, AR parameter 0.4 and MA parameter 0.2.

The specification of the bilinear model in (19.18) is easily extended for longer lags, alternative bilinear product terms and higher dimensions. An example of a bivariate bilinear model with two lags is

$$\begin{aligned} y_{1,t} &= \phi_{1,1}y_{1,t-1} + \phi_{1,2}y_{1,t-2} + v_{1,t} + \psi_{1,1}v_{1,t-1} + \gamma_{2,1}y_{1,t-1}v_{2,t-1} \\ y_{2,t} &= \phi_{2,1}y_{2,t-1} + \phi_{2,2}y_{2,t-2} + v_{2,t} + \psi_{2,1}v_{2,t-1} + \gamma_{2,2}y_{1,t-1}v_{1,t-1}. \end{aligned}$$

The terms $y_{1,t-1}v_{2,t-1}$ and $y_{1,t-1}v_{1,t-1}$ capture alternative types of spillovers that may represent a way to model contagion (Dungey and Martin, 2007).

19.5.2 Estimation

Estimation of a bilinear time series model is easily handled by maximum likelihood methods using the conditional log-likelihood function. In the case of the bilinear model in (19.18) where the disturbance terms is $N(0, \sigma^2)$, the number of lags is $p = q = 1$ and the unknown parameters are $\theta =$

$\{\phi, \psi, \gamma, \sigma^2\}$, the conditional log-likelihood function for a sample of $t = 1, 2, \dots, T$ observations is

$$\ln L_T(\theta) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma^2 - \frac{1}{2\sigma^2(T-1)} \sum_{t=2}^T (y_t - \phi y_{t-1} - \psi v_{t-1} - \gamma y_{t-1} v_{t-1})^2.$$

This function is maximized with respect to θ using one of the algorithms discussed in Chapter 3. As an alternative iterative procedure, the Gauss-Newton algorithm discussed in Chapter 6 can also be used because the log-likelihood function is expressed in terms of a sum of squared disturbances.

19.5.3 Testing

A special case of the bilinear model is the ARMA(p, q) model. This suggests that the hypothesis testing methods discussed in Chapter 4 can be used to construct general tests of bilinearity. In the case where there is no moving average term, the regression form of the LM given in Chapter 6 is particularly simple to implement. For example, consider the bilinear model

$$y_t = \phi y_{t-1} + v_t + \gamma y_{t-1} v_{t-1}, \quad (19.19)$$

where v_t is *iid* $N(0, \sigma^2)$, a test of bilinearity is based on the following hypotheses

$$H_0 : \gamma = 0, \quad H_1 : \gamma \neq 0.$$

To implement an LM test of bilinearity, the steps are as follows.

Step 1: Rewrite (19.19) as

$$v_t = y_t - \phi y_{t-1} - \gamma y_{t-1} v_{t-1},$$

and compute the derivatives

$$\begin{aligned} z_{1,t} &= -\frac{\partial v_t}{\partial \phi} = y_{t-1} \gamma + y_{t-1} \frac{\partial v_{t-1}}{\partial \phi} \\ z_{2,t} &= -\frac{\partial v_t}{\partial \gamma} = y_{t-1} v_{t-1} + \gamma y_{t-1} \frac{\partial v_{t-1}}{\partial \gamma}. \end{aligned}$$

Step 2: Evaluate the expressions in Step 1 under the null, $\gamma = 0$

$$\begin{aligned} v_t &= y_t - \phi y_{t-1} \\ z_{1,t} &= y_{t-1} \\ z_{2,t} &= y_{t-1} v_{t-1}. \end{aligned} \quad (19.20)$$

- Step 3:** Estimate the model under the null hypothesis by regressing y_t on y_{t-1} and extract the OLS residuals $\hat{v}_t$.
- Step 4:** Regress $\hat{v}_t$ on $\{y_{t-1}, y_{t-1}\hat{v}_{t-1}\}$ and compute R^2 .
- Step 4:** Form the LM statistic $LM = TR^2$, where T is the sample size, and compare with χ_1^2 distribution. The null hypothesis is rejected for large values of the test statistic because this result implies that the bilinear component, $y_{t-1}v_{t-1}$, has been excluded from the first-stage regression.

19.6 Markov Switching Model

The Markov switching model is widely used to model nonlinearities in economic and financial processes. Examples include business cycles (Hamilton, 1989), the term structure of interest rates (Hamilton, 1988), exchange rates (Engel and Hamilton, 1990), stock returns (Hamilton and Susmel, 1994) and business cycle factor models (Kim, 1994). This model has much in common with the threshold time series models discussed previously, except in those models switching is deterministic whereas in the Markov switching model it is stochastic. This model is also similar to the latent factor models analyzed in Chapter 15 with the minor difference that the factor is discrete in the present context.

Let w_t , represent a stochastic weighting variable that switches between regimes according to

$$w_t = \begin{cases} 1 & : \text{Regime 1} \\ 0 & : \text{Regime 2.} \end{cases} \quad (19.21)$$

Unlike the threshold time series models where the weighting function is identified by expressing it in terms of lagged observed variables, w_t is not observable. Moreover, w_t is binary for the Markov switching model compared to the weighting function in the threshold models where it takes on intermediate values in the unit circle. The model of y_t is specified as

$$\begin{aligned} y_t &= \alpha + \beta w_t + u_t \\ \sigma_t^2 &= \gamma + \delta w_t \\ u_t &\sim N(0, \sigma_t^2) \\ p &= P[w_t = 1 \mid w_{t-1} = 1, y_{t-1}, y_{t-2}, \dots] \\ q &= P[w_t = 0 \mid w_{t-1} = 0, y_{t-1}, y_{t-2}, \dots], \end{aligned} \quad (19.22)$$

where $\theta = \{\alpha, \beta, \gamma, \delta, p, q\}$ are unknown parameters and u_t is a heteroskedastic disturbance. In regime 1 the mean is $E[y_t] = \alpha + \beta$ with variance $\gamma + \delta$, whereas in regime 2 the mean is $E[y_t] = \alpha$ with variance γ . The weighting

variable is specified in terms of the parameters p and q which represent the conditional probabilities of staying in each regime.

The model is easily extended to allow for p lags by writing the first equation in (19.22) as

$$y_t = \alpha + \beta w_t + \sum_{i=1}^p \phi_i (y_{t-i} - \alpha - \beta w_{t-i}) + u_t.$$

Further extensions consist of specifying y_t as a vector of variables, and allowing for time-varying conditional probabilities p and q by expressing these probabilities as functions of explanatory variables. This extension is investigated in Exercise 6.

Estimation of the model by maximum likelihood requires the joint conditional distribution $f(y_t, w_t | y_{t-1}, y_{t-2}, \dots; \theta)$, which is complicated by the fact that w_t is not observed. To circumvent this problem, it is necessary to integrate, or more correctly sum, out w_t , thereby expressing the likelihood just in terms of the observable variable y_t . A similar strategy is adopted for the stochastic frontier model in Chapter 6. As w_t is binary this strategy is straightforward as the summation involves just two terms. From the rules of probability, the marginal distribution of y_t is given by

$$\begin{aligned} f(y_t) &= P[w_t = 1, y_t] + P[w_t = 0, y_t] \\ &= P[w_t = 1] f(y_t | w_t = 1) + P[w_t = 0] f(y_t | w_t = 0), \end{aligned}$$

which shows that the marginal is obtained by summing over the joint distribution which, in turn, is expressed in terms of the product of the marginal and conditional distributions. This expression is re-expressed by making conditioning on past values of y_t explicit

$$f(y_t | y_{t-1}, y_{t-2}, \dots) = w_{1,t|t-1} f_{1,t|t-1} + w_{0,t|t-1}, \quad (19.23)$$

where

$$w_{i,t|t-1} = P[w_t = i | y_{t-1}, y_{t-2}, \dots], f_{i,t|t-1} = f(y_t | w_t = i, y_{t-1}, y_{t-2}, \dots), i = 1, 2$$

Equation (19.23) is the likelihood function at time t which now excludes w_t and has the form of a mixture distribution with weights that vary over the sample.

To evaluate the likelihood function in (19.23), from equation (19.22)

$$\begin{aligned} f_{1,t|t-1} &= \frac{1}{\sqrt{2\pi(\gamma + \delta)}} \exp \left[-\frac{(y_t - \alpha - \beta)^2}{2(\gamma + \delta)} \right] \\ f_{0,t|t-1} &= \frac{1}{\sqrt{2\pi\gamma}} \exp \left[-\frac{(y_t - \alpha)^2}{2\gamma} \right]. \end{aligned} \quad (19.24)$$

To derive expressions for $w_{1,t|t-1}$ and $w_{0,t|t-1}$ from equation (19.23) the updated conditional probabilities based on observing the dependent variable at time t , are

$$\begin{aligned} w_{1,t|t} &= P[w_t = 1 | y_t, y_{t-1}, y_{t-2}, \dots] = \frac{w_{1,t|t-1} f_{1,t|t-1}}{w_{1,t|t-1} f_{1,t|t-1} + w_{0,t|t-1}} \\ w_{0,t|t} &= P[w_t = 0 | y_t, y_{t-1}, y_{t-2}, \dots] = \frac{w_{0,t|t-1} f_{0,t|t-1}}{w_{1,t|t-1} f_{1,t|t-1} + w_{0,t|t-1}}. \end{aligned} \quad (19.25)$$

Using the definition of p and q in equation (19.22), it follows that

$$\begin{bmatrix} w_{1,t+1|t} \\ w_{0,t+1|t} \end{bmatrix} = \begin{bmatrix} p & 1-q \\ 1-p & q \end{bmatrix} \begin{bmatrix} w_{1,t|t} \\ w_{0,t|t} \end{bmatrix}. \quad (19.26)$$

For a sample of $t = 1, 2, \dots, T$ observations, combining equations (19.23) and (19.24) the log-likelihood function is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \ln f(y_t | y_{t-1}, y_{t-2}, \dots; \theta) = \frac{1}{T} \sum_{t=1}^T \ln(w_{1,t|t-1} f_{1,t|t-1} + w_{0,t|t-1}), \quad (19.27)$$

where $w_{i,t|t-1}$ and $f_{i,t|t-1}$ are defined in equations (19.26) and (19.24), respectively. This function is maximized with respect to θ using the iterative algorithms presented in Chapter 3. At $t = 1$, it is necessary to be able to evaluate $w_{1,1|0}$ and $w_{0,1|0}$, which are set equal to their respective stationary probabilities

$$w_{1,1|0} = \frac{1-q}{1-p+1-q}, \quad w_{0,1|0} = \frac{1-p}{1-p+1-q}.$$

Example 19.9 Markov Switching Model of the Business Cycle

A Markov switching model of the business cycle is specified where

$$w_t = \begin{cases} 1 & : \text{Expansionary phase of the business cycle} \\ 0 & : \text{Contractionary phase of the business cycle.} \end{cases}$$

Using percentage quarterly growth rates on U.S. GNP from June 1951 to December 1984, given in Figure 19.9, the log-likelihood function in (19.27) is maximized with respect to θ , with the parameter estimates reported in Table 19.4. To ensure that the conditional probabilities p and q are in the unit interval these parameters are restricted using the logistic transformation

$$p = \frac{1}{1 + \exp(-\kappa)}, \quad q = \frac{1}{1 + \exp(-\lambda)},$$

with (19.27) maximized with respect to $\theta = \{\alpha, \beta, \gamma, \delta, \kappa, \lambda\}$. The unrestricted parameter estimates are then obtained by maximizing (19.27) with respect to $\theta = \{\alpha, \beta, \gamma, \delta, p, q\}$.

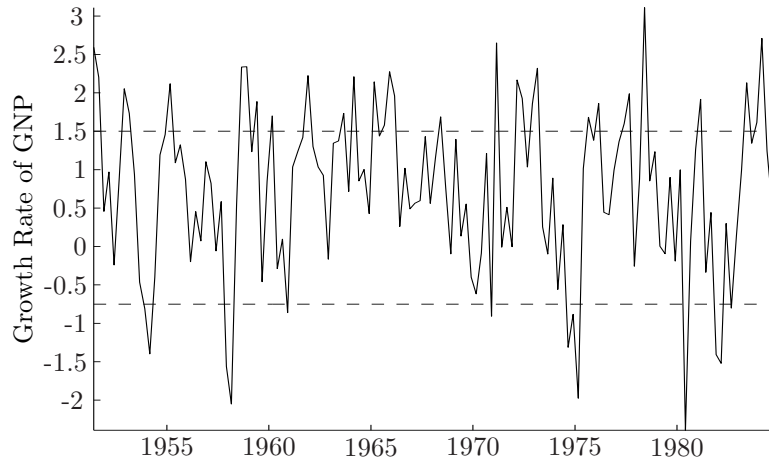


Figure 19.9 Quarterly percentage growth rate of U.S. output, June 1951 to December 1984. Also given are maximum likelihood estimates of expansionary and contractionary average growth rates. Source: Hamilton (1989).

Table 19.4

Maximum likelihood parameter estimates of the Markov switching model of the U.S. business cycle. Quasi-maximum likelihood standard errors with $P = 0$ lags.

Parameter	Estimate	Std error	p-value
α	-0.224	0.417	0.591
β	1.401	0.283	0.000
γ	0.942	0.223	0.000
δ	-0.323	0.275	0.241
p	0.892	0.055	0.000
q	0.753	0.130	0.000

The average growth rate in the expansionary phase of the business cycle is $\hat{\alpha} + \hat{\beta} = -0.224 + 0.283 = 0.059\%$ per quarter, compared to the average growth rate in the contractionary phase of $\hat{\alpha} = -0.224\%$ per quarter. The expansionary phase exhibits smaller variance $\hat{\gamma} + \hat{\delta} = 0.942 - 0.323 = 0.620$, compared to $\hat{\gamma} = 0.942$ in the contractionary period. Expansions tend to

last longer than contractions with $\hat{p} = 0.892$ greater than $\hat{q} = 0.753$. An estimate of the average duration of each phase is, respectively, (Hamilton, 1989, p.374)

$$\begin{aligned}\text{DURATION(Expansion)} &= \frac{1}{1 - \hat{p}} = \frac{1}{1 - 0.892} = 9.270 \text{ quarters} \\ \text{DURATION(Contraction)} &= \frac{1}{1 - \hat{q}} = \frac{1}{1 - 0.753} = 4.050 \text{ quarters.}\end{aligned}$$

□

19.7 Nonparametric Autoregression

The nonlinear time series models discussed in this chapter are based on parametric specifications which are primarily designed to capture specific types of nonlinearities. A nonparametric approach allows for general nonlinear structures by specifying the model

$$y_t = m(y_{t-k}) + v_t, \quad (19.28)$$

where $m(y_{t-k})$ is the conditional mean with $k > 0$ and v_t is an *iid* disturbance term. The Nadaraya-Watson kernel regression estimator, discussed in Chapter 11, of the conditional mean is given by

$$\hat{m}(y_{t-k}) = \frac{1}{T} \sum_{t=1}^T y_t w_t, \quad (19.29)$$

where w_t is the weight at time t , which, in the case of a Gaussian kernel with bandwidth h , is

$$w_t = \frac{K\left(\frac{y - y_{t-k}}{h}\right)}{\sum_{t=1}^T K\left(\frac{y - y_{t-k}}{h}\right)} = \frac{\frac{1}{\sqrt{2\pi h^2}} \exp\left[-\left(\frac{y - y_{t-k}}{h}\right)^2\right]}{\sum_{t=1}^T \frac{1}{\sqrt{2\pi h^2}} \exp\left[-\left(\frac{y - y_{t-k}}{h}\right)^2\right]}, \quad k \geq 1. \quad (19.30)$$

By computing equation (19.29) for each k , a nonparametric autocorrelation function of a nonlinear time series model is constructed. Parametric information can also be included by specifying a parametric form for the distribution of v_t in (19.28), or for the conditional mean, $m(y_{t-k})$. Chapter 11 provides an example of a semi-parametric estimator of a nonlinear time series model.

Example 19.10 Estimating a Nonlinear AR(1) Model

The nonlinear autoregressive model

$$y_t = \theta |y_{t-1}| + \sqrt{1 - \theta^2} z_t, \quad z_t \sim N(0, 1),$$

where z_t is an *iid* disturbance, is simulated for $T = 5000$ observations with $\theta = 0.8$. Figure 19.10 shows the nonparametric estimate of the conditional mean based on a Gaussian kernel using a rule-of-thumb bandwidth. The true conditional mean given by $m(y_{t-1}) = \theta |y_{t-1}|$ is also shown together with the conditional mean of a linear AR(1) model, θy_{t-1} . The nonparametric estimator tracks the linear conditional mean for $y_{t-1} > 0$, but also captures the nonlinearity of the model for $y_{t-1} < 0$. $\square$

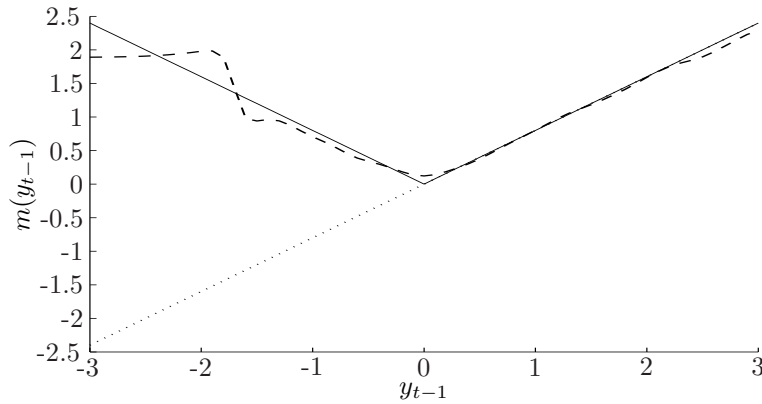


Figure 19.10 Nonparametric estimate of a nonlinear AR(1) model. The nonparametric estimate (dashed line) is compared to a linear AR(1) model (dotted line) and the true conditional mean (solid line).

19.8 Nonlinear Impulse Responses

The impulse response functions discussed in Chapter 13 show how a shock in a variable, y_t , is propagated through all of the variables in the model. Because the impulse response functions are discussed in the context of linear time series models, the impact of a shock on y_t has two key properties.

- (i) The shock has a symmetric effect: apart from the sign, positive and negative shocks have the same impact.
- (ii) The effect of the shock is independent of the history of the process.

To highlight the properties of impulse responses of linear time series models, consider a linear univariate AR(1) model

$$y_t = \phi_1 y_{t-1} + v_t, \quad (19.31)$$

where v_t is an *iid* disturbance term. Writing the model out at time $t + 1$ gives

$$y_{t+1} = \phi_1 (\phi_1 y_{t-1} + v_t) + v_{t+1} = \phi_1^2 y_{t-1} + \phi_1 v_t + v_{t+1}. \quad (19.32)$$

Choosing a shock at time t of size $v_t = \delta$, setting $v_{t+1} = 0$ and conditioning on information at time $t - 1$, yields the conditional expectations with and without the shock as

$$\begin{aligned} E_{t-1} [y_{t+1} | v_{t+1} = 0, v_t = \delta] &= \phi_1^2 y_{t-1} + \phi_1 \times \delta + 0 \\ E_{t-1} [y_{t+1} | v_{t+1} = 0, v_t = 0] &= \phi_1^2 y_{t-1} + \phi_1 \times 0 + 0, \end{aligned} \quad (19.33)$$

respectively. The difference of the two conditional expectations gives the impulse response at $t + 1$

$$\begin{aligned} IRF_1 &= E_{t-1} [y_{t+1} | v_{t+1} = 0, v_t = \delta] - E_{t-1} [y_{t+1} | v_{t+1} = 0, v_t = 0] \\ &= \phi_1 \delta. \end{aligned} \quad (19.34)$$

Inspection of (19.34) shows that the impulse response function is both symmetric, because the impulse response is a linear function of the shock, δ , and independent of the history of y_t .

The properties of symmetry and independence that characterize impulse response functions of linear time series models do not, in general, arise in nonlinear time series models. This point is highlighted in Figure 19.10 of Example 19.10 where the relationship between y_t and y_{t-1} is characterized by the absolute function $|y_{t-1}|$. A positive shock in y results in an increase in y the next period if $y_{t-1} > 0$ whereas the effect of the same shock yields a decrease in y the next period if $y_{t-1} < 0$. In contrast, for the linear model in Figure 19.10 a positive shock in y results in an increase in y regardless of the value of y when the shock occurs.

To demonstrate the features of impulse responses of nonlinear time series models more formally, consider the nonlinear model

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-1}^2 + v_t, \quad (19.35)$$

which reduces to the linear time series model in (19.31) for $\phi_2 = 0$. Proceeding as before, express the model at time $t + 1$

$$y_{t+1} = \phi_1 (\phi_1 y_{t-1} + \phi_2 y_{t-1}^2 + v_t) + \phi_2 (\phi_1 y_{t-1} + \phi_2 y_{t-1}^2 + v_t)^2 + v_{t+1}.$$

Choosing a shock of $v_t = \delta$, setting $v_{t+1} = 0$ and conditioning on information at time $t - 1$, yields the conditional expectations with and without the shock

respectively, as

$$\begin{aligned} E_{t-1} [y_{t+1} | v_{t+1} = 0, v_t = \delta] &= \phi_1 (\phi_1 y_{t-1} + \phi_2 y_{t-1}^2 + \delta) \\ &\quad + \phi_2 (\phi_1 y_{t-1} + \phi_2 y_{t-1}^2 + \delta)^2 + 0 \\ E_{t-1} [y_{t+1} | v_{t+1} = 0, v_t = 0] &= \phi_1 (\phi_1 y_{t-1} + \phi_2 y_{t-1}^2 + 0) \\ &\quad + \phi_2 (\phi_1 y_{t-1} + \phi_2 y_{t-1}^2 + 0)^2 + 0. \end{aligned}$$

The impulse response function is therefore

$$\begin{aligned} IRF_1 &= E_{t-1} [y_{t+1} | v_{t+1} = 0, v_t = \delta] - E_{t-1} [y_{t+1} | v_{t+1} = 0, v_t = 0] \\ &= \phi_1 \delta + \phi_2 \left((\phi_1 y_{t-1} + \phi_2 y_{t-1}^2 + \delta)^2 - (\phi_1 y_{t-1} + \phi_2 y_{t-1}^2 + 0)^2 \right) \\ &= \underbrace{\phi_1 \delta}_{\text{linear}} + \underbrace{\delta^2 \phi_2 + 2\delta \phi_1 \phi_2 y_{t-1} + 2\delta \phi_2^2 y_{t-1}^2}_{\text{nonlinear}}. \end{aligned} \tag{19.36}$$

This expression encapsulates all of the key issues in generating impulse responses of nonlinear time series models. The first term in equation (19.36) is identical to the impulse response for the linear model in equation (19.34). The second term in equation (19.36) reflects the nonlinear features of the model. It is clear from inspection of this term that the impulse response is now no longer symmetrical in the size or the sign of the shock δ . Positive and negative shocks of the same magnitude yield impulses that are not opposite in sign as they are in the case of the linear model. The impulse response in equation (19.36) is also dependent on the history of the process, as it contains y_{t-1} . Furthermore, in contrast to the linear time series model, a shock occurring when $y_{t-1} < 0$, does not necessarily have the same effect on y_t as a shock occurring when $y_{t-1} > 0$. A typical example is in nonlinear models of the business cycle where a shock during a recession results in a different response in output to the same shock in an expansion period.

To compute impulse responses over $n+1$ horizons for nonlinear time series models, it is necessary to compute the conditional expectations in (19.36) over all possible histories of y_t . One way to proceed is to use simulation methods where the conditional expectations are calculated as sample averages across a range of histories obtained from the data. The steps involved are as follows.

Step 1: Estimate the model and obtain the residuals $\hat{v}_t$, $t = 1, 2, \dots, T$.

Step 2: Simulate the estimated model for $n+1$ periods where the $n+1$ residuals are randomly drawn from $\{\hat{v}_1, \hat{v}_2, \dots, \hat{v}_T\}$. Repeat the simulations R times and take the sample average of the R impulse re-

sponses for each of the $n+1$ horizons to compute the base conditional expectation.

Step 3: Repeat Step 2 with the same randomly drawn residuals except replace the initial condition by $\hat{v}_1 + \delta$ where $\hat{v}_1$ is the first residual and δ is the shock. Again repeat the simulations R times and take the sample average of the R impulse responses for each of the $n+1$ horizons to compute the conditional expectation corresponding to the shock δ .

Step 4: Repeat Steps 2 and 3 for a new set of random draws and where the initial condition in Step 3 is now $\hat{v}_2 + \delta$, where $\hat{v}_2$ is the second residual.

Step 5: Repeating Step 4 results in T impulse responses of length $n+1$ for the base and the shock conditional expectations. Averaging across these T impulses for each horizon and taking the difference in the two sample averages, yields the nonlinear impulse response functions.

The process of random sampling from $\{\hat{v}_1, \hat{v}_2, \dots, \hat{v}_T\}$ to construct the impulse responses is known as bootstrapping. There are two averaging processes going on here. The first is averaging across sample paths for alternative forecast horizons. The second is averaging across the initial conditions. Koop, Pesaran and Potter (1996) call impulses computed this way generalized impulse response functions (GIRF).

Example 19.11 Impulse Responses of a Threshold Autoregression

Consider the threshold autoregressive model

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-1} w_t + v_t, \quad v_t \sim N(0, 1)$$

$$w_t = \begin{cases} 1 & y_{t-1} \geq 0 \\ 0 & y_{t-1} < 0. \end{cases},$$

where v_t is an *iid* disturbance. The model is simulated for $T = 1000$ observations for parameter values $\phi_1 = 0.25$ and $\phi_2 = 0.50$, which for simplicity are directly used to compute the impulse responses without estimating the model. The number of bootstrapped samples is set at T . The simulated impulse responses are given in Table 19.5 for horizons of $n = 10$ and for shocks of size $\delta = \{1, -1\}$. For comparison the analytical impulse response of an AR(1) model are also given by setting $\phi_2 = 0$. Reversing the shock from $\delta = 1$ to $\delta = -1$ results in asymmetrical impulse responses for the TAR model, with the impulse responses decaying at a relatively faster rate for the negative shock. In contrast, the linear AR model impulse responses exhibit symmetry with the rate of decay being the same for $\delta = 1$ and $\delta = -1$. $\square$

Table 19.5

Impulse responses of a TAR model and an AR(1) model for two different shocks.

Horizon	TAR(1)		AR(1)	
	$\delta = 1$	$\delta = -1$	$\delta = 1$	$\delta = -1$
0	1.001	-0.999	1.000	-1.000
1	0.593	-0.406	0.750	-0.750
2	0.357	-0.214	0.563	-0.563
3	0.220	-0.125	0.422	-0.422
4	0.137	-0.076	0.316	-0.316
5	0.086	-0.047	0.237	-0.237
6	0.054	-0.030	0.178	-0.178
7	0.034	-0.019	0.133	-0.133
8	0.022	-0.012	0.100	-0.100
9	0.014	-0.007	0.075	-0.075
10	0.009	-0.005	0.056	-0.056

The construction of the impulse responses presented in the previous example is obtained by averaging across all of the shocks in the data. An alternative refinement is to average across histories where the initial condition is a particular sign, as would be appropriate in nonlinear business cycle models that distinguish between recessions and expansions. Finally, whilst the approach is presented for single equation nonlinear time series models, it easily generalizes to multivariate nonlinear time series model. For example, Koop, Pesaran and Potter (1996) provides an example based on a bivariate TAR model of output and unemployment.

19.9 Applications

19.9.1 A Multiple Equilibrium Model of Unemployment

In this application, an LSTAR model of the U.S. monthly unemployment rate given in Figure 19.5, is specified and estimated. Skalin and Teräsvirta (2002) The specified model represents a two-regime error correction model with the change in the unemployment rate Δy_t , specified as

$$\Delta y_t = \mu_1 + \alpha_1 y_{t-1} + (\mu_2 + \alpha_2 y_{t-1}) w_t + v_t, \quad \mu_i > 0, \quad (19.37)$$

where v_t is an *iid* disturbance term. The weighting function w_t is

$$w_t = \frac{1}{1 + \exp[-\gamma(\Delta_{12} y_{t-1} - c)/s]}, \quad \gamma > 0,$$

where $\Delta_{12}y_{t-1} = y_{t-1} - y_{t-13}$ is the lagged annual change in the unemployment rate, c is the threshold and s is the sample standard deviation of $\Delta_{12}y_t$, which is used as a scaling factor (Teräsvirta, 1994). Rewriting equation (19.37) as

$$\Delta y_t = \alpha_1 \left(\frac{\mu_1}{\alpha_1} + y_{t-1} \right) + \alpha_2 \left(\frac{\mu_2}{\alpha_2} + y_{t-1} \right) w_t + v_t, \quad (19.38)$$

expresses the model in error correction form, as discussed in Chapter 18, where the parameters α_1 and α_2 represent the error correction parameters. As a result of this nonlinear specification, two unemployment rates satisfy the equilibrium condition $\Delta y_t = 0$, namely,

$$y_{\text{low}} = -\frac{\mu_1}{\alpha_1}, \quad y_{\text{high}} = -\frac{\mu_1 + \mu_2}{\alpha_1 + \alpha_2}, \quad (19.39)$$

which correspond, respectively, to the low state where $w_t = 0$ and to the high state where $w_t = 1$. The error correction parameters associated with these two equilibrium states are respectively, α_1 and $\alpha_1 + \alpha_2$, which need to be negative to ensure that the low and high multiple equilibria exist and satisfy the stability restriction $-1 < \alpha_1 + \alpha_2 < 0$.

Table 19.6

Maximum likelihood parameter estimates of the LSTAR multiple equilibrium model of U.S. unemployment in (19.38), with standard errors based on the Hessian matrix in parentheses.

Parameter	Estimate	s.e.
μ_1	0.077	0.040
α_1	-0.021	0.007
μ_2	0.282	0.083
α_2	-0.017	0.012
γ	6.886	3.272
c	0.420	0.115

The results from estimating the LSTAR model in (19.37) by maximum likelihood are given in Table 19.6. The value of the log-likelihood function at the maximum likelihood estimates is $\ln L_T(\hat{\theta}) = 0.160$ while $T \ln L_T(\hat{\theta}) = 734 \times 0.160 = 117.507$. The parameter estimates of the error-correction parameters are negative, which is consistent with the presence of multiple equilibria. The second error correction parameter estimate, $\hat{\alpha}_2$, is statistically insignificant suggesting that the speed of adjustment to both low and

high equilibria are the same and equal to $\hat{\alpha}_1$. The point estimates also satisfy the stability restriction as $-1 < \hat{\alpha}_1 + \hat{\alpha}_2 < 0$. Finally, the estimates of the low and high equilibrium unemployment rates are, respectively,

$$\begin{aligned}\hat{y}_{\text{low}} &= -\frac{\hat{\mu}_1}{\hat{\alpha}_1} = -\frac{0.077}{-0.021} = 3.691 \\ \hat{y}_{\text{high}} &= -\frac{\hat{\mu}_1 + \hat{\mu}_2}{\hat{\alpha}_1 + \hat{\alpha}_2} = -\frac{0.077 + 0.282}{-0.021 - 0.017} = 9.431.\end{aligned}$$

Inspection of Figure 19.5 shows that these estimates represent sensible estimates for the means of the different states of the U.S. unemployment rate.

19.9.2 Bivariate Threshold Models of G7 Countries

Anderson, Anthansopoulos and Vahid (2007) investigate the relationship between the real output growth rate and the interest rate spread of G7 countries using a range of nonlinear times series models. The variables are

$$y_{1,t} = 100 \times (\ln RGDP_t - \ln RGDP_{t-1}), \quad y_{2,t} = R_{10yr,t} - R_{3mth,t},$$

where $RGDP_t$ is real GDP, and $R_{10yr,t}$ and $R_{3mth,t}$ are, respectively, the 10-year and 3-month interest rates, both expressed in percentages. The sample periods vary for each country and are as follows:

Canada	: June 1961 to December 1999
France	: March 1970 to September 1998
Germany	: June 1960 to December 1999
Italy	: June 1971 to December 1999
Japan	: March 1970 to June 1999
U.K.	: June 1960 to December 1999
U.S.	: June 1960 to December 1999.

For each G7 country a generic bivariate threshold time series model is specified

$$\begin{aligned}y_{1,t} &= \phi_{1,0} + \sum_{i=1}^2 \phi_{i,1,1} y_{1,t-i} + \sum_{i=1}^2 \phi_{i,1,2} y_{2,t-i} \\ &\quad + (\beta_{1,0} + \sum_{i=1}^2 \beta_{i,1,1} y_{1,t-i} + \sum_{i=1}^2 \beta_{i,1,2} y_{2,t-i}) w_{1,t-2} + v_{1,t} \\ y_{2,t} &= \phi_{2,0} + \sum_{i=1}^2 \phi_{i,2,1} y_{1,t-i} + \sum_{i=1}^2 \phi_{i,2,2} y_{2,t-i} \\ &\quad + (\beta_{2,0} + \sum_{i=1}^2 \beta_{i,2,1} y_{1,t-i} + \sum_{i=1}^2 \beta_{i,2,2} y_{2,t-i}) w_{2,t-0} + v_{2,t},\end{aligned}\tag{19.40}$$

where the disturbance vector $v_t = [v_{1,t}, v_{2,t}]'$ is distributed as $v_t \sim N(0, V)$ with

$$V = \begin{bmatrix} \sigma_{1,1} & \sigma_{1,2} \\ \sigma_{2,1} & \sigma_{2,2} \end{bmatrix},$$

and the weighting functions are given by the LSTAR specifications

$$w_{1,t-2} = \frac{1}{1 + \exp(-\gamma_1(y_{1,t-2} - c_1))} \quad (19.41)$$

$$w_{2,t-1} = \frac{1}{1 + \exp(-\gamma_2(y_{1,t-1} - c_2))}. \quad (19.42)$$

The specifications of the weighting functions show that both output and spread adjustments between regimes are governed by lagged output growth. Other specifications of the model can be entertained including longer lag structures in each regime and alternative threshold specifications. For example, Anderson, Anthansopoulos and Vahid (2007) choose longer lag structures and threshold specifications that vary for each country. They also delete parameter estimates that are statistically insignificant.

Table 19.7

Log-likelihood function values and residual variance-covariance matrices of G7 threshold models given in equations (19.40) to (19.42).

Country	$\ln L$	$\hat{V}$
Canada	-322.274	$\begin{bmatrix} 0.587 & -0.003 \\ -0.003 & 0.417 \end{bmatrix}$
France	-233.199	$\begin{bmatrix} 0.318 & -0.042 \\ -0.042 & 0.725 \end{bmatrix}$
Germany	-440.403	$\begin{bmatrix} 1.326 & -0.035 \\ -0.035 & 0.760 \end{bmatrix}$
Italy	-295.208	$\begin{bmatrix} 0.489 & 0.070 \\ 0.070 & 1.439 \end{bmatrix}$
Japan	-275.014	$\begin{bmatrix} 0.801 & 0.044 \\ 0.044 & 0.535 \end{bmatrix}$
U.K.	-373.531	$\begin{bmatrix} 0.942 & -0.038 \\ -0.038 & 0.452 \end{bmatrix}$
U.S.	-298.734	$\begin{bmatrix} 0.613 & -0.054 \\ -0.054 & 0.269 \end{bmatrix}$

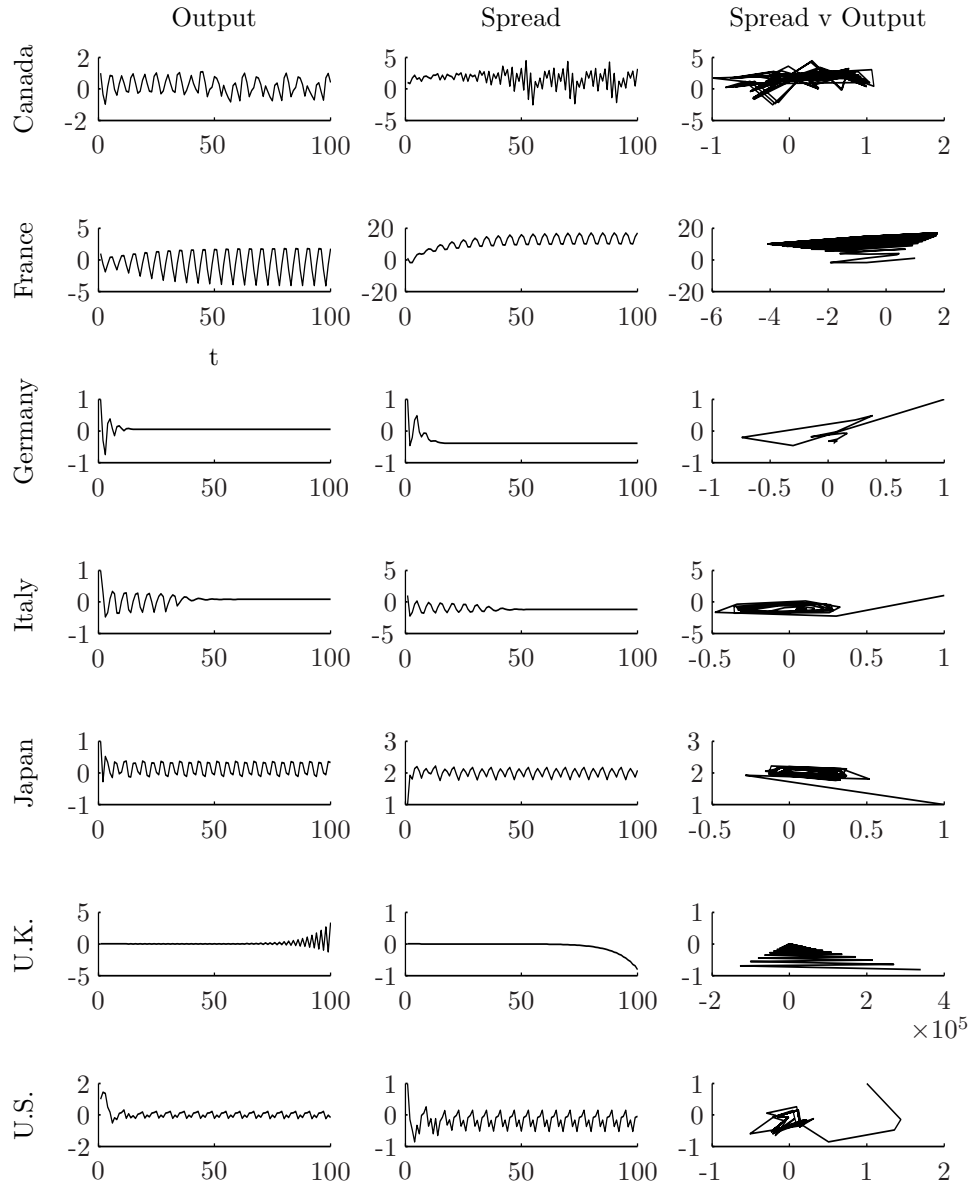


Figure 19.11 Simulated TAR models of G 7 countries based on the maximum likelihood parameter estimates.

The log-likelihood function is

$$\ln L_T(\theta) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |V| - \frac{1}{2T} v_t' V^{-1} v_t, \quad (19.43)$$

with $N = 2$. The full log-likelihood function for a sample of size T is maximized using the BFGS algorithm discussed in Chapter 3. When estimating the model, the adjustment parameters in equations (19.41) and (19.42) are fixed at $\gamma_1 = \gamma_2 = 100$. This has the effect of forcing the LSTAR model to become a TAR model. The log-likelihood values and residual variance-covariance matrices are reported in Table 19.7.

The results of simulating the bivariate threshold models for each G7 country with parameter values replaced by their maximum likelihood estimates are given in Figure 19.11. The dynamic properties of the simulated models vary across the countries. Canada tends to exhibit chaotic behaviour with the dynamics of output and the spread characterized by a strange attractor. Germany and Italy converge to a fixed point. France, Japan and the U.S. converge to stable limit cycles. The U.K. exhibits an unstable cycle.

19.10 Exercises

(1) Features of Nonlinear Models

Gauss file(s)	<code>nlm_features.g</code> , <code>usunemp.dat</code>
Matlab file(s)	<code>nlm_features.m</code> , <code>usunemp.dat</code>

Consider the AR(2) process

$$y_t = 0.9 + 0.7y_{t-1} - 0.6y_{t-2}.$$

- Simulate the model where the initial value is $y_0 = 0.5$. Show that the model converges to the equilibrium point $y = 0.9/(1 - 0.7 + 0.6) = 1.0$.
- Show that the following process converges to a 9-period cycle.

$$y_t = \begin{cases} 0.8023 + 1.0676y_{t-1} - 0.2099y_{t-2} + 0.1712y_{t-3} & : y_{t-2} \leq 3.05 \\ -0.4528y_{t-4} + 0.2237y_{t-5} - 0.0331y_{t-6} & \\ 2.2964 + 1.4246y_{t-1} - 1.0795y_{t-2} - 0.0900y_{t-3} & : y_{t-2} > 3.05. \end{cases}$$

- Consider the nonlinear business cycle model of Kaldor

$$\begin{aligned} y_{t+1} - y_t &= \alpha(i_t - s_t) && \text{[Output]} \\ k_{t+1} - k_t &= i_t - \delta k_t && \text{[Capital]} \\ i_t &= c2^{-(dy_t + \varepsilon)^{-2}} + ey_t + a(fk_t^{-1})^g && \text{[Investment]} \\ s_t &= sy_t, && \text{[Savings]} \end{aligned}$$

where y_t is output, k_t is capital, i_t is investment, s_t is savings and $\theta = \{\alpha, \delta, c, d, \varepsilon, e, a, f, g, s\}$ are parameters. Using the parameter

values given in the program file, simulate the model and show that the dynamics represent a strange attractor.

- (d) Consider the nonlinear error correction model of the logarithm of real money, m_t , which is a simplification of the model estimated in Ericsson, Hendry and Prestwich (1998, p.296)

$$\Delta m_t = 0.45\Delta m_{t-1} - 0.10\Delta^2 m_{t-2} - 2.54(m_{t-1} - 0.1)m_{t-1}^2 + v_t$$

$$v_t \sim N(0, 0.005^2),$$

where $\Delta m_t = m_t - m_{t-1}$ and $\Delta^2 m_{t-2} = m_{t-2} - 2m_{t-3} + m_{t-4}$ and v_t is an *iid* disturbance term. Simulate the model and show that the majority of the mass is around the stable equilibrium value of 0.1 with smaller mass at the inflexion point of 0.0.

- (e) Plot the U.S. monthly unemployment rate from January 1948 to March 2010. Describe its statistical properties including potential differences in the duration of expansion and contraction periods.

(2) Sampling Properties of the LST Linearity Test

Gauss file(s)	nlm_tarsim.g
Matlab file(s)	nlm_tarsim.m

This exercise is based on the Monte Carlo experiments of Luukkonen, Saikkonen and Teräsvirta (1988) where the DGP is

$$y_t = -0.5y_{t-1} + \beta_1 y_{t-1} w_t + v_t$$

$$w_t = (1 + \exp(-\gamma(y_{t-1} - 0)))^{-1}$$

$$v_t \sim N(0, 25),$$

where v_t is an *iid* disturbance term, and the parameters are $\beta_1 = \{0.0, 0.5, 1.0\}$ and $\gamma = \{0.5, 1.0, 5.0\}$. The number of replications is set at 10000.

- (a) For $T = 100$, compute the size of the LST linearity test in (19.10), assuming a nominal size of 5%, and the size-adjusted power function. Compare the results with Table 19.2.
- (b) Repeat part (a) for $T = 250$ and discuss the consistency property of the test.
- (c) Show that improvements in the power of the test are obtained by imposing the restrictions $\{\pi_{3,i} = 0, \forall i\}$ in the auxiliary equation in (19.10). Interpret this result.

- (d) Replace the LSTAR model with a STAR model in the data generating process and redo parts (a) and (b). Compare the two power functions.
- (e) Let the weighting function w_t be given by the ESTAR model and redo parts (a) and (b). Show that for this model

$$\left. \frac{\partial w_t}{\partial z_t} \right|_{z_t=0} = 0; \quad \left. \frac{\partial^2 w_t}{\partial z_t^2} \right|_{z_t=0} = 2; \quad \left. \frac{\partial^3 w_t}{\partial z_t^3} \right|_{z_t=0} = 0,$$

and that improvements in the power for this data generating process are obtained by imposing the restrictions $\{\pi_{3,i} = 0, \forall i\}$ in the auxiliary equation (19.10).

- (f) Now choose as the data generating process

$$\begin{aligned} y_t &= 1 - 0.5y_{t-1} + \beta_0 w_t + v_t \\ w_t &= (1 + \exp(-\gamma(y_{t-1} - 0)))^{-1} \\ v_t &\sim N(0, 1), \end{aligned}$$

where v_t is an *iid* disturbance term and the parameters are $\beta_0 = \{0.0, -1, -2, -3, -4\}$ and $\gamma = \{2\}$. Show that the linearity test based on (19.10) has power in the direction of β_0 , but that the version of the test with the restrictions $\{\pi_{2,i} = \pi_{3,i} = 0, \forall i\}$ in (19.10) has low power.

(3) Artificial Neural Networks

Gauss file(s)	n1m_ann.g, n1m_neural.g
Matlab file(s)	n1m_ann.m, n1m_neural.m

- (a) Simulate the ANN model

$$y_t = \phi y_{t-1} + \gamma \left[\frac{1}{1 + \exp(-(\delta_0 + \delta_1 y_{t-1}))} \right] + v_t, \quad v_t \sim N(0, \sigma^2),$$

where v_t is an *iid* disturbance. The parameter values are $\phi = 0.2$, $\gamma = 2.0$, $\delta_0 = -2$, $\delta_1 = 2$, $\sigma^2 = 0.1$ and the initial condition is $y_0 = 0.0$. Choose a sample size of $T = 2100$ observations and discard the first 100 observations.

- (b) Estimate the unknown parameters $\theta = \{\phi, \gamma, \delta_0, \delta_1, \sigma^2\}$ by maximum likelihood using the Newton-Raphson algorithm.
- (c) Estimate the unknown parameters $\theta = \{\phi, \gamma, \delta_0, \delta_1, \sigma^2\}$ using the neural algorithm where the weighting function is

$$w_t^{(i)} = \left[\frac{1}{1 + \exp(-(\delta_0^{(i)} + \delta_1^{(i)} y_{t-1}))} \right],$$

with $\delta_0^{(i)}$ and $\delta_1^{(i)}$ taken as the i^{th} draw from the rectangular distribution $[-5, 5]$. Let the total number of draws be 100.

(4) Neural Network Test

Gauss file(s)	<code>nlm_lgwtest.g</code>
Matlab file(s)	<code>nlm_lgwtest.m</code>

- (a) Consider the linear model

$$y_t = 0.6y_{t-1} + v_t, \quad v_t \sim N(0, 1),$$

where v_t is an *iid* disturbance. Simulate the model 5000 times for a sample of size $T = 250$. Determine the size properties of the neural network test for the case of $q = 1, 2$ activation functions.

- (b) Simulate the nonlinear model

$$y_t = \gamma y_{t-1} v_{t-2} + v_t, \quad v_t \sim N(0, 1),$$

where v_t is an *iid* disturbance, with $\gamma = 0.7$, 5,000 times for a sample of size $T = 250$. Determine the size-adjusted power properties of the neural network test for the case of $q = 1, 2$ activation functions.

- (c) Extend part (b) by computing the power of the test for $\gamma = 0.8, 0.9$.

(5) Bilinearity

Gauss file(s)	<code>nlm_bilinear.g, nlm_blinest.g</code>
Matlab file(s)	<code>nlm_bilinear.m, nlm_blinest.m</code>

- (a) Simulate the bilinear model

$$y_t = 0.4y_{t-1} + v_t + 0.2v_{t-1} + \gamma y_{t-1}v_{t-1}, \quad v_t \sim N(0, 1),$$

for $\gamma = \{0.0, 0.4, 0.8, 1.2\}$, choosing as starting values $y_0 = 0.0$ and $u_0 = 0.0$. Choose the sample size to be $T = 300$ and then discard the first 100 observations. Describe the properties of the four time series.

- (b) Simulate the bilinear model

$$y_t = 0.1 + 0.4y_{t-1} + v_t + \gamma y_{t-1}v_{t-1}, \quad v_t \sim N(0, 0.1),$$

with $\gamma = 0.4$. Choose a sample size of $T = 1100$ observations and discard the first 100 observations. Let the initial conditions be $y_0 = 0.0$ and $u_0 = 0.0$. Estimate the parameters by maximum likelihood.

- (c) Using the results in part (b), test for bilinearity using a LR test, a Wald test and a LM test.
- (d) Repeat parts (b) and (c) where the bilinearity parameter is $\gamma = 0.0$.

(6) **A Markov Switching Model of the U.S. Business Cycle**

Gauss file(s) `nlm_bcycle.g, gnp.dat`

Matlab file(s) `nlm_bcycle.m, gnp.mat`

The data file contains $T = 136$ quarterly observations on U.S. GNP from March 1951 to December 1984. Consider the following model of the business cycle

$$\begin{aligned}
 y_t &= \alpha + \beta w_t + u_t \\
 \sigma_t^2 &= \gamma + \delta w_t \\
 u_t &\sim N(0, \sigma_t^2) \\
 p &= P[w_t = 1 \mid w_{t-1} = 1, y_{t-1}, y_{t-2}, \dots] \\
 q &= P[w_t = 0 \mid w_{t-1} = 0, y_{t-1}, y_{t-2}, \dots],
 \end{aligned}$$

where y_t is the quarterly percentage growth rate in output, w_t is a stochastic binary variable representing the business cycle, u_t is a disturbance and $\theta = \{\alpha, \beta, \gamma, \delta, p, q\}$ are unknown parameters.

- (a) Estimate the unknown parameters θ by maximum likelihood and interpret the parameter estimates. Compare these estimates to the values reported in Hamilton (1989, Table 1, p.372).
- (b) A measure of duration of an expansion is given by $(1 - p)^{-1}$ and of a contraction by $(1 - q)^{-1}$. Estimate the duration of the two phases of the business cycle together with their standard errors.
- (c) Test the restriction $\delta = 0$ and interpret the result.
- (d) A potential test of Markov switching is based on the joint restriction $\beta = \delta = 0$. Briefly discuss the statistical issues associated in performing this test.
- (e) Re-estimate the model by allowing the conditional probabilities to be time-varying functions of the unemployment rate, em_t ,

$$p_t = \frac{1}{1 + \exp(-\kappa_0 - \kappa_1 em_t)}, \quad q_t = \frac{1}{1 + \exp(-\lambda_0 - \lambda_1 em_t)}.$$

Perform a LR test of time-invariant transitional probabilities by testing the joint restriction $\kappa_1 = \lambda_1 = 0$.

(7) Nonparametric Autoregression

Gauss file(s)	<code>nlm_linear.g</code> , <code>nlm_tar.g</code>
Matlab file(s)	<code>nlm_linear.m</code> , <code>nlm_tar.m</code>

- (a) Simulate $T = 5,000$ observations from the linear AR(k) model

$$y_t = \mu + \phi y_{t-k} + \sigma z_t, \quad z_t \sim N(0, 1),$$

for $k = 1, 2, 3, 4, 5$ lags and with parameter values $\mu = 0.0$, $\phi = 0.8$ and $\sigma^2 = 1$. Estimate the conditional mean $m(y_{t-k})$ for $-3 \leq k \leq 3$ using the Nadaraya-Watson kernel regression estimator with a Gaussian kernel and the rule-of-thumb bandwidth. Compare the estimated conditional mean and the true conditional mean given by $m(y_{t-k}) = \phi^k y$.

- (b) Repeat part (a) for the nonlinear autoregressive model

$$y_t = \theta |y_{t-k}| + \sqrt{1 - \theta^2} z_t, \quad z_t \sim N(0, 1),$$

for $k = 1, 2, 3, 4, 5$ lags with parameter $\theta = 0.8$. For each k , compare the nonparametric estimate of the conditional mean with the linear conditional mean in part (a). Interpret the results.

(8) Impulse Response Function of Nonlinear Time Series Models

Gauss file(s)	<code>nlm_girf.g</code>
Matlab file(s)	<code>nlm_girf.m</code>

Simulate $T = 1,000$ observations from the nonlinear model

$$\begin{aligned} y_t &= \phi_1 y_{t-1} + \phi_2 y_{t-1} w_t + v_t, \\ w_t &= \begin{cases} 1 & y_{t-1} \geq 0 \\ 0 & y_{t-1} < 0. \end{cases} \\ v_t &\sim N(0, 1), \end{aligned}$$

with parameter values $\phi_1 = 0.25$ and $\phi_2 = 0.50$.

- (a) Compute the generalized impulse responses for horizons of $n = 10$ and shocks of $\delta = \{2, 1, -1, -2\}$. Interpret the results.
- (b) Compare the results in part (a) to the impulse responses of a linear model by setting $\phi_2 = 0.0$.
- (c) Repeat parts (a) and (b) for alternative values of ϕ_1 and ϕ_2 .

(9) LSTAR Model of U.S. Unemployment

Gauss file(s)	<code>nlm_usurate.g,usunemp.dat</code>
Matlab file(s)	<code>nlm_userate.m,usunemp.dat</code>

The data are observations on the monthly percentage unemployment rate in the U.S., y_t , beginning January 1948 and ending March 2010.

- Estimate an AR(1) model with a constant of y_t , and estimate the long-run equilibrium level of unemployment.
- Perform the LST linearity test on the unemployment rate.
- Estimate the LSTAR model

$$\Delta y_t = \mu_1 + \alpha_1 u_{t-1} + (\mu_2 + \alpha_2 u_{t-1}) w_t + v_t$$

$$w_t = \frac{1}{1 + \exp(-\gamma(\Delta_{12}y_{t-1} - c)/s)},$$

where v_t is *iid* $N(0, \sigma^2)$, and s is a scale factor equal to the standard deviation of $\Delta_{12}y_{t-1}$. Interpret the parameter estimates.

- Estimate the long-run equilibrium level of unemployment in the ‘low’ state and the ‘high’ state and compare these estimates with the estimate obtained in part (a) based on the linear model. Discuss the results in the light of the linearity test conducted in part (b).

(10) LSTAR Model of Australian Unemployment

Gauss file(s)	<code>nlm_ozurate.g,ausu.xls</code>
Matlab file(s)	<code>nlm_userate.m,ausu.xls</code>

The data are observations on the quarterly percentage unemployment rate in Australia, y_t , beginning March 1971 and ending December 2008.

Redo the previous exercise for the Australian unemployment rate.

(11) Bivariate LSTAR Model of G7 Countries

Gauss file(s)	<code>nlm_g7.g, G7Data.xlsx</code>
Matlab file(s)	<code>nlm_g7.m, G7Data.xlsx</code>

The data are percentage quarterly growth rates of output, $y_{1,t}$, and spreads, $y_{2,t}$, for the G7 countries, Canada, France, Germany, Italy, Japan, U.K. and the U.S. All sample periods end December 1999, but the starting dates vary which are given in the data file.

- Estimate a bivariate VAR of $y_{1,t}$ and $y_{2,t}$ with a constant and two lags. Describe the dynamic properties of this model.

- (b) A bivariate analogue of the LST test is based on the regression equation

$$\begin{aligned}
 v_{j,t} = & \pi_0 + \sum_{i=1}^p \pi_{0,i,1} y_{1,t-i} + \sum_{i=1}^p \pi_{0,i,2} y_{2,t-i} \\
 & + \sum_{i=1}^p \pi_{1,i,1} y_{1,t-i} y_{k,t-d} + \sum_{i=1}^p \pi_{1,i,2} y_{2,t-i} y_{k,t-d} \\
 & + \sum_{i=1}^p \pi_{2,i,1} y_{1,t-i} y_{k,t-d}^2 + \sum_{i=1}^p \pi_{2,i,2} y_{2,t-i} y_{k,t-d}^2 \\
 & + \sum_{i=1}^p \pi_{3,i,1} y_{1,t-i} y_{k,t-d}^3 + \sum_{i=1}^p \pi_{3,i,2} y_{2,t-i} y_{k,t-d}^3 + e_t,
 \end{aligned}$$

where $v_{j,t}$ is the residual of variable j from estimating the linear VAR in part (a). Apply this test for $j, d = 1, 2$. Suggest alternative variants of this test based on the univariate variants discussed above.

- (c) Estimate the following bivariate LSTAR model by maximum likelihood for each of the 7 countries

$$\begin{aligned}
 y_{1,t} = \phi_{1,0} & + \sum_{i=1}^2 \phi_{i,1,1} y_{1,t-i} + \sum_{i=1}^2 \phi_{i,1,2} y_{2,t-i} \\
 & + (\beta_{1,0} + \sum_{i=1}^2 \beta_{i,1,1} y_{1,t-i} + \sum_{i=1}^2 \beta_{i,1,2} y_{2,t-i}) w_{1,t-2} + v_{1,t} \\
 y_{2,t} = \phi_{2,0} & + \sum_{i=1}^2 \phi_{i,2,1} y_{1,t-i} + \sum_{i=1}^2 \phi_{i,2,2} y_{2,t-i} \\
 & + (\beta_{2,0} + \sum_{i=1}^2 \beta_{i,2,1} y_{1,t-i} + \sum_{i=1}^2 \beta_{i,2,2} y_{2,t-i}) w_{2,t-0} + v_{2,t},
 \end{aligned}$$

and

$$\begin{aligned}
 w_{1,t-2} &= \frac{1}{1 + \exp(-\gamma_1(y_{1,t-2} - c_1))}, \\
 w_{2,t-1} &= \frac{1}{1 + \exp(-\gamma_2(y_{1,t-1} - c_2))},
 \end{aligned}$$

where the disturbance vector $v_t = [v_{1,t}, v_{2,t}]'$ is *iid* distributed as $v_t \sim N(0, V)$ and the threshold parameters are restricted to be $\gamma_1 = \gamma_2 = 100$.

- (d) Simulate each of the 7 estimated models and discuss the dynamic properties of these models. Compare the results with the time paths given in Figure 19.11.

(12) **Sunspots**

Gauss file(s)	n1m_sunspots.g, sunspots.xlsx
Matlab file(s)	n1m_sunspots.m, sunspots.xlsx

The data, y_t , are the monthly averages of daily sunspot numbers from January 1749 to June 2009, compiled by the Solar Influences Data Analysis Centre in Belgium. Sunspots are magnetic regions on the sun with magnetic field strengths thousands of times stronger than the earth's magnetic field. They appear as dark spots on the surface of the sun and typically last for several days. They were first observed in the early 1600s following the invention of the telescope.

- Perform the LST linearity test on the sunspots data.
- Estimate the following LSTAR model of sunspots by maximum likelihood methods

$$y_t = \phi_0 + \sum_{i=1}^6 \phi_i y_{t-i} + (\beta_0 + \sum_{i=1}^3 \beta_i y_{t-i}) w_t + v_t,$$

$$w_t = \frac{1}{1 + \exp(-\gamma(y_{t-2} - c))},$$

where the delay parameter is fixed at $d = 2$, and the threshold parameter is fixed at the following values $\gamma = \{1, 5, 10, 50, 100\}$.

- Repeat part (b) for alternative values of the delay parameter d .

(13) **Relationship with Time-varying Parameter Models**

Gauss file(s)	n1m_tvarying.g
Matlab file(s)	n1m_tvarying.m

Granger (2008) uses White's theorem to argue that nonlinear models can be approximated using time-varying parameter models. To understand the theorem consider the regression equation

$$y_t = E[y_t | y_{t-1}, y_{t-2}, \dots] + v_t,$$

where v_t is a disturbance term. Assuming that $y_{t-1} > 0$, this equation is rewritten as

$$y_t = (E[y_t | y_{t-1}, y_{t-2}, \dots] / y_{t-1}) y_{t-1} + v_t = s_t y_{t-1} + v_t,$$

where $s_t = (E[y_t | y_{t-1}, y_{t-2}, \dots] / y_{t-1})$. This is an AR(1) model with a time-varying parameter given by s_t , which can be used to approximate

$E[y_t|y_{t-1}, y_{t-2}, \dots]$ by either a Kalman filter (Chapter 15) or a nonparameter regression estimator (Chapter 11).

- (a) To highlight the approximation properties of the time-varying model for the simplest case of a linear model, simulate the AR(2) model for $T = 200$ observations

$$y_t = 10 + 0.6y_{t-1} + 0.3y_{t-2} + v_t, \quad v_t \sim N(0, 3).$$

where v_t is an *iid* disturbance.

- (b) Estimate the time-varying Kalman filter model

$$\begin{aligned} y_t &= \lambda_t s_t + v_t \\ s_t &= \phi s_{t-1} + \eta_t, \\ v_t &\sim N(0, \sigma^2) \\ \eta_t &\sim N(0, 1), \end{aligned}$$

for the unknown parameters $\theta = \{\sigma^2, \phi\}$, with $\lambda_t = y_{t-1}$, where v_t and η_t are *iid* disturbances. Approximate the conditional mean of y_t by $s_{t|T}y_{t-1}$, where $s_{t|T}$ is the smoothed estimate of the latent factor s_t .

- (c) Estimate the conditional mean of y_t using a nonparametric estimator.
- (d) Compare the approximating conditional means obtained in parts (b) and (c) with the true conditional mean $E[y_t|y_{t-1}] = 10 + 0.6y_{t-1} + 0.3y_{t-2}$.
- (e) Repeat parts (a) to (d) where the true model is nonlinear based on the bilinear specification

$$y_t = 10 + 0.6y_{t-1} + 0.3y_{t-2} + 0.8y_{t-1}v_{t-1} + v_t, \quad v_t \sim N(0, 3),$$

where v_t is an *iid* disturbance.

(14) GENTS

Gauss file(s)	<code>n1m_gents.g,ftse.xls</code>
Matlab file(s)	<code>n1m_gents.m,ftse.xls</code>

The data are the daily returns on the FTSE, y_t , beginning 20 November 1973 and ending 23 July 2001, $T = 7000$ observations. The generalized exponential nonlinear time series (GENTS) model of Lye and Martin (1994) consists of specifying a generalized Student t distribution with

time-varying parameters

$$f(y|y_{t-1}) = \exp \left[\theta_{1,t} \tan^{-1} \left(\frac{y}{\gamma} \right) + \theta_{2,t} \ln(\gamma^2 + y^2) + \theta_{3,t}y + \theta_{4,t}y^2 - \eta_t \right],$$

where

$$\theta_{1,t} = \alpha y_{t-1}, \quad \theta_{2,t} = -\frac{(1-\gamma^2)}{2}, \quad \theta_{3,t} = \beta y_{t-1}, \quad \theta_{4,t} = -\frac{1}{2},$$

and

$$\eta_t = \ln \int_{-\infty}^{\infty} \exp \left[\theta_{1,t} \tan^{-1} \left(\frac{y}{\gamma} \right) + \theta_{2,t} \ln(\gamma^2 + y^2) + \theta_{3,t}y + \theta_{4,t}y^2 - \eta_t \right] dy,$$

is the normalizing constant. This model not only allows for a time-varying conditional mean, but also time-varying higher order moments, as well as endogenous jumping during periods where the distribution exhibits multimodality.

- (a) Estimate the parameters $\theta = \{\gamma, \alpha, \beta\}$ by maximum likelihood. In computing the log-likelihood function all integrals are computed numerically.
- (b) Estimate the conditional mean

$$E[y_t|y_{t-1}] = \int_{-\infty}^{\infty} y f(y|y_{t-1}) dy.$$

20

Nonlinearities in Variance

20.1 Introduction

The previous chapter focussed on using maximum likelihood methods to estimate and test models specified with nonlinearities in the mean. This chapter addresses models that are nonlinear in the variance. It transpires that the variance of the returns of financial assets, commonly referred to as the volatility, is a crucial aspect of much of modern finance theory, because it is a key input to areas such as portfolio construction, risk management and option pricing. In this chapter, the particular nonlinear variance specification investigated is the autoregressive conditional heteroskedasticity (ARCH) class of models introduced by Engle (1982).

As in the case with nonlinear models in the mean, however, a wide range of potential nonlinearities can be entertained when modelling the variance. There are two important approaches to modelling the variance of financial asset returns that are not discussed in this chapter. The first is the stochastic volatility model, introduced by Taylor (1982) and discussed briefly in Chapters 9 and 12, and the second is realized volatility proposed by Anderson, Bollerslev, Diebold and Labys (2001, 2003).

20.2 Statistical Properties of Asset Returns

Panel (a) of Figure 20.1 provides a plot of the returns of the daily percentage returns, y_t , on the FTSE from 5 January 1989 to 31 December 2007, $T = 4952$. At first sight, the returns appear to be random, a point highlighted in panel (c), which shows that the autocorrelation function of returns is flat. Closer inspection of the returns reveals periods when returns hardly change (market tranquility) and others where large movements in returns are followed by further large changes (market turbulence). This property is

demonstrated in panel (b) of Figure 20.1, which gives a time series plot of the squares of returns, y_t^2 . This volatility clustering is a commonly-observed empirical characteristic of financial returns and it gives rise to autocorrelation in the squared returns, as is demonstrated in panel (d) of Figure 20.1.

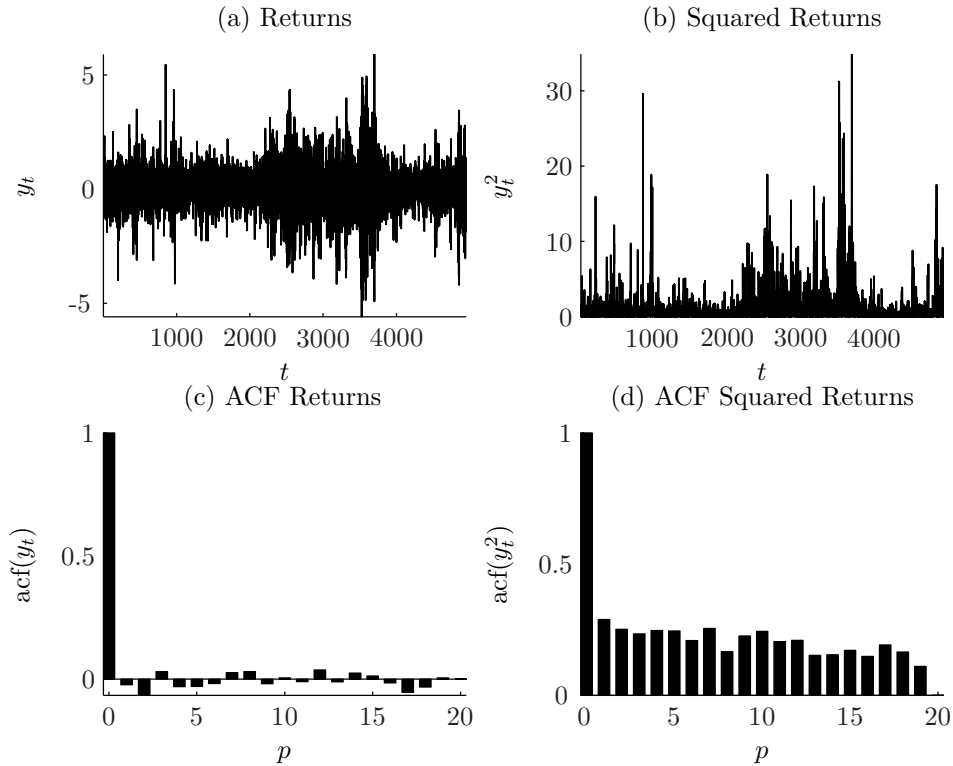


Figure 20.1 Statistical properties of daily percentage returns, y_t , on the FTSE from 5 January 1989 to 31 December 2007.

The properties observed in Figure 20.1 for the returns to the FTSE are commonly observed in most financial returns and are now outlined in more detail.

Property 1: No Autocorrelation in Returns

The autocorrelation in the levels of returns demonstrated in Figure 20.1 shows that predicting the direction of asset returns is not possible. This suggests the following model of y_t

$$y_t = \rho y_{t-1} + u_t, \quad u_t \sim N(0, \sigma^2).$$

Throughout this chapter, the distribution of u_t with time-varying variance

σ_t^2 should be interpreted to be a conditional normal distribution based on information at $t-1$. The restriction of no autocorrelation requires that $\rho = 0$ and in this case the unconditional mean and variance of y_t are, respectively,

$$\begin{aligned} \text{Mean} & : E[y_t] = E[u_t] = 0 \\ \text{Variance} & : E[y_t^2] - E[y_t]^2 = E[y_t^2] = E[u_t^2] = \sigma^2. \end{aligned} \quad (20.1)$$

Also, the conditional mean at time t , based on information at time $t-1$, is

$$E_{t-1}[y_t] = E_{t-1}[\rho y_{t-1} + u_t] = \rho E_{t-1}[y_{t-1}] + E_{t-1}[u_t] = \rho y_{t-1} = 0, \quad (20.2)$$

since $\rho = 0$.

Property 2: Autocorrelation in Squared Returns

The autocorrelation in the squares of returns demonstrated in Figure 20.1 shows that while predicting the direction of returns is not possible, predicting their volatility is. This suggests a model of squared returns, y_t^2 , of the following form

$$y_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + v_t, \quad (20.3)$$

where v_t is a disturbance term and α_0 and α_1 are parameters.

An alternative expression of the unconditional variance of y_t given in (20.1) is obtained by taking unconditional expectations of (20.3)

$$E[y_t^2] = E[\alpha_0 + \alpha_1 y_{t-1}^2 + v_t] = \alpha_0 + \alpha_1 E[y_{t-1}^2] + E[v_t]. \quad (20.4)$$

Using the fact that $E[y_t^2] = E[y_{t-1}^2] = \sigma^2$ and given that $E_{t-1}[y_t]$ from (20.2), expression (20.4) is rewritten as

$$\sigma_u^2 = \alpha_0 + \alpha_1 \sigma_u^2 + 0,$$

or

$$\sigma_u^2 = \frac{\alpha_0}{1 - \alpha_1}. \quad (20.5)$$

It follows that, for the unconditional variance to be positive, the restrictions $\alpha_0 > 0$ and $|\alpha_1| < 1$ are needed. Another important implication is that if $\alpha_1 = 1$ the unconditional variance is undefined.

Now consider the conditional variance of y_t based on information at $t-1$

$$\sigma_t^2 = E_{t-1}[y_t^2] - E_{t-1}[y_t]^2.$$

Using (20.2) reduces this expression to

$$\sigma_t^2 = E_{t-1}[y_t^2].$$

It follows from (20.3) that the conditional variance is

$$\sigma_t^2 = E_{t-1}[\alpha_0 + \alpha_1 y_{t-1}^2 + v_t] = \alpha_0 + \alpha_1 y_{t-1}^2 + E_{t-1}[v_t] = \alpha_0 + \alpha_1 y_{t-1}^2. \quad (20.6)$$

Unlike the conditional mean in (20.2), which is time invariant, the conditional variance does change over time as a result of changes in y_{t-1}^2 .

Property 3: Volatility Clustering

The volatility clustering property shows that small movements in returns tend to be followed by small returns in the next period, whereas large movements in returns tend to be followed by large returns in the next period. These movements imply that the autocorrelation of squared returns is positive, a property demonstrated in panel (d) of Figure 20.1. The expression of the conditional variance in (20.6) shows that volatility clustering requires the tighter restriction $0 < \alpha_1 < 1$.

Property 4: Conditional Normality

The conditional distribution of returns is normal with conditional mean given by (20.2) and conditional variance given by (20.6)

$$f(y|y_{t-1}) \sim N(0, \alpha_0 + \alpha_1 y_{t-1}^2). \quad (20.7)$$

For small values of y_{t-1} , the conditional variance is drawn from a relatively compact distribution with mean of zero and approximate variance α_0 . This indicates a high probability of drawing another small value of y in the next period. By contrast, for larger values of y_{t-1} the conditional variance is drawn from a more dispersed distribution with mean zero and variance $\alpha_0 + \alpha_1 y_{t-1}^2$. There is, therefore, a high probability of drawing another large value of y in the next period.

Property 5: Unconditional Leptokurtosis

The unconditional distribution is derived by averaging over all T conditional distributions. Even though the conditional distribution is normal, the unconditional distribution is not. For the relatively low-volatility conditional distributions, the normal distributions are relatively compact with high peaks, whereas, for the relatively high-volatility conditional distributions, the normal distributions are relatively more dispersed with low peaks. Averaging across the conditional distributions yields a nonnormal unconditional distribution, $f(y)$, that has fat-tails and a sharp peak compared to the normal distribution. A distribution with these two properties is said to exhibit leptokurtosis.

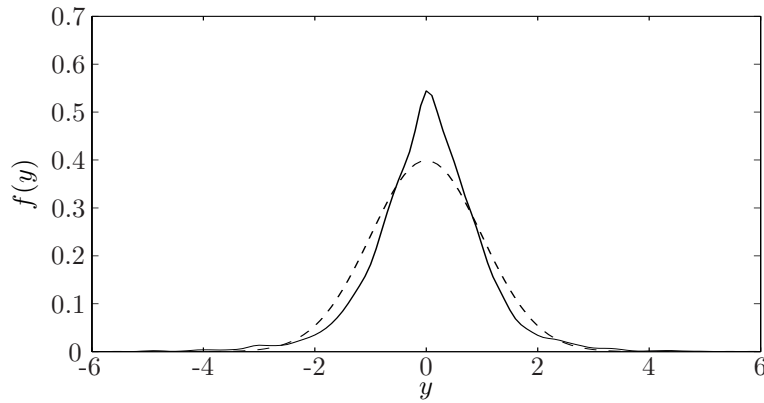


Figure 20.2 Unconditional (empirical) distribution of FTSE percentage returns (solid line) compared to the conditional (standardized normal) distribution (dashed line).

The unconditional distribution of the FTSE returns estimated using a nonparametric kernel estimator with a rule-of-thumb bandwidth is given in Figure 20.2 and is compared to the conditional distribution given by the standard normal distribution. The unconditional distribution exhibits leptokurtosis since it has a sharper peak and fatter tails than the standard normal distribution. The peak of the empirical distribution is 0.522 and the estimated kurtosis is 6.077. These values are to be compared to their standard normal counterparts of 0.399 and 3 respectively.

20.3 The ARCH Model

The analysis of the empirical characteristics of asset returns in the previous section suggests that modelling the autocorrelation structure of the variance is relatively more important than modelling the autocorrelation structure in the mean. This section discusses the ARCH class of model, which captures the autocorrelation properties in the variance.

20.3.1 Specification

The specification of the ARCH model is motivated by the discussion in the previous section and, in particular, equation (20.6) in which the conditional

variance is expressed as a function of lagged squared returns. The model is

$$\begin{aligned} y_t &= u_t \\ u_t &\sim N(0, \sigma_t^2) \\ \sigma_t^2 &= \alpha_0 + \sum_{i=1}^q \alpha_i u_{t-i}^2. \end{aligned} \quad (20.8)$$

This model is referred to as ARCH(q), where q refers to the order of the lagged squared returns included in the model. The conditional variance is given by

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \dots + \alpha_q u_{t-q}^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \dots + \alpha_q y_{t-q}^2,$$

where $\theta = \{\alpha_0, \alpha_1, \dots, \alpha_q\}$ is a vector of parameters to be estimated.

A special case of (20.8) is the ARCH(1) model given by

$$\begin{aligned} y_t &= u_t \\ u_t &\sim N(0, \sigma_t^2) \\ \sigma_t^2 &= \alpha_0 + \alpha_1 u_{t-1}^2 = \alpha_0 + \alpha_1 y_{t-1}^2, \end{aligned} \quad (20.9)$$

with the conditional distribution of y_t given by

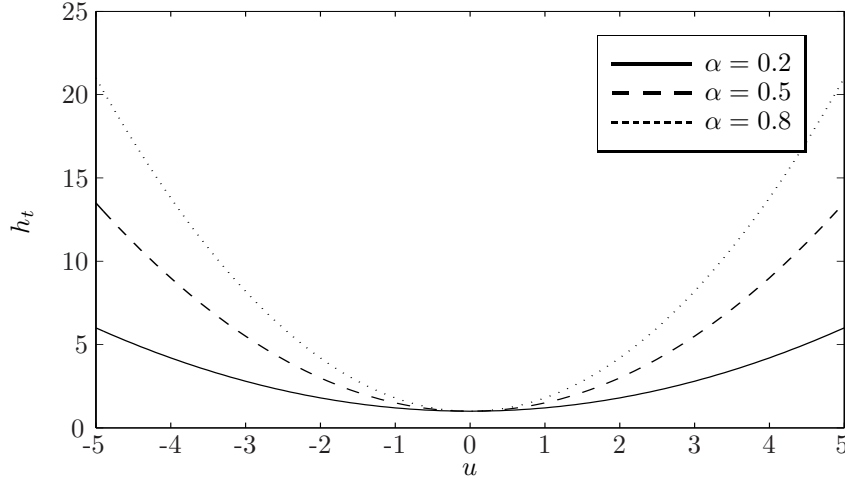
$$\begin{aligned} f(y_t | y_{t-1}; \theta) &= \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp \left[-\frac{y_t^2}{2\sigma_t^2} \right] \\ &= \frac{1}{\sqrt{2\pi(\alpha_0 + \alpha_1 y_{t-1}^2)}} \exp \left[-\frac{y_t^2}{2(\alpha_0 + \alpha_1 y_{t-1}^2)} \right]. \end{aligned} \quad (20.10)$$

Clearly the shape of this distribution changes over time as y_{t-1} changes.

Example 20.1 The News Impact Curve (NIC)

A property of the ARCH(1) model in (20.9) is that the conditional variance, σ_t^2 , changes as the disturbance in the previous period changes. This relationship is presented in Figure 20.3, which shows that σ_t^2 increases as the absolute value of u_{t-1} increases. This figure also shows that the relationship between σ_t^2 and u_{t-1} is symmetric since a positive disturbance has the same effect on σ_t^2 as does a negative disturbance of the same magnitude. This figure is commonly referred to as the news impact curve (NIC) because the disturbance term u_{t-1} being the unexpected portion of the conditional mean is usually interpreted as the news. The symmetry of the NIC curve is inconsistent with empirical research, a point that is taken up again in Section 20.4. $\square$

Figure 20.3 The news impact curve (NIC) for alternative parameterizations of the ARCH(1) model given in equation (20.9).



20.3.2 Estimation

For ease of exposition, the details of the estimation of ARCH models are presented for the ARCH(1) specification in equation (20.9). The model can be estimated by maximum likelihood using a standard iterative algorithm discussed in Chapter 3. For a sample of $t = 1, 2, \dots, T$ observations, the log-likelihood function is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \ln L_t(\theta) = \frac{1}{T} \sum_{t=1}^T \ln f(y_t|y_{t-1}; \theta) + \frac{1}{T} \ln f(y_0), \quad (20.11)$$

where $f(y_t|y_{t-1}; \theta)$ is the conditional distribution in (20.10) and $f(y_0)$ is the marginal distribution of y_0 . In practice, y_0 is taken as given, usually chosen from the unconditional distribution. Formally, this means that $f(y_0) = 1$ resulting in the log-likelihood function simplifying to

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \ln f(y_t|y_{t-1}; \theta), \quad (20.12)$$

where, for the ARCH model in equation (20.10), the log-likelihood function is

$$\begin{aligned}\ln L_T(\theta) &= \frac{1}{T} \sum_{t=1}^T \ln f(y_t|y_{t-1}; \theta) \\ &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2T} \sum_{t=1}^T \ln(\sigma_t^2) - \frac{1}{2T} \sum_{t=1}^T \frac{y_t^2}{\sigma_t^2} \\ &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2T} \sum_{t=1}^T \ln(\alpha_0 + \alpha_1 y_{t-1}^2) - \frac{1}{2T} \sum_{t=1}^T \frac{y_t^2}{\alpha_0 + \alpha_1 y_{t-1}^2}.\end{aligned}\quad (20.13)$$

The first and second derivative of the log-likelihood function in equation (20.13) are required if the parameters $\theta = \{\alpha_0, \alpha_1\}$ are to be estimated by means of a gradient algorithm. Analytical derivatives are easily computed. The first derivatives are

$$\begin{aligned}\frac{\partial \ln L_T(\theta)}{\partial \alpha_0} &= \frac{\partial \ln L_T(\theta)}{\partial \sigma_t^2} \frac{\partial \sigma_t^2}{\partial \alpha_0} = \frac{1}{T} \sum_{t=1}^T \left[-\frac{1}{2\sigma_t^2} + \frac{y_t^2}{2\sigma_t^4} \right] \frac{\partial \sigma_t^2}{\partial \alpha_0} = \frac{1}{T} \sum_{t=1}^T \frac{1}{2\sigma_t^2} \left[\frac{y_t^2}{\sigma_t^2} - 1 \right], \\ \frac{\partial \ln L_T(\theta)}{\partial \alpha_1} &= \frac{\partial \ln L_T(\theta)}{\partial \sigma_t^2} \frac{\partial \sigma_t^2}{\partial \alpha_1} = \frac{1}{T} \sum_{t=1}^T \left[-\frac{1}{2\sigma_t^2} + \frac{y_t^2}{2\sigma_t^4} \right] \frac{\partial \sigma_t^2}{\partial \alpha_1} = \frac{1}{T} \sum_{t=1}^T \frac{1}{2\sigma_t^2} \left[\frac{y_t^2}{\sigma_t^2} - 1 \right] y_{t-1}^2,\end{aligned}$$

so that the gradient vector is

$$G_T(\theta) = \frac{\partial \ln L_T(\theta)}{\partial \theta} = \frac{1}{T} \sum_{t=1}^T \frac{1}{2\sigma_t^2} \left[\frac{y_t^2}{\sigma_t^2} - 1 \right] \begin{bmatrix} 1 \\ y_{t-1}^2 \end{bmatrix}. \quad (20.14)$$

The second derivatives are

$$\begin{aligned}\frac{\partial^2 \ln L_T(\theta)}{\partial \alpha_0^2} &= \frac{1}{T} \sum_{t=1}^T \left(\frac{1}{2\sigma_t^4} - \frac{y_t^2}{\sigma_t^6} \right) \\ \frac{\partial^2 \ln L_T(\theta)}{\partial \alpha_0 \partial \alpha_1} &= \frac{1}{T} \sum_{t=1}^T \left(\frac{1}{2\sigma_t^4} - \frac{y_t^2}{\sigma_t^6} \right) y_{t-1}^2 \\ \frac{\partial^2 \ln L_T(\theta)}{\partial \alpha_1^2} &= \frac{1}{T} \sum_{t=1}^T \left(\frac{1}{2\sigma_t^4} - \frac{y_t^2}{\sigma_t^6} \right) y_{t-1}^4,\end{aligned}$$

which yields the Hessian matrix

$$H_T(\theta) = \frac{\partial^2 \ln L_T(\theta)}{\partial \theta \partial \theta'} = \frac{1}{T} \sum_{t=1}^T \left(\frac{1}{2\sigma_t^4} - \frac{y_t^2}{\sigma_t^6} \right) \begin{bmatrix} 1 & y_{t-1}^2 \\ y_{t-1}^2 & y_{t-1}^4 \end{bmatrix}. \quad (20.15)$$

Beginning with some starting values of the parameters, say $\theta_{(0)}$, the Newton-Raphson algorithm can be used to update the parameter values using the iterative scheme

$$\theta_{(k)} = \theta_{(k-1)} - H_{(k-1)}^{-1} G_{(k-1)}, \quad (20.16)$$

where $G_{(k-1)} = G_T(\theta_{(k-1)})$ using equation (20.14), $H_{(k-1)} = H_T(\theta_{(k-1)})$ using equation (20.15), and σ_t^2 is evaluated at $\theta_{(k-1)}$

The scoring algorithm can also be used. In deriving the information matrix, however, it is convenient to modify the algorithm by using the conditional expectation instead of the usual unconditional expectation. By definition $\sigma_t^2 = E_{t-1}[y_t^2]$ so that, from equation (20.15), the information matrix is

$$\begin{aligned} I(\theta) &= -E_{t-1}[H_T(\theta)] \\ &= \frac{1}{T} \sum_{t=1}^T \left(\frac{1}{2\sigma_t^4} - \frac{E_{t-1}[y_t^2]}{\sigma_t^6} \right) \begin{bmatrix} 1 & y_{t-1}^2 \\ y_{t-1}^2 & y_{t-1}^4 \end{bmatrix} \\ &= -\frac{1}{T} \sum_{t=1}^T \left(\frac{1}{2\sigma_t^4} - \frac{\sigma_t^2}{\sigma_t^6} \right) \begin{bmatrix} 1 & y_{t-1}^2 \\ y_{t-1}^2 & y_{t-1}^4 \end{bmatrix} \\ &= \frac{1}{T} \sum_{t=1}^T \frac{1}{2\sigma_t^4} \begin{bmatrix} 1 & y_{t-1}^2 \\ y_{t-1}^2 & y_{t-1}^4 \end{bmatrix}. \end{aligned} \quad (20.17)$$

The modified scoring algorithm proceeds as follows

$$\theta_{(k)} = \theta_{(k-1)} + I_{(k-1)}^{-1} G_{(k-1)}, \quad (20.18)$$

where both $G_{(k-1)}$, defined in (20.14), and $I_{(k-1)}$, defined in (20.17), are evaluated at $\theta_{(k-1)}$.

The maximization of the log-likelihood function in equation (20.13) requires a starting value for σ_1^2

$$\sigma_1^2 = \alpha_0 + \alpha_1 y_0^2,$$

which, in turn, requires a starting value for y_0 . The simplest solution is to choose σ_1^2 as the sample variance of y_t , which represents an estimate of the unconditional variance of y_t . Another approach is to compute σ_1^2 as immediately above, by setting $y_0 = 0$, which is the unconditional mean of y_t for the model in equation (20.9).

The specification of the variance in equation (20.9) does not necessarily ensure that σ_t^2 is always non-negative. For example, a negative estimate of the conditional variance may arise if α_0 and/or α_1 become negative during the iterations. If this happens, even for just one observation, the optimization

algorithm will break down because $\ln \sigma_t^2$ in (20.13) cannot be computed. One possible solution is to follow the suggestion made in Chapter 3 and transform α_0 and α_1 using an appropriate mapping that ensures positive estimates. A popular choice is exponential tilting in which the algorithm estimates the transformed parameters δ_i where $\alpha_i = \exp \delta_i$. Standard errors of $\hat{\alpha}_i$ can be computed by the delta method or the model can be iterated one more time with $\exp(\hat{\delta}_i)$ replaced by $\hat{\alpha}_i$. From the invariance property of maximum likelihood estimators, the estimates of α_i obtained using this strategy are also the maximum likelihood estimates.

It is also usual to confine attention to cases in which the process generating the disturbances is variance stationary, that is, the unconditional variance of u_t is unchanging over time. This requires that $\alpha_1 < 1$ in equation (20.9), which can also be imposed by using an appropriate transformation of the parameter as discussed in Chapter 3.

Example 20.2 Estimating an ARCH Model

Consider the daily percentage returns, y_t , on the FTSE from 5 January 1989 to 31 December 2007. The maximum likelihood estimates of the parameters of the ARCH(1) model in equation (20.9) applied to this data are

$$\begin{aligned} y_t &= \hat{u}_t \\ \hat{\sigma}_t^2 &= \hat{\alpha}_0 + \hat{\alpha}_1 \hat{u}_{t-1}^2 = 0.739 + 0.255 y_{t-1}^2. \end{aligned}$$

The value of the log-likelihood function is $\ln L_T(\theta) = -1.378$, and $T \ln L_T(\theta) = -4952 \times 1.378 = -6824.029$. An estimate of the theoretical unconditional variance using (20.5) is

$$\hat{\sigma}^2 = \frac{0.739}{1 - 0.255} = 0.993,$$

which is consistent with the sample variance of 0.984 obtained for y_t . $\square$

20.3.3 Testing

A test of ARCH is given by testing that $\alpha_1 = 0$ in (20.9) so that the model under the null hypothesis reduces to a normal distribution with zero mean and constant variance $\sigma_t^2 = \alpha_0$,

$$f(y_t) \sim N(0, \alpha_0).$$

The null and alternative hypotheses are

$$H_0 : \alpha_1 = 0 \quad [\text{No ARCH}], \quad H_1 : \alpha_1 \neq 0 \quad [\text{ARCH}].$$

Let the parameters of the restricted model under the null hypothesis be

given by $\hat{\theta}_0 = \{\hat{\alpha}_0, 0\}$, where $\hat{\alpha}_0$ is the maximum likelihood estimator of α_0 under the null hypothesis. For the model in equation (20.9) the maximum likelihood estimator is simply

$$\hat{\alpha}_0 = \frac{1}{T} \sum_{t=1}^T y_t^2. \quad (20.19)$$

The test of ARCH can be performed by using either the LR, Wald or LM tests. In practice the LM test is commonly used since it simply involves estimating an OLS regression equation and performing a goodness-of-fit test.

The information matrix version of the LM test is based on the test statistic

$$LM = TG_T(\hat{\theta}_0)' I(\hat{\theta}_0)^{-1} G_T(\hat{\theta}_0), \quad (20.20)$$

where $G_T(\hat{\theta}_0)$ and $I(\hat{\theta}_0)$ are the gradient vector and information matrix, respectively, evaluated at the parameter estimates under the null hypothesis. From equations (20.14) and (20.17), the gradient vector and information matrix evaluated at $\hat{\theta}_0$ are

$$G_T(\hat{\theta}_0) = \frac{1}{T} \sum_{t=1}^T \frac{1}{2\hat{\alpha}_0} \left[\frac{y_t^2}{\hat{\alpha}_0} - 1 \right] \begin{bmatrix} 1 \\ y_{t-1}^2 \end{bmatrix}$$

$$I_t(\hat{\theta}_0) = \frac{1}{T} \sum_{t=1}^T \frac{1}{2\hat{\alpha}_0^2} \begin{bmatrix} 1 & y_{t-1}^2 \\ y_{t-1}^2 & y_{t-1}^4 \end{bmatrix}.$$

Using these expressions in (20.20) and simplifying gives

$$LM = \frac{1}{2} \begin{bmatrix} \sum_{t=1}^T v_t \\ \sum_{t=1}^T v_t y_{t-1}^2 \end{bmatrix}' \begin{bmatrix} T & \sum_{t=1}^T y_{t-1}^2 \\ \sum_{t=1}^T y_{t-1}^2 & \sum_{t=1}^T y_{t-1}^4 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T v_t \\ \sum_{t=1}^T v_t y_{t-1}^2 \end{bmatrix}, \quad (20.21)$$

where

$$v_t = \frac{y_t^2}{\hat{\alpha}_0} - 1. \quad (20.22)$$

Alternatively, since under the null hypothesis

$$z_t = \frac{y_t}{\sqrt{\hat{\alpha}_0}} \sim N(0, 1),$$

it follows that v_t in equation (20.22) has the property

$$\text{plim} \left(\frac{1}{T} \sum_{t=1}^T v_t^2 \right) = \text{plim} \left(\frac{1}{T} \sum_{t=1}^T (z_t^2 - 1)^2 \right) = \text{plim} \left(\frac{1}{T} \sum_{t=1}^T (z_t^4 - 2z_t^2 + 1) \right) = 2,$$

because $\text{plim}(T^{-1} \sum_{t=1}^T z_t^2) = 1$ and $\text{plim}(T^{-1} \sum_{t=1}^T z_t^4) = 3$ from the properties of the normal distribution. Consequently, another asymptotic form for the LM test in equation (20.21) is to replace the $1/2$ in equation (20.21) by $T / \sum_{t=1}^T v_t^2$ which is the inverse of the variance of v_t . This yields the test statistic

$$LM = \frac{T}{\sum_{t=1}^T v_t^2} \begin{bmatrix} \sum_{t=1}^T v_t \\ \sum_{t=1}^T v_t y_{t-1}^2 \end{bmatrix}' \begin{bmatrix} T & \sum_{t=1}^T y_{t-1}^2 \\ \sum_{t=1}^T y_{t-1}^2 & \sum_{t=1}^T y_{t-1}^4 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T v_t \\ \sum_{t=1}^T v_t y_{t-1}^2 \end{bmatrix}. \quad (20.23)$$

This form of the LM test may be computed as TR^2 , where R^2 is the coefficient of determination from regressing v_t on $\{1, y_{t-1}^2\}$ because under the null hypothesis $\bar{v} = T^{-1} \sum_{t=1}^T v_t = 0$. An even more simple form is to replace v_t in equation (20.22) by y_t^2 , since R^2 is invariant to linear transformations. The ARCH test now involves regressing y_t^2 on $\{1, y_{t-1}^2\}$ and computing the R^2 from this regression. This form of the ARCH test corresponds to the regression equation initially proposed in equation (20.3) as a model of squared returns. Under the null hypothesis, the LM statistic is distributed as χ_1^2 .

Generalizing the LM test to the ARCH(q) model in (20.8) is straightforward. The null and alternative hypotheses are now

$$\begin{aligned} H_0 : & \alpha_1 = \alpha_2 = \cdots \alpha_q = 0 & [\text{No ARCH}] \\ H_1 : & \text{at least one of the restrictions is violated} & [\text{ARCH}]. \end{aligned}$$

The ARCH test is implemented using the following steps.

Step 1: Estimate the regression equation

$$y_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i y_{t-i}^2 + v_t, \quad (20.24)$$

by least squares, where v_t is an *iid* disturbance term.

Step 2: Compute TR^2 from this regression and the corresponding p-value using the χ_q^2 distribution. A p-value less than 0.05 is evidence of ARCH in y_t at the 5% level.

Example 20.3 Testing for ARCH

Daily percentage returns to the FTSE, Dow Jones Industrial Average (DOW) and the NIKKEI indices for the period 5 January 1989 to 31 December 2007 are used to test for ARCH(1) and ARCH(2) and the results are summarized in Table 20.1. As the LM test is based on the model in equation (20.8) has zero mean, the returns are first transformed to have

zero mean by subtracting the sample mean, $\bar{y}$. All p-values given in the last column are clearly less than 0.05, showing strong evidence of first-order and second-order ARCH in all equity returns. $\square$

Table 20.1

LM test for ARCH(1) and ARCH(2) in equity returns for the period 5 January 1989 to 31 December 2007, $T = 4952$. The parameter estimates, $\hat{\alpha}_i$, are the least squares estimates for the specification in equation (20.24) and R^2 is the coefficient of determination from this regression.

Index	$\hat{\alpha}_0$	$\hat{\alpha}_1$	$\hat{\alpha}_2$	T	R^2	$LM = TR^2$	pv
<i>ARCH(1)</i>							
FTSE	0.765	0.223		4951	0.050	245.306	0.000
DOW	0.774	0.158		4951	0.025	123.382	0.000
NIKKEI	1.716	0.108		4951	0.012	57.840	0.000
<i>ARCH(2)</i>							
FTSE	0.573	0.167	0.251	4950	0.109	541.888	0.000
DOW	0.674	0.137	0.130	4950	0.041	204.805	0.000
NIKKEI	1.503	0.095	0.124	4950	0.027	133.006	0.000

20.4 Univariate Extensions

A vast number of extensions to the ARCH model have been proposed in the literature and applied to modelling financial data. In this section, some of the more popular extensions are discussed briefly.

20.4.1 GARCH

The ARCH(q) model in (20.8) has the property that the memory in the variance stops at lag q . This suggests that, for processes that exhibit long memory in the variance, it would be necessary to specify and estimate a high dimensional model. A natural way to circumvent this problem is to specify the conditional variance as a function of its own lags. The equation for the conditional variance then becomes

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i u_{t-i}^2 + \sum_{i=1}^p \beta_i \sigma_{t-i}^2, \quad (20.25)$$

which is known as GARCH(p, q) where the p and the q identify the lags of the model and the ‘G’ stands for generalized ARCH. Once again, without

loss of generality, it is convenient to work with the GARCH(1,1) model, which is

$$y_t = u_t, \quad u_t \sim N(0, \sigma_t^2) \quad (20.26)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{t-1}^2. \quad (20.27)$$

To highlight the long memory properties of this model, using the lag operator $L^k y_t = y_{t-k}$ (see Appendix B) rewrite the expression for the conditional variance, σ_t^2 , as

$$(1 - \beta_1 L) \sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2.$$

Assuming that $|\beta_1| < 1$ and using the properties of the lag operator, the conditional variance can be expressed as

$$\begin{aligned} \sigma_t^2 &= (1 - \beta_1 L)^{-1} \alpha_0 + \alpha_1 (1 - \beta_1 L)^{-1} y_{t-1}^2 \\ &= \frac{\alpha_0}{1 - \beta_1} + \alpha_1 y_{t-1}^2 + \alpha_1 \sum_{i=1}^{\infty} \beta_1^i y_{t-1-i}^2, \end{aligned} \quad (20.28)$$

which is instantly recognizable as an ARCH(∞) model. The third term in equation (20.28) captures the effects of higher order lags $\{y_{t-2}^2, y_{t-3}^2, y_{t-4}^2, \dots\}$ on the conditional variance, which are now controlled by a sole parameter, β_1 . The GARCH model is therefore an attractive specification to use in modelling financial data because it provides a parsimonious representation of the memory characteristics commonly observed in the variance of financial returns.

Another way to highlight the memory characteristics of the GARCH conditional variance is to define the (forecast) error

$$v_t = y_t^2 - \sigma_t^2, \quad (20.29)$$

which has the property $E_{t-1}[v_t] = 0$. Rearranging this expression and using (20.27) gives

$$\begin{aligned} y_t^2 &= \sigma_t^2 + v_t \\ y_t^2 &= \alpha_0 + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + v_t \\ y_t^2 &= \alpha_0 + \alpha_1 y_{t-1}^2 + \beta_1 (y_{t-1}^2 - v_{t-1}) + v_t \\ y_t^2 &= \alpha_0 + (\alpha_1 + \beta_1) y_{t-1}^2 - \beta_1 v_{t-1} + v_t, \end{aligned} \quad (20.30)$$

which is an ARMA(1,1) model in terms of y_t^2 . The memory of this process is determined by the autoregressive parameter $\alpha_1 + \beta_1$. The closer is $\alpha_1 + \beta_1$ to unity the longer is the effect of a shock on volatility.

The effect of a shock on volatility in the long-run is obtained from the

unconditional variance of y_t , defined as $\sigma^2 = E[y_t^2]$. Taking unconditional expectations of (20.30) and using $E[v_t] = E[v_{t-1}] = 0$, it follows that

$$\begin{aligned} E[y_t^2] &= E[\alpha_0 + (\alpha_1 + \beta_1)y_{t-1}^2 - \beta_1 v_{t-1} + v_t], \\ \sigma^2 &= \alpha_0 + (\alpha_1 + \beta_1)\sigma^2, \end{aligned}$$

which gives, upon rearranging, an expression of the unconditional, or long-run, variance

$$\sigma^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}. \quad (20.31)$$

Example 20.4 ACF of a GARCH(1,1) Model

The autocorrelation function (ACF) of the GARCH(1,1) model in (20.27) is simulated for $T = 1000$ observations. Two sets of parameter values are used:

$$\begin{aligned} \text{Model 1 (short memory)} &: \alpha_0 = 0.10, \alpha_1 = 0.70, \beta_1 = 0.20 \\ \text{Model 2 (long memory)} &: \alpha_0 = 0.05, \alpha_1 = 0.15, \beta_1 = 0.80, \end{aligned}$$

where the memory classification of the model is determined by the relative size of β_1 . The ACFs of the two models are given in Figure 20.4. The short memory feature of model 1 is highlighted by the ACF approaching zero at lag 10, compared to the long memory model where the decay is slower. $\square$

As with the ARCH model, the GARCH model can be estimated by maximizing the log-likelihood function in (20.12) using a gradient algorithm. For the GARCH(1,1) model the unknown parameters are $\theta = \{\alpha_0, \alpha_1, \beta_1\}$ and the log-likelihood function at observation t is

$$\begin{aligned} \ln l_t(\theta) &= \ln f(y_t|y_{t-1}) \\ &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\sigma_t^2) - \frac{1}{2} \frac{y_t^2}{\sigma_t^2} \\ &= -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln(\alpha_0 + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{t-1}^2) \\ &\quad - \frac{1}{2} \frac{y_t^2}{\alpha_0 + \alpha_1 y_{t-1}^2 + \beta_1 \sigma_{t-1}^2}. \end{aligned} \quad (20.32)$$

In estimating this model, as with the ARCH model, it may be necessary to restrict the parameters α_0 , α_1 and β_1 to be positive in order to ensure that the conditional variance is positive for all t .

The LR, Wald and LM testing methods can also be used to test the GARCH model. The LM test derived previously in the context of the ARCH model is still relevant for the GARCH model. It should be noted that finding

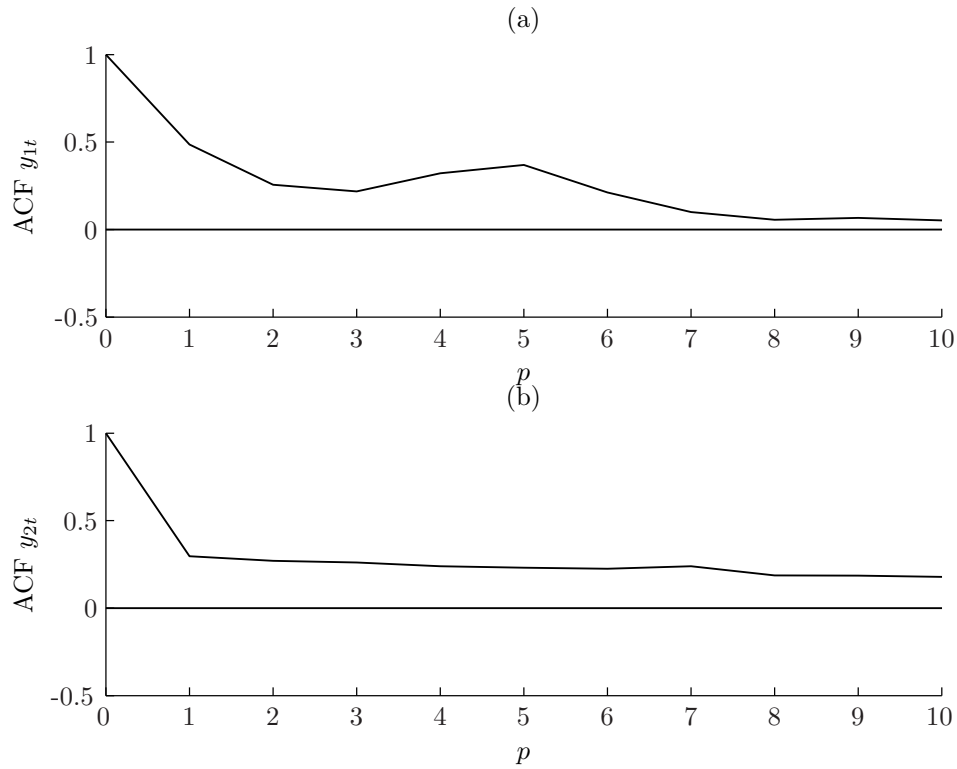


Figure 20.4 Autocorrelation function of two GARCH(1,1) models. The first model has parameter values $\alpha_0 = 0.10$, $\alpha_1 = 0.70$ and $\beta_1 = 0.20$. The second model has parameter values $\alpha_0 = 0.05$, $\alpha_1 = 0.15$, and $\beta_1 = 0.8$.

ARCH at relatively longer lags is evidence of a GARCH(1,1) or, possibly, a GARCH(2,2) specification. Another natural test is given by the hypotheses

$$H_0 : \beta_1 = 0 \quad [\text{No GARCH}], \quad H_1 : \beta_1 \neq 0 \quad [\text{GARCH}],$$

which may be tested using either the Wald or LR test.

Example 20.5 A GARCH Model of Equity Returns

Consider again the daily percentage returns to the FTSE, DOW and NIKKEI used to test for ARCH in Example 20.3. A GARCH(1,1) model of the zero-mean equity returns, y_t , in equation (20.27) is estimated by maximum likelihood for these equity indexes. For comparative purposes, the results for the GARCH(0,1) or ARCH(1) model are also computed. The parameter estimates are reported in Table 20.2, with standard errors based on the Hessian in parentheses.

An important feature of the empirical results is the consistency of the

Table 20.2

Maximum likelihood estimates of GARCH(1,1) and GARCH(0,1) = ARCH(1) models of equity returns. Standard errors in parentheses are based on the Hessian. The sample size is $T = 4952$.

Index	α_0	α_1	β_1	$T \ln L$	Unconditional Variance	
					(Theoretical)	(Empirical)
FTSE	0.013	0.079	0.907	-6348.796	0.964	0.984
	(0.003)	(0.008)	(0.012)			
	0.740	0.255	0.000	-6824.029	0.993	0.984
	(0.020)	(0.023)				
DOW	0.009	0.051	0.940	-6316.263	0.975	0.919
	(0.002)	(0.006)	(0.008)			
	0.748	0.195	0.000	-6712.500	0.929	0.919
	(0.019)	(0.022)				
NIKKEI	0.026	0.088	0.903	-8187.337	2.728	1.924
	(0.005)	(0.008)	(0.008)			
	1.587	0.182		-8561.058	1.940	1.924
	(0.040)	(0.020)				

parameter estimates across all three equity returns, with the estimates of β_1 being around 0.9 and the estimates of α_1 being between 0.05 and 0.10. Also given in Table 20.2 are the theoretical variance computed using equation (20.31) and the empirical variance. There is good agreement between these two statistics for the FTSE and the DOW, but less so for the NIKKEI in the case of the GARCH(1,1) model.

Figure 20.5 provides a plot of the estimated conditional variance of the FTSE. The first three observations of $\hat{h}_t$ are computed recursively as

$$\begin{aligned}\hat{\sigma}_1^2 &= \frac{1}{T} \sum_{t=1}^T y_t^2 = 0.984, \\ \hat{\sigma}_2^2 &= \hat{\alpha}_0 + \hat{\alpha}_1 y_1^2 + \hat{\beta}_1 \hat{h}_1 = 0.013 + 0.079 \times 0.628^2 + 0.907 \times 0.984 = 0.937, \\ \hat{\sigma}_3^2 &= \hat{\alpha}_0 + \hat{\alpha}_1 y_2^2 + \hat{\beta}_1 \hat{h}_2 = 0.013 + 0.079 \times 1.083^2 + 0.907 \times 0.937 = 0.956,\end{aligned}$$

where $y_1 = 0.628$ and $y_2 = 0.083$ are the first two observations of the zero-mean returns.

From Table 20.2, in the case of the FTSE, $T \ln L_T(\hat{\theta}_1) = -6348.796$ for the GARCH(1,1) model and $\ln L_T(\hat{\theta}_0) = -6824.029$ for the ARCH(1) model.

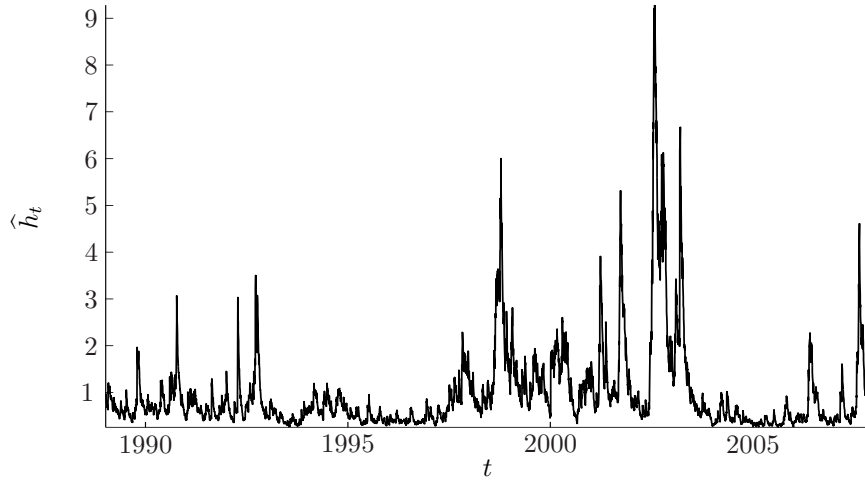


Figure 20.5 Conditional variance estimate of the zero-mean percentage daily returns to FTSE from 5 January 1989 to 31 December 2007.

The LR statistic is

$$LR = -2(T \ln L(\hat{\theta}_0) - T \ln L(\hat{\theta}_1)) = -2 \times (-6824.029 + 6348.796) = 950.467,$$

which is distributed as χ_1^2 under the null hypothesis. The p-value is 0.000 indicating a strong rejection of the ARCH(1) model at the 5% level in favour of the GARCH(1,1) model. $\square$

20.4.2 Integrated GARCH

A feature of the volatility estimates of the GARCH(1,1) model of equity returns in Table 20.2 is that $\hat{\alpha}_1 + \hat{\beta}_1 \simeq 1$. For example, for the FTSE

$$\hat{\alpha}_1 + \hat{\beta}_1 = 0.079 + 0.907 = 0.986.$$

From (20.30) this suggests that the volatility series has a (near) unit root. GARCH models with a unit root are referred to as IGARCH, where the ‘I’ stands for integrated following the discussion of nonstationary models in Part FIVE.

One way to proceed in estimating IGARCH models is to impose the unit root restriction. In the case of the GARCH(1,1) model, the restriction is $\alpha_1 + \beta_1 = 1$, in which case β_1 is replaced by $1 - \alpha_1$ so the conditional variance in equation (20.27) becomes

$$\sigma_t^2 = \alpha_0 + \alpha_1 y_{t-1}^2 + (1 - \alpha_1) \sigma_{t-1}^2. \quad (20.33)$$

The log-likelihood function is now maximized with respect to the variance parameters $\{\alpha_0, \alpha_1\}$.

A generalization of the IGARCH model is the fractional integrated GARCH model (FIGARCH) where now the memory in the GARCH model is controlled by a fractional differencing parameter.

20.4.3 Additional Variables

A further extension of the ARCH class of model is to include additional variables in both the mean and the variance. An example, using the GARCH(1,1) conditional variance specification, is given by

$$y_t = \gamma_0 + \gamma_1 x_t + u_t, \quad u_t \sim N(0, \sigma_t^2) \quad (20.34)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + \lambda w_t, \quad (20.35)$$

where x_t and w_t are additional explanatory variables which are important in explaining the mean and the variance, respectively. Examples of w_t include trade volume, policy announcements and day-of-the-week and holiday effects. Han and Park (2008) allow for nonstationary variables in the variance to model the IGARCH properties discussed earlier.

Example 20.6 LR Test of Day-of-the-Week Effects

Consider the GARCH(1,1) model of financial returns

$$\begin{aligned} y_t &= u_t, \quad u_t \sim N(0, \sigma_t^2) \\ \sigma_t^2 &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\ &\quad + \lambda_1 TUE_t + \lambda_2 WED_t + \lambda_3 THUR_t + \lambda_4 FRI_t, \end{aligned}$$

where TUE_t , WED_t , $THUR_t$ and FRI_t are dummy variables taking the value 1 on the appropriate day and 0 otherwise. A test of day-of-the-week effects is given by the hypotheses

$$\begin{aligned} H_0 : \quad & \lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 0 \quad [\text{No Day-of-the-Week Effects}] \\ H_1 : \quad & \lambda_i \neq 0 \text{ for at least one } i \quad [\text{Day-of-the-Week Effects}]. \end{aligned}$$

When the FTSE data from Example 20.5 are used, the unconstrained model yields a value of $T \ln L_T(\hat{\theta}_1) = -6347.691$, whereas the constrained value is $T \ln L_T(\hat{\theta}_0) = -6348.796$, which is obtained from Table 20.2. The LR statistic is

$$LR = -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)) = -2 \times (-6348.796 + 6347.691) = 2.210,$$

which is distributed under the null as χ_4^2 . The p-value is 0.697 resulting in a failure to reject the null of no day-of-the-week effects at the 5% level. $\square$

To construct a LM test of ARCH where explanatory variables are included in the model, the assumption of normality enables the test to be implemented using two least squares regressions because the information matrix separates into two blocks. The first block contains the mean parameters $\{\gamma_0, \gamma_1\}$ and the second block contains the variance parameters $\{\alpha_0, \alpha_1, \beta_1, \lambda\}$. The steps to test for ARCH of order q in the model

$$y_t = \gamma_0 + \gamma_1 x_t + u_t, \quad u_t \sim N(0, \sigma_t^2) \quad (20.36)$$

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i u_{t-i}^2, \quad (20.37)$$

are as follows.

Step 1 Regress y_t on $\{1, x_t\}$ and compute the least squares residuals $\hat{u}_t$.

Step 2 Regress $\hat{u}_t^2$ on $\{1, \hat{u}_{t-1}^2, \hat{u}_{t-2}^2, \dots, \hat{u}_{t-q}^2\}$.

Step 3 Compute TR^2 from Step 2 and the corresponding p-value using the χ_q^2 distribution. A p-value of less than 0.05 is evidence of ARCH in y_t at the 5% level.

20.4.4 Asymmetries

The GARCH model assumes that negative and positive shocks have the same impact on volatility, an assumption which ensures that the NIC of the GARCH model, like that of the ARCH model in Figure 20.3, is symmetric. A natural extension of the GARCH model is to allow the effects of negative and positive shocks on the conditional variance to differ. This is especially important in modelling equity markets where negative shocks are expected to have a relatively bigger effect on volatility than positive shocks as a result of leverage effects.

One way to model asymmetries in volatility is the threshold GARCH model (TARCH). The variance is now specified as

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i u_{t-i}^2 + \sum_{i=1}^p \beta_i \sigma_{t-i}^2 + \phi u_{t-1}^2 d_{t-1}, \quad (20.38)$$

where d_{t-1} is a dummy variable given by

$$d_{t-1} = \begin{cases} 1 & : u_{t-1} < 0 \\ 0 & : u_{t-1} \geq 0. \end{cases}$$

If $\phi > 0$, then good news, given by $u_{t-1} \geq 0$, has an effect on volatility equal to α_1 , whereas bad news, given by $u_{t-1} < 0$, has an effect on volatility equal to $\alpha_1 + \phi$. If falls in the equity market are followed by relatively higher

volatility, then $\phi > 0$. For the case of $p = q = 1$ lags, these features of the model are conveniently summarized as

$$\begin{aligned}\sigma_t^2(\text{good}) &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\ \sigma_t^2(\text{bad}) &= \alpha_0 + (\alpha_1 + \phi) u_{t-1}^2 + \beta_1 \sigma_{t-1}^2.\end{aligned}$$

The effect of $\phi \neq 0$ is to make the NIC asymmetric. The TARCH model can be estimated using maximum likelihood methods and a test of the model can be conducted using a Wald test of the hypothesis $\phi = 0$.

An alternative asymmetric model of the variance is the exponential GARCH (EGARCH) model

$$\ln \sigma_t^2 = \alpha_0 + \alpha_1 \frac{u_{t-1}}{\sigma_{t-1}} + \phi \left| \frac{u_{t-1}}{\sigma_{t-1}} \right| + \beta_1 \ln \sigma_{t-1}^2. \quad (20.39)$$

In this simple EGARCH(1, 1) model, the asymmetry in the conditional variance is governed by the parameter ϕ . As it is the logarithm of the conditional variance that is being modelled, the EGARCH conditional variance is constrained to be non-negative for all observations, regardless of the sign of the parameter estimates. There is, therefore, no need to constrain the parameter estimates to be positive when estimating the EGARCH model.

20.4.5 Garch-in-Mean

An important application of the ARCH class of volatility models is in modelling the trade-off between the expected return, μ , and risk, σ , of an asset

$$\mu = \gamma_0 + \varphi \sigma^\rho, \quad (20.40)$$

where γ_0, φ, ρ are parameters. In general, the higher the risk, the higher the expected return needed to compensate the asset holder. However, the increase in the expected return required to compensate for a given increase in risk varies according to the attitudes toward risk of asset holders. Three categories of risk behaviour are identified assuming that $\rho > 0$:

1. Risk averter ($\varphi > 0$) : μ increases as σ increases.
2. Risk neutral ($\varphi = 0$) : μ does not change as σ increases.
3. Risk lover ($\varphi < 0$) : μ decreases as σ increases.

These differences are highlighted in Figure 20.6 which shows the trade-off between μ and σ in equation (20.40) where $\rho > 0$.

The ARCH model provides a natural and convenient way to model the dynamic tradeoff between expected return and risk by simply including the

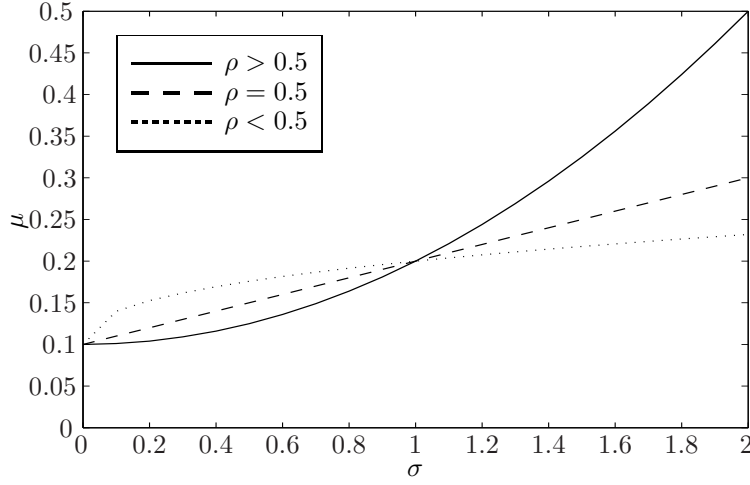


Figure 20.6 Alternative risk-return trade-offs for different risk preferences.

conditional standard deviation σ_t , into the conditional mean, μ_t . This model is known as the GARCH-M model and is given by

$$\begin{aligned} y_t &= \mu_t + u_t, & u_t &\sim N(0, \sigma_t^2) \\ \mu_t &= \gamma_0 + \varphi \sigma_t^\rho \\ \sigma_t^2 &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2. \end{aligned} \quad (20.41)$$

The second equation gives the relationship between the expected return (μ_t) and the risk (σ_t) of an asset, while the third equation gives the dynamics of the conditional variance (σ_t^2) assuming a GARCH(1,1) process. The alternative types of risk aversion are classified as

$$\begin{aligned} \text{Risk aversion} &: \varphi > 0 \\ \text{Risk neutral} &: \varphi = 0 \\ \text{Risk lover} &: \varphi < 0. \end{aligned} \quad (20.42)$$

As illustrated in Figure 20.6, the relationship between expected return and risk is nonlinear for $\rho \neq 1.0$, but linear for $\rho = 1.0$.

The parameters $\theta = \{\gamma_0, \varphi, \rho, \alpha_0, \alpha_1, \beta_1\}$ can be estimated by maximum likelihood methods as usual and hypothesis tests can be performed to test the hypotheses embodied, for example, in equation (20.42).

Example 20.7 Risk in the Term Structure of Interest Rates

A GARCH-M model is estimated using daily U.S. interest rate data from 10 October 1988 to 28 December 2001 ($T = 3307$) for the 10-year yield, $r_{10y,t}$, and the 3-month yield on U.S. zero coupon bonds, $r_{3m,t}$. The estimated

model is

$$\begin{aligned} r_{10,t} &= \underset{(0.018)}{2.259} + \underset{(0.003)}{0.776} r_{3m,t} + \underset{(0.020)}{0.117} \overset{2.071}{\underset{(0.211)}{\hat{\sigma}_t}} + \hat{u}_t \\ \hat{\sigma}_t^2 &= \underset{(0.004)}{0.049} + \underset{(0.019)}{0.962} \hat{u}_{t-1}^2 + \underset{(0.050)}{0.275} \hat{\sigma}_{t-1}^2. \end{aligned}$$

Standard errors based on the Hessian are shown in parentheses. The point estimate of $\hat{\varphi} = 0.117$ is positive, providing evidence of risk aversion. A formal test is based on the hypotheses

$$\begin{aligned} H_0 : \quad & \varphi = 0 \quad [\text{Risk neutral}] \\ H_1 : \quad & \varphi > 0 \quad [\text{Risk aversion}]. \end{aligned}$$

A Wald test based on the t-statistic is

$$t = \frac{0.117 - 0.0}{0.020} = 9.816.$$

This yields a p-value of 0.000 providing strong evidence against the null of risk neutrality. $\square$

20.4.6 Diagnostics

A summary of diagnostic tests commonly employed to test the adequacy (G)ARCH models will complete this section. These tests, which are designed to ascertain whether or not the estimated model captures the nonlinearity in the conditional variance, are usually implemented using the standardized residuals

$$z_t = \frac{u_t}{\sigma_t},$$

where σ_t is the estimated conditional standard deviation obtained from the model. The null and alternative hypotheses are respectively

$$\begin{aligned} H_0 : \quad & [\text{No ARCH}] \quad \text{model specified correctly} \\ H_1 : \quad & [\text{ARCH}] \quad \text{model not specified correctly.} \end{aligned}$$

Under the null hypothesis, there is no evidence of ARCH in the standardized residuals, suggesting that the model is specified correctly. If the statistical evidence favours the alternative hypothesis, the model needs to be re-estimated with a different specification for the conditional variance and the testing procedure repeated.

(1) ARCH test on standardized residuals

Apply a test for ARCH(q) to the standardized residuals.

(2) *Overfitting*

Estimate a GARCH(p, q) model and compute the values of one or more commonly used information criteria, such as the Akaike Information Criterion (AIC) and the Schwarz Information Criterion (SIC).

Now systematically re-estimate the model by increasing the lag order for both p and q and compute the associated values of the AIC or SIC. If none of the estimated values of these information criteria are smaller than the value returned by the original model, then there is strong statistical evidence that the original model is correctly specified. If, however, a model with a larger lag order returns a lower value of the AIC or SIC than the original model, then the efficacy of the original model is brought into question.

(3) *Conditional distribution test*

Use a Kolmogorov-Smirnov test to compare the empirical distribution of z_t with the theoretical conditional distribution used to specify the log-likelihood function. Under the null hypothesis of no specification error, the two distributions should match.

(4) *Unconditional distribution test*

Use a Kolmogorov-Smirnov test to compare the empirical distribution of y_t with the theoretical unconditional distribution of the model. Under the null hypothesis of no specification error, the two distributions should match. As no analytical solution exists for the unconditional distribution of the ARCH model, one way to make this comparison is to simulate the model several (say 1000) steps ahead and repeat this many times (say 10000). Choosing the last simulated value from each of the replications will yield 10000 draws from the unconditional distribution of the estimated ARCH model and a nonparametric estimate of this distribution can then be obtained using a kernel density estimator.

20.5 Conditional Nonnormality

An important feature of the volatility models discussed so far is the role of conditional normality. The combination of conditional normality and GARCH variance yields an unconditional distribution of financial returns that is leptokurtotic. In practice, a simple GARCH model specified with normal disturbances is not able to model all of the leptokurtosis in the data. Consequently, there are three common approaches to generalize the GARCH model to account for leptokurtosis.

20.5.1 Parametric

The parametric solution to leptokurtosis is to replace the conditional normality specification by a parametric distribution that also exhibits kurtosis. Consider the GARCH(1,1) model

$$\begin{aligned} y_t &= u_t, & u_t &= \sigma_t z_t, \\ \sigma_t^2 &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \end{aligned} \quad (20.43)$$

where the conditional distribution of z_t is chosen as the standardized Student t distribution, $St(0, 1, \nu)$, with ν degrees of freedom and with the property that $E[z_t] = 0$ and $E[z_t^2] = 1$. To derive this distribution, consider the (non-standardized) Student t distribution

$$f(e) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu} \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{e^2}{\nu}\right)^{-\left(\frac{\nu+1}{2}\right)},$$

where $E[e] = 0$ and $E[e^2] = \nu/(\nu-2)$. Defining the standardized random variable $z = e/\sqrt{\nu/(\nu-2)}$, it follows from the transformation of variable technique that

$$f(z) = f(e) \left| \frac{de}{dz} \right| = f\left(z \sqrt{\frac{\nu}{\nu-2}}\right) \left| \sqrt{\frac{\nu}{\nu-2}} \right|.$$

which yields the standardized Student t distribution

$$f(z) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi(\nu-2)} \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{z^2}{\nu-2}\right)^{-\left(\frac{\nu+1}{2}\right)}. \quad (20.44)$$

The additional parameter ν in (20.44) provides the additional flexibility required to model the nonnormality in the data. Using equations (20.43) and (20.44), the log-likelihood function is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T \ln l_t(\theta), \quad (20.45)$$

where

$$\begin{aligned}
 \ln l_t(\theta) &= \ln f(y_t; \theta) \\
 &= -\frac{1}{2} \ln \sigma_t^2 + \ln f(z_t) \\
 &= -\frac{1}{2} \ln \sigma_t^2 + \ln \left(\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi(\nu-2)} \Gamma\left(\frac{\nu}{2}\right)} \right) - \frac{(\nu+1)}{2} \ln \left(1 + \frac{z_t^2}{\nu-2} \right),
 \end{aligned} \tag{20.46}$$

and $\theta = \{\alpha_0, \alpha_1, \beta_1, \nu\}$. Maximizing (20.45) may require constraining ν to be positive. In fact, inspection of the log-likelihood function shows that this parameter may have to be constrained to be greater than 2.

To derive the covariance matrix of the maximum likelihood estimator corresponding to (20.46) consider focussing on the GARCH parameters by conditioning on the degrees of freedom parameter, so now $\theta = \{\alpha_0, \alpha_1, \beta_1\}$. The first derivative at observation t is

$$g_t = \frac{\partial \ln l_t}{\partial \theta} = -\frac{1}{2} \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2}{\partial \theta} \left(1 - \frac{(\nu+1)y_t^2}{(\sigma_t^2(\nu-2) + y_t^2)} \right), \tag{20.47}$$

where

$$\frac{\partial \sigma_t^2}{\partial \theta} = \begin{bmatrix} \frac{\partial \sigma_t^2}{\partial \alpha_0} \\ \frac{\partial \sigma_t^2}{\partial \alpha_1} \\ \frac{\partial \sigma_t^2}{\partial \beta_1} \end{bmatrix} = \begin{bmatrix} 1 + \beta_1 \frac{\partial \sigma_{t-1}^2}{\partial \alpha_0} \\ u_{t-1}^2 + \beta_1 \frac{\partial \sigma_{t-1}^2}{\partial \alpha_1} \\ \sigma_{t-1}^2 + \beta_1 \frac{\partial \sigma_{t-1}^2}{\partial \beta_1} \end{bmatrix}. \tag{20.48}$$

The covariance matrix is based on $J(\theta) = E[g_t g_t']$, which is consistently estimated by

$$J_T(\hat{\theta}) = \frac{1}{T} \sum_{t=1}^T g_t g_t' = \frac{1}{4} \frac{1}{T} \sum_{t=1}^T \left[\left(\frac{1}{\sigma_t^4} \left(1 - \frac{(\nu+1)y_t^2}{(\sigma_t^2(\nu-2) + y_t^2)} \right)^2 \right) \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'} \right], \tag{20.49}$$

where σ_t^2 in (20.43) and $\partial \sigma_t^2 / \partial \theta$ in (20.48) are evaluated at $\theta = \hat{\theta}$. The estimated covariance matrix

$$\hat{\Omega} = J_T^{-1}(\hat{\theta}), \tag{20.50}$$

with $J_T(\hat{\theta})$ given by (20.49).

20.5.2 Semi-Parametric

A semi-parametric approach to dealing with leptokurtosis is to specify the conditional mean and conditional variance equations parametrically but to use a nonparametric density estimator, as discussed in Chapter 11, for the distribution of the disturbance term. Consider extending the GARCH(1, 1) model in (20.43) to allow for a regressor in the mean

$$\begin{aligned} y_t &= \gamma_0 + \gamma_1 x_t + u_t, & u_t &= \sigma_t z_t \\ \sigma_t^2 &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \end{aligned}$$

where $f(z_t)$ is to be estimated using nonparametric methods. Estimation would proceed as follows:

Step 1: Choose starting values for the parameters $\theta = \{\gamma_0, \gamma_1, \alpha_0, \alpha_1, \beta_1\}$.

Step 2: For each t construct the conditional mean $\gamma_0 + \gamma_1 x_t$, the residual $u_t = y_t - \gamma_0 - \gamma_1 x_t$ and the conditional variance $\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2$.

Step 3: For each t construct the standardized residual

$$z_t = \frac{y_t - \gamma_0 - \gamma_1 x_t}{\sigma_t}.$$

Step 4: Use a kernel estimator to obtain a nonparametric estimate of the density of the standardized residuals, $f_{np}(z)$. Evaluate $f_{np}(z)$ at each observation and hence compute $f_{np}(z_t)$.

Step 5: For each t compute

$$\ln l_t(\theta) = \ln f_{np}(z_t) - 0.5 \ln \sigma_t^2,$$

and maximize the log-likelihood function for the entire sample using an iterative algorithm.

20.5.3 Nonparametric

A simpler approach to dealing with the problem of leptokurtosis is to recognize that the assumption of normally distributed disturbances results in the misspecification of the log-likelihood function. Despite the shape of the distribution being incorrect, the mean and variance of this distribution are, however, correctly specified. From the analysis of Chapter 9, maximum likelihood estimation in terms of the misspecified log-likelihood function yields quasi-maximum likelihood estimators of θ that are consistent. The standard

errors of θ , known as Bollerslev-Wooldridge standard errors in this context in this context (Bollerslev and Wooldridge, 1992). are given by

$$\hat{\Omega} = H_T^{-1}(\hat{\theta}) J_T(\hat{\theta}) H_T^{-1}(\hat{\theta}), \quad (20.51)$$

where

$$H_T(\hat{\theta}) = \frac{1}{T} \sum_{t=1}^T h_t, \quad J_T(\hat{\theta}) = \frac{1}{T} \sum_{t=1}^T g_t g_t', \quad (20.52)$$

with g_t and h_t respectively representing the gradient and the Hessian at time t of the misspecified model corresponding to the normal distribution. It is understood that all derivatives are evaluated at $\hat{\theta}$. The vector $\hat{\theta}$ represents the quasi-maximum likelihood parameter estimates at the final iteration.

In the case where the true distribution is the Standardized Student t distribution in (20.44), an analytical expression of the quasi maximum likelihood estimator covariance matrix in (20.51) is available. As the misspecified model is normal, the log-likelihood function at observation t is

$$\ln l_t(\theta) = \ln f(y_t; \theta) = -\frac{1}{2} \ln 2\pi - \frac{1}{2} \ln \sigma_t^2 - \frac{1}{2} \frac{y_t^2}{\sigma_t^2}. \quad (20.53)$$

For the model in (20.43) where conditioning is again on the degrees of freedom parameter so the parameter vector is $\theta = \{\alpha_0, \alpha_1, \beta_1\}$, the first derivatives and second derivatives in (20.52) are respectively

$$g_t = -\frac{1}{2} \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2}{\partial \theta} \left(1 - \frac{y_t^2}{\sigma_t^2}\right),$$

$$h_t = -\frac{1}{2} \left(-\frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'} \left(1 - \frac{y_t^2}{\sigma_t^2}\right) + \frac{1}{\sigma_t^2} \frac{\partial^2 \sigma_t^2}{\partial \theta \partial \theta'} \left(1 - \frac{y_t^2}{\sigma_t^2}\right) + \frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'} \frac{y_t^2}{\sigma_t^2} \right).$$

To derive the quasi-maximum likelihood estimator covariance matrix of $\hat{\theta}$, expectations of the derivatives are taken initially with respect to the conditional expectations operator. In the case of the first derivative, it follows that

$$E_{t-1}[g_t] = E_{t-1} \left[-\frac{1}{2} \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2}{\partial \theta} \left(1 - \frac{y_t^2}{\sigma_t^2}\right) \right] = -\frac{1}{2} \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2}{\partial \theta} \left(1 - \frac{E_{t-1}[y_t^2]}{\sigma_t^2}\right) = 0,$$

because $E_{t-1}[y_t^2]/\sigma_t^2 = E_{t-1}[z_t^2] = 1$ by definition. Using the law of iterated expectations, it immediately follows that $E[E_{t-1}[g_t]] = 0$, which from Chapter 9 establishes that the quasi-maximum likelihood estimator, $\hat{\theta}$, is a consistent estimator of the true population parameter, θ_0 .

For the second derivative

$$\begin{aligned} E_{t-1}[h_t] &= E_{t-1} \left[-\frac{1}{2} \left(-\frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'} \left(1 - \frac{y_t^2}{\sigma_t^2} \right) + \frac{1}{\sigma_t^2} \frac{\partial^2 h_t}{\partial \theta \partial \theta'} \left(1 - \frac{y_t^2}{\sigma_t^2} \right) \right. \right. \\ &\quad \left. \left. \left(\frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'} \frac{y_t^2}{\sigma_t^2} \right) \right) \right] \\ &= E_{t-1} \left[-\frac{1}{2} \left(\frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'} \right) \right], \end{aligned} \quad (20.54)$$

which again uses the property $E_{t-1}[z_t^2] = E_{t-1}[y_t^2]/\sigma_t^2 = 1$, so the first two terms are zero. Finally, the outer product of gradients matrix is given by

$$\begin{aligned} E_{t-1}[g_t g_t'] &= E_{t-1} \left[\left(-\frac{1}{2} \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2}{\partial \theta} \left(1 - \frac{y_t^2}{\sigma_t^2} \right) \right) \left(-\frac{1}{2} \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2}{\partial \theta'} \left(1 - \frac{y_t^2}{\sigma_t^2} \right) \right) \right] \\ &= \frac{1}{4} \frac{1}{\sigma_t^4} E_{t-1} \left[\left(1 - \frac{y_t^2}{\sigma_t^2} \right)^2 \right] \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'} \\ &= \frac{1}{4} \frac{1}{\sigma_t^4} \left(2 + \frac{6}{\nu - 4} \right) \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'}, \end{aligned} \quad (20.55)$$

because

$$\begin{aligned} E_{t-1}[(1 - y_t^2 \sigma_t^{-2})^2] &= E_{t-1}[(1 - z_t^2)^2] = E_{t-1}[1 + z_t^4 - 2z_t^2] \\ &= 1 + E_{t-1}[z_t^4] - 2E_{t-1}[z_t^2] = 2 + 6/(\nu - 4), \end{aligned}$$

which uses the properties of the standardized Student t distribution that $E_{t-1}[z_t^2] = 1$ and $E_{t-1}[z_t^4] = 3 + 6/(\nu - 4)$. By using the law of iterated expectations, the unconditional expectations under the true model of (20.54) and (20.55) are respectively

$$\begin{aligned} H(\theta_0) &= E[E_{t-1}[h_t]] = E \left[-\frac{1}{2} \left(\frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'} \right) \right] \\ J(\theta_0) &= E[E_{t-1}[g_t g_t']] = E \left[\frac{1}{4} \frac{1}{\sigma_t^4} \left(\frac{\partial \sigma_t^2}{\partial \theta} \right)^2 \left(2 + \frac{6}{\nu - 4} \right) \right], \end{aligned}$$

where $\partial \sigma_t^2 / \partial \theta$ is given by (20.48). Consistent estimates of these matrices are obtained as

$$\begin{aligned} H_T(\hat{\theta}) &= -\frac{1}{2} \frac{1}{T} \sum_{t=1}^T \left(\left(\frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'} \right) \right) \\ J_T(\hat{\theta}) &= \frac{1}{4} \left(2 + \frac{6}{\nu - 4} \right) \frac{1}{T} \sum_{t=1}^T \left(\frac{1}{\sigma_t^4} \left(\frac{\partial \sigma_t^2}{\partial \theta} \right)^2 \right), \end{aligned} \quad (20.56)$$

where σ_t^2 in (20.43) and $\partial \sigma_t^2 / \partial \theta$ in (20.48) are evaluated at $\theta = \hat{\theta}$. Substi-

tuting these expressions into (20.51) yields the estimated quasi-maximum likelihood covariance matrix estimator.

Example 20.8 Quasi-maximum Likelihood Estimation

Engle and González-Rivera (1991) investigate the relative efficiency properties of the quasi-maximum likelihood estimator in GARCH models where the true conditional distribution is Student *t* and the misspecified distribution is normal. The results are given in Table 20.3 where the relative efficiency is computed by simulating the GARCH(1,1) model in (20.43) for $T = 1000000$ observations subject to the restriction $\alpha_0 = 1 - \alpha_1 - \beta_1$ and forming the ratio of the diagonal elements of (20.50) and (20.51). The population parameters are $\alpha_1 = 0.1$ and $\beta_1 = 0.8$ and the GARCH intercept is restricted to $a_0 = 1 - \alpha_1 - \beta_1$. In computing the variances the derivative $\partial\sigma_t^2/\partial\theta$ is computed recursively as

$$\frac{\partial\sigma_t^2}{\partial\theta} = \begin{bmatrix} \frac{\partial\sigma_t^2}{\partial\alpha_1} \\ \frac{\partial\sigma_t^2}{\partial\beta_1} \end{bmatrix} = \begin{bmatrix} -1 + u_{t-1}^2 + \beta_1 \frac{\partial\sigma_{t-1}^2}{\partial\alpha_1} \\ -1 + \sigma_{t-1}^2 + \beta_1 \frac{\partial\sigma_{t-1}^2}{\partial\beta_1} \end{bmatrix}.$$

Table 20.3

Relative efficiency of the maximum likelihood estimator and the quasi-maximum likelihood estimator for the parameters of a GARCH(1,1) model.

ν	Simulated		Analytical	
	α_1	β_1	α_1	β_1
5	0.400	0.400	0.400	0.400
8	0.787	0.787	0.787	0.787
12	0.911	0.911	0.909	0.909

For comparative purposes the analytical efficiency ratios taken from Table 1 in González-Rivera and Drost (1999). are also reported. These analytical results are transformed from the original where they are quoted as the percentage change in the quasi-maximum likelihood variance over the maximum likelihood variance. For example, González-Rivera and Drost report a value of 150 for the variance parameters where $\nu = 5$. Transforming the result to $1/(1 + 150/100) = 0.400$ gives the result reported in the table.

The simulated and analytical relative efficiency results are in strong agreement. For relatively small values of ν there is a large efficiency loss from using the quasi-maximum likelihood estimator in the presence of misspecification.

In the limit as $\nu \rightarrow \infty$ the two variances approach each other signifying no efficiency loss as the extent of the misspecification diminishes. $\square$

20.6 Multivariate GARCH

The models discussed so far are univariate since they focus on modelling the variance of the returns on a single financial asset. A natural extension is to consider a multivariate model and specify both conditional variances as well as conditional covariances. A simple multivariate specification is

$$y_t = u_t, \quad u_t \sim iid N(0, V_t),$$

where y_t and u_t are now $(N \times 1)$ vectors of time series and V_t is a $(N \times N)$ symmetric positive definite matrix containing the variances $(\sigma_{i,i,t})$ on the main diagonal and covariances $(\sigma_{i,j,t})$ on the off diagonals. For $N = 2$, V_t is

$$V_t = \begin{bmatrix} \sigma_{1,1,t} & \sigma_{1,2,t} \\ \sigma_{2,1,t} & \sigma_{2,2,t} \end{bmatrix},$$

where $\sigma_{1,1,t}$ and $\sigma_{2,2,t}$ are the conditional variances and $\sigma_{1,2,t} = \sigma_{2,1,t}$ is the conditional covariance.

Assuming conditional normality, the log-likelihood function at time t is

$$\ln l_t(\theta) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |V_t| - \frac{1}{2} u_t' V_t^{-1} u_t, \quad (20.57)$$

which is to be maximized. There are, however, two computational problems that arise in estimating MGARCH models. First, the covariance matrix, V_t , must be positive definite for all t . This requires that the conditional variances are positive, $\sigma_{i,i,t}^2 > 0$, and also that $|V_t| > 0$. This restriction is necessary from a statistical perspective since V_t represents a covariance matrix and, from a computational perspective, violation of this condition will result in numerical error when computing the term $\ln |V_t|$ in equation (20.57). Indeed, it takes just one observation with a negative definite V_t for the computation to fail. Second, the number of unknown parameters governing the behaviour of the variances and covariances in MGARCH models increases exponentially as the dimension N increases.

Four multivariate models are discussed each of which, to a greater or lesser degree, deal with these two problems.

- (i) The VECH model, which is a direct generalization of the univariate GARCH model to multi dimensions.

- (ii) The BEKK model of Engle and Kroner (1995), which reduces the parameter dimension of the VEC model and has the advantage that V_t is restricted to be positive definite at each t .
- (iii) The DCC model of Engle (2002) and Engle and Sheppard (2001), which reduces the dimension of the unknown parameters of the BEKK model further, thus making estimation of higher dimensional MGARCH models more feasible.
- (iv) The DECO model of Engle and Kelly (2009), which further simplifies the DCC model by restricting all correlations to be equal contemporaneously with the result that estimation is simplified dramatically.

20.6.1 VEC

The VEC specification is the simplest and natural multivariate analogue of the univariate GARCH model. For the case of $N = 2$ variables, V_t is specified as

$$\begin{aligned} \text{vech}(V_t) &= C + A \text{vech}(u_{t-1}u'_{t-1}) + D \text{vech}(V_{t-1}), \\ \begin{bmatrix} \sigma_{1,1,t} \\ \sigma_{1,2,t} \\ \sigma_{2,2,t} \end{bmatrix} &= \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} + \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \begin{bmatrix} u_{1,t-1}^2 \\ u_{1,t-1}u_{2,t-1} \\ u_{2,t-1}^2 \end{bmatrix} \\ &\quad + \begin{bmatrix} d_{1,1} & d_{1,2} & d_{1,3} \\ d_{2,1} & d_{2,2} & d_{2,3} \\ d_{3,1} & d_{3,2} & d_{3,3} \end{bmatrix} \begin{bmatrix} \sigma_{1,1,t-1} \\ \sigma_{1,2,t-1} \\ \sigma_{2,2,t-1} \end{bmatrix}, \end{aligned} \quad (20.58)$$

where $\text{vech}(\cdot)$ represents the column stacking of the unique elements of a symmetric matrix. The VEC model does not guarantee that V_t will be positive definite at each t and, even in the bivariate case, there are a large number of parameters (21 in total) that need to be estimated.

The total number of parameters for an N dimensional VEC model is made up as follows:

$$\begin{aligned} C \text{ matrix} &: \# \text{ parameters} = \frac{N(N+1)}{2} \\ A \text{ matrix} &: \# \text{ parameters} = \left(\frac{N(N+1)}{2}\right)^2 \\ D \text{ matrix} &: \# \text{ parameters} = \left(\frac{N(N+1)}{2}\right)^2, \end{aligned}$$

so that the total number of parameters is

$$\frac{N(N+1)}{2} + \left(\frac{N(N+1)}{2}\right)^2 + \left(\frac{N(N+1)}{2}\right)^2 = \frac{N^4 + 2N^3 + 2N^2 + N}{2}.$$

For example, extending the model to $N = 3$ variables, the number of unknown parameters is

$$\frac{N^4 + 2N^3 + 2N^2 + N}{2} = \frac{3^4 + 2 \times 3^3 + 2 \times 3^2 + 3}{2} = 78.$$

For $N = 4$ variables, this number grows to 210.

20.6.2 BEKK

To impose the positive definiteness restriction on V_t while simultaneously reducing the dimension of the parameter vector, the BEKK specification is

$$V_t = CC' + Au_{t-1}u_{t-1}'A' + DV_{t-1}D', \quad (20.59)$$

where C is a $(N \times N)$ lower (Choleski) triangular matrix and A and D are $(N \times N)$ matrices that need not be symmetric.

For illustrative purposes, the matrices of the BEKK model in equation (20.59) are now spelled out for the cases of $N = 1$ and $N = 2$. In the univariate case, the matrices are

$$\begin{aligned} V_{t-1} &= \begin{bmatrix} h_{1,1,t-1} \end{bmatrix}, \quad u_{t-1} = \begin{bmatrix} u_{1,t-1} \end{bmatrix}, \\ C &= \begin{bmatrix} c_{1,1} \end{bmatrix}, \quad A = \begin{bmatrix} a_{1,1} \end{bmatrix}, \quad D = \begin{bmatrix} d_{1,1} \end{bmatrix}, \end{aligned}$$

with V_t having the form

$$\begin{aligned} V_t &= \begin{bmatrix} c_{1,1} \end{bmatrix} \begin{bmatrix} c_{1,1} \end{bmatrix}' + \begin{bmatrix} a_{1,1} \end{bmatrix} \begin{bmatrix} u_{1,t-1} \end{bmatrix} \begin{bmatrix} u_{1,t-1} \end{bmatrix}' \begin{bmatrix} a_{1,1} \end{bmatrix}' \\ &\quad + \begin{bmatrix} d_{1,1} \end{bmatrix} \begin{bmatrix} \sigma_{1,1,t-1}^2 \end{bmatrix} \begin{bmatrix} d_{1,1} \end{bmatrix}' \\ &= c_{1,1}^2 + a_{1,1}^2 u_{1,t-1}^2 + d_{1,1}^2 \sigma_{1,1,t-1}^2. \end{aligned}$$

This is just a univariate GARCH model with 3 parameters where the parameters are restricted to be positive by specifying them in terms of squares.

With $N = 2$, the matrices of the BEKK model in equation (20.59) are

$$\begin{aligned} V_{t-1} &= \begin{bmatrix} \sigma_{1,1,t-1}^2 & \sigma_{1,2,t-1}^2 \\ \sigma_{1,2,t-1}^2 & \sigma_{2,2,t-1}^2 \end{bmatrix}, \quad u_{t-1} = \begin{bmatrix} u_{1,t-1} \\ u_{2,t-1} \end{bmatrix}, \\ C &= \begin{bmatrix} c_{1,1} & 0 \\ c_{2,1} & c_{2,2} \end{bmatrix}, \quad A = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix}, \quad D = \begin{bmatrix} d_{1,1} & d_{1,2} \\ d_{2,1} & d_{2,2} \end{bmatrix}, \end{aligned}$$

with V_t having the form

$$\begin{aligned}
 V_t &= \begin{bmatrix} c_{1,1} & 0 \\ c_{2,1} & c_{2,2} \end{bmatrix} \begin{bmatrix} c_{1,1} & 0 \\ c_{2,1} & c_{2,2} \end{bmatrix}' \\
 &+ \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \begin{bmatrix} u_{1,t-1} \\ u_{2,t-1} \end{bmatrix} \begin{bmatrix} u_{1,t-1} \\ u_{2,t-1} \end{bmatrix}' \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix}' \\
 &+ \begin{bmatrix} d_{1,1} & d_{1,2} \\ d_{2,1} & d_{2,2} \end{bmatrix} \begin{bmatrix} \sigma_{1,1,t-1}^2 & \sigma_{1,2,t-1}^2 \\ \sigma_{1,2,t-1}^2 & \sigma_{2,2,t-1}^2 \end{bmatrix} \begin{bmatrix} d_{1,1} & d_{1,2} \\ d_{2,1} & d_{2,2} \end{bmatrix}' \\
 &= \begin{bmatrix} c_{1,1} & 0 \\ c_{2,1} & c_{2,2} \end{bmatrix} \begin{bmatrix} c_{1,1} & c_{2,1} \\ 0 & c_{2,2} \end{bmatrix} \\
 &+ \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \begin{bmatrix} u_{1,t-1}^2 & u_{1,t-1}u_{2,t-1} \\ u_{1,t-1}u_{2,t-1} & u_{2,t-1}^2 \end{bmatrix} \begin{bmatrix} a_{1,1} & a_{2,1} \\ a_{1,2} & a_{2,2} \end{bmatrix} \\
 &+ \begin{bmatrix} d_{1,1} & d_{1,2} \\ d_{2,1} & d_{2,2} \end{bmatrix} \begin{bmatrix} \sigma_{1,1,t-1}^2 & \sigma_{1,2,t-1}^2 \\ \sigma_{1,2,t-1}^2 & \sigma_{2,2,t-1}^2 \end{bmatrix} \begin{bmatrix} d_{1,1} & d_{2,1} \\ d_{1,2} & d_{2,2} \end{bmatrix}.
 \end{aligned}$$

The number of unknown parameters is 11 compared to 21 parameters in the VEC specification given in equation (20.58).

The number of unknown parameters may be further reduced by restricting the A and D parameter matrices to be symmetric

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{1,2} & a_{2,2} \end{bmatrix}, \quad D = \begin{bmatrix} d_{1,1} & d_{1,2} \\ d_{1,2} & d_{2,2} \end{bmatrix}. \quad (20.60)$$

These restrictions can be tested by performing a LR test of

$$a_{1,2} = a_{2,1}, \quad d_{1,2} = d_{2,1}.$$

Another class of models is to assume that the covariance is constant. The restrictions to be tested for the asymmetric BEKK model are

$$a_{1,2} = a_{2,1} = d_{1,2} = d_{2,1} = 0.$$

Example 20.9 Bivariate BEKK Model of Interest Rates

Consider the bivariate asymmetric BEKK model

$$\begin{aligned}
 y_{1,t} &= \gamma_1 + u_{1,t} \\
 y_{2,t} &= \gamma_2 + u_{2,t} \\
 V_t &= CC' + Au_{t-1}u_{t-1}'A' + DV_{t-1}D' \\
 u_t &\sim iid N(0, V_t)
 \end{aligned}$$

The model is estimated using the same data as in Example 20.7, so that $y_{1,t}$ and $y_{2,t}$ are, respectively, 3307 daily observations on the yields of 3-month and 1-year U.S. zero coupon bonds, expressed in basis points, over the period

10 October 1988 to 28 December 2001. The estimated parameter matrices are

$$\begin{aligned}\hat{\gamma} &= \begin{bmatrix} \hat{\gamma}_1 \\ \hat{\gamma}_2 \end{bmatrix} = \begin{bmatrix} 0.060 \\ 0.059 \end{bmatrix}, & \hat{C} &= \begin{bmatrix} 1.071 & 0.000 \\ 0.285 & 0.567 \end{bmatrix}, \\ \hat{A} &= \begin{bmatrix} 0.385 & 0.049 \\ 0.008 & 0.224 \end{bmatrix}, & \hat{D} &= \begin{bmatrix} 0.881 & 0.027 \\ -0.006 & 0.973 \end{bmatrix},\end{aligned}$$

and $T \ln L_T(\hat{\theta}) = -19197.918$. Time series plots of the two conditional variances and covariance are given in Figures 20.7 and the conditional correlation is

$$r_{1,2,t} = \frac{\sigma_{1,2,t}^2}{\sqrt{\sigma_{1,1,t}^2 \sigma_{2,2,t}^2}}.$$

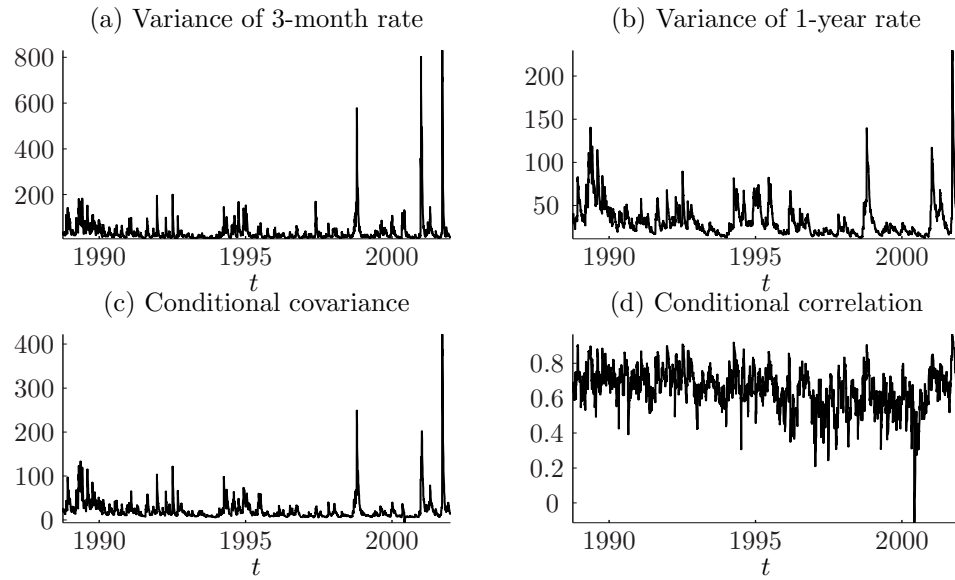


Figure 20.7 BEKK estimates of the conditional variances, the covariance and the correlation of the yields on 3-month and 1-year U.S. zero coupon bonds.

The restriction of constant covariance in interest rates may now also be tested. There are four restrictions given by

$$H_0 : a_{1,2} = a_{2,1} = d_{1,2} = d_{2,1} = 0.$$

The constrained model yields $T \ln L_T(\hat{\theta}_0) = -19220.161$. The LR test statis-

tic is

$$LR = -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)) = -2(-19220.161 + 19197.918) = 44.485,$$

which is distributed as χ_4^2 under the null hypothesis. The p-value is 0.000, leading to a strong rejection of the null hypothesis of a constant covariance. $\square$

20.6.3 DCC

The Dynamic Conditional Correlation (DCC) model imposes positive definiteness on V_t and achieves a large reduction in the number of parameters by specifying univariate conditional GARCH variances $\sigma_{i,t}^2$ together with a parsimonious conditional correlation matrix, R_t . The conditional covariance matrix is recovered from

$$V_t = S_t R_t S_t, \quad (20.61)$$

where R_t is a $(N \times N)$ conditional correlation matrix and S_t is a diagonal matrix containing the conditional standard deviations along the main diagonal

$$S_t = \begin{bmatrix} \sigma_{1,t} & & 0 \\ & \ddots & \\ 0 & & \sigma_{N,t} \end{bmatrix}, \quad (20.62)$$

and the conditional variances have univariate GARCH representations

$$\sigma_{i,t}^2 = \alpha_{0,i} + \alpha_{1,i} u_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2. \quad (20.63)$$

The specification of the conditional correlation matrix R_t is given by

$$R_t = \text{diag}(Q_t)^{-1/2} Q_t \text{diag}(Q_t)^{-1/2}, \quad (20.64)$$

where Q_t has the GARCH(1,1) specification

$$Q_t = (1 - \alpha - \beta) \bar{Q} + \alpha z_{t-1} z'_{t-1} + \beta Q_{t-1}, \quad (20.65)$$

which is a function of just two unknown scalar parameters α and β , and the unconditional covariance matrix of the standardized residuals

$$z_{i,t} = \frac{u_{i,t}}{\sigma_{i,t}},$$

is given by

$$\overline{Q} = \frac{1}{T} \sum_{t=1}^T \begin{bmatrix} z_{1,t}^2 & z_{1,t}z_{2,t} & \cdots & z_{1,t}z_{N,t} \\ z_{2,t}z_{1,t} & z_{2,t}^2 & \cdots & z_{2,t}z_{N,t} \\ \vdots & \vdots & \ddots & \vdots \\ z_{N,t}z_{1,t} & z_{N,t}z_{2,t} & \cdots & z_{N,t}^2 \end{bmatrix}. \quad (20.66)$$

A special case of the DCC model is the constant correlation matrix that arises when $\alpha = \beta = 0$. These restrictions can be tested using a LR test.

Example 20.10 Bivariate DCC Model

For illustrative purposes, it is useful to spell out the specifics of the construction of the conditional covariance matrix of the DCC model for the 2 asset case.

Step 1: Estimate the univariate GARCH(1, 1) models for the two assets

$$\begin{aligned} \sigma_{1,t}^2 &= \alpha_{01} + \alpha_{11}u_{1,t-1}^2 + \beta_1\sigma_{1,t-1}^2 \\ \sigma_{2,t}^2 &= \alpha_{02} + \alpha_{12}u_{2,t-1}^2 + \beta_2\sigma_{2,t-1}^2, \end{aligned}$$

and construct the matrix S_t as follows

$$S_t = \begin{bmatrix} \sigma_{1,t} & 0 \\ 0 & \sigma_{2,t} \end{bmatrix}.$$

Step 2: Compute the standardised residuals

$$\begin{bmatrix} z_{1,t} & z_{2,t} \\ z_{2,t} & z_{1,t} \end{bmatrix} = S_t^{-1}u_t = \begin{bmatrix} \frac{1}{\sigma_{1,t}} & 0 \\ 0 & \frac{1}{\sigma_{2,t}} \end{bmatrix} \begin{bmatrix} u_{1,t} & u_{2,t} \\ u_{2,t} & u_{1,t} \end{bmatrix} = \begin{bmatrix} \frac{u_{1,t}}{\sigma_{1,t}} & \frac{u_{2,t}}{\sigma_{2,t}} \\ \frac{u_{2,t}}{\sigma_{2,t}} & \frac{u_{1,t}}{\sigma_{1,t}} \end{bmatrix}.$$

Step 3: Compute $\overline{Q}$

$$\overline{Q} = \begin{bmatrix} \overline{\rho}_{1,1} & \overline{\rho}_{1,2} \\ \overline{\rho}_{1,2} & \overline{\rho}_{2,2} \end{bmatrix} = \frac{1}{T} \sum_{t=1}^T \begin{bmatrix} z_{1,t}^2 & z_{1,t}z_{2,t} \\ z_{2,t}z_{1,t} & z_{2,t}^2 \end{bmatrix},$$

and use the result to construct the elements of the matrix Q using a simple GARCH(1, 1) specification for the conditional correlations. The relevant equations are, respectively,

$$\begin{aligned} q_{1,1,t} &= \overline{\rho}_{1,1}(1 - \alpha - \beta) + \alpha(z_{1,t-1}z_{1,t-1}) + \beta(q_{1,1,t-1}) \\ q_{2,2,t} &= \overline{\rho}_{2,2}(1 - \alpha - \beta) + \alpha(z_{2,t-1}z_{2,t-1}) + \beta(q_{2,2,t-1}) \\ q_{1,2,t} &= \overline{\rho}_{1,2}(1 - \alpha - \beta) + \alpha(z_{1,t-1}z_{2,t-1}) + \beta(q_{1,2,t-1}). \end{aligned}$$

Step 4: Construct the matrix of conditional correlations, R_t ,

$$R_t = \begin{bmatrix} \frac{1}{\sqrt{q_{1,1,t}}} & 0 \\ 0 & \frac{1}{\sqrt{q_{2,2,t}}} \end{bmatrix} \begin{bmatrix} q_{1,1,t} & q_{1,2,t} \\ q_{1,2,t} & q_{2,2,t} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{q_{1,1,t}}} & 0 \\ 0 & \frac{1}{\sqrt{q_{2,2,t}}} \end{bmatrix} \\ = \begin{bmatrix} 1 & \frac{q_{1,2,t}}{\sqrt{q_{1,1,t}q_{2,2,t}}} \\ \frac{q_{1,2,t}}{\sqrt{q_{1,1,t}q_{2,2,t}}} & 1 \end{bmatrix}.$$

Step 5: The estimate of conditional covariance matrix is then

$$V_t = \begin{bmatrix} \sqrt{\sigma_{1,t}^2} & 0 \\ 0 & \sqrt{\sigma_{2,t}^2} \end{bmatrix} \begin{bmatrix} 1 & \rho_{12,t} \\ \rho_{12,t} & 1 \end{bmatrix} \begin{bmatrix} \sqrt{\sigma_{1,t}^2} & 0 \\ 0 & \sqrt{\sigma_{2,t}^2} \end{bmatrix} \\ = \begin{bmatrix} \sigma_{1,t}^2 & \rho_{12,t} \sqrt{\sigma_{1,t}^2} \sqrt{\sigma_{2,t}^2} \\ \rho_{12,t} \sqrt{\sigma_{1,t}^2} \sqrt{\sigma_{2,t}^2} & \sigma_{2,t}^2 \end{bmatrix}.$$

□

Maximum Likelihood Estimation: Small N

Let the parameter vector be partitioned into $\theta = \{\theta_1, \theta_2\}$, where θ_1 are the volatility parameters and $\theta_2 = \{\alpha, \beta\}$ are the correlation parameters. The log-likelihood function at observation t is given by using (20.61) for V_t in (20.57)

$$\ln l_t(\theta) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |S_t R_t S_t| - \frac{1}{2} u_t' [S_t R_t S_t]^{-1} u_t. \quad (20.67)$$

Using the results $|AB| = |A| |B|$ and $[AB]^{-1} = B^{-1} A^{-1}$, this expression can be rearranged into two components

$$\ln l_t(\theta) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} (2 \ln |S_t| + \ln |R_t|) - \frac{1}{2} z_t' R_t^{-1} z_t \\ = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |S_t|^2 - \frac{1}{2} \ln |R_t| - \frac{1}{2} z_t' R_t^{-1} z_t,$$

where $z_t' = u_t' S_t^{-1}$ is the standardized variable. Since $z_t' z_t = u_t' S_t^{-1} S_t^{-1} u_t$,

adding and subtracting $z_t' z_t$ gives

$$\begin{aligned}\ln l_t(\theta) &= -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |S_t|^2 - \frac{1}{2} \ln |R_t| - \frac{1}{2} z_t' R_t^{-1} z_t - \frac{1}{2} (u_t' S_t^{-1} S_t^{-1} u_t - z_t' z_t) \\ &= -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |S_t|^2 - \frac{1}{2} u_t' S_t^{-1} S_t^{-1} u_t - \frac{1}{2} \ln |R_t| - \frac{1}{2} z_t' R_t^{-1} z_t + \frac{1}{2} z_t' z_t \\ &= \ln l_{1,t}(\theta_1) + \ln l_{2,t}(\theta_2),\end{aligned}\quad (20.68)$$

where

$$\ln l_{1,t}(\theta_1) = -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |S_t|^2 - \frac{1}{2} u_t' S_t^{-2} u_t, \quad (20.69)$$

contains just the GARCH parameters and

$$\ln l_{2,t}(\theta_2) = -\frac{1}{2} \ln |R_t| - \frac{1}{2} z_t' R_t^{-1} z_t + \frac{1}{2} z_t' z_t, \quad (20.70)$$

contains the GARCH and correlation parameters.

From (20.68), the full log-likelihood function is

$$\begin{aligned}\ln L_T(\theta) &= \frac{1}{T} \sum_{t=1}^T \ln l_t(\theta) = \frac{1}{T} \sum_{t=1}^T \ln l_{1,t}(\theta_1) + \frac{1}{T} \sum_{t=1}^T \ln l_{2,t}(\theta_1, \theta_2) \\ &= \ln L_1(\theta_1) + \ln L_2(\theta_1, \theta_2).\end{aligned}\quad (20.71)$$

The gradient is

$$G_T(\theta) = \begin{bmatrix} \frac{\partial \ln L_T(\theta)}{\partial \theta_1} \\ \frac{\partial \ln L_T(\theta)}{\partial \theta_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial \ln L_1(\theta_1)}{\partial \theta_1} + \frac{\partial \ln L_2(\theta_1, \theta_2)}{\partial \theta_1} \\ \frac{\partial \ln L_2(\theta_1, \theta_2)}{\partial \theta_2} \end{bmatrix}, \quad (20.72)$$

and the Hessian is

$$H_T(\theta) = \begin{bmatrix} \frac{\partial^2 \ln L_T(\theta)}{\partial \theta_1 \partial \theta_1'} & \frac{\partial^2 \ln L_T(\theta)}{\partial \theta_1 \partial \theta_2'} \\ \frac{\partial^2 \ln L_T(\theta)}{\partial \theta_2 \partial \theta_1'} & \frac{\partial^2 \ln L_T(\theta)}{\partial \theta_2 \partial \theta_2'} \end{bmatrix} = \begin{bmatrix} \frac{\partial^2 \ln L_1}{\partial \theta_1 \partial \theta_1'} + \frac{\partial^2 \ln L_2}{\partial \theta_1 \partial \theta_1'} & \frac{\partial^2 \ln L_2}{\partial \theta_1 \partial \theta_2'} \\ \frac{\partial^2 \ln L_2}{\partial \theta_2 \partial \theta_1'} & \frac{\partial^2 \ln L_2}{\partial \theta_2 \partial \theta_2'} \end{bmatrix}. \quad (20.73)$$

As with the BEKK model, for relatively small values of N , standard iterative algorithms can be used to maximize (20.71) with respect to θ . The total number of unknown parameters is $3N + 2$, consisting of the $3N$ univariate GARCH parameters in (20.63) and the 2 correlation parameters α and β in (20.65). For larger values of N this approach may not be feasible.

Two-step Estimation: Large N

For large N , a two-step estimator has significant computational advantages. This estimator proceeds as follows.

Step 1: *Volatility parameters* (θ_1)

In the first step the log-likelihood function in (20.69) is maximized with respect to the GARCH parameters by solving

$$\left. \frac{\partial \ln L_1(\theta_1)}{\partial \theta_1} \right|_{\theta_1 = \hat{\theta}_1} = 0.$$

Since S_t is a diagonal matrix, $|S_t|$ is simply the product of the diagonal elements, the log-likelihood function in equation (20.69) decomposes into N separate components

$$\ln l_{1,t}(\theta_1) = -\frac{1}{2} \sum_{i=1}^N \left(\ln(2\pi) + \ln \sigma_{i,t}^2 + \frac{u_{i,t}^2}{\sigma_{i,t}^2} \right), \quad (20.74)$$

suggesting that the volatility parameters in each GARCH equation can be estimated individually.

Step 2: *Correlation parameters* (θ_2)

The second step involves maximizing (20.70) to obtain consistent estimates of the conditional correlation parameters α and β , given the estimates of the GARCH parameters in the first step, by solving

$$\left. \frac{\partial \ln L_2(\hat{\theta}_1, \theta_2)}{\partial \theta_2} \right|_{\theta_2 = \hat{\theta}_2} = 0.$$

As z_t does not contain θ_2 , the expression to be maximized is simply

$$\ln L_{2,t}(\theta_2) = -\frac{1}{2} \ln |R_t| - \frac{1}{2} z_t' R_t^{-1} z_t. \quad (20.75)$$

The two-step parameter estimator is consistent, but not asymptotically efficient because the full log-likelihood function is not maximized with respect to all parameters. This occurs as the second step ignores the term $\partial \ln L_2 / \partial \theta_1$ in the first element of the gradient vector in (20.72) by treating the volatility parameters as fixed. Moreover, the standard errors of $\hat{\theta}$ obtained from estimating θ_1 and θ_2 separately are not correct as a result of treating $\hat{\theta}_1$ as fixed at the second step. Actually, as shown below, it is just the standard errors on $\hat{\theta}_2$ that are not correct, whereas the standard errors on the volatility parameters $\hat{\theta}_1$, from estimating the GARCH equations separately in the first step, are correct.

To derive the correct standard errors for the two-step estimator, the gra-

dients of this estimator are written as

$$\begin{aligned} G_1(\theta_1) &= \frac{\partial \ln L_1(\theta_1)}{\partial \theta_1} \\ G_2(\theta_1, \theta_2) &= \frac{\partial \ln L_2(\theta_1, \theta_2)}{\partial \theta_2}, \end{aligned} \quad (20.76)$$

where $\hat{\theta}_1$ is replaced by θ_1 in the expression for G_2 so as to derive the correct standard error. The corresponding Hessian is block-triangular

$$H_T(\theta) = \begin{bmatrix} \frac{\partial G_1}{\partial \theta_1} & \frac{\partial G_1}{\partial \theta_2'} \\ \frac{\partial G_2}{\partial \theta_1} & \frac{\partial G_2}{\partial \theta_2'} \end{bmatrix} = \begin{bmatrix} H_{11} & 0 \\ H_{21} & H_{22} \end{bmatrix}.$$

From the properties of partitioned inverses it follows that

$$H_T(\theta)^{-1} = \begin{bmatrix} H_{11} & 0 \\ H_{21} & H_{22} \end{bmatrix}^{-1} = \begin{bmatrix} H_{11}^{-1} & 0 \\ -H_{22}^{-1}H_{21}H_{11}^{-1} & H_{22}^{-1} \end{bmatrix} = \begin{bmatrix} H_{11}^{-1} & 0 \\ -H_{22}^{-1}\Psi & H_{22}^{-1} \end{bmatrix},$$

where $\Psi = H_{21}H_{11}^{-1}$. Defining $J_T(\theta)$ as the outer product of gradients matrix associated with (20.76), the quasi-maximum likelihood covariance matrix is

$$\begin{aligned} \hat{\Omega} &= H_T(\hat{\theta})^{-1} J_T(\hat{\theta}) H_T(\hat{\theta})^{-1'} \\ &= \frac{1}{T} \begin{bmatrix} H_{11}^{-1} & 0 \\ -H_{22}^{-1}\Psi & H_{22}^{-1} \end{bmatrix} \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix} \begin{bmatrix} H_{11}^{-1} & 0 \\ -H_{22}^{-1}\Psi & H_{22}^{-1} \end{bmatrix}' \\ &= \frac{1}{T} \begin{bmatrix} H_{11}^{-1}J_{11} & H_{11}^{-1}J_{12} \\ -H_{22}^{-1}\Psi J_{11} + H_{22}^{-1}J_{21} & -H_{22}^{-1}\Psi J_{12} + H_{22}^{-1}J_{22} \end{bmatrix} \begin{bmatrix} H_{11}^{-1} & -\Psi' H_{22}^{-1} \\ 0 & H_{22}^{-1} \end{bmatrix} \\ &= \frac{1}{T} \begin{bmatrix} H_{11}^{-1}J_{11}H_{11}^{-1} & -H_{11}^{-1}J_{11}\Psi' H_{22}^{-1} + H_{11}^{-1}J_{12}H_{22}^{-1} \\ -H_{22}^{-1}\Psi J_{11}H_{11}^{-1} + H_{22}^{-1}J_{21}H_{11}^{-1} & H_{22}^{-1}(J_{22} - J_{21}\Psi' - \Psi J_{12} + \Psi J_{11}\Psi')H_{22}^{-1} \end{bmatrix}. \end{aligned} \quad (20.77)$$

From the block-diagonal components of (20.77), the correct covariance matrices of the two-step estimator are

$$\frac{1}{T}\hat{\Omega}(\hat{\theta}_1) = \frac{1}{T}H_{11}^{-1}J_{11}H_{11}^{-1} \quad (20.78)$$

$$\frac{1}{T}\hat{\Omega}(\hat{\theta}_2) = \frac{1}{T}H_{22}^{-1}(J_{22} - J_{21}\Psi' - \Psi J_{12} + \Psi J_{11}\Psi')H_{22}^{-1}. \quad (20.79)$$

From (20.74), $\hat{\Omega}(\hat{\theta}_1)$ is also block diagonal and the quasi-maximum likelihood covariance matrices from estimating each GARCH equation separately in the first step are correct. Also note that only in the special case of

$$\hat{\Omega}(\hat{\theta}_1, \hat{\theta}_2) = -H_{11}^{-1}J_{11}\Psi' H_{22}^{-1} + H_{11}^{-1}J_{12}H_{22}^{-1} = 0_2,$$

is the quasi-maximum likelihood covariance matrix at the second step correct.

20.6.4 DECO

The key assumption underlying the Dynamic Equicorrelation (DECO) model of Engle and Kelly (2009) is that the unconditional correlation matrix of systems of financial returns has entries of roughly similar magnitude. The correlations of the DCC model in (20.64) are therefore assumed to be equal contemporaneously across all N variables, but not over time. The pertinent restrictions on the correlation matrix are

$$R_t = \begin{bmatrix} 1 & r_{1,2,t} & \cdots & r_{1,N,t} \\ r_{2,1,t} & 1 & \cdots & \vdots \\ \vdots & \vdots & \ddots & r_{N-1,N,t} \\ r_{N,1,t} & \cdots & r_{N,N-1,t} & 1 \end{bmatrix} = \begin{bmatrix} 1 & \bar{r}_t & \cdots & \bar{r}_t \\ \bar{r}_t & 1 & \cdots & \vdots \\ \vdots & \vdots & \ddots & \bar{r}_t \\ \bar{r}_t & \cdots & \bar{r}_t & 1 \end{bmatrix}, \quad (20.80)$$

where $\bar{r}_t$ represents the average of the $N(N-1)/2$ correlations at time t in (20.64)

$$\bar{r}_t = \frac{2}{N(N-1)} \sum_{i>j} r_{i,j,t}. \quad (20.81)$$

The advantage of the restriction (20.80) is that, as N increases, it becomes relatively easier to maximize the log-likelihood function corresponding to the correlation component of the DECO model in (20.71). Rewriting the restriction in (20.80) shows that

$$R_t = (1 - \bar{r}_t) \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & 1 \end{bmatrix} + \bar{r}_t \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & \vdots \\ \vdots & \vdots & \ddots & 1 \\ 1 & 1 & \cdots & 1 \end{bmatrix} = (1 - \bar{r}_t)I_N + \bar{r}_t O_N,$$

where I_N is the identity matrix and O_N is a $(N \times N)$ matrix of ones. This form of the correlation matrix, R_t , has the following analytical properties

$$|R_t| = (1 - \bar{r}_t)^{N-1} (1 + (N-1)\bar{r}_t)$$

$$R_t^{-1} = \frac{1}{1 - \bar{r}_t} I_N - \frac{\bar{r}_t}{(1 - \bar{r}_t)(1 + (N-1)\bar{r}_t)} O_N.$$

Substituting these analytical expressions in equation (20.70) simplifies the computation of the log-likelihood function since now no large determinants or inverses need to be computed at each t .

20.7 Applications

In this section, two applications based on multivariate methods for volatility modelling are presented. The first is a simple implementation of the DCC and DECO models using U.S. data, and the second illustrates how the SVAR models of Chapter 14 can be augmented to allow the shocks to have a time-varying conditional variance.

20.7.1 DCC and DECO Models of U.S. Zero Coupon Yields

Let $y_{1,t} \cdots y_{5,t}$ be daily observations on the changes in U.S. zero coupon yields with maturities 1, 5, 10, 15 and 20 years, respectively. The data are for the period 3 January 2000 to 21 August 2006.

Table 20.4

Maximum likelihood estimates of the DCC and DECO models of daily changes in U.S. zero coupon yields, expressed in basis points, with maturities 1, 5, 10, 15 and 20 years. The data period is 3 January 2000 to 21 August 2006. Quasi-maximum likelihood standard errors in parentheses based on no lags.

Yield	DCC			DECO		
	α_0	α_1	β_1	α_0	α_1	β_1
1-year	0.771 (0.209)	0.108 (0.021)	0.873 (0.022)	1.006 (0.329)	0.152 (0.047)	0.843 (0.036)
5-year	1.514 (0.337)	0.069 (0.014)	0.898 (0.016)	0.792 (0.351)	0.065 (0.019)	0.912 (0.025)
10-year	1.548 (0.308)	0.059 (0.012)	0.899 (0.014)	0.631 (0.265)	0.054 (0.013)	0.920 (0.020)
15-year	1.633 (0.324)	0.054 (0.010)	0.895 (0.013)	0.814 (0.359)	0.055 (0.014)	0.913 (0.024)
20-year	1.446 (0.323)	0.046 (0.010)	0.902 (0.014)	0.796 (0.370)	0.055 (0.015)	0.912 (0.026)
Correlation		0.074 (0.024)	0.942 (0.014)		0.180 (0.253)	0.953 (0.031)
	$T \ln L = -14347.624$			$T \ln L = -20257.098$		

Table 20.4 gives parameter estimates of the DCC and DECO models.

- (i) The parameter estimates for the individual GARCH models for each yield, α_0 , α_1 and β_1 correspond to the specification given in (20.63).

The α_1 and β_1 parameters are initially transformed to be constrained in the unit circle and then untransformed at the final iteration to obtain the appropriate standard errors.

- (ii) The parameters corresponding to the correlation specification in (20.65) taken to be α and β are given in the second last row of Table 20.4.

The results indicate that the correlation parameters are statistically significant. Setting the correlation parameters to zero yields $T \ln L_T(\hat{\theta}_0) = -15045.233$. A joint test of constant correlation is given by the LR statistic $LR = -2(T \ln L_T(\hat{\theta}_0) - T \ln L_T(\hat{\theta}_1)) = -2(-15045.233 + 14347.624) = 1395.218$,

which is distributed as χ_2^2 under the null hypothesis. The p-value is 0.000, leading to a strong rejection of the null hypothesis of a constant correlation matrix. A comparison of the DCC and DECO parameter estimates indicates a large degree of similarity between the two sets of estimates. A plot of the average correlation of the five interest rates given in Figure 20.8 shows that it varies between 0.85 and 0.95 over the sample.



Figure 20.8 DECO estimate of the average correlation of U.S. zero-coupon yields, 3 January 2000 to 21 August 2006.

20.7.2 A Time-Varying Volatility SVAR Model

The following application uses an SVAR with short-run restrictions to estimate a model for the 3-month U.K. LIBOR rate. The data are for the period 1 January 2004 to 28 May 2008. An important feature of the model is an allowance for the structural shocks to have time-varying conditional volatilities thereby providing an alternative method for modelling multivariate GARCH models.

Let y_t represent the following 6 variables: volatility represented by the VIX index constructed from European put and call option prices such that at any given time it represents the risk-neutral expectation of the integrated variance averaged over the next 30 calendar days (or 22 trading days); liquidity measured by the on-off run U.S. 5-year spread; the GBP/USD swap rate; credit spreads as measured by the REPO spread; a measure of the default risk for the U.K.; and the 3-month U.K. LIBOR rate spread.

Consider the following dynamic structural model of y_t

$$B_0 y_t = B_1 y_{t-1} + B_2 y_{t-2} + \cdots + B_k y_{t-k} + u_t \quad (20.82)$$

$$E[u_t] = 0, \quad E[u_t u'_t] = D_t, \quad E[u_t u'_s] = 0, t \neq s \quad (20.83)$$

$$d_{i,t} = \delta_i + \alpha_i u_{i,t-1}^2 + \beta_i d_{i,t-1}, \quad (20.84)$$

where B_i , $i = 0, 1, \dots$ u_t are matrices of structural parameters, with B_0 having coefficients of 1 down the main diagonal to represent the usual normalization, k represents the order of the lags and u_t is a vector of independent structural disturbances with separate GARCH(1,1) conditional variances with the conditional variances in (20.84) given by the diagonal elements of D_t .

The VAR is

$$\begin{aligned} y_t &= B_0^{-1} B_1 y_{t-1} + B_0^{-1} B_2 y_{t-2} + \cdots + B_0^{-1} B_k y_{t-k} + B_0^{-1} u_t \\ &= \Phi_1 y_{t-1} + \Phi_2 y_{t-2} + \cdots + \Phi_k y_{t-k} + v_t, \end{aligned} \quad (20.85)$$

where the VAR parameters are related to the structural parameters by

$$\Phi_i = B_0^{-1} B_i, \forall i, \quad (20.86)$$

and the VAR disturbances are

$$v_t = B_0^{-1} u_t = B_0^{-1} D_t^{1/2} z_t = S_t z_t, \quad (20.87)$$

with

$$S_t = B_0^{-1} D_t^{1/2}.$$

The VAR disturbances have time-varying covariances as

$$E[v_t v'_t] = E[B_0^{-1} u_t u'_t B_0^{-1}] = B_0^{-1} E[u_t u'_t] B_0^{-1} = B_0^{-1} D_t B_0^{-1} = S_t S'_t = V_t.$$

Unlike the structural disturbances, this matrix is not diagonal, provided that B_0 and hence S are not diagonal, in which case the volatility of all factors has an effect on all the variables in the VAR.

To identify the parameters of the SVAR, the following short-run restrictions are imposed on B_0^{-1}

$$B_0^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ b^{2,1} & 1 & b^{2,3} & 0 & 0 & 0 \\ b^{3,1} & 0 & 1 & 0 & 0 & 0 \\ b^{4,1} & 0 & 0 & 1 & 0 & 0 \\ b^{5,1} & b^{5,2} & b^{5,3} & b^{5,4} & 1 & 0 \\ b^{6,1} & b^{6,2} & b^{6,3} & b^{6,4} & b^{6,5} & 1 \end{bmatrix}. \quad (20.88)$$

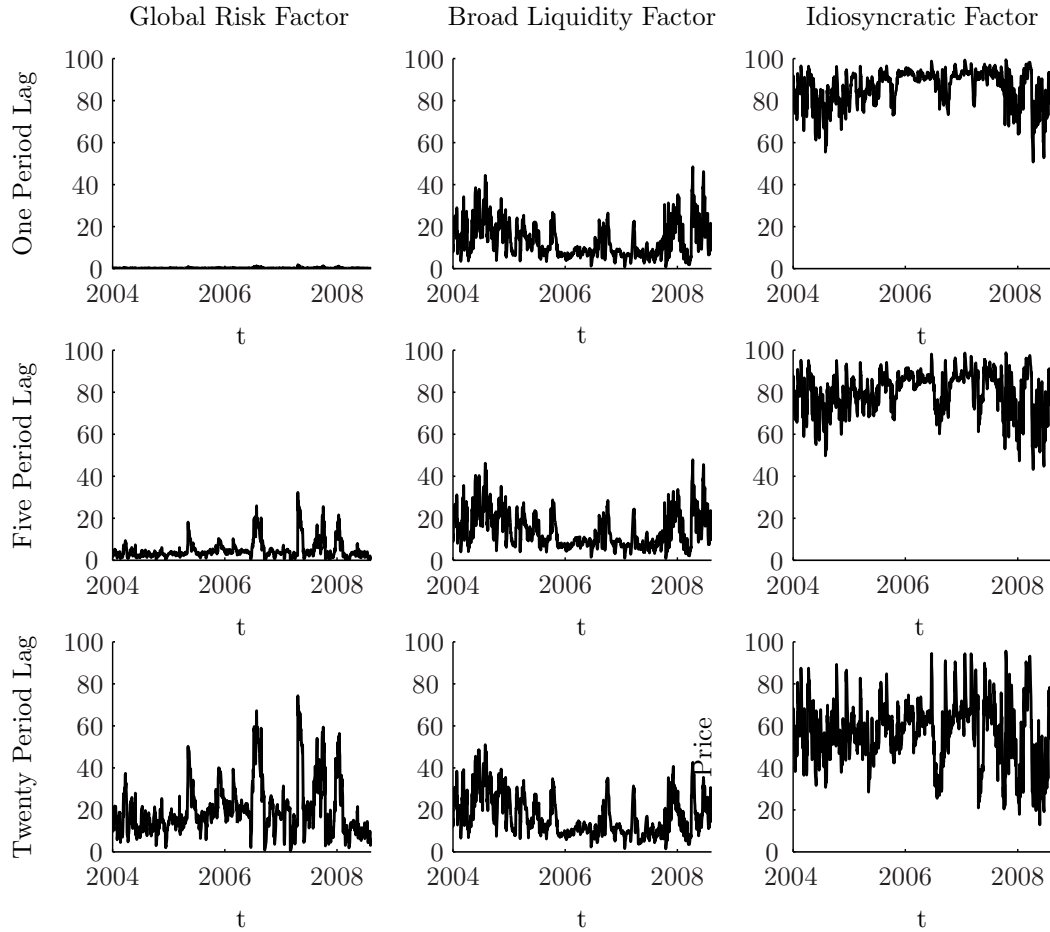


Figure 20.9 Time-varying variance instantaneous (first row), 5-period (second row) and 20-period (third row) decompositions of the U.K. LIBOR spread.

The interpretation of the structural shocks, u_t , is

$$u_t = \begin{bmatrix} \text{Global risk factor} \\ \text{Narrow liquidity factor} \\ \text{Broad liquidity factor} \\ \text{Credit factor} \\ \text{Default factor} \\ \text{Idiosyncratic factor} \end{bmatrix}.$$

Notice that all factors are designed to impact on the LIBOR, as given by the last row in the matrix S_t .

The variance decompositions are given in Figure 20.9, for instantaneous shocks, 5-period shocks and 20-period shocks respectively. These decompositions are time-varying as $S D_t^{1/2}$ is used to compute the orthogonalized impulse responses which is time-varying through the conditional variance matrix D_t . The results of the instantaneous variance decompositions over time for the U.K. LIBOR spread given in the first row of Figure 20.9 show that the key factors are the idiosyncratic factor and the broader liquidity factor. The time-varying variance decompositions after 5 periods (second row of Figure 20.9) show that the global risk factor has become relatively more important. The global risk factor becomes even more important once the analysis is extended to 20 periods (third row of Figure 20.9).

20.8 Exercises

(1) Statistical Properties of Equity Returns

Gauss file(s)	<code>garch_statistic.g, equity.xlsx</code>
Matlab file(s)	<code>garch_statistic.m, equity.mat</code>

The data are daily observations on the FTSE, DOW and NIKKEI stock indexes from 5 January 1989 to 31 December 2007, a total of $T = 4,953$ sample observations.

- Compute the daily percentage return on the FTSE index, y_t .
- Plot y_t and y_t^2 and compare the results with Figures 20.1(a) and 20.1(b). Interpret the plots.
- Compute the ACF of y_t and y_t^2 , for 20 lags and compare the results with ACFs reported in Figures 20.1, panels (c) and (d). Interpret the plots.
- Compute the skewness and kurtosis coefficients of y_t . Compare these

values with the corresponding moments of the standardized normal distribution, namely 0 and 3, respectively.

- (e) Estimate the unconditional distribution of y_t using a nonparametric kernel density estimator and compare the result with the conditional distribution based on the standardized normal distribution.
- (f) Repeat parts (a) to (e) for the DOW and the NIKKEI.

(2) Testing for ARCH in Equity Returns

Gauss file(s)	<code>garch_test.g</code> , <code>equity.xlsx</code>
Matlab file(s)	<code>garch_test.m</code> , <code>equity.mat</code>

Use the same data as in Exercise 1.

- (a) Compute the daily percentage return on the FTSE index, y_t .
- (b) Test y_t for ARCH(1), ARCH(2) and ARCH(3) using the LM test and compare the results with those presented in Table 20.1.
- (c) Repeat parts (a) and (b) for the DOW and the NIKKEI.

(3) GARCH Model Properties

Gauss file(s)	<code>garch_simulate.g</code>
Matlab file(s)	<code>garch_simulate.m</code>

- (a) Simulate the following GARCH(1,1) model

$$y_t = u_t, \quad u_t \sim N(0, \sigma_t^2)$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

for a sample of size $T = 1000$, using the parameters $\alpha_0 = 0.10$, $\alpha_1 = 0.70$ and $\beta_1 = 0.20$.

- (b) Compute the ACF of y_t^2 and compare the results with the ACF reported in Figure 20.4, panel (a).
- (c) Repeat parts (a) and (b) using the parameters $\alpha_0 = 0.05$, $\alpha_1 = 0.15$ and $\beta_1 = 0.80$. Compare the results with the ACF reported in Figure 20.4, panel (b).
- (d) Repeat parts (a) and (b) using the parameters $\alpha_0 = 0.05$, $\alpha_1 = 0.05$, $\beta_1 = 0.90$ and compare the memory characteristics of the GARCH models in parts (b) and (c).

(4) Estimating a GARCH Model of Equity Returns

Gauss file(s)	<code>garch_equity.g</code> , <code>equity.xlsx</code>
Matlab file(s)	<code>garch_equity.m</code> , <code>equity.mat</code>

Use the same data as in Exercise 1.

- (a) Estimate the following GARCH(1,1) model

$$\begin{aligned} y_t &= u_t, & u_t &\sim N(0, \sigma_t^2) \\ \sigma_t^2 &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \end{aligned}$$

where y_t is the zero-mean percentage return on the FTSE. Compare the results with those presented in Table 20.2.

- (b) Plot the conditional variance and interpret the time series patterns of the series.
- (c) Compare the quasi-maximum likelihood standard errors with $p = 0$ lags with the standard errors based on the Hessian and outer product of the gradients, respectively.
- (d) Compute the unconditional variance using (20.31) and compare the value to the empirical estimate.
- (e) Test a GARCH(1,1) model against a GARCH(0,1) = ARCH(1) model using a LR test.
- (f) Re-estimate the model with an IGARCH conditional variance by imposing the restriction $\beta_1 = 1 - \alpha_1$ given in (20.33).
- (g) Repeat parts (a) to (f) for the DOW and the NIKKEI.

(5) **Testing for Day-of-the Week Effects in Equity Returns**

Gauss file(s)	<code>garch_seasonality.g, equity.xlsx</code>
Matlab file(s)	<code>garch_seasonality.m, equity.mat</code>

Use the same data as for Exercise 1.

- (a) Consider the GARCH(1,1) model

$$\begin{aligned} y_t &= u_t, & u_t &\sim N(0, \sigma_t^2) \\ \sigma_t^2 &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + \lambda_1 TUE_t + \lambda_2 WED_t + \lambda_3 THUR_t + \lambda_4 FRI_t, \end{aligned}$$

where y_t is the zero-mean percentage return on the FTSE. Perform a LR test of seasonality.

- (b) Extend the variance to include a holiday dummy variable, and perform a joint test of day-of-the-week and holiday effects.
- (c) Repeat parts (a) and (b) for the DOW and the NIKKEI.

(6) Modelling Risk in the Term Structure of Interest Rates

Gauss file(s)	<code>garch_m.g, yields_us.dat</code>
Matlab file(s)	<code>garch_m.m, yields_us.mat</code>

The data consist of daily yields, r_t , on U.S. zero coupon bonds, expressed as a percentage, over the period 10 October 1988 to 28 December 2001, a total of 3307 observations. The maturities of the bonds are 3 months, 1 year, 3 year, 5 year, 7 year and 10 year.

- (a) Estimate the following GARCH-M model

$$\begin{aligned} r_{10y,t} &= \mu_t + u_t, & u_t &\sim N(0, \sigma_t^2) \\ \mu_t &= \gamma_0 + \gamma_1 r_{3m,t} + \varphi \sigma_t^\rho \\ \sigma_t^2 &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \end{aligned}$$

where $r_{10y,t}$ is the 10-year yield and $r_{3m,t}$ is the 3-month yield on U.S. zero coupon bonds.

- (b) Perform a test of risk neutrality by testing that $\varphi = 0$.
 (c) Perform a test of a linear risk-return relationship by testing that $\rho = 1.0$.
 (d) Repeat parts (a) to (c) by choosing the 1-year, 3-year, 5-year and 7-year bond yields as the dependent variables, respectively.

(7) GARCH Model with Conditional Student t

Gauss file(s)	<code>garch_student.g, equity.xlsx</code>
Matlab file(s)	<code>garch_student.m, equity.mat</code>

Use the same data as for Exercise 1.

- (a) Estimate the following GARCH(1,1) model

$$\begin{aligned} y_t &= \gamma_0 + u_t, & u_t &\sim St(0, \sigma_t^2, \nu) \\ \sigma_t^2 &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \end{aligned}$$

where y_t is the percentage return on the FTSE. Compare the GARCH point estimates with those presented in Table 20.2, which are based on conditional normality.

- (b) Interpret the estimate of the degrees of freedom parameter ν and discuss its implications for the existence of the moments of the conditional distribution of y_t .

- (c) Re-estimate the model assuming a constant conditional variance by imposing the restrictions $\alpha_1 = \beta_1 = 0$. Compare the estimate of the degrees of freedom parameter in part (b) and hence discuss the effects of the GARCH conditional variance on modelling the distribution of u_t .
- (d) A limiting case of the Student t distribution occurs when the degrees of freedom approach infinity, $\nu \rightarrow \infty$. Construct a Wald test of normality. *Hint: under the null $\nu^{-1} = 0$.*
- (e) Repeat parts (a) to (d) for the DOW and the NIKKEI.

(8) **Quasi-maximum Likelihood Estimation**

Gauss file(s)	<code>garch_studt.g</code> , <code>garch_gam.g</code>
Matlab file(s)	<code>garch_studt.m</code> , <code>garch_gam.m</code>

Consider the GARCH(1,1) model

$$y_t = u_t, \quad u_t = \sigma_t z_t$$

$$\sigma_t^2 = 1 - \alpha_1 - \beta_1 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

The aim of this exercise is to investigate the efficiency of the quasi-maximum likelihood estimator relative to the maximum likelihood estimator of the parameters $\theta = \{\alpha_1, \beta_1\}$ of the model.

- (a) Let the true distribution of z_t be Student t with ν_0 degrees of freedom. Write the log-likelihood function for observation t and derive an expression for the covariance matrix of the maximum likelihood estimator, $(\hat{\theta})$, of the true model.
- (b) Let the misspecified distribution of z_t be $N(0, 1)$. Write the log-likelihood function for observation t and derive an expression for the covariance matrix of the quasi-maximum likelihood estimator, $(\hat{\theta})$, of the misspecified.
- (c) Simulate $T = 1000000$ observations from the true model with the degrees of freedom parameter $\nu_0 = \{5, 8, 12\}$. For each value of ν_0 evaluate

$$\frac{\text{cov}_0(\hat{\alpha}_1)}{\text{cov}_1(\hat{\alpha}_1)}, \quad \frac{\text{cov}_0(\hat{\beta}_1)}{\text{cov}_1(\hat{\beta}_1)}.$$

Compare the simulation results to the analytical results of González-Rivera and Drost (1999).

- (d) Repeat parts (a) to (c) where the true distribution of z_t is the gamma

distribution,

$$f(z_t) = \frac{\sqrt{c}}{\Gamma(c)} (\sqrt{c}z_t)^{c-1} \exp[-\sqrt{c}z_t - c],$$

with shape parameter $c = 50$.

(9) BEKK Model of U.S. Yields

Gauss file(s)	mgarch_bekk.g, mgarch_student.g, yields_us.dat
Matlab file(s)	mgarch_bekk.m, mgarch_student.m, yields_us.mat

Use the same data as for Exercise 6.

- (a) Construct the variables

$$y_{1,t} = 100(r_{3m,t} - r_{3m,t-1}), \quad y_{2,t} = 100(r_{1y,t} - r_{1y,t-1}),$$

where $r_{3m,t}$ is the 3-month yield and r_{1y} is the 1-year yield.

- (b) Estimate the following symmetric BEKK specification

$$\begin{aligned} y_{1,t} &= \gamma_1 + u_{1,t} \\ y_{2,t} &= \gamma_2 + u_{2,t} \\ u_t &\sim N(0, V_t) \\ V_t &= CC' + Au_{t-1}u_{t-1}'A' + DV_{t-1}D', \end{aligned}$$

where C , A and D are defined in (20.60).

- (c) Perform a LR test of the constant covariance BEKK model by testing the restrictions $a_{1,2} = a_{2,1} = d_{1,2} = d_{2,1} = 0$.
- (d) Perform a LR test of the symmetric BEKK model by testing the restrictions $a_{1,2} = a_{2,1}$ and $d_{1,2} = d_{2,1}$.
- (e) Re-estimate the model using the standardized multivariate Student t distribution

$$f(u_t) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi(\nu-2)}\Gamma(\frac{\nu}{2})} |V_t|^{-1/2} \left(1 + \frac{u_t V_t^{-1} u_t}{\nu-2}\right)^{-\frac{\nu+1}{2}},$$

where ν is the degrees of freedom parameter. Interpret the estimate of the degrees of freedom parameter.

(10) DCC and DECO Model of U.S. Yields

Gauss file(s)	mgarch_dcc.g, yields_us.dat
Matlab file(s)	mgarch_dcc.m, yields_us.mat

The data file contains daily yields, r_t , on U.S. zero coupon bonds, expressed as a percentage, over the period 3 January 2000 to 21 August 2006, a total of 1658 observations. The maturities of the bonds are 1, 5, 10, 15 and 20 years.

- (a) Construct the variables

$$y_{i,t} = 100(r_{i,t} - r_{i,t-1}), \quad i = 1, 2, \dots, 5.$$

- (b) Compute the maximum likelihood estimates of a DCC model for $y_{i,t}$ and interpret the values.
- (c) Perform a LR test of the constant correlation model by testing $\alpha = \beta = 0$ in equation (20.65).
- (d) Compute the maximum likelihood estimates of a DECO model for $y_{i,t}$ and compare the estimated parameters with the estimates obtained using the DCC model. Plot the average correlation of the five interest rates and interpret the result.

(11) International CAPM with Time-varying Beta

Gauss file(s)	mgarch_icapm.g, icapm.dat
Matlab file(s)	mgarch_icapm.m, icapm.mat

The data are excess returns on the NYSE, r_t , and MSCI, m_t , which represents the excess returns on a world equity index. The sample period is 3 February 1988 to 29 December 1995, a total of $T = 2000$ observations.

- (a) Estimate the following international CAPM using the symmetric BEKK multivariate conditional variance specification with conditional normality

$$\begin{aligned} r_t &= \alpha + \beta_t m_t + u_{r,t} \\ m_t &= u_{m,t} \\ \beta_t &= \frac{\sigma_{r,m,t}^2}{\sigma_{m,m,t}^2}. \end{aligned}$$

- (b) Plot the estimate of β_t over the sample period.
- (c) Estimate the constant beta risk model ($\beta_t = \beta$) by regressing r_t on a constant and m_t , and compare this estimate to the plot of β_t given in part (b).

(12) Modelling Libor Rates with Time-Varying Volatility

Gauss file(s)	<code>svar_liboruk.g, libor_data.xls</code>
Matlab file(s)	<code>svar_liboruk.m, libor_data.xls</code>

The data, y_t , are daily, beginning 1 January 2004 and ending 28 May 2008, and correspond to the 6 variables listed in Section 20.7.2, namely, volatility, liquidity, the GBP/USD swap rate, the REPO spread, default risk for the U.K. and the 3-month U.K. LIBOR rate spread. Consider the dynamic structural model of y_t with GARCH conditional variances

$$\begin{aligned}
 B_0 y_t &= B_1 y_{t-1} + B_2 y_{t-2} + \cdots + B_k y_{t-k} + u_t \\
 E[u_t] &= 0, \quad E[u_t u'_t] = D_t, \quad E[u_t u'_s] = 0, t \neq s \\
 d_{i,t} &= \delta_i + \alpha_i u_{i,t-1}^2 + \beta_i d_{i,t-1},
 \end{aligned}$$

where identification is based on the restrictions in (20.88).

- (a) Estimate the model and interpret the factors.
- (b) Compute the variance decompositions at lags 0, 5 and 20 and interpret the results.

(13) ARCH-NNS Model

Gauss file(s)	<code>garch_nns.g, han_park.xls</code>
Matlab file(s)	<code>garch_nns.m, han_park.xls</code>

The exercise is based on Han and Park (2008). The data consist of S&P500 stock returns, y_t , and interest rate spreads between AAA and BBB bonds, w_t , which are recorded monthly, weekly and daily.

- (a) Estimate the GARCH(1,1) model

$$\begin{aligned}
 y_t &= \mu + u_t \quad u_t \sim N(0, \sigma_t^2) \\
 \sigma_t^2 &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2,
 \end{aligned}$$

by maximum likelihood using the monthly, weekly and daily data. Discuss the time series properties of the estimate of σ_t^2 .

- (b) Consider the nonstationary ARCH model with nonlinear heteroskedasticity (ARCH-NNS)

$$\begin{aligned}
 y_t &= \mu + u_t, \quad u_t \sim N(0, \sigma_t^2) \\
 \sigma_t^2 &= \alpha_1 u_{t-1}^2 + \lambda |w_{t-1}|^\phi,
 \end{aligned}$$

where w_{t-1} represents a nonstationary variable. Estimate the parameters $\theta = \{\mu, \alpha_1, \lambda, \phi\}$ by maximum likelihood for the monthly, weekly and daily data. Test the restriction $\phi = 1$.

- (c) Estimate the GARCH-NNS model

$$\begin{aligned}y_t &= \mu + u_t, & u_t &\sim N(0, \sigma_t^2) \\ \sigma_t^2 &= \alpha_0 + \alpha_1 u_{t-1}^2 + \beta_1 \sigma_{t-1}^2 + \lambda |w_{t-1}|^\phi.\end{aligned}$$

Compare the parameter estimates in parts (a) and (b) and discuss the ability of the nonstationary variable w_{t-1} to capture the nonstationarity in the conditional variance.

- (d) Consider using a Wald test of the restriction $\lambda = 0$ in part (c). Briefly discuss the problem(s) in implementing this test.

21

Discrete Time Series Models

21.1 Introduction

In most of the models previously discussed, the dependent variable, y_t , is assumed to be a continuous random variable. However, since the continuity assumption is sometimes inappropriate, alternative classes of models must be specified to explain the time series features of discrete random variables. This chapter reviews the important class of discrete time series models commonly used in microeconometrics namely the probit, ordered probit and Poisson regression models. It also discusses some recent advances in the modelling of discrete random variables with particular emphasis on the binomial thinning model of Steutel and Van Harn (1979) and the Autoregressive Conditional Duration (ACD) model of Engle and Russell (1998), together with some of its extensions.

21.2 Motivating Examples

Recent empirical research in financial econometrics has emphasized the importance of discrete random variables. Here, data on the number of trades and the duration between trades are recorded at very high frequencies. The examples which follow all highlight the need for econometric models that deal with discrete random variables by preserving the distributional characteristics of the data.

Example 21.1 Transactions Data on Trades

Table 21.1 provides a snapshot of transactions data recorded every second, on the U.S. stock AMR, the parent company of American Airlines, on 1 August 2006. Three examples of discrete random variables can be obtained from the data in Table 21.1.

Binary data

Table 21.1

Snapshot of transactions data on AMR: 1 August 2006, 9.30am to 4.00pm.

9:42:Seconds	Trade 1 sec. interval y_t	Trade 3 sec. interval y_t	Duration u_t	Price p_t
5	1		1	21.58
6	0		1	21.58
7	0	1	1	21.58
8	0		1	21.58
9	0		1	21.58
10	0	0	1	21.58
11	1		6	21.59
12	1		1	21.59
13	0	2	1	21.59
14	1		2	21.59
15	1		1	21.59
16	0	2	1	21.59
17	1		2	21.59
18	1		1	21.59
19	0	2	1	21.59

Data on whether a trade on AMR stock occurs within a second are given in the second column in Table 21.1, with the binary variable defined as

$$y = \begin{cases} 1 & : \text{Trade occurs} \\ 0 & : \text{Trade does not occur.} \end{cases}$$

The data in this column represents a binary random variable which is modelled using a Bernoulli distribution

$$f(y; \theta) = \theta^y (1 - \theta)^{1-y}, \quad 0 < \theta < 1.$$

Count data

In the third column of Table 21.1, the number of trades in every three second interval is recorded. The frequency of trades now ranges from 0 to 2 and represents a discrete counting variable. A potential model of the random variable y now is the Poisson distribution

$$f(y; \theta) = \frac{\theta^y \exp[-\theta]}{y!}, \quad \theta > 0.$$

Duration data

The time interval between trades is given in the forth column of Table 21.1. If the trade counts represents a Poisson random variable, then the duration between trades, u , is modelled as an exponential random variable

$$f(u; \theta) = \frac{1}{\theta} \exp \left[-\frac{u}{\theta} \right], \quad \theta > 0.$$

□

Example 21.2 U.S. Monetary Policy

The key instrument of monetary policy in the U.S. is the Federal funds target rate. A plot of the monthly target rate over the period 1984 to 2009 given in Figure 21.1, shows that the series in general, moves in discrete steps of $\pm 0.25\%$. The change in the target rate ranges from extreme easing of interest rates (changes of -1%), as experienced during the Global Financial Crisis of 2007 to 2009, to extreme tightening (changes of 1%). This variable represents an ordered polychotomous variable. □

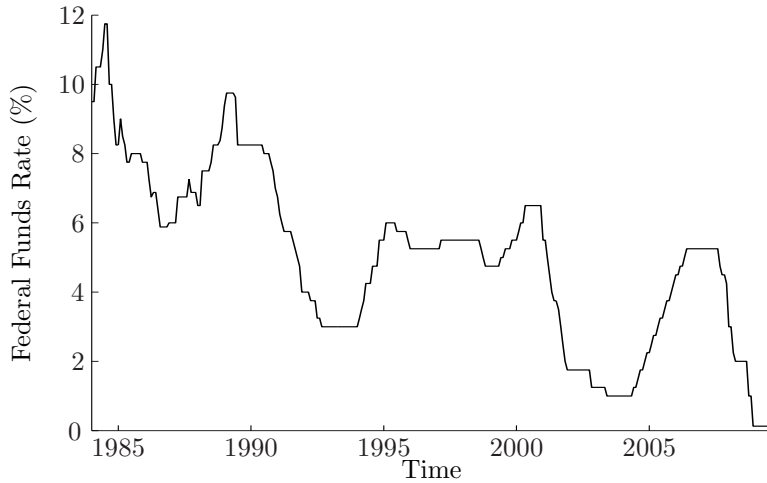


Figure 21.1 U.S. Federal funds target rate (%) from 1984 to 2009

Example 21.3 Interest Rate Floors and Money Demand

The demand for money expresses the interest rate, r_t , as a function of real income, y_t , and the real stock of money, m_t ,

$$r_t = \beta_0 + \beta_1 y_t + \beta_2 m_t + u_t,$$

where u_t is a disturbance term. If the monetary authorities set an interest

rate floor of r_{FLOOR} , then r_t is given by

$$r_t = \begin{cases} r_t & : r_t > r_{\text{FLOOR}} \\ r_{\text{FLOOR}} & : r_t \leq r_{\text{FLOOR}} \end{cases}.$$

Even though the money demand equation without the floor is linear, with the floor it is nonlinear. This is highlighted in Figure 21.2 where the floor interest rate is 4%. Even if u_t in the demand for money equation follows a normal distribution, it does not follow that the distribution r_t is also normal because it is now a censored random variable. $\square$

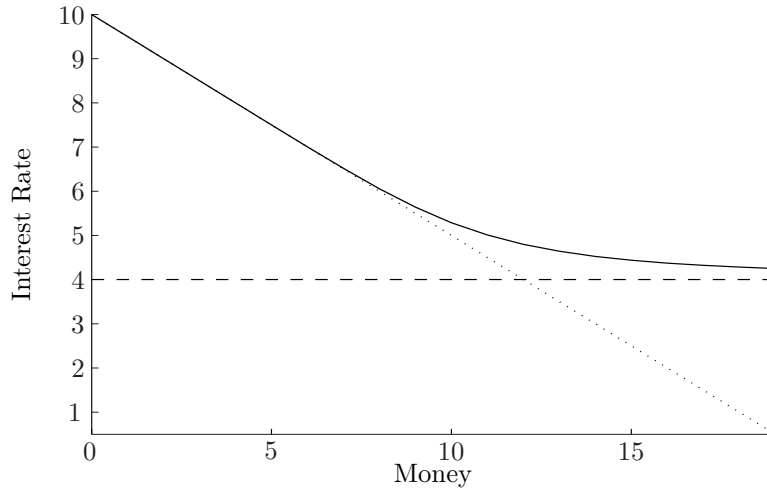


Figure 21.2 Money demand equation with a floor interest rate of 4% (solid line). The demand equation without the floor is linear (dashed line).

21.3 Qualitative Data

Before recent developments in dynamic discrete time series models are investigated, traditional methods commonly used in microeconometrics for modelling and estimating qualitative response models will be presented.

21.3.1 Specification

Consider the following normal linear regression model

$$\begin{aligned} y_t^* &= \beta_0 + \beta_1 x_t^* + u_t^* \\ u_t^* &\sim N(0, \sigma^2), \end{aligned} \tag{21.1}$$

where y_t^* and x_t^* are the quantitative dependent and exogenous variables, respectively, and u_t^* is the disturbance term. The $*$ notation is introduced to distinguish between quantitative (not filtered) and qualitative (filtered) variables. The fundamental difference between the different kinds of qualitative response models concerns the way that the data are transformed, or filtered. The models will now be demonstrated with reference to the sample of $T = 10$ simulated realizations from the model in equation (21.1) for y_t^* and x_t^* . The simulated observations are tabulated in Table 21.2.

Table 21.2

Data used to compare alternative filtering methods; $T = 10$.

Full		Binary		Censored		Truncated	
$x_t = x_t^*$	$y_t = y_t^*$	x_t	y_t	x_t	y_t	x_t	y_t
5	-1	5	0	5	0	n.a.	n.a.
4	3	4	1	4	3	4	3
2	-4	2	0	2	0	n.a.	n.a.
1	-2	1	0	1	0	n.a.	n.a.
5	2	5	1	5	2	5	2
6	0	6	0	6	0	n.a.	n.a.
7	1	7	1	7	1	7	1
3	2	3	1	3	2	3	2
3	-3	3	0	3	0	n.a.	n.a.
9	4	9	1	9	4	9	4

In the first two columns of Table 21.2 headed Full, there is no filtering so that $y_t = y_t^*$ and $x_t = x_t^*$. In each of the other columns, Binary, Censored and Truncated, the data are subject to application of a filter so that the quantitative and qualitative data are not identical. The qualitative data y_t and x_t are observed and the underlying quantitative data y_t^* and x_t^* are unobserved. Simply ignoring the filter and estimating the equation using the observed qualitative variables y_t and x_t

$$y_t = \alpha_0 + \alpha_1 x_t + u_t, \quad (21.2)$$

by ordinary least squares results in biased estimates of the unknown parameters $\theta = \{\beta_0, \beta_1, \sigma^2\}$ as

$$\text{plim}(\hat{\alpha}_0) \neq \beta_0, \quad \text{plim}(\hat{\alpha}_1) \neq \beta_1.$$

The maximum likelihood estimator, however, offers a means of still obtaining a consistent estimator of θ . It surmounts the potential statistical problems

caused from filtering by explicitly building into the likelihood the form of the filter. Each of the cases represented in Table 21.2 is now discussed in more detail.

Full Information: Linear Regression Model

The data are in the columns headed Full in Table 21.2 where $y_t = y_t^*$ and $x_t = x_t^*$. This case corresponds to the simple linear regression model. The distribution of y_t is

$$f(y|x_t; \theta) = \frac{1}{\sigma} \phi_t,$$

where ϕ_t is the standard normal distribution

$$\phi_t = \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{1}{2\sigma^2} (y - \beta_0 - \beta_1 x_t)^2 \right]. \quad (21.3)$$

The log-likelihood function for a sample of T observations is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T [-\ln \sigma + \ln \phi_t].$$

Binary Information: Probit Regression Model

The binary data for the probit regression model are created by applying the filter

$$\begin{aligned} y_t &= \begin{cases} 1 & : y_t^* > 0 \\ 0 & : y_t^* \leq 0 \end{cases} \\ x_t &= x_t^*, \end{aligned} \quad (21.4)$$

with the results recorded in the columns headed Binary in Table 21.2. Because y_t is a binary random variable, it has a Bernoulli distribution

$$f(y|x_t; \theta) = \Phi_t^y (1 - \Phi_t)^{1-y}, \quad (21.5)$$

where $\Phi_t = \Pr(y = 1)$ and $1 - \Phi_t = \Pr(y = 0)$. As u_t in (21.1) is normally distributed, Φ_t is defined in terms of the cumulative normal distribution

$$\Phi_t = \int_{-\infty}^{\frac{\beta_0 + \beta_1 x_t}{\sigma}} \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{s^2}{2} \right] ds. \quad (21.6)$$

This represents the probit regression model. The log-likelihood function for a sample of T observations is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T [y_t \ln \Phi_t + (1 - y_t) \ln(1 - \Phi_t)].$$

For the probit regression model as the upper limit of the integration in equation (21.6) is

$$\frac{\beta_0 + \beta_1 x_t}{\sigma} = \frac{\beta_0}{\sigma} + \frac{\beta_1}{\sigma} x_t,$$

it is only possible to identify the ratios β_0/σ and β_1/σ and not the individual terms. It is common practice therefore to adopt the normalization $\sigma = 1$.

Censored Information: Censored Regression Model

In the case of the censored model, the data are constructed using the filter

$$\begin{aligned} y_t &= \begin{cases} y_t^* & : y_t^* > 0 \\ 0 & : y_t^* \leq 0 \end{cases} \\ x_t &= x_t^*, \end{aligned} \quad (21.7)$$

with the results recorded in the columns headed Censored in Table 21.2. This model is a mixture of the full information model where $y_t^* > 0$, and the binary information model where $y_t^* > 0$. The distribution of y_t thus consists of two parts

$$f(y|x_t; \theta) = \left(\frac{1}{\sigma}\right)^{d_t} \phi_t^{d_t} (1 - \Phi_t)^{1-d_t}, \quad (21.8)$$

where ϕ_t and Φ_t are defined in (21.3) and (21.6), respectively, and d_t is a dummy variable

$$d_t = \begin{cases} 1 & : y_t^* > 0 \\ 0 & : y_t^* \leq 0. \end{cases} \quad (21.9)$$

This is the censored regression model, or the Tobit model. The log-likelihood function for a sample of T observations is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T [-d_t \ln \sigma + d_t \ln \phi_t + (1 - d_t) \ln(1 - \Phi_t)].$$

Truncated Information: Truncated Regression Model

Finally, the data for the truncated model are constructed using the filter

$$y_t = \begin{cases} y_t^* & : y_t^* > 0 \\ n.a & : y_t^* \leq 0, \end{cases} \quad x_t = \begin{cases} x_t^* & : y_t^* > 0 \\ n.a & : y_t^* \leq 0, \end{cases} \quad (21.10)$$

with the results recorded in the Truncated columns in Table 21.2. The distribution of the dependent variable is the truncated normal distribution

$$f(y|x_t; \theta) = \frac{\phi_t}{\sigma \Phi_t}, \quad (21.11)$$

where ϕ_t and Φ_t are defined in (21.3) and (21.6), respectively. This is the

truncated regression model. The log-likelihood function for a sample of T observations is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T [-d_t \ln \sigma + d_t \ln \phi_t - d_t \ln \Phi_t],$$

where d_t is as defined in equation (21.9).

The main differences between the models presented is the degree of information lost from filtering the data. To quantify this loss of information a Monte Carlo experiment is now used to demonstrate the finite sample properties of the maximum likelihood estimator of these models.

Example 21.4 Finite Sample Properties of Estimators

The model is specified as in equation (21.1) where $x_t^* \sim N(0, 1)$, $u_t^* \sim N(0, \sigma^2)$, the parameters are $\theta = \{\beta_0 = 1, \beta_1 = 1, \sigma^2 = 1\}$ and truncation is at zero. The results obtained by simulating the model using 5000 replications with samples of size $T = 100$ are given in Table 21.3. The maximum likelihood estimator shows little bias for the probit, Tobit and truncated regression models, demonstrating that the estimator is able to correct for the different data filters. The root mean square errors satisfy the relationship

$$\text{RMSE}(\text{Censored}) < \text{RMSE}(\text{Truncated}) < \text{RMSE}(\text{Probit}),$$

showing that the greatest loss of efficiency, not surprisingly occurs for the probit regression model. In comparison, least squares estimation using data on values of $y_t > 0$ only, yields biased parameter estimates with the size of the bias being approximately 30%. $\square$

21.3.2 Estimation

With the exception of the linear regression model, the probit, censored and truncated regression models are nonlinear in the parameters. The parameters of these models must, therefore, be estimated using one of the numerical algorithms discussed in Chapter 3. In practice, it is convenient to compute the gradient vector and Hessian numerically, although for all of these models analytical solutions are easily derived.

The probit regression model with just a constant in the regression model is an example of a case in which an analytical expression is available for the maximum likelihood estimator. In this situation, estimation of the model reduces to estimating the parameter of the Bernoulli distribution.

Example 21.5 Bernoulli Distribution

Table 21.3

Finite sample properties of the maximum likelihood estimator of alternative models. The restricted model is estimated using data on values of $y_t > 0$ only. The sample size is $T = 100$ and the number of repetitions is 5000.

Statistics	Linear	Probit	Censored	Truncated	Restricted
True (β_0)	1.000	1.000	1.000	1.000	1.000
Mean	0.999	1.039	0.996	0.990	1.372
Bias	-0.001	0.039	-0.004	-0.010	0.372
RMSE	0.103	0.211	0.110	0.170	0.384
True (β_1)	1.000	1.000	1.000	1.000	1.000
Mean	1.000	1.057	1.005	1.006	0.704
Bias	0.000	0.057	0.005	0.006	-0.296
RMSE	0.102	0.254	0.116	0.172	0.317

Let y_t , $t = 1, 2, \dots, T$ be *iid* observations from the Bernoulli distribution

$$f(y; \theta) = \theta^y (1 - \theta)^{1-y}, \quad y_t = 0, 1,$$

where $\theta > 0$, is an unknown parameter (probability). The log-likelihood is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T y_t \ln \theta + \frac{1}{T} \sum_{t=1}^T (1 - y_t) \ln(1 - \theta).$$

The gradient and Hessian are, respectively,

$$\begin{aligned} G_T(\theta) &= \frac{d \ln L_T(\theta)}{d\theta} = \frac{1}{\theta T} \sum_{t=1}^T y_t - \frac{1}{(1 - \theta)T} \sum_{t=1}^T (1 - y_t) \\ H_T(\theta) &= \frac{d^2 \ln L_T(\theta)}{d\theta^2} = -\frac{1}{\theta^2 T} \sum_{t=1}^T y_t - \frac{1}{(1 - \theta)^2 T} \sum_{t=1}^T (1 - y_t). \end{aligned}$$

The maximum likelihood estimator is given as the solution of $G_T(\hat{\theta}) = 0$,

$$0 = \frac{1}{\hat{\theta} T} \sum_{t=1}^T y_t - \frac{1}{(1 - \hat{\theta}) T} \sum_{t=1}^T (1 - y_t) = \frac{1}{\hat{\theta}(1 - \hat{\theta}) T} \sum_{t=1}^T (y_t - \hat{\theta}),$$

which yields the sample mean of y_t as the maximum likelihood estimator

$$\hat{\theta} = \frac{1}{T} \sum_{t=1}^T y_t = \bar{y}.$$

As the data are binary, the sample mean also corresponds to the sample proportion. Evaluating the Hessian at $\hat{\theta}$ gives

$$\begin{aligned} H_T(\hat{\theta}) &= -\frac{1}{\hat{\theta}^2 T} \sum_{t=1}^T y_t - \frac{1}{(1-\hat{\theta})^2 T} \sum_{t=1}^T (1-y_t) \\ &= -\frac{\hat{\theta}}{\hat{\theta}^2} - \frac{(1-\hat{\theta})}{(1-\hat{\theta})^2} = -\frac{1}{\hat{\theta}(1-\hat{\theta})}. \end{aligned}$$

The variance of $\hat{\theta}$ is

$$\text{var}(\hat{\theta}) = -\frac{1}{T} H_T^{-1}(\hat{\theta}) = \frac{\hat{\theta}(1-\hat{\theta})}{T},$$

which is the usual variance reported for testing the sample proportion. $\square$

Example 21.6 BHHH Estimation of the Probit Model

Consider the binary variable, y_t that has a Bernoulli distribution

$$f(y|x_t; \theta) = \Phi_t^y (1 - \Phi_t)^{1-y},$$

where Φ_t is defined by (21.6), x_t is an exogenous variable and $\theta = \{\beta_0, \beta_1\}$. The log-likelihood function of the probit regression model is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T y_t \ln \Phi_t + \frac{1}{T} \sum_{t=1}^T (1-y_t) \ln(1-\Phi_t).$$

The first derivatives are

$$\begin{aligned} \frac{\partial \ln L_T(\theta)}{\partial \beta_0} &= \frac{1}{T} \sum_{t=1}^T y_t \frac{\phi_t}{\Phi_t} - \frac{1}{T} \sum_{t=1}^T (1-y_t) \frac{\phi_t}{1-\Phi_t} = \frac{1}{T} \sum_{t=1}^T \frac{(y_t - \Phi_t) \phi_t}{\Phi_t(1-\Phi_t)} \\ \frac{\partial \ln L_T(\theta)}{\partial \beta_1} &= \frac{1}{T} \sum_{t=1}^T y_t x_t \frac{\phi_t}{\Phi_t} - \frac{1}{T} \sum_{t=1}^T (1-y_t) x_t \frac{\phi_t}{1-\Phi_t} = \frac{1}{T} \sum_{t=1}^T \frac{(y_t - \Phi_t) \phi_t}{\Phi_t(1-\Phi_t)} x_t, \end{aligned}$$

where ϕ_t is defined by (21.3). The BHHH algorithm amounts to updating using a weighted least squares regression with weights given by $w_t = \sqrt{\phi_t / (\Phi_t(1-\Phi_t))}$. The approach is to regress the weighted residual $(y_t - \Phi_t)w_t$ on the weighted explanatory variables $\{w_t, x_t w_t\}$, evaluated at the starting value $\theta_{(0)}$, to get Δ_θ . The updated parameter values are computed as

$$\theta_{(1)} = \theta_{(0)} + \Delta_\theta.$$

$\square$

Example 21.7 A Probit Model of Monetary Policy

Consider the Federal funds rate data introduced in Example 21.2. Changes

in the target rate, Δy_t , are assumed to be a function of the spread between the 6-month Treasury bill rate and the Federal funds rate, s_t , the 1-month lagged inflation rate, π_{t-4} , and the growth rate in output, gdp_t . If just the direction of monetary policy is known, the model is specified as

$$y_t = \begin{cases} 1 & : \Delta y > 0 \quad [\text{Monetary tightening}] \\ 0 & : \Delta y \leq 0 \quad [\text{Monetary easing or no change}], \end{cases}$$

where

$$f(y; \theta) = \Phi_t^{y_t} (1 - \Phi_t)^{1-y_t},$$

$$\Phi_t = \Phi(\beta_0 + \beta_1 s_t + \beta_2 \pi_{t-4} + \beta_3 gdp_t).$$

Table 21.4

Descriptive statistics of movements in the Federal funds target rate on Federal Open Market Committee meeting days, February 1984 to June 1997.

Target change (%)	Number	Proportion
-0.50	3	0.028
-0.25	11	0.104
0.00	82	0.774
0.25	6	0.057
0.50	4	0.034
	106	

Table 21.5

Maximum likelihood estimates of the probit model of monetary policy, with calculations based on the BFGS algorithm with derivatives computed numerically.

Parameter	Unconstrained		Constrained	
	Estimate	p-value	Estimate	p-value
β_0	-3.859	0.024	-1.315	0.000
β_1	2.328	0.001		
β_2	58.545	0.136		
β_3	0.277	0.098		
$T \ln L_T(\hat{\theta})$	-20.335		-33.121	

Summary statistics of the movements in the Federal funds target rate, y_t ,

given in Table 21.4, show that, of the $T = 106$ meeting days from 1984 to 1997, 10 of these days resulted in monetary tightening ($\Delta y > 0$) and 96 resulted in either monetary easing or no change in policy ($\Delta y \leq 0$). The parameter estimates given in Table 21.5 show that s_t is significant at the 5% level of significance, gdp_t is significant at the 10% level of significance and π_{t-4} is not statistically significant. The constrained model given by the restrictions $\beta_1 = \beta_2 = \beta_3 = 0$ yields an estimate of the intercept of this model of -1.315 . Substituting this value into the cumulative normal distribution gives

$$\int_{-\infty}^{-1.315} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{s^2}{2}\right] ds = 0.094,$$

which equals the observed proportion of FOMC days of monetary tightening, that is 10 out of 106. $\square$

21.3.3 Testing

The LR, Wald and LM testing procedures can be used to perform hypothesis tests on the parameters $\theta = \{\beta_0, \beta_1\}$. Consider testing the hypotheses for the model in (21.1) in the case of the probit regression model

$$H_0 : \beta_1 = 0, \quad H_1 : \beta_1 \neq 0.$$

Choosing the LM test has the advantage that, under the null hypothesis of $\beta_1 = 0$, the maximum likelihood estimator $\hat{\theta}_0 = \{\hat{\beta}_0, 0\}$, has an analytical solution where $\Phi(\hat{\beta}_0) = \bar{y}$. The LM statistic is

$$LM = TG'_T(\hat{\theta}_0)I^{-1}(\hat{\theta}_0)G_T(\hat{\theta}_0). \quad (21.12)$$

In the case of the probit regression model, the gradient vector and information matrix under the null are, respectively,

$$G_T(\hat{\theta}_0) = \frac{1}{T} \sum_{t=1}^T \hat{u}_t \frac{\hat{\phi}}{\hat{\Phi}(1 - \hat{\Phi})} \begin{bmatrix} 1 \\ x_t \end{bmatrix} \quad (21.13)$$

$$I(\hat{\theta}_0) = -E[H_T(\hat{\theta}_0)] = \frac{1}{T} \sum_{t=1}^T \frac{\hat{\phi}^2}{\hat{\Phi}(1 - \hat{\Phi})} \begin{bmatrix} 1 & x_t \\ x_t & x_t^2 \end{bmatrix}, \quad (21.14)$$

where $\hat{u}_t = y_t - \hat{\Phi}$ is the residual and

$$\hat{\phi} = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(y_t - \hat{\beta}_0)^2\right], \quad \hat{\Phi} = \int_{-\infty}^{\hat{\beta}_0} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{s^2}{2}\right] ds,$$

and the normalization $\sigma = 1$ is adopted. Substituting into the LM statistic in (21.12) and rearranging gives

$$\begin{aligned}
 LM &= \frac{1}{\widehat{\Phi}(1 - \widehat{\Phi})} \begin{bmatrix} \sum_{t=1}^T u_t \\ \sum_{t=1}^T u_t x_t \end{bmatrix}' \begin{bmatrix} T & \sum_{t=1}^T x_t \\ \sum_{t=1}^T x_t & \sum_{t=1}^T x_t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T u_t \\ \sum_{t=1}^T u_t x_t \end{bmatrix} \\
 &= \frac{T}{\sum_{t=1}^T u_t^2} \begin{bmatrix} \sum_{t=1}^T u_t \\ \sum_{t=1}^T u_t x_t \end{bmatrix}' \begin{bmatrix} T & \sum_{t=1}^T x_t \\ \sum_{t=1}^T x_t & \sum_{t=1}^T x_t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T u_t \\ \sum_{t=1}^T u_t x_t \end{bmatrix} \\
 &= TR^2.
 \end{aligned} \tag{21.15}$$

The second step is based on the property of the Bernoulli distribution that $\text{var}(y_t) = \Phi(1 - \Phi)$, which suggests that under the null hypothesis $\widehat{\Phi}(1 - \widehat{\Phi})$ can be replaced by the sample variance

$$s^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \bar{y})^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \widehat{\Phi})^2 = \frac{1}{T} \sum_{t=1}^T u_t^2.$$

The LM statistic in (21.15) shows that small values of R^2 suggest that x_t does not contribute to explaining the difference between y_t and its sample mean, that is its sample proportion. Consequently, a LM test of the overall explanatory power of the model is implemented as a two-stage regression procedure as follows.

Step 1: Regress y_t on a constant and obtain the restricted residuals, $\widehat{u}_t = y_t - \bar{y}$.

Step 2: Regress $\widehat{u}_t$ on $\{1 x_t\}$.

Step 3: Compute TR^2 from this second-stage regression and the corresponding p-value from the χ_K^2 distribution, where K is the number of variables in x_t . A p-value of less than 0.05 represents rejection of the null hypothesis at the 5% level.

Example 21.8 Joint LM Test in a Probit Model

Consider the probit model of monetary policy in Example 21.7. The estimated regression equation at the second stage is

$$\widehat{u}_t = -0.013 + 0.278 s_t + 4.022 \pi_{t-4} + 0.030 gdp_t + \widehat{v}_t,$$

with coefficient of determination of $R^2 = 0.231$. The LM statistic of the joint

test that $\beta_1 = \beta_2 = \beta_3 = 0$ is

$$LM = TR^2 = 106 \times 0.231 = 24.486,$$

which is distributed under the null hypothesis as χ_3^2 . The p-value is 0.000 showing strong rejection of the null hypothesis at the 5% level. $\square$

21.3.4 Binary Autoregressive Models

In the models discussed in this chapter so far, the distribution of the dependent variable y_t is conditional on a set of explanatory variables x_t . Without loss of generality, this set can also include lagged values of the x_t variables. However, extending the set of explanatory variables to include lagged values of the dependent variable requires more careful modelling. One possible specification is the AR class of models investigated in Chapter 13. In the case of an AR(1) model the dynamics are represented by

$$y_t = \phi_0 + \phi_1 y_{t-1} + v_t, \quad (21.16)$$

where $-1 < \phi_1 < 1$ is a parameter and v_t is a continuous *iid* disturbance term. As y_t represents binary integers, the proposed model needs to preserve this property. This is not the case with the model in equation (21.16) as (i) the conditional expectation $E_{t-1}[y_t] = \phi_1 y_{t-1}$ is unlikely to be a binary integer even if y_{t-1} is; (ii) the disturbance term v_t is continuous suggesting that y_t is also continuous. An alternative strategy is needed for the present case of a binary random variable.

Consider the case where y_t is a binary random variable corresponding to one of two states

$$y_t = \begin{cases} 1 & : \text{State 1} \\ 0 & : \text{State 0.} \end{cases}$$

In the next period, y_{t+1} is also a binary random variable taking on the values 1, 0 depending upon its value in the previous state. The dynamics of y_t are represented by the AR(1) model

$$y_t = (1 - q) + (p + q - 1)y_{t-1} + u_t. \quad (21.17)$$

The parameters p and q represent the conditional probabilities of staying in each respective state

$$\begin{aligned} p &= P[y_t = 1 \mid y_{t-1} = 1] \\ q &= P[y_t = 0 \mid y_{t-1} = 0]. \end{aligned} \quad (21.18)$$

As y_t is binary, the disturbance term u_t also needs to be binary in order

to preserve the characteristics of the dependent variable. The disturbance term has the following form depending upon the conditional value of y_{t-1} . If $y_{t-1} = 1$

$$v_t = \begin{cases} 1 - p & : \text{ with probability } p \\ -p & : \text{ with probability } 1 - p, \end{cases} \quad (21.19)$$

whereas, if $y_{t-1} = 0$

$$v_t = \begin{cases} -(1 - q) & : \text{ with probability } q \\ q & : \text{ with probability } 1 - q. \end{cases} \quad (21.20)$$

To show that the model given by equations (21.17) to (21.20) preserves the binary characteristics of y_t suppose that $y_{t-1} = 1$. From (21.17)

$$y_t = (1 - q) + (p + q - 1) \times 1 + v_t = p + u_t,$$

and, from (21.18), $u_t = 1 - p$ with probability p , which means that

$$y_t = p + (1 - p) = 1,$$

with probability p , which is the definition of p in (21.18). Alternatively, $u_t = -p$ with probability $1 - p$, which means that

$$y_t = p + (-p) = 0,$$

with probability $1 - p$. A similar result occurs for the case of $y_{t-1} = 0$.

An alternative way of summarizing these transition probabilities is to define the transition matrix

$$\begin{bmatrix} p & 1 - q \\ 1 - p & q \end{bmatrix}, \quad (21.21)$$

where the first row is used to compute the probability of ending in state $y_t = 1$, while the second row gives the probability of ending in state $y_t = 0$. Thus,

$$\begin{bmatrix} P(y_t = 1) \\ P(y_t = 0) \end{bmatrix} = \begin{bmatrix} p & 1 - q \\ 1 - p & q \end{bmatrix} \begin{bmatrix} P(y_{t-1} = 1) \\ P(y_{t-1} = 0) \end{bmatrix}.$$

Estimation of the parameters $\theta = \{p, q\}$ in equation (21.17) can be accomplished using the EMM estimator discussed in Chapter 12. A natural choice of the auxiliary model is the continuous time AR(1) model in equation (21.16)

$$\begin{aligned} y_t &= \phi_0 + \phi_1 y_{t-1} + v_t \\ v_t &\sim N(0, \sigma_v^2). \end{aligned} \quad (21.22)$$

The algorithm proceeds as follows. Estimate the auxiliary model in (21.22)

using the actual data by regressing y_t on a constant and y_{t-1} to obtain $\hat{\phi}_0$ and $\hat{\phi}_1$, as well as the residual variance $\hat{\sigma}_v^2$. Simulate the model in equations (21.17) to (21.20) for starting values of $\theta = \{p, q\}$ and generate simulated data $y_{s,t}$. Evaluate the (2×1) gradient function of the auxiliary model, replacing the actual data by the simulated data

$$G_S(\theta) = \begin{bmatrix} \frac{1}{S\hat{\sigma}_v^2} \sum_{t=2}^S (y_{s,t} - \hat{\phi}_0 - \hat{\phi}_1 y_{s,t-1}) \\ \frac{1}{S\hat{\sigma}_v^2} \sum_{t=2}^S (y_{s,t} - \hat{\phi}_0 - \hat{\phi}_1 y_{s,t-1}) y_{s,t-1} \end{bmatrix},$$

where $S = T \times K$ and K is the number of simulated paths. The EMM estimator is the solution of

$$\hat{\theta} = \arg \max_{\theta} G'_S(\theta) W_T^{-1} G_S(\theta), \quad (21.23)$$

where W_T is the optimal weighting matrix previously encountered in Chapters 9, 10 and 12. An example of this class of model is the Markov switching model of Hamilton (1989), discussed previously in Chapter 19.

21.4 Ordered Data

In the probit model of monetary policy discussed in Example 21.7, the dependent variable is defined in terms of two regimes that capture the sign of movements in the target rate. Inspection of Table 21.4 reveals that there is also additional information available about the size of the change in terms of monetary tightening and easing. It is possible to characterise monetary policy in terms of five regimes over the period 1984 to 1997, ranging from -0.50% to 0.50% in steps of 0.25% . As the dependent variable is an ordered discrete random variable, a simple probit model is no longer appropriate and an ordered probit model is required.

To specify the ordered probit model, the dependent variable y_t is defined in terms of the dummy variable d_t as

$$d_t = \begin{cases} 0 & : y_t = -0.50 & [\text{Extreme easing}] \\ 1 & : y_t = -0.25 & [\text{Easing}] \\ 2 & : y_t = 0.00 & [\text{No change}] \\ 3 & : y_t = 0.25 & [\text{Tightening}] \\ 4 & : y_t = 0.50 & [\text{Extreme tightening}]. \end{cases}$$

The probabilities associated with each regime are

$$\begin{aligned}\Phi_{0,t} &= \Pr(d_t = 0) = \Phi(c_0 - x_t\beta) \\ \Phi_{1,t} &= \Pr(d_t = 1) = \Phi(c_1 - x_t\beta) - \Phi(c_0 - x_t\beta) \\ \Phi_{2,t} &= \Pr(d_t = 2) = \Phi(c_2 - x_t\beta) - \Phi(c_1 - x_t\beta) \\ \Phi_{3,t} &= \Pr(d_t = 3) = \Phi(c_3 - x_t\beta) - \Phi(c_2 - x_t\beta) \\ \Phi_{4,t} &= \Pr(d_t = 4) = 1 - \Phi(c_3 - x_t\beta),\end{aligned}$$

where x_t is the set of explanatory variables with parameter vector β and $c_0 < c_1 < c_2 < c_3$ are parameters corresponding to the intercepts of each regime. The order restriction on the c parameters is needed to ensure that the probabilities of each regime are positive. In the special case of two regimes, $c_0 = -\beta_0$ is the intercept of the probit regression model whereby

$$y_t = \begin{cases} 0 & : \text{ with probability } \Phi(-\beta_0 - x_t\beta) = 1 - \Phi(\beta_0 + x_t\beta) \quad [\text{Easing or no change}] \\ 1 & : \text{ with probability } 1 - \Phi(-\beta_0 - x_t\beta) = \Phi(\beta_0 + x_t\beta) \quad [\text{Tightening}] \end{cases}$$

The log-likelihood function for a sample of size T is

$$\ln L_T(\theta) = \frac{1}{T} \sum_{t=1}^T [d_{0,t} \ln \Phi_{0,t} + d_{1,t} \ln \Phi_{1,t} + d_{2,t} \ln \Phi_{2,t} + d_{3,t} \ln \Phi_{3,t} + d_{4,t} \ln \Phi_{4,t}],$$

where the unknown parameters are $\theta = \{\beta, c_0, c_1, c_2, c_3\}$. This function is maximized with respect to θ using the iterative algorithms discussed in Chapter 3.

Example 21.9 An Ordered Probit Model of Monetary Policy

Table 21.6 gives the results of estimating the ordered probit model of U.S. monetary policy, with the same explanatory variables as used in the probit version of this model given in Example 21.7: namely, the spread between the Federal funds rate and the 6-month Treasury bill rate, s_t , the 1-month lagged inflation rate, π_{t-4} , and the growth rate in output, gdp_t . Similar to the empirical results reported for the probit model in Table 21.7, s_t is statistically the most important explanatory variable of monetary policy. The constrained model is estimated by imposing the restrictions $\beta_1 = \beta_2 = \beta_3 = 0$. Substituting these estimates into the cumulative normal distribution recovers the empirical probabilities of each regime given in Table 21.4. For

Table 21.6

Maximum likelihood estimates of the ordered probit model of monetary policy: calculations based on the BFGS algorithm with derivatives computed numerically.

Parameter	Unconstrained		Constrained	
	Estimate	p-value	Estimate	p-value
c_0	-2.587	0.003	-1.906	0.000
c_1	-1.668	0.048	-1.117	0.000
c_2	1.542	0.063	1.315	0.000
c_3	2.321	0.008	1.778	0.000
β_1	1.641	0.000		
β_2	4.587	0.823		
β_3	0.125	0.109		
$T \ln L_T$	-20.335		-33.121	

example

$$\Pr(d_t = 0) = \int_{-\infty}^{-1.906} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{s^2}{2}\right] ds = 0.028$$

$$\Pr(d_t = 1) = \int_{-1.906}^{-1.117} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{s^2}{2}\right] ds = 0.132 - 0.028 = 0.104,$$

correspond to 3/106 and 10/106, respectively.

□

21.5 Count Data

Count data measure the number of times that an event occurs within a given time period. An example was given in Table 21.1 where the number of trades y , in a three second interval was

$$y = \begin{cases} 0 \\ 1 \\ 2. \end{cases}$$

The data are positive integer counts and, therefore, represent a discrete random variable.

A natural way to model a dependent variable that measures counts is to assume that y has a Poisson distribution

$$f(y; \theta) = \frac{\theta^y \exp(-\theta)}{y!}, \quad y = 0, 1, 2, \dots, \quad \theta > 0, \quad (21.24)$$

where θ is an unknown parameter, which represents the mean of the distribution $\theta = E[y]$.

Example 21.10 Poisson Distribution

For a sample of T observations on y_t assumed to be independent drawings from the Poisson distribution, the log-likelihood function is

$$\begin{aligned} \ln L_T(\theta) &= \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta) \\ &= \frac{1}{T} \sum_{t=1}^T \ln \left(\frac{\theta^{y_t} \exp(-\theta)}{y_t!} \right) = \ln \theta \frac{1}{T} \sum_{t=1}^T y_t - \theta - \frac{1}{T} \sum_{t=1}^T \ln y_t!. \end{aligned}$$

The gradient and Hessian are, respectively,

$$G_T(\theta) = \frac{d \ln L}{d\theta} = \frac{1}{\theta T} \sum_{t=1}^T y_t - 1, \quad H_T(\theta) = \frac{d^2 \ln L}{d\theta^2} = -\frac{1}{\theta^2 T} \sum_{t=1}^T y_t.$$

Setting the gradient to zero $G_T(\hat{\theta}) = 0$ gives $0 = \hat{\theta}^{-1} T^{-1} \sum_{t=1}^T y_t - 1$, yielding the sample mean as the maximum likelihood estimator

$$\hat{\theta} = \frac{1}{T} \sum_{t=1}^T y_t.$$

Evaluating the Hessian at $\hat{\theta}$ shows that the maximum likelihood estimator satisfies the second-order condition

$$H_T(\hat{\theta}) = -\frac{1}{\hat{\theta}^2 T} \sum_{t=1}^T y_t = -\frac{1}{\hat{\theta}^2 T} T \hat{\theta} = -\frac{1}{\hat{\theta}} < 0.$$

The variance of $\hat{\theta}$ is

$$\text{var}(\hat{\theta}) = -\frac{1}{T} H^{-1} = \frac{\hat{\theta}}{T},$$

which is the usual formula given for computing the variance of the sample mean from the Poisson distribution. $\square$

21.5.1 The Poisson Regression Model

The Poisson regression model is obtained by specifying the mean of the Poisson distribution in equation (21.24) as a function of the exogenous variable x_t ,

$$f(y; \theta) = \frac{\mu_t^y \exp(-\mu_t)}{y!}, \quad y_t = 0, 1, 2, \dots \quad (21.25)$$

$$\mu_t = \exp(\beta_0 + \beta_1 x_t), \quad \mu_t > 0, \quad (21.26)$$

where $\theta = \{\beta_0, \beta_1\}$ are the unknown parameters and the nonlinear function $\exp(\beta_0 + \beta_1 x_t)$ ensures that the conditional mean is positive for all t .

For a sample of T observations, the log-likelihood function is

$$\begin{aligned} \ln L_T(\theta) &= \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta) = \frac{1}{T} \sum_{t=1}^T (y_t \ln \mu_t - \mu_t - \ln y_t!) \\ &= \frac{1}{T} \sum_{t=1}^T (y_t(\beta_0 + \beta_1 x_t) - \exp(\beta_0 + \beta_1 x_t) - \ln y_t!). \end{aligned}$$

The gradient is

$$G_T(\theta) = \begin{bmatrix} \frac{\partial \ln L}{\partial \beta_0} \\ \frac{\partial \ln L}{\partial \beta_1} \end{bmatrix} = \frac{1}{T} \sum_{t=1}^T (y_t - \mu_t) \begin{bmatrix} 1 \\ x_t \end{bmatrix}.$$

The Hessian is

$$H_T(\theta) = \begin{bmatrix} \frac{\partial^2 \ln L}{\partial \beta_0^2} & \frac{\partial^2 \ln L}{\partial \beta_0 \partial \beta_1} \\ \frac{\partial^2 \ln L}{\partial \beta_0 \partial \beta_1} & \frac{\partial^2 \ln L}{\partial \beta_1^2} \end{bmatrix} = -\frac{1}{T} \sum_{t=1}^T \mu_t \begin{bmatrix} 1 & x_t \\ x_t & x_t^2 \end{bmatrix},$$

yielding the information matrix

$$I(\theta) = -E[H_T(\theta)] = \frac{1}{T} \sum_{t=1}^T \mu_t \begin{bmatrix} 1 & x_t \\ x_t & x_t^2 \end{bmatrix}.$$

As there is no analytical solution for $\hat{\theta} = \{\hat{\beta}_0, \hat{\beta}_1\}$, it is necessary to use an iterative algorithm.

Hypothesis tests are performed using the LR, Wald and LM testing frameworks. As with the probit model, the LM statistic

$$LM = T G_T'(\hat{\theta}_0) I^{-1}(\hat{\theta}_0) G_T(\hat{\theta}_0),$$

where $\hat{\theta}_0$ is the maximum likelihood estimator under the null hypothesis, is often a convenient testing strategy. Consider a test of the hypothesis $\beta_1 = 0$ in equation (21.26). The maximum likelihood estimator is $\hat{\mu} = \exp(\hat{\beta}_0) = \bar{y}$. The gradient and information matrix under the null hypothesis are, respectively,

$$G_T(\hat{\theta}_0) = \frac{1}{T} \sum_{t=1}^T (y_t - \bar{y}) \begin{bmatrix} 1 \\ x_t \end{bmatrix} = \frac{1}{T} \sum_{t=1}^T u_t \begin{bmatrix} 1 \\ x_t \end{bmatrix}, \quad I(\hat{\theta}_0) = \frac{1}{T} \sum_{t=1}^T \bar{y} \begin{bmatrix} 1 & x_t \\ x_t & x_t^2 \end{bmatrix},$$

where $\hat{u}_t = y_t - \bar{y}$. Substituting the expressions for $G_T(\hat{\theta}_0)$ and $I(\hat{\theta}_0)$ into the LM statistic and rearranging gives

$$\begin{aligned} LM &= \frac{1}{\bar{y}} \begin{bmatrix} \sum_{t=1}^T \hat{u}_t \\ \sum_{t=1}^T \hat{u}_t x_t \end{bmatrix}' \begin{bmatrix} T & \sum_{t=1}^T x_t \\ \sum_{t=1}^T x_t & \sum_{t=1}^T x_t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T \hat{u}_t \\ \sum_{t=1}^T \hat{u}_t x_t \end{bmatrix} \\ &= \frac{T}{\sum_{t=1}^T \hat{u}_t^2} \begin{bmatrix} \sum_{t=1}^T \hat{u}_t \\ \sum_{t=1}^T \hat{u}_t x_t \end{bmatrix}' \begin{bmatrix} T & \sum_{t=1}^T x_t \\ \sum_{t=1}^T x_t & \sum_{t=1}^T x_t^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{t=1}^T \hat{u}_t \\ \sum_{t=1}^T \hat{u}_t x_t \end{bmatrix} \\ &= TR^2, \end{aligned}$$

where the second step is based on the property of the Poisson distribution that $\mu = \text{var}(y_t)$, which suggests that under the null hypothesis $\bar{y}$ can be replaced by the sample variance

$$s^2 = \frac{1}{T} \sum_{t=1}^T (y_t - \bar{y})^2 = \frac{1}{T} \sum_{t=1}^T u_t^2.$$

The LM test of the overall explanatory power of the model is implemented as a two-stage regression procedure as follows:

Step 1: Regress y_t on a constant to obtain the restricted residuals, $\hat{u}_t = y_t - \bar{y}$.

Step 2: Regress $\hat{u}_t$ on $\{1, x_t\}$.

Step 3: Compute TR^2 from this second-stage regression and the corresponding p-value from the χ_K^2 distribution, where K is the number of variables in x_t . A p-value of less than 0.05 represents rejection of the null hypothesis at the 5% level.

21.5.2 Integer Autoregressive Models

Consider specifying a Poisson autoregressive model where y_t is a Poisson random variable representing count data assumed to be a function of its own lag y_{t-1} . As discussed in Section 21.3.4, the continuous time AR(1) model in equation (21.16) is not appropriate to model autoregressive discrete random variables as (i) the conditional expectation $E_{t-1}[y_t]$ is unlikely to be an integer; (ii) the disturbance term v_t is continuous suggesting that y_t is also continuous. The second problem is easily rectified by replacing the continuous distribution of v_t by a discrete distribution such as the Poisson distribution. The first problem is solved by replacing the multiplication operator ρy_{t-1} in equation (21.16) by the binomial thinning operator introduced by Steutel and Van Harn (1979) and extended by McKenzie (1988) and Al-Osh and Alzaid (1987). See also Jung, Ronning and Tremayne (2005) for a review.

A first-order Poisson autoregressive model is specified as

$$y_t = \underbrace{\rho \circ y_{t-1}}_{\text{departures (Bernoulli)}} + \underbrace{u_t}_{\text{arrivals (Poisson)}}, \quad (21.27)$$

where

$$\begin{aligned} \rho \circ y_{t-1} &= \sum_{s=1}^{y_{t-1}} e_{s,t-1} \\ e_{s,t-1} &= \begin{cases} 1 : & \text{Prob} = \rho \\ 0 : & \text{Prob} = 1 - \rho \end{cases} \\ u_t &\sim P(\lambda). \end{aligned} \quad (21.28)$$

The notation $\circ$ is referred to as the binomial thinning operator because it sums y_{t-1} Bernoulli random variables, $e_{s,t-1}$, each with probability of success equal to ρ . The disturbance term u_t is a Poisson random variable with parameter λ , given by $P(\lambda)$. This model has two sources of randomness

Thinning: Departures at each t are determined by the thinning operator assuming Bernoulli random variables.

Disturbance: Arrivals at each t are determined by a Poisson random variable.

Example 21.11 Simulating a Poisson AR(1) Model

The parameters of the Poisson autoregressive model are $\rho = 0.3$ and $\lambda = 3.5$, with the initial value of y_t given by $y_1 = 5$. To simulate y_t at $t = 2$, $y_1 = 5$ uniform random numbers are drawn with values $\{0.509, 0.244, 0.262, 0.512, 0.590\}$. As 2 of these random variables are less than $\rho = 0.3$, the value of the thinning

operator is

$$\rho \circ y_1 = \sum_{s=1}^{y_1} e_{s,1} = \sum_{s=1}^5 e_{s,1} = 0 + 1 + 1 + 0 + 0 = 2.$$

Drawing from the Poisson distribution with parameter $\lambda = 3.5$ yields the disturbance term $u_2 = 2$. The updated value of y_2 is

$$y_2 = \rho \circ y_1 + u_2 = 2 + 2 = 4.$$

To simulate y_t at $t = 3$, $y_2 = 4$ new uniform random numbers are drawn with values $\{0.297, 0.844, 0.115, 0.600\}$, resulting in the thinning operation

$$\rho \circ y_2 = \sum_{s=1}^{y_2} e_{s,1} = \sum_{s=1}^4 e_{s,1} = 1 + 0 + 1 + 0 = 2.$$

Drawing from the Poisson distribution with parameter $\lambda = 3.5$ yields the disturbance term $u_3 = 1$, so the updated value of y_3 is

$$y_3 = \rho \circ y_2 + u_3 = 2 + 1 = 3.$$

These calculations are repeated for $t = 4$ with the initial value now $y_2 = 3$. $\square$

The parameters of the first-order Poisson autoregressive model, $\theta = \{\rho, \lambda\}$ in equations (21.27) and (21.28), can be estimated by maximum likelihood methods. The conditional distribution of y_t is shown by Al-Osh and Alzaid (1987) to be

$$f(y|y_{t-1}; \theta) = \sum_{k=0}^{\min(y, y_{t-1})} \frac{y_{t-1}! \rho^k (1-\rho)^{y_{t-1}-k}}{k!(y_{t-1}-k)!} \frac{\lambda^{y-k} \exp(-\lambda)}{(y-k)!}. \quad (21.29)$$

This distribution represents a mixture of a binomial distribution and a Poisson distribution which naturally follows from the two sources of randomness underlying the model. Conditioning on the first observation, the log-likelihood for a sample of $t = 2, 3, \dots, T$ observations is

$$\begin{aligned} \ln L_T(\theta) &= \frac{1}{T-1} \sum_{t=2}^T \ln f(y_t|y_{t-1}; \theta) \\ &= -\lambda + \frac{1}{T-1} \sum_{t=2}^T \ln \left(\sum_{k=0}^{\min(y_t, y_{t-1})} \frac{y_{t-1}! \rho^k (1-\rho)^{y_{t-1}-k}}{k!(y_{t-1}-k)!} \frac{\lambda^{y_t-k}}{(y_t-k)!} \right). \end{aligned}$$

The numerical optimization algorithms presented in Chapter 3 are used to maximize this function with respect to θ . Maximum likelihood estimation of

higher-order integer autoregressive models are discussed by Bu, McCabe and Hadiri (2008). For ARMA structures Martin, Tremayne and Jung (2011) adopt an EMM estimator. The sampling properties of this estimator are investigated in Section 21.7.

The testing procedures discussed in Chapter 4 can be used to conduct hypothesis tests on θ . For example, a test of the hypothesis $\rho = 0$, provides a test of independence. Imposing this restriction on the conditional distribution in (21.29) gives

$$\begin{aligned}
 f(y|y_{t-1}; \rho = 0, \lambda) &= \sum_{k=0}^{\min(y, y_{t-1})} \frac{y_{t-1}! 0^k (1-0)^{y_{t-1}-k}}{k! (y_{t-1}-k)!} \frac{\lambda^{y-k} \exp(-\lambda)}{(y-k)!} \\
 &= \left(\frac{y_{t-1}! 0^0 (1-0)^{y_{t-1}-0}}{0! (y_{t-1}-0)!} \frac{\lambda^{y-0} \exp(-\lambda)}{(y-0)!} \right. \\
 &\quad \left. + \frac{y_{t-1}! 0^1 (1-0)^{y_{t-1}-1}}{1! (y_{t-1}-1)!} \frac{\lambda^{y-1} \exp(-\lambda)}{(y-1)!} + \dots \right) \\
 &= \frac{\lambda^y \exp(-\lambda)}{y!},
 \end{aligned}$$

which is the unconditional Poisson distribution of y . Under the null hypothesis, the consequence of $\rho = 0$ is the suppression of the randomness of the thinning operator thus leaving the Poisson component as the only source of error.

Example 21.12 Sampling Properties

This example reproduces the results of Jung, Ronning and Tremayne (2005, Table 1) for the case of maximum likelihood estimation. The sample size is $T = 100$ and the number of draws is 5000. The results are given in Table 21.7 where the population parameter values are $\rho = 0.3$ and $\lambda = 3.5$ and $E[y] = \lambda/(1 - \rho) = 3.5/(1 - 0.3) = 5$. Also reported are the sampling properties of the conditional least squares (CLS) estimator of Klimko and Nelson (1978), where ρ and λ are estimated, respectively, from a least squares regression of y_t on y_{t-1} and a constant. The maximum likelihood estimator has superior sampling properties to the CLS estimator, producing smaller bias and lower variances. A measure of the asymptotic relative efficiency (ARE) is given by the ratio of the MSE of the maximum likelihood estimator to the CLS estimator

$$\begin{aligned}
 \text{ARE}(\hat{\rho}) &= \frac{\text{MSE}(\hat{\rho}_{MLE})}{\text{MSE}(\hat{\rho}_{CLS})} = \frac{0.009}{0.010} = 0.894 \\
 \text{ARE}(\hat{\lambda}) &= \frac{\text{MSE}(\hat{\lambda}_{MLE})}{\text{MSE}(\hat{\lambda}_{CLS})} = \frac{0.264}{0.293} = 0.900,
 \end{aligned}$$

Table 21.7

Finite sample properties of the maximum likelihood estimator of the parameters of the Poisson first-order autoregressive model. The sample size is $T = 100$ and the number of draws is 5000.

Statistics	MLE		CLS	
	ρ	λ	ρ	λ
True	0.300	3.500	0.300	3.500
Mean	0.293	3.542	0.277	3.620
Bias	-0.007	0.042	-0.023	0.120
St. dev.	0.096	0.512	0.099	0.528
RMSE	0.096	0.514	0.102	0.542
MSE	0.009	0.264	0.010	0.293

which show efficiency gains of around 10% in the maximum likelihood estimator compared to the CLS estimator. $\square$

Bayesian methods for dealing with time series counts are discussed in McCabe and Martin (2005) and Fr  hwirth-Schnatter and Wagner (2006).

21.6 Duration Data

In the previous section, the Poisson distribution with parameter $\theta = \{\mu\}$ is used to model the number of counts within a given time period. From the properties of the Poisson distribution, the durations between counts have an exponential distribution with the same parameter μ .

If y is now taken to represent the duration time between events, the exponential distribution is specified as

$$f(y; \theta) = \frac{1}{\mu_t} \exp\left(-\frac{y}{\mu_t}\right), \quad \mu_t > 0, \quad (21.30)$$

where μ_t is the conditional mean of the exponential distribution, which is a function of the explanatory variable x_t . To ensure that μ_t is strictly positive for all t , the following restriction is imposed

$$\mu_t = \exp(\beta_0 + \beta_1 x_t). \quad (21.31)$$

For a sample of T observations, the log-likelihood function for the duration

model is

$$\begin{aligned}\ln L_T(\theta) &= \frac{1}{T} \sum_{t=1}^T \ln f(y_t; \theta) = \frac{1}{T} \sum_{t=1}^T \left(-\ln \mu_t - \frac{y_t}{\mu_t} \right) \\ &= \frac{1}{T} \sum_{t=1}^T \left(-(\beta_0 + \beta_1 x_t) - \frac{y_t}{\exp(\beta_0 + \beta_1 x_t)} \right),\end{aligned}\quad (21.32)$$

which is maximized with respect to the parameters $\theta = \{\beta_0, \beta_1\}$.

Example 21.13 The Engle-Russell ACD Model A class of models used in empirical finance to model the duration between trades, y , is the autoregressive conditional duration model (ACD) of Engle and Russell (1998), where y is distributed exponentially as in (21.30) and the conditional mean in (21.31) is specified as

$$\mu_t = \alpha_0 + \sum_{j=1}^q \alpha_j y_{t-j} + \sum_{j=1}^p \beta_j \mu_{t-j}.$$

To ensure that the conditional mean is stationary, the following condition needs to hold

$$\sum_{j=1}^q \alpha_j + \sum_{j=1}^p \beta_j < 1.$$

The parameters β_j control the memory of the process, being a weighted sum of lagged durations. $\square$

An important concept in modelling duration data is the hazard rate which represents the probability of an event occurring in the next time interval

$$h_t = \frac{f(y_t)}{S(y_t)} = \frac{f(y_t)}{1 - F(y_t)}, \quad (21.33)$$

where $f(y_t)$ is the probability density of y_t , $S(y_t)$ is the survival function and $F(y_t)$ is the corresponding cumulative probability distribution. In the case where $f(y_t)$ is the exponential distribution given in (21.30), the survival function is $S(y_t) = \exp(-y_t \mu_t^{-1})$ and the hazard rate reduces to

$$h_t = \frac{f(y_t)}{S(y_t)} = \frac{\mu_t^{-1} \exp(-y_t \mu_t^{-1})}{\exp(-y_t \mu_t^{-1})} = \frac{1}{\mu_t}. \quad (21.34)$$

As the hazard rate represents a probability it needs to satisfy the restriction

$$0 \leq h_t \leq 1.$$

Example 21.14 The Hamilton-Jorda ACH Model of Trades

Let y be a binary variable with $y = 1$ representing an event occurs and $y = 0$ that no event occurs. The autoregressive conditional hazard model (ACH) of Hamilton and Jordà (2002) is given by

$$\begin{aligned} f(y; \theta) &= h_t^y (1 - h_t)^{1-y} \\ h_t &= \frac{1}{1 + \exp(\psi_t)} \\ \psi_t &= \alpha_0 + \alpha_1 u_{t-1} + \beta_1 \psi_{t-1}, \end{aligned}$$

where h_t is the hazard rate, ψ_t is the conditional duration and u_t is the observed duration between events. The specification of the hazard rate ensures that $0 < h_t < 1$. This is a dynamic probit regression model where the probability represents the hazard rate which is time-varying according to durations in the previous period u_{t-1} , and the lagged hazard rate as $\psi_{t-1} = \ln(h_{t-1}^{-1} - 1)$. As $\alpha_1 > 0$, an increase in u_{t-1} decreases the probability (hazard rate) of an event occurring the next period. The equation for ψ_t corresponds to the conditional mean equation in the ACD model. The difference in the two models is that the dependent variable of the ACD model is the duration time between events at time t , whereas it is the binary variable that an event occurs at time t in the ACH model. An application of the ACH model is given next. $\square$

21.7 Applications**21.7.1 An ACH Model of U.S. Airline Trades**

Nowak (2008) uses a ACH model to investigate the effects of firm-specific and macroeconomic-specific news announcements on transactions of U.S. airline stocks. Let y be a binary variable at time t that identifies whether a trade has occurred

$$y = \begin{cases} 1 & : \text{Trade occurs in a second} \\ 0 & : \text{No trade occurs.} \end{cases}$$

The model is specified as

$$\begin{aligned} f(y; \theta) &= h_t^y (1 - h_t)^{1-y} \\ h_t &= \frac{1}{1 + \exp(\psi_t + \delta_1 Firm_{t-1} + \delta_2 Macro_{t-1})} \\ \psi_t &= \alpha_0 + \alpha_1 u_{t-1} + \beta_1 \psi_{t-1}, \end{aligned}$$

where h_t is the hazard rate, ψ_t is the conditional duration and u_t is the observed duration between trades. The variables $Firm_{t-1}$ and $Macro_{t-1}$

are dummy variables representing, respectively, lagged news announcements of the firm and the macroeconomy. The log-likelihood function for a sample of $t = 1, 2, \dots, T$ observations is

$$\ln L_T(\theta) = \frac{1}{T-1} \sum_{t=2}^T \left(y_t \ln(h_t) + (1 - y_t) \ln(1 - h_t) \right),$$

which is maximized with respect to $\theta = \{\alpha_0, \alpha_1, \beta_1, \delta_1, \delta_2\}$.

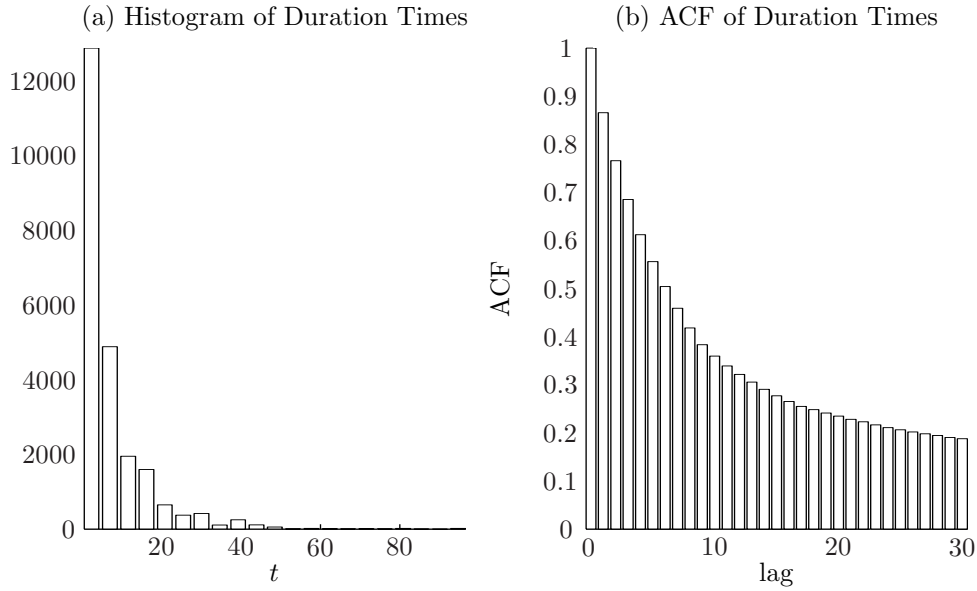


Figure 21.3 Histogram and ACF of the durations between AMR trades.

The data consist of trades per second on the U.S. company AMR, which is the parent company of American Airlines. The data are recorded on 1 August 2006, from 9.30am to 4.00pm. The total number of time periods is 23,367 second intervals, of which 3,641 intervals correspond to a trade ($y_t = 1$). A snapshot of the data is given in Table 21.1. The average time between trades is $\bar{u} = 7.28$ seconds. This corresponds to an unconditional hazard rate of

$$\bar{h} = \frac{1}{\bar{u}} = \frac{1}{7.28} = 0.137,$$

showing that the (unconditional) probability of a trade occurring in the next second is 0.137. A histogram of duration times given in panel (a) of Figure 21.3 shows that the distribution has an exponential shape which is consistent

with the assumptions of the hazard rate in the ACH model. A plot of the autocorrelation function of u_t in panel (b) of Figure 21.3 shows that the series exhibits long-memory characteristics with the ACF decaying slowly.

Table 21.8

Maximum likelihood estimates of the ACH model of trades for the AMR data recorded on 1 August 2006. Estimation uses the BFGS algorithm with numerical derivatives. Quasi-maximum likelihood standard errors are given in parentheses.

Variable	Parameter	Unconstrained		Constrained	
		Estimate	Std. Error	Estimate	Std. Error
<i>Const</i>	α_0	0.020	0.004	0.020	0.004
u_{t-1}	α_1	0.001	0.0001	0.001	0.0001
ψ_{t-1}	β_1	0.984	0.002	0.984	0.002
<i>Firm</i>	δ_1	0.503	1.123		
<i>Macro</i>	δ_2	-1.747	1.294		
	$(T-1) \ln L$	-10024.460		-10025.172	

The maximum likelihood estimates of the model are given in Table 21.8 for the unconstrained model as well as for the constrained model where the news announcement variables are excluded. The estimate of

$$\hat{\alpha}_1 + \hat{\beta}_1 = 0.001 + 0.984 = 0.985,$$

is consistent with the long-memory property of the autocorrelation structure of durations identified in panel (b) of Figure 21.3.

To identify the effect of the news variables on trades, compute the hazard rate with and without the two news dummy variables as follows

$$\bar{h}(Firm = 0, Macro = 0) = \frac{1}{1 + \exp(1.722)} = 0.152$$

$$\bar{h}(Firm = 1, Macro = 1) = \frac{1}{1 + \exp(1.722 + 0.503 - 1.747)} = 0.383,$$

where $\bar{\psi} = 1.722$ is the sample average of the estimated conditional duration function $\hat{\psi}_t$, obtained from the last iteration of the maximum likelihood algorithm. With the inclusion of the two news variables, the probability of a trade occurring in the next second immediately after a news announcement more than doubles. The estimate of the average length of time to the next trade is $1/0.152 = 6.594$ seconds when there is no news announcement and $1/0.383 = 2.613$ seconds when there is a news announcement. Whilst

these results suggest that the news variables are economically significant in terms of the overall effect on trades of AMR stock, the individual t-statistics reported in Table 21.8 show that the news variables are nonetheless not statistically significant.

21.7.2 EMM Estimator of Integer Models

An important feature of the maximum likelihood estimator of the Poisson binomial thinning model presented in Section 21.5.2, is that while the likelihood function of an AR(1) model is tractable, it becomes increasingly more complicated for higher-order integer autoregressive models. Expanding the class of models further to allow for integer moving average processes as well as mixed integer ARMA processes, the log-likelihood function becomes intractable. As demonstrated in Example 21.12, conditional least squares (CLS) offers a solution in the case of autoregressive models as it is simple to implement and has relatively good sampling properties relative to the maximum likelihood estimator, but not in the case of integer MA models.

Martin, Tremayne and Jung (2011) propose a variant of the efficient method of moments estimator (EMM) introduced in Chapter 12, to compute the parameters of Poisson binomial thinning models which include AR(p), MA(q) and mixed ARMA(p, q) models. This choice of the EMM estimator is motivated by the following considerations.

- (1) The binomial thinning model is easily simulated for integer ARMA models of arbitrary lag orders.
- (2) There exists a natural auxiliary model that provides a suitable approximation to the likelihood of the true binomial thinning model.

To simulate a mixed process, consider the integer ARMA(1,1) specification proposed by McKenzie (1988)

$$\begin{aligned}x_t &= \rho \circ x_{t-1} + u_t \\ y_t &= x_{t-1} + \psi \circ u_t,\end{aligned}\tag{21.35}$$

where y_t is observable, x_t is unobservable, u_t is an *iid* Poisson random variable with parameter λ , and ρ and ψ are the autoregressive and moving average parameters respectively. Given starting parameters for $\theta = \{\rho, \lambda, \psi\}$ as well as a starting value for the latent variable x_t , simulated values of y_t are easily obtained following the approach of Example 21.11.

The choice of the auxiliary model is based on a linear AR model where the lag structure is chosen to be of sufficient length to identify the parameters

in (21.35)

$$y_t = \phi_0 + \sum_{i=1}^p \phi_i y_{t-i} + v_t, \quad (21.36)$$

where v_t is *iid* $N(0, \sigma_v^2)$. For the model in (21.35) a minimum choice of the lag structure is $p = 2$ so that ϕ_1 and ϕ_2 are used to identify the parameters ρ and ψ . The disturbance variance σ_v^2 and the intercept ϕ_0 are used to identify λ . The moment conditions of the auxiliary model are

$$G_T(\theta) = \frac{1}{T-p} \sum_{t=p+1}^T \left[\begin{array}{cccc} \frac{v_t}{\sigma_v^2} & \frac{v_t y_{t-1}}{\sigma_v^2} & \frac{v_t y_{t-2}}{\sigma_v^2} & \cdots & \left(\frac{v_t^2}{\sigma_v^2} - 1 \right) \frac{1}{2\sigma_v^2} \end{array} \right]'. \quad (21.37)$$

The EMM estimator is based on solving

$$\hat{\theta} = \arg \min_{\theta} G'_S(\theta) W_T^{-1} G_S(\theta), \quad (21.38)$$

where $G_s(\theta)$ is the gradient vector in (21.37) with y_t replaced by the simulated data and the auxiliary parameters ϕ_i s in equation (21.35) estimated by least squares. The matrix W is the optimal weighting matrix which is based on the outer product of the gradient vector of the auxiliary model evaluated at y_t (see Chapter 12). As a result of the discrete nature of the underlying process, the objective function in expression (21.38) is not continuous in θ . To circumvent this problem a grid search algorithm is used instead of a gradient algorithm to maximise the log-likelihood function. Under certain regularity conditions the EMM estimator has the same asymptotic properties as the maximum likelihood estimator and, in some instances, can yield better small sample properties, as the following simulation experiment demonstrates.

The finite sample results of the EMM estimator for the INARMA(1,1) model in (21.35) are given in Table 21.9. Two auxiliary models are chosen with the lag structure of the auxiliary model in (21.36) varying from $p = 2$ (just-identified) to $p = 3$ (over-identified). The true parameter values are set at

$$\theta_0 = \{\rho = 0.3, \lambda = 3.5, \psi = 0.7\}.$$

The sample sizes are $T = \{50, 100, 200\}$, the length of a simulation run is fixed at $N = 500T$ and the number of replications is 5000.

Inspection of the results in Table 21.9 shows that the biases are relatively small and that they decrease as the sample size increases. The percentage bias of $\hat{\rho}$, in the case of the auxiliary model with $p = 2$ and a sample of size

Table 21.9

Mean and Root Mean Square Error (RMSE) of the EMM estimator for the INARMA(1,1) model in finite samples. Population parameters are $\rho = 0.3$, $\lambda = 3.5$ and $\psi = 0.7$. The number of draws is 5000.

Auxiliary lag (p)	Mean (ρ)			RMSE (ρ)		
	$T = 50$	$T = 100$	$T = 200$	$T = 50$	$T = 100$	$T = 200$
2	0.280	0.287	0.290	0.183	0.162	0.145
3	0.306	0.309	0.306	0.190	0.162	0.140

	Mean (λ)			RMSE (λ)		
	$T = 50$	$T = 100$	$T = 200$	$T = 50$	$T = 100$	$T = 200$
2	3.561	3.537	3.519	0.673	0.508	0.405
3	3.485	3.487	3.483	0.744	0.545	0.417

	Mean (ψ)			RMSE (ψ)		
	$T = 50$	$T = 100$	$T = 200$	$T = 50$	$T = 100$	$T = 200$
2	0.642	0.653	0.663	0.276	0.254	0.232
3	0.620	0.629	0.651	0.278	0.260	0.218

$T = 50$, is just $-100 \times 0.020/0.3 = -6.67\%$. Increasing the sample size to $T = 200$ halves the bias to $-100 \times 0.010/0.3 = -3.33\%$. A comparison of the root mean square error for alternative values of p in the auxiliary model suggests that for smaller samples of size $T = 50$ choosing a smaller lag length for the auxiliary model yields marginally more efficient parameter estimates. The opposite appears to be true for larger samples of size $T = 200$.

21.8 Exercises

(1) Finite Sample Properties of Qualitative Response Estimators

Gauss file(s)	<code>discrete_simulation.g</code>
Matlab file(s)	<code>discrete_simulation.m</code>

The model is

$$y_t^* = \beta_0 + \beta_1 x_t^* + u_t^*$$

$$u_t^* \sim N(0, \sigma^2),$$

where $x_t^* \sim N(0, 1)$ and the parameters are $\theta = \{\beta_0 = 1, \beta_1 = 1, \sigma^2 = 1\}$.

(a) Simulate 5000 replications of the model with sample size $T = 100$.

(i) Estimate the linear regression model where $y_t = y_t^*$ and $x_t = x_t^*$.

- (ii) Estimate the probit regression model where y_t is transformed as

$$\begin{aligned} y_t &= \begin{cases} 1 & : y_t^* > 0 \\ 0 & : y_t^* \leq 0 \end{cases} \\ x_t &= x_t^*. \end{aligned}$$

- (iii) Estimate the censored regression model where

$$\begin{aligned} y_t &= \begin{cases} y_t^* & : y_t^* > 0 \\ 0 & : y_t^* \leq 0 \end{cases} \\ x_t &= x_t^*. \end{aligned}$$

- (iv) Estimate the truncated regression model where

$$y_t = \begin{cases} y_t^* & : y_t^* > 0 \\ n.a & : y_t^* \leq 0 \end{cases}, \quad x_t = \begin{cases} x_t^* & : y_t^* > 0 \\ n.a & : y_t^* \leq 0. \end{cases}$$

- (v) Estimate the restricted regression model where $y_t > 0$ is regressed on a constant and the corresponding values of x_t .
- (vi) For each estimator compute the bias and root mean square error generated by Monte Carlo experiment and compare the results with those presented in Table 21.7.
- (b) Repeat part (a) for $T = 200, 500, 1000$ and discuss the asymptotic properties of the various estimation procedures and their ability to correct for alternative filtering procedures.

(2) A Probit Model of Monetary Policy

Gauss file(s)	<code>discrete_probit.g, usmoney.xls</code>
Matlab file(s)	<code>discrete_probit.m, usmoney.mat</code>

The data file contains 693 weekly observations for the U.S. starting in the first week of February 1984 and ending in the first week of June 1997 for the Federal funds rate, r_t , the spread between the Federal funds rate and the 6-month Treasury bill rate, s_t , the 1-month lagged inflation rate, π_{t-4} and the growth rate in output, gdp_t .

- (a) Using just FOMC meeting days, estimate a probit model of monetary policy where

$$y_t = \begin{cases} 1 & : \Delta r_t > 0 \quad [\text{Monetary tightening}] \\ 0 & : \Delta r_t \leq 0 \quad [\text{Monetary easing or no change}], \end{cases}$$

and $\{const, s_t, \pi_{t-1}, gdp_t\}$ are the explanatory variables. Compare the results with the estimates given in Table 21.5.

- (b) Interpret the parameter estimates.
- (c) Test the restriction that the parameters on the explanatory variables are all zero, with the exception of the intercept.

(3) An Ordered Probit Model of Monetary Policy

Gauss file(s)	discrete_ordered.g, usmoney.xls
Matlab file(s)	discrete_ordered.m, usmoney.mat

This exercise uses the same data as Exercise 2.

- (a) Using just FOMC meeting days, estimate an ordered probit model of monetary policy where

$$d_t = \begin{cases} 0 & : \Delta r_t = -0.50 & \text{[Extreme easing]} \\ 1 & : \Delta r_t = -0.25 & \text{[Easing]} \\ 2 & : \Delta r_t = 0.00 & \text{[No change]} \\ 3 & : \Delta r_t = 0.25 & \text{[Tightening]} \\ 4 & : \Delta r_t = 0.50 & \text{[Extreme tightening]}, \end{cases}$$

and $\{const, s_t, \pi_{t-4}, gdp_t\}$ are the explanatory variables. Compare the results with the estimates given in Table 21.6.

- (b) Interpret the parameter estimates.
- (c) Test the restriction that the parameters on the explanatory variables are all zero, with the exception of the intercept.

(4) Hamilton-Jorda Ordered Probit Model of Monetary Policy

Gauss file(s)	discrete_hamilton_jorda.g, usmoney.xls
Matlab file(s)	discrete_hamilton_jorda.m, usmoney.mat

This exercise uses weekly U.S. data from the first week in February 1984 to the last week of April 2001, being an extended version of the data used in Exercises 2 and 3.

- (a) Using just event days, estimate the ordered probit model in Exercise 3 using data from the first week in February 1984 to the first week of June 1997, where the explanatory variables are the magnitude of the last target change as of the previous week and the spread between the 6-month Treasury bill rate and the Federal funds rate. Compare the parameter estimates reported in Hamilton and Jordà (2002, Table 6).
- (b) Re-estimate the model in part (a), extending the sample period to April 2001. Check the robustness properties of the parameter estimates over the two sample periods.

(5) Simulating the Binomial Thinning Model

Gauss file(s)	<code>discrete_thinning.g</code>
Matlab file(s)	<code>discrete_thinning.m</code>

Consider the binomial thinning model

$$y_t = \rho \circ y_{t-1} + u_t,$$

where

$$\begin{aligned} \rho \circ y_{t-1} &= \sum_{s=1}^{y_{t-1}} e_{s,t-1} \\ e_{s,t-1} &= \begin{cases} 1 : & \text{Prob} = \rho \\ 0 : & \text{Prob} = 1 - \rho \end{cases} \\ u_t &\sim P(\lambda). \end{aligned}$$

- (a) Simulate the model for $t = 2, 3, 4, 5$, with parameters $\rho = 0.3$ and $\lambda = 3.5$, and initial value $y_1 = 5$.
- (b) Repeat part (a) with parameters $\rho = 0.8$ and $\lambda = 3$.
- (c) Repeat part (a) with parameters $\rho = 0.3$ and $\lambda = 3$.

(6) Finite Sample Properties of the Binomial Thinning Model

Gauss file(s)	<code>discrete_poissonauto.g</code>
Matlab file(s)	<code>discrete_poissonauto.m</code>

This exercise uses Monte Carlo experiments to investigate the finite sample properties of the maximum likelihood estimator of the binomial thinning model in Exercise 5, where the parameters are $\rho = 0.5$ and $\lambda = 3.0$. Simulate the model with 5000 replications and a sample of size $T = 100$.

- (a) Compute the maximum likelihood estimator and construct summary statistics of the finite sample distribution of the estimators. Compare the results with those reported in Table 21.7 as well as with the results in Jung, Ronning and Tremayne (2005, Table 1).
- (b) Repeat part (a) for the conditional least squares (CLS) estimator of Klimko and Nelson (1978).
- (c) Repeat parts (a) and (b) for the alternative parameterizations given in Jung, Ronning and Tremayne (2005).

(7) A Duration Model of Strikes

Gauss file(s)	<code>discrete_strike.g, strike.dat</code>
Matlab file(s)	<code>discrete_strike.m, strike.mat</code>

The data file contains 62 observations on the duration of strikes, y_t , expressed in days, in the U.S. from 1968 to 1976. Also given are annual data on unanticipated output, $output_t$. All data are based on Kennan (1985). Consider the Weibull model of durations

$$\ln y_t = \beta_0 + \beta_1 output_t + \sigma z_t$$

$$f(z_t) = \exp(z_t - e^{z_t}).$$

- (a) Estimate the parameters $\theta = \{\beta_0, \beta_1, \sigma^2\}$ by maximum likelihood. As starting values for β_0 and β_1 , use the least squares parameter estimates from regressing $\ln y_t$ on a constant and $output_t$ and for σ^2 choose 1.0.
- (b) Re-estimate the exponential model by imposing the restriction $\sigma^2 = 1$. Perform a LR test of this restriction.

(8) **An ACH Model of U.S. Airline Trades**

Gauss file(s)	<code>discrete_trade.g</code> , <code>amr_aug1.dat</code>
Matlab file(s)	<code>discrete_trade.m</code> , <code>amr_aug1.mat</code>

The data file contains trades per second on trades of the U.S. airline AMR on 1 August 2006, from 9.30am to 4.00pm, a total of $T = 23367$ observations.

- (a) Estimate the following ACH model

$$f(y; \theta) = h_t^y (1 - h_t^{1-y}), \quad y = 0, 1,$$

$$h_t = \frac{1}{1 + \exp(\psi_t + \delta_1 Firm_{t-1} + \delta_2 Macro_{t-1})}$$

$$\psi_t = \alpha_0 + \alpha_1 u_{t-1} + \beta_1 \psi_{t-1},$$

where

$$y_t = \begin{cases} 1 & : \text{Trade occurs in a second} \\ 0 & : \text{No trade occurs,} \end{cases}$$

u_t is the length of time between the last trades and the variables $Firm_{t-1}$ and $Macro_{t-1}$ are dummy variables representing, respectively, lagged news announcements of the firm and the macroeconomy. Compare the estimates with those reported in Table 21.8.

- (b) Compute the effect on the hazard rate h_t of the two news announcement variables.

(9) **EMM Estimator of Integer Models**

Gauss file(s)	<code>discrete_emm.g</code>
Matlab file(s)	<code>discrete_emm.m</code>

Consider the integer ARMA(1,1) specification

$$x_t = \rho \circ x_{t-1} + u_t$$

$$y_t = x_{t-1} + \psi \circ u_t,$$

where y_t is observable, x_t is unobservable, u_t is an *iid* Poisson random variables with parameters $\{\rho = 0.3, \lambda = 3.5, \psi = 0.7\}$.

- (a) Derive the finite sample properties of the EMM estimator for the integer AR(1) model by setting $\psi = 0$. Choose sample sizes of $T = \{50, 100, 200\}$ and 5,000 replications.
- (b) Repeat part (a) for the integer MA(1) model by setting $\rho = 0.3$.
- (c) Repeat part (a) for the integer ARMA(1,1) model by setting $\psi = 0.7$.

Appendix A

Change of Variable in Probability Density Functions

Let X be a continuous random variable with pdf $f(x)$. Define a new random variable Y by means of the relation $Y = g(X)$ where the g is a monotonic one-to-one mapping, so that its inverse function exists. The pdf of the continuous random variable Y , $h(y)$ is given by

$$h(y) = f(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right| = f(x) \left| \frac{dx}{dy} \right|,$$

where dx/dy is known as the Jacobian of the transformation.

This result generalizes to functions of more than one variable. Let $X_1, \dots, X_n$ be random variables whose joint pdf is given by $f(x_1, \dots, x_n)$. Once again define the functions g_i to be monotonic bijections so that

$$\begin{aligned} Y_i &= g_i(X_1, \dots, X_n) \\ X_i &= g_i^{-1}(Y_1, \dots, Y_n), \end{aligned}$$

and assume that the derivatives $\partial g_i^{-1}(x_1, \dots, x_n)/\partial y_j$ exist for all i, j .

The Jacobian of the transformation is now defined as the determinant of the matrix of partial derivatives

$$J = \begin{vmatrix} \frac{\partial g_1^{-1}(x_1, \dots, x_n)}{\partial y_1} & \dots & \frac{\partial g_1^{-1}(x_1, \dots, x_n)}{\partial y_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_n^{-1}(x_1, \dots, x_n)}{\partial y_1} & \dots & \frac{\partial g_n^{-1}(x_1, \dots, x_n)}{\partial y_n} \end{vmatrix} = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \dots & \frac{\partial x_1}{\partial y_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial y_1} & \dots & \frac{\partial x_n}{\partial y_n} \end{vmatrix}.$$

The joint pdf of $Y_1, \dots, Y_n$ is now given by

$$\begin{aligned} h(y_1, \dots, y_n) &= f(g_1^{-1}(y_1, \dots, y_n), \dots, g_n^{-1}(y_1, \dots, y_n)) |J| \\ &= f(x_1, \dots, x_n) |J|. \end{aligned}$$

Appendix B

The Lag Operator

B.1 Basics

The lag operator, L , takes the time series y_t and lags it once

$$Ly_t = y_{t-1} .$$

From this definition, it follows that

$$\begin{aligned} L^2 y_t &= L Ly_t = Ly_{t-1} = y_{t-2} \\ L^p y_t &= y_{t-p} \\ L^{-p} y_t &= y_{t+p} . \end{aligned}$$

It is also possible to define polynomials in the lag operator. Two polynomials in the lag operator that are often encountered are of the form

$$\begin{aligned} \phi(L) &= \phi_0 + \phi_1 L + \phi_2 L^2 + \phi_3 L^3 + \dots \\ \psi(L) &= 1 - \psi_1 L - \psi_2 L^2 - \psi_3 L^3 - \dots \end{aligned}$$

or, in the multivariate case,

$$\begin{aligned} \Phi(L) &= \Phi_0 + \Phi_1 L + \Phi_2 L^2 + \Phi_3 L^3 + \dots \\ \Psi(L) &= I - \Psi_1 L - \Psi_2 L^2 - \Psi_3 L^3 - \dots , \end{aligned}$$

where Φ_i and Ψ_i are now square matrices of parameters. Without loss of generality, this appendix presents the univariate case, but all the results extend naturally to the multivariate case.

B.2 Polynomial Convolution

Consider the two infinite polynomials in the lag operator

$$\begin{aligned}\phi(L) &= \phi_0 + \phi_1 L + \phi_2 L^2 + \phi_3 L^3 + \cdots \\ \psi(L) &= \psi_0 + \psi_1 L + \psi_2 L^2 + \psi_3 L^3 + \cdots\end{aligned}$$

The multiplication or convolution of the two polynomials gives

$$\varphi(L) = \phi(L)\psi(L)$$

where

$$\begin{aligned}\varphi_0 &= \phi_0 \psi_0 \\ \varphi_1 &= \phi_0 \psi_1 + \phi_1 \psi_0 \\ \varphi_2 &= \phi_0 \psi_2 + \phi_1 \psi_1 + \phi_2 \psi_0 \\ \varphi_3 &= \phi_0 \psi_3 + \phi_1 \psi_2 + \phi_2 \psi_1 + \phi_3 \psi_0 \\ &\vdots = \vdots \\ \varphi_n &= \phi_0 \psi_n + \phi_1 \psi_{n-1} + \phi_2 \psi_{n-2} + \cdots + \phi_n \psi_0 \\ &\vdots = \vdots\end{aligned}$$

Therefore, for the convolution of two infinite-order polynomials, the elements of the resultant polynomial $\varphi(L)$ are given by the simple rule

$$\varphi_k = \sum_{j=0}^k \phi_j \psi_{k-j}.$$

The situation is not quite as straightforward for the finite-order polynomials

$$\begin{aligned}\phi_p(L) &= \phi_0 + \phi_1 L + \phi_2 L^2 + \cdots + \phi_p L^p \\ \psi_q(L) &= \psi_0 + \psi_1 L + \psi_2 L^2 + \cdots + \psi_q L^q.\end{aligned}$$

Simple experimentation will show that that the resultant polynomial is of order $p + q + 1$

$$\phi_p(L)\psi_q(L) = \varphi_k(L) \quad k = (p + 1) + (q + 1) - 1 = p + q + 1$$

so that simple application of the rule for infinite-order polynomials be problematic if k is set greater than $p + q + 1$. In the case of a finite-order convolution, it turns out that the rule for generating the coefficients of $\varphi_k(L)$

is

$$\varphi_k = \sum_{j=\max(0; k-p)}^{\min(k; q+1)} \phi_j \psi_{k-j}.$$

B.3 Polynomial Inversion

In some cases, it is possible to invert polynomials in the lag operator. Consider the infinite series

$$1 + \lambda + \lambda^2 + \lambda^3 + \cdots = \frac{1}{1 - \lambda} \quad |\lambda| < 1.$$

This series suggests that if $|\phi L| < 1$ then

$$\frac{1}{1 - \phi L} = 1 + \phi L + \phi^2 L^2 + \phi^3 L^3 + \cdots.$$

Simple multiplication verifies this conjecture as

$$\begin{aligned} (1 - \phi L) \times (1 + \phi L + \phi^2 L^2 + \phi^3 L^3 + \cdots) \\ = (1 + \phi L + \phi^2 L^2 + \phi^3 L^3 + \cdots) - (\phi L + \phi^2 L^2 + \phi^3 L^3 + \phi^4 L^4 + \cdots) \\ = 1. \end{aligned}$$

The situation is more complex when trying to find the inverse of higher-order polynomials in the lag operator. Consider, for example, the second-order polynomial

$$1 - \phi_1 L - \phi_2 L^2.$$

Factorizing this expression gives

$$1 - \phi_1 L - \phi_2 L^2 = (1 - \gamma L)(1 - \rho L).$$

with

$$\gamma + \rho = \phi_1 \quad \gamma \rho = -\phi_2.$$

This expression may be inverted using the previous result for a first-order polynomial

$$(1 - \gamma L)^{-1} (1 - \rho L)^{-1} = (1 + \gamma + \gamma^2 L^2 + \gamma^3 L^3 + \cdots) (1 + \rho + \rho^2 L^2 + \rho^3 L^3 + \cdots).$$

The result is, therefore, a convolution of two polynomials in the lag operator of infinite order, and it is straightforward to evaluate this convolution using the results on polynomial convolution presented earlier.

B.4 Polynomial Decomposition

There is an important decomposition of a lag polynomial. Define the operator

$$a(L) = 1 - a_1L - a_2L^2 - a_3L^3 - \cdots - a_nL^n = 1 - \sum_{k=1}^n a_kL^k$$

and let $a(1) = 1 - \sum_{k=1}^n a_k$ (the sum of the coefficients). In this case,

$$a(L) - a(1) = \left(1 - \sum_{k=1}^n a_kL^k\right) - \left(1 - \sum_{k=1}^n a_k\right) = \sum_{k=1}^n a_k(1 - L^k).$$

It is a fact that $1 - x^k = (1 - x)(1 + x + x^2 + \cdots + x^{k-1})$ (just check it by multiplication), or,

$$1 - x^k = (1 - x) \sum_{r=0}^{k-1} x^r.$$

Therefore,

$$a(L) - a(1) = \sum_{k=1}^n a_k(1 - L) \left(\sum_{r=0}^{k-1} L^r \right) = (1 - L) \sum_{k=1}^n a_k \left(\sum_{r=0}^{k-1} L^r \right).$$

The task is now to re-index the summation on the right-hand side of the previous expression. The result is

$$a(L) - a(1) = (1 - L) \sum_{r=0}^{n-1} L^r \left(\sum_{k=r+1}^n a_k \right) = (1 - L) \sum_{r=0}^{n-1} b_r L^r,$$

where

$$b_r = \sum_{k=r+1}^n a_k.$$

Appendix C

FIML Estimation of a Structural Model

C.1 Log-likelihood Function

Consider the bivariate simultaneous equation model discussed in Chapter 5

$$\begin{aligned}y_{1,t} &= \beta y_{2,t} + u_{1,t} \\ y_{2,t} &= \gamma y_{1,t} + \alpha x_t + u_{2,t},\end{aligned}$$

where the disturbance term $u_t = (u_{1,t}, u_{2,t})'$ has a bivariate normal distribution with variance parameters σ_{11} and σ_{22} , respectively. The log-likelihood function is

$$\begin{aligned}\ln L_T(\theta) &= -\frac{N}{2} \ln(2\pi) - \frac{1}{2} \ln |\sigma_{11}\sigma_{22}| + \ln |1 - \beta\gamma| - \frac{1}{2\sigma_{11}T} \sum_{t=1}^T (y_{1,t} - \beta y_{2,t})^2 \\ &\quad - \frac{1}{2\sigma_{22}T} \sum_{t=1}^T (y_{2,t} - \gamma y_{1,t} - \alpha x_t)^2.\end{aligned}$$

The model imposes the restriction that the disturbances are not contemporaneously correlated, that is $\sigma_{12} = \sigma_{21} = 0$.

C.2 First-order Conditions

To derive the maximum likelihood estimators of the parameter vector $\theta = \{\beta, \gamma, \alpha, \sigma_{11}, \sigma_{22}\}$, set the first-order derivatives of the log-likelihood function

equal to zero to yield

$$\frac{\partial \ln L}{\partial \beta} = \frac{\hat{\gamma}}{1 - \hat{\beta}\hat{\gamma}} + \frac{1}{\hat{\sigma}_{11}T} \sum_{t=1}^T (y_{1,t} - \hat{\beta}y_{2,t})y_{2,t} = 0 \quad (\text{C.1})$$

$$\frac{\partial \ln L}{\partial \gamma} = -\frac{\hat{\beta}}{1 - \hat{\beta}\hat{\gamma}} + \frac{1}{\hat{\sigma}_{22}T} \sum_{t=1}^T (y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)y_{1,t} = 0 \quad (\text{C.2})$$

$$\frac{\partial \ln L}{\partial \alpha} = \frac{1}{\hat{\sigma}_{22}T} \sum_{t=1}^T (y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)x_t = 0 \quad (\text{C.3})$$

$$\frac{\partial \ln L}{\partial \sigma_{11}} = -\frac{1}{2\hat{\sigma}_{11}} + \frac{1}{2\hat{\sigma}_{11}^2T} \sum_{t=1}^T (y_{1,t} - \hat{\beta}y_{2,t})^2 = 0 \quad (\text{C.4})$$

$$\frac{\partial \ln L}{\partial \sigma_{22}} = -\frac{1}{2\hat{\sigma}_{22}} + \frac{1}{2\hat{\sigma}_{22}^2T} \sum_{t=1}^T (y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)^2 = 0. \quad (\text{C.5})$$

C.3 Solution

Solving these first-order conditions proceeds as follows. Self-evidently, the equations for $\partial \ln L / \partial \sigma_{11}$ and $\partial \ln L / \partial \sigma_{22}$ may be simplified directly to yield

$$\begin{aligned} \hat{\sigma}_{11} &= \frac{1}{T} \sum_{t=1}^T (y_{1,t} - \hat{\beta}y_{2,t})^2 \\ \hat{\sigma}_{22} &= \frac{1}{T} \sum_{t=1}^T (y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)^2. \end{aligned}$$

A number of useful results need to be derived before the solutions for $\hat{\beta}$, $\hat{\gamma}$ and $\hat{\alpha}$ become apparent. Rewrite the equation for $\partial \ln L / \partial \gamma$ as

$$-\frac{1}{\hat{\beta}^{-1} - \hat{\gamma}} + \sum_{t=1}^T \frac{(y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)y_{1,t}}{\hat{\sigma}_{22}T} = 0,$$

and multiply both sides by $(\hat{\beta}^{-1} - \hat{\gamma})$ to give

$$\begin{aligned} -1 + \sum_{t=1}^T \frac{(y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)y_{1,t}(\hat{\beta}^{-1} - \hat{\gamma})}{\hat{\sigma}_{22}T} &= 0 \\ -1 + \sum_{t=1}^T \frac{(y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)(y_{1,t}\hat{\beta}^{-1} - y_{1,t}\hat{\gamma})}{\hat{\sigma}_{22}T} &= 0. \end{aligned} \quad (\text{C.6})$$

Subtract equation (C.3) from equation (C.6) to yield

$$\begin{aligned}
-1 + \sum_{t=1}^T \frac{(y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)(y_{1,t}\hat{\beta}^{-1} - y_{1,t}\hat{\gamma})}{\hat{\sigma}_{22}T} - \sum_{t=1}^T \frac{(y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)\hat{\alpha}x_t}{\hat{\sigma}_{22}T} &= 0 \\
-1 + \sum_{t=1}^T \frac{(y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)(y_{1,t}\hat{\beta}^{-1} - y_{1,t}\hat{\gamma} - \hat{\alpha}x_t)}{\hat{\sigma}_{22}T} &= 0 \\
-1 + \sum_{t=1}^T \frac{(y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)(y_{2,t} - y_{1,t}\hat{\gamma} - \hat{\alpha}x_t - y_{2,t} + y_{1,t}\hat{\beta}^{-1})}{\hat{\sigma}_{22}T} &= 0 \\
-1 + \sum_{t=1}^T \frac{(y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)^2}{\hat{\sigma}_{22}T} + \sum_{t=1}^T \frac{(y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)(-y_{2,t} + y_{1,t}\hat{\beta}^{-1})}{\hat{\sigma}_{22}T} &= 0.
\end{aligned}$$

This expression may be simplified by substituting for $\hat{\sigma}_{22}$ and rearranging to yield

$$\sum_{t=1}^T (y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)(-y_{2,t} + y_{1,t}\hat{\beta}^{-1}) = 0,$$

or

$$\sum_{t=1}^T (y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)(y_{1,t} - \hat{\beta}y_{2,t}) = 0. \quad (\text{C.7})$$

Using the definition of the residuals reveals that equation (C.7) is equivalent to

$$\sum_{t=1}^T \hat{u}_{2,t}\hat{u}_{1,t} = 0,$$

which of course comes from the restriction of the model that the disturbances are contemporaneously independent $\sigma_{12} = 0$.

Now rewrite equation (C.1) as

$$-\frac{1}{\hat{\gamma}^{-1} - \hat{\beta}} + \sum_{t=1}^T \frac{(y_{1,t} - \hat{\beta}y_{2,t})y_{2,t}}{\hat{\sigma}_{11}T} = 0,$$

and multiply both sides by $(\hat{\gamma}^{-1} - \hat{\beta})$ to give

$$\begin{aligned} -1 + \sum_{t=1}^T \frac{(y_{1,t} - \hat{\beta}y_{2,t})(y_{2,t}\hat{\gamma}^{-1} - y_{2,t}\hat{\beta})}{\hat{\sigma}_{11}T} &= 0 \\ -1 + \sum_{t=1}^T \frac{(y_{1,t} - \hat{\beta}y_{2,t})(y_{1,t} - y_{2,t}\hat{\beta} - y_{1,t} + y_{2,t}\hat{\gamma}^{-1})}{\hat{\sigma}_{11}T} &= 0 \\ -1 + \sum_{t=1}^T \frac{(y_{1,t} - \hat{\beta}y_{2,t})^2}{\hat{\sigma}_{11}T} + \sum_{t=1}^T \frac{(y_{1,t} - \hat{\beta}y_{2,t})(-y_{1,t} + y_{2,t}\hat{\gamma}^{-1})}{\hat{\sigma}_{11}T} &= 0. \end{aligned}$$

This equation may be simplified by substituting for $\hat{\sigma}_{11}$ and rearranging to yield

$$\sum_{t=1}^T (y_{1,t} - \hat{\beta}y_{2,t})(-y_{1,t} + y_{2,t}\hat{\gamma}^{-1}) = 0,$$

or

$$\sum_{t=1}^T (y_{1,t} - \hat{\beta}y_{2,t})(y_{2,t} - y_{1,t}\hat{\gamma}) = 0. \quad (\text{C.8})$$

Now subtract (C.7) from (C.8) to give

$$\begin{aligned} \sum_{t=1}^T (y_{1,t} - \hat{\beta}y_{2,t}[(y_{2,t} - \hat{\gamma}y_{1,t}) - (y_{2,t} - \hat{\gamma}y_{1,t} - \hat{\alpha}x_t)]) &= 0 \\ -\hat{\alpha} \sum_{t=1}^T (y_{1,t} - \hat{\beta}y_{2,t})x_t &= 0, \end{aligned}$$

which yields the final preliminary result

$$\sum_{t=1}^T (y_{1,t} - \hat{\beta}y_{2,t})x_t = 0. \quad (\text{C.9})$$

The results in expressions (C.7)-(C.9) may now be used to solve for $\hat{\beta}$, $\hat{\gamma}$ and $\hat{\alpha}$ in the following sequence. Expression (C.9) is a function of $\hat{\beta}$ only and may be solved directly to give

$$\hat{\beta} = \frac{\sum_{t=1}^T y_{1,t}x_t}{\sum_{t=1}^T y_{2,t}x_t}.$$

This solution for $\hat{\beta}$ can then be used in (C.8) to solve for $\hat{\gamma}$

$$\hat{\gamma} = \frac{\sum_{t=1}^T y_{2,t}\hat{u}_{1,t} \sum_{t=1}^T x_t^2 - \sum_{t=1}^T x_t\hat{u}_{1,t} \sum_{t=1}^T y_{2,t}x_t}{\sum_{t=1}^T y_{1,t}\hat{u}_{1,t} \sum_{t=1}^T x_t^2 - \sum_{t=1}^T x_t\hat{u}_{1,t} \sum_{t=1}^T y_{1,t}x_t}.$$

Finally, given solutions for $\hat{\beta}$ and $\hat{\gamma}$, equation (C.7) can be solved for $\hat{\alpha}$ to give

$$\hat{\alpha} = \frac{\sum_{t=1}^T y_{1,t} \hat{u}_{1,t} \sum_{t=1}^T y_{2,t} x_t - \sum_{t=1}^T y_{1,t} x_t \sum_{t=1}^T y_{2,t} \hat{u}_{1,t}}{\sum_{t=1}^T y_{1,t} \hat{u}_{1,t} \sum_{t=1}^T x_t^2 - \sum_{t=1}^T x_t \hat{u}_{1,t} \sum_{t=1}^T y_{1,t} x_t}.$$

Appendix D

Additional Nonparametric Results

D.1 Mean

The kernel estimate of a density is given by

$$\hat{f}(y) = \frac{1}{Th} \sum_{t=1}^T K\left(\frac{y - y_t}{h}\right) = \frac{1}{T} \sum_{t=1}^T w_t, \quad w_t = \frac{1}{h} K\left(\frac{y - y_t}{h}\right),$$

where h is the kernel bandwidth and $K(\cdot)$ is the kernel function. Consider taking expectations of $\hat{f}(y)$ to derive its mean,

$$\begin{aligned} E[\hat{f}(y)] &= E\left[\frac{1}{T} \sum_{t=1}^T w_t\right] \\ &= \frac{1}{T} \sum_{t=1}^T E[w_t]. \end{aligned}$$

That is, the mean is derived by averaging out the randomness in the data. Because of the identically distributed assumption

$$E[w_1] = E[w_2] = \cdots = E[w_T],$$

the mean reduces to

$$\begin{aligned} E[\hat{f}(y)] &= E[w_t] \\ &= E\left[\frac{1}{h} K\left(\frac{y - y_t}{h}\right)\right] \\ &= \frac{1}{h} \int_{-\infty}^{\infty} K\left(\frac{y - y_t}{h}\right) f(y_t) dy_t, \end{aligned}$$

where w_t is taken as a “representative” random variable.

Now define the standardized random variable

$$z_t = \frac{y - y_t}{h}.$$

The mean is rewritten in terms of z_t (by using the change of variable technique)

$$\begin{aligned} E[\hat{f}(y)] &= \frac{1}{h} \int_{-\infty}^{\infty} K\left(\frac{y - y_t}{h}\right) f(y_t) \left| \frac{dy_t}{dz_t} \right| dz_t \\ &= \frac{1}{h} \int_{-\infty}^{\infty} K(z_t) f(y - h z_t) |-h| dz_t \\ &= \int_{-\infty}^{\infty} K(z_t) f(y - h z_t) dz_t. \end{aligned} \quad (\text{D.1})$$

This expression shows that the mean of the kernel density estimator at y is related to the unknown distribution, $f(y)$, in a nonlinear way. To highlight this relationship, consider expanding $f(y - h z_t)$ in a Taylor series expansion for small h

$$f(y - h z_t) = f(y) - \frac{h z_t}{1} f^{(1)}(y) + \frac{(h z_t)^2}{2} f^{(2)}(y) - \frac{(h z_t)^3}{6} f^{(3)}(y) + O(h^4),$$

where

$$f^{(k)}(y) = \frac{d^k f(y)}{dy^k},$$

represents the k^{th} derivative of $f(y)$. Substituting into the expression for $E[\hat{f}(y)]$ and rearranging gives

$$\begin{aligned} E[\hat{f}(y)] &= \int_{-\infty}^{\infty} K(z_t) \left(f(y) - h z_t f^{(1)}(y) + \frac{(h z_t)^2}{2} f^{(2)}(y) \right. \\ &\quad \left. - \frac{(h z_t)^3}{6} f^{(3)}(y) + O(h^4) \right) dz_t \\ &= f(y) \int_{-\infty}^{\infty} K(z_t) dz_t - h f^{(1)}(y) \int_{-\infty}^{\infty} z_t K(z_t) dz_t \\ &\quad + \frac{h^2}{2} f^{(2)}(y) \int_{-\infty}^{\infty} z_t^2 K(z_t) dz_t - \frac{h^3}{6} f^{(3)}(y) \int_{-\infty}^{\infty} z_t^3 K(z_t) dz_t + O(h^4) \\ &= f(y) - 0 + \frac{h^2}{2} f^{(2)}(y) \mu_2 - 0 + O(h^4) \\ &= f(y) + \frac{h^2}{2} f^{(2)}(y) \mu_2 + O(h^4), \end{aligned} \quad (\text{D.2})$$

by using the properties of the kernel functions (see Chapter 11) and particularly the property that higher odd-order moments are zero by virtue of the fact that the kernel is a symmetric function. This expression is an example of Jensen's inequality since it shows that the expectation of a nonlinear function is not equal to the nonlinear function of the expectation. The term $\frac{h^2}{2}f^{(2)}(y)\mu_2$ is simply a Jensen inequality correction of order $O(h^2)$ that controls the bias

$$\begin{aligned}\text{bias}(\hat{f}(y)) &= E[\hat{f}(y)] - f(y) \\ &= \frac{h^2}{2}f^{(2)}(y)\mu_2 + O(h^4).\end{aligned}\tag{D.3}$$

D.2 Variance

The variance of $\hat{f}(y)$ is given by

$$\begin{aligned}\text{var}(\hat{f}(y)) &= E[(\hat{f}(y) - E[\hat{f}(y)])^2] \\ &= E\left[\left(\frac{1}{T}\sum_{t=1}^T(w_t - E[w_t])\right)^2\right] \\ &= \frac{1}{T^2}E\left[\left(\sum_{t=1}^T(w_t - E[w_t])\right)^2\right] \\ &= \frac{1}{T^2}\left[\sum_{t=1}^T\text{var}(w_t) + 2\text{cov}(w_1, w_2) + 2\text{cov}(w_1, w_3) + \dots\right] \\ &= \frac{1}{T^2}\sum_{t=1}^T\text{var}(w_t),\end{aligned}$$

where the covariances are all zero because of the independence assumption. When the identically distributed assumption is invoked, the variance expression reduces to

$$\text{var}[\hat{f}(y)] = \frac{1}{T}\text{var}(w_t),$$

where, once again, w_t is taken as the representative random variable. Now, by definition

$$\begin{aligned}\text{var}(w_t) &= E[(w_t - E[w_t])^2] \\ &= E[w_t^2] - (E[w_t])^2.\end{aligned}$$

Consider

$$\begin{aligned} E[w_t^2] &= E \left[\left(\frac{1}{h} K \left(\frac{y - y_t}{h} \right) \right)^2 \right] \\ &= \frac{1}{h^2} E \left[\left(K \left(\frac{y - y_t}{h} \right) \right)^2 \right] \\ &= \frac{1}{h^2} \int_{-\infty}^{\infty} \left(K \left(\frac{y - y_t}{h} \right) \right)^2 f(y_t) dy_t. \end{aligned}$$

Rewriting in terms of the standardised random variable z_t , gives

$$\begin{aligned} E[w_t^2] &= \frac{1}{h^2} \int_{-\infty}^{\infty} \left(K \left(\frac{y - y_t}{h} \right) \right)^2 f(y_t) \left| \frac{dy_t}{dz_t} \right| dz_t \\ &= \frac{1}{h^2} \int_{-\infty}^{\infty} K^2(z_t) f(y - h z_t) | -h | dz_t \\ &= \frac{1}{h} \int_{-\infty}^{\infty} K^2(z_t) f(y - h z_t) dz_t. \end{aligned}$$

Substituting into the variance expression of $\text{var}(w_t)$ gives

$$\text{var}(w_t) = \frac{1}{h} \int_{-\infty}^{\infty} K^2(z_t) f(y - h z_t) dz_t - \left(\int_{-\infty}^{\infty} K(z_t) f(y - h z_t) dz_t \right)^2,$$

which, in turn, yields an expression for the variance of the kernel estimator

$$\text{var}(\hat{f}(y)) = \frac{1}{T} \left[\frac{1}{h} \int_{-\infty}^{\infty} K^2(z_t) f(y - h z_t) dz_t - \left(\int_{-\infty}^{\infty} K(z_t) f(y - h z_t) dz_t \right)^2 \right].$$

A Taylor series expansion of $f(y - h z_t)$ around a small value of h , gives

$$\begin{aligned} \text{var}(\hat{f}(y)) &= \frac{1}{Th} \int_{-\infty}^{\infty} K^2(z_t) \left(f(y) - h z_t f^{(1)}(y) + \frac{(h z_t)^2}{2} f^{(2)}(y) - \dots \right) dz_t \\ &\quad - \frac{1}{T} \left(\int_{-\infty}^{\infty} K(z_t) \left(f(y) - h z_t f^{(1)}(y) + \frac{(h z_t)^2}{2} f^{(2)}(y) - \dots \right) dz_t \right)^2 \\ &= \frac{1}{Th} f(y) \int_{-\infty}^{\infty} K^2(z_t) dz_t - 0 + \frac{1}{Th} \frac{h^2}{2} f^{(2)}(y) \int_{-\infty}^{\infty} z_t^2 K^2(z_t) dz_t - \dots \\ &\quad - \frac{1}{T} \left(f(y) \int_{-\infty}^{\infty} K(z_t) dz_t - 0 + \frac{h^2}{2} f^{(2)}(y) \int_{-\infty}^{\infty} z_t^2 K(z_t) dz_t - \dots \right)^2. \end{aligned}$$

Noting that when h is small, $h^2/Th = h/T$ is of a smaller order of magnitude than $(Th)^{-1}$, this expression is simplified to yield

$$\text{var}(\hat{f}(y)) = \frac{1}{Th} f(y) \int_{-\infty}^{\infty} K^2(z_t) dz_t + o\left(\frac{1}{Th}\right). \quad (\text{D.4})$$

This expression shows that the variance of the kernel estimator is inversely related to the bandwidth h .

D.3 Mean Square Error

$$\text{AMISE} = \int_{-\infty}^{\infty} \left(\text{bias}^2 + \text{var}(\hat{f}(y)) \right) dy. \quad (\text{D.5})$$

From (D.2), the aggregate squared bias is

$$\begin{aligned} \int_{-\infty}^{\infty} \text{bias}^2 dy &= \int_{-\infty}^{\infty} \left(\frac{h^2}{2} f^{(2)}(y) \mu_2 + O(h^4) \right)^2 dy \\ &= \int_{-\infty}^{\infty} \left(\frac{h^4}{4} (f^{(2)}(y))^2 \mu_2^2 + O(h^8) + h^2 f^{(2)}(y) \mu_2 O(h^4) \right) dy \\ &= \int_{-\infty}^{\infty} \left(\frac{h^4}{4} (f^{(2)}(y))^2 \mu_2^2 + O(h^6) \right) dy \\ &= \int_{-\infty}^{\infty} \left(\frac{h^4}{4} (f^{(2)}(y))^2 \mu_2^2 \right) dy + O(h^6) \\ &= \frac{h^4}{4} \mu_2^2 \int_{-\infty}^{\infty} (f^{(2)}(y))^2 dy + O(h^6) \\ &= \frac{h^4}{4} \mu_2^2 R(f^{(2)}(y)) + O(h^6), \end{aligned} \quad (\text{D.6})$$

where

$$R(f^{(2)}(y)) = \int_{-\infty}^{\infty} \left[f^{(2)}(y) \right]^2 dy,$$

is the roughness operator of the density function.

Similarly, from (D.4), the aggregate squared variance is

$$\begin{aligned} \int_{-\infty}^{\infty} \text{var}(\hat{f}(y)) dy &= \int_{-\infty}^{\infty} \left(\frac{1}{Th} f(y) \int_{-\infty}^{\infty} K^2(z_t) dz_t + o\left(\frac{1}{Th}\right) \right) dy \\ &= \frac{1}{Th} \int_{-\infty}^{\infty} f(y) \int_{-\infty}^{\infty} K^2(z_t) dz_t dy + o\left(\frac{1}{Th}\right) \\ &= \frac{1}{Th} \int_{-\infty}^{\infty} f(y) dy \int_{-\infty}^{\infty} K^2(z_t) dz_t + o\left(\frac{1}{Th}\right) \\ &= \frac{1}{Th} \int_{-\infty}^{\infty} K^2(z_t) dz_t + o\left(\frac{1}{Th}\right) \\ &= \frac{1}{Th} R(K) + o\left(\frac{1}{Th}\right) \end{aligned} \quad (\text{D.7})$$

as $\int_{-\infty}^{\infty} f(y)dy = 1$ from the property that it is a density and

$$R(K) = \int_{-\infty}^{\infty} K^2(z_t)dz_t,$$

is the roughness of the kernel density $K(z_t)$ (see below). Combining (D.6) and (D.7) in (D.5) yields the integrated mean squared error. Retaining terms up to order h gives the asymptotic equivalent

$$\text{AMISE} = \frac{h^4}{4}\mu_2^2 R(f^{(2)}(y)) + \frac{1}{Th}R(K). \quad (\text{D.8})$$

D.4 Roughness

Many of the properties of the kernel density estimator and the algorithms designed for the optimal selection of the bandwidth rely on results relating to the roughness operator R , defined by

$$R(g) = \int_{-\infty}^{\infty} g^2(y) dy.$$

D.4.1 Roughness Results for the Gaussian Distribution

If $f(y; \mu, \sigma^2)$ is the probability density function of the Gaussian distribution then

$$\begin{aligned} R(f^{(2)}(y)) &= \int_{-\infty}^{\infty} [f^{(2)}(y)]^2 dy \\ &= \int_{-\infty}^{\infty} \left[\frac{d^2}{dy^2} \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(y-\mu)^2}{2\sigma^2} \right) \right) \right]^2 dy \\ &= \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi\sigma^2}} \left(\frac{y-\mu}{\sigma^2} \right)^2 \exp \left(-\frac{(y-\mu)^2}{2\sigma^2} \right) \right]^2 dy \\ &= \frac{1}{2\pi\sigma^6} \int_{-\infty}^{\infty} \left(\frac{y-\mu}{\sigma} \right)^4 \exp \left(-\frac{(y-\mu)^2}{\sigma^2} \right) dy. \end{aligned}$$

This expression is simplified (by using the change of variable technique) with the transformation $u/\sqrt{2} = (y - \mu)/\sigma$

$$\begin{aligned}
 R(f^{(2)}(y)) &= \frac{1}{2\pi\sigma^6} \int_{-\infty}^{\infty} \left(\frac{y-\mu}{\sigma}\right)^4 \exp\left(-\frac{(y-\mu)^2}{\sigma^2}\right) \left|\frac{dy}{du}\right| du \\
 &= \frac{1}{2\pi\sigma^6} \int_{-\infty}^{\infty} \left(\frac{u}{\sqrt{2}}\right)^4 \exp\left(-\frac{u^2}{2}\right) \left|\frac{\sigma}{\sqrt{2}}\right| du \\
 &= \frac{1}{\sqrt{\pi}8\sigma^5} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} u^4 \exp\left(-\frac{u^2}{2}\right) du \\
 &= \frac{1}{\sqrt{\pi}8\sigma^5} E[u^4] \\
 &= \frac{3}{\sqrt{\pi}8\sigma^5}, \tag{D.9}
 \end{aligned}$$

where the result that $E[u^4] = 3$ for a standardised normal random variable is used. In addition, it may be shown that

$$R(f^{(1)}) = \frac{1}{4\sigma^3\sqrt{\pi}}, \quad R(f^{(2)}) = \frac{3}{8\sigma^5\sqrt{\pi}}, \quad R(f^{(3)}) = \frac{15}{16\sigma^7\sqrt{\pi}}, \quad R(f^{(4)}) = \frac{105}{32\sigma^9\sqrt{\pi}}.$$

D.4.2 Roughness Results for the Gaussian Kernel

The roughness of a Gaussian kernel is

$$\begin{aligned}
 R(K) &= \int_{-\infty}^{\infty} K^2(z_t) dz_t \\
 &= \int_{-\infty}^{\infty} \left(\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z_t^2}{2}\right)\right)^2 dz_t \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp(-z_t^2) dz_t \\
 &= \frac{1}{\pi} \int_0^{\infty} \exp(-z_t^2) dz_t.
 \end{aligned}$$

This expression is simplified by using the change of variable technique with the transformation $u = z_t^2$, to give

$$\begin{aligned}
 R(K) &= \frac{1}{\pi} \int_0^\infty \exp(-u) \left| \frac{dz_t}{du} \right| du \\
 &= \frac{1}{\pi} \int_0^\infty \exp(-u) \left| \frac{1}{2u^{1/2}} \right| du \\
 &= \frac{1}{2\pi} \int_0^\infty u^{-1/2} \exp(-u) du \\
 &= \frac{\Gamma(1/2)}{2\pi} \\
 &= \frac{\sqrt{\pi}}{2\pi} \\
 &= \frac{1}{2\sqrt{\pi}}.
 \end{aligned} \tag{D.10}$$

In addition,

$$R(K^{(1)}) = \frac{1}{4\sqrt{\pi}}, \quad R(K^{(2)}) = \frac{3}{8\sqrt{\pi}}, \quad R(K^{(3)}) = \frac{15}{16\sqrt{\pi}}.$$

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